Prepladder Version X Qbank

Subject: Microbiology

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Scientists, Stains, and Microscopes

1. A 8-year-old boy presents with a 5-day history of severe sore throat, high fever, hoarseness of voice, and swollen neck. Physical examination reveals a thick greyish pseudomembrane in the throat and significant cervical lymphadenopathy. A throat swab was sent for microbiological analysis and revealed the following. Which staining technique is used to identify the causative organism in this case?



A. Albert stain

B. Gram stain

C. Zeil Neelsen stain

D. Gabbet stain

2. Which of the following statements is true regarding the differences between a light and an electron microscope?

A. Light microscope has a magnification of 10,000x, while that of an electron microscope is 1,00,000x.

B. The resolution of a light microscope is 0.2 μm and that of a transmission electron microscope is 0.2 nm.

C. Light microscope uses a single lens while an electron microscope uses an electromagnet.

D. Both operate in a vacuum medium.

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3. Which of the following statements is true regarding an electron microscope?
TEM- Transmission Electron Microscopy SEM- Scanning Electron Microscopy

A. Thick sections are to be used to study the internal structures of an organism in a TEM.

B. A fixed magnification of 1,00,000x can be obtained using an SEM.

C. The whole system of an electron microscope functions in a vacuum.

D. TEM has a resolution of 0.2 μm.

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4. Which of the following is a common application of phase contrast microscopy?

A. Visualizing fluorescently labelled proteins in fixed cells.

B. Studying the ultrastructure of cellular components at high resolution.

C. Observing live cells and microorganisms without the need for staining.

D. To detect cell surface antigens or antibodies bound to cell surface antigens

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5. Which of the following best describes the process of fluorochroming in fluorescence microscopy?

A. Using brightfield illumination to visualize unstained samples

B. Staining cells with dyes that absorb and re-emit light at specific wavelengths

C. Applying heavy metals to samples for better contrast

D. Enhancing image contrast by manipulating light paths with prisms and polarizers

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6. Which of the following statements accurately describes the visualization of organelles using interference contrast microscopy?

A. It requires the use of fluorescent dyes to enhance contrast.

B. It produces high-contrast images by manipulating light paths using polarizers, analyzers, and prisms.

C. It is less effective than brightfield microscopy for observing live cells.

D. It provides better contrast for transparent samples than phase contrast microscopy but is invasive to the sample.

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7. Which microscope among the following is best suited for observing microbial motility?

A. Scanning electron microscopy

B. Darkfield microscopy

C. Fluorescence microscopy

D. Compound light microscopy

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8. Which part of a light microscope plays a crucial role in determining the resolution of the microscope and is defined by the product of the refractive index of the medium between the lens and the specimen and the sine of half the angle of the cone of light entering the objective?

A. Objective lens

B. Condenser

C. Eyepiece lens

D. Stage

9. Which of the following statements is false regarding a light microscope?

A. Magnification is 10-1000 times.

B. The limit of resolution for a light microscope is about 0.2 μm.

C. Contrast is only achieved through staining.

D. All of the above

10. During the microscopic examination of a sample from a patient suspected of having tetanus, the microbiologist uses a staining method to visualize bacterial spores. Which stain is primarily used to stain spores in this method?

A. Leifson Ryu Stain

B. Malachite Green

C. Silver Stain

D. Crystal violet

11. In the 19th century, a scientist conducted experiments to develop a vaccine against rabies. He used a weakened form of the rabies virus to inoculate animals and successfully protect them from the disease. Name the scientist who laid the foundation for modern immunology and vaccination.

A. Robert Koch

B. Joseph Lister

C. Louis Pasteur

D. Edward Jenner

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12. A 50-year-old patient presents with a 6-week history of chronic cough, weight loss, night sweats, and low-grade fever. Physical examination and imaging reveal right upper lobe consolidation with cavitary lesions. Which staining technique would help identify the bacilli in the patient's sputum sample?

A. Gram stain

B. Albert stain

C. Ziehl Neelsen stain

D. Neisser stain

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13. A microbiology resident is teaching medical students about Gram staining. She prepares slides with pure cultures of Escherichia coli and Staphylococcus aureus. After completing the Gram staining procedure, which of the following observations would correctly differentiate these two organisms?

A. E. coli appears purple, S. aureus appears pink

B. E. coli appears pink, S. aureus appears purple

C. Both organisms appear purple

D. Both organisms appear pink

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14. A 45-year-old man presents with a painless genital ulcer. Dark-field microscopy is negative for Treponema pallidum. The clinician suspects a non-syphilitic treponematosis and orders a special stain to visualize the spirochetes. Which of the following is an impregnation stain that can be used to demonstrate these organisms?

A. Gram stain

B. Giemsa stain

C. Fontana-Masson stain

D. Ziehl-Neelsen stain

15. A 35-year-old HIV-positive man presents with headache, fever, and altered mental status. India ink preparation of CSF shows rounded yeast cells with a clear halo. Which of the following stains can be used as a negative stain to visualize the capsule of the microorganism responsible for symptoms in this patient?

A. Gram stain

B. Ziehl-Neelsen stain

C. Nigrosin stain

D. Giemsa stain

16. A 34-year-old presented to the OPD with complaints of a productive cough with blood in sputum and fever for 2 weeks and also has a history of loss of weight and decreased appetite. Chest X-ray shows apical nodules and sputum test was advised. Identify the causative organism, along with the fixation and staining method used.

A. Mycobacterium Tuberculosis, Gram stain with heat fix fixation

B. Corynebacterium diphtheriae, Albert stain with methanol fixation

C. Streptococcus Pneumoniae, Acid fast stain with methanol fixation

D. Mycobacterium Tuberculosis, Acid fast stain with Heat fixation

17. A genetic researcher is studying CRISPR cas9 technology. He came across the scientists who were awarded the Nobel prize for their work in CRISPR cas9 in COVID, identifying the scientists.

A. Emmanuelle Charpentier and Jennifer A Doudna

B. Luc Montangnier and F Barre-Sinoussi

C. Stanley B Prusiner

D. J Michael Bishop and Harold E Varmus

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18. A medical historian explaining infection control in surgical procedures and the decrease in mortality and morbidity due to surgical sepsis comes across the scientist who explained the importance of sterilizing surgical tools. Name the scientist who introduced the importance of sterilizing surgical tools.

A. Paul Ehrlich

B. Ernst Ruska

C. Edward Jenner

D. Joseph Lister

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19. A 50-year-old patient presents with a chronic cough and night sweats. A sputum sample was collected and sent to the laboratory for testing. Acid-fast staining reveals red-coloured bacilli consistent with Mycobacterium tuberculosis. Who is credited with the original development of the acid-fast stain technique?

A. Paul Ehrlich

B. Robert Koch

C. Louis Pasteur

D. Ziehl-Neelsen

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20. A researcher suspects that a novel virus is causing a respiratory illness outbreak in a community. They collected samples from several affected individuals and isolated viruses from these samples. The virus was successfully cultured in a laboratory setting. The researcher then introduced the cultured virus into healthy laboratory animals. However, the animals do not show any signs of respiratory illness. What does this result indicate about fulfilling Koch's postulates?

A. Koch's postulates have been fulfilled because the virus was isolated and cultured successfully

B. Koch's postulates were not fulfilled because the virus did not cause disease in healthy laboratory animals.

C. Koch's postulates can be fulfilled if the researcher tried a higher dose of the virus in the animals.

D. Koch's postulates can be fulfilled if the researcher observes the virus under an electron microscope.

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	3
Question 4	3
Question 5	2
Question 6	2
Question 7	2
Question 8	1
Question 9	3
Question 10	2
Question 11	3
Question 12	3
Question 13	2
Question 14	3
Question 15	3
Question 16	4
Question 17	1
Question 18	4

Question 19	1
Question 20	2

Solution for Question 1:

Correct Answer: A) Albert stain

Explanation:

The above image shows Albert stain of corynebacterium diphtheria showing metachromatic granules.

Albert Stain

Principle:

Albert's stain causes the metachromatic granules of Corynebacterium diphtheriae to appear bluish-purple upon staining.

Type	Components	Function
Albert Solution 1	Malachite Green for organism staining Toluidine Blue for metachromatic staining	Enhances contrast and specificity in staining.
Albert Solution 2	Iodine	Specifically targets volutin or Babes-Ernst granules.

Procedure:

- Stain the smear with Albert I stain, then drain and wash it
- Apply Albert II stain to cover the smear, then drain it

Application:

- Corynebacterium diphtheriae are slender bacilli arranged in V or L shapes, characterized by metachromatic granules (polar bodies, volutin, or Babes-Ernst granules) at the poles of the bacilli
- Albert's, Neisser's, and Ponder's stains help stain these distinctive granules in the bacteria
- These granules are also Found in Spirillum, Gardnerella, Yersinia pestis, yeast, and Mycobacterium tuberculosis (MTB)

Gram stain (Option B) : It helps identify rod-shaped Corynebacterium diphtheriae but does not help stain the metachromatic granules as stated in the scenario.

Zeil Neelsen stain (Option C) : It cannot be used in this case as it only stains acid-fast bacilli.

Gabbet stain (Option D) : It is also known as Cold Zeil Neelsen stain, cannot be used in this case as it only stains acid-fast bacilli.

Solution for Question 2:

Correct Answer: B) The resolution of a light microscope is 0.2 µm and that of a transmission electron microscope is 0.2 nm.

Explanation:

The resolution of-

- A light microscope is 0.2 µm.
- A transmission electron microscope is 0.2 nm.
- A scanning electron microscope is 0.5 to 4 nm.

Light Microscope:	Electron Microscope:	
Light from the sources passes through the specimen after being focused by the condenser, and the lenses collect it.	Electron microscopes use electrons to visualise the specimens.	
It is set up in an air medium.	It is set up in a vacuum medium.	(Option D ruled out)
Types: bright-field, phase-contrast, differential interference contrast, dark-field, and fluorescence.	Types- TEM- Transmission Electron Microscope and SEM- Transmission Electron Microscopy	
A compound light microscope contains two types of lenses- objective and ocular- that combine to magnify the image.	Electromagnets function as lenses.	(Option C ruled out)

Light Microscope:	Electron Microscope:	
Magnification : Ocular lenses- 10 to 30x; Objective lenses- 10 to 100x. The product of the two gives the total magnification (1000x.)	Magnification: 15x, up to about 1,00,000x with an SEM and up to 50 million times with a TEM.	(Option A ruled out)
Resolution: 0.2 µm.	Resolution: The resolving power of TEM is 0.2 nm and SEM ranges between 0.5 and 4 nm.	(Option B)
Structures are visualised using contrast. Staining, phase contrast, differential interference contrast, dark field, and fluorescence improve contrast.	TEM- osmic acid, permanganate, uranium, lanthanum, or lead salts are used for staining. SEM- the specimen is coated with a thin film of a heavy metal like gold.	

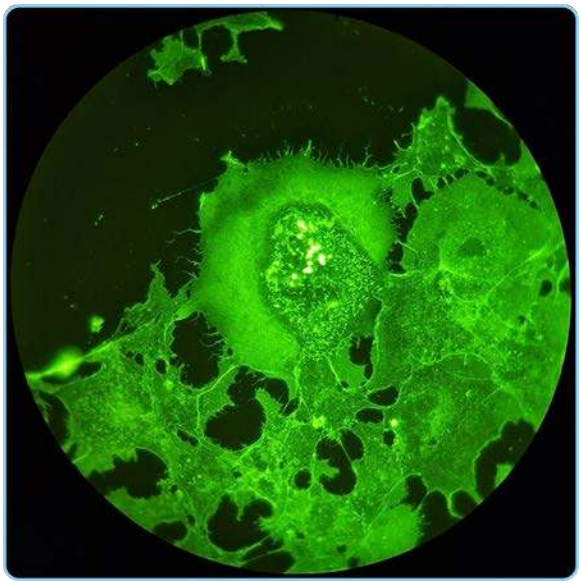
Structures are visualised using contrast in Light microscopy:

Phase contrast

Differential interference contrast



Dark-field



Fluorescence improve contrast

Solution for Question 3:

Correct Answer: C) The whole system of an electron microscope functions in a vacuum.

Explanation:

In the electron microscope, electromagnets function as lenses, and the whole system operates in a vacuum.

- Electron microscopes use electrons to visualise cells or organisms, unlike a light microscope that uses visible light
- In the electron microscope, electromagnets function as lenses, and the whole system operates in a vacuum (Option C)
- Electron microscopes have cameras that capture an electron micrograph
- Two types of electron microscopy are used in microbiology:

Transmission Electron Microscopy	Scanning Electron Microscopy
Cells and cell structures are visualised at very high magnification and resolution The resolving power of a TEM is much greater than that of a light microscope. Therefore, one can view structures at the molecular level. Uses a wavelength much shorter than visible light The resolving power is 0.2 nm (light microscope = 0.2 µm.) (Option D ruled out) As the penetrating power of an electron is very low, thin sections are needed, unless only external structures of an organism are to be studied. (Option A ruled out) Sections should be stabilised and stained to be examined Stains used - osmic acid, permanganate, uranium, lanthanum, or lead salts Intact cells or cell components can be observed directly in the TEM by a technique called negative staining	Used for optimal three-dimensional examination of cells Magnification ranges from as low as 15x, up to about 1,00,000x, but only the object's surface can be visualized (Option B ruled out) The specimen is coated with a thin film of a heavy metal like gold Electrons are scattered from the metal coating. They are collected and projected on a monitor, producing an image Large specimens can be studied with an excellent focus

Solution for Question 4:

Correct Answer: C) Observing live cells and microorganisms without the need for staining.

Explanation:

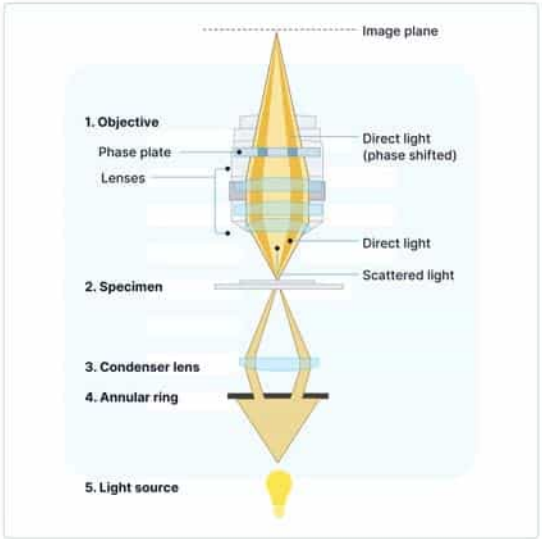
- Phase contrast microscopy is widely used to observe live cells and microorganisms in their natural state without the need for staining
- This technique enhances the contrast of transparent and colourless specimens, by converting phase shifts in the light passing through the specimen into differences in light intensity and allowing the proper study of cellular morphology, behaviour, and dynamic processes such as motility and cell division
- Principle : It is achieved using a phase plate in the objective lens and an annular diaphragm in the condenser. As light waves go through different parts of the specimen, they experience different refractive indices and optical path lengths, causing phase shifts, which are then transformed into variations in brightness, allowing detailed visualization of structures such as organelles in live cells.
- It is achieved using a phase plate in the objective lens and an annular diaphragm in the condenser.
- As light waves go through different parts of the specimen, they experience different refractive indices and optical path lengths, causing phase shifts, which are then transformed into variations in brightness, allowing detailed visualization of structures such as organelles

in live cells.

• It is achieved using a phase plate in the objective lens and an annular diaphragm in the condenser.

• As light waves go through different parts of the specimen, they experience different refractive indices and optical path lengths, causing phase shifts, which are then transformed into variations in brightness, allowing detailed visualization of structures such as organelles in live cells.





Visualizing fluorescently labelled proteins in fixed cells (Option A) : It describes an application of fluorescence microscopy.

Studying the ultrastructure of cellular components at high resolution (Option B) : It is related to electron microscopy, which provides high-resolution images of cellular ultrastructure.

Detecting cell surface antigens or antibodies bound to cell surface antigens (Option D) refers to immunofluorescence.

Reference:

Solution for Question 5:

Correct Answer: B) Staining cells with dyes that absorb and re-emit light at specific wavelengths.

Explanation:

- Fluorochroming is the process of staining cells or tissues with fluorescent dyes (fluorochromes) that absorb light at a specific wavelength and then emit light at a longer wavelength.
- The re-emitted light is used to create high-contrast images of specific cellular components under a fluorescence microscope.
- A fluorescent microscope uses light of a high-intensity source which excites a fluorescent agent, which in turn emits a low-energy light of a longer wavelength that produces the image.

Fluorochromes and their uses in fluorescence microscopy :

- FITC (Fluorescein Isothiocyanate): Labels antibodies for detecting specific antigens.
- Auramine O : Stains mycolic acid in Mycobacterium tuberculosis (acid-fast bacilli).
- Rhodamine B : Stains mycolic acid in Mycobacterium tuberculosis and other mycobacteria.
- Calcofluor White : Stains chitin in fungal cell walls.
- Acridine Orange : Stains DNA (dsDNA in green, ssDNA/RNA in orange/red); used in microhaematocrit tubes for detecting parasites.

Using brightfield illumination to visualize unstained samples (Option A) : Brightfield illumination is a different technique used for visualizing stained or unstained samples without the use of fluorescence.

Applying heavy metals to samples for better contrast (Option C) : Heavy metals are used in electron microscopy to enhance contrast but are not related to fluorescence microscopy.

Enhancing image contrast by manipulating light paths with prisms and polarizers (Option D) : It describes the principle of interference contrast microscopy , not fluorochroming.

Solution for Question 6:

Correct Answer: B) It produces high-contrast images by manipulating light paths using polarizers, analyzers, and prisms.

Explanation:

Interference contrast microscopy, particularly Differential Interference Contrast (DIC) microscopy, enhances the contrast of unstained, transparent samples by using optical components such as polarizers, analyzers, and prisms.

Interference contrast microscope:

- Polarizers, analyzers, and prisms manipulate the light paths to create high-contrast images with a pseudo-three-dimensional effect , which is particularly useful for visualizing fine cellular structures and organelles
- This technique is non-invasive , preserving the natural state of the sample, and it is more effective for live cell imaging compared to brightfield microscopy (Options C & D ruled out)
- It provides better contrast and detail than phase contrast microscopy without requiring fluorescent dyes (Option A ruled out)

Solution for Question 7:

Correct Answer: B) Dark field microscopy

Explanation:

Dark field microscopy is particularly effective for observing microbial motility because it enhances the contrast of transparent or unstained specimens against a dark background.

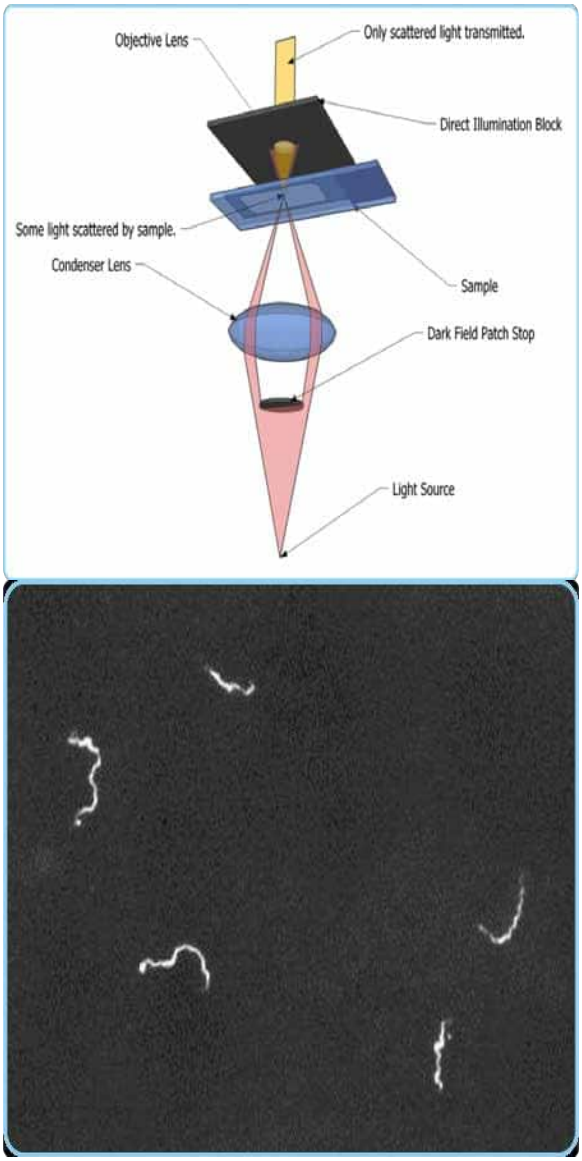
Dark Field microscopy:

• In dark field microscopy, illumination avoids passing directly through the specimen. Instead, light is angled from the sides, allowing only scattered light to reach the lens.

• This results in the specimen appearing bright against a dark background

• Darkfield microscopy frequently offers superior resolution compared to traditional light microscopy and can reveal details that are not observable through bright field or phase contrast techniques

• It is especially effective for observing microbial motility, such as the discernible bundles of flagella responsible for swimming motion (Option B)



Scanning electron microscopy (Option A) : It provides high-resolution, three-dimensional surface images of specimens but requires samples to be fixed and coated, making it unsuitable for

observing live microbial motility.

Fluorescence microscopy (Option C) :

- It involves staining or tagging specific parts of the cell with fluorescent dyes.
- It is commonly used to observe specific cellular structures or molecules rather than live motility in unstained cells, as the dyes may interfere with cell movement.



Compound light microscopy (Option D) :

- It is the standard microscopy method using visible light.
- However, it provides limited contrast without staining, making it difficult to observe unstained live cells and their motility.



Solution for Question 8:

Correct Answer: A) Objective lens

Explanation:

- The numerical aperture (NA) of the objective lens is a critical factor in determining the resolution of a light microscope.
- It is defined by the refractive index of the medium between the lens and the specimen (usually air or oil) and the sine of half the angle of the cone of light entering the objective
- A higher numerical aperture allows the objective to gather more light and resolve finer details of the specimen.

Part of the Light Microscope	Function
Source of Illumination	Visible Light
Stage (Option D ruled out)	Slides are placed on it
Condenser (Option B ruled out)	Regulates the passage of light
Big Knob	Coarse adjustment
Small Knob	Fine adjustment

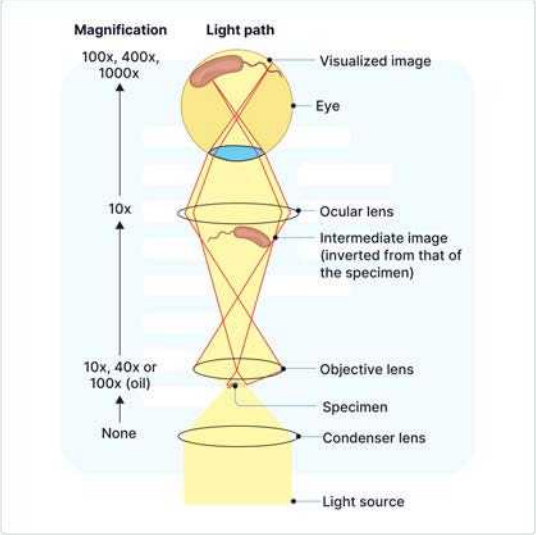
Part of the Light Microscope	Function
Eyepiece Lens (Option C ruled out)	10x magnification
Objective Lens	Scanner: 4x Low Power: 10x High Power: 40x Oil Immersion: 100x

• The resolution limit of a light microscope depends on both the wavelength of light employed and the light-collecting capacity of the objective lens, which is termed its numerical aperture.

• Objective lenses of greater magnification generally exhibit higher numerical apertures.

• The size of the smallest discernible object through any lens equals 0.5*λ divided by the numerical aperture, where λ represents the wavelength of light used.





Solution for Question 9:

Correct Answer: C) Contrast is only achieved through staining.

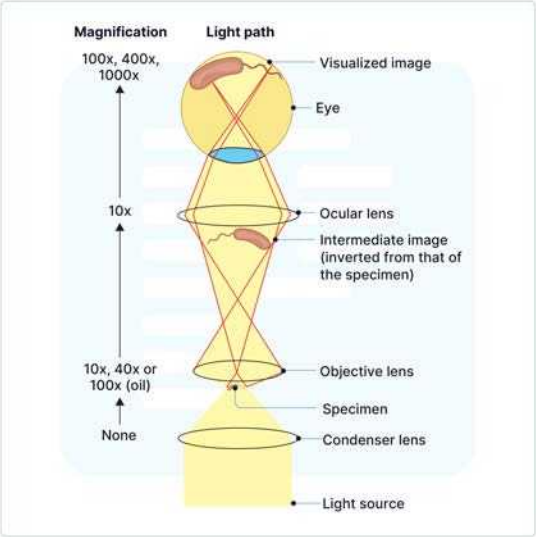
Explanation:

In addition to staining, other methods of light microscopy have been developed to improve contrast, such as phase contrast, differential interference contrast, and dark field microscopy.

Features of a Microscope:

Magnification	Resolution	Contrast
Magnification describes the capacity to enlarge an image. All microscopes employ lenses that improve magnification. However, the limiting factor for distinguishing is not magnification. Maximum magnification is 1000x. (Option A)	Resolution is the ability to distinguish two adjacent objects as distinct and separate. The limit of resolution for a light microscope is nearly 0.2 µm. (Option B) Two objects near less than 0.2 µm from each other cannot be resolved as distinct and separate.	Contrast is necessary to distinguish microorganisms from their surroundings. Bright-field microscopy uses staining to visualise bacteria with a good contrasting effect. Advances have been made in microscopy and new contrast methods, such as phase contrast, differential interference contrast, dark field, and fluorescence. (Option C)





Solution for Question 10:

Correct Answer: B) Malachite Green

Explanation:

Special techniques for endospore staining include the Schaeffer–Fulton and Moeller stains . Malachite Green is the primary stain in these methods , as it penetrates the harsh spore coat to stain the spores specifically.

Spore stain- Modified Ashby / Schaeffer-Fulton:

- Principle : The aim is to differentiate bacterial spores, which are highly resistant structures, from vegetative cells. This method involves staining with a primary agent that selectively binds to the spore's tough outer coat
- The aim is to differentiate bacterial spores, which are highly resistant structures, from vegetative cells.
- This method involves staining with a primary agent that selectively binds to the spore's tough outer coat
- Primary agent : Malachite green - penetrates the spore’s resistant coat and helps stain bacterial spores
- Mordant : Heat
- Decolorizer : water
- Counter stain : safranin red
- The aim is to differentiate bacterial spores, which are highly resistant structures, from vegetative cells.
- This method involves staining with a primary agent that selectively binds to the spore's tough outer coat

The Moeller stain is another well-established method for visualizing bacterial spores.

- It has very similar steps to Schaeffer-Fulton but can be considered a variant.

Flagella stain-Leifson and Ryu stain (Option A ruled out)

- Used to stain the flagella
- Dyes used : Basic Fuchsin for flagella (Red) and Methylene blue for cells (blue)
- Mordant: Tannic acid
- Silver stains are also used to stain the flagella in the impregnation method (Option C ruled out).

Crystal violet (Option D) is mainly used in gram staining and does not stain the bacterial spores.

Solution for Question 11:

Correct Answer: C) Louis Pasteur

Explanation:

- Louis Pasteur, also known as the father of microbiology, laid the foundation for modern immunology and vaccination.
- He developed the first rabies vaccine.



His contributions to microbiology are as follows:

- Proposed the principles of fermentation for the preservation of food
- Introduced the sterilization techniques and developed steam sterilizer, hot air oven and autoclave
- Described the method of pasteurization of milk
- Contributed for the vaccine development against several diseases, such as anthrax, fowl cholera and rabies
- He disproved the theory of spontaneous generation of disease and postulated the ‘germ theory of disease’. He stated that disease cannot be caused by bad air or vapour, but it is produced by the microorganisms present in the air
- Liquid media concept: He used the nutrient broth to grow microorganisms
- He was the founder of the Pasteur Institute, Paris.

Robert Koch (Option A) :

- Developed the Koch’s postulates , establishing the link between specific microorganisms and specific diseases

- Discovered the causative agents of tuberculosis, cholera, and anthrax .
-

Joseph Lister (Option B) :

- Father of Modern Surgery (Also Father of modern surgery)
- Pioneered aseptic surgical techniques by promoting the use of carbolic acid (phenol) to disinfect wounds, greatly reducing infections in surgery.

Edward Jenner (Option D) :

- Father of Immunology & Father of Vaccination
- Developed the first smallpox vaccine using material from cowpox lesions, laying the foundation for modern immunology and vaccination.

Solution for Question 12:

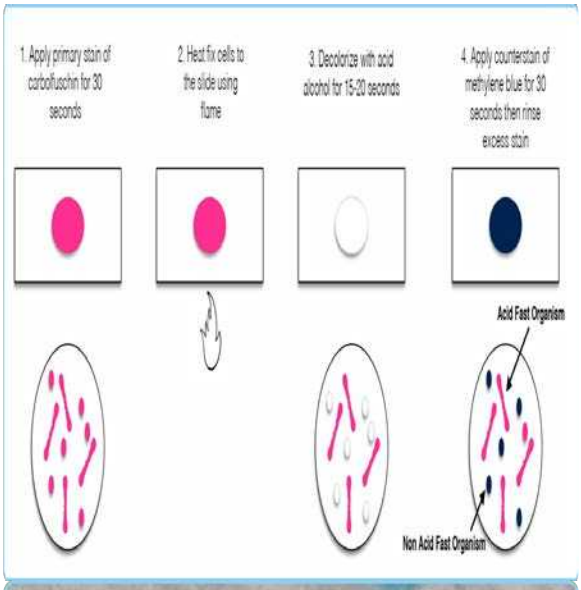
Correct Answer: C) Ziehl Neelsen stain

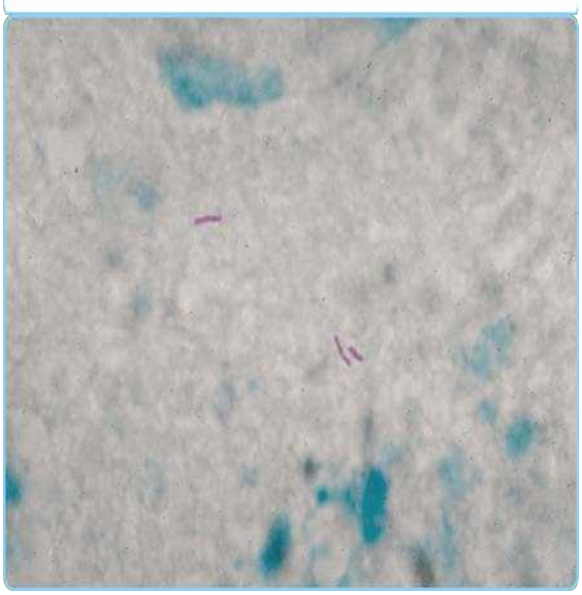
Explanation:

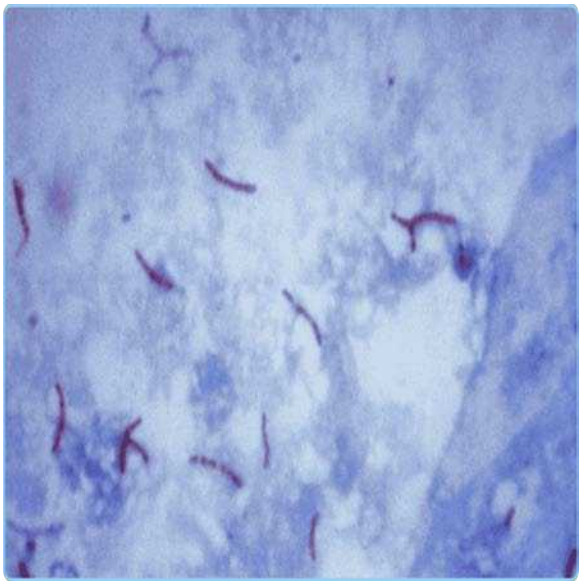
- The above scenario points to a diagnosis of Pulmonary tuberculosis caused by Mycobacterium tuberculosis.
- Zeil Neelsen staining of the sputum helps confirm the presence of acid-fast bacilli, aiding in the diagnosis.

Ziehl Neelsen stain (Acid Fast stain):

- Principle: Acid fastness is due to the high content and variety of lipids, fatty acids, and higher alcohols in tubercle bacilli.
- Primary Stain: Carbol Fuschin (red colour)
- Mordant: Heat
- Decolorizing Agent: Acid/Acid-alcohol (commonly sulphuric acid). The concentration of sulfuric acid may vary depending on the acid fastness of the structure to be demonstrated.
- Counterstaining: The smear is counterstained with a contrasting dye such as methylene blue.
- Result: Acid-fast bacteria retain the fuchsin (red) colour. Non-acid-fast bacteria take the counterstain colour (blue).


1. Apply primary stain of carbolfuschin for 30 seconds
2. Heat fix cells to the slide using flame
3. Decolorize with acid alcohol for 15-20 seconds
4. Apply counterstain of methylene blue for 30 seconds then rinse excess stain
Acid Fast Organism
Non Acid Fast Organism





COLD ZN Stain (Kinyoun Stain/Gabbet Stain): Modified ZN stain where intermittent heating is not required with increased phenol concentration in carbol fuchsin, used for coccidian parasites.

Gram stain (Option A) : It cannot be used in this case because Mycobacterium tuberculosis has a waxy cell wall that resists the Gram stain, requiring acid-fast staining techniques instead.

Albert stain (Option B) : Albert stain cannot be used in this case because it is used to stain the volutin granules present in the organism and is not specific for Mycobacterium tuberculosis.

Neisser stain (Option D) : It cannot be used in this case because it is used to stain the volutin granules present in the organism and is not specific for Mycobacterium tuberculosis.

Solution for Question 13:

Correct Answer: B) E. coli appears pink, S. aureus appears purple

Explanation:

- In Gram staining, Gram-negative bacteria like Escherichia coli appear pink (or red) because they do not retain the primary crystal violet stain due to their thin peptidoglycan layer and outer membrane.
- Instead, they take up the counterstain (safranin or fuchsin), resulting in a pink color.

• Gram-positive bacteria like Staphylococcus aureus retain the crystal violet-iodine complex due to their thick peptidoglycan layer, appearing purple.

• Steps of Gram staining: Fixation : involved making the smear Crystal Violet Staining : Flood the slide with crystal violet for 1 minute to stain all cells purple. Iodine Treatment : Add iodine solution for 1 minute, which forms a complex with the crystal violet, helping it stick to the cells. Decolorization : Rinse with alcohol or acetone to remove the stain from Gram-negative cells (while Gram-positive cells retain the dye). Counterstaining : Flood the slide with safranin for 30 seconds to stain Gram-negative cells pink/red.

• Fixation : involved making the smear

• Crystal Violet Staining : Flood the slide with crystal violet for 1 minute to stain all cells purple.

• Iodine Treatment : Add iodine solution for 1 minute, which forms a complex with the crystal violet, helping it stick to the cells.

• Decolorization : Rinse with alcohol or acetone to remove the stain from Gram-negative cells (while Gram-positive cells retain the dye).

• Counterstaining : Flood the slide with safranin for 30 seconds to stain Gram-negative cells pink/red.

• Fixation : involved making the smear

• Crystal Violet Staining : Flood the slide with crystal violet for 1 minute to stain all cells purple.

• Iodine Treatment : Add iodine solution for 1 minute, which forms a complex with the crystal violet, helping it stick to the cells.

• Decolorization : Rinse with alcohol or acetone to remove the stain from Gram-negative cells (while Gram-positive cells retain the dye).

• Counterstaining : Flood the slide with safranin for 30 seconds to stain Gram-negative cells pink/red.

• Interpretation: Gram-positive bacteria appear purple . Gram-negative bacteria appear pink .

• Gram-positive bacteria appear purple .

• Gram-negative bacteria appear pink .

• Gram-positive bacteria appear purple .

• Gram-negative bacteria appear pink .

E. coli appears purple, S. aureus appears pink (Option A) : This is the reverse of the correct staining pattern and would indicate an error in the staining procedure or interpretation.

Both organisms appear purple (Option C) : If both appeared purple, it would suggest a problem with the decolourization step, where the acetone-alcohol solution failed to remove the crystal violet from the Gram-negative bacteria.

Both organisms appear pink (Option D) : If both appeared pink, it would indicate over-decolorization, where the crystal violet was removed from both Gram-positive and Gram-negative bacteria.

Solution for Question 14:

Correct Answer: C) Fontana-Masson stain

Explanation:

The Fontana-Masson stain is an impregnation stain that uses metallic salts such as silver nitrate to deposit silver on certain cellular structures, making them appear black against a lighter background.

- It is particularly useful for demonstrating spirochetes, melanin, and some fungi.
- In this case, it would be an appropriate choice for visualizing non-syphilitic treponemes that dark-field microscopy may not detect.
- Other stains like Levaditi (for Treponema pallidum in tissue) and Fontana (for melanin and spirochetes in smears) are used for specific detection of microorganisms and cellular structures through silver impregnation.

Gram stain (Option A) : It differentiates bacteria based on their cell wall composition but is not suitable for visualizing spirochetes.

Giemsa stain (Option B) : It is a Romanowsky-type stain used primarily for blood smears and parasites.

Ziehl-Neelsen stain (Option D) : It is an acid-fast stain used primarily for mycobacteria.

Solution for Question 15:

Correct Answer: C) Nigrosin stain

Explanation:

• The symptoms described—headache, fever, altered mental status, and findings of rounded yeast cells with a clear halo in the cerebrospinal fluid (CSF) India ink preparation—suggest an infection with Cryptococcus neoformans.

• Nigrosin stain is a negative stain commonly used to visualize the capsule of Cryptococcus neoformans.

• This method enhances the visibility of the capsule, which is a key identifying feature of C. neoformans.

Negative stain:

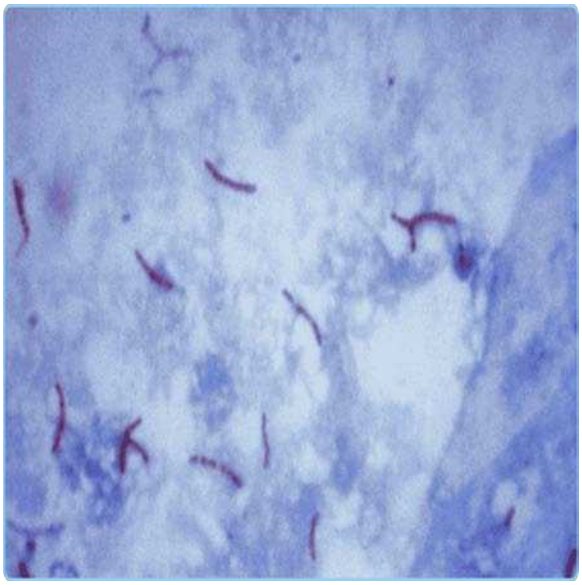
• Dyes used: Indian ink, nigrosin

• Application: Demonstration of bacterial and yeast capsules

• In this technique, the background is stained dark while the yeast cells and their capsules remain unstained, appearing as clear halos around the cells.

Gram stain (Option A): While Gram stain can visualize C. neoformans (which appears as gram-positive yeast), it is not a negative stain and does not specifically highlight the capsule. This option is incorrect.

Ziehl-Neelsen stain (Option B): Ziehl-Neelsen stain is primarily used for acid-fast bacilli like Mycobacterium tuberculosis and is not suitable for visualizing fungal capsules.



Giemsa stain (Option D): Giemsa is used to stain blood and bone marrow smears, as well as some parasites. It is not a negative stain and is not typically used for capsule visualization of C. neoformans.

Solution for Question 16:

Correct Answer: D) Mycobacterium Tuberculosis, Acid fast stain with heat fixation

Explanation:

- The above clinical manifestation is most likely due to pulmonary tuberculosis.
- It is caused by Mycobacterium tuberculosis.
- Acid fast stain with heat fixation is preferred for sputum examination.
- Fixation is done to protect internal structures of cells in fixed position during the staining process, it is two types. Heat fixation-used for Bacterial smears. Fixation is done by coagulation of proteins. Methanol fixation-used for blood smears
- Heat fixation-used for Bacterial smears. Fixation is done by coagulation of proteins.
- Methanol fixation-used for blood smears
- Heat fixation-used for Bacterial smears. Fixation is done by coagulation of proteins.
- Methanol fixation-used for blood smears

Staining techniques	Dyes used	Features
Simple stain	Methylene blue,basic fuschin	Provide colour contrast but impart the same colour to bacteria
Negative stain	Indian ink, nigrosin	Demonstration of bacterial and yeast capsules
Impregnation methods	Silver	Demonstration of cells and structures which are very thin and not visible under light microscope. Eg- bacterial flagella, spirochetes
Differential stain	Different colours used	Gram stain differentiates gram-positive and gram-negative bacteria Acid-fast stain - differentiates acid-fast and non-acid-fast bacteria Albert stain - differentiate metachromatic and non-metachromatic bacteria

Mycobacterium Tuberculosis, Gram stain with heat fixation (Option A) :

- Gram stain does not effectively stain Mycobacterium tuberculosis due to mycolic acid in the cell wall.

• Heat fixation is commonly used with acid-fast stains, not for Gram staining in this context. This option is incorrect.

• Acid-fast staining (Ziehl-Neelsen) involves carbol fuchsin , decolorizing with acid-alcohol , and counterstaining with methylene blue to identify Mycobacterium tuberculosis as red against a blue background.

• Corynebacterium diphtheriae, Albert stain with methanol fixation (Option B): Corynebacterium diphtheriae causes diphtheria, presenting with a pseudomembrane in the throat and not typically with the pulmonary symptoms described here.

• Albert stain is used to visualize Corynebacterium diphtheriae. This option is incorrect.

Streptococcus Pneumoniae, acid-fast stain with methanol fixation (Option C) :

• Streptococcus pneumoniae is a Gram-positive bacterium that does not require acid-fast staining, as it does not possess mycolic acids in its cell wall.

• It typically causes pneumonia with symptoms like fever, chest pain, and productive cough but without the characteristic weight loss and prolonged cough.

Solution for Question 17:

Correct Answer: A) Emmanuelle Charpentier and Jennifer A Dounda

Explanation:

Emmanuelle Charpentier and Jennifer A. Doudna were awarded the Nobel Prize in Chemistry in 2020 for their pioneering work on the CRISPR-Cas9 gene-editing technology.



- CRISPR cas9 is used in editing parts of the genome by removing, adding or altering sections of the DNA sequence.
- It is currently the simplest, most versatile and precise method of genetic manipulation.

C -clustered

R -regulatory

I -interspaced

S -short

P -palindromic

R -repeats

In 2024 , the Nobel Prize in Physiology or Medicine was awarded to Katalin Karikó and Drew Weissman for their groundbreaking work on mRNA technology , which was pivotal in the development of COVID-19 vaccines.

Luc Montangnier and F Barre-Sinoussi (Option B) : Discovered HIV.

Stanley B Prusiner (Option C) : Discovered double helix structure of DNA.

J Michael Bishop and Harold E Varmus (Option D) : Discovered origin of retroviral oncogenes.

Solution for Question 18:

Correct Answer: D) Joseph Lister

Explanation:

Joseph Lister is the scientist who explained the importance of sterilization of surgical tools.

Antonie Philips van Leeuwenhoek	First to observe bacteria and other microorganisms using a single-lens microscope, constructed by him
Edward Jenner (Option C ruled out)	Developed smallpox vaccine
Louis Pasteur	Father of microbiology Proposed principles of fermentation Introduced sterilization techniques Described pasteurization of milk Postulated germ theory of diseases Liquid media concept
Robert Koch	Father of Modern Microbiology Introduced solid media Described hanging drop method for testing motility Discovered bacteria such as anthrax bacilli, tubercle bacilli and cholera bacilli Proposed Koch's phenomenon and Koch postulates
Joseph Lister (Option D)	Father of antiseptic surgery 1st used carbolic acid during surgery, to sterilize his instruments and to clean wounds which reduced postoperative infections
Paul Ehrlich (Option A ruled out)	Father of chemotherapy First to report acid fast nature of tubercle bacilli. Proposed side-chain theory.
Hans Christian Gram	Gram staining
Ernest Ruska (Option B ruled out)	Founder of electron microscope
Alexander Fleming	Discovered Penicillin
Good Pasture	Described viral culture technique
Barbara McClintock	Described transposons

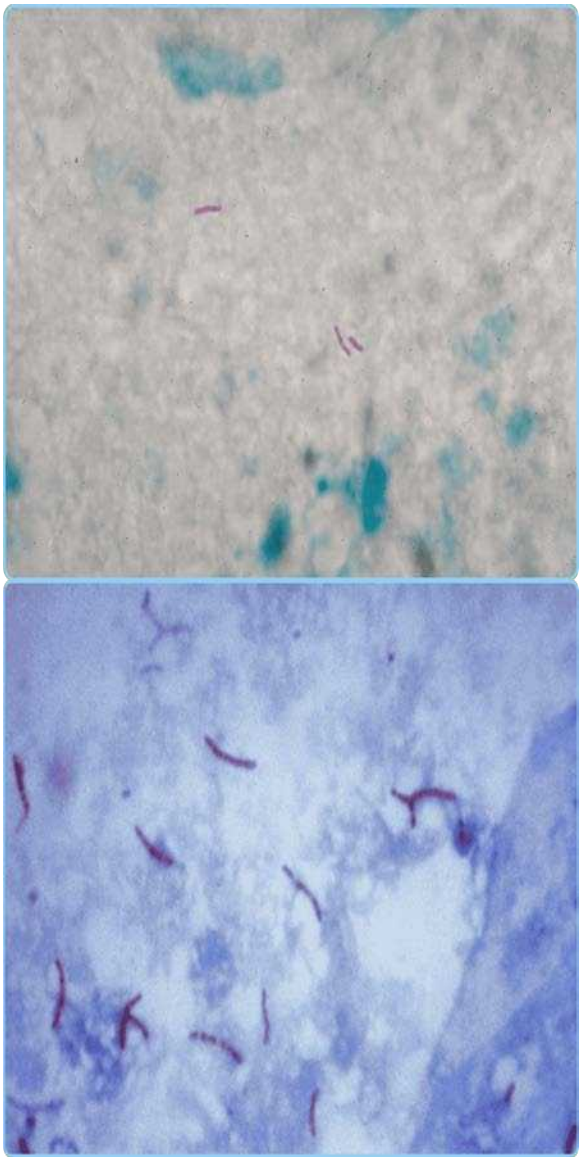
Antonie Philips van Leeuwenhoek	First to observe bacteria and other microorganisms using a single-lens microscope, constructed by him
Frederick Sanger and Walter Gilbert	DNA sequencing
Karry B Mullis	Discovered PCR

Solution for Question 19:

Correct Answer: A) Paul Ehrlich

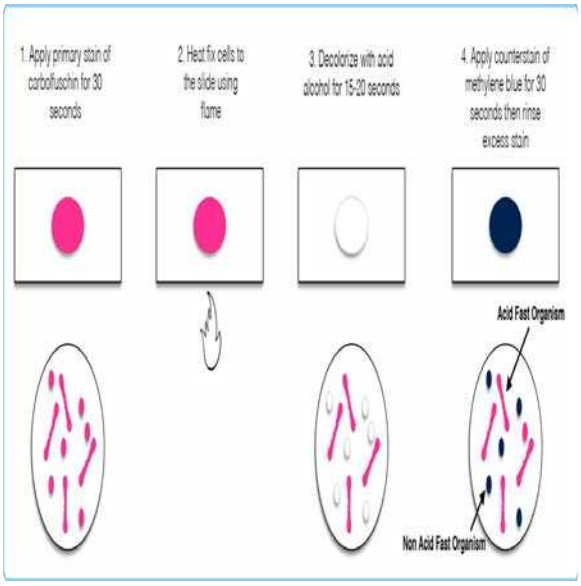
Explanation:

- The acid-fast stain was discovered by Paul Ehrlich, who found that after staining with aniline dyes, tubercle bacilli resist decolourisation with acids.
- Acid Fastness property is due to the presence of unique lipids called mycolic acids , found only in species of the genus Mycobacterium, on the surface of the mycobacterial cell.
- The extent of acid fastness depends on the integrity and amount of the mycolic acids within the organism's cell wall.
- Cells of acid-fast organisms stain red, whereas the background and non-acid-fast organisms appear blue.



Modifications of Acid-Fast Staining :

- Hot method (Ziehl-Neelsen technique): It is the most commonly done acid-fast staining technique (Option D ruled out)
- Cold method (Kinyoun's method): Intermittent heating is not required here
- Acid-alcohol can be used as a decolourizer alternatively
- Malachite green can be used as counterstain
- The concentration of sulfuric acid may vary depending on the acid fastness of the structure to be demonstrated.



Solution for Question 20:

Correct Answer: B) Koch's postulates were not fulfilled because the virus did not cause disease in healthy laboratory animals.

Explanation:

According to Koch’s third postulate, ‘cells from a pure culture of the suspected pathogen must cause disease in a healthy animal’, therefore in the above clinical scenario since the virus was not able to cause disease in a healthy laboratory animal, Koch's postulate cannot be fulfilled.

- Koch formulated a set of criteria known as Koch’s postulates, for linking cause and effect in an infectious disease.

Koch’s postulates are as follows:

An additional fifth criterion was introduced which states that antibody to the causative organism should be demonstrable in the patient's serum.

Exceptions to Koch's postulates: Some bacteria do not satisfy one or more of the four criteria of Koch's postulates. They are :

- Mycobacterium leprae and Treponema pallidum cannot be grown in vitro; however, can be maintained in experimental animals.

- Neisseria gonorrhoeae: there is no animal model; however, it can be grown in vitro.

Koch's postulates have been fulfilled because the virus was isolated and cultured successfully (Option A).

- However, Koch's postulates require that the pathogen cause disease in a healthy, susceptible host (postulate 3).
- Since the virus did not cause illness in the animals, the postulates are not fully satisfied.

Koch's postulates can be fulfilled if the researcher tried a higher dose of the virus in the animals (Option C):

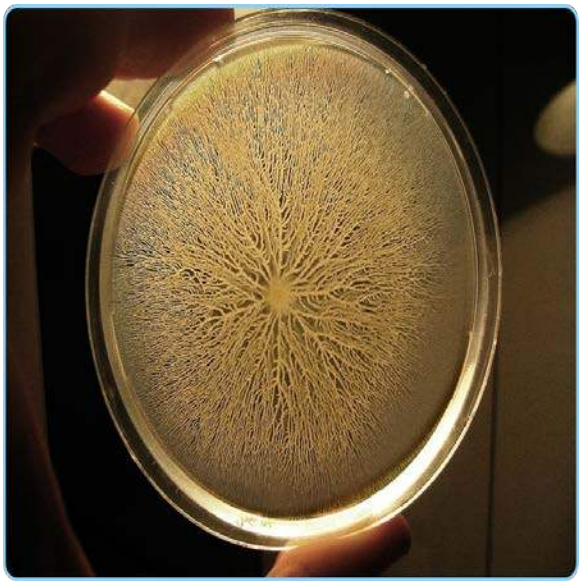
- Koch's postulates don't rely on increasing the dose to fulfil them. If the virus doesn't cause disease in one dose, it may not be the causative agent, or the animal model may not be suitable.

Koch's postulates can be fulfilled if the researcher observes the virus under an electron microscope (Option D):

- Electron microscopy helps identify the virus but doesn't fulfill Koch's postulates. Postulate 3 requires the cultured pathogen to cause disease in a healthy host, which microscopy cannot demonstrate.

Bacterial Anatomy

1. Which among the given options exhibits swarming behaviour?



A. Salmonella enterica

B. Vibrio cholerae

C. Bacillus cereus

D. Proteus mirabilis

---

2. All of the following are true regarding eukaryotic cells except?

A. Eukaryotic cells have a true nucleus, while prokaryotic cells do not.

B. Eukaryotic cells have membrane-bound organelles, whereas prokaryotic cells lack them.

C. Eukaryotic cells have linear DNA, whereas prokaryotic cells have circular DNA.

D. In eukaryotic cells, transcription and translation occur simultaneously, while in prokaryotic cells, they are separated by the nuclear membrane.

3. Which of the following spores are commonly used for sterilization control in hot air ovens?

A. Bacillus subtilis

B. Clostridium tetani

C. Geobacillus stearothermophilus

D. Bacillus atrophaeus

4. Arrange the stages of spore formation (sporulation) in correct order:

A. Axial filament formation ➔ Septum formation ➔ Asymmetric division ➔ Engulfment ➔ Cortex formation ➔ Maturation and lysis

B. Engulfment ➔ Cortex formation ➔ Septum formation ➔ Asymmetric division ➔ Axial filament formation ➔ Maturation and lysis

C. Septum formation ➔ Asymmetric division ➔ Engulfment ➔ Cortex formation ➔ Maturation and lysis ➔ Axial filament formation

D. Asymmetric division ➔ Engulfment ➔ Cortex formation ➔ Septum formation ➔ Axial filament formation ➔ Maturation and lysis

5. Which of the following organisms is known for terminal spore formation?

A. Clostridium tetani

B. Bacillus anthracis

C. Clostridium perfringens

D. Bacillus cereus

6. Arrange the layers of a spore from outside to inside in the correct order:

A. Exosporium → Cortex → Coat → Core

B. Coat → Exosporium → Cortex → Core

C. Exosporium → Coat → Cortex → Core

D. Coat → Exosporium → Core → Cortex

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7. Identify the correct statements regarding fimbriae/pili from the list below: Fimbriae are made of subunits called fibrillins. Pili are involved in bacterial conjugation. Pili can be involved in motility through a process known as " twitching." Fimbriae play a role in pellicle formation at the liquid-surface interface. Pili are longer and more numerous than flagella.

A. 1, 2, and 4

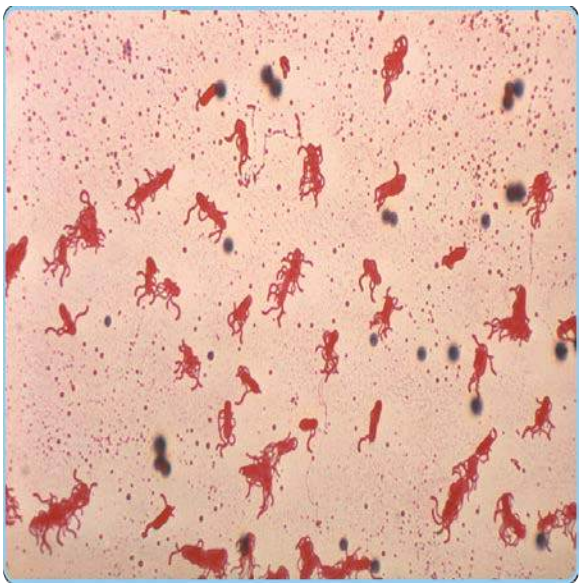
B. 2, 3, and 4

C. 2, 3, and 5

D. 1, 2, and 5

---

8. Leifson and Ryu's stain is used for staining which of the following?



A. Fungi

B. Mycobacteria

C. Flagella

D. Capsules

---

9. Which of the following is not used for the demonstration of bacterial motility?

A. Direct observation under dark-ground illumination

B. Special staining techniques using mordant

C. Electron microscopy

D. Gram staining

---

10. Which of the following contains a polysaccharide capsule?

A. Haemophilus influenzae

B. Pseudomonas aeruginosa

C. Klebsiella pneumoniae

D. Bacillus anthracis

---

11. Match the types of motility with their respective organism Motility Organism i) Twitching motility a) Vibrio ii) Falling leaf motility b) Listeria iii) Tumbling motility c) Giardia lamblia iv) Darting motility d) Pseudomonas aeruginosa

Motility	Organism
i) Twitching motility	a) Vibrio
ii) Falling leaf motility	b) Listeria
iii) Tumbling motility	c) Giardia lamblia
iv) Darting motility	d) Pseudomonas aeruginosa

A. i-a ; ii-b ; iii-c ; iv-d

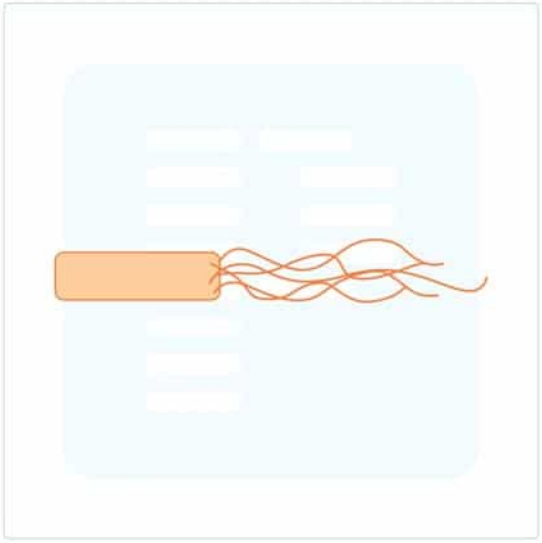
B. i-d ; ii-c ; iii-b ;iv-a

C. ii-a ; iii-d ; i-b ; iv-c

D. ii-d ; iv-b ; iii-a ; i-c

---

12. Which of the following organisms corresponds to the morphology of the image given?



A. Escherichia coli

B. Spirillum volutans

C. Vibrio cholerae

D. Bacillus subtilis

---

13. Which of the following receptors is primarily responsible for recognising and binding to endotoxin?

A. RIG-I-like receptor (RLR)

B. Toll-like receptor 4 (TLR4)

C. C-type lectin receptor (CLR)

D. NOD-like receptor 1 (NLR1)

---

14. Which of the following regarding lipopolysaccharides (LPS) is true?

A. LPS is primarily found in gram-positive bacterial cell walls.

B. LPS triggers an immune response through its lipid A component.

C. LPS is composed solely of proteins and does not contain any lipid or carbohydrate components.

D. LPS is composed of lipid A, lipoteichoic acid and O antigen

---

15. A 23-year-old woman presents to the emergency department with a sudden high fever, low blood pressure, and an erythematous rash over her body. She reports nausea, vomiting, and muscle aches. On further questioning, she recalls using tampons during her recent menstrual cycle but forgot to remove one for several days. Laboratory investigations showed catalase-positive organisms arranged in clusters. All of the following are true about the cell wall of this bacterium except:

A. Thick peptidoglycan layer

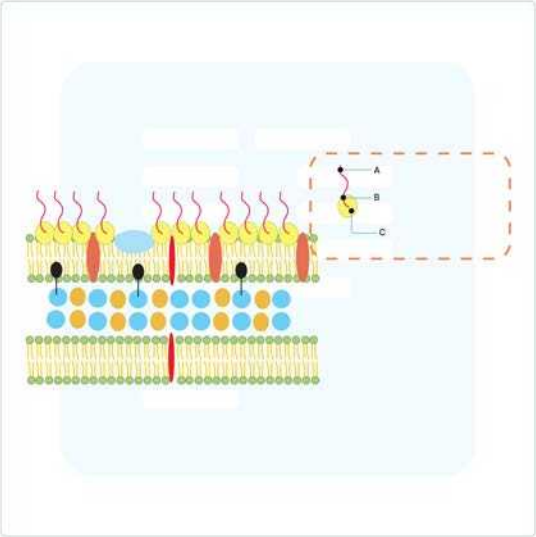
B. Contains teichoic acids

C. Pentaglycine bridge is present

D. Contains lipopolysaccharides

---

16. Identify ‘A’ in the following image.



A. Lipoteichoic acid

B. O antigen

C. Lipid A

D. Teichoic acid

---

17. A 65-year-old male, who underwent mitral valve replacement 3 months ago, presents with persistent fever, fatigue, and a new heart murmur. Blood cultures reveal gram-positive cocci, and echocardiography shows vegetations on the prosthetic valve. What is the most likely causative organism?

A. Staphylococcus epidermidis

B. Escherichia coli

C. Streptococcus mutans

D. cStreptococcus pyogenes

---

18. A 45-year-old male presents with fever and an eschar on his arm after handling animal hides. Which of the following methods is used to demonstrate the capsule of the causative agent?

A. India ink staining

B. Quelleng reaction

C. McFadyean reaction

D. Latex agglutination test

Correct Answers

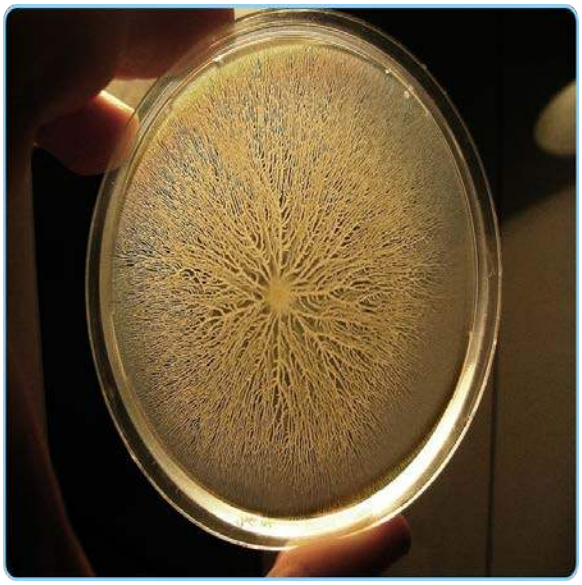
Question	Correct Answer
Question 1	4
Question 2	4
Question 3	1
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	3
Question 9	4
Question 10	4
Question 11	2
Question 12	2
Question 13	2
Question 14	2
Question 15	4
Question 16	2
Question 17	1
Question 18	3

Solution for Question 1:

Correct Answer: D) Proteus mirabilis

Explanation:

- The image shows swarming behavior , characterized by the rapid movement and spreading of bacteria across a solid surface.
- Proteus mirabilis is known for its distinctive swarming motility, where it forms concentric rings or bull's-eye patterns due to the coordinated movement of peritrichous flagella.



- This motility allows Proteus mirabilis to migrate across surfaces, which is essential for its pathogenesis and survival in various environments.
- Other examples: Vibrio parahaemolyticus, Pseudomonas aeruginosa, Bacillus subtilis and Serratia liquefaciens

Solution for Question 2:

Correct Answer: D) In eukaryotic cells, transcription and translation occur simultaneously, while in prokaryotic cells, they are separated by the nuclear membrane.

Explanation:

Feature	Prokaryotic Cells	Eukaryotic Cells
Nucleus	No true nucleus, nucleoid region with circular DNA	True nucleus enclosed by a nuclear membrane
Size	Generally smaller (0.1-5.0 µm in diameter)	Generally larger (10-100 µm in diameter)
Organelles	Few organelles, no membrane-bound organelles .	Numerous membrane-bound organelles (e.g., mitochondria, ER).
Cell Wall	Present (peptidoglycan in bacteria, pseudo peptidoglycan in archaea)	Present in plants (cellulose) and fungi (chitin), absent in animals
Ribosomes	70S ribosomes	80S ribosomes in cytoplasm, 70S in mitochondria and chloroplasts
Genetic Material	Single circular DNA molecule , no histones, rarely introns present	Multiple linear chromosomes with histones, commonly introns, present
Gene Expression	Transcription and translation occur simultaneously in cytoplasm	Transcription occurs in the nucleus, translation in the cytoplasm
Reproduction	Asexual (binary fission)	Sexual and asexual (mitosis and meiosis)
Flagella	Simple, made of flagellin	Complex, 9+2 microtubule arrangement
Cytoskeleton	Absent or very simple	Complex cytoskeleton (microtubules, microfilaments)
Intracellular Movement	Generally absent	Present (vesicle transport, cytoplasmic streaming)
Plasma Membrane	No sterols (except in Mycoplasma)	Sterols (cholesterol in animals)
Protein Synthesis	Occurs in the cytoplasm	Occurs in the cytoplasm and on the rough ER
Metabolic Pathways	Various simple pathways, often anaerobic	Mostly aerobic pathways, complex metabolic processes
Examples	Bacteria, Archaea	Protists, fungi, plants, animals

Solution for Question 3:

Correct Answer: A) Bacillus subtilis

Explanation:

Bacillus subtilis is commonly used as a biological indicator for hot air ovens because its spores are heat-resistant and can indicate whether the sterilization process effectively kills spores.

Application	Bacterial Species
Autoclave sterilization	Clostridium tetani
Hot air oven sterilization	Bacillus subtilis
Steam sterilization	Bacillus stearothermophilus Geobacillus stearothermophilus
Ethylene oxide sterilization	Bacillus atrophaeus
Sterility testing in pharmaceuticals and canned foods	Clostridium sporogenes

Solution for Question 4:

Correct Answer: A) Axial filament formation ➔ Septum formation ➔ Asymmetric division ➔ Engulfment ➔ Cortex and coat formation ➔ Maturation and lysis

Explanation:

spore formation" data-author="" data-hash="9921" data-license="" data-source="" data-tags="" height="359" src="https://image.prepladder.com/notes/C2c1YHzVuHk8NTzPHkLl1744449387.png" width="359" />

- Axial filament formation: The bacterial cell prepares for sporulation by forming an axial filament, which helps organise the process.
- Septum formation: Triggered by environmental conditions such as nutrient limitation or other stressors, the bacterium divides asymmetrically, forming a septum that separates the mother cell from the forespore.
- Asymmetric division: One compartment becomes the smaller forespore (develops into the spore), and the other becomes the larger mother cell.

• Engulfment: The mother cell engulfs the forespore within a second membrane.

• Cortex formation : A thick layer of peptidoglycan forms between the two membranes of the spore.

• Coat formation: Various proteins assemble around the spore, forming the coat.

• Maturation and lysis: The spore matures, becoming highly resistant to environmental stresses. Eventually, the mother cell lyses, releasing the mature spore into the environment.

• Eventually, the mother cell lyses, releasing the mature spore into the environment.

• Eventually, the mother cell lyses, releasing the mature spore into the environment.

Solution for Question 5:

Correct Answer: A) Clostridium tetani

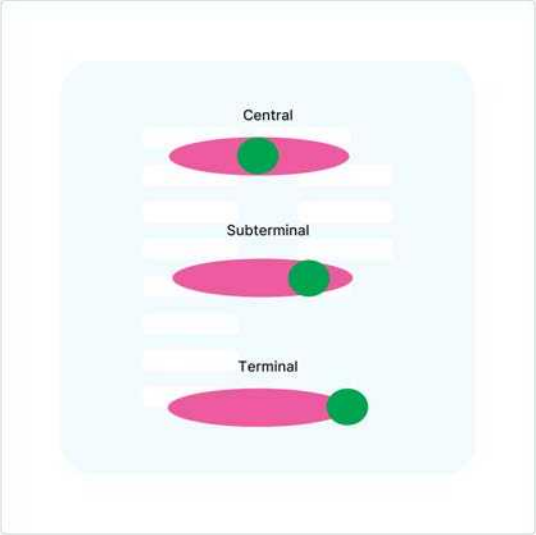
Explanation:

• Clostridium tetani is a bacterium known for forming terminal spores.

• Terminal spores are spores formed at the end of the bacterial cell,

• These spores are resistant to heat, desiccation, and chemical agents, allowing the bacterium to survive adverse conditions.

• Clostridium tetani specifically forms terminal spores as part of its life cycle, facilitating its ability to persist in the environment and cause tetanus in humans when spores enter wounds.



Spore Location	Examples
Central	Bacillus anthracis, Bacillus cereus
Terminal	Clostridium tetani
Subterminal	Bacillus thuringiensis, Bacillus subtilis, Clostridium botulinum, Clostridium perfringens

Bacillus anthracis (Option B) is a spore-forming bacterium but forms central spores, which are located more centrally within the bacterial cell.

The causative agent of anthrax.

Clostridium perfringens (Option C) is another spore-forming bacterium.

- Instead, it forms subterminal spores, which are located near the end but not exactly at the terminal end of the bacterial cell.
- It can cause food poisoning and gas gangrene.

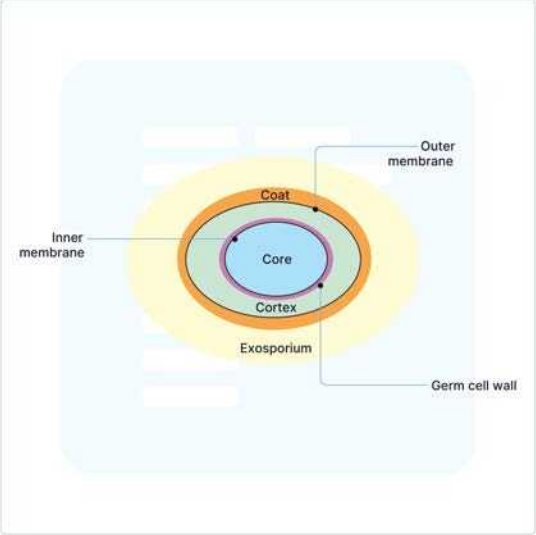
Bacillus cereus (Option D) is a spore-forming bacterium that forms central spores that are not located at the terminal end of the bacterial cell.

- It is known for causing food poisoning through the production of enterotoxins in contaminated food.

Solution for Question 6:

Correct Answer: C) Exosporium → Coat → Cortex → Core .

Explanation:



• Exosporium: This is the outermost layer of a spore, consisting of proteins and sometimes lipids, providing protection against environmental factors.

• Coat: Beneath the exosporium lies the spore coat, which is a thick layer made of peptidoglycan and proteins. It provides structural support and resistance to chemicals.

• It provides structural support and resistance to chemicals.

• Cortex: The cortex is located beneath the coat and consists of peptidoglycan. It plays a role in maintaining spore dehydration and resistance.

• It plays a role in maintaining spore dehydration and resistance.

• Core: The innermost layer of a spore is the core, containing the spore's genetic material (DNA), ribosomes, and enzymes necessary for germination and metabolic activity.

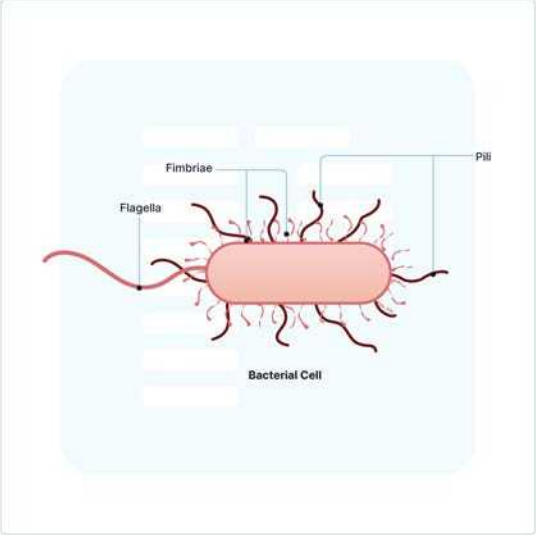
• It provides structural support and resistance to chemicals.

• It plays a role in maintaining spore dehydration and resistance.

Solution for Question 7:

Correct Answer: B) 2, 3, and 4

Explanation:



Definition	Fimbriae are thin, hair-like appendages found on the surface of many bacteria, primarily Gram-negative bacteria.
Structure	Fimbriae are composed of protein subunits called pilins (Statement 1). They are usually shorter and more numerous than flagella. (Statement 5)
Types	Common (Type I) Fimbriae: Adhesive fimbriae that bind to mannose-containing receptors on host cells. Sex Pili: Involved in bacterial conjugation and DNA transfer. (Statement 2)
Functions	Adhesion: Fimbriae allow bacteria to adhere to surfaces, cells, and tissues, which is crucial for colonization and infection. Biofilm Formation: Contribute to biofilm formation, protecting bacteria from environmental stresses. Gene Transfer: Sex pili facilitates the transfer of genetic material between bacteria during conjugation, promoting genetic diversity and the spread of antibiotic resistance. Motility: Usually not involved except Type IV Pili, particularly in Neisseria species, enabling a type of movement known as twitching motility. (Statement 3)

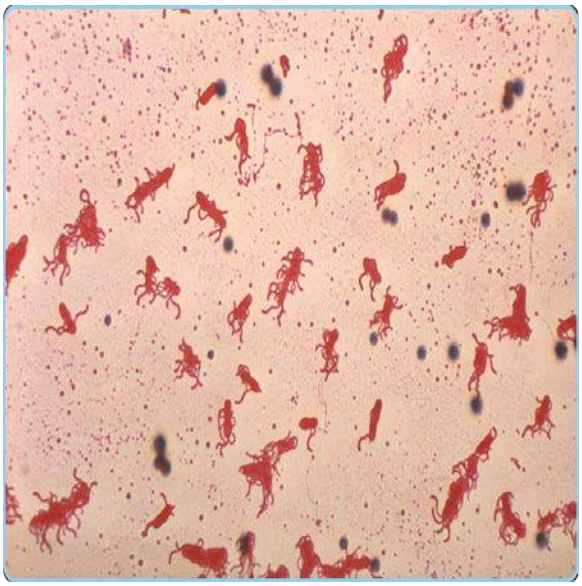
Definition	Fimbriae are thin, hair-like appendages found on the surface of many bacteria, primarily Gram-negative bacteria.
Uses in Medicine	Target for Vaccines: Components of fimbriae can be used as antigens in vaccines to prevent bacterial infections. Antimicrobial Targets: Inhibiting fimbrial function can prevent bacterial adhesion and biofilm formation, making it a potential target for new antibiotics.
Clinical Importance	Urinary Tract Infections (UTIs): Uropathogenic E. coli (UPEC) use Type 1 fimbriae to adhere to the urinary tract, causing UTIs. Respiratory Infections: Bacteria like Bordetella pertussis (whooping cough) use fimbriae to adhere to the respiratory tract. Gastrointestinal Infections: Enteropathogenic E. coli (EPEC) use fimbriae to adhere to the intestinal mucosa, causing diarrhea. Pseudomonas aeruginosa : Uses Type IV pili for surface attachment and biofilm formation. Neisseria gonorrhoeae: Uses fimbriae for attachment to mucosal surfaces in the genitourinary tract.
Laboratory Identification	Hemagglutination Assay: Detects Type 1 fimbriae by their ability to agglutinate red blood cells. Electron Microscopy: Visualizes fimbriae on the surface of bacteria. Pellicle Formation : Observing pellicle formation on liquid culture surfaces can indicate the presence of fimbriae. (Statement 4)

Solution for Question 8:

Correct Answer: C) Flagella

Explanation:

Leifson and Ryu's stain is specifically used for staining flagella, allowing them to be visualized under a dark field microscope.



• Fungi (Option A): Fungi are typically stained using special fungal stains like lactophenol cotton blue or potassium hydroxide (KOH) preparations.

• Mycobacteria (Option B): Mycobacteria are stained using acid-fast stains like Ziehl-Neelsen or Kinyoun stains due to their high lipid content in the cell wall.

• Capsules (Option D): Capsules are often stained using India ink or special stains like Maneval's capsule stain to highlight their presence around bacterial cells.

Solution for Question 9:

Correct Answer : D) Gram staining

Explanation :

- Gram staining is a technique used to differentiate bacterial species into two groups based on the structural differences in their cell walls.
- It does not provide information about bacterial motility.

Methods of demonstrating motility

- Dark Field / Dark Ground microscopy : Flagella may sometimes be seen under dark-ground illumination, though this is unreliable due to their small size.
- Special Staining Techniques: Mordanting techniques increase flagella thickness, making them visible under a light microscope. Eg: Leifson or Ryu's flagella staining techniques.
- Electron Microscopy : Allows high-resolution visualization of flagella.
- Semi-Solid Agar Medium : Motile bacteria exhibit spreading growth away from the inoculation site, unlike non-motile bacteria.
- Hanging Drop Method : Observing a drop of bacterial suspension under a microscope can show motile bacteria moving.

Solution for Question 10:

Correct Answer: D) Bacillus anthracis

Explanation:

Most of the bacterial capsules are polysaccharide in nature except in Bacillus anthracis where it is polypeptide in nature.

- Bacteria often secrete a viscid material around their cell surface, called glycocalyx.
- When the glycocalyx is organized into a defined structure, it is called a capsule , as in Streptococcus pneumoniae.
- When the glycocalyx layer is in the form of loose, undefined material, it is known as a slime layer , as seen in Leuconostoc.
- Very thin capsules are termed microcapsules .
- Certain bacteria, such as Streptococcus salivarius, possess both a capsule and a slime layer.

Capsulated Organisms	
Polysaccharide Capsule	Exceptions
Pneumococcus Meningococcus Haemophilus influenzae (Option A) Klebsiella pneumoniae (Option C) Pseudomonas aeruginosa (Option B) Bacteroides fragilis Streptococcus pneumoniae	Bacillus anthracis - Polypeptide (glutamate) (Option D) Streptococcus pyogenes (some strains) - Hyaluronic acid

Solution for Question 11:

Correct Answer: B) i-d; ii-c; iii-b;iv-a

Explanation:

Motility	Organism
Twitching motility	Pseudomonas aeruginosa
Falling leaf motility	Giardia lamblia
Tumbling motility	Listeria
Darting motility	Vibrio, Campylobacter
Lashing motility	Spirochetes
Gliding motility	Mycoplasma
Swarming motility	Proteus, Clostridum
Stately motility	Clostridium

Solution for Question 12:

• Escherichia coli (Option A): This bacterium typically has peritrichous flagella, meaning flagella are distributed all over the cell surface.

• Vibrio cholerae (Option C) : This bacterium typically has a monotrichous flagellum, meaning a single flagellum at one end.

• Proteus (Option D) : This bacterium also typically has peritrichous flagella, with flagella distributed around the entire cell surface.

• Monotrichous: A single flagellum at one end (e.g., Vibrio cholerae).

• Lophotrichous: A tuft of flagella at one or both ends (e.g., Spirilla).

• Amphitrichous: Flagella at both poles of the cell. (e.g., Campylobacter)

• Peritrichous: Flagella arranged all around the cell (e.g., Proteus).

flagella Answer" data-author="" data-hash="9908" data-license="" data-source="" data-tags="" height="380" src="https://image.prepladder.com/notes/dBs924eR405vmUSniDZQ1744448490.png" width="380" />

Solution for Question 13:

Correct Answer: B) Toll-like receptor 4 (TLR4)

Explanations:

• Endotoxins bind to Toll-like receptor 4 (TLR4) on the surface of immune cells.

• This binding initiates a signaling cascade that leads to the production of pro-inflammatory cytokines.

• Upon activation, TLRs recruit adaptor proteins (proteins that mediate other protein-protein interactions) within the cytosol of the immune cell to propagate the antigen-induced signal transduction pathway .

• These recruited proteins are then responsible for the subsequent activation of other downstream proteins, including protein kinases (IKKi, IRAK1 , IRAK4 , and TBK1).

• These lead to cytokine production, proliferation , and survival while also leading to greater adaptive immunity.

RIG-I-like receptor (RLR) (Option A) :

• Intracellular receptors that detect viral RNA in the cytoplasm, play a crucial role in antiviral responses.

- They do not recognize bacterial lipopolysaccharides (LPS).

C-type lectin receptor (CLR) (Option C) :

- Receptors that identify carbohydrate structures on pathogens like fungi and some bacteria.
- They do not primarily recognize lipopolysaccharides (LPS).

NOD-like receptor 1 (NLR1) (Option D) :

- Intracellular receptors that detect bacterial peptidoglycan and other danger signals within the cytoplasm.
- They are not the main receptors for lipopolysaccharides (LPS).

Endotoxins

- Nature: Heat-stable lipopolysaccharides (LPS) from Gram-negative bacteria.
- Toxicity: Depends on the lipid component (lipid A).
- Release: Released only by cell wall disintegration.
- Action: Effect nonspecific; action common to all endotoxins
- Tissue Affinity: No specific tissue affinity
- Antigenicity: Weakly antigenic
- Neutralization: Neutralization by antibody ineffective
- Clinical Symptoms: Fever, leucopenia, thrombocytopenia, fall in blood pressure, circulatory collapse, and bloody diarrhea leading to death.

Solution for Question 14:

Correct Answer: B) LPS triggers an immune response through its lipid A component.

Explanation:

Lipid A, a component of the lipopolysaccharide (LPS) unit, is responsible for eliciting strong immune responses, including fever and inflammation.

Lipopolysaccharide (LPS):

- LPS is an important outer membrane component of gram-negative bacteria (Option A) .
- These molecules are also known as lipogylcans due to the presence of lipid and sugar molecules (Option C) .

• It consists of 3 parts (Option D) : O side chain (or O antigen or somatic antigen) (hydrophilic) : A polysaccharide chain which greatly varies in composition between bacterial strains and hence this is used for serotyping for bacteria (e.g., Salmonella typhi - widal test). It is also a major surface antigen, induces antibody formation. Core polysaccharide (hydrophilic) : It is projected from lipid A region. It is composed of 10–12 sugar moieties. Lipid A or the endotoxin (hydrophobic) : Main virulence factor, responsible for endotoxic activities such as pyrogenicity, lethal effect, tissue necrosis, anticomplementary activity, B cell mitogenicity, adjuvant property and antitumor activity.

• O side chain (or O antigen or somatic antigen) (hydrophilic) : A polysaccharide chain which greatly varies in composition between bacterial strains and hence this is used for serotyping for bacteria (e.g., Salmonella typhi - widal test). It is also a major surface antigen, induces antibody formation.

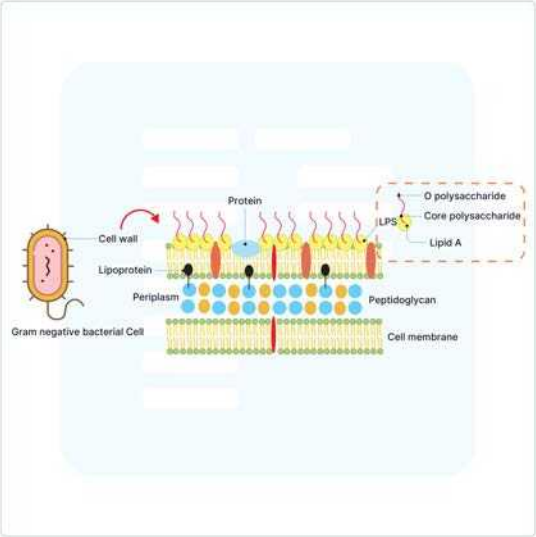
• Core polysaccharide (hydrophilic) : It is projected from lipid A region. It is composed of 10–12 sugar moieties.

• Lipid A or the endotoxin (hydrophobic) : Main virulence factor, responsible for endotoxic activities such as pyrogenicity, lethal effect, tissue necrosis, anticomplementary activity, B cell mitogenicity, adjuvant property and antitumor activity.

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• Core polysaccharide (hydrophilic) : It is projected from lipid A region. It is composed of 10–12 sugar moieties.

• Lipid A or the endotoxin (hydrophobic) : Main virulence factor, responsible for endotoxic activities such as pyrogenicity, lethal effect, tissue necrosis, anticomplementary activity, B cell mitogenicity, adjuvant property and antitumor activity.



Solution for Question 15:

Correct Answer: D) Contains lipopolysaccharides

Explanation:

The given clinical history of retained tampons leading to high fever and rash is suggestive of toxic shock syndrome (TSS), most likely caused by Staphylococcus aureus, as the laboratory investigations revealed catalase positive organisms arranged in clusters. S. aureus is a gram positive bacteria, and gram positive bacteria does not contain lipopolysaccharides (LPS).

CELL WALL OF GRAM-POSITIVE BACTERIA:

- Cell wall of gram-positive bacteria is simpler than that of gram-negative bacteria.

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- Peptidoglycan layer: Much thicker (50–100 layers thick, 16–80 nm) than gram-negative cell wall (Option A) . Composed of a mucopeptide (murein) chain, with alternate units of NAM and NAG molecules. A tetrapeptide side chain ascended from a NAM molecule is composed of L-alanine—D-glutamine—L-lysine—D-alanine. (Mnemonic- AGLA) The L-lysine of one tetrapeptide chain is covalently linked to the terminal D-alanine of the adjacent chain via a

pentaglycine bridge (Option C) .

• Much thicker (50–100 layers thick, 16–80 nm) than gram-negative cell wall (Option A) .

• Composed of a mucopeptide (murein) chain, with alternate units of NAM and NAG molecules.

• A tetrapeptide side chain ascended from a NAM molecule is composed of L-alanine—D-glutamine—L-lysine—D-alanine. (Mnemonic- AGLA)

• The L-lysine of one tetrapeptide chain is covalently linked to the terminal D-alanine of the adjacent chain via a pentaglycine bridge (Option C) .

• Much thicker (50–100 layers thick, 16–80 nm) than gram-negative cell wall (Option A) .

• Composed of a mucopeptide (murein) chain, with alternate units of NAM and NAG molecules.

• A tetrapeptide side chain ascended from a NAM molecule is composed of L-alanine—D-glutamine—L-lysine—D-alanine. (Mnemonic- AGLA)

• The L-lysine of one tetrapeptide chain is covalently linked to the terminal D-alanine of the adjacent chain via a pentaglycine bridge (Option C) .

Gram Positive bacteria" data-author="" data-hash="9888" data-license="" data-source="" data-tags="" height="378" src="https://image.prepladder.com/notes/b9KlKZD5DQtWVnqqKM9r1744446496.png" width="378" />

• Teichoic Acid: Teichoic acids are polymers which are present only in gram-positive bacteria (Option B) . These molecules help maintain the structure of the cell wall. They are of two types: (i) Cell wall teichoic acid and (ii) Lipoteichoic acid .

• Teichoic acids are polymers which are present only in gram-positive bacteria (Option B) .

• These molecules help maintain the structure of the cell wall.

• They are of two types: (i) Cell wall teichoic acid and (ii) Lipoteichoic acid .

• Teichoic acids are polymers which are present only in gram-positive bacteria (Option B) .

• These molecules help maintain the structure of the cell wall.

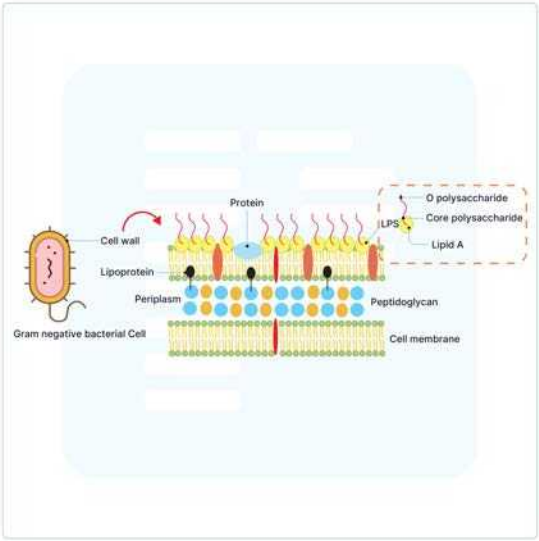
• They are of two types: (i) Cell wall teichoic acid and (ii) Lipoteichoic acid .

Solution for Question 16:

Correct Answer: B) O antigen

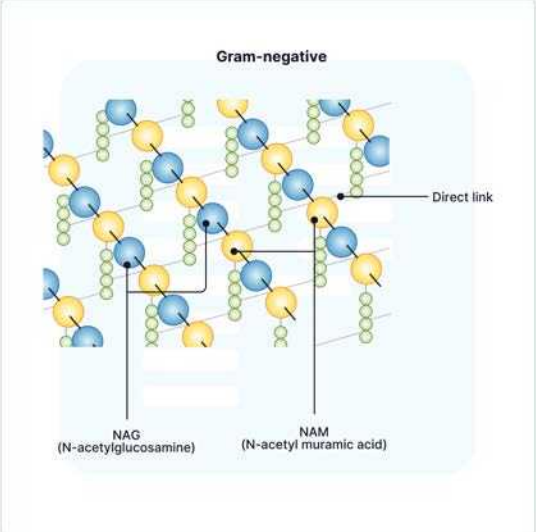
Explanation

The diagram shows the cell wall of Gram-negative bacteria, and the labeled area ‘A’ refers to the O antigen (O polysaccharide).



CELL WALL OF GRAM-NEGATIVE BACTERIA:

- Peptidoglycan layer: It is thinner (1–2 layers, 2 nm) and more complex than the gram-positive cell wall. Composed of a mucopeptide chain, with alternate NAM and NAG molecules. The pentaglycine bridge is absent . The tetrapeptide chains are covalently linked betweenthe D-alanine of one chain with meso-diaminopimelic acid of the adjacent chain.
- It is thinner (1–2 layers, 2 nm) and more complex than the gram-positive cell wall.
- Composed of a mucopeptide chain, with alternate NAM and NAG molecules.
- The pentaglycine bridge is absent . The tetrapeptide chains are covalently linked betweenthe D-alanine of one chain with meso-diaminopimelic acid of the adjacent chain.
- It is thinner (1–2 layers, 2 nm) and more complex than the gram-positive cell wall.
- Composed of a mucopeptide chain, with alternate NAM and NAG molecules.
- The pentaglycine bridge is absent . The tetrapeptide chains are covalently linked betweenthe D-alanine of one chain with meso-diaminopimelic acid of the adjacent chain.



• Outer membrane: It is a phospholipid layer, attached to the thin peptidoglycan layer by a membrane protein called Braun's lipoprotein . Contains outer membrane proteins (OMP), which help in transport of small molecules.

• It is a phospholipid layer, attached to the thin peptidoglycan layer by a membrane protein called Braun's lipoprotein .

• Contains outer membrane proteins (OMP), which help in transport of small molecules.

• Lipopolysaccharide (LPS): This layer is unique to gram-negative bacteria, consists of 3 parts (mnemonic - OPA): O side chain (or O antigen or somatic antigen): A polysaccharide chain which greatly varies in composition between bacterial strains and hence this is used for serotyping for bacteria (e.g., Salmonella typhi - widal test). It is also a major surface antigen, induces antibody formation. Core polysaccharide: It is projected from lipid A region. It is composed of 10–12 sugar moieties. Lipid A or the endotoxin: Responsible for endotoxic activities such as pyrogenicity, tissue necrosis, anti-tumor activity, etc.

• O side chain (or O antigen or somatic antigen): A polysaccharide chain which greatly varies in composition between bacterial strains and hence this is used for serotyping for bacteria (e.g., Salmonella typhi - widal test). It is also a major surface antigen, induces antibody formation.

• Core polysaccharide: It is projected from lipid A region. It is composed of 10–12 sugar moieties.

• Lipid A or the endotoxin: Responsible for endotoxic activities such as pyrogenicity, tissue necrosis, anti-tumor activity, etc.

• Periplasmic Space: It is the space between the inner cell membrane and outer membrane. It encompasses the peptidoglycan layer.

• Cell membrane (plasma membrane): The semipermeable membrane acts as an osmotic barrier and is composed of bilayered phospholipids. Lacks sterols, such as cholesterol (except in Mycoplasma). Site for metabolic processes like respiration and synthesis of lipids and cell walls.

• The semipermeable membrane acts as an osmotic barrier and is composed of bilayered phospholipids.

• Lacks sterols, such as cholesterol (except in Mycoplasma).

• Site for metabolic processes like respiration and synthesis of lipids and cell walls.

• It is a phospholipid layer, attached to the thin peptidoglycan layer by a membrane protein called Braun's lipoprotein .

• Contains outer membrane proteins (OMP), which help in transport of small molecules.

• O side chain (or O antigen or somatic antigen): A polysaccharide chain which greatly varies in composition between bacterial strains and hence this is used for serotyping for bacteria (e.g., Salmonella typhi - widal test). It is also a major surface antigen, induces antibody formation.

• Core polysaccharide: It is projected from lipid A region. It is composed of 10–12 sugar moieties.

• Lipid A or the endotoxin: Responsible for endotoxic activities such as pyrogenicity, tissue necrosis, anti-tumor activity, etc.

• The semipermeable membrane acts as an osmotic barrier and is composed of bilayered phospholipids.

• Lacks sterols, such as cholesterol (except in Mycoplasma).

• Site for metabolic processes like respiration and synthesis of lipids and cell walls.

Teichoic acids are polymers which are present only in gram-positive bacteria .

• These molecules help maintain the structure of the cell wall.

• They are of two types: (i) Cell wall teichoic acid and (ii) Lipoteichoic acid (Options A & D) .

Solution for Question 17:

Correct Answer: A) Staphylococcus epidermidis

Explanation:

The clinical history is suggestive of endocarditis and the most likely causative organism in this case is Staphylococcus epidermidis, due to the formation of biofilm over the prosthetic valve.

- A biofilm is a complex structure made up of millions of bacteria that stick to surfaces and are encased in a self-produced layer of extracellular polymers, often called a polysaccharide slime layer.
- Persistent biofilms with pathogenic bacteria can attach to damaged tissues and medical devices, like catheters and pacemakers.
- This attachment is the first step in bacterial colonization, which can sometimes lead to infections, such as prosthetic valve endocarditis and catheter-associated urinary tract infections.

Biofilm advantages: Helps in adherence Anti- phagocytic Antibiotic resistance	Examples: Streptococcus mutans In mouth - cause dental caries Forms a biofilm over the teeth. Staphylococcus epidermidis - causes prosthetic heart valve endocarditis. Biofilm is also shown by Pseudomonas aeruginosa

Escherichia coli (Option B): This is a gram-negative bacterium typically associated with urinary tract infections and abdominal infections.

Streptococcus mutans (Option C): While S. mutans is part of the normal oral flora and can occasionally cause endocarditis, it is more commonly linked with native valve endocarditis after dental procedures rather than prosthetic valve infections.

Streptococcus pyogenes (Option D): Although S. pyogenes (Group A Streptococcus) can cause a range of infections, including pharyngitis and skin infections, it is not a common cause of prosthetic valve endocarditis.

Solution for Question 18:

Correct Answer: C) McFadyean reaction

Explanation

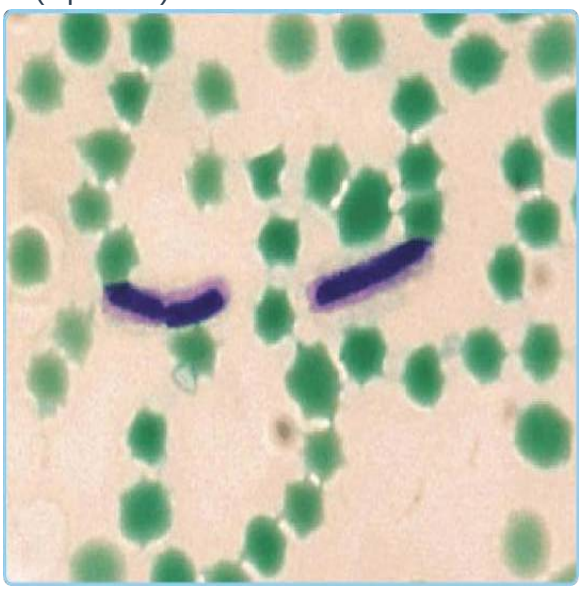
- The symptoms of eschar on arms after handling animal hides is consistent with cutaneous anthrax disease.
- It is caused by Bacillus anthracis, which has a polypeptide capsule that can be demonstrated using the McFadyean reaction.

Methods for Detecting Capsules

- Negative Staining with India Ink and Nigrosin Stain (Option A):

The capsule is seen as clear halos around the bacteria, against a black background (image of cryptococcus).

- McFaydean Reaction (Option C):

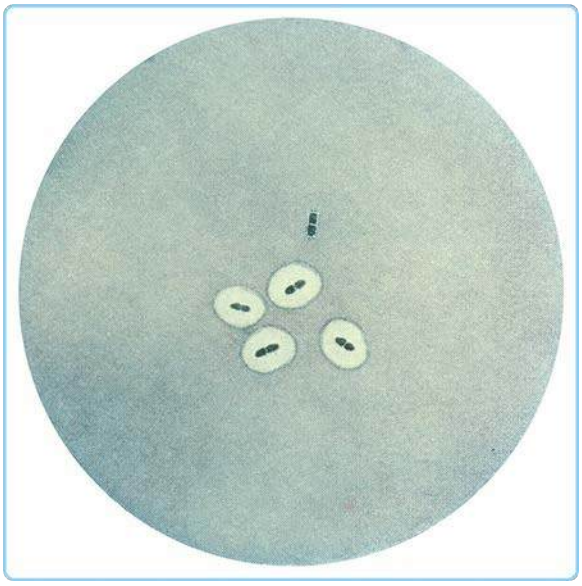


This method uses polychrome methylene blue stain to demonstrate the capsule of Bacillus anthracis (image amorphous purple capsule surrounding blue bacilli of B. anthracis).

- Serological Test:

The capsular material, being antigenic, can be detected by mixing it with specific anticapsular serum.

- Quellung Reaction (Option B):



Capsular serotypes of Streptococcus pneumoniae can be identified by adding antisera mixed with methylene blue, causing the capsule to swell, become refractile, and be clearly delineated (image of S. pneumoniae).

- Capsular Antigen Detection:

Capsular antigens can be identified in samples (e.g., CSF) using a latex agglutination test (Option D) with specific anticapsular antibodies coated on latex particles, applicable for pathogens like pneumococcus, Cryptococcus, Haemophilus influenzae, and meningococcus.

Bacterial genetics

1. Which bacterial species is not naturally capable of undergoing transformation?

A. Streptococcus pneumoniae

B. Escherichia coli

C. Neisseria gonorrhoeae

D. Haemophilus influenzae

---

2. Transduction is the most common mode of resistance in which of the following organisms?

A. Escherichia coli

B. Staphylococcus aureus

C. Mycobacterium tuberculosis

D. Pseudomonas aeruginosa

---

3. Transduction in bacteria involves the transfer of genetic material via bacteriophages. Which of the following statements is true about specialized transduction?

A. It can transfer any segment of the donor bacterium's DNA.

B. It occurs due to packaging errors during the lytic cycle.

C. It transfers specific genes located near the prophage's integration site.

D. It destroys the host cell.

---

4. Match the bacterial cell type with its correct description A) Hfr 1) A bacterial cell that lacks the F plasmid and can act only as a recipient in conjugation. B) F+ 2) A bacterial cell with an F plasmid that has excised from the chromosome and carries some chromosomal genes C) F■ 3) A bacterial cell that can act as a donor in conjugation, having the F plasmid incorporated into its chromosomal DNA D) F' 4) A bacterial cell that has an F plasmid separate from its chromosomal DNA and can initiate conjugation

A) Hfr	1) A bacterial cell that lacks the F plasm
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C) F■	3) A bacterial cell that can act as a donor in conjugation, having the
D) F'	4) A bacterial cell that has an F plasmid separate from its chromoso

A. A - 1; B - 2; C - 3; D - 4

B. A - 2; B - 1; C - 4; D - 3

C. A - 4; B - 3; C - 1; D - 2

D. A - 3; B - 4; C - 1; D - 2

---

5. Which of the following statements are false regarding transferable drug resistance?

A. Shows a high degree of resistance

B. Cannot be overcome by drug combinations

C. Virulence not decreased

D. Exhibits resistance to one drug at a time

---

6. Which statement about restriction endonucleases(RE) is false?

A. They cleave double-stranded DNA at specific sequences

B. They require divalent metal ions like Mg2+

C. Bacteria protect their own genome from RE by Methylation

D. They only produce blunt ends

---

7. What is the main function of the Cas9 enzyme in the CRISPR-Cas9 gene-editing system?

A. To replicate DNA

B. To repair DNA

C. To cut DNA at specific locations

D. To transcribe RNA

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	4
Question 5	4
Question 6	4
Question 7	3

Solution for Question 1:

Correct Answer: B) Escherichia coli

Explanation:

- Escherichia coli is not capable of natural transformation and does not undergo transformation under normal conditions.
- It requires specific laboratory techniques (such as chemical or electroporation treatments) to artificially induce transformation competence.
- Natural genetic competence , the ability of bacteria to take up exogenous DNA, is common and plays a key role in horizontal gene transfer.
- This process is controlled by specialized proteins that form a DNA-uptake complex.
- Competence development is regulated by cell-cell signalling and environmental factors, such as nutrient availability.

Bacterial Transformation:

Definition	Uptake of free DNA fragments by a bacterial cell, leading to heritable changes.
Growth Phase	Log phase of bacterial growth.
Naturally Capable Bacteria	Streptococcus pneumoniae (Option A ruled out) Haemophilus influenzae (Option D ruled out) Neisseria gonorrhoeae (Option C ruled out) Bacillus Acinetobacter Pseudomonas
Historical Significance	Griffith's Experiment (1928): Demonstrated transformation using Streptococcus pneumoniae in mice. Avery, MacLeod, and McCarty (1944): Identified DNA as the transforming principle.

Solution for Question 2:

Correct Answer: B) Staphylococcus aureus

Explanation:

Transduction is a significant mode of resistance transfer in S. aureus, particularly for genes like penicillin resistance.

Bacterial Transduction

- Definition : Transduction is a process where bacteriophages (viruses that infect bacteria) transfer genetic material between bacteria.
- Types : Generalised and specialized.
- Occurrence : Due to errors during the lytic cycle of virulent and sometimes temperate phages, which ends with the destruction of the host cell.
- Process : Any part of the donor bacterium's genome can be transferred when a phage accidentally packages host DNA instead of its own.
- Outcomes : Abortive Transduction : Donor DNA does not integrate and disappears. Stable Gene Transfer : Donor DNA integrates into the recipient's chromosome, causing a permanent change. Unstable Gene Transfer : Donor DNA is broken down and does not cause a lasting change.
- Abortive Transduction : Donor DNA does not integrate and disappears.

• Stable Gene Transfer : Donor DNA integrates into the recipient's chromosome, causing a permanent change.

• Unstable Gene Transfer : Donor DNA is broken down and does not cause a lasting change.

• Abortive Transduction : Donor DNA does not integrate and disappears.

• Stable Gene Transfer : Donor DNA integrates into the recipient's chromosome, causing a permanent change.

• Unstable Gene Transfer : Donor DNA is broken down and does not cause a lasting change.

• Occurrence : Linked to temperate phages that integrate their DNA into the host's chromosome (prophage).

• Process : Specific genes are transferred when a prophage incorrectly excises from the host chromosome, taking nearby bacterial genes during the transition from the lysogenic to the lytic cycle.

• Example : Lambda phage in E. coli, transferring genes for galactose utilization or biotin synthesis.

• Antibiotic Resistance : Transduces plasmids carrying antibiotic resistance genes (e.g., penicillin resistance in Staphylococcus aureus).

• Toxin Production : Bacteria gain virulence by acquiring toxin genes (e.g., Corynebacterium diphtheriae acquires a diphtheria toxin gene via lysogenic conversion).

Escherichia coli (Option A): While E. coli can acquire resistance through various mechanisms, transduction is not its most common mode of resistance.

Mycobacterium tuberculosis (Option C) primarily acquires resistance through mutations and horizontal gene transfer mechanisms other than transduction.

Pseudomonas aeruginosa (Option D) commonly acquires resistance through plasmid-mediated conjugation and mutations, rather than transduction.

Solution for Question 3:

Correct Answer: C) It transfers specific genes located near the prophage's integration site .

Explanations:

Specialized transduction involves transferring specific genes adjacent to the prophage integration site.

• Lytic Cycle : Virulent cycle. Bacteriophage replicates inside the host. Results in host cell death and release of new phages. (Option D ruled out)

• Virulent cycle.

• Bacteriophage replicates inside the host.

• Results in host cell death and release of new phages. (Option D ruled out)

• Lysogenic Cycle : Phage DNA integrates into the host's chromosome. Replicates with the host without harming it. Integrated phage DNA is called a prophage. The process is known as lysogenic conversion. There are two types of phages: Virulent phage: follow only the lytic cycle Temperate phage: follow the lytic and lysogenic cycle

• Phage DNA integrates into the host's chromosome.

• Replicates with the host without harming it.

• Integrated phage DNA is called a prophage.

• The process is known as lysogenic conversion.

• There are two types of phages: Virulent phage: follow only the lytic cycle Temperate phage: follow the lytic and lysogenic cycle

• Virulent phage: follow only the lytic cycle

• Temperate phage: follow the lytic and lysogenic cycle

• Virulent cycle.

• Bacteriophage replicates inside the host.

• Results in host cell death and release of new phages. (Option D ruled out)

• Phage DNA integrates into the host's chromosome.

• Replicates with the host without harming it.

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• There are two types of phages: Virulent phage: follow only the lytic cycle Temperate phage: follow the lytic and lysogenic cycle

• Virulent phage: follow only the lytic cycle

• Temperate phage: follow the lytic and lysogenic cycle

• Virulent phage: follow only the lytic cycle

• Temperate phage: follow the lytic and lysogenic cycle

• Generalized Transduction : (Options A&B; ruled out) Usually utilizes the lytic cycle Can transfer any segment of the donor bacterium's DNA. Occurs due to packaging errors during the lytic cycle. Fragments of host DNA are incorporated into phage particles. Phages inject donor DNA into a new recipient bacterium.

• Usually utilizes the lytic cycle

• Can transfer any segment of the donor bacterium's DNA.

• Occurs due to packaging errors during the lytic cycle.

• Fragments of host DNA are incorporated into phage particles.

• Phages inject donor DNA into a new recipient bacterium.

• Specialized Transduction : Usually utilizes the lysogenic cycle Transfers specific genes near the prophage's integration site. (Option C) Occurs when the prophage is excised from the host chromosome. Phage carries adjacent bacterial genes along with it. Example: Lambda phage in E. coli integrates between genes for galactose utilization and biotin synthesis, transducing only these genes.

• Usually utilizes the lysogenic cycle

• Transfers specific genes near the prophage's integration site. (Option C)

• Occurs when the prophage is excised from the host chromosome.

• Phage carries adjacent bacterial genes along with it.

• Example: Lambda phage in E. coli integrates between genes for galactose utilization and biotin synthesis, transducing only these genes.

• Usually utilizes the lytic cycle

• Can transfer any segment of the donor bacterium's DNA.

• Occurs due to packaging errors during the lytic cycle.

• Fragments of host DNA are incorporated into phage particles.

• Phages inject donor DNA into a new recipient bacterium.

• Usually utilizes the lysogenic cycle

• Transfers specific genes near the prophage's integration site. (Option C)

• Occurs when the prophage is excised from the host chromosome.

• Phage carries adjacent bacterial genes along with it.

• Example: Lambda phage in E. coli integrates between genes for galactose utilization and biotin synthesis, transducing only these genes.

Solution for Question 4:

Correct answer: D) A - 3; B - 4; C - 1; D - 2

Explanation:

• Hfr -A bacterial cell that can act as a donor in conjugation, having the F plasmid incorporated into its chromosomal DNA.

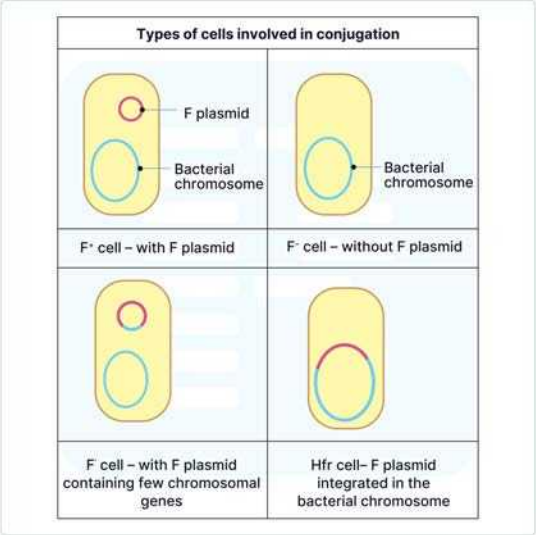
• F+ : A bacterial cell that has an F plasmid separate from its chromosomal DNA and can initiate conjugation.

• F■ : A bacterial cell that lacks the F plasmid and can act only as a recipient in conjugation.

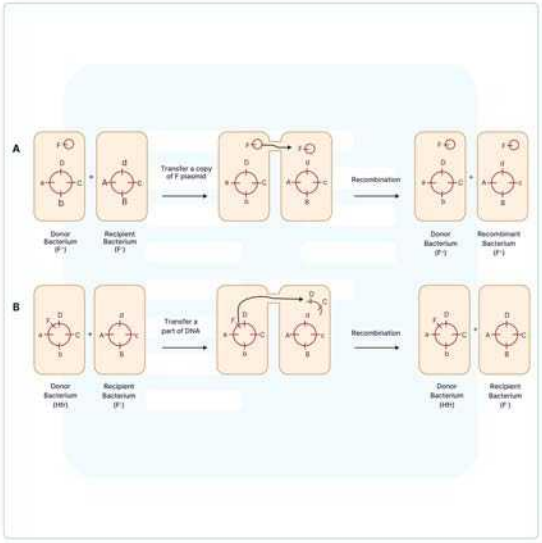
• F' : A bacterial cell with an F plasmid that has excised from the chromosome and carries some chromosomal genes.

Conjugation:

Transfer of genetic material from one bacterium (donor or male) to another bacterium (recipient or female) by mating or contact with each other and forming the conjugation tube.



F+xF- Mating	Hfr conjugation	F' conjugation
The F+ cell donor or male bacterium contains the F factor plasmid and encodes for sex pilus formation and plasmid transfer. During conjugation, the F pilus forms a bridge to transfer a copy of the F factor plasmid to the F■ recipient cell, converting it into an F+ cell. This process allows the new F+ cell also to initiate conjugation with other F■ cells. F+ x F■ = F+	The F factor plasmid can integrate into the bacterial chromosome and act as an episome, forming Hfr cells (high-frequency recombination cells). During conjugation, only a portion of the chromosomal genes and part of the F factor are transferred to the F■ recipient cell. F■ recipient does not become an F+ cell because the complete F factor is not transferred. Hfr with F■ = rF■	F factor can revert to a free state and carry adjacent chromosomal DNA, forming an F' factor. When an F' cell conjugates with an F■ recipient, it transfers both the F factor and the incorporated chromosomal DNA, converting the recipient into an F' cell. F' x F = F' and F'. This process is known as sexduction.



Solution for Question 5:

Correct answer: D) Exhibits resistance to one drug at a time

Explanation: Transferable drug resistance is transferable to other organisms by horizontal spread (conjugation).

- Transferable drug resistance often involves plasmids or transposons that can carry genes for resistance to multiple drugs simultaneously, not just one drug at a time.
- This is a hallmark of multidrug-resistant organisms (MDR).

Drug Resistance:

The ability of a bacterium to resist a class of antimicrobial agents due to its inherent structural or functional characteristics.

Types of drug resistance

Mutational Drug Resistance	Transferrable Drug Resistance
Resistance to one drug at a time	Multiple drug resistance at the same time (Option D)
Low-degree resistance	High-degree resistance (Option A ruled out)

Mutational Drug Resistance	Transferrable Drug Resistance
Resistance can be overcome by a combination of drugs	Cannot be overcome by drug combinations (Option B ruled out)
The virulence of resistance mutants may be lowered	Virulence not decreased (Option C ruled out)
Resistance is not transferable to other organisms. Spread to off-springs by vertical spread only	Resistance is transferable to other organisms. Spread by: Horizontal spread (conjugation, or rarely by transduction/transformation)
Eg. Mycobacterium tuberculosis resistance to Rifampicin	Eg. R plasmid in bacteria- E.coli resistance to ampicillin, tetracycline, and chloramphenicol

Solution for Question 6:

Correct Answer: D) They only produce blunt ends

Explanation:

Restriction digestion of double-stranded DNA produces two kinds of ends -Blunt ends and sticky ends

- Blunt ends- possess a 5’-phosphate group that promotes ligation. Compatible with other blunt-ended DNA.
- Sticky ends- small stretches of single-stranded DNA capable of self-ligation or ligation with a complementary region from another DNA molecule.
- The sticky ends possess 3’- or 5’-overhangs of 1–4 nucleotides

Restriction endonucleases (Restriction Enzymes)

- Microbial enzymes that cleave ds-DNA at specific oligonucleotide sequences (Option A ruled out) .
- They identify a particular site of palindrome called the Restriction site and cut the DNA strands.
- For example, both the strands, i.e. the strand going from 5' to 3' and its complementary strand from 3' to 5' must be complementary.
- A typical nucleotide palindromic sequence would be TATA (5' to 3') and ATAT's complementary sequence from 3' to 5'.

• Restriction enzymes split DNA strands into fragments of varying lengths, which can be separated by gel electrophoresis. Eg. Eco RI, Hind III, Taq I

• The bacteria methylate their DNA at these identical sequences so that the restriction enzymes do not cut the DNA in their cell. (Option C ruled out)

• Most type II restriction endonucleases show an absolute requirement for divalent metal ions as cofactors for DNA cleavage - Mg2+ natural cofactor (Option B ruled out)

Types and activities of restriction enzymes:

Type	Description
I	Cleaves DNA at random sites far from its recognition sequence
II	Cleaves DNA at defined positions close to or within its recognition sequence
IIG	Cleaves outside its recognition sequence with both restriction endonuclease and methyltransferase enzymatic activities in the same protein
IIP	Cleaves symmetric targets and cleavage sites
IIS	Recognizes asymmetric sequences
III	Cleaves outside its recognition sequence and require two sequences in opposite directions within the same DNA
IV	Cleaves modified DNA (e.g. methylated DNA)

Solution for Question 7:

Correct Answer: C) To cut DNA at specific locations

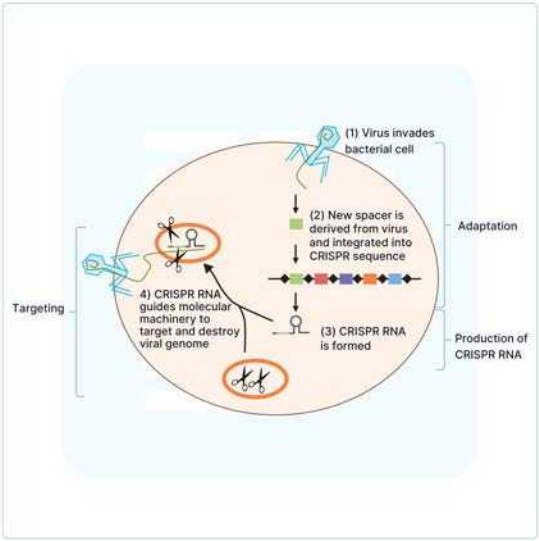
Explanation:

CRISPR system consists of an enzyme Cas9 that is used to cut through DNA at specific locations

CRISPR-Cas9

- Clustered Regularly Interspaced Short Palindromic Repeat.
- Unique organization of short, partially palindromic repeated DNA sequences found in the genomes of bacteria and other microorganisms.
- CRISPR's immune system can counter the attack of a viral infection by destroying the genome of the invading virus.
- The CRISPR immune system protects bacteria from ongoing viral infection by inhibiting its replication by destroying the viral genome.

Mechanism:



The CRISPR system has two components :

- Targeting device (a small strand of RNA programmed to look for a specific DNA sequence)
- Cutting device (an enzyme called Cas9 that can cut through DNA) (Options A, B & D ruled out)

The CRISPR immune system works via three basic steps:

Step 1: Adaptation – Viral DNA is cut into short pieces and added to the CRISPR sequence.

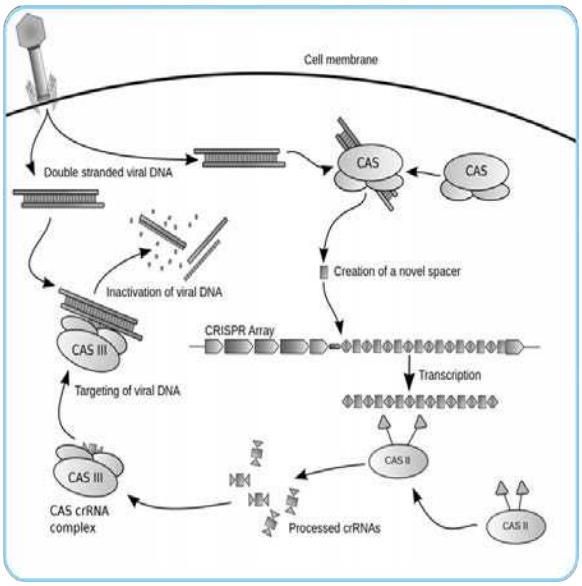
Step 2 : Production of CRISPR RNA – The CRISPR sequence is transcribed into RNA, which is then cut into smaller CRISPR RNAs.

Step 3 : Targeting – These CRISPR RNAs guide bacterial proteins to destroy the matching viral DNA.

CRISPR Defense Against Viruses:

(a) Immunization

(b) Interference:



This dual process of immunization and interference enables CRISPR to provide an adaptive immune defense against viral infections.

Culture Media

1. Which culture method is useful in motility testing?

A. Streak Culture

B. Liquid Culture

C. Lawn Culture

D. Stab Culture

---

2. Which of the following describes the correct process used in the pour plate technique?

A. Mix the sample with molten agar and pour it onto a solidified agar plate.

B. Adding the sample onto the solidified agar surface and allowing colonies to grow.

C. Use a glass spreader to evenly distribute the sample on the surface of molten agar.

D. Colonies grow primarily on the surface.

---

3. Which of the following statements about agar is TRUE?

A. Agar has a high nutritional value for bacteria.

B. Agar melts at 42°C and solidifies at 98°C

C. Agar is derived from seaweed and is used to solidify culture media.

D. Solid media typically use a concentration of approximately 5% agar.

---

4. All of the following are examples of enriched media except

A. Blood agar

B. MacConkey agar

C. Chocolate agar

D. Brain-heart infusion (BHI) broth

---

5. Which of the following enrichment broths is used to enhance the growth of group B streptococci?

A. Selenite F medium

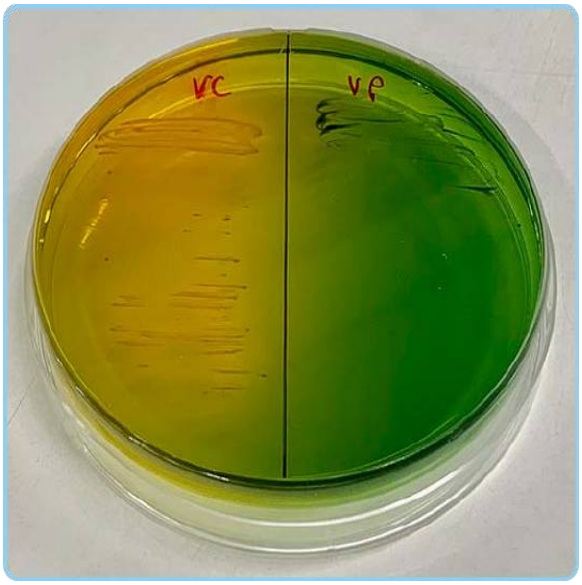
B. Tetrathionate broth

C. Lim broth

D. Alkaline peptone water

---

6. Which of the following selective media is used to isolate and differentiate Vibrio species based on their ability to ferment sucrose?



A. Mannitol Salt Agar (MSA)

B. MacConkey Agar (MAC)

C. Thiosulfate Citrate Bile Salts Sucrose (TCBS) Agar

D. Potassium Tellurite Media

---

7. Which medium is selectively used for isolating Pseudomonas aeruginosa from specimens with mixed microbiota?

A. PPLO agar

B. DCA (Desoxycholate Citrate Agar)

C. Thayer-Martin Medium

D. Cetrimide Agar

---

8. Which medium is specifically used for processing urine specimens and differentiates between lactose fermenters (LF) and non-lactose fermenters (NLF) producing a yellow colour with LF?

A. MacConkey Agar

B. CLED Agar

C. Blood Agar

D. Chocolate Agar

---

9. Match the transport medium with the organism it is commonly used for: 1. Stuart medium a) Vibrio cholerae 2. VR medium b) Shigella, Salmonella 3. Buffered glycerol saline c) Neisseria

1. Stuart medium	a) Vibrio cholerae

2. VR medium	b) Shigella, Salmonella
3. Buffered glycerol saline	c) Neisseria

A. 1-a, 2-b, 3-c

B. 1-c, 2-b, 3-a

C. 1-c, 2-a, 3-b

D. 1-b, 2-a, 3-c

---

10. Which medium is specifically designed with an oxygen indicator and supports growth at different levels based on oxygen requirements?

A. Robertson’s cooked meat (RCM) broth

B. Thioglycollate broth

C. BHIS agar

D. Anaerobic blood agar

---

11. Which is the most commonly used method for anaerobiosis, especially for laboratories with less sample load that uses reduced methylene blue as a chemical indicator?

A. Anoxomat

B. GasPak system

C. Anaerobic Glove Box

D. McIntosh and Filde’s anaerobic jar

---

12. What is the Minimal Inhibitory Concentration (MIC) of an antibiotic?

A. The highest concentration of the antibiotic that inhibits bacterial growth

B. The lowest concentration of the antibiotic that inhibits bacterial growth

C. The lowest concentration of the antibiotic that kills the bacteria

D. The highest concentration of the antibiotic that kills the bacteria

---

13. In the Broth Dilution Method, what medium is typically used for serial dilutions of the antimicrobial agent?

A. Molten agar

B. Mueller-Hinton broth

C. Sabouraud dextrose agar

D. Blood agar

---

14. Which statement is NOT true regarding the Kirby-Bauer method?

A. Apply 4-6 antibiotic discs per 9 cm plate

B. Factors such as the nature and composition of the medium can influence the zone diameter.

C. Incubation is typically done overnight

D. Disc concentration does not affect the zone of inhibition diameter.

---

15. In antibiotic susceptibility testing using the Stokes Method, bacteria susceptible to the drug at higher than normal dosages are termed as:

A. Susceptible

B. Resistant

C. Intermediately susceptible

D. Extremely susceptible

---

16. In a 65 y/o male with COPD showing signs of a bacterial respiratory infection, the sputum sample undergoes culture and sensitivity testing using the Epsilometer (E-test). Following incubation, an elliptical zone of inhibition is observed around the antibiotic strip. What is the primary advantage of using the E-test in this clinical context?

A. It identifies the specific bacterium causing the infection.

B. It determines the bacterium's resistance to all known antibiotics.

C. It provides a visual and precise measurement of the minimum inhibitory concentration of the antibiotic.

D. It allows for the immediate administration of antibiotics without further testing.

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	2
Question 5	3
Question 6	3
Question 7	4
Question 8	2
Question 9	3
Question 10	2
Question 11	2
Question 12	2
Question 13	2
Question 14	4
Question 15	3
Question 16	3

Solution for Question 1:

Correct Answer: D) Stab Culture

Explanation:

Stab Culture is made by stabbing the semisolid agar butt with a straight wire.

It is used for motility testing using

- Mannitol motility medium and
- Triple sugar iron agar test.

CULTURE TECHNOLOGIES		
Culture Type	Uses	Picture
Streak Culture (Option A)	Primary inoculum smeared onto solid media. Secondary and tertiary inoculum formed by streaking parallel lines. Intermittent heating to obtain isolated colonies.	{{caption_text}}
Stroke Culture	Zigzag streaking on agar slopes or slants. Used for biochemical tests like urease test.	
Lawn Culture (Option C)	Uniform bacterial growth is obtained by swabbing or flooding. Used for antimicrobial susceptibility testing (AST) by disk diffusion method.	{{caption_text}}
Stab Culture (Option D)	Agar butt stabbed with straight wire. Used for motility testing and triple sugar iron agar test.	{{caption_text}}
Liquid Culture (Option B)	Used for blood or body fluids. Bacterial growth is detected by turbidity or surface pellicles. Advantages: Suitable for specimens with antibiotics, large yields. Disadvantages: It doesn't provide pure culture from mixed inoculum, and no visible colonies.	{{caption_text}}

Solution for Question 2:

Correct Answer: B) Adding the sample onto the solidified agar surface and allowing colonies to grow.

Explanation:

• In the pour plate method, the sample is added onto the solidified agar surface allowing colonies to grow.

• This allows for the colonies to grow both within the agar medium and on its surface, facilitating the enumeration of various types of bacteria including aerobes, anaerobes, and facultative anaerobes.

• Advantages : Provides quantitative data on viable bacterial count per ml of the original sample. Allows visualization and counting of individual colonies.

• Provides quantitative data on viable bacterial count per ml of the original sample.

• Allows visualization and counting of individual colonies.

• Provides quantitative data on viable bacterial count per ml of the original sample.

• Allows visualization and counting of individual colonies.

Feature	Pour Plate Technique	Spread Plate Technique
Process	The sample is added onto the solidified agar surface.	The sample is mixed with molten agar and poured. (Option A)
Accuracy	Generally considered more accurate.	Slightly less accurate compared to the pour plate.
Sample Spreading	The sample is not spread; colonies grow within agar.	The sample is spread using a glass spreader. (Option C)
Types of Bacteria	Aerobes, anaerobes, facultative anaerobes.	Aerobes only.
Medium State	Solidified agar plates are necessary.	Molten agar medium is used.
Visibility of Colonies	Colonies can grow both within and on the surface of agar.	Colonies grow primarily on the surface. (Option D)
Image	{{caption_text}}	{{caption_text}}

Solution for Question 3:

Correct Answer: C) Agar is derived from seaweed and is used to solidify culture media.

Explanation:

Properties of Agar:

- Agar is extracted from seaweed and is commonly used to solidify culture media for bacterial growth.
- Agar has negligible nutritional value for bacteria; it is mainly used as a solidifying agent. (Option A ruled out)
- Agar melts at 98°C and solidifies at around 42°C. (Option B ruled out)
- Solid media typically use a concentration of approximately 2% agar. (Option D ruled out)

Simple media, also known as basal media, is a type of culture media used in laboratories to grow bacteria.

Simple Media	
Nutrient Broth Peptone NaCl Water Meat Extract {{caption_text}}	Nutrient Agar Peptone NaCl Water Meat Extract 2% Agar {{caption_text}}

Solution for Question 4:

Correct answer: B) MacConkey agar

Explanation:

- MacConkey agar is a selective and differential medium , not an enriched medium. It is used to isolate Gram-negative bacteria and differentiate them based on lactose fermentation.
- Enriched media contain growth enhancers added to nonselective agar to support the growth of fastidious organisms.

Enriched Media		
Name	Characteristics	Image
Blood agar (Option A ruled out)	Base medium enriched with 5-10% defibrinated sheep blood, used to grow moderately fastidious bacteria and observe hemolytic reactions.	{{caption_text}}
Chocolate agar (Option C ruled out)	This is an enriched medium made by heating blood agar, which lyses red blood cells and releases nutrients such as haemoglobin and NAD. It supports the growth of fastidious bacteria such as Haemophilus and Neisseria species.	{{caption_text}}
Brain-heart infusion (BHI) broth (Option D ruled out)	This is an enriched medium containing nutrients from brains and beef hearts, along with peptones, glucose, sodium chloride, and buffers. It supports the growth of non-fastidious and moderately fastidious microorganisms.	infusion (BHI) broth" data-author="Ajpolino" data-hash="9481" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:BHI_media.png" data-tags="April2025" height="429" src="https://image.prepladder.com/notes/OVRbuoQF6OhaH7nvgUZX1744385843.png" width="200" /> {{caption_text}}
Loeffler serum slope	An enriched medium with high serum content, prepared with eggs. Primarily used for the recovery and identification of Corynebacterium diphtheriae from nose and throat specimens.	serum slope" data-author="Albaraa Mehdar" data-hash="9482" data-license="CC BY NC 2.0" data-source="https://www.flickr.com/photos/albaraamehdar/4599370854" data-tags="April2025" height="142" src="https://image.prepladder.com/notes/TRC67Pq2K2uB2yvBYtKN1744385881.png" width="200" /> {{caption_text}}
Lowenstein-Jensen medium:	Used specifically for cultivating Mycobacterium species, this enriched, selective medium contains glycerol, mineral salts, aspargine, malachite green, and hen’s eggs.	{{caption_text}}

Solution for Question 5:

Correct answer: C) Lim broth

Explanation:

Enrichment Broth:

- Liquid media with inhibitory agents selectively promote the growth of specific organisms and inhibit others.
- Used for isolating pathogens from clinical specimens containing normal flora (e.g., stool, sputum).

Examples:

Lim Broth is an enrichment broth specifically used to enhance the growth of group B streptococci. It provides the necessary nutrients and conditions for the selective multiplication of these organisms.

Selenite F medium (Option A): This enrichment medium is used primarily for stool samples to suppress gut bacteria growth while allowing diarrheagenic bacteria (Salmonella, Shigella) to grow. It is not specific to group B streptococci.

Tetrathionate broth (Option B): This enrichment medium inhibits coliforms while allowing the growth of typhoid-paratyphoid bacilli.

Alkaline peptone water (Option D): This enrichment medium is used to recover Vibrio and Aeromonas species from stool samples.

Solution for Question 6:

Correct answer: C) Thiosulfate Citrate Bile Salts Sucrose (TCBS) Agar

Explanation:

- TCBS is a selective medium that inhibits gram-positive bacteria and many gram-negative rods with bile salts and other components.
- It differentiates Vibrio species based on their ability to ferment sucrose , with sucrose fermenters producing yellow colonies and non-fermenters producing blue-green colonies.

Selective Media:

They are solid media containing inhibitory substances that inhibit the normal flora present in the specimen and allow the pathogens to grow.

Selective Media		
Media	Property	Image
Mannitol Salt Agar (MSA) (Option A ruled out)	The high salt concentration (7.5%) inhibits most bacteria, except Staphylococcus. Staphylococcus aureus ferments mannitol, producing yellow colonies with a yellow zone, while other Staphylococci and Micrococcus sp. do not ferment mannitol and produce reddish colonies.	Agar (MSA)" data-author="Ajay Kumar Chaurasiya" data-hash="9530" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Mannitol_salt_agar_(MSA)_with_growth_of_S._aureus,_CoNS_and_no_growth_of_E._coli.jpg" data-tags="April2025" height="181" src="https://image.prepladder.com/notes/wVJcM3Lnro8ebBcbGnIg1744388775.png" width="200" /> {{caption_text}}
MacConkey Agar (MAC) (Option B ruled out)	MacConkey Agar (MAC) consists of peptone, lactose, agar, neutral red, and taurocholate. It selectively and differentially isolates gram-negative enteric bacilli via lactose fermentation: Lactose fermenters produce pink/red colonies, and Non-fermenters produce colourless/transparent colonies.	Agar (MAC)" data-author="Medimicro" data-hash="9533" data-license="OPEN ACCESS" data-source="https://commons.wikimedia.org/wiki/File:MacConkey_agar_with_LF_and_LF_colonies.jpg" data-tags="April2025" height="150" src="https://image.prepladder.com/notes/L4rnhNtxxsAAhyVvGjgP1744388917.png" width="200" /> {{caption_text}}
Potassium Tellurite Media (Option D ruled out)	Potassium tellurite media, like potassium tellurite blood agar and Tinsdale agar, selectively isolate Corynebacterium diphtheriae by inhibiting most other bacteria. C. diphtheriae reduces tellurite, forming grey to black colonies.	{{caption_text}}
CLED Medium	A type of differential medium, akin to MacConkey agar, distinguishes between lactose fermenters (LF) and non-lactose fermenters (NLF). It serves as a substitute for using both blood agar and MacConkey agar when processing urine samples .	{{caption_text}}

Medium	Indicator	Colour
Mannitol Salt agar S. aureus	Phenol red	MF (yellow)
TCBS V. cholera	Bromothymol blue	Sucrose fermentation (yellow)
CLED	Bromothymol blue	Lactose fermentation (yellow)
Mac Conkey	Neutral red	Lactose fermentation (pink)

Solution for Question 7:

Correct answer D) Cetrimide Agar

Explanation:

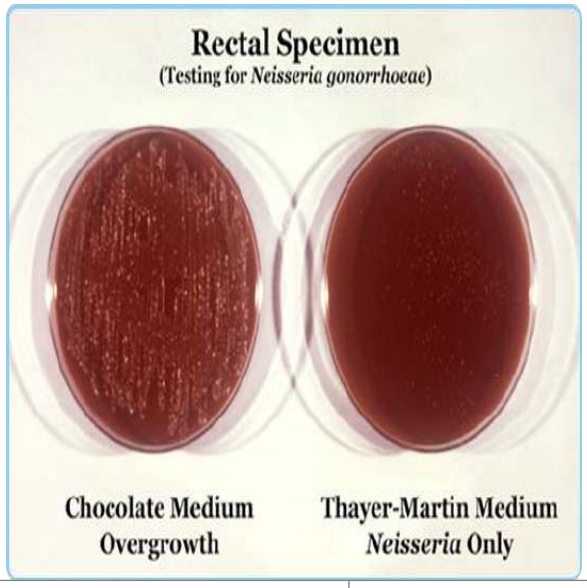
- Cetrimide agar is specifically formulated to select Pseudomonas aeruginosa by inhibiting most gram-positive and gram-negative bacteria except for P. aeruginosa and P. fluorescens.
- It contains cetrimide, magnesium chloride, and potassium sulfate to enhance the growth and pigment production (pyocyanin) characteristic of Pseudomonas aeruginosa.



PPLO Agar (Option A): This medium isolates Mycoplasma species. Antimicrobials can be added to inhibit the growth of other bacteria.

DCA (Desoxycholate Citrate Agar) (Option B): They are employed to isolate gastrointestinal pathogens like Salmonella and Shigella from faecal specimens.

Thayer-Martin Medium (Option C): This selective medium is used to isolate Neisseria gonorrhoeae from clinical specimens. The medium contains the following antimicrobials, which suppress the growth of other organisms: vancomycin, colistin, and nystatin. Thayer-Martin plates are incubated in an atmosphere containing 3-10% CO2.



	Other differential media & their characteristics
1. Polymyxin-Lysozyme-EDTA-Thallous Acetate (PLET) Medium: Acetate (PLET) Medium" data-author="Jesus Hernandez" data-hash="9578" data-license="CC BY SA 3.0" data-source="https://www.wikidoc.org/index.php/File:Bacillus_anthracis10.jpeg" data-tags="April2025" height="149" src="https://image.prepladder.com/notes/BiLZzRJQ5MZ3kiXD1N411744391760.png" width="200" /> {{caption_text}}	Purpose: Isolates Bacillus anthracis from mixtures with other spore-bearing bacilli. Key Components: Polymyxin, lysozyme, EDTA, thallous acetate, heart infusion agar.
2. Mannitol, Egg Yolk, Polymyxin B, Phenol Red, and Agar (MYPA): Polymyxin B, Phenol Red, and Agar (MYPA)" data-author="Haque, Md & Quan, Hongkun & Zuo, Zonghui & Khan, Ahrar & Siddique, Naila." data-hash="9547" data-license="NA" data-source="https://www.researchgate.net/figure/Bacillus-cereus-on-MYPA-agar_fig2_344672283" data-tags="April2025" height="155" src="https://image.prepladder.com/notes/QSTMmcS5zvsCoU6OC7XZ1744389481.png" width="300" /> {{caption_text}}	Purpose: Isolates Bacillus cereus from faeces. Key Components: Mannitol, egg yolk, polymyxin B, phenol red, agar.

	Other differential media & their characteristics
3. Bacillus Cereus Agar (PEMBA): Agar (PEMBA)" data-author="Haque, Md & Quan, Hongkun & Zuo, Zonghui & Khan, Ahrar & Siddique, Naila" data-hash="9550" data-license="NA" data-source="https://www.researchgate.net/figure/Bacillus-cereus-on-PEMBA-agar_fig3_344672283" data-tags="April2025" height="213" src="https://image.prepladder.com/notes/56ZsRyJRkliM5hvfLtYR1744389549.png" width="200" /> {{caption_text}}	Key Purpose: Selective isolation and enumeration of Bacillus cereus from food, animal feed, and environmental samples. Characteristics of Bacillus Cereus: Turquoise-blue colonies (2-5 mm) with an opaque egg yolk precipitation zone (up to 5 mm wide). Confirm by hemolysis test on sheep blood agar or microscopic examination.
4. Xylose-Lysine-Desoxycholate (XLD) Agar: {{caption_text}}	Purpose: Isolates Salmonella and Shigella from the stool and mixed biota. Key Components: Sodium desoxycholate, sucrose, lactose, xylose, phenol red, lysine, sodium thiosulfate, ferric ammonium citrate. Colony Types: Yellow colonies (e.g., E. coli) Yellow with black centres (e.g., Citrobacter, some Proteus spp.) Colourless/red colonies (e.g., Shigella, Providencia) Red with black centres (e.g., Salmonella, Edwardsiella)

Solution for Question 8:

Correct Answer: B) CLED Agar

Explanation:

- Differential media differentiate between two groups of bacteria by using an indicator, which changes the colour of the colonies of a particular group of bacteria but not the other group.
- CLED Agar (Cysteine Lactose Electrolyte-Deficient Agar) is used for processing urine specimens and differentiates between lactose fermenters (LF) and non-lactose fermenters (NLF) based on colony colour change.

Differential media		
Medium	Composition & Use	Differentiation

Differential media		
MacConkey (Option A) Agar (MAC)" data-author="Medimicro" data-hash="9533" data-license="OPEN ACCESS" data-source="https://commons.wikimedia.org/wiki/File:MacConkey_agar_with_LF_and_LF_colonies.jpg" data-tags="April2025" height="150" src="https://image.prepladder.com/notes/L4rnhNtxxsAAhyVvGjgP1744388917.png" width="200" /> {{caption_text}}	A differential and selective medium primarily used for isolating enteric gram-negative bacteria. Peptone, lactose, agar, neutral red (indicator), taurocholate	Lactose Fermenters (LF): Pink colonies (e.g., Escherichia coli); Non-Lactose Fermenters (NLF): Colorless colonies (e.g., Shigella species)
CLED Agar (Option B) {{caption_text}}	Cysteine lactose electrolyte-deficient agar Processing of urine specimens	LF produces a yellow color and NLF differentiated based on colony color change
Blood Agar (Option C) {{caption_text}}	Agar, blood, cysteine, hemin Growth of most aerobic and anaerobic bacteria; indicates hemolysis	Beta hemolysis: Complete lysis (clear zone, e.g., Group A Streptococci); Alpha hemolysis: Partial lysis (greenish discolouration, e.g., Viridans Streptococci); No hemolysis: No change (e.g., Enterococci)

Chocolate agar (Option D) is an enriched medium made by heating blood agar, which lyses red blood cells and releases nutrients such as haemoglobin and NAD. It supports the growth of fastidious bacteria such as Haemophilus and Neisseria species. It is not a differential media.

Solution for Question 9:

Correct Answer: C) 1-c, 2-a, 3-b

Explanation:

- Transport Media are designed for transporting specimens suspected to contain fastidious or delicate organisms, especially when there may be a delay in transporting the specimens from the collection site to the laboratory.
- Characteristics : Non-nutrient, soft agar gels Contain reducing agents to prevent oxidation Include substances like charcoal to neutralize bacterial inhibitors Bacteria do not multiply in

these media but remain viable

• Non-nutrient, soft agar gels

• Contain reducing agents to prevent oxidation

• Include substances like charcoal to neutralize bacterial inhibitors

• Bacteria do not multiply in these media but remain viable

• Non-nutrient, soft agar gels

• Contain reducing agents to prevent oxidation

• Include substances like charcoal to neutralize bacterial inhibitors

• Bacteria do not multiply in these media but remain viable

Organism	Transport media
Neisseria	Amies medium Stuart's medium
Vibrio cholerae	VR (Venkatraman-Ramakrishnan) medium Autoclaved seawater Cary Blair medium
Shigella, Salmonella	Buffered glycerol saline Cary Blair medium

Solution for Question 10:

Correct Answer: B) Thioglycollate broth

Explanation:

• Anaerobic media are used to grow anaerobic organisms in the presence of reducing substances or the absence of oxygen

• Thioglycollate broth contains an oxygen indicator and supports growth at different levels within the medium based on oxygen requirements: obligate aerobes at the top, microaerophiles in the middle, and anaerobes at the bottom.

ANAEROBIC MEDIA

Media	Uses	Picture
Robertson’s cooked meat broth (Option A)	Contains chopped beef hearts providing glutathione and unsaturated fatty acids. Widely used for anaerobic culture and stock cultures.	{{caption_text}}
Thioglycollate broth (Option B)	Supports the growth of organisms with varying oxygen needs. Boiled to remove oxygen; an indicator shows oxygen levels (pink or blue). Layers for obligate aerobes (top)-D, microaerophiles (middle)-C, and anaerobes (bottom)-B. Contains yeast extract, casitone, sodium chloride, l-cystine, thioglycollic acid, agar, and methylene blue.	{{caption_text}}

Media	Uses	Picture
BHI (Brain Heart Infusion) Agar (Option C)	Nutritious medium with calf brain and cow heart infusion. Contains peptone, dextrose, vitamin K, and hemin. Cultivation of fastidious organisms.	{{caption_text}}
Other Anerobic media; Anaerobic blood agar (Option D) Neomycin blood agar Egg yolk agar Phenyl ethyl agar Bacteroides bile esculin agar (BBE agar)		

Solution for Question 11:

Correct Answer: B) GasPak system

Explanation:

- Anaerobic methods involve creating oxygen-free environments using techniques like evacuation and replacement with inert gases (e.g., hydrogen) and chemical reactions to remove residual oxygen, essential for culturing obligate anaerobic bacteria in laboratories.
- GasPak System uses chemical reactions to remove oxygen. It is the most commonly used method for anaerobiosis, especially for laboratories with less sample load using reduced

methylene blue as a chemical indicator.

ANAEROBIC METHODS

Method	Uses	Image
McIntosh and Filde's Anaerobic Jar (Option D)	Manual Method of Evacuation and replacement of air with hydrogen gas. Catalyst : Aluminum pellets coated with palladium. Usage : Less common now due to automation but was historically popular	Anaerobic Jar" data-author="Ajay Kumar Chaurasiya" data-hash="9610" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:McIntosh_and_Fildes%E2%80%99_Anaerobic_Jar.jpg" data-tags="April2025" height="314" src="https://image.prepladder.com/notes/zWSL21tN36sJxXhTa2aH1744393704.png" width="200" /> {{caption_text}}
GasPak System (Option B)	Principle : Uses chemical reactions to remove oxygen. Components : Includes a sachet with sodium borohydride and sodium bicarbonate. Catalyst : Aluminum pellets coated with palladium. Indicator : Uses reduced methylene blue as a chemical indicator.	{{caption_text}}
Other methods: Anoxomat Anaerobic Glove Box and Anaerobic Work Station Pre-reduced Anaerobically Sterilized (PRAS) Media GENbag		

Anoxomat (Option A) automatically replaces air with hydrogen gas from a cylinder. Claims high effectiveness in creating anaerobic conditions.

Anaerobic Glove Box (Option C) provides a controlled environment for handling specimens without oxygen exposure.

Solution for Question 12:

Correct Answer: B) The lowest concentration of the antibiotic that inhibits bacterial growth

Explanation:

• Antimicrobial Susceptibility Testing (AST) helps clinicians choose appropriate antibiotics by determining which drugs are effective against the specific bacteria causing the infection. AST is conducted only on pathogenic bacteria isolated from clinical specimens, not on commensal bacteria.

• MIC is defined as the lowest concentration of an antibiotic that prevents visible growth of the microorganism in a standardized growth medium. (Options A & D ruled out)

Minimal Inhibitory Concentration (MIC)	The lowest concentration of the antibiotic that inhibits the growth of the pathogen in the medium.
Minimal Bactericidal Concentration (MBC)	The lowest concentration of antibiotic that kills the bacteria in a medium.
Methods	Dilution method Disc diffusion method E-Test (combo of first two methods) Automated methods

Clinical significance of MIC:

• If MIC > breakpoint value, then there is resistance

• If MIC < breakpoint value, then there is sensitivity

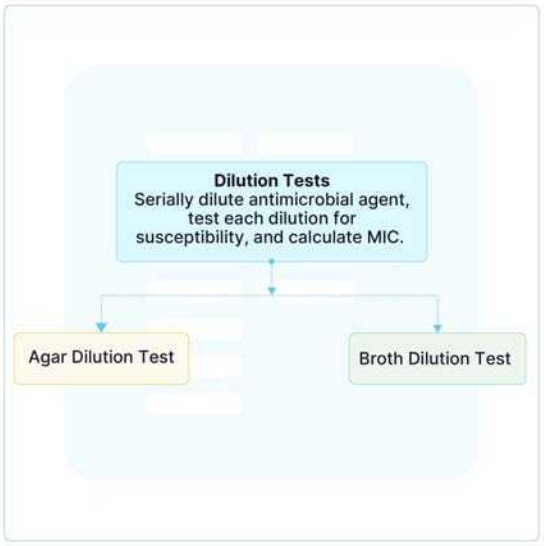
The lowest concentration of the antibiotic that kills the bacteria (Option C): This definition describes the minimal bactericidal concentration (MBC), not the MIC.

Solution for Question 13:

Correct Answer: B) Mueller-Hinton broth

Explanation:

Mueller-Hinton broth is used in the Broth Dilution Method for serial dilutions of the antimicrobial agent.



Minimum Inhibitory Concentration (MIC): Lowest concentration of antimicrobial agent that inhibits visible growth after overnight incubation.

Minimal bactericidal concentration (MBC): The lowest concentration of antibiotic that kills the bacteria in a medium.

1. Agar Dilution Method:

- Serial dilutions in molten agar (Option A ruled out) , poured into Petri dishes, spot inoculates test strain.
- Advantages: Test several strains simultaneously on the same plate. Directly measures MBC without the need for subculturing.
- Test several strains simultaneously on the same plate.
- Directly measures MBC without the need for subculturing.
- Test several strains simultaneously on the same plate.
- Directly measures MBC without the need for subculturing.

2. Broth Dilution Method:

- Types: Macro broth dilution (tubes) and micro broth dilution (microtiter plate).
- Procedure: Serial dilutions in Mueller-Hinton broth , inoculate with test organism, incubate at 37°C for 18 hours. Determine MIC: by lowest concentration with no visible growth (clear broth). Determine MBC: In Subculture tubes with no growth onto nutrient agar, the lowest concentration with no growth on subculture is MBC.
- Serial dilutions in Mueller-Hinton broth , inoculate with test organism, incubate at 37°C for 18 hours.
- Determine MIC: by lowest concentration with no visible growth (clear broth).

• Determine MBC: In Subculture tubes with no growth onto nutrient agar, the lowest concentration with no growth on subculture is MBC.

• Serial dilutions in Mueller-Hinton broth , inoculate with test organism, incubate at 37°C for 18 hours.

• Determine MIC: by lowest concentration with no visible growth (clear broth).

• Determine MBC: In Subculture tubes with no growth onto nutrient agar, the lowest concentration with no growth on subculture is MBC.

Sabouraud dextrose agar (Option C) and Blood agar (Option D) are not used in Diffusion methods.

Solution for Question 14:

Correct answer: D) Disc concentration does not affect the zone of inhibition diameter.

Explanation:

Disc concentration is one of the Factors Influencing Zone Diameter

KIRBY-BAUER METHOD	
Step	Description
Inoculation	Use a standard dilution of a broth culture of the test bacterium. Inoculate the surface of a solid medium (Cation-adjusted Mueller-Hinton agar (CAMHA) as a lawn culture.
Application of Antibiotic Discs	Dry the plate at 37°C for 30 minutes. Apply 4-6 antibiotic discs per 9 cm plate with sterile forceps. (Option A ruled out)
Incubation	Incubate overnight. (Option C ruled out) Measure the zones of inhibition around the discs to determine sensitivity. Growth inhibition occurs around discs with effective antibiotics.
Factors Influencing Zone Diameter	Diffusibility of the drug Disc concentration Nature and composition of the medium (Option B ruled out) Medium thickness Presence of inhibitory or stimulatory substances pH Incubation time

KIRBY-BAUER METHOD	
Potency Check	Periodically check the potency of discs using standard ATCC strains: S. aureus ATCC 25923 E. coli ATCC 25922 P. aeruginosa ATCC 27853
Interpretation of Results	Record zones of inhibition around the discs. Interpret using zone diameters from CLSI guidelines or EUCAST guidelines.



Zone of inhibition around antibiotic discs in Kirby Bauer Test

Solution for Question 15:

Correct Answer: C) Intermediately susceptible

Explanation:

Intermediately Susceptible bacteria are Susceptible to the drugs at higher dosages or when the drug has clinical efficacy at the body site where the drugs are physiologically concentrated.

STOKES METHOD :

- Comparison of zones of inhibition: between standard & test bacteria.

• Indicates sensitivity/resistance of bacteria.

Bacterial Sensitivity:

• Susceptible : Inhibited by drug concentration achieved in blood at the recommended dosage. (Option A ruled out)

• Intermediately Susceptible : Susceptible at higher dosages or effective at physiologically concentrated sites.

• Resistant : Not inhibited; specific resistance mechanisms present. (Option B ruled out)

Choice of Antibiotic Disk:

• Tests clinically relevant antibiotics.

• Example: Chloramphenicol is unnecessary for urinary pathogens (mostly excreted inactive). Nitrofurantoin is relevant only for urinary pathogens.

• Chloramphenicol is unnecessary for urinary pathogens (mostly excreted inactive).

• Nitrofurantoin is relevant only for urinary pathogens.

• Chloramphenicol is unnecessary for urinary pathogens (mostly excreted inactive).

• Nitrofurantoin is relevant only for urinary pathogens.

Extremely susceptible (Option D) is not defined under Stoke's Method

Solution for Question 16:

Correct answer: C) It provides a visual and precise measurement of the minimum inhibitory concentration of the antibiotic.

Explanation:

The primary advantage of the E-test is that it provides a clear, visual indication of the MIC, which is the lowest concentration of an antibiotic that inhibits the growth of the bacterium. This helps clinicians choose the most effective antibiotic and dosage for treatment.



Image showing E-test

Step	Description
Setup	An absorbent strip with a gradient of antibiotic concentrations is placed on a culture plate inoculated with the bacterium.
Process	The strip releases the antibiotic, creating a gradient in the medium. An elliptical zone of inhibition forms around the strip after incubation.
Determination	The MIC is identified at the point where the edge of the ellipse intersects with the strip, representing the lowest concentration of the antibiotic that inhibits bacterial growth (MIC). (Option C)
Advantages	Allows for the simultaneous testing of multiple antibiotics on a single plate and provides a visual, precise measurement of the MIC, enhancing accuracy compared to traditional disk diffusion methods.

It identifies the specific bacterium causing the infection (Option A): The E-test assesses bacterial susceptibility to antibiotics but does not identify the specific bacterium. Additional tests like culture characteristics, biochemical tests, or molecular methods are needed for bacterial identification.

It determines the bacterium's resistance to all known antibiotics (Option B): The E-test measures the MIC for specific antibiotics tested, not for all known antibiotics . Multiple E-tests with different antibiotics can be performed to get a broader resistance profile, but each test is specific to one antibiotic.

It allows for the immediate administration of antibiotics without further testing (Option D): While the E-test provides valuable information, it still requires incubation and interpretation time. Immediate administration of antibiotics is typically guided by clinical judgment and empirical

therapy, not solely by the results of the E-test.

Sterilization and Disinfection

1. A second year medical student attending his viva in the microbiology department, was asked to identify the given device and name any one item that can be sterilised by it. Which of the following options is correct?



A. Hot air oven and culture media

B. Autoclave and glassware

C. Hot air oven and glassware

D. Autoclave and culture media

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2. Match the following sterilisation techniques with their respective control methods. Sterilisation technique Control method 1. Hot air oven a. Bacillus pumilus 2. Autoclave b. Serratia marcescens 3. Membrane filtration c. Bacillus atrophaeus/subtilis spores 4.Ionizing Radiation d. Geobacillus stearothermophilus spores

Sterilisation technique	Control method
1. Hot air oven	a. Bacillus pumilus
2. Autoclave	b. Serratia marcescens
3. Membrane filtration	c. Bacillus atrophaeus/subtilis spores
4.Ionizing Radiation	d. Geobacillus stearothermophilus spores

A. 1-a, 2-c, 3-b, 4-d

B. 1-c, 2-d, 3-a, 4-b

C. 1-c, 2-d, 3-b, 4-a

D. 1-d, 2-c, 3-a, 4-b

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3. During a community health meeting, a health inspector discusses the importance of preventing consumption of raw milk from cattle. What is the false statement regarding the treatment process of raw milk before consumption?

A. It is a type of disinfection method that uses moist heat.

B. Holder method is used to disinfect milk from Coxiella burnetii.

C. Presence of the enzyme phosphatase in the milk post-treatment suggests inadequate efficacy.

D. It helps to extend the shelf life of milk.

---

4. Which of the following sterilisation/disinfection techniques acts on the principle of moist heat at 100oC?

A. Inspissation

B. Incineration

C. Tyndallization

D. Pasteurization

5. As an intern in the orthopaedic department, you have been tasked to deliver the surgical instruments planned for the operating theatre to the CSSD. What temperature settings are kept in the following sterilisation machine at the CSSD? CSSD - Central Sterile Supply Department



A. 121 o C for 15 minutes at 15 psi pressure

B. 160 o C for 2 hours

C. 180 o C for 30 minutes

D. 80 o C for 20-30 minutes for 3 consecutive days

6. Which of the following substances can be sterilised by membrane filters?

A. Suture material

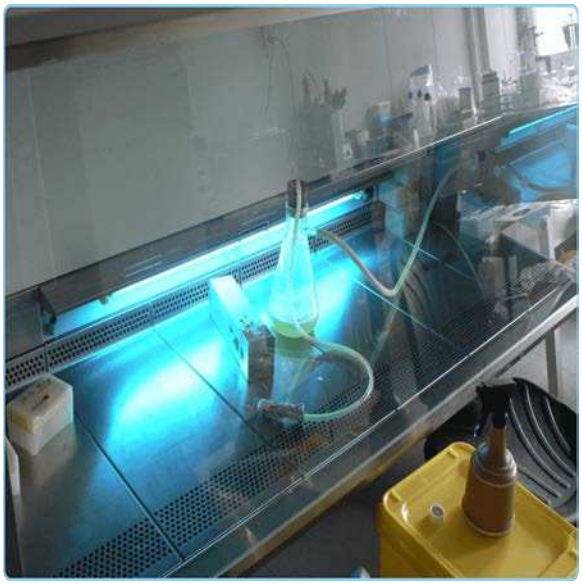
B. Liquid paraffin

C. Plastic syringe

D. Antibiotic solution

---

7. What is the quality control agent used in the shown method of disinfection?



A. Bacillus atrophaeus niger

B. Bacillus atrophaeus globigii

C. Bacillus pumilus

D. Geobacillus stearothermophilus

---

8. What is the ideal agent of choice for sterilizing the shown instrument?



A. 2% Glutaraldehyde

B. Ethylene oxide

C. 35% Hydrogen peroxide

D. Isopropyl alcohol

---

9. What is the mechanism of action of the most commonly used oral antiseptic?

A. Oxidisation

B. DNA dimer formation

C. Alkylation

D. Protein denaturation

---

10. Which among the following is NOT true regarding Gas sterilisation?

A. Mechanism of bactericidal action is Alkylation

B. Used to sterilise cardio-pulmonary machines

C. Refers to Ethylene oxide ideally used at 55-60°C for 4-12 hours

D. Suitable for fumigation of OT complexes

---

11. What is the ideal agent of choice for managing blood spill from a fall of 5 mL vacutainer?



A. 1:10 dilution 5.25% Sodium hypochlorite

B. 1:100 dilution 15.25% Sodium hypochlorite

C. 1:10 dilution 15.25% Sodium hypochlorite

D. 1:100 dilution 5.25% Sodium hypochlorite

---

12. Which among the following tests gives a real-time assessment of microbiological contamination of a disinfectant in use?

A. Rideal and Walker test

B. Chick Martin test

C. Kelsey-Sykes test

D. Kelsey-Maurer test

---

13. A chronic smoker with persistent cough was found to have a lung mass on imaging. A bronchoscope was used to obtain a lung biopsy. Based on the Spaulding classification, what category does this medical device fall under, and what is the appropriate sterilisation or disinfection agent?

A. Critical device and Autoclave

B. Semi-critical device and Autoclave

C. Critical device and Glutaraldehyde

D. Semi-critical device and Glutaraldehyde

---

14. To which group of disinfectants does the commonly used cetrimide solution belong to?

A. Phenols

B. Alcohols

C. Surface acting agents

D. Aldehydes

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	3
Question 5	1
Question 6	4
Question 7	3
Question 8	1
Question 9	4
Question 10	4
Question 11	4
Question 12	4
Question 13	4
Question 14	3

Solution for Question 1:

Correct Answer : C) Hot air oven and glassware

Explanation:

The device in the given image is that of a hot air oven and it can be used to sterilise glassware .

- Hot air oven is a sterilisation method that works on the principle of dry heat .
- Dry heat causes its killing effect by protein denaturation , damage by oxidising molecules , destroying cell constituents and toxic effects of elevated levels of electrolytes.
- The items to be sterilised are kept inside the oven for 2 hours and a holding period of 160 o C .
- Sterilisation is achieved through direct conduction of heat to the items kept inside the oven.
- Items that can be sterilised are mainly glassware , metallic instruments (both sharps and non sharps), some heat-resistant oils and waxes that are immiscible in water such as dusting powder and liquid paraffin .



Hot air oven

Hot air oven and culture media (Option A) : The given image is that of a hot air oven but culture media in general cannot be sterilised by hot air oven . They are sterilised by autoclave (moist heat) and some culture media that contain egg like LJ media and serum like Loeffler’s serum slope are sterilised by inspissation . (Options A, B and D ruled out)



Autoclave

Solution for Question 2:

Correct Answer: C) 1-c, 2-d, 3-b, 4-a

Explanation:

Sterilisation technique	Control method
1. Hot air oven	c. Bacillus atrophaeus/subtilis spores
2. Autoclave	d. Geobacillus stearothermophilus spores
3. Membrane filtration	b. Serratia marcescens
4.Ionizing Radiation	a. Bacillus pumilus

These control methods are biological indicators used to check the effectiveness of the sterilisation method.

- Spores of bacillus atrophaeus/subtilis are used as a control method for hot air ovens and ethylene oxide sterilisation (used mainly for heart-lung machines).

• Spores of geobacillus stearothermophilus are used as a control method for autoclave and plasma sterilisation .

• Bacillus globigi is also used as a control method for ethylene oxide sterilisation .

• Serratia marcescens and Brevundimonas diminuta are used as control methods for membrane filters .

Solution for Question 3:

Correct answer: B) Holder method is used to disinfect milk from Coxiella burnetii.

Explanation:

• The treatment process of raw milk discussed is pasteurisation and the method used to disinfect milk from Coxiella burnetii is Flash method and not Holder method.

• Pasteurisation is a moist heat based disinfection method used to destroy food spoiling organisms in milk, thereby extending their shelf-life. It was found and described by Louis Pasteur . (Options A and D ruled out)

• There are three methods of pasteurisation:

Presence of the enzyme phosphatase in the milk post-treatment suggests inadequate efficacy (Option C):

• Phosphatase test is used to test the efficacy of pasteurisation as the process of pasteurisation destroys the enzyme phosphatase present in milk.

• If present after the process, then it indicates contamination or inadequate pasteurisation .

Coliform test is another test used to check for efficacy of pasteurisation.

Solution for Question 4:

Correct Answer: C) Tyndallization

Explanation:

Tyndallization is the sterilisation/disinfection technique that acts on the principle of moist heat at 100 o C.(BAT-boling at 100 is tyndallisation.

- Tyndallization is also called intermittent sterilisation . It involves exposure to moist heat at 100 o C for 20 minutes on three successive days .
- The first exposure kills all vegetative bacteria . Spores which survive the first heat treatment, are allowed to germinate in the favourable medium. Subsequent exposures kill the newly formed vegetative cells from the germinated spores .
- Mainly used for sterilising media containing sugars or gelatin like TCBS, XLD, DCA, Selenite F broth . And other culture media where traditional autoclaving may affect the composition.
- However, its limitations are that it may not effectively kill spores of certain anaerobic bacteria and thermophiles (heat-loving bacteria) may also survive.

Inspissation (Option A):

- It is also a sterilisation technique using moist heat but at 80–85°C for 30 minutes on 3 successive days.
- It is used to sterilise egg containing and serum containing culture media such as Lowenstein–Jensen medium for Mycobacterium species and Loeffler's serum slope for diphtheria respectively.

Incineration (Option B):

- It is a sterilisation technique that uses dry heat at very high temperatures in the range of 870°C to 980°C.
- This is a safe method t o treat large solid infectious waste, particularly anatomical waste like amputated limbs, animal carcasses and chemical wastes such as expired medications and chemicals.
- It converts the material into ash, leaving about a tenth of the original volume .
- However, it is expensive and is generally used only by very large establishments.
- It is the safest method to ensure that no infective materials remain , including prions(infective proteins).

Pasteurization (Option D) : It is an intermediate-level disinfection technique that uses moist heat at temperatures <100°C to prevent various milk-borne infections and increase the shelf life of milk and fruit juices.

Solution for Question 5:

Correct Answer: A) 121 o C for 15 minutes at 15 psi pressure.

Explanation:

The given image is of an autoclave machine. The temperature settings kept in the machine for sterilisation are 121 o C for 15 minutes at 15 psi pressure .

- An autoclave is a sterilisation technique that uses moist heat at a temperature > 100 o C.
- The autoclave (steam steriliser) works on the principle of steam production under pressure .
- When this steam meets a cooler surface, it condenses, transferring its heat to that surface. The process continues until the surface temperature matches that of the steam.
- The resulting condensed water provides moist heat, effectively killing microbes on the materials by protein denaturation .
- Its temperature settings are 121 o C for 15 minutes at 15 psi pressure.
- Materials that can be sterilised include linen , surgical instruments (non-sharps only) , laboratory ware, pharmaceuticals , suture material except catgut (cold sterilisation) and culture media except egg and serum containing media(inspissation).
- Sterilisation control indicator - Geobacillus stearothermophilus

Types of Autoclaves in Healthcare:

- Laboratory autoclaves
- Hospital dressing sterilisers
- Instrument sterilisers
- Rapid cooling sterilisers

160 o C for 2 hours (Option B): It is the temperature setting used in a hot air oven . It uses dry heat as its sterilising agent. Other settings used in a hot air oven are:

- 170 o C for 1 hour
- 180 o C for 30 minutes (Option C ruled out)



Hot air oven

80 o C for 20-30 minutes for 3 consecutive days (Option D): IIt is the temperature setting used in the case of an inspissator . It uses moist heat to sterilise egg and serum containing culture media such as Lowenstein-Jensen medium and Loeffler's serum slope respectively.

Solution for Question 6:

Correct answer: D) Antibiotic solution

Explanation:

Out of all the options, antibiotic solutions can be sterilised only by membrane filters.

- Membrane filtration is considered as one of the sterilisation techniques although only filtration of microorganisms occurs and not killing .
- Filtration is of mainly two types - membrane filtration and depth filtration .
- Membrane filtration is widely used in healthcare mainly in the filtration of air and water .

Air filtration	Water filtration
It is done by using high-efficiency particulate air (HEPA) filters designed to remove organisms larger than 0.3 μm from isolation rooms, operating rooms , and biological safety cabinets (BSCs) . ULPA filters (Ultra-low particulate/penetration air) can remove from the air at least 99.9% of dust and any airborne particles with a size of 0.12 μm or larger. Surgical (3-ply) masks and respirators are the simplest examples of filters being used for purification of air. They remove microorganisms based on their pore size.	Used for bacteriological examination of water in hospital settings , especially dialysis water. Also used to remove bacteria from pharmaceutical fluids that are heat labile and cannot be purified by any other means e.g. sera , sugar, toxin , vaccine and antibiotic solutions .

- The sterilisation control of membrane filters are Brevundimonas diminuta and Serratia marcescens .

Suture material (Option A): In general, all suture materials are sterilised by autoclave except catgut which is sterilised by ionising radiation .(cold sterilisation)

Liquid paraffin (Option B) is sterilised by a hot air oven, not membrane filters.

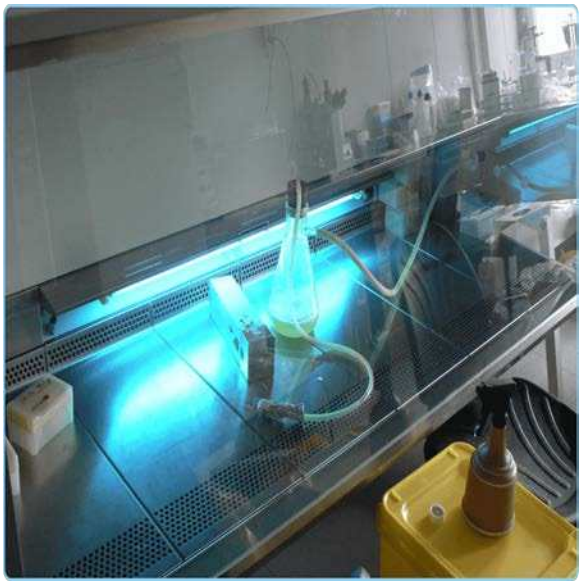
Plastic syringe (Option C): Ionizing radiation, such as gamma radiation, is used to sterilise plastic disposables such as plastic syringes, catheters, or gloves before use.

Solution for Question 7:

Correct answer: C) Bacillus pumilus

Explanation:

Shown method (in the image) is disinfection by Ultraviolet (UV) radiation . Quality control agent used for Radiation is Bacillus pumilus .



Radiation:

Ionising	Features: High penetration power Sporicidal action + Known as Cold radiation (do not increase surface heat)
Examples: Gamma rays, X-rays, Cosmic rays	
Uses: Plastics, syringes, catheters Catgut sutures - Gamma (γ) rays	
Non-ionising	Features: Low penetration power Only bactericidal action +
Examples: UV rays, infrared rays	
Uses: Plastics, syringes, catheters	
Mechanism of action	DNA damage by Dimer formation
Quality control agent	Bacillus pumilus (Option C)

Other options:

Bacillus atrophaeus niger (Option A): used for Hot air oven

Bacillus atrophaeus globigii (Option B): used for Ethylene oxide (Gas plasma)

Geobacillus stearothermophilus (Option D): used for Autoclave

Solution for Question 8:

Correct Answer: A) 2% Glutaraldehyde

Explanation:

- Flexible endoscopes and Laryngoscopes (shown in the image) are categorized under semi-critical items in Spaulding's Classification . Aldehydes are the preferred agents for sterilization of semi-critical items.
- Spaulding’s classification of medical devices is used to classify medical devices into three categories according to the degree of risk for infection according to their use and the subsequent level of reprocessing required to render the devices safe for reuse .

Risk category	Recommended method	Medical device/Equipment
1. Critical device (high risk): Items that enter a normally sterile site .	Sterilisation If heat resistant : Hot air oven or autoclave If heat-sensitive : Ionising radiation or ethylene oxide or plasma sterilisation.	Surgical instruments , implants /prostheses, rigid endoscopes , syringes, needles, CuT, Urinary catheter
2. Semi-critical device (intermediate risk): Items in contact with mucous membranes or body fluids .	High-level disinfection (HLD) : Such as glutaraldehyde , ortho-phthalaldehyde, formaldehyde and hydrogen peroxide .	Respiratory equipment , non-invasive flexible endoscopes, bedpans , urine bottles
3. Non-critical (low-risk) Items in contact with intact skin.	Intermediate or Low-level disinfection (ILD or LLD): Such as phenols , chlorhexidine, isopropyl alcohol , povidone-iodine , etc	Patient items include blood pressure cuffs , ECG electrodes, thermometer and stethoscopes . Environmental surfaces include computers, bed rails , food utensils, bedside tables , patient furniture and floor.



Aldehydes:

Examples	2% Glutaraldehyde (Option A) : Activated by alkalinisation 0.55% Orthophthaldehyde: More stable, more active Formaldehyde
Mechanism of action	Alkylation + DNA damage by cross linking
Uses [Semi-critical: in contact with mucous membrane]	Flexible endoscopes, Laryngoscope (Option A) , Vaginal speculum, Anesthetic equipment. Specific uses: Formaldehyde: OT fumigation Glutaraldehyde: Bronchoscopes and laryngoscopes Orthophthaldehyde: Cystoscopes

Alcohols:

Examples	70% Ethyl alcohol (spirit), Isopropyl alcohol, Methyl alcohol
Mechanism of action	Membrane damage + denaturation of bacterial proteins
Uses [Non-critical: in contact with intact skin]	Skin antiseptic, Thermometers, Stethoscopes Methyl alcohol: Fungal spores, incubators

Ethylene oxide (Option B): Method of Gas sterilization, used for Heart-Lung machines, respirators.

35% Hydrogen peroxide (Option C): Useful in sterilization of Rigid endoscopes (Arthroscopes)

Isopropyl alcohol (Option D): Used in disinfection of Stethoscopes, Thermometers, skin antiseptic.

Solution for Question 9:

Correct Answer: D) Protein denaturation

Explanation:

The most commonly used oral antiseptic is Chlorhexidine, a type of Phenol that causes protein Denaturation and membrane damage, leading to bactericidal action.

Phenols:

Examples	Cresol: Rideal-walker coefficient- 10 , Stools disinfectant Lysol: Hospital disinfectant Chloroxylenol: Less action against Pseudomonas Chlorhexidine: non-toxic
Mechanism of action	Membrane damage + Protein denaturation (Option D)
Uses	Skin antiseptics, wound dressing Hospital floors and surroundings disinfectant Chlorhexidine: Oral antiseptic

Halogens:

Examples	Tincture Iodine (2% Iodine): bactericidal + sporicidal Povidone Iodine (Betadine): has residual action Sodium hypochlorite (5.25%): Both bactericidal + virucidal
Mechanism of action	Oxidisation (Option A)
Uses	Skin antiseptic, wound dressing (Betadine preferred) Town water supply, Swimming pools (Chlorine) Sodium hypochlorite: Blood spills (1:10/1:100 dilution of 5.25% solution used)

DNA dimers formation (Option B): MOA in Radiation sterilisation

Alkylation (Option C): MOA in Gas sterilisation

Solution for Question 10:

Correct Answer: D) Suitable for fumigation of OT complexes

Explanation:

Ethylene oxide is the most commonly used gas steriliser. Due to its explosive nature, it is NOT suitable for fumigating OT complexes.

Gas sterilisation:

Most commonly used	Ethylene oxide (ETO)
Concentration	450-700 mg ETO at 55-60°C for 4-12 hours (Option C)
Mechanism of action	Alkylation of DNA (Option A) and associated microbial proteins
Uses	Cardio-pulmonary (Heart-Lung) machine (Option B) , Respirators [Critical class under Spaulding's classification]
Disadvantages	Irritant: workers must use protective wears while using Explosive & Inflammable: Not suitable for fumigation (Option D)

Solution for Question 11:

Correct Answer: D) 1:100 dilution 5.25% Sodium hypochlorite

Explanation:

Small blood spillage (<10 mL) should be managed by 1:100 dilution of 5.25% Sodium hypochlorite after cleaning the area with an absorbable material.



Blood spills management:

- Block off the area to be disinfected.
- Wear a pair of non-sterile gloves.
- Use brush and tongs to clean up broken glass pieces (never use bare hands) and discard the pieces onto a Sharps container (White).
- Use an absorbable material (paper/gauze/cloth) to wipe off body fluids/blood.
- Saturate the area with Sodium hypochlorite: Dose: 5.25% Sodium hypochlorite (Option D) Strength: Small spills (<10 mL) - 1:100 (Option D) , Large spills (>10 mL) - 1:10 dilution
- Dose: 5.25% Sodium hypochlorite (Option D)
- Strength: Small spills (<10 mL) - 1:100 (Option D) , Large spills (>10 mL) - 1:10 dilution
- Rinse off the instruments used with running water and place to dry, discard absorbable material into the yellow bin.
- Remove and discard gloves into the red bin.
- Wash hands carefully with soap and water and dry using single-use towels.
- Record the incidence in the register.
- Dose: 5.25% Sodium hypochlorite (Option D)
- Strength: Small spills (<10 mL) - 1:100 (Option D) , Large spills (>10 mL) - 1:10 dilution

Solution for Question 12:

Correct Answer: D) Kelsey-Maurer test

Explanation:

Real-time assessment of microbiological contamination is done using the Kelsey-Maurer test .

Testing of disinfectants:

Rideal-Walker test	Also known as the Phenol coefficient test . Tests the efficacy of a phenolic disinfectants (Lysol, Cresol) to kill Salmonella Typhi. Coefficient > 1 signifies that the disinfectant is efficacious. Disadvantage: Efficacy of organic matter is unknown
Chick Martin test	Modification of the Rideal-Walker test Tests the efficacy of phenolic disinfectant in presence of organic matter.
Kelsey-Sykes test	Tests the capacity of disinfectant to retain its activity with increasing microbiological load when used repeatedly.
Kelsey-Maurer test	Real-time assessment of microbiological contamination of in-use disinfectants in hospital

Solution for Question 13:

Correct Answer: D) Semi-critical device and Glutaraldehyde

Explanation:

- According to Spaulding's classification of medical devices, the bronchoscope comes under semi-critical medical devices and it requires high-level disinfectants such as glutaraldehyde, ortho-phthalaldehyde and formaldehyde. (Options A, B and C ruled out)
- Spaulding's classification of medical devices is used to classify medical devices into three categories according to the degree of risk for infection according to their use and the subsequent level of reprocessing required to render the devices safe for reuse .

Risk category	Recommended method	Medical device/Equipment
1. Critical device (high risk): Items that enter a normally sterile site .	Sterilisation If heat resistant : Hot air oven or autoclave If heat sensitive : Ionising radiation or ethylene oxide or plasma sterilisation.	Surgical instruments , implants /prosthesis, rigid endoscopes , syringes, needles, CuT, Urinary catheter
2. Semi-critical device (intermediate risk): Items in contact with mucous membranes or body fluids .	High-level disinfection (HLD) : such as glutaraldehyde , ortho-phthalaldehyde, formaldehyde and hydrogen peroxide .	Respiratory equipment , non-invasive flexible endoscopes, bedpans , urine bottles
3. Non-critical (low-risk) Items in contact with intact skin.	Intermediate or Low-level disinfection (ILD or LLD): Such as phenols , chlorhexidine, isopropyl alcohol , povidone iodine , etc	Patient items include blood pressure cuffs , ECG electrodes, thermometers and stethoscopes . Environmental surfaces include computers, bed rails , food utensils, bedside tables , patient furniture and floor.

Solution for Question 14:

Correct Answer: C) Surface acting agents

Explanation:

Cetrimide is chemically acetyl trimethyl ammonium bromide which is classified under the group of surface acting agents of disinfectants. (Options A, B and D are ruled out)

- Surface-active agents are substances that act on the phosphate group of cell membranes by reducing the surface tension . After entering the cell, protein denaturation occurs.
- They act on bacteria, but have no action on spores , tubercle bacilli and most viruses .
- They are widely used as wetting agents, detergents and emulsifiers.
- The common compounds are acetyl trimethyl ammonium bromide (Cetavlon or Cetrimide) and benzalkonium chloride .
- These are most active at alkaline pH and acid inactivates them.
- Organic matter reduces their action and anionic surface-active agents, like ordinary soaps, render them inactive.

• Soaps prepared from saturated fatty acids (such as coconut oil) are more effective against Gram-negative bacilli, while those prepared from unsaturated fatty acids (oleic acid) have greater action against Gram-positive and the Neisseria group of organisms.

Previous Year Questions

1. ■Disposable pipette tips are sterilized by which of the following methods? a. Autoclave b. Hot air oven c. Microwave d. 70% ethanol

A. a and b only

B. c only

C. c and d only

D. a only

---

2. In which of the following is the disc diffusion method for antibiotic susceptibility testing is applied?

A. Agar dilution method

B. Kirby-Bauer method

C. Broth dilution method

D. PCR based assay

---

3. Biological indicator for determining efficacy of autoclaving is?

A. Pseudomonas aeruginosa

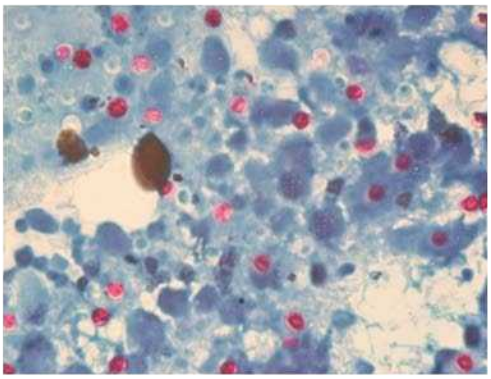
B. Clostridium perfringenes

C. Bacillus stearothermophilus

D. Salmonella typhi

---

4. An HIV patient with fever and chronic diarrhea presented to OPD. His stool was examined, and it showed the following oocysts. Identify the components used for staining the stool.



A. Crystal violet, iodine, acid alcohol, kinyoun carbol fuchsin

B. Kinyoun carbol fuchsin, acid alcohol, methylene blue

C. Methanol, dilute carbol fuchsin, acid alcohol, crystal violet

D. Methylene blue, iodine, acid alcohol, kinyoun carbol fuchsin

---

5. Which of the following options is considered a high-level disinfectant?

A. Isopropyl alcohol

B. Glutaraldehyde

C. Iodophores

D. Phenol

---

6. If a sample is cultured in LJ media, what is the likely staining characteristic of the cultured organism?

A. Negative staining

B. Gram negative

C. Acid fast

D. Non-acid fast

---

7. Which of the following is specifically found in gram-negative bacteria?

A. Lipoproteins

B. Peptidoglycan

C. Lipopolysaccharide

D. Teichoic acid

---

8. Which of the organisms listed below are capable of being stained with acid-fast staining? Nocardia Mycobacterium leprae Actinomyces Cryptosporidium parvum Isospora belli

A. 1, 2, 3, 4, 5

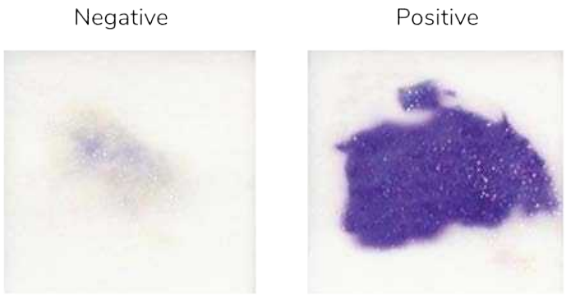
B. 1, 2, 4, 5

C. 1, 2, 3

D. 3, 4, 5

---

9. A patient admitted with 50% burns develops an infection at the burn site. The swab was cultured, the isolate is a strict aerobe and the test shown in the image is positive. What is the likely etiology of the burn infection?



A. Klebsiella

B. Escherichia coli

C. Pseudomonas aeruginosa

D. Salmonella

---

10. What types of organisms are capable of surviving in temperatures as high as 100 degrees Celsius?

A. Pseudomonas

B. Geobacillus stearothermophilus

C. Bacillus anthrax

D. Staphylococcus epidermidis

---

11. Which of the options listed below assists in bacterial adhesion?

A. Cytoplasmic membrane

B. Mesosomes

C. Fimbriae

D. Lipopolysaccharides

---

12. A 45-year-old man presents with fever, headache, photophobia, and neck rigidity for the last 2 days. He had undergone a heart transplant one year back and is on immunomodulatory drugs. On microscopy, differential motility was seen, non-motile at 37 degrees, and motile at 25 degrees. What is the most likely organism causing this?

A. Listeria

B. Streptococcus

C. Staphylococcus

D. Giardia lamblia

---

13. Which of the following stains is used for the diagnosis of mycobacterium tuberculosis infection?

A. Gomori Methenamine silver

B. Perl’s Prussian blue

C. Ziehl-Neelsen stain

D. Masson-Fontana stain

---

14. Which one of the following is an enrichment medium?

A. Alkaline peptone water

B. Monsur's taurocholate tellurite peptone water

C. Selenite F broth

D. All of the above

---

15. Which of the following organisms can thrive in an acidic pH?

A. Vibrio

B. Lactobacilli

C. Salmonella

D. Shigella

---

16. What is the diagnostic method depicted in the image?



A. Darkfield microscopy

B. Phase contrast microscopy

C. Fluorescent microscopy

D. Electron microscopy

---

17. A patient has complaints of fever with chills. A throat examination reveals swollen glands and a thick, whitish membrane over his tonsils. Albert staining is to be done. What are the constituents of Albert's stain?

A. Carbol fuschin and methylene blue

B. Carbol fuschin and malachite green

C. Toluidine and malachite green

D. Toluidine and methylene blue

---

18. DNA transfer in bacteria via phage is?

A. Conjugation

B. Transduction

C. Transformation

D. Translation

---

19. Sporulation occurs in this phase of bacterial growth curve:

A. Stationary phase

B. Lag phase

C. Log phase

D. Decline phase

---

20. Which of the following organisms is considered a strict anaerobe?

A. Burkholderia cepacian

B. Yersinia enterocoliticia

C. Stenotrophomonas maltophilia

D. Prevotella

---

21. Which of the organisms listed below are capable of producing urease?

A. Helicobacter pylori

B. Salmonella

C. Escherichia coli

D. Shigella

---

22. Vibrio cholerae disrupts which extracellular cell adhesion?

A. Integrin

B. Cadherin

C. Occludens

D. None of the above

---

23. The toughness of bacterial cell walls is due to:

A. Teichoic acid

B. Peptidoglycan

C. Lipopolysaccharides

D. Outer membrane

---

24. For the diagnosis of typhus, an antigen of Proteus is used. What is this type of antigen called?

A. Sequestered antigen

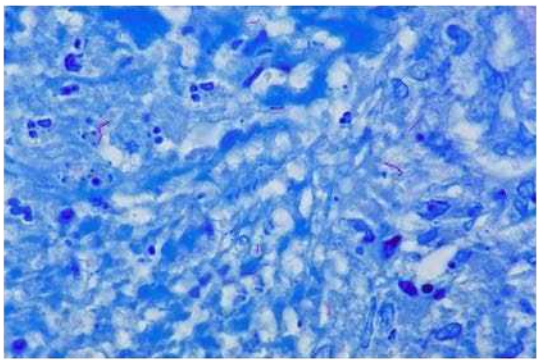
B. Super antigen

C. Iso antigen

D. Heterophile antigen

---

25. A 32-year-old laborer working at a construction site presented with fever and hemoptysis. The sputum sample collected for examination showed the following. The smear will be stained by which of the following sequences?



A. Gentian violet - Iodine - alcohol - safranin

B. Malachite green - heat – water - safranin

C. Basic fuschin - tannic acid - methylene blue

D. Carbol fuschin - acid - methylene blue

---

26. Which of the following organisms cannot be grown on an artificial culture media?

A. Mycobacterium leprae

B. Staphylococcus aureus

C. Mycobacterium tuberculosis

D. Streptococcus pneumoniae

---

27. Which culture medium is employed for distinguishing lactose fermenters from non-lactose fermenters?

A. Bile salt agar

B. Thayer-Martin agar

C. Sabouraud Dextrose Agar

D. MacConkey agar

---

28. Which of the following is an acid-fast organism?

A. Streptococcus pneumoniae

B. Escherichia coli

C. Mycobacterium avium

D. Actinomyces

---

29. The stain used for identifying Cryptococcus is?

A. Albert's stain

B. Silver stain

C. India ink stain

D. Auramine-rhodamine stain

---

30. Plasma sterilization is done by

A. Ethylene oxide

B. Gamma rays

C. Hydrogen peroxide

D. Ultraviolet radiation

---

31. Which among the following culture media is used in the identification of Salmonella?

A. PLET medium

B. BCYE medium

C. Wilson & Blair medium

D. Stuart's medium

---

32. Which of the following agents can be used as a disinfectant but not as a sanitizer?

A. Hydrogen peroxide

B. Penicillin

C. Methyl paraben

D. Alcohol

---

33. What microscopy technique is utilized to generate a high contrast image of transparent, viable bacteria?

A. Phase contrast microscopy

B. Dark field microscopy

C. Electron microscopy

D. Polarized microscopy

---

34. A 3-year-old child presents with fever, loose stools, and blood in the stool. Which enrichment medium is used to isolate the organism from the stool sample?

A. Robertson cooked meat broth

B. Alkaline peptone water

C. Selenite F broth

D. Brain heart infusion broth

---

35. Which type of microscope can be used to detect the motility of the probable causative organism in a truck driver presenting with painless indurated genital ulcer following unprotected sexual activity?

A. Dark-field microscope

B. Electron microscope

C. Light microscope

D. Fluorescent microscope

---

36. E.coli can be grown best in which of the following media?

A. Chocolate agar

B. Thayer-Martin agar

C. Sabouraud dextrose agar

D. MacConkey agar

---

37. Which of the following methods is used to sterilize a glass petri dish?

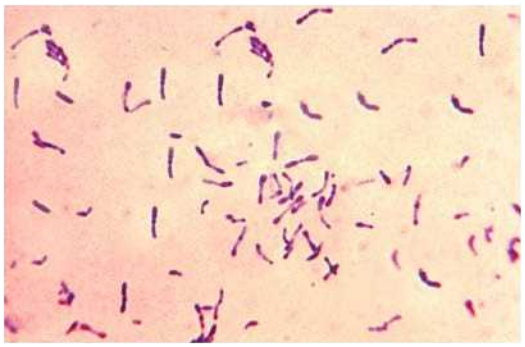
A. Autoclaving at 121 degrees Celsius for 30-60 minutes

B. Hot air oven at 160 degrees Celsius for 120 minutes

C. Autoclaving at 134 degrees Celsius for 3 minutes

D. Hot air oven at 121 degrees Celsius for 30-60 minutes

38. A 6-year-old boy complains of cough and throat pain. On examination of the oral cavity, a white pseudomembrane was visualized. A swab was taken from it and stained, as given in the image below. What is the procedure for the following staining? Albert's stain – I Acid Alcohol Iodine Safranin Methylene blue Crystal violet Acetone



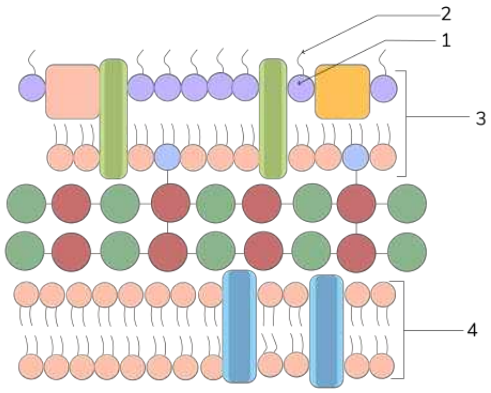
A. 1, 2, 5, 7

B. 6, 3, 7, 4

C. 7, 6, 4, 3

D. 3, 2, 1, 7

39. Identify the component responsible for endotoxic shock in gram-negative bacteria.



A. 2

B. 4

C. 1

D. 3

---

40. While collecting a blood sample for investigations, you accidentally dropped a 5 ml tube of blood and spilled blood. Which among the following is the best solution to clean it?

A. 1:10 sodium hypochlorite

B. Isopropyl alcohol

C. Hydrogen peroxide

D. 1:1000 sodium hypochlorite

---

41. Which of the following instruments are sterilized by gamma radiation?

A. Syringes, needles, and catheters

B. Contaminated clothes and animal carcasses

C. Gloves and aprons

D. Bacterial vaccines and intravenous solutions

42. When would you choose to utilize a sanitizer? Prior to conducting a patient examination Following a patient examination After coming into contact with the patient's environment On open wound

A. 1, 2 & 3 only

B. 1 & 2 only

C. 1, 2, 3 & 4

D. 3 & 4 only

43. Pasteurization is done at .....

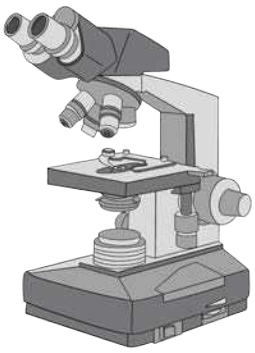
A. 73 degree Celsius for 20 minutes

B. 63 degree Celsius for 30 minutes

C. 72 degree Celsius for 30 seconds

D. 63 degree Celsius for 30 seconds

44. Identify the condenser in the following image.



A. Above the stage

B. Below the stage

C. Near the eyepiece

D. Above the objective lenses

---

45. Plasma sterilization makes use of ……

A. Hydrogen peroxide

B. Ethylene oxide

C. Gamma rays

D. UV rays

---

46. DNA transfer in bacteria via phage is:

A. Conjugation

B. Transduction

C. Transformation

D. Translation

---

47. Sporulation occurs in which phase of bacterial growth curve:

A. Stationary phase

B. Lag phase

C. Log phase

D. Dead phase

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	3
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	1
Question 13	3
Question 14	4
Question 15	2
Question 16	1
Question 17	3
Question 18	2
Question 19	1
Question 20	4
Question 21	1
Question 22	3

Question 23	2
Question 24	4
Question 25	4
Question 26	1
Question 27	4
Question 28	3
Question 29	3
Question 30	3
Question 31	3
Question 32	1
Question 33	1
Question 34	3
Question 35	1
Question 36	4
Question 37	2
Question 38	2
Question 39	3
Question 40	1
Question 41	1
Question 42	1
Question 43	2
Question 44	2
Question 45	1
Question 46	2
Question 47	1

Solution for Question 1:

Correct Option D - a only:

- Disposable pipettes are sterilized by autoclave
- All plastic and rub materials are sterilized by autoclave.

Disposable pipettes are sterilized by autoclave

All plastic and rub materials are sterilized by autoclave.

Incorrect Options:

Option A, B and C: These are incorrect options

- Hot air oven: Sharpes, powdery and oily materials are sterilized by hot air oven.
- Microwave: Another useful method of sterilization of small-volume waste at the point of generation, but ideally not used for animal or human body parts, metal items or toxic or radioactive material.
- 70% ethanol: It is used in disinfection and it is a low level disinfectant.

Hot air oven: Sharpes, powdery and oily materials are sterilized by hot air oven.

Microwave: Another useful method of sterilization of small-volume waste at the point of generation, but ideally not used for animal or human body parts, metal items or toxic or radioactive material.

70% ethanol: It is used in disinfection and it is a low level disinfectant.

Solution for Question 2:

Correct Option B - Kirby-Bauer Method:

- The disc diffusion method is applied in the Kirby-Bauer method, a widely used technique for antibiotic susceptibility testing. It is applied to determine the susceptibility or resistance of bacteria to different antibiotics.
- It involves placing antibiotic discs on an agar plate that has been uniformly inoculated with the test organism, this establishes a gradient with concentration being highest near the site of the disc and decreasing with distance. After incubation, the zone of inhibition around each

disc is measured, indicating the bacteria's susceptibility or resistance to the antibiotics.

Incorrect Options:

Option A - Agar Dilution Method: The agar dilution method employs a dilution method where drugs are prepared and inoculated with the test bacterium into the agar.

Option C - Broth Dilution Method: Broth dilution uses a dilution method; in this, serial dilutions of antibiotics are prepared in liquid broth, against the suspension of the bacterium.

Option D - PCR-Based Assay: PCR-based assays are molecular techniques used for the detection of specific genetic material, such as DNA or RNA, of pathogens. They are not used for antibiotic susceptibility testing but rather for identifying the presence or absence of specific genetic markers or genes associated with antibiotic resistance.

Solution for Question 3:

Correct Option C - Bacillus stearothermophilus:

- Bacillus stearothermophilus is a spore forming heat-resistant bacterium commonly used as a biological indicator to assess the effectiveness of autoclaving (steam sterilization) processes. Its resistance to heat makes it a suitable indicator for determining if autoclaving has achieved sufficient sterilization.
- Spore-forming organisms used as a biological indicator for sterilisation are: Bacillus stearothermophilus is used as a control in autoclave and plasma sterilization. Bacillus subtilis is used as a control in a hot-air oven. Bacillus pumilis is used as a control in sterilization by radiation. Bacillus golbigii is used as a control in gas sterilization.
- Bacillus stearothermophilus is used as a control in autoclave and plasma sterilization.
- Bacillus subtilis is used as a control in a hot-air oven.
- Bacillus pumilis is used as a control in sterilization by radiation.
- Bacillus golbigii is used as a control in gas sterilization.
- Bacillus stearothermophilus is used as a control in autoclave and plasma sterilization.
- Bacillus subtilis is used as a control in a hot-air oven.
- Bacillus pumilis is used as a control in sterilization by radiation.
- Bacillus golbigii is used as a control in gas sterilization.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 4:

Correct Option B - Kinyoun carbol fuchsin, acid alcohol, methylene blue:

- The presentation of diarrhea in an immunocompromised individual and the image showing an acid-fast oocyst suggestive of infection by coccidian parasites like cryptosporidium, cyclospora, and isosopora.
- The stain used is a modification of the Ziehl Neelsen staining, in which heat is not used as a mordant, instead an increased concentration of phenol in the primary stain, i.e., carbol fuschin is used as a mordant, hence known as Cold Ziehl Neelsen staining or Gabbet's stain or Kinyoun staining
- Procedure of Kinyoun staining: Kinyoun's carbol fuchsin: This is a modified acid-fast stain that is used to stain the Cryptosporidium oocysts. It helps visualize the oocysts under the microscope. Acid alcohol: Acid alcohol (5% H2SO4 and 95% ethanol) is used as a decolorizing agent to remove excess stain from the sample. Methylene blue: Methylene blue is a counterstain that is applied after the decolorization step. It helps in enhancing the contrast and visualization of the stained oocysts.
- Kinyoun's carbol fuchsin: This is a modified acid-fast stain that is used to stain the Cryptosporidium oocysts. It helps visualize the oocysts under the microscope.
- Acid alcohol: Acid alcohol (5% H2SO4 and 95% ethanol) is used as a decolorizing agent to remove excess stain from the sample.
- Methylene blue: Methylene blue is a counterstain that is applied after the decolorization step. It helps in enhancing the contrast and visualization of the stained oocysts.
- Kinyoun's carbol fuchsin: This is a modified acid-fast stain that is used to stain the Cryptosporidium oocysts. It helps visualize the oocysts under the microscope.
- Acid alcohol: Acid alcohol (5% H2SO4 and 95% ethanol) is used as a decolorizing agent to remove excess stain from the sample.
- Methylene blue: Methylene blue is a counterstain that is applied after the decolorization step. It helps in enhancing the contrast and visualization of the stained oocysts.

Incorrect Option:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 5:

Correct Option B - Glutaraldehyde:

- Glutaraldehyde is a potent disinfectant that is effective against a wide range of microorganisms, including bacteria, viruses, fungi, and even some spores. It is commonly used in healthcare settings to disinfect medical instruments and equipment that are not heat resistant or cannot be sterilized.

High-Level Disinfectant	Intermediate Level Disinfectant	Low-Level Disinfectant
Glutaraldehyde Formaldehyde Hydrogen Peroxide Chlorine	Isopropyl alcohol Phenol Iodophore	Chlorhexidine Quaternary ammonium compound

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 6:

Correct Option C - Acid Fast:

- LJ (Lowenstein-Jensen) media is used for the cultivation of Mycobacterium species, which are acid-fast bacteria.
- In acid-fast staining, the bacteria are stained with a carbol fuchsin dye and then treated with acid and alcohol.
- Acid-fast bacteria retain the red color of the dye even after acid and alcohol treatment, while non-acid-fast bacteria are decolorized and appear blue when counterstained with methylene blue.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

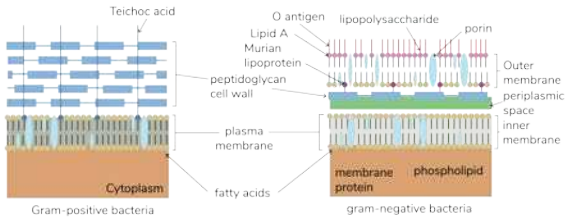
Solution for Question 7:

Correct Option C - Lipopolysaccharide:

- Gram-negative bacteria have a distinctive cell wall structure that includes an outer membrane, a thin peptidoglycan layer, and an inner plasma membrane.
- Lipopolysaccharides serves as an endotoxin and is responsible for the toxic effects associated with gram-negative bacterial infections.

Difference between gram positive and negative bacteria cell wall are:

Features	Gram-positive	Gram-negative
Thickness	100 layer thick	1-2 layer thickness
Lipid content	2-5%	15-50%
Composition	Peptidoglycans Lipoproteins Teichoic acid	Lipopolysaccharide Lipoproteins Peptidoglycan
Periplasmic space and Porins	Absent	Present
Teichoic acid	Present	Absent
Pentapeptide cross bridge	Present	Absent
NAM-NAG	Present	Present



Incorrect Options:

Option A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 8:

Correct Option B - 1, 2, 4, 5:

- Among the given list, the acid-fast organisms are: Cryptosporidium. Mycobacterium leprae. Nocardia. Isospora belli.
- Cryptosporidium.
- Mycobacterium leprae.
- Nocardia.
- Isospora belli.
- Actinomyces are gram-positive filamentous non-acid fast anaerobic to microaerophilic bacteria that typically colonize the human mouth, urogenital tract, and gastrointestinal tract but can cause an infection known as actinomycosis.
- Cryptosporidium.
- Mycobacterium leprae.
- Nocardia.
- Isospora belli.

Cryptosporidium.

Mycobacterium leprae.

Nocardia.

Isospora belli.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 9:

Correct Option C - Pseudomonas Aeruginosa:

- The test shown in the image is a positive oxidase test. It detects the presence of cytochrome oxidase enzyme in bacteria, which catalyzes the oxidation of reduced cytochrome by atmospheric oxygen, resulting in deep purple color of the smear.

• Oxidase-positive organisms are : Pseudomonas Vibrio Neisseria Bacillus anthrax Haemophilus

• Pseudomonas

• Vibrio

• Neisseria

• Bacillus anthrax

• Haemophilus

• Pseudomonas aeruginosa is a gram-negative bacterium that is a common cause of burn infections. It is known for its ability to thrive in moist environments and can easily colonize burn wounds. It is often resistant to multiple antibiotics, making it a significant concern in burn units.

• Pseudomonas

• Vibrio

• Neisseria

• Bacillus anthrax

• Haemophilus

Incorrect Options:

Options A, B, and D are incorrect. These organisms are oxidase negative.

Solution for Question 10:

Correct Option B - Geobacillus stearothermophilus:

• Geobacillus stearothermophilus is a thermophile that can survive between the temperature range of 55°C to over 100°C.

Incorrect Options:

Option A - Pseudomonas: Pseudomonas is a psychrophile with an optimum growth below 20° C.

Option C - Bacillus Anthrax: It is a mesophile with an optimum growth rate between 25° C - 45° C.

Option D - Staphylococcus Epidermidis: It is also a mesophile with an optimum growth rate between 25° C - 45° C.

Solution for Question 11:

Correct Option C - Fimbriae:

- Fimbriae, also known as pili, are hair-like appendages found on the surface of many bacteria.
- They play a crucial role in bacterial adhesion, allowing bacteria to attach to host cells or receptors, facilitating colonization and the initiation of infection.

Incorrect Options:

Option A - Cytoplasmic Membrane: Cytoplasmic membrane is essential for the integrity and functioning of the bacterial cell.

Option B - Mesosomes: Mesosomes are irregular invaginations of the cytoplasmic membrane, a respiratory unit in bacteria.

Option D - Lipopolysaccharides (LPS): Lipopolysaccharides are components of the outer membrane of Gram-negative bacteria. It is responsible for the production of endotoxin.

Solution for Question 12:

Correct Option A - Listeria:

- The clinical presentation of the patient is suggestive of meningitis.
- The most common bacterium responsible for meningitis are Streptococcus pneumoniae, Neisseria meningitidis, H. influenzae, Listeria, streptococcus agalactiae, and E. coli.
- On microscopy, the organism shows differential motility, i.e., motile (tumbling motility) at 25° C and non-motile at 37° C, which suggests the causative organism is Listeria monocytogens.
- Other organisms showing differential motility are Yersinia enterocolitica and Yersinia pseudotuberculosis.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 13:

Correct Option C - Ziehl-Neelsen stain:

- Mycobacterium is an acid-fast bacteria due to the presence of mycolic acid, which resists the decolorization of aniline dyes. The stain used to demonstrate the bacilli is Ziehl-Neelsen staining.
- The stain involves the use of a primary stain called carbol fuchsin, which stains the acid-fast bacteria red. After staining, a decolorizing agent (acid-alcohol) is applied, and then a counterstain, such as methylene blue, is used to stain non-acid-fast bacteria.

Incorrect Option:

Option A - Gomori Methenamine Silver: This stain is commonly used for the identification of fungi and pneumocytis jirovecii.

Option B - Prussian Blue: This stain is used to detect iron (hemosiderin) in tissue sections. For example, hemosiderin heart failure cells and ferruginous bodies in asbestosis.

Option D - Masson-Fontana Stain: This stain is utilized to detect melanin in tissues and to stain the pigmented fungi.

Solution for Question 14:

Correct Option D - All of the above:

- Enrichment media are used for the incubation of the organism during transport before inoculating it into a selective medium.
- Examples: Alkaline peptone water used for the pre-enrichment of vibrio cholera at a pH of 8.6. Monsur's taurocholate tellurite peptone water is used for the incubation of the vibrio. Selenite F broth is used for the incubation of the Salmonella and Shigella species.

Tetrathionate broth is used for the incubation of Salmonella species.

• Alkaline peptone water used for the pre-enrichment of vibrio cholera at a pH of 8.6.

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• Tetrathionate broth is used for the incubation of Salmonella species.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 15:

Correct Option B - Lactobacilli:

• Most pathogenic bacteria grow best between pH 7.2 and pH 7.6, but there are some exceptions.

• Lactobacilli can grow at acidic pH below pH 4. They are commonly found in various acidic environments, such as the vagina, gastrointestinal tract, and fermented foods like yogurt and sauerkraut.

• Vibrio cholerae can grow at alkaline pH (8.2-8.9).

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 16:

Correct Option A - Darkfield Microscopy:

- Darkfield microscopy is a specialized technique used for visualizing thin structures that are difficult to see using conventional brightfield microscopy. In darkfield microscopy, the specimen appears bright against a dark background, which enhances the contrast and visibility of certain structures as seen in the image.
- In darkfield microscopy, a special condenser is used to block the direct light from reaching the objective lens. Instead, oblique or angled illumination i.e., reflected light is employed, causing light to be scattered or refracted by the specimen. This scattered light enters the objective lens and forms the image against a dark background.
- Darkfield microscopy is particularly useful for visualizing thin structures like flagella, treponema, and spirochetes.

Incorrect Options:

Option B - Phase contrast Microscopy: Phase contrast microscopy is a different optical technique used to enhance the contrast of transparent or low-contrast specimens. It converts slight differences in refractive index and cell density into easily detectable variations in light intensity. It is useful to study microbial motility and detecting microbial internal cellular components.

Option C - Fluorescent Microscopy: Fluorescent microscopy is a method that utilizes fluorophores to label specific structures or molecules within a specimen. It requires the use of fluorescent dyes or fluorescently labeled antibodies. In fluorescent microscopy, the specimen emits fluorescence when illuminated by specific wavelengths of light, resulting in a bright, colorful image against a dark background.

Option D - Electron Microscopy: Electron microscopy uses a beam of accelerated electrons instead of light to image the specimen. It provides extremely high-resolution images, allowing for the visualization of ultrastructural details of cells and tissues.

Solution for Question 17:

Correct Option C - Toluidine and Malachite Green:

- Albert's stain is used for staining the volutin granules of Corynebacterium diphtheriae, the bacterium responsible for diphtheria.

• The constituents of Albert's stain are as follows: Albert solution 1: Green: Malachite green Purple: Toluidine Blue- shows metachromasia. Albert solution 2: Has iodine First add Toluidine blue then Iodine and lastly Malachite green On Albert's staining: organism-blue color and volutin granules-purple color.

• Albert solution 1: Green: Malachite green Purple: Toluidine Blue- shows metachromasia.

• Green: Malachite green

• Purple: Toluidine Blue- shows metachromasia.

• Albert solution 2: Has iodine

• First add Toluidine blue then Iodine and lastly Malachite green

• On Albert's staining: organism-blue color and volutin granules-purple color.

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• On Albert's staining: organism-blue color and volutin granules-purple color.

• Green: Malachite green

• Purple: Toluidine Blue- shows metachromasia.

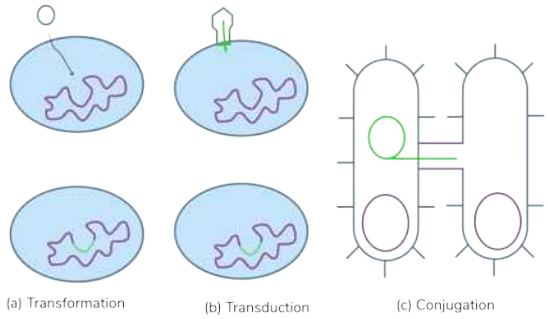
Incorrect Options:

Options A, B, and D are incorrect. These are not the constituents of Albert stain.

Solution for Question 18:

Correct Option B - Transduction:

• DNA transfer in bacteria mediated by phages (bacteriophages) is called transduction. During transduction, a bacteriophage infects a bacterial cell and incorporates bacterial DNA into its phage particles. When these phage particles infect other bacteria, they deliver the bacterial DNA, resulting in the transfer of genetic material between bacteria.



Incorrect Options:

Option A - Conjugation: It refers to the direct transfer of genetic material between bacteria through physical contact.

Option C - Transformation: It refers to the uptake of free DNA from the environment by bacteria.

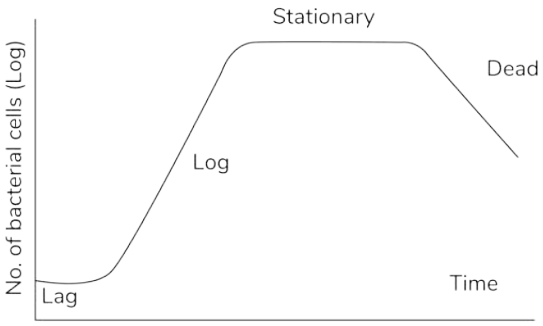
Option D - Translation: Translation is the process by which the genetic information encoded in mRNA is used to synthesize proteins. It does not involve the transfer of DNA between bacteria.

Solution for Question 19:

Correct Option A - Stationary Phase:

Bacterial Growth Curve:

- It has 4 phases



Lag Phase	Log Phase	Stationary Phase	Dead Phase

No increase in the number of bacteria Cell adaptation Increase in size, enzymes and metabolite The maximum size of bacteria is seen in end of the lag phase	Cell division Smaller in size Maximum metaboli c activity More susceptible to antibiotics	Some bacteria remain viable and some are dead Bacterial growth ceases due to depletion of nutrients Mnemonic STAB (STAT from stationary) S in sporulation T in Exo Toxin production A in antibiotic production B in Bacteriocin production Antibiotic production happens in this phase	Decline phase Cell death or involution

Incorrect Options: Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 20:

Correct Option D - Prevotella:

- Numerous species of bacteria in the genus Prevotella are recognized as strict anaerobes.
- Strict anaerobes can only exist and develop in settings free of oxygen because they cannot tolerate the presence of oxygen.
- Other anaerobic organisms are C. perfringens, C. tetani, C.botulinum, Clostridiodes difficile, Peptostreptococcus, Bacteroids, Prevotella, Porohyromonas, and actinomyces

Incorrect Options:

Option A - Burkholderia cepacia: It is an obligate aerobe.

Option B - Yersinia enterocolitica: It is a facultative anaerobic bacteria, which means it can survive and grow in both the presence and absence of oxygen.

Option C - Stenotrophomonas maltophilia: Stenotrophomonas maltophilia is also a strictly aerobic organism.

Solution for Question 21:

Correct Option A - Helicobacter pylori:

Urease Positive Organisms (Mnemonic Punch Kiss)

- Proteus mirabilis
- Ureaplasma urealyticum
- Nocardia
- Cryptococcus neoformans
- Helicobacter pylori
- Staphylococcus Saprophyticus
- Staphylococcus Epidermidis

Incorrect Options:

Options B, C, and D are incorrect. They are not urease positive organisms.

Solution for Question 22:

Correct Option C- Occludens:

- Function: Occludens refers to tight junction proteins (occludin, claudins) involved in forming tight junctions between epithelial cells.
- Role in Cholera: Vibrio cholerae secretes cholera toxin, which disrupts the tight junctions (occludens) in the intestinal epithelium. This disruption leads to increased secretion of water and electrolytes, a characteristic feature of cholera.

Incorrect Options:

Option A- Integrin:

- Function: Integrins are cell adhesion receptors involved in cell-to-cell and cell-to-extracellular matrix interactions.
- Role in Cholera: Vibrio cholerae does not primarily disrupt integrin-mediated adhesion

Option B- Cadherin:

- Function: Cadherins are proteins involved in calcium-dependent cell adhesion.

• Role in Cholera: While cadherins play essential roles in various cellular adhesion processes, they are not the primary targets of disruption by Vibrio cholerae.

Solution for Question 23:

Correct Answer: B) Peptidoglycan

Explanation: Toughness of bacterial cell wall is due to Peptidoglycan.

Bacterial Cell Wall:

- Tough and rigid structure surrounding the bacterium.
- 10-25 nm in thickness and constitutes 20-25% of the dry weight of the cell.
- Functions of Cell Wall: Provides protection to the cell against osmotic lysis. Confers rigidity upon bacteria due to the presence of a peptidoglycan layer in the cell wall. Protect a cell from toxic substances and is the site of action of several antibiotics. Contains certain virulence factors (e.g., endotoxin), which contribute to bacterial pathogenicity. Antibody raised against specific cell wall antigens (e.g., antibody to LPS) may provide immunity against some bacterial infections.
- Provides protection to the cell against osmotic lysis.
- Confers rigidity upon bacteria due to the presence of a peptidoglycan layer in the cell wall.
- Protect a cell from toxic substances and is the site of action of several antibiotics.
- Contains certain virulence factors (e.g., endotoxin), which contribute to bacterial pathogenicity.
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- Contains certain virulence factors (e.g., endotoxin), which contribute to bacterial pathogenicity.
- Antibody raised against specific cell wall antigens (e.g., antibody to LPS) may provide immunity against some bacterial infections.

Difference between Gram-positive and Gram-negative Bacterial Cell Walls:

Gram-positive	Features	Gram-negative
Thick layer	Peptidoglycan layer	Thin layer
100 layer thick	Thickness	1–2-layer thickness
2-5 %	Lipid content	15-50%
Peptidoglycans+++ Lipoproteins Teichoic acid	Composition	Lipopolysaccharide Lipoproteins Peptidoglycan+
Absent	Outer membrane	Present
Absent	Periplasmic space + Porins	Present
Present	Teichoic acid (used for attachment)	Absent
Present	Pentapeptide cross bridge	Absent
Present	NAM-NAG	Present

Teichoic acid (Option A) is present in Gram-positive bacteria, but it mainly contributes to cell wall stability and ion transport, not toughness.

Lipopolysaccharides (LPS) (Option C) are found in Gram-negative bacteria, but they form the outer membrane, playing a role in immune evasion and permeability, not rigidity.

The outer membrane (Option D) is present in Gram-negative bacteria, providing selective permeability and protection, but the structural strength comes from peptidoglycan.

Solution for Question 24:

Correct Answer: D) Heterophile antigen

Explanation: Proteus antigens are classified as heterophile antigens because they cross-react with antibodies produced against Rickettsia in the diagnosis of typhus.

Weil-Felix Test:

- Weil-Felix Test is a heterophile agglutination test that works on the principle of rickettsial antibodies showing antigenic cross-reactivity with Proteus antigens (OX 19, OX 2 and OX K strains).

• It is a tube agglutination test where serial dilutions of a patient's serum are treated with non-motile strains of P. vulgaris OX 19 and OX 2 and P. mirabilis OX K.

Disease	Agglutination Test Result
Epidemic and endemic typhus	Mainly agglutinates with OX19; sometimes with OX2
Tick-borne Spotted Fever	Antibodies to both OX19 and OX2 elevated.
Scrub Typhus	Raised antibodies to OXK
Rickettsial Pox, Q Fever, Ehrlichiosis, Bartonellosis	Negative test

• False-positive titer - Due to underlying Proteus infection. Hence, fourfold rise of antibody titer in paired sera is better than a single high titer.

• False-negative titer - Due to excess antibodies in the patient's sera (prozone phenomenon). This can be obviated by testing with serial dilutions of the patient's serum.

Sequestered antigens (Option A) are normally hidden from the immune system due to anatomical barriers (e.g., myelin basic protein, lens proteins of the eye, sperm-associated antigens). When exposed after trauma, infection, or inflammation, they may trigger autoimmune reactions (e.g., sympathetic ophthalmia, multiple sclerosis).

Superantigens (Option B) are highly potent immune activators that bypass normal antigen presentation by directly binding to MHC class II molecules and T-cell receptors, leading to massive cytokine release and systemic immune activation. Examples include:

• Staphylococcal enterotoxins (food poisoning)

• Toxic shock syndrome toxin (TSST-1)

• Streptococcal pyrogenic exotoxins (scarlet fever, toxic shock syndrome)

Isoantigens (or alloantigens) (Option C) are antigens that vary among individuals of the same species, such as blood group antigens (ABO, Rh) and HLA antigens. These play a role in blood transfusion compatibility and organ transplantation but are not relevant in the diagnosis of typhus.

Solution for Question 25:

Correct Option D - Carbol fuschin - acid - methylene blue:

• Based on the clinical history given and the basis of the picture of microscopy showing bright red acid-fast bacilli, the patient is suffering from tuberculosis, and the stain used is Ziehl Nielsen stain.

• The procedure of staining using ZN stain is Carbol fuschin - acid - methylene blue.

• Procedure: Carbol fuchsin: The first step in ZN staining is the application of carbol fuchsin. Carbol fuchsin is a red-colored dye containing basic fuchsin and phenol. The slide with the bacterial sample is flooded with carbol fuchsin, which stains both acid-fast and non-acid-fast bacteria. Heat: Mordant. Acid: The next step is the application of an acid-alcohol solution, often referred to as the decolorizing agent. This acid-alcohol solution to stain M tuberculosis typically contains sulphuric acid (20%) and 95% ethanol. The acid decolorization step removes the carbol fuchsin from non-acid-fast bacteria. Methylene blue: The final step involves the application of a contrasting counterstain, typically methylene blue. Methylene blue stains non-acid-fast bacteria that were decolorized during the previous steps. This step helps differentiate acid-fast bacteria, which retain the initial red color of carbol fuchsin, from non-acid-fast organisms that take up the counterstain.

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• Heat: Mordant.

• Acid: The next step is the application of an acid-alcohol solution, often referred to as the decolorizing agent. This acid-alcohol solution to stain M tuberculosis typically contains sulphuric acid (20%) and 95% ethanol. The acid decolorization step removes the carbol fuchsin from non-acid-fast bacteria.

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• Heat: Mordant.

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Incorrect Options:

Option A - Gentian violet – Iodine – alcohol - safranin: This staining method is a gram stain. It is used to differentiate between gram-positive and gram-negative bacteria. The sequence of gram stains is Gentian violet – Iodine – alcohol safranin.

Option B - Malachite green - heat – water - safranin: The staining technique is Schaeffer-Fulton`s method used for staining bacterial endospores.

Option C - Basic fuschin - tannic acid - methylene blue: The staining technique is Leifson and Ryu stain to demostrate flagella.

Solution for Question 26:

Correct Option A - Mycobacterium leprae:

- Mycobacterium leprae is an obligate intracellular pathogen causing leprosy, aka Hansen's disease, cannot be grown on artificial culture media.
- It can be grown in cell lines like the footpads of mice or nine-banded armadillo.
- Other bacteria which are not cultivated on artificial media are: Rickettsia. Chlamydia.
- Rickettsia.
- Chlamydia.
- Rickettsia.
- Chlamydia.

Incorrect Options:

Option B - Staphylococcus aureus: It is a Gram-positive bacterium that can grow on various artificial culture media such as blood agar, chocolate agar, and MacConkey agar.

Option C - Mycobacterium tuberculosis: It is a slow-growing bacteria that requires special culture media such as Lowenstein-Jensen medium or Middlebrook 7H10 agar for growth.

Option D - Streptococcus pneumoniae: It is a Gram-positive bacterium that can grow on various culture media such as blood agar, chocolate agar, and MacConkey agar.

Solution for Question 27:

Correct Option D - MacConkey agar:

- The MacConkey agar medium is most commonly used to differentiate between lactose fermenters and non-lactose fermenters. This is because of the ability of the lactose fermenters to produce acid as a result of fermentation, which changes the color of the medium to pink.
- MacConkey agar acts as a selective medium, differential medium, and indicator medium.
- Contents of MacConkey agar: (Mnemonic: PLANT) Peptone Lactose (differential medium: lactose fermentation +/-) Agar Neutral red: Acts as an indicator The fermentation of lactose gives a pink color. If there is no fermentation, it looks pale. Taurocholate (selective medium).
- Peptone
- Lactose (differential medium: lactose fermentation +/-)
- Agar
- Neutral red: Acts as an indicator The fermentation of lactose gives a pink color. If there is no fermentation, it looks pale.
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- Taurocholate (selective medium).
- The fermentation of lactose gives a pink color.
- If there is no fermentation, it looks pale.



Lactose fermenter	E. coli Klebsiella Enterobacter citrobacter
Late lactose fermenter	Shigella sonnei
Non lactose fermenter	Salmonella, shigella, proteus, yersinia

Incorrect Options:

Option A - Bile salt Agar: It is a plating medium with a pH of 8.2 used to grow Vibrio cholerae. It does not help differentiate between lactose fermenters and non-lactose fermenters.

Option B - Thayer-Martin agar: This medium is used for the isolation of Neisseria species. It contains antimicrobials (vancomycin, colistin, and nystatin) which inhibit the growth of microbes that inhibit the growth of N. gonorrhoeae.

Option C - Sabouraud Dextrose Agar: It is used for the isolation of the yeast and most filamentous fungi. It contains 2% dextrose, antibiotics (gentamicin and chloramphenicol) to kill bacteria, cycloheximide to kill saprophytic fungi, and a pH of 5.6 (acidic).

Solution for Question 28:

Correct Option C - Mycobacterium avium:

- Mycobacterium avium is an acid-fast organism.
- The acid-fast stain is a differentiating stain that uses the principle of resistance to decolorization with acids after being stained by aniline dyes.

• This occurs due to the presence of mycolic acid present in the cell wall of the mycobacterial genus.

• Procedure: Smear is poured with a primary stain, i.e., carbol fuchsin (1%), for 5 minutes. Intermittent heating is done by flaming. Heat (mordant). Acid/ Acid-alcohol is added to decolorize - Sulphuric acid (most commonly) Methylene blue or Malachite green (secondary stain) added -Background color

• Smear is poured with a primary stain, i.e., carbol fuchsin (1%), for 5 minutes. Intermittent heating is done by flaming.

• Heat (mordant).

• Acid/ Acid-alcohol is added to decolorize - Sulphuric acid (most commonly)

• Methylene blue or Malachite green (secondary stain) added -Background color

• Acid fast organisms takes up the primary stain (red) and the other organisms takes up the secondary stain (background blue).

• Acid-fast organisms/structures and percentage of sulfuric acid suitable for staining.

• Smear is poured with a primary stain, i.e., carbol fuchsin (1%), for 5 minutes. Intermittent heating is done by flaming.

• Heat (mordant).

• Acid/ Acid-alcohol is added to decolorize - Sulphuric acid (most commonly)

• Methylene blue or Malachite green (secondary stain) added -Background color

Acid-fast organisms/structures	Sulfuric acid (%) needed for decolorization
Mycobacterium tuberculosis	25%
Mycobacterium leprae	5%
Nocardia	1%
Acid-fast parasites such as Cryptosporidium, Cyclospora, and Microsporidia	0.5%
Bacterial spore	0.25%-0.5

Incorrect Options:

Option A - Streptococcus pneumoniae: It is a gram-positive organism. It contains a thick peptidoglycan layer in its cell wall, causing its gram positivity.

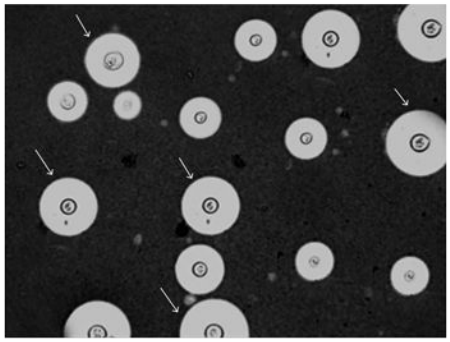
Option B - Escherichia coli: It is a gram-negative organism. It contains high amounts of lipid in its cell wall, making it permeable to secondary dye after decolourisation with organic solvents like acetone.

Option D - Actinomyces: It is a gram-positive filamentous bacterium. It contains a thick peptidoglycan layer in its cell wall, causing its gram positivity.

Solution for Question 29:

Correct Option C - India ink stain:

- India ink staining is a negative staining used to visualise cryptococcus fungi.
- Cryptococcus capsule does not take up any color, and thus, fungi stand out over the dark background caused by the stain.



Incorrect Options:

Option A - Albert stain: It is used to visualize volutin granules, aka Babes-Ernst granules, of Corynebacterium diphtheriae. The granules take on a bluish-purple color, hence called metachromatic granules.

Option B - Silver staining: Silver stains are used to visualize tin structures like flagella and treponema. It stains the structure black.

Option D - Auramine-rhodamine stain: It is a fluorescent stain that helps differentiate tubercular bacteria from the others as these appear as bright rods against a dark background when examined under UV illumination.

Solution for Question 30:

Correct Option C - Hydrogen peroxide:

- Plasma sterilization is a low-temperature sterilization technique that utilizes the fourth state of matter.
- It uses plasma, plasma refers to a gaseous state consisting of ions, photons, free electrons, and free radicals (such as O and OH).
- The plasma is generated by exposing the chemical hydrogen peroxide to radiofrequency waves.
- The plasma can penetrate small crevices and cavities, making it an effective sterilization method for complex and delicate instruments like arthroscopes and ureteroscopes.
- Control is Bacillus stearothermophilus.
- Sterrad is a machine used for plasma sterilization.

Incorrect Options:

Option A - Ethylene oxide: It is a colorless liquid that disinfects by alkylating the amino, carboxy, hydroxyl, and sulfhydryl groups in protein molecules. It is highly inflammable, highly irritant, and highly explosive and is not used in plasma sterilization.

Option B - Gamma rays: Gamma rays are ionizing radiations that causes ionization of the molecules in organisms leading to breakage of DNA.

Option D - Ultraviolet radiation: It is a nonionizing radiation that causes the destruction of nucleic acid through the induction of thymine dimers. Used for surface sterilization of air, water, and surfaces.

Solution for Question 31:

Correct Option C - Wilson & Blair medium:

- Wilson & Blair medium or SS (Salmonella Shigella) agar is a selective and differential medium used for the isolation of salmonella.
- Salmonella typhi are facultative aerobic bacteria that grow readily in simple media over a pH range of 6-8 and a temperature range between 15 and 41 degrees C.
- On this medium, they appear as jet-black colonies with a metallic sheen due to the production of H2S.

• The medium contains bile salts and dyes that inhibit the growth of most gram-positive bacteria and many gram-negative bacteria, allowing for the selective growth of Salmonella.

• The paratyphi and other species produce green colonies as they do not produce H2S.



Incorrect Options:

Option A - PLET medium: Polymyxin-Lysozyme-EDTA-Thallous acetate medium is used to culture and isolate bacillus anthracis.

Option B - BCYE medium: Buffered charcoal yeast extract medium is used to isolate and culture Legionella pneumophila. It contains L-cysteine and antibiotic supplements, with 5% CO2 at a pH of 6.9.

Option D - Stuart's medium: It is a non-selective transport medium that is used for the short-term storage and transport of Neisseria bacteria.

Solution for Question 32:

Correct Option A - Hydrogen Peroxide:

• Hydrogen peroxide is a powerful disinfectant. H2O2 works by producing destructive hydroxyl free radicals that can attack various cell components. Hydrogen peroxide is an effective disinfectant for inanimate objects and surfaces, but on animal surface it produces skin irritation, vapors may irritate the eyes and mucous membranes. Hence, it is not used as a sanitizer.

Incorrect Options:

Option B - Penicillin: Penicillin is an antibiotic, not a sanitizing agent or a disinfectant.

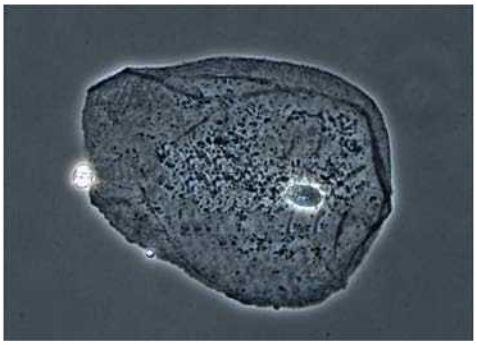
Option C - Methyl paraben: Methyl paraben is used as antimicrobial preservative in cosmetic products, not a disinfectant.

Option D - Alcohol: Alcohol can be used as a disinfectant as well as sanitizer.

Solution for Question 33:

Correct Option A - Phase contrast microscopy:

- Phase contrast microscopy is a technique that produces high-contrast images. This microscope creates a contrast between living cells and water by converting slight differences in refractive index and cell density into detectable variations in light intensity, allowing for visualization of unstained cells. It enables background light to be separated from specimen-diffracted light.



Incorrect Options:

Option B - Dark field microscopy: It is used for the demonstration of thin structures like flagella and spirochetes.

Option C - Electron microscopy: It is used to study the surface and internal details of a cell.

Option D - Polarized microscopy: This technique is used to detect amyloidosis and crystal deposits (silica in silicosis).

Solution for Question 34:

Correct Option C - Selenite F broth:

- The clinical presentation of the patient suggests the diagnosis of dysentery most commonly caused by salmonella, shigella species, or enteroinvasive coli.
- Among the options, Selenite F broth is an enrichment media for detecting Salmonella and Shigella species.

Incorrect Options:

Option A - Robertson cooked meat broth: It isolates aerobes, anaerobes, and microaerophilic organisms. However, this broth is commonly used for the isolation of Clostridium species.

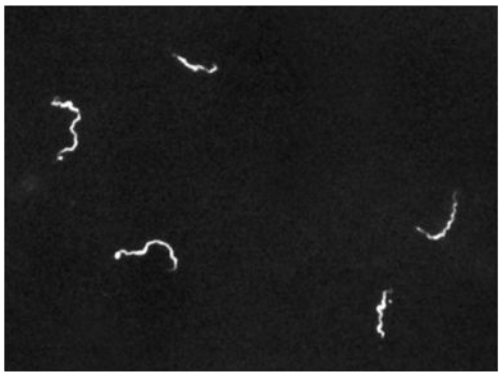
Option B - Alkaline peptone water broth: It is a medium that allows the enrichment of Vibrio cholera. The main feature of Vibrio cholera is rice water stool.

Option D - Brain heart infusion broth: It is used to grow fastidious and non-fastidious bacteria. Hence, the diagnosis would take longer if the clinical specimen is inoculated into this broth.

Solution for Question 35:

Correct Option A - Dark-field microscope:

- The clinical presentation of painless indurated genital ulcer and occupation of the patient is suggestive of syphilis caused by treponema pallidum.
- Treponema can be visualized under dark field microscope, it appears as slender, flexible, spirally coiled bacilli with tapering ends and slow movements.
- Spiral-shaped bacteria with axial filaments, includes leptospira, treponema, and borrelia. Only borrelia can be visualized using aniline dyes (Wright or Giemsa stain) in light microscopy due to size. Treponema is visualized by dark-field microscopy or direct fluorescent antibody (DFA) microscopy.



Incorrect Options:

Options B, C and D are incorrect. Refer to Option A for an explanation.

Solution for Question 36:

Correct Option D - MacConkey agar:

- MacConkey agar is a widely used medium for identifying and isolating enteric bacteria, including E. coli.
- It contains: Peptone Lactose (differential medium: lactose fermentation +/-) Agar Neutral red: Acts as an indicator The fermentation of lactose gives a pink color. If there is no fermentation, it looks pale. Taurocholate (bile salts making it a selective medium).
- Peptone
- Lactose (differential medium: lactose fermentation +/-)
- Agar
- Neutral red: Acts as an indicator The fermentation of lactose gives a pink color. If there is no fermentation, it looks pale.
- The fermentation of lactose gives a pink color.
- If there is no fermentation, it looks pale.
- Taurocholate (bile salts making it a selective medium).
- Bile salts prevent the growth of gram-positive bacteria, giving preference to gram-negative bacteria like E. coli. Moreover, the medium is enriched with lactose and neutral red, which is a pH indicator that helps distinguish between bacteria that can digest lactose and those that

cannot.

• Other culture mediums used to isolate E.coli are: Deoxycholate citrate agar (DCA). Xylose lysine deoxycholate agar (XLD). Hektoen enteric agar.

• Deoxycholate citrate agar (DCA).

• Xylose lysine deoxycholate agar (XLD).

• Hektoen enteric agar.

• Peptone

• Lactose (differential medium: lactose fermentation +/-)

• Agar

• Neutral red: Acts as an indicator The fermentation of lactose gives a pink color. If there is no fermentation, it looks pale.

• The fermentation of lactose gives a pink color.

• If there is no fermentation, it looks pale.

• Taurocholate (bile salts making it a selective medium).

• The fermentation of lactose gives a pink color.

• If there is no fermentation, it looks pale.

• Deoxycholate citrate agar (DCA).

• Xylose lysine deoxycholate agar (XLD).

• Hektoen enteric agar.

Incorrect Options:

Option A - Chocolate agar: It is a non-selective medium used for the growth of fastidious organisms like H.influenzae.

Option B - Thayer-Martin agar: This medium is used for the isolation of Neisseria species. It contains antimicrobials (vancomycin, colistin, and nystatin) which inhibit the growth of microbes that inhibit the growth of N. gonorrhoeae.

Option C - Sabouraud Dextrose Agar: It is used for the isolation of the yeast and most filamentous fungi. It contains 2% dextrose, antibiotics (gentamicin and chloramphenicol) to kill bacteria, cycloheximide to kill saprophytic fungi, and a pH of 5.6 (acidic).

Solution for Question 37:

Correct Option B - Hot air oven at 160 degrees Celsius for 120 minutes:

- The method used to sterilize a glass petri dish is a hot air oven at 160 degrees Celsius for 120 minutes.
- Hot air oven uses dry heat for the sterilization of materials that may be damaged by moist heat or that are impenetrable to moist heat.
- This method involves heating the petri dish at a high temperature in a dry oven, which leads to the oxidation of the cell constituents.
- The most common cycles used are: 170°C for 60 minutes, 160°C for 120 minutes, 180°C for 30 minutes.
- 170°C for 60 minutes,
- 160°C for 120 minutes,
- 180°C for 30 minutes.
- Materials sterilized are: Glassware Cotton swabs Gel, oil, wax, Dusting powder. Greasy paraffin wax Metallic instruments.
- Glassware
- Cotton swabs
- Gel, oil, wax, Dusting powder.
- Greasy paraffin wax
- Metallic instruments.
- 170°C for 60 minutes,
- 160°C for 120 minutes,
- 180°C for 30 minutes.
- Glassware
- Cotton swabs
- Gel, oil, wax, Dusting powder.
- Greasy paraffin wax
- Metallic instruments.

Incorrect Options:

- Options A, C and D are incorrect. Refer to Option B for an explanation.

Solution for Question 38:

Correct Option B - 6, 3, 7, 4:

• The clinical presentation and given picture show a gram-positive bacilli arranged in a V and L shape, giving a Chinese letter or cuneiform appearance, which is suggestive of Corynebacterium diphtheriae.

• The correct sequence of gram staining is Crystal violet → Gram’s iodine → Acetone → Safranin. Crystal Violet: The first step in gram staining is applying pararosaniline dyes like crystal violet, methyl violet, or gentian violet dye to the bacterial smear. These dye stains both gram-positive and gram-negative bacteria purple. Gram's Iodine: After applying the crystal violet, Gram's iodine solution, also known as the mordant, is added. Gram's iodine forms a complex with the crystal violet, creating a larger molecule that is trapped within the bacterial cell. Acetone: The next step involves decolorization, which is the critical step in differentiating between gram-positive and gram-negative bacteria. Acetone or ethanol is commonly used as a decolorizing agent. In the case of gram-positive bacteria, the crystal violet-iodine complex is retained, and the purple color persists. However, in gram-negative bacteria, the decolorizing agent disrupts the outer membrane, allowing the crystal violet-iodine complex to wash out, leading to color loss. Safranin: The final step is the application of a counterstain, usually safranin, carbol fuschin, or neutral red. These dye stains gram-negative bacteria, which have been decolorized, allowing them to be visualized. Gram-positive bacteria retain the purple crystal violet stain and do not take up the counterstain.

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Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

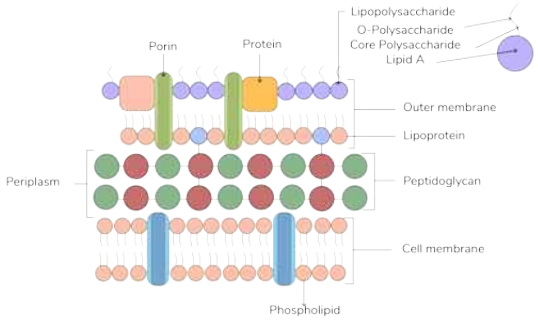
Solution for Question 39:

Correct Option C - 1:

• Lipid A is responsible for the endotoxic activity of a gram-negative bacilli.

• It is present in the lipopolysaccharide (LPS) layer of gram-negative bacilli. LPS is found exclusively in the cell wall of Gram-negative bacteria.

• The correct labeling of the image is given below.



Incorrect Options:

Option A - 2: It is a O-antigen a component of lipopolysaccharide (LPS).

Option B - 4: It is the cell membrane. The cell membrane, also known as the plasma membrane, is a vital component of all cells, including bacteria.

Option D - 3: It is the outer membrane of Gram-negative bacteria containing proteins like porins which form trans membrane proteins that serves as diffusion channel.

Solution for Question 40:

Correct Option C - 1:10 sodium hypochlorite:

Blood spill management:

- Block off the area to be disinfected: Use a caution board.
- Wear a pair of non-sterile gloves.
- Use disposable or paper towels to absorb as much of the body fluids as possible.
- Saturate the area again with sodium hypochlorite (1:10 dilution): Minimum 20 minutes.
- Rinse off the tongs, brush, and pan under running water and place to dry.
- Remove gloves and discard them.
- Wash hands carefully with soap and water, and dry thoroughly with single-use towels.
- Record the incident in the incident book if a specimen was lost or persons were exposed to blood and body fluids.

Incorrect Options: Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 41:

Correct Option A - Syringes, needles, and catheters:

- Gamma rays are ionizing radiations. It is a low-temperature sterilization method that has been used for disinfection of catheters, syringes, swabs, and plastic. It causes ionization of the molecules in organisms, leading to the breakage of DNA.

Incorrect Options:

Option B - Contaminated clothes and animal carcasses: These are sterilized through Incineration.

Option C - Gloves and aprons: These are usually sterilized by autoclaving processes.

Option D - Bacterial vaccines and intravenous solutions: These are usually sterilized by autoclaving at 60° C for one hour.

Solution for Question 42:

Correct Option A - 1, 2 & 3 only:

- Sanitizers are used for hand hygiene to reduce the microbial load on the hands and prevent the transmission of pathogens.
- Before examining the patient: It is important to use a sanitizer to ensure that your hands are clean and reduce the risk of introducing harmful microorganisms to the patient.
- After examining the patient: Using a sanitizer helps remove any potential pathogens that may have been acquired during the examination.
- After touching the patient's surroundings: Sanitizing hands after touching the patient's surroundings, such as medical equipment or surfaces, helps to prevent the spread of pathogens from one patient to another or from the environment to the healthcare provider.
- It should not be used on open wounds as it might cause a severe burning sensation.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 43:

Correct Option B - 63 degrees Celsius for 30 minutes:

- Pasteurization is done to kill or inactivate harmful microorganisms, such as bacteria, viruses, and parasites while preserving the flavor and nutritional value of the product.
- Methods of pasteurization: Holder Method: 63° C for 30 minutes. Flash Method: 72° C for 20 seconds. The Flash method is better as it kills Coxiella burnetii (Q fever)
  Ultra-high-temperature method: 149° C for 1-2 secs
- Holder Method: 63° C for 30 minutes.
- Flash Method: 72° C for 20 seconds. The Flash method is better as it kills Coxiella burnetii (Q fever)
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- Ultra-high-temperature method: 149° C for 1-2 secs

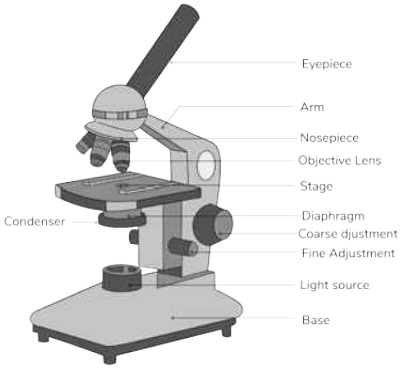
Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 44:

Correct Option B - Below the stage:

- In a compound microscope, the condenser is present below the stage in a compound microscope. It consists of an iris, which regulates the amount of light falling on the slide.



Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 45:

Correct Option C - Hydrogen peroxide:

- Plasma sterilization is a low-temperature sterilization technique that utilizes the fourth state of matter.
- It uses plasma, plasma refers to a gaseous state consisting of ions, photons, free electrons, and free radicals (such as O and OH).
- The plasma is generated by exposing the chemical hydrogen peroxide to radiofrequency waves.
- The plasma can penetrate small crevices and cavities, making it an effective sterilization method for complex and delicate instruments like arthroscopes and ureteroscopes.
- Control is Bacillus stearothermophilus.
- Sterrad is a machine used for plasma sterilization.

Incorrect Options:

Option A - Ethylene oxide: It is a colorless liquid that disinfects by alkylating the amino, carboxy, hydroxyl, and sulfhydryl groups in protein molecules. It is highly inflammable, highly irritant, and highly explosive and is not used in plasma sterilization.

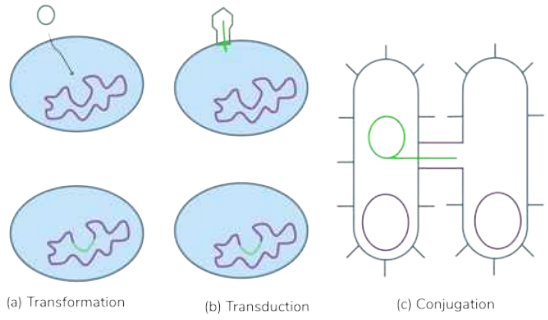
Option B - Gamma rays: Gamma rays are ionizing radiations that causes ionization of the molecules in organisms leading to breakage of DNA.

Option D - Ultraviolet radiation: It is a nonionizing radiation that causes the destruction of nucleic acid through the induction of thymine dimers. Used for surface sterilization of air, water, and surfaces.

Solution for Question 46:

Correct Option B - Transduction:

• DNA transfer in bacteria mediated by phages (bacteriophages) is called transduction. During transduction, a bacteriophage infects a bacterial cell and incorporates bacterial DNA into its phage particles. When these phage particles infect other bacteria, they deliver the bacterial DNA, resulting in the transfer of genetic material between bacteria.



Incorrect Options:

Option A - Conjugation: It refers to the direct transfer of genetic material between bacteria through physical contact.

Option C - Transformation: It refers to the uptake of free DNA from the environment by bacteria.

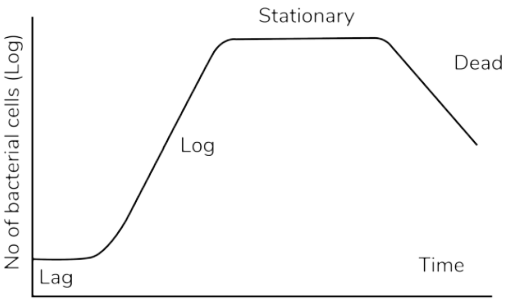
Option D - Translation: Translation is the process by which the genetic information encoded in mRNA is used to synthesize proteins. It does not involve the transfer of DNA between bacteria.

Solution for Question 47:

Correct Option A - Stationary phase:

Bacterial Growth Curve:

- It has 4 phases



Lag Phase	Log Phase	Stationary Phase	Dead Phase
No increase in the number of bacteria Cell adaptation Increase in size, enzymes and metabolite The maximum size of bacteria is seen in end of the lag phase	Cell division Smaller in size Maximum metaboli c activity More susceptible to antibiotics	Some bacteria remain viable and some are dead Bacterial growth ceases due to depletion of nutrients Mnemonic STAB (STAT from stationary) S in sporulation T in Exo Toxin production A in antibiotic production B in Bacteriocin production Antibiotic production happens in this phase	Decline phase Cell death or involution

Incorrect Options: Options B, C, and D are incorrect. Refer to Option A for an explanation.

Staphylococcus, Streptococcus, and Neisseria

1. A 7-year-old child presents with a sore throat, fever, and tender cervical lymphadenopathy. A rapid antigen detection test for Group A Streptococcus is positive. What is the most appropriate next step in management?

A. Prescribe oral penicillin V for 10 days

B. Perform a throat culture to confirm the diagnosis

C. Administer a single dose of intramuscular benzathine penicillin G

D. Prescribe oral azithromycin for 5 days

---

2. A 40-year-old man presents with a persistent cough and purulent sputum. He has a history of chronic obstructive pulmonary disease (COPD) and frequent respiratory infections. A sputum sample is sent for culture, which grows a gram-negative diplococcus that is oxidase positive and butyrate esterase positive. What is the most likely pathogen?

A. Moraxella lacunata

B. Neisseria meningitidis

C. Moraxella catarrhalis

D. Haemophilus influenzae

---

3. A 25-year-old male presents to the clinic with a 5-day history of dysuria and purulent urethral discharge. He reports having unprotected sexual intercourse with a new partner two weeks ago. Gram stain of the urethral discharge reveals gram-negative diplococci within polymorphonuclear leukocytes. What is the most appropriate initial treatment for this patient?

A. Ceftriaxone

B. Azithromycin

C. Doxycycline

D. Ciprofloxacin

---

4. A 19-year-old college student presents to the emergency department with high fever, severe headache, neck stiffness, and a petechial rash. Lumbar puncture is performed, and the CSF analysis shows elevated white blood cell count with a predominance of neutrophils, elevated protein, and decreased glucose levels. Gram stain reveals gram-negative diplococci, confirming Neisseria meningitidis. What is the most appropriate initial empirical antibiotic therapy for this patient?

A. Ceftriaxone

B. Vancomycin

C. Ampicillin

D. Ciprofloxacin

---

5. Which of the following findings in the cerebrospinal fluid (CSF) would most strongly suggest a diagnosis of Neisseria meningitidis infection?

A. Low Neutrophil count, normal protein, normal glucose

B. High Neutrophil count, high protein, low glucose

C. High Neutrophil count, low protein, normal glucose

D. Low Neutrophil count, high protein, high glucose

---

6. Which of the following tests would best help to differentiate Neisseria meningitidis from Neisseria gonorrhoeae in this clinical context?

A. Oxidase test

B. Gram staining

C. Bile esculin test

D. Carbohydrate utilization test

---

7. A clinical microbiology laboratory is performing biochemical tests to identify a strain of gamma-hemolytic streptococci isolated from a blood culture. Which of the following tests would be most appropriate to differentiate Streptococcus bovis from other gamma-hemolytic streptococci?

A. Optochin susceptibility test

B. PYR test

C. Bile esculin test

D. CAMP test

---

8. A 65-year-old woman with a history of chronic obstructive pulmonary disease (COPD) presents to her primary care physician for an annual check-up. She asks about vaccines recommended for adults to prevent pneumonia. Which of the following vaccines would be most appropriate to recommend to this patient?

A. Pneumococcal Conjugate Vaccine (PCV)

B. Varicella (chickenpox) vaccine

C. Haemophilus influenzae type b (Hib) vaccine

D. Pneumococcal Polysaccharide Vaccine (PPSV)

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9. A 60-year-old man presents to the emergency department with high fever, chills, productive cough with rust-colored sputum, and pleuritic chest pain. His medical history is significant for chronic obstructive pulmonary disease (COPD). A chest

X-ray reveals a lobar consolidation. Sputum Gram stain shows gram-positive, lancet-shaped diplococci. Which of the following is the most appropriate next step in confirming the diagnosis?

A. Throat swab culture

B. Blood culture

C. Polymerase chain reaction (PCR) test on the sputum

D. Bile solubility test on the sputum sample

---

10. A 62-year-old female with a history of rheumatic heart disease and a prosthetic mitral valve undergoes a tooth extraction. Two weeks later, she presents with fever, night sweats, and malaise. Blood cultures reveal alpha-hemolytic Streptococcus viridans. Transesophageal echocardiography shows vegetations on the mitral valve. Which of the following is the best next step in management?

A. Start oral amoxicillin and schedule follow-up in one week.

B. Admit to the hospital and start intravenous ceftriaxone and gentamicin.

C. Discharge with a prescription for oral clindamycin and instruct her to return if symptoms worsen.

D. Schedule urgent valve replacement surgery.

---

11. A 36-week pregnant woman presents for routine prenatal care. Her GBS culture from 35 weeks gestation is positive. Which of the following is the most appropriate management strategy for this patient during labor?

A. Administer intravenous ampicillin every 4 hours until delivery

B. Prescribe oral penicillin for 7 days prior to expected delivery date

C. Administer a single dose of intramuscular ceftriaxone at the onset of labor

D. No antibiotic prophylaxis needed; treat the newborn empirically after birth

---

12. A 28-year-old pregnant woman at 36 weeks gestation undergoes routine screening for Group B Streptococcus (GBS). The vaginal-rectal swab culture shows beta-hemolytic colonies on blood agar. Which of the following is the most appropriate next step to confirm GBS?

A. Perform a CAMP test

B. Conduct a latex agglutination test for group B antigen

C. Observe for orange pigment production

D. Perform a bacitracin sensitivity test

---

13. A full-term newborn develops respiratory distress, lethargy, and poor feeding within 24 hours of birth. The mother had no prenatal care. Blood culture grows gram-positive cocci in chains. Which of the following is the most likely diagnosis?

A. Early-onset Group B Streptococcal disease

B. Late-onset Group B Streptococcal disease

C. Listeria monocytogenes sepsis

D. Escherichia coli sepsis

---

14. A 48-year-old woman has a purulent discharge from a surgical wound after an abdominal surgery performed two weeks ago. An analysis of the discharge, reveals Gram-positive cocci. Further testing shows that the organism is catalase-positive and coagulase-negative. What is the most likely organism?

A. Staphylococcus aureus

B. Staphylococcus epidermidis

C. Enterococcus

D. Streptococcus pyogenes

15. A 6-year-old boy presents with fever, sore throat, and rash that started on the neck and chest and then spread to the trunk and extremities. Examination findings are given below. Which of the following is the most likely diagnosis?





A. Kawasaki disease

B. Scarlet fever

C. Toxic shock syndrome

D. Erysipelas

16. A 7-year-old boy presents with fever, sore throat, and enlarged cervical lymph nodes. Throat culture grows beta-hemolytic colonies on blood agar, and the isolate is identified as Streptococcus pyogenes. Which of the following virulence factors of S. pyogenes is primarily responsible for its ability to resist phagocytosis and contribute to invasive infections?

A. Streptolysin O

B. M protein

C. Streptokinase

D. Hyaluronidase

---

17. A 60-year-old diabetic patient presents with a persistent fever and septicemia. Blood cultures show alpha-hemolytic growth on blood agar. Which of the following organisms is most likely responsible for this patient's condition?

A. Streptococcus pyogenes

B. Streptococcus agalactiae

C. Streptococcus pneumoniae

D. Streptococcus bovis

---

18. A 26-year-old sexually active female comes with complaints of burning micturition and haematuria. On investigation, the causative agent is a gram-positive coccus which is coagulase-negative. What is the most likely organism?

A. Staphylococcus epidermidis

B. Staphylococcus aureus

C. Staphylococcus saprophyticus

D. Staphylococcus haemolyticus

---

19. A 45-year-old male patient presented with a skin infection and he was started on benzylpenicillin. A culture was done when the patient failed to respond to the given treatment and methicillin resistance was detected. Which of the following drugs can be given to this patient?

A. Ceftazidime

B. Vancomycin

C. Ceftriaxone

D. Amoxicillin

---

20. Consider the following statements. Statement A: Differentiates Staphylococcus aureus from other staphylococci species. Statement B: Differentiates tubercle bacilli from atypical mycobacteria and indicates isoniazid sensitivity.

A. Statement A is true for a catalase test.

B. Statement B is true for a coagulase test.

C. Statement A is true for a coagulase test.

D. Both statements are true for a catalase test.v

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21. A 45-year-old diabetic patient presents with a chronic foot ulcer showing signs of infection. Which laboratory test is most appropriate for identifying the causative organism, suspected to be Staphylococcus aureus?

A. Gram staining

B. PCR for mecA gene

C. Culture on blood agar

D. Antimicrobial susceptibility testing

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22. A 21-year-old woman presents with high fever, diffuse macular rash, hypotension, purulent vaginal discharge, and desquamation of the skin on her palms and soles. She has used tampons during her recent menstrual cycle and forgot to remove them on time. Laboratory tests reveal a Gram-positive, catalase-positive, coagulase-positive coccus. Which toxin is primarily responsible for the condition observed in this patient?

A. Toxic shock syndrome toxin

B. Exfoliative toxin

C. Beta-hemolysin

D. Enterotoxin type A

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23. Bronchopneumonia due to Staphylococcus aureus infection is suspected in a 40-year-old patient. Which diagnostic test utilizes protein A to detect the presence of this bacteria?

A. Coagulase test

B. Co agglutination test

C. Oxidase test

D. Catalase test

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24. A 3-year-old child presents with fever, and widespread peeling of the skin, especially around the face and trunk. The causative organism is identified as a Gram-positive, catalase-positive, and coagulase-positive coccus. Which virulence factor of this organism is responsible for the clinical manifestation observed in this child?

A. Exfoliative toxin

B. Coagulase

C. Hemolysin

D. Toxic shock syndrome toxin

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25. A 28-year-old male presents with a purulent discharge from the wound. The laboratory identifies the causative bacteria to be gram-positive cocci, non-motile, non-spore-forming, catalase-positive, and coagulase-positive. Which notable characteristic of this organism is seen when cultured on nutrient agar helps in diagnosis?

A. Pink colonies

B. Blue-green pigment

C. Matt-finely granular colony

D. Golden pigment production

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	1
Question 4	1
Question 5	2
Question 6	4
Question 7	3
Question 8	4
Question 9	2
Question 10	1
Question 11	1
Question 12	1
Question 13	1
Question 14	2
Question 15	2
Question 16	2
Question 17	3
Question 18	3
Question 19	2
Question 20	1
Question 21	3
Question 22	1

Question 23	2
Question 24	1
Question 25	4

Solution for Question 1:

Answer: A) Prescribe oral penicillin V for 10 days

Explanation:

- The most likely diagnosis in this case is Group A Streptococcal pharyngitis.
- Oral penicillin V for 10 days is the first-line treatment for Group A Streptococcal pharyngitis in most cases. It helps prevent rheumatic fever. The 10-day course is recommended to ensure complete eradication of the organism.

Gram Stain	Gram-positive cocci in chains
Appearance on Blood Agar	Pinpoint colonies with a wide zone of hemolysis Indicates beta-hemolysis
Transport Medium	Pike's media
Selective Media	PNF (Phenylethyl alcohol agar), Crystal Violet Blood Agar (CVBA)
Liquid Media	Granular turbidity Suggests the presence of streptococci, as these organisms can cause cloudiness in liquid media.
Biochemical Tests	Result
Catalase Test	Negative
Oxidase Test	Negative
PYR Test	Positive
Bacitracin Sensitivity	Sensitive

Antibiotic	Dosage (Children)	Dosage (Adolescents/ Adults)	Special Notes
Penicillin V (Oral)	250 mg 2–3 times daily for 10 days	250 mg 4 times daily or 500 mg twice daily for 10 days	First-line oral treatment; is effective and inexpensive.
Amoxicillin (Oral)	50 mg/kg once daily (max 1000 mg) or 25 mg/kg	N/A	Preferred for younger children due to better

	twice daily (max 500 mg)for 10 days		palatability; similarefficacy to penicillin.
Benzathine Penicillin G (IM)	600,000 U (for children <27 kg) 1 dose 1,200,000 U (for children ≥27 kg) 1 dose	N/A	For patients who cannot tolerate oral therapy or have poor adherence to a 10-day regimen.

Other options:

Perform Throat Culture(Option B): for Group A Streptococcus has high specificity. A positive Rapid Antigen Detection Test result in a symptomatic patient is sufficient for diagnosis and treatment initiation without the need for throat culture confirmation.

Intramuscular benzathine penicillin G (Option C): is not the first-line option for uncomplicated cases. Usually reserved for patients who cannot tolerate oral medication or where compliance is a concern.

Azithromycin (Option D): is an alternative for patients with penicillin allergy. However, it is not the first-line treatment due to increasing resistance rates and should be reserved for specific situations.

Solution for Question 2:

Answer: C) Moraxella catarrhalis

Explanation:

Moraxella catarrhalis is a gram-negative diplococcus that is oxidase positive and butyrate esterase positive. It is commonly associated with respiratory tract infections, especially in individuals with chronic lung conditions like COPD.

Moraxella Catarrhalis and Lacunata:

Characteristic	Moraxella Catarrhalis	Moraxella Lacunata (axenfeld)
Gram Stain	Gram-negative diplococci	Gram-negative coccobacilli
Growth on Blood Agar and Chocolate Agar	Smooth, opaque, grayish-white colonies	Smooth, spreading colonies with a pit in the center
Catalase Test	Positive	Positive
Oxidase Test	Positive	Positive

DNase Test	Positive	Negative
Butyrate Esterase Test	Positive	Negative
Nitrate Reduction Test	Positive	Positive
Sugar Fermentation	Does not ferment sugars	Does not ferment sugars
Common Infections	Otitis media, sinusitis, bronchitis, pneumonia	Conjunctivitis (not a common pathogen)
Virulence Factors	Beta-lactamase production, immune evasion mechanisms	Less characterized
Typical Symptoms	Ear pain, nasal congestion, cough, fever, respiratory distress	Redness, itching, discharge from the eyes
Transmission	Respiratory droplets	Direct contact with contaminated surfaces or secretions
Antibiotic Susceptibility	Generally susceptible to beta-lactamase inhibitors, macrolides, and fluoroquinolones	Typically sensitive to similar antibiotics but less commonly treated due to lower pathogenicity

M. Lacunata (Option A) is generally associated with ocular infections, such as conjunctivitis, and is not typically found in respiratory infections. Additionally, Moraxella lacunata is DNase negative and butyrate esterase negative

N. Meningitidis (Option B) is a gram-negative diplococcus and oxidase positive, it is primarily associated with meningitis and septicemia, not respiratory infections. It does not produce butyrate esterase.

H. Influenzae (Option D) is a gram-negative coccobacillus, not a diplococcus. While it can cause respiratory infections, it does not match the described gram stain and biochemical characteristics.

Solution for Question 3:

Answer: A) Ceftriaxone

Explanation:

Ceftriaxone is the first-line treatment for gonococcal infections due to its efficacy and low resistance rates. It is recommended to administer it intramuscularly at a dose of 500 mg for

uncomplicated gonorrhea.

Neisseria Gonorrhoeae:

Clinical Features	
Men	Urethritis, purulent urethral discharge, dysuria, epididymitis, prostatitis.
Women	Cervicitis, vaginal discharge, pelvic inflammatory disease (PID), dysuria, and intermenstrual bleeding. Rarely can cause Fitz-Hugh-Curtis syndrome.
Pharyngeal Infection	Often asymptomatic, sore throat, pharyngitis.
Rectal Infection	Proctitis, rectal discharge, pain, and tenesmus, may be asymptomatic.
Disseminated Gonococcal Infection (DGI)	Fever, dermatitis, polyarthralgia, septic arthritis.
Neonates	Ophthalmia neonatorum, conjunctivitis, potential for blindness if untreated.
Diagnosis	
Gram Stain	Gram-negative diplococci within polymorphonuclear leukocytes (particularly useful in men).
Culture	Thayer-Martin or other selective media, oxidase-positive, and carbohydrate utilization tests.
NAAT (Nucleic Acid Amplification Test)	Highly sensitive and specific, recommended for urogenital, pharyngeal, and rectal specimens.
Serological Tests	Not commonly used due to lower sensitivity and specificity compared to NAAT and culture.
Treatment- Green STD kit	
First-Line Therapy	Ceftriaxone 500 mg IM as a single dose PLUS doxycycline 100 mg orally twice daily for 7 days (if chlamydial co-infection is not excluded).
Alternative Therapy	Gentamicin 240 mg IM as a single dose PLUS azithromycin 2 g orally as a single dose (if ceftriaxone is not available).

Azithromycin (Option B) is an alternative treatment if ceftriaxone is unavailable or if the patient has a severe allergy to cephalosporins.

Doxycycline (Option C) is used to treat co-infection with Chlamydia trachomatis but is not effective as monotherapy for Neisseria gonorrhoeae. It should be used in conjunction with ceftriaxone for comprehensive treatment.

Ciprofloxacin (Option D): Fluoroquinolones, including ciprofloxacin, were previously used for gonococcal infections, but due to widespread resistance, they are no longer recommended as a first-line treatment.

Solution for Question 4:

Answer: A) Ceftriaxone

Explanation:

Ceftriaxone is a third-generation cephalosporin that is highly effective against Neisseria meningitidis. It penetrates the blood-brain barrier well and provides broad-spectrum coverage against other possible bacterial pathogens in cases of bacterial meningitis. Cefotaxime is an alternative with similar efficacy.

Empirical Antibiotic Therapy:

- Ceftriaxone or Cefotaxime: First-line treatment for suspected bacterial meningitis, including Neisseria meningitidis.
- Vancomycin: Added empirically to cover potential penicillin-resistant Streptococcus pneumoniae until the pathogen is identified. (Option B ruled out)
- Ampicillin: Added if Listeria monocytogenes is suspected, especially in immunocompromised patients and the elderly. (Option C ruled out)

Definitive Therapy:

- Penicillin G: The antibiotic of choice for confirmed Neisseria meningitidis, given intravenously.
- Alternative Antibiotics: Ceftriaxone or cefotaxime can be continued if the pathogen is confirmed to be susceptible.
- Duration: Treatment typically lasts 7 days.

Dexamethasone: May be used to reduce inflammation and neurological complications, particularly if started before or with the first dose of antibiotics.

Chemoprophylaxis for Close Contacts:

- Rifampin: 600 mg every 12 hours for 2 days (adults); 10 mg/kg every 12 hours for 2 days (children).

• Ciprofloxacin: A single dose of 500 mg (adults); not typically used in children. (Option D ruled out)

• Ceftriaxone: A single intramuscular dose of 250 mg (adults) or 125 mg (children).

Vaccination Prophylaxis:

Meningococcal polysaccharide vaccines (serogroups A, C, Y, and W135) administered as 2 doses to children aged 3-18 months or a single dose to older children and adults.

It is not given to children less than 2-3 years of age.

Indications-

Solution for Question 5:

Answer: B) High white blood cell count, high protein, low glucose

Explanation:

These findings are characteristic of bacterial meningitis, including meningitis caused by Neisseria meningitidis. A high Neutrophil count indicates an inflammatory response, high protein levels indicate increased permeability of the blood-brain barrier, and low glucose levels indicate bacterial consumption of glucose in the CSF.

Lab Investigations of N. Meningitidis:

Investigation	Findings
Cerebrospinal Fluid (CSF) Analysis	
Appearance	Cloudy or purulent
White Blood Cell Count (WBC)	Elevated (pleocytosis), typically >1,000 cells/µL, with predominance of neutrophils
Protein	Elevated
Glucose	Decreased
Gram Stain	Presence of gram-negative diplococci
Culture	Growth of Neisseria meningitidis on chocolate agar or Thayer-Martin agar

Polymerase Chain Reaction (PCR)	Detection of Neisseria meningitidis DNA
Latex Agglutination Test	Positive for Neisseria meningitidis antigens
Blood Tests	
Culture	Growth of Neisseria meningitidis
White Blood Cell Count	Elevated leukocytosis with left shift

CSF findings in Meningitis:

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Solution for Question 6:

Answer: D) Carbohydrate utilization test

Explanation:

The carbohydrate utilization test differentiates N. meningitidis from N. gonorrhoeae based on their ability to oxidize different sugars. N. meningitidis oxidizes both glucose and maltose, while N. gonorrhoeae oxidizes only glucose.

Differences between N. Gonorrhoeae and N. Meningitidis:

Characteristic	Neisseria gonorrhoeae	Neisseria meningitidis
Morphology	Gram-negative diplococci Kidney shaped	Gram-negative diplococci with capsule
Oxidase Test	Positive	Positive
Catalase Test	Positive	Positive
Habitat	Human genitourinary tract	Human nasopharynx
Transmission	Sexual contact	Respiratory droplets, close contact
Diseases Caused	Gonorrhea, pelvic inflammatory disease, conjunctivitis in newborns	Meningitis, septicemia (meningococcemia)
Virulence Factors	Pili, outer membrane proteins, lipooligosaccharides (LOS)	Polysaccharide capsule, pili, outer membrane proteins, LOS
Culture Media	Thayer-Martin agar, Chocolate agar	Thayer-Martin agar, Chocolate agar

Carbohydrate Utilization	Glucose positive, maltose negative	Glucose positive, maltose positive
Superoxol Test	Strong positive with 30% hydrogen peroxide	Weak or negative reaction
Antibiotic Resistance	Increasing resistance, especially to penicillin and fluoroquinolones	Generally susceptible to penicillin, but resistance can occur
Vaccination	No vaccine available	Vaccines available for various serogroups (A, C, W, Y, B)
Epidemiology	Common STD worldwide	Causes outbreaks, especially in crowded settings

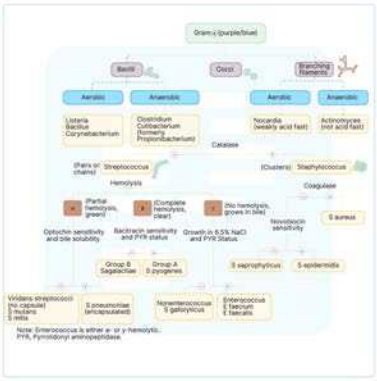
Oxidase test (Option A): Both Neisseria meningitidis and Neisseria gonorrhoeae are oxidase positive.

Gram stain morphology (Option B): Both organisms appear as gram-negative diplococci on Gram stain. This test helps to identify the general appearance of the bacteria.

Bile esculin test (Option C): It is typically used to differentiate enterococci and group D streptococci, which can hydrolyze esculin in the presence of bile.

Solution for Question 7:

Answer: C) Bile esculin test



Explanation:

The bile esculin test is used to differentiate enterococci (e.g., Enterococcus faecalis, Enterococcus faecium) from other streptococci. Streptococcus bovis is positive in this test due to its ability to hydrolyze esculin in the presence of bile.

Gamma-hemolytic streptococci do not cause hemolysis on blood agar plates. They can grow in the presence of 40% bile, 6.5% NaCl, pH 9.6, and 45 degrees Celsius and can be further classified into different groups based on their biochemical and serological characteristics

- Includes groups such as Streptococcus bovis, Streptococcus equinus, and others within the Enterococcus genus.
- These organisms are often part of the normal flora of the gastrointestinal tract but can also cause infections in humans, particularly in immunocompromised individuals.

Clinical Relevance	Infections: Can cause a range of infections including urinary tract infections, bacteremia, intra-abdominal infections, and wound infections. Endocarditis: Streptococcus bovis is particularly associated with infective endocarditis, often in patients with underlying gastrointestinal malignancies or colon disorders.
Laboratory Diagnosis	Gram Stain: Gram-positive oval cocci are often arranged in chains or pairs. Culture: Growth on blood agar without hemolysis. Growth on Mac Conkey medium- tiny magenta colony Biochemical Tests: Differentiation within the group using tests like bile esculin agar (positive for Enterococcus), PYR test (positive for Enterococcus), and other biochemical markers.
Treatment	Antibiotics: Generally susceptible to penicillin and ampicillin, although some strains, especially within the Enterococcus genus, may exhibit resistance to multiple antibiotics. Treatment of Endocarditis: Combination therapy with aminoglycosides (e.g., gentamicin) is often used for synergy, particularly in cases of enterococcal endocarditis.
Epidemiology	Common in the gastrointestinal tract, especially in elderly or immunocompromised patients Streptococcus bovis is associated with colorectal cancer and inflammatory bowel disease, warranting screening for these conditions in patients with the bacterium

Optochin susceptibility test (Option A) is used to differentiate Streptococcus pneumoniae (susceptible to optochin) from other alpha-hemolytic streptococci.

PYR test (Option B) is used to identify group A streptococci (Streptococcus pyogenes), which are beta-hemolytic. It is also used to further classify gamma haemolytic Streptococcus.

CAMP test (Option D) is used to identify group B streptococci (Streptococcus agalactiae) by demonstrating enhanced hemolysis when plated adjacent to a Staphylococcus aureus streak.

Solution for Question 8:

Answer: D) Pneumococcal Polysaccharide Vaccine (PPSV)

Explanation:

PPSV is recommended for adults over 65 years old and younger adults with certain medical conditions, such as COPD, chronic heart disease, or diabetes. It covers a broader range of pneumococcal serotypes compared to PCV and helps prevent pneumococcal pneumonia and invasive diseases in adults.

Treatment for Strep Pneumoniae	Vaccines for Strep Pneumoniae
Penicillin and Penicillin Derivatives: Effective for susceptible strains of Streptococcus pneumoniae. High-dose penicillin G or amoxicillin is the treatment of choice for non-severe infections. Cephalosporins: Such as ceftriaxone or cefotaxime are used for more severe infections or in patients with penicillin allergy. Macrolides: Such as erythromycin or azithromycin, may be used in cases of resistance or intolerance to beta-lactams. Supportive care including antipyretics, oxygen therapy, and fluids as needed.	Pneumococcal Conjugate Vaccine (PCV): Contains polysaccharide antigens from the most common serotypes of Streptococcus pneumoniae conjugated to a carrier protein. Recommended for infants and young children to prevent invasive pneumococcal disease (IPD), pneumonia, and otitis media.(Option A ruled out) It's a seven-valent conjugate vaccine. Pneumococcal Polysaccharide Vaccine (PPSV): Covers a broader range of pneumococcal serotypes compared to PCV. Recommended for adults over 65 years old, and younger adults with certain medical conditions (e.g., immunocompromised, chronic heart or lung disease).

Varicella vaccine (Option B) is used to prevent chickenpox caused by the Varicella zoster virus. It is recommended for the elderly but is not relevant for preventing pneumococcal infections in adults.

Hib vaccine (Option C) is used to prevent infections caused by Haemophilus influenzae type b, such as meningitis, pneumonia, and epiglottitis, primarily in children under 5 years old.

Solution for Question 9:

Answer: B) Blood culture

Explanation:

Blood cultures are important in diagnosing invasive pneumococcal diseases, including pneumonia, as they can identify the presence of bacteria in the bloodstream and confirm the diagnosis.

Table rendering failed

Lab investigations-

Gram Stain:	Gram-positive, lancet-shaped diplococci. Useful in the initial identification of clinical specimens like sputum, blood, or cerebrospinal fluid (CSF)
Culture	Grows well on blood agar, producing alpha-hemolytic (greenish) colonies. Optochin sensitivity test: S. pneumoniae is sensitive to optochin, differentiating it from other alpha-hemolytic streptococci. Bile solubility test: S. pneumoniae is bile soluble, confirming its identity Blood Culture Critical for diagnosing invasive pneumococcal disease and guiding antibiotic therapy.
Antigen Detection:	Latex agglutination or immunochromatographic tests for pneumococcal antigens in urine or CSF. Rapid diagnostic tool, especially useful in meningitis cases.
Polymerase Chain Reaction (PCR)	Detection of pneumococcal DNA in clinical specimens. Highly sensitive and specific, useful for diagnosing invasive pneumococcal disease
Quellung Reaction:	Capsular swelling test where specific antisera cause the capsule of S. pneumoniae to swell, confirming its presence and serotype

Throat swab culture (Option A) is not appropriate for diagnosing pneumonia. They are primarily used for diagnosing streptococcal pharyngitis, which is typically caused by Streptococcus pyogenes.

Polymerase chain reaction (PCR) test on the sputum (Option C): While PCR is highly sensitive and specific for detecting pneumococcal DNA, it is not typically the first step in standard diagnostic protocols

Bile solubility test on the sputum sample (Option D) is used to confirm the identity of Streptococcus pneumoniae from isolated colonies, not directly from sputum samples. It is a secondary confirmatory test once the organism has been cultured.

Solution for Question 10:

Answer: B) Admit to the hospital and start intravenous ceftriaxone and gentamicin

Explanation:

Intravenous ceftriaxone and gentamicin are often used together in treating infective endocarditis caused by Streptococcus viridans, especially in patients with prosthetic valves, as combination therapy can be synergistic and more effective in eradicating the infection.

Most commonly affecting prosthetic valve <12 months after replacement- Staph epidermidis

More than 12 months after replacement- Strep viridans

Alpha Hemolytic Streptococcus:

- Part of the normal flora of the human mouth, throat, and gastrointestinal tract.
- Exhibits alpha-hemolysis on blood agar, leading to a partial or greenish discoloration around colonies.
- Includes multiple species such as S. mitis, S. mutans, S. sanguinis (leads to SABE), and S. salivarius.

Epidemiology:	Dental procedures can introduce Streptococcus viridans into the bloodstream, potentially leading to infective endocarditis. Patients with pre-existing heart conditions, such as valvular heart disease or prosthetic valves, are at higher risk.
Infection Mechanism:	Capable of producing biofilms, aiding in adherence to heart valves, and resistance to phagocytosis and antibiotics. Production of extracellular polysaccharides enhances its ability to colonize and persist in host tissues.
Clinical Presentation:	Symptoms can include fever, chills, heart murmur, fatigue, and signs of embolic events such as petechiae or splinter hemorrhages. Complications may involve heart failure, systemic embolization, and severe valve dysfunction.

Diagnosis:	Blood cultures are essential for identifying the causative organism. Echocardiography (especially transesophageal echocardiography) is used to visualize vegetations on the heart valves
Treatment:	Prolonged antibiotic therapy, often involves penicillin or ceftriaxone combined with an aminoglycoside. In some cases, surgical intervention may be necessary to repair or replace damaged valves.

Start oral amoxicillin and schedule follow-up in one week (Option A): Oral antibiotics are not appropriate for treating endocarditis due to the need for high, sustained bactericidal concentrations in the bloodstream.

Discharge with a prescription for oral clindamycin and instruct her to return if symptoms worsen (Option C): Oral antibiotics are not appropriate for treating endocarditis due to the need for high, sustained bactericidal concentrations in the bloodstream.

Schedule urgent valve replacement surgery (Option D): While valve replacement surgery (Option D) may eventually be necessary, the immediate next step is to control the infection with appropriate intravenous antibiotics. Surgery is considered if there is severe valve dysfunction, heart failure, or persistent infection despite antibiotics.

Solution for Question 11:

Answer: A) Administer intravenous ampicillin every 4 hours until delivery

Explanation:

Intravenous ampicillin (or penicillin) administered every 4 hours during labor until delivery is the recommended intrapartum antibiotic prophylaxis (IAP) for GBS-positive mothers. This approach significantly reduces the risk of early-onset GBS disease in newborns.

GBS prevention:

1. Screening: Universal screening of pregnant women at 35-37 weeks gestation is recommended in many countries.

2. Indications for IAP:

- Positive GBS screening culture during current pregnancy

- GBS bacteriuria during current pregnancy
- Previous infant with invasive GBS disease
- Unknown GBS status with risk factors (preterm labor, prolonged rupture of membranes, intrapartum fever)

3. Antibiotics:

- First-line: Penicillin G or Ampicillin
- For penicillin-allergic patients: Cefazolin (if low risk for anaphylaxis) or Clindamycin/Vancomycin (if high risk for anaphylaxis and isolate susceptible)

In case the newborn gets infected, antibiotics like penicillin and other supportive management are used for treatment.

Prescribe oral penicillin for 7 days prior to expected delivery date (Option B): Oral antibiotics are not effective for GBS prophylaxis. The colonization often recurs after oral treatment, and the timing of labor onset is unpredictable.

Administer a single dose of intramuscular ceftriaxone at the onset of labor (Option C): While ceftriaxone is active against GBS, a single intramuscular dose is not the recommended regimen. Continuous intravenous therapy during labor is necessary to maintain adequate antibiotic levels.

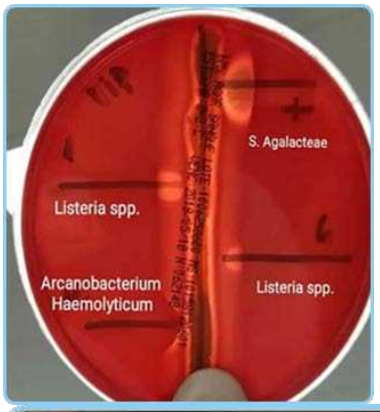
No antibiotic prophylaxis needed; treat the newborn empirically after birth (Option D): No antibiotic prophylaxis needed; treat the newborn empirically after birth: This approach is not recommended as it fails to prevent vertical transmission of GBS during labor and delivery.

Solution for Question 12:

Answer: A) Perform a CAMP test

Explanation:

The CAMP (Christie, Atkins, Munch-Petersen) test is a simple and specific test for presumptive identification of Streptococcus agalactiae (GBS). It detects the CAMP factor, a protein produced by GBS that enhances the hemolytic activity of Staphylococcus aureus beta-lysin.





A positive CAMP reaction is indicated by an arrowhead-shaped zone of enhanced hemolysis where the GBS streak approaches the S. aureus streak.

GBS Diagnosis:

1. Culture on blood agar (beta-hemolytic colonies)

2. CAMP test (presumptive identification)

3. Latex agglutination or nucleic acid amplification tests for confirmation

4. Hippurate hydrolysis test (positive for GBS)

5. Resistance to bacitracin and optochin

GBS Treatment:

1. For neonatal sepsis: Ampicillin plus gentamicin empirically; adjust based on susceptibility

2. For maternal colonization: Intrapartum antibiotic prophylaxis with penicillin G or ampicillin

3. For penicillin-allergic patients: Cefazolin (if low risk for anaphylaxis) or clindamycin/vancomycin (if high risk for anaphylaxis and isolate susceptible)

Conduct a latex agglutination test for group B antigen (Option B): Latex agglutination test for GBS is typically used for rapid detection from clinical specimens or for confirming isolates. The

CAMP test is more cost-effective for initial presumptive identification.

Observe for orange pigment production (Option C): While some strains of GBS can produce an orange pigment, this characteristic is not consistent across all strains and is not a reliable method for identification.

Perform a bacitracin sensitivity test (Option D): The Bacitracin sensitivity test is primarily used for differentiating Group A Streptococcus (S. pyogenes), which is typically bacitracin-sensitive. It is not useful for identifying GBS.

Solution for Question 13:

Answer: A) Early-onset Group B Streptococcal disease

Explanation:

- Early-onset Group B Streptococcal (GBS)/ Streptococcus agalactiae disease typically presents within the first 24-48 hours of life.
- The clinical features of respiratory distress, lethargy, and poor feeding are classic signs of neonatal sepsis.
- The gram-positive cocci in chains from the blood culture are consistent with GBS. The unknown GBS status of the mother increases the risk for this condition.
- Prevention strategies focus on screening pregnant women at 35-37 weeks gestation and providing intrapartum antibiotic prophylaxis to GBS-positive mothers.

Conditions caused by GBS in neonates:

Condition	Onset	Clinical Features	Diagnosis	Treatment	Complications
Neonatal Sepsis	Early-Onset (0-6 days)/ Late-Onset (7-89 days)	Fever, irritability, lethargy, poor feeding, respiratory distress	Blood cultures, CSF analysis, PCR	Ampicillin + gentamicin or cefotaxime	Meningitis, pneumonia, death
Neonatal Meningitis	Late-Onset (7-89 days)	Fever, irritability, poor feeding, bulging fontanelle	CSF analysis, blood cultures, PCR	Ampicillin + gentamicin or cefotaxime	Neurological sequelae, hydrocephalus, death
Neonatal Pneumonia	Early-Onset (0-6 days)	Respiratory distress, tachypnea, grunting,	Chest X-ray, blood cultures, PCR	Ampicillin + gentamicin or cefotaxime	Respiratory failure, sepsis, death

		cyanosis			

In pregnant women:

- Urinary tract infections
- Chorioamnionitis
- Postpartum endometritis

In adults:

- Skin and soft tissue infections
- Bacteremia
- Pneumonia
- Osteomyelitis
- Endocarditis

Late-onset Group B Streptococcal disease (Option B): typically occurs between 1 week and 3 months of age.

Listeria monocytogenes sepsis (Option C): While Listeria can cause neonatal sepsis, it is less common than GBS. Listeria typically appears as gram-positive rods.

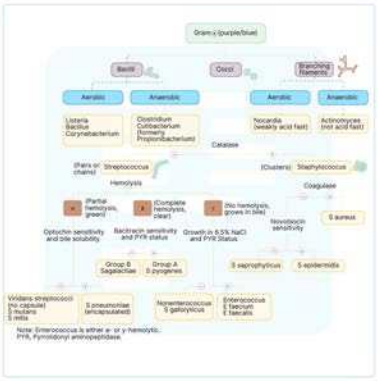
Escherichia coli sepsis (Option D): E. coli is a common cause of neonatal sepsis, but it is a gram-negative rod.

Solution for Question 14:

Correct Answer: B) Staphylococcus epidermidis

Explanation:

Staphylococcus epidermidis is present on normal human skin and is the most common cause of CoNS infections in humans. When the host's defenses are breached (in this scenario it is the surgery), stitch abscess frequently results from it.



(Options A, C, and D)

CONS (Coagulase negative staphylococcus):

- Normal flora of the human body
- S. epidermidis, S.haemolyticus, and S.lugdunensis cause device-associated and healthcare-associated infections.
- The most frequent cause of CoNS infections in people is S. epidermidis.
- It causes biofilm formation and prosthetic heart valve infection(<12 months post-replacement)
- It is normal skin flora and causes illness when the host's defenses are compromised. It frequently results in stitch abscesses.

Solution for Question 15:

Correct Answer: B) Scarlet fever

Explanation:

The clinical presentation is classic for scarlet fever. This disease is caused by the Streptococcus pyogenes strain that produces pyrogenic exotoxins. The characteristic features include:

- Fever and sore throat (streptococcal pharyngitis)
- Sandpaper-like rash with Pastia’s lines that blanches on pressure



- Rash typically starts on neck and chest, then spreads
- "Strawberry tongue" (red, swollen tongue with prominent papillae)



Diseases caused by Streptococcus pyogenes:

Disease	Clinical Features	Diagnosis	Treatment	Complications
Pharyngitis	Sore throat, fever, swollen lymph nodes	Throat culture, rapid antigen detection test	Penicillin, amoxicillin, cephalosporins	Rheumatic fever, acute glomerulonephritis
Scarlet Fever (Option B)	Rash, fever, strawberry tongue	Clinical presentation, throat culture	Penicillin, erythromycin	Rheumatic fever, septicemia
Impetigo- Non-bullous	Honey-crusted lesions on the face and limbs	Clinical presentation, culture of lesions	Topical mupirocin, oral antibiotics	Post-streptococcal glomerulonephritis
Erysipelas (Option D)	Raised, well-demarcated erythematous lesions	Clinical presentation, blood cultures	Penicillin, erythromycin	Bacteremia, septicemia
Cellulitis	Diffuse, spreading infection of the skin and subcutaneous tissue	Clinical presentation, blood cultures	Penicillin, clindamycin, cephalosporins	Necrotizing fasciitis, sepsis

Necrotizing Fasciitis	Severe pain, swelling, fever, rapid spread of infection	Clinical presentation, imaging studies	Surgical debridement, broad-spectrum antibiotics	Septic shock, multi-organ failure
Streptococcal Toxic Shock Syndrome (STSS) (Option C)	Fever, hypotension, multi-organ involvement	Clinical presentation, blood cultures	IV fluids, vasopressors, broad-spectrum antibiotics	Acute respiratory distress syndrome (ARDS)
Acute Rheumatic Fever	Migratory arthritis, carditis, chorea, erythema marginatum	Clinical criteria (Jones criteria), throat culture	Penicillin, anti-inflammatory drugs (aspirin)	Chronic rheumatic heart disease
Acute glomerulonephritis	Hematuria, proteinuria, edema, hypertension	Clinical presentation, urine analysis	Supportive care, antihypertensives	Chronic kidney disease

Kawasaki disease (Option A): can cause a rash and fever, but it also typically presents with conjunctivitis, cervical lymphadenopathy, and swelling of extremities.

Solution for Question 16:

Answer: B) M protein

Explanation:

- M protein is a major virulence factor of Streptococcus pyogenes (Group A Streptococcus) that plays a crucial role in resisting phagocytosis.
- It's an anti-phagocytic surface protein that interferes with complement activation and opsonization, allowing the bacteria to evade the host's immune defenses.
- The M protein also plays a role in adherence to epithelial cells and is implicated in the post-streptococcal sequelae due to molecular mimicry with human tissues

Virulence Factor	Description	Mechanism of Action	Associated Diseases
M Protein (Option B)	Surface protein	Resists phagocytosis by inhibiting opsonization, interacts with host cell receptors	Pharyngitis, rheumatic fever
Streptolysin O and S (Option A)	Hemolysins	Lyse red and white blood cells, and platelets; cause	Pharyngitis, skin infections

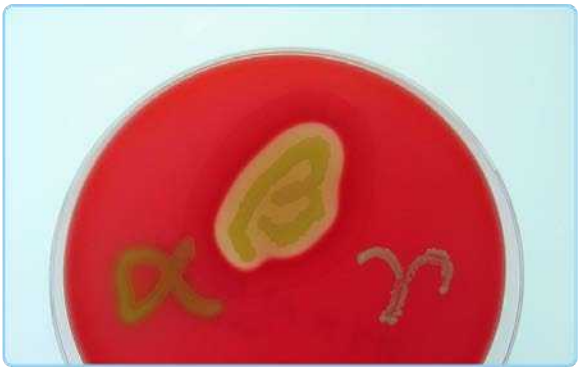
		tissue damage	
Streptococcal Pyrogenic Exotoxins (SPEs)	Exotoxins	Act as superantigens, causing a massive immune response; associated with scarlet fever and streptococcal toxic shock syndrome	Scarlet fever, toxic shock-like syndrome (TSLS)
Hyaluronidase (Option D)	Enzyme	Degrades hyaluronic acid in the host's connective tissue, facilitating bacterial spread	Cellulitis, necrotizing fasciitis
Streptokinase (Option C)	Enzyme	Converts plasminogen to plasmin, dissolving blood clots and facilitating spread	Deep tissue infections, pharyngitis
DNases (A, B, C, D)	Enzymes	Depolymerize free DNA in pus, reducing viscosity and aiding the spread	Abscess formation, skin infections
C5a Peptidase	Enzyme	Degrades the C5a component of the complement system, inhibiting chemotaxis of phagocytes	Immune evasion, systemic infections
Capsule	Polysaccharide capsule	Composed of hyaluronic acid; inhibits phagocytosis	Severe invasive diseases, immune evasion

Solution for Question 17:

Answer: C) Streptococcus pneumoniae

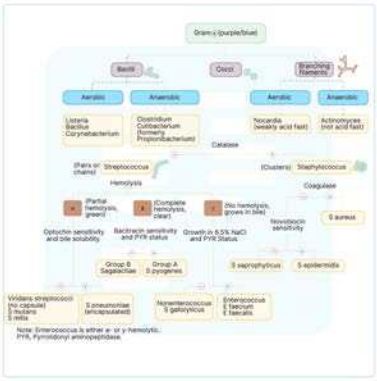
Explanation:

Streptococcus pneumoniae is an alpha-hemolytic streptococcus commonly causing pneumonia, septicemia, and meningitis, especially in diabetic patients and those with other underlying health issues.



Classification of Streptococcus Based on Hemolysis:

Hemolytic Activity	Example Species	Characteristics
Alpha-hemolytic	S. pneumoniae, (Option C) S. viridans	Partial hemolysis, greenish color
Beta-hemolytic	S. pyogenes, (Option A) S. agalactiae (Option B)	Complete hemolysis, clear zone
Gamma-hemolytic	S. bovis, (Option D) Enterococcus	No hemolysis



Lancefield Classification of Beta-hemolytic Streptococci:

- Based on specific carbohydrate antigens on cell walls
- Groups: A-H, K-V (excluding I and J)

Lancefield Group	Key Species	Associated Infections
Group A	Streptococcus pyogenes	Strep throat, impetigo, erysipelas, cellulitis, necrotizing fasciitis, rheumatic fever, glomerulonephritis
Group B	Streptococcus agalactiae	Neonatal meningitis, sepsis, infections in immunocompromised adults

Group C	Streptococcus equisimilis	Pharyngitis, endocarditis
Group F	Various Streptococcus species	Rare, "minute streptococci", infections in various body sites
Group G	Various Streptococcus species	Tonsillitis, endocarditis, urinary tract infections

Solution for Question 18:

Answer: C) Staphylococcus saprophyticus

Explanation:

The causative agent is a gram-positive coccus and is coagulase-negative. In a sexually active, the most common cause of a UTI, with the above features, is Staphylococcus saprophyticus.

Coagulase-negative staphylococci (CoNS) are part of the human body's normal flora.

- Significant species include: Staphylococcus epidermidis Staphylococcus saprophyticus Staphylococcus haemolyticus Staphylococcus lugdunensis
- Staphylococcus epidermidis
- Staphylococcus saprophyticus
- Staphylococcus haemolyticus
- Staphylococcus lugdunensis
- Staphylococcus epidermidis
- Staphylococcus saprophyticus
- Staphylococcus haemolyticus
- Staphylococcus lugdunensis

Staphylococcus saprophyticus:

- Commonly found on human skin and periurethral area
- Causes urinary tract infections- second most common cause, especially in sexually active young women
- Generally susceptible to most antibiotics
- Resistant to nalidixic acid

- Identifiable by its novobiocin resistance

Infections Caused by CoNS:

Staphylococcus epidermidis:

- The most prevalent cause of CoNS infections
- Commonly found on human skin

Conditions caused by Staphylococcus epidermidis:	
Bacteremia	Implanted foreign bodies
Cystitis	Compromised defenses
Bloodstream infections	Central lines
Endocarditis	Drug addicts

Condition caused by Staphylococcus saprophyticus:	
Urinary tract infections	Sexually active young women
Antibiotic resistance	Nalidixic acid, novobiocin

Staphylococcus epidermidis (Option A): is also coagulase-negative, but it is more commonly associated with infections related to indwelling medical devices or prosthetic materials.

Staphylococcus aureus (Option B): is a coagulase-positive organism

Staphylococcus haemolyticus (Option D): causes severe infections in several body systems including meningitis, endocarditis, prosthetic joint infections, and bacteremia, and is prevalent in the hospital environment

Solution for Question 19:

Answer: B) Vancomycin.

Explanation:

The given scenario depicts a case of methicillin-resistant Staphylococcus aureus and the suitable drug for the same is Vancomycin.

- Definition: MRSA is a type of Staphylococcus aureus resistant to methicillin and other β-lactam antibiotics.

• Common Habitats: Found on human skin and mucous membranes; typically harmless unless entering through damaged skin.

• Cause: Resistance arises from the Staphylococcal chromosomal cassette for methicillin resistance (SCCmec).

• Key Proteins: MecA: An alternative penicillin-binding protein encoded by SCCmec, not recognized by β-lactam antibiotics. MecI and MecR1: Regulate MecA synthesis upon exposure to methicillin or other β-lactams.

• MecA: An alternative penicillin-binding protein encoded by SCCmec, not recognized by β-lactam antibiotics.

• MecI and MecR1: Regulate MecA synthesis upon exposure to methicillin or other β-lactams.

• MecA: An alternative penicillin-binding protein encoded by SCCmec, not recognized by β-lactam antibiotics.

• MecI and MecR1: Regulate MecA synthesis upon exposure to methicillin or other β-lactams.

Diagnostic Methods:

Diagnostic Method	Description
Culture-based techniques	Use of chromogenic agar media where MRSA colonies appear pink, and other bacteria appear blue.
Blood Sample Testing	10–20 ml of venous blood is drawn and injected into two culture bottles, incubated aerobically and anaerobically at 35°C. Automated systems detect growth via turbidity, fluorescence, O2 consumption, or CO2 production. Most bacteria are recovered within 2 days. Gram staining and specialized media are used for further identification.
CSF Sample Testing	For suspected meningitis, a lumbar puncture is performed to obtain 3–5 ml of CSF. Turbidity and high leukocyte counts indicate infection. CSF is Gram-stained and inoculated onto selective culture media.

Antibiotic Susceptibility Testing- Confirmation of infection is followed by testing to determine the best treatment strategy.

• Treatment Options: Vancomycin: Effective but challenged by vancomycin-resistant strains (VRSA) (Option B). Linezolid: Primary treatment for VRSA especially Lung infections for other infections Daptomycin is used. Concerns: Emergence of vancomycin-intermediate Staphylococcus aureus (VISA) with reduced vancomycin susceptibility and thickened cell walls Tetracyclines Sulpha-drugs.

• Vancomycin: Effective but challenged by vancomycin-resistant strains (VRSA) (Option B).

• Linezolid: Primary treatment for VRSA especially Lung infections for other infections Daptomycin is used.

• Concerns: Emergence of vancomycin-intermediate Staphylococcus aureus (VISA) with reduced vancomycin susceptibility and thickened cell walls

• Tetracyclines

• Sulpha-drugs.

• Vancomycin: Effective but challenged by vancomycin-resistant strains (VRSA) (Option B).

• Linezolid: Primary treatment for VRSA especially Lung infections for other infections Daptomycin is used.

• Concerns: Emergence of vancomycin-intermediate Staphylococcus aureus (VISA) with reduced vancomycin susceptibility and thickened cell walls

• Tetracyclines

• Sulpha-drugs.

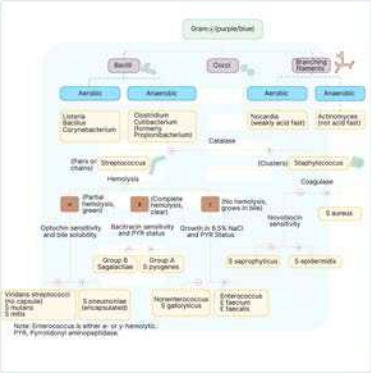
Feature	VISA	VRSA
Full Name	Vancomycin-Intermediate Staphylococcus aureus	Vancomycin-Resistant Staphylococcus aureus
Resistance Level	Intermediate resistance to vancomycin	High resistance to vancomycin
MIC (Minimum Inhibitory Concentration)	4-8 µg/mL	≥ 16 µg/mL
Mechanism of Resistance	Thickened cell wall leading to reduced penetration of vancomycin	Acquisition of vanA gene from Enterococcus, altering the target site of vancomycin
Treatment Options	Higher doses of vancomycin, linezolid, daptomycin, or alternative agents	Linezolid, daptomycin, quinupristin-dalfopristin, or alternative agents
Prevalence	More common than VRSA, but still relatively rare	Very rare compared to VISA
Clinical Significance	This can lead to treatment failure with standard vancomycin dosing	Represents a significant therapeutic challenge due to high-level resistance

Other Options (Ceftazidime, Ceftriaxone, and Amoxicillin): are β-lactam antibiotics and therefore are ineffective against MRSA.

Solution for Question 20:

Answer: C) Statement A is true for a coagulase test.

Explanation: The coagulase test differentiates Staphylococcus aureus from other staphylococci species.



Coagulase Test:	Catalase Test:
Purpose: Differentiates Staphylococcus aureus from other staphylococci species. Methods: Tube Test: Detects free coagulase (extracellular enzyme produced by S. aureus). Bacterial culture mixed with plasma in a tube. Positive if plasma clots. Slide Test: Detects bound coagulase (clumping factor, bound to bacterial cell wall). Bacteria mixed with plasma on a slide. Positive if bacteria clump together.	Purpose: Tests for production of catalase enzyme. All Staphylococci produce catalase. Catalase neutralises hydrogen peroxide's antibacterial properties. It differentiates staphylococci form streptococci

Solution for Question 21:

Answer: C) Culture on blood agar

Explanation:

• The use of blood agar culture is considered the primary method for isolating and identifying bacterial pathogens from clinical samples.

• It is the gold standard method for isolating and identifying bacterial pathogens from clinical specimens.

• Staphylococcus aureus typically forms creamy white colonies on blood agar plates, which helps in its identification

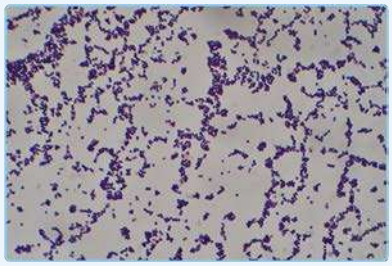
Step:	Description:
Specimen Collection	Pus samples from suppurative lesions, sputum for respiratory infections, nasal swabs for carriers
Direct Microscopy	 Gram staining reveals Gram-positive cocci in clusters, indicating staphylococci presence (Option A) Grape bunch- appearance
Culture	Culturing on blood agar to isolate and identify S. aureus colonies (Option C) Blood agar: beta hemolysis Nutrient agar: golden colonies
Selective Media	Salt-milk agar, Ludlam's medium, or Roberts' cooked meat medium to isolate S. aureus from mixed samples Mannitol salt agar- yellow color from S. aureus fermenting mannitol
Biochemical Reactions	Coagulase test: Detects S. aureus by its ability to clot plasma (positive result indicates S. aureus) Catalase test: Positive
Antimicrobial Susceptibility Testing	Determines antibiotic sensitivity to guide treatment decisions (Option D)
Molecular Methods	PCR to detect specific genes like mecA, associated with antibiotic resistance (Option B)

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Solution for Question 22:

Answer: A) Toxic shock syndrome toxin

Explanation:

• In this scenario, the woman is suffering from Toxic shock syndrome due to tampon use, which is often associated with Staphylococcus aureus.

• Bacteria produce Toxic shock syndrome toxin, a superantigen that can trigger a massive immune response causing high fever, rash, hypotension, and desquamation.

Disease caused by staphylococcus	Key features
Primary infection	
Furuncle	Superficial skin infection develops in hair follicles, sebaceous gland, or sweat gland.
Carbuncle	Development of one or more abscesses in adjacent subcutaneous tissues due to the spread of infection from a furuncle.
Impetigo	The exfoliative toxin produced causes bullous impetigo, characterized by blisters. Non-bullous impetigo is caused by streptococcus.
Acute osteomyelitis	Bacterial growth usually concentrates in a terminal blood artery of the metaphysis of a long bone, resulting in bone necrosis and persistent suppuration.
Necrotizing pneumonia	Caused by strains producing the PVL leukocidin. Localized, destructive abscesses
Endocarditis	In drug addicts Due to bacteremia
Toxins mediated	
Scalded skin syndrome	Erythema and intraepidermal desquamation in neonates caused by exfoliative toxin (Option B)
Toxic shock syndrome	TSST-1 (Toxic Shock Syndrome Toxin) Early findings: Fever, vomiting, diarrhea, muscle pain Later: Shock, renal, and hepatic injury, MODS Tampon-associated
Staphylococcal food poisoning	Vomiting is prominent without fever. Ingestion of enterotoxin-contaminated food causing acute vomiting and diarrhea within 1 to 5 hours of ingestion (Option D)

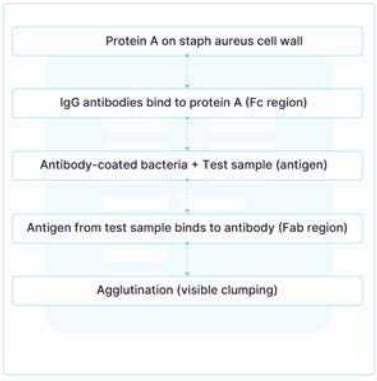
Beta-hemolysin is not responsible for Toxic shock syndrome (Option C)

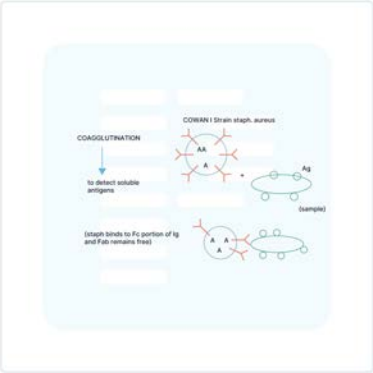
Solution for Question 23:

Answer: B) Co agglutination test

Explanation:

Protein A binds to the Fc region of IgG, allowing Staphylococcus aureus to be coated with antibodies. The coated bacteria are then used to detect the presence of Staphylococcus aureus antigens in the test sample through visible clumping.





Coagulase test (Option A): Coagulase produced by Staphylococcus aureus causes the clotting of blood plasma by converting fibrinogen to fibrin. There are two types of coagulase tests: the slide coagulase test and the tube coagulase test.

Oxidase test (Option C): is used to determine the presence of cytochrome oxidase in bacteria

Catalase test (Option D): is used to identify the presence of the enzyme catalase in bacteria, which breaks down hydrogen peroxide (H■O■) into water and oxygen.

Solution for Question 24:

Answer: A) Exfoliative toxin

Explanation:

In this scenario, the child is suffering from Staphylococcal scalded skin syndrome (SSSS), which is caused due to the Epidermolytic/Exfoliative toxin of Staphylococcus aureus.

Staphylococcal Scalded Skin Syndrome (SSSS) is a skin condition caused by Staphylococcus aureus bacteria that produce exfoliative toxins. These toxins cause the skin to blister and peel, resembling burns.

- Affects: Primarily infants and young children, but can also occur in immunocompromised adults.
- Symptoms: Fever, red, tender skin that blisters and peels, especially on the face and neck. Nikolsky's sign (skin sloughing with pressure) is often positive.
- Diagnosis: Clinical, supported by bacterial cultures or detection of exfoliative toxins.
- Treatment: Antibiotics (e.g., nafcillin, clindamycin) and supportive care (fluids, wound care).
- Prognosis: Good with prompt treatment, though severe cases can be life-threatening in infants or immunocompromised individuals.

Virulence factors	Key features	
Cell wall-associated polymers	Peptidoglycan	Activates complement-induced release of inflammatory cytokines
Teichoic acid	Adherence to the host cell surface, and shields them from complement-mediated opsonization	
Capsular polysaccharide	Inhibits opsonization	
Cell surface	Protein A	Anti-complementary Anti-phagocytic Co-agglutination

Clumping factor	Responsible for the slide coagulase test	
Toxins	Hemolysin (Option C)	Alpha hemolysin- Lyse sheep and human red blood cells. Toxic to macrophages, lysosomes, muscle tissues, the renal cortex, and the circulatory system Beta-hemolysins or sphingomyelinase Lyse sheep red blood cells only
Panton-Valentine (PVL) Leucocidin	Synergohymenotropic toxin Attacks neutrophils, platelets Produced by Methicillin-Resistant Staph. aureus (MRSA)	
Enterotoxin (superantigen)	Food poisoning Most cases are caused by type A. It acts directly on the autonomic nervous system causing symptoms. Vomiting is stimulated in the brain stem.	
Toxic shock syndrome toxin(TSST) (Option D) (superantigen)	Fatal multisystem disease Tampon-related TSS	
Epidermolytic/Exfoliative toxin (Option A)	Staphylococcal scalded skin syndrome (SSSS): the outer layer of the epidermis gets separated from the underlying tissues. Severe form of SSSS: Ritter's disease in the newborn and toxic epidermal necrolysis in older patients. Milder forms: pemphigus neonatorum and bullous impetigo	
Enzymes	Coagulase (Option B)	Clotting of human or rabbit plasma
Lipid hydrolases	Helps in skin and subcutaneous tissue infections	
Hyaluronidase	Breaks down the connective tissues Staphylokinase, protease	
Nuclease	Heat-stable DNase	

Solution for Question 25:

Answer: D) Golden pigment production

Explanation:

On nutrient agar, after 24 hours of incubation, Staphylococcus aureus produces golden-yellow pigment.

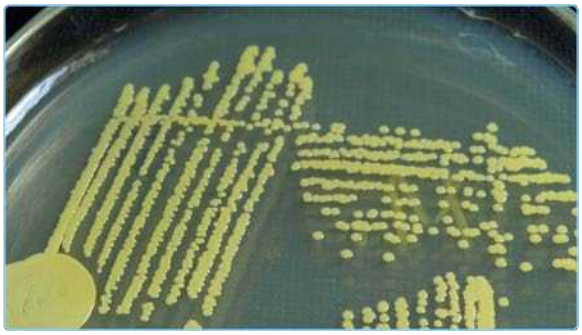
Key features of staphylococcus aureus:

Morphology

- Gram-positive bacteria
- Spherical cocci, arranged in grape-like clusters
- Non-motile
- Non-spore forming

Cultural characteristics

- Nutrient agar: Large colonies (2-4 mm in diameter), convex, smooth, shiny, opaque, and easily emulsifiable. They produce golden-yellow pigment (Option D), which does not diffuse into the medium.



- Blood agar: Marked hemolysis
- MacConkey agar: Pink colonies due to lactose fermentation (Option A)

Biochemical characteristics:

- Coagulase test positive
- Catalase test positive
- Oxidase negative

Incorrect options:

Blue-green pigment (Option B): (Pyocyanin) is produced by Pseudomonas aeruginosa.

Matt-finely granular colony (Option C): is characteristic of virulent strains of streptococcus pyogenes when grown in blood agar.

Corynebacterium & Filamentous bacterium

1. A 10 y/o girl presents with a sore throat, low-grade fever, and swollen neck glands. Examination reveals a thick greyish membrane covering her tonsils. Which among the following is the most suitable culture medium for confirming the diagnosis at the earliest stage?

A. Potassium Tellurite Agar

B. Loeffler’s serum slope

C. McLeod’s media

D. Hoyle’s media

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2. A 6 y/o child presents with a sore throat, low-grade fever, and a greyish membrane covering the tonsils. The doctor suspects diphtheria. Which diagnostic test is used for confirming the production of diphtheria toxin by C.diptheriae?

A. Blood culture

B. Throat swab culture

C. Gram stain

D. Elek's gel test

---

3. Which of the following statements regarding Nocardia and Actinomyces is false?

A. Actinomyces infections are endogenous, whereas Nocardia infections are exogenous.

B. Actinomyces is nonacid-fast, while Nocardia is partially acid-fast.

C. Penicillin is the preferred treatment for Nocardia infections.

D. Nocardia is an obligate aerobe, whereas Actinomyces is anaerobic.

4. A microbiologist cultivates a sample from a patient suspected of having a Corynebacterium infection. The bacteria grow slowly on standard media but grow well in a medium containing lipids. Which Corynebacterium species is likely being cultured?

A. C. striatum

B. C. urealyticum

C. C. pseudotuberculosis

D. C. ulcerans

5. Which of the following statements accurately describes diphtheria toxin (DT) based on the provided information?

A. DT, a DNA virus, integrates its tox gene into Corynebacterium diphtheriae, allowing toxin production.

B. The iron concentration in the blood affects DT production.

C. DT enters host cells by binding to DNA, which triggers direct cell lysis.

D. C.diptheriae spreads throughout the bloodstream, causing both bacteremia and toxaemia.

6. A child presented with a sore throat, fever, and swollen lymph nodes in the neck. Throat culture was subjected to Albert's stain and revealed bacilli arranged in a Chinese letter pattern. The patient's condition deteriorates rapidly, with signs of severe respiratory distress. Which of the following mechanisms is primarily responsible for the clinical manifestations seen in this patient?

A. Cleavage of nicotinamide adenine dinucleotide by Diphtheria Toxin (DT).

B. Inactivation of elongation factor 2 (EF-2) by Exotoxin A.

C. Binding of fragment B to human cell receptors by Diphtheria Toxin (DT).

D. Secretion of toxins by Pseudomonas aeruginosa.

---

7. A patient presents with severe swelling of the neck, along with difficulty swallowing. O/E there was a grey-white pseudomembrane in the throat. Which of the following bacteria is most likely responsible for these clinical findings? {{caption_text}}



A. Streptococcus pyogenes

B. Corynebacterium diphtheriae

C. Haemophilus influenza

D. Neisseria meningitidis

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8. A 7 y/o presents with a sore throat, low-grade fever, and difficulty swallowing. O/E, he has a greyish membrane covering his tonsils and swollen cervical lymph nodes. Which investigation helps to identify the presence of volutin granules in a case of suspected diphtheria?

A. Throat swab culture

B. Gram stain

C. Elek Test

D. Albert stain

---

9. A 2-month-old infant is brought in for vaccination. The paediatrician recommends administering the first dose of the DPT vaccine. Which among the following statements is false regarding the DPT vaccine?

A. DPT is given at the anterolateral aspect of the thigh..

B. Thiomersal is used as a preservative.

C. DPT should be kept at 2-8 Celsius.

D. The pediatric dose of diphtheria toxoid contains 2 Lf units, whereas the adult dose contains 25 Lf units

---

10. A 7-year-old child presents with severe throat pain, difficulty swallowing, and a thick grey membrane extending to the nasopharynx. Given these clinical features, what would be the initial dose of diphtheria antitoxin to administer?

A. 20000-40000

B. 40000-60000

C. 80000-120000

D. 5000-10000

---

11. Match the type of diphtheria bacilli biotypes with their characteristic colonies on Tellurite Blood Agar. 1) Gravis a) Poached egg 2) Intermedius b) Daisy's head 3) Mitis c) Frog’s egg

1) Gravis	a) Poached egg

2) Intermedius	b) Daisy's head
3) Mitis	c) Frog’s egg

A. 1-a, 2-b, 3-c

B. 1-b, 2-c, 3-a

C. 1-c, 2-b, 3-a

D. 1-a, 2-c, 3-b

12. Which of the following clinical forms of actinomycosis best describes the two given statements? Statement 1- The most common clinical form of actinomycosis Statement 2- Abscesses in bone and soft tissue, with chronic draining sinuses to the exterior

A. 1)- Pelvic form, 2)- Cervicofacial

B. 1)- Thoracic, 2)- Abdominal

C. 1)- Abdominal, 2)- Thoracic

D. 1)- Cervicofacial, 2)- Pelvic

13. Which statement about clinical manifestations of Listeria monocytogenes infection is False?

A. Pregnant women infected with Listeria monocytogenes may experience flu-like symptoms.

B. Neonatal listeriosis can present as early-onset or late-onset disease.

C. Listeria monocytogenes infection in adults very rarely leads to meningitis.

D. Immunocompromised individuals are at higher risk of severe Listeria monocytogenes infections.

14. Which of the following statements regarding laboratory diagnosis and characteristics of Listeria monocytogenes is false?

A. Shows the tumbling type of motility at 37°C.

B. Growth on PALCAM agar is enhanced by the presence of antibiotics.

C. The cold enrichment technique involves storing clinical materials at 4°C.

D. Experimental inoculation in rabbits causes marked monocytosis.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	2
Question 8	4
Question 9	4
Question 10	2
Question 11	2
Question 12	4
Question 13	3
Question 14	1

Solution for Question 1:

Correct Answer: B) Loeffler’s serum slope

Explanation: Corynebacterium diphtheria colonies can be visualized as early as 6-8 hours in Loeffler’s serum slope

Isolation of C.diptheriae

- Fastidious

• Aerobe and facultative anaerobe; do not grow on ordinary medium

• Grows best in an enriched medium such as blood agar, chocolate agar, and Loeffler's serum slope

Isolation in Different Culture Media

Culture Media	Colony appearance	Evaluation
Loeffler's serum slope	Small Circular opaque disks Glistening, White with a yellow tinge	Advantage: Growth can be detected as early as 6–8 hours. The best medium for metachromatic granule production Disadvantage: Supports the growth of other throat commensals also
Potassium tellurite agar (McLeod's and Hoyle's-modified PTA) (Options A, C&D ruled out)	Grey or black colour	Advantage: Throat commensals are inhibited Disadvantage: Colonies appear only after 48 hours of incubation
Sheep blood agar	Small Circular White Sometimes hemolytic (mitis biotype)	Differentiate colonies from streptococcal or staphylococcal pharyngitis, which may simulate diphtheria. The serum slope may show growth in 4-8 hours but, if negative, will have to be incubated for 24 hours

Identification by biochemical tests is done by Sugar fermentation test using Hiss's serum sugar media

• It is necessary to use Hiss's serum sugars for fermentation tests.

• Diphtheria ferments glucose, galactose, maltose, and dextrin

• Produces acid but without gas

• Do not ferment lactose, mannitol or sucrose.

Solution for Question 2:

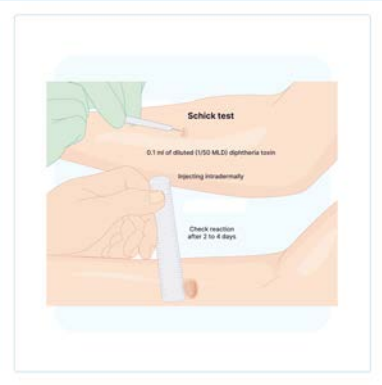
Correct Answer: D) Elek's gel test

Explanation: Elek's gel test is used to demonstrate the diphtheria toxin produced by C.diptheriae.

Demonstration of toxicity

In-vivo test- subcutaneous and intracutaneous tests done on guinea pigs

In vitro test-

Test	Features
Elek’s gel precipitation test gel precipitation test" data-author="" data-hash="9889" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/duP7wodTOR1trhtzKtWV1744446503.png" />	Immunodiffusion in gel. The strain isolated is streaked onto a media containing a filter paper soaked with antitoxin. If the strain is toxigenic, it liberates the toxin which diffuses in the agar and meets with the antitoxin to produce an arrow-shaped precipitation band. Used to know the relatedness between the strains isolated during an outbreak.
 ... (content truncated)	Based on the ability of circulating antitoxin to neutralize the diphtheria toxin given intradermally Indicates immunity or susceptibility to the disease.
Tissue culture test	The toxigenicity can be demonstrated by incorporating the strains in the agar overlay of cell culture monolayers. The toxin produced diffuses into the cells below and causes lysis of the cells
PCR	Detection of Toxin gene (tox) has been developed to detect the presence of genes coding for the toxin, in clinical isolates
ELISA	Detect toxins from the patient's isolate using an antitoxin and enzyme-substrate system.

Blood culture (Option A) and throat swab culture (Option B) are used to isolate and identify Corynebacterium diphtheriae, the bacterium that causes diphtheria, but it does not specifically identify volutin granules.

Gram stain (Option C) helps identify rod-shaped Corynebacterium diphtheriae but does not help stain the metachromatic granules

Solution for Question 3:

Correct Answer: C) Penicillin is the preferred treatment for Nocardia infections.

Explanation:

Penicillin is the preferred treatment for Actinomyces, whereas Nocardia is treated with cotrimoxazole.

Features	Actinomycetes	Nocardia
Acid-fastness (Option B ruled out)	Nonacid-fast	Partially acid-fast
O2 Requirement (Option D ruled out)	Anaerobe	Obligate aerobe
Habitat (Option A ruled out)	Found as oral flora Infections occur endogenously	The usual habitat is the soil Infections occur exogenously
Risk factors	Immunocompetent also	Usually only in immunocompromised
Clinical forms	Cervicofascial, abdominal, and others.	Pulmonary, CNS forms, Actinomycetoma
Granules	Sulfur granules are hard and not emulsifiable, consist of branching filamentous bacilli, and are surrounded by clubs (sun-ray appearance)	Granules are soft and lobulated and also show a sun-ray appearance Commonly found in mycetoma, rare in other conditions
Culture	Spidery molar teeth colony in solid media Fluffy ball at the bottom of the liquid medium	Colonies are creamy, wrinkled, and pink Isolation is done in Selective media Paraffin bait technique LJ medium
Treatment (Option C)	Penicillin	Cotrimoxazole

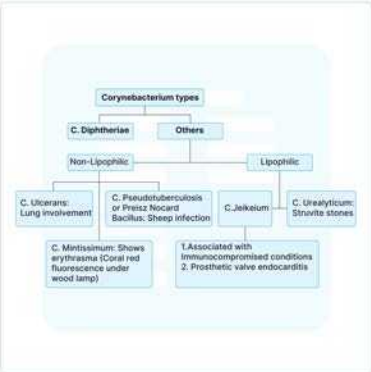
Solution for Question 4:

Correct Answer: B) C. urealyticum

Explanation:

Corynebacterium urealyticum is a lipophilic Corynebacterium species.

Group	Growth Characteristics	Example Species
Lipophilic	Grow slowly on standard media; enhanced by lipids	C. jeikeium C. urealyticum (Option B)
Non-lipophilic	Grow well on standard laboratory media	C. striatum (Option A ruled out) C. amycolatum C. pseudodiphtheriticum C. pseudotuberculosis (Option C ruled out) C. ulcerans (Option D ruled out) C. minutissimum



Some Important types of Corynebacteria and their important characteristics:

Species	Lipophilicity	Characteristics
C. pseudotuberculosis	Non-Lipophilic	Infection in cows, transmitted via milk. Erythromycin sensitivity Diphtheria antitoxin is a protective
C. minutissimum	Non-Lipophilic	Erythrasma rash in axilla and groin These lesions demonstrate coral red fluorescence on Wood's lamp evaluation

C.jeikeium	Lipophilic	Infections of the cutaneous and vascular system. High mortality rate Immunocompromised host Multidrug resistance and responds only to vancomycin

Solution for Question 5:

Correct Answer: B) Iron concentration of blood affects DT production.

Explanation:

DT production is regulated by the availability of iron; low iron concentrations trigger the expression of the tox gene. High production of DT with low iron availability (0.1 mg/l); iron addition inhibits production (0.5mg/l).

Diphtheria Toxin:

Aspect	Details
Production and Structure	DT is a protein exotoxin produced by the toxigenic Corynebacterium diphtheriae. Consists of two fragments(Bipartite), A and B, with a molecular weight of approximately 62,000 daltons. (Option C ruled out)
Role of Lysogenic β-phage	The tox gene is carried by a lysogenic β-phage. Only phage-infected strains produce toxin; nontoxigenic strains can become toxigenic if infected. Corynebacterium diphtheriae remains localized at the infection site, with only the toxin entering circulation, causing toxaemia but not bacteremia.(Option D ruled out)
Mechanism of Action	Fragment B binds host cell receptors, facilitating fragment A entry into the cytoplasm. Fragment A catalyses ADP-ribosylation of EF-2, halting protein synthesis.
Potency	The lethal dose for a 250g guinea pig: is 0.0001 mg. The estimated lethal dose for humans: 130 ng/kg body weight.
Systemic Effects	DT spreads via blood, causing myocarditis and neuritis. Myocarditis is potentially fatal.

Factors Affecting Production	Influenced by environmental factors: alkaline pH (7.8-8.0), oxygen, and iron concentration in the growth medium. High production with low iron availability (0.1 mg/l); iron addition inhibits production (0.5 mg/l).
Toxoid Formation	Treatment with formalin (0.2-0.4%), and incubation at 37°C for 4-6 weeks converts DT to toxoid. Toxoid lacks toxicity but induces an immune response; used in vaccines.
Immunity	Immunity is primarily mediated by antitoxin antibodies neutralizing DT.

DT, a DNA virus, integrates its tox gene into Corynebacterium diphtheriae, allowing toxin production (Option A): Diphtheria toxin (DT) is not a DNA virus. It is a protein exotoxin produced by Corynebacterium diphtheriae when infected by a lysogenic β-phage carrying the tox gene.

DT enters host cells by binding to DNA, which triggers direct cell lysis (Option C):DT does not bind to DNA to enter cells or cause cell lysis directly; instead, Fragment B binds to host cell receptors, facilitating the entry of Fragment A, which inhibits protein synthesis by ADP-ribosylation of EF-2.

DT primarily causes localized skin lesions, via direct contact with infected individuals (Option D): Diphtheria is primarily transmitted through direct contact or respiratory droplets. Its hallmark symptoms include respiratory distress and a pseudomembrane in the throat, rather than primarily causing localized skin lesions. While cutaneous diphtheria is possible, it's not the main presentation of diphtheria toxin effects.

Solution for Question 6:

Correct Answer: C) Binding of fragment B to human cell receptors by Diphtheria Toxin (DT).

Explanation:

The child's symptoms of sore throat , fever and swollen lymph nodes along with Albert’s stain revealing bacilli arranged in Chinese letter pattern points to a diagnosis of Diptheria caused by C.diptheriae

- In diphtheria caused by C. diphtheriae, Diphtheria Toxin (DT) is primarily responsible for the severe clinical manifestations.

- Fragment B of DT binds to specific receptors on human cells (Option C) → facilitating the entry of fragment A into the cytoplasm (Option B ruled out)→ fragment A ADPribosylates elongation factor 2 (EF-2), halting protein synthesis → leading to cell death.
- This mechanism contributes significantly to the respiratory distress and other severe symptoms observed in diphtheria patients.

Cleavage of nicotinamide adenine dinucleotide by Diphtheria Toxin (DT) (Option A): The primary toxic action involves the ADP-ribosylation of elongation factor 2 (EF-2), which results from the toxin's enzymatic activity using NAD as a substrate.

Secretion of toxins by Pseudomonas aeruginosa (Option D): It refers to a different bacterium and a different set of toxins. Pseudomonas aeruginosa is not the causative agent of diphtheria, and its toxins are not involved in the pathogenesis of this disease.

Solution for Question 7:

Correct Answer: B) Corynebacterium diphtheriae

Explanation:

- The patient’s symptoms and clinical findings are suggestive of diphtheria.
- Primarily caused by toxigenic strains of Corynebacterium diphtheriae.
- Characterized by the hallmark "bull's neck" presentation due to severe neck swelling from infection and inflammatory response to the diphtheria toxin.

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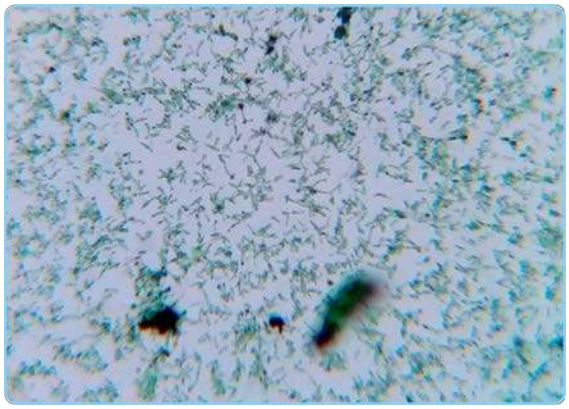
Streptococcus pyogenes (Option A) may cause infections resulting in sore throat & fever, but it never presents with a Bull’s neck appearance or a grey-white pseudomembrane.

Haemophilus influenza & Neisseria meningitidis (Options C & D) are usually associated with pneumonia and meningitis symptoms and are also not associated with the characteristic bull-neck appearance and a pseudomembrane in the throat.

Solution for Question 8:

Correct Answer: D) Albert Stain

Explanation:



Albert stain of Corynebacterium diphtheria shows bacilli with metachromatic granules.

Albert Stain

Principle: Albert's stain causes the metachromatic granules of C. diphtheriae to appear bluish-purple upon staining.

- Corynebacterium diphtheria shows chinese letter/ cuneiform appearance
- Diphtheroids show regular and palisading appearance with lesser number of granules

Type	Components	Function
Albert Solution 1	Malachite Green for organism staining Toluidine Blue for metachromatic staining	Enhances contrast and specificity in staining.
Albert Solution 2	Iodine	Specifically targets volutin or Babes-Ernst granules.

Procedure:

Application:

- Corynebacterium diphtheriae are slender bacilli arranged in V or L shapes, characterized by metachromatic granules (polar bodies, volutin, or Babes-Ernst granules) at the poles of the bacilli.
- Albert's, Neisser's, and Ponder's stains help stain these distinctive granules in the bacteria.
- These granules are also Found in Spirillum, Gardnerella, Yersinia pestis, yeast, and Mycobacterium tuberculosis (MTB)

A throat swab culture (Option A) is used to isolate and identify Corynebacterium diphtheriae, the bacterium that causes diphtheria, but it does not specifically identify volutin granules.

Gram stain (Option B) helps identify rod-shaped Corynebacterium diphtheriae but does not help stain the metachromatic granules

Elek's test (Option C) is used to detect the production of diphtheria toxin by Corynebacterium diphtheriae, confirming the bacterium's pathogenicity

Solution for Question 9:

Correct Answer: D) The pediatric dose of diphtheria toxoid contains 2 Lf units, whereas the adult dose contains 25 Lf units.

Explanation: The pediatric dose of diphtheria toxoid contains 25Lf units, whereas the adult dose contains 2Lf units.

Types of vaccine Single vaccine: Diphtheria toxoid (DT) is prepared by incubating the toxin with formalin and then adsorbed onto alum. Alum acts as an adjuvant and increases the immunogenicity of toxoid. Combined vaccine:

Type	Composition
DPT	Dt (Diphtheria Toxoid) Pertussis whole cell TT (Tetanus toxoid) The pertussis component acts as an adjuvant and increases the immunogenicity of DT & TT
DaPT	DT TT Acellular pertussis(aP)
Td	Tetanus toxoid Adult dose 2Lf of diphtheria toxin
Pentavalent	DPT Haemophilus influenza B Hepatitis B

Schedule

Children	7 doses are given- 3 doses of pentavalent vaccine at 6, 10, and 14 weeks of birth +

	2 booster doses of DPT at 16–24 months and 5 years + another 2 booster doses of Td at 10 years and 16 years
Pregnant women	2 doses of Td at 1 month interval
Adults >18 years -completed primary vaccination >18 years completed primary vaccination	Td booster dose is indicated every 10 years till 65 years. 3 doses of Td given at 0, 1 month, and 1 year

• Site: DPT is given deep intramuscularly (IM) at the anterolateral aspect of the thigh (Option A ruled out)

• Preservative: Thiomersal (0.01%) is used (Option B ruled out)

• Storage: DPT should be kept at 2–8C (Option C ruled out)

• Dose: The usual pediatric dose of diphtheria toxoid contains 25 Lf units, whereas the adult dose contains 2 Lf units

• Adverse Reaction: Fever: Common, occurring in nearly 50% of cases. Neurotoxic Reactions: Inconsolable crying lasting more than 3 hours. Hypotonic-hyporesponsive episode (HHE). Febrile seizures and encephalopathy.

• Fever: Common, occurring in nearly 50% of cases.

• Neurotoxic Reactions:

• Inconsolable crying lasting more than 3 hours.

• Hypotonic-hyporesponsive episode (HHE).

• Febrile seizures and encephalopathy.

• Fever: Common, occurring in nearly 50% of cases.

• Neurotoxic Reactions:

• Inconsolable crying lasting more than 3 hours.

• Hypotonic-hyporesponsive episode (HHE).

• Febrile seizures and encephalopathy.

Contraindications: Avoid in progressive neurological conditions (e.g., active epilepsy) but permissible in non-progressive conditions (e.g., cerebral palsy).

Solution for Question 10:

Correct Answer: B) 40000-60000

Explanation: 40000-60000 units of diphtheria antitoxin are administered if the disease extends up to the nasopharynx

Treatment

- Passive immunity with Antidiphtheric serum- ADS (antitoxin) is the treatment of choice as it neutralizes the toxin
- A test dose is given to check hypersensitivity
- It is given IV or IM depending upon the stage of illness:

Type of Disease	Antitoxin
Early stage <48hrs	20,000-40,000 IM or IV
Nasopharyngeal Disease	40,000-60,000 IM or IV
Extensive disease with neck disease (bull neck) > 3 days	80,000-120,000 half IM or half IV

Antibiotics:

- Penicillin or Erythromycin is the drug of choice
- Useful only if given early disease (<6hrs)-before the toxin is released.
- Prevents further release of toxins by killing bacteria
- Treatment of carriers and for cutaneous diphtheria

Solution for Question 11:

Correct Answer: B) 1-b, 2-c, 3-a

Explanation:

Corynebacterium produces characteristic colonies on Tellurite Blood Agar differentiating into 3 biotypes, namely

- C. gravis - Daisy head colonies
- C. intermedius - Frog's egg colonies
- C. mitis - Poached egg colonies

Aspect	Gravis	Intermedius	Mitis

Morphology	-Usually short rods, with uniform staining, few or no granules. -Some degree of pleomorphism, with irregularly barred, snow-shoe and tear-drop forms	Long barred forms with clubbed ends; poor granulation, very pleomorphic	Long, curved, pleomorphic rods with prominent granulation
Colony on tellurite blood agar	-In 18 hours, the colony is 1-2 mm in size, with a greyish-black centre, paler, semitranslucent periphery and commencing crenation of edge. -In 2-3 days, 3-5 mm in size, flat colony with raised dark centre and crenated edge with radial striation—'daisy head' colony	-18-hour colony, small, 1 mm in size, misty. -Does not enlarge in 48 hours, smoother, more glistening periphery and a lighter ring near the edge—'frog's egg' colony	-Size variable, shiny black. -In 2-3 days, colonies become flat, with a central elevation—'poached egg' colony
Consistency of colonies	Like 'cold margarine', brittle, moves as a whole on the plate, not easily picked out or emulsifiable	Intermediate between gravis and mitis	Soft, buttery, easily emulsifiable
Hemolysis	Variable	Non-hemolytic	Usually hemolytic
Growth in broth	Surface pellicle, granular deposit, little or no turbidity	Turbidity in 24 hours, clearing in 48 hours, with fine granular sediment	Diffuse turbidity with soft pellicle later
Glycogen and starch fermentation	Positive	Negative	Negative

Solution for Question 12:

Correct Answer: D) 1)- Cervicofacial, 2)- Pelvic

Explanation:

• The cervicofacial form is the most common type, usually presenting as a painless, slow-growing hard mass with cutaneous fistulas, commonly known as a lumpy jaw or woody jaw.

• The pelvic form is often associated with the use of intrauterine devices, producing abscesses in bone and soft tissue, with chronic draining sinuses to the exterior.

Clinical Form	Description
Cervicofacial	The most common form- usually presents as a painless, slow-growing hard mass with cutaneous fistulas, commonly known as a lumpy jaw. Indurated lesions on the cheek and submaxillary regions.
Thoracic	Lesions in the lung that may involve the pleura and pericardium, spreading outwards through the chest wall
Abdominal	Lesions usually around the cecum involving neighbouring tissues and the abdominal wall. Infection can spread to the liver via the portal vein.
Pelvic	Often associated with the use of intrauterine devices, producing abscesses in bone and soft tissue, with chronic draining sinuses to the exterior.

Solution for Question 13:

Correct Answer: C) Listeria monocytogenes infection in adults rarely leads to meningitis.

Explanation:

Listeria monocytogenes infection in adults can lead to meningitis, particularly in elderly individuals or those who are immunocompromised.

Listeria monocytogenes	
Mode of Transmission	Contaminated food Vertical transmission (mother to fetus, during birth)
Common Food Sources	Contaminated coleslaw Milk, Soft cheeses "Ready-to-eat" foods (delicatessen meat, uncooked hotdogs)

Age Groups at Risk	Neonates Elderly individuals
Increased Risk Factors	Use of proton pump inhibitors Pregnant women Immunocompromised individuals
Survival in Food	Survives refrigeration (4°C) Found in stored foods (aged soft cheeses, packaged meats, milk, cold salads)
Infection Type	Sporadic Outbreaks can occur
Psychrotrophic	L. monocytogenes is a psychrotrophic organism with an optimum growth temperature of 30ºC to 37ºC but has the ability to grow at a wide range of temperatures from 1ºC to 45ºC.
Motility	Shows Tumbling and Differential motility(motile at 25.C by forming actin rockets, non motile at 37.C)
Pathogenesis	Intracellular survival: lysis of phagosome (listeriolysin O) Host cell actin polymerization (cell membrane approach) Direct cell-to-cell spread (listeriopods)
Clinical Manifestations	Neonatal listeriosis: Early-onset and late-onset disease (Option B ruled out) Pregnant women: Granulomatosis infantiseptica in baby in 3rd trimester Fetal complications (abortion, preterm delivery) Maternal complications (flu-like symptoms, bacteremia, rarely meningitis) (Option A ruled out) Adults: Bacteremia and meningitis (elderly, >60 years) Immunosuppression- higher risk (steroid therapy, HIV, diabetes, malignancy) (Option D ruled out) Post-kidney transplant meningitis (1-month post-transplant) Gastroenteritis (consumption of contaminated milk, meat, salads)

Solution for Question 14:

Correct Answer: A) Shows the tumbling type of motility at 37°C.

Explanation:

Listeria shows tumbling type of motility at 25°C and non-motile at 37°C, which is known as differential motility.

Laboratory Diagnosis of Listeria monocytogenes:

- Motility: Shows tumbling type of motility at 25°C. Spreads from cell to cell by forming actin rockets Nonmotile at 37°C, which is known as differential motility. Temperature-dependent flagella expression controls this differential motility.
- Shows tumbling type of motility at 25°C.
- Spreads from cell to cell by forming actin rockets
- Nonmotile at 37°C, which is known as differential motility.
- Temperature-dependent flagella expression controls this differential motility.
- Culture: Grows on blood agar, producing β-hemolytic colonies. Grows well on chocolate agar, which supports its growth. Selective media - PALCAM agar, containing a mixture of antibiotics. (Option B ruled out)
- Grows on blood agar, producing β-hemolytic colonies.
- Grows well on chocolate agar, which supports its growth.
- Selective media - PALCAM agar, containing a mixture of antibiotics. (Option B ruled out)
- Isolation:
- Shows tumbling type of motility at 25°C.
- Spreads from cell to cell by forming actin rockets
- Nonmotile at 37°C, which is known as differential motility.
- Temperature-dependent flagella expression controls this differential motility.
- Grows on blood agar, producing β-hemolytic colonies.
- Grows well on chocolate agar, which supports its growth.
- Selective media - PALCAM agar, containing a mixture of antibiotics. (Option B ruled out)
- Isolation from - cervical and vaginal secretions, lochia (postpartum discharge), meconium (infant's first stool), cord blood, blood, and cerebrospinal fluid.
- Incubation is done under a 5% CO2 environment on blood agar.
- The cold enrichment technique- effective method for isolation. It involves storing materials in tryptose phosphate or thioglycollate broth at 4°C. (Option C ruled out)
- Subcultures are performed at weekly intervals for 1-6 months to enhance isolation.
- Identification: Identification is primarily through biochemical tests- catalase positive. Antibody detection against listeriolysin O can aid in diagnosis. cAMP positive Experimental Inoculation in Rabbits causes marked Monocytosis which is also observed in human listeriosis. (Option D ruled out) Anton Test: Instillation into the eyes of rabbits produces keratoconjunctivitis.

• Identification is primarily through biochemical tests- catalase positive.

• Antibody detection against listeriolysin O can aid in diagnosis.

• cAMP positive

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• Experimental Inoculation in Rabbits causes marked Monocytosis which is also observed in human listeriosis. (Option D ruled out)

• Anton Test: Instillation into the eyes of rabbits produces keratoconjunctivitis.

Sereny's test done for enteroinvasive E coli and to check for invasiveness in Listeria

Spore Forming Bacteria

1. Which of the following statements about Bacillus anthracis is wrong?

A. B. anthracis colonies give a frosted glass appearance on blood agar

B. B. anthracis forms "strings of pearls" on agar with penicillin

C. B. anthracis shows more growth at the bottom of gelatin stab cultures

D. PLET medium is used to isolate B. anthracis

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2. A 65 y/o male presents with intense pain, swelling, and a foul-smelling discharge from a rapidly worsening wound on his left foot, accompanied by fever and hypotension. Examination reveals crepitus, skin discolouration, and necrosis. Which of the following is most likely the causative organism?



A. C. tertium

B. C. difficile

C. C. sordelli

D. C. novyi

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3. A 65 y/o female presents with severe diarrhoea, abdominal pain, and fever. She has been on clindamycin for the past 10 days to treat a dental infection. O/E, her colonoscopy reveals microabscesses. Stool samples are sent for further testing. Which diagnostic method is considered the gold standard for confirming the suspected infection?

A. Colonoscopy

B. Stool Culture

C. ELISA for Toxin Detection

D. Histopathology

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4. A 70 y/o patient with a h/o recent hospitalization for hip replacement surgery presents with watery diarrhoea and abdominal cramping. Physical examination reveals mild abdominal tenderness, and laboratory tests show leukocytosis. Stool examination reveals the presence of Clostridioides difficile toxins. What is the most likely risk factor contributing to this patient's condition?

A. Recent Ingestion of Contaminated Food

B. History of Smoking

C. Prolonged Antibiotic Use

D. Allergic Reaction to Medications

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5. Which of the following mechanisms primarily contributes to the pathogenicity of Clostridium difficile by disrupting the actin cytoskeleton in cells?

A. Spore Production

B. Toxins A and B

C. Gene Acquisition

D. Use of Antibiotics

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6. A 35 y/o man presents to the emergency with a 24-hour history of double vision, difficulty swallowing, and generalised weakness.O/E, has diplopia, a diminished gag reflex, and symmetric descending paralysis. He reports that he recently consumed canned vegetables. Which type of botulism is most likely responsible for his symptoms?

A. Infant Botulism

B. Wound Botulism

C. Foodborne Botulism

D. Adult Intestinal Botulism

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7. What is the primary mechanism by which Clostridium botulinum acts?

A. Invasion and destruction of muscle tissue

B. Production of a toxin that inhibits acetylcholine release at neuromuscular junctions

C. Induction of an inflammatory response in peripheral nerves

D. Production of a toxin that stimulates excessive acetylcholine release

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8. A 30-year-old farmer presents with a deep laceration from a farm tool. His last documented tetanus vaccination was 8 years ago. According to current guidelines, what is the appropriate tetanus prophylaxis for this patient?

A. Administer Td vaccine only

B. Administer Td vaccine and HTIG

C. Administer Td complete series and HTIG

D. No tetanus prophylaxis is required

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9. A 35-year-old male presents to the ED with muscle stiffness and spasms. He injured himself with a deep puncture wound while gardening 2 weeks ago; the wound looks necrotic, and he hasn't had a tetanus shot. Lab receives tissue for testing. Which lab finding best indicates Clostridium tetani?

A. Gram-positive cocci with clusters and no spore formation on microscopy

B. Alpha hemolysis on blood agar, with no concentric swarming movement

C. Drumstick-shaped terminal spores with black colour and gas production in Robertson’s cooked meat broth

D. Rapid liquefaction of gelatin and no proteolytic reaction in Robertson’s cooked meat broth

---

10. A 50 y/o woman presents with severe muscle pain and stiffness that began 10 days ago after a minor gardening injury. Initially, she had trouble opening her mouth and swallowing, and now she has generalized muscle spasms with an arched back posture. Her vitals show normal BP but increased HR and sweating. Which of the following is NOT a complication of this condition?

A. Risus sardonicus

B. Opisthotonos

C. Acute renal failure

D. Muscle weakness in the extremities

---

11. Which of the following statements about tetanospasmin is INCORRECT?

A. Tetanospasmin is a neurotoxin responsible for the manifestations of tetanus

B. Tetanospasmin is antigenic and can be neutralized by its antitoxin

C. Tetanospasmin prevents the release of inhibitory neurotransmitters glycine and GABA

D. Tetanospasmin is used in its active form for vaccine preparation

12. A 50 y/o male presents with severe pain, swelling, and crepitus in his left thigh following a traumatic injury. Laboratory tests, including a Nagler's reaction, return positive, indicating a clostridial infection. Which of the following organisms does not test positive for Nagler's reaction?

A. C. tetani

B. C. baratti

C. C. sordelli

D. C. bifermentans

13. A 35 y/o male presents with severe abdominal pain, vomiting, and bloody diarrhoea after consuming a large meal of pork. Imaging and laboratory results suggest enteritis necroticans. Which of the following organisms cause enteritis necroticans?

A. C. perferingens type A

B. C. difficile

C. C. perferingens type C

D. C. tetani

14. Which of the following is NOT a characteristic of Bacillus anthracis?

A. It is non-motile

B. It produces beta-hemolytic colonies on blood agar

C. It is generally susceptible to penicillin

D. It produces anthrax toxin, which includes protective antigen, oedema factor, and lethal factor

15. A 60 y/o male presents with severe pain, swelling, and a foul-smelling discharge from a wound on his leg sustained three days ago.O/E the affected area has signs of necrosis, raising suspicion of gas gangrene. Which of the following toxins is responsible for toxaemia in gas gangrene?

A. Alpha toxin

B. Beta toxin

C. Gamma toxin

D. Theta toxin

16. A 58 y/o farmer presented to the casualty with ■ high fever and chills for the last 5 days.O/E there are no obvious sources of infection. A blood culture was obtained and the patient was diagnosed with PUO. Gram staining revealed characteristic “Tennis racket appearance”. Which of the following is the likely organism causing this infection?

A. C. perferingens

B. C. tertium

C. C. biofermentans

D. C. tetani

17. Which of the following statements about Clostridium species is TRUE?

A. Most Clostridium species are encapsulated and non-motile

B. C. perfringens is both encapsulated and non-motile

C. C. butyricum is non-motile

D. Clostridium species generally exhibit rapid movement due to peritrichate flagella

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18. A 30 y/o man arrives at the ER with symptoms of food poisoning, including severe nausea and vomiting, after eating fried rice from a local restaurant 5 hours ago. Which statement about Bacillus cereus food poisoning is TRUE?

A. Symptoms of emetic form typically manifest within 8-16 hours after ingestion

B. The emetic form of B. cereus food poisoning is associated with consuming contaminated meat and poultry

C. Diagnosis of B. cereus can be confirmed by finding more than >105 cells per gram of food or stool culture

D. B. cereus grows anaerobically at 37°C on sheep blood agar

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19. A 38 y/o biologist returns from a field expedition in an anthrax-endemic area, displaying symptoms and skin lesions indicative of cutaneous anthrax. Lab tests confirm the diagnosis. Which statement about the treatment of cutaneous anthrax is FALSE?

A. Oral administration of fluoroquinolones or doxycycline for 7-14 days is recommended

B. Combined therapy with antibiotics and anthrax immunoglobulin is necessary

C. Human Anthrax Immunoglobulin (Anthrasil) is used as an antitoxin option

D. Treatment includes a combination of at least three antimicrobial drugs, including clindamycin

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20. A 45 y/o veterinarian presents to the clinic for a routine check-up. He has been working in an area with a recent outbreak of anthrax among livestock. He discusses the precautions he takes and the vaccines available to him. Which of the

following statements is FALSE regarding the precautions and vaccines for Bacillus anthracis?

A. Using a biosafety level II/III cabinet is essential.

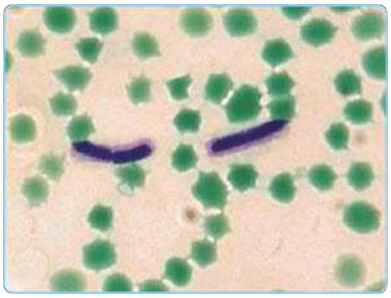
B. Autopsies on animals suspected of anthrax are recommended

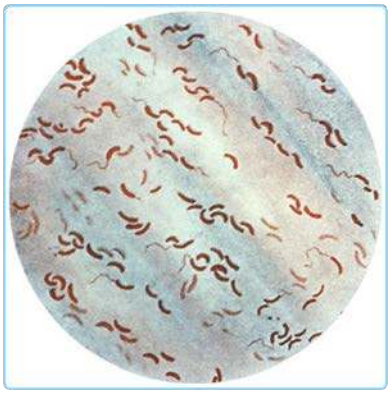
C. Workers in industries handling animal products should wear protective clothing

D. The BioThrax vaccine is considered safe and effective for human use

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21. Match the following and choose the correct option: Name Microscopy a. Bacillus anthracis 1. b.Vibrio cholerae 2. c. Corynebacterium diphtheria 3. d. McFadyean's Reaction 4.

Name	Microscopy
a. Bacillus anthracis	 1.
b.Vibrio cholerae	 2.

c. Corynebacterium diphtheria	 ... (content truncated)
d. McFadyean’s Reaction	 ... (content truncated)

A. a-4, b-2, c-3, d-1

B. a-1, b-3, c-2, d-4

C. a-4, b-3, c-2, d-1

D. a-2, b-3, c-4, d-1

22. A patient presents with a black necrotic lesion on their arm, surrounded by swelling and secondary vesicles. They report handling animal hides recently. What is the most likely diagnosis and the route of transmission?

A. Cutaneous anthrax, spores entered through damaged skin

B. Pulmonary anthrax, inhalation of spores

C. Intestinal anthrax, ingestion of spores

D. Bacillus cereus infection; food poisoning

23. A patient infected with Bacillus anthracis presents with severe oedema and systemic symptoms. Which combination of virulence factors is primarily responsible for these clinical manifestations?

A. Polysaccharide capsule and endotoxin

B. Protective antigen, oedema factor, and lethal factor

C. Poly-D-glutamic acid capsule and enterotoxin

D. Hemolysin and cytotoxin

24. Which of the following statements about Bacillus anthracis are true? Bacillus anthracis was the first pathogenic bacterium to be seen under a microscope It was the first communicable disease demonstrated to be transmitted by the inoculation of infected blood It was the first bacillus to be isolated in pure culture and shown to possess spores The anthrax vaccine was the first live attenuated bacterial vaccine prepared.

A. 1, 2

B. 2 only

C. 1, 2, 3

D. All statements are correct

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	3
Question 4	3
Question 5	2
Question 6	3
Question 7	2
Question 8	1
Question 9	3
Question 10	4
Question 11	4
Question 12	1
Question 13	3
Question 14	2
Question 15	1
Question 16	2
Question 17	2
Question 18	3
Question 19	2
Question 20	2
Question 21	3
Question 22	1

Question 23	2
Question 24	4

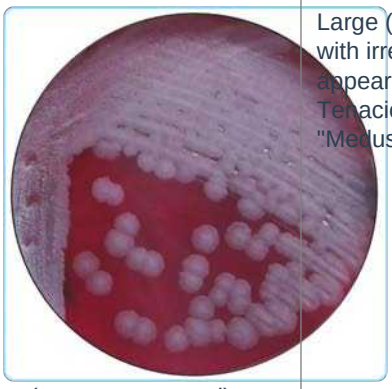
Solution for Question 1:

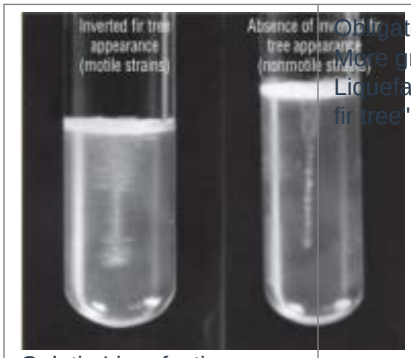
Correct answer: C) B. anthracis shows more growth at the bottom of gelatin stab cultures.

Explanation:

B. anthracis shows more growth at the surface of gelatin stab cultures due to higher oxygen availability, not at the bottom.

Blood Culture Morphology:

Characteristic	Description
 ... (content truncated)	Large (2 to 5 mm), grey, flat, non-hemolytic colonies with irregular margins, giving a frosted glass appearance. (Option A ruled out) Tenacious consistency firmly adheres to agar. "Medusa head" appearance under microscope.
 Agar with Penicillin ... (content truncated)	Susceptible to penicillin. Grows large, spherical cells in 3-6 hours on media with 0.05-0.50 units of penicillin/ml. Forms "string of pearls" chains. (Option B ruled out)

 Gelatin Liquefaction	Obligate aerobe. More growth and energy production at the surface. Liquefaction starts from the top, resembling an "inverted fir tree".
PLET Medium	Contains polymyxin, lysozyme, EDTA, and thallous acetate in heart infusion agar (PLET) Selective for isolating B. anthracis from samples with other spore-forming bacilli. (Option D ruled out)
Serology	Ascoli's Thermoprecipitin Test: Detects anthrax antigens in tissue extracts. DFA: Detects antibodies against B. anthracis; confirms presence via cell wall polysaccharide and capsule antigens.
Laboratory Safety Note	High risk of infection from spores. Caution is advised when handling B.anthracis isolates.

Solution for Question 2:

Correct Answer:D) C. novyi

Explanation: C.novyi is an established agent that causes gas gangrene (20-40%)

Gas Gangrene

- Definition: A rapidly spreading, edematous myonecrosis, occurs in association with severely crushed wounds contaminated with pathogenic clostridia, particularly with C. perfringens.
- Etiological agents Established agents-C.perfirengens (60%), C.novyi and C.septicum(20-40%) (Mnemonic P>S>N) Probable agents- less common- C.bifermentans, C.sordellii (Option C ruled out) C.aerofoetidum, and C.tertium (Option A ruled out)
- Established agents-C.perfirengens (60%), C.novyi and C.septicum(20-40%) (Mnemonic P>S>N)

- Probable agents- less common- C.bifermentans, C.sordellii (Option C ruled out)
- C.aerofoetidum, and C.tertium (Option A ruled out)
- Established agents-C.perfirengens (60%), C.novyi and C.septicum(20-40%) (Mnemonic P>S>N)
- Probable agents- less common- C.bifermentans, C.sordellii (Option C ruled out)
- C.aerofoetidum, and C.tertium (Option A ruled out)

Pathogenesis:

gangrene -Pathogenesis" data-author="" data-hash="10201" data-license="" data-source="" data-tags="April2025" height="450" src="https://image.prepladder.com/notes/zcDGQxW5qcQPclXXTSSo1744481632.png" width="450" />

Clinical Manifestations

Organism	Incubation period
C.perferingens	10-48hrs
C.septicum	2-3 days
C.novyi	5-6 days

- Excruciating pain at the affected site
- Foul-smelling thin serosanguineous discharge
- Gas bubbles (crepitus) in the muscle planes
- Brawny edema and induration
- Shock and organ failure develop later

Treatment

- Surgical Debridement - Wide resection of Devitalized tissue
- Antibiotics- Penicillin + Clindamycin for 10-14 days
- Hyperbaric oxygen
- Passive Immunization with anti-alpha toxin antiserum

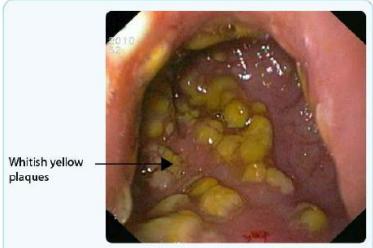
C.difficile causes antibiotic-associated diarrhoea and pseudomembranous colitis. (Option B ruled out)

Solution for Question 3:

Correct Answer: C) ELISA for Toxin Detection

Explanations:

ELISA for detecting toxins A and/or B in stool samples is considered the gold standard for diagnosing C. difficile infection. This method directly identifies the presence of toxins responsible for the symptoms.

Diagnosis	
1. Colonoscopy	 Colonoscopic view Examination of the colon whitish-yellow plaques on the colonic mucosa known as microabscesses. Although it is highly specific, it is not as sensitive as ELISA, hence it is not considered the Gold standard for diagnosis. (Option A ruled out) ... (content truncated)
2. Culture	Stool samples cultured on selective media to identify C. difficile; less sensitive than toxin detection. (Option B ruled out)
3. Toxin Detection	ELISA: Detects toxins A and/or B in stool samples, the gold standard. Cell Culture: Toxins produce characteristic effects on Hep-2 or human diploid cell cultures, neutralized by C. sordellii antitoxin.
4. Molecular Methods	PCR: Detects genes coding for toxins A and B in stool samples.
5. Histopathology	This can be performed with hematoxylin and eosin stains Highly specific but low sensitivity (Option D ruled out)
Treatment	
Antibiotic Discontinuation	Stop the antibiotic suspected of contributing to the infection.
Antibiotic Therapy	Oral vancomycin or metronidazole. Severe cases: Combination of oral vancomycin, intravenous metronidazole, and rectal vancomycin enemas.

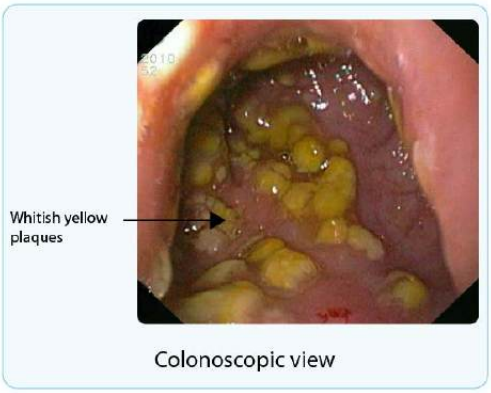
Prevention	
Antibiotic Stewardship	Limit the use of broad-spectrum antibiotics like clindamycin, ampicillin, and fluoroquinolones; prevent disruption of normal gut flora.
Infection Control Measures	Meticulous hand hygiene and contact precautions in healthcare settings.
Recent Treatment Option	
Fecal Transplants	Replenishing gut flora with donated faeces from screened healthy donor

Solution for Question 4:

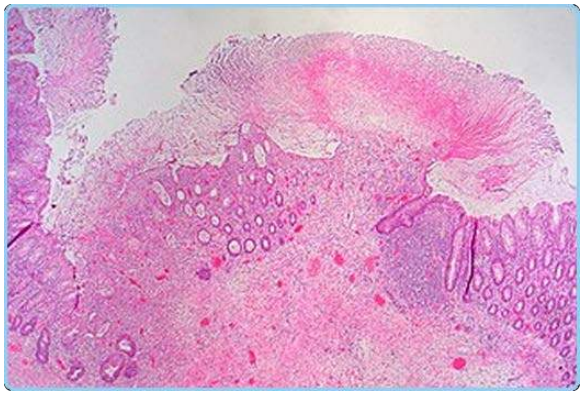
Correct Answer: C) Prolonged Antibiotic Use

Explanation:

Prolonged antibiotic use disrupts the normal gut flora, creating an environment conducive to Clostridioides difficile colonization and infection. This patient must have been put on multiple courses of broad-spectrum antibiotics for surgical prophylaxis as well as post-surgery. This is a well-established risk factor for developing pseudomembranous colitis.



Colonoscopic view



Pseudomembranous colitis, a severe manifestation of C. difficile infection (CDI), involves:

- Colon inflammation
- Formation of whitish-yellow plaques on the colonic mucosa. Sized - 1-2mm, or covering extensive areas of the colon. These plaques consist of necrotic leukocytes. Fibrin, Mucus, and Cellular debris
- Sized - 1-2mm, or covering extensive areas of the colon.
- These plaques consist of necrotic leukocytes. Fibrin, Mucus, and Cellular debris
- Fibrin,
- Mucus, and
- Cellular debris
- Sized - 1-2mm, or covering extensive areas of the colon.
- These plaques consist of necrotic leukocytes. Fibrin, Mucus, and Cellular debris
- Fibrin,
- Mucus, and
- Cellular debris
- Fibrin,
- Mucus, and
- Cellular debris

Recent Ingestion of Contaminated Food (Option A): While ingestion of contaminated food can lead to foodborne illnesses, it is not a known risk factor specifically for CDI. CDI is primarily associated with disruptions in gut microbiota due to antibiotic use.

History of Smoking (Option B): Smoking is not directly associated with an increased risk of Clostridioides difficile infection. Risk factors for CDI include advanced age, recent hospitalization, and antibiotic use.

Allergic Reaction to Medications (Option D): While allergic reactions can occur with medications, they do not predispose individuals to Clostridioides difficile infection. CDI is primarily linked to

antibiotic-associated disruption of gut microbiota.

Solution for Question 5:

Correct Answer: B) Toxins A and B

Explanation:

Toxins A (enterotoxin) and B (cytotoxin) are the main toxins produced by C. difficile. They disrupt the actin cytoskeleton by glycosylating GTP-binding proteins, which leads to cell shape loss, reduced cell adhesion, and a weakened epithelial barrier. This disruption is a key factor in the pathogenicity of C. difficile.

Spore Production (Option A): While the ability to form spores helps C. difficile survive in the environment and spread, it does not directly disrupt the actin cytoskeleton. This mechanism is more related to the bacterium's durability and ability to infect new hosts.

Gene Acquisition (Option C): Gene acquisition, or horizontal gene transfer, refers to obtaining genes from other organisms but, this process itself does not directly disrupt the actin cytoskeleton.

Use of Antibiotics (Option D): Long-term antibiotic use disrupts the normal gut microbiota, allowing C. difficile to grow and produce toxins. While this creates a conducive environment for infection, it is not the direct mechanism by which C. difficile disrupts the actin cytoskeleton.

Solution for Question 6:

Correct Answer: C) Foodborne Botulism

Explanation:

The patient has a history of consuming canned vegetables and presents with typical symptoms of foodborne botulism, including double vision, difficulty swallowing, and descending paralysis.

Foodborne botulism	Wound botulism

Bottled and canned food (Heat Labile toxins) IP: 12-36 hrs Results in Diplopia Dysphagia Dysarthria Descending Flaccid Paralysis Dilated pupils. GI symptoms are also present.	IP: 7-10 days No GI symptoms
Infant botulism	Iatrogenic botulism
Bottle-fed/honey/baby food IP: 1-2 days The first symptom is Constipation. Floppy baby syndrome: flaccid muscles.	Surgeries

Infant Botulism (Option A): This affects infants under one year old and presents with symptoms like poor feeding, lethargy, and constipation. The patient's age and symptoms do not match this type.

Wound Botulism (Option B): This type results from C. botulinum spores contaminating a wound. It presents similarly to foodborne botulism but lacks gastrointestinal symptoms. The patient reports ingestion of home-canned vegetables, which suggests foodborne botulism.

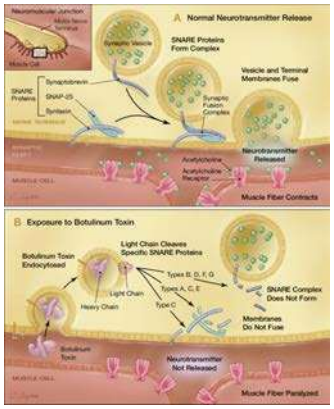
Adult Intestinal Botulism (Option D): This is a rare form affecting adults with compromised gut flora, presenting similarly to infant botulism. The patient's history of recent food consumption and acute onset of symptoms makes foodborne botulism more likely.

Solution for Question 7:

Correct Answer: B) Production of a toxin that inhibits acetylcholine release at neuromuscular junctions

Explanation:

Botulinum toxin binds to acetylcholine receptors at neuromuscular junctions, preventing the release of acetylcholine, which is essential for muscle contraction. This leads to flaccid paralysis, a hallmark of botulism.



Invasion and destruction of muscle tissue (Option A): Clostridium botulinum is a non-invasive pathogen, it does not directly destroy muscle tissue. Instead, it exerts its effects through the production of a potent neurotoxin (Botulinum Toxin).

Induction of an inflammatory response in peripheral nerves (Option C): Botulinum toxin does not cause an inflammatory response. Its primary action is to block neurotransmitter release, leading to paralysis.

Production of a toxin that stimulates excessive acetylcholine release (Option D): The botulinum toxin produced by Clostridium botulinum does not stimulate the release of acetylcholine, rather, inhibits it. If the toxin stimulated the release of acetylcholine, it would result in continuous muscle contraction (spastic paralysis), which is contrary to the clinical presentation of botulism.

Solution for Question 8:

Correct Answer: A) Administer Td vaccine only

Explanation:

All types of wounds need surgical toilet followed by immunization which depends on the wound type and immunization status of the individual.

Tetanus-prone wound:

- Age of wound > 6 hours,
- Penetrating wounds (missiles, crush, burn, bite),
- Depth of wound >1 cm,
- Presence of devitalized tissue and contaminants (dirt, saliva, etc.)

PREVENTION OF TETANUS AFTER INJURY

Complete course of vaccine	Not tetanus-prone wound	Tetanus-prone wound*
Taken within 5 years	Nothing required	Nothing required
Taken within >5 to <10 year	Td 1 dose	Td 1 dose
Taken >10 years back	Td 1 dose	Td 1 dose + HTIG
Unknown/incomplete immunization	Td complete dose	Td complete dose + HTIG

Td- Tetanus toxoid and adult diphtheria toxoid

HTIG- Human tetanus immunoglobulin

Solution for Question 9:

Correct Answer: C) Drumstick-shaped terminal spores with black colour and gas production in Robertson's cooked meat broth

Explanation:

Clostridium tetani is known for its drumstick-shaped terminal spores, which are a hallmark of its identification. Robertson's cooked meat broth showing black colour and gas production is also characteristic of C. tetani.

DIAGNOSIS OF C. TETANI	
Specimen	Necrotic Tissue
Microscope examination Drumstick appearance" data-author="CDC/ Dr. Holdeman" data-hash="10213" data-license="OPEN ACCESS" data-source="https://commons.wikimedia.org/wiki/File:12056_lores.jpg" data-tags="April2025" src="https://image.prepladder.com/notes/Pjv4C4OaDvXN1NtprKVB1744483469.png" /> {{caption_text}}	Gram +ve Bacilli Spores with Drumstick appearance- Terminal Spores (Option A ruled out)

 ... (content truncated)	Alpha hemolysis is produced, which later develops into Beta hemolysis, due to the production of hemolysin (tetanolysin). Concentric movement is seen- Swarming Growth (Option B ruled out)
 ... (content truncated)	Black colour, gas production Proteolytic reaction (Option D ruled out)
Gelatin stab	Fir tree appearance Slow liquefaction
Fildes Technique	To obtain pure colonies of C. tetani A test tube is filled with Nutrient broth, and pure colonies can be isolated from the surface.

Solution for Question 10:

Correct Answer: D) Muscle weakness in the extremities

Explanation:

The case scenario given points to the diagnosis of Tetanus.

Muscle weakness, particularly in the extremities, is not typically associated with tetanus. Instead, tetanus is characterized by muscle stiffness and spasms, with hands and feet generally spared

Table rendering failed

Solution for Question 11:

Correct Answer: D) Tetanospasmin is used in its active form for vaccine preparation.

Explanation:

Tetanospasmin is not used in its active form for vaccine preparation; it is toxoided (rendered non-virulent) by formaldehyde for use in vaccines.

Exotoxin	Type	Characteristics
Tetanolysin	Hemolysin	Hemolytic activity No role in the Pathogenesis of Tetanus
Tetanospasmin	Neurotoxin	Responsible for tetanus (Option A ruled out) Antigenic; Neutralized by antitoxin (Option B ruled out) Plasmid coded Toxin gets toxoided spontaneously or by formaldehyde. The toxoid form is antigenic but loses its virulence property, hence, it is used for vaccine preparation. Mechanism of Action of Tetanus Toxin: Binds to receptors (polysialoganglioides) on motor nerve terminals, leading to internalization. Transported retrogradely to GABA and glycine-producing inhibitory neuron terminals. Prevents the presynaptic release of inhibitory neurotransmitters (glycine and GABA). (Option C ruled out) Results in spastic muscle contraction.

Solution for Question 12:

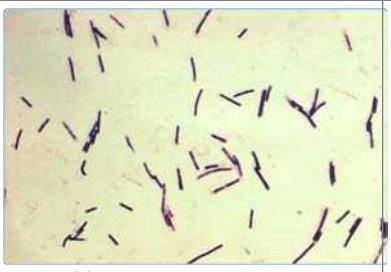
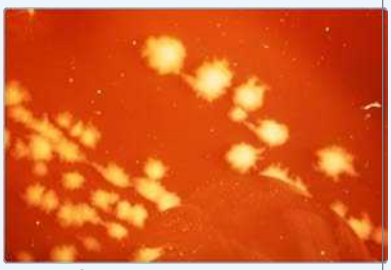
Correct Answer: A) C.tetani

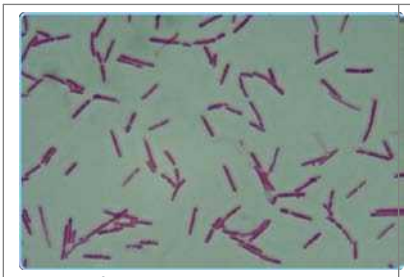
Explanation: C.tetani produces tetanoplasmin toxin and does not produce alpha toxin. Hence it does not give a positive Nagler reaction

Lab diagnosis of C.perferingens

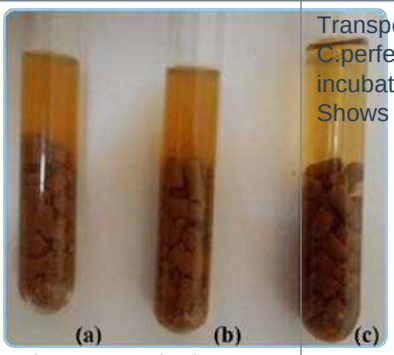
1. Confirmation of clinical diagnosis:

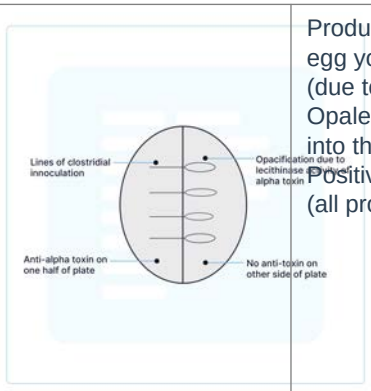
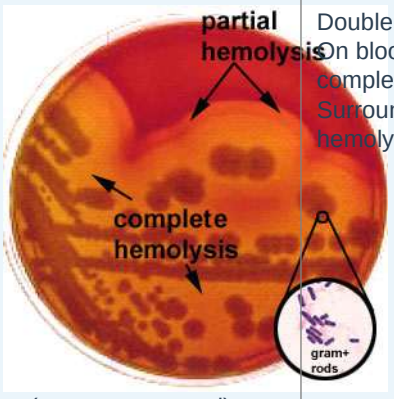
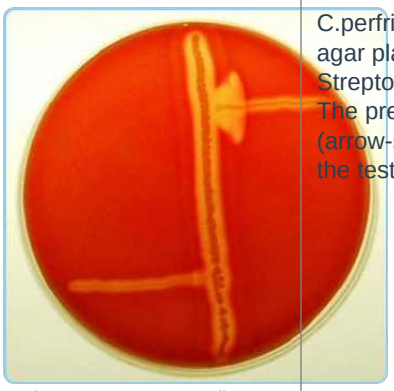
- Specimen-necrotic tissue, muscle fragments, exudates from deeper parts of the wound
- Blood culture may be positive for C. perfringens and C. septicum.
- Direct Microscopy-Absence of neutrophils in the infected tissues is a characteristic feature

Organism	Microscopic appearance
 C. Perfringens {{caption_text}}	Thick, stubby, boxcar-shaped, gram-positive bacilli without spore
 C. septicum {{caption_text}}	Citron bodies (boat or leaf-shaped pleomorphic irregularly stained bacilli with spores)

 C. novyi {{caption_text}}	Large rods with oval sub-terminal spores

2. Species Identification

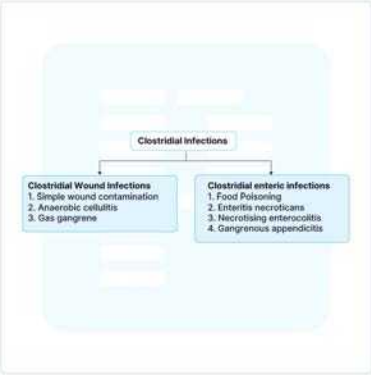
Method	Properties
 Robertson cooked meat medium {{caption_text}}	Transport as well as culture media C.perferingens tolerates heat and can grow when incubated at 45°C for 4-6 hrs Shows a pink and turbid appearance
 ... (content truncated)	Produces “stormy clot reaction” Fermentation of lactose-producing acid and vigorous gas

 ... (content truncated)	Produces an opalescence surrounding the streak line on egg yolk agar or media containing 20% human serum (due to lecithinase activity of α-toxin). Opalescence is inhibited by incorporating anti-α-toxin into the medium. Positive for C. bifermentans, C. baratti, and C. sordellii (all produce α-toxin) (Option B, C, D ruled out)
 ... (content truncated)	Double zone hemolysis On blood agar, it produces an inner narrow zone of complete hemolysis (due to θ-toxin) Surrounded by a much wider zone of incomplete hemolysis (due to the alpha-toxin)
 ... (content truncated)	C.perfringens is streaked over the centre of the blood agar plate Streptococcus agalactiae is streaked perpendicular to it The presence of an enhanced zone of hemolysis (arrow-shaped) pointing toward C. perfringens indicates the test is positive

Solution for Question 13:

Correct Answer: C) C. perferingens type C

Explanation: Enteritis necroticans is caused by C.perfringens type C strains



Clostridial enteric infections:

Type of infection	Features
Food poisoning	Caused by C. perfringens type A enterotoxin Consumption of improperly cooked meat Requires 10^8 viable vegetative bacilli for infection to occur. Diagnosis by detection of enterotoxin in faeces
Enteritis necroticans	Caused by C. perfringens type C strains, producing β toxin Gas gangrene of the bowel Ischemic necrosis of the jejunum and gas in the tissue plane PIGBEL (papa new guinea)- abdominal pain and diarrhoea following unaccustomed feasting on pig meat along with trypsin inhibitors like sweet potatoes.
Necrotising enterocolitis	C. perfringens type A (Option A ruled out) Resembles enteritis necroticans
Gangrenous appendicitis	C. perfringens type A

Other Clostridial infections

- Bacteremia: C.perfringens >C.tertium and C.septicum
- Skin and soft-tissue infections: C.perfringens, C.histolyticum, C.septicum, C.novyi, and C.sordellii cause necrotizing infections of the skin and soft tissues
- Infection of the endometrium leading to toxic shock syndrome— C. sordellii
- Meningitis and brain abscess
- Panophthalmitis (due to C. sordellii or C. perfringens)

C.difficle (Option B) causes antibiotic-associated diarrhoea and pseudomembranous colitis.

Clostridium tetani (Option D): It produces a powerful neurotoxin tetanospasmin causing tetanus.

Solution for Question 14:

Correct answer: B) It produces beta-hemolytic colonies on blood agar.

Explanation:

Bacillus anthracis forms nonhemolytic or weakly hemolytic colonies on blood agar. In contrast, Bacillus cereus produces strongly beta-hemolytic colonies.

Characteristics	Bacillus anthracis	Bacillus cereus
Hemolysis on Sheep Blood Agar	Non-hemolytic	Hemolytic (Option B)
Motility	Non-motile (Option A ruled out)	Motile
Penicillin susceptibility	Sensitive (Option C ruled out)	Resistant
Lecithinase Production	+	+
Gelatin hydrolysis	-	+
Growth in Penicillin agar (10 U/mL)	-	+
Growth on phenyl ethyl alcohol agar	-	+

Bacillus anthracis produces anthrax toxin (Option D), consisting of three components: protective antigen (PA), oedema factor (EF), and lethal factor (LF). This combination of factors is specific to Bacillus anthracis and is not produced by other Bacillus species.

Some Bacillus species include:

- Bacillus cereus
- Bacillus thuringiensis
- Bacillus subtilis
- Bacillus licheniformis
- Bacillus circulans
- Bacillus pumilus

- Bacillus sphaericus
- Geobacillus stearothermophilus
- Bacillus atrophaeus

Solution for Question 15:

Correct Answer: A) Alpha toxin

Explanation: The alpha toxin released by C.perferingens is lethal, dermonecrotic, and hemolytic and is hence responsible for toxaemia in gas gangrene.

Virulence Factors - Clostridium

- Most prolific toxin-producing bacteria
- Forms 12 different toxins
- 4 major toxins- alpha, beta, epsilon, and iota and 8 minor toxins
- It also has enzymes

Major Toxin	Uses
1. Alpha	Most important toxin Responsible for toxemia in gas gangrene Dermonecrotic and hemolytic Naglers reaction -Phospholipase in the presence of Ca2+ and Mg2+ ions split into phosphorylcholine and diglyceride-opalescence in Serum or egg yolk media
2. Beta (Option B ruled out)	Lethal and necrotising properties
3. Epsilon	
4. Iota	

Minor toxin	Uses
1. Gamma (Option C ruled out)	Minor lethal action
2. Eta	
3. Delta	Lethal effect Hemolytic for the red cells of even-toed ungulates (sheep, goats, pigs, cattle)
4. Theta (Option D ruled out)	Oxygen-labile hemolysin antigenically related to streptolysin 0

	Lethal and a general cytolytic toxin
5. Kappa	Collagenase
6. Lamba	Proteinase Gelatinase
7. Mu	Hyaluronidase
8. Nu	Deoxyribonuclease

Enzymes	Uses
Neuraminidase	Destroys myxovirus receptors on RBCs
Bursting Factor	Specific action on muscle tissue Muscle lesions in gas gangrene
Circulating Factor	Increase in adrenaline sensitivity of capillary bed Inhibit phagocytosis
Haemagglutinin	Active against the red blood cells of human beings and most animals
Other enzymes include	Histamine, Fibrinolysin, hemolysin

Solution for Question 16:

Correct Answer: B) C.tertium

Explanation:

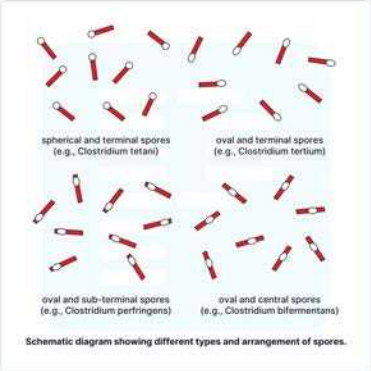
Clostridium tertium on gram staining reveals oval and terminal spores, resembling a tennis racket.

Spore formation in Clostridium

- Spores wider than bacillary body- swollen appearance/spindle-shaped
- Spore formation occurs in unfavourable conditions
- Shape and position vary in different species, which helps in the identification of species
- All species are killed by autoclaving at 121■ for 20 minutes

Species	Spore position &shape
Clostridium perfringens (Option A ruled out)	Subterminal -club-shaped

Clostridium tetani (Option D ruled out)	Spherical and terminal -drumstick
Clostridium tertium (Option B)	Oval and terminal - tennis racket
Clostridium bifermentans (Option C ruled out)	Central and oval-spindle-shaped



Solution for Question 17:

Correct Answer: B) C. perfringens is both encapsulated and non-motile.

Explanation:

C. perfringens is both encapsulated and non-motile.

Characteristic	Details
Unencapsulated and Motile	Most Clostridium species fall into this category (Option A ruled out)
Encapsulated Species	C. perfringens C. butyricum
Non-motile Species	C. perfringens (Option C ruled out) C. tetani type VI
Motility	The majority are motile by peritrichate flagella (flagella attached in multiple locations). (Option D ruled out) This includes C. butyricum. (Option B) Movement described as "stately" due to its slowness (Option D)

Solution for Question 18:

Correct Answer: C) Diagnosis of Bacillus cereus can be confirmed by finding more than >105 cells per gram of food or stool culture.

Explanation:

Diagnosis of Bacillus cereus food poisoning involves culturing the implicated food or performing quantitative stool cultures, confirming the presence of more than >105 bacteria per gram in the absence of other pathogens.

Table rendering failed

Diagnosis:

Diagnostic Method	Details
Food Poisoning	Culture of implicated food essential. >105 B. cereus cells per gram confirms the diagnosis. Quantitative stool cultures with at least 105 cells per gram support diagnosis.
Culture Characteristics	Aerobic growth at 37°C on sheep blood agar. (Option D ruled out) β-hemolytic colonies with frosted glass appearance. Motile, lecithinase positive, fermenting salicin
Selective Media	MYPA aids in isolating B. cereus from faeces and other sources
Microscopy	Spore-forming, Gram-positive bacilli. Motile

Solution for Question 19:

Correct Answer: B) Combined therapy with antibiotics and anthrax immunoglobulin is necessary.

Explanation:

While combined therapy with antibiotics and anthrax immunoglobulin is recommended for pulmonary and intestinal anthrax due to their higher fatality risk, it is not typically used for uncomplicated cutaneous anthrax, which is often self-limiting and does not require immunoglobulin therapy.

Treatment of Anthrax:

Clinical Presentation	Recommended Treatment
Uncomplicated Cutaneous Anthrax	Oral fluoroquinolones (e.g., ciprofloxacin) or doxycycline for 7-14 days (Option A ruled out)
Pulmonary and Intestinal Anthrax	Combined therapy with antibiotics and anthrax immunoglobulin due to higher fatality risk
Systemic Anthrax	Antitoxin treatment alongside antimicrobial drugs to neutralize B. anthracis toxin effects.
Antitoxin Options	Human Anthrax Immunoglobulin (Anthrasil). (Option C ruled out) Monoclonal Antibodies (Obiltoxaximab, Raxibacumab)
Anthrax Meningitis	Combination therapy with at least three antimicrobial drugs including clindamycin for 60 days (Option D ruled out)
Post-exposure prophylaxis (PEP)	Ciprofloxacin and doxycycline for 60 days. AVA BioThrax vaccine to enhance immune response

Solution for Question 20:

Correct Answer: B) Autopsies on animals suspected of anthrax are recommended.

Explanation:

Autopsies on animals suspected of anthrax are discouraged to prevent soil contamination with spores. Instead, safer alternatives like blood samples or a cut ear are recommended, and proper disposal methods include deep burial in quicklime or cremation.

Precautions for Bacillus anthracis:

Category	Details
Biosafety Level	Use biosafety level II/III cabinets to handle specimens. (Option A ruled out)
Handling Suspected Carcasses	Discouraged to prevent soil contamination with spores.

	Safer alternatives: blood samples or a cut ear. Proper disposal: deep burial in quicklime or cremation
Decontamination of Animal Products	Decontaminate suspected products via autoclaving. Workers should wear protective clothing and gloves. (Option C ruled out)
Duckering	Disinfect wool: 2% formaldehyde solution at 30-40°C for 20 minutes. Animal hair and bristles: 0.25% formaldehyde solution at 60°C for six hours

Vaccines for Bacillus anthracis:

Types of Vaccines	Details
Live Attenuated, Non-encapsulated Spore Vaccine (Sterne Vaccine)	Used in animals. Effective for about one year after a single injection. Not safe for human use
Adsorbed (alum-precipitated) Toxoid Vaccine	Uses protective antigens to trigger immune response. Safe and effective for human use Eg. BioThrax (Option D ruled out)
Pasteur's Anthrax Vaccine	Historically significant, Developed by Louis Pasteur in 1881
Mazzucchi Vaccine	Uses spores from the attenuated Carbazoo strain combined with 2% saponin

Indications and Administration for Bacillus anthracis Vaccination:

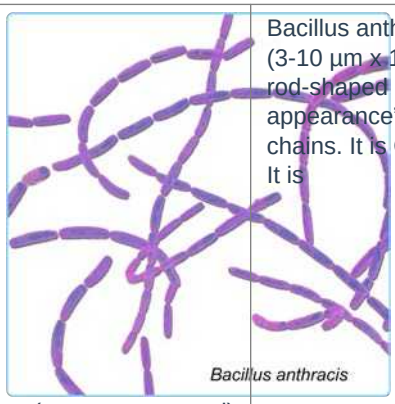
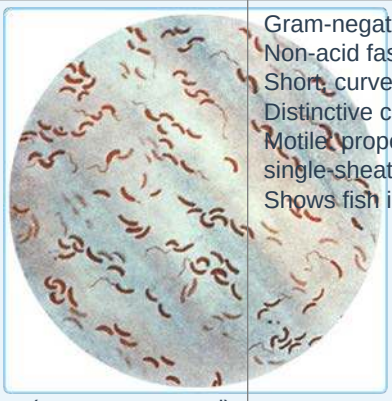
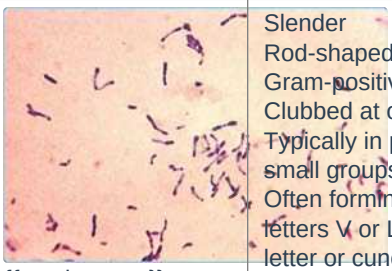
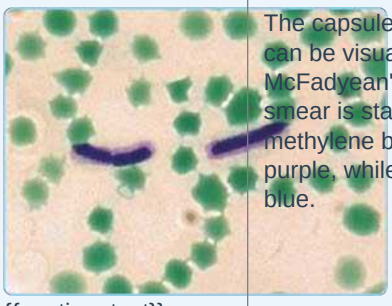
Category	Details
Pre-exposure	High-risk individuals (e.g., veterinarians, lab personnel). Five-dose regimen followed by boosters
Post-exposure	Administer alongside antibiotic treatment. For suspected or confirmed inhalation anthrax exposure

Solution for Question 21:

Correct answer: C) a-4, b-3, c-2, d-1

Explanation:

Name	Microscopy	Features

a. Bacillus Anthracis	 ... (content truncated)	Bacillus anthracis appears as large (3-10 µm x 1-1.6 µm), rectangular, rod-shaped bacteria (“box-car appearance”), often arranged in chains. It is Gram-positive. It is
b.Vibrio Cholerae	 ... (content truncated)	Gram-negative Non-acid fast Short, curved cylindrical rod. Distinctive comma shape. Motile, propelled by a single-sheathed polar flagellum. Shows fish in stream appearance.
c. Corynebacterium diphtheria	 {{caption_text}}	Slender Rod-shaped Gram-positive bacterium Clubbed at one or both ends. Typically in pairs, palisades, or small groups, Often forming angles resembling the letters V or L, known as a Chinese letter or cuneiform arrangement
d. McFadyean’s Reaction	 {{caption_text}}	The capsule of Bacillus Anthracis can be visualized using McFadyean's reaction, in which a smear is stained with polychrome methylene blue. The capsule stains purple, while the bacterial cells stain blue.

Solution for Question 22:

Correct answer: A) Cutaneous anthrax, spores entered through damaged skin

Explanation:

- The presentation of a black necrotic lesion (eschar) surrounded by swelling and secondary vesicles is characteristic of cutaneous anthrax, which occurs when Bacillus anthracis spores enter through damaged skin. MO
- Commonly seen in dock workers carrying hides and skins on their bare backs. Also known as Hide-Porter’s disease.



Pulmonary anthrax, inhalation of spores (Option B): Pulmonary anthrax, or Woolsorter's disease, is caused by inhaling spores from contaminated material. It presents as hemorrhagic pneumonia, not a localised skin lesion.

Intestinal anthrax, ingestion of spores (Option C): Intestinal anthrax is characterized by severe enteritis with bloody diarrhoea and is associated with consuming contaminated meat. It does not present with a black necrotic skin lesion.

Bacillus cereus infection; food poisoning (Option D): Bacillus cereus typically causes food poisoning or other infections like ocular infections, not a black necrotic skin lesion. The patient's recent contact with animal hides and the lesion's characteristics point to cutaneous anthrax.

Solution for Question 23:

Correct answer B) Protective antigen, edema factor, and lethal factor

Explanation:

The clinical manifestations of severe oedema and systemic symptoms are due to the combined action of protective antigen (PA), oedema factor (EF), and lethal factor (LF). PA facilitates the entry of EF and LF into cells, EF causes oedema, and LF contributes to lethality.

Polysaccharide capsule and endotoxin (Option A): Bacillus anthracis has a poly-D-glutamic acid (polypeptide) capsule, not a polysaccharide capsule, and does not produce endotoxin.

Poly-D-glutamic acid capsule and enterotoxin (Option C): While the poly-D-glutamic acid capsule is a virulence factor, Bacillus anthracis does not produce enterotoxin. The relevant toxin is the anthrax toxin, composed of PA, EF, and LF.

Hemolysin and cytotoxin (Option D): Bacillus anthracis does not primarily rely on hemolysin and cytotoxin for its virulence. Its main virulence factors are the poly-D-glutamic acid capsule and the anthrax toxin (PA, EF, and LF).

Virulence Factor of Bacillus Anthracis	
Capsule	Exotoxin
The capsule is composed of poly-D-glutamic acid, which is different from the polysaccharide capsules found in most other bacteria. It protects against phagocytosis. The capsule is encoded by a plasmid. If this plasmid is lost, the strain will become avirulent.	It is composed of protective antigen (PA), oedema factor (EF), and lethal factor (LF). These proteins are nontoxic individually but work together to cause harmful effects. The genes encoding anthrax toxin are carried on a plasmid. PA binds to cell receptors and forms a membrane channel, which allows for the entry of EF and LF into the cell. EF is an adenylate cyclase that causes oedema. LF, when combined with PA, forms a lethal toxin. Lethal toxin is a major virulence factor and is the cause of death in infected people and animal.

Solution for Question 24:

Correct answer: D) All statements are correct

Explanation:

• Bacillus anthracis was the first pathogenic bacterium observed under a microscope by Pollender in 1849.

• Davaine demonstrated in 1850 that it was the first communicable disease transmitted through inoculation with infected blood.

• Koch was the first to isolate this bacillus in pure culture and show it possessed spores in 1876.

• Pasteur created the first live attenuated bacterial vaccine using B. anthracis in 1881.

Mycobacterium

1. A patient with a persistent cough and fever undergoes diagnostic testing, revealing acid-fast bacilli in a sputum sample stained with Ziehl-Neelsen stain. Which of the following characteristics of the bacteria contributes to its ability to retain the stain despite acid-alcohol decolourisation?

A. Lipopolysaccharide (LPS) in the cell wall

B. Mycolic acids in the cell wall

C. Biofilm formation

D. Capsular polysaccharides

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2. A 30-year-old patient presents with a cough for 3 weeks, weight loss, fever, and blood in sputum. Chest X-ray reveals a small, calcified granuloma in the mid-lung region along with enlarged hilar lymph nodes. Which of the following terms best describes these findings in the context of primary pulmonary tuberculosis?

A. Rhanke complex

B. Primary complex

C. Ghon's focus

D. Miliary tuberculosis

---

3. Which of the following diagnostic tests on sputum is most appropriate as an initial test for a case of suspected tuberculosis?

A. Interferon-gamma release assay

B. Nucleic acid (NAAT) methods

C. Mantoux test

D. Ziehl-Neelsen staining technique

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4. A sputum sample from a patient suspected of having tuberculosis is stained using the Ziehl-Neelsen (ZN) method as per the Revised National Tuberculosis Control Programme (RNTCP) guidelines. On microscopic examination 12 acid-fast bacilli are counted in 100 fields. According to the RNTCP ZN staining grading system, how would this sample be classified?

A. Positive, 3+

B. Positive, 1+

C. Positive, 2+

D. Doubtful positive/repeat

---

5. A sputum sample from a patient suspected of having tuberculosis is cultured on Lowenstein-Jensen media. Which of the following best describes the characteristics of the colonies after 6 weeks of incubation?

A. Rough, buff-coloured colonies

B. Golden-yellow colonies

C. Blue-green colonies

D. Matt-finely granular colonies

---

6. Which of the following devices uses the fluorometric technique to monitor microbial growth in culture bottles?

A. Bac T/alert

B. Bactec machine

C. Middlebrook’s medium

D. Versatrek

7. A microbiology laboratory wants to implement an automated system for the detection of mycobacteria in clinical specimens. Which of the following is the primary detection method used by the BacT/ALERT system to identify bacterial growth?

A. Detection of fluorescence

B. Detection of CO■ production using colorimetric sensors

C. Detection of pressure changes due to gas production

D. Detection of oxygen consumption

8. A 45-year-old HIV-positive patient presents with a persistent cough with blood in sputum, fever, and weight loss. The chest X-ray shows infiltrates in the upper lobes. The clinician wants a rapid and reliable test for diagnosing. Which of the following statements about the GeneXpert MTB/RIF assay is correct?

A. It can detect Mycobacterium tuberculosis complex and rifampicin resistance within 2 hours

B. It can differentiate between M. tuberculosis and non-tuberculous mycobacteria (NTM)

C. It requires a biosafety level 3 laboratory for processing

D. It has low sensitivity in smear-negative, culture-positive samples

9. A 28-year-old healthcare worker is being screened for latent tuberculosis infection (LTBI) as part of an occupational health program. Which of the following tests is considered the most specific for diagnosing LTBI in a person who has received BCG vaccination?

A. Tuberculin Skin Test (TST)

B. QuantiFERON-TB Gold In-Tube test

C. BacT / ALERT

D. Chest X-ray

---

10. A 35-year-old is being screened for latent tuberculosis infection (LTBI). Which of the following statements about Interferon-Gamma Release Assays (IGRAs) is correct?

A. IGRAs use PPD as the stimulating antigen

B. IGRAs can differentiate between LTBI and active TB

C. IGRAs measure interferon-gamma released by T cells in response to specific M. tuberculosis antigens

D. IGRAs can detect MPT64 antigen in patient samples

---

11. A newborn child in a high TB-burden country is being considered for BCG vaccination. Which of the following statements about the BCG vaccine is correct?

A. It provides 100% protection against all forms of tuberculosis

B. It is a live attenuated vaccine derived from Mycobacterium bovis

C. It is contraindicated in HIV-positive infants regardless of their immune status

D. It is most effective in preventing tuberculous meningitis in adults

---

12. A 30-year-old man presents with a non-healing ulcer on his left leg and hand. He gives a history of frequently swimming in a local pool. The ulcer grew in size gradually and developed into a chronic, painless granuloma as shown in the image. Which organism is most likely responsible for his condition? {{caption_text}}



A. Mycobacterium kansasii

B. Mycobacterium marinum

C. Mycobacterium avium complex

D. Mycobacterium fortuitum

---

13. A 45-year-old man with a history of chronic obstructive pulmonary disease (COPD) presents with chronic cough, weight loss, and night sweats. Sputum culture grows colonies within two weeks, producing no pigment regardless of light exposure. Which of the following is the most likely causative organism?

A. Mycobacterium kansasii

B. Mycobacterium fortuitum

C. Mycobacterium avium complex

D. Mycobacterium marinum

---

14. Which type of leprosy is most likely affecting a patient who presents with localized, well-defined skin lesions and minimal bacterial presence?

A. Tuberculoid leprosy

B. Lepromatous leprosy

C. Borderline leprosy

D. Indeterminate leprosy

15. A 35-year-old male presents with hypoesthetic skin lesions on his arms, legs, and face, along with palpable thickening of the peripheral nerves. There are no visible nodules. Skin smears reveal a low bacterial index. Which treatment regimen is most appropriate for this patient?

A. Rifampicin and dapsone daily for 6 months

B. Rifampicin monthly, daily dapsone, and daily clofazimine for 6 months

C. Rifampicin monthly and daily dapsone for 6 months

D. Rifampicin, dapsone, and clofazimine for 12 months

16. A patient with chronic leprosy develops sudden onset of red, swollen, and tender lesions on existing skin patches. Which of the following reactions is most likely occurring in this patient?

A. Type 1 Reaction (Reversal Reaction)

B. Type 2 Reaction (Erythema Nodosum Leprosum - ENL)

C. Lepromatous transformation

D. Tuberculoid transformation

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	2
Question 5	1
Question 6	2
Question 7	2
Question 8	1
Question 9	2
Question 10	3
Question 11	2
Question 12	2
Question 13	3
Question 14	1
Question 15	3
Question 16	2

Solution for Question 1:

Answer: B) Mycolic acids in the cell wall

Explanation:

Mycolic acids are long-chain fatty acids found in the cell wall of Mycobacterium tuberculosis. They contribute to the hydrophobic nature of the cell wall, which retains stains like Ziehl-Neelsen despite acid-alcohol decolourisation.

Mode of transmission of TB

- Spread via respiratory droplets from coughs or sneezes of infected individuals.
- Other modes are: Inoculation Ingestion
- Inoculation
- Ingestion
- Inoculation
- Ingestion

Antigenic Structure

- Cell wall antigens: Peptidoglycan layer Arabinogalactan layer Mycolic acid layer: Principal constituent and is responsible for the acid fastness and also reduces the entry of most antibiotics Outermost lipid layer Protein layer Lipoarabinomannan layer in the plasma membrane, which is used as a target antigen for TB diagnosis
- Peptidoglycan layer
- Arabinogalactan layer
- Mycolic acid layer: Principal constituent and is responsible for the acid fastness and also reduces the entry of most antibiotics
- Outermost lipid layer
- Protein layer
- Lipoarabinomannan layer in the plasma membrane, which is used as a target antigen for TB diagnosis
- Cytoplasmic antigens: Antigen 5 Antigen 60
- Antigen 5
- Antigen 60
- Peptidoglycan layer
- Arabinogalactan layer
- Mycolic acid layer: Principal constituent and is responsible for the acid fastness and also reduces the entry of most antibiotics
- Outermost lipid layer
- Protein layer
- Lipoarabinomannan layer in the plasma membrane, which is used as a target antigen for TB diagnosis

- Antigen 5
- Antigen 60

Cell wall of M. Tuberculosis

Lipopolysaccharide (LPS) in the cell wall (Option A): LPS is typically found in gram-negative bacteria and is not a component of the cell wall of Mycobacterium tuberculosis. This option is incorrect.

Biofilm formation (Option C): Mycobacterium tuberculosis can form biofilms in certain conditions, but this characteristic does not directly contribute to its ability to retain acid-fast staining. This option is incorrect.

Capsular polysaccharides (Option D): Mycobacterium tuberculosis does not produce a capsule that would affect acid-fast staining. This option is incorrect.

Solution for Question 2:

Answer: B) Primary Complex

Explanation: In primary pulmonary tuberculosis, granuloma forms in the lung parenchyma known as a Ghon focus. The combination of this granuloma and the adjacent hilar lymph node involved is referred to as a Ghon /primary complex

Clinical manifestations of Mycobacterium tuberculosis:

Pulmonary tuberculosis is most common in India.

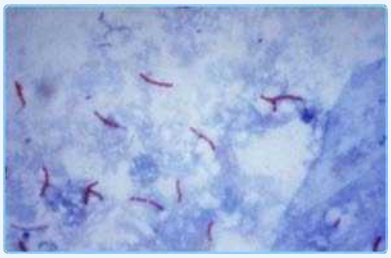
Primary tuberculosis	Post-primary or secondary tuberculosis	Miliary tuberculosis (option D ruled out)

Initial infection by the bacteria in the host. Ghon's focus (option C ruled out): the subpleural focus of tuberculous pneumonia representing the initial site of infection and the primary response of the host. Common location: lower lobe or the lower part of the upper lobe primary complex (option B): Ghon's focus together with the enlarged hilar lymph node. Ranke complex (option A ruled out): Healed calcified nodules from primary infection.	Post-primary progression or reactivation of latent infection, or exogenous reinfection. Common location: upper lobes of the lungs. Lesions undergo necrosis and destruction of tissues leading to cavitation. Lymph node involvement is uncommon.	It is caused in immunocompromised individuals where the bacteria in primary or secondary infection can spread causing dissemination to other organs. Genitourinary tract Lymph nodes (cervical lymphadenitis) Central nervous system (meningitis) Bone and joint (arthritis and osteomyelitis) Peritoneum Gastrointestinal system Pericardium Larynx Pleural lining (pleuritis)

Solution for Question 3:

Answer: D) Ziehl-Neelsen staining technique

Explanation: Ziehl-Neelsen staining technique is less expensive than NAAT and useful in rapid detection and determining infectiousness as the incidence of false-positive smears is very low with the maintenance of good quality control.

Diagnostic test	Key concepts
Microscopy (useful in rapid detection and determining infectiousness)	
Ziehl-Neelsen technique (ZN) (hot stain procedure) (option D)	 Smears are dried, heat-fixed and stained Carbol fuschin, sulphuric acid, methylene blue {{caption_text}}
Kinyoun acid-fast stain (cold stain procedure)	No heating.

Fluorochrome Stain	Smears are stained with auramine phenol or auramine rhodamine fluorescent dyes Examined under ultraviolet illumination
Culture	
Solid media	Löwenstein-Jensen media Cultures are examined weekly for growth Negative cultures (no growth) are reported, and the cultures are discarded only after 8 weeks of incubation
Liquid media	Middlebrook 7H9 broth, BACTEC 12B medium, MGIT, BacT Alert Reduces the isolation turnaround time down to about ten days
Animal inoculation	Inoculated intramuscularly
Immunodiagnosis	
Mantoux test (option C ruled out)	Demonstration of hypersensitivity to tuberculoprotein. 0.1 ml of PPD containing 5 TU (tuberculosis unit) is injected intradermally, induration of 10mm or more is positive.
Interferon-gamma release assay (IGRAs) (option A ruled out)	Mycobacterium TB antigen CFP10 is used in the test, and it reacts with the patient's T cells to release γ interferons.
Molecular Tests	
Genetic Sequencing and Nucleic Acid Amplification (option B ruled out)	PCR amplification of mycobacterial DNA with genus-specific primers. Xpert MTB/RIF (also detects the mutations associated with rifampin resistance)
DNA Microarrays	Quick analysis of many DNA sequences using just one hybridization step.

Solution for Question 4:

Answer: B) Positive, 1+

Explanation: According to the RNTCP ZN staining grading system, 12 AFB/100 fields is graded as positive 1+.

RNTCP ZN staining grading (using 100x oil immersion objective & 10x	Auramine O fluorescent staining grading (using 20 or 25x objective &	Reporting/grading

eyepiece	10x eyepiece	
>10 AFB/field after examination of 20 fields	>100 AFB/field after examination of 20 fields	Positive, 3+
1-10 AFB/field after examination of 50 fields	11-100 AFB/field after examination of 50 fields	Positive, 2+
10-99 AFB/100 fields	1-10 AFB/field after examination of 100 fields	Positive, 1+
1-9 AFB/100 fields	1-3 AFB/100 fields	Doubtful positie/repeat
No AFB per 100 fields	No AFB per 100 fields	Negative

Solution for Question 5:

Answer: A) Rough, buff-colored colonies

Explanation: Rough, buff (cream-coloured) and tough colonies on Lowenstein-Jensen media are characteristic of Mycobacterium tuberculosis.

Culture of M. tuberculosis

Solid medium	
Löwenstein-Jensen media	6-8 weeks to grow Composition Coagulated hen's egg Mineral salt solution Asparagine Malachite green Morphology of colony: Rough, buff (cream-coloured) and tough. (option A) Cultures are examined weekly for growth Negative cultures (those showing no growth) are reported, and the cultures are discarded only after 8 weeks of incubation.
Tharshi medium	Blood-based culture
Loeffler's medium	Serum-based culture
Pawlowsky medium	Potato-based

Liquid medium	
Middlebrook's medium	Contains 2% glycerol
Bactec MGIT	Newer automated method Uses Middlebrook 7H9 broth



Golden-yellow colonies (option B): It is characteristic of staphylococcus aureus on nutrient agar. This option is incorrect.

Blue-green pigment (Pyocyanin) (option C): It is produced by Pseudomonas aeruginosa. This option is incorrect.

Matt-finely granular colonies (option D): It is characteristic of virulent strains of streptococcus pyogenes when grown in blood agar. This option is incorrect.

Solution for Question 6:

Answer: B) Bactec machine

Explanation: The Bactec machine uses the fluorometric technique, where the mycobacterial growth uses oxygen, leading to fluorescence.

Newer automated methods:

Bactec MGIT 960 (option B)	 MGIT: Mycobacterial Growth Indicator Tube Middlebrook 7h9 Broth is used Principle: ... (content truncated)
Bac T/alert (option A ruled out)	Colorimetric detection of CO2 produced by growing bacteria.
Versatrek (option D ruled out)	Detects pressure changes in media due to gas production.

Middlebrook's medium (option C): It is a liquid medium culture containing 2% glycerol. This option is incorrect.

Solution for Question 7:

Answer: B) Detection of CO■ production using colorimetric sensors

Explanation:

The BacT/ALERT system uses colorimetric sensors to detect CO■ production from metabolizing bacteria. As bacteria grow and produce CO■, the system senses a colour change, indicating bacterial presence and growth.

Newer automated methods

Bactec MGIT 960	MGIT: Mycobacterial Growth Indicator Tube Middlebrook 7h9 Broth is used Principle: In the presence of oxygen, fluorescence is quenched Bacteria use oxygen while growing causing fluorescence (Options A & D ruled out) It can detect both Mycobacterium tuberculosis complex and nontuberculous mycobacteria. Incubation temperature: 37 °C

Bac T/alert	Colorimetric detection of CO2 produced by growing bacteria. (Option B)
Versatrek	Detects pressure changes in media due to gas production. (Option C ruled out)

Solution for Question 8:

Answer: A) It can detect Mycobacterium tuberculosis complex and rifampicin resistance within 2 hours

Explanation:

The GeneXpert MTB/RIF assay is a rapid, cartridge-based, automated nucleic acid amplification test that can detect both M. tuberculosis complex and rifampicin resistance in approximately 2 hours making it particularly valuable in high-burden, resource-limited settings.

The GeneXpert MTB/RIF assay offers several advantages:

- Rapid results: Provides results in about 2 hours
- Simultaneous detection of M. tuberculosis and rifampicin resistance
- High specificity: >99% specific for M. tuberculosis complex
- Minimal biosafety requirements: Can be used outside of reference laboratories
- Automated interpretation: Reduces operator-dependent errors

It can differentiate between M. tuberculosis and non-tuberculous mycobacteria (NTM) (Option B): While the GeneXpert MTB/RIF assay is highly specific for the M. tuberculosis complex, it cannot differentiate between M. tuberculosis and other members of the complex, nor can it identify NTM. This option is incorrect.

It requires a biosafety level 3 laboratory for processing (Option C): One of the advantages of the GeneXpert system is that it allows for safe handling of specimens and can be used in lower biosafety level settings due to its closed-cartridge system. This option is incorrect.

It has low sensitivity in smear-negative, culture-positive samples (Option D): While the sensitivity of GeneXpert is higher in smear-positive samples, it still maintains good sensitivity in smear-negative, culture-positive samples (60-80%), outperforming smear microscopy. This option is incorrect.

Solution for Question 9:

Answer: B) QuantiFERON-TB Gold In-Tube test

Explanation:

QuantiFERON-TB Gold In-Tube test is an Interferon-Gamma Release Assay (IGRA) that uses M. tuberculosis-specific antigens ESAT-6, CFP-10, and TB7.7 not present in BCG or most non-tuberculous mycobacteria, making it more specific for LTBI diagnosis in BCG-vaccinated individuals.

Latent TB can be diagnosed by two tests:

- Tuberculin test /Mantoux test
- Interferon-Gamma Release Assay (IGRA)

Tuberculin test /Mantoux test

- Antigen: Purified protein derivative (PPD)
- Procedure:
- 0.1 ml of PPD containing 5 TU (tuberculosis unit) is injected intradermally
- Examined after 48-72 hours
- Interpretation:

Width of induration (measurement done perpendicular to the long axis of the forearm)	Inference
≥ 10mm	Positive
6-9mm	Equivocal
≤ 5mm	Negative

- Equivocal is considered positive in: HIV / Immunosupressed Known close contact H/o Prior TB
- HIV / Immunosupressed
- Known close contact
- HIV / Immunosupressed
- Known close contact

Note: Erythema is not taken into consideration for the interpretation of results.

• Lower specificity in BCG-vaccinated individuals due to cross-reactivity with antigens present in the BCG vaccine and can yield false-positive results (Option A ruled out)

Interferon-Gamma Release Assay (IGRA)

• Principle: T-lymphocytes release interferon-gamma when exposed to M. tuberculosis antigens.

• IGRAs like QuantiFERON-TB Gold In-Tube offer several advantages for LTBI diagnosis: Higher specificity in BCG-vaccinated individuals (negative in BCG-vaccinated individuals) No cross-reactivity with most non-tuberculous mycobacteria Requires only a single patient visit Results are available within 24-48 hours No boosting effect with repeated testing

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• Disadvantage: Cannot differentiate between an active and inactive infection

• Higher specificity in BCG-vaccinated individuals (negative in BCG-vaccinated individuals)

• No cross-reactivity with most non-tuberculous mycobacteria

• Requires only a single patient visit

• Results are available within 24-48 hours

• No boosting effect with repeated testing

BacT / ALERT (Option C): It is an automated blood culture system designed to detect M. tuberculosis in clinical samples containing viable bacteria. It is not used for detecting latent tuberculosis. This option is incorrect.

Chest X-ray (Option D): It is not a specific test for LTBI. It may show changes suggestive of past TB exposure or active TB, but cannot definitively diagnose LTBI in BCG vaccinated person. This option is incorrect.

Solution for Question 10:

Answer: C) IGRAs measure interferon-gamma released by T cells in response to specific M. tuberculosis antigens

Explanation:

IGRAs measure interferon-gamma released by T cells in response to specific M. tuberculosis antigens, providing a measure of the cell-mediated immune response to TB infection.

Interferon-Gamma Release Assay (IGRA)

- Principle: T-lymphocytes release interferon-gamma when exposed to M. tuberculosis antigens.
- IGRAs like QuantiFERON-TB Gold In-Tube offer several advantages for LTBI diagnosis: Higher specificity in BCG-vaccinated individuals (negative in BCG-vaccinated individuals) No cross-reactivity with most non-tuberculous mycobacteria Requires only a single patient visit Results are available within 24-48 hours No boosting effect with repeated testing
- Higher specificity in BCG-vaccinated individuals (negative in BCG-vaccinated individuals)
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- Results are available within 24-48 hours
- No boosting effect with repeated testing
- Disadvantage: Cannot differentiate between an active and inactive infection
- Higher specificity in BCG-vaccinated individuals (negative in BCG-vaccinated individuals)
- No cross-reactivity with most non-tuberculous mycobacteria
- Requires only a single patient visit
- Results are available within 24-48 hours
- No boosting effect with repeated testing

LAM assay:

- Lipoarabinomannan (LAM) is a cell wall component of M. tuberculosis
- It can be detected in urine especially
- It can be useful in diagnosing active TB in HIV-positive patients with low CD4 counts

MPT 64 patch test: (Option D ruled out)

- Antigen: Patch containing purified MPT64 protein
- Helps differentiate Mycobacterium.Tb from Atypical Mycobaterium
- Procedure: The patch is placed on the ventral surface of the forearm The skin reaction is read after 4 days
- The patch is placed on the ventral surface of the forearm
- The skin reaction is read after 4 days
- Inference: Positive reaction: Erythema, induration or small red vesicles at the patch site

- The patch is placed on the ventral surface of the forearm
- The skin reaction is read after 4 days

IGRAs use PPD as the stimulating antigen (Option A): IGRAs do not use PPD (Purified Protein Derivative) as the stimulating antigen. Instead, they use M. tuberculosis-specific antigens like ESAT-6 and CFP-10. This option is incorrect.

IGRAs can differentiate between LTBI and active TB (Option B): While IGRAs are excellent for detecting M. tuberculosis infection, they cannot distinguish between latent infection and active disease. This option is incorrect.

Solution for Question 11:

Correct Answer: B) It is a live attenuated vaccine derived from Mycobacterium bovis

Explanation:

BCG vaccine is a live attenuated vaccine derived from Mycobacterium bovis. It was first developed by Calmette and Guérin in 1921 through serial passages of a virulent M. bovis strain.

BCG Strains:

Different BCG strains are used worldwide, which may contribute to variations in vaccine efficacy.

Route of Administration:

- Intradermally, usually in the left deltoid region
- Correct technique is crucial for optimal immune response and minimizing adverse effects

Duration of Protection:

It is not well-defined and may wane over time.

It provides 100% protection against all forms of tuberculosis (Option A): The BCG vaccine does not provide complete protection against all forms of tuberculosis. Its efficacy varies widely (0-80%) depending on the form of TB and geographical location. It is most effective against severe forms of childhood TB, such as miliary TB and tuberculous meningitis. This option is incorrect.

It is contraindicated in HIV-positive infants regardless of their immune status (Option C): While caution is needed, BCG is not universally contraindicated in all HIV-positive infants. The decision depends on the infant's immune status and local TB prevalence. It is contraindicated in

symptomatic HIV infection but may be given to asymptomatic HIV-positive infants in highTB-burden settings. This option is incorrect.

It is most effective in preventing tuberculous meningitis in adults (Option D): While BCG can provide some protection against tuberculous meningitis, it is most effective in preventing severe forms of TB in children, not adults. This option is incorrect.

Solution for Question 12:

Answer: B) Mycobacterium marinum

Explanation:

M. marinum is a photochromogen associated with "swimming pool granuloma," resulting from exposure to contaminated water. The chronic, painless ulcer described depicts the case of an M. marinum infection.

Non-tuberculous Mycobacteria (NTM)

- Definition: NTM are diverse mycobacteria found in soil, water, and animals, primarily affecting those with weakened immune systems
- Transmission: Not spread person-to-person; typically contracted from contaminated water or soil

Group	Characteristics	Examples
Group I: Photochromogens	Produce pigment in light; slow growth.	M. kansasii (lung disease), (Option A ruled out) M. marinum ("swimming pool granuloma"). (Option B)
Group II: Scotochromogens	Produce pigment regardless of light; varied growth rates.	M. scrofulaceum (swollen lymph nodes), M. gordonae (water contamination).
Group III: Non-photochromogens	No pigment production; diverse characteristics.	M. avium complex (lung infections), (Option C ruled out) M. ulcerans (Buruli ulcers).
Group IV: Rapid Growers	Grow quickly; early colony formation.	M. fortuitum (skin abscesses), (Option D ruled out) M. chelonae (injection site infections).

• Microscopy: Acid-fast staining was used initially but was not conclusive for NTM diagnosis (shared with M. tuberculosis).

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• Culture: Essential for definitive identification. Various growth rates help classify NTM species:

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• Acid-fast staining was used initially but was not conclusive for NTM diagnosis (shared with M. tuberculosis).

• Essential for definitive identification.

• Various growth rates help classify NTM species:

NTM Group	Growth Characteristics
Rapid Growers	Form colonies within days (e.g., M.fortuitum).
Slow Growers	May take weeks to form colonies (e.g., M.kansasii).

Treatment:

• Specific to species and drug susceptibility.

• Includes prolonged multi-drug therapy.

• Surgery may be necessary for localized cases.

Solution for Question 13:

Answer: C) Mycobacterium avium complex

Explanation:

M. avium complex is a non-photochromogen. It does not produce pigment regardless of light exposure. It is known for causing lung infections, particularly in immunocompromised individuals. It is a slow grower, forming colonies within a few weeks.

Group	Characteristics	Examples

Group I: Photochromogens	Produce pigment in light; slow growth.	M. kansasii (lung disease), (Option A ruled out) M. marinum ("swimming pool granuloma"). (Option D ruled out)
Group II: Scotochromogens	Produce pigment regardless of light; varied growth rates.	M. scrofulaceum (swollen lymph nodes), M. gordonae (water contamination).
Group III: Non-photochromogens	No pigment production; diverse characteristics.	M. avium complex (lung infections), (Option C) M. ulcerans (Buruli ulcers).
Group IV: Rapid Growers	Grow quickly; early colony formation.	M. fortuitum (skin abscesses), (Option B ruled out) M. chelonae (injection site infections).

Solution for Question 14:

Answer: A) Tuberculoid leprosy

Explanation:

Tuberculoid leprosy is characterized by high host resistance, limited, well-defined skin lesions, and minimal bacterial presence.

- Source of Infection: Leprosy is solely a human disease. Infected individuals are the only source.
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- Infected individuals are the only source.
- Transmission Route: The precise route is unclear. Significant shedding of bacilli through nasal secretions. Possible transmission through the respiratory tract or skin.
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- Significant shedding of bacilli through nasal secretions.
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- Leprosy is solely a human disease.
- Infected individuals are the only source.
- The precise route is unclear.
- Significant shedding of bacilli through nasal secretions.

• Possible transmission through the respiratory tract or skin.

Type	Characteristics
Lepromatous Leprosy	Low host resistance, abundant bacilli in lesions, widespread lesions, higher transmission likelihood (Option B ruled out).
Tuberculoid Leprosy	High resistance, limited well-defined skin lesions, minimal bacilli in the lesions (Option A).
Dimorphous (Borderline)	Characteristics of both lepromatous and tuberculoid forms, potential to shift between the two (Option C ruled out).
Indeterminate Leprosy	Early, unstable stage, may resolve spontaneously or progress to other forms (Option D ruled out).

Ridley-Jopling Classification:

Type	Characteristics
Tuberculoid (TT)	High resistance, localized lesions
Borderline Tuberculoid (BT)	Intermediate between TT and BB, some features of both
Borderline (BB)	Intermediate, with features of both tuberculoid and lepromatous forms
Borderline Lepromatous (BL)	Intermediate between BB and LL, some features of both
Lepromatous (LL)	Low resistance, disseminated disease

Category	Lepromatous Leprosy (LL)	Tuberculoid Leprosy (TT)
Bacillary Load	Multi-bacillary	Pauci-bacillary
Bacteriological Index	Higher load (4-6+)	Lower load (0-1+)
Skin Lesions	Leonine facies, Alopecia	Few lesions
Nerve Lesions	Late	Early
Cell-mediated Immunity	Low	Normal
Lepromin Test	Negative	Positive
CD4/CD8 Ratio	1:2	2:1
Humoral Immunity	Increased	Normal
Macrophages	Foamy type	Epithelioid type
Giant Cells	Negative	Present

This system provides a spectrum from TT (high resistance, localized lesions) to LL (low resistance, disseminated disease), characterizing the disease's progression.

Classification	Characteristics	Associated Type	Clinical Features
Multibacillary Leprosy	High bacterial load, clusters of bacilli ("globi") in lesions	Lepromatous leprosy	Low host resistance, widespread lesions, higher transmission risk
Paucibacillary Leprosy	Minimal bacterial presence	Tuberculoid leprosy	High host resistance, limited skin lesions

Solution for Question 15:

Answer: C) Rifampicin monthly and daily dapsone for 6 months

Explanation:

The patient's presentation with hypoesthetic skin lesions and thickened peripheral nerves without nodules is consistent with a paucibacillary form of leprosy. For paucibacillary leprosy, the WHO recommends the following regimen:

- Rifampicin: once a month
- Dapsone: daily
- Clofazimine: daily
- Clinical Classification: Classified into multibacillary (MB) and paucibacillary (PB) forms based on bacterial load.
- Diagnostic Methods: Microscopy: Mainstay for detecting acid-fast bacilli (AFB) in nasal and skin smears by Ziehl-Neelsen staining. Nasal smears target shed bacilli; skin smears from lesion edges. Slit-skin smear enhances bacillus detection from small cuts. Grading Systems: Quantify bacilli to determine bacterial index (BI) Histopathology: Nodule or thickened nerve biopsy Culture: Mouse footpad or armadillo inoculation. Serological Tests and Molecular Methods.
- Microscopy: Mainstay for detecting acid-fast bacilli (AFB) in nasal and skin smears by Ziehl-Neelsen staining. Nasal smears target shed bacilli; skin smears from lesion edges. Slit-skin smear enhances bacillus detection from small cuts.
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- Grading Systems: Quantify bacilli to determine bacterial index (BI)
- Histopathology: Nodule or thickened nerve biopsy

• Culture: Mouse footpad or armadillo inoculation.

• Serological Tests and Molecular Methods.

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• Slit-skin smear enhances bacillus detection from small cuts.

• Grading Systems: Quantify bacilli to determine bacterial index (BI)

• Histopathology: Nodule or thickened nerve biopsy

• Culture: Mouse footpad or armadillo inoculation.

• Serological Tests and Molecular Methods.

Age group	Drug	Duration
Adult	Rifampicin - 600mg/month + Clofazimine - 300mg once a month & 50mg daily + Dapsone - 100mg daily	Multibacillary - 12 months Paucibacillary - 6 months
Children (10-14 years)	Rifampicin - 450mg/month + Clofazimine - 150mg once a month & 50mg on alternate days + Dapsone - 50mg daily	Multibacillary - 12 months Paucibacillary - 6 months

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Solution for Question 16:

Answer: B) Type 2 Reaction (Erythema Nodosum Leprosum - ENL)

Explanation:

• The clinical presentation of sudden onset red, swollen, and tender lesions in a patient with chronic leprosy is characteristic of Type 2 Reaction, also known as Erythema Nodosum Leprosum (ENL).

• ENL is an inflammatory response triggered by antigens released from dead M. leprae cells, leading to neutrophil infiltration and the formation of tender subcutaneous nodules. This reaction typically occurs in lepromatous and borderline lepromatous cases, often a few months into chemotherapy.

Character	Lepra Reaction Type I	Lepra Reaction Type II
Hypersensitivity	Type IV (Delayed Hypersensitivity)	Type III (Immune Complex-Mediated)
Seen with	Borderline leprosy (BL), Lepromatous leprosy (LL)	Lepromatous variety (LL)
Manifests as	Inflammation of previous lesions, new skin lesions, and neuritis	Crops of painful erythematous papules which become nodular, called ENL (Erythema Nodosum Leprosum)
Progresses as	If it occurs before treatment, progresses towards LL (down-grading reaction) If it occurs after treatment, it progresses towards TT (reversal reaction)	Usually occurs following the start of chemotherapy
T-helper Response	TH1 predominates, leading to increased IFN-γ and IL-2	TH2 predominates, leading to increased IL-6 and IL-8. TNF-α plays a central role
Other organs affected	Usually not affected	Eyes, testes, and kidneys are affected
Treatment	Glucocorticoid	Glucocorticoid, thalidomide, clofazimine, and antipyretics

Lepromatous Transformation (Option C): It doesn't refer to a sudden reaction but rather to a shift toward the lepromatous end of the leprosy spectrum. It is characterized by a gradual increase in bacterial load and spread of lesions, rather than the acute onset of tender lesions. This option is incorrect.

Tuberculoid Transformation (Option D): It doesn't indicate an acute reaction. It refers to a shift toward the tuberculoid form, usually indicating an improved immune response. However, such transformation does not cause acute inflammation or red, tender lesions as seen in the patient. This option is incorrect.

E coli and shigella

1. A 50-year-old man presents with fever, abdominal pain, and diarrhea. Stool cultures reveal the presence of a gram-negative, rod-shaped bacterium that is oxidase-negative, ferments glucose, and reduces nitrate to nitrite. Which of the following bacteria is most likely the causative agent?

A. Escherichia coli

B. Clostridium difficile

C. Staphylococcus aureus

D. Vibrio cholerae

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2. A 35-year-old patient is suspected to have a urinary tract infection. After culturing her urine sample on MacConkey agar, pink colonies of E. coli are isolated. Which component in the agar is responsible for changing the color to pink?

A. Bromothymol blue

B. Neutral red

C. Phenol red

D. Methylene Blue

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3. A 45-year-old male presents with severe watery diarrhea following a recent trip to Thailand. Stool analysis reveals the presence of an E. coli enterotoxin. Which of the following tests is most indicative of a positive result for heat-stable toxin?

A. Ligated rabbit ileal loop test read at 6 hours

B. Elongation of Chinese hamster ovary (CHO) cells

C. Eiken's test

D. Passive agglutination tests

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4. Which strain of E. coli is known for its ability to adhere to intestinal epithelial cells in a distinctive "stacked-brick" arrangement, and is frequently linked to persistent diarrhea in children and individuals with HIV?

A. Enteropathogenic E. coli (EPEC)

B. Enteroaggregative E. coli (EAEC)

C. Enterotoxigenic E. coli (ETEC)

D. Diffuse-adhering E. coli (DAEC)

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5. Which test is utilized to distinguish E. coli O157 from other E. coli strains by assessing their capacity to ferment sorbitol?

A. Sereny Test

B. SMAC Test

C. Voges-Proskauer Test

D. Methyl Red Test

---

6. Which combination of biochemical tests is best suited for confirming the identification of Escherichia coli in a clinical laboratory?

A. Indole Test, Methyl Red (MR) Test, Voges-Proskauer (VP) Test, and Citrate Utilization Test

B. Gram Staining, Urease Test, Catalase Test, and Oxidase Test

C. Triple Sugar Iron (TSI) Agar, Urease Test, Citrate Utilization Test, and Methyl Red (MR) Test

D. Indole Test, Urease Test, Voges-Proskauer (VP) Test, and Oxidase Test

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7. Which toxin produced by Shigella inhibits protein synthesis in host cells and has the potential to cause hemolytic uremic syndrome (HUS)?

A. ShET-1

B. Shiga Toxin (Stx)

C. Ipa Proteins

D. IcsA (VirG)

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8. A patient presents with dysentery symptoms, including bloody diarrhea and abdominal cramps. Stool culture shows lactose-negative colonies on MacConkey agar, and the laboratory notes non-motile, Gram-negative rods that are oxidase-negative. What additional test would be most appropriate to confirm the identity of the pathogen?

A. Serotyping

B. Indole Test PCR for ipaH

C. PCR for ipaH gene

D. Citrate Utilization Test

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	2
Question 5	2
Question 6	1
Question 7	2
Question 8	3

Solution for Question 1:

Correct Answer: A) Escherichia coli

Explanation:

E. coli is a gram-negative, rod-shaped bacterium that ferments glucose and is oxidase-negative. It can reduce nitrate to nitrite and is a common cause of gastrointestinal infections, particularly diarrhea, in adults. Given the symptoms of fever, abdominal pain, and diarrhea, along with the culture results, E. coli is the most likely causative agent.

Edwards-Ewing Classification of the Enterobacteriaceae Family

Tribes and Genera:

Tribe	Genus
Escherichiae	Escherichia (Option A), Shigella
Edwardsiellae	Edwardsiella
Salmonellae	Salmonella

Citrobacteriaceae	Citrobacter
Klebsielleae	Klebsiella, Enterobacter, Serratia, Hafnia, Pantoea
Proteae	Proteus, Providencia, Morganella
Yersinieae	Yersinia
Erwinieae	Erwinia

Characteristics of the Enterobacteriaceae Family (Option A)

• Microscopy: Gram-negative bacilli, appearing as rod-shaped bacteria. Non-sporing, indicating they do not form spores for survival in harsh environments.

• Gram-negative bacilli, appearing as rod-shaped bacteria.

• Non-sporing, indicating they do not form spores for survival in harsh environments.

• Oxygen Requirements: Aerobic and facultatively anaerobic, meaning they can grow in the presence or absence of oxygen, providing them with versatility in various environments.

• Aerobic and facultatively anaerobic, meaning they can grow in the presence or absence of oxygen, providing them with versatility in various environments.

• Motility: Most members are motile due to peritrichous flagella, which are flagella distributed all over the surface of the bacteria. Notable non-motile genera include Shigella and Klebsiella, which lack flagella.

• Most members are motile due to peritrichous flagella, which are flagella distributed all over the surface of the bacteria.

• Notable non-motile genera include Shigella and Klebsiella, which lack flagella.

• Biochemical Properties: Ferments glucose, producing both gas and acid as byproducts, a characteristic that can be useful in distinguishing these bacteria in the lab. Reduces nitrates to nitrites, an important metabolic process that aids in the identification of specific species. Produces catalase, which helps break down hydrogen peroxide, protecting the bacteria from oxidative damage. Does not produce oxidase, distinguishing it from other similar families of bacteria.

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• Reduces nitrates to nitrites, an important metabolic process that aids in the identification of specific species.

• Produces catalase, which helps break down hydrogen peroxide, protecting the bacteria from oxidative damage.

• Does not produce oxidase, distinguishing it from other similar families of bacteria.

• Gram-negative bacilli, appearing as rod-shaped bacteria.

• Non-sporing, indicating they do not form spores for survival in harsh environments.

• Aerobic and facultatively anaerobic, meaning they can grow in the presence or absence of oxygen, providing them with versatility in various environments.

• Most members are motile due to peritrichous flagella, which are flagella distributed all over the surface of the bacteria.

• Notable non-motile genera include Shigella and Klebsiella, which lack flagella.

• Ferments glucose, producing both gas and acid as byproducts, a characteristic that can be useful in distinguishing these bacteria in the lab.

• Reduces nitrates to nitrites, an important metabolic process that aids in the identification of specific species.

• Produces catalase, which helps break down hydrogen peroxide, protecting the bacteria from oxidative damage.

• Does not produce oxidase, distinguishing it from other similar families of bacteria.

Clostridium difficile (Option B): Clostridium difficile is gram-positive bacillus.

Staphylococcus aureus (Option C): Staphylococcus aureus is gram-positive cocci.

Vibrio cholerae (Option D): Vibrio cholerae is gram-negative curved bacilli with polar flagellum.

Solution for Question 2:

Correct Answer: B) Neutral red

Explanation:

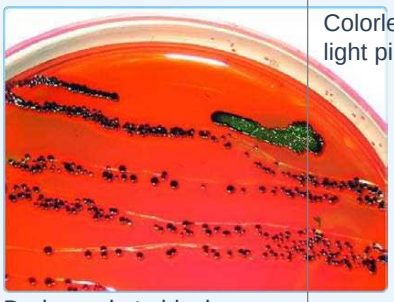
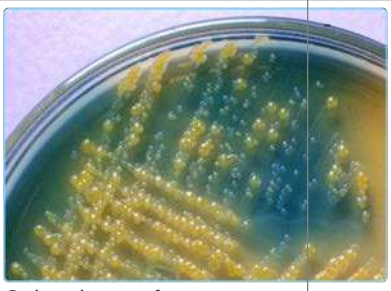
Neutral red is the pH indicator used in MacConkey agar. It turns pink in the presence of acid produced by lactose fermentation, which occurs when E. coli ferments lactose. The acid lowers the pH, causing the neutral red to change color, resulting in the pink colonies observed.

Classification Based on Lactose Fermentation

Lactose Fermenters	Escherichia coli Klebsiella Enterobacter Citrobacter
Late Lactose Fermenters	Shigella sonnei

Non-Lactose Fermenters	Salmonella Shigella Proteus Yersinia

Tests for Lactose Fermentation

Test	Indicator	Lactose Fermenters	Non-Lactose Fermenters
MacConkey Agar	Neutral Red (Option B)	 Pink or red colonies	Colorless colonies
Eosin Methylene Blue Agar (EMB)	Eosin Y and Methylene Blue (Option D ruled out)	 Dark purple to black, or with a green metallic sheen	Colorless or light pink
Cystine Lactose Electrolyte-Deficient (CLED)	Bromothymol Blue (Option A ruled out)	 Color change from green to yellow	

Phenol red (Option C) Phenol red is a pH indicator and changes the color from red to yellow if mannitol is fermented to produce acids. Coagulase-positive staphylococci ferment mannitol to produce yellow colonies.

Solution for Question 3:

Correct Answer: A) Ligated rabbit ileal loop test read at 6 hours

Explanation:

The ligated rabbit ileal loop test is a classic method used to detect enterotoxins, including heat-stable toxins produced by certain strains of E. coli. In this test, a segment of the rabbit's ileum is ligated, and the response to the toxin can be observed. A positive result would indicate fluid accumulation in the loop, consistent with heat-stable enterotoxin activity.

Types of Toxins:

Type of Toxin	Description	Mechanism of Action (MOA)	Role in Pathogenesis
Heat Labile Toxin (LT)	A complex of polypeptide subunits	Binds to intestinal epithelial cells, activates adenyl cyclase, increases cAMP, leading to water and electrolyte loss into the gut lumen	Not entirely clear
Heat-Stable Toxin (ST)	Small peptide toxin	Activates cyclic guanosine monophosphate (cGMP) in the intestine	Contributes to diarrhea
Verotoxin (Shiga-like Toxin)	Cytotoxic and enterotoxic, affects Vero and HeLa cells	Inhibits protein synthesis by inactivating ribosomal RNA, causing cell death	Causes severe dysentery and damage to intestinal epithelium

Testing for Toxins:

Assay	Heat Labile Toxin	Heat Stable Toxin
In Vivo Tests		
Ligated Rabbit Ileal Loop (6 hours)	Positive (Option A)	Negative
Ligated Rabbit Ileal Loop (18 hours)	Positive	Negative
In Vitro Tests		

Elongation of Chinese Hamster Ovary (CHO) Cells (Option B ruled out)	Positive	Negative
ELISA	Positive	Positive (ST ELISA with monoclonal antibody)
Passive Agglutination Tests (Option D ruled out)	Positive	Negative
Passive Immune Hemolysis	Positive	Negative
Precipitin Test (Eiken's Test) (Option C ruled out)	Positive	Negative
PCR	Positive	Not Available

Solution for Question 4:

Correct Answer: B) Enteroaggregative E. coli (EAEC)

Explanation:

EAEC is known for its unique ability to adhere to the intestinal epithelial cells in a "stacked-brick" configuration, which is indicative of its aggregative adherence. This strain is frequently associated with persistent diarrhea, especially in children and immunocompromised individuals, such as those with HIV.

Types of E. coli

Type of E. coli	Clinical Features	Toxins Produced
Enterotoxigenic E. coli (ETEC)	Watery diarrhea, Abdominal cramps, Low-grade fever, Nausea and vomiting Most common cause of traveller's diarrhea	Causes watery diarrhea through attachment via CFA (colonization factor antigen) and toxin production (heat-labile toxin ↑cAMP, heat-stable toxin ↑cGMP).
Enteropathogenic E. coli (EPEC)	Watery diarrhea (especially in infants), Fever, Vomiting, Dehydration	None (mechanism involves attaching and effacing lesions)
Enterohemorrhagic E. coli (EHEC)	Bloody diarrhea, Severe abdominal cramps, Hemolytic uremic syndrome (HUS)	Shiga-like toxins/Verocytotoxin

Enteroinvasive E. coli (EIEC)	Watery diarrhea followed by dysentery-like symptoms, Fever, Severe abdominal pain	None (mechanism involves cell invasion and intracellular multiplication)
Enteroaggregative E. coli (EAEC)	Persistent watery diarrhea (duration > 14 days), Abdominal pain, Nausea, Low-grade fever	Plasmid-encoded toxin (PET) I"stacked-brick" pattern of bacterial aggregation when adhering to HEp-2 cells.
Diffuse-adhering E. coli (DAEC)	Watery diarrhea (usually mild), Abdominal cramps, Low-grade fever	None (mechanism involves diffuse adherence to epithelial cells)

Clinical Features of E. coli Infections

- E. coli accounts for a significant majority of UTIs.
- Approximately 5-7% of pregnant women may have asymptomatic bacteriuria, which can lead to symptomatic infections later.
- P pili-positive E. coli strains are generally uropathogenic.
- Various types of E. coli can lead to different forms of diarrhea, ranging from watery to bloody.
- E. coli can cause peritonitis and abscesses.
- It is also an important cause of neonatal meningitis.
- Bloodstream invasion can lead to septic shock and systemic inflammatory response syndrome (SIRS).

EPEC (Option A) adheres to intestinal cells forming attaching and effacing lesions, mainly causing watery diarrhea in infants.

ETEC (Option C) produces heat-labile and/or heat-stable enterotoxins causing watery diarrhea, but does not adhere in a "stacked-brick" pattern.

DAEC (Option D) adheres diffusely to intestinal epithelial cells, typically causing mild watery diarrhea in young children, but lacks the "stacked-brick" pattern of adhesion.

Solution for Question 5:

Correct Answer: B) SMAC Test

Explanation:

E. coli O157 is a non-sorbitol fermenter, meaning it does not produce acid from sorbitol fermentation, resulting in colorless colonies on SMAC agar. In contrast, most other E. coli strains ferment sorbitol and produce pink colonies. Thus, this test is crucial for identifying E. coli O157.

SMAC Test (Sorbitol-MacConkey Agar)

- Purpose: Isolation and differentiation of Enterohemorrhagic E. coli (EHEC), particularly E. coli O157. Contains sorbitol as the fermentable sugar instead of lactose. Includes bile salts and crystal violet to inhibit the growth of Gram-positive bacteria. Uses neutral red as a pH indicator.
- Isolation and differentiation of Enterohemorrhagic E. coli (EHEC), particularly E. coli O157.
- Contains sorbitol as the fermentable sugar instead of lactose.
- Includes bile salts and crystal violet to inhibit the growth of Gram-positive bacteria.
- Uses neutral red as a pH indicator.
- Interpretation:
- Isolation and differentiation of Enterohemorrhagic E. coli (EHEC), particularly E. coli O157.
- Contains sorbitol as the fermentable sugar instead of lactose.
- Includes bile salts and crystal violet to inhibit the growth of Gram-positive bacteria.
- Uses neutral red as a pH indicator.

Organism Type	Fermentation of Sorbitol	Colony Color
E. coli O157	Does not ferment	Colorless colonies
Non-O157 E. coli	Ferments	Pink/red colonies

Sereny Test (Option A ruled out)

- Purpose: Determines the invasiveness of enteric bacteria, particularly Shigella and Enteroinvasive E. coli (EIEC).
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- Procedure: A suspension of the bacterial culture is instilled into the conjunctival sac of a guinea pig. Observe for signs of keratoconjunctivitis.
- A suspension of the bacterial culture is instilled into the conjunctival sac of a guinea pig.
- Observe for signs of keratoconjunctivitis.
- Interpretation:
- Determines the invasiveness of enteric bacteria, particularly Shigella and Enteroinvasive E. coli (EIEC).
- A suspension of the bacterial culture is instilled into the conjunctival sac of a guinea pig.
- Observe for signs of keratoconjunctivitis.

Result	Interpretation
Positive Result	Development of keratoconjunctivitis (indicates invasiveness, e.g., Shigella, EIEC)
Negative Result	No signs of keratoconjunctivitis (indicates non-invasive bacteria)



Sereny test showing Rabbit eye

Voges-Proskauer Test (Option C): Used to detect acetoin production from glucose fermentation, not related to sorbitol fermentation.

Methyl Red Test (Option D): Measures the production of stable acid end products from glucose fermentation, not specific to sorbitol fermentation.

Solution for Question 6:

Correct Answer: A) Indole Test, Methyl Red (MR) Test, Voges-Proskauer (VP) Test, and Citrate Utilization Test

Explanation:

This combination of tests is commonly used to identify E. coli.

- Indole Test: E. coli is typically indole positive.
- Methyl Red Test: E. coli ferments glucose to produce stable acid, resulting in a positive MR test.
- Voges-Proskauer Test: E. coli is usually VP negative.

• Citrate Utilization Test: E. coli does not utilize citrate as a sole carbon source, leading to a negative result.



Results of an IMViC series done on Escherichia coli after 24-hour incubation at 37°C.

Tube A shows a positive indole test in tryptone broth. The positive result is indicated by the red layer at the top of the tube after adding Kovács reagent.

Tube B shows a positive methyl red test as indicated by the red color after the addition of methyl red reagent.

Tube C shows a negative Voges-Proskauer test as indicated by the lack of colour change after the addition of Barritt's A and Barritt's B reagents. Tube D shows a negative citrate result as indicated by the lack of growth and color change in the tube.

Specimen Collection

• Types of Specimens: Stool Samples: For gastrointestinal infections. Urine Samples: For urinary tract infections (UTIs). Blood Samples: For septicemia. Cerebrospinal Fluid (CSF): For meningitis.

• Stool Samples: For gastrointestinal infections.

• Urine Samples: For urinary tract infections (UTIs).

• Blood Samples: For septicemia.

• Cerebrospinal Fluid (CSF): For meningitis.

• Stool Samples: For gastrointestinal infections.

• Urine Samples: For urinary tract infections (UTIs).

• Blood Samples: For septicemia.

• Cerebrospinal Fluid (CSF): For meningitis.

Laboratory Techniques

Microscopy

• Gram Staining: E. coli appears as Gram-negative rods. Useful for preliminary identification in cases of septicemia and meningitis.

• E. coli appears as Gram-negative rods.

• Useful for preliminary identification in cases of septicemia and meningitis.

• E. coli appears as Gram-negative rods.

• Useful for preliminary identification in cases of septicemia and meningitis.

Culture Methods

• General Media: MacConkey Agar: E. coli typically produces pink colonies due to lactose fermentation.

• MacConkey Agar: E. coli typically produces pink colonies due to lactose fermentation.

• E. coli typically produces pink colonies due to lactose fermentation.

• Selective and Differential Media: Eosin Methylene Blue (EMB) Agar: E. coli shows a characteristic green metallic sheen. Sorbitol-MacConkey (SMAC) Agar: Differentiates E. coli O157 (colorless colonies) from other E. coli strains (pink colonies).

• Eosin Methylene Blue (EMB) Agar: E. coli shows a characteristic green metallic sheen.

• E. coli shows a characteristic green metallic sheen.

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• Sorbitol-MacConkey (SMAC) Agar: Differentiates E. coli O157 (colorless colonies) from other E. coli strains (pink colonies).

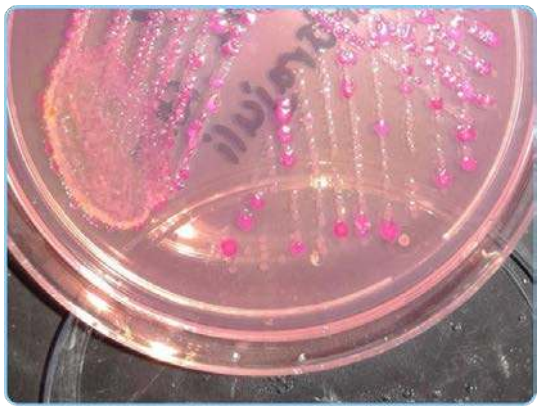
• Differentiates E. coli O157 (colorless colonies) from other E. coli strains (pink colonies).

• E. coli shows a characteristic green metallic sheen.

• Differentiates E. coli O157 (colorless colonies) from other E. coli strains (pink colonies).



E. coli on EMB agar



Sorbitol fermenting and non-fermenting colonies of Escherichia coli on sorbitol MacConkey agar.

Biochemical Tests

IMViC Tests

Test	Result
Indole Production	E. coli is typically indole positive.
Methyl Red (MR) Test	E. coli shows a positive result.
Voges-Proskauer (VP) Test	E. coli is typically negative.
Citrate Utilization	E. coli usually does not utilize citrate (negative result).
Triple Sugar Iron (TSI) Agar	Shows fermentation of glucose, lactose, and production of gas.
Urease Test	E. coli is urease negative.

Solution for Question 7:

Correct Answer: B) Shiga Toxin (Stx)

Explanation:

Shiga toxin, particularly produced by Shigella dysenteriae type 1, inhibits protein synthesis in host cells by cleaving a specific adenine residue from the 28S rRNA of the 60S ribosomal subunit. This inhibition can cause cell death and is linked to severe complications, including hemolytic uremic syndrome (HUS).

Features of Shigella

Pathogenic Species

- Shigella dysenteriae
- Shigella flexneri
- Shigella boydii
- Shigella sonnei

Transmission

- Primarily via the fecal-oral route.
- Commonly spread through contaminated food and water or direct person-to-person contact.

Infective Dose

- Very low; as few as 10-100 bacteria can cause infection.

Invasion

- Invades and multiplies within colonic epithelial cells, leading to cell death and inflammation.

Shigella Toxins

Type of Toxin	Produced By	Mechanism	Effects
Shiga Toxin (Stx)	Shigella dysenteriae type 1	Inhibits protein synthesis in host cells by cleaving a specific adenine residue from the 28S rRNA of the 60S ribosomal subunit.	Causes cell death, contributes to intestinal damage, and can lead to hemolytic uremic syndrome (HUS) in severe cases.
Enterotoxins (ShET-1 & ShET-2) (Option A ruled out)	Various Shigella species	Induce fluid secretion in the intestine, contributing to diarrhea.	Promote diarrhea by causing fluid accumulation in the intestinal lumen.

Ipa Proteins (Option C ruled out)	-	Facilitate entry into host epithelial cells and escape from phagosomes.	Enhance bacterial invasion and survival within host cells.
IcsA (VirG) (Option D ruled out)	-	Promotes actin polymerization for intracellular movement and spread to adjacent cells.	Aids in the spread of Shigella within the intestinal epithelium.
Injected by Type III Secretion System (T3SS)	-	Proteins like IpaB, IpaC, and IpaD disrupt host cell processes, aiding in bacterial invasion and evasion of the host immune response.	Enhance bacterial survival and replication by evading host immune defenses.
Endotoxins	-	Due to the lipopolysaccharides (LPS) of the Gram-negative cell wall.	Contributes to inflammatory response and symptoms associated with infection.

Solution for Question 8:

Correct Answer: C) PCR for ipaH gene

Explanation:

The clinical presentation and laboratory findings of lactose-negative colonies on MacConkey agar, non-motile, Gram-negative rods, and oxidase-negative characteristics strongly suggest Shigella species. The ipaH gene is specific to Shigella and is associated with its pathogenicity. Performing a PCR test to detect the ipaH gene provides a rapid and accurate method for confirming the presence of Shigella in the stool sample, making it the most appropriate choice.

Clinical Presentation of Shigella:

Diarrhea:

- Typically acute and often bloody (dysentery).
- Incubation Period: 1-3 days after exposure.
- Prodromal Period: 1-2 days after the incubation period.

Abdominal Pain:

- Severe cramping.

Fever:

- Often high-grade.

Tenesmus:

- Painful straining with defecation.

Systemic Symptoms:

- Malaise, headache, and sometimes vomiting.

Complications

- Hemolytic Uremic Syndrome (HUS): Particularly associated with Shigella dysenteriae type 1.
- Reactive Arthritis: More commonly seen after infection with Shigella flexneri.
- Neurological Complications: Including seizures, especially in young children.

Diagnosis

Clinical Evaluation:

- Based on symptoms such as bloody diarrhea, abdominal pain, and fever, along with a recent history of exposure (e.g., travel to endemic areas).

Laboratory Tests:

- Stool Culture: Gold standard for diagnosis; Shigella species identified on selective media like MacConkey agar or XLD (xylose lysine deoxycholate) agar.
- Biochemical Tests: Confirmatory tests include lactose non-fermentation and indole production, among other specific biochemical characteristics.
- Serotyping: Identification of specific Shigella species and serotypes using agglutination tests with specific antisera.
- PCR (Polymerase Chain Reaction): Rapid detection of Shigella-specific genes and virulence factors.
- Stool Microscopy: Presence of leukocytes and erythrocytes in stool samples indicative of invasive diarrhea.

Treatment

Fluid Replacement:

- Oral Rehydration Therapy (ORT): Recommended for mild to moderate dehydration.
- Intravenous Fluids: Necessary for severe dehydration or if the patient cannot tolerate oral fluids.

Antibiotic Therapy:

- Recommended for severe cases, young children, the elderly, and immunocompromised patients.
- First-line Antibiotics: Include fluoroquinolones (e.g., ciprofloxacin), third-generation cephalosporins (e.g., ceftriaxone), or azithromycin based on susceptibility testing.
- Increasing antibiotic resistance necessitates periodic updates on local resistance patterns.

Symptomatic Treatment:

- Antipyretics for managing fever.
- Avoidance of antimotility agents like loperamide, which can worsen the infection.

Serotyping (Option A): Serotyping is used to identify specific serotypes of Shigella based on their O (somatic) and H (flagellar) antigens, but it is not the initial test for confirming the identity of the organism.

Indole Test (Option B): While useful for differentiating Enterobacteriaceae, it does not specifically confirm Shigella.

Citrate Utilization Test (Option D): Shigella species are typically citrate-negative, but this test does not confirm the identity of Shigella.

Klebsiella, Proteus, Yersinia, Salmonella

1. A 60-year-old male presents with symptoms of a respiratory infection. A sputum sample is taken and cultured. The isolated organism shows pink, mucoid colonies on MacConkey agar. Biochemical tests are performed, and the organism is found to be urease-positive, indole-negative, and citrate-positive. Which of the following is the most likely causative organism?

A. Escherichia coli

B. Klebsiella pneumoniae

C. Proteus mirabilis

D. Enterobacter cloacae

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2. A microbiologist observes samples of bacteria and notes that the bacteria exhibit motility at 22°C but become non-motile at 37 °C. Additionally, the bacteria is better isolated when kept at 4°C, and weekly subcultures are done. Which of the following organisms is least likely to be in the sample?

A. Yersinia enterocolitica

B. Yersinia pseudotuberculosis

C. Listeria monocytogenes

D. Yersinia pestis

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3. Which statement is correct regarding the currently available vaccines for typhoid fever?

A. The Vi-CPS vaccine can be given to children under 2 years of age.

B. The Typhoral vaccine is given via intramuscular injection.

C. The Vi-rEPA vaccine can be administered to children under 2 years of age.

D. The Typhoral vaccine requires only a single dose for effective protection.

---

4. In the context of interpreting the Widal test results for typhoid fever, which of the following statements is not accurate?

A. The presence of high titers of O-antibodies generally indicates a later stage of infection.

B. A significant rise in H-antibody titers between acute and convalescent phases can confirm a diagnosis of typhoid fever.

C. The Widal test is most accurate when conducted within the first week of illness

D. False-positive results in the Widal test can occur due to prior vaccination or exposure to non-typhoidal Salmonella strains.

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5. A 30-year-old man presents to the emergency department with a 10-day history of high fever, headache, and abdominal discomfort. He reports feeling increasingly fatigued and has developed a dry cough. On physical examination, his temperature is 39.5°C (103.1°F), his pulse is 60 beats/min, and he has rose-colored spots on his abdomen. Considering the timing of his symptoms, which diagnostic method is most appropriate for confirming the diagnosis at this stage?

A. Blood culture

B. Antibody test

C. Stool culture

D. Urine culture

---

6. A patient who is working as a cook in a restaurant had symptoms such as prolonged fever, abdominal pain, and diarrhea 8 months ago, was diagnosed with typhoid fever and has been asymptomatic since then. Several customers who

dined at the restaurant recently developed similar symptoms. The cook in this scenario is an example of a:

A. Convalescent carrier

B. Temporary carrier

C. Chronic carrier

D. Acute carrier

---

7. A 25-year-old male presents with a 7-day history of high-grade fever, chills, and generalized weakness. The fever has been persistent and gradually increasing in intensity, reaching 39.5°C (103.1°F) and, often rising in the afternoon and evening. The patient also reports abdominal pain and has experienced several episodes of diarrhea, characterized by intermittent loose, greenish stools which started a few days after the onset of fever. On examination, pulse is 60/min, and BP is 110/70 mmHg. He appears ill with a coated tongue and mild hepatosplenomegaly. Based on the clinical scenario above, what is the most likely diagnosis for this patient?

A. Malaria

B. Typhoid Fever

C. Appendicitis

D. Dengue Fever

---

8. Given the various types of antigenic variations associated with Salmonella, which of the following statements is incorrect?

A. Phase variation involves flagellar antigens in which Phase 1 antigens are species-specific and Phase 2 antigens are non-specific and widely shared among different species.

B. V-W variation refers to the presence of a surface layer of Vi antigen that masks the O antigen, which can be lost after multiple subcultures.

C. H-0 variation is associated with the permanent loss of flagella when salmonellae are grown on agar containing phenol.

D. S-R variation is associated with a change in colony morphology and the loss of the O antigen and virulence.

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9. Which of the following statements about prophylaxis against Yersinia pestis is incorrect?

A. Sokhey's Modification of Haffkine Vaccine is a formalin-killed vaccine administered subcutaneously.

B. Chemoprophylaxis of Yersinia involves administration of Cotrimoxazole or Tetracycline orally for at least five days.

C. Sokhey's Modification of Haffkine Vaccine provides protection against all clinical types of plague.

D. The killed vaccine provides protection for up to 6 months.

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10. A 35-year-old man presents to the clinic with a high fever, chills, and painful lymphadenopathy in the inguinal region. He mentions recent travel to a rural area known for plague outbreaks. Suspecting Yersinia pestis, the physician orders various diagnostic tests, including biochemical reactions and serology. Which of the following findings would support the diagnosis of Yersinia pestis?

A. Positive catalase test, Positive oxidase test, and Passive hemagglutination against F1 antigen with a titer> 64 A in a single sample

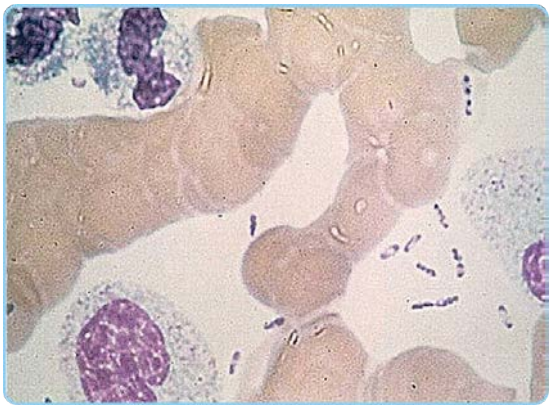
B. Negative catalase test, Positive oxidase test, Passive hemagglutination against F1 antigen with a titer> 64 A in a single sample

C. Positive catalase test, Negative oxidase test, and Passive hemagglutination against F1 antigen with a titer> 128 A in a single sample

D. Negative catalase test, Negative oxidase test, and Passive hemagglutination against F1 antigen with a titer> 128 A in a single sample

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11. A 29-year-old man presents to the emergency department with high fever, chills, and a painful swelling in his axillary region. He reports that he recently returned from a trip to Himachal Pradesh. On physical examination, a large, tender lymph node is noted in the axillary region. Aspirate when stained with methylene blue is shown in the image below. Which of the following organisms does not show a similar typical appearance as seen in the image?



A. Burkholderia pseudomallei

B. Vibrio parahaemolyticus

C. Clostridium tetani

D. Klebsiella granulomatis

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12. Match the following Klebsiella species with the diseases they commonly cause: 1. Klebsiella pneumoniae A. Granuloma inguinale 2. Klebsiella oxytoca B. Pneumonia 3. Klebsiella granulomatis C. Rhinoscleroma 4. Klebsiella rhinoscleromatis D. Hemorrhagic colitis

1. Klebsiella pneumoniae	A. Granuloma inguinale
2. Klebsiella oxytoca	B. Pneumonia
3. Klebsiella granulomatis	C. Rhinoscleroma
4. Klebsiella rhinoscleromatis	D. Hemorrhagic colitis

A. 1-B, 2-D, 3-A, 4-C

B. 1-C, 2-A, 3-D, 4-B

C. 1-D, 2-B, 3-C, 4-A

D. 1-A, 2-C, 3-B, 4-D

---

13. In a microbiological study, two different strains of Proteus are inoculated at separate points on the same agar plate containing a non-inhibitory medium. Upon observation, the areas of growth from each strain remain distinctly separate with a line of demarcation between them. What phenomenon is being demonstrated in this experiment?

A. Biofilm formation

B. Quellung reaction

C. Dienes phenomenon

D. Kirby-Bauer test

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14. Which of the following bacteria exhibits a characteristic pattern as shown below?



A. Proteus

B. Vibrio parahaemolyticus

C. Clostridium tetani

D. All of the above

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15. Which of the following statements is false?

A. All members of Tribe Proteeae test positive for the phenylalanine deaminase (PPA) test

B. Morganella tests positive for Urease test

C. Proteus lowers urine pH and contributes to the formation of bladder stones.

D. Providencia tests negative for Urease test

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16. A 35-year-old male patient, hospitalized for two weeks following a motor vehicle accident, presents with fever, chills, and difficulty breathing. He has a central venous catheter in place. Blood cultures grow a gram-negative, facultative anaerobic bacillus that produces a red pigment on nutrient agar. Which of the following is NOT a feature of this organism?

A. Organisms commonly found in healthcare facilities

B. Respiratory infection characterized by bloody sputum

C. Cause keratitis in contact lens wearers

D. Prodigiosin is a red pigment seen in culture

---

17. A 45-year-old female presents with a urinary tract infection. A urine sample is cultured on MacConkey agar. The colonies appear pink and mucoid. A string test is performed, and the colonies form a viscous string longer than 5 mm. Which of the following is the most likely causative organism?

A. Escherichia coli

B. Klebsiella pneumoniae

C. Pseudomonas aeruginosa

D. Proteus mirabilis

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18. Which of the following infections is not typically caused by the recent-hyperviscous, hypervirulent strain of Klebsiella pneumoniae?

A. Community acquired pneumonia

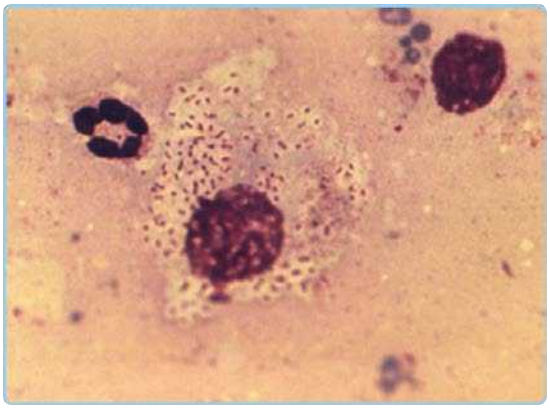
B. Meningitis

C. Gastroenteritis

D. Sepsis

---

19. A 35-year-old male presents with painless, progressive genital ulcers. A tissue biopsy is taken, and a Wright-Giemsa stain is performed. The microscopic examination reveals Donovan bodies within macrophages as shown below. What is the most likely diagnosis?



A. Syphilis

B. Gonorrhea

C. Chlamydia

D. Granuloma inguinale

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	3
Question 5	2
Question 6	2
Question 7	2
Question 8	3
Question 9	3
Question 10	3
Question 11	3
Question 12	1
Question 13	3
Question 14	4
Question 15	3
Question 16	2
Question 17	2
Question 18	3
Question 19	4

Solution for Question 1:

Correct Answer: B) Klebsiella pneumoniae

Explanation: The following tests are used to identify Klebsiella pneumoniae

- MacConkey Agar: Pink, mucoid colonies indicate lactose fermentation and a capsule-producing bacterium, both characteristic of Klebsiella pneumoniae.
- Urease-positive: Klebsiella pneumoniae produces urease, which breaks down urea into ammonia and carbon dioxide.
- Indole-negative: Klebsiella pneumoniae does not produce indole from tryptophan, differentiating it from Escherichia coli, which is indole-positive.
- Citrate-positive: Klebsiella pneumoniae can utilize citrate as its sole carbon source.

Laboratory diagnosis of Klebsiella pneumoniae:

- Gram staining: It is short, plump, straight capsulated gram-negative rods
- Culture: On MacConkey agar, it produces large dome-shaped mucoid (due to capsule) sticky, pink color, lactose fermenting colonies.
- Identification: Identification of K. pneumoniae from colonies is made either by automated identification systems such as MALDI-TOF or VITEK; or by conventional biochemical tests as described below It is catalase positive and oxidase negative ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.
- It is catalase positive and oxidase negative
- ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.
- It is catalase positive and oxidase negative
- ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.

Escherichia coli (Option A): Although it can appear pink on MacConkey agar due to lactose fermentation, E. coli is indole-positive and urease-negative, ruling it out.



Proteus mirabilis (Option C): Proteus species are typically non-lactose fermenting and would not appear pink on MacConkey agar. They also do not appear mucoid.



Enterobacter cloacae (Option D): While this organism is citrate-positive and a lactose fermenter, it generally lacks the mucoid, highly capsulated appearance of Klebsiella pneumoniae colonies and is urease-negative.



Solution for Question 2:

Correct Answer: D) Yersinia pestis

Explanation:

- Differential motility refers to the ability of bacteria to be motile at lower temperatures (22°C) and non-motile at higher temperatures (37°C).
- Cold enrichment refers to the process of enhanced isolation of certain bacteria at lower temperatures (4°C) where other bacteria fail to grow.

Organism	Differential Motility (22°C vs 37°C)	Cold Enrichment Isolation
Yersinia enterocolitica (Option A)	Motile at 22°C, Non-motile at 37°C	Yes
Yersinia pseudotuberculosis (Option B)	Motile at 22°C, Non-motile at 37°C	Yes
Listeria monocytogens (Option C)	Motile at 22°C, Non-motile at 37°C	Yes
Yersinia pestis	Non-motile	No

Solution for Question 3:

Correct Answer: C) The Vi-rEPA vaccine can be administered to children under 2 years of age.

Explanation:

- The Vi-rEPA vaccine can be administered to children under 2 years of age.
- The Vi-rEPA vaccine, which is the Vi antigen conjugated with recombinant Pseudomonas aeruginosa Exotoxin A, has increased immunogenicity due to the conjugation.
- This enhancement allows the vaccine to be effective in younger children, including those under 2 years of age.

Properties of Salmonella Vaccines:

Vaccine	Vi-CPS Vaccine	Typhoral Vaccine	Parenteral TAB Vaccine
Type	Vi antigen capsular polysaccharide	Oral live attenuated S. Typhi Ty21a	Heat-killed whole cell S. Typhi/S. Paratyphi A and B

Composition	Purified Vi capsular polysaccharide antigen from S. Typhi strain Ty2	Stable live attenuated mutant of S. Typhi strain Ty21a	Whole cell of S. Typhi and S. Paratyphi A and B, killed by heating at 50-60°C and preserved in 0.5% phenol.
Dosage	Single dose containing 25 µg of Vi antigen	4 doses taken orally on days 1, 3, 5, and 7	Two doses of 0.5 ml subcutaneously at an interval of 4-6 weeks; alternatively, 0.1 ml intradermally
Administration Route	Intramuscular (IM) or subcutaneous	Oral (enteric-coated capsules) (Option B)	Subcutaneous or intradermal
Age Indication	Only after 2 years of age (Option A)	Only after 6 years of age	Recommended for troops, medical and paramedical personnel in non-endemic areas; for all children in endemic areas
Booster	Every 2 years	Every 5 years with a full 4-dose series (Option D)	Not applicable
Duration of Protection	2 years	4 years	3 years
Mechanism of Action	Induces immune response to Vi antigen	Multiplies and initiates immune response but self-destructs	Induces immune response to whole bacterial cells
Special Considerations	Poorly immunogenic to children under 2 years	No antibiotics should be given during the vaccination period	Significant side effects led to discontinuation
Current Status	In use	In use	No longer in use due to side effects

The Vi-CPS vaccine can be given to children under 2 years of age (Option A): The Vi-CPS (Vi capsular polysaccharide) vaccine is not recommended for children under 2 years of age due to its lower immunogenicity in younger children.

The Typhoral vaccine is given via intramuscular injection (Option B): Typhoral (Ty21a), a live attenuated vaccine, is administered orally, not via injection.

The Typhoral vaccine requires only a single dose for effective protection (Option D): The Typhoral (Ty21a) vaccine requires multiple doses (typically 3 or 4 doses taken on alternating days) to achieve effective immunity.

Solution for Question 4:

Correct Answer: C) The Widal test is most accurate when conducted within the first week of illness due to the lag in antibody production.

Explanation:

- The Widal test is generally more reliable when performed after the first week of illness.
- During the early stage, antibody levels may be too low to detect, making the test less reliable.
- The antibodies usually become detectable around the second week of illness, so testing too early can lead to false negatives.

WIDAL Test: It is a highly sensitive but poorly specific serological test used to diagnose typhoid fever caused by Salmonella typhi.

- It detects specific antibodies in the patient's serum that react with the antigens of Salmonella typhi.
- The test is based on the agglutination reaction where patient serum is mixed with antigens of Salmonella typhi.
- If the antibodies are present, they will cause the antigens to clump together (agglutinate), indicating a positive result.
- The test primarily measures: O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi. H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.
- O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi.
- H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.
- O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi.
- H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.

Interpretation of WIDAL Test				
	TO	TH	AH	BH
S. Typhi	+	+	-	-
S. Paratyphi A	+	-	+	-
S. Paratyphi B	+	-	-	+

Early infection	+	-	-	-
Late infection	-	+	-	-
Vaccine	-	+	+	+
Where, TO : O Antigen, TH : S.Typhi H Antigen, AH : S.Paratyphi AH Antigen, BH : S. Paratyphi B H Antigen.				

Option A: High titers of O-antibodies are often associated with a more advanced stage of infection rather than the acute stage. In the early stages of typhoid fever, O-antibody levels may still be low. It is typically during the second or third week of illness that high O-antibody titers become detectable.

Option B: A significant rise in H-antibody titers between acute and convalescent phases can confirm a diagnosis of typhoid fever.

Option D: False positives can occur if the patient has been vaccinated against typhoid fever or has been exposed to non-typhoidal Salmonella strains, which can also produce cross-reacting antibodies that interfere with the test results.

Solution for Question 5:

Correct Answer: B) Antibody test

Explanation:

- Blood culture is most effective during the first week of the illness but becomes less reliable as the disease progresses■.
- The antibody test (Widal test) is most useful in the second week when the body has had time to develop antibodies against the Salmonella Typhi antigens.
- Stool culture becomes more reliable in the third to fifth weeks of the illness■.
- Urine culture is less commonly used and less reliable compared to the other methods■.

Diagnosis (mnemonic: BASU):

Week 1(B)	Blood culture

Week 2(A)	Antibody test (WIDAL)
Week 3(S)	Stool culture
Week 4(U)	Urine culture

Solution for Question 6:

Correct Answer: B) Temporary carrier

Explanation:

- The term "carrier" refers to an individual who continues to shed typhoid bacilli in feces. In this case, despite being asymptomatic for 8 months, the cook is still able to transmit the pathogen (Salmonella) to others, making them a temporary carrier.
- Temporary carriers are individuals who carry and transmit the pathogen for a period of more than 3 months but less than a year.

Convalescent carrier (Option A) refers to a person who has recovered from a disease but continues to shed the infectious agent for a period of 3 weeks to 3 months after recovery.

Chronic carriers (Option C) are individuals who continue to carry and shed the pathogen for a period of more than a year.

Acute carriers (Option D) are individuals who are currently infected with the pathogen and actively shedding it, typically during the acute phase of the illness when symptoms are present.

Solution for Question 7:

Correct Answer: B) Typhoid Fever

Explanation:

Based on the provided scenario, the clinical presentation is consistent with Typhoid fever, which is typically characterized by a high-grade fever that progressively worsens over several days. Other common symptoms include abdominal pain, diarrhea (often with greenish stools),

weakness, and hepatosplenomegaly. A coated tongue is also a characteristic sign.

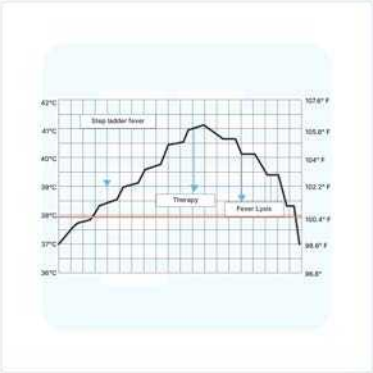
Typhoid fever is caused by Salmonella Typhi, as well as Salmonella Paratyphi A, B, and C.

Clinical features:

1. Fever Pattern

    • Stepwise Fever: Typhoid fever is often characterized by a "stepwise" or progressively increasing fever that rises a little each day, peaking in the afternoon or evening.

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    • Sustained High Fever: As the disease progresses, the fever becomes continuous, often reaching temperatures of 39–40°C.

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2. Abdominal Symptoms

    • Abdominal Pain: Patients often experience diffuse or vague abdominal pain.

    • Diarrhea or Constipation: Early in the illness, adults may experience constipation, while children more commonly present with diarrhea. Typhoid diarrhea is often described as “pea-soup” greenish stools.

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3. Relative Bradycardia

    • Despite high fever, patients may have a relatively lower pulse rate than expected, a phenomenon called "relative bradycardia." This is unusual in febrile illnesses and can be a

clue in diagnosing typhoid.

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4. Systemic Symptoms

- Fatigue and Weakness: Patients may feel very weak and fatigued, often feeling progressively worse over days.
- Headache: A common feature that can be severe, typically frontal or generalized.
- Myalgia: Muscle pain is common and contributes to the overall sense of malaise.

Fatigue and Weakness: Patients may feel very weak and fatigued, often feeling progressively worse over days.

Headache: A common feature that can be severe, typically frontal or generalized.

Myalgia: Muscle pain is common and contributes to the overall sense of malaise.

5. Gastrointestinal Findings

- Hepatosplenomegaly: Enlargement of the liver (hepatomegaly) and spleen (splenomegaly) can occur, typically palpable on physical examination.
- Coated Tongue: Patients often present with a dry, coated tongue, which may appear white or “furry.”

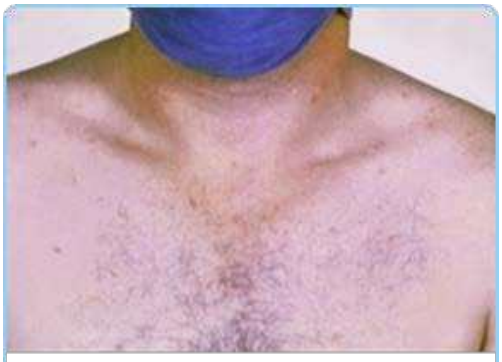
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Coated Tongue: Patients often present with a dry, coated tongue, which may appear white or “furry.”

6. Rose Spots

- Rash: In some cases, patients develop rose-colored maculopapular spots on the chest or abdomen. These “rose spots” appear for a short time, lasting about 2–3 days, and are easier to see on lighter skin.

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Rose spots on the chest

• Confusion or Altered Consciousness: In more severe cases, especially untreated, patients may show signs of delirium or altered mental status, sometimes termed “typhoid state” in severe cases.

Malaria (Option A) typically presents with high-grade fever, chills, and sweats, often in a cyclical pattern (every 48-72 hours) depending on the Plasmodium species.

Appendicitis (Option C) usually presents with acute onset of abdominal pain, starting around the umbilicus and later localizing to the right lower quadrant (McBurney's point). It may be accompanied by nausea, vomiting, and a low grade fever. Diarrhea is not a typical symptom of appendicitis, and the described fever pattern does not align with appendicitis.

Dengue (Option D) presents with high fever, severe headache, retro-orbital pain, myalgia, arthralgia, and sometimes a rash. Dengue often shows a biphasic fever pattern and can have signs of hemorrhage or plasma leakage in severe cases. Abdominal pain and diarrhea can occur, but they are not predominant symptoms.

Solution for Question 8:

Correct Answer: C) H-0 variation is associated with the permanent loss of flagella when salmonellae are grown on agar-containing phenol.

Explanation: H-0 variation in Salmonella is related to the loss of flagella which occurs when the bacteria are grown on agar containing phenol. It is a temporary phenotypic change and not a permanent loss; flagella can reappear when the strain is subcultured on media without phenol.

Antigenic Variati	Description	Mechanism

on		
H-0 Variation	Loss of flagella	Phenotypic and temporary change when grown on agar containing phenol.
Phase Variation	Alternation between two sets of flagellar antigens (Phase 1 and Phase 2)	Involves two sets of genes in the bacterial genome. Phase 1 antigens are species-specific or shared by a few species, while Phase 2 antigens are more widely shared. (Option A)
V-W Variation	Presence of Vi antigen (V form) and loss of Vi antigen (W form)	Fresh isolates carry Vi antigen masking the O antigen (V form). Vi antigen is lost after several subcultures, exposing the O antigen (W form). (Option B)
S-R Variation	Change from smooth (S) to rough (R) colonies	Loss of O antigen and virulence, often through mutation. (Option D)

Solution for Question 9:

• Sokhey's Modification of Haffkine Vaccine is a formalin-killed vaccine administered subcutaneously, and it provides protection specifically against bubonic plague for up to 6 months.

• It does not provide protection against other subtypes of plague, such as pneumonic or septicemic plague.

Prophylaxis against Yersinia pestis

Vaccine	Notes
Sokhey's Modification of Haffkine Vaccine	Formalin-killed vaccine, administered subcutaneously. Protects against bubonic plague. Provides protection for up to 6 months. (Option A and D)
Live vaccine	No longer recommended
Chemoprophylaxis	Cotrimoxazole/Tetracycline for 5 days (Option B)

Solution for Question 10:

Explanation:

Serology and Biochemical Reactions of Yersinia pestis

Biochemical Reactions	Fermentation [+ve acid production -ve gas production] - Glucose, Maltose, Mannitol Catalase Test - Positive Oxidase Test - Negative IMViC - + - -
Serology	Passive hemagglutination by antibody against F1 antigen Titre > 128 Å in a single sample (1:128)

Solution for Question 11:

Explanation:

- The clinical picture (high fever, chills, tender axillary lymph node) and morphological appearance on staining with new methylene blue (Waysons stain) points to the case of Bubonic plague, caused by Yersinia pestis.
- Yersinia pestis characteristically shows safety pin appearance.

Safety pin appearance is shown by the following organisms:

- Yersinia pestis
- Vibrio parahaemolyticus (Option B)
- Burkholderia mallei
- Burkholderia pseudomallei (Option A)
- Haemophilus ducreyi
- Klebsiella granulomatis (Option D)

Clostridium tetani is a Gram-positive, spore-forming anaerobic bacillus. It typically appears as a " drumstick" or "tennis racket" shape under the microscope due to the terminal spores.



Plague (Yersinia pestis)	
Endemic Foci	Kolar (Tamil Nadu), Beed Latur (Maharashtra) Rhoru (Himachal)
Reservoir	Wild rodents
Vector	Rat flea (Xenopsylla cheopis, Xenopsylla astia)
Types of Plague	Bubonic Plague: 2-7 days incubation period; Lymph node painful enlargement and fever Pneumonic Plague: 1-3 days incubation period; Respiratory symptoms; Associated with bio-terrorism Septicemic Plague: 2-7 days incubation period; A/k/a Black death; DIC, shock and multi-organ failure
Microscopic Findings	Safety pin appearance (Bipolar staining) using Wayson stain, Wright stain, Giemsa stain pestis Histology" data-author="U.S. Center for Disease Control" data-hash="10238" data-license="Open Access" data-source="https://commons.wikimedia.org/wiki/File:Yersinia_pestis_wayson.jpg" data-tags="April2025" height="295" src="https://image.prepladder.com/notes/7HfpCPt32y2qo2fUKt7o1744502315.png" width="400" /> {{caption_text}}

Culture	 ... (content truncated)
Biochemical Reactions	Fermentation [+ve acid production -ve gas production] Catalase positive Oxidase negative IMViC = - - + -
Vaccine	Recombinant R1F1v Vaccine: Pre-exposure Prophylaxis Sokhey's Modification of Haffkine Vaccine- Protects only against bubonic plague,
Treatment	Streptomycin Fluoroquinolones: For CNS and cardiovascular involvement

Solution for Question 12:

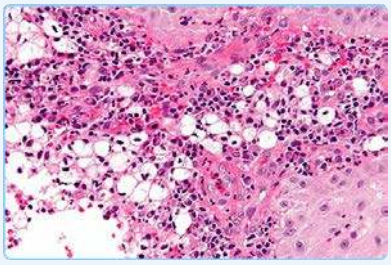
Correct Answer: A) 1-B, 2-D, 3-A, 4-C

Explanation:

- Klebsiella species: They are usually found as commensals in human intestines and as saprophytes in soil.
- It has three species—K. pneumoniae, K. oxytoca and K. granulomatis. K. pneumoniae further comprises three subspecies.

The diseases caused by them can be summarized as follows:

Klebsiella Species	Disease Caused	Symptoms

Klebsiella pneumoniae	Pneumonia	 Productive cough with "currant jelly" sputum, fever, chills, chest pain, difficulty breathing ... (content truncated)
Klebsiella oxytoca	Hemorrhagic colitis	Bloody diarrhea, abdominal pain and cramping, nausea, vomiting
Klebsiella granulomatis	Granuloma inguinale	 Painless, beefy-red genital or anal ulcers, tissue granulation, bleeding from lesions ... (content truncated)
Klebsiella rhinoscleromatis	Rhinoscleroma	 ... (content truncated)

Solution for Question 13:

Correct Answer: C) Dienes phenomenon

Explanation:

Dienes phenomenon helps in identifying strains of Proteus in different patients.

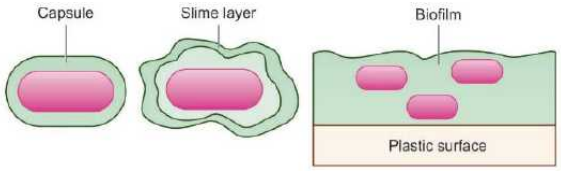
Two strains of Proteus bacteria inoculate at separate points on the agar plate and their interaction is noted.

- If the two strains of Proteus are identical or closely related, their growth patterns will exhibit a tendency to merge or coalesce.

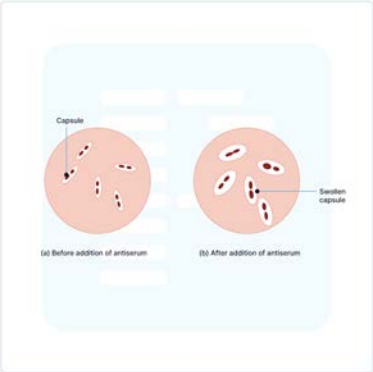
If the two strains are different or genetically distinct, their growth patterns will remain separate and distinctive line of demarcation seen on the agar plate.



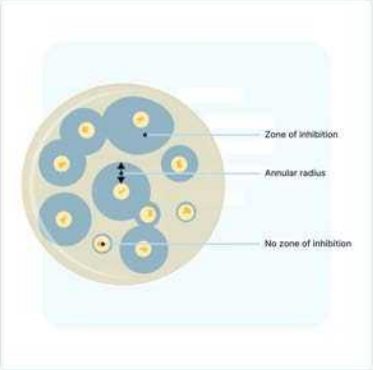
Biofilm formation (Option A): Biofilms refers to the ability of bacteria to adhere to surfaces and form slimy layers for survival.



Quellung reaction (Option B) Quellung reaction: This test involves the swelling of the bacterial capsule in the presence of specific antisera and is used to identify encapsulated bacteria like Streptococcus pneumoniae.



Kirby-Bauer test (Option D) Kirby-Bauer test is used to determine the antibiotic susceptibility of bacteria. Antibiotic discs are placed on an agar plate inoculated with bacteria, and the zone of inhibition around each disc indicates whether the bacteria are susceptible, intermediate, or resistant to the antibiotic.



Solution for Question 14:

Correct Answer: D) All of the above

Explanation:

The image above indicates swarming motility. Swarming motility is the ability of certain bacteria to move rapidly over a semi-solid surface, resulting in a characteristic concentric growth pattern.

- Swarming can be avoided by Increasing agar concentration(5-6% agar): makes it harder for motile bacteria to spread across the surface, thus reducing swarming. Normal agar

concentration is 2% Chemicals: like Chloral hydrate, sodium azide, alcohol, sulphonamides,surface-active agents, or boric acid inhibit bacterial swarming MacConkey agar: this containsbile salts like taurocholate, which are known to inhibit swarming.

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• MacConkey agar: this contains bile salts like taurocholate, which are known to inhibit swarming.

Most common bacteria showing swarming motility:

Other organisms that cause swarming motility are:

• Vibrio Alginolyticus

• Bacillus Cereus

• Serratia

• Salmonella

Solution for Question 15:

Correct Answer: C) Proteus lowers urine pH and contributes to the formation of bladder stones

Explanation:

• Proteus, a member of Tribe Proteeae, produces the enzyme urease, which breaks down urea and raises urine pH.

• This can contribute to the formation of bladder stones.

Tribe Proteeae classification:



Proteus	Urease positive Phenyl Pyruvic Acid (PPA) positive
Providencia	Urease negative PPA positive
Morganella	Urease positive PPA positive

All members of Tribe Proteeae test positive for the phenylalanine deaminase (PPA) test (Option A) Members of the Tribe Proteeae, which includes Proteus, Morganella, and Providencia, are positive for the phenylalanine deaminase test. This test differentiates them from other Enterobacteriaceae.

Morganella tests positive for the Urease test (Option B): This statement is true. Morganella species are urease-positive, meaning they produce urease, which hydrolyzes urea to ammonia and increases the pH.

Providencia tests negative for the Urease test (Option D): Providencia species generally test negative for urease, differentiating them from Proteus and Morganella in the Proteeae tribe.

Solution for Question 16:

Correct Answer: B) Respiratory infection characterized by bloody sputum

Explanation:

- The organism responsible for the patient's symptoms and the growth of a gram-negative, facultative anaerobic bacillus that produces a red pigment on nutrient agar is Serratia marcescens.
- It is responsible for the phenomenon known as "pseudohemoptysis" where sputum samples turn red, mimicking the appearance of blood in the sputum.
- Serratia marcescens emerged as a significant cause of infections in healthcare settings, particularly among patients with compromised immune systems. (Option A)

Causes:

- Pneumonia (Pseudohemoptysis- Red color sputum)
- Contact lens keratitis (Option C)
- Meningitis, Cardiovascular problems, Septicemia in immunocompromised

Culture:

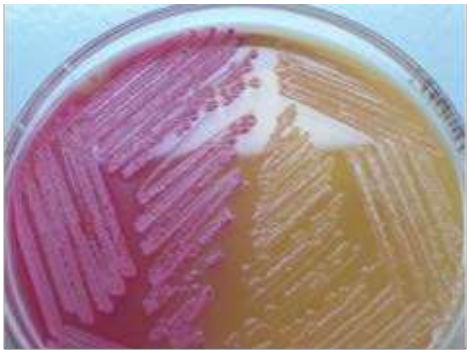
- Presents as a non-diffusible red pigment at room temperature.
- Prodigiosin is responsible for Pseudohemoptysis (red pigmented sputum) (Option D)

Solution for Question 17:

Correct Answer: B) Klebsiella pneumoniae

Explanation: The combination of pink, mucoid colonies on MacConkey agar and a positive string test strongly suggests that the causative organism is Klebsiella pneumoniae. The described characteristics of the bacterial culture help distinguish between different organisms:

- MacConkey Agar: This is a selective and differential medium that indicates lactose fermentation. Both Escherichia coli and Klebsiella pneumoniae ferment lactose, producing pink colonies.



MacConkey agar with LF and NLF colonies.

- Mucoid Colonies: Klebsiella pneumoniae is known for producing mucoid colonies due to its polysaccharide capsule, which is more prominent than that of Escherichia coli.
- String Test: The string test involves touching a colony with an inoculating loop and pulling it away slowly. If a string longer than 5 mm forms, it indicates a hyperviscous strain, which is characteristic of Klebsiella pneumoniae.



String test

Laboratory diagnosis of Klebsiella pneumoniae:

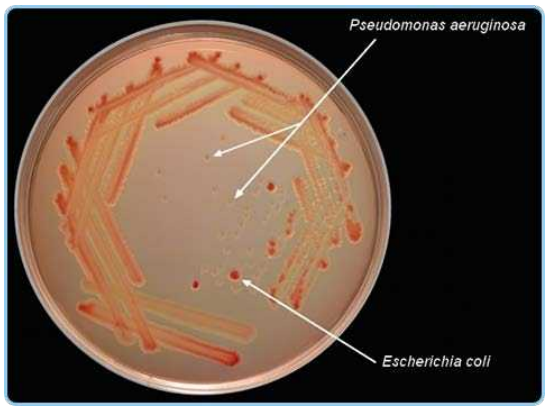
- Gram staining: It is short, plump, straight capsulated gram-negative rods
- Culture: On MacConkey agar, it produces large dome-shaped mucoid (due to capsule) sticky, pink color, lactose fermenting colonies
- Identification: Identification of K. pneumoniae from colonies is made either by automated identification systems such as MALDI-TOF or VITEK; or by conventional biochemical tests as described below It is catalase positive and oxidase negative ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.
- It is catalase positive and oxidase negative
- ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.
- It is catalase positive and oxidase negative
- ICUT tests: Indole test (negative), citrate test (positive), urease test (positive) and TSI (triple sugar iron agar) test shows acid/acid, gas present, H2S absent.

Thus, the combination of pink, mucoid colonies on MacConkey agar and a positive string test strongly suggests that the causative organism is Klebsiella pneumoniae.

Escherichia coli (Option A): While it also ferments lactose and may appear pink on MacConkey agar, it does not typically have a mucoid appearance or give a positive string test.



Pseudomonas aeruginosa (Option C): Non-lactose fermenting, typically appears colorless on MacConkey agar and does not have a mucoid appearance.



Proteus mirabilis (Option D): Non-lactose fermenting and does not appear mucoid. It typically exhibits swarming motility on non-selective media and does not give a positive string test.



Solution for Question 18:

Correct Answer: C) Gastroenteritis

Explanation: The recent strain of Klebsiella pneumoniae is known to cause community-acquired pneumonia, meningitis and sepsis. Hence gastroenteritis is the correct answer.

Hyper-viscous, hyper-virulent strain of Klebsiella pneumoniae:

- Hyperviscous, hypervirulent
- Resistant to treatment

Clinical manifestations:

- ■■Community-acquired pneumonia: Often accompanied by lung abscesses and necrotizing infection. This type of pneumonia can rapidly progress, leading to complications like empyema or sepsis. (Option A)


Bulging fissure sign

- Meningitis: Although less common than pneumonia or sepsis, hvKp can invade the central nervous system, leading to meningitis. Typical K. pneumoniae strains do not frequently cause meningitis, especially in otherwise healthy individuals. (Option B)
- Sepsis: Hypervirulent K. pneumoniae is frequently implicated in cases of bacteremia and sepsis. This strain can spread through the bloodstream, leading to a widespread systemic infection that can cause multiorgan failure, even in immunocompetent patients. (Option D)

Gastroenteritis: While K. pneumoniae may occasionally be isolated in cases of gastroenteritis, this is generally not due to the hypervirulent strain and is not a common infection associated with hvKp. Gastrointestinal infections are typically linked to pathogens like Escherichia coli, Salmonella, and Shigella, rather than Klebsiella species.

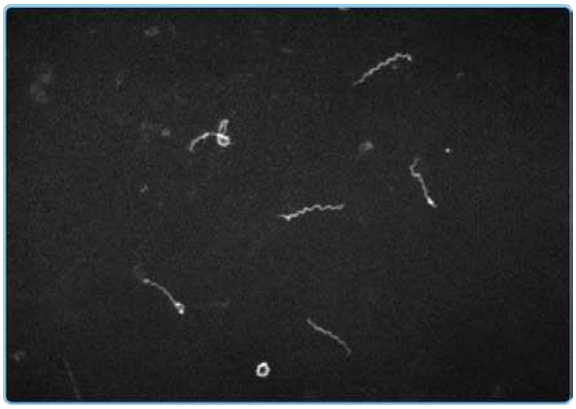
Solution for Question 19:

Correct Answer: D) Granuloma inguinale

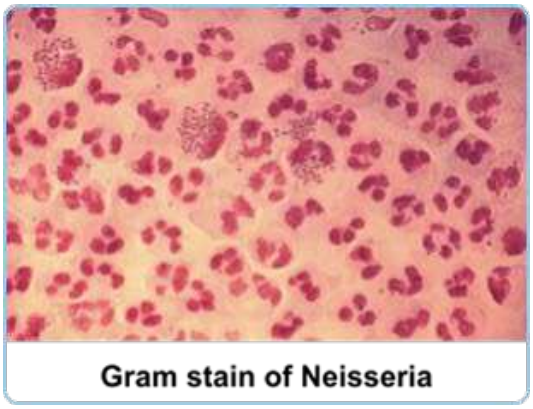
Explanation:

- The clinical presentation of painless, progressive genital ulcers, along with the presence of c on Wright-Giemsa stain, is characteristic of granuloma inguinale (donovanosis), caused by Klebsiella granulomatis.
- Donovan bodies are intracytoplasmic inclusions found in macrophages and are a diagnostic feature of this condition.

Syphilis (Option A): Typically presents with painless genital ulcers but is diagnosed by serologic tests and darkfield microscopy, not by finding Donovan bodies.



Gonorrhea (Option B): Generally presents with purulent discharge and painful urination; it does not produce Donovan bodies.



Gram stain of Neisseria

Chlamydia (Option C): Often asymptomatic or presents with discharge and dysuria, but also does not produce Donovan bodies.



The intracellular inclusion body represents the replication phase of Chlamydia, where, the reorganized reticulate body (RB) multiplies through binary fission into 100-500 new RBs, which mature into elementary bodies (EB).

Vibrio and non-fermenters

1. A 45-year-old fisherman presents with a painful wound infection following an injury sustained while handling seafood. Upon evaluation, a Gram-negative, salt-tolerant bacterium is identified. Which of the following organisms is not classified as a non-fermenting bacterium?

A. Pseudomonas

B. Acinetobacter baumannii

C. Burkholderia pseudomallei

D. Vibrio alginolyticus

---

2. A patient with watery diarrhea tests positive for Vibrio cholerae O1. Which of the following laboratory findings is useful in differentiating the El Tor biotype from the classical biotype of Vibrio cholerae?

A. The classical biotype is resistant to polymyxin B.

B. El Tor biotype shows negative chick erythrocyte agglutination..

C. The classical biotype does not cause hemolysis of sheep RBCs.

D. El Tor biotype is susceptible to group IV phage

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3. A 32-year-old man presents with severe, watery diarrhea resembling "rice-water stools" and signs of dehydration. Laboratory tests confirm a Vibrio cholerae infection. Which of the following best explains the mechanism by which cholera toxin induces watery diarrhea?

A. Causes direct intestinal mucosal damage, leading to inflammation and diarrhea

B. Activates adenylate cyclase, resulting in increased cyclic AMP levels

C. Increases cyclic GMP levels in intestinal cells

D. B subunit binds to GM1 ganglioside receptors on intestinal cells

---

4. A 34-year-old patient presents with severe, watery diarrhea and dehydration after consuming food while traveling in a region known for cholera outbreaks. He is initiated on oral rehydration solution (ORS). Which of the following diagnostic tests is the most specific for confirming an infection with Vibrio cholerae in this patient?

A. Stool culture on thiosulfate-citrate-bile salts-sucrose (TCBS) agar

B. Complement-dependent vibriocidal antibody test

C. Susceptibility to cholera phage IV

D. Stool culture on Cary-Blair medium

---

5. Which virulence factor of Pseudomonas aeruginosa is known to inhibit protein synthesis in host cells?

A. Elastase

B. Exotoxin A

C. Phospholipase C

D. Biofilm

---

6. Which biochemical test is most frequently employed to confirm the presence of Pseudomonas aeruginosa by detecting an enzyme characteristic of this organism?

A. Indole test

B. Urease test

C. Oxidase test

D. Coagulase test

---

7. A patient presents with pneumonia and septicemia after returning from a tropical region. Which organism is most likely responsible, and which specialized culture medium should be used for its isolation?

A. Burkholderia pseudomallei - Ashdown's medium

B. Salmonella typhi - Xylose lysine deoxycholate (XLD) agar

C. Leptospira interrogans - EMJH medium

D. Moraxella catarrhalis - Chocolate agar

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	2
Question 4	1
Question 5	3
Question 6	3
Question 7	1

Solution for Question 1:

Vibrio alginolyticus is a fermenting Gram-negative bacterium. It is capable of fermenting carbohydrates, particularly in saline environments, and is commonly associated with seafood and marine environments.

Non-Fermenting Bacteria

- Metabolism: Primarily oxidative metabolism, as these organisms do not ferment sugars.
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- Examples: Pseudomonas (Option A ruled out) Burkholderia pseudomallei (Option C ruled out) Burkholderia mallei Acinetobacter baumannii (Option B ruled out) Vibrio parahaemolyticus:
- Pseudomonas (Option A ruled out)
- Burkholderia pseudomallei (Option C ruled out)
- Burkholderia mallei
- Acinetobacter baumannii (Option B ruled out)
- Vibrio parahaemolyticus:
- Primarily oxidative metabolism, as these organisms do not ferment sugars.

- Pseudomonas (Option A ruled out)
- Burkholderia pseudomallei (Option C ruled out)
- Burkholderia mallei
- Acinetobacter baumannii (Option B ruled out)
- Vibrio parahaemolyticus:

Organisms showing bipolar safety pin appearance on microscopy:

- Klebsiella granulomatis
- Haemophilus ducreyi
- Burkholderia pseudomallei(causing melioidosis)
- Yersinia pestis
- Vibrio parahaemolyticus

Vibrio alginolyticus

- Characteristics: Halophilic (salt-loving) Capable of fermenting sucrose (Option D)
- Halophilic (salt-loving)
- Capable of fermenting sucrose (Option D)
- Clinical Associations: Linked to infections of the eyes, ears, and wounds in humans exposed to seawater.
- Linked to infections of the eyes, ears, and wounds in humans exposed to seawater.
- Halophilic (salt-loving)
- Capable of fermenting sucrose (Option D)
- Linked to infections of the eyes, ears, and wounds in humans exposed to seawater.

Solution for Question 2:

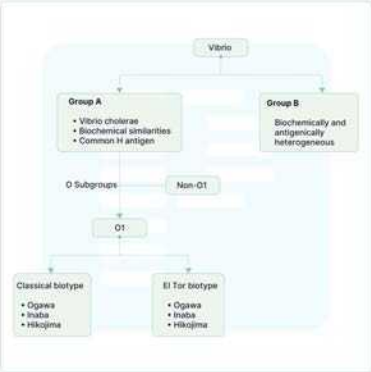
The El Tor biotype of Vibrio cholerae O1 shows hemolysis of sheep red blood cells (RBCs), while the classical biotype does not. Hemolysis of sheep RBCs is one of the distinguishing features of the El Tor biotype, which helps differentiate it from the classical biotype in the laboratory.

Vibrio classification:

Vibrio Classification	Halophilic Vibrio	Non-Halophilic Vibrio

Salt Requirement	Requires salt for growth	Cannot grow in salt
Examples	V. parahaemolyticus, V. alginolyticus, V. vulnificus	Vibrio cholerae

Gardner and Venkatraman introduced a serological classification:



Differences between classical cholera and El Tor vibrios:

Tests/Characteristics	Classical	El Tor
Tests:		
Hemolysis of sheep RBCs (Option C)	Negative	Positive
Voges-Proskauer	Negative	Positive
Chick erythrocyte agglutination (Option B ruled out)	Negative	Positive
Polymyxin B sensitivity (Option A ruled out)	Sensitive	Resistant
Group IV phage susceptibility (Option D ruled out)	Susceptible	Not susceptible
El Tor phage 5 susceptibility	Not susceptible	Susceptible
Others:		
Mortality	Higher	Lower
Carrier rate	Low	High

Solution for Question 3:

Cholera toxin binds to the GM1 ganglioside receptors on intestinal epithelial cells. Once internalized, the A subunit of the toxin activates adenylate cyclase, leading to an increase in cyclic AMP (cAMP) levels. Elevated cAMP disrupts ion transport, causing a significant efflux of chloride ions and water into the intestinal lumen, resulting in profuse watery diarrhea.

Pathogenesis

- Natural Host: Only humans.
- Transmission: Orally through contaminated water or food.
- Incubation Period: 1-5 days.

Mechanism of Action

- Mucus Crossing: Hemagglutinin-protease cleaves mucus and fibronectin.
- Epithelial Adhesion: Toxin coregulated pilus (TCP) helps the bacteria adhere to the epithelial surface without invading or damaging cells, remaining attached throughout the infection. (Option A ruled out)

Enterotoxin

- Structure: Consists of one A (active) subunit and five B (binding) subunits.
- Dissociation in Enterocytes: A subunit splits into A1 and A2 fragments. A1 Fragment Action: Activates cellular adenylate cyclase, leading to increased cyclic AMP (cAMP) levels. This results in an outflow of large quantities of water and electrolytes into the intestinal lumen, causing watery diarrhea. (Option B)
- A subunit splits into A1 and A2 fragments.
- A1 Fragment Action: Activates cellular adenylate cyclase, leading to increased cyclic AMP (cAMP) levels. This results in an outflow of large quantities of water and electrolytes into the intestinal lumen, causing watery diarrhea. (Option B)
- Receptor Binding: B subunits attach to GM1 ganglioside receptors on the surface of jejunal epithelial cells. (Option D ruled out)
- A subunit splits into A1 and A2 fragments.
- A1 Fragment Action: Activates cellular adenylate cyclase, leading to increased cyclic AMP (cAMP) levels. This results in an outflow of large quantities of water and electrolytes into the intestinal lumen, causing watery diarrhea. (Option B)

Clinical Features

- Stool Characteristics: "Rice water" stools with a mild sweet odor.

• Diarrhea Progression: Initially mild diarrhea and vomiting, rapidly progressing to sudden massive diarrhea.

• Electrolyte Imbalance: Hypokalemia, base-deficit acidosis.

• Shock: Due to fluid loss and electrolyte imbalance.

Complications

• Muscular Cramps

• Renal Failure

• Pulmonary Edema

• Cardiac Arrhythmias

• Paralytic Ileus

Increases cyclic GMP levels in intestinal cells (Option C) Heat-stable toxin (ST) of E.coli increases intracellular cGMP levels causing watery diarrhea.

Solution for Question 4:

TCBS agar is a selective culture medium specifically designed for isolating Vibrio species, including Vibrio cholerae. It allows for the growth of alkaline-tolerant bacteria and inhibits the growth of most other enteric bacteria. Colonies of Vibrio cholerae typically appear as yellow on this medium due to sucrose fermentation, making this method highly specific for confirming the diagnosis.

Diagnosis of Vibrio cholerae

Specimen Collection:

• Preferred Specimen: Stool (mucus flakes)

• Alternative: Rectal swab if stool is not available

• Enrichment broth : selenite F broth and alkaline peptone water

Transport Media:

• VR (Venkatraman Ramakrishnan) media - specific transport media for V. cholera (Option D ruled out)

• Cary-Blair medium

Microscopy:

• Morphology: Comma or curved-shaped, reddish Gram-negative bacilli.

• Appearance: Fish-in-stream appearance

• Motility: Demonstrated by dark field or phase contrast microscopy showing darting motility.

Culture:

• Culture Media and Colony Characteristics: MacConkey Agar: Colorless colonies that may later turn pink due to late fermentation of lactose Blood Agar: Green colonies indicating hemolysis Nutrient Agar: Translucent, round discs with a bluish tinge under transmitted light Gelatin Stab: Infundibuliform (funnel-shaped) or napiform (turnip-shaped) liquefaction

• MacConkey Agar: Colorless colonies that may later turn pink due to late fermentation of lactose

• Blood Agar: Green colonies indicating hemolysis

• Nutrient Agar: Translucent, round discs with a bluish tinge under transmitted light

• Gelatin Stab: Infundibuliform (funnel-shaped) or napiform (turnip-shaped) liquefaction

• MacConkey Agar: Colorless colonies that may later turn pink due to late fermentation of lactose

• Blood Agar: Green colonies indicating hemolysis

• Nutrient Agar: Translucent, round discs with a bluish tinge under transmitted light

• Gelatin Stab: Infundibuliform (funnel-shaped) or napiform (turnip-shaped) liquefaction

Selective Media:

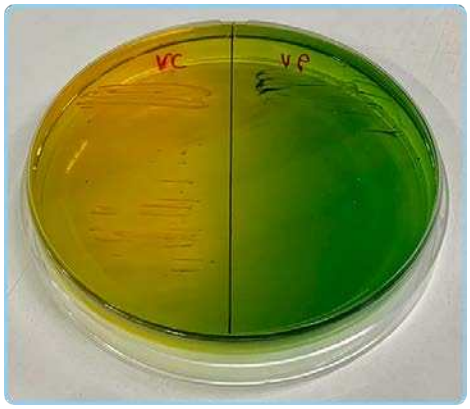
• Thiosulfate-Citrate-Bile Salts-Sucrose (TCBS) Agar: (Option A) Differentiates between sucrose fermenters (like Vibrio cholerae) and non-sucrose fermenters (such as Vibrio parahaemolyticus). Vibrio cholerae produces yellow colonies due to sucrose fermentation.

• Differentiates between sucrose fermenters (like Vibrio cholerae) and non-sucrose fermenters (such as Vibrio parahaemolyticus).

• Vibrio cholerae produces yellow colonies due to sucrose fermentation.

• Differentiates between sucrose fermenters (like Vibrio cholerae) and non-sucrose fermenters (such as Vibrio parahaemolyticus).

• Vibrio cholerae produces yellow colonies due to sucrose fermentation.



Biochemical Reactions:

- Fermentation: Glucose and sucrose
- Indole Test: Positive
- Nitrate Reduction: Nitrates are reduced to nitrites
- Cholera Red Reaction: Due to nitroso-indole formation
- Catalase: Positive
- Oxidase: Positive
- String Test: A loopful of growth mixed with 0.5% deoxycholate results in string formation
- A loopful of growth mixed with 0.5% deoxycholate results in string formation
- A loopful of growth mixed with 0.5% deoxycholate results in string formation

Serological Tests (Not useful for diagnosis)

- Indirect Hemagglutination
- Complement-Dependent Vibriocidal Antibody Test (Option B ruled out)

Treatment:

- Rehydration: Oral Rehydration Solution (ORS) Intravenous (IV) fluids
- Oral Rehydration Solution (ORS)
- Intravenous (IV) fluids
- Antibiotics: Secondary treatment
- Oral Rehydration Solution (ORS)
- Intravenous (IV) fluids

Susceptibility to Cholera Phage IV (Option C): Susceptibility to Cholera Phage IV differentiates between classical cholera and El Tor vibrios.

Solution for Question 5:

Exotoxin A is a potent virulence factor of Pseudomonas aeruginosa that inhibits protein synthesis in host cells. It does this by ADP-ribosylating elongation factor-2 (EF-2), which is crucial for the translation process during protein synthesis. By interfering with this process, Exotoxin A leads to cell death and contributes to the pathogenicity of the bacterium.

Pseudomonas Pigments: Pseudomonas aeruginosa produces several pigments that diffuse freely into their surroundings, inhibiting other bacteria and mediating tissue injury:

- Pyocyanin: Blue-green color
- Pyoverdine: Yellow-green color, fluorescent
- Pyorubin: Red-brown color
- Pyomelanin: Brown-black color

Virulence Factors: Pseudomonas aeruginosa has multiple virulence factors that contribute to its pathogenicity:

- Exotoxin A: Inhibits protein synthesis by ADP-ribosylating elongation factor-2, leading to cell death.
- Inhibits protein synthesis by ADP-ribosylating elongation factor-2, leading to cell death.
- Elastases (LasA and LasB): Degrade elastin, collagen, and other host tissues, facilitating spread and invasion.
- Degrade elastin, collagen, and other host tissues, facilitating spread and invasion.
- Alkaline Protease: Degrades host proteins, aiding in tissue invasion and immune evasion.
- Degrades host proteins, aiding in tissue invasion and immune evasion.
- Phospholipase C: Degrades phospholipids in host cell membranes, contributing to cell lysis and tissue damage.
- Degrades phospholipids in host cell membranes, contributing to cell lysis and tissue damage.
- Type III Secretion System (T3SS): Injects toxins directly into host cells, disrupting cellular processes and evading immune responses.
- Injects toxins directly into host cells, disrupting cellular processes and evading immune responses.
- Biofilm Formation: Protects bacteria from antibiotics and host immune responses, promoting chronic infection and persistence.

• Protects bacteria from antibiotics and host immune responses, promoting chronic infection and persistence.

• Non-Diffusible Toxins: Exotoxins S, U, T, and Y allow the bacteria to evade phagocytic cells and induce tissue injuries.

• Exotoxins S, U, T, and Y allow the bacteria to evade phagocytic cells and induce tissue injuries.

• Inhibits protein synthesis by ADP-ribosylating elongation factor-2, leading to cell death.

• Degrade elastin, collagen, and other host tissues, facilitating spread and invasion.

• Degrades host proteins, aiding in tissue invasion and immune evasion.

• Degrades phospholipids in host cell membranes, contributing to cell lysis and tissue damage.

• Injects toxins directly into host cells, disrupting cellular processes and evading immune responses.

• Protects bacteria from antibiotics and host immune responses, promoting chronic infection and persistence.

• Exotoxins S, U, T, and Y allow the bacteria to evade phagocytic cells and induce tissue injuries.

Clinical Features: Pseudomonas aeruginosa can cause a variety of infections, with the following clinical features based on infection type and affected populations:

Infection Type	Affected Population	Symptoms and Clinical Features
Pulmonary Infections	Cystic fibrosis patients, individuals with chronic lung diseases	Persistent cough, sputum production, shortness of breath, fever, chest pain, difficulty breathing
Urinary Tract Infections (UTIs)	Catheterized patients, individuals with anatomical abnormalities	Dysuria, frequency, urgency, cloudy or foul-smelling urine, lower abdominal pain
Skin and Soft Tissue Infections	Burn patients, individuals with wounds	Redness, swelling, pus formation, pain, delayed wound healing
Bacteremia and Septicemia	Immunocompromised patients	Fever, chills, hypotension, multi-organ failure, altered mental status
Otitis Externa (Swimmer's Ear)	Swimmers, individuals with ear injuries	Ear pain, itching, discharge, hearing loss, redness, swelling of the outer ear
Endocarditis	Intravenous drug users, individuals with prosthetic heart valves	Fever, heart murmur, fatigue, chest pain, signs of heart failure
Gastrointestinal Infections	Immunocompromised individuals, children	Diarrhea, abdominal pain, fever, nausea, vomiting

Keratitis and Conjunctivitis	Contact lens wearers, individuals with eye trauma	Eye pain, redness, discharge, blurred vision, potential vision loss

Other features associated with P.aeruginosa:

- Hot tub folliculitis
- Shanghai fever
- Ecthyma gangrenosum

Elastase (Option A):

- Elastase is a proteolytic enzyme produced by Pseudomonas aeruginosa that breaks down elastin and other proteins in host tissues.
- While it contributes to tissue damage and inflammation, it does not directly inhibit protein synthesis in host cells.

Phospholipase C (Option C):

- Phospholipase C is an enzyme that hydrolyzes phospholipids, leading to the production of lysophospholipids and fatty acids.
- While it can contribute to tissue damage and inflammatory responses, it does not inhibit protein synthesis in host cells.

Biofilm (Option D):

- Biofilm formation is a survival strategy that allows Pseudomonas aeruginosa to adhere to surfaces and resist host immune responses and antibiotic treatment.
- Although biofilms enhance the virulence of the bacteria by providing a protective environment, they do not directly inhibit protein synthesis in host cells.

Solution for Question 6:

The oxidase test is used to detect the presence of cytochrome c oxidase, an enzyme that is a key characteristic of Pseudomonas aeruginosa. This organism is oxidase-positive, making this test a reliable method for confirming its identification.

Diagnostic Tests for Pseudomonas aeruginosa:

Diagnostic Method	Description	Specific Features for Pseudomonas aeruginosa

Sample Collecti on	Collect specimens such as sputum, urine, wound swabs, blood, ear discharge, eye discharge, or cerebrospinal fluid (CSF).	Specimens vary based on the site of infection.
Microscopy	Gram stain	Gram-negative rods; motile with flagella.
Culture	Growth on common laboratory media:	
Blood agar	Produces beta-hemolytic gray, moist colonies.	
MacConkey agar	Produces pale, non-lactose fermenting colonies.	
Nutrient agar	Produces large, irregular colonies with a metallic sheen.	
Peptone water	Forms uniform turbidity with surface pellicle.	
Cetrimide Agar	Selective medium that inhibits other Gram-negative bacteria; specifically selected for Pseudomonas aeruginosa; pigment production (e.g., pyocyanin).	
	King’s medium	used to increase pigment production by pseudomonas
Biochemical Tests	Various biochemical tests to identify specific metabolic properties:	
Oxidase test	Positive	
Catalase test	Positive	
ICUT Test:		
Indole test	Negative	
Citrate test	Positive	
Urease test	Negative	
Triple sugar test	Shows alkaline slant with no gas and hydrogen sulfide.	
Molecular Methods	PCR (Polymerase Chain Reaction), 16S rRNA sequencing	Provides specific and sensitive detection of Pseudomonas aeruginosa DNA.
Antimicrobial Susceptibility Testing	Disk diffusion method, Minimum Inhibitory Concentration (MIC) testing	Determines the susceptibility of Pseudomonas aeruginosa to various antibiotics.

Indole test (Option A): This test is used to identify bacteria that can produce indole from tryptophan, but it is not typically used for identifying Pseudomonas aeruginosa.

Urease test (Option B): This test detects the presence of urease enzyme, which breaks down urea into ammonia and carbon dioxide. Pseudomonas aeruginosa is typically urease negative.

Coagulase test (Option D): This test is used to identify Staphylococcus aureus based on the production of coagulase enzyme and is not relevant for Pseudomonas aeruginosa.

Solution for Question 7:

Burkholderia pseudomallei is the causative agent of melioidosis, which can present with pneumonia and septicemia, particularly in individuals returning from tropical regions, especially Southeast Asia. Ashdown's medium is a selective culture medium used specifically for isolating B. pseudomallei from clinical specimens. It inhibits the growth of contaminating flora while promoting the growth of this organism.

Characteristics of Burkholderia Species

Feature	Burkholderia pseudomallei	Burkholderia mallei	Burkholderia cepacia
Disease	Melioidosis	Glanders	Burkholderia cepacia complex (BCC) infections
Habitat	Soil and water in tropical and subtropical regions	Primarily found in equines (horses, donkeys, mules)	Soil, water, and moist environments
Transmission	Inhalation, ingestion, or through skin abrasions from contaminated soil or water	Direct contact with infected animals or contaminated materials	Inhalation, ingestion, or skin contact with contaminated surfaces
Clinical Features	Pneumonia, bronchitis, abscesses in various organs, septicemia	Ulceration of skin and mucous membranes, lymphadenitis, pneumonia	Respiratory infections, chronic lung disease, urinary tract infections, septicemia, opportunistic infections in cystic fibrosis patients, nosocomial pathogens
Virulence Factors	Type III secretion system, capsule, lipopolysaccharides (LPS)	Type III secretion system, capsule, lipopolysaccharides (LPS)	Biofilm formation, lipopolysaccharides (LPS), elastase

Diagnostic Methods	Culture on Ashdown's agar, Gram stain (Gram-negative rods), serological tests (e.g., ELISA), PCR	Latex agglutination test, Gram stain (Gram-negative rods), serological tests (e.g., ELISA), PCR	Culture (e.g., blood, sputum), Gram stain (Gram-negative rods), biochemical tests, PCR
Treatment	Ceftazidime, carbapenems, long-term antibiotic therapy due to relapse potential	Doxycycline, ceftazidime, combination therapy (e.g., doxycycline + ceftazidime)	Cotrimoxazole, ceftazidime, meropenem
Extra Features	Considered a potential bioterrorism agent, endemic in Southeast Asia and Northern Australia	Considered a potential bioterrorism agent, primarily affects equines but can infect humans	Part of the Burkholderia cepacia complex (BCC), major concern for cystic fibrosis patients

Salmonella typhi (Option B): Xylose lysine deoxycholate (XLD) agar: Salmonella typhi, causing typhoid fever, is typically isolated on XLD agar, which inhibits the growth of Gram-positive organisms and aids in the differentiation of Salmonella species.

Leptospira interrogans (Option C): EMJH medium: Leptospira interrogans, causing leptospirosis, is best cultured in EMJH (Ellinghausen-McCullough-Johnson-Harris) medium under specific conditions suitable for the growth of spirochetes.

Moraxella catarrhalis (Option D): Chocolate agar: Moraxella catarrhalis, causing respiratory infections, is cultured on chocolate agar, which provides enriched growth conditions.

Hemophillus, Bordatella, and Brucella

1. Which of the following organisms is not oxidase positive?

A. Aeromonas hydrophila

B. Helicobacter pylori

C. Moraxella catarrhalis

D. Escherichia coli

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2. A 3-year-old unvaccinated child presents with high fever, irritability, and neck stiffness. Gram stain of CSF reveals small gram-negative coccobacilli. Which of the following statements regarding the suspected pathogen and its management is correct?

A. Requires X and V factors; type b capsule most virulent in young children

B. PCR of bexA gene confirms diagnosis; culture on chocolate agar is unnecessary

C. Non-typeable strains' lipooligosaccharide is more potent than type b capsule

D. Third-generation cephalosporins are contraindicated; use chloramphenicol

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3. Gram stain of a painful genital ulcer exudate shows gram-negative coccobacilli in a "school of fish" arrangement. Which statement about the suspected pathogen is correct?

A. Requires X factor but not V factor; grows on vancomycin-supplemented chocolate agar

B. "Pushing test" is diagnostic; further laboratory confirmation unnecessary

C. Multiplex PCR is less sensitive than culture for diagnosis

D. Single-dose azithromycin; partner treatment optional if asymptomatic

4. Which of the following organisms is not typically considered a commensal of the human oral cavity?

A. Aggregatibacter actinomycetemcomitans

B. Cardiobacterium hominis

C. Moraxella catarrhalis

D. Eikenella corrodens

5. A 6-month-old unvaccinated infant presents with paroxysmal cough followed by whooping inspiration and post-tussive vomiting for the past 2 weeks. Which of the following is the most sensitive and specific method for confirming the diagnosis in this case?

A. Gram staining of nasopharyngeal swab

B. Culture on Bordet-Gengou medium

C. Polymerase Chain Reaction (PCR) of nasopharyngeal aspirate

D. Serological testing for anti-pertussis toxin IgG

6. A 35-year-old veterinarian presents with intermittent fever, sweating, and joint pain. Blood cultures are performed, but the initial results are negative. Further laboratory tests show a significant titer of 640 in the Standard Agglutination Test (SAT). Which of the following characteristics is least likely associated with the causative organism?

A. Non-motile, aerobic, Gram-negative coccobacilli

B. Urease positive

C. Ferments carbohydrates

D. Destroyed by heat at 60°C in 10 minutes

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	1
Question 4	3
Question 5	3
Question 6	3

Solution for Question 1:

Correct Answer: D) Escherichia coli

Explanation:

- Escherichia coli (Enterobacteriaceae family) is a Gram-negative rod that is oxidase-negative.
- The oxidase test is used to identify bacteria that produce cytochrome c oxidase, an enzyme involved in the electron transport chain.
- E. coli does not produce this enzyme and, therefore, does not turn the oxidase reagent purple, indicating a negative result.

Oxidase-positive organisms-

Organism	Morphology	Associated Diseases
Neisseria	Gram-negative diplococci	Gonorrhea, Meningitis
Pseudomonas	Gram-negative rods	Opportunistic infections, Pneumonia, UTI
Vibrio	Gram-negative comma-shaped rods	Cholera, Gastroenteritis

Moraxella (Option C ruled out)	Gram-negative diplococci	Respiratory infections (e.g., otitis media, sinusitis)
Campylobacter	Gram-negative curved rods	Gastroenteritis (Campylobacter jejuni)
Aeromonas (Option A ruled out)	Gram-negative rods	Wound infections, Gastroenteritis
Helicobacter (Option B ruled out)	Gram-negative spiral rods	Peptic ulcers, Gastritis (H. pylori)
Legionella	Gram-negative rods	Legionnaires' disease
Brucella	Gram-negative coccobacilli	Brucellosis
Haemophilus	Gram-negative coccobacilli	Pneumonia, Otitis media, Epiglottitis, Meningitis, Sinusitis (H. Influenzae) Chancroid (H. Ducreyi)
Bordetella	Gram-negative coccobacilli	Pertussis

All Enterobacteriaceae are oxidase-negative and catalase-positive, except Plesiomonas.

Solution for Question 2:

Correct Answer A) Requires X and V factors; type b capsule most virulent in young children.

Explanation:

- Haemophilus influenzae requires both X (hemin) and V (NAD) factors for growth.
- The type b capsular polysaccharide (polyribosylribitol phosphate or PRP) is the most virulent serotype, particularly in unvaccinated children under 5 years old, often causing invasive diseases like meningitis.

Clinical features:

- Meningitis: Fever, irritability, neck stiffness (as described in the question)
- Other invasive diseases: Epiglottitis, pneumonia, septic arthritis
- Nontypeable strains often cause mucosal infections like otitis media, sinusitis

Diagnosis:

- CSF analysis in meningitis: elevated protein, low glucose, pleocytosis

- Gram stain: small gram-negative coccobacilli
- Culture on chocolate agar
- Antigen detection tests (latex agglutination) for rapid diagnosis
- PCR for definitive identification and serotyping

Treatment:

- Empiric therapy: Third-generation cephalosporins (ceftriaxone or cefotaxime)
- Alternative: Chloramphenicol (if cephalosporin allergy)
- Duration: Usually 10 days for meningitis

PCR of bexA gene confirms diagnosis; culture on chocolate agar is unnecessary (Option B):

- While PCR for the bexA gene can detect H. influenzae type b, culture remains crucial for antibiotic susceptibility testing.
- Moreover, chocolate agar (which provides X and V factors) is the preferred medium for culturing H. influenzae.

Non-typeable strains' lipooligosaccharide is more potent than type b capsule (Option C):

- The capsule of type b strains is actually more potent in inducing meningeal inflammation than the lipooligosaccharide of non-typeable strains.
- The capsule is a major virulence factor in invasive disease.

Third-generation cephalosporins are contraindicated; use chloramphenicol (Option D):

- Third-generation cephalosporins (like ceftriaxone) are actually the first-line treatment for H. influenzae meningitis, even in β-lactamase-producing strains.
- Chloramphenicol is no longer considered first-line due to resistance concerns and potential toxicity.

Solution for Question 3:

Correct Answer: A) Requires X factor but not V factor; grows on vancomycin-supplemented chocolate agar

Explanation:

H. ducreyi requires X factor (hemin) but not V factor. Chocolate agar supplemented with vancomycin is the optimal culture medium, providing necessary nutrients and inhibiting normal

flora.

Features of H. ducreyi:

- Gram-negative coccobacillus
- Fastidious, requiring X factor but not V factor
- Forms characteristic "school of fish" or "railroad track" arrangements in Gram stains
- Safety pin appearance of bipolar staining.

Clinical features of chancroid:

- Painful genital ulcers with ragged edges and a purulent base
- Painful inguinal lymphadenopathy (buboes)
- Incubation period of 3-10 days

Diagnosis:

Treatment:

- Azithromycin 1g orally single dose, OR
- Ceftriaxone 250mg IM single dose, OR
- Ciprofloxacin 500mg orally twice daily for 3 days, OR
- Erythromycin 500mg orally three times daily for 7 days
- Treatment of sexual partners is crucial

"Pushing test" is diagnostic; further laboratory confirmation unnecessary (Option B):

- The "pushing test" is suggestive but not diagnostic for chancroid. Laboratory confirmation is still necessary.
- The push test is a method used to identify Haemophilus ducreyi colonies on an agar surface by gently pushing them with a toothpick:
- H. ducreyi colonies are cohesive and remain intact when pushed across the agar surface.
- The colonies are typically translucent to semi-opaque, yellow-gray in color, and pinpoint in size.

Multiplex PCR is less sensitive than culture for diagnosis (Option C): Multiplex PCR is more sensitive than culture for H. ducreyi diagnosis, not less.

Single-dose azithromycin; partner treatment optional if asymptomatic (Option D): While single-dose azithromycin is a recommended treatment, partner treatment is necessary regardless of symptoms.

Solution for Question 4:

Correct Answer: C) Moraxella catarrhalis

Explanation:

- Moraxella catarrhalis is not typically considered an oral commensal.
- It is primarily found in the upper respiratory tract, particularly the nasopharynx, but is not a typical member of the oral microbiota. It is not part of the HACEK group.
- Understanding the normal microbiota of different body sites is crucial for interpreting culture results and identifying potential pathogens.
- While HACEK organisms are oral commensals, they can cause opportunistic infections, particularly endocarditis.

The HACEK group consists of:

- Haemophilus species (particularly H. parainfluenzae)
- Aggregatibacter species (including A. actinomycetemcomitans) (Option A ruled out)
- Cardiobacterium hominis (Option B ruled out)
- Eikenella corrodens (Option D ruled out)
- Kingella kingae

Body Site	Predominant Commensal Organisms
Skin	Staphylococcus epidermidis, Propionibacterium acnes
Nose	Staphylococcus aureus, Staphylococcus epidermidis
Oropharynx	Viridans streptococci, Neisseria species
Dental Plaque	Streptococcus mutans, Streptococcus sanguinis
Colon	Bacteroides fragilis, Escherichia coli
Vagina	Lactobacillus species, Gardnerella vaginalis, Group B Streptococcus
Urethra	Staphylococcus epidermidis, Corynebacterium species

Solution for Question 5:

Correct Answer: C) Polymerase Chain Reaction (PCR) of nasopharyngeal aspirate

Explanation:

- PCR of nasopharyngeal aspirate is currently the most sensitive and specific method for diagnosing pertussis, especially in the later stages of the disease.
- It can detect even small numbers of organisms and provides rapid results.

Bordetella features:

- Small, aerobic, gram-negative coccobacilli
- B. pertussis is the main cause of whooping cough
- Other species include B. parapertussis, B. bronchiseptica

Virulence factors:

- Pertussis toxin (increase insulin release; hypoglycemia)
- Filamentous hemagglutinin (binds bacteria to epithelial cells)
- Pertactin
- Adenylate cyclase toxin (increases intracellular cAMP)

Clinical features of pertussis:

1. Catarrhal stage: Mild upper respiratory symptoms

2. Paroxysmal stage: Characteristic paroxysmal cough with inspiratory whoop and post-tussive vomiting

3. Convalescent stage: Gradual recovery

Treatment:

- Macrolides (azithromycin, clarithromycin) are the first-line antibiotics
- Early treatment (within 1-2 weeks of symptom onset) can reduce severity and duration
- Supportive care is crucial, especially for infants

Prevention:

- DTaP vaccine for infants and children
- Tdap booster for adolescents and adults

Gram staining of nasopharyngeal swab (Option A):

- Gram staining is not reliable for diagnosing pertussis.
- B. pertussis is a small gram-negative coccobacillus that is difficult to visualize and differentiate from other respiratory flora.

Culture on Bordet-Gengou medium (Option B): While culture on Bordet-Gengou medium is specific, it has low sensitivity, especially after the first 2 weeks of illness or if antibiotics have been started. It also takes 7-10 days for results.

Serological testing for anti-pertussis toxin IgG (Option D): Serological testing for anti-pertussis toxin IgG can be useful for late diagnosis or epidemiological studies, but it's not ideal for acute diagnosis, especially in infants who may not have mounted a significant antibody response yet.

Solution for Question 6:

Correct Answer: C) Ferments carbohydrates

Explanation:

- Brucella does not typically ferment carbohydrates.
- Therefore, it is the characteristic least likely associated with Brucellae.

Brucella features:

Brucella melitensis is the most pathogenic species for humans.

Characteristics of Brucella:

Morphology	Biochemical Properties	Growth & Susceptibility
Small, non-motile, aerobic, Gram-negative coccobacilli (Option A ruled out)	Catalase positive	Do not grow well on standard laboratory media
Non-capsulated and non-spore forming	Oxidase positive (except B. neotomae and B. ovis)	Destroyed by heat at 60°C in 10 minutes (Option D ruled out)
Often exhibit a bipolar staining appearance	Urease positive (Option B ruled out)	Killed by 1% phenol in 15 minutes
	Utilize oxygen but do not typically ferment carbohydrates	Killed by pasteurization

Zoonotic Nature: Primarily affects goats, sheep, cattle, buffaloes, and pigs. Humans can become infected through contact with infected animals or consumption of contaminated animal products

Clinical Presentation in Humans:

- Symptoms: Intermittent fever, loss of appetite, nausea, muscular and articular pains, fatigue, sweating, and joint pain

- Also known as Undulant fever, Mediterranean fever, Maltese fever
- Diagnosis is challenging due to nonspecific symptoms and relies on high clinical suspicion

Diagnosis of Brucellosis:

Laboratory Testing Methods:

- Blood Culture: Most definitive diagnostic method Requires trypticase soy broth or brucella broth in biphasic blood culture bottles (e.g., Castaneda method) Large blood volume (5 ml) recommended due to low bacterial load Incubation at 37°C under 5-10% CO2 for at least 6-8 weeks Positive in approximately 30-50% of cases Bone marrow cultures may have a higher yield
- Most definitive diagnostic method
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- Large blood volume (5 ml) recommended due to low bacterial load
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- Serological Tests: Used when culture methods are unsuccessful
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- Positive in approximately 30-50% of cases
- Bone marrow cultures may have a higher yield
- Used when culture methods are unsuccessful

Serological Test Types:

Test	Description	Key Points
Standard Agglutination Test (SAT)	Tube agglutination test using serial dilutions of patient's serum and killed B. abortus antigen	Significant titer: 160 or more. Acute brucellosis titer: 640 or more within 3-4 weeks. Prozone phenomenon common. False positives with cholera, tularemia, yersinia infection, or immunization

Complement Fixation Test	Detects IgG antibodies	Valuable in chronic cases
ELISA (Enzyme-Linked Immunosorbent Assay)	Distinguishes between IgM and IgG antibodies	Useful for differentiating acute from chronic infections
Rapid Tests	Includes rapid dipstick tests and Rose Bengal card test	Quick results. Can be used for initial screening

Treatment of Brucellosis:

• Antibiotics: Standard treatment for adults: Doxycycline for 45 days and intramuscular streptomycin daily for the first two weeks Children: Cotrimoxazole with rifampicin or gentamycin

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Syphilis, Chlamydia and Rickettsiae

1. Which of the following tests is most specific for confirming a diagnosis of syphilis?

A. VDRL

B. TP-PA

C. RPR

D. EIA

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2. Which of the following statements correctly describes the two forms of Chlamydia?

A. The Elementary Body (EB) is the replicative form, while the Reticulate Body (RB) is the infectious form.

B. The Elementary Body (EB) is the infectious form that enters host cells, while the Reticulate Body (RB) is the replicative form.

C. The Elementary Body (EB) and Reticulate Body (RB) are extracellular forms.

D. The Reticulate Body (RB) is the metabolically inactive form.

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3. Which of the following statements is true regarding Ehrlichiosis?

A. It primarily affects the monocytes

B. It primarily affects the granulocytes

C. Transmitted by mites

D. Aminoglycosides are used for treatment

---

4. A 30 year old female patient presented with high fever, severe headache, myalgia and anemia on returning from Brazil recently. A suspicion of Oroya fever was made. What is the causative organism and vector involved in this disease?

A. Bartonella henselae and cat scratch/bite

B. Bartonella quintana and louse

C. Bartonella bacilliformis and sandfly

D. Ehrlichia chaffeensis and tick

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5. A veterinarian who had been working in an animal research facility came was suspected to have Q fever. What is the false statement regarding the disease?

A. The infective organism shows affinity to the vascular endothelium

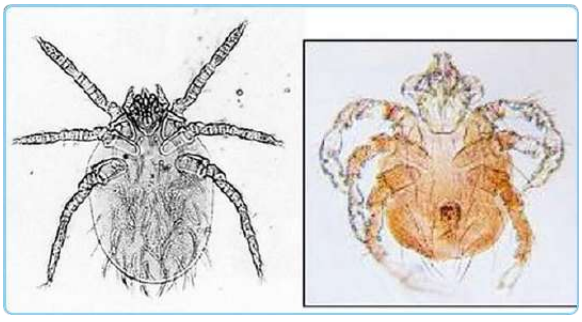
B. Produces donut shaped granulomas in histopathology

C. Chronic infection can present as infective endocarditis

D. Transmission is by inhalation of aerosols

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6. A hiker had recently returned from a forest expedition with complaints of high grade fever along with a painless blackish crusted lesion at the site of insect bite.The following insect was identified by him. What will a Weil-Felix test show for this patient?



A. Only OX 19+

B. OX 2+ and OX 19+

C. OX K and OX 19+

D. Only OX K+

---

7. A 35 year old male patient was admitted in the medicine ward with complaints of high grade fever, maculopapular rash and a skin lesion at the site of insect bite as shown below. He gives a recent history of hiking in a scrub forest. What is the false statement regarding the probable disease?



A. Causative organism is Orientia tsutsugamushi

B. Maculopapular rash involves the palms and soles

C. Transovarial transmission of the causative organism occurs through chiggers

D. The drug of choice is doxycycline

---

8. A middle aged female patient presented with high grade fever, maculo-papular rashes primarily affecting the trunk extending to the extremities sparing the palms and soles. She also mentions that there are a lot of rats in her area. A Weil-Felix test showed positive OX 19 and OX 2 antibodies. What is the probable diagnosis?

A. Bubonic Plague

B. Epidemic typhus

C. Endemic typhus

D. Hymenolepiasis

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9. During a medical camp at a newly established refugee centre, a medical officer observes several patients presenting with high-grade fever, headache, and macular rashes primarily affecting the trunk and extending to the extremities. He also learns about a widespread louse infestation within the centre. What might be the diagnosis and the likely causative agent in this situation?

A. Endemic typhus and Rickettsia typhi

B. Epidemic typhus and Rickettsia prowazekii

C. Rickettsial pox and Rickettsia akari

D. Rocky Mountain spotted fever and Rickettsia rickettsia

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10. Match the following diseases with their respective vectors. Disease Vector 1. Scrub typhus a) Hard Tick 2. Epidemic typhus b) Trombiculid mite 3. Rocky Mountain Spotted Fever c) Rat flea 4. Endemic typhus d) Louse

Disease	Vector
1. Scrub typhus	a) Hard Tick
2. Epidemic typhus	b) Trombiculid mite
3. Rocky Mountain Spotted Fever	c) Rat flea
4. Endemic typhus	d) Louse

A. 1-b, 2-d, 3-c, 4-a

B. 1-b, 2-d, 3-a, 4-c

C. 1-a, 2-c, 3-b, 4-d

D. 1-a, 2-b, 3-d, 4-c

11. Which of the following rickettsial bacteria specifically target granulocytes in their pathogenesis?

A. Anaplasma phagocytophilum

B. Orientia tsutsugamushi

C. Ehrlichia chaffeensis

D. Bartonella henselae

12. Which of the following statements is true regarding Chlamydia pneumoniae and Chlamydia psittaci?

A. Both are primarily transmitted through respiratory droplets from person to person.

B. Chlamydia psittaci causes infectious atherosclerosis.

C. C.pneumoniae has only TWAR strain.

D. The incubation period for Chlamydia pneumoniae is longer than that of Chlamydia psittaci.

13. Which of the following statements about Lymphogranuloma Venereum (LGV) is true?

A. LGV is caused by serotypes A, B, C of Chlamydia trachomatis.

B. The primary lesion in LGV is typically a painful genital ulcer.

C. Doxycycline is the first-line treatment for LGV.

D. LGV is primarily transmitted through blood transfusions.

14. Which of the following serotypes of Chlamydia trachomatis is primarily associated with causing trachoma?

A. D, E, F, G, H, I, J, K

B. A, B, Ba, C

C. L1, L2, L3

D. G, H, I

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15. Match the following species of Treponema with the respective diseases they cause. Treponema pallidum I. Pinta Treponema endemicum II. Venereal syphillis Treponema pertenue III. Yaws Treponema carateum IV. Endemic syphillis (Bejel)

Treponema pallidum	I. Pinta
Treponema endemicum	II. Venereal syphillis
Treponema pertenue	III. Yaws
Treponema carateum	IV. Endemic syphillis (Bejel)

A. 1:I; 2:II; 3:III; 4:IV;

B. 1:II; 2:III; 3:I; 4:IV;

C. 1:I; 2:III; 3:II; 4:IV;

D. 1:II; 2:IV; 3:III; 4:II;

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16. Which of the following features is false regarding Chlamydia?

A. It is an obligate intracellular organism

B. Cannot be cultured

C. Cell wall rich in peptidoglycan

D. Basophilic bacterial inclusions

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17. What is the preferred diagnostic test for confirming a case of leptospirosis?

A. Cold Agglutination Test

B. Standard Agglutination Test

C. Dark-Field Microscopy

D. Microscopic Agglutination Test (MAT)

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18. Which of the following statements on leptospirosis is false?

A. Zoonotic disease

B. Weil’s disease is icteric illness

C. Caused by Leptospira interrogans

D. Majority develop into Weil’s disease

---

19. Match the Borellia species with its mode of transmission Species Transmission 1.Borrelia vincentii a.Direct contact 2.Borrelia recurrentis b.Tick-borne 3.Borrelia hermsii c.Louse-borne

Species	Transmission
1.Borrelia vincentii	a.Direct contact
2.Borrelia recurrentis	b.Tick-borne
3.Borrelia hermsii	c.Louse-borne

A. 1b,2c,3a

B. 1a,2c,3b

C. 1a,2b,3c

D. 1c,2a,3b

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20. Which of the following is the pathogen responsible for Lyme disease?

A. Borrelia garinii

B. Borrelia afzelii

C. Borrelia recurrentis

D. Borrelia burgdorferi

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21. Which of the following statements on the Rapid Plasma Reagin test is false?

A. Uses the VDRL antigen

B. Results are easily seen with the naked eye

C. Unheated serum or plasma can be used

D. Can be used for testing CSF

---

22. Which of the following statements on the VDRL test is false?

A. Uses cardiolipin as antigen

B. Detects reagin antibody

C. Can be false positive in leprosy

D. Reaction is observed under naked eye

---

23. What is the appearance of Treponema pallidum in dark field microscopy?

A. Slender

B. Spiral

C. Slow movement

D. All of the above

---

24. A 2-year-old infant is brought to the pediatric clinic for evaluation of persistent nasal congestion and a rash. On examination, the infant has a bloody nasal discharge, a diffuse rash with scaly lesions on the trunk, and notable dental abnormalities, including peg-shaped incisors. Which of the following findings is the most characteristic feature to identify the disease?

A. Respiratory distress with wheezing

B. Hutchinson's teeth and bone abnormalities

C. Microcephaly and developmental delay

D. Hyperbilirubinemia and hepatosplenomegaly

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25. A 30-year-old female presents to the clinic with complaints of a widespread rash and flu-like symptoms that have been present for the past week. She reports that the rash started on her trunk and has now spread to her arms and legs, along with lesions on her oral mucosa. She also mentions experiencing mild sore throat and fatigue. She has a history of multiple sexual partners in the past year. What is the most likely diagnosis?

A. Genital herpes

B. Secondary syphilis

C. Psoriasis

D. Drug reaction

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26. A 45-year-old man presents to the clinic with complaints of progressive weakness and difficulty walking over the past few months. He has a history of untreated syphilis 15 years ago, which he believed had resolved. On examination, the patient exhibits significant degeneration in his right knee. Neurological examination reveals decreased proprioception and vibratory sensation in the lower extremities. Which diagnostic test would be most useful in confirming the suspicion of tertiary syphilis in this patient?

A. Polymerase Chain Reaction (PCR) for Treponema pallidum

B. CSF Venereal Disease Research Laboratory (VDRL) test

C. Lumbar puncture for cerebrospinal fluid (CSF) analysis

D. Rapid Plasma Reagin (RPR) test

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27. A 28-year-old male presents to the clinic with painless ulcers on his genital region that have been present for about three weeks as shown below. He denies any associated symptoms such as fever, malaise, or lymphadenopathy. His sexual history reveals multiple partners in the past year. What is the most likely diagnosis based on the clinical presentation?



A. Genital herpes

B. Primary syphilis

C. Chancroid

D. Lymphogranuloma venereum

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	3
Question 5	1
Question 6	4
Question 7	2
Question 8	2,3
Question 9	2
Question 10	2
Question 11	1
Question 12	3
Question 13	3
Question 14	2
Question 15	4
Question 16	3
Question 17	4
Question 18	4
Question 19	2
Question 20	4
Question 21	4
Question 22	4

Question 23	4
Question 24	2
Question 25	2
Question 26	2
Question 27	2

Solution for Question 1:

TP-PA is a treponemal test that detects antibodies specific to Treponema pallidum, making it highly specific and ideal for confirming syphilis.

Diagnosis is often made through clinical examination of the chancre and serological tests that detect antibodies to Treponema pallidum.

Serological tests:

- Non-Treponemal Tests: VDRL (Venereal Disease Research Laboratory) and RPR (Rapid Plasma Reagin): These tests are used for screening. They measure nonspecific antibodies that may indicate syphilis but can also be positive in other conditions (like pregnancy, lupus, or infections).
- VDRL (Venereal Disease Research Laboratory) and RPR (Rapid Plasma Reagin): These tests are used for screening. They measure nonspecific antibodies that may indicate syphilis but can also be positive in other conditions (like pregnancy, lupus, or infections).
- Treponemal Tests: TP-PA (Treponema pallidum particle agglutination), EIA (Enzyme Immunoassays), and FTA-ABS (Fluorescent Treponemal Antibody Absorption): These tests are more specific to syphilis and confirm the diagnosis. They detect antibodies specific to the syphilis bacterium.
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syphilis and confirm the diagnosis. They detect antibodies specific to the syphilis bacterium.

• CSF Analysis

• Darkfield Microscopy: In the presence of a chancre or other lesions, a sample may be taken to observe the organism directly under a microscope, which can confirm syphilis.

• Treatment in this stage typically involves a course of antibiotics, most commonly penicillin. Benzathine penicillin G is the recommended treatment. A single intramuscular injection (2.4 million units) is typically sufficient.

• Alternative for Penicillin Allergies: Doxycycline (100 mg orally twice daily for 14 days) or tetracycline can be used for those allergic to penicillin, but penicillin is preferred.

• For pregnant patients allergic to penicillin, desensitization may be considered as it is the only effective proven treatment effective for preventing transmission to the fetus.

Early diagnosis and treatment of primary syphilis are vital. If left untreated, the infection can progress to secondary, latent, and tertiary stages.

Solution for Question 2:

Chlamydia has two main forms in its life cycle and a unique replication process:

	Elementary Body (EB)	Reticulate Body (RB)
Infectious form	The small, extracellular, and metabolically inactive form.	The larger, intracellular, and metabolically active form.
Role	It enters the host cell by endocytosis.	Once inside the host cell, the EB transforms into the RB.
Characteristics	It is designed to survive outside the host cell and initiate infection.	RBs are responsible for replication within a host cell.

Replication Process:

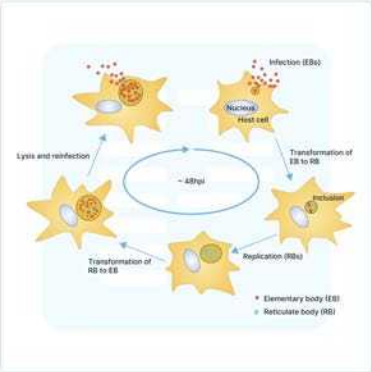
• Attachment and Entry: The EB attaches to a host cell and enters through endocytosis.

• Transformation: Inside the host cell, the EB converts into an RB.

• Replication: The RB replicates by binary fission within a membrane-bound vacuole.

• Maturation and Release: After replication, RBs reorganize back into EBs. The host cell eventually releases EBs (either by lysis or extrusion), which can go on to infect new cells.

• This biphasic life cycle helps Chlamydia evade host defenses and sustain infection.

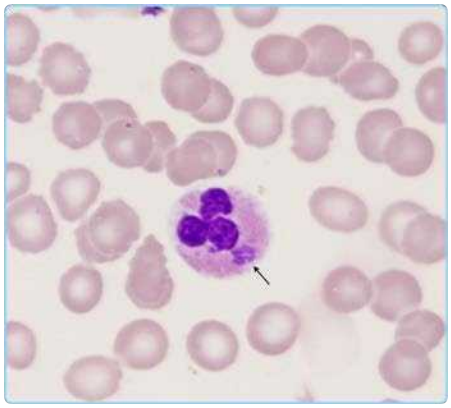


Solution for Question 3:

• Ehrlichiosis is a disease caused by Ehrlichia chaffeensis transmitted by ticks (Lone star tick). They primarily affect the monocytes. (Option B and C ruled out)

• They cause an acute febrile disease characterised by headache, myalgia, arthralgia, cough, pharyngitis, lymphadenopathy, diarrhoea, nausea, vomiting, abdominal pain and changes in mental status.

• They show inclusion bodies in the phagosome and multiply to produce the following three stages of growth— elementary body, initial body, and mulberry-like inclusions called Morula.



• The drug of choice for ehrlichiosis is doxycycline. (Option D ruled out)

Solution for Question 4:

The causative organism and vector involved in Oroya fever are Bartonella bacilliformis and sandfly respectively.

Oroya fever refers to infection with Bartonella bacilliformis presenting as an acute hemolytic bacteremia.It is due to intravascular hemolysis of erythrocytes infected with B. bacilliformis. It is predominantly seen in South American countries.Symptoms include fever, malaise, myalgia, headache, tachycardia, jaundice, and hepatomegaly

The chronic form of this infection is referred to as Carrion disease. It is a chronic vasoproliferative disease resulting in the appearance of cutaneous nodular angioproliferative lesions called as verruga.

Bartonella henselae and cat scratch/bite (Option A): It causes cat scratch disease in the immunocompetent and causes bacillary angiomatosis in the immunocompromised.



Bartonella quintana and louse (Option B): It causes Trench fever and not Oroya fever.

Ehrlichia chaffeensis and tick (Option D): It causes Ehrlichiosis and not Oroya fever.

Solution for Question 5:

The false statement regarding Q fever is that the infective organism shows affinity to the vascular endothelium instead of to the reticuloendothelial system especially pulmonary alveolar macrophages. (Option A)

Q fever	
Causative agent	Coxiella burnetii (Zoonosis transmitted from animals to humans)

Mode of spread	By inhalation of aerosols from infected animals or their products. (Option D is ruled out) Other modes include consumption of unpasteurized milk and ingestion of other infected animals or their products. High risk groups include farmers, veterinarians, and workers involved in slaughtering. High-risk areas include animal research facilities that do not have adequate personnel protection.
Pathophysiology	C.burnetii shows affinity to the reticuloendothelial system, especially the alveolar macrophages. They are known to grow at low pH and even resist lysosomal enzymes. In its growth cycle Coxiella includes a form that is resistant to drying and other environmental conditions much like a bacterial spore(but not actual spores as in Bacillus or Clostridium species). These spores are inhaled as aerosols. They are known to produce donut shaped granuloma/ fibrin ring granuloma in histopathology sections. (Option B is ruled out)
Clinical features	It usually begins at an average of 20 days after inhalation with abrupt onset of fever, chills, and headache without rash. A mild, dry, hacking cough and patchy interstitial pneumonia may occur as like in atypical pneumonia. Hepatosplenomegaly and abnormal liver function tests are common. Complications such as myocarditis, pericarditis, and encephalitis are rare. Chronic infection is also rare, but particularly important when it takes the form of culture negative infective endocarditis. (Option C is ruled out)
Investigations	The diagnosis of Q fever is usually made by demonstrating high or rising titers of antibodies to the Q fever antigen by enzyme immunoassay. Single positive blood culture for Coxiella burnetii or phase I IgG antibody titer of >1:800 to diagnose infective endocarditis. Molecular techniques such as PCR for C.burnetii DNA.
Treatment	Acute Q fever: Doxycycline is the drug of choice; given for 14 days. Quinolones are also effective. Chronic Q fever: Hydroxychloroquine is added to alkalinize the phagolysosome and to render doxycycline bactericidal against the organism. Vaccines have been shown to have a protective effect for heavily exposed workers. (Q-vax in Australia)

Solution for Question 6:

The given clinical scenario of a hiker who had recently returned from a forest expedition with complaints of high grade fever along with a painless blackish crusted lesion at the site of insect bite(black eschar) with an image showing a chigger (larva of trombiculid mite) is suggestive of scrub typhus.



Black Eschar

The Weil-Felix reaction seen in the case of scrub typhus is only OX K+ antibodies.

- Weil–Felix Test is a heterophile agglutination test that works on the principle of rickettsial antibodies showing antigenic cross reactivity with Proteus antigens(OX 19, OX 2 and OX K strains).
- It is a tube agglutination test where serial dilutions of a patient's serum are treated with non-motile strains of P. vulgaris OX 19 and OX 2 and P. mirabilis OX K.

Disease	Agglutination Test Result
Epidemic and endemic typhus	Mainly agglutinates with OX19; sometimes with OX2 (Option A ruled out)
Tick-borne Spotted Fever	Antibodies to both OX19 and OX2 elevated.(Option B ruled out)
Scrub Typhus	Raised antibodies to OXK(Option D)
Rickettsial Pox, Q Fever, Ehrlichiosis, Bartonellosis	Negative test

.

- False-positive titer - Due to underlying Proteus infection. Hence, fourfold rise of antibody titer in paired sera is better than a single high titer.

• False-negative titer - Due to excess antibodies in patient's sera (prozone phenomenon). This can be obviated by testing with serial dilutions of the patient's serum.

OX K and OX 19+ (Option C): This combination is not seen in any rickettsial infection.

Solution for Question 7:

• The given clinical scenario of a 35 year old male patient with complaints of high grade fever, maculopapular rash and a black eschar at the site of insect bite with a recent history of hiking in a scrub forest is suggestive of Scrub typhus.

• The false statement regarding scrub typhus is that the rash involves the palms and soles.(Option B)

• Rash involving the palms and soles are typically seen in the following diseases: Rocky Mountain spotted fever (Rickettsia rickettsii) Hand-foot-mouth disease (CoxSackie A virus) Secondary syphilis

• Rocky Mountain spotted fever (Rickettsia rickettsii)

• Hand-foot-mouth disease (CoxSackie A virus)

• Secondary syphilis

• Rocky Mountain spotted fever (Rickettsia rickettsii)

• Hand-foot-mouth disease (CoxSackie A virus)

• Secondary syphilis

Scrub typhus	
Causative agent	Orientia tsutsugamushi - It differs from Rickettsia both genetically and by its cell wall composition i.e. it lacks lipopolysaccharide layer. (Option A ruled out)
Vector	Chiggers (larvae of trombiculid mite)
Pathophysiology	Scrub typhus is so named because it occurs in scrub vegetation i.e. low-lying trees and bushes. The bite of infected trombiculid mites transmits it. Among all stages of mite, the larvae (called chiggers) are the only stage that feed on humans. Hence, scrub typhus is also called chiggerosis. Mites can maintain the organisms through transovarial transmission. (Option C ruled out) Orientia tsutsugamushi infects mononuclear cells but still produces fever and rash.

Clinical features	The classic presentation of scrub typhus consists of triad of an eschar (at the site of bite), regional lymphadenopathy and maculopapular rash sparing palms and soles. Non-specific manifestations may appear early such as fever, headache, myalgia, cough, and gastrointestinal symptoms. Complications such as encephalitis and interstitial pneumonia may occur rarely in the late stage (due to vascular injury).
Investigations	The Weil-Felix test shows OX K+ only.
Treatment	Doxycycline is the drug choice. (Option D ruled out) Azithromycin is given alternatively and the drug of choice in children and pregnant women.

Solution for Question 8:

The given clinical scenario of a middle-aged female patient with high-grade fever, maculopapular rashes primarily affecting the trunk sparing the palms and soles, a lot of rats in her area, and a Weil-Felix test showing positive OX 19 and OX 2 antibodies is suggestive of endemic typhus.

Endemic typhus	
Causative agent	Rickettsia typhi (It is endemic worldwide, especially in warm coastal areas having high rat infestations).
Vector	Rat flea (Xenopsylla cheopis)
Pathophysiology	It is transmitted by rubbing or scratching on the skin or inhalation of flea's dried faeces, less frequently by the flea bite. The incubation period is 1–2 weeks. They also show tropism for vascular endothelium. Focal areas of endothelial proliferation cause the rash and petechial lesions, perivascular infiltration leading to thrombosis and leakage of red blood cells into the surrounding tissues.
Clinical features	Symptoms are similar to epidemic typhus but milder and rarely fatal. Fever, headache, and rash begin 1 to 2 weeks after the bite. Rash involves the trunk and primarily spreads to the extremities, sparing the palms and soles.

Investigations	The diagnostic test of choice is serology i.e. rise in antibody titre. Weil-Felix test showing both Ox19+ and Ox2+.
Treatment	Doxycycline is the drug of choice.

Neil-Mooser (or tunica) reaction occurs when a male guinea pig inoculated intraperitoneally with blood from a patient infected with R. typhi develops fever and scrotal swelling, with the testes becoming fixed due to adhesions. This reaction is absent with R. prowazekii infections.

Bubonic plague (Option A):

- It is a zoonotic disease caused by Yersinia pestis transmitted by rat flea(Xenopsylla cheopis) resulting in enlarged and tender lymphadenopathy.
- Pneumonic and septicemic forms are more systemic variants of plague.
- Rashes and Weil-felix test positivity are not seen in plague.

Epidemic plague (Option B):

- It is a rickettsial disease caused by Rickettsia prowazekii resulting in a clinical picture and Weil-Felix test(Ox19 and Ox2 positive) similar to endemic typhus but more severe.
- The vector involved in epidemic typhus is louse whereas in endemic typhus is rat flea.

Hymenolepiasis (Option D):

- Hymenolepiasis is a protozoan infection caused by consumption of contaminated food and water with Hymenolepis nana eggs.
- At times accidental ingestion of rat flea containing the cysticercoid larvae which subsequently develop into adult worms causes the disease.
- They are usually asymptomatic and rarely patients develop symptoms like anorexia, abdominal pain, and mucus diarrhoea.

Solution for Question 9:

The given clinical scenario of a newly established refugee centre with several patients presenting with high-grade fever, headache, and macular rashes primarily affecting the trunk with widespread louse infestation suggests a diagnosis of epidemic typhus and the causative agent is Rickettsia prowazekii.

Epidemic typhus	
Causative agent	Rickettsia prowazekii (It is the only rickettsial disease that can occur as an epidemic)
Vector	Louse
Pathophysiology	It occurs during conditions favourable to human body lice infestation such as in refugee and famine camps.(crowding, infrequent bathing) The cycle starts with R.prowazekii circulating in a patient's blood during an acute febrile infection. After 5 to 10 days of incubation, large numbers of rickettsiae appear in its infective faeces. Since the louse defecates when it feeds, the organisms can be rubbed into the louse bite wounds when the host scratches the site. Dried louse faeces are also infectious through the mucous membranes of the eye or respiratory tract. The louse dies of its infection in 1 to 3 weeks. Most of Rickettsiae, including Rickettsia prowazekii, shows tropism for vascular endothelium. They multiply in the endothelial cells, leading to increased vascular permeability, hypovolemia, and hypotension. Focal areas of endothelial proliferation cause the rash and petechial lesions, perivascular infiltration that leads to thrombosis, and leakage of red blood cells into the surrounding tissues.
Clinical features	Fever, headache, and rash begin 1 to 2 weeks after the bite. In the majority of cases, a maculopapular rash appears first on the trunk and then spreads centrifugally to the extremities, sparing the palms and soles. Eschar is absent at the site of entry. Headache, malaise, and myalgia are prominent components of the illness. Complications include myocarditis and central nervous system dysfunction. In untreated disease, the fatality rate increases with age as much as 60%.
Investigations	The diagnostic test of choice is serology i.e. rise in antibody titre. Weil-Felix test showing both OX 19+ and OX 2+.
Treatment	Treatment with doxycycline is effective. Louse control is the best means of prevention and is particularly important in controlling epidemics. No effective vaccine is available.

Endemic typhus and Rickettsia typhi (Option A):

• Although the clinical features and Weil Felix test are similar to those of epidemic typhus, it does not occur as an epidemic.

• Another key point of endemic typhus is that the vector involved is rat flea, not lice.

Rickettsial pox and Rickettsia akari (Option C):

• It is a benign rickettsial illness caused by Rickettsia akari and transmitted by a rodent mite.

• Distinguishing features of the disease include an eschar at the site of the bite, later followed by fever and a vesicular rash distributed randomly sparing the palms and soles.

Rocky Mountain spotted fever (RMSF) and Rickettsia rickettsia(Option D):

• RMSF is an acute febrile illness associated with residential and recreational exposure to wooded areas where infected ticks exist.

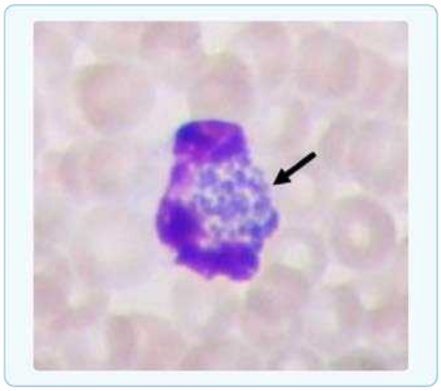
Unlike other rickettsial diseases, the rash here spreads first from palms and soles, then from the extremities to the trunk. As transovarial transmission occurs in ticks, it is essentially a parasite of ticks.

Solution for Question 10:

Disease	Vector
1. Scrub typhus	b) Trombiculid mite
2. Epidemic typhus	d) Louse
3. Rocky Mountain Spotted Fever	a) Hard Tick
4. Endemic typhus	c) Rat flea

Solution for Question 11:

Anaplasma phagocytophilum specifically targets granulocytes and results in human granulocytic anaplasmosis.(HGA)



Intra granulocytic morulae with Anaplasma phagocytophilum

Orientia tsutsugamushi (Option B) and Ehrlichia chaffeensis (Option C): Both of them target monocytes and produce Scrub typhus and human monocytic ehrlichiosis (HME) respectively.

Bartonella henselae (Option D): They target vascular endothelial cells and RBCs.

Solution for Question 12:

Chlamydia pneumoniae	Chlamydia psittaci
Only one strain (TWAR strain) (Option C)	Many serotypes
Respiratory infections (pneumonia, bronchitis, sinusitis)	Psittacosis (a zoonotic infection)
Primarily through respiratory droplets and person-to-person contact.	Primarily from infected birds (mainly parrots and pigeons) through inhalation of aerosolised particles. (Option A)
1 to 3 weeks of incubation period.(Option D)	2 to 4 weeks of incubation
Doxycycline is the first line treatment.	Doxycycline is the first-line treatment.
It also causes asthma, sarcoidosis and infectious atherosclerosis. (Option B)	Mainly cause only pneumonia.

Solution for Question 13:

Lymphogranuloma Venereum (LGV)

- Etiology: Caused by specific serotypes of Chlamydia trachomatis (specifically serotypes L1, L2, and L3). (Option A)
- Transmission: Primarily transmitted through sexual contact, particularly in anal or genital intercourse. It can also be spread through direct contact with infected lesions. (Option D)

Pathogenesis:

- The bacterium infects epithelial cells of the genital tract, leading to local inflammation.
- The initial infection often goes unnoticed; the bacteria can spread to regional lymph nodes, causing severe lymphadenopathy.
- Systemic spread may occur, leading to complications such as abscess formation and systemic symptoms.

Clinical Features:

Primary Stage:

- Genital Ulcer: A small, painless ulcer at the site of infection, often overlooked. (Option B)



Secondary Stage:

- Buboes: Swollen, painful lymph nodes in the groin or pelvic region (inguinal or femoral), which may become fluctuant and form abscesses.
- Systemic Symptoms: Fever, malaise, and headache.

Late Stage/Tertiary Stage:

- If untreated, LGV can lead to chronic complications, including strictures and rectovaginal fistulas (Esthiomene).
- Fitz Hugh Curtis syndrome (Perihepatitis)

Diagnosis:

Sample collection:

- Urine sample
- Urethral scrapping
- Blood sample
- Sputum

Laboratory Tests:

- Nucleic Acid Amplification Test (NAAT): The preferred test for detecting Chlamydia trachomatis.
- Antibody testing: Complement fixation test
- Antigen testing: ELISA

Treatment:

Antibiotics: Mainstay of treatment

- Doxycycline: 100 mg orally twice daily for 21 days is the first-line treatment.(Option C)
- Azithromycin (1 g orally once) can be used in some cases, but doxycycline is preferred due to efficacy.

Partner Notification and Treatment: All sexual partners should be informed and tested to prevent reinfection.

Solution for Question 14:

Serotypes A, B, Ba, and C are primarily trachoma-associated.

Serotypes of Chlamydia trachomatis:

Serotypes	Associated Infections
A, B, Ba, C (Option B)	Trachoma: Eye infections leading to blindness. M/C form is B
D, E, F, G, H, I, J, K (Option A)	Genital Infections: Non gonococcal Urethritis, cervicitis, and pelvic inflammatory disease (PID). Infant pneumonia.

L1, L2, L3 (Option C)	Lymphogranuloma Venereum (LGV): A more severe STI characterized by swollen lymph nodes and genital ulcers. M/C form is L2

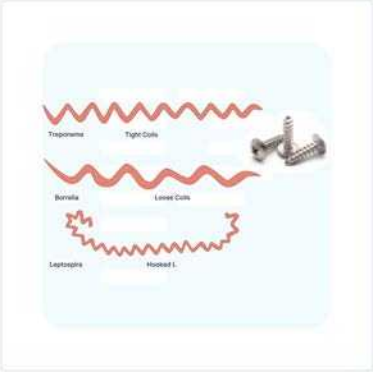
• Chlamydia psittaci consists of serotypes A, B, C, D, E which is usually found in bird species.

Chlamydia pneumonia consists of one strain(TWAR) causing community acquired pneumonia.

Solution for Question 15:

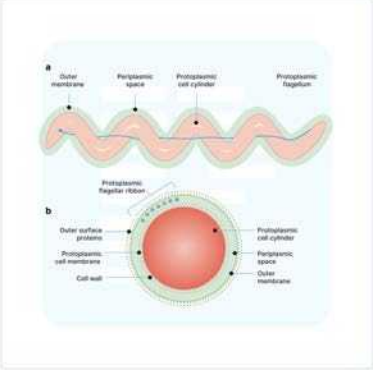
Treponema is a genus of spiral-shaped bacteria that are pathogenic to humans.

Spirochetes are spiral hair-like organisms. They are classified as follows.



Shape- Treponemes are corkscrew-shaped cells 6 to 15 µm long and 0.1 to 0.2 µm wide.

Structure- Treponemes have an outer membrane, a peptidoglycan-cytoplasmic membrane complex, and a protoplasmic cylinder.



Movement—Treponemes have flagella that extend from motors at the pole, producing undulating movement through fluids: corkscrew motility.

- Treponema pallidum: The causative organism of Venereal syphilis. It is generally acquired by close sexual contact.
- Treponema pertenue: Causes yaws. It is acquired via skin-to-skin contact.
- Treponema carateum: Causes pinta. It is acquired via skin-to-skin contact.
- Treponema endemicium: Causes endemic syphilis. It is acquired via kissing.
- Treponema paraluiscuniculi: Associated with syphilis in rabbits.
- Treponema phagedenis- Normal GUT flora- Reiter's strain
- Treponema refringens- Normal GUT flora- Noguchi's strain

Solution for Question 16:

Chlamydia has a cell wall with minimal to no peptidoglycan, unlike most other bacteria, which helps it evade certain host immune responses.

Chlamydia Bacteria:

- Classification: Chlamydia is the genus, with species like Chlamydia trachomatis (causes human infections), Chlamydia pneumoniae(respiratory infections), and Chlamydia psittaci (zoonotic infections, often from birds).

Morphology:

- Small, gram-poor bacteria.
- Unique biphasic life cycle, alternating between an infectious, extracellular form (elementary body) and a replicative, intracellular form (reticulate body).

Life Cycle:

- Elementary Body (EB): The infectious, metabolically inactive form that can survive outside the host cell and initiates infection by entering host cells.
- Reticulate Body (RB): The intracellular, metabolically active form that replicates by binary fission within a host cell vacuole.
- Pathogenicity: Evades immune detection by residing within host cells.
- ATP dependent: Lacks specific metabolic pathways for ATP production, making it dependent on the host for energy.

- It is an obligate intracellular organism which requires a host cell to survive and replicate.(Option A)
- Cannot be cultured: Cannot grow on standard culture media and requires cell cultures.(Option B)

Basophilic bacterial inclusions which forms basophilic inclusions visible in infected cells.(Option D)

Solution for Question 17:

Microscopic Agglutination Test (MAT) is the gold standard test for leptospirosis, detects specific antibodies against Leptospira in serum, providing high sensitivity and specificity.

Leptospira:

Culture Media:

- Ellinghausen-McCullough-Johnson-Harris (EMJH) medium is commonly used for culturing Leptospira species showing Dinger's ring.



- It is a semi-solid or liquid medium that supports the slow-growing leptospires, often requiring incubation for weeks.
- Specimen Sources: Blood, urine, or cerebrospinal fluid (CSF) samples are often used, depending on the stage of infection.
- Other culture media are Korthoff and Fletcher.

Diagnosis:

• Microscopic Agglutination Test (MAT): Considered the gold standard for Leptospira detection, MAT measures antibodies in serum samples.

• The Modified Faine's Criteria used to aid in the diagnosis of leptospirosis, especially in endemic areas.This scoring system combines clinical, epidemiological and laboratory data to increase diagnostic accuracy.

• PCR (Polymerase Chain Reaction): Provides a faster diagnosis by detecting leptospira DNA in blood or urine, especially effective in the early stage.

• ELISA: Used for antibody detection and is helpful for confirming infection in the later stages.

Treatment:

• Antibiotics: Early antibiotic therapy is crucial and often involves: Doxycycline: Typically used for mild cases and prophylaxis. Penicillin or Ampicillin: Preferred for severe cases.

• Doxycycline: Typically used for mild cases and prophylaxis.

• Penicillin or Ampicillin: Preferred for severe cases.

• Doxycycline: Typically used for mild cases and prophylaxis.

• Penicillin or Ampicillin: Preferred for severe cases.

Cold Agglutination Test (Option A): Primarily used to detect antibodies for Mycoplasma pneumoniae.

Standard Agglutination Test (Option B): Commonly used for diagnosing brucellosis.

Dark-Field Microscopy (Option C): Allows visualization of Leptospira in fresh samples but is less sensitive and not commonly used due to difficulty in interpreting results.

Solution for Question 18:

Only 10% of leptospirosis develops into Weil's disease.

Leptospirosis:

Causative Agent	Leptospira interrogans (bacteria) (Option C ruled out)
Transmission	Zoonotic disease (Option A ruled out) Indirect: Contact with water, moist soil, or surfaces contaminated with animal urine. Direct: Contact with urine or parturition products from infected animals.

Pathogenesis Phases	Septicemic Phase: Entry through mucosa or skin, dissemination via bloodstream to organs. Immune Phase: Antibodies develop, spirochetes disappear from blood; antigen-antibody complexes form, and renal colonization occurs.
Clinical Syndromes	Mild Anicteric Illness: Seen in 90% of cases, flu-like symptoms with fever, headache, conjunctival suffusion, nausea, vomiting, and myalgia. Weil's Disease: Severe icteric illness (Option B ruled out) in 10% of cases(Option D); rapid progression, fatality rate of 5-15%.

Solution for Question 19:

Species	Disease	Transmission
Borrelia recurrentis	Relapsing fever (epidemic)-illnesses characterized by recurrent fevers, chills, and malaise.	Louse-borne
Borrelia duttonii	Relapsing fever (endemic)	Tick-borne
Borrelia hermsii	Relapsing fever (endemic)	Tick-borne
Borrelia vincentii	Vincent's angina- acute necrotizing ulcerative gingivostomatitis	Direct contact

Solution for Question 20:

Lyme disease is caused by Borrelia burgdorferi

Lyme disease:

Cause	Borrelia burgdorferi

Stages of Disease	 ... (content truncated)
Vector	 ... (content truncated)
Laboratory Diagnosis	Culture: Requires modified Kelley's (BSK) medium, 33°C incubation for 2+ weeks (difficult to use in routine diagnosis). Serology: ELISA and IF tests, with immunoblotting for confirmation; IgM appears first, followed by IgG.
Cross-Reactivity	False-positive reactions may occur in syphilis serology
Treatment	Doxycycline, amoxicillin, and cefuroxime.

Solution for Question 21:

Rapid Plasma Reagin test cannot be used for CSF testing

Rapid Plasma Reagin (RPR):



- Popular modification of the VDRL test.
- Uses the VDRL antigen with fine carbon particles to enhance the visibility of results.
- Results are easily seen with the naked eye.
- Unheated serum or plasma can be used for the RPR test.
- Not suitable for CSF testing.

Solution for Question 22:

VDRL reaction is observed under a low-power microscope.

VDRL (Venereal Disease Research Laboratory) test:

Antigen Used	Purified lipid extract of beef heart (cardiolipin), with added lecithin and cholesterol (Option A ruled out)

Test Procedure	 The inactivated serum (heated at 56°C for 30 minutes) is mixed with cardiolipin antigen on a special slide. The slide is rotated for 4 minutes. Cardiolipin remains uniform in normal serum but clumps with reagin antibody.(Option B ruled out) Reaction observed under a low-power microscope.(Option D)
False positives found in	Infections, injuries, or inflammation.(Acute) SLE, leprosy(Option C ruled out), malaria, relapsing fever, mononucleosis, hepatitis, tropical eosinophilia, and other collagen diseases.(Chronic)

Solution for Question 23:

The appearance of Treponema pallidum under dark ground microscopy is a slender spiral structure with slow movement

Table rendering failed

Solution for Question 24:

Congenital syphilis is a serious infection in infants caused by the transmission of Treponema pallidum from an infected mother during pregnancy or childbirth. Congenital syphilis occurs when the bacteria cross the placenta from the mother to the fetus. This can happen at any stage of pregnancy, but the risk is highest in the later stages.

Clinical Manifestations

Congenital syphilis can be present in two main forms: early congenital syphilis and late congenital syphilis.

• Early Congenital Syphilis (occurs within the first 2 years of life): Symptoms may include: Rash: Diffuse rash, often with scaly lesions, particularly on the trunk and extremities. Nasal Symptoms: Nasal discharge (often bloody), referred to as "snuffles." Hepatosplenomegaly: Enlarged liver and spleen. Bone Abnormalities: Osteitis or osteochondritis, leading to deformities. Anemia: May present as pallor or fatigue. Fever: This can be a non-specific sign of infection. Wimberger sign-refers to localized bilateral metaphyseal destruction of the medial proximal tibias. It is a pathognomonic sign of congenital syphilis.

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• Nasal Symptoms: Nasal discharge (often bloody), referred to as "snuffles."

• Hepatosplenomegaly: Enlarged liver and spleen.

• Bone Abnormalities: Osteitis or osteochondritis, leading to deformities.

• Anemia: May present as pallor or fatigue.

• Fever: This can be a non-specific sign of infection.

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Metaphyseal destruction of medial proximal tibia

• Late Congenital Syphilis (occurs after 2 years of age): Hutchinson’s triad Hutchinson's Teeth: Notched, peg-shaped incisors characteristic of congenital syphilis. Hearing difficulty: Sensorineural hearing loss Ocular Problems: Interstitial keratitis, which can lead to vision issues. Other symptoms Saddle Nose Deformity: Collapse of the nasal bridge. Bone Abnormalities: Saber shins or other skeletal deformities. Neurological Issues: These may include developmental delays or seizures.

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• Hutchinson's Teeth: Notched, peg-shaped incisors characteristic of congenital syphilis.

• Hearing difficulty: Sensorineural hearing loss

• Ocular Problems: Interstitial keratitis, which can lead to vision issues.

• Other symptoms Saddle Nose Deformity: Collapse of the nasal bridge. Bone Abnormalities: Saber shins or other skeletal deformities. Neurological Issues: These may include developmental delays or seizures.

• Saddle Nose Deformity: Collapse of the nasal bridge.

• Bone Abnormalities: Saber shins or other skeletal deformities.

• Neurological Issues: These may include developmental delays or seizures.

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• Neurological Issues: These may include developmental delays or seizures.

• Hutchinson's Teeth: Notched, peg-shaped incisors characteristic of congenital syphilis.

• Hearing difficulty: Sensorineural hearing loss

• Ocular Problems: Interstitial keratitis, which can lead to vision issues.

• Saddle Nose Deformity: Collapse of the nasal bridge.

• Bone Abnormalities: Saber shins or other skeletal deformities.

• Neurological Issues: These may include developmental delays or seizures.

Diagnosis

• Non-treponemal tests (e.g., RPR, VDRL)

• Treponemal tests (e.g., FTA-ABS)

Treatment

• Penicillin: This antibiotic is the first-line treatment effective at all stages. Dosage and duration depend on the infant's age and the severity of the disease.

• Supportive Care: Infants may require additional support for symptoms such as anemia or organ involvement.

Prevention

• Screening: Routine screening for syphilis during prenatal visits, particularly in high-risk populations.

• Prompt Treatment: Ensuring timely and appropriate treatment of syphilis in pregnant women to reduce the risk of transmission to the fetus.

Respiratory distress with wheezing (Option A) is not specifically characteristic of congenital syphilis and can be seen in various other respiratory conditions.

Microcephaly and developmental delay(Option B) are not primary features of congenital syphilis. These findings are more commonly associated with other congenital infections (e.g., Zika).

Hyperbilirubinemia and hepatosplenomegaly (Option D) can occur due to various causes, including other congenital infections, but are not specific to congenital syphilis.

Solution for Question 25:

1. Rash: The most characteristic feature is a diffuse, non-pruritic rash that can appear anywhere on the body. It often presents as reddish-brown spots (macules) and may also include raised lesions (papules). The rash is polymorphous, non-infectious, and never vesicular



2. Mucous Membrane Lesion- Condylomata Lata: These are broad, wart-like lesions that can appear in moist areas, such as the oral cavity, genital region, and perineum. They are highly

infectious.The oral features of secondary syphilis can be painless or painful erythematouslesions, grayish-white mucous patches, or irregular linear erosions termed snail track ulcers

3.Moth eaten alopecia-nonscarring, patchy hair loss with a "moth-eaten" appearance, most commonly affecting the scalp. It is considered a pathognomonic sign of the disease but can also affect other hair-bearing areas.

4.Blushke olendroff sign-The Buschke-Ollendorff sign in secondary syphilis and cutaneous vasculitis involves deep dermal tenderness when pressing on lesions, such as syphilitic papules, with a pinhead.

5. Lymphadenopathy: Generalized painless lymphadenopathy,characteristic epitrochlear node involvement

6. Systemic Symptoms: Patients may experience flu-like symptoms, including mild fever, malaise, and fatigue.

7. Rarely, secondary syphilis can lead to neurological symptoms or ocular involvement, though this is more common in tertiary syphilis.

Diagnosis: Serological Testing:

Treatment:

- The standard treatment for secondary syphilis is Benzathine penicillin G. Dosage: 2.4 million units intramuscularly in a single dose.
- For those allergic to penicillin, alternatives such as doxycycline or tetracycline can be used.

Complications: If left untreated, secondary syphilis can progress to the latent stage and eventually tertiary syphilis, which can lead to serious complications, including cardiovascular, neurological, and systemic damage.

- Genital herpes(Option A) usually presents with painful vesicular lesions rather than a widespread rash and systemic symptoms.
- Psoriasis(Option C) is a chronic skin condition that does not present with mucous membrane lesions or flu-like symptoms.
- Drug reaction(Option D) could cause a rash but typically involves additional symptoms related to the drug exposure and would not explain the oral lesions or the patient's sexual history

Solution for Question 26:

This patient exhibits symptoms consistent with tertiary syphilis, specifically the neurological manifestations and joint issues indicative of neurosyphilis and Charcot's joint.

The CSF VDRL test is highly specific and, when reactive, is considered diagnostic of neurosyphilis. Involvement of the CNS in syphilis is detected by examination of CSF for:

- Mononuclear pleocytosis (>5 WBC/μL).
- Increased protein concentration (>45 mg/dL).
- CSF VDRL reactivity.

Tertiary syphilis is the late stage of syphilis It can develop years after the initial infection if left untreated.

Symptoms:



Diagnosis of tertiary syphilis involves:

Treatment

- Penicillin: The standard treatment for all stages of syphilis. Dosing may vary based on the stage and severity.
- Supportive care: For complications, such as pain management for gummas or treatment for neurological issues.

Polymerase Chain Reaction (PCR) for Treponema pallidum(Option A): While PCR can diagnose syphilis, it is not the standard test for neurosyphilis, where CSF analysis is preferred.

Lumbar puncture (Option C) is a procedure that involves collecting CSF. While it is necessary to obtain CSF for analysis, it doesn't provide a definitive diagnosis. For neurosyphilis, a specific test on the CSF sample is required—the CSF VDRL (Venereal Disease Research Laboratory) test, which directly detects antibodies to Treponema pallidum in the CSF.

Rapid Plasma Reagin (RPR) test (Option D): The RPR test is helpful for initial screening, but it does not provide information about neurological involvement, making CSF analysis the better

choice in this case.

Solution for Question 27:

Primary syphilis is the initial stage of syphilis infection, typically occurring about three weeks after exposure to the bacterium Treponema pallidum. The hallmark of this stage is the development of a painless ulcer known as a chancre, which appears at the site where the bacteria entered the body—commonly on the genitals, anus, or mouth.

- The chancre is usually round, firm, and painless. It can vary in size, often resembling a small, raised sore.
- It is highly infectious and typically lasts between three to six weeks. Even though it heals on its own, the absence of symptoms does not mean the infection is gone; the bacteria remain in the body.

Syphilis is primarily transmitted through sexual contact, but it can also spread from mother to child during pregnancy or childbirth, leading to congenital syphilis.

- The chancre is the only symptom in primary syphilis, but regional lymph nodes may become swollen and painless.

Diagnosis is often made through clinical examination of the chancre and serological tests that detect antibodies to Treponema pallidum.

Genital herpes(Option A): Usually presents with painful ulcers and is often associated with systemic symptoms like fever and malaise.

Chancroid(Option C): Typically causes painful ulcers and is associated with swollen lymph nodes (lymphadenopathy).

Lymphogranuloma venereum(Option D): Characterized by an ulcer, but typically presents with significant lymphadenopathy and systemic symptoms.

Miscellaneous bacteria

1. A 28-year-old man presents with symptoms of tracheobronchitis and a persistent cough. After testing, Mycoplasma pneumoniae is identified as the causative agent. Which of the following is a notable feature of Mycoplasma pneumoniae infection that helps in its diagnosis?

A. The presence of a characteristic "fried egg" colony appearance in blood agar culture

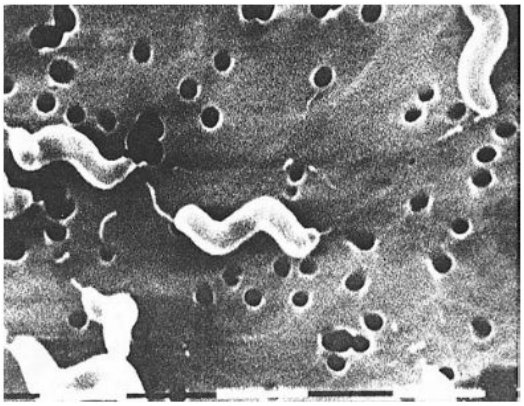
B. Detection of mycoplasma DNA through polymerase chain reaction (PCR)

C. High titers of cold agglutinins in the blood at 37°C

D. Indirect Coombs test detects red blood cells coated with antigens

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2. A 25-year-old patient presents with fever, abdominal pain, and watery diarrhoea. The stool examination reveals the presence of leukocytes and blood. The patient has a history of consuming undercooked poultry. Direct microscopic examination of the stool shows darting motility of spiral rods. The laboratory report indicates that the pathogen is a Gram-negative rod that grows optimally at 42°C in microaerophilic conditions. Which of the following is the most likely causative agent of this infection?



A. Salmonella enterica

B. Shigella dysenteriae

C. Campylobacter jejuni

D. Escherichia coli

---

3. A 55-year-old patient with chronic gastritis undergoes diagnostic testing for Helicobacter pylori infection. The tests include both invasive and non-invasive methods. Which of the following non-invasive tests is specifically known for its ability to detect active H. pylori infection and provide rapid results?

A. Endoscopic Biopsy with Microscopy

B. Urea Breath Test

C. Rapid Urease Test

D. Culture analysis on nutrient-rich medium

---

4. A 55-year-old man presents with fever, non-productive cough, and shortness of breath. A chest X-ray shows signs of pneumonia. Suspecting Legionnaires' disease, a sputum sample is cultured on Buffered Charcoal Yeast Extract (BCYE) agar. Which of the following findings on BCYE agar is indicative of Legionella growth?

A. Blue-green pigment

B. Yellow colonies

C. Black or grayish colonies

D. Hemolytic zones around colonies

---

5. A microbiologist is studying Klebsiella pneumoniae and its microscopic characteristics. Which of the following statements about the microscopy of

Klebsiella pneumoniae is true?

A. It is Gram-positive and appears purple under Gram staining

B. Its capsule is visible using the Warthin-Starry method.

C. They are observed as elongated, slender rods that are encapsulated

D. There is an absence of biofilm when examined with a scanning electron microscope (SEM).

---

6. Which of the following statements regarding the diagnosis of Francisella tularensis infection (tularemia) is true?

A. Culturing F. tularensis is easily performed on standard blood agar.

B. Serological tests are always reliable and do not require confirmation with other diagnostic methods

C. Direct microscopy of patient samples is the primary method for diagnosis

D. Isolation of the organism from clinical specimens is essential for confirming tularemia

---

7. Match the following features/diseases with their respective organisms: 1. Klebsiella pneumoniae a. Chronic granulomatous hypertrophy of nose 2. Klebsiella ozaenae b. Foul-smelling nasal discharge 3. Klebsiella rhinoscleromatis c. Healthcare-associated infections, UTI, septicemia

1. Klebsiella pneumoniae	a. Chronic granulomatous hypertrophy of nose
2. Klebsiella ozaenae	b. Foul-smelling nasal discharge
3. Klebsiella rhinoscleromatis	c. Healthcare-associated infections, UTI, septicemia

A. 1-a ; 2-b ; 3-c

B. 1-c ; 2-b ; 3-a

C. 1-b ; 2-c ; 3-a

D. 1-a ; 2-c ; 3-b

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	3
Question 5	2
Question 6	4
Question 7	2

Solution for Question 1:

Answer: B) Detection of mycoplasma DNA through polymerase chain reaction (PCR)

Explanation:

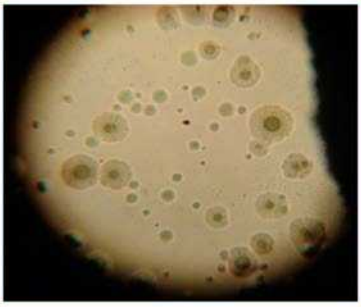
PCR-based detection of mycoplasma DNA is a specific and effective method for diagnosing Mycoplasma pneumoniae infections.

Features of Mycoplasma:

- Smallest free-living organism
- Lack of cell wall (therefore Resistant to antibiotic interfering cell wall synthesis for ex:
- B lactams), leading to pleomorphic nature
- Cell membrane contains sterols
- Ability to pass through standard bacterial filters
- Historically misclassified as viruses (Joker of the microbiological park)
- Chemoorganotrophs
- Energy production via fermentation
- Main energy sources: glucose or arginine

Clinical Features:

Pulmonary Infections:	Genital Infections:	Extrapulmonary Infections:
Causative Agent: Mycoplasma pneumoniae (mc cause of atypical pneumonia) Common Name: Primary atypical pneumonia Presentation: Tracheobronchitis, Acute pharyngitis (less common), Gradual onset of fever, malaise, headache, sore throat, Persistent cough, potentially blood-tinged sputum Incidence: About one-third develop pneumonia Incubation Period: 1 to 3 weeks Diagnosis: Radiological assessments reveal patchy consolidation in lower lobes of lungs	Causative Agents: Mycoplasma hominis, Ureaplasma urealyticum Transmission: Sexual contact Conditions in Men: Urethritis, proctitis, balanoposthitis, Reiter's syndrome Conditions in Women: Acute salpingitis, pelvic inflammatory disease, cervicitis, vaginitis, Infertility, abortion, postpartum fever, chorioamnionitis, low birth weight in infants	GI: nausea, vomiting, diarrhea Cardiovascular: myocarditis, pericarditis Neurological: meningitis, encephalitis, optic neuritis Musculoskeletal/skin: Erythema multiforme, SJS, Maculopapular/vesicular rash Others: cold AIHA, bullous myringitis, vasculitis

Method	Description	Details
Culture	Requires specialized media and has slow growth (Option A ruled out)	 Takes 1 to 3 weeks for visible colonies Colonies have a "fried egg" appearance (In PPLO agar, with Dienes stain) ... (content truncated)
Molecular Methods	Detects mycoplasma DNA	PCR is used for faster and more specific detection of DNA, particularly useful for slow-growing organisms

Serology	Detects antibodies in response to Mycoplasma infection	Specific Tests: Immunofluorescence, hemagglutination inhibition, growth inhibition Complement fixation, indirect hemagglutination (less sensitive) Non-Specific Tests: Streptococcus MG test: Agglutination with titre of 1:20 or higher Cold agglutination test: where clumping of red blood cells is observed after the mixture is incubated at 4°C and disappears upon warming to 37°C (Option C ruled out); titre of 1:32 or greater Indirect Coombs test: Detects red blood cells coated with antibodies. (Option D ruled out)

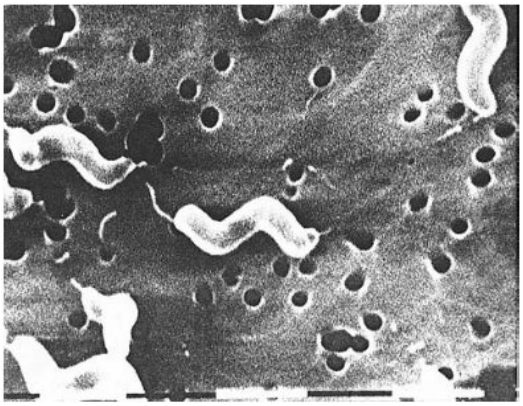
- Antibiotics: Tetracyclines (e.g., doxycycline) Macrolides (e.g., erythromycin)
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- Concerns: Emerging resistance, especially among certain Ureaplasma strains
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- Macrolides (e.g., erythromycin)
- Emerging resistance, especially among certain Ureaplasma strains

Solution for Question 2:

Answer: C) Campylobacter jejuni

Explanation:

Campylobacter jejuni is known to cause diarrheal disease with symptoms including fever, abdominal pain, and watery diarrhoea. The presence of darting motility in stool, growth at 42°C, and a history of consuming undercooked poultry are characteristic of Campylobacter jejuni. It is a Gram-negative, spiral-shaped rod and is strongly oxidase-positive.



Scanning electron micrograph of the single polar flagellum and corkscrew shape of Campylobacter jejuni. These morphologic characteristics contribute to the characteristic darting motility of C jejuni.

Feature	Description
Morphology and Structure	Slender, spirally curved, Gram-negative rods; comma-shaped, may form 'S' or multispiral chains; non-sporing; single unsheathed polar flagellum
Growth Conditions	Microaerophilic; 5% oxygen concentration; thermophilic species grow at 42°C; strongly oxidase positive
Pathogenic Significance	Campylobacter jejuni is the most common cause of bacterial diarrheal disease in developed countries

Clinical Features:

Aspect	Description
Infection Pathway	Ingestion; primary sites: jejunum and ileum; may spread to colon and rectum
Pathogenesis	Invasive pathogen; incubation period: 1-7 days; damages small intestine cells, disrupting fluid absorption
Symptoms	Fever, abdominal pain, watery diarrhea; stool may contain leukocytes and blood; typically self-limiting with possible continued shedding

Diagnosis:

Method	Description
Direct Microscopic Examination	Phase-contrast or dark field microscopy for motility; stained smears for detection of small curved rods
Culture Techniques	Plated on selective media; use transport medium if delayed; commonly used media include Skirrow's, Butzler's, Campy BAP

Incubation Conditions	42°C with 5% oxygen, 10% CO■, 85% N■; also at 37°C; suppresses normal fecal flora
Colony Appearance	Non-hemolytic, grey/colorless, moist, flat/convex; appears in 48 hours
Identification	Gram staining, motility, oxidase tests; biochemical tests for confirmation (positive catalase and nitrate reduction)

Treatment

Treatment Type	Description
Primary Treatment	Fluid and electrolyte replacement
Antibiotic Therapy	Erythromycin is the preferred antibiotic

Salmonella enterica (Option A): Although it is associated with poultry, Salmonella typically does not display darting motility or a spiral shape and does not grow optimally at 42°C in microaerophilic conditions.

Shigella dysenteriae (Option B): Shigella is a Gram-negative rod but does not have darting motility or a spiral shape, nor does it require microaerophilic conditions or high temperatures for growth.

Escherichia coli (Option D): While certain strains can cause bloody diarrhea, E. coli is not spiral-shaped and does not exhibit darting motility or a growth preference at 42°C under microaerophilic conditions.

Solution for Question 3:

Answer: B) Urea Breath Test

Explanation:

The Urea Breath Test is a non-invasive test that involves ingesting carbon-labelled urea, which is broken down by H. pylori to release carbon dioxide that is measured in the breath. It is specifically known for detecting active infection and provides rapid and reliable results. The other options are invasive modalities.

- Genetic Diversity and Pathogenesis H. pylori exhibits significant genetic diversity, impacting its virulence. Key genes: cag (cytotoxin-associated gene) and vac (vacuolating cytotoxin gene). Contributes to pathogenesis via urease production and motility.

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• The infection triggers a complex immune response involving IgM, IgG, IgA, and cellular immunity.

• Immune response does not effectively protect against infection.

• Endoscopic Biopsy: Tissue sample from the gastric mucosa. (Option A rule out) Rapid Urease Test: Detects urease in the biopsy; results available within minutes. (Option C ruled out) Microscopy: Identifies H. pylori through staining (Warthin starry stain) Culture: Most sensitive method; involves growing bacteria from a biopsy on a nutrient-rich medium under specific conditions (37°C, microaerophilic environment). Results in 3-7 days. Further characterization includes biochemical tests (catalase and oxidase positive). (Option D ruled out)

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• Serology (ELISA): Detects antibodies against H. pylori; useful for epidemiological studies, less reliable for diagnosing active infection or monitoring treatment.

• Urea Breath Test: Involves ingestion of carbon-labeled urea; breakdown by H. pylori releases carbon dioxide, measured in breath. Sensitive and reliable, but requires specialized equipment.

• Antibiotic and Bismuth Salt Regimen Standard treatment: Bismuth subsalicylate + two antibiotics (tetracycline or amoxicillin + metronidazole) for two weeks. Alternative: Proton pump inhibitor (e.g., omeprazole) + clarithromycin.

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• Recommended for H. pylori-associated gastric or duodenal ulcers.

• Asymptomatic colonization does not require treatment.

• Increasing antibiotic resistance.

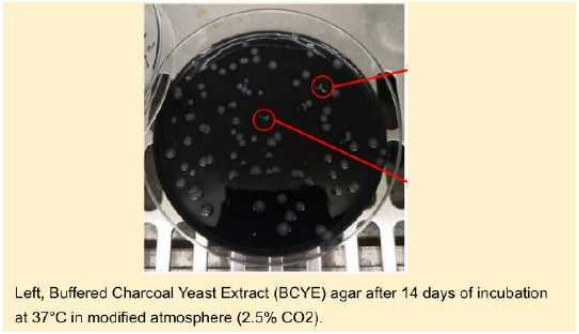
• High rate of infection recurrence.

Solution for Question 4:

Answer: C) Black or grayish colonies

Explanation:

On Buffered Charcoal Yeast Extract (BCYE) agar, Legionella species typically form small, grayish or black colonies due to the reduction of iron salts in the medium. This color change is distinctive for Legionella and aids in its identification.



Left, Buffered Charcoal Yeast Extract (BCYE) agar after 14 days of incubation at 37°C in modified atmosphere (2.5% CO2).

Legionella

Features

- Thin, Gram-negative, non-capsulated bacilli (2-5 x 0.1-0.3 µm).
- Poor Gram staining; best visualized by direct fluorescent antibody (DFA) staining.
- Requires buffered charcoal yeast extract (BCYE) agar for growth.

Clinical Features

Condition	Characteristics	Severity
Legionnaires' Disease	Fever, non-productive cough, dyspnea, progressing to pneumonia, diarrhea, encephalopathy, causes SIADH (hyponatremia, seizures) Causes both typical and atypical pneumonia. Mode of infection: microaspiration and aerosol inhalation There is no human to human transmission Associated with: AC, showers, coolers, cruise Seen in : Smokers, immunocompromised patients, older patients	High (15-20% fatality)

Pontiac Fever	Mild, non-fatal symptoms: fever, chills, myalgia, headache	Mild

Diagnosis Methods

Method	Sample Type	Notes
Direct Fluorescent Antibody (DFA)	Sputum, bronchial aspirate, lung biopsy	Detects Legionella in specimens
Culture	Clinical specimens	Buffered Charcoal Yeast Extract (BCYE) Agar- Gold standard for Legionella culture; contains yeast extract, iron, L-cysteine, and charcoal; optimal pH ~6.9 Small colonies, often grayish or black, due to iron reduction by bacteria. (Option C) L-cysteine Presence: Required for growth, confirming presumptive Legionella colonies.
Antigen Detection in Urine	Urine	Latex agglutination or ELISA
Serum Antibody Detection	Blood	ELISA or indirect immunofluorescence

Treatment

- First-line options: Macrolides, ciprofloxacin, tetracyclines.
- Severe cases: Rifampicin may be added.
- Ineffective antibiotics: Beta-lactamase antibiotics and aminoglycosides.

Blue-green pigment (Option A): It is characteristic of Pseudomonas aeruginosa.

Yellow colonies (Option B): Legionella does not produce yellow colonies on BCYE agar.

Hemolytic zones (Option D): Legionella does not typically cause hemolysis on BCYE agar.

Solution for Question 5:

Answer: B) Its capsule is visible using the Warthin-Starry method.

Explanation:

The Warthin-Starry method is a specific staining technique that highlights the polysaccharide capsule of Klebsiella pneumoniae, which plays a crucial role in its virulence by helping it evade phagocytosis.

Klebsiella:

Features	Non-motile, capsulated, short, plump rods (1-2 µm x 0.5-1.8 µm). Common as intestinal commensals and saprophytes in soil and water. Ferment glucose, lactose, sucrose, and mannitol, producing acid and gas
Clinical Features	Major healthcare-associated pathogen, commonly causing: Pneumonia (upper lobe pneumonia): Severe, often affecting middle-aged/older adults with risk factors (e.g., alcoholism, diabetes). UTIs, particularly catheter-associated (CAUTI). Lung abscess, red current jelly sputum, bulging fissure sign Neonatal sepsis: High mortality due to multidrug resistance. Septicemia and other pyogenic infections.
Morphological Characteristics	Appears as short, plump, encapsulated rods with visible halos in Gram-stained smears. (Option C ruled out)
Gram Staining	Gram-negative; appears pink after counterstaining with safranin. (Option A ruled out) Typically rod-shaped (bacilli).
Capsule Visualization	Encapsulation visualized using staining techniques (e.g., Warthin-Starry method, India ink preparation), highlighting the polysaccharide capsule, crucial for evading phagocytosis.
Fluorescence Microscopy	Used to assess antibiotic efficacy (e.g., meropenem) and monitor morphological changes post-treatment, showing enlarged or distorted cells.
Biofilm Structure	Studied using scanning electron microscopy (SEM) to visualize complex biofilm architecture, essential for understanding bacterial adherence and interaction with surfaces. (Option D ruled out)
Biochemical tests:	
Treatment	Empiric therapy for bloodstream infections often needed: Options include piperacillin-tazobactam, cefoperazone-sulbactam, or amikacin. Carbapenems (imipenem, meropenem) are preferred for resistant or critical cases.

Klebsiella Infections:

Aspect	Description
Common Infections	Pneumonia, UTIs, septicemia, neonatal sepsis, pyogenic infections
High-Risk Groups	Middle-aged/older adults with comorbidities (alcoholism, diabetes), neonates, ICU patients
Mortality Factors	High for pneumonia and neonatal sepsis, often due to multidrug-resistant strains

Treatment:

Infection Type	Initial Empiric Treatment	Resistant/Critical Cases Treatment
Bloodstream Infections	Piperacillin-tazobactam, cefoperazone-sulbactam, amikacin	Carbapenems (imipenem, meropenem)

Solution for Question 6:

Answer: D) Isolation of the organism from clinical specimens is essential for confirming tularemia.

Explanation:

Isolation of Francisella tularensis from clinical specimens, such as blood or tissue samples, is the gold standard for diagnosing tularemia. This method provides definitive evidence of the pathogen's presence, while other methods, like serological tests, may not be reliable in all cases and could lead to false negatives. Therefore, culture confirmation is crucial for accurate diagnosis and appropriate treatment.

Francisella tularensis

- Characteristics: Minute, capsulated, non-motile, Gram-negative bacillus. Filterable and capable of filament formation and budding, similar to Mycoplasma.
- Minute, capsulated, non-motile, Gram-negative bacillus.
- Filterable and capable of filament formation and budding, similar to Mycoplasma.
- Strains are classified into biotypes based on virulence and epidemiological behavior: Highly virulent strains are exclusive to North America. Low-virulence strains are found in Europe and Asia.

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- Highly virulent strains are exclusive to North America.
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Clinical Features

- Francisella tularensis causes tularemia, a zoonotic infection transmitted to humans from rabbits and rodents.
- Transmission occurs via: Tick bites Ingestion of contaminated meat or water Inhalation of infectious aerosols
- Tick bites
- Ingestion of contaminated meat or water
- Inhalation of infectious aerosols
- Clinical manifestations include: Tularemia also called ulceroglandular fever: Local ulceration with lymphadenopathy Typhoid-like fever with glandular swelling Influenza-like respiratory symptoms
- Tularemia also called ulceroglandular fever: Local ulceration with lymphadenopathy
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- Tularemia also called ulceroglandular fever: Local ulceration with lymphadenopathy
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- Influenza-like respiratory symptoms

Disease progression:

Ticket bite during rabbit killing → hand ulcer → axillary lymphadenitis

Diagnosis

- Diagnosis of tularemia can be achieved through several methods:

Method	Description
Culture	Requires specialized media (e.g., Francis' blood dextrose cystine agar); small transparent colonies appear after 3-5 days of incubation. (Option A ruled out)

Animal Inoculation	Identified by inoculating guinea pigs or mice with patient samples.
Serology	Detection of agglutinating antibodies in patient sera. Can yield false negatives and therefore require further confirmation with cultures.(Option B ruled out)

Treatment

- An attenuated vaccine can be given by scarification to high-risk individuals.

Direct microscopy of patient samples is the primary method for diagnosis (Option C): Direct microscopy is not the primary method for diagnosing tularemia. While it can be used to visualize organisms in tissue samples, it is not sufficiently sensitive or specific to serve as the main diagnostic tool.

Solution for Question 7:

Correct answer: B) 1-c; 2-b; 3-a

Explanation:

1. Klebsiella pneumoniae	c. Healthcare-associated infections, UTI, septicemia
2. Klebsiella ozaenae	b. Foul-smelling nasal discharge
3. Klebsiella rhinoscleromatis	a. Chronic granulomatous hypertrophy of nose

1. Klebsiella pneumoniae (Friedlander's bacillus, Bacillus mucosus capsulatus) :

- They grow well on ordinary media, forming large, dome-shaped, mucoid colonies of varying degrees of stickiness.
- On MacConkey agar, they grow as lactose-forming colonies
- Common infections associated with K. pneumoniae are pneumonia, urinary infection, septicemia, several pyogenic infections and, very rarely, diarrhea

Pneumonia due to Klebsiella is a serious disease with high case fatality:

- It occurs in middle-aged or older persons with risk factors such as alcoholism, chronic bronchopulmonary disease or diabetes mellitus.
- The disease is characterized by massive mucoid inflammatory exudate of lobar or lobular distribution, involving one or more lobes of the lung.

• Necrosis and abscess formation are more frequent than in pneumococcal pneumonia. Serotypes 1, 2 and 3 are usually responsible for pneumonia.

K.pneumoniae is a frequent cause of :

• Urinary infection, often causing catheter-associated UTI (CAUTI).

• As most strains are resistant to antibiotics, treatment poses serious problems.

• It also causes pyogenic infections such as abscesses, meningitis and septicemia.

• Mortality is high in neonatal sepsis caused by Klebsiella pneumoniae, as mostly multidrug-resistant strains cause these infections.

2. K.ozaenae :

• It is a bacillus associated with ozena, a disease characterized by foul-smelling nasal discharge.

• Identification is difficult due to wide variations in the biochemical reactions of individual strains.

• K.ozaenae belongs to capsular types 3-6.

3. K.rhinoscleromatis:

• It causes rhinoscleroma, a chronic granulomatous hypertrophy of the nose.

• The bacilli are seen intracellularly in lesions.

• It can be identified by biochemical reactions and belongs to capsular type 3

Biochemical reaction differentiating Klebsiella species:

Klebsiella species" data-author="" data-hash="9875" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/XonorHWxanBA82UzmI6G1744445885.png" />

• Klebsiella ferment carbohydrates (glucose, lactose, sucrose and mannitol) with the production of acid and abundant gas.

• The IMViC reaction is - -+ +.

Treatment:

• Empiric treatment is done for bloodstream infections. 4.5 g of Piperacillin + Tazobactam every six hours OR

• 3 g of Cefoperazone-sulbactam every 12 hours or Aminoglycoside, e.g., amikacin 20 mg/kg/ day must be given.

• Carbapenems like imipenem and Meropenem must be given in resistant isolates and critically ill patients.

Previous Year Questions

1. A militant presents with fever and rash all over his body, which started in the trunk first. On examination, a rash was all over the body, sparing palms and soles, along with a louse noted. Which of the following is responsible for his condition?

A. Rickettsia typhi

B. Rickettsia prowazekii

C. Rickettsia akari

D. Rickettsia conorii

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2. A newly joined teacher develops rice water stools. The causative agent acts on which of the following receptors?

A. GM 1 ganglioside receptor

B. GM 2 ganglioside receptor

C. Sphingomyelin

D. Cerebroganglioside

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3. Which of the following is considered to be the best for the detection of Clostridium difficile?

A. Aerobic culture of stool

B. Toxin assay

C. Glutamate dehydrogenase assay

D. Colonoscopy

---

4. A patient has presented with fever and blister on the hand, followed by the development of axillary lymphadenopathy. What will you ask the patient with regard to history?

A. History of rat bite

B. History of exposure to cat

C. History of similar complaints among family members

D. History of louse bite

---

5. A 38-year-old IV drug abuser presents with a 1-week history of malaise, fatigue, and rigors. His temperature on admission was 38.5° C. Examination revealed a pansystolic murmur, loudest at the left sternal edge. Three sets of blood cultures were taken. Which organism is most likely to be found in his blood culture?

A. Streptococcus sanguis

B. Enterococcus faecium

C. Coxiella burnetii

D. Staphylococcus aureus

---

6. A patient presented with vomiting to medicine OPD. He mentions eating fried rice recently. What is the most likely causative organism?

A. Clostridium perfringens

B. Bacillus cereus

C. Staphylococcus aureus

D. Bacillus anthracis

---

7. A patient presents with abdominal pain, diarrhea, and a step-ladder pattern fever. He also complains of a headache and malaise. His symptoms gradually worsened over the past 2 weeks. Which of the following is the best diagnostic test?

A. Widal test

B. Stool culture

C. Blood culture

D. Urine culture

---

8. A truck driver comes with complaints of a painless indurated genital ulcer after unsafe intercourse. Which of the following investigations can be used to diagnose the causative organism?



A. Darkfield microscopy

B. HSV 2 antibody

C. Albert stain

D. Acid-fast stain

---

9. A 45-year-old man with multiple sexual partners presents with a palmoplantar rash, as seen below. On examination, mucocutaneous papules on the scrotum and lymphadenopathy are observed. Which of the following is the likely diagnosis?



A. Psoriasis

B. Secondary syphilis

C. Primary syphilis

D. Tertiary syphilis

---

10. Which investigation would be the most appropriate for a sewage worker who has been experiencing abdominal pain, jaundice, conjunctival injection, and blood in the urine for the last 5 days?

A. Widal test

B. Microscopic agglutination test

C. Weil-Felix reaction

D. Paul-Bunnell test

---

11. A child was brought with complaints of high fever, multiple seizures, headache, and neck rigidity. CSF analysis showed low glucose, high protein, and many

polymorphs. The CSF culture showed pleomorphic gram-negative bacilli. What is true about the causative organism?

A. It is bacitracin sensitive.

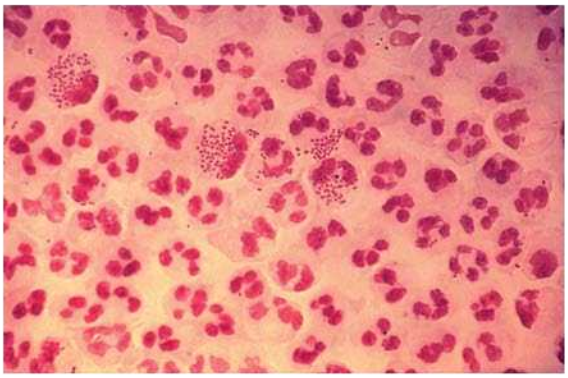
B. Produces non-lactose fermenting colonies on MacConkey agar

C. Exhibits satellitism around Staphylococcus aureus colonies

D. It exhibits alpha hemolysis on sheep blood agar

---

12. A man presents with a history of dysuria and urethral discharge after having unprotected sexual intercourse. Gram-staining of the purulent discharge is shown in the following image. What is the treatment of choice for this infection?



A. Erythromycin

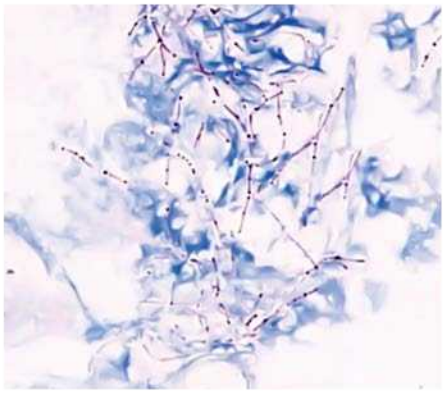
B. Ciprofloxacin + Azithromycin

C. Ceftriaxone + Azithromycin

D. Penicillin G

---

13. Which of the following is a gram-positive organism that shows the following appearance on Ziehl-Neelsen staining?



A. Nocardia

B. Mycobacterium tuberculosis

C. Mycobacterium leprae

D. Actinomyces

---

14. Which of the options provided is an instance of mollicutes?

A. Ureaplasma urealyticum

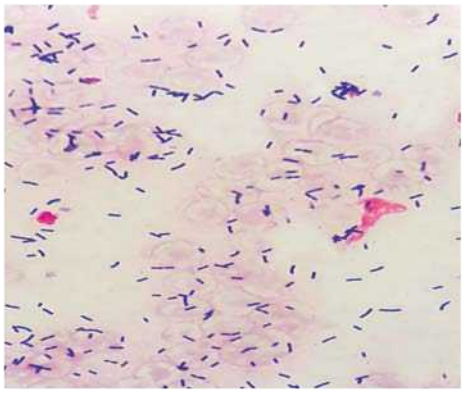
B. Orientia tsutsugamushi

C. Chlamydophilia pneumoniae

D. Chlamydia trachomatis

---

15. An elderly male presented with meningitis, and the CSF sample was subjected to gram staining and microscopy was shown in the image given. Which of the following will be characteristics of the likely organism?



A. Motile at 22° C and non-motile at 37° C

B. No motility at 22° C and motile at 37° C

C. Motile at both 37° C and 22° C

D. Non motile at both 37° C and 22° C

---

16. What is the probable causative agent involved in the case of two individuals from a group of 20 friends who experienced nausea and vomiting in the early morning after consuming pastries from an external source during a late-night party?

A. Enterotoxin of staphylococcus

B. Emetic toxin of B. cereus

C. Verocytotoxin from E. coli

D. Shiga toxin from shigella

---

17. One week after having unprotected sexual intercourse, a 25-year-old male develops urethral discharge as shown in the image below. What is the most likely causative organism for this condition?



A. Neisseria gonorrheae

B. Ureplasma urealyticum

C. Herpes simplex virus

D. Trichomonas vaginalis

---

18. Which organism is the most probable cause of the symptoms described in a 14-year-old boy who experienced vomiting within 3 hours after eating at a party?

A. Staphylococcus aureus

B. Clostridium botulinum

C. Clostridium perfringes

D. Salmonella typhi

---

19. A 22-year-old presented with high-grade fever, confused state and purpuric rash as shown in the image below. CSF sample shows gram-negative diplococci. When of the following is the most likely etiology?



A. Neisseria meningitidis

B. Pseudomonas aeruginosa

C. Streptococcus pneumoniae

D. E.coli

---

20. Which one of the following is not associated with urethritis?

A. Trichomonas

B. Chlamydia

C. Neisseria gonorrheae

D. H. ducreyi

---

21. Which of the following culture media is used to differentiate other E.coli species from E.coli O157 : H7?

A. Mannitol salt agar

B. Xylose lysine deoxycholate agar

C. Sorbitol MacConkey agar

D. CLED medium

22. Which of the following agents is classified as a category A biothreat in bioterrorism?

A. Nipah virus

B. Anthrax

C. Coxiella

D. Brucella

23. A patient presented with complaints of fever for 3 days and general fatigue. Echocardiography revealed vegetations on the valves and blood culture was done. The organism in the culture showed alpha hemolysis on blood agar and was resistant to optochin and bile. Which of the following bacteria is most commonly implicated in this condition?

A. Mycoplasma

B. Streptococcus sanguis

C. Enterococcus

D. Streptococcus agalactiae

24. A patient has complaints of fever with a mild headache associated with some difficulty in swallowing. A throat examination reveals swollen glands and a thick, whitish membrane over his tonsils which bleeds on scraping the lesion. A sample is taken from the exudate and a gram stain is done, which is as shown below. What is the organism associated with the disease?



A. Parvovirus

B. HHV6

C. Corynebacterium diphtheria

D. Beta hemolytic streptococci

---

25. A patient was diagnosed with TB and while testing for resistance he was found to be resistant to Isoniazid, Rifampicin, Amikacin, and levofloxacin. What is this type of TB called?

A. Poly-drug resistant TB

B. Multi drug sensitive TB

C. Extensively drug resistant TB

D. Rifampicin resistant TB

---

26. In a resource-limited setting, what diagnostic method can be utilized to identify the condition when there is a visible painless hardened genital ulcer following unprotected sexual activity, as reported by a truck driver?

A. Dark field microscopy

B. Fluorescent treponema antibody absorption

C. Venereal disease research laboratory test

D. Treponema pallidum immobilization test

---

27. A 21-year-old girl who presented with multiple beefy-red painless ulcers over the vulval region is diagnosed with granuloma inguinale. What is the causative organism?

A. Chlamydia trachomatis

B. Klebsiella granulomatis

C. Haemophilus ducreyi

D. Herpes simplex virus 2

---

28. A farmer with a history of flea bite in a godown presents with the features of high fever, painful inguinal lymphadenopathy, vomiting and diarrhea and hypotension. Smears prepared from the discharge of these swellings revealed bacteria with bipolar staining. Which stain will help in the diagnosis?

A. Neisser stain

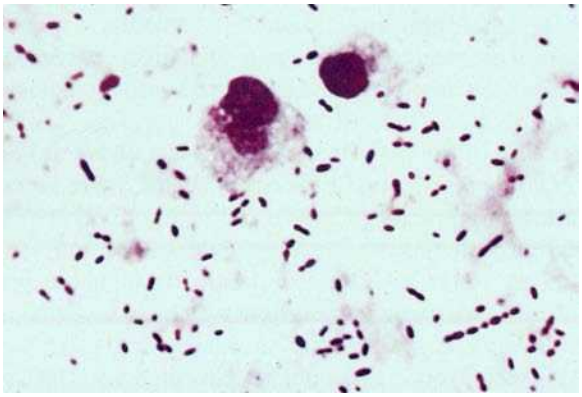
B. Wayson’s stain

C. Albert’s stain

D. McFadyeans reaction

---

29. A patient presented with meningitis, and the CSF sample was subjected to gram staining, which appeared in the microscopic image below. Which of the following is true of the organism?



A. Oxidase and catalase negative

B. No capsule

C. β-hemolytic and lanceolate-shaped

D. α-hemolytic and lanceolate-shaped

---

30. A 55-year-old patient had a tooth extraction three days back. He now presents with a mass in the submandibular region, which is otherwise asymptomatic. There is no ulcer on the oral mucosa. On drainage of the discharge, it was foul-smelling with a lot of yellow granules. What is the most likely implicated pathogen?

A. Fonsecaea pedrosoi

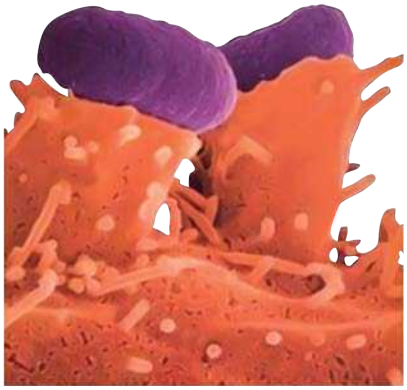
B. Mucor

C. Actinomyces israelli

D. Histoplasma capsulatum

---

31. Which pathogen causes attachment & effacement lesions in the intestinal mucosa, as shown in the image?



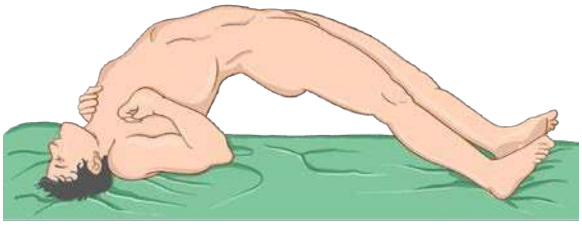
A. Entertoxigenic Esherichia coli

B. Enteropathogenic Escherichia coli

C. Enteroinvasive Eshcerichia coli

D. Diffusely adherent Escherichia coli

32. A 45-year-old man developed generalized muscle spasm with arching of his back, as shown in the image, a few days after sustaining a puncture wound to his leg. The toxin responsible for this type of abdominal posturing acts by which of the following mechanisms?



A. Presynaptic inhibitor of GABA and glycine release

B. Presynaptic inhibitor of Ach receptors

C. Postsynaptic inhibitor of GABA

D. Cleavage of SNARE proteins preventing Ach release.

33. BCYE medium is used to culture:

A. Leptospira

B. Bacillus cereus

C. Campylobacter

D. Legionella

---

34. A 16-year-old boy is admitted with fever, icterus, conjunctival injection, and hematuria for 15 days. Which serological test should be done for diagnosis?

A. Widal test

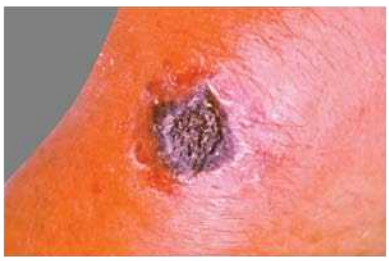
B. Microscopic agglutination test

C. Weil felix reaction

D. Paul Bunnell test

---

35. A veterinarian developed an ulcer with a central black eschar over the forearm, as shown in the image. Which of the following is the most probable causative organism?



A. Pseudomonas aeruginosa

B. Yersinia pestis

C. Bacillus anthracis

D. Francisella tularensis

---

36. Which of the following is not used as an immediate test in the screening of urinary tract infections (UTI)?

A. Pour plate culture

B. Griess Nitrite test

C. Leukocyte esterase test

D. Wet mount examination

---

37. What is the investigation of choice for neurosyphilis?

A. Venereal disease research laboratory (VDRL) test

B. FTA-ABS-Fluorescent Treponemal Antibody Absorption test

C. RPR – Rapid Plasma Reagin test

D. TPI – Treponema pallidum immobilization test

---

38. Contact based isolation is employed for which of the following diseases?

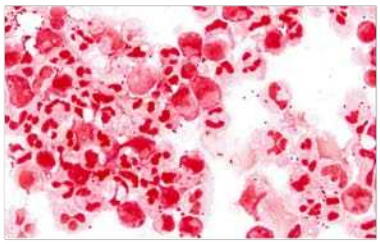
A. Diphtheria

B. MRSA

C. Mumps

D. Asthma

39. An 18 year old female presents with a severe headache and high fever of 101°F. On examination, papilloedema and nuchal rigidity was present. Gram staining of the CSF is shown below. Causative organism is



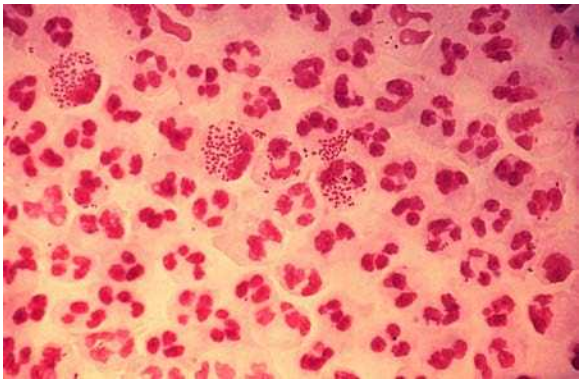
A. Haemophilus influenzae type B

B. Neisseria meningitidis

C. Stretptococcus pneumoniae

D. Listeria monocytogenes

40. A 12-year-old girl presents with a severe headache, photophobia, and a high fever of 101 degrees F. On examination, papilledema and nuchal rigidity was present. Kernig' and Brudzinski's signs were positive. CSF examination shows reduced glucose, increased protein, and increased leukocytes with the predominance of neutrophils. Gram staining of the CSF is shown below. The causative organisms are.



A. Haemophilus influenzae type B

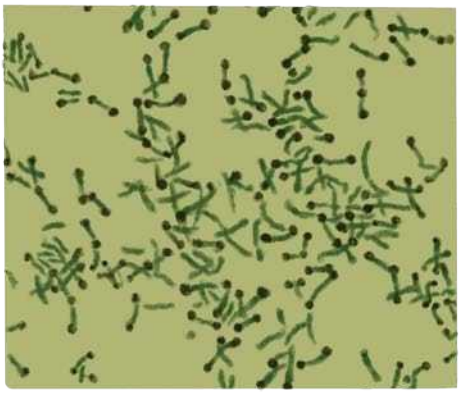
B. Neisseria meningitidis

C. Streptococcus pneumoniae

D. Listeria monocytogenes

---

41. A 4-year-old boy presents with complaints of fever and sore throat. On examination, he has extensive cervical swelling and a well-demarcated white-gray exudative substance adherent to his oropharynx and attempts to remove this membrane result in bleeding. The parents report that all his childhood vaccines were skipped due to religious beliefs. A throat swab under Albert's staining is as below. Which of the following is the most likely diagnosis?



A. Streptococcal pharyngitis

B. Diphtheria

C. Tetanus

D. Pertussis

---

42. Which of the following is the mechanism of action of cholera toxin?

A. Inhibit elongation factor -2

B. Toxin causes increased guanyl cyclase activity

C. Activates adenylate cyclase

D. Inhibit 60S subunit

---

43. Safety pin appearance is seen in:

A. Haemophilus influenzae

B. Burkholderia pseudomallei

C. Pseudomonas aeruginosa

D. Burkholderia cepacia

---

44. What is the causative organism of bull neck?

A. Corynebacterium diphtheriae

B. Corynebacterium jeikeium

C. Corynebacterium minutissimum

D. Corynebacterium urealyticum

---

45. A 34-year-old HIV-positive male presented with low-grade fever, cough, night sweats, and weight loss for the past 3 weeks. In view of TB investigations, the Mantoux test was negative; chest X-ray showed non-specific findings, and an adequate amount of sputum could not be collected for slide examination. Which of the following is used to diagnose latent TB infection in this patient?

A. CBNAAT

B. MGIT system

C. Interferon-gamma release assay

D. Line probe assay

---

46. A middle-aged man presents with a painless genital ulcer, as shown below. A Microscopy of a sample from the lesion is also shown below. Which of the following is the most likely diagnosis?





A. Granuloma inguinale

B. Lymphogranuloma venereum

C. Chancroid

D. Gonorrhoea

---

47. An El Tor vibrio was identified as the causative agent of a diarrhea epidemic. Which of the following is not a characteristic feature of this organism?

A. Resistant to El Tor phage V

B. Intrinsic resistance to polymyxin B

C. Voges-Proskauer test is positive

D. Not susceptible to group IV phage

---

48. GeneXpert detects mutations in which of the following genes?

A. InhA

B. rpoB

C. KatG

D. pncA

---

49. Which of the following statements about scrub typhus is false?

A. Transmitted by chiggers

B. Lymphadenopathy is a feature

C. Produces NLF producing colonies in MacConkey agar

D. Man is an accidental host

50. Pneumonia due to a gram-negative pleomorphic bacillus growing on chocolate agar, which is catalase and oxidase–positive is due to:

A. E. coli

B. Pseudomonas

C. Haemophilus influenzae

D. Actinomyces

51. A 30-year-old male presented to the OPD with painful genital ulcers. Which of the following can cause a painful ulcer on the glans penis ?

A. Treponema pallidum

B. Chlamydia trachomatis

C. H. ducreyi

D. Lymphogranuloma venereum

52. Match the following E. Coli with their respective pathogenesis: E.coli Mechanism of action 1. ETEC 2. EIEC 3. EHEC 4. EPEC a. Adheres to intestinal mucosa via EAC and causes effacement of microvilli b. Invades intestinal mucosal cell via virulence marker antigen c. Produces a verocytotoxin/Shiga toxin against vascular endothelium cells d. Invades intestinal mucosal cell via CFA

E.coli	Mechanism of action

1. ETEC 2. EIEC 3. EHEC 4. EPEC	a. Adheres to intestinal mucosa via EAC and causes effacement of microvilli b. Invades intestinal mucosal cell via virulence marker antigen c. Produces a verocytotoxin/Shiga toxin against vascular endothelium cells d. Invades intestinal mucosal cell via CFA

A. 1-a, 2-c, 3-2, 4-d

B. 1-b, 2-c, 3-a, 4-d

C. 1-c, 2-d, 3-a, 4-b

D. 1-d, 2-b, 3-c, 4-a

---

53. What does Burkholderia cepacia exhibit resistance to among the options provided?

A. Ceftazidime

B. Trimethoprim – sulfamethoxazole

C. Temocillin

D. Cefotetan

---

54. What is the causative agent for scombroid food poisoning among the following organisms?

A. Salmonella

B. Staphylococcus

C. Pseudomonas aeruginosa

D. Peptostreptococcus

---

55. Bacterial transduction occurs by

A. Plasmids

B. Sex pilli

C. Bacteriophage

D. Uptake of genetic material by other bacteria

---

56. TB test done only on whole blood sample is _____.

A. TrueNat TB

B. GeneXpert

C. Interferon gamma release assay

D. MGIT system

---

57. What causes the formation of multiple abscesses with discharging sinuses that show the presence of sulphur granules in the pus?

A. Actinomyces israelii

B. Nocardia

C. Salmonella

D. Tularemia

---

58. Which antibiotics are Mycoplasma naturally resistant to?

A. Beta lactam antibiotics

B. Macrolides

C. Tetracycline

D. Fluoroquinolones

59. What is the incubation period of lymphogranuloma venereum ?

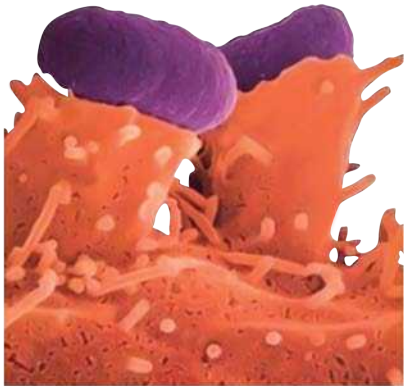
A. 1-5 days

B. 7-10 days

C. 3-35 days

D. 30-60 days

60. Identify the organism shown below, which causes effacement and attachment of intestinal cells.



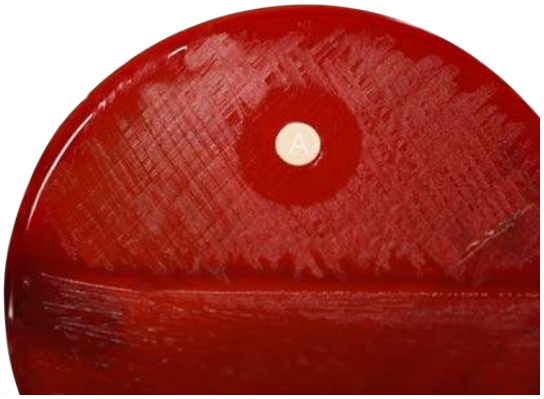
A. ETEC

B. EIEC

C. EPEC

D. EAEC

61. The test shown below, which uses a bacitracin disc, is useful in identifying ……



A. Staphylococcus aureus

B. Streptococcus pyogenes

C. Staphylococcus epidermidis

D. Streptococci agalctiae

---

62. A cook prepares sandwiches for 10 people going for picnic. Eight out of them develop severe gastroenteritis within 4-6 hrs of consumption of the sandwiches. It is likely that on investigations the cook is found to be the carrier of:

A. Salmonella typhi

B. Vibrio cholrea

C. Entamoeba histolytica

D. Staphylococcus aureus

---

63. Which of the following drugs is not used for typhoid fever in adults ?

A. Azithromycin

B. Ciprofloxacin

C. Ceftriaxone

D. Amikacin

---

64. A child presented to the outpatient department (OPD) with an inflamed and reddened knee joint, which is sensitive to touch. The child had experienced a fever and bloody diarrhea two weeks prior. Which of the following organisms could potentially be responsible for these symptoms?

A. E. coli

B. Shigella

C. Giardia

D. Rotavirus

---

65. A patient has complains of fever with chills. A throat examination reveals swollen glands and a thick, whitish membrane over his tonsils. Albert staining is to be done. What are the constituents of Albert's stain ?

A. Toluidine blue and malachite green

B. Carbol fuschin and methylene blue

C. Carbol fuschin and malachite green

D. Toluidine and methylene blue

---

66. Which is correct about Diphtheria membrane?

A. Grey membrane that bleeds on removal

B. Grey membrane that does not bleed on removal

C. Grey pseudomembrane that bleeds on removal

D. Grey pseudomembrane that does not bleed on removal

---

67. Bacillus Anthrax is:

A. Gram positive cocci in cluster

B. Gram-positive rods with square ends

C. Gram-positive cocci in chains

D. Gram negative cocci in cluster

---

68. What is the most probable cause in a case of granuloma with acid-fast bacilli positivity?

A. Cat scratch disease

B. Trench fever

C. Leprosy

D. Syphilis

---

69. Legionnaires disease is implicated in causation of which among the following?

A. Respiratory disease

B. U.T.I

C. Retroperitoneal fibrosis

D. Acute gastroenteritis

---

70. When should you perform a Widal test in a case of Typhoid?

A. 1st week

B. 2nd week

C. 3rd week

D. 4th week

---

71. Hemolytic uremic syndrome is associated with?

A. E. coli 0157

B. Malaria

C. Parvovirus B19

D. Bartonella henselae

---

72. Bacillus Anthrax is:

A. Gram positive cocci in cluster

B. Gram positive rods with square ends

C. Gram-positive cocci in chains

D. Gram negative cocci in cluster

---

73. HUS in children is caused by:

A. E. coli 0157: H7

B. Malaria

C. Parvovirus B19

D. Bartonella henselae

---

74. A 55-year-old female comes to the outpatient department complaining of fever, cough, muscle pain and headache. On taking occupational history, she reveals working for a centrally airconditioned corporate company. The doctor sent a sputum sample for culture sensitivity. The microbiologist used buffered charcoal yeast extract to culture the suspected organism. Which of the following is the most likely diagnosis?

A. Legionnaire disease

B. Brucellosis

C. Pneumonia caused by Haemophilus

D. Acute gastroenteritis

---

75. When should you perform Widal test in a case of typhoid?

A. 1st week

B. 2nd week

C. 3rd week

D. 4th week

---

76. The disruption of which of the following is responsible for watery diarrhea in cholera caused by Vibrio?

A. Intercalated discs

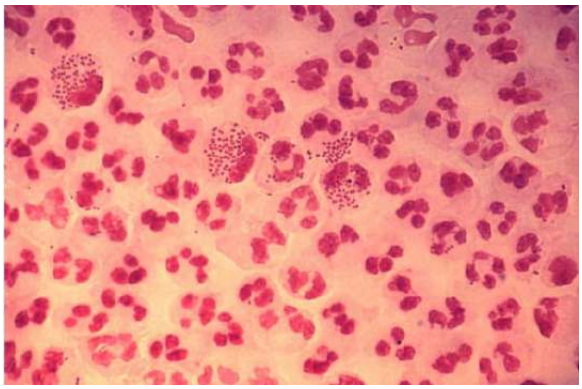
B. Macula densa

C. Zonula occludens

D. Hemidesmosomes

---

77. A patient is exhibiting symptoms of urethral discharge. A sample of the discharge is used to create a gram stain smear, revealing the provided microscopic image. Based on this information, what is the most probable organism responsible for the symptoms?



A. Trichomonas

B. Chlamydia trachomatis

C. Neisseria gonorrheae

D. Gardnerella

---

78. Yaws is caused by:

A. T. Pertunae

B. T. endemicum

C. T. Carateum

D. Hemophilus

---

79. Which bacteria show Quellung reaction?

A. Streptococcus pneumoniae

B. Clostridium tetani

C. Vibrio cholerae

D. Staphylococcus

---

80. In the small intestine, cholera toxin acts by:

A. ADP ribosylation of the G protein

B. Inhibition of adenyl cyclase

C. Activation of GTPase

D. Active absorption of NaCl

---

81. A hospital has reported an outbreak of MRSA infection. On investigation, it was found out that a staff nurse had nasal carriage of MRSA. Which of the following drugs helps in the removal of colonization of bacteria?

A. Oral vancomycin

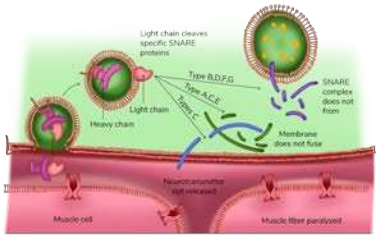
B. Inj cephalosporin

C. Topical mupirocin

D. Inhaled colistin

---

82. Which toxin acts via the mechanism depicted in the image below ?



A. Botulinum

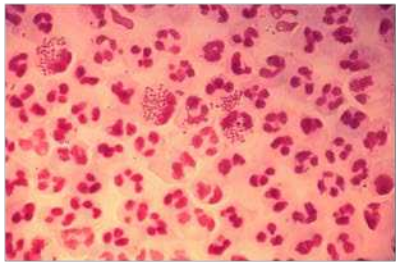
B. Tetanus

C. Diphtheria

D. Pertussis

---

83. A 25-year-old man presents with a history of urethral discharge for the last three days. A gram stain smear of the discharge is shown below. All of the following are true regarding the likely etiology except?



A. Catalase and oxidase positive

B. Ferments glucose

C. Ferments maltose

D. Shows twitching motility

84. What is the likely pathogen responsible for an 8-year-old boy's continuous diarrhea, starting 20 hours ago, which is rice watery in consistency, causing drowsiness, unresponsiveness, sunken eyes, and poor skin turgor?

A. Vibrio cholera

B. Entamoeba coli

C. Escherichia

D. Shigella sonnei

85. A 4-year-old boy developed bloody diarrhea and abdominal cramps. The bacteria isolated from the stool specimen was non motile and did not ferment lactose or mannitol. Which of the following organisms is implicated here?

A. Shigella dysenteriae

B. E. coli

C. Salmonella

D. Shigella sonnei

86. Mw vaccine is extracted from which bacteria, mentioned below :

A. M. indicus pranii

B. M. ulcerans

C. M.leprae

D. M. bovis

87. The toxin produced by the organism shown below acts via ?



A. Increase in cAMP

B. Increase in cGMP

C. ADP ribosylation of EF2

D. 60s ribosome

---

88. Yaws is caused by:

A. T. Pertunae

B. T. endemicum

C. T. Carateum

D. Hemophilus

---

89. Assertion: WHO recommends Typhoid vaccine for disaster management in endemic areas Reason: Vaccination is a cost – effective measure of disaster control

A. Both assertion and reason are true and the reason is the correct explanation for the assertion

B. Both assertion and reason are true, but the reason is not the correct explanation for the assertion

C. Assertion is true, but the reason is false

D. Both assertion and reason are false

---

90. A 12-year-old boy presents with right upper quadrant pain, calf pain, conjunctival suffusion, icterus, and fever. What is the most likely diagnosis?

A. Chikungunya

B. Leptospirosis

C. Dengue hemorrhagic fever

D. Encephalopathy caused by hepatitis A

---

91. Regarding Clostridium difficile infection, which of the following statements is false in the case of a woman prescribed a broad-spectrum antibiotic for recurrent diarrhea?

A. Oral fidaxomicin is used for treatment

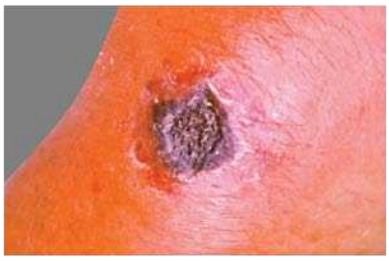
B. It is toxin mediated

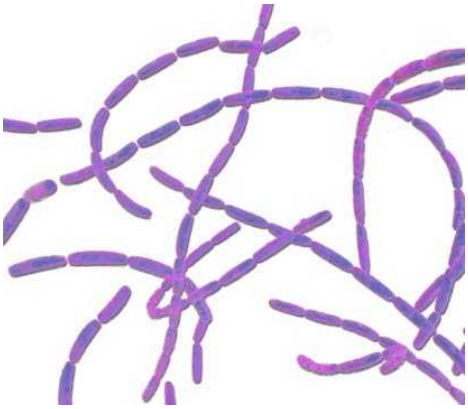
C. No risk of Fulminant infection

D. Pseudomembrane is a layer of inflammatory cells and debris

---

92. A cattle farmer presented with a lesion on the arm as shown below. Gram staining showed gram-positive rods as shown below. Identify the organism.





A. Bacillus cereus

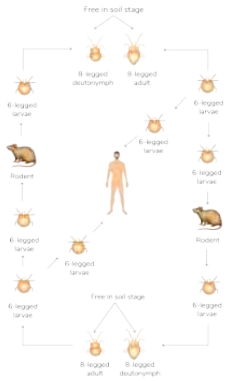
B. Bacillus anthracis

C. Pseudomonas aeruginosa

D. Brucella abortus

---

93. The life cycle of an organism is given below. Identify the vector involved



A. Hard tick

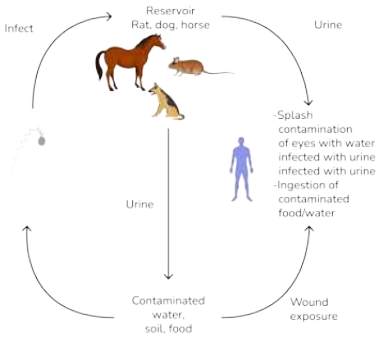
B. Soft tick

C. Trombiculid mite

D. Louse

---

94. Which of the following organisms belongs to the given life cycle?



A. Leptospira

B. Hantavirus

C. Yersinia pestis

D. Salmonella typhi

---

95. A patient has a fever, rash, and eschar formation due to a mite bite. What is the diagnosis?

A. Anthrax

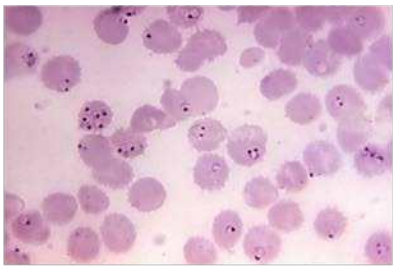
B. Scrub typhus

C. African tick typhus

D. Rocky mountain spotted fever

---

96. A 28-year-old woman with a recent travel history to Africa presents with fever, confusion and neurological deficits. Laboratory analysis reveals the following findings. This organism exhibits a specific property that facilitates sequestration in vascular beds to avoid splenic clearance. What is the phenomenon called?



A. Rosetting

B. Agglutination

C. Cytoadherence

D. Schizogony

---

97. A patient with meningitis was examined. His CSF shows lymphocytosis, decreased glucose, and increased protein. What is the etiological agent?

A. Streptococcus pneumoniae

B. Hemophilus influenza

C. Neisseria meningitidis

D. Mycobacterium tuberculosis

---

98. Which of the following is the primary identification made by a CBNAAT/ gene expert?

A. Mycobacterium tuberculosis

B. MTB+ Rifampicin resistance

C. MTB+ Isoniazid

D. MTB+ Ethambutol

---

99. A patient with acute rheumatic fever was presented in the OPD. Examination shows- gram-positive cocci in chains. The etiological agent is?



A. Streptococcus viridans

B. Streptococcus pyogenes

C. Streptococcus pneumoniae

D. Enterococcus

---

100. A farmer presents with a lesion on his arm after working with cattle on his farm. Gram staining of a sample from the lesion reveals gram-positive rods with square ends and central spores. What is the most likely organism causing this lesion?

A. Brucella abortus

B. Pseudomonas aeruginosa

C. Bacillus anthracis

D. Bacillus cereus

---

101. A 68-year-old woman was hospitalized for three days for left leg cellulitis. She was treated with clindamycin. Two days after her discharge, abdominal pain, and watery diarrhea. Lab results showed leukocytes 17,500/mm3, elevated ESR and CRP. What is the causative agent?

A. Escherichia coli

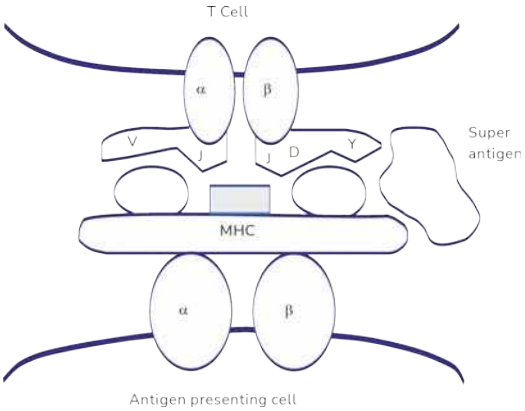
B. Norovirus

C. Clostridioides difficile

D. Rotavirus

---

102. Which of the following acts by the following MOA?



A. Vibrio cholerae

B. Streptococcus pyogenes

C. Clostridium difficile

D. Clostridium botulinum

---

103. A farmer presents to the outpatient department with complaints of fever. On physical examination examination, you note the findings shown in the following image. The patient's serum creatinine was 1.4, and platelets were 88000. What is your diagnosis?



A. Scrub typhus

B. Anthrax

C. Leptospirosis

D. Brucellosis

---

104. Which bacterial toxin acts by interfering with EF-2 function?

A. Diphtheria

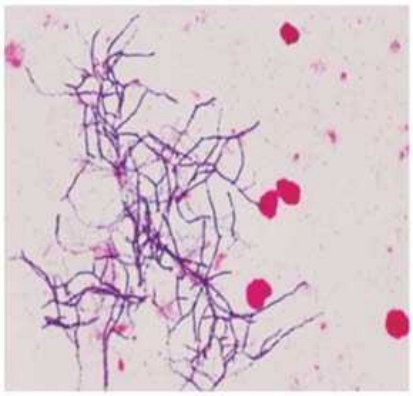
B. Clostridium

C. Bacillus

D. Vibrio

---

105. Identify the organism and staining



A. Nocardia and acid-fast stain

B. Streptococcus and gram stain

C. Cryptococcus and India ink

D. Coccidioidomycosis and PAS stain

---

106. Organism showing movement transcellular by ACTIN FILAMENTS?

A. Listeria

B. EIEC

C. Salmonella

D. None of above

---

107. A 26-year-old male presents to the physician with complaints of fever, abdominal pain and diarrhoea for 3 days. He further gives the history of consumption of raw fish 3 days back. The patient is otherwise vitally stable, and other general physical examinations reveal no abnormalities. Further investigation reveals that there is no pus in the stool. Culture grew in Wagatsuma agar. Which causative agent is the most likely to have caused this illness?

(or)

A 26-year-old male presents with fever, abdominal pain and diarrhoea for 3 days after consumption of raw fish 3 days back. Culture grew in Wagatsuma agar. Which causative agent is the most likely?

A. Vibrio cholera

B. Salmonella typhi

C. Shigella sonnei

D. Vibrio parahaemolyticus

---

108. A 54-year-old female with a history of rheumatic heart disease underwent prosthetic valve replacement surgery 6 months ago. He presents to the cardiology clinic with complaints of fever, fatigue, and worsening shortness of breath over the past few weeks. Physical examination reveals a new heart murmur and splinter hemorrhages on the nail beds. Based on the clinical presentation, what is the most likely causative organism responsible for this patient's infective endocarditis?

A. Staphylococcus epidermidis

B. Streptococcus viridans

C. Enterococcus faecalis

D. Staphylococcus saprophyticus

---

109. A 10-year-old child presents to the pediatric clinic with complaints of joint pain, swelling, and difficulty in walking. On examination, the child is found to have a tender, swollen knee joint and a history of recent sore throat. Laboratory investigations reveal elevated antistreptolysin O (ASO) titers. A renal biopsy is performed, and it shows glomerular changes consistent with PSGN. Which of the following organisms is most likely responsible for both the acute rheumatic fever and PSGN in this patient?

A. Streptococcus pneumoniae

B. Streptococcus pyogenes

C. Streptococcus agalactiae

D. Streptococcus mutans

---

110. A neonate admitted with the features of meningitis. Bacteria isolated from CSF is bacitracin resistant. Neonatal meningitis is caused by which organism?

A. Pneumococcus

B. Streptococcus pyogenes

C. Streptococcus agalactiae

D. Meningococci

---

111. A 28 year old patient presented with watery diarrhoea and dysentery. After taking history, it is noted that he had eaten Oysters on his recent trip. Which of the

following can be responsible for his symptoms?

A. Vibrio Parahemolyticus

B. Staph aureus

C. E. coli

D. Vibio Cholerae

---

112. Among the following conditions, which one does not result from exotoxins?

A. Meningitis

B. HUS

C. Toxic shock syndrome

D. Pseudomembranous colitis

---

113. A patient from Uttarakhand presented in the month of September with fever and myalgia after a bite with a mite. The patient was started on IV ceftriaxone, but it did not help; later, the patient noticed cutaneous lesions on examination (black eschar), as given below in the figure. What is your diagnosis ?



A. Scrub typhus

B. Murine typhus

C. Q fever

D. Louse borne typhus

---

114. A patient presented with severe watery diarrhea resembling that of rice gruel stool. Damage of which structure is likely to have resulted in this diarrhea?

A. Zona occludens

B. Gap junction

C. Adherens junction

D. Hemidesmosomes

---

115. Match the following? 1. Patch test A. Leprosy 2. KOH mount B. Lupus Vulgaris 3. Biopsy C. Contact dermatitis 4. Slit skin smear D. Tinea

1. Patch test	A. Leprosy
2. KOH mount	B. Lupus Vulgaris
3. Biopsy	C. Contact dermatitis
4. Slit skin smear	D. Tinea

A. 1- C, 2- D, 3- B, 4- A

B. 1- A, 2- D, 3- B, 4- C

C. 1- D, 2- C, 3- B, 4- A

D. 1- B, 2- D, 3- C, 4- A

---

116. Assertion and Reason: Assertion: Myocarditis is a common complication of faucial diphtheria Reason: It is due to bacteremia by diphtheria

A. Assertion is true, reason is not the correct explanation for the assertion

B. Assertion is false, reason is false

C. Assertion is true and reason is false

D. Assertion is false and reason is true

---

117. A patient with a history of penicillin allergy has been diagnosed with Enterococcus infective endocarditis. Which of the following antibiotics is the most appropriate treatment option?

A. Vancomycin

B. Ceftriaxone

C. Piperacillin

D. Aztreonam

---

118. In which of the following bacteria is increased penicillin drug resistance is noted?

A. Donovanosis

B. Neisseria gonorrhoeae

C. Chlamydia

D. Syphilis

---

119. Which among the following is the most specific disease of Streptococcus agalactiae?

A. Neonatal sepsis

B. Infective endocarditis

C. Dental caries

D. Pharyngitis with or without tonsillitis

---

120. A patient came to the OPD with an injury caused by a rusted metal. What is the mechanism of action of Tetanus?

A. Inhibition of protein synthesis

B. Inhibit the release of Ach

C. Inhibit the release of GABA and Glycine

D. Inhibit the release of Norepinephrine

---

121. Which of the following diseases is spread via all components of blood?

A. Syphilis

B. Toxoplasmosis

C. Malaria

D. Filarial infection

---

122. A female patient presents with swollen Bartholin glands. Discharging pus showed gram negative intracellular diplococci. Which is the organism?

A. Neisseria gonorrhoeae

B. Chlamydia

C. Chancroid

D. Syphilis

---

123. Match the following types of Maltoma with their associated microbial etiology: 1. Gastric Maltoma A. Chlamydia 2. Ocular Maltoma B. Borrelia 3. Skin Maltoma C. Campylobacter 4. Small bowel Maltoma D. Helicobacter

1. Gastric Maltoma	A. Chlamydia
2. Ocular Maltoma	B. Borrelia
3. Skin Maltoma	C. Campylobacter
4. Small bowel Maltoma	D. Helicobacter

A. 1- B, 2- A, 3- D, 4- C

B. 1- A, 2- C, 3- D, 4- B

C. 1- C, 2- A, 3- D, 4- B

D. 1- D, 2- A, 3- B, 4- C

---

124. Relative bradycardia is seen in

A. Dengue

B. Brucella

C. Bordetella

D. Leptospirosis

---

125. A patient presents with a history of watery diarrhea and vomiting. The pathogen exhibited darting motility. Which is the best selective media for the involved pathogen?

A. Nutrient media

B. TCBS

C. Cetrimide agar

D. Potassium Tellurite Agar

---

126. Which of the following is the mechanism of action of diphtheria toxin?

A. ADP phosphorylation of EF-2 - Decreasing protein synthesis

B. Increasing cAMP

C. Increasing cGMP

D. Decreasing 60s ribosome

---

127. A 40-year-old farmer presents with retro-orbital pain, burning red eyes, conjunctival suffusion, fever with chills, calf tenderness, and joint pain. What is the most likely diagnosis?

A. Dengue

B. Malaria

C. Leptospirosis

D. Filariasis

---

128. A patient from a military background presents with fever, a rash, and neck rigidity. Laboratory investigations reveal a lack of membrane-attacking complex. What is the most likely causative agent?

A. Neisseria meningitidis

B. Neisseria gonorrhoea

C. Klebsiella

D. CMV

129. A female patient with a urinary tract infection (UTI) has more than 10^4 colonies of novobiocin-resistant Staphylococcus. What is the appropriate course of action?

A. Consider UTI

B. Take three urine samples

C. No treatment is needed

D. Contaminated sample

130. A child presents with chorioretinitis, hydrocephalus, and cerebral calcifications. What is the most likely causative organism?

A. Toxoplasma

B. CMV

C. Rubella

D. Zika

131. A patient has a history of infection by Enterohemorrhagic Escherichia coli (EHEC) 0157, H7. What does the "H" represent?

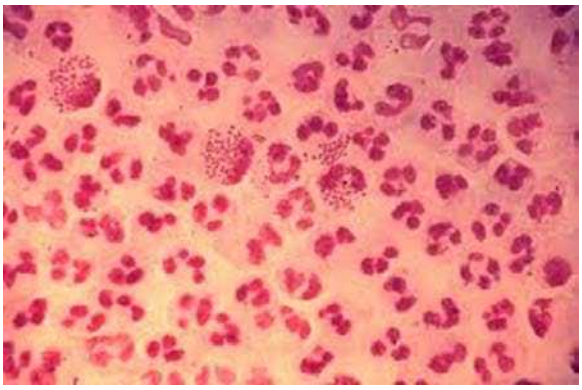
A. Capsule

B. Flagella

C. Fimbriae

D. Lipopolysacharides

132. An adult man presents with a purulent discharge from the urethra with a history of unprotected sexual intercourse. On microscopy, gram-negative intracellular cocci are noticed. What is the drug of choice (DOC) for this patient?



A. Tab Cefixime 400mg

B. Tab Doxycycline 200mg

C. Tab Acyclovir

D. Inj. Benzathine Penicillin

---

133. A regular contact lens user presents with corneal ulcers and diminished vision. He was very careless about his lens fluid. On microscopy, an organism with spikes is identified. What is the causative organism?

A. Acanthamoeba

B. Naegleria

C. Balamuthia

D. Giardia lamblia

---

134. A traveler with a history of oyster consumption presents with diarrhea, abdominal pain, and vomiting. What is the most likely causative agent?

A. S. Typhi

B. V. parahemolyticum

C. Staphylococcus

D. Cholera

---

135. A farmer suffered a wound on the dorsum of his foot. It later involved the subcutaneous tissue and became a discharging ulcer. Microscopy reveals gram-positive branching filaments. Identify the organism.

A. Nocardia

B. Sporothrix

C. Histoplasma

D. Candida

---

136. Patient with HIV presenting with bouts of cough and fever. Which medium is appropriate for diagnosis?

A. Tinsdale media

B. Lowenstein-Jensen (LJ) medium

C. MacConkey agar

D. Loeffler serum slope (LSS)

---

137. A firefighter was admitted with fever, cough, culture shown in the image reveals gram-negative, oxidase-positive bacilli. Colonies transilluminating while passing under UV light. Choose the correct agent.



A. Klebsiella

B. Staphylococcus aureus

C. Pseudomonas aeruginosa

D. Serratia

---

138. A group of students consumed fried rice in a restaurant. Six hours later, they start vomiting. What is the etiological agent?

A. Bacillus cereus

B. Clostridium botulinum

C. Escherichia coli

D. Shigella dysentriae

---

139. A farmer presents with fever and subconjunctival hemorrhage. The microscopic agglutination test turns to be positive. What is the diagnosis?

A. Brucella

B. Leptospira

C. E coli

D. Staph aureus

140. A 70-year-old man was admitted with a murmur and blood culture positive for Gram-positive cocci. Choose the common agent causing prosthetic valve infective endocarditis.

A. Streptococcus viridans

B. Staphylococcus epidermidis

C. Staphylococcus aureus

D. Enterococci

141. A patient presents with fever, neck rigidity, headache, vomiting, and petechial rashes on the lower limbs. What is the most likely causative organism?

A. Neisseria meningitidis

B. Clostridium perfringens

C. Cryptococcus neoformans

D. Clostridium difficile

142. A patient's laboratory results were positive for malarial antigen, and the Widal test findings were as follows: O ag >1:300, H ag >1:20, A >1:20, B >1:20. What is the most likely diagnosis?

A. Recent infection by Salmonella spp

B. Patient has taken TAB vaccine

C. Past infection

D. Reconvalescence

143. MOA of botulinum toxin is

A. Inhibits Elongation factor 2

B. Inhibits the release of GABA

C. Inhibits the release of Acetylcholine

D. Increases cAMP

---

144. A person with the H/o long-term sexual relationship presented with painful ulcers on his genitals with tender lymphadenopathy. What is the diagnosis?

A. Gonorrhoea

B. Chlamydia

C. LGV

D. Chancroid

---

145. A patient admitted with fever and nuchal rigidity. CSF analysis shows decreased glucose and increased protein and neutrophils. Choose the correct treatment

A. Ampicillin

B. Ceftriaxone

C. Penicillin

D. Vancomycin

---

146. KAT G mutation associated with:

A. Isoniazid

B. Rifampicin

C. Levofloxacin

D. Ethambutol

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	2
Question 4	2
Question 5	4
Question 6	2
Question 7	1
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	3
Question 13	1
Question 14	1
Question 15	1
Question 16	1
Question 17	1
Question 18	1
Question 19	1
Question 20	4
Question 21	3
Question 22	2

Question 23	2
Question 24	3
Question 25	3
Question 26	1
Question 27	2
Question 28	2
Question 29	4
Question 30	3
Question 31	2
Question 32	1
Question 33	4
Question 34	2
Question 35	3
Question 36	1
Question 37	1
Question 38	2
Question 39	2
Question 40	2
Question 41	2
Question 42	3
Question 43	2
Question 44	1
Question 45	3
Question 46	1
Question 47	1
Question 48	2

Question 49	3
Question 50	3
Question 51	3
Question 52	4
Question 53	4
Question 54	3
Question 55	3
Question 56	3
Question 57	1
Question 58	1
Question 59	3
Question 60	3
Question 61	2
Question 62	4
Question 63	4
Question 64	2
Question 65	1
Question 66	3
Question 67	2
Question 68	3
Question 69	1
Question 70	2
Question 71	1
Question 72	2
Question 73	1
Question 74	1

Question 75	2
Question 76	3
Question 77	3
Question 78	1
Question 79	1
Question 80	1
Question 81	3
Question 82	1
Question 83	3
Question 84	1
Question 85	1
Question 86	1
Question 87	3
Question 88	1
Question 89	4
Question 90	2
Question 91	3
Question 92	2
Question 93	3
Question 94	1
Question 95	2
Question 96	3
Question 97	4
Question 98	2
Question 99	2
Question 100	3

Question 101	3
Question 102	2
Question 103	1
Question 104	1
Question 105	1
Question 106	1
Question 107	4
Question 108	1
Question 109	2
Question 110	3
Question 111	1
Question 112	1
Question 113	1
Question 114	1
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Question 125	2
Question 126	1

Question 127	3
Question 128	1
Question 129	1
Question 130	1
Question 131	2
Question 132	1
Question 133	1
Question 134	2
Question 135	1
Question 136	2
Question 137	3
Question 138	1
Question 139	2
Question 140	2
Question 141	1
Question 142	1
Question 143	3
Question 144	4
Question 145	2
Question 146	1

Solution for Question 1:

Correct Option B - R. prowazekii:

• The clinical presentation of a febrile patient with rash sparing palms and soles and lice infestation is suggestive of epidemic typhus, which is caused by the bacterium Rickettsia

prowazekii.

• Rickettsia prowazekii is a gram-negative, obligate intracellular bacterium that is transmitted to humans through the bites of infected lice.

• The bacterium infects endothelial cells, causing vasculitis and capillary leakage, which leads to the characteristic rash and other systemic manifestations.

• Treatment for epidemic typhus involves the use of antibiotics such as doxycycline or chloramphenicol. Supportive care may also be necessary, including management of hypotension and fluid resuscitation.

Incorrect Options:

Option A - Rickettsia typhi: Rickettsia typhi causes endemic or murine typhus, which is typically transmitted to humans through the bites of infected fleas. The clinical presentation of endemic typhus is similar to epidemic typhus, but the rash typically appears on the trunk and spreads to the extremities, including the palms and soles.

Option C - Rickettsia akari: Rickettsia akari causes rickettsial pox, which is typically transmitted to humans through the bites of infected mites. The clinical presentation of rickettsialpox includes fever, headache, and a papulovesicular rash that typically starts at the site of the mite bite and spreads to other areas of the body.

Option D - Rickettsia conorii: Rickettsia conorii causes Indian tick typus, which is typically transmitted to humans through the bites of infected ticks. The clinical presentation includes fever, headache, myalgias, and a maculopapular rash that typically starts on the extremities and spreads to the trunk. The palms and soles may be involved in some cases.

Solution for Question 2:

Correct Option A - GM 1 Ganglioside Receptor:

• The presence of rice water stools suggests the causative agent is Vibrio cholerae.

• Vibrio cholerae produces cholera toxin, which is coded by the CTx gene, and it is bipartite.

• Cholera toxin has 2 fragments, A and B.

• Fragment B has 5 subunits responsible for the attachment to the jejunal epithelial cells through GM 1 ganglioside.

• Fragment A has 2 subunits, A1 and A2. A2 fragments link the active subunit A1 with the B fragment. Fragment A1 is biologically active subunit responsible for toxic effect, Activate

adenyl ate cyclase. Phosphorylation of cystic fibrosis transmembrane protein. Ions secretedin lumen, responsible for rice water stools.

• Activate adenyl ate cyclase.

• Phosphorylation of cystic fibrosis transmembrane protein.

• Ions secreted in lumen, responsible for rice water stools.

• Activate adenyl ate cyclase.

• Phosphorylation of cystic fibrosis transmembrane protein.

• Ions secreted in lumen, responsible for rice water stools.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 3:

Correct Option B - Toxin assay:

• Clostridium difficile is a bacterium that can cause pseudomembranous enterocolitis characterized by watery diarrhea.

• The most common method for diagnosis of C. difficile is the detection of the toxins A or B in stool by ELISA.

• Other diagnosing methods are:

• Colonoscopy: 100% specificity, 50% sensitivity

• Media: CCFA: Cefoxitin Cycloserine Fructose Agar CCYA: Cefoxitin Cysteine Yeast Extract Agar

• CCFA: Cefoxitin Cycloserine Fructose Agar

• CCYA: Cefoxitin Cysteine Yeast Extract Agar

• GDH assay: It is present in both toxigenic and non-toxigenic strains, therefore called as C. difficile common antigen. Detection of GDH, in the absence of toxin A/B, indicates the presence of non-toxigenic strains of C. difficile in the specimen.

• CCFA: Cefoxitin Cycloserine Fructose Agar

• CCYA: Cefoxitin Cysteine Yeast Extract Agar

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 4:

Correct Option B - History of exposure to cat:

- The presentation of a blister on the hand, followed by axillary lymphadenopathy, is highly suggestive of cat scratch disease among the given options
- It occurs when an infected cat licks a person's open wound or bites or scratches a person hard enough to break the surface of the skin. It is caused by the bacteria Bartonella henselae, characterized by a febrile illness with lymphadenopathy.
- In immunocompromised patients, it causes bacillary angiomatosis, in which vascular nodules appear on skin and mucosa.
- B. henselae can be demonstrated in lymph node biopsy with Warthin-Starry staining.

Incorrect Options:

Option A - History of rat bite: Rat bite can cause rabies, but it does not present with regional lymphadenopathy.

Option C - History of similar complaints among family members: A history of similar illness among family members is not relevant in the case of blister with regional lymphadenopathy.
Option D - History of louse bite: A louse bite can cause epidemic typhus characterized by fever and a rash starting from the trunk and spreading to the limbs sparing palms and sole without regional lymphadenopathy.

Solution for Question 5:

Correct Option D - Staphylococcus Aureus:

- The clinical presentation and history of intravenous drug abuser is suggestive of infective endocarditis (IE) with valvular involvement.

• In IV drug abusers, the most common organism causing infective endocarditis is Staphylococcus aureus.

Condition	Most common organism
Most common organism (overall)	Staphylococcus aureus
Community-acquired native valve endocarditis	Streptococci
Healthcare-associated native valve endocarditis	Staphylococcus aureus
Endocarditis in injection drug users (Right side)	Staphylococcus aureus
Endocarditis in injection drug users (Left side)	Enterococci and Staphylococcus
Single most common organism causing prosthetic valve endocarditis	Staphylococcus aureus
Most common group of organisms causing early prosthetic valve endocarditis (<12 months after surgery)	Coagulase-negative staphylococci (CoNS)

Incorrect Options:

Options A, B, and C are incorrect: These pathogens do not cause infective endocarditis in IV drug abusers.

Solution for Question 6:

Correct Option B - Bacillus Cereus:

• Bacillus cereus is a gram-positive, spore-forming bacteria that causes food poisoning.

• It causes two types of food poisoning emetic type and diarrheal type depending on the toxins formed

Features	Emetic type	Diarrheal type
IP	<6 hours	8-16 hours
Food	Chinese fried rice	Meat and milk
Heat	Heat stable	Heat labile
MOA	Increased cGMP	Increased cAMP
Toxins	Cerulide: vagomimetic action	Nhe toxins: secretory diarrhea

Incorrect Options:

Option A - Clostridium Perfringens: Clostridium perfringens is also a gram-positive, spore-forming bacteria that can cause food poisoning. It is associated with cold or warmed up meat dishes.

Option C - Staphylococcus Aureus: Staphylococcus aureus is a gram-positive bacterium that can cause food poisoning. It is associated with dairy products, meat, and egg dishes that have been left at room temperature for extended periods.

Option D - Bacillus Anthracis: Bacillus anthracis is a gram-positive, spore-forming bacteria that causes anthrax. It is not associated with food poisoning.

Solution for Question 7:

Correct Option A - Widal Test:

- The patient presents with symptoms of enteric fever, a systemic illness caused by Salmonella typhi or Salmonella paratyphi. A step-ladder pattern fever is a characteristic feature.
- The best diagnostic method for enteric fever in the second week of symptom onset is the Widal test, which detects antibodies against S. typhi and S. paratyphi in the patient's serum.
- It is a serological test that detects antibodies against O (surface) and H (flagellar) antigens.
- An antibody titer of greater than 1:60 for O antigen and greater than 1:180 for H antigen, is considered a cutoff level to predict recent infection of typhoid fever in an endemic area.

Incorrect Options:

Option B - Stool Culture: This is helpful in diagnosing enteric fever in the 3rd week of infection.

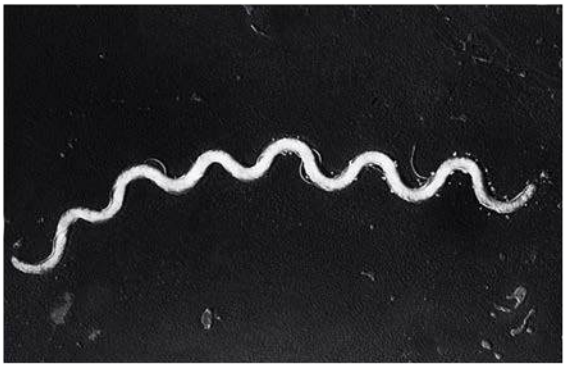
Option C - Blood Culture: Blood culture remains the primary mechanism of confirmation of typhoid fever in the 1st week of the infection.

Option D - Urine Culture: Urine culture is helpful in diagnosing enteric fever in the 4th week of the infection.

Solution for Question 8:

Correct Option A - Dark Field Microscopy:

- The patient has a painless indurated genital ulcer after having unsafe intercourse, which is a significant finding of primary syphilis and is called a chancre.
- Syphilis is caused by Treponema pallidum, which is a spirochete that is sexually transmitted.
- Syphilis is diagnosed by dark field microscopy.
- Exudates from the chancre are observed under dark field microscopy where spiral-shaped slowly moving Treponema pallidum are identified.



Incorrect Options:

Option B - HSV 2 Antibody: Herpes simplex virus-2 causes herpetic ulcers in the genital area that are extremely painful to touch.

Option C - Albert Stain: It is used to stain volutin granules of Corynebacterium diphtheriae. C. diphtheriae causes the disease diphtheria which is a disease of the upper respiratory tract.

Option D - Acid-fast Stains: It is used to differentiate between acid-fast and non-acid fast bacteria. Acid-fast bacteria include the bacteria of the mycobacteria genus. The most common mycobacteria are M. tuberculosis and M. leprae, infections of which do not present as painless ulcers in the genital region.

Solution for Question 9:

Correct Option B - Secondary syphilis:

• The patient presents a rash involving the palms of the hand, maculopapular rash on the scrotum (condylomata lata), and generalized lymphadenopathy with a history of high-risk sexual are suggestive of secondary syphilis. It occurs after 1 - 3 months of primary syphilis.

Incorrect Options:

Option A - Psoriasis: It is a dermatological condition that causes itchy, scaly patches over the bony prominences of the body, most commonly. It is not caused by unsafe sexual practices. The lesions present on the patient do not correspond with those of psoriasis.

Option C - Primary syphilis: Primary syphilis is characterized by an indurated, painless genital ulcer with inguinal lymphadenopathy.

Option D - Tertiary syphilis: Tertiary syphilis is characterized by formation of gummas on the skin, neurological symptoms like general paresis and tabes dorsalis,

Solution for Question 10:

Correct Option B - Microscopic Agglutination Test:

• The symptoms described are suggestive of Weil's disease aka hepatorenal syndrome, a bacterial infection caused by pathogenic Leptospira interrogans characterized by two stages, i.e.,

	First Stage/septicemic phase	Second Stage/immune phase
Features	High grade fever Liver: Jaundice & increased Liver enzymes Hemorrhages Pulmonary Skin Conjunctival Kidney: Raised creatinine level and impending RF	All features of septicaemic phase can be present
Isolation	Blood & CSF No IgM	From urine IgM present

• Leptospirosis is commonly associated with occupational exposures, such as contact with contaminated water or soil, which is relevant to a sewage worker's job.

• The microscopic agglutination test (MAT) is the most appropriate and specific serological test for diagnosing leptospirosis. It detects specific antibodies produced by the patient in response to Leptospira infection. The test involves exposing a patient's serum sample to a panel of live Leptospira serovars and observing for agglutination (clumping) of the bacteria.

• The highest dilution of the serum that shows agglutination is reported as the titer and helps determine the serovar(s) of Leptospira involved in the infection.

Incorrect Options:

Option A - Widal test: The Widal test is used to diagnose typhoid fever. It detects antibodies against Salmonella typhi, the causative agent of typhoid fever, in a patient's serum. The symptoms and occupational history described in the case are not suggestive of typhoid fever.

Option C - Weil-Felix reaction: The Weil-Felix reaction is a serological test historically used to diagnose rickettsial infections, particularly typhus. It involves testing a patient's serum Proteus strains and observing for agglutination.

Option D - Paul-Bunnell test: The Paul-Bunnell test, also known as the Monospot test, is used to diagnose infectious mononucleosis, primarily caused by Epstein-Barr virus (EBV) infection.

Solution for Question 11:

Correct Option C - Exhibits satellitism around Staphylococcus aureus colonies:

• The child is suffering from meningitis. Common causes of meningitis in a child is due to Haemophilus influenza, Neisseria meningitidis, and Streptococcus pneumoniae. Among these organisms, the only pleomorphic, gram-negative bacilli is Haemophilus influenzae.

• Satellitism refers to a phenomenon where one bacterium grows around or near another bacterium, usually relying on the nutrients produced by the other bacterium.

• Haemophilus influenzae exhibits satellitism around Staphylococcus aureus colonies.

• Haemophilus influenzae requires factors like factor V and factor X produced by Staphylococcus aureus due to hemolysis of the RBCs. Therefore, it grows around the colonies of Staphylococcus aureus on culture media.

Incorrect Options:

Option A - It is bacitracin sensitive: Bacitracin sensitivity is typically used to differentiate different strains of Streptococcus, particularly in the context of respiratory infections such as pharyngitis.

Option B - Produces non-lactose fermenting colonies on MacConkey agar: This characteristic is more indicative of Enterobacteriaceae, which are gram-negative bacteria that typically ferment lactose and produce pink/red colonies on MacConkey agar.

Option D - It exhibits alpha hemolysis on sheep blood agar: Alpha hemolysis refers to a partial breakdown of red blood cells, resulting in a greenish discoloration of the agar around the colonies. This characteristic is commonly seen in Streptococcus pneumoniae, a gram-positive bacterium and one of the leading causes of bacterial meningitis.

Solution for Question 12:

Correct Option C - Ceftriaxone + Azithromycin:

- The patient's presentation suggests urethritis, which is most commonly caused by Neisseria gonorrhoeae. The gram-staining shows gram-negative diplococci with kidney bean shapes suggestive of N. gonorrhoeae.
- A combination of ceftriaxone and azithromycin is highly effective against N. gonorrhoeae. Single dose of 125 mg IM ceftriaxone, and a single dose of 1 gram azithromycin are considered the treatment of choice for urethritis in men.

Incorrect Options:

Options A, B, and D are incorrect. They are not the preferred regimen for the treatment of Neisseria gonorrhoeae.

Solution for Question 13:

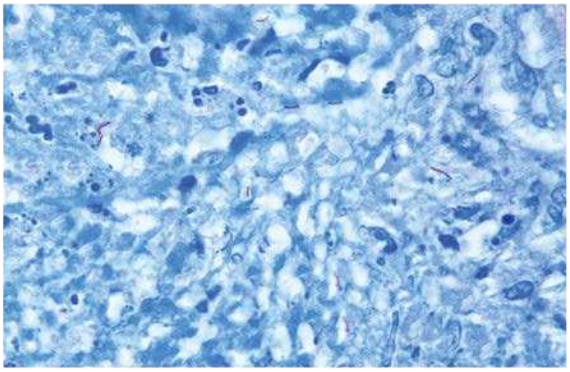
Correct Option A - Nocardia:

- The gram-positive organism on Ziehl-Neelsen staining, as shown in the image, is of Nocardia.
- It is seen as branching filamentous rods with a beaded appearance on modified Ziehl-Neelsen staining as shown in the image. Nocardia is decolorised using 1% acid

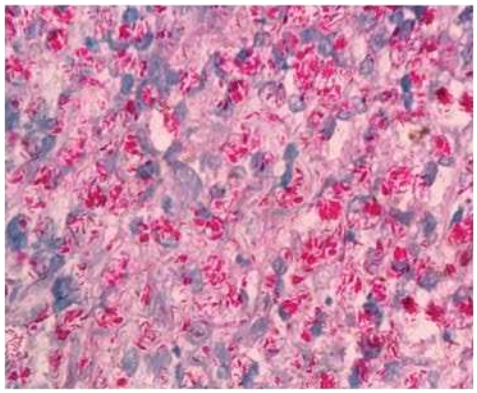
alcohol. Nocardia are weakly acid-fast following staining with the modified Ziehl-Neelsen.

Incorrect Options:

Option B - Mycobacterium tuberculosis: It is seen as small, rod-shaped bright red bacilli on ZN staining.



Option C - Mycobacterium leprae: Leprae bacilli appears as arranged in parallel bundles inside macrophages (lepra cell).



Option D - Actinomyces: It is strict anaerobe, non-acid fast and shows antler like branching/filamentous branching.

Solution for Question 14:

Correct Option A - Ureaplasma urealyticum:

• Mollicutes are a class of bacteria characterized by the absence of a cell wall. They can lead to disease in humans by attaching to cells of respiratory or urogenital tracts, particularly species of Mycoplasma and Ureaplasma.

Incorrect Options:

Options B, C, and D are not Mollicutes.

Solution for Question 15:

Correct Option A - Motile at 22° C and non-motile at 37° C:

• The most common cause of meningitis in an elderly patient more than 60 years are Streptococcus agalactiae and Listeria monocytogenes.

• The image shows gram-positive bacilli; therefore, Listeria monocytogenes is the causative agent of meningitis in this elderly patient.

• It shows differential motility, i.e., motile at room temperature (20 - 25°C) but not at 37°C. This characteristic can be helpful in distinguishing Listeria from other non-motile bacteria. The motility of Listeria monocytogenes is due to the presence of peritrichous flagella, which are whip-like structures

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 16:

Correct Option A - Staphylococcus aureus:

• The history of taking pastry and presentation of nausea and vomiting early in the morning is suggestive of food poisoning caused by enterotoxin of the Staphylococcus aureus.

• Staphylococcus aureus can produce enterotoxins in dairy items and meat that cause rapid-onset symptoms of nausea and vomiting, typically occurring within a few hours (within

6 hours) of ingesting contaminated food.

Incorrect Options:

Option B - Emetic toxin of B. Cereus: Emetic toxin is a heat-stable enterotoxin known for causing vomiting. This Emetic toxin is present mostly in Chinese fried rice.

Option C - Verocytotoxin from E. Coli: Verocytotoxin is a toxin that resembles Shiga toxin in structure and function. It causes Hemolytic Uremic Syndrome, characterized by watery diarrhea that develops into bloody stools.

Option D - Shiga toxin from shigella: Shiga toxin is produced by Shigella causing dysentery.

Solution for Question 17:

Correct Option A - Neisseria gonorrheae:

- Urethral seed discharge (semen-like yellow color) one week after unprotected sexual intercourse is suggestive of urethritis most commonly caused by Neisseria gonorrhoeae.
- N. gonorrheae infects the mucous membranes of the reproductive tract, including the cervix, uterus, fallopian tubes, and urethra in women, and the urethra and prostatitis in men.

Incorrect Options:

Option B - Ureaplasma urealyticum: It causes the majority of the non-gonococcal urethritis. The discharge seen in usually purulent not yellow.

Option C - Herpes simplex virus: It is a common infection that can cause painful blisters or ulcers. It primarily spreads by skin-to-skin contact.

- HSV-1 mostly spreads by oral contact and causes infections in or around the mouth (oral herpes or cold sores).
- HSV-2 spreads by sexual contact and causes genital herpes.

Option D - Trichomonas vaginalis: It is a sexually transmitted disease and causes strawberry cervix, urethritis, and vaginitis characterized by greenish discharge.

Solution for Question 18:

Correct Options A - Staphylococcus Aureus:

- Staphylococcus aureus is a bacterium known to produce toxins that can cause food poisoning.
- Staphylococcus aureus can produce enterotoxins in dairy items and meat that cause rapid-onset symptoms of nausea and vomiting, typically occurring within a few hours (within 6 hours) of ingesting contaminated food.

Incorrect Options:

Option B - Clostridium Botulinum: It is associated with food poisoning caused due to preform toxins in preserved food known as botulism. The symptoms of botulism typically take longer to develop, usually between 12 and 36 hours after ingestion.

Option C - Clostridium Perfringens: It is another bacterium that can cause food poisoning, but its onset of symptoms usually occurs within 8 to 24 hours after ingestion.

Option D - Salmonella Typhi: It is the causative agent of typhoid fever, which is characterized by high fever, abdominal pain, constipation or diarrhea, headache, and other systemic symptoms. It typically has a more gradual onset, usually appearing after several days to weeks of exposure to contaminated food or water.

Solution for Question 19:

Correct Option A - Neisseria Meningitidis:

The presentation of high-grade fever, confused state, and purpuric rash along with CSF smear showing gram-negative diplococci is suggestive of meningitis caused by Neisseria meningitidis.

Incorrect Options:

Option B - Pseudomonas Aeruginosa: This bacterium is a gram-negative rod associated with pneumonia and sepsis.

Option C - Streptococcus Pneumoniae: While Streptococcus pneumoniae is a common cause of bacterial meningitis, it typically presents as gram-positive cocci on Gram staining.

Option D - E. Coli: Although E. coli can cause urinary tract infections, diarrhea, sepsis, and neonatal meningitis, it typically presents as gram-negative rod shape bacteria on gram satining.

Solution for Question 20:

Correct Option D - H. Ducreyi:

- Haemophilus ducreyi is the bacterium that causes the sexually transmitted infection known as chancroid or soft sore characterized by tender, non-indurated irregular ulcer on the genitalia. It does not cause urethritis.
- Organisms causing urethritis are: Gonococcal urethritis: Nesseria gonorrhoeae. Non-gonococcal urethritis: caused by Chlamydia trachomatis (D-K) Genital mycoplasms: Ureaplasma urealyticum, Mycoplasma genitalium, M, hominis. Herpes simplex virus Candida albicans Trichomonas vaginalis
- Gonococcal urethritis: Nesseria gonorrhoeae.
- Non-gonococcal urethritis: caused by Chlamydia trachomatis (D-K) Genital mycoplasms: Ureaplasma urealyticum, Mycoplasma genitalium, M, hominis. Herpes simplex virus Candida albicans Trichomonas vaginalis
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- Herpes simplex virus
- Candida albicans
- Trichomonas vaginalis

- Chlamydia trachomatis (D-K)
- Genital mycoplasms: Ureaplasma urealyticum, Mycoplasma genitalium, M, hominis.
- Herpes simplex virus
- Candida albicans
- Trichomonas vaginalis

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 21:

Correct Option C - Sorbitol MacConkey agar:

- Sorbitol MacConkey agar is a selective and differential culture medium commonly used to isolate and differentiate E. coli O157:H7 from other E. coli species. E. coli O157:H7 is a pathogenic strain that does not ferment sorbitol, whereas most other E. coli strains do.
- The agar contains sorbitol as the fermentable carbohydrate source, along with bile salts and crystal violet, which inhibit the growth of Gram-positive bacteria and most Gram-negative bacteria other than E. coli.
- When E. coli O157:H7 is inoculated on Sorbitol MacConkey agar, it appears as colorless colonies, indicating that it does not ferment sorbitol. In contrast, other E. coli species that can ferment sorbitol will produce pink colonies due to the acid production from sorbitol fermentation.

Incorrect Options:

Option A - Mannitol salt agar: It is used to isolate and differentiate Staphylococcus aureus (yellow color) from S. epidermidis (no color change).

Option B - Xylose lysine deoxycholate agar: It is used to isolate and differentiate Salmonella and Shigella species based on their ability to ferment xylose and lactose and produce characteristic colonies.

Option D - CLED medium: Cystine Lactose Electrolyte Deficient is used to isolate and differentiate urinary microbes.

Solution for Question 22:

Correct Option B - Anthrax:

• Bioterrorism agents are categorized into three levels based on their potential threat to public health and national security.

• Category A agents are considered the highest priority due to their high potential for mass casualties, ability to spread easily, and the potential for major public health impact.

• Organisms include in bioterrorism category A are:

Category	Characteristics	Examples
Category A	High priority due to high potential for mass casualties, easy dissemination, and major public health impact	Anthrax (Bacillus anthracis), Smallpox (Variola virus), Botulism (Clostridium botulinum toxin), Plague (Yersinia pestis), Tularemia (Francisella tularensis), Viral hemorrhagic fevers (e.g., Ebola, Marburg)
Category B	Second-highest priority, moderate morbidity and low mortality rates	Brucellosis (Brucella species), Q fever (Coxiella burnetii), Glanders (Burkholderia mallei), Melioidosis (Burkholderia pseudomallei), Psittacosis (Chlamydia psittaci)
Category C	Third-highest priority, emerging pathogens that could be engineered for mass dissemination	Nipah virus, Hantaviruses, Tickborne hemorrhagic fever viruses (e.g., Crimean-Congo hemorrhagic fever virus), Drug-resistant tuberculosis

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 23:

Correct Option B - Streptococcus Sanguinis:

• The patients presentation is suggestive of infective endocarditis. Organisms causing infective endocarditis are: Staphylococcus aureus Coagulase-negative staphylococci (e.g., Staphylococcus epidermidis) Streptococci (Viridans streptococci and others) Enterococci Pneumococci Fastidious gram-negative coccobacilli (HACEK group) Enterobacteriaceae Pseudomonas spp. (usually in drug users) Candida species Diphtheroids

• Staphylococcus aureus

• Coagulase-negative staphylococci (e.g., Staphylococcus epidermidis)

• Streptococci (Viridans streptococci and others)

• Enterococci

• Pneumococci

• Fastidious gram-negative coccobacilli (HACEK group)

• Enterobacteriaceae

• Pseudomonas spp. (usually in drug users)

• Candida species

• Diphtheroids

• Cuture showing alpha-hemolysis on blood agar with resistance to optochin and bile is suggestive of the streptococcus viridans group, a normal flora of mouth and upper respiratory tract.

• Streptococcus sanguinis, formerly known as Streptococcus sanguis, is a Gram-positive facultative anaerobic coccus species of bacteria and a member of the Viridans Streptococcus group.

• S. sanguinis is a normal inhabitant of the healthy human mouth, particularly in dental plaque. It can cause infective endocarditis (IE) when it gains access to the bloodstream, typically through dental procedures. IE is a serious infection that can damage the heart valves. The patient, in this case, has several risk factors for IE, including fever, fatigue, and vegetation on the heart valves.

• Other members of streptococcus viridans are S. mitis, S. mutans, and S. salivarus.

• Staphylococcus aureus

• Coagulase-negative staphylococci (e.g., Staphylococcus epidermidis)

• Streptococci (Viridans streptococci and others)

• Enterococci

• Pneumococci

• Fastidious gram-negative coccobacilli (HACEK group)

• Enterobacteriaceae

- Pseudomonas spp. (usually in drug users)
- Candida species
- Diphtheroids

Incorrect Options:

Option A- Mycoplasma: Mycoplasma are causative organism of meningitis, encephalitis, GBS, Steven Johnson's syndrome, and carditis. They are cultured on PPLO agar and show a fried egg colonies.

Option C - Enterococcus: Enterococcus are gram-positive facultative anaerobic cocci that show alpha-hemolysis on blood agar.

Option D - Streptococcus Agalactiae: The gram-positive coccus Streptococcus agalactiae, generally known as group B streptococcus or GBS is a facultative anaerobe that shows beta-hemolysis on agar.

Solution for Question 24:

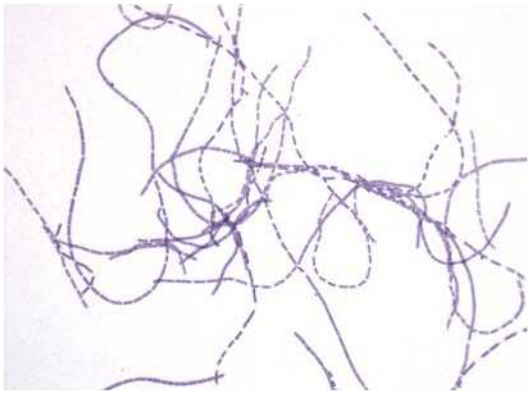
Correct Option C - Corynebacterium Diphtheria:

- The patient's presentation is suggestive of diphtheria caused by Corynebacterium diphtheria.
- The gram staining shows gram-positive bacilli with a characteristic appearance resembling a Chinese letter (V or L shaped) or cuneiform arrangement, and the ends show metachromatic granules, giving it a club-shaped appearance.
- Corynebacterium diphtheria primarily infects the throat and produces a toxin (diphtheria toxin) that causes exudative pharyngitis and membranous tonsillitis. It is a pseudomembrane because it bleeds on, peeling it off.

Incorrect Options:

Option A and B - Parvovirus and HHV6: Both the options represent the virus, lacking the cell wall and do not show gram staining.

Option D - Beta hemolytic streptococci: Streptococci pyogenes is a beta-hemolytic streptococci, it is gram-positive cocci that occur in pairs or chains. It can cause tonsillopharyngitis, acute otitis media, pneumonia, skin and soft-tissue infections.



Solution for Question 25:

Correct Option C - Extensively drug-resistant TB:

- This kind of TB is called Extensively drug-resistant TB (XDR-TB).
- It is resistant to first-line drugs isoniazid and rifampicin, along with resistance to one of the fluoroquinolones (levofloxacin) and one injectable (amikacin).

Incorrect Options:

Option A - Poly-drug resistant TB: This includes resistance to more than 1 first-line drug other than both rifampicin and isoniazid.

Option B - Multi-drug sensitive TB: This type of TB is resistant to Isoniazid and Rifampicin with or without resistance to other first-line drugs.

Option D - Rifampicin-resistant TB: As the name suggests, this type of TB is resistant to Rifampicin.

Solution for Question 26:

Correct Option A - Dark Field Microscopy:

• The patient's presentation of a painless, indurated genital ulcer with a history of unprotected sexual intercourse is suggestive of syphilis caused by Treponema pallidum.

• It can be seen in a low-resource setting by examining clinical specimens from the ulcer under dark field microscopy. It enables the direct viewing of the spirochetes in the samples, which is especially helpful when an active ulcer is present.

• Using a dark field microscope, which illuminates the material with tangential or oblique light, the motile spirochetes can be observed as dazzling, silver-white, or golden-yellow corkscrew-like structures on a dark backdrop.

Incorrect Option:

Option B - Fluorescent treponema antibody absorption (FTA-ABS): FTA-ABS is a serological test used to detect antibodies against Treponema pallidum. It is not used for direct visualization of the organism and requires a laboratory setup for testing.

Option C - Venereal disease research laboratory (VDRL) test: The VDRL test is also used to detect syphilis. It works by detecting antibodies that the body makes in case of Treponema pallidum infection and requires a laboratory setup for the test.

Option D - Treponema pallidum immobilization (TPI) test: A specialized serological test called the TPI test is performed to validate the presence of Treponema pallidum antibodies in the blood and requires a laboratory setup for the test.

Solution for Question 27:

Correct Option B - Klebsiella Granulomatis:

• Klebsiella granulomatis, a gram-negative bacteria, is the causative organism for granuloma inguinale, also known as Donovanosis.

• Granuloma inguinale is a chronic sexually transmitted disease in which painless, beefy-red ulcers develop in the genital and inguinal regions.

Incorrect Options:

Option A - Chlamydia Trachomatis: Chlamydia trachomatis can causes diseases like trachoma, ophthalmia neonatorum, non-gonococcal urethritis, and lymphogranuloma venereum.

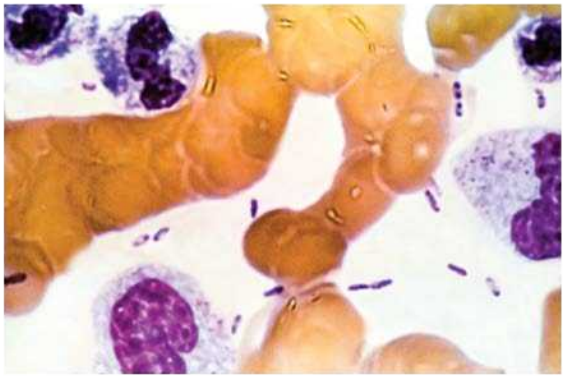
Option C - Haemophilus Ducreyi: Haemophilus ducreyi causes chancroid, which is painful genital ulcers.

Option D - Herpes Simplex Virus 2: Herpes simplex virus 2 causes painful genital ulcers with painful inguinal lymphadenopathy, meningitis, and encephalitis..

Solution for Question 28:

Correct Option B - Wayson's Stain:

- The patient's presentation of high fever, severe inguinal lymphadenopathy, nausea, vomiting, and hypotension, as well as a history of a flea bite suggestive of bubonic plague caused by Yersinia pestis.
- Yersinia pestis is a gram-negative bacteria that has the ability to exhibit bipolar staining, giving it a safety pin appearance when stained with Wayson stain.



Incorrect Options:

Option A - Neisser Stain: It is used for staining the volutin granules of Corynebacterium diphtheriae, spirillum, and Gardnerella.

Option C - Albert's Stain: It is also used for staining the volutin granules of Corynebacterium diphtheriae, spirillum, and Gardnerella.

Option D. McFadyean's Reaction: It is a reaction characterized by an amorphous purple capsule surrounding blue bacilli (polychrome methylene blue stain)used for the diagnosis of anthrax.

Solution for Question 29:

Correct Option D - α-hemolytic and lanceolate-shaped:

- The most common bacterium responsible for meningitis are Streptococcus pneumoniae, Neisseria meningitidis, H. influenzae, Listeria, streptococcus agalactiae, and E. coli.
- The above given microscopic image shows gram-positive cocci arranged in the chain. Hence, they are likely to be Streptococci pneumoniae.
- Streptococcus pneumoniae is a lanceolate-shaped, gram-positive, capsulated diplococci that is bile soluble.
- It shows α-hemolysis on blood agar and is a catalase and oxidase-negative organism.
- It ferments inulin and is optochin sensitive, differentiating from streptococcus viridans.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 30:

Correct Option C - Actinomyces Israelli:

- The patient had a recent tooth extraction, and the subsequent mass in the submandibular region suggests an infection spreading from the oral cavity.
- Actinomyces israelii is a commensal bacterium in the oral flora, and it can cause infection when there is a breach in the mucosal barrier, such as after dental procedures or trauma.
- The presence of foul-smelling discharge and yellow granules is a characteristic finding in actinomycosis.
- Actinomyces form aggregates called sulfur granules, which are composed of bacterial colonies surrounded by host tissue reaction. These granules are typically yellowish and contain tangled filamentous structures seen on microscopy.

Incorrect Options:

Option A - Fonsecaea Pedrosoi: It causes chromoblastomycosis, a chronic, granulomatous infectious skin disease also known as verrucous dermatitis seen in barefoot walkers.

Option B - Mucor: It causes mucormycosis, which is often referred to as zygomycosis because the fungi that cause it reproduce by forming zygospores. It commonly causes pneumonia, sinusitis, and CNS infection in immunocompromised people.

Option C - Histoplasma Capsulatum: Histoplasmosis is a fungal infection caused by the fungus Histoplasma capsulatum caused due to bat or bird droppings. Histoplasmosis commonly causes pneumonia and meningitis in immunocompromised people.

Solution for Question 31:

Correct Option B - Enteropathogenic Escherichia Coli:

- Attachment - effacement lesion in intestinal mucosa is caused by EPEC (Enteropathogenic Escherichia coli).
- Enteropathogenic E coli are non-invasive and do not produce enterotoxins.
- EPEC has two factors: EPEC adherence factor (EAF) The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).
- EPEC adherence factor (EAF)
- The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).
- After attachment, there is a loss of microvilli (effacement) and formation of filamentous actin pedestals or cuplike structures, causing entry of the EPEC into the mucosal cells. This attachment-effacement lesion disrupts the normal absorptive function of the intestinal epithelium and can lead to diarrhea.
- EPEC adherence factor (EAF)
- The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).

Incorrect Options:

Option A - Enterotoxigenic Escherichia Coli: Enterotoxigenic Escherichia coli (ETEC) ETEC is another type of Escherichia coli bacteria that causes traveler's diarrhea. It produces toxins that act on the intestinal mucosa and cause watery diarrhea by adhering to the intestinal mucosa by fimbriae called colonization factor antigens.

Option C - Enteroinvasive Escherichia Coli: Enteroinvasive E. coli (EIEC) is a type of Escherichia coli bacteria that invades and damages the intestinal epithelial cells, leading to an inflammatory response. It can cause dysentery-like symptoms but does not specifically cause effacement and attachment of intestinal cells like EPEC.

Option D - Diffusely adherent Escherichia Coli: Diffusely adherent E. coli forms a diffuse adherence pattern on the intestinal epithelium but does not cause attachment-effacement lesions.

Solution for Question 32:

Correct Option A - Presynaptic inhibitor of GABA and glycine release:

- The presentation described, with generalized muscle spasm and arching of the back after a puncture wound to his leg, is characteristic of tetanus. Tetanus is a condition caused by the neurotoxin produced by the bacterium Clostridium tetani.
- The toxin responsible for tetanus, called tetanospasmin, acts by inhibiting the release of inhibitory neurotransmitters, specifically gamma-aminobutyric acid (GABA) and glycine, at the presynaptic level in the central nervous system.
- GABA and glycine are neurotransmitters that have inhibitory effects on motor neurons. Their release normally helps to regulate muscle activity and prevent excessive muscle contractions.
- By inhibiting the release of GABA and glycine, the toxin disrupts the balance between excitatory and inhibitory signals in the nervous system. This leads to hyperreflexia, muscle spasms, and spastic paralysis.

Incorrect Options:

Options B and C are incorrect. Refer to Option A for an explanation.

Option D - Cleavage of SNARE proteins preventing Ach release: This mechanism is seen in botulism caused by botox toxin. It is produced by C.botulinum.

Solution for Question 33:

Correct Option D - Legionella:

- BCYE (Buffered Charcoal Yeast Extract) medium is a specialized culture medium that supports the growth of Legionella bacteria. Legionella is a pathogenic bacterium responsible for causing a severe respiratory illness known as Legionnaires' disease.
- BCYE medium contains yeast extract, which provides essential growth factors, and activated charcoal, which helps to absorb and neutralize inhibitory substances, buffered salts maintain the appropriate pH. Legionellae grow slowly; visible colonies are usually present after 3 days of incubation. Colonies are round or flat with entire edges. They vary in color from colorless to iridescent pink or blue and are translucent or speckled.

Incorrect Options:

Option A - Leptospira: Leptospira is a spirochete bacterium that causes an infectious disease known as leptospirosis. It is typically cultured using specific media such as the Ellinghausen-McCullough-Johnson-Harris (EMJH) medium.

Option B - Bacillus cereus: Bacillus cereus is a gram-positive bacterium that can cause food poisoning. It is cultured on mannitol egg yolk phenol red polymyxin (MYPA) agar.

Option C - Campylobacter: Campylobacter is a genus of bacteria that includes species responsible for gastrointestinal infections, such as Campylobacter jejuni. Campylobacter is commonly cultured using selective media like Skirrow's, Butzlers, or Campy BAP selective medium.

Solution for Question 34:

Correct Option B - Microscopic Agglutination Test:

- The patient's presentation of fever, jaundice, conjunctival injection, and hematuria are classic symptoms of leptospirosis, aka hepatorenal syndrome, aka Weil's disease, caused by leptospira interrogans.
- Microscopic Agglutination Test (MAT) is the specific and most commonly used serological test for diagnosing Leptospirosis. It detects the presence of antibodies against Leptospira bacteria in the patient's serum. The test involves agglutination reactions between the

patient's serum and a panel of different Leptospira serovars. A positive result indicates thepresence of specific antibodies and confirms the diagnosis of Leptospirosis.

Incorrect Options:

Option A - Widal test: It is primarily used for diagnosing typhoid fever caused by Salmonella typhi and Salmonella paratyphi.

Option C - Weil-Felix reaction: The Weil-Felix reaction is a serological test that uses cross-reacting antigens from different Proteus strains to detect antibodies against rickettsial infections.

Option D - Paul-Bunnell test: The Paul-Bunnell test, also known as the Monospot test, is used for the diagnosis of infectious mononucleosis, which is caused by the Epstein-Barr virus (EBV).

Solution for Question 35:

Correct Option C - Bacillus Anthracis:

- The image shows an ulcer with a central black eschar on the forearm, and the occupation of the patient is highly suggestive of cutaneous anthrax, which is caused by the bacterium Bacillus anthracis.

Incorrect Options:

Option A - Pseudomonas Aeruginosa: Pseudomonas aeruginosa is a common cause of various hospital-acquired infections like meningitis, endocarditis, and infection of burns.

Option B - Yersinia Pestis: Yersinia pestis is the causative organism of plague, which typically presents with symptoms such as fever, lymphadenopathy, respiratory symptoms, and shock.

Option D - Francisella Tularensis: Francisella tularensis is the bacterium responsible for tularemia, also known as rabbit fever. It primarily causes ulceroglandular tularemia, which presents with skin ulcers and swollen lymph nodes.

Solution for Question 36:

Correct Option A - Pour Plate Culture:

- Pour plate culture is not used as an immediate test in the screening of urinary tract infections (UTIs). Pour plate culture is a quantitative method used for bacterial culture and enumeration, it is not a screening test for UTI.

Incorrect Options:

Option B - Griess Nitrite Test: The Griess Nitrite test is a rapid screening test used to detect the presence of nitrites in the urine. Nitrites are produced by certain bacteria, such as Escherichia coli, commonly associated with UTIs. The presence of nitrites indicates the possibility of a UTI.

Option C - Leukocyte Esterase Test: The leukocyte esterase test is a rapid screening test used to detect the leukocyte esterase secreted by pus cells in the urine. The presence of leukocyte esterase suggests inflammation and a possible UTI.

Option D - Wet Mount Examination: The wet mount examination is a microscopic examination of urine sediment, which is used for the detection of pus cells in urine. Pyuria of more than 8 pus cells/mm3 is taken as significant for UTI.

Solution for Question 37:

Correct Option A - Venereal Disease Research Laboratory (VDRL) Test:

- Neurosyphilis is a form of syphilis that affects the central nervous system. The VDRL test is the most preferred screening test for syphilis, including neurosyphilis.
- The VDRL test is a non-specific test that detects reagin antibodies in the serum or cerebrospinal fluid (CSF) of individuals with syphilis by using cardiolipin antigen extracted from beef heart. Positive VDRL results indicate the presence of antibodies, suggesting an active infection.

Incorrect Options:

Option B - FTA-ABS-Fluorescent Treponemal Antibody Absorption Test: The FTA-ABS test is a specific test for syphilis. It detects antibodies specifically against Treponema pallidum, the bacterium that causes syphilis. It is highly sensitive and specific for syphilis diagnosis, including neurosyphilis. However, the FTA-ABS test is not the investigation of choice specifically for neurosyphilis.

Option C - RPR – Rapid Plasma Reagin Test: The RPR test is another non-specific test used for the screening and diagnosis of syphilis. It detects antibodies against cardiolipin, a substance found in the plasma. It is more expensive than VDRL.

Option D - TPI – Treponema Pallidum Immobilization Test: The TPI test is a specific test used to detect antibodies against Treponema pallidum. It involves mixing patient serum with live Treponema pallidum organisms and observing for immobilization.

Solution for Question 38:

Correct Option B - MRSA:

- Contact-based isolation precautions are employed for MRSA (Methicillin Resistance Staphylococcus Aureus). These precautions prevent the transmission of the bacteria from an infected or colonized individual to others.
- It includes hand washing, isolation of patients with open staphylococcal lesions, and detection of staphylococcal lesions among surgeons.

Incorrect Options:

Option A - Diphtheria: It requires droplet precautions because it is primarily transmitted through respiratory droplets. Droplet precautions involve wearing a mask and other appropriate measures to prevent the spread of infectious droplets.

Option C - Mumps: It is also transmitted through respiratory droplets, and droplet precautions are necessary to prevent its transmission.

Option D -Asthma: It is a chronic respiratory condition and not an infectious disease. It does not require isolation precautions since it is not transmissible from person to person.

Solution for Question 39:

Correct Option B - Neisseria Meningitidis:

- The clinical presentation of the patient is suggestive of meningitis.
- The most common bacterium responsible for meningitis are Streptococcus pneumoniae, Neisseria meningitidis, H. influenzae, Listeria, streptococcus agalactiae, and E. coli.
- Gram staining of the cerebrospinal fluid (CSF) showing gram-negative diplococci suggests the diagnosis of Neisseria meningitidis.

Incorrect Options :

Option A - Haemophilus Influenzae Type B: While Haemophilus influenzae can cause meningitis, it is a gram-negative present as clusters of coccobacillus and would not appear as gram-positive diplococci on Gram staining.

Option C - Streptococcus Pneumoniae: Streptococcus pneumoniae is a gram-positive bacterium that can cause bacterial meningitis, but it appears as gram-positive diplococci, giving a lanceolate shape. Option D - Listeria Monocytogenes: Listeria monocytogenes is a gram-positive bacterium associated with meningitis, but it is characterized by its rod-shaped and exhibits differential motility.

Solution for Question 40:

Correct Option B - Neisseria Meningitidis:

- Severe headache, photophobia, high fever along with papilledema, nuchal rigidity, kernig's sign and brudzinski sign being positive points towards bacterial meningitis.
- Reduced glucose, increased protein, and leukocytosis with neutrophilic predominance point towards a bacterial cause of meningitis.
- Streptococcus pneumoniae, Neisseria meningitidis, and Haemophilus influenzae are the most common bacterial causes of meningitis in children and teenagers.
- Gram staining of the cerebrospinal fluid (CSF) showing gram-negative diplococci suggests the diagnosis of Neisseria meningitidis.

Incorrect Options:

Option A - Haemophilus Influenzae: It is a gram-negative coccobacilli arranged in clusters.

Option C - Streptococcus Pneumoniae: Streptococcus pneumoniae is a gram-positive bacterium that can cause bacterial meningitis, but it appears as gram-positive diplococci, giving a lanceolate shape.

Option D - Listeria Monocytogens: Listeria is known to cause meningitis in extremes of age i.e. meningitis in infants and elderly. It is a gram-positive rod-shaped bacteria and exhibits differential motility.

Solution for Question 41:

Correct Option B - Diphtheria:

- The clinical presentation and Albert staining showing volutin granules are suggestive of Diphtheria caused by Corynebacterium diphtheria.
- It is characterized by the fever and formation of white-gray exudate that adheres tightly to the mucous membrane of the tonsils, soft palate, mouth, and sometimes larynx and bleeds on peeling, hence called pseudomembranous which can cause difficulty in breathing and swallowing.
- Corynebacterium diphtheriae on Albert stain demonstrates the volutin granules, aka Babe-Ernst granule aka metachromatic granules, are stained at the end of the bacilli as shown in the image

Incorrect Options:

Option A - Streptococcal pharyngitis: It is caused by Streptococcus pyogenes presents with fever, cervical lymphadenopathy, erythema and swelling of pharyngeal mucosa with purulent exudate formation. Volutin granules are not characteristic of Streptococcus pyogenes.

Option C - Tetanus: Tetanus is caused by the bacterium Clostridium tetani and is characterized by muscle stiffness and spasms after a penetrating injury.

Option D - Pertussis: Pertussis, also known as whooping cough, is caused by the bacterium Bordetella pertussis. It is characterized by severe coughing spells and a whooping sound during inhalation.

Solution for Question 42:

Correct Option C - Activates adenylate cyclase:

• Cholera toxin is produced by Vibrio cholerae bacteria and is responsible for the characteristic clinical features of cholera, including profuse, watery diarrhea ("rice water stools") and severe dehydration.

• The cholera toxin is a bipartite consisting of two main fragments: A and B.

• Fragment A (action): This fragment is responsible for the intracellular actions of the toxin. It has two fragments, A1 and A2. The A2 fragment only links the biologically active A1 fragment to the B subunit. A1 fragment activates adenylate cyclase, which is an enzyme responsible for converting ATP to cyclic AMP (cAMP). Increased levels of cAMP in the intestinal epithelial cells lead to phosphorylation of the cystic fibrosis transmembrane regulatory protein (CFTR), leading to increased secretion of Cl- and HCO3- ions into the intestinal lumen. This disrupts the normal osmotic balance and results in a massive efflux of water, leading to the profuse, watery diarrhea seen in cholera.

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• A1 fragment activates adenylate cyclase, which is an enzyme responsible for converting ATP to cyclic AMP (cAMP). Increased levels of cAMP in the intestinal epithelial cells lead to phosphorylation of the cystic fibrosis transmembrane regulatory protein (CFTR), leading to increased secretion of Cl- and HCO3- ions into the intestinal lumen. This disrupts the normal osmotic balance and results in a massive efflux of water, leading to the profuse, watery diarrhea seen in cholera.

• Fragment B (binding): It has 5 subunits, responsible for binding to the GM1 ganglioside on the surface of intestinal epithelial cells. The binding allows the toxin to be internalized into the cells, where the A fragment exerts its effects.

• This fragment is responsible for the intracellular actions of the toxin. It has two fragments, A1 and A2.

• The A2 fragment only links the biologically active A1 fragment to the B subunit.

• A1 fragment activates adenylate cyclase, which is an enzyme responsible for converting ATP to cyclic AMP (cAMP). Increased levels of cAMP in the intestinal epithelial cells lead to phosphorylation of the cystic fibrosis transmembrane regulatory protein (CFTR), leading to increased secretion of Cl- and HCO3- ions into the intestinal lumen. This disrupts the normal osmotic balance and results in a massive efflux of water, leading to the profuse, watery diarrhea seen in cholera.

Incorrect Options:

Option A - Inhibit elongation factor-2: This mechanism is associated with the action of diphtheria toxin and Pseudomonas exotoxin decreasing protein synthesis.

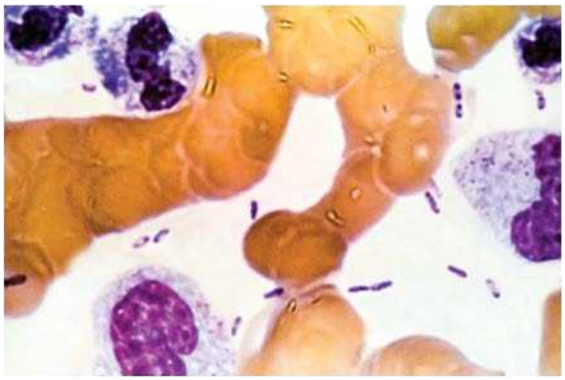
Option B - Toxin causes increased guanyl cyclase activity: This mechanism is associated with the action of heat-stable enterotoxins produced by certain strains of Escherichia coli (E. coli).

Option D - Inhibit 60S subunit: This mechanism is associated with the action of shiga like toxin, decreasing protein synthesis.

Solution for Question 43:

Correct Option B - Burkhuloderia pseudomallei:

- Safety pin appearance is due to bipolar staining shown by some organisms, in this the terminal parts of the organism is darkly stained and centre is poorly satined giving safety pin appearance.
- Organisms showing safety pin appearance are: Burkholderia pseudomallei Burkholderia mallei Yersinia pestis Vibrio parahemolyticus Haemophilus ducreyi Klebsiella granulomatis
- Burkholderia pseudomallei
- Burkholderia mallei
- Yersinia pestis
- Vibrio parahemolyticus
- Haemophilus ducreyi
- Klebsiella granulomatis
- Burkholderia pseudomallei
- Burkholderia mallei
- Yersinia pestis
- Vibrio parahemolyticus
- Haemophilus ducreyi
- Klebsiella granulomatis



Incorrect Options:

Options A, C , and D are incorrect. Refer to Option B for an explanation.

Solution for Question 44:

Correct Option A - Corynebacterium diphtheriae:

• Bull’s neck diphtheria, also called respiratory/faucial diphtheria, is caused by Corynebacterium Diphtheriae. It is characterized by massive tonsillar swelling and neck edema. Patients present with foul breath, thick speech, and stridor (noisy breathing).



Incorrect Options:

Option B - Corynebacterium jeikeium: It is known to cause cutaneous and bloodstream infection (endocarditis) in immunocompromised individuals.

Option C - Corynebacterium minutissimum: It causes erythrasma, a localized infection of the stratum corneum usually affecting the groin and axilla.

Option D - Corynebacterium urealyticum: It is mostly known to cause urinary tract infections.

Solution for Question 45:

Correct Option C - Interferon-gamma release assay:

- In an HIV-positive individual with suspected latent tuberculosis (TB) infection, the Mantoux test may yield false-negative results due to the impaired immune response. Therefore, the alternative method used to diagnose latent TB infection in such cases is IGRA (interferon-gamma release assay).
- Interferon-gamma release assay measured using ELISA. T-lymphocytes are exposed to antigens: ESAT6 TB 7.7 CFP 10 Exposure stimulates the release of interferon-gamma released by CD 4 T cells in response - measured by ELISA. Advantage: Negative in BCG vaccination.
- T-lymphocytes are exposed to antigens: ESAT6 TB 7.7 CFP 10
- ESAT6
- TB 7.7
- CFP 10
- Exposure stimulates the release of interferon-gamma released by CD 4 T cells in response - measured by ELISA.
- Advantage: Negative in BCG vaccination.
- T-lymphocytes are exposed to antigens: ESAT6 TB 7.7 CFP 10
- ESAT6
- TB 7.7
- CFP 10
- Exposure stimulates the release of interferon-gamma released by CD 4 T cells in response - measured by ELISA.
- Advantage: Negative in BCG vaccination.
- ESAT6
- TB 7.7

- CFP 10

Incorrect Options:

Option A - CBNAAT: It is used to diagnose active tuberculosis and detection of resistance to rifampicin.

Option B - MGIT system: It is an automated culture method used for the diagnosis of active tuberculosis and to detect resistance against antitubercular drugs.

Option D - Line probe assay: It is used for the diagnosis of active tuberculosis and detection of resistance to 1st and 2nd line antitubercular drugs.

Solution for Question 46:

Correct Option A - Granuloma Inguinale:

- The presentation of a painless genital ulcer and lesion on microscopy showing bipolar staining is suggestive of Granuloma inguinale (donovanosis) caused by a gram-negative bacterium Klebsiella granulomatis. It causes a painless papule that ulcerates throughout the disease.
- The first image shows a red beefy ulcer that bleeds profusely on touch or contact is characteristic of granuloma inguinale.
- The second image is a microscopy image showing Donovan's body, large cyst-like macrophages filled with deeply stained capsulated bacilli having a safety-pin (bipolar) appearance.

Incorrect Options:

Option B - Lymphogranuloma venereum: It is caused by Chlamydia trachomatis, characterized by a painless genital ulcer with painful lymphadenopathy(Inguinal bubos). The inguinal lymph nodes above and below the inguinal ligament swell up and appear like groove sign.

Option C - Chancroid: Also called oft Chancre/chancroid, is caused by Haemophilus ducreyi characterized by painful genital ulcer.

Option D - Gonorrhea: It is caused by Neisseria Gonorrhoeae. It usually presents as cervicitis, prostatitis, urethritis, or vaginitis associated with purulent discharge out of the urethral or vaginal meatus.

Solution for Question 47:

Correct Option A - Resistant to EI Tor phage V:

Differences between classical and EI Tor V. Cholerae		
Biotypes of V. Cholerae O1	Classical biotype	EI Tor biotype
β -hemolysis on sheep blood agar	Negative	Positive
Chick erythrocyte agglutination	Negative	Positive
Polymyxin B (50 IU)	Susceptible	Resistant
Group IV phage susceptibility	Susceptible	Resistant
EI Tor Phage V susceptibility	Resistant	Susceptible
VP (Voges-Proskauer) test	Negative	Positive
Cholera toxin gene	CTX-1	CTX-2

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 48:

Correct Option B - rpoB:`

- GeneXpert is a molecular diagnostic tool used for the rapid detection of Mycobacterium tuberculosis and rifampicin resistance targeting five different sequences of the rpoB (RNA polymerase subunit B) gene.
- GeneXpert is a CBNAAT system recommended by WHO as the initial test for the diagnosis of extrapulmonary tuberculosis.

Incorrect Options:

Options A and C - inha & KatG: inhA (Enoyl ACP reductase) and KatG (Catalase-peroxidase) gene is associated with resistance to isoniazid and is detected by line probe assay.

Option D - pncA: The pncA (Pyrazinamidase) gene is associated with resistance to the drug pyrazinamide and is detected by line probe assay.

Solution for Question 49:

Correct Option C - Produces NLF-producing colonies in MacConkey agar:

- The Scrub typhus is caused by orientia tsutsugamushi. Among all stages of mites, the larvae (called chiggers) are the only stage that feeds on humans. Hence also called chiggerosis.
- The classic presentation of scrub typhus consists of a triad of an eschar (at the site of the bite), regional lymphadenopathy, and maculopapular rash.
- Humans are considered accidental hosts of Orientia tsutsugamushi. The natural hosts of the bacterium are small rodents, such as mice and rats, which serve as reservoirs.
- It is an obligate intracellular pathogen and does not grow on standard laboratory culture media like MacConkey agar. It requires specialized cell culture like the Yolk sac of an embryonated hen's egg (or) Cell lines - Varo, Hela, Hep2, Detroit 6.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 50:

Correct Option C - Haemophilus Influenzae:

- Haemophilus influenzae is a gram-negative pleomorphic (coccobacilli) bacillus that can cause respiratory tract infections, including pneumonia. It grows well on chocolate agar, which contains factors that support its growth. It is catalase and oxidase-positive, which helps differentiate it from other bacteria.

Incorrect Options:

Option A - E. Coli: E. coli is a gram-negative, catalase-positive, and oxidase-negative bacilli. It may cause urinary tract infections, gastrointestinal infections, sepsis, or pyogenic infections.

Option B - Pseudomonas: Pseudomonas is a gram-negative, catalase, and oxidase-positive bacillus that can cause pneumonia. It is grown on nutrient agar, MacConkey agar, and blood agar.

Option D - Actinomyces: Actinomyces is a gram-positive, filamentous bacterium that can cause chronic granulomatous infection characterized by indurated swellings with sulfur granule discharge.

Solution for Question 51:

Correct Option C - H. Ducreyi:

• Chancroid is a sexually transmitted infection caused by the bacterium Haemophilus ducreyi. It is characterized by the development of painful genital ulcers, which can occur on the glans penis, shaft, or in the pubic area.

Painful genital ulcer	Painless genital ulcer
Chancroid (soft) – Haemophilus ducreyi Herpes simplex virus 2	Syphilis – Treponema pallidum Granuloma inguinale – Klebsiella granuolomatis Lymphogranuloma venereum – Chlamydia trachomatis

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 52:

Correct Option D - 1-d, 2-b, 3-c, 4-a:

• ETEC - Invades intestinal mucosal cells via CFA (colonization factor antigens) and produces heat labile and heat stable toxin. LT activates adenylate cyclase, leading to increased cAMP levels and electrolyte secretion. ST activates guanylate cyclase, leading to increased cGMP levels and electrolyte secretion.

• LT activates adenylate cyclase, leading to increased cAMP levels and electrolyte secretion.

• ST activates guanylate cyclase, leading to increased cGMP levels and electrolyte secretion.

• EIEC - Invades intestinal mucosal cells via virulence marker antigen.

• EHEC - Produces a verocytotoxin/Shiga toxin against vascular endothelium cells.

• EPEC - Adheres to intestinal mucosa via EAC and causes effacement of microvilli.

• LT activates adenylate cyclase, leading to increased cAMP levels and electrolyte secretion.

• ST activates guanylate cyclase, leading to increased cGMP levels and electrolyte secretion.

Incorrect Options:

Options A, B, and C are incorrect.

Solution for Question 53:

Correct Option D - Cefotetan:

• Burkholderia cepacia is known to exhibit intrinsic resistance to cefotetan.

• This resistance is primarily attributed to the presence of various mechanisms, including efflux pumps, enzymatic degradation, and alterations in target sites.

• Burkholderia cepacia is resistant to aminoglycosides, polymyxine, 1st and 2nd generation cephalosporins, and carboxypenicillins.

• Burkholderia cepacia is susceptible to Ceftazidime, Trimethoprim-sulfamethoxazole, Temocillin, doxycycline, meropenem, and piperacillin.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 54:

Correct Option C - Pseudomonas Aeruginosa:

- Pseudomonas aeruginosa is a gram-negative bacterium that can cause various infections in humans.
- Scombroid poisoning is caused by the consumption of fish.
- Organisms responsible for causing Scombroid poisoning are Morganella morganii, Escherichia coli, Klebsiella species, and Pseudomonas aeruginosa.
- They are commensal in seafish and can break down histidine present in fish into histamine, which can cause food poisoning following seafood intake.

Incorrect Options:

Option A- Salmonella: Salmonella is a group of bacteria that causes typhoid fever characterized by step ladder pattern fever, bradycardia, and pea soup diarrhea.

Option B- Staphylococcus: It causes food poisoning with the consumption of milk and meat products with an incubation period of 1 to 6 hours.

Option D- Peptostreptococcus: Peptostreptococcus is a genus of anaerobic bacteria that can be part of skin, mouth, intestine, and vagina. It is associated with puerperal sepsis and skin and soft tissue infection.

Solution for Question 55:

Correct Option C - Bacteriophage:

- Bacterial transduction occurs through the action of bacteriophages, which transfer bacterial genetic material from one bacterium to another.

Types of Bacteriophages	
Virulent phage	Temperate phage
Dangerous Follow only lytic cycle Ex: T1 and T2 phage of E.coli	Less dangerous Follow both lytic and lysogenic cycle Ex: Lambda phage of E. coli

Incorrect Options:

Option A - Plasmids: Plasmids are circular, extrachromosomal DNA molecules that can be found in bacteria. While plasmids can carry genetic information and can be transferred between bacteria through a process called conjugation where plasmids are transferred between bacteria through direct physical contact.

Option B - Sex pilli: Sex pilli, or pili, are appendages found on the surface of certain bacteria that facilitate the transfer of genetic material during conjugation.

Option D - Uptake of genetic material by other bacteria: This refers to the gene transfer by transformation, a mechanism of horizontal gene transfer in bacteria, where bacteria can uptake free extracellular DNA from the environment and incorporate it into their own genome.

Solution for Question 56:

Correct Option C - Interferon-gamma release assay:

- Interferon-gamma release assay (IGRA) is a type of tuberculosis (TB) test that is performed on whole blood samples.
- IGRA tests measure the release of interferon-gamma by T-cells in response to specific TB antigens (ESAT, 6TB 7.7, and CFP 10).

Incorrect Options:

Option A - TrueNat TB: TrueNat TB is a molecular diagnostic test for the detection of tuberculosis. It utilizes a cartridge-based platform to perform real-time PCR (polymerase chain reaction) to detect Mycobacterium tuberculosis DNA in sputum samples. TrueNet TB is not specifically limited to whole blood samples, as it is primarily used for sputum-based testing.

Option B - GeneXpert: GeneXpert is a molecular diagnostic platform that uses real-time PCR technology to detect tuberculosis and resistance against rifampicin. The test is typically performed on sputum samples or other respiratory specimens rather than whole blood.

Option D - MGIT system: It is a molecular diagnostic platform that uses probe-based detection of amplified DNA in the specimen to detect tuberculosis and resistance 1st line and 2nd line antitubercular drugs. The test is typically performed on sputum samples or other respiratory specimens.

Solution for Question 57:

Correct Option A - Actinomyces Israelii:

- Multiple abscesses with discharging sinuses and the presence of sulfur granules in pus are characteristic features of actinomycosis caused by actinomyces israelii a commensal bacterium in the oral flora.
- Actinomyces form aggregates called sulfur granules, which are composed of bacterial colonies surrounded by host tissue reaction. These granules are typically yellowish and contain tangled filamentous structures seen on microscopy.

Incorrect Options:

Option B - Nocardia: Nocardia can cause pulmonary infections in immunocompromised individuals and cutaneous lesions in immunocompetent individuals.

Option C - Salmonella: Salmonella bacteria are primarily associated with gastrointestinal infections, such as dysentery characterized by bloody diarrhea, abdominal pain, and fever.

Option D - Tularemia: Tularemia, also known as rabbit fever, is caused by the bacterium Francisella tularensis. It typically presents as a flu-like illness with symptoms such as fever, fatigue, and swollen lymph nodes.

Solution for Question 58:

Correct Option A - Beta-lactam Antibiotics:

- Beta-lactam antibiotics: Mycoplasma is inherently resistant to beta-lactam antibiotics.
- This is because mycoplasmas lack a cell wall, which is the target of penicillins (like β-lactamase inhibitors) and cephalosporins.
- The absence of a cell wall makes mycoplasmas naturally resistant to these types of antibiotics.

Incorrect Options:

Option B - Macrolides: Mycoplasma is susceptible to macrolide antibiotics such as erythromycin, azithromycin, and clarithromycin. Macrolides are commonly used in the treatment of mycoplasma infections.

Option C - Tetracycline: Mycoplasma is susceptible to tetracycline antibiotics. Tetracyclines, including doxycycline and minocycline, are often effective in treating mycoplasma infections.

Option D - Fluoroquinolones: Mycoplasma is susceptible to fluoroquinolone antibiotics such as ciprofloxacin and levofloxacin. Fluoroquinolones are commonly used in the treatment of mycoplasma infections.

Solution for Question 59:

Correct Option C - 3-35 days:

- The incubation period of lymphogranuloma venereum (LGV) is typically between 3 days to 5 weeks.
- Lymphogranuloma venereum is a sexually transmitted infection caused by L1, L2, and L3 serovars of the bacterium Chlamydia trachomatis, after which a small, painless, papulovesicular lesion appears on the genitalia.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 60:

Correct Option C - EPEC:

- Attachment - effacement lesion in intestinal mucosa is caused by EPEC (Enteropathogenic Escherichia coli).
- Enteropathogenic E coli are non-invasive and do not produce enterotoxins.

• EPEC has two factors: EPEC adherence factor (EAF) The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).

• EPEC adherence factor (EAF)

• The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).

• After attachment, there is a loss of microvilli (effacement) and formation of filamentous actin pedestals or cuplike structures, causing entry of the EPEC into the mucosal cells. This attachment-effacement lesion disrupts the normal absorptive function of the intestinal epithelium and can lead to diarrhea.

• EPEC adherence factor (EAF)

• The chromosomal locus of enterocyte effacement (LEE) pathogenicity island that promotes the tight adherence characteristic of EPEC (attachment and effacement).

Incorrect Options:

Option A - ETEC (Enterotoxigenic Escherichia coli): ETEC is another type of Escherichia coli bacteria that causes traveler's diarrhea. It produces toxins that act on the intestinal mucosa and cause watery diarrhea by adhering to the intestinal mucosa by fimbriae called colonization factor antigens.

Option B - EIEC (Enteroinvasive Escherichia coli): EIEC is a type of Escherichia coli bacteria that invades and damages the intestinal epithelial cells, leading to an inflammatory response. It can cause dysentery-like symptoms but does not specifically cause effacement and attachment of intestinal cells like EPEC.

Option D - EAEC (Enteroaggregative Escherichia coli): EAEC is a type of Escherichia coli bacteria associated with persistent diarrhea, particularly in young children. It forms aggregative adherence patterns on the surface of intestinal cells but does not cause the effacement and attachment observed in EPEC infections.

Solution for Question 61:

Correct Option B - Streptococcus Pyogenes:

• The test shown, which uses a bacitracin disc on a blood agar culture plate with a zone of inhibition around the disc, is useful in identifying Streptococcus pyogenes.

• The test being referred to is the bacitracin susceptibility test. In this test, a disc containing a standardized concentration of bacitracin is placed on a blood agar culture plate that has been inoculated with the bacterial isolate. Streptococcus pyogenes is highly sensitive to bacitracin than other streptococci, showing wide range of inhibition as shown in the image.

Incorrect Options:

Option A - Staphylococcus aureus: Staphylococcus aureus is resistant to bacitracin, so it would not show a zone of inhibition around the disc in this test.

Option C - Staphylococcus epidermidis: Staphylococcus epidermidis is resistant to bacitracin, so it would not show a zone of inhibition around the disc in this test.

Option D - Streptococcus agalactiae: Streptococcus agalactiae is resistant to bacitracin, so it would not show a zone of inhibition around the disc in this test.

Solution for Question 62:

Correct Option D - Staphylococcus Aureus:

• Staphylococcus aureus is a bacterium that can cause food poisoning when contaminated food, particularly carbohydrate and protein-rich food, milk, meat, and fish, is consumed.

• Staphylococcus aureus can produce heat-stable enterotoxins that can withstand 100°C for 10 to 40 minutes and cause rapid-onset symptoms of nausea and vomiting, typically occurring within a few hours (within 6 hours is characteristic) of ingesting contaminated food.

Incorrect Options:

Option A - Salmonella typhi: Salmonella typhi is the bacterium responsible for causing typhoid fever. It is transmitted through the ingestion of contaminated food or water. However, the incubation period for typhoid fever is usually much longer, ranging from 7 to 14 days.

Option B - Vibrio cholerae: Vibrio cholerae is the bacterium that causes cholera, a severe diarrheal disease. Cholera is typically associated with the ingestion of contaminated water or food, usually in settings with poor sanitation. The incubation period for cholera is less than 24 hours to 5 days.

Option C - Entamoeba histolytica: Entamoeba histolytica is a protozoan parasite that causes amoebic dysentery, a form of severe diarrhea. The incubation period for amoebic dysentery is typically longer, ranging from 2 weeks to 4 weeks.

Solution for Question 63:

Correct Option D - Amikacin:

- Amikacin is not commonly used for the treatment of typhoid fever in adults. Typhoid fever is a systemic bacterial infection caused by Salmonella typhi, and appropriate antibiotic therapy is essential for effective treatment.
- The currently recommended drugs are as follows: Third-generation cephalosporins: Ceftriaxone is the drug of choice for empirical treatment; given 1–2 g/day, IV, for 10–14 Days Azithromycin: Alternative drug for empirical therapy. Given orally, 1g/day for 5 days. Fluoroquinolones, e.g., ciprofloxacin: It can be given only if it is found susceptible. It is given as 500 mg twice a day orally for 5 days.
- Third-generation cephalosporins: Ceftriaxone is the drug of choice for empirical treatment; given 1–2 g/day, IV, for 10–14 Days
- Azithromycin: Alternative drug for empirical therapy. Given orally, 1g/day for 5 days.
- Fluoroquinolones, e.g., ciprofloxacin: It can be given only if it is found susceptible. It is given as 500 mg twice a day orally for 5 days.
- Third-generation cephalosporins: Ceftriaxone is the drug of choice for empirical treatment; given 1–2 g/day, IV, for 10–14 Days
- Azithromycin: Alternative drug for empirical therapy. Given orally, 1g/day for 5 days.
- Fluoroquinolones, e.g., ciprofloxacin: It can be given only if it is found susceptible. It is given as 500 mg twice a day orally for 5 days.

Incorrect Options:

Option A, B, and C are incorrect. Refer to Option D of an explanation.

Solution for Question 64:

Correct Option B - Shigella:

• The clinical presentation of a swollen, red knee joint that is tender to touch, along with a history of fever and bloody diarrhea, raises suspicion for reactive arthritis, a postinfectious disease caused after infection by Shigella species.

• Postinfectious phase caused after shigella and chlamydia infection: Reiter's syndrome Urethritis. Reactive arthritis. Conjunctivitis.

• Urethritis.

• Reactive arthritis.

• Conjunctivitis.

• Urethritis.

• Reactive arthritis.

• Conjunctivitis.

Incorrect Option:

Option A - E. coli: While certain strains of E. coli can cause gastrointestinal infections, they are not typically associated with reactive arthritis.

Option C - Giardia: Giardia infection, caused by the parasite Giardia lamblia, primarily affects the gastrointestinal tract and is associated with symptoms such as diarrhea, abdominal cramps, and bloating.

Option D. Rotavirus: Rotavirus is a common cause of viral gastroenteritis in children, characterized by symptoms such as vomiting and watery diarrhea.

Solution for Question 65:

Correct Option A - Toluidine Blue and Malachite Green:

• Albert's stain is used for staining the volutin granules of Corynebacterium diphtheriae, the bacterium responsible for diphtheria.

• The constituents of Albert's stain are as follows: Albert solution 1: Green: Malachite green Purple: Toluidine Blue- shows metachromasia. Albert solution 2: Has iodine First add Toluidine blue, then Iodine, and lastly, Malachite green On Albert's staining: organism-blue color and volutin granules-purple color.

• Albert solution 1: Green: Malachite green Purple: Toluidine Blue- shows metachromasia.

• Green: Malachite green

- Purple: Toluidine Blue- shows metachromasia.
- Albert solution 2: Has iodine
- First add Toluidine blue, then Iodine, and lastly, Malachite green
- On Albert's staining: organism-blue color and volutin granules-purple color.
- Albert solution 1: Green: Malachite green Purple: Toluidine Blue- shows metachromasia.
- Green: Malachite green
- Purple: Toluidine Blue- shows metachromasia.
- Albert solution 2: Has iodine
- First add Toluidine blue, then Iodine, and lastly, Malachite green
- On Albert's staining: organism-blue color and volutin granules-purple color.
- Green: Malachite green
- Purple: Toluidine Blue- shows metachromasia.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 66:

Correct Option C - Grey pseudomembrane that bleeds on removal:

- The characteristic diphtheria membrane is a grey pseudomembrane that adheres firmly to the affected mucous membranes, and it does tend to bleed on removal. This pseudomembrane is composed of fibrinous exudates with the disintegrating epithelial cells, leucocytes, erythrocytes, and bacteria.

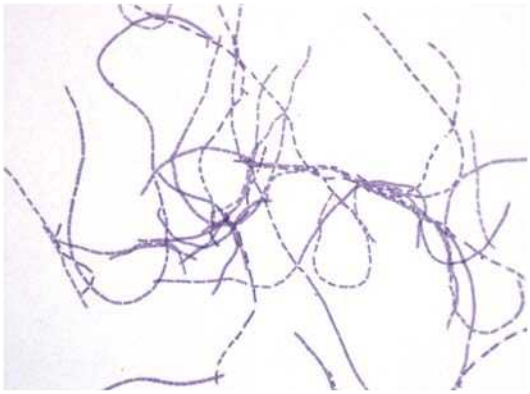
Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 67:

Correct Option B - Gram-positive rods with square ends:

- Bacillus anthracis, the bacterium responsible for anthrax, is a Gram-positive bacterium that appears as rods under a microscope. These rods have square or boxy ends arranged in a chain, giving a bamboo stick or box car appearance.



Incorrect Options: Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 68:

Correct Option C - Leprosy:

- When encountering a case of granuloma with positive acid-fast bacilli (AFB), leprosy is the most likely cause. Leprosy, also known as Hansen's disease, is a chronic infectious disease caused by the bacteria Mycobacterium leprae. It primarily affects the skin, peripheral nerves, mucosa of the upper respiratory tract, and the eyes.
- Leprosy is characterized by the formation of granulomas in the affected tissues, which are clusters of immune cells, including macrophages, that surround and attempt to contain the mycobacteria. Acid-fast staining is a diagnostic test used to visualize mycobacteria, such as Mycobacterium leprae, which retain the stain despite being exposed to acid.

Incorrect options:

Option A - Cat scratch disease: Cat scratch disease is caused by the bacterium Bartonella henselae, and it typically presents with swollen lymph nodes and a papule or pustule at the site of a cat scratch or bite. It does not cause granulomas with positive AFB staining.

Option B - Trench fever: Trench fever is caused by the bacterium Bartonella quintana and is transmitted by body lice. It is characterized by recurrent fever, headache, and bone pain. Trench fever does not typically present with granulomas or positive AFB staining.

Option D - Syphilis: Syphilis is a sexually transmitted infection caused by the bacterium Treponema pallidum. It presents with painless genital ulcer and inguinal lymph node, condylomata lata, and neurosyphilis. However, syphilis does not commonly present with granulomas or positive AFB staining.

Solution for Question 69:

Correct Option A - Respiratory Disease:

- Legionnaires disease is a severe form of pneumonia caused by the bacterium Legionella pneumophila. It primarily affects the respiratory system, leading to respiratory disease symptoms like mild fever, myalgia, non-productive cough, and respiratory failure.
- The bacteria Legionella pneumophila are transmitted through inhalation of contaminated water droplets, typically from sources such as air conditioning systems, hot tubs, or showers.

Incorrect Options:

Options B, C, and D are incorrect. These conditions are not associated with legionnaire's disease.

Solution for Question 70:

Correct Option B - 2nd week:

- The Widal test is a serological test used for the diagnosis of typhoid fever caused by the bacterium Salmonella typhi. It detects antibodies in the patient's blood against specific antigens of the Salmonella typhi bacteria.
- Performing the Widal test in the 2nd week of illness is considered the most appropriate timing for obtaining reliable and accurate results. During the 2nd week of typhoid fever, the

patient's immune response has had sufficient time to produce detectable levels of antibodiesagainst Salmonella typhi antigens.

• Tests performed for the diagnosis of typhoid during the course of infection.

Week	Test performed
Week 1	Blood culture
Week 2	Antibody test (Widal)
Week 3	Stool culture
Week 4	Urine culture

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an example.

Solution for Question 71:

Correct Option A - E. Coli 0157:

• Hemolytic Uremic Syndrome (HUS) is a condition characterized by the combination of hemolytic anemia, thrombocytopenia, and acute kidney injury. HUS can have several causes, but the most common and well-known association is with infection by the bacterium Escherichia coli (E. coli) serotype O157:H7 and shigella.

• E. coli O157: H7 is a strain of E. coli that produces toxins called Shiga toxins. These toxins can cause damage to the lining of blood vessels, leading to red blood cell destruction, platelet clumping, and kidney injury, which are characteristic features of HUS.

Incorrect Options:

Option B - Malaria: Malaria is caused by Plasmodium parasites transmitted by infected mosquitoes. It is characterized by fever, and severe cases of cerebral malaria caused by P. falciparum can be seen. It is not typically associated with HUS.

Option C - Parvovirus B19: Parvovirus B19 is a viral infection that primarily affects children and can cause a condition called erythema infectiosum (fifth disease).

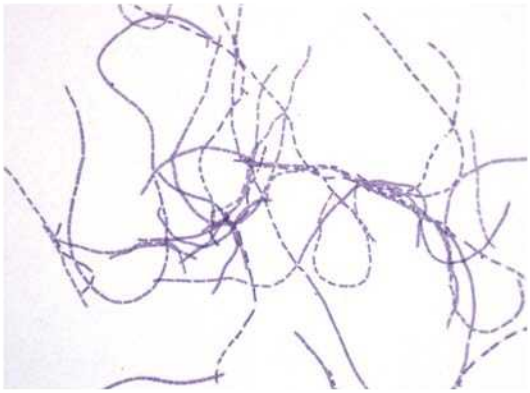
Option D - Bartonella Henselae: It is a bacterium that causes cat scratch disease, a usually benign infection characterized by swollen lymph nodes and fever. Bartonella infections can have various manifestations, including neurologic and cardiac complications, they are not typically

associated with HUS.

Solution for Question 72:

Correct Option B - Gram-positive rods with square ends:

• Bacillus anthracis, the bacterium responsible for anthrax, is a Gram-positive bacterium that appears as rods under a microscope. These rods have square or boxy ends arranged in a chain, giving a bamboo stick or box car appearance.



Incorrect Options: Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 73:

Correct Option A - E. coli 0157:

• Hemolytic Uremic Syndrome (HUS) is a condition characterized by the combination of hemolytic anemia, thrombocytopenia, and acute kidney injury. HUS can have several causes, but the most common and well-known association is with infection by the bacterium Escherichia coli (E. coli) serotype O157:H7 and shigella.

• E. coli O157:H7 is a strain of E. coli that produces toxins called Shiga toxins. These toxins can cause damage to the lining of blood vessels, leading to red blood cell destruction,

platelet clumping, and kidney injury, which are characteristic features of HUS.

Incorrect Options:

Option B - Malaria: Malaria is caused by Plasmodium parasites transmitted by infected mosquitoes. It is characterized by fever, and severe cases of cerebral malaria caused by P. falciparum can be seen. It is not typically associated with HUS.

Option C - Parvovirus B19: Parvovirus B19 is a viral infection that primarily affects children and can cause a condition called erythema infectiosum (fifth disease).

Option D - Bartonella henselae: It is a bacterium that causes cat scratch disease, a usually benign infection characterized by swollen lymph nodes and fever. Bartonella infections can have various manifestations, including neurologic and cardiac complications, they are not typically associated with HUS.

Solution for Question 74:

Correct Option A - Legionnaire Disease:

- The symptoms and occupational history described are consistent with Legionnaire's disease, caused by the bacterium Legionella pneumophila. The use of buffered charcoal yeast extract medium is specific for culturing Legionella species.
- Legionella inhabits aquatic bodies, which could be either: Natural water sources, such as rivers, streams, or even inside amoeba Artificial aquatic sources, such as air conditioners, water coolers, shower heads, large plumbing systems, and decorative fountains.
- Natural water sources, such as rivers, streams, or even inside amoeba
- Artificial aquatic sources, such as air conditioners, water coolers, shower heads, large plumbing systems, and decorative fountains.
- Natural water sources, such as rivers, streams, or even inside amoeba
- Artificial aquatic sources, such as air conditioners, water coolers, shower heads, large plumbing systems, and decorative fountains.

Incorrect Options:

Option B - Brucellosis: Brucellosis is caused by Brucella species and is typically associated with exposure to infected animals or their products.

Option C - Pneumonia caused by Haemophilus: Haemophilus influenzae is a common causative organism of meningitis, pneumonia, croup, and suppurative lesions and requires Factor X and V for growth.

Option D - Acute gastroenteritis: Acute gastroenteritis is characterized by fever, nausea, vomiting or diarrhea.

Solution for Question 75:

Correct Option B - 2nd week:

- The Widal test is a serological test used for the diagnosis of typhoid fever caused by the bacterium Salmonella typhi. It detects antibodies in the patient's blood against specific antigens of the Salmonella typhi bacteria.
- Performing the Widal test in the 2nd week of illness is considered the most appropriate timing for obtaining reliable and accurate results. During the 2nd week of typhoid fever, the patient's immune response has had sufficient time to produce detectable levels of antibodies against Salmonella typhi antigens.
- Tests performed for the diagnosis of typhoid during the course of infection.

Week of Infection	Test performed
Week 1	Blood culture
Week 2	Antibody test (Widal)
Week 3	Stool culture
Week 4	Urine culture

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an example.

Solution for Question 76:

Correct Option C - Zonula occludens:

- Vibrio cholerae causing watery diarrhea occurs due to the disruption of the zonula occludens, also known as tight junctions, in the intestinal epithelial cells. The zonula occludens form a barrier between the cells, regulating the movement of ions, water, and solutes across the intestinal lining.
- When Vibrio cholerae infects the small intestine, it produces a toxin called Zonula toxin. This toxin binds to the surface of intestinal cells and interferes with the function of the tight junctions. It leads to increased secretion of chloride ions and water into the intestinal lumen, resulting in the characteristic watery diarrhea seen in cholera.

Incorrect Option:

Option A - Intercalated discs: Intercalated discs are specialized junctions found in cardiac muscle cells that help coordinate the contraction of the heart.

Option B - Macula densa: Macula densa is a structure located in the distal convoluted tubules of the kidney and is involved in the regulation of renal blood flow and filtration.

Option D - Hemidesmosomes: Hemidesmosomes are structures found in epithelial cells that help anchor them to the underlying basement membrane.

Solution for Question 77:

Correct Option C - Neisseria Gonorrheae:

- The clinical presentation of urethral discharge is suggestive of urethritis, and the presence of gram-negative intracellular diplococci in the microscopic image suggests the likely causative organism is Neisseria gonorrhea.
- This bacterium is known to cause sexually transmitted infections, including urethritis with semen-like discharge (most commonly), prostatitis in males, vaginitis, cervicitis, urethritis in females, and ophthalmia neonatorum in infants contracted during birth.

Incorrect Option:

Option A - Trichomonas: Trichomonas is a protozoan parasite that causes a sexually transmitted infection and non-gonococcal urethritis, prostatitis, and cystitis in males. It is typically observed as pear-shaped or oval flagellated organisms.

Option B - Chlamydia trachomatis: Chlamydia trachomatis is a bacterial pathogen known to cause non-gonococcal urethritis. They are gram-negative bacteria, and elementary bodies are seen on a microscope.

Option D - Gardnerella: Gardnerella vaginalis, which is associated with bacterial vaginosis and a few cases of urethritis. It is gram-negative coccobacilli with metachromatic granules.

Solution for Question 78:

Correct Option A - T. Pertunae:

- Yaws is a tropical disease caused by the bacterium Treponema pertenue. It is a subspecies of Treponema pallidum, which causes syphilis. Yaws primarily affects the skin, bones, and joints.

Incorrect Options:

Option B - T. endemicum: Treponema endemicum causes endemic syphilis, not yaws.

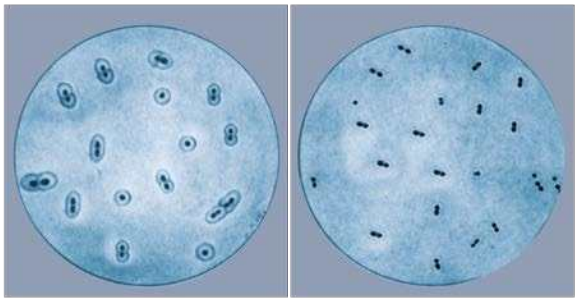
Option C - T. Carateum: Treponema carateum causes pinta.

Option D - Hemophilus: Hemophilus is associated with pneumonia, meningitis, croup, and suppurative lesions.

Solution for Question 79:

Correct Option A - Streptococcus Pneumoniae:

- The Quellung reaction is a phenomenon in which the capsule surrounding bacteria swells and becomes more visible when exposed to specific antibodies. Streptococcus pneumoniae is known to exhibit the Quellung reaction.



A. Positive reaction showing swollen capsules B. Negative reaction (Capsules not swollen)

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 80:

Correct Option A - ADP ribosylation of the G protein:

- Cholera toxin produced by Vibrio cholerae acts by ADP ribosylation of the G protein in the intestinal cells by the A1 subunit of the toxin, leading to increased production of cyclic AMP (cAMP) and subsequent activation of the cystic fibrosis transmembrane conductance regulator (CFTR) ion channel, resulting in massive secretion of electrolytes and water into the lumen of the small intestine.

Incorrect Options :

Option B - Inhibition of adenyl cyclase: Cholera toxin actually stimulates adenyl cyclase, leading to increased production of cAMP.

Option C - Activation of GTPase: Cholera toxin does not activate GTPase. It acts by ADP ribosylation of the G protein.

Option D - Active absorption of NaCl: Cholera toxin disrupts the normal absorption of electrolytes and water, resulting in excessive fluid secretion rather than active absorption of NaCl.

Solution for Question 81:

Correct Option C - Topical mupirocin:

MRSA Spread

- The most common form of spread is via the hands of healthcare workers.

Precaution

- Perform rigorous hand washing.
- Elimination involves: Administration of ointment mupirocin for nasal carriers Chlorhexidine washes for other sites of the body.
- Administration of ointment mupirocin for nasal carriers
- Chlorhexidine washes for other sites of the body.
- Administration of ointment mupirocin for nasal carriers
- Chlorhexidine washes for other sites of the body.

Treatment of MRSA

- Vancomycin
- Daptomycin.

Incorrect Option:

Options A, B, and D are incorrect. Refer to option C for an explanation.

Solution for Question 82:

Correct Option A - Botulinum:

- Botulinum toxin is produced by the bacterium Clostridium botulinum and has a specific mechanism of action
- Mechanism of action: Binding: Botulinum toxin binds to presynaptic nerve terminals at the neuromuscular junction. Internalization: The toxin is internalized into the nerve terminal via receptor-mediated endocytosis. Cleavage: Inside the nerve terminal, the toxin undergoes enzymatic cleavage, resulting in the formation of two subunits: a light chain (LC) and a heavy chain (HC). Neurotransmitter release inhibition: The light chain (LC) of the botulinum toxin is

a protease that specifically cleaves SNARE proteins involved in neurotransmitter release.SNARE proteins play a crucial role in the fusion of synaptic vesicles containing acetylcholinewith the neuronal membrane, allowing the release of acetylcholine into the synaptic cleft. Bycleaving SNARE proteins, botulinum toxin prevents the release of acetylcholine, leading tomuscle paralysis. Temporary effects: The effects of botulinum toxin are reversible, as newSNARE proteins can be synthesized by the affected neuron over time. The duration ofparalysis depends on the type of botulinum toxin used and the individual's metabolism.

• Binding: Botulinum toxin binds to presynaptic nerve terminals at the neuromuscular junction.

• Internalization: The toxin is internalized into the nerve terminal via receptor-mediated endocytosis.

• Cleavage: Inside the nerve terminal, the toxin undergoes enzymatic cleavage, resulting in the formation of two subunits: a light chain (LC) and a heavy chain (HC).

• Neurotransmitter release inhibition: The light chain (LC) of the botulinum toxin is a protease that specifically cleaves SNARE proteins involved in neurotransmitter release. SNARE proteins play a crucial role in the fusion of synaptic vesicles containing acetylcholine with the neuronal membrane, allowing the release of acetylcholine into the synaptic cleft. By cleaving SNARE proteins, botulinum toxin prevents the release of acetylcholine, leading to muscle paralysis.

• Temporary effects: The effects of botulinum toxin are reversible, as new SNARE proteins can be synthesized by the affected neuron over time. The duration of paralysis depends on the type of botulinum toxin used and the individual's metabolism.

• Binding: Botulinum toxin binds to presynaptic nerve terminals at the neuromuscular junction.

• Internalization: The toxin is internalized into the nerve terminal via receptor-mediated endocytosis.

• Cleavage: Inside the nerve terminal, the toxin undergoes enzymatic cleavage, resulting in the formation of two subunits: a light chain (LC) and a heavy chain (HC).

• Neurotransmitter release inhibition: The light chain (LC) of the botulinum toxin is a protease that specifically cleaves SNARE proteins involved in neurotransmitter release. SNARE proteins play a crucial role in the fusion of synaptic vesicles containing acetylcholine with the neuronal membrane, allowing the release of acetylcholine into the synaptic cleft. By cleaving SNARE proteins, botulinum toxin prevents the release of acetylcholine, leading to muscle paralysis.

• Temporary effects: The effects of botulinum toxin are reversible, as new SNARE proteins can be synthesized by the affected neuron over time. The duration of paralysis depends on the type of botulinum toxin used and the individual's metabolism.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 83:

Correct Option C - Ferments Maltose:

• The clinical presentation of urethral discharge is suggestive of urethritis, and the presence of gram-negative intracellular diplococci in the microscopic image suggests the likely causative organism is Neisseria gonorrhea.

Features	Neisseria meningitidis	Neisseria Gonorrhoeae
Habitat	Nasopharynx	Genitals
Disease	Meningitis	Gonorrhoea
Fermentation	Maltose and glucose	Only glucose
Shape	Lens shape with Capsule	Bean – shaped organism
Catalase and oxidase	Positive	Positive
	Strict Aerobe	Facultative Anaerobe
Virulence factor	Capsular polysaccharide A, B, C, Y, W135	Not present

Incorrect Option:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 84:

Correct Option A - Vibrio Cholera:

• Vibrio cholerae is a bacterium that causes cholera, a severe diarrheal disease. The symptoms of cholera include continuous rice watery diarrhea, which is consistent with the description provided. The drowsiness and unresponsiveness may be due to severe

dehydration and electrolyte imbalances resulting from profuse diarrhea.

Incorrect Options:

Option B - Entamoeba Coli: It is a non-pathogenic amoeba commonly found in the large intestine.

Option C - Escherichia: E. coli species can cause diarrhea and dysentery leading to dehydration, but rice water stools are not associated with E.coli.

Option D - Shigella Sonnei: It is a bacterium that causes shigellosis, a type of dysentery. While dysentery can cause bloody diarrhea and some degree of dehydration, but the rice water stool is characteristic of V. cholera..

Solution for Question 85:

Correct Option A - Shigella Dysenteriae:

• Shigella dysenteriae is a bacterium known to cause bacillary dysentery, characterized by symptoms such as bloody diarrhea and abdominal cramps. Shigella species (sonnei) are non-motile and do not ferment lactose or mannitol, which matches the findings described in the stool specimen.

Member	Motility	Lactose
E. coli	Motile	Lactose fermenting
Klebsiella	Non – motile	Lactose fermenting
Enterobacter	Motile	Lactose fermenting
Citrobacter	Motile	Lactose fermenting
Shigella spp. (sonnei)	Non – motile	Non-lactose fermenting
Shigella sonnei	Non – motile	Late lactose fermenting
Salmonella	Motile	Non - lactose fermenting

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 86:

Correct Option A - M. indicus Pranii:

- M. indicus pranii: Mw vaccine, also known as the Mw immunotherapeutic vaccine, is extracted from Mycobacterium indicus pranii. It is a heat-killed whole-cell preparation of this non-pathogenic mycobacterium.

Incorrect Option:

Option B - M. Ulcerans: M. ulcerans causes Buruli ulcer, which produces exotoxin myolactone.

Option C - M. Leprae: M. leprae is the causative agent of leprosy, not used for the production of the Mw vaccine.

Option D - M. Bovis: M. bovis is a pathogenic bacterium that causes bovine tuberculosis. It is not involved in the production of the Mw vaccine.

Solution for Question 87:

Correct Option C - ADP ribosylation of EF2:

- The gram staining shows gram-positive bacilli with a characteristic appearance resembling a Chinese letter (V or L shaped) or cuneiform arrangement, and the ends show metachromatic granules, giving it a club-shaped appearance suggestive of Corynebacterium diphtheriae.
- The toxin produced by Corynebacterium diphtheriae, known as diphtheria toxin, acts by ADP ribosylation of EF2.
- It is a bipartite. Fragment B helps in binding, and fragment A inhibits polypeptide chain elongation in the presence of nicotinamide adenine dinucleotide, inactivating the elongation factor EF-2.

Incorrect Options:

Option A - Increase in cAMP: This mechanism is associated with toxins such as cholera toxin and pertussis toxin, not the toxin produced by Corynebacterium diphtheriae.

Option B - Increase in cGMP: This mechanism is associated with toxins such as heat-stable enterotoxins of E. coli.

Option D - 60s ribosome: This mechanism is associated with toxins such as Shiga toxin and verocytotoxin.

Solution for Question 88:

Correct Option A - T. Pertunae:

- Yaws is a tropical disease caused by the bacterium Treponema pertenue. It is a subspecies of Treponema pallidum, which causes syphilis. Yaws primarily affects the skin, bones, and joints.

Incorrect Options:

Option B - T. endemicum: Treponema endemicum causes endemic syphilis, not yaws.

Option C - T. Carateum: Treponema carateum causes pinta.

Option D - Hemophilus: Hemophilus is associated with pneumonia, meningitis, croup, and suppurative lesions.

Solution for Question 89:

Correct Option D - Both assertion and reason are false:

- Assertion: The statement that WHO recommends Typhoid vaccine for disaster management in endemic areas is false. WHO does not specifically recommend Typhoid vaccine as a measure of disaster management in endemic areas. While vaccination can be an important public health intervention, the specific recommendation of Typhoid vaccine for disaster management in endemic areas is not supported.
- Reason: The reason provided, stating that vaccination is a cost-effective measure of disaster control, is also false. While vaccination can be cost-effective in certain situations, its effectiveness and cost-effectiveness depend on various factors such as the specific disease, population, vaccine availability, and infrastructure. The reason does not specifically support the assertion mentioned in the question.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 90:

Correct Option B - Leptospirosis:

- Leptospirosis is a bacterial infection that can occur in humans and animals. It is caused by the spirochete bacterium Leptospira interrogans.
- The clinical presentation of this patient, including right upper quadrant pain, calf pain, conjunctival suffusion, icterus, and fever, is consistent with the diagnosis of a severe form of leptospirosis, aka Weil's disease, aka hepato-renal-hemorrhagic syndrome.

Incorrect Options:

Option B: Chikungunya: It presents with fever, joint pain, headache, muscle pain, joint swelling, and rash. While fever is a common symptom in both leptospirosis and chikungunya, the other symptoms are not consistent with the presentation of this patient.

Option C: Dengue hemorrhagic fever: It presents with fever, headache, joint and muscle pain, and rash. Dengue hemorrhagic fever is a severe form of the disease that can lead to bleeding, shock, and death. While fever is a common symptom in both leptospirosis and dengue fever, the other symptoms are not consistent with the presentation of this patient.

Option D. Encephalopathy caused by hepatitis A: It presents with fever, fatigue, nausea, vomiting, abdominal pain, dark urine, and jaundice. While jaundice is a common symptom in both leptospirosis and hepatitis A, the other symptoms are not consistent with the presentation of this patient.

Solution for Question 91:

Correct Option C - No risk of Fulminant Infection:

• Clostridium difficile is a gram-negative bacteria responsible for causing acute colitis with bloody or watery diarrhea and pseudomembrane colitis in a patient on long-term usage of broad spectrum antibiotics like clindamycin, ampicillin, fluoroquinolones, and 3rd generation cephalosporins.

• It is responsible for the production of enterotoxin A and cytotoxin B, responsible for the disruption of the epithelial cell barrier causing diarrhea and pseudomembrane colitis.

• Pseudomembrane is composed of necrotic leukocytes, fibrin, mucus, and cellular debris.

• It can lead to fulminant colitis characterized by hypotension or shock, ileus, and megacolon.

• Treatment for Clostridium difficile infection typically involves discontinuing the offending antibiotic and initiating treatment with a fidaxomicin (drug of choice) and oral vancomycin.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 92:

Correct Option B - Bacillus Anthracis:

• The image shows an ulcer with a central black eschar on the arm, and gram staining shows gram-positive rods in chains, giving a bamboo-stick appearance due to concave or swollen ends of the bacilli is suggestive of anthrax (cutaneous) caused by bacillus anthracis.

Incorrect Options:

Option A - Bacillus Cereus: It is a gram-positive, rod-shaped bacterium that is found in soil and food. It can cause emetic or diarrheal type of food poisoning.

Option C - Pseudomonas Aeruginosa: It is a gram-negative, rod-shaped bacterium that is commonly found in soil and water.

Option D - Brucella Abortus: It is a gram-negative coccobacilli that causes brucellosis, which is a zoonotic disease. Its human form is an intracellular form that affects the reticuloendothelial system leading to hepatosplenomegaly. Patients present with irregular fevers, muscular pains, exhaustion, constipation, chills and many a times neurological complications.

Solution for Question 93:

Correct Option C - Trombiculid Mite:

This is the life cycle of Orientia tsutsugamushi, the causative agent of Scrub typhus, aka chigger-borne disease. Rodents and birds act as reservoirs of the bacteria and are transmitted to humans by the bite of 6-legged larvae, aka chigger of the tombiculid mite, when they trespass into scrub vegetation. Adult trombiculid mite has 4 pairs of legs.

Incorrect Options:

Option A - Hard Tick: Lyme disease, caused by Borrelia burdorferi, is transmitted by an 8-legged adult hard tick. Rodents and deer act as reservoir for the bacteria.

Option B - Soft Tick: Endemic/tick-borne relapsing fever caused by Borrelia recurrentis is transmitted by an 8-legged adult soft tick. It is spread transovarially; hence, soft ticks act as a vector as well as a reservoir.

Option D - Louse: Epidemic/louse-borne relapsing fever caused by Borrelia recurrentis is transmitted by an 8-legged adult louse (Pediculus humanus corporis).

Solution for Question 94:

Correct Option A - Leptospira:

- Leptospira is a genus of bacteria that causes the infectious disease leptospirosis. It has a complex life cycle involving both vertebrate and invertebrate hosts. Leptospira grows in the convoluted tubules of the rodent-like rats, which are then shed in the urine of infected animals like rats, mice, and dogs and can survive in water. Humans can be infected by direct contact of abraded skin or mucosa with contaminated water or soil or indirectly through contact with infected animals.

Incorrect Options:

Option B - Hantavirus: Hantaviruses are primarily transmitted to humans through inhalation of aerosolized urine, droppings, or saliva of infected rodents.

Option C - Yersinia Pestis: Yersinia pestis, the bacterium that causes plague, it is primarily transmitted to humans through the bite of infected fleas.

Option D - Salmonella Typhi: Salmonella typhi is the bacterium that causes typhoid fever. It is transmitted through ingestion of contaminated food or water.

Solution for Question 95:

Correct Option B - Scrub typhus:

Scrub typhus:

- Clinical presentation given in the question is consistent with Scrub typhus.
- Organism causing: Orientia tsutsugamushi
- Vector: Mite
- Clinical presentation:Fever,Nausea,Vomiting,Eschars (black lesion)
- Diagnosis: Weil Felix reaction: OX K is +ve only in scrub typhus.
- Treatment: Doxycycline or tetracycline are given
- Complication:Pneumonia,ARDS,Shock

Incorrect Options:

Option A - Anthrax:

- Anthrax is caused by Bacillus anthracis which presents with lesions on neck referred as Malignant Eschar.
- This is not the diagnosis, as anthax is not transmitted via mite bite

Option C - African tick typhus:

- African tick typhus is caused by R.Africae which is transmitted via tick bite.

Option D - Rocky mountain spotted fever:

- Rocky mountain spotted fever is caused by R.Rickettsii which is transmitted via tick bite.

Solution for Question 96:

Correct Option C - Cytoadherence:

- The phenomenon described, where infected red blood cells adhere to endothelial cells, is known as cytoadherence. This process is particularly associated with Plasmodium falciparum infection and is mediated by specialised proteins, such as PFEMP1 (Plasmodium falciparum Erythrocyte Membrane Protein 1), which are present on the surface of infected red blood cells.
- At times, inflammatory cytokines, notably IFN-gamma, generated by the malaria parasite, enhance the expression of endothelial cytoadherence receptors such as thrombospondin, E-selectin, VCAM-1, and ICAM-1 in brain capillaries, chondroitin sulfate B in the placenta, and CD36 in various organs. This leads to the adherence of infected red blood cells within capillaries and venules, eventually causing blockage.

Incorrect Options:

Option A - Rosetting: Rosetting refers to the clustering of uninfected red blood cells around an infected red blood cell, often seen in Plasmodium falciparum infection. These results in clinical manifestation of cerebral malaria.

Option - B Agglutination: Agglutination involves the clumping together of particles or cells, often mediated by antibodies.

Option D - Schizogony: Schizogony is the process of asexual reproduction in Plasmodium parasites, leading to the release of merozoites.

Solution for Question 97:

Correct Option D - Mycobacterium tuberculosis:

- Tuberculous meningitis manifests extrapulmonary tuberculosis caused by seeding of the meninges with the bacilli of Mycobacterium tuberculosis.
- Clinical presentation: Low-grade fever, malaise, headache, personality changes, confusion, varying presentation of cranial nerve and long tract signs, seizures, hemiparesis, and coma.
- Diagnosis: CSF analysis: Shows lymphocytosis, decreased glucose, and increased protein PCR MRI
- CSF analysis: Shows lymphocytosis, decreased glucose, and increased protein
- PCR

- MRI
- CSF analysis: Shows lymphocytosis, decreased glucose, and increased protein
- PCR
- MRI

Incorrect Options: CSF findings in Options A, B, and C show polymorphonucelar neutrophils and not lymphocytes.

Solution for Question 98:

Correct Option B - MTB + Rifampicin resistance:

- Gene xpert is a type of CBNAAT (Cartridge based nucleic acid amplification test)
- Useful for diagnosis of MTB and rifampicin resistance (based on Rpo B gene)
- Sensitivity : 88%
- Specificity: 99%
- Detection limit: 131 organism/ml

Incorrect Options:

- Gene xpert is not effective in identifying following conditions MTB + Isoniazid MTB +Ethambutol.
- MTB + Isoniazid
- MTB +Ethambutol.
- MTB + Isoniazid
- MTB +Ethambutol.

Solution for Question 99:

Correct Option B - Streptococcus pyogenes:

• Streptococcus pyogenes is group A beta-hemolytic streptococci

• Clinical features

Suppurative C/F	Non-suppurative C/F
Necrotizing fasciitis Impetigo Pneumonia Pharyngitis Lymphangitis Erysipelas Streptococcal toxic shock syndrome Scarlet fever	Post-streptococcal glomerulonephritis (PSGN) Rheumatic heart disease (RHD) Paediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococcal infection (PANDAS)

• Diagnosis Stain : gram positive cocci in chain Blood agar findings: pin point colonies with wide zone hemolysis Biochemical test: Catalase negative Oxidase negative PYR test positive Bacitracin sensitive

• Stain : gram positive cocci in chain

• Blood agar findings: pin point colonies with wide zone hemolysis

• Biochemical test: Catalase negative Oxidase negative PYR test positive Bacitracin sensitive

• Catalase negative

• Oxidase negative

• PYR test positive

• Bacitracin sensitive

• Treatment Penicillin Erythromycin in case of resistance

• Penicillin

• Erythromycin in case of resistance

• Stain : gram positive cocci in chain

• Blood agar findings: pin point colonies with wide zone hemolysis

• Biochemical test: Catalase negative Oxidase negative PYR test positive Bacitracin sensitive

• Catalase negative

• Oxidase negative

• PYR test positive

• Bacitracin sensitive

• Catalase negative

• Oxidase negative

• PYR test positive

• Bacitracin sensitive

• Penicillin

• Erythromycin in case of resistance

Incorrect Options:

Option A - Streptococcus viridans:

• Streptococcus viridans is gram positive cocci cause infective endocaridtis.

Option C - Streptococcus pneumoniae:

• Streptococcus pneumoniae is Lanceolate shaped gram positive diplococci causes Meningitis, Otitis media, Sinusitis and Community acquired pneumonia

Option D - Enterococcus:

• Enterococcus is spectacle shaped gram positive cocci presents with clinical features of catheter associated UTI,Septicemia.

Solution for Question 100:

Correct Option C- Bacillus anthracis:

• Bacillus anthracis is the causative agent of anthrax, which commonly affects individuals exposed to infected animals such as cattle.

• The characteristic gram-positive rods with square ends and central spores, known as " boxcar" appearance, are typical of Bacillus anthracis.

• The clinical presentation of a lesion with central necrosis, erythema, and raised borders is consistent with cutaneous anthrax, often seen in individuals handling infected animal products.

Incorrect Options:

Option A- Brucella abortus: Brucella species are gram-negative bacteria responsible for brucellosis, a zoonotic disease transmitted from animals to humans. The gram-staining appearance and clinical presentation of brucellosis, such as fever, arthralgia, and hepatosplenomegaly, differ significantly from the findings in the presented scenario.

Option B- Pseudomonas aeruginosa: This gram-negative bacterium is known for causing opportunistic infections, particularly in immunocompromised individuals.

Option D- Bacillus cereus: Bacillus cereus is a gram-positive, spore-forming bacterium that can cause food poisoning after consumption of Chinese fried rice. It does not produce skin lesions.

Solution for Question 101:

Correct Option C- Clostridioides difficile:

- Clostridioides difficile is a gram-positive bacterium that is the cause most implicated in antibiotic-associated diarrhea.
- Antibiotic use is the leading risk factor for C. difficile infection. Several classes of antibiotics, such as penicillins, cephalosporins, fluoroquinolone, and clindamycin, are implicated.
- Apart from antibiotic usage, other risk factors involved in the causation include advanced age, chemotherapy, use of proton pump inhibitors, chronic renal disease, chronic liver disease, and malnutrition.
- Watery diarrhea with mucus or occult blood, anorexia, nausea, vomiting, low-grade fever, and lower abdominal pain are the symptoms.

Incorrect Options:

Option A- Escherichia coli: While E. coli can cause diarrhea, it is less likely in this scenario because the patient developed symptoms after being discharged and being on clindamycin, not due to contaminated food or water sources commonly associated with E. coli infections.

Option B- Norovirus: Norovirus is a common cause of viral gastroenteritis. It is less likely in this case because the patient's symptoms and history of antibiotic use are more consistent with a bacterial infection.

Option D- Rotavirus: Rotavirus is another viral cause of gastroenteritis, primarily affecting children. It presents with symptoms like fever, vomiting, and watery diarrhea. It is less likely in an adult patient like the one described here.

Solution for Question 102:

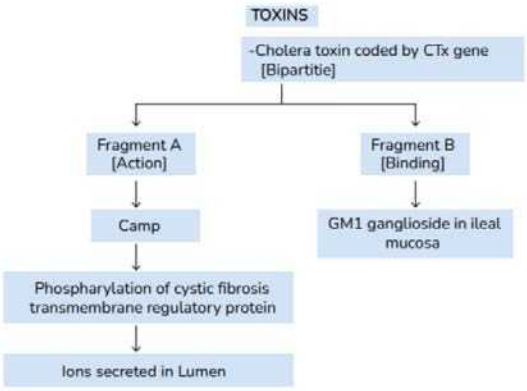
Correct Option B - Streptococcus pyogenes:

• Superantigens, in contrast to conventional antigens, don't require processing and presentation by antigen-presenting cells (APCs) to activate T cells. Instead, they interact laterally with the variable beta (Vβ) part of the T-cell receptor (TCR) outside the peptide-binding groove.

Superantigens, in contrast to conventional antigens, don't require processing and presentation by antigen-presenting cells (APCs) to activate T cells. Instead, they interact laterally with the variable beta (Vβ) part of the T-cell receptor (TCR) outside the peptide-binding groove.

Incorrect Options:

Option A - Vibrio cholerae: Cholera toxin is produced by Vibrio cholerae, the bacterium responsible for cholera. The toxin acts by ADP-ribosylation of the Gs protein in intestinal cells. This leads to increased intracellular cAMP levels, resulting in the continuous activation of the CFTR chloride channel and the efflux of chloride ions into the intestinal lumen. The osmotic imbalance causes massive diarrhea, leading to dehydration and potentially severe electrolyte imbalances.



Option C - Clostridium difficile: Clostridium difficile is a bacterium that can cause antibiotic-associated diarrhea and colitis. It produces toxins, particularly toxin A (enterotoxin) and toxin B (cytotoxin).The toxins produced by Clostridium difficile rapidly target and attack the GTP-binding proteins, specifically Rho, Rac, and CDC 42. This attack results in considerable damage to the actin cytoskeleton, ultimately leading to cellular death.

Option D -Clostridium botulinum: Clostridium botulinum produces botulinum toxin, which is one of the most potent toxins known. Subtypes A, B, and E cause infections, with A being the most severe. The toxin induces flaccid paralysis by blocking acetylcholine release, targeting Snare proteins. In neurons, it cleaves Snap 25, Syntaxin 1, and Synaptobrevin, preventing muscle contraction and causing floppy muscles. Clinical manifestations include the absence of muscle contraction, leading to flaccidity.

Solution for Question 103:

Correct Option A - Scrub typhus:

- The clinical presentation of a farmer with fever, thrombocytopenia (platelet count of 88,000), elevated serum creatinine levels (1.4), and the presence of a black eschar is suggestive of a diagnosis of scrub typhus.

The clinical presentation of a farmer with fever, thrombocytopenia (platelet count of 88,000), elevated serum creatinine levels (1.4), and the presence of a black eschar is suggestive of a diagnosis of scrub typhus.

Mite larvae

↓

Seek the moist area of body(axilla, groin)

After 1-3 weeks

Eschars (black lesion), Fever, Nausea, Vomiting

Compilation: Pneumonia, ARDS, Shock

Incorrect Options:

Option B - Anthrax: Anthrax usually presents with fever, malaise, and characteristic skin lesions (cutaneous anthrax) Thrombocytopenia and elevated creatinine levels are less commonly associated with anthrax.

Option C - Leptospirosis: Leptospirosis also known as "Weil's disease "usually presents with fever, headache, muscle aches, and jaundice. Cutaneous black eschar is not seen in leptospirosis Infection is likely to occur among rice/paddy field workers who are exposed to water contaminated with rat urine, especially during rainy conditions.

Option D - Brucellosis: Brucellosis is a zoonotic disease, potentially transmitted through direct contact with infected animals or the consumption of raw milk. The disease is characterized by hepatosplenomegaly, attributed to the invasion of reticuloendothelial cells, particularly macrophages. Symptoms include night sweats, arthritis, and a low platelet count. The incubation period typically spans from 1 week to 1 month. The black eschar is not seen with brucellosis.

Solution for Question 104:

Correct Option A- Diphtheria Toxin:

- Produced by Corynebacterium diphtheriae, the bacterium that causes diphtheria.
- Mechanism of Action: Diphtheria toxin inhibits protein synthesis by ADP-ribosylation of EF-2. This modification prevents EF-2 from participating in the translocation step during protein synthesis, leading to the inhibition of elongation and ultimately causing cell death.

Incorrect Options:

Option B- Clostridium Toxins (Botulinum and Tetanus Toxins):

- Clostridium botulinum toxin and Clostridium tetani toxin act on different targets. Botulinum toxin inhibits the release of acetylcholine at neuromuscular junctions, leading to muscle paralysis.
- Tetanus toxin interferes with neurotransmitter release, causing spastic paralysis.

Option C- Bacillus anthracis Toxin:

- Bacillus anthracis produces a toxin known as anthrax toxin.
- Anthrax toxin does not directly interfere with EF-2 function but exerts its effects through the disruption of cellular signaling pathways.

Option D- Vibrio cholerae Toxin (Cholera Toxin):

- Cholera toxin primarily affects the regulation of ion transport in host cells by increasing intracellular levels of cyclic AMP (cAMP).
- It does not directly interfere with EF-2 function.

Solution for Question 105:

Correct Option A- Nocardia and acid-fast stain:

- Nocardia is a Gram-positive bacterium that appears purple after Gram staining, forms filaments , and is detectable using the Ziehl-Neelsen staining method.

Incorrect Options:

Option B- Streptococcus gram stain:

• Streptococcus bacteria stain purple in the Gram staining method, indicating their Gram-positive nature. Streptococcus bacteria typically have a spherical or ovoid shape and can form chains or pairs. Unlike Nocardia, they do not display filamentous structures.

Option C- Cryptococcus india ink:

• Cryptococcus yeast cells are round to oval and may be encapsulated. The capsule is the key feature that will be visible in the India ink preparation, creating a halo around the stained cells. Unlike Nocardia, they do not display filamentous structures.

Option D- Coccidioidomycosis PAS stain:

• Coccidioides species will appear magenta or pink due to the PAS stain, and characteristic structures like spherules may be visible. Unlike Nocardia, they do not display filamentous structures.

Solution for Question 106:

Correct Option A- Listeria:

• Listeria monocytogenes is a bacterium that can move within and between cells by exploiting the host cell's actin cytoskeleton.

• It uses host cell machinery to polymerize actin filaments at one end of the bacterium, pushing it into the cell membrane. This allows Listeria to move within the host cell and from cell to cell.

Incorrect Options:

Option B- EIEC (Enteroinvasive Escherichia coli):

• EIEC does invade and multiply within the epithelial cells of the colon, but its mechanism of movement is not primarily based on actin polymerization like Listeria.

Option C- Salmonella:

• Salmonella also invades host cells, but it uses a different mechanism involving a type III secretion system. It induces rearrangement of the host cell cytoskeleton, but its movement is not primarily transcellular by actin filaments

Solution for Question 107:

Correct option D - Vibrio parahaemolyticus:

- This is a case of acute gastroenteritis as the patient presents with complaints of fever, diarrhoea and abdominal pain for 3 days.
- Further history reveals that he had consumed raw fish 3 days back.
- The most likely agent to have caused this illness is Vibrio parahaemolyticus.
- Vibrio parahaemolyticus causes gastroenteritis. It is also found in raw fish. The selective media is Wagatsuma agar.

Incorrect options:

Option A - Vibrio cholerae:

- Vibrio cholerae causes cholera.

Option B - Salmonella typhi:

- Salmonella typhi is the causative agent of enteric fever.
- The selective media for Salmonella is Wilson Blair Media.

Option C - Shigella sonnei:

- Shigella infections are always invasive and generally will have a little pus in the stool.

Solution for Question 108:

Correct Option A - Staphylococcus epidermidis

- Staphylococcus epidermidis a coagulase-negative staphylococci is a common cause of prosthetic valve endocarditis, especially within the first year after surgery

Incorrect Options:

Option B - Streptococcus viridans: This is a common cause of native valve endocarditis, not prosthetic valve endocarditis. Streptococcus viridans is a part of the normal flora of the mouth and throat and can cause endocarditis after dental procedures or in people with poor oral hygiene.

Option C - Enterococcus faecalis: This bacterium is a common cause of healthcare-associated infections, including urinary tract infections, wound infections, and, less commonly, endocarditis. While it can cause endocarditis, it is not the most likely cause in this patient who has a prosthetic heart valve.

Option D - Staphylococcus saprophyticus: Staphylococcus saprophyticus is a coagulase-negative staphylococci associated with urinary tract infections commonly.

Solution for Question 109:

Correct Option B - Streptococcus pyogenes

- Streptococcus pyogenes, also known as group A Streptococcus, is the correct answer. This bacterium is responsible for both acute rheumatic fever (ARF) and post-streptococcal glomerulonephritis (PSGN). ARF is a systemic immune-mediated inflammatory disorder that can develop after an untreated or inadequately treated streptococcal pharyngitis (sore throat). It can lead to joint pain, inflammation, and other clinical manifestations. PSGN, on the other hand, is a kidney disease that can occur as a result of an immune response to a streptococcal infection, particularly in the skin or throat. The elevated antistreptolysin O (ASO) titers and the glomerular changes seen on renal biopsy in the given scenario are indicative of a streptococcal infection-related condition.

Incorrect Options:

Option A - Streptococcus pneumoniae: Streptococcus pneumoniae is a common cause of bacterial pneumonia and other respiratory infections. It is not typically associated with acute rheumatic fever or post-streptococcal glomerulonephritis (PSGN).

Option C - Streptococcus agalactiae: Streptococcus agalactiae, also known as group B Streptococcus, is a common cause of neonatal infections and can also cause infections in adults with underlying medical conditions. It is not associated with acute rheumatic fever or PSGN.

Option D - Streptococcus mutans: Streptococcus mutans is primarily associated with dental caries and oral infections. It is not a common cause of acute rheumatic fever or PSGN.

Solution for Question 110:

Correct Option C - Streptococcus agalactiae (Group B Streptococcus)

- This is the correct answer. Group B Streptococcus is a known cause of neonatal meningitis, particularly in the first few days after birth. It is often found in the birth canal of asymptomatic carriers and can be transmitted to the newborn during delivery. Bacitracin resistance is a characteristic feature of Group B Streptococcus.

Incorrect Options:

Option A - Pneumococcus (Streptococcus pneumoniae): Pneumococcus is a common cause of bacterial meningitis, but it is not typically associated with neonatal meningitis. Pneumococcus is generally sensitive to bacitracin.

Option B - Streptococcus pyogenes (Group A Streptococcus): Streptococcus pyogenes is more commonly associated with conditions like strep throat and skin infections. It is not a common cause of neonatal meningitis, and it is also sensitive to bacitracin.

Option D - Meningococci (Neisseria meningitidis): Meningococcal meningitis is more common in older children and adults. It is not a common cause of neonatal meningitis. Neisseria meningitidis is typically sensitive to bacitracin.

Solution for Question 111:

Correct Option A - Vibrio Parahemolyticus

- Vibrio parahemolyticus is a bacterium commonly associated with consuming raw or undercooked seafood, particularly oysters. It causes gastroenteritis with symptoms like watery diarrhea, abdominal cramps, nausea, vomiting, and sometimes low-grade fever. The incubation period is usually short, ranging from a few hours to a couple of days after consuming contaminated seafood.

Incorrect Options:

Option B - Staph aureus: Staphylococcus aureus is a bacterium that can cause food poisoning when certain strains produce toxins. However, symptoms usually manifest within a few hours of consuming contaminated food and involve rapid onset of nausea, vomiting, abdominal cramps, and occasionally diarrhea. This bacterium is not typically associated with dysentery (bloody

diarrhea) as described in the patient's case.

Option C - E. coli: Certain strains of Escherichia coli (E. coli) can cause foodborne illnesses, including those leading to diarrhea. However, these infections are not usually associated with dysentery. E. coli strains like enterotoxigenic E. coli (ETEC) or enteropathogenic E. coli (EPEC) tend to cause watery diarrhea rather than dysentery.

Option D - Vibrio Cholerae: Vibrio cholerae is responsible for causing cholera, which leads to profuse watery diarrhea, rapid dehydration, and electrolyte imbalances. Cholera typically results in a "rice-water" appearance of stools due to its watery nature. While cholera can lead to severe diarrhea, it does not usually cause dysentery with bloody stools.

Solution for Question 112:

Correct Option A - Meningitis:

- Meningitis is typically the result of a direct invasion of the meninges by bacteria or viruses, and it is not primarily caused by the production of exotoxins.

Incorrect Options:

Option B - HUS: HUS is often associated with Shiga toxin-producing E. coli (STEC) infections. Shiga toxin is an example of an exotoxin, and it plays a role in the pathogenesis of HUS.

Option C - Toxic shock syndrome: TSS is typically caused by exotoxins produced by Staphylococcus aureus or Streptococcus pyogenes. Exotoxins are a primary factor in the development of TSS.

Option D - Pseudomembranous colitis: Pseudomembranous colitis is primarily caused by Clostridium difficile infection. The main virulence factor in C. difficile infection is the production of toxins (exotoxins), particularly toxins A and B, which contribute to the characteristic symptoms of this condition.

Solution for Question 113:

Correct Option A - Scrub Typhus:

- The patient presenting with fever, myalgia, and cutaneous lesions (black eschar) after a bite with a mite in Uttarakhand during September is suggestive of Scrub typhus.
- Scrub typhus is a bacterial infection caused by the Orientia tsutsugamushi bacterium. It is transmitted to humans through the bite of infected chiggers (larva of trombiculid mites) found in areas with dense vegetation. Scrub typhus is commonly seen in rural areas, particularly in regions with heavy vegetation, such as forests or grasslands.
- The clinical presentation of scrub typhus includes fever, headache, myalgia (muscle pain), and a characteristic eschar at the site of the chigger bite. The eschar is a black, necrotic lesion surrounded by an area of inflammation.

Incorrect Options:

Option B - Murine Typhus: Murine typhus is caused by the bacteria Rickettsia typhi, transmitted to humans by rubbing or inhalation of the feces of infected fleas. It typically presents with fever, headache, myalgia, anorexia, and rash (involving the trunk more often than the extremities).

Option C - Q Fever: Q fever is caused by the bacteria Coxiella burnetii, transmitted to humans through ingestion of contaminated milk or by handling wools, hides or meat. It can cause a flu-like illness, hepatitis, meningoencephalitis or endocarditis.

Option D - Louse Borne Typhus: Louse-borne typhus also known as epidemic typhus, is caused by the bacteria Rickettsia prowazekii and is transmitted to humans through rubbing of louse (Pediculus humanus corporis) feces on an abraded skin. It presents with high fever, headache, and rash starting from trunk and spreading to limbs.

Solution for Question 114:

Correct Option A - Zona occludens:

- Rice watery stool is a characteristic of cholera caused by vibrio cholerae. A toxin against the Zona occludens, or tight junctions is produced in the cholerae disrupting the junction leading to increased permeability, leakage of fluid and electrolytes resulting in rice water stools.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 115:

Correct Option A - 1- C, 2- D, 3- B, 4- A:

- Patch test is done to detect the hypersensitivity with allergen in contact dermatitis.
- KOH mount is done to detect Tinea infections which shows a presence of branching hyaline septate hyphae.
- Biopsy is the preferred investigation in case of Lupus Vulgaris.
- Slit skin smear is used in the diagnosis of Leprosy by taking skin samples over the ear lobes

Incorrect Options:

- Option B, C & D: These options are incorrect.

Solution for Question 116:

Correct Option C - Assertion is true and reason is false:

- Toxemia is the reason for myocarditis caused by faucial diphtheria
- The bacilli remain confined to the site of entry, where they multiply and form the toxin, which is absorbed and produces toxic damage to the heart (myocarditis), kidney (tubular necrosis), liver and adrenal glands.

Toxemia is the reason for myocarditis caused by faucial diphtheria

The bacilli remain confined to the site of entry, where they multiply and form the toxin, which is absorbed and produces toxic damage to the heart (myocarditis), kidney (tubular necrosis), liver and adrenal glands.

Incorrect Options:

- Option A, B and D: These options are incorrect.

Option A, B and D: These options are incorrect.

Solution for Question 117:

Correct Option A - Vancomycin:

- In Enterococcus infective endocarditis patients with penicillin allergy or penicillin resistant strains, the drug of choice in the treatment of such cases is vancomycin.
- Enterococci are intrinsically resistant to a wide array of antimicrobial agents, and they are generally resistant to killing by any of the single agents (e.g., ampicillin or vancomycin) that are bactericidal for most other gram-positive cocci.
- Effective bactericidal activity can only be achieved with the combination of a cell wall–active agent, such as ampicillin or vancomycin, and an aminoglycoside, such as gentamicin or streptomycin.

Incorrect Options:

Option B - Ceftriaxone:

- Ceftriaxone is 3rd generation cephalosporins used in the treatment of community acquired gram negative infections.

Option C - Piperacillin:

- Piperacillin is frequently used in treating gram negative infections in neutropenic/immunocompromised or burn patients.

Option D - Aztreonam:

- Aztreonam is Beta-lactam antibiotic that inhibits enteric bacilli, H.influenzae, and pseudomonas.

Solution for Question 118:

Correct Option B - Neisseria gonorrhoeae:

- The Gonococcal Isolate Surveillance Project (GISP) is a national collaborative surveillance project in the United States. The program was established in 1986 to analyze and monitor

antimicrobial resistance trends among N. gonorrhoeae.

• Meningococci were uniformly sensitive to penicillin and other antibiotics, but resistant strains have emerged and become common in many areas.

• This penicillin resistance is due to the ability of the Neisseria to produce penicillinase.

• They are also resistant to fluoroquinolones.

Incorrect Options:

• Option A, C and D: Penicillin resistance is not noted in these diseases.

Solution for Question 119:

Correct Option A - Neonatal sepsis:

• Streptococcus agalactiae is a group B ■- hemolytic streptococci which is the single most common cause of neonatal meningitis.

• Early onset type- Occurs within a week of birth

• Late onset type- Develops between second and twelfth weeks of life.

• The more common early onset type presents as meningitis or septicemia and is often fatal.

Incorrect Options:

Option B - Infective endocarditis: Infective endocarditis is caused by Staph. aureus, Staph. epidermidis, Streptococcus viridans.

Option C: - Dental caries: Dental caries is caused by other species of streptococcus like Streptococcus mitis.

Option D - Pharyngitis with or without tonsillitis: Pharyngitis with or without tonsillitis is caused by Group A streptococcus.

Solution for Question 120:

Correct Option C - Inhibit the release of GABA and Glycine:

- Tetanospasmin is the toxin responsible for tetanus
- Tetanospasmin resembles strychnine in its effects. The tetanus toxin specifically blocks synaptic inhibition in the spinal cord, presumably at inhibitory terminals that use glycine and GABA as neurotransmitters.
- The abolition of spinal inhibition causes uncontrolled spread of impulses initiated anywhere in the central nervous system.
- This results in muscle rigidity and spasms due to the simultane - ous contraction of agonists and antagonists, in the absence of reciprocal inhibition.

Incorrect Options:

Option A - Inhibition of protein synthesis: Shiga toxin or verotoxin which is similar to EHEC verotoxin acts by inhibition of protein synthesis.

Option B - Inhibit the release of Ach: Botulinum toxin acts by inhibiting the release of ACh.

Option D - Inhibit the release of Norepinephrine: Release of norepinephrine is inhibited by certain drugs like alpha-2 agonists.

Solution for Question 121:

Correct Option A - Syphilis:

- Syphilis is transmitted through blood transfusions and its antigen can be transmitted through all the components of the blood.

Incorrect Options:

Option B - Toxoplasmosis: Toxoplasma is transmitted through nucleated cells in the blood, especially WBC's.

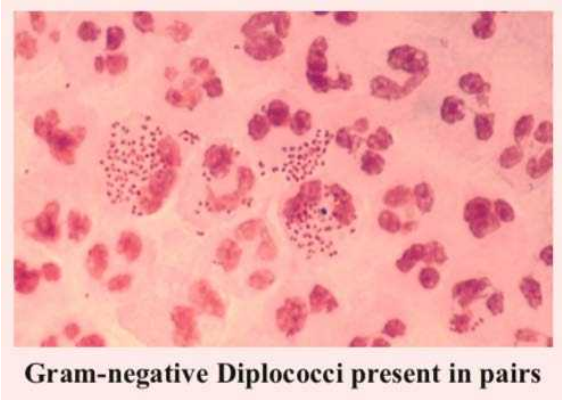
Option C - Malaria: Malaria is transmitted through the RBC's in the blood.

Option D - Filarial infection: Filarial infection is mainly transmitted through the plasma of the blood.

Solution for Question 122:

Correct Option A - Neisseria gonorrhoeae:

- In women, the initial infection involves the urethra and the cervix uteri and it can extend to Bartholin's glands, the endometrium and the fallopian tubes.
- On microscopy the Gram negative diplococci present in pairs can be seen.



Gram-negative Diplococci present in pairs

Incorrect Options:

Option B - Chlamydia: Chlamydia appears as a cluster of spores like on microscopy.

Option C - Chancroid: The bacilli may be arranged in small groups or whorls or in parallel chains, giving a 'school of fish' or 'railroad track' appearance.

Option D - Syphilis: Syphilis on dark microscopy shows thin forward and backward moving organisms.

Solution for Question 123:

Correct Option D - 1- D, 2- A, 3- B, 4- C:

- Gastric Maltoma is the most frequent malt lymphoma that is caused by H. pylori.
- Chlamydia psittaci is involved in causing malt lymphoma of the eyes and tear ducts.
- Campylobacter jejuni, commonly involved in food poisoning, is a causative agent of small bowel maltoma.
- Borrelia burgdorferi, causes lyme disease which is involved in causing skin Mlatoma.

Incorrect Options:

Option A, B, and C: These are incorrect options.

Solution for Question 124:

Correct Option B – Brucella:

- Relative bradycardia is seen in Brucella

Incorrect Options:

Option A, D:

- These are also the organisms associated with relative bradycardia. But the better answer here will be brucella.

Option C - Bordetella:

- This organism is not associated with relative bradycardia

Solution for Question 125:

Correct Option B - TCBS agar:

- TCBS agar is the selective media used for Vibrio cholerae
- Bile salt makes it selective.
- Sucrose fermentation gives yellow color

Incorrect Options:

Option A - Mannitol salt agar:

- It is used to grow Staphylococcus aureus

Option C - Cetrimide agar:

- Cetrimide agar is used for the growth of Pseudomonas

Option D - Potassium Tellurite Agar:

- Potassium Tellurite Agar is used for the growth of C. Diphtheriae

Solution for Question 126:

Correct Option A - ADP phosphorylation of EF-2 - Decreasing protein synthesis:

- The mechanism of action of Diphtheria toxin (Fragment A) is ADP phosphorylation of EF-2 - Decreasing protein synthesis

Toxin	Target/Mechanism
Diphtheria Toxin	ADP-ribosylates elongation factor-2 (EF-2), halting protein elongation.
Exotoxin A of Pseudomonas	ADP-ribosylates EF-2, similar to diphtheria toxin, blocking protein elongation.
Shiga Toxin and Shiga like toxin /Verocytotoxin(EHEC)	Cleaves a specific adenine residue in 28S rRNA (60S subunit), disrupting ribosomal function.
Bacillus anthracis	Lethal factor cleaves MAPKKs (not directly inhibiting ribosomes but leads to downstream effects on protein synthesis).
Diarrheal type of Bacillus cereus poisoning & heat labile toxin of ETEC	Increasing cAMP
Emetic type of Bacillus cereus poisoning & heat stable toxin of ETEC	Increasing cGMP

Incorrect Options

Option B - Increasing cAMP:

- The mechanism of action of Diarrheal type of Bacillus cereus poisoning is by increasing cAMP

Option C -Increasing cGMP:

- The mechanism of action of Emetic type of Bacillus cereus poisoning is by increasing cGMP

Option D - Decreasing 60S ribosome:

- The mechanism of action of Shiga toxin is by decreasing 60S ribosome

Solution for Question 127:

Correct Option C - Leptospirosis:

- Leptospirosis is a bacterial infection caused by Leptospira species. The given clinical features, especially conjunctival suffusion, are characteristic of leptospirosis.

Incorrect Options:

- Options A, B, and D do not show signs of conjunctival suffusion and calf tenderness.

Solution for Question 128:

Correct Option A - Neisseria meningitidis:

- Neisseria meningitidis is the likely cause of bacterial meningitis. The symptoms of fever, rash, and neck rigidity, combined with the deficiency in the membrane-attacking complex, are consistent with this diagnosis.

Incorrect Options:

Option B - Neisseria gonorrhoea: Neisseria gonorrhea primarily affects the genital tract and anorectal region and is not associated with meningitis.

Option C - Klebsiella: Klebsiella causes community-acquired pneumonia, catheter-associated UTI, and sepsis.

Option D - CMV: Cytomegalovirus typically causes hepatosplenomegaly, jaundice, and purpuric rashes.

Solution for Question 129:

Correct Option A - Consider UTI:

- The presence of novobiocin-resistant Staphylococcus (likely Staphylococcus saprophyticus) in significant numbers (>10^4 colonies) suggests a urinary tract infection.

Incorrect Options:

- Options B, C, and D are incorrect, as explained above.

Solution for Question 130:

Correct Option A - Toxoplasma:

- The triad of chorioretinitis, hydrocephalus, and cerebral calcifications is characteristic of congenital toxoplasmosis caused by Toxoplasma gondii.

Incorrect Options:

- Options B, C, and D do not cause cerebral calcifications.

Solution for Question 131:

Correct Option B - Flagella:

- The "H" in EHEC 0157, H7, refers to the flagella of Enterohemorrhagic Escherichia coli.

Incorrect Options:

- Options A, C, and D are not denoted by “H”.

Solution for Question 132:

Correct Option A - Tab Cefixime 400mg:

- Cefixime or ceftriaxone is the drug of choice for treating gonorrhea, which is the likely cause of purulent urethral discharge after unprotected sexual intercourse and as gram-negative diplococci in pairs are seen in the image.

Incorrect Options:

- Options B, C, and D are not preferred in the management of gonorrhea.

Solution for Question 133:

Correct Option A - Acanthamoeba:

- Acanthamoeba is most likely responsible for this patient's condition. Acanthamoeba keratitis is associated with improper use of contact lenses, and spikes are seen on microscopy.

Incorrect Options:

- Options B, C, and D are not responsible for causing this patient’s condition, as explained above.

Solution for Question 134:

Correct Option B - V. Parahemolyticum:

- Vibrio parahaemolyticus is a common cause of gastroenteritis linked to the consumption of raw or undercooked seafood, especially oysters. The typical symptoms include diarrhea, abdominal pain, and vomiting, often appearing within 24 hours of ingestion.

Incorrect Options:

Option A - S. Typhi: S. Typhi causes enteric fever.

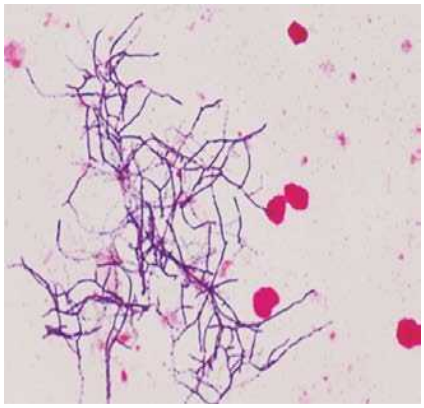
Option C - Staphylococcus: Staphylococcus aureus can cause food poisoning after consumption of stale meats and dairy products.

Option D - Cholera: Vibrio cholerae causes cholera, characterized by severe, watery diarrhea ("rice water stools").

Solution for Question 135:

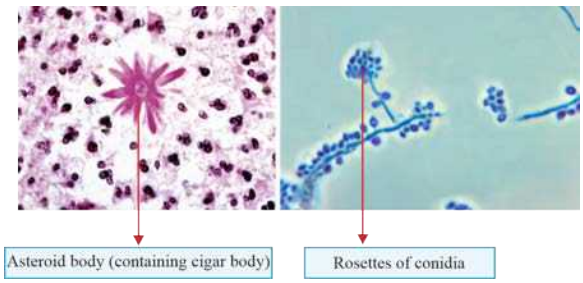
Correct Option A - Nocardia:

- Nocardia is a gram-positive bacterium that forms branching filaments and is often associated with causing cutaneous infections in immunocompetent individuals.

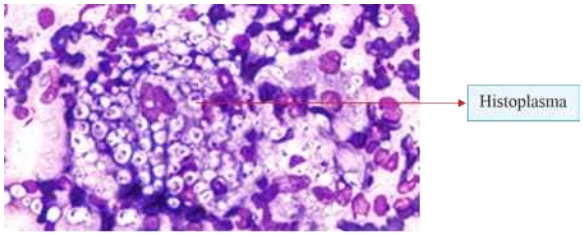


Incorrect Options:

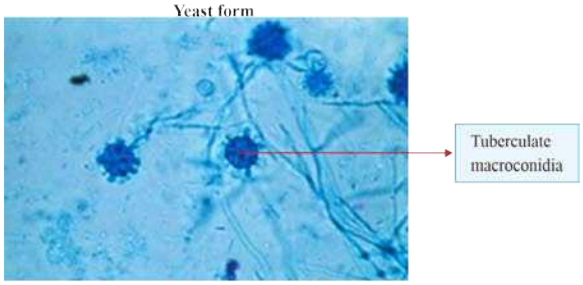
Option B - Sporothrix: Patients with sporotrichosis will usually present with a history of thorn prick followed by ulcers/pustules formation at the trauma site and lymphadenopathy.



Option C - Histoplasma: Histoplasmosis is caused by bat or bird droppings and affects the reticuloendothelial system.

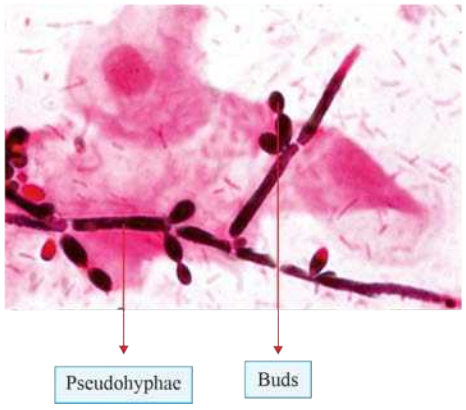


Yeast form



LPCB stain on SDA agar

Option D - Candida: On microscopy, candida will appear as both buds and pseudohyphae forms.



Solution for Question 136:

Correct Answer: B) Lowenstein-Jensen (LJ) medium

Explanation: A patient with HIV presenting with cough and fever raises suspicion of tuberculosis (TB), which is common in immunocompromised individuals. Lowenstein-Jensen (LJ) medium is a specialized culture medium used to isolate Mycobacterium tuberculosis.

Lab Investigations for TB:

Investigations	Purpose	Advantages	Limitations
Sputum smear	Initial screening for pulmonary TB. Samples are stained using the Ziehl-Neelsen method or Auramine-O stain to visualize acid-fast bacilli (AFB) under a microscope.	Quick and inexpensive. Can provide results within a few hours.	Low sensitivity, especially in cases of extrapulmonary TB or in patients with HIV. Cannot differentiate between Mycobacterium tuberculosis and non-tuberculous mycobacteria (NTM).
Culture Methods	Confirm the diagnosis of TB and assess drug susceptibility. Solid Media: Sputum is inoculated onto solid culture media like Löwenstein-Jensen or Middlebrook agar. Liquid Media: Automated systems like BACTEC or MGIT are used to detect growth faster.	The gold standard for TB diagnosis. Allows for drug susceptibility testing.	Time-consuming; results can take weeks. Requires specialized laboratory facilities.
Xpert MTB/ RIF	Detects Mycobacterium tuberculosis and rifampicin resistance. Method: A cartridge-based system that performs nucleic acid amplification (NAAT). Sputum or other specimens are processed directly in the cartridge.	Rapid results (within 2 hours). High sensitivity and specificity. Simultaneously detects rifampicin resistance.	More expensive than smear microscopy. Does not provide full drug susceptibility testing.
LAMP (Loop-Mediated Isothermal Amplification)	Detects DNA of Mycobacterium tuberculosis. Method: Amplifies DNA at a constant temperature without the need for thermal cycling. Sputum or other specimens are used.	Rapid results (within 1-2 hours). High specificity and sensitivity. Can be performed at the point of care. Used for patients with HIV with CD4+ counts of less than 100 cells/μL irrespective of signs and symptoms of TB.	Less widely available than PCR-based tests. Cannot detect drug resistance directly.

TrueNat	Point-of-care test for TB detection. Method: Uses a micro-PCR technology to amplify DNA. Results are available within 60 minutes.	Portable and easy to use in peripheral health facilities. Provides results quickly, facilitating immediate treatment decisions.	
Line Probe Assay (LPA)	Detects TB and drug resistance (rifampicin and isoniazid). Method: Sputum samples are processed, and DNA is extracted. Amplified DNA is hybridized to specific probes on a membrane strip.	Rapid results (within 1-2 days). Can detect multiple mutations associated with drug resistance.	Requires a laboratory setup and trained personnel. Not suitable for all strains of TB.
RFLP	IS6110 restriction fragment length polymorphism (RFLP) analysis is widely used for molecular epidemiological studies of tuberculosis. Restriction enzymes Nru I, EcoN I, Pst I, and Pvu II were used to analyze IS6110 fingerprints in Mycobacterium tuberculosis strains H37Rv (virulent) and H37Ra (avirulent).		

Tinsdale medium (Option A) is used for Corynebacterium diphtheriae, which causes diphtheria.

MacConkey agar (Option C) is used for gram-negative bacteria, particularly enteric pathogens like E. coli and Klebsiella.

Loeffler serum slope (LSS) (Option D) enhances the growth of Corynebacterium diphtheriae.

Solution for Question 137:

Correct Answer: C) Pseudomonas aeruginosa

Explanation: The firefighter with fever and cough has a culture showing gram-negative, oxidase-positive bacilli with fluorescent colonies under UV light, which is characteristic of Pseudomonas aeruginosa.

Diagnostic Tests for Pseudomonas aeruginosa:

Diagnostic Method	Description	Specific Features for Pseudomonas aeruginosa
Sample Collecti on	Collect specimens such as sputum, urine, wound swabs, blood, ear discharge, eye discharge, or cerebrospinal fluid (CSF).	Specimens vary based on the site of infection.
Microscopy	Gram stain	Gram-negative rods; motile with flagella.
Culture	Growth on common laboratory media:	
Blood agar	Produces beta-hemolytic gray, moist colonies.	
MacConkey agar	Produces pale, non-lactose fermenting colonies, Fluorescence under UV light.	
Nutrient agar	Produces large, irregular colonies with a metallic sheen.	
Peptone water	Forms uniform turbidity with surface pellicle.	
Cetrimide Agar	Selective medium that inhibits other Gram-negative bacteria; specifically selected for Pseudomonas aeruginosa; pigment production (e.g., pyocyanin).	
	King’s medium	used to increase pigment production by pseudomonas
Biochemical Tests	Various biochemical tests to identify specific metabolic properties:	
Oxidase test	Positive	
Catalase test	Positive	
ICUT Test:		
Indole test	Negative	
Citrate test	Positive	
Urease test	Negative	

Triple sugar test	Shows alkaline slant with no gas and hydrogen sulfide.	
Molecular Methods	PCR (Polymerase Chain Reaction), 16S rRNA sequencing	Provides specific and sensitive detection of Pseudomonas aeruginosa DNA.
Antimicrobial Susceptibility Testing	Disk diffusion method, Minimum Inhibitory Concentration (MIC) testing	Determines the susceptibility of Pseudomonas aeruginosa to various antibiotics.

Klebsiella (Option A): Although Gram-negative, it is oxidase-negative. It does not produce fluorescent pigments under UV light and typically forms mucoid colonies due to its capsule. Staphylococcus aureus (Option B) is Gram-positive, whereas the given organism is Gram-negative. It does not produce fluorescent pigments under UV light and forms golden-yellow colonies, not greenish ones. Serratia (Option D) is Gram-negative and oxidase-negative. Some Serratia species produce red pigment (prodigiosin), but they do not fluoresce under UV light like Pseudomonas aeruginosa.

Solution for Question 138:

Correct Answer: A) Bacillus cereus

Explanation: Bacillus cereus causes rapid-onset vomiting (within 1–6 hours) after consuming contaminated fried rice.

Clinical Features of Bacillus cereus:

Table rendering failed

Diagnosis of Bacillus cereus:

Diagnostic Method	Details
Food Poisoning	Culture of implicated food essential. >105 B. cereus cells per gram confirms the diagnosis. Quantitative stool cultures with at least 105 cells per gram support diagnosis.
Culture Characteristics	Aerobic growth at 37°C on sheep blood agar. β-hemolytic colonies with frosted glass appearance. Motile, lecithinase positive, fermenting salicin
Selective Media	MYPA aids in isolating B. cereus from faeces and other sources

Microscopy	Spore-forming, Gram-positive bacilli. Motile

Clostridium botulinum (Option B) causes botulism, presenting with descending paralysis and no vomiting, with symptoms appearing 12–36 hours after ingestion.

Escherichia coli (Option C): ETEC (Enterotoxigenic E. coli) causes watery diarrhea, not isolated vomiting, and has a longer incubation period (12–72 hours).

Shigella dysenteriae (Option D): Causes bloody diarrhea (dysentery) with fever, not acute vomiting.

Solution for Question 139:

Correct Answer: B) Leptospira

Explanation: A farmer with fever and subconjunctival hemorrhage along with a positive microscopic agglutination test points toward Leptospira infection (Leptospirosis).

Leptospirosis:

Causative Agent	Leptospira interrogans (bacteria)
Transmission	Zoonotic disease Indirect: Contact with water, moist soil, or surfaces contaminated with animal urine. Direct: Contact with urine or parturition products from infected animals.
Pathogenesis Phases	Septicemic Phase: Entry through mucosa or skin, dissemination via bloodstream to organs. Immune Phase: Antibodies develop, spirochetes disappear from blood; antigen-antibody complexes form, and renal colonization occurs.
Clinical Syndromes	Mild Anicteric Illness: Seen in 90% of cases, flu-like symptoms with fever, headache, conjunctival suffusion, nausea, vomiting, and myalgia. Weil's Disease: Severe icteric illness in 10% of cases; rapid progression, fatality rate of 5-15%.

Culture Media	 ... (content truncated)
Diagnosis	Microscopic Agglutination Test (MAT): Considered the gold standard for Leptospira detection, MAT measures antibodies in serum samples. The Modified Faine's Criteria used to aid in the diagnosis of leptospirosis, especially in endemic areas.This scoring system combines clinical, epidemiological and laboratory data to increase diagnostic accuracy. PCR (Polymerase Chain Reaction): Provides a faster diagnosis by detecting leptospira DNA in blood or urine, especially effective in the early stage. ELISA: Used for antibody detection and is helpful for confirming infection in the later stages.
Treatment	Antibiotics: Early antibiotic therapy is crucial and often involves: Doxycycline: Typically used for mild cases and prophylaxis. Penicillin or Ampicillin: Preferred for severe cases.

Brucella (Option A) causes undulant fever and hepatosplenomegaly but does not cause subconjunctival hemorrhage. The standard test is the serum agglutination test (SAT), not the microscopic agglutination test (MAT). Escherichia coli (Option C) primarily causes gastrointestinal and urinary tract infections and is not associated with conjunctival hemorrhage, or a positive MAT. Staphylococcus aureus (Option D) causes skin infections, pneumonia, and endocarditis but does not cause subconjunctival hemorrhage or a positive MAT, which is specific for Leptospira.

Solution for Question 140:

Correct Answer: B) Staphylococcus epidermidis

Explanation: Staphylococcus epidermidis, a coagulase-negative Staphylococcus (CoNS), is the most common cause of prosthetic valve infective endocarditis.

Prosthetic Valve Endocarditis:

- It occurs following cardiac valve replacement.
- Early prosthetic valve endocarditis (occurs within 12 months of valve replacement): Generally nosocomial. Results from intra-operative contamination of the prosthesis or a bacteremic postoperative complication. Common agents: CoNS (e.g., S. epidermidis) and S. aureus.
- Generally nosocomial.
- Results from intra-operative contamination of the prosthesis or a bacteremic postoperative complication.
- Common agents: CoNS (e.g., S. epidermidis) and S. aureus.
- Late prosthetic valve endocarditis (occurs after 12 months of valve replacement): Usually community-acquired. Common agents: Viridans streptococci.
- Usually community-acquired.
- Common agents: Viridans streptococci.
- Overall: Regardless of the time of onset after surgery, CoNS are the most common agents (at least 68-85%). Majority are methicillin-resistant CoNS.
- Regardless of the time of onset after surgery, CoNS are the most common agents (at least 68-85%).
- Majority are methicillin-resistant CoNS.
- Generally nosocomial.
- Results from intra-operative contamination of the prosthesis or a bacteremic postoperative complication.
- Common agents: CoNS (e.g., S. epidermidis) and S. aureus.
- Usually community-acquired.
- Common agents: Viridans streptococci.
- Regardless of the time of onset after surgery, CoNS are the most common agents (at least 68-85%).
- Majority are methicillin-resistant CoNS.

Streptococcus viridans (Option A) is a common cause of native valve endocarditis, particularly following dental procedures or in patients with pre-existing valvular disease. However, it is not the primary cause of prosthetic valve endocarditis (PVE), where Staphylococcal species

predominate.

Staphylococcus aureus (Option C) is a virulent pathogen that causes rapidly progressive (acute) infective endocarditis, affecting both native and prosthetic valves. It is a leading cause of healthcare-associated IE, often seen in intravenous drug users (IVDU) and patients with catheters or implanted devices. While it can cause prosthetic valve IE, it is more common in late-onset cases rather than early PVE.

Enterococci (Option D) (e.g., Enterococcus faecalis) are opportunistic pathogens that cause subacute endocarditis, particularly in elderly patients with genitourinary or gastrointestinal infections. Though they can infect prosthetic valves, they are not the primary causative agent.

Solution for Question 141:

Correct Answer: A) Neisseria meningitidis

Explanation: The patient's clinical features of fever, neck rigidity, headache, vomiting, and petechial rashes on the lower limbs point toward meningococcal meningitis caused by Neisseria meningitidis.

ClinicalFeatures of Neisseria meningitidis:

- Rashes: Non-blanching rash (petechial or purpuric) develops in more than 80% of cases.
- Septicemia: Attributed to endotoxin-induced endothelial injury. Leads to increased vascular permeability and intravascular thrombosis.
- Attributed to endotoxin-induced endothelial injury.
- Leads to increased vascular permeability and intravascular thrombosis.
- Waterhouse-Friderichsen syndrome: Severe form of fulminant meningococcemia. Characterized by large purpuric rashes (purpura fulminans), shock, disseminated intravascular coagulation (DIC), bilateral adrenal hemorrhage, and multiorgan failure.
- Severe form of fulminant meningococcemia.
- Characterized by large purpuric rashes (purpura fulminans), shock, disseminated intravascular coagulation (DIC), bilateral adrenal hemorrhage, and multiorgan failure.
- Pyogenic meningitis: Commonly affects young children (3-5 years of age). Presentation includes fever, vomiting, headache, and neck stiffness. Similar to other bacterial meningitis but with the presence of rashes.

• Commonly affects young children (3-5 years of age).

• Presentation includes fever, vomiting, headache, and neck stiffness.

• Similar to other bacterial meningitis but with the presence of rashes.

• Chronic meningococcemia: Rare condition. Characterized by repeated episodes of petechial rashes, fever, arthritis, and splenomegaly.

• Rare condition.

• Characterized by repeated episodes of petechial rashes, fever, arthritis, and splenomegaly.

• Postmeningococcal reactive disease: Develops 4-10 days later due to immune complexes (capsular antigens and their antibodies). Manifestations include arthritis, rash, iritis, pericarditis, polyserositis, and fever.

• Develops 4-10 days later due to immune complexes (capsular antigens and their antibodies).

• Manifestations include arthritis, rash, iritis, pericarditis, polyserositis, and fever.

• Attributed to endotoxin-induced endothelial injury.

• Leads to increased vascular permeability and intravascular thrombosis.

• Severe form of fulminant meningococcemia.

• Characterized by large purpuric rashes (purpura fulminans), shock, disseminated intravascular coagulation (DIC), bilateral adrenal hemorrhage, and multiorgan failure.

• Commonly affects young children (3-5 years of age).

• Presentation includes fever, vomiting, headache, and neck stiffness.

• Similar to other bacterial meningitis but with the presence of rashes.

• Rare condition.

• Characterized by repeated episodes of petechial rashes, fever, arthritis, and splenomegaly.

• Develops 4-10 days later due to immune complexes (capsular antigens and their antibodies).

• Manifestations include arthritis, rash, iritis, pericarditis, polyserositis, and fever.

Lab Investigations of Neisseria meningitidis:

Investigation	Findings
Cerebrospinal Fluid (CSF) Analysis	
Appearance	Cloudy or purulent
White Blood Cell Count (WBC)	Elevated (pleocytosis), typically >1,000 cells/µL, with predominance of neutrophils
Protein	Elevated

Glucose	Decreased
Gram Stain	Presence of gram-negative diplococci
Culture	Growth of Neisseria meningitidis on chocolate agar or Thayer-Martin agar
Polymerase Chain Reaction (PCR)	Detection of Neisseria meningitidis DNA
Latex Agglutination Test	Positive for Neisseria meningitidis antigens
Blood Tests	
Culture	Growth of Neisseria meningitidis
White Blood Cell Count	Elevated leukocytosis with left shift

Clostridium perfringens (Option B) causes gas gangrene and food poisoning, not meningitis with petechial rash.

Cryptococcus neoformans (Option C) is typically seen in immunocompromised patients (e.g., HIV) and presents with chronic meningitis without petechial rash.

Clostridium difficile (Option D) causes antibiotic-associated diarrhea and pseudomembranous colitis, not meningitis

Solution for Question 142:

Correct Answer: A) Recent infection by Salmonella spp

Explanation: A significantly elevated O antigen titer (>1:300) only in the Widal test suggests a recent Salmonella infection.

Widal Test:

- It is a highly sensitive but poorly specific serological test used to diagnose typhoid fever caused by Salmonella typhi.
- It detects specific antibodies in the patient's serum that react with the antigens of Salmonella typhi.
- The test is based on the agglutination reaction where patient serum is mixed with antigens of Salmonella typhi.

• If the antibodies are present, they will cause the antigens to clump together (agglutinate), indicating a positive result.

• The test primarily measures: O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi. H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.

• O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi.

• H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.

• Antigens Used: O antigen of S. Typhi (TO) H antigen of S. Typhi (TH) H antigen of S. Paratyphi A (AH) H antigen of S. Paratyphi B (BH)

• O antigen of S. Typhi (TO)

• H antigen of S. Typhi (TH)

• H antigen of S. Paratyphi A (AH)

• H antigen of S. Paratyphi B (BH)

• Interpretation: Cut-off values in most parts of India: H agglutinin titer >200. O agglutinin titer >100. Low titers: Considered baseline in endemic areas and should be ignored.

• Cut-off values in most parts of India: H agglutinin titer >200. O agglutinin titer >100.

• H agglutinin titer >200.

• O agglutinin titer >100.

• Low titers: Considered baseline in endemic areas and should be ignored.

• O-antibodies: These are antibodies against the somatic (O) antigens of Salmonella typhi.

• H-antibodies: These are antibodies against the flagellar (H) antigens of Salmonella typhi.

• O antigen of S. Typhi (TO)

• H antigen of S. Typhi (TH)

• H antigen of S. Paratyphi A (AH)

• H antigen of S. Paratyphi B (BH)

• Cut-off values in most parts of India: H agglutinin titer >200. O agglutinin titer >100.

• H agglutinin titer >200.

• O agglutinin titer >100.

• Low titers: Considered baseline in endemic areas and should be ignored.

• H agglutinin titer >200.

• O agglutinin titer >100.

Widal Test Result	Suggestive of
Rise of TO and TH antibody	Enteric fever due to S. Typhi
Rise of TO and AH antibody	Enteric fever due to S. Paratyphi A

Rise of TO and BH antibody	Enteric fever due to S. Paratyphi B
Rise of only TO antibody	Recent infection due to S. Typhi or S. Paratyphi A or B
Rise of only TH antibody	Convalescent stage / Anamnestic response
Rise of all TH, AH, BH	Post-TAB vaccination antibodies (Option B)

Past infection (Option C): A past infection usually results in elevated H antigen titers, whereas in this case, the O antigen titer is significantly high, indicating a recent infection.

Reconvalescence (Option D): In the recovery phase, both O and H antigen titers decline, whereas in this case, the O antigen titer remains markedly high, supporting a recent infection instead.

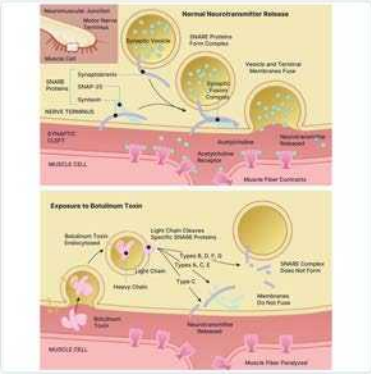
Solution for Question 143:

Correct Answer: C) Inhibits the release of acetylcholine

Explanation: The mechanism of action of botulinum toxin is inhibition of acetylcholine release at the neuromuscular junction.

Mechanism of Action of Botulinum Toxin:

Botulinum toxin binds to acetylcholine receptors at neuromuscular junctions, preventing the release of acetylcholine, which is essential for muscle contraction. This leads to flaccid paralysis, a hallmark of botulism.



Inhibits Elongation Factor 2 (Option A): This describes the MOA of Diphtheria toxin and Pseudomonas exotoxin A, which inhibit protein synthesis by ADP-ribosylation of Elongation

Factor-2 (EF-2), leading to cell death.

Inhibits the release of GABA (Option B): This describes the action of Tetanus toxin, which blocks inhibitory neurotransmitters (GABA and glycine) in the spinal cord, causing spastic paralysis.

Increases cAMP (Option D): Toxins that increase cyclic AMP (cAMP) include cholera toxin and pertussis toxin, which cause fluid secretion (diarrhea) and immune modulation, respectively.

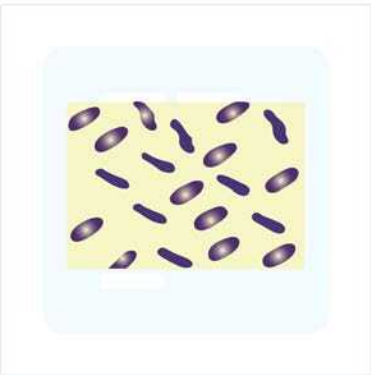
Solution for Question 144:

Correct Answer: D) Chancroid

Explanation: The clinical scenario of a person with a history of a long-term sexual relationship presenting with painful ulcers on the genitals and tender lymphadenopathy points towards a diagnosis of Chancroid, typically caused by Haemophilus ducreyi.

Haemophilus ducreyi:

- Gram-negative coccobacillus
- Fastidious, requiring X factor but not V factor
- Forms characteristic "school of fish" or "railroad track" arrangements in Gram stains
- Safety pin appearance of bipolar staining.



- Clinical features of chancroid: Painful genital ulcers with ragged edges and a purulent base Painful inguinal lymphadenopathy (buboes) Incubation period of 3-10 days
- Painful genital ulcers with ragged edges and a purulent base
- Painful inguinal lymphadenopathy (buboes)
- Incubation period of 3-10 days

• Diagnosis: Clinical presentation Gram stain of ulcer exudate Culture on selective media (challenging and low sensitivity) PCR (most sensitive method) Differential diagnosis includes syphilis, herpes simplex, and lymphogranuloma venereum

• Clinical presentation

• Gram stain of ulcer exudate

• Culture on selective media (challenging and low sensitivity)

• PCR (most sensitive method)

• Differential diagnosis includes syphilis, herpes simplex, and lymphogranuloma venereum

• Painful genital ulcers with ragged edges and a purulent base

• Painful inguinal lymphadenopathy (buboes)

• Incubation period of 3-10 days

• Clinical presentation

• Gram stain of ulcer exudate

• Culture on selective media (challenging and low sensitivity)

• PCR (most sensitive method)

• Differential diagnosis includes syphilis, herpes simplex, and lymphogranuloma venereum

• Treatment: Azithromycin 1g orally single dose, OR Ceftriaxone 250mg IM single dose, OR Ciprofloxacin 500mg orally twice daily for 3 days, OR Erythromycin 500mg orally three times daily for 7 days Treatment of sexual partners is crucial

• Azithromycin 1g orally single dose, OR

• Ceftriaxone 250mg IM single dose, OR

• Ciprofloxacin 500mg orally twice daily for 3 days, OR

• Erythromycin 500mg orally three times daily for 7 days

• Treatment of sexual partners is crucial

• Azithromycin 1g orally single dose, OR

• Ceftriaxone 250mg IM single dose, OR

• Ciprofloxacin 500mg orally twice daily for 3 days, OR

• Erythromycin 500mg orally three times daily for 7 days

• Treatment of sexual partners is crucial

Gonorrhea (Option A) typically causes urethritis and discharge, not painful ulcers or lymphadenopathy.

Chlamydia (Option B) can cause urethritis or cervicitis, but ulcers are rare.

Lymphogranuloma venereum (LGV) (Option C) presents with painless ulcers and later, painful lymphadenopathy, unlike the painful ulcers seen in chancroid.

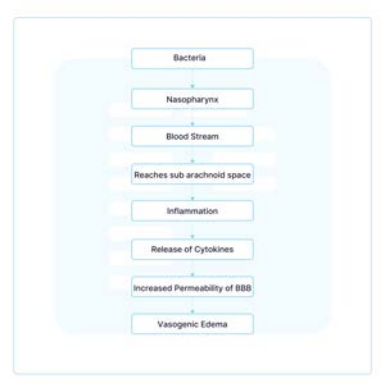
Solution for Question 145:

Correct Answer: B) Ceftriaxone

Explanation:

The clinical scenario of a patient with fever, nuchal rigidity, and CSF findings of decreased glucose, increased protein, and neutrophilic pleocytosis points towards a diagnosis of bacterial meningitis.

Ceftriaxone is the first-line antibiotic for treatment of bacterial meningitis.

Bacterial Meningitis in Adults and Prevention	
Causes	Streptococcus pneumoniae and Neisseria meningitides are the primary causes. Listeria monocytogenes is a significant risk in adults over 50, especially in those with diabetes, alcoholism, or weakened immune systems.
Pathophysiological Process	
Classic Triad of Symptoms	Fever Headache Nuchal rigidity (stiff neck)
CSF Findings	Opening pressure: Elevated Predominant cell type: Polymorphonuclear neutrophils (PMNs) Protein: Elevated Glucose: Decreased Gross appearance: Turbid

Treatment	Immediate Management: Start with intravenous broad-spectrum antibiotics to cover pathogens. First-line antibiotic of choice: Ceftriaxone Treatment is then adjusted based on the identified organism and its susceptibility Adjunctive Therapy: Dexamethasone may be given to reduce inflammation and prevent complications, particularly if administered early. Supportive Care: Manage complications like raised intracranial pressure and seizures.
Prevention	Vaccines for H. influenzae type b, N. meningitidis, and S. pneumoniae. No vaccine currently exists for group B meningococcal infections. Chemoprophylaxis: Close contacts of individuals with meningococcal meningitis should receive prophylactic antibiotics like rifampicin, ciprofloxacin, azithromycin or ceftriaxone to reduce the risk of secondary infections. Rifampicin - 600 mg 12 hourly for 2 days Ciprofloxacin - 500 mg single dose Azithromycin 500 mg single dose Ceftriaxone - 250 mg Intramuscular single dose

Ampicillin (Option A) is primarily used in Listeria monocytogenes meningitis, which is more common in neonates, elderly, and immunocompromised patients. It is not first-line for typical bacterial meningitis in healthy adults.

Penicillin (Option C) was historically used for Neisseria meningitidis but is not sufficient as monotherapy due to increasing resistance. Empirical therapy requires broader coverage.

Vancomycin (Option D) is used in combination with Ceftriaxone when penicillin-resistant Streptococcus pneumoniae is suspected, but it is not sufficient alone.

Solution for Question 146:

Correct Answer: A) Isoniazid

Explanation: The katG mutation causes Isoniazid resistance by preventing its activation inside Mycobacterium tuberculosis, making the drug ineffective.

Mechanisms of Isoniazid Resistance	

KatG gene mutation	INH is a prodrug, converted to the active form by catalase-peroxidase produced by the Kat G gene. The mutated catalase enzyme does not form the active drug
KasA gene mutation	Component of cell wall mycolic acid
InhA gene overexpression	Causes cross-resistance to Ethionamide (second-line drug)



Rifampicin (Option B) resistance is associated with mutations in the rpoB gene (RNA polymerase β-subunit), not KAT G.

Levofloxacin (Option C) resistance is linked to mutations in the gyrA and gyrB genes, which encode DNA gyrase, a key enzyme for bacterial DNA replication.

Ethambutol (Option D) resistance is associated with mutations in the embB gene, which affects arabinosyl transferase, an enzyme involved in mycobacterial cell wall synthesis.

Virology - General properties

1. Based on the properties of viruses, which of the following sentences is not true?

A. A virus can either possess DNA or RNA, but never both.

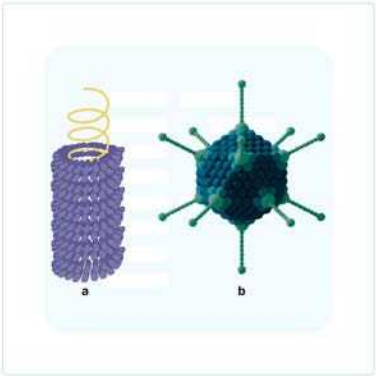
B. All animal viruses are spherical in shape.

C. Viruses are obligate intracellular and can only replicate inside a host cell.

D. All viruses have a protein coat that encases their genetic material.

---

2. Based on the image given below, which of the following statements is true regarding the symmetry of the viruses? Poxvirus has a symmetry similar to structure A Adenovirus has a symmetry similar to structure B Rhabdovirus has a symmetry similar to structure A Structure B has two types of capsomers The helical tube in structure A is rigid in all viruses with a similar structure.



A. 1, 2 and 5

B. 2 and 4

C. 2, 3 and 4

D. 1,3 and 5

---

3. A 34-year-old researcher accidentally exposes a viral sample to alcohol during a lab procedure. He recalls that alcohol can disrupt certain viral structures. Which of the following statements is true regarding the structure of viral envelopes and capsids?

A. Hemagglutinin is a protein subunit found exclusively in non-enveloped viruses.

B. Non-enveloped viruses are more alcohol labile than enveloped viruses.

C. Viral envelope is composed of lipids derived from the host cell membrane.

D. Capsid in enveloped viruses gives the virus its pleomorphic shape.

---

4. Three patients come to the hospital, each presenting with three different infectious diseases. Patient A has Creutzfeldt-Jakob disease. Patient B has Potato spindle tuber viroid disease. Patient C has an Influenza type B infection. Which option correctly describes the composition of each pathogen?

A. Patient A: Nucleic acids only Patient B: Protein and nucleic acids Patient C: RNA only

B. Patient A: Infectious protein only Patient B: Naked circular ssRNA without a capsid Patient C: Enveloped RNA

C. Patient A: Enveloped RNA Patient B: Infectious protein only Patient C: Naked circular ssRNA without a capsid

D. Patient A: Enveloped RNA Patient B: RNA only Patient C: Protein and nucleic acids

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5. Identify the mismatched pair of viruses and their site of viral replication. Adenovirus - Cytosol Poxvirus - Cytosol Parvovirus - Nucleus HIV - Mixed

A. Poxvirus - Cytosol

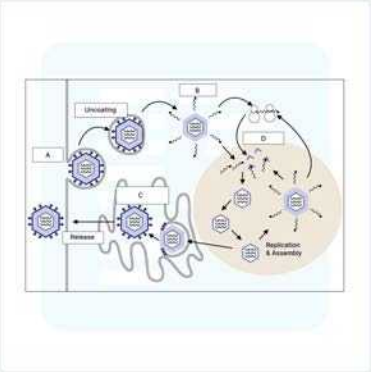
B. Adenovirus - Cytosol

C. Parvovirus - Nucleus

D. HIV - Both Cytosol and Nucleus

---

6. Based on the steps of viral replication, label the following diagram



A. A - Attachment, B - Translation, C- Transcription, D- budding

B. A - Penetration, B - Translation, C- Transcription, D- budding

C. A - Penetration, B - Transcription, C- Translation, D - Maturation

D. A - Attachment, B - Transcription, C- Translation, D- Maturation

---

7. A virology lab receives samples from a patient with a suspected viral infection presenting with fever, cough, and body aches. Testing confirms the presence of an RNA virus. Which of the following is an example of an RNA virus?

A. Epstein-Barr virus

B. JC virus

C. Adenovirus

D. Respiratory syncytial virus

---

8. Which of the following non-enveloped viruses contain DNA as their genetic material?

A. Picornavirus

B. Papovavirus

C. Corona virus

D. Poxvirus

---

9. A 45-year-old patient is diagnosed with a viral infection after presenting with symptoms such as fever, chills, and body aches. Laboratory analysis reveals a helical virus with a negative-sense RNA genome. Which of the following viruses is most likely responsible for this patient's condition?

A. Coronavirus

B. Rabies virus

C. Dengue virus

D. Hepatitis C virus

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10. During the production of a viral vaccine using embryonated hen's eggs, which site is specifically utilised for the cultivation of yellow fever virus?

A. Yolk Sac

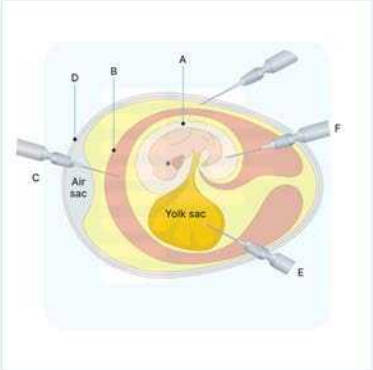
B. Amniotic Sac

C. Allantoic Sac

D. Chorioallantoic Membrane

---

11. Which of the following structures can be used to cultivate the Japanese encephalitis virus?



A. D

B. B

C. E

D. F

---

12. A researcher needs to isolate a virus using a cell line that can divide a maximum of 10 to 50 times before senescence. Which of the following cell lines should they choose?

A. HeLa Cell Line

B. MRC-5 Cell Line

C. Vero Cell Line

D. Chick embryo cell line

---

13. A patient's cell culture exhibits distinct cytopathic effects characterised by the formation of syncytia (multinucleated giant cells). Based on these observations, which virus is most likely responsible for the infection?

A. Adenovirus

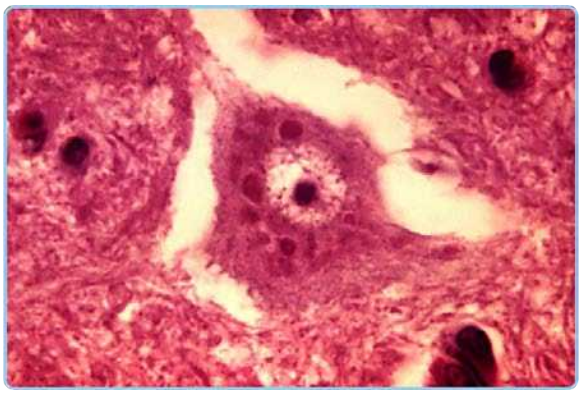
B. Enterovirus

C. Measles virus

D. Simian virus 40

---

14. A 45-year-old patient presents with neurological symptoms, and a brain biopsy is performed, revealing the presence of inclusion bodies as shown in the image. What are these inclusion bodies commonly referred to as?



A. Paschen Bodies

B. Negri Bodies

C. Bollinger Bodies

D. Molluscum Bodies

---

15. A new respiratory virus has emerged in a rural community, with early reports indicating a high transmission rate and increased hospitalisations. Public health officials must evaluate the virus's infectivity to implement effective control measures. Which of the following assays is most appropriate for quantifying the virus's infectivity?

A. Endpoint Assay

B. Electron microscopy

C. Plaque Assay

D. Haemagglutination

---

16. Match the following interferons with their respective sources, 1. Alpha interferon a. Fibroblasts 2. Beta interferon b. T-lymphocytes 3. Gamma interferon c. Leucocytes

1. Alpha interferon	a. Fibroblasts
2. Beta interferon	b. T-lymphocytes
3. Gamma interferon	c. Leucocytes

A. 1-b, 2-c, 3-a

B. 1-a, 2-c, 3-b

C. 1-b, 2-a, 3-c

D. 1-c, 2-a, 3-b

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	4
Question 8	2
Question 9	2
Question 10	3
Question 11	3
Question 12	2
Question 13	3
Question 14	2
Question 15	3
Question 16	4

Solution for Question 1:

Correct Answer: B) All animal viruses are spherical in shape

Explanation:

Animal viruses have different shapes, but most are spherical.

General properties of viruses	
Characteristics	They do not have a cellular organisation. Obligate intracellular parasites - cannot replicate outside a host. (Option C) Lack enzymes necessary for protein and nucleic acid synthesis. Multiply via a complex process. They cannot be cultured in cell-free culture media but can grow in embryonated eggs or tissue culture. They can pass through filters.
Morphology	All viruses have a protein coat that encases their genetic material. Either possess DNA or RNA but never both. (Option A) Nucleic acid is enclosed by a protein capsid, which may or may not be enclosed by a lipid envelope. (Option D)
Shape	Most are spherical. (Option B) Some are irregular and pleomorphic Different shapes of animal viruses are Rabies virus - bullet-shaped Ebola virus - filamentous Pox virus - brick shaped Tobacco mosaic virus - rod-shaped
Size	Ranges from 300 nm - Poxvirus to 20 nm - Parvovirus

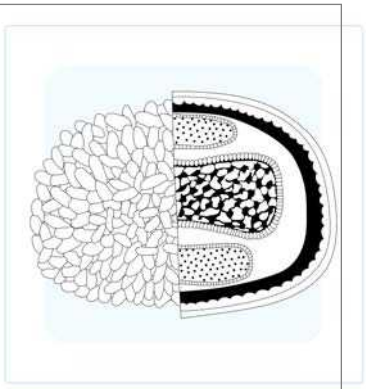
Solution for Question 2:

Correct Answer: C) 2, 3 and 4

Explanation:

The symmetry of the virus depends on the arrangement of the capsomers surrounding the nucleic acid.

Symmetry of virus		
Icosahedral	Helical	Complex

Polygon with 12 vertices and 20 sides Consist of two types of capsomer (Statement 4) Structure is rigid All DNA viruses have this symmetry except for Poxvirus Eg Adenovirus (Statement 2) Herpesvirus Hepadnavirus Papovavirus	 ... (content truncated)	 ... (content truncated)

Solution for Question 3:

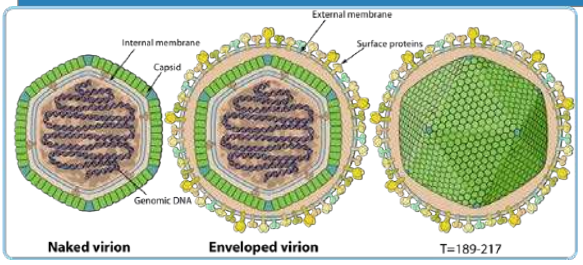
Correct Answer: C) Viral envelope is composed of lipids derived from the host cell membrane.

Explanation:

The viral envelope is composed of lipids derived from the host cell membrane when the progeny virus is released by budding.

STRUCTURE OF VIRUS
CAPSID: Protein coat surrounding the viral nucleic acid.
It is composed of small morphological units called capsomeres The polypeptide molecules are arranged symmetrically It forms an impenetrable shell around the nucleic core It is adsorbed by the host cell membrane readily, thus introducing the viral genome into the host
ENVELOPED: Viral capsid is surrounded by a lipid membrane

It is derived from the host cell membrane by viral progeny during the budding process (Option C) They have protein subunits that project on the envelope surface (Peplomers), which are viral-coded The shape of the virus depends on the lipid envelope. (Option D) Example (Option A) Influenza virus has two types of peplomers: Hemagglutinin: Triangular spike. Neuraminidase: Mushroom-shaped structure. Functions: Confers chemical, antigenic, and biological properties. Facilitates attachment to host cell surfaces. Aids in processes such as hemagglutination Enveloped viruses are sensitive to lipid solvents, including Ether, Chloroform and Bile salts. Specific neutralisation of virus infectivity is mediated by antibodies targeting surface antigens. ... (content truncated)
NON-ENVELOPED: Lack the lipid membrane, and viral capsid is naked
They lack the lipid membrane More resistant to lipid solvents (Option B) The shape of the virus depends on the shape of the capsid (Option D)
Most viruses are enveloped except: Naked DNA viruses: Parvovirus, Papavovirus and Adenovirus Naked RNA viruses: Picornaviruses, reovirus, calicivirus, astroviridae, Hep A and Hep E



Solution for Question 4:

Correct Answer: B) Patient A: Infectious protein only; Patient B: Naked circular ssRNA without a capsid; Patient C: Enveloped RNA

Explanation:

- Patient A (Creutzfeldt-Jakob disease) is caused by a prion, an infectious protein.
- Patient B (Potato spindle tuber viroid) involves a viroid, a naked circular ssRNA without a capsid.
- Patient C (Influenza type B) involves an enveloped RNA virus.

	VIRUS	VIROID	PRION
Definition	Infectious agent composed of nucleic acid enclosed in a protein coat with or without a lipid envelope. (Patient C)	Infectious molecule composed of small, circular ssRNA without a capsid. (Patient B)	Infectious agent, composed of only infectious proteins. (Patient A)
Composition	DNA/RNA + protein +/- Lipid.	RNA only.	Protein only.
Replication	Replicates using host cellular machinery	Replicates using host cellular machinery.	Replicates by inducing protein misfolding in the host.
Stability	Stability varies Enveloped viruses are sensitive to lipid solvents, while non-enveloped are more resistant.	Resistant to heat and organic solvent.	Resistant to heat and proteolytic enzymes and heat, it can remain infectious for years.
Examples	Influenza virus, Poxvirus, Reovirus	Potato spindle tuber viroid, Plant diseases	Creutzfeldt-jakob disease
Virion: Complete virus particle that includes a genome and a protein coat, which exists outside of host cells, representing one of the physical forms of a virus during a specific phase of its replication cycle.			

Solution for Question 5:

Correct Answer: B) Adenovirus - Cytosol

Explanation:

Adenovirus - Cytosol is a mismatched pair because adenovirus is a DNA virus that replicates in the nucleus, similar to most DNA viruses.

- Most DNA viruses replicate within the nucleus of the host cells, with a few exceptions, such as poxvirus, which replicates in the cytosol (Option A).

• Examples of DNA viruses: Hepadna virus, Reovirus, Parvovirus (Option C), Adenovirus (Option B) and Papovavirus

• Most RNA viruses replicate in the cytosol of the host cells except for orthomyxovirus and retroviruses(HIV) (Option D), which synthesise their RNA in the nucleus.

Solution for Question 6:

Correct Answer: C) A - Penetration, B - Transcription, C- Translation, D - Maturation

Explanation:

The above diagram shows the steps of viral replication within the host cell, and the labelling is A - Penetration, B - Transcription, C- Translation, D - Maturation.

VIRAL REPLICATION
Attachment: Receptor interaction between viral surface antigen and host cell surface antigens. HIV: Viral surface gp120 binds to CD 24 on host cells. Influenza: Viral hemagglutinin binds to glycoprotein receptors on respiratory epithelium
Penetration: Viral particle enters the host cells via - Phagocytosis/viropexis (receptor mediated endocytosis occurs) Membrane fusion (some enveloped viruses enter via fusion of their membrane with the host cell membrane; thus, only the nucleocapsid enters). Injection of nucleic acid by bacteriophages.
Uncoating: Host cell lysosomal enzymes separate the viral capsid and nucleic acid.
Biosynthesis Nucleic acid replication DNA viruses replicate within the nucleus except for the Pox virus, which replicates in the cytosol. RNA viruses replicate within the cytosol except retroviruses and orthomyxoviruses Transcription of Viral nucleic material DNA viruses: DNA is transcribed into RNA in the mRNA in the cytosol. + sense RNA viruses: Directly replicated/transcribed or translated - sense RNA viruses: Transcribed into +ve sense RNA first, which is then replicated dsRNA viruses +ve strand is directly replicated/transcribed or translated. -ve strand is transcribed into +ve sense RNA first. Both strands can serve as templates for complementary strands In retroviruses, reverse transcription converts viral single-stranded RNA (ssRNA) into double-stranded DNA (dsDNA), which is then integrated into the host genome. ... (content truncated)

Assembly: Viral nucleic acid and protein are packed together to form new viral progeny. DNA viruses are assembled in the nucleus except Poxvirus and Hepadnavirus. RNA viruses are assembled in the cytosol.
Maturation: The daughter virions mature in the nucleus/cytosol/membranes (Golgi Apparatus or endoplasmic reticulum or plasma membrane.
RELEASE Lysis of cells: Usually occurs in non-enveloped viruses and bacteriophages. Budding: occurs in enveloped viruses. During budding, the virus acquires a part of the host membrane, leading to envelope formation.

Solution for Question 7:

Correct Answer: D) Respiratory syncytial virus

Explanation:

Respiratory syncytial virus (RSV) is an RNA virus. Epstein-Barr virus, JC virus, and Adenovirus are DNA viruses.

DNA virus	RNA virus
Herpesviridae: Herpes simplex virus (HSV-1, HSV-2) Varicella-zoster virus Epstein-Barr virus (Option A) Cytomegalovirus (CMV) HHV-6,7,8 Hepadnaviridae: Hepatitis B virus Parvoviridae: Parvovirus B19 Papovaviridae: Human papillomavirus (HPV) JC virus (Option B) ... (content truncated)	Picornaviridae: Poliovirus Coxsackievirus Rhinovirus Hepatitis A virus Enterovirus Caliciviridae: Hepatitis E virus Togaviridae: Rubella virus Eastern equine encephalitis virus Western equine encephalitis virus Flaviviridae: ... (content truncated)

Types of RNA Viruses:

- Positive-sense RNA Viruses have genomes that can be directly translated into proteins by the host cell's ribosomes.
- Negative-Sense RNA Viruses have genomes that cannot be directly translated into proteins. Instead, their RNA must first be converted into a complementary positive-sense RNA by an RNA-dependent RNA polymerase before translation can occur.



Solution for Question 8:

Correct Answer: B) Papovavirus

Explanation

Papovavirus is a non-enveloped virus that contains DNA as its genetic material.

- Non- enveloped viruses

DNA	RNA
Parvovirus Adenovirus Polyoma virus Papovavirus (Option B)	Picornavirus (Option A) Astrovirus Reovirus Calicivirus Hepatitis A Hepatitis E

- Enveloped viruses All other viruses except from above have an envelope, few of which are
- All other viruses except from above have an envelope, few of which are
- All other viruses except from above have an envelope, few of which are

DNA	RNA

Pox virus (Option D) Herpes virus Hepadenovirus	Deltaviridae Arena viridae Bunyaviridae Paramyxoviridae Orthomyxoviridae Filoviridae Rhabdoviridae Flaviviridae Togaviridae Matonaviridae Coronaviridae (Option C) Retroviridae

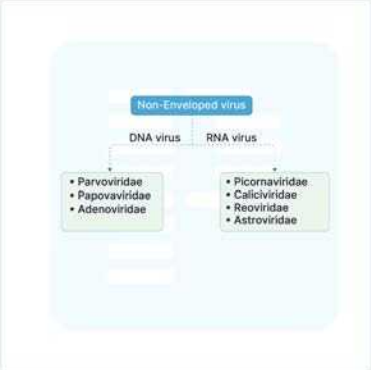
Solution for Question 9:

Explanation:

Rabies virus belongs to the Rhabdoviridae family and has a helical structure with a negative-sense RNA genome.

General characteristics of viruses:

- Virus exhibiting Helical Symmetry: (MNEMONIC: MR FAB has Corona)
- Rhabdoviridae
- Filoviridae
- Myxoviridae
- Coronaviridae
- Bunyaviridae
- Arenaviridae
- All other RNA viruses and DNA viruses (Except Poxviridae - Complex symmetry) exhibit icosahedral symmetry
- Enveloped and Non-Enveloped virus:



• All other viruses have an envelope.

• Nucleic Acid:

• All DNA viruses have dsDNA except Parvoviridae

• All RNA viruses have a single copy of ssRNA (Unsegmented) Exception: 2 Copies of RNA: HIV dsRNA: Reoviridae Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• Exception: 2 Copies of RNA: HIV dsRNA: Reoviridae Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• 2 Copies of RNA: HIV

• dsRNA: Reoviridae

• Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• Orthomyxoviridae (7/8 segments)

• Reoviridae (10-12 segments)

• Bunyaviridae (3 segments)

• Arenaviridae (2 segments)

• Exception: 2 Copies of RNA: HIV dsRNA: Reoviridae Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• 2 Copies of RNA: HIV

• dsRNA: Reoviridae

• Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• Orthomyxoviridae (7/8 segments)

• Reoviridae (10-12 segments)

• Bunyaviridae (3 segments)

• Arenaviridae (2 segments)

• 2 Copies of RNA: HIV

• dsRNA: Reoviridae

• Segmented: Orthomyxoviridae (7/8 segments) Reoviridae (10-12 segments) Bunyaviridae (3 segments) Arenaviridae (2 segments)

• Orthomyxoviridae (7/8 segments)

• Reoviridae (10-12 segments)

• Bunyaviridae (3 segments)

• Arenaviridae (2 segments)

• Orthomyxoviridae (7/8 segments)

• Reoviridae (10-12 segments)

• Bunyaviridae (3 segments)

• Arenaviridae (2 segments)

• Shape of viruses: Rabies virus: Bullet-shaped Ebola virus: Filamentous Poxvirus: Brick-shaped Adenovirus: Space vehicle-shaped Rotavirus: Wheel-shaped Tobacco mosaic virus: Rod-shaped

• Rabies virus: Bullet-shaped

• Ebola virus: Filamentous

• Poxvirus: Brick-shaped

• Adenovirus: Space vehicle-shaped

• Rotavirus: Wheel-shaped

• Tobacco mosaic virus: Rod-shaped

• Rabies virus: Bullet-shaped

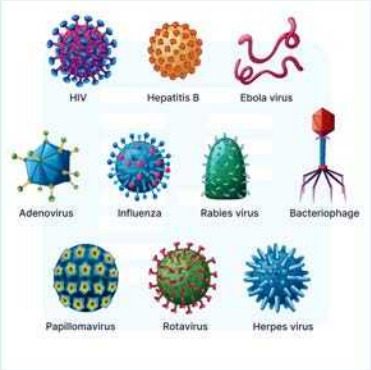
• Ebola virus: Filamentous

• Poxvirus: Brick-shaped

• Adenovirus: Space vehicle-shaped

• Rotavirus: Wheel-shaped

• Tobacco mosaic virus: Rod-shaped



- Size: Largest: Poxvirus (400 nm) Smallest: Parvovirus (20 nm)
- Largest: Poxvirus (400 nm)
- Smallest: Parvovirus (20 nm)
- Largest: Poxvirus (400 nm)
- Smallest: Parvovirus (20 nm)

Solution for Question 10:

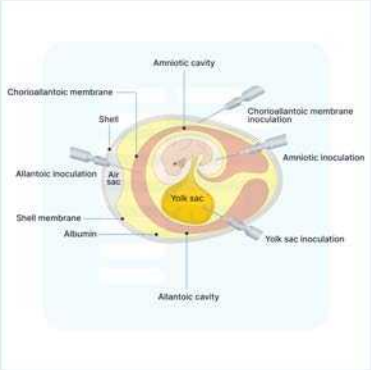
Correct Answer: C) Allantoic Sac

Explanation:

The Allantoic sac is the site specifically used for cultivating the yellow fever virus, providing a larger cavity for better viral yield.

Isolation of Viruses: Viruses cannot be grown on artificial culture media but are cultivated through various methods, including animal inoculation, embryonated egg inoculation, or tissue cultures.

- Animal Inoculation: Used for the primary isolation of certain viruses, such as:
- Arboviruses
- Coxsackieviruses
- Both these are isolated in the brain of suckling mice.
- Egg Inoculation:



• Yolk Sac: Useful for cultivating: Arboviruses (e.g., Japanese encephalitis virus) Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).

• Arboviruses (e.g., Japanese encephalitis virus)

• Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).

• Amniotic Sac: Used for isolating the Influenza virus

• Allantoic Sac: A larger cavity that provides better yields for viral vaccines, including: Influenza Yellow fever (17D)

• Influenza

• Yellow fever (17D)

• Chorioallantoic Membrane: Employed for isolating poxviruses (Eg, vaccinia and variola) & herpes simplex viruses (HSV-1, HSV-2), which produce visible lesions known as pocks on the membrane.

• Arboviruses (e.g., Japanese encephalitis virus)

• Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).

• Influenza

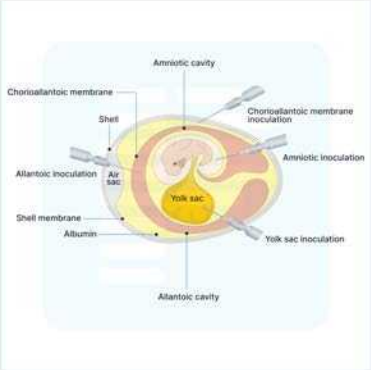
• Yellow fever (17D)

Solution for Question 11:

Correct Answer: C) E

Explanation

The Japanese encephalitis virus (an arbovirus) can be cultured using the yolk sac.



Egg inoculation and the viruses associated with the specific site of inoculation:

- Yolk Sac: Useful for cultivating: Arboviruses (e.g., Japanese encephalitis virus) Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).
- Arboviruses (e.g., Japanese encephalitis virus)
- Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).
- Amniotic Sac: Used for isolating the Influenza virus
- Allantoic Sac: Provides better yields for viral vaccines, including: Influenza Yellow fever (17D)
- Influenza
- Yellow fever (17D)
- Chorioallantoic Membrane: Employed for isolating poxviruses (Eg, vaccinia and variola) & herpes simplex viruses (HSV-1, HSV-2), which produce visible lesions known as pocks on the membrane.
- Arboviruses (e.g., Japanese encephalitis virus)
- Bacteria (e.g., Rickettsia, Chlamydia, Haemophilus ducreyi).
- Influenza
- Yellow fever (17D)

Solution for Question 12:

Correct Answer: B) MRC-5 Cell Line

Explanation:

• The MRC-5 cell line is a secondary cell line that can divide 10 to 50 times before senescence.

• HeLa and Vero cell lines are continuous cell lines, while the chick embryo cell line has a very limited division capacity.

Tissue Culture: Laboratory technique used to grow cells or tissues in a controlled environment outside their natural context.

• Types of Tissue Culture: Organ Culture Explant Culture Cell Line Culture

• Organ Culture

• Explant Culture

• Cell Line Culture

• Currently, Cell line cultures are the only widely used method for virus isolation.

• Organ Culture

• Explant Culture

• Cell Line Culture

Types of Cell Lines:

Primary Cell Lines	Secondary/Diploid Cell Lines	Continuous Cell Lines
5-10 divisions (Limited growth capacity)	10-50 divisions	Indefinite growth (Continuous divisions)
Maintain a diploid karyotype	Maintain a diploid karyotype	Altered haploid chromosomes
Monkey kidney cell line (for myxoviruses, enteroviruses, adenoviruses). Human amnion cell line. Chick embryo cell line.	Human fibroblast cell line (CMV) MRC-5 and WI-38 (Used for viral vaccines like rabies, chickenpox, and MMR, and virus isolation such as HSV)	HeLa cell line (human cervical carcinoma) HEp-2 cell line (for RSV, adenoviruses, HSV) KB cell line (human nasopharynx carcinoma) McCoy cell line (for virus isolation and Chlamydia) Vero cell line (for rabies vaccine production). BHK cell line (baby hamster kidney).

Solution for Question 13:

Correct Answer: C) Measles virus

Explanation:

- Syncytia formation is a characteristic cytopathic effect associated with measles virus infection, distinguishing it from other viruses.
- This occurs due to the fusion of infected cells, resulting in multinucleated giant cells.
- While the Herpes simplex virus can also cause syncytia, measles is more characteristically associated with this effect in cell cultures.
- Adenovirus and enterovirus typically do not exhibit syncytia formation.

Detection of Virus Growth in Cell Cultures:

- Cytopathic Effect (CPE) refers to the morphological changes induced by a virus in a cell line, which can be observed using a light microscope.
- While not all viruses cause CPE, those are known as cytopathic viruses.
- Each virus has a unique type of CPE that helps to identify: Enteroviruses (Option B): Rapid CPE with cell crenation and degeneration. Measles virus: Syncytium formation (RSV or respiratory syncytial virus). Herpesvirus: Discrete focal degeneration, ballooning and syncitium formation. Adenovirus (Option A): Large granular clumps resembling bunches of grapes. SV40 or Simian virus 40 (Option D): Prominent cytoplasmic vacuolation.
- Enteroviruses (Option B): Rapid CPE with cell crenation and degeneration.
- Measles virus: Syncytium formation (RSV or respiratory syncytial virus).
- Herpesvirus: Discrete focal degeneration, ballooning and syncitium formation.
- Adenovirus (Option A): Large granular clumps resembling bunches of grapes.
- SV40 or Simian virus 40 (Option D): Prominent cytoplasmic vacuolation.
- Metabolic Inhibition: Cellular metabolism produces acid in normal cultures, causing a colour change in the medium's pH indicator (phenol red).
- When viruses grow, metabolism gets inhibited, leading to reduced acid production and a colour change.
- Hemadsorption: Hemagglutinating viruses, such as influenza, can be detected by adding guinea pig erythrocytes to the culture. If the viruses replicate, the erythrocytes will adsorb onto the cell surfaces.
- If the viruses replicate, the erythrocytes will adsorb onto the cell surfaces.
- Interference: The presence of a non-cytopathogenic virus inhibits the growth of a subsequent challenge with a known cytopathogenic virus.
- Transformation: Oncogenic viruses can induce cell transformation, leading to loss of contact inhibition and the formation of microtumors in culture.

- Enteroviruses (Option B): Rapid CPE with cell crenation and degeneration.
- Measles virus: Syncytium formation (RSV or respiratory syncytial virus).
- Herpesvirus: Discrete focal degeneration, ballooning and syncitium formation.
- Adenovirus (Option A): Large granular clumps resembling bunches of grapes.
- SV40 or Simian virus 40 (Option D): Prominent cytoplasmic vacuolation.
- If the viruses replicate, the erythrocytes will adsorb onto the cell surfaces.

Other Methods for Detecting Viral Growth:

- Direct Immunofluorescence Assay detects viral antigens on the surface of cells.
- Polymerase Chain Reaction (PCR) is used to detect viral genes.
- Electron Microscopy allows visualisation of viruses within infected cell lines.
- Shell Vial Technique: For viruses like CMV, which require several weeks to exhibit CPE, the shell vial technique offers a quicker detection method (1-2 days).

Solution for Question 14:

Correct Answer: B) Negri Bodies

Explanation:

Negri bodies are characteristic intracytoplasmic eosinophilic inclusions found in the brain cells of animals infected with rabies, which causes neurological symptoms.

Inclusion bodies:

- Characteristic histological features observed in virus-infected cells exhibiting distinct sizes, shapes, locations and staining properties.
- Types: Cytoplasmic inclusion bodies Nuclear inclusion bodies Both cytoplasmic and nuclear
- Cytoplasmic inclusion bodies
- Nuclear inclusion bodies
- Both cytoplasmic and nuclear
- Inclusion bodies are generally acidophilic and appear as pink structures when stained with the Giemsa method or eosin methylene blue.
- Certain viruses, such as adenovirus, may produce basophilic inclusions.

• Intracytoplasmic inclusion bodies: Negri bodies: Intracytoplasmic eosinophilic inclusions found in the brain cells of animals, indicative of rabies.

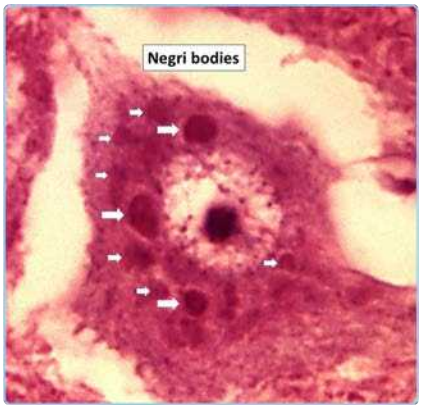
• Negri bodies: Intracytoplasmic eosinophilic inclusions found in the brain cells of animals, indicative of rabies.

• Cytoplasmic inclusion bodies

• Nuclear inclusion bodies

• Both cytoplasmic and nuclear

• Negri bodies: Intracytoplasmic eosinophilic inclusions found in the brain cells of animals, indicative of rabies.



• Paschen body (Option A): Variola virus

• Guarnieri bodies: Smaller, multiple inclusions in cells infected by vaccinia virus

• Bollinger bodies (Option C): Large inclusions observed in fowlpox.

• Molluscum bodies (Option D): HP bodies (Henderson Patterson Bodies) - Very large inclusion bodies (20-30 µm) found in molluscum contagiosum.

• Perinuclear cytoplasmic body: Reovirus

• Classification of Intranuclear Inclusion Bodies: Cowdry type A inclusions: Variable in size with a granular appearance & associated with: Torres body: Yellow fever virus Lipschultz body: Herpes simplex virus Cowdry type B inclusions: More circumscribed and often multiple, seen in: Adenovirus (basophilic) Poliovirus

• Cowdry type A inclusions: Variable in size with a granular appearance & associated with: Torres body: Yellow fever virus Lipschultz body: Herpes simplex virus

• Torres body: Yellow fever virus

• Lipschultz body: Herpes simplex virus

• Cowdry type B inclusions: More circumscribed and often multiple, seen in: Adenovirus (basophilic) Poliovirus

• Adenovirus (basophilic)

• Poliovirus

• Intracytoplasmic and intranuclear inclusion bodies: Owl's eye appearance: Cytomegalovirus (CMV) Measles virus (Warthin Finkeldey Giant Cell)

• Owl's eye appearance: Cytomegalovirus (CMV)

• Measles virus (Warthin Finkeldey Giant Cell)

• Cowdry type A inclusions: Variable in size with a granular appearance & associated with: Torres body: Yellow fever virus Lipschultz body: Herpes simplex virus

• Torres body: Yellow fever virus

• Lipschultz body: Herpes simplex virus

• Cowdry type B inclusions: More circumscribed and often multiple, seen in: Adenovirus (basophilic) Poliovirus

• Adenovirus (basophilic)

• Poliovirus

• Torres body: Yellow fever virus

• Lipschultz body: Herpes simplex virus

• Adenovirus (basophilic)

• Poliovirus

• Owl's eye appearance: Cytomegalovirus (CMV)

• Measles virus (Warthin Finkeldey Giant Cell)

Solution for Question 15:

Correct Answer: C) Plaque Assay

Explanation:

• The Plaque assay is the most suitable method for quantifying viral infectivity, as it allows for the determination of the number of infectious viral particles based on the formation of plaques in a cell culture.

• Each plaque represents an area of localised infection caused by a single infectious unit of the virus, providing a direct count of viral infectivity.

• The endpoint assay (Option A) measures the highest dilution, which still produces a specific effect but does not provide a direct count.

• Electron microscopy (Option B) is used to visualise viral particles.

• Haemagglutination (Option D) detects viral activity but does not quantify it.

Viral assays:

• Viral enumeration Electron microscopy: The ratio of a virus suspension mixed with latex particles, as observed under an electron microscope (negative staining), indicates the virus count.

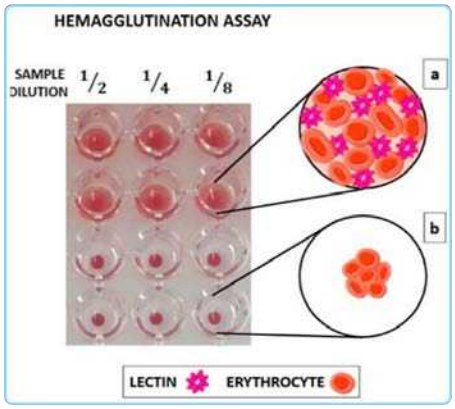
• Electron microscopy: The ratio of a virus suspension mixed with latex particles, as observed under an electron microscope (negative staining), indicates the virus count.

• Electron microscopy: The ratio of a virus suspension mixed with latex particles, as observed under an electron microscope (negative staining), indicates the virus count.

• Hemagglutination: RBCs are added to a serial dilution of a virus sample (with H spikes) to determine the hemagglutination titer; the highest dilution showing RBC agglutination indicates the number of virions.

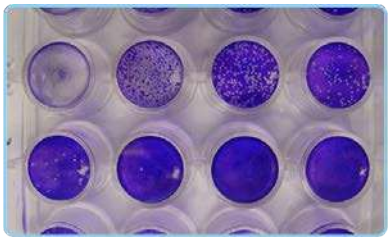
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• Hemagglutination: RBCs are added to a serial dilution of a virus sample (with H spikes) to determine the hemagglutination titer; the highest dilution showing RBC agglutination indicates the number of virions.



• Viral infective assays

Assay	Description
Endpoint assay	They are used for infectivity titration, for example, animal mortality, hemagglutinin production in allantoic fluid, and the appearance of cytopathic effects (CPE) in cell cultures.

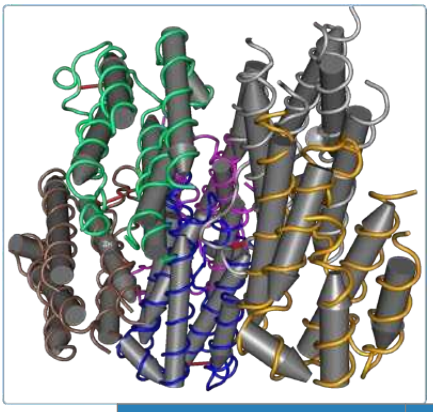
Plaque assay	 Agar is poured over a viral specimen on a cell line, revealing visible plaques (localised focus) of infected cells. The number of plaques corresponds to the infective virions. {{caption_text}}
Pox assay	Assessed by counting the number of pocks formed on the chorioallantoic membrane (CAM) from the appropriate viral inocula (Eg, Vaccinia)

Solution for Question 16:

Correct Answer: D) 1-c, 2-a, 3-b

Explanation:

Interferons are host-coded proteins produced in response to viral or non-viral inducers that help other cells of the same species to resist viral infections.



Classification of Interferons			
Aspects	Alpha interferon (IFN-α)	Beta interferon (IFN-β)	Gamma interferon (IFN-γ)

Structure	Non-glycosylated protein	Glycoprotein	Glycoprotein
Source	Leukocytes	Fibroblasts and epithelial cells	Th1 (T-helper type 1) lymphocytes
Function	Antiviral properties	Antiviral properties Used for multiple sclerosis.	Immunomodulatory and anti-proliferative (granuloma formation). Treatment of chronic granulomatous disease.
Chromosome (interferon)	9	9	12
Chromosome (interferon receptor)	21	21	6

Herpesviridae

1. A 28-year-old woman with multiple sexual partners presents with painful genital sores that started two days ago, accompanied by flu-like symptoms, fever, and headache. Examination shows shallow ulcers (as shown in the image) and tender inguinal lymphadenopathy. A swab of the ulcer has been sent for testing. What is the most likely diagnosis?



A. Haemophilus ducreyi

B. Treponema pallidum

C. Klebsiella granulomatis

D. Herpes simplex virus

---

2. A 27-year-old woman is brought to the emergency department with sudden confusion, fever, and seizures over the past 24 hours. Her partner reports memory loss and difficulty recognising faces. Examination reveals a fever of 102°F, disorientation, and mild aphasia. PCR is positive for HSV. Which feature is not associated with the causative organism?

A. Gingivostomatitis

B. Herpes labialis

C. Mollaret meningitis

D. Dendritic corneal ulcers

---

3. A 6-month-old infant presents with a 1-week history of fever, irritability, decreased appetite, and lethargy. Examination reveals high temperature, moderate hepatomegaly, and purpura/petechiae on the trunk and extremities. CBC shows thrombocytopenia and leukopenia. ELISA detects antibodies for antigen PP65. What is the cause of the condition?

A. HHV 1

B. HHV 2

C. HHV 4

D. HHV 5

---

4. A 28-year-old pregnant woman in her second trimester reports mild flu-like symptoms and mild lymphadenopathy during a routine check-up. An ultrasound shows signs of intrauterine growth restriction (IUGR). She is diagnosed with the infection caused by the largest member of the human herpesviridae (HHV) family. Which subfamily does it belong to?

A. Alphaherpesviruses

B. Betaherpesviruses

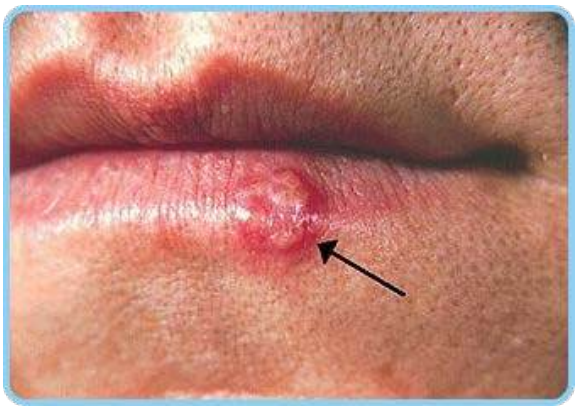
C. Gammaherpesviruses

D. Deltaherpesviruses

---

5. A 25-year-old man visits urgent care with painful lip sores and a fever, which began after a recent cold and increased work stress. Examination reveals vesicular lesions around his lips (as shown below) and swollen lymph nodes in his neck,

suggesting infection by an enveloped double-stranded DNA virus. Which feature is associated with this organism?



A. Transmitted sexually

B. Can result in neonatal herpes

C. Latency occurs in trigeminal ganglion

D. More common in adults

---

6. A 24-year-old male wrestler presents with a painful rash on his forehead and neck, initially small red bumps that developed into fluid-filled blisters (shown in the image). He reports fever and malaise for the past few days. What is the most likely diagnosis?



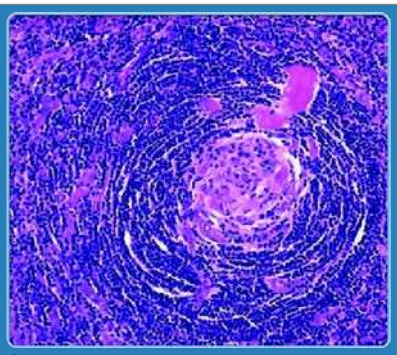
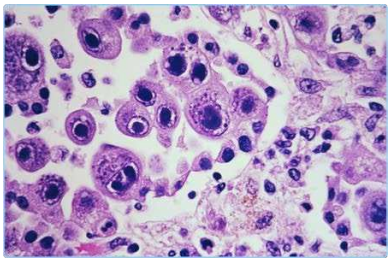
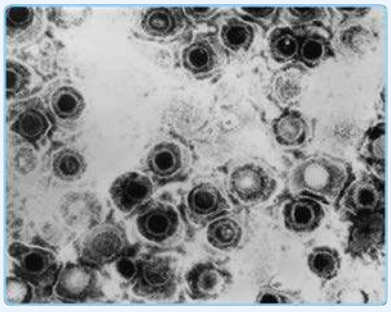
A. Herpetic whitlow

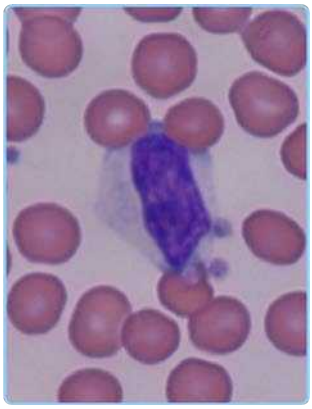
B. Herpes gladiatorum

C. Herpes zoster

D. Eczema herpeticum

7. Match the following herpes virus with its corresponding microscopic findings. 1. Herpes Simplex Virus A. 2. Epstein Barr Virus B. 3. Cytomegalovirus C. 4. Human herpes virus type 8 D.

1. Herpes Simplex Virus	 A.
2. Epstein Barr Virus	 B.
3. Cytomegalovirus	 C.

4. Human herpes virus type 8	 ... (content truncated)

A. 1-D, 2-A, 3-B, 4-C

B. 1-C, 2-D, 3-B, 4-A

C. 1-B, 2-A, 3-D, 4-C

D. 1-D, 2-A, 3-B, 4-C

---

8. A 32-year-old woman at 20 weeks gestation reports a mild rash, fever, and flu-like symptoms after exposure to a child with a rash. Ultrasound shows cicatricial lesions, limb hypoplasia, microcephaly, and intrauterine growth restriction. What is the most likely diagnosis?

A. Congenital rubella syndrome

B. Congenital varicella syndrome

C. Congenital toxoplasmosis

D. Congenital CMV infection

---

9. A 55-year-old male with a history of diabetes presents with a painful rash on the right side of his torso. He reports that the pain began as a burning sensation a few days ago, and he has since developed red, blistering, and tender lesions, as shown

below. What is the most common complication of this condition?



A. Postherpetic neuralgia

B. Zoster ophthalmicus

C. Meningoencephalitis

D. Ramsay hunt syndrome

---

10. A 25-year-old female presents with severe fatigue, a sore throat, and swollen lymph nodes for the past week, noting a similar illness in her partner. Examination shows low-grade fever, mild hepatosplenomegaly, and cervical lymphadenopathy. A complete blood count indicates absolute lymphocytosis with atypical lymphocytes. What is the most likely diagnosis?

A. Influenza

B. Toxoplasmosis

C. CMV infection

D. Glandular fever

---

11. A 21-year-old male presents with extreme fatigue, a severe sore throat, and swollen lymph nodes for two weeks, with similar recent symptoms in his partner. Examination reveals hepatosplenomegaly and cervical lymphadenopathy, and lab

results showed a high percentage of atypical lymphocytes. All the following malignancies are associated with it except?

A. Angiosarcoma

B. Burkitts lymphoma

C. Gastric carcinoma

D. Nasopharyngeal carcinoma

---

12. A 22-year-old male college student comes in with a 1-week history of fever, significant fatigue, sore throat, and swollen cervical lymph nodes. Examination shows elevated temperature, bilateral tender lymphadenopathy, and tonsillar enlargement with exudate. His CBC reveals leukocytosis with atypical lymphocytes. How does the heterophile agglutination test aid in diagnosing his condition?

A. Antibodies react with early antigen

B. Antibodies react with EBNA

C. Antibodies react with VCA

D. Antibodies react with sheep RBC

---

13. A 7-year-old boy has been brought to the clinic with a painful blistering rash around his mouth for the past three days, accompanied by a low-grade fever and irritability. Examination reveals lesions as below and cervical lymphadenopathy. The mother mentions a history of recurrent cold sores. What is the most appropriate management for this child?



A. Valganciclovir

B. Ganciclovir

C. Cidofovir

D. Acyclovir

---

14. A 2-year-old child presents with a sudden high fever for 3 days, followed by a rose-coloured maculopapular rash that starts on the trunk and spreads to the face and extremities. The child is well-hydrated and active. Which of the following statements is incorrect regarding the diagnosis of this condition?

A. HHV-6 shows limited antigenic cross-reactivity with HHV-7.

B. HHV-6 can cause symptoms similar to infectious mononucleosis.

C. HHV 6 and 7 both bind to T- cells.

D. HHV-6 causes the above condition, and HHV-7 is not related to it.

---

15. A 35-year-old man with HIV presents with fever, weight loss, and reddish-purple raised skin lesions. A biopsy shows spindle-shaped cells with mixed inflammatory infiltrate, and serological testing reveals antibodies to Human Herpesvirus 8. Which of the following statements regarding HHV 8 is false?

A. HHV 8 can cause primary effusion lymphoma

B. Associated with Kaposi sarcoma

C. HHV 8 causes Castleman's disease

D. HHV 8 does not spread through oral secretions

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	4
Question 4	2
Question 5	3
Question 6	2
Question 7	3
Question 8	2
Question 9	1
Question 10	4
Question 11	1
Question 12	4
Question 13	4
Question 14	4
Question 15	4

Solution for Question 1:

Correct Answer: D) Herpes simplex virus

Explanation:

The scenario of painful genital ulcers (as shown in the image above), flu-like symptoms, tender inguinal lymphadenopathy, and a history of multiple sexual partners is most likely indicative of a

Herpes Simplex Virus Type 2 (HSV 2) infection.

Genital lesions of HSV

- Mostly seen in HSV type 2.
- Bilateral, painful genital vesiculo-ulcerative lesions.
- Tender inguinal lymphadenopathy.
- Site: Lesions are mainly seen on, Males: The penis or in the urethra (urethritis).
- Males: The penis or in the urethra (urethritis).
- Males: The penis or in the urethra (urethritis).



- Females: The cervix, vagina, vulva and perineum
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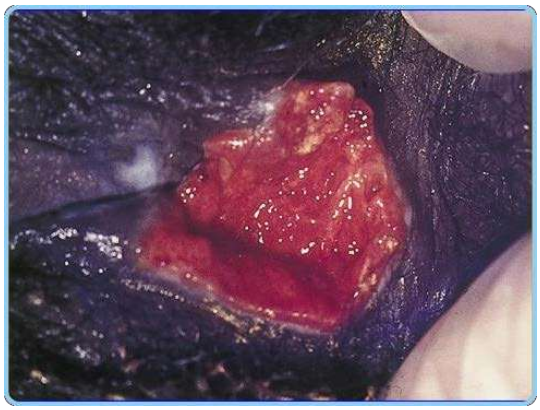


- It is associated with systemic features like fever (flu-like) and malaise.
- In pregnant females, a cesarean is preferred over normal labour to prevent vertical transmission.

Haemophilus ducreyi (Option A) is a gram-negative bacillus that causes painful, non-indurated irregular ulcers on the genitalia (chancroid or soft sore) (which is not vesicular as seen in HSV 2, and there is no association of flu-like symptoms).

Treponema pallidum (Option B) is a spirochete that causes syphilis, resulting in the characteristic painless, relatively avascular, circumscribed, indurated, superficially ulcerated lesion (hard chancre).

Klebsiella granulomatis (Option C) is a gram-negative bacteria that causes a painless papule on the genitalia, which leads to a slowly progressive, auto-inoculable ulcer known as donovanosis or granuloma inguinale.



Solution for Question 2:

Correct Answer: C) Mollaret meningitis

Explanation:

- The scenario presenting with seizures, aphasia, disorientation, confusion, fever, and positive PCR testing for HSV indicates a diagnosis of herpes simplex virus encephalitis caused by HSV-1.
- Mollaret meningitis is recurrent aseptic meningitis associated with HSV-2, not HSV-1.

Clinical manifestations of Herpes simplex virus

- Cutaneous: Site: Face (cheeks, chin, around the mouth or on the forehead) Buttocks (infants as napkin rash) 'Fever blister' or herpes febrile or cold sores (viral reactivation in febrile patients)
- Site: Face (cheeks, chin, around the mouth or on the forehead) Buttocks (infants as napkin rash)
- Face (cheeks, chin, around the mouth or on the forehead)

• Buttocks (infants as napkin rash)

• 'Fever blister' or herpes febrile or cold sores (viral reactivation in febrile patients)

• Site: Face (cheeks, chin, around the mouth or on the forehead) Buttocks (infants as napkin rash)

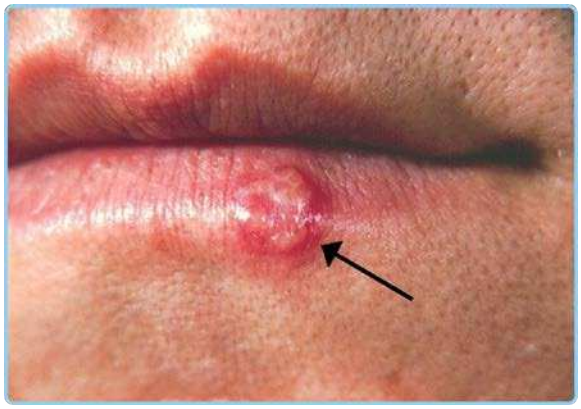
• Face (cheeks, chin, around the mouth or on the forehead)

• Buttocks (infants as napkin rash)

• 'Fever blister' or herpes febrile or cold sores (viral reactivation in febrile patients)

• Face (cheeks, chin, around the mouth or on the forehead)

• Buttocks (infants as napkin rash)



• Herpetic whitlow: Occupational variety seen in doctors, dentists, and nurses.

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• Eczema herpeticum: Generalised crops of vesicles appear in the affected area with widespread ulceration, as seen in children with eczema.

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• Herpes gladiatorum (Wrestler’s herpes): In athletes who play skin-to-skin contact sports like wrestlers.

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• Herpes gladiatorum (Wrestler’s herpes): In athletes who play skin-to-skin contact sports like wrestlers.



• Herpes multiforme (red, raised, target-like lesions on skin)

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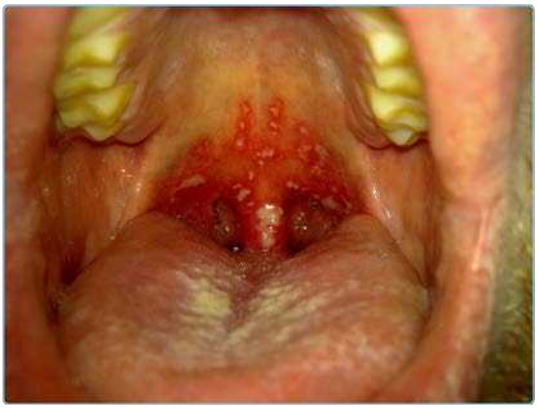
• Mucosal Site: Buccal mucosa (most common) Gingivostomatitis and pharyngitis (primary infection) (Option A)

• Site: Buccal mucosa (most common)

• Gingivostomatitis and pharyngitis (primary infection) (Option A)

• Site: Buccal mucosa (most common)

• Gingivostomatitis and pharyngitis (primary infection) (Option A)



• Recurrent herpes labialis (Option B)

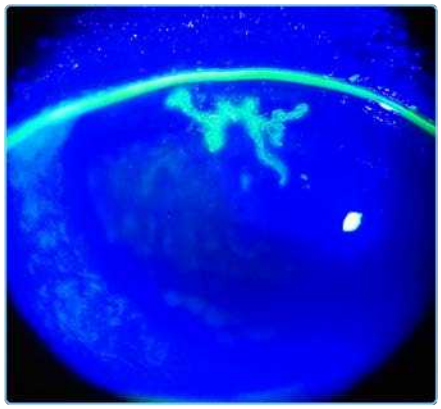
• Recurrent herpes labialis (Option B)

• Recurrent herpes labialis (Option B)

• Ophthalmic Dendritic corneal ulcers (Option D)

• Dendritic corneal ulcers (Option D)

• Dendritic corneal ulcers (Option D)



• Corneal blindness and conjunctivitis are also observed Chorioretinitis and acute necrotising retinitis are rare.

• Corneal blindness and conjunctivitis are also observed

• Chorioretinitis and acute necrotising retinitis are rare.

• Nervous system Encephalitis (HSV Encephalitis) Caused by HSV-1 Most common cause of sporadic acute viral encephalitis Affected site: Temporal lobe Meningitis Most commonly caused by HSV-2 Mollaret meningitis (recurrent lymphocytic meningitis) (Option C) HSV can result in Bell's palsy.

• Encephalitis (HSV Encephalitis) Caused by HSV-1 Most common cause of sporadic acute viral encephalitis Affected site: Temporal lobe

• Caused by HSV-1

• Most common cause of sporadic acute viral encephalitis

• Affected site: Temporal lobe

• Meningitis Most commonly caused by HSV-2 Mollaret meningitis (recurrent lymphocytic meningitis) (Option C) HSV can result in Bell's palsy.

• Most commonly caused by HSV-2

• Mollaret meningitis (recurrent lymphocytic meningitis) (Option C)

• HSV can result in Bell's palsy.

• Visceral Symptoms like dysphagia, substernal pain and weight loss. Tracheobronchitis and pneumonitis (lung involvement)

• Symptoms like dysphagia, substernal pain and weight loss.

• Tracheobronchitis and pneumonitis (lung involvement)

• Genital infections Mostly seen in HSV type 2 Bilateral, painful genital vesiculo-ulcerative lesions Tender inguinal lymphadenopathy Site: Lesions are mainly seen on, Male: The penis or in the urethra (urethritis). Female: The cervix, vagina, vulva and perineum.

• Mostly seen in HSV type 2

• Bilateral, painful genital vesiculo-ulcerative lesions

• Tender inguinal lymphadenopathy

• Site: Lesions are mainly seen on, Male: The penis or in the urethra (urethritis). Female: The cervix, vagina, vulva and perineum.

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• Encephalitis (HSV Encephalitis) Caused by HSV-1 Most common cause of sporadic acute viral encephalitis Affected site: Temporal lobe

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• Most common cause of sporadic acute viral encephalitis

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• Meningitis Most commonly caused by HSV-2 Mollaret meningitis (recurrent lymphocytic meningitis) (Option C) HSV can result in Bell's palsy.

• Most commonly caused by HSV-2

• Mollaret meningitis (recurrent lymphocytic meningitis) (Option C)

• HSV can result in Bell's palsy.

• Caused by HSV-1

• Most common cause of sporadic acute viral encephalitis

• Affected site: Temporal lobe

• Most commonly caused by HSV-2

• Mollaret meningitis (recurrent lymphocytic meningitis) (Option C)

• HSV can result in Bell's palsy.

• Symptoms like dysphagia, substernal pain and weight loss.

• Tracheobronchitis and pneumonitis (lung involvement)

• Mostly seen in HSV type 2

• Bilateral, painful genital vesiculo-ulcerative lesions

• Tender inguinal lymphadenopathy

• Site: Lesions are mainly seen on, Male: The penis or in the urethra (urethritis). Female: The cervix, vagina, vulva and perineum.

• Male: The penis or in the urethra (urethritis).

• Female: The cervix, vagina, vulva and perineum.

• Male: The penis or in the urethra (urethritis).

• Female: The cervix, vagina, vulva and perineum.



• Congenital If a pregnant woman has genital lesions during labour, the risk of infection can be mitigated by performing a cesarean section. Neonatal herpes: A disseminated disease involving multiple organs, with or without encephalitis. The mortality rate is high, and survivors may have neurological impairments.

• If a pregnant woman has genital lesions during labour, the risk of infection can be mitigated by performing a cesarean section.

• Neonatal herpes: A disseminated disease involving multiple organs, with or without encephalitis. The mortality rate is high, and survivors may have neurological impairments.

• If a pregnant woman has genital lesions during labour, the risk of infection can be mitigated by performing a cesarean section.

• Neonatal herpes: A disseminated disease involving multiple organs, with or without encephalitis. The mortality rate is high, and survivors may have neurological impairments.



• Transplacental infections leading to congenital malformations are rare.

• Transplacental infections leading to congenital malformations are rare.

• Transplacental infections leading to congenital malformations are rare.

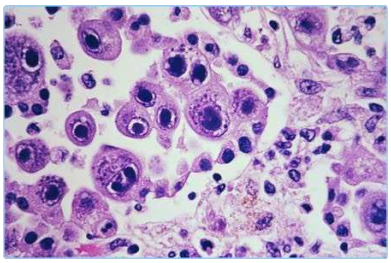
Solution for Question 3:

Correct Answer: D) HHV 5

Explanation:

The presentation suggests cytomegalovirus (CMV) infection, caused by Human Herpesvirus 5 (HHV-5), indicated by fever, hepatomegaly, petechiae/purpura, thrombocytopenia, leukopenia, and the detection of antibodies to the specific CMV antigen PP65 via ELISA.

CMV infection		
Modes of transmission	Oral route, respiratory & body secretions (most common) Vertical transmission Sexual transmission Organ transplantation Blood transfusion	
Host cells	Salivary gland Kidney	
Clinical features	Congenital infection	Most common intrauterine infection with congenital defects. Common presentation: Petechiae Hepatosplenomegaly Jaundice Less common presentation: Microcephaly Cerebral calcifications Intrauterine growth retardation Prematurity Other defects: Sensorineural hearing loss.
Perinatal infection	Commonly asymptomatic. Occasionally, premature infants present with interstitial pneumonitis.	

Immunocompromised	Seen in AIDS, HIV, organ transplant patients and cancer patients undergoing chemo Presentations in AIDS and HIV CMV chorioretinitis Progressive dementia, Cranial nerve defects, Presentations in Organ transplant Bilateral interstitial pneumonia Febrile leukopenia Obliterative bronchiolitis Graft atherosclerosis Rejection of renal allografts.	
Immunocompetent	Usually, it occurs following a blood transfusion. It presents similar to infectious mononucleosis, but heterophile antibody tests would be negative.	
Investigations	Peripheral smear	 Owl's eye appearance of inclusion bodies {{caption_text}}
Virus isolation	It is done by Human fibroblast cell line culture Shell vial culture	
Antigen detected	PP65	
Antibody detection	It is done by ELISA, and the antibodies detected are CMV IgM antibodies CMV IgG antibodies Against PP150 and PP65, Against the PP52	
PCR	Used in detecting CMV DNA Quantitative nucleic acid testing (QNAT) by PCR is the most sensitive and specific testing.	

Treatment	Ganciclovir (Drug of choice) Foscarnet (in Ganciclovir resistance) CMV immunoglobulins and Ganciclovir are given to bone marrow transplant patients.	

HHV 1 (Option A) is Herpes simplex virus 1, which causes oral herpes (cold sores) and can commonly lead to encephalitis or herpes keratitis.

- It is not typically associated with the systemic symptoms, hepatomegaly, thrombocytopenia, or the presence of the PP65 antigen seen in the infant's case.

HHV 2 (Option B), also known as Herpes simplex virus 2, primarily causes genital herpes.

- While it can cause severe disease in newborns, especially during delivery (neonatal herpes), it does not typically present with the systemic symptoms described in this infant's presentation.
- Like HHV 1, it is not associated with the PP65 antigen.

HHV 4 (Option C) is the Epstein-Barr virus (EBV), primarily associated with infectious mononucleosis and certain lymphoproliferative disorders.

- It does not show antibodies against the PP65 antigen.

Solution for Question 4:

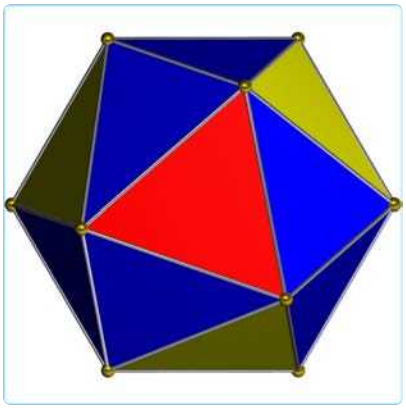
Correct Answer: B) Betaherpesviruses

Explanation:

- The infection described is likely caused by cytomegalovirus (CMV), a member of the Betaherpesvirinae subfamily of the Herpesviridae family.
- CMV is the largest human herpesvirus and can cause congenital infection, leading to intrauterine growth restriction (IUGR) and other fetal complications.
- It often presents with mild, flu-like symptoms and lymphadenopathy in pregnant individuals.

General Properties of Herpesviridae:

- Double-stranded DNA viruses (enveloped)
- Icosahedral symmetry



• eplication site: Host cell nucleus

• Size: 150-200 nm (2nd largest virus)

• Largest virus: Pox virus (300nm)

• Smallest virus: Parvovirus (20nm)

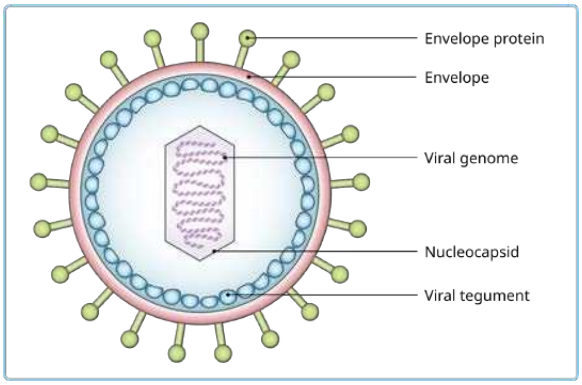
• Characteristics of infection: Latent infection (ability to persist within the infected host) Periodic reactivation

• Latent infection (ability to persist within the infected host)

• Periodic reactivation

• Latent infection (ability to persist within the infected host)

• Periodic reactivation



• Classification

Subfamily	Human herpesvirus (HHV) type	Common name	Cytopathology	Site of latency
Alpha (α) or Alphaherpesviruses (Option A)	HHV 1	Herpes simplex virus 1 (HSV1)	Cytolytic	Neurons
HHV 2	Herpes simplex virus 2 (HSV2)			

HHV 3	Varicella zoster virus			
Beta (β) or Betaherpesviruses (Option B)	HHV 5	Cytomegalovirus (or salivary gland virus)	Cytomegalic	Secretory glands (salivary), kidneys, and many other cells of the body.
HHV 6	-	Lymphoproliferative	Lymphoid tissues	
HHV 7	-			
Gamma (γ) or Gammaherpesviruses (Option C)	HHV 4	Epstein-Barr virus		
HHV 8	-	-	-	

Deltaherpesviruses (Option D): There is no such subfamily in the Herpesviridae family.

Solution for Question 5:

Correct Answer: C) Latency occurs in trigeminal ganglion

Explanation:

The scenario, marked by orofacial lesions, lymphadenopathy, a recent cold, and work-related stress, indicates an infection with Herpes simplex virus type 1 (HSV-1), an enveloped, double-stranded DNA virus that establishes latency in the trigeminal ganglion.

Aspects	Herpes simplex virus type 1 (HSV 1)	Herpes simplex virus type 2 (HSV 2)
Transmission	Mucosal membrane or abraded skin	Sexual or vertical (mother to baby) (Option A)
Site of latency	Trigeminal ganglia (Option C)	Sacral ganglia
Age group	Children	Adults (Option D)

Clinical features	Skin lesions above the waist (oral lesions like gingivostomatitis and labialis) gingivostomatitis children" data-author="Dr. Syed Wali Peeran" data-hash="9929" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Oral_Ulcerative_Lesions.jpg" data-tags="April2025" height="222" src="https://image.prepladder.com/notes/taAOU1aeM2nbNQlL2YIj1744449643.png" width="288" /> {{caption_text}} Encephalitis Meningitis	 ... (content truncated)
Growth on chick embryo fibroblasts	Poor growth	Good growth (larger pox resembling variola)
Neurovirulence	Less	More
Drug resistance	Less	More
Temperature sensitivity	Less	More

Solution for Question 6:

Correct Answer: B) Herpes gladiatorum

Explanation:

The scenario describes a wrestler with painful fluid-filled blisters and flu-like symptoms, which is diagnosed as Herpes gladiatorum, also known as Wrestler's Herpes.

Herpes gladiatorum:

- A cutaneous lesion observed in infections caused by the Herpes simplex virus.
- Transmission: Contact sports, like wrestling, due to skin-to-skin contact in athletes.
- Feature: Painful, fluid-filled blisters on the skin, accompanied by symptoms like fever and malaise.



- Treatment: Antiviral medications
- Prevention: Good hygiene practices during sports.

Herpetic whitlow (Option A) is another type of cutaneous lesion associated with HSV infection.

- It commonly occurs on the fingers of doctors and healthcare workers who have been exposed to infected patients.



Herpes zoster (Option C), or shingles, is caused by the Varicella zoster virus (HHV 3).

- It occurs when the virus reactivates and travels along sensory nerves, producing zoster lesions in the affected skin area, often accompanied by neuritic pain.

Eczema herpeticum (Option D) is another cutaneous lesion linked to HSV infection.

- It presents as generalised clusters of vesicles on the affected area, often accompanied by widespread ulceration, and is typically seen in children with eczema.

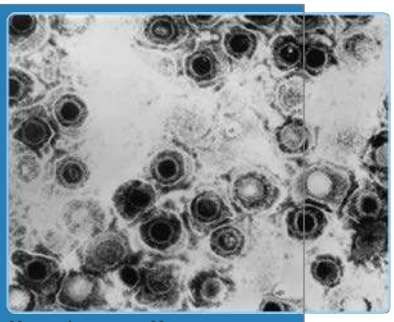
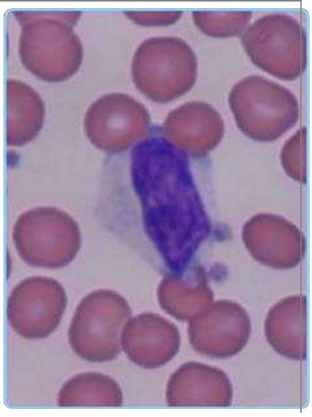


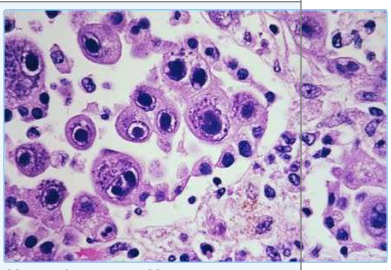
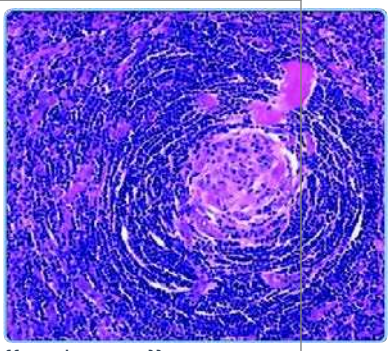
Solution for Question 7:

Correct Answer: B) 1-C, 2-D, 3-B, 4-A

Explanation:

Microscopic features of the herpes virus:

Herpes simplex virus Tzanck smear: Multinucleated giant cells with faceted nuclei and homogeneously stained 'ground-glass' chromatin (Tzanck cells). Formation of Lipschutz bodies (Cowdry type A).	 {{caption_text}}
EBV (Epstein Barr virus) Downey cells: Atypical lymphocytes (large, vacuolated) are observed. Ballerina skirt appearance.	

	... (content truncated)
CMV (Cytomegalovirus) The presence of inclusion bodies gives the characteristic “Owl-eye appearance”.	 {{caption_text}}
HHV 8 (human herpes virus type 8) Castleman disease: Onion skinning of lymphocytes and blood vessels, giving the appearance of “Lollipop follicles”.	 {{caption_text}}

Solution for Question 8:

Correct Answer: B) Congenital varicella syndrome

Explanation:

- The scenario describes a pregnant woman at 20 weeks of gestation with flu-like symptoms, fever, and a rash, while the fetus shows limb hypoplasia, microcephaly, and cicatricial lesions, indicating a diagnosis of Congenital (or fetal) Varicella Syndrome.

Congenital Varicella Syndrome: If a pregnant woman contracts the varicella-zoster virus during the first 20 weeks of gestation, the virus may be transmitted to the fetus through the placenta, leading to congenital (or fetal) varicella syndrome.

- Clinical features (fetus): Cicatricial skin lesion Limb hypoplasia Microcephaly Cortical atrophy Chorioretinitis Some babies may carry latent infection without exhibiting any defects.
- Cicatricial skin lesion
- Limb hypoplasia

- Microcephaly
- Cortical atrophy
- Chorioretinitis
- Some babies may carry latent infection without exhibiting any defects.
- Cicatricial skin lesion
- Limb hypoplasia
- Microcephaly
- Cortical atrophy
- Chorioretinitis
- Some babies may carry latent infection without exhibiting any defects.

Congenital rubella syndrome (Option A) occurs when a mother transmits rubella to her fetus, resulting in miscarriage or classic triad of cataract, patent ductus arteriosus and sensorineural hearing loss and also various complications such as hepatosplenomegaly, thrombocytopenic purpura, myocarditis, and bone lesions.

Congenital toxoplasmosis (Option C) occurs when non-immune mothers are infected with Toxoplasma gondii during pregnancy.

- This can lead to stillbirths, chorioretinitis, intracranial calcifications, psychomotor disturbances, and conditions such as hydrocephaly or microcephaly.
- The classic triad of neurological findings consists of chorioretinitis, hydrocephalus, and intracranial calcifications.



Congenital CMV infection (Option D) results from a cytomegalovirus infection during pregnancy and can lead to fetal death or cytomegalic inclusion disease in newborns, often characterised by hepatosplenomegaly, jaundice, thrombocytopenic purpura, hemolytic anaemia, and a "blueberry muffin rash" with purplish-blue spots due to extramedullary hematopoiesis, where blood cell production occurs outside the bone marrow.

- Survivors may experience mental retardation and developmental delays.

Solution for Question 9:

Correct Answer: A) Postherpetic neuralgia

Explanation:

• The scenario depicts a diabetic (poor immunity) male with a painful, blistering rash on one side of his torso, diagnosed as Herpes Zoster (Shingles).

• The most common complication associated with this condition is postherpetic neuralgia.

• It occurs in individuals who previously had chickenpox, allowing the virus to remain latent in the sensory ganglia and reactivate during immunocompromised states, such as old age, diabetes, or HIV.

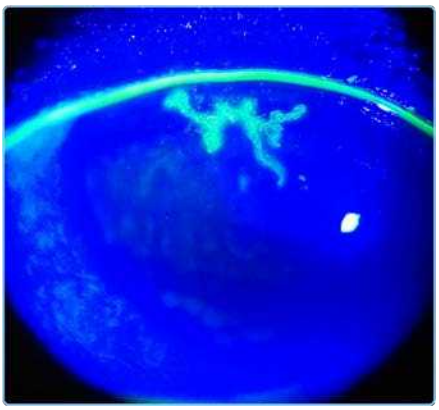
• Complications Postherpetic neuralgia (Option A): Pain and paresthesia at the affected site Zoster ophthalmicus (Option B): The reactivation of the virus in the trigeminal ganglion is marked by symptoms like pain, blisters, fever, and malaise, and it can also lead to the formation of dendritic corneal ulcers.

• Postherpetic neuralgia (Option A): Pain and paresthesia at the affected site

• Zoster ophthalmicus (Option B): The reactivation of the virus in the trigeminal ganglion is marked by symptoms like pain, blisters, fever, and malaise, and it can also lead to the formation of dendritic corneal ulcers.

• Postherpetic neuralgia (Option A): Pain and paresthesia at the affected site

• Zoster ophthalmicus (Option B): The reactivation of the virus in the trigeminal ganglion is marked by symptoms like pain, blisters, fever, and malaise, and it can also lead to the formation of dendritic corneal ulcers.



• Ramsay Hunt syndrome (Option D) Reactivation of the virus involving the geniculate ganglion of the facial nerve can cause lesions to erupt on the tympanic membrane and the external auditory canal (otalgia). This condition may also present with symptoms such as facial palsy, Bell's palsy, and loss of taste in the anterior 2/3rd of the tongue.

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• Reactivation of the virus involving the geniculate ganglion of the facial nerve can cause lesions to erupt on the tympanic membrane and the external auditory canal (otalgia).

• This condition may also present with symptoms such as facial palsy, Bell's palsy, and loss of taste in the anterior 2/3rd of the tongue.

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• Other complications like meningoencephalitis (Option C) and lower motor neuron paralysis might also occur

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Solution for Question 10:

Correct Answer: D) Glandular fever

Explanation:

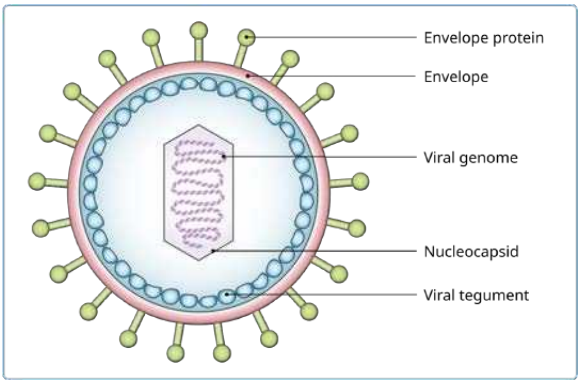
The clinical presentation suggests an Epstein-Barr virus (EBV) or HHV 4 (human herpes virus type 4) infection, diagnosed as Infectious Mononucleosis (also known as Kissing Disease or Glandular Fever).

Epstein-Barr virus (EBV):

• Route of infection: Close contact, kissing, sexual intercourse, blood transfusion, bone marrow transplant.

• Site of latency: B cells

• Family: Herpesviridae (gamma (γ) or gammaherpesviruses)



• Clinical conditions: Infectious mononucleosis EBV-associated malignancies: Burkitt's lymphoma, lymphomas in immunodeficient (AIDS and transplant recipients), nasopharyngeal

carcinoma.

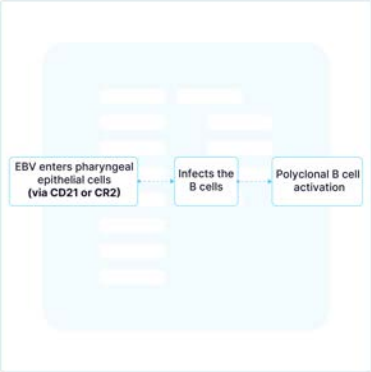
• Infectious mononucleosis

• EBV-associated malignancies: Burkitt's lymphoma, lymphomas in immunodeficient (AIDS and transplant recipients), nasopharyngeal carcinoma.

• Pathogenesis: Polyclonal B cell activation

• Infectious mononucleosis

• EBV-associated malignancies: Burkitt's lymphoma, lymphomas in immunodeficient (AIDS and transplant recipients), nasopharyngeal carcinoma.



• Infectious mononucleosis (Kissing disease or Glandular fever) Incubation period: 4-8 weeks Clinical features: Splenomegaly, hepatitis, encephalitis, fever, sore throat, lymphadenopathy, arthralgia, lethargy. Microscopy: Downey cells: Atypical lymphocytes (large, vacuolated) are observed. Appearance: Ballerina skirt appearance.

• Incubation period: 4-8 weeks

• Clinical features: Splenomegaly, hepatitis, encephalitis, fever, sore throat, lymphadenopathy, arthralgia, lethargy.

• Microscopy: Downey cells: Atypical lymphocytes (large, vacuolated) are observed. Appearance: Ballerina skirt appearance.

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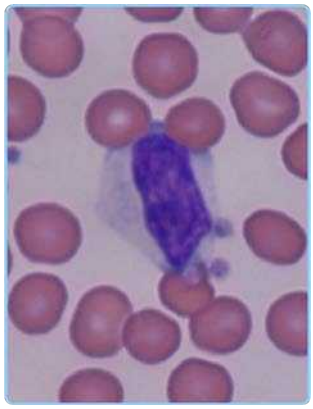
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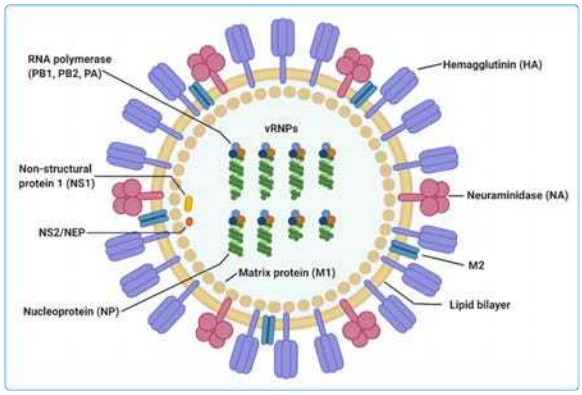
• Downey cells: Atypical lymphocytes (large, vacuolated) are observed.

• Appearance: Ballerina skirt appearance.



Influenza (Option A), commonly known as the flu, is a contagious respiratory illness caused by influenza viruses.

• It can affect people of all ages and is characterised by sudden onset of fever, body aches, and respiratory symptoms (hepatosplenomegaly and atypical lymphocytes are not characteristic of influenza).

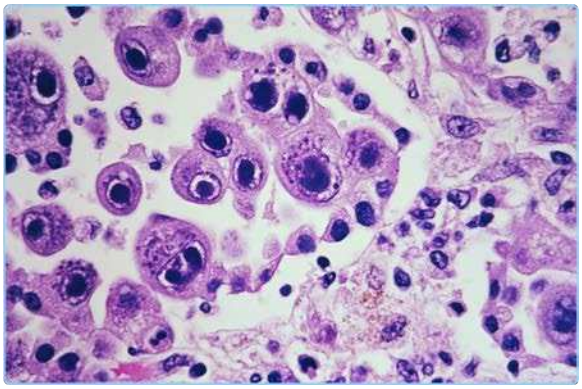


Toxoplasmosis (Option B) is an infection caused by the parasite Toxoplasma gondii.

• It is one of the most prevalent parasitic infections in humans and presents with symptoms such as mild flu-like signs (fever, muscle aches), swollen lymph nodes, fatigue, and headaches (without atypical lymphocytes).

CMV (Cytomegalovirus) infection (Option C) is often asymptomatic; however, in some cases, it can present with flu-like symptoms such as fever, fatigue, and sore throat.

• Microscopy reveals the characteristic "owl-eye" appearance of the infected cells.



Solution for Question 11:

Correct Answer: A) Angiosarcoma

Explanation:

- The patient's presentation is consistent with infectious mononucleosis (kissing disease or glandular fever), likely caused by Epstein-Barr Virus (EBV).
- Angiosarcoma is a rare malignancy of endothelial cells and is not associated with EBV infection.

Malignancies associated with EBV:

Types of malignancy	Examples
Carcinoma	Gastric carcinoma (most common) (Option C) Nasopharyngeal carcinoma (Option D)
Lymphoma	Hodgkin's lymphoma (most common lymphocyte depletion and mixed cellularity) Non-Hodgkin's lymphoma Burkitt's lymphoma (Option B) DLBCL (diffuse large B cell lymphoma) Post-transplant lymphoma T or NK cell lymphoma
Sarcoma	Leiomyosarcoma

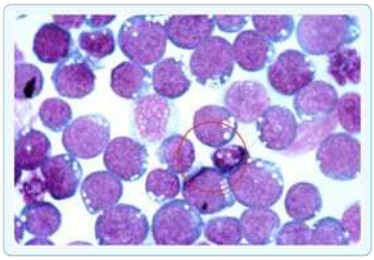
Solution for Question 12:

Correct Answer: D) Antibodies react with sheep RBC

Explanation:

• The heterophile agglutination (Paul-Bunnell test) test, also known as the Monospot test, is used to diagnose infectious mononucleosis, typically caused by Epstein-Barr Virus (EBV).

• It detects heterophile antibodies that react with antigens on non-human red blood cells, such as sheep or horse RBCs.

Investigations for EBV		
Peripheral smear	 Initially, leukopenia is noted, often followed by leukocytosis as the condition progresses. The presence of atypical mononuclear cells (derived from cytotoxic T cells) exhibits basophilic vacuolated cytoplasm and kidney-shaped nuclei (Downey cells). Ballerina skirt appearance: Characteristic morphology where atypical lymphocytes have cytoplasm that extends and flares, resembling a ballerina's skirt. {{caption_text}}	
Serology	Heterophile antibody test (Non-specific antibody test)	Paul-Bunnell test: Heterophile antibodies react with Sheep RBC. Monospot test: Heterophile antibodies react with Horse RBC.
Specific antibody test (ELISA or Immunofluorescence)	lgM antibody against Viral Capsid antigen (earliest antigen) (Option C). Antibody against the EB nuclear antigen (EBNA) (Option B). Antibodies against early antigens (Option A).	

Culture	For Virus isolation	Not commonly used.
Other tests	PCR Nucleic acid hybridisation test (Most sensitive & specific)	

Solution for Question 13:

Correct Answer: D) Acyclovir

Explanation:

- The child's presentation of a painful blistering rash around the mouth, low-grade fever, irritability, and a history of recurrent cold sores suggests an outbreak of Herpes simplex virus (HSV) infection.
- Acyclovir is the first-line antiviral treatment for herpes simplex virus infections, including oral herpes in children.
- It is ineffective against CMV because it targets thymidine kinase, an enzyme in herpes simplex virus (HSV) and varicella-zoster virus (VZV), but it is absent in CMV.
- Valacyclovir and Famciclovir are also effective for managing HSV 1 and 2.
- Foscarnet is used in conditions that are resistant to the above antivirals.

Valganciclovir (Option A) is a prodrug of Ganciclovir.

- It is mostly used to manage CMV, especially in the case of CMV retinitis.

Ganciclovir (Option B) is the drug of choice to manage CMV and is also used to manage EBV; in case of resistance, Foscarnet can be used.

Cidofovir (Option C) can be used to manage both CMV and HSV infections. It is effective against many acyclovir-resistant HSV and ganciclovir-resistant CMV cases.

- However, it is nephrotoxic, so it should be used under caution.

Solution for Question 14:

Correct Answer: D) HHV-6 causes the above condition, and HHV-7 is not related to it.

Explanation:

- The child's sudden high fever, followed by a rose-coloured maculopapular rash, indicates roseola infantum, primarily caused by human herpesvirus 6 (HHV-6).
- While HHV-6 causes the condition, HHV-7 is related and can cause similar illnesses.
- HHV-6 shows some antigenic cross-reactivity with HHV-7, and both bind to T-cells via CD46 (Options A and C).
- HHV-6 has 2 variants, i.e. A and B.
- Both HHV 6 & 7 spreads through saliva.

Conditions caused	
HHV 6	In children: Sixth disease/ Roseola/ Exanthem Subitum (sudden high fever followed by a distinctive rash) In adults: Infectious mononucleosis-like symptoms and focal encephalitis (Option B) In Immunodeficient - Pneumonia and Disseminated disease.
HHV 7	Causes Sixth disease/ Roseola/ Exanthem Subitum in children

Solution for Question 15:

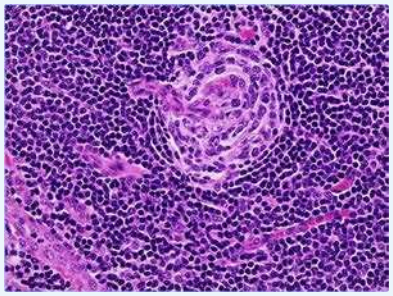
Correct Answer: D) HHV 8 does not spread through oral secretions

Explanation:

HHV-8 is found in oral secretions and can spread through saliva (the most common route of transmission). This can also be transmitted through

- Sexual contact
- Vertical transmission
- Blood transfusion
- Organ transplants

HHV 8	

Investigations	PCR Serological tests: Indirect immunofluorescence, Western blot and ELISA.
Clinical conditions	 Kaposi sarcoma: (Option B) Commonly affected sites are lower limbs, followed by lymph nodes. It commonly affects the immunocompromised people. ... (content truncated)

Parvoviridae, Poxviridae, Adenovirus

1. A 9-year-old boy presents with fever, malaise, and a bright red "slapped cheek" rash on his face, which later spreads to his trunk and extremities. Blood tests reveal mild anaemia. The clinical diagnosis of erythema infectiosum is made. What are the characteristics of the virus responsible for this condition?



A. Non-enveloped single-stranded DNA (ssDNA) virus

B. Enveloped single-stranded DNA (ssDNA) virus

C. Enveloped double-stranded DNA (dsDNA) virus

D. Non-enveloped double-stranded DNA (dsDNA) virus

---

2. A 35-year-old woman with sickle cell disease presents with pallor, fatigue, and weakness. Her hemoglobin is significantly decreased, with a low reticulocyte count. She reports recent exposure to her child, who had a finding as shown in the image below. Given the virus's specific tropism, which maternal or fetal tissue is primarily affected in this scenario?

A. Fetal liver, targeting erythroid progenitor cells

B. Fetal bone marrow, affecting myeloid progenitor cells

C. Fetal spleen, targeting immune cells

D. Fetal placenta, affecting trophoblasts

---

3. A 14-year-old boy presents with a 5-day history of painful, itchy swelling and erythema of his hands and feet, following mild flu-like symptoms that resolved a few days ago. Examination reveals the following findings, as shown in the image below. What syndrome is this child exhibiting, and which pathogen is it associated with?



A. Erythema infectiosum; Parvovirus B19

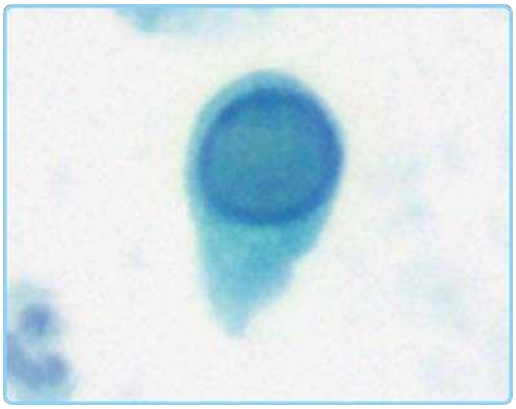
B. Papular-purpuric gloves and socks syndrome; Parvovirus B19

C. Hand, foot, and mouth disease; Coxsackievirus

D. Scarlet fever; Streptococcus pyogenes

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4. A 55-year-old kidney transplant recipient presents with worsening renal function. Urinalysis shows the following finding in the image below. PCR confirms the presence of viral DNA. Which virus is most likely responsible for this patient's nephropathy?



A. JC virus

B. Cytomegalovirus

C. BK virus

D. Epstein-Barr virus

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5. A 30-year-old female is diagnosed with cervical dysplasia, and HPV testing reveals a high-risk strain. A histopathological examination of her cervical biopsy shows the following findings. Which viral protein is associated with the formation of these findings?

A. E1 protein

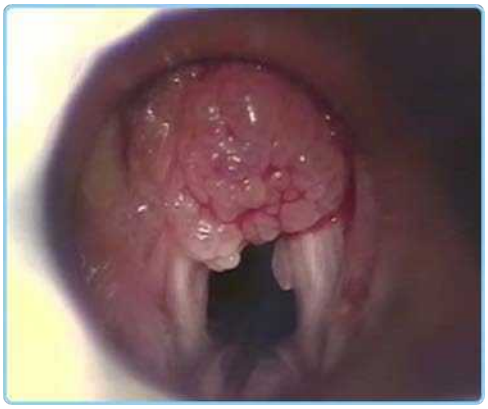
B. E6 protein

C. E7 protein

D. E5 protein

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6. A 25-year-old woman presents with recurrent hoarseness and difficulty breathing. Laryngoscopy reveals findings, as shown in the image below. What is the most likely HPV serotype associated with these lesions?



A. HPV 5

B. HPV 11

C. HPV 16

D. HPV 31

---

7. A 16-year-old girl is brought to the clinic for her HPV vaccination with the nine-valent HPV vaccine (Gardasil 9). Which of the following HPV types does this vaccine protect against?

A. 6, 11, 16, 18, 20, 24, 31, 33, 45

B. 6, 11, 16, 18, 31, 33, 45, 52, 58

C. 9, 11, 16, 18, 30, 31, 33, 45, 52

D. 6, 11, 14, 16, 18, 31, 33, 52, 58

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8. A 28-year-old male traveller returns from Central Africa with symptoms of fever and mild skin rashes. On examination, he has lymphadenopathy. He reports having direct contact with wild animals during his trip. Which of the following viruses causes these symptoms?

A. Variola

B. Vaccinia

C. Monkeypox virus

D. Varicella

---

9. A 6-year-old child is brought to the physician as he developed lesions on the body, as below. Which of the following statements would be false about this condition?



A. These are caused by molluscum contagiosum virus

B. The lesions can be seen all over the body

C. This is transmitted through direct and indirect contact

D. These typically resolve in 3-4 months

---

10. Smallpox was the first infectious disease to be eradicated globally. All the following factors helped eradicate smallpox except.

A. Carrier states did not exist

B. Use of highly effective vaccine

C. Variola has no animal carriers

D. Cross-infection from animal pox viruses

11. A researcher is studying the vaccinia virus. She notices certain characteristics in the virus that enable it to be used for developing vaccines. In the context of vaccine development, what is a significant advantage of using the vaccinia virus as a vector?

A. It has a high pathogenicity, making it suitable for inducing strong immune responses.

B. Vaccinia virus can accommodate a substantial amount of foreign DNA.

C. It is an entirely natural virus found commonly in the human population.

D. The vaccinia virus is only effective against a limited range of diseases.

12. A 14-year-old child is brought by her mother with complaints of redness of the eyes with watery discharge. She also complains of a sore throat and fever and gives a history of being in a swimming pool at school. Which serotype of the virus causing this condition is responsible for these symptoms?

A. 1,2, and 5

B. 8, 19, and 37

C. 3, 7, and 14

D. 3, 7, and 21

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	2
Question 4	3
Question 5	4
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	4
Question 11	2
Question 12	3

Solution for Question 1:

orrect Answer: A) Non-enveloped single-stranded DNA (ssDNA) virus

Explanation:

The clinical scenario describes erythema infectiosum, a disease caused by parvovirus B19.

Characteristics of Parvovirus B19:

- Genome: Single-stranded DNA (ssDNA).
- Envelope: Non-enveloped, making it resistant to environmental conditions.
- Size: It is among the smallest DNA viruses.

• Tropism: Targets erythroid progenitor cells in the bone marrow, causing the transient aplastic crisis in patients with hemolytic disorders.

Enveloped ssDNA virus (Option B): No known ssDNA viruses are enveloped.

Enveloped dsDNA virus (Option C) includes herpesviruses, which cause different clinical syndromes (e.g., herpes simplex, chickenpox).

Non-enveloped dsDNA virus (Option D) includes adenoviruses, which typically cause respiratory or gastrointestinal infections, not erythema infectiosum.

Solution for Question 2:

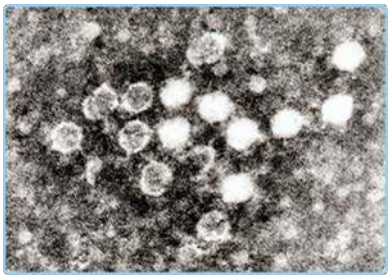
Correct Answer: A) Fetal liver, targeting erythroid progenitor cells

Explanation:

• This case suggests Parvovirus B19 infection due to recent exposure to a child with a “slapped cheek” rash and the patient’s anemia with low reticulocytes, indicating a transient aplastic crisis.

• Parvovirus B19 specifically targets erythroid progenitor cells by binding to the P antigen, halting red blood cell production.

• In the fetus, this viral tropism affects the fetal liver.

Parvovirus	
 {{caption_text}}	Morphology Simplest animal viruses infecting humans, responsible for erythema infectiosum (fifth disease). Smallest viruses (18–26 nm in size). Non-enveloped with icosahedral symmetry. Only DNA viruses with single-stranded DNA. Dependent on host cell enzymes for replication.
Pathogenesis	

Exclusively infects humans. Transmission: Primarily through the respiratory route. Blood transfusion. Transplacental transmission. Target Cells: Tropism for erythroid progenitor cells in adult bone marrow and fetal liver. (Option A) Receptor Binding: Binds to blood group P antigen on RBC surface. Effects: Leads to RBC destruction and inhibition of erythropoiesis, potentially causing anemia.	

• Parvovirus B19 targets erythroid progenitor cells, not myeloid progenitor cells (Option B), leading to anemia rather than affecting white blood cell function.

• The virus does not specifically affect the fetal spleen (Option C); its primary pathology occurs in the bone marrow and fetal liver.

• Parvovirus B19 does not target trophoblasts in the placenta (Option D); its main impact is on erythropoiesis in the fetal liver, not direct placental damage.

Solution for Question 3:

Correct answer: Papular-purpuric gloves and socks syndrome; Parvovirus B19

Explanation:

In this scenario, painful, itchy swelling and erythema of the hands and feet, along with B/L papular lesions on his palms and soles as shown in the figure, indicate papular-purpuric gloves and socks syndrome associated with Parvovirus B19.

• This syndrome typically follows mild viral symptoms, aligning with the boy's recent flu-like illness.

• Parvovirus B19 targets erythroid progenitor cells, leading to the characteristic rash.

Clinical Manifestations of Parvovirus infections:	

 ... (content truncated)	Erythema Infectiosum (Fifth Disease) (Option A) Children: "Slapped cheek" facial rash. Adult women: Symmetrical polyarthropathy, often in hand and knee joints. Transient Aplastic Crisis: Severe acute anemia in patients with pre-existing hemolytic anemia. Pure Red Cell Aplasia: Chronic anemia in immunocompromised individuals due to persistent B19 infection. Non-Immune Hydrops Fetalis: Fatal fetal anemia and death; occurs via transplacental transmission (30% of cases) Highest risk in the second trimester. Papular-Purpuric Gloves and Socks Syndrome: Painful, itchy, and symmetric swelling and erythema of distal hands and feet; may mimic acute meningococcemia. (Option B)
Laboratory Diagnosis	
PCR: Highly sensitive, detects viral DNA (VP1, VP2 genes) in serum, tissue, or respiratory secretions. ELISA: Detects IgM antibody (early indicator, elevated 2–3 months) and IgG against VP1 and VP2 antigens. Immunohistochemistry: Detects viral antigens in fetal tissues and bone marrow. Bone marrow cytology shows Giant pro-erythroblasts with multiple cytoplasmic projections that look like "dog ears" or "pseudopodia". marrow aspiration Giant pro-erythroblasts" data-author=""Yadav Kamlesh Pallav Gupta Manjula Murari Rohan Malik"" data-hash="10022" data-license="CC BY 3.0" data-source="https://www.researchgate.net/figure/Bone-marrow-aspiration-showing-giant-proerythroblast-with-prominent-nucleoli-and_fig2_51494153" data-tags="April2025" height="226" src="https://image.prepladder.com/notes/BgvvZUBfKCz11744457298.png" width="511" /> {{caption_text}}	

• Hand, foot, and mouth disease (Option C) typically features vesicular lesions on the palms and soles , and around mouth which do not match the boy’s papular lesions and swelling.

• It is caused by Coxsackie A16 virus and enterovirus 71.



Scarlet fever (Option D) is marked by a diffuse, sandpaper-like rash following a sore throat, inconsistent with the boy's symptoms.



Solution for Question 4:

Correct Answer: C) BK virus

Explanation:

This case points to BK virus nephropathy, common in kidney transplant recipients, indicated by renal dysfunction and "decoy cells" with enlarged nuclei and ground glass appearance of cytoplasm with basophilic viral inclusions in the urine.

- Decoy cells are infected renal tubular epithelial cells found in urine and are a sign of polyomavirus (PV) replication.

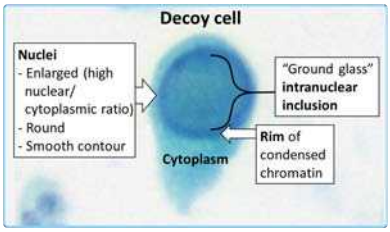
• BK virus reactivates in immunocompromised hosts, and PCR confirmation of viral DNA aligns with its role in post-transplant nephropathy.

Papovaviridae:

Papovaviridae includes 2 main genera:

• Polyomavirus: Contains simian vacuolating virus (SV40) & human polyomaviruses (JC and BK).

• Papillomavirus

Polyomavirus		
MCV	Merkel cell polyomavirus. Causes merkel cell cancer. CK7+ & CK 20-ve.	
SV40	Simian vacuolating virus (SV40) Associated with mesothelioma.	
BK virus (Option C)	 {{caption_text}}	BK virus is a polyomavirus with double-stranded DNA, typically causing: Nephropathy in kidney transplant recipients. The time period of nephropathy is after 6 months post-transplant. Hemorrhagic cystitis in hematopoietic stem cell transplant recipients. Infections in immunocompetent individuals are often asymptomatic. Cytologic examination of urine can reveal “decoy cells”. These characteristic cells have enlarged nuclei with a single, basophilic, intranuclear inclusion, which mimics cancer cells on pap smear.
JC Virus (John Cunningham) (Option A)	Causes Progressive Multifocal Leukoencephalopathy (PML Characterized by multifocal demyelination across the brain. Involves the deformation of astrocytes and oligodendrocytes. Primarily occurs in immunocompromised patients and patients taking natalizumab in multiple sclerosis treatment.	

Cytomegalovirus (Option B) causes systemic infections in transplant patients but doesn’t produce "decoy cells" or nephropathy.

• The most common time period of CMV systemic infection is 1-6 months post-transplant.

Epstein-Barr virus (Option D) is associated with lymphoproliferative disorders in immunocompromised hosts but does not typically cause nephropathy or produce decoy cells.

Solution for Question 5:

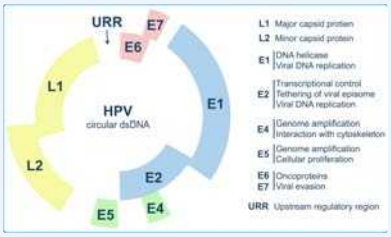
Correct Answer: D) E5 protein

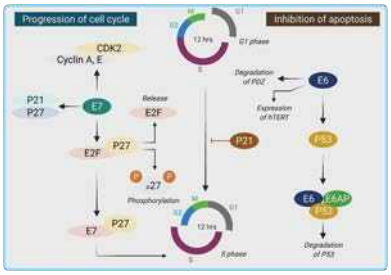
Explanation:

• The E5 protein is responsible for the formation of perinuclear vacuoles in infected cells, as shown in the figure, leading to cytoplasmic "halos."

• These halos indicate a cytopathic effect associated with E5, as it localizes to the membranes of the endoplasmic reticulum.

• The nuclear changes accompanying these vacuoles are referred to as koilocytic atypia.

HPV is a DNA virus belonging to the Papillomaviridae family. Tropism: Selectively infects epithelial cells of skin and mucous membranes. Morphology Non-enveloped virus, 50–55 nm in size. Icosahedral capsid containing circular double-stranded DNA (dsDNA).	
Viral Genome	
 {{caption_text}}	Early (E) Region: Includes E1-E7 genes coding for nonstructural proteins. E1 and E2 Proteins: Regulate viral DNA replication. (Option A) E4 Protein: Disrupts cytokeratins to facilitate the release of the virus. E5 Protein: contributes to the formation of perinuclear vacuoles, leading to cytoplasmic “halos.” (Option D) E6 Protein: Promotes degradation of the p53 tumor-suppressor protein. (Option B) E7 Protein: inhibits retinoblastoma protein, contributing to oncogenesis. (Option C) Late (L) Region: L1 and L2 genes code for structural capsid proteins. Types: Over 100 HPV types, classified by DNA sequences in the L1 region. L1 capsid protein is used for making vaccines.

 HPV Mechanism of Oncogenesis {{caption_text}}	
E6 Protein: Promotes degradation of p53, blocking apoptosis and inhibiting p53-driven activation of tumor suppressor gene p21. E7 Protein: Inhibits the retinoblastoma (RB) tumor suppressor gene through direct RB inhibition and by interfering with p53 and p21 functions. HPV infection alone is insufficient for malignancy; factors such as co-transfection with v-fos DNA, genetic mutations (e.g., mutated RAS gene), & environmental influences (e.g., smoking, microbial co-infections, dietary deficiencies, hormonal changes) enhance cancer risk.	

Solution for Question 6:

Correct Answer: B) HPV 11

Explanation:

- Recurrent laryngeal papillomas are benign lesions primarily associated with HPV serotypes 6 and 11.
- These serotypes are known to cause condyloma acuminatum and are commonly associated with laryngeal lesions, particularly in children and young adults

Types of HPV		
Cutaneous involvement	HPV 1, 2, 3 (low risk)	Filiform wart Verruca Plantaris
HPV 5, 8 (high risk)	Epidermodysplasia Verruciformis	

Mucosal involvement	HPV 6, 11 (low risk)	Condyloma Acuminatum Recurrent laryngeal papillomas
HPV 16, 18, 31, 33, 35 (high risk)	Squamous cell carcinoma	

Cutaneous involvement		
Benign Warts	 Filiform wart {{caption_text}} Common Skin Warts (Verruca Vulgaris): Small, hard, rough growths on the skin; commonly seen in children (associated with serotypes 2, 4, 27, 57). ... (content truncated)	
Epidermodysplasia Verruciformis	 {{caption_text}}	Rare autosomal recessive conditions may progress to squamous cell carcinoma, especially in sun-exposed areas. Associated with serotypes 5 (Option A), 8, 9, 12, 17, 20, 36, and 47.
Mucosal involvement		
Anogenital Warts	 ... (content truncated)	

Head and Neck Lesions	 {{caption_text}} Benign: Recurrent laryngeal papillomas, especially in children (serotypes 6, 11). (Option B) ... (content truncated)	
Carcinoma Cervix	Mainly squamous cell carcinoma linked to high-risk serotypes like 16, 18, 31, 33, and others. (Option D) Serotype 16 alone accounts for ~50% of cases, while serotypes 16 & 18 together cause ~70% of cervical cancers.	

Solution for Question 7:

Correct Answer: B) 6, 11, 16, 18, 31, 33, 45, 52, 58

Explanation:

The nine-valent HPV vaccine (Gardasil 9) provides protection against nine HPV types, including high-risk types linked to cervical cancer (16, 18, 31, 33, 45, 52, 58) and low-risk types associated with genital warts (6, 11).

Laboratory Diagnosis of HPV	

 {{caption_text}} resinoid nucleus" data-author="Jr, Marián & Mezencev, Roman & kašpírková, Jana & Kacerovská, Denisa & Kazakov, Dmitry & Ondic, Ondrej. " data-hash="10053" data-license="CC BY 4.0" data-source="https://www.researchgate.net/figure/Koilocyte-like-change-in-the-superficial-part-of-dysplastic-epithelium-characterized-by_fig6_304723215" data-tags="April2025" height="283" src="https://image.prepladder.com/notes/PEWE8HzNywtjDAZnyPMJ1744460350.png" width="307" /> {{caption_text}}	Molecular Methods: PCR (gold standard) and hybrid capture assays detect HPV DNA and identify types by targeting E6 and E7 genes. Lesion Detection: Most HPV lesions are visible; 5% acetic acid enhances lesion visibility. Cytologic Evidence: Detected via Papanicolaou (Pap) smears from cervical or anal scrapings show perinuclear cytoplasmic vacuolization termed koilocytosis. Histopathological staining of biopsy samples shows koilocytes with resinoid nuclei.

Prevention (HPV Vaccine):

Subunit Vaccine: Virus-like particles with HPV L1 proteins produced via DNA recombinant technology.

Bivalent Vaccine (Cervarix)	Covers high-risk types 16 and 18; administered as a single IM dose.
Quadrivalent Vaccine	Includes HPV types 6, 11, 16, and 18
Nine-Valent Vaccine (Gardasil 9)	Covers 6, 11, 16, 18, 31, 33, 45, 52, 58. (Option B) Doses: IM at 9-14 months, with 2 doses (6–12 months apart). After age 15, 3 doses (0, 2, and 6 months). Recommended up to age 26 if unvaccinated; better immune response if administered before age 14.

Solution for Question 8:

Correct Answer: C) Monkeypox virus

Explanation:

- The patient has monkeypox as per his symptoms of fever, skin rashes, and, most importantly lymphadenopathy after a visit and contact with wild animals.

• The monkeypox virus causes monkeypox disease, and the clinical symptom of lymphadenopathy differentiates it from other pox virus infections and chickenpox.

Poxviruses	
Poxviruses are the largest viruses that infect vertebrates.	
 Morphology {{caption_text}}	Poxviruses measure approximately 400 nm in length and 230 nm in diameter, making them visible under a light microscope. Poxviruses are typically brick-shaped or ellipsoid. They possess an envelope consisting of two lipoprotein membranes (outer and inner), and the outer membrane features ridges. The envelope encases a core, along with two structures of unknown function known as lateral bodies. The core or nucleocapsid is biconcave and shaped like a dumbbell. Poxviruses contain a single linear double-stranded DNA (dsDNA) genome. They are unique among DNA viruses for replicating in the cytoplasm.
Human-Important poxviruses	Variola is the causative agent of smallpox, the first disease to be eradicated globally. Vaccinia: Used as a vaccine for smallpox. Molluscum contagiosum Virus: Causes warty lesions known as molluscum contagiosum.
Genera classification	Orthopoxvirus: Smallpox, monkeypox, vaccinia. Molluscipoxvirus: Molluscum contagiosum. Parapoxvirus: Milker's nodules, bovine papular stomatitis viruses, etc. Capripoxvirus: Sheeppox, goatpox, lumpy skin disease, etc. Avipoxvirus: Fowlpox, pigeon pox, etc. Leporipoxvirus: Myxoma, hare fibroma, etc. Suipoxvirus: Swinepox virus. Yatapoxvirus: Tanapox, Yabapox viruses.

Monkeypox infections	
The monkeypox virus belongs to the Orthopoxvirus genus. Monkeypox is a rare zoonotic infection first recognized in captive monkeys in 1958.	
Transmission	Infection likely occurs through direct contact with wild animals. The exact primary reservoir host remains unidentified, but species such as squirrels, rabbits, and rodents can be infected.

 Clinical features {{caption_text}}	Symptoms resemble those of smallpox but are generally milder. Fever and rash. Lymphadenopathy is a prominent feature in most cases, unlike in smallpox or chickenpox.
Complications	Respiratory distress Secondary bacterial infections
Vaccination	The vaccinia vaccine can offer protection against monkeypox or lessen its severity.
Human-to-Human transmission	Transmission between humans is considered limited. Estimates suggest that only about 15% of close contacts of infected individuals become infected.

Variola (Option A):

- This virus causes smallpox, characterized by highly contagious and severe rashes, which is a disease eradicated in 1980.
- It is caused by two strains of the virus, variola major and minor. Lymphadenopathy is not a sign of this disease.

Vaccinia (Option B):

- This virus is used in the development of vaccines for smallpox and is even effective in preventing monkeypox.
- This is an artificially made virus and does not have any pathogenicity in humans.

Varicella (Option D):

- This is a human herpes virus that causes chickenpox, a viral infection that has symptoms of fever, malaise, and a rash.
- Skin rashes are characteristic of the disease that develop from maculopapules to vesicles to scabs as the disease progresses.

Solution for Question 9:

Correct Answer: B) The lesions can be seen all over the body

Explanation:

The lesions given in the image are of the disease molluscum contagiosum, which is a viral disease caused by a poxvirus and presents with dome-shaped, pink, pearly wart-like lesions that are not found in the palms and soles of the body.

Molluscum contagiosum	
Molluscum contagiosum virus is an obligate human poxvirus known for producing characteristic skin lesions. (Option A)	
 Clinical manifestations {{caption_text}}	 Lesions: Dome-shaped, pink, pearly wart-like lesions (2–5 mm in size) with a dimple at the center. Lesions can appear singly or in clusters, located anywhere on the body except on the palms and soles. ... (content truncated)
Transmission	Primarily affects children, transmitted through direct and indirect contact (e.g., barbers, shared towels, swimming pools). (Option C) Rare instances of sexual transmission have been reported in young adults.
 Histology of skin scrapings of Molluscum contagiosum ... (content truncated)	Molluscum bodies: Characteristic intracytoplasmic eosinophilic inclusions were observed in skin scrapings stained with histopathological stains. Confirmation methods: Electron microscopy PCR Cultivation: The virus cannot be propagated in tissue culture, embryonated eggs, or animals.
Treatment	Surgical removal of the lesions by ablation (by cryotherapy or laser therapy) is the mainstay of

	treatment. Cidofovir has been shown to have some efficacy.

Solution for Question 10:

Correct Answer: D) Cross-infection from animal pox viruses

Explanation:

Cross-infection from animal pox viruses is not a factor that helped in the eradication of smallpox even though they produce similar and milder illnesses.

Smallpox virus	
Smallpox was the first infectious disease eradicated globally, characterized by highly contagious and severe rashes. The disease was caused by two strains: variola major and variola minor.	
Transmission	The virus enters through the mucous membranes of the upper respiratory tract via aerosol transmission.
 ... (content truncated)	Incubation period: Typically lasts 12 days (ranging from 7 to 17 days) before the appearance of rashes. Rash characteristics: Unique rashes that are distinguishable from chickenpox. Deep-seated rashes that appear simultaneously in affected areas with a slow evolution. Centrifugal distribution: rashes initially affect the palms, soles, and extensor surfaces. The fever subsides with the appearance of the rash. Exanthem progression: the disease featured a single crop of centrifugal exanthems that progressed through several stages: Macular stage: Flat, discolored lesions. Papular stage: Raised bumps. Vesicular stage: Fluid-filled blisters. Pustular stage: Pus-filled lesions.

Laboratory Diagnosis	Direct detection: Intracytoplasmic inclusion bodies known as Paschen bodies can be identified in rash scrapings. Electron microscopy reveals a brick-shaped appearance with a biconcave DNA core. Egg inoculation: Characteristic pock formation observed on the chorioallantoic membrane (CAM) of chick embryos.
Eradication of smallpox	The World Health Organization declared the eradication of smallpox nearly three years after the last recorded case, on May 8, 1980.
Factors that helped the eradication	Variola had no animal reservoirs. (Option C) Patients were the only source, and no carriers existed (subclinical disease). (Option A) Characteristic rashes made identification straightforward. Asymptomatic cases did not spread the disease. Global Eradication Program: Launched by WHO in 1967, leading to eradication in about 10 years through international cooperation. Effective vaccination: (Option B) Utilized a freeze-dried form of the live vaccinia vaccine for better stability. Administered via a simple and cost-effective bifurcated needle using the multiple puncture technique.

Solution for Question 11:

Correct Answer: B) Vaccinia virus can accommodate a substantial amount of foreign DNA.

Explanation:

Vaccinia is an artificial virus that is non-pathogenic to humans and can accommodate a substantial amount of foreign DNA (~25,000 foreign base pairs), allowing for the insertion of multiple vaccine antigens and, hence, a very good choice for use as a vaccine.

Features	Variola virus	Vaccinia virus
Causative agent	Responsible for smallpox (variola major and minor)	Used as the smallpox vaccine

Pathogenicity	High pathogenicity (Option A) Infects only monkeys in animal experiments	Non-pathogenic to humans May produce mild skin lesions Can be used for studies as its safer to work with Infects monkeys, calves, sheep, and rabbits
Eradication	Eradicated in 1977 WHO declared global eradication on May 8, 1980	The virus is not naturally occurring and is artificially made (Option C)
Antigenic properties	Antigenically identical variants (major and minor)	Changed from cowpox during human passage
Research applications	Limited to historical study; No current vaccine use	Extensively studied and serves as a vector for many recombinant vaccines (Option D)
Genetic capacity	N/A	Can accommodate ~25,000 foreign base pairs
Histopathology (Intracytoplasmic inclusion bodies)	Inclusion bodies named Paschen bodies are seen	Inclusion bodies named Guarnieri bodies are seen

Solution for Question 12:

Correct Answer: C) 3, 7, and 14

Explanation:

The diagnosis here is pharyngoconjunctival fever, also known as swimming pool conjunctivitis, which presents with ocular and throat symptoms with fever, and it is caused by adenovirus serotypes 3, 7, and 14.

Adenovirus	
Adenoviruses are non-enveloped DNA viruses characterized by icosahedral symmetry. They feature fiber proteins extending from each vertex, creating a distinctive appearance reminiscent of a space vehicle.	

Morphology	Adenoviruses range from 70 to 75 nm in diameter. It is composed of 252 capsomers forming an icosahedron. Capsomer Types: Hexons and Pentons. Each penton includes: A penton base anchored in the capsid. A fiber projection consisting of a rod-like portion with a knob at the end.
Clinical manifestations	
While many serotypes can cause disease, types 1–7 are the most prevalent.	
Respiratory infections	Upper respiratory tract infections: Common in children Primarily caused by serotypes 1, 2, and 5. (Option A) In adolescents, serotypes 3, 4, and 7 are responsible for mild infections. Pneumonia: Types 3, 7, and 21 account for 10-20% of childhood pneumonia cases. (Option D) Serotype 14 is linked to severe pneumonia in healthy young adults. Acute respiratory disease: Occur as outbreaks, especially in military recruits. Associated with types 4, 7, and occasionally type 3. Transplant recipients: Serotypes 34 and 35 are frequently isolated from these patients.
Ocular Infections	Pharyngoconjunctival fever: Outbreaks typically occur in children's summer camps. Often referred to as swimming pool conjunctivitis. Associated with serotypes 3, 7, and 14. Epidemic keratoconjunctivitis: Known as shipyard eye. It is highly contagious in adults. Caused by types 8, 19, and 37. (Option B)
Gastrointestinal Infections	Infantile gastroenteritis: Serotypes 40 and 41 are responsible for 5-15% of viral gastroenteritis cases in young children.
Urinary Tract Infections	Acute hemorrhagic cystitis: Primarily affects children, especially boys. Caused by serotypes 11 and 21.

Hepatitis virus

1. A 28-year-old man presents to the emergency department with jaundice, fatigue, and nausea after returning from a trip to a rural area in India. Laboratory findings reveal elevated serum bilirubin, ALT, and AST. Which of the following can be used for a definitive detection of the causative agent?

A. Serum IgM detection

B. Serum IgG detection

C. Electron microscopy to detect virions

D. HEV PCR for viral RNA

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2. A 32-year-old male presents to the clinic with fatigue, jaundice, and elevated liver enzymes. He reports recent unprotected sexual encounters and has a history of intravenous drug use. During further investigation of his blood sample, a specific particle associated with this viral infection is identified under the electron microscope. Which of the following structures, when transmitted, is capable of leading to infection in another person?

A. Spherical particles

B. Tubular structures

C. Filamentous structures

D. Dane particles

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3. In the genome of the hepatitis B virus, which region codes for the marker indicating high viral replication during infection?

A. S gene

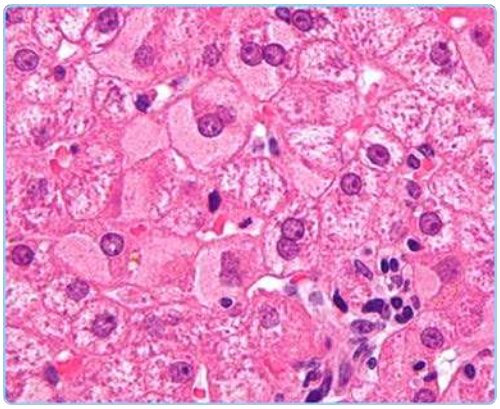
B. X gene

C. C gene

D. P gene

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4. A 45-year-old male presents with hepatomegaly and mild jaundice. A liver biopsy shows the following appearance. What does this appearance of hepatocytes indicate?



A. Acute liver injury

B. Chronic hepatitis B infection

C. Autoimmune hepatitis

D. Chronic hepatitis C infection

---

5. A 32-year-old man presents with fever, jaundice, and dark urine. He is unvaccinated for hepatitis, and liver enzymes are elevated. A healthcare worker uses a detergent-based disinfectant to clean the room. Which of the following viruses is most likely to survive the cleaning process due to its resistance to detergents and solvents?

A. Hepatitis A

B. Hepatitis B

C. Hepatitis C

D. Hepatitis D

6. Which of the following statements about Hepatitis A is incorrect?

A. It belongs to the family Hepadnaviridae.

B. The virus can be cultured in primate diploid cell lines.

C. Epidemiology is similar to that of poliovirus.

D. Outbreaks are more common during the summer months.

7. A 9-year-old boy presents to the clinic with a 2-week history of fever, fatigue, loss of appetite, and jaundice. He recently attended a picnic with school friends. Laboratory tests reveal elevated liver enzymes (AST and ALT) and elevated bilirubin levels. During the follow-up visit, which serological marker would you expect to be markedly raised in the convalescent phase of the viral infection?

A. IgM anti-viral

B. IgG anti-viral

C. Australian antigen

D. Core antigen

8. All of the following statements regarding the prophylactic vaccine for Hepatitis A are true, except.

A. The vaccine is an inactivated vaccine

B. Two doses are recommended at an interval of six months.

C. It provides lifelong immunity.

D. It is recommended for individuals under one year of age.

---

9. A 32-year-old patient presents with fatigue and mild jaundice. The physician suspects an acute Hepatitis B infection. Which of the following findings is indicative of active viral replication and high infectivity in Hepatitis B infection?

A. Presence of Hepatitis B Precore Antigen (HBeAg)

B. Presence of Hepatitis B Surface Antigen (HBsAg)

C. Presence of Hepatitis B Surface Antibody (Anti-HBs)

D. Presence of Hepatitis B Core Antibody IgM (Anti-HBc IgM)

---

10. A 28-year-old patient presents for follow-up after a Hepatitis B infection. Laboratory results indicate that HBsAg is negative, HBeAg is negative, Anti-HBc is positive for IgG, and Anti-HBs are positive. What is the most likely interpretation of these findings?

A. Acute Hepatitis B infection

B. Recovery from Hepatitis B infection

C. Post-vaccination immunity to Hepatitis B

D. Chronic Hepatitis B infection, low infectivity

---

11. A 45-year-old patient with chronic hepatitis B presents with worsening liver function tests. Laboratory analysis reveals negative HBeAg, although other HBV markers are still positive. The HBV DNA load is >2000 copies/mL. Which of the following mutations is most likely associated with these findings?

A. Escape mutants

B. Precore mutants

C. YMDD mutation

D. Antigenic shift

---

12. A newborn is delivered to a mother with chronic Hepatitis B who tests positive for HBeAg. What is the most appropriate prevention and estimated risk of perinatal transmission to the newborn?

A. HBIG only; 25% risk

B. Hepatitis B vaccine only; 5% risk

C. HBIG and hepatitis B vaccine; 70–90% risk

D. HBIG and hepatitis B vaccine; 10–20% risk

---

13. A healthcare worker suffers a needle-stick injury from a patient who is HBsAg positive. The healthcare worker's anti-HBs titer is 7 IU/mL. What is the appropriate post-exposure prophylaxis?

A. HBIG only

B. HBV vaccine series only

C. HBIG and start HBV vaccine series

D. No treatment needed

---

14. A 40-year-old patient with chronic hepatitis B presents with sudden worsening of liver function and jaundice. Laboratory tests show elevated liver enzymes and the presence of IgG anti-HBc along with mixed IgM and IgG anti-HDV. What type of hepatitis D infection does this patient likely have?

A. Co-infection

B. Superinfection

C. Acute hepatitis D

D. Chronic hepatitis D

---

15. A 34-year-old man with a history of IV drug use presents with fatigue, nausea, and mild jaundice. He reports a history of similar symptoms about 5 years ago, which resolved spontaneously. Laboratory testing shows elevated liver enzymes, and an ELISA test is positive for anti-HCV antibodies. Which of the following tests would be most appropriate to confirm an active HCV infection in this patient?

A. HCV Core Antigen Assay

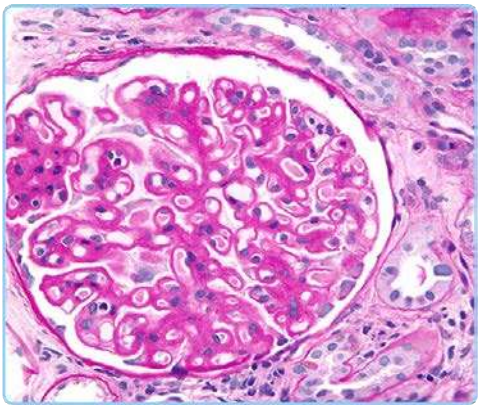
B. Liver biopsy

C. RT-PCR for HCV RNA

D. Fibroscan

---

16. A 60-year-old man with HBsAg positivity and anti-HBc IgG positivity develops renal symptoms, including edema and heavy proteinuria. His lab tests show low serum albumin and elevated cholesterol levels. His histopathology is attached below, which glomerular condition does he likely have?



A. Minimal change disease

B. Focal segmental glomerulosclerosis

C. Membranoproliferative glomerulonephritis

D. Membranous nephropathy

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	2
Question 5	1
Question 6	1
Question 7	2
Question 8	4
Question 9	1
Question 10	2
Question 11	2
Question 12	3
Question 13	3
Question 14	2
Question 15	3
Question 16	4

Solution for Question 1:

Correct Answer: D) HEV PCR for viral RNA

Explanation: The symptoms of the patient correlate with the diagnosis of hepatitis E, which can definitively be detected using a polymerase chain reaction to detect the genome (RNA) of the

virus.

Hepatitis E Virus (HEV)

Epidemiology:

- Hepatitis E virus is a zoonotic pathogen that can infect several animals, including monkeys, cats, pigs, and dogs.
- Most non-A, non-B (NANB) hepatitis cases are transmitted through fecal contamination of drinking water (referred to as enterically transmitted NANB or E-NANB).
- Epidemics are common and hence are also called epidemic NANB.
- In India, HEV infection is responsible for the majority of sporadic acute hepatitis cases and epidemic hepatitis, accounting for 30-60% of occurrences.
- Young adults aged 20 to 40 are more commonly affected by HEV infection than children are by HAV infection.
- Transmission:
- Infection sources include fecally contaminated water and the environment.
- The secondary attack rate (transmission from infected persons to their close contacts) among household contacts is low (2-3%), compared to 10-20% for hepatitis A.

Morphology:

- HEV is a spherical, non-enveloped virus measuring 32-34 nm in diameter and featuring a single-stranded RNA genome.
- The virion surface has indentations and spikes, and the virus is labile.
- It has a specific antigen, HEV-Ag.
- It is morphologically similar to Caliciviruses, such as the Norwalk virus, which is classified provisionally in the Hepeviridae family.

Clinical features:

- Incubation period: Ranges from 2 to 9 weeks, averaging around six weeks.
- Ranges from 2 to 9 weeks, averaging around six weeks.
- Affected population: Primarily young to middle-aged adults (ages 15-40).
- Primarily young to middle-aged adults (ages 15-40).
- Severity: Generally mild and self-limited, with a low case fatality rate of about 1%.
- Generally mild and self-limited, with a low case fatality rate of about 1%.
- Pregnant women: Higher risk, with a 20-40% case fatality rate in the last trimester. Fulminant hepatitis can occur at a significantly higher risk, with a rate of around 20%.
- Higher risk, with a 20-40% case fatality rate in the last trimester.

• Fulminant hepatitis can occur at a significantly higher risk, with a rate of around 20%.

• There is no chronic infection or carrier state associated with this condition.

• Ranges from 2 to 9 weeks, averaging around six weeks.

• Primarily young to middle-aged adults (ages 15-40).

• Generally mild and self-limited, with a low case fatality rate of about 1%.

• Higher risk, with a 20-40% case fatality rate in the last trimester.

• Fulminant hepatitis can occur at a significantly higher risk, with a rate of around 20%.

Laboratory diagnosis:

• HEV RNA can be identified in stool and serum using reverse transcriptase PCR, even before clinical symptoms appear.

• Serum Antibody Detection by ELISA: IgM anti-HEV: Appears in serum simultaneously with elevated liver enzyme levels, indicating an acute infection. IgG anti-HEV: Replaces IgM within 2 to 4 weeks after symptoms resolve and can persist for years, indicating recovery or a past infection. ELISA kits are available for detecting IgG and IgM antibodies using recombinant and synthetic peptide antigens. These have a high rate of false negatives and positives and hence cannot be used as a definitive method. (Options A & B)

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• IgG anti-HEV: Replaces IgM within 2 to 4 weeks after symptoms resolve and can persist for years, indicating recovery or a past infection.

• ELISA kits are available for detecting IgG and IgM antibodies using recombinant and synthetic peptide antigens.

• These have a high rate of false negatives and positives and hence cannot be used as a definitive method. (Options A & B)

• HEV virions can be detected through immuno-electron microscopy (IEM) of bile, serum, and feces during the incubation or acute phase. This is not a feasible or easily accessible method and hence, cannot be done in an acute setup. (Options C)

• This is not a feasible or easily accessible method and hence, cannot be done in an acute setup. (Options C)

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• These have a high rate of false negatives and positives and hence cannot be used as a definitive method. (Options A & B)

• This is not a feasible or easily accessible method and hence, cannot be done in an acute setup. (Options C)

Treatment:

• There are no specific antivirals against it and only supportive therapy is necessary.

Prevention:

• China has developed and licensed the first hepatitis E vaccine, known as 'HEV 239,' based on recombinant HEV proteins.

• It is not available worldwide yet.

Solution for Question 2:

Correct Answer: D) Dane particles

Explanation:

• Dane particles are the complete virions of the Hepatitis B virus, measuring approximately 42 nm.

• Dane particles contain the viral genome and are essential for establishing infection.

• They consist of a double-shelled structure that includes both surface antigens (Small HbsAg and Medium HbsAg) and are capable of transmitting Hepatitis B from one individual to another, leading to infection.

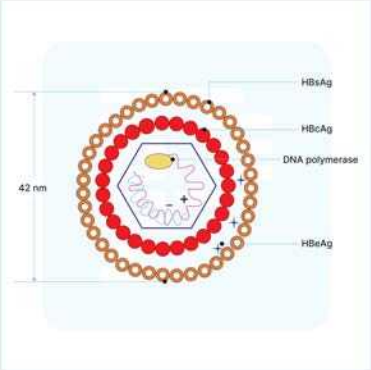
Hepatitis B virus (HBV)

Characteristics

• Family: Hepadnaviridae

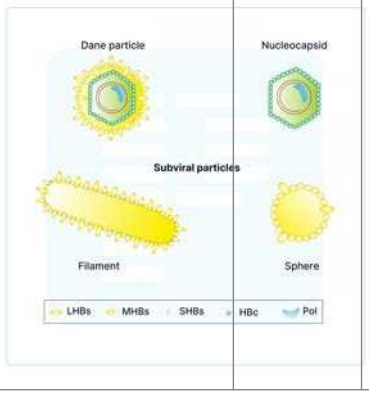
• Infectivity: Highly infectious; HBV is 50-100 times more infectious than HIV.

• Impact: Can result in both acute and chronic liver disease, potentially leading to cirrhosis and hepatocellular carcinoma (HCC).



Morphology

There are four particle types associated with Hepatitis B, visible under electron microscopy:

Particle Type	Size (nm)	Genome Presence	Infectious	Composition
Dane Particles (Option D)	42	Yes	Yes	Complete virions (Genome + HBsAg)
Spherical Particles (Option A)	22	No	No	Surface proteins (Small HBsAg)
Tubular Particles (Option B)	22	No	No	Surface proteins (Medium HBsAg)
Filamentous Particles (Option C)	22	No	No	Surface proteins (HBsAg)
				

Genotypes

HBV includes 10 recognized genotypes, labeled A through J:

- Genotype C: Common in Asia; associated with an increased risk of cirrhosis and hepatocellular carcinoma.
- Genotype D: Common in India.

Solution for Question 3:

Correct Answer: C) C gene

Explanation:

- High levels of HBV replication are associated with the presence of HBeAg, which is derived from the pre-core region of the C gene.
- The presence of HBeAg in the serum indicates active viral replication and correlates with higher levels of viral DNA, making it a significant marker of infectivity.

Replication Process of Hepatitis B Virus

Attachment and Entry:

- HBV attaches to hepatocytes using the sodium taurocholate co-transporting polypeptide (NTCP) receptor.
- The virus enters the cell via endocytosis.

↓

Uncoating and Nuclear Transport:

- Post-entry, the viral capsid is transported to the nucleus.
- The relaxed circular, partially double-stranded DNA (rcDNA) is released and repaired by host enzymes to form covalently closed circular DNA (cccDNA), which serves as the transcription template.

Transcription and Translation:

- cccDNA is transcribed by host RNA polymerase II, producing viral RNAs, including pregenomic RNA (pgRNA) and subgenomic RNAs encoding HBV proteins (surface antigen, core protein, polymerase, and X protein).

Genome

Genes	Gene Product	Function/Description
S (Surface) (Option A)	Surface Antigen (HBsAg)	Encodes the surface antigen, essential for viral entry and immune response induction. It is commonly detected as a marker in serological

		testing.
C (Core)	Core Antigen (HBcAg)	Forms the viral nucleocapsid and plays a role in genome packaging and immune evasion.
HBeAg	Secretory form of the core antigen, aids in immune tolerance, and is associated with viral replication and infectivity levels.	
P (Polymerase) (Option D)	Viral Polymerase	Functions in reverse transcription, crucial for converting RNA intermediate back to DNA for viral replication.
X (Option B)	Regulatory Protein (HBx)	Involved in viral replication regulation; may play a role in oncogenesis by affecting cell growth and survival pathways.

- These RNAs move to the cytoplasm for translation, resulting in viral protein production.

Reverse Transcription:

- pgRNA is encapsidated within new capsids.
- Within the capsid, HBV polymerase reverse transcribes pgRNA into partially double-stranded DNA, a unique feature for HBV.

Assembly and Release:

- DNA-containing capsids are enveloped with surface antigens and exit the cell via exocytosis or direct virion release.
- Some capsids return to the nucleus to maintain cccDNA, enabling chronic infection.

Mode of Transmission

Highly Infectious, easily spreads through bodily fluids.

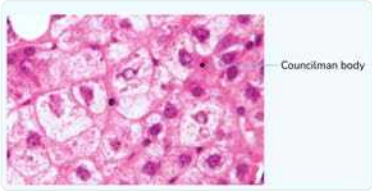
- Blood Contact: Direct blood contact transmits HBV.
- Sexual Contact: Unprotected sex is a key transmission route.
- Perinatal: Mother-to-child transmission during birth.
- Needle Sharing: High risk via shared needles (IV drug use).
- Occupational Exposure: Healthcare workers at risk from bodily fluids.
- Other Fluids: Also transmits through saliva and semen.

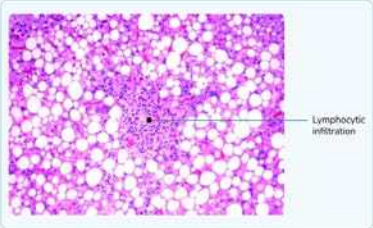
Solution for Question 4:

Correct Answer: B) Chronic hepatitis B infection

Explanation:

The liver biopsy shows hepatocytes with a "ground-glass" appearance. This appearance is specifically associated with chronic hepatitis B due to the accumulation of viral proteins in the endoplasmic reticulum of hepatocytes. This finding is a key histopathological feature indicative of chronic hepatitis B infection.

Acute hepatitis	
hepatitis Features" data-author="" data-hash="9857" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/jYlvhOdhEmIVpwJM3fba1744444940.png" />	 Acinar inflammation (may include or exclude portal inflammation). Ballooning degeneration of hepatocytes Apoptotic bodies, also known as acidophil or Councilman bodies, are large cell fragments or intact apoptotic cells found free in the sinusoids. Kupffer cell hypertrophy. Hepatocyte dropout and focal or spotty necrosis. Severe cases may show zonal or panacinar necrosis. ... (content truncated)
Chronic hepatitis	
hepatitis Features" data-author="" data-hash="9855" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/jpkh6RzWz4253WUnhqGX1744444757.png" />	Inflammation and Injury: Portal lymphocytic or lymphoplasmacytic inflammation, often breaching the limiting plate and causing periportal hepatocyte injury (interface activity), with variable lobular inflammation. Fibrosis and Progression: Begins as portal and periportal fibrosis, advancing to fibrous septa formation with porto-portal bridging fibrosis, and potentially progressing to cirrhosis. Chronic Hepatitis B: (Option B) Hepatocytes show a "ground-glass" appearance due to HBsAg accumulation in the swollen endoplasmic reticulum. Shikata Orcein Antigen Stain used for HBsAg Chronic Hepatitis C: (Option D) Portal tracts contain prominent lymphoid aggregates or fully formed lymphoid follicles. Steatosis is common, especially with genotype 3. Focal bile duct injury may occur, occasionally resembling biliary disease.

 hepatitis B Groundglass hepatocytes" data-author="Nephron" data-hash="9858" data-license="CC BY SA 3.0" data-source="https://commons.wikimedia.org/wiki/File:Ground_glass_hepatocytes_high_mag_cropped_2.jpg" data-tags="April2025" height="145" src="https://image.prepladder.com/notes/qbVIpvujCrfkCOaAN0GW1744445005.png" width="178" /> {{caption_text}} {{caption_text}}	

Acute liver injury (Option A) is characterized by hepatocyte necrosis and inflammation, typically showing ballooning degeneration and necrosis, not "ground-glass" hepatocytes.

Autoimmune hepatitis (Option C) is characterized by plasma cell infiltration, interface hepatitis, and emperipolesis, but it lacks the ground-glass appearance typical of chronic hepatitis B. Rosette formation may be present but is not a defining feature.

Solution for Question 5:

Correct Answer: A) Hepatitis A

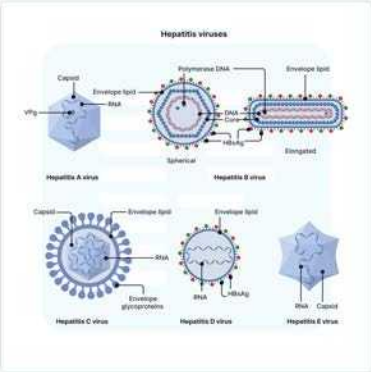
Explanation:

Hepatitis A virus (HAV) is a non-enveloped RNA virus, which makes it more resistant to inactivation by detergents and solvents compared to the enveloped viruses Hepatitis B, C, and D.

Hepatitis Viruses and Their Features

Feature	Hepatitis A (HAV)	Hepatitis B (HBV)	Hepatitis C (HCV)	Hepatitis D (HDV)	Hepatitis E (HEV)
Common Name	Infectious Hepatitis	Serum Hepatitis	Post-transfusion Hepatitis	Delta Virus	Enteric Hepatitis

Family	Picornaviridae	Hepadnaviridae	Flaviviridae	Unclassified (Viroid-like)	Unclassified (Caliciviridae-like)
Virion Structure	27 nm, icosahedral shape	42 nm, spherical shape	60 nm, spherical shape	35 nm, spherical shape	30-32 nm, icosahedral shape
Envelope	Absent	Present (HBsAg) (Option B)	Present (Option C)	Present (HBsAg) (Option D)	Absent
Genome Type	ssRNA	Partially dsDNA	ssRNA	Negative sense, ssRNA	ssRNA
Age Group Affected	Primarily children	Mainly young adults	All ages	All ages	Common in young adults (20-40 years)
Transmission Route	Fecal-oral	Bloodborne (most common), sexual, and mother-to-child	Bloodborne (most common), sexual, vertical transmission	Bloodborne (most common), sexual, vertical transmission	Fecal-oral
Incubation Period (IP)	15-45 days	30-180 days	15-160 days	30-180 days	14-60 days
Fulminant Disease	Rare (0.1%)	Rare (1%)	Rare	Frequent (5-20%)	Common in pregnancy (20-40%)
Carrier State	None	Present (0.1-30%)	Present (1.5-3.2%)	Dependent on HBV co-infection	None
Chronicity	Absent	Occasional (1-10%); higher in neonates	Common (85%)	Dependent on HBV co-infection	Absent
Cancer Risk	No	Yes, linked to liver cancer (HCC)	Yes, linked to cirrhosis and liver cancer	Yes, increases risk of liver cancer and cirrhosis	No



Hepatitis G virus

- Hepatitis G virus (HGV) is a bloodborne virus from the Flaviviridae family, closely related to Hepatitis C.
- Primarily transmitted through blood transfusions, shared needles, and less commonly through sexual contact or from mother to child.
- HGV usually doesn't cause liver inflammation, and most individuals are asymptomatic.
- Some studies suggest HGV may slow HIV progression in co-infected patients.
- Diagnosis is through PCR testing of HGV RNA.
- Specific antiviral treatment isn't necessary due to its generally benign course.

Solution for Question 6:

Correct Answer: A) It belongs to the family Hepadnaviridae.

Explanation:

- Hepatitis A virus (HAV) is classified under the family Picornaviridae, not Hepadnaviridae, which is the family associated with the Hepatitis B virus.
- Hepatitis A shares a similar epidemiology with poliomyelitis, as both viral infections are transmitted through the fecal-oral route.

Hepatitis A Virus (HAV)

Family:

- Picornaviridae family (Option A)

- Genus: Hepatovirus
- Also known as Enterovirus-72
- Single serotype and six genotypes (I, II, III affecting humans)

Genotype	Common Locations
I	Worldwide (most prevalent genotype)
II	Found in specific regions (less common)
IIIa	Predominant in India and some Asian countries

Viral Morphology:

- Small, non-enveloped virus
- Icosahedral symmetry
- Single-stranded, positive-sense RNA genome (+ssRNA)
- Encodes four capsid proteins (VP1–VP4)

Transmission:

- Fecal-oral route, especially in areas with poor sanitation
- Spreads through direct contact, contaminated food or water, fomites, or person-to-person contact with inadequate hygiene
- Rare transmission through blood transfusion and sexual contact
- In India, over 90% of individuals are seropositive by age 10, indicating widespread early exposure

Epidemiology:

- Similarities with poliovirus (fecal-oral transmission) (Option C)
- Seasonal trends, peaking during warmer months due to increased exposure to contaminated food (Option D)

Cultivability:

Can be cultivated in specific cell cultures, such as human or primate diploid cell lines (unlike most hepatitis viruses) (Option B)

Solution for Question 7:

Correct Answer: B) IgG anti-viral

Explanation:

In the convalescent phase, IgG antibodies are produced as part of the immune response, indicating that the body has cleared the virus and is developing long-term immunity.

Hepatitis A: Clinical Presentation and Investigation

Clinical Presentation

Hepatitis A is a viral infection caused by the Hepatitis A virus (HAV) and is primarily transmitted via the fecal-oral route, often through contaminated food or water.

Incubation Period: 15 to 50 days (average 28 days).

Symptoms:

- Prodromal Phase: Initial symptoms may include fatigue, malaise, low-grade fever, nausea, vomiting, and abdominal discomfort, typically lasting a few days.
- Mild Hepatitis: Progression may lead to jaundice, dark urine, and pale stools due to impaired bilirubin excretion.
- Icterus (Jaundice): Occurs in ~70% of patients.
- Hepatomegaly
- Other Symptoms: pruritus and arthralgia may occur.

Recovery: Slow spontaneous recovery typically occurs within 1 to 2 months; complete resolution is common.

Rare Complications: Cholestatic jaundice and fulminant hepatitis, which occurs in <0.1% of patients

Investigations

Serological Markers:

- IgM Anti-HAV: Indicates recent infection; detectable within the first week of illness and lasts for several months.
- IgG Anti-HAV: Indicates past infection or vaccination; provides lifelong immunity and is detectable for years.



Detection of HAV particles:

- HAV appears in stool from –2 to +2 weeks of jaundice.
- It can also be detected in the liver, bile, and blood by immunoelectron microscopy.

Liver Function Tests (LFTs):

- ALT and AST: Elevated
- Alkaline Phosphatase: May be mildly elevated, especially in cholestatic cases.

Bilirubin Levels: Elevated total bilirubin (conjugated and unconjugated).

Serology for Other Hepatitis Viruses: To rule out co-infections with Hepatitis B and C

Solution for Question 8:

Correct Answer: D) It is recommended for individuals under one year of age.

Explanation:

The Hepatitis A vaccine is generally recommended for children starting at age 1 (12 months), not for those under one year of age.

Hepatitis A Vaccine

Type of Vaccine: Inactivated (killed) form

Indications:

Routine Vaccination for Children: Recommended for all children aged 1 year (typically between 12-23 months).

High-Risk Groups:

- Travelers to regions with high rates of HAV infection.
- Individuals with chronic liver disease or clotting factor disorders.
- People who use injection or non-injection illicit drugs.

Outbreak Control: Used to help control HAV outbreaks within communities.

Two types of vaccines are available for human use:

Age for Vaccination

Children: The recommended starting age is 1 year (12-23 months).

Immunogenicity and Efficacy

Effectiveness:

- Provides 95% protection after the first dose.
- Near-complete protection is achieved after the second dose.

Duration of Immunity:

Offers long-lasting immunity, with many individuals maintaining protective antibody levels for decades post-vaccination.

Post-exposure prophylaxis (PEP)

For Unvaccinated Individuals Exposed to HAV:

- Within 2 weeks of exposure: A single dose of the Hepatitis A vaccine or immunoglobulin (IG) is recommended.
- Preference for IG: Immunoglobulin may be preferred for infants (<12 months), older adults, and individuals with weakened immune systems.

Solution for Question 9:

Correct Answer: A) Presence of Hepatitis B Precore Antigen (HBeAg)

Explanation:

The presence of HBeAg indicates active intrahepatic viral replication and high viral infectivity.

Antigen and Antibodies in Hepatitis B Infection:

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Antigens	Features	Time of appearance in serum
Hepatitis B Surface Antigen (HBsAg) (Option B)	First marker to elevate after infection. Indicates onset of infectivity. Detected during the incubation period, 2-6 weeks before clinical and biochemical signs of hepatitis. Remains elevated throughout acute hepatitis. Becomes undetectable 1-2 months after jaundice onset. Persists > 6 months if the disease progresses to chronic hepatitis or carrier state.	Appears within 1-12 weeks (usually between 8 and 12 weeks of infection)
Hepatitis B Precore Antigen (HBeAg) (Option A)	Indicates active intrahepatic viral replication and high viral infectivity.	Appear in serum concurrently with or shortly after HBsAg.
Hepatitis B Core Antigen (HBcAg)	Considered a hidden antigen due to its surrounding HBsAg coat.	Non-secretory in nature, so it cannot be detected in blood. Can be detected in hepatocytes using an immunofluorescence test.

Antibodies	Features	Time of appearance in serum
Anti-HBc IgM (Hepatitis B Core Antibody, IgM) (Option D)	First antibody elevated after infection. Indicates recent infection. Often the only marker (sometimes along with anti-HBc IgG) present during the window period between HBsAg disappearance and anti-HBs appearance.	Appears within 1-2 weeks after HBsAg and lasts for 3-6 months.
Anti-HBc IgG (Hepatitis B Core Antibody, IgG)	Indicates remote infection	Appears after 6 months.
Anti-HBe (Hepatitis B Precore Antibody)	Indicates reduced viral replication and decreased infectivity.	Appears after clearance of HBeAg.
Anti-HBs (Hepatitis B Surface Antibody) (Option C)	Indicates recovery, immunity, and non-infectivity (no further transmission risk). Only marker of hepatitis B vaccination.	Appears after clearance of HBsAg.

Solution for Question 10:

Correct Answer: B) Recovery from Hepatitis B infection

Explanation:

The presence of Anti-HBc IgG and Anti-HBs, along with negative HBsAg and HBeAg, indicates that the patient has recovered from Hepatitis B infection.

Stages and Associated Markers of Hepatitis B Virus (HBV):

HBsAg	HBeAg	Anti-HBc	Anti-HBs	Anti-HBe	Interpretation
+	+	IgM			Acute HBV infection Highly infectious (Option A)
+	+	IgG		-	Chronic HBV infection, highly infectious
		IgG		-/+	Chronic HBV infection, low infectivity (Option D)
-		IgG	-/+	-/+	Recovery (Option B)
-		IgM	-	-	Window period
					Post-vaccination (Option C)

Window Period: The period between the disappearance of HBsAg and the appearance of Anti-HBs is known as the "window period". During this time, other viral serologies may also be undetectable. Only marker detected in this period is Anti-HBc IgM.

Solution for Question 11:

Correct Answer: B) Precore mutants

Explanation:

Precore mutants abolish HBeAg production and are often associated with more severe chronic hepatitis, leading to rapid progression to cirrhosis. Although the virus continues to replicate, it fails to produce functional HBeAg in this condition.

Hepatitis B Virus Mutants:

- Mutation occurs in the precore region (abolishes HBeAg production) or the basal core promoter region (downregulates HBeAg production).
- Often diagnosed late in infected patients.
- Associated with more severe chronic hepatitis that progresses rapidly to cirrhosis.
- Patients do not produce HBeAg, although other HBV markers are present.
- Mutation occurs in the common "a" determinant in the S gene of HBsAg, which prevents neutralization by anti-HBsAg antibodies.
- Situations where escape mutants are observed: Infants born to HBeAg-positive mothers. Liver transplant recipients who received combined immunization with anti-HBV immunoglobulin and vaccine.
- Infants born to HBeAg-positive mothers.
- Liver transplant recipients who received combined immunization with anti-HBV immunoglobulin and vaccine.
- Clinical Implication: If escape mutants become more prevalent, they may pose challenges for hepatitis B prophylaxis.
- Infants born to HBeAg-positive mothers.
- Liver transplant recipients who received combined immunization with anti-HBV immunoglobulin and vaccine.
- Occurs in the YMDD locus within the HBV reverse transcriptase region of the polymerase gene.
- Clinical Implication: HBV-infected patients on lamivudine therapy may develop resistance due to this mutation.

Antigenic shift (Option D): This specifically refers to significant changes in influenza A virus antigens, not applicable to Hepatitis B and HBeAg production.

Solution for Question 12:

Correct Answer: C) HBIG and hepatitis B vaccine; 70–90% risk

Explanation:

In newborns of HBeAg-positive mothers, the perinatal transmission risk is 70–90%. Immediate administration of HBIG and the hepatitis B vaccine series is recommended to reduce this risk.

Prevention of Perinatal Transmission:

Babies Born to Carrier Mothers:

- 0.5 ml of HBIG was given IM immediately after birth.
- Followed by the full vaccine course at a different anatomical site, starting within 12 hours of birth.
- 95% of infected newborns become chronic carriers.
- Breastfeeding is not contraindicated.

Antigen in Mothers	Risk of Perinatal Transmission
HBeAg-positive	70–90% (Option C)
HBeAg-negative/HBeAb-negative	25% (Option A)
HBeAg-negative/anti-HBe-positive	12% (Option B)

Hepatitis B Vaccine:

- Recombinant subunit vaccine.
- Prepared in baker's yeast via DNA recombinant technology by cloning the S gene.
- Route of Administration: Intramuscular injection in the deltoid region (anterolateral thigh for infants).
- Dosage: 10–20 µg per dose (Half dose for children under 10 years).
- Schedule: For Adults: Three doses at 0, 1, and 6 months. National Immunization Schedule: Given at 6, 10, and 14 weeks along with the DPT vaccine.
- For Adults: Three doses at 0, 1, and 6 months.
- National Immunization Schedule: Given at 6, 10, and 14 weeks along with the DPT vaccine.
- Minimum interval between doses: 4 weeks.
- For Adults: Three doses at 0, 1, and 6 months.
- National Immunization Schedule: Given at 6, 10, and 14 weeks along with the DPT vaccine.
- Marker of Protection: Seroconversion with anti-HBsAg antibody titer ≥10 mIU/mL.

• Revaccination: For healthcare workers with anti-HBs titer <10 mIU/mL after the first vaccine series, a second series (3 doses at 0, 1, and 6 months) is given.

HBIG and hepatitis B vaccine; 10–20% risk (Option D): While HBIG and the hepatitis B vaccine series are recommended, the transmission risk for newborns of HBeAg-positive mothers is actually much higher, at 70–90%, not 10–20%.

Solution for Question 13:

Correct Answer: C) HBIG and start HBV vaccine series

Explanation:

For individuals with a history of inadequate vaccine response (anti-HBs titer of <10 IU/mL), the recommended treatment is to administer HBIG and start a complete 3-dose vaccine series or a second dose of HBIG after one month.

Post-Exposure Prophylaxis:

Healthcare Worker Status	Source		
HBsAg Positive	HBsAg Negative	Status not known	
Unvaccinated	HBIG + 3 doses of HBV vaccine series	3 doses of HBV vaccine series	
Previously vaccinated with complete HBV series; with protective titre (anti-HBs titer is >10 IU/m)	No treatment		
Previously vaccinated with complete HBV series; unprotective titre (anti-HBs titer is <10 IU/m)	HBIG + 3 doses of HBV vaccine series	HBV Vaccine: second series with 3 doses	HBIG + 3 doses of HBV vaccine series/administer second dose of HBIG 1 month later
Previously vaccinated with complete HBV series; unknown antibody responds	Check titre if <10 mIU/mL: HBIG + Complete 3 dose vaccine series >10 mIU/mL: No treatment	No treatment	Check titre if <10 mIU/mL: HBIG + Complete 3 dose vaccine series >10 mIU/mL: No treatment

Undergoing vaccination at time of exposure	HBIG + Complete 3 dose vaccine series	Complete 3 dose vaccine series	
Nonresponders (exposed person who is vaccinated for 2 series i.e, 6 doses and has unprotective titre)	Administer HBIG, 2 doses, 1 month apart	No treatment	

HBIG only (Option A): This option is inadequate as it does not provide the necessary vaccination to prevent infection.

HBV vaccine series only (Option B): This option is not sufficient without the addition of HBIG, especially given the exposure to an HBsAg-positive patient.

No treatment needed (Option D): This option is unsafe and does not address the risk of infection after exposure to a known source of hepatitis B.

Solution for Question 14:

Correct Answer: B) Superinfection

Explanation:

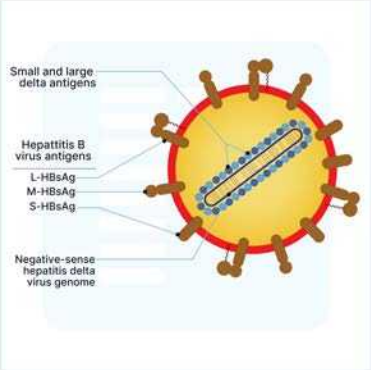
The presence of IgG anti-HBc and mixed IgM and IgG anti-HDV indicates that the patient is experiencing a superinfection with HDV on top of a chronic HBV infection.

Hepatitis D:

- Characteristics of Hepatitis D Virus (HDV): Small (35 nm) with circular, negative-sense ssRNA. Protein coat composed of hepatitis D antigen (HDAg). Enveloped with hepatitis B surface antigen (HBsAg) from HBV, hence called defective virus Dependent on HBV for replication and expression; cannot exist independently.
- Small (35 nm) with circular, negative-sense ssRNA.
- Protein coat composed of hepatitis D antigen (HDAg).
- Enveloped with hepatitis B surface antigen (HBsAg) from HBV, hence called defective virus
- Dependent on HBV for replication and expression; cannot exist independently.
- Small (35 nm) with circular, negative-sense ssRNA.
- Protein coat composed of hepatitis D antigen (HDAg).

• Enveloped with hepatitis B surface antigen (HBsAg) from HBV, hence called defective virus

• Dependent on HBV for replication and expression; cannot exist independently.



• Types of Infection: Co-infection: (Option A) HDV and HBV are transmitted together. HBV infection occurs first, allowing HBsAg to support HDV. Presents as acute hepatitis B, with symptoms from mild to fulminant. Usually self-limited; rarely progresses to chronic stage (1-10%). Superinfection: (Option B) HDV infects a chronic HBV carrier, often leading to severe chronic illness No link with hepatocellular carcinoma. M/c cause of fulminant hepatitis: HDV superinfection

• Co-infection: (Option A) HDV and HBV are transmitted together. HBV infection occurs first, allowing HBsAg to support HDV. Presents as acute hepatitis B, with symptoms from mild to fulminant. Usually self-limited; rarely progresses to chronic stage (1-10%).

• HDV and HBV are transmitted together.

• HBV infection occurs first, allowing HBsAg to support HDV.

• Presents as acute hepatitis B, with symptoms from mild to fulminant.

• Usually self-limited; rarely progresses to chronic stage (1-10%).

• Superinfection: (Option B) HDV infects a chronic HBV carrier, often leading to severe chronic illness No link with hepatocellular carcinoma. M/c cause of fulminant hepatitis: HDV superinfection

• HDV infects a chronic HBV carrier, often leading to severe chronic illness

• No link with hepatocellular carcinoma.

• M/c cause of fulminant hepatitis: HDV superinfection

• Co-infection: (Option A) HDV and HBV are transmitted together. HBV infection occurs first, allowing HBsAg to support HDV. Presents as acute hepatitis B, with symptoms from mild to fulminant. Usually self-limited; rarely progresses to chronic stage (1-10%).

• HDV and HBV are transmitted together.

• HBV infection occurs first, allowing HBsAg to support HDV.

• Presents as acute hepatitis B, with symptoms from mild to fulminant.

• Usually self-limited; rarely progresses to chronic stage (1-10%).

• Superinfection: (Option B) HDV infects a chronic HBV carrier, often leading to severe chronic illness No link with hepatocellular carcinoma. M/c cause of fulminant hepatitis: HDV superinfection

• HDV infects a chronic HBV carrier, often leading to severe chronic illness

• No link with hepatocellular carcinoma.

• M/c cause of fulminant hepatitis: HDV superinfection

• HDV and HBV are transmitted together.

• HBV infection occurs first, allowing HBsAg to support HDV.

• Presents as acute hepatitis B, with symptoms from mild to fulminant.

• Usually self-limited; rarely progresses to chronic stage (1-10%).

• HDV infects a chronic HBV carrier, often leading to severe chronic illness

• No link with hepatocellular carcinoma.

• M/c cause of fulminant hepatitis: HDV superinfection

• Diagnosis: Co-infection: IgM anti-HBc + IgM anti-HDV IgM anti-HDV appears late and is short-lived. Superinfection: IgG anti-HBc + mixed IgM and IgG anti-HDV HBeAg indicating active HBV replication may be present.

• Co-infection: IgM anti-HBc + IgM anti-HDV IgM anti-HDV appears late and is short-lived.

• IgM anti-HBc + IgM anti-HDV

• IgM anti-HDV appears late and is short-lived.

• Superinfection: IgG anti-HBc + mixed IgM and IgG anti-HDV HBeAg indicating active HBV replication may be present.

• IgG anti-HBc + mixed IgM and IgG anti-HDV

• HBeAg indicating active HBV replication may be present.

• Co-infection: IgM anti-HBc + IgM anti-HDV IgM anti-HDV appears late and is short-lived.

• IgM anti-HBc + IgM anti-HDV

• IgM anti-HDV appears late and is short-lived.

• Superinfection: IgG anti-HBc + mixed IgM and IgG anti-HDV HBeAg indicating active HBV replication may be present.

• IgG anti-HBc + mixed IgM and IgG anti-HDV

• HBeAg indicating active HBV replication may be present.

• IgM anti-HBc + IgM anti-HDV

• IgM anti-HDV appears late and is short-lived.

- IgG anti-HBc + mixed IgM and IgG anti-HDV
- HBeAg indicating active HBV replication may be present.

Acute hepatitis D (Option C): While the patient shows worsening symptoms, the presence of IgG anti-HBc suggests that this is not an acute infection but rather a superinfection on top of a chronic HBV infection.

Chronic hepatitis D (Option D): This is inaccurate as the presence of mixed IgM and IgG anti-HDV signifies a superinfection with HDV, which typically leads to more severe liver disease rather than chronic infection.

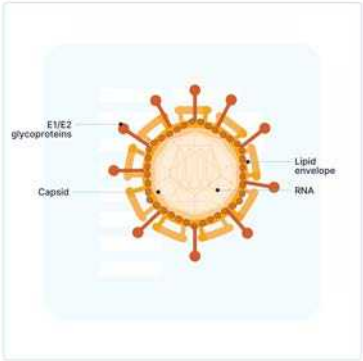
Solution for Question 15:

Correct Answer: C) RT-PCR for HCV RNA

Explanation:

- RT-PCR for HCV RNA is the gold standard for confirming active HCV infection.
- While an ELISA test detects antibodies, it does not differentiate between past and active infections.
- The presence of HCV RNA indicates ongoing viral replication and active infection.
- The core antigen assay, though helpful for initial diagnosis and monitoring, is less sensitive than RT-PCR.

Hepatitis C Virus (HCV)	
Classification	HCV belongs to the Flaviviridae family, genus Hepacivirus.

Structure and Genome	 ... (content truncated)
Transmission	Primarily bloodborne: Transfusions, organ transplants, needle-sharing. Vertical transmission (~6%) & rarely sexual transmission. Not spread by breast milk, food, or casual contact.
Clinical Manifestation	Incubation: 15–160 days (average 50 days). Asymptomatic in over 75% of cases. Acute hepatitis in ~20%: usually mild or anicteric Chronic Progression Chronic Disease: 75–85% of cases progress to chronic disease. Chronic Hepatitis: 60–70% of chronic cases. Cirrhosis: 5–20% of cases with chronic infection. Hepatocellular Carcinoma (HCC): 1–5% of cases; HCV accounts for ~25% of liver cancers.
Extrahepatic Manifestations	Immune complex deposition can cause: Mixed cryoglobulinemia Glomerulonephritis Arthritis and joint pain.
Laboratory Diagnosis	Antibody Detection Tests: ELISA is standard for initial screening with high sensitivity and specificity. Cannot differentiate active vs. past infections. HCV Core Antigen Assay: Automated, cost-effective for active infection diagnosis and treatment monitoring. Less sensitive than RT-PCR. (Option A) Molecular Methods (RT-PCR): Gold standard for confirming active infection; detects HCV RNA 2–3 weeks post-infection, used for viral load, treatment response, and genotype. (Option C) Biopsy (Option B) is the gold standard for staging; Fibroscan (Option D) is a non-invasive alternative for assessing liver fibrosis.
Treatment	Goals: Prevent complications (e.g., cirrhosis, liver cancer) and achieve sustained viral response (SVR). Sofosbuvir/velpatasvir is effective against all genotypes. Duration: Typically 12–24 weeks, depending on genotype and liver status.

Solution for Question 16:

Correct Answer: D) Membranous nephropathy

Explanation:

The presentation of heavy proteinuria, edema, and elevated cholesterol levels along with the histological findings is indicative of membranous nephropathy, which is commonly associated with HBsAg and anti-HBc IgG positivity in patients with chronic Hepatitis B infection.

Extrahepatic manifestation of Hepatitis B and C	
Hepatitis B	Hepatitis C
Serum sickness like syndrome Membranous glomerulonephritis (Option D) Polyarteritis nodosa Gianotti-Crosti syndrome	■r■oglobulinemi■ Porphyria cutanea tarda Lichen planus Necrolytic acral erythema Sjögren's disease/sicca syndrome Membranoproliferative glom■r■l■■■■hritis (Option C)

Minimal change disease (Option A): Primarily seen in children, usually idiopathic, and lacks association with Hepatitis B markers (HBsAg, anti-HBc IgG).

Focal segmental glomerulosclerosis (Option B): Can present with proteinuria and edema but is often linked to other conditions like HIV infection and heroin abuse rather than Hepatitis B infection.

Myxovirus

1. During an outbreak investigation in Kerala, India, several patients presented with fever and encephalitis. The outbreak is linked to contact with infected pigs, and fruit bats are abundant in the area. Laboratory analysis of patient samples reveals a paramyxovirus. The mortality rate is approximately 70%. Which of the following is the most likely diagnosis?

A. Japanese Encephalitis Virus

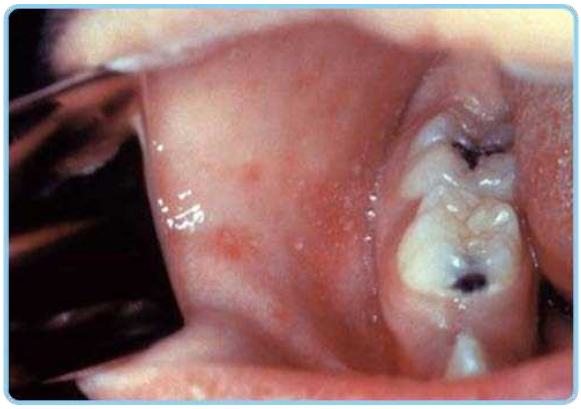
B. Nipah Virus

C. Rabies Virus

D. Eastern Equine Encephalitis

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2. A 4-year-old child presents with fever, cough, conjunctivitis, and spots on the buccal mucosa as shown below after a recent family trip. The mother reports no vaccinations. Which statement best describes the incubation period and transmission mode of the suspected illness? {{caption_text}}



A. Incubation period: 1-2 days; bloodborne transmission

B. Incubation period: 10-14 days; respiratory droplets

C. Incubation period: 3-5 days; contaminated surfaces only

D. Incubation period: 7-8 days; direct skin contact

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3. A 75-year-old female presents to the clinic with symptoms of high fever, chills, cough, body aches, and fatigue. She has a history of hypertension and chronic kidney disease. She reports she has never received an influenza vaccine. The patient is worried about her symptoms and potential complications. Which of the following statements about influenza vaccination is most appropriate for this patient?

A. She should receive the live attenuated influenza vaccine (LAIV) because it is more effective for elderly individuals.

B. She should avoid the influenza vaccine because of her chronic kidney disease.

C. She should receive the high-dose inactivated influenza vaccine (HD-IIV) to provide better protection.

D. Influenza vaccination is not recommended for individuals above 65 years old.

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4. A 3-year-old child presents with acute onset of barking cough, hoarseness, and mild respiratory distress that worsens at night. Physical examination reveals inspiratory stridor but normal lung fields. The child attends preschool where several other children have similar symptoms. The illness started with mild upper respiratory symptoms 2 days ago. No other significant clinical findings are seen. Which of the following viruses is most likely responsible for this presentation?

A. Adenovirus

B. Respiratory Syncytial Virus

C. Parainfluenza Virus Type 2

D. Measles Virus

---

5. A 3-month-old infant is brought to the emergency department with progressive respiratory distress, cough, and rhinorrhea during winter. The child attends daycare, and several other infants there have similar symptoms. Physical examination reveals tachypnea, wheezing, and bilateral crackles. Further testing is done, and the patient is diagnosed with Respiratory Syncytial Virus. Which of the following statements regarding RSV infection in this case is most accurate?

A. PCR testing is insensitiv,e but viral culture is the gold standard

B. Maternal antibodies provide complete protection at this age

C. The peak incidence occurs at 2-3 months of age with increased hospitalization risk

D. Live-attenuated vaccine is recommended for prevention

---

6. A 6-year-old boy presents with bilateral painful swelling in the region of his parotid glands that began 2 days ago. His mother notes that the pain worsens when he drinks orange juice. He had a fever of 101.2°F (38.4°C) before the swelling appeared. The patient's vaccination status is unknown. During examination, bilateral parotid enlargement is noted. Which of the following is the most specific diagnostic test to confirm the suspected infection in this patient? {{caption_text}}



A. Serum IgG antibody test

B. Viral culture from parotid secretions

C. RT-PCR from buccal swab

D. Physical examination findings alone

7. A 15-year-old boy presents with a 3-day history of fever, headache, and bilateral parotid gland swelling. He reports no recent vaccinations. Physical examination reveals painful, swollen parotid glands and lab tests show elevated serum amylase. Which complication is most likely to develop if the condition progresses?

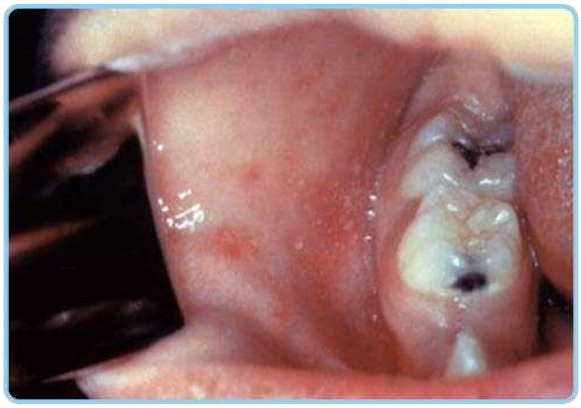
A. Orchitis

B. Encephalitis

C. Hearing loss

D. Pancreatitis

8. A 9-month-old infant presents with high fever, cough, conjunctivitis, rash over the buccal mucosa, and a spreading maculopapular rash. What is the most appropriate next step in management? {{caption_text}}



A. Administer the vaccine

B. Prescribe antiviral therapy

C. Administer immunoglobulin therapy

D. Educate on supportive care and schedule the vaccine at 12 months

9. A 3-year-old unvaccinated child presents with high fever, cough, conjunctivitis, and a maculopapular rash that began on the face and has spread to the trunk and limbs. Oral examination reveals small, white lesions on the buccal mucosa. Which complication is most likely to develop in the acute phase of this illness?

A. Acute encephalitis

B. Otitis media

C. Subacute sclerosing panencephalitis (SSPE)

D. Pneumonia

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10. A patient comes to the clinic with symptoms of a viral infection. The physician is reviewing potential viruses based on their classification into myxovirus families. Which of the following viruses is NOT a member of the Paramyxoviridae family?

A. Measles virus

B. Parainfluenza virus

C. Influenza virus

D. Mumps virus

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11. A 30-year-old healthcare worker presents with sudden onset fever, chills, sore throat, and muscle aches. She reports exposure to multiple patients with similar symptoms. The physician suspects influenza and requests a diagnostic test for confirmation. A nasopharyngeal swab is collected, and a Hemagglutination Inhibition (HI) test is performed. The results show a significant increase in antibody titers in the second sample compared to the first sample, taken 10 days apart. Which of the following best explains the findings and indicates a confirmed diagnosis of influenza?

A. The HI test shows no change in antibody titers between the first and second samples.

B. The HI test shows a fourfold increase in antibody titers between the first and second samples.

C. The HI test shows a twofold decrease in antibody titers between the first and second samples.

D. The HI test shows a single positive result with no subsequent titer testing.

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12. A 67-year-old patient with COPD presents with high fever, chills, myalgia, and dry cough, which started suddenly. Influenza A is confirmed by a rapid test. He missed his annual influenza vaccine this year. What is the best next step in management?

A. Start Oseltamivir immediately

B. Prescribe supportive care only, as antivirals are not effective beyond 48 hours.

C. Administer high-dose azithromycin

D. Recommend the influenza vaccine now to prevent the progression of symptoms

---

13. A 50-year-old patient presents to the clinic with flu-like symptoms during the winter months. The physician discusses the different types of influenza viruses with the patient, emphasizing their characteristics and impacts on public health. Which of the following statements about influenza viruses discussed by the physician is FALSE?

A. Influenza A has 18 HA and 11 NA subtypes.

B. Influenza B is classified into subtypes based on its HA and NA proteins.

C. Influenza A is known to cause both seasonal epidemics and pandemics.

D. Influenza C rarely causes severe illness in humans and has a low public health impact.

---

14. Which statement correctly describes antigenic drift and antigenic shift in influenza viruses?

A. Drift causes pandemics; shift causes minor epidemics.

B. Drift involves minor mutations; shift involves genetic reassortment.

C. Shift occurs in Influenza A and B; drift occurs only in Influenza A.

D. Drift produces a new strain; shift evades immune recognition.

---

15. A 45-year-old man visits the emergency department with complaints of high fever, severe headache, and muscle aches that started 24 hours ago. He mentions he has been feeling increasingly fatigued and has developed a dry cough. Which of the following statements about the virus causing his symptoms is true?

A. Amantadine acts against both A and B strains of the virus whereas Oseltamivir acts against strain A only.

B. The incubation period for the virus is 7 to 14 days.

C. The immune response mainly involves antibodies against the hemagglutinin protein.

D. Once infected, individuals develop lifelong immunity against all strains of the virus.

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16. Match each structure of the Orthomyxovirus (Influenza Virus) with its corresponding function. 1. Hemagglutinin (HA) A) Forms ion channels in the envelope 2. Neuraminidase (NA) B) Responsible for RNA transcription and replication 3. Polymerase proteins (PB1, PB2, PA) C) Aids in viral release and prevents virion aggregation 4. Nucleoprotein D) Binds sialic acid receptors for viral entry 5. Matrix protein - M1 E) Forms a shell beneath the envelope 6. Matrix protein - M2 F) Forms a ribonucleoprotein complex with viral RNA

1. Hemagglutinin (HA)	A) Forms ion channels in the envelope
2. Neuraminidase (NA)	B) Responsible for RNA transcription and replication
3. Polymerase proteins (PB1, PB2, PA)	C) Aids in viral release and prevents virion aggregation

4. Nucleoprotein	D) Binds sialic acid receptors for viral entry
5. Matrix protein - M1	E) Forms a shell beneath the envelope
6. Matrix protein - M2	F) Forms a ribonucleoprotein complex with viral RNA

A. 1-D, 2-C, 3-B, 4-E, 5-F, 6-A

B. 1-B, 2-D, 3-C, 4-F, 5-E, 6-A

C. 1-D, 2-C, 3-B, 4-F, 5-E, 6-A

D. 1-C, 2-B, 3-A, 4-D, 5-F, 6-E

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17. A medical student is studying viral classifications and comes across various RNA virus families. The student needs to identify misconceptions regarding the properties of these viruses. Which of the following statements is FALSE?

A. Coronaviridae and Flaviviridae are examples of positive-sense RNA viruses.

B. Bunyavirus has segmented RNA with 7 segments.

C. Orthomyxovirus contains segmented RNA with 7-8 segments.

D. Rhabdovirus and Myxovirus are examples of negative-sense RNA viruses.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	3
Question 5	3
Question 6	3
Question 7	1
Question 8	3
Question 9	4
Question 10	3
Question 11	2
Question 12	1
Question 13	2
Question 14	2
Question 15	3
Question 16	3
Question 17	2

Solution for Question 1:

Correct Answer: B) Nipah Virus

Explanation: The clinical presentation and history are indicative of the diagnosis of the NIPAH virus based on:

NIPAH Virus

Feature	Details
Key Facts (NIPAH)	N - Nervous system affected I - Infection mortality 60-70% P - Pigs are amplifiers A - Animal-to-human transmission H - History of the outbreak in Kerala 2018
Transmission Cycle	Vector: Fruit bats (natural reservoir) Intermediate host: Pigs (amplifiers) Human infection: Through contaminated food/water Patient secretions can spread disease
Clinical Features	Encephalitis (primary manifestation) Fever High mortality rate (60-70%) Neurological symptoms
Diagnosis	RT-PCR is the primary diagnostic test Sample: CSF or brain tissue Serology can be supportive
Prevention	Avoid contact with infected animals Proper handling of infected pig carcasses Avoid consuming contaminated fruits Protection from fruit bat secretions

Japanese Encephalitis (Option A) has lower mortality and is mosquito-borne.

Rabies (Option C) has a different transmission pattern and 100% mortality if untreated.

EEE (Option D) has a different geographic distribution and vectors involved.

Solution for Question 2:

Correct Answer: B) Incubation period: 10-14 days; respiratory droplets

Explanation:

The incubation period of measles typically ranges from 10 to 14 days and spreads primarily through respiratory droplets, making it highly contagious.

Incubation Period and Transmission of Measles:

Incubation Period:

- Average Duration: 10 days in children 21 days in adults.
- 10 days in children
- 21 days in adults.
- 10 days in children
- 21 days in adults.

Transmission:

- Primarily transmitted through the respiratory route by: Droplet Inhalation: Commonly over short distances, in close settings. Aerosol Transmission: Small-particle aerosols can remain airborne for extended periods, especially in enclosed public spaces like schools and hospitals, facilitating longer-range transmission.
- Droplet Inhalation: Commonly over short distances, in close settings.
- Aerosol Transmission: Small-particle aerosols can remain airborne for extended periods, especially in enclosed public spaces like schools and hospitals, facilitating longer-range transmission.

Pathogenesis and Spread:

- Begins with local replication in the respiratory tract.
- Primary Viremia: Virus moves to regional lymph nodes.
- Secondary Viremia: Virus enters the bloodstream, leading to widespread distribution throughout the body.

Measles does not spread through blood, and the incubation is much longer than 1-2 days. (Option A)

Although surfaces can carry the virus briefly, respiratory droplets are the primary transmission route. (Option C)

Direct skin contact is not a transmission method for measles; the incubation period is usually longer than 7-8 days. (Option D)

Solution for Question 3:

Correct Answer: C) She should receive the high-dose inactivated influenza vaccine (HD-IIV) to provide better protection.

Explanation:

Influenza vaccines, commonly known as flu shots, are designed to protect against infection by influenza viruses.

Types and important features of Influenza vaccines:

- Trivalent Vaccines: These vaccines protect against three different influenza viruses (two influenza A viruses- H1N1 and H3N2 and one influenza B virus).
- Quadrivalent Vaccines: These vaccines protect against four different influenza viruses (two influenza A viruses and two influenza B viruses).
- Annual Vaccination: Influenza vaccines are updated each year to match the strains predicted to be most common during the upcoming flu season.
- Formulations: Inactivated Influenza Vaccine (IIV): Injectable, the most widely used, suitable for anyone older than six months. Cell Culture-Based Vaccine (ccIIV): Produced in cell cultures rather than eggs, making it suitable for those with severe egg allergies. Recombinant Influenza Vaccine (RIV): Contains recombinant hemagglutinin proteins and is completely egg-free. Live Attenuated Influenza Vaccine (LAIV): A nasal spray form containing a weakened virus, suitable for healthy individuals aged 2-49, not recommended for high-risk groups.
- Inactivated Influenza Vaccine (IIV): Injectable, the most widely used, suitable for anyone older than six months.
- Cell Culture-Based Vaccine (ccIIV): Produced in cell cultures rather than eggs, making it suitable for those with severe egg allergies.
- Recombinant Influenza Vaccine (RIV): Contains recombinant hemagglutinin proteins and is completely egg-free.
- Live Attenuated Influenza Vaccine (LAIV): A nasal spray form containing a weakened virus, suitable for healthy individuals aged 2-49, not recommended for high-risk groups.
- Inactivated Influenza Vaccine (IIV): Injectable, the most widely used, suitable for anyone older than six months.
- Cell Culture-Based Vaccine (ccIIV): Produced in cell cultures rather than eggs, making it suitable for those with severe egg allergies.
- Recombinant Influenza Vaccine (RIV): Contains recombinant hemagglutinin proteins and is completely egg-free.
- Live Attenuated Influenza Vaccine (LAIV): A nasal spray form containing a weakened virus, suitable for healthy individuals aged 2-49, not recommended for high-risk groups.
- Reduces Risk of Illness

- Prevents Complications
- Community Protection.

Solution for Question 4:

Correct Answer: C) Parainfluenza Virus Type 2

Explanation: Parainfluenza virus type 2 is the correct diagnosis based on:

Parainfluenza Virus	
Clinical Features	Croup (Types 1 and 2) Lower respiratory disease (Type 3) Mild respiratory illness (Type 4) Adults: mild URI, hoarseness Rarely causes parotitis
Epidemiology	Type 3: Common in the first year of life Types 1 & 2: Preschool children First infections more serious 50% of infants become seropositive by 12 months
Laboratory Diagnosis	Specimen: Throat/nasal swabs Virus isolation: Monkey kidney cells Serology: Paired sera samples PCR: Most rapid and sensitive
Transmission	Air/respiratory droplets Through fingers Nosocomial spread uncommon No vaccine available

Adenovirus typically causes pharyngitis, fever, and conjunctivitis rather than croup (Option A)

RSV typically causes bronchiolitis with wheezing rather than croup (Option B)

Measles presents with characteristic rash and Koplik spots (Option D)

Solution for Question 5:

Correct Answer: C) The peak incidence occurs at 2-3 months of age with increased hospitalization risk

Explanation: RSV shows peak incidence at 2-3 months of age when infants are particularly vulnerable to severe disease requiring hospitalization.

Respiratory Syncytial Virus	
Epidemiology	Global distribution Winter epidemics in temperate regions Peak age: 2-3 months Common in daycares, nurseries Incubation period: 4-6 days
Transmission	Close Contact Contaminated fingers and fomites Respiratory droplets Virus shedding: 1-3 weeks
Clinical Features	Most infections symptomatic Range from mild to severe respiratory illness 25-40% develop lower respiratory involvement Hospitalization rate: ~1% Recovery usually in 1-2 weeks
Diagnosis	Specimen: Nasopharyngeal swabs PCR: Sensitive and rapid(Option A) Immunofluorescence tests Serology: Paired samples for antibody titers
Treatment	Primarily supportive care Ribavirin for severe cases Palivizumab is a respiratory syncytial virus f protein monoclonal antibody and is used in severe cases in children. No effective vaccine available(Option D)
Risk Factors	Age < 6 months, Since maternal antibodies provide only partial protection and decline by 2-3 months (Option B) Prematurity Immunodeficiency Cardiac defects

Solution for Question 6:

Correct Answer: C) RT-PCR from buccal swab

Explanation:

• The case presents classic symptoms of mumps: bilateral parotitis with pain that increases with citrus juice consumption (stimulation of salivary glands), along with fever as a prodromal symptom.

• RT-PCR from a buccal swab is the most specific diagnostic test for confirming acute mumps infection, especially when collected early in the course of illness.

Diagnosis Methods	
RT-PCR	Buccal swab Most specific test Collect within first 3 days
Antigen	ELISA method Used for rapid screening
Clinical Features	Bilateral parotitis Pain increases with citrus juices Fever and malaise
Culture	Only in cell lines
Vaccine Details- MMR	
Recommended Strain- Mumps	Jeryl Lynn Strain
Measles Strain	Edmonston/Zagreb
Rubella Strain	RA 27/3
Type	MMR-V (with varicella)

Serum IgG antibody test (Option A): IgG antibodies indicate past infection or immunity rather than a current, acute infection. It is less specific for diagnosing an active mumps infection since it cannot distinguish between recent and past exposure to the virus.

Viral culture from parotid secretions (Option B): Although a viral culture from parotid secretions can confirm mumps, it is less sensitive than RT-PCR and requires more time, which makes it less practical for timely diagnosis.

Physical examination findings alone (Option D): Clinical signs like parotid swelling are suggestive but not confirmatory of mumps, as other conditions (such as bacterial parotitis) can also present similarly.

Solution for Question 7:

Correct Answer: A) Orchitis

Explanation:

- The diagnosis of mumps is supported by the patient's presentation of bilateral parotitis, fever, and elevated serum amylase levels, along with a lack of recent vaccinations.
- The most likely complication is orchitis, as it occurs in 20-30% of post-pubertal males with mumps and is more common than other potential complications such as encephalitis or pancreatitis.

Mumps:

It is a viral infection primarily affecting the parotid glands, with the mumps virus being the most common cause of parotid gland enlargement in children. Severe cases can lead to complications such as orchitis and aseptic meningitis.

Epidemiology:

- Mumps is endemic worldwide, peaks in winter and spring, and is often associated with unvaccinated individuals in overcrowded areas.
- Approximately 500,000 cases are reported globally each year. Immunization has significantly reduced the incidence.
- Primarily affects children aged 5–9 years. However, adults without immunity can also contract the disease, with potentially more severe outcomes.
- Lifelong immunity usually follows either natural infection or vaccination, with a high secondary attack rate of 86% in close contacts.

Transmission:

Respiratory Route	Spread occurs through respiratory droplets, saliva, and fomites contaminated by the virus.
Incubation Period	About 19 days, with a range from 7 to 23 days.
Infectivity	Patients are contagious from 1 week before symptoms to 1 week after. The peak infective period is 1–2 days before symptoms begin.
Source	Both clinical and subclinical cases are infectious, with subclinical cases (30–40% of infections) helping sustain transmission.
Reservoir	Humans are the only known reservoir, with no carrier state identified.

Clinical Features:

Up to 50% of infected individuals show mild or no symptoms.

Parotitis	The most common specific feature, presenting in 70–90% of cases, is usually bilateral parotid swelling.
Orchitis/Epididymo-orchitis	Seen in 15–30% of postpubertal males, usually unilateral; infertility is rare due to unilateral involvement.
Aseptic Meningitis	Occurs in less than 10% of cases, typically self-limiting, but can cause permanent deafness if cranial nerves are affected.
Oophoritis	Found in about 5% of infected women.
Pancreatitis	Affects approximately 4% of cases, leading to diabetes in rare cases.
Atypical Presentation	Some patients may present with aseptic meningitis without parotitis (in 10% of cases).

Encephalitis (Option B) is rare, affecting less than 0.3% of mumps cases, mainly in younger children.

Hearing loss (Option C) can occur but is extremely rare (1 in 20,000 cases).

Pancreatitis (Option D) is possible but less common in adolescents compared to orchitis.

Solution for Question 8:

Correct Answer: C) Administer immunoglobulin therapy

Explanation:

- The infant is symptomatic and unvaccinated; immunoglobulin within 6 days of exposure can reduce measles severity.
- Vaccination is ineffective post-exposure, and there is no specific antiviral therapy.
- Deferring until 12 months without prophylaxis increases risk.

Diagnosis of Measles:

- Preferred samples include nasopharyngeal swab, conjunctival swab, blood, respiratory secretions, and urine.

• Detectable by using direct immunofluorescence with anti-nucleoprotein antibodies applied to infected cells.

• Highly specific for diagnosing acute measles infections.

• Monkey or human kidney cells are the standard cell lines used.

• Viral growth is characterized by the presence of Warthin-Finkeldey cells—multinucleated giant cells containing intranuclear and intracytoplasmic inclusions.

• Shell vial culture is a quicker alternative, allowing early detection within 2-3 days.

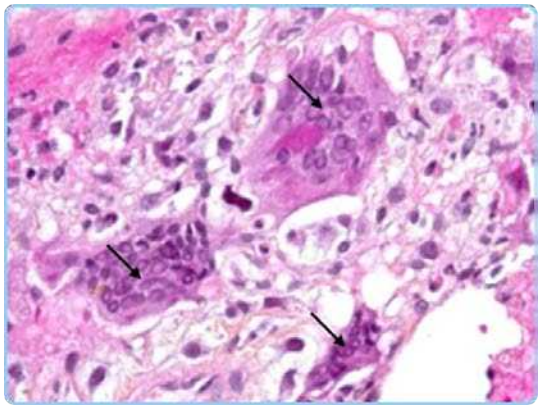
• The presence of measles-specific IgM in serum or oral fluid, or a four-fold rise in IgG titer between acute and convalescent-phase sera is diagnostic.

• Anti-measles antibody levels in cerebrospinal fluid (CSF) are diagnostic for subacute sclerosing panencephalitis (SSPE).

• Reverse-transcription PCR can detect measles RNA (e.g., nucleoprotein gene) with high sensitivity and specificity.

• Useful for tracking measles strains and conducting epidemiological studies.

• Detectable RNA persists longer than the virus itself (10–14 days post-rash).



Vaccination:

• Live attenuated vaccine: Strains used: Schwartz, Edmonston-Zagreb, Moraten strains. Prepared in chick embryo cell lines.

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• Available as MR (measles-rubella), MMR (measles, mumps, rubella), and MMR-V (measles, mumps, rubella, varicella) vaccines.

• Stored at –20°C due to its thermolabile nature.

- India’s Immunization Schedule: First dose of MR at 9-12 months, accompanied by vitamin A supplementation. Second dose at 16-24 months.
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- Second dose at 16-24 months.
- Post-Exposure Prophylaxis: If exposed, administering a measles vaccine within three days can prevent the disease in children 9-12 months old.
- First dose of MR at 9-12 months, accompanied by vitamin A supplementation.
- Second dose at 16-24 months.
- Severe allergic reactions to prior MMR doses, severe immunodeficiency, pregnancy, and active tuberculosis.

Administer the measles vaccine (Option A): Vaccination is preventive and does not provide therapeutic benefits once the infection is established. Additionally, live vaccines like MMR are typically not administered during an active infection due to the risk of exacerbating symptoms.

Prescribe antiviral therapy (Option B): There is currently no specific antiviral treatment for measles. Management of measles is generally supportive, including hydration, antipyretics, and in some cases, high-dose vitamin A supplementation to reduce complications.

Educate on supportive care and schedule the vaccine at 12 months (Option D):

- While supportive care education is essential, deferring intervention until 12 months is inadequate for this high-risk infant.
- Waiting without immunoglobulin administration increases the risk of complications from measles in a young, unvaccinated infant. Immediate post-exposure prophylaxis is warranted given the high risk associated with active measles in this age group.

Solution for Question 9:

Correct Answer: D) Pneumonia

Explanation:

- Pneumonia is the leading cause of morbidity and mortality in acute measles and is the most likely immediate complication in this case, especially in an unvaccinated child.
- Measles is an acute, highly contagious viral infection caused by the measles virus, a member of the Paramyxoviridae family.

• It primarily affects children and is transmitted through respiratory droplets or small-particle aerosols.

Clinical Stages of Measles:

Stage	Duration	Symptoms
Incubation Period	10 days (up to 21 days)	Asymptomatic phase; the virus multiplies in the respiratory tract, followed by primary and secondary viremia.
Prodromal Stage	4 days	Fever (day 10), Koplik's spots (day 12), non-specific symptoms (cough, coryza, conjunctivitis).
Eruptive Stage	4 days post fever	The maculopapular rash appears first behind the ears, then spreads to the face, trunk, and limbs, fading in the same order.
Post-measles Stage	Variable	Characterized by weight loss, weakness, and a risk of progressing to chronic conditions.

Complications of Measles:

A. Secondary Bacterial Infections

• Immune suppression following measles increases susceptibility to bacterial infections.

• Otitis media and bronchopneumonia are the most frequent bacterial complications.

• Worsening of underlying tuberculosis with a possible false-negative Mantoux test.

B. Complications Due to Measles Virus Itself

• Giant-cell pneumonitis (Hecht's pneumonia), primarily in immunocompromised and HIV-infected individuals.

• Acute laryngotracheobronchitis (croup).

• Diarrhea leading to malnutrition and vitamin A deficiency.

C. Central Nervous System (CNS) Complications

• Subacute sclerosing panencephalitis (SSPE): A rare but severe late complication characterized by progressive CNS deterioration.

• Other CNS complications include post-measles encephalomyelitis and measles inclusion body encephalitis.

Acute encephalitis (Option A): Occurs in approximately 1 in 1,000 cases, often presenting later with neurological symptoms, not during the acute phase.

Otitis media (Option B): It is common but typically less severe than pneumonia in terms of immediate risk.

SSPE (Option C): It is a delayed complication, developing years after initial infection, thus not relevant to the acute phase.

Solution for Question 10:

Correct Answer: C) Influenza virus

Explanation:

The influenza virus belongs to the Orthomyxoviridae Family, which is distinct from Paramyxoviridae. Influenza viruses are responsible for seasonal flu and can lead to severe respiratory illness.

Myxovirus

Myxoviruses bind to mucin receptors on RBCs, causing hemagglutination.

Most myxoviruses are respiratory pathogens.

Divided into two families:

- Contains segmented RNA.
- Important human pathogen: Influenza viruses (cause URTIs, rare pneumonia).
- Includes several viruses: Parainfluenza virus: causes laryngotracheobronchitis and URTIs. (Option B) Mumps virus: causes parotitis in children; rare complications include meningitis. (Option D) Measles virus: causes exanthematous lesions; rare CNS complications. (Option A) Respiratory syncytial virus: causes acute bronchiolitis in infants. Metapneumovirus: causes URTIs. Zoonotic viruses: Nipah and Hendra (mainly cause encephalitis).
- Parainfluenza virus: causes laryngotracheobronchitis and URTIs. (Option B)
- Mumps virus: causes parotitis in children; rare complications include meningitis. (Option D)
- Measles virus: causes exanthematous lesions; rare CNS complications. (Option A)
- Respiratory syncytial virus: causes acute bronchiolitis in infants.
- Metapneumovirus: causes URTIs.
- Zoonotic viruses: Nipah and Hendra (mainly cause encephalitis).
- Parainfluenza virus: causes laryngotracheobronchitis and URTIs. (Option B)

• Mumps virus: causes parotitis in children; rare complications include meningitis. (Option D)

• Measles virus: causes exanthematous lesions; rare CNS complications. (Option A)

• Respiratory syncytial virus: causes acute bronchiolitis in infants.

• Metapneumovirus: causes URTIs.

• Zoonotic viruses: Nipah and Hendra (mainly cause encephalitis).

Properties	Orthomyxoviridae	Paramyxoviridae
Size	80-120 nm	100-300 nm
Shape	Spherical; Rarely filamentous	Pleomorphic
Nucleic acid	Negative sense ssRNA, Segmented; eight pieces	Negative sense ssRNA Unsegmented; single piece
Genetic recombination	Seen	Not seen
Antigenic variation	Seen	Not seen
Site for RNA Replication	Nucleus	Cytoplasm
Important human pathogens	Influenza virus	Parainfluenza virus, Mumps virus Respiratory syncytial virus Measles virus, Metapneumovirus
Antigens: Hemagglutinin (HA) and Neuraminidase (NA)	Both HA and NA spikes are present, So hemagglutination is reversible (Elution seen)	HA spike present in Parainfluenza, Mumps, Measles, NA spike present in Parainfluenza, Mumps Inclusion body: Intracytoplasmic (only for measles, it is both intracytoplasmic and intranuclear)

Solution for Question 11:

Correct Answer: B) The HI test shows a fourfold increase in antibody titers between the first and second samples.

Explanation:

• The Hemagglutination Inhibition (HI) test is a specific serological test used to detect antibodies against influenza viruses. This test measures the ability of patient antibodies to inhibit the agglutination (clumping) of red blood cells by the influenza virus.

• For a confirmed influenza infection, a fourfold or greater rise in antibody titers between acute (initial) and convalescent (second) serum samples taken 7–14 days apart is indicative of an active or recent infection. The increase in titers reflects the body's immune response to the infection.

Lab Diagnosis of Influenza:

The laboratory diagnosis of influenza involves various methods for detecting the virus or its antibodies. These methods include:

• Ideal specimens include nasopharyngeal swabs, nasal aspirates, or lavage fluids.

• Swabs with synthetic tips (polyester or Dacron) are recommended. Specimens are placed in viral transport media, stored at 4°C if processed within 4 days, and at -70°C if stored longer.

• Influenza virus can be inoculated in embryonated eggs. Growth detection occurs through hemadsorption or hemagglutination.

• Primary monkey kidney cell lines are also used to grow influenza viruses for detection.

• This rapid method involves staining viral antigens on epithelial cells with fluorescent antibodies. However, it has lower sensitivity than virus isolation.

• Reverse Transcriptase PCR (RT-PCR): Highly sensitive and specific, detecting influenza RNA quickly.

• Real-Time RT-PCR: The preferred diagnostic method due to its quantitative capability and ability to detect multiple seasonal strains, including A/H1N1, A/H3N2, and influenza B.

• Hemagglutination Inhibition Test (HAI): Previously used for influenza, HAI measures specific antibodies by detecting their ability to inhibit virus-induced hemagglutination.

• Enzyme-Linked Immunosorbent Assay (ELISA) and Neutralization Test: ELISA is commonly used to detect serum antibodies against influenza antigens, particularly for epidemiological studies.

Hemagglutination Inhibition (HAI) Test:

The HAI test is a serological method used primarily in epidemiology to detect antibodies against influenza viruses.

• Principle: Influenza viruses possess hemagglutinin (HA) proteins, allowing them to agglutinate red blood cells (RBCs). Specific antibodies in a patient's serum can inhibit this agglutination. The test measures these antibodies' ability to prevent hemagglutination.

• Procedure: Step 1: Diluted serum samples are added to standardized influenza virus suspensions. Step 2: Red blood cells are added to the mixture. Step 3: If antibodies are present, they bind to the HA antigens, preventing them from agglutinating the RBCs. Lack of agglutination indicates a positive HAI result, suggesting prior exposure or immunity.

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- Interpretation: Positive HAI: No agglutination indicates antibodies that neutralize the virus's HA protein, preventing it from binding RBCs. Negative HAI: Agglutination implies an absence of specific antibodies in the serum.
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The HI test shows no change in antibody titers between the first and second samples (Option A): suggesting either the absence of infection or past exposure without recent immune activation. In cases of acute influenza infection, we would expect to see a significant increase in titers due to the active immune response.

The HI test shows a twofold decrease in antibody titers between the first and second samples (Option C): A decrease in titers is generally not seen in acute infection and suggests either a resolving past infection or a laboratory error. This does not confirm active or recent influenza infection, as we would expect a rise in antibody levels if the patient is actively fighting the virus.

The HI test shows a single positive result with no subsequent titer testing (Option D): A single positive titer cannot confirm a recent infection, as it does not provide a baseline for comparison. A single sample only indicates previous exposure or vaccination and cannot distinguish between current and past infections without paired testing.

Solution for Question 12:

Correct Answer: A) Start Oseltamivir immediately

Explanation:

Oseltamivir is indicated for influenza treatment, especially in patients at high risk for complications, such as older adults and those with COPD.

- Prompt initiation within 48 hours is ideal but may also be effective if started later in severe cases.
- Antiviral therapy in high-risk groups reduces symptom duration and the risk of complications, like pneumonia.

Clinical Features of Influenza:

- Incubation Period: 18–72 hours, depending on viral load and host immunity.
- Symptoms: Starts with minor upper respiratory symptoms like chills, dry cough, and headache. Progresses to high-grade fever, myalgia, fatigue, and appetite loss. Most cases are self-limiting and resemble other upper respiratory infections, such as the common cold.
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- Complications: Secondary Bacterial Pneumonia: The most common complication, caused by bacteria like Staphylococcus, Streptococcus pneumoniae, and Haemophilus influenzae. Primary Influenza Pneumonia: Less common but more severe, especially in high-risk groups. Exacerbation of Chronic Conditions: Can worsen conditions like COPD and asthma. Reye's Syndrome: A rare but serious condition in children causing liver damage and encephalopathy, due to aspirin use during viral infections. Other Complications: Include myocarditis, encephalitis, myositis, and Guillain-Barré syndrome, mainly in severe cases■.
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• Secondary Bacterial Pneumonia: The most common complication, caused by bacteria like Staphylococcus, Streptococcus pneumoniae, and Haemophilus influenzae.

• Primary Influenza Pneumonia: Less common but more severe, especially in high-risk groups.

• Exacerbation of Chronic Conditions: Can worsen conditions like COPD and asthma.

• Reye's Syndrome: A rare but serious condition in children causing liver damage and encephalopathy, due to aspirin use during viral infections.

• Other Complications: Include myocarditis, encephalitis, myositis, and Guillain-Barré syndrome, mainly in severe cases■.

Treatment:

• Antiviral Drugs: Neuraminidase Inhibitors: Oseltamivir (Tamiflu): Oral; for both Influenza A and B; recommended within 48 hours of symptom onset. Zanamivir: Inhaled formulation, suitable for both Influenza A and B; effective for patients who can use inhalation. Peramivir: Intravenous option for severe cases when other drugs are not viable. Mechanism: Neuraminidase inhibitors block the viral enzyme neuraminidase, which prevents the release of new viral particles, slowing down virus spread in the respiratory tract.

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• Peramivir: Intravenous option for severe cases when other drugs are not viable.

• Mechanism: Neuraminidase inhibitors block the viral enzyme neuraminidase, which prevents the release of new viral particles, slowing down virus spread in the respiratory tract.

• Symptomatic Treatment: Antipyretics (acetaminophen) for fever. Adequate hydration is to manage fluid loss from fever and respiratory symptoms■.

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• Adequate hydration is to manage fluid loss from fever and respiratory symptoms■.

Vaccination:

• Types of Vaccines: Trivalent Vaccine (TIV): Protects against three strains—H1N1, H3N2, and one B strain. Quadrivalent Vaccine (QIV): Covers an additional B strain, broadening protection.

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• Formulations: Inactivated Influenza Vaccine (IIV): Injectable, the most widely used, suitable for anyone older than six months. Cell Culture-Based Vaccine (ccIIV): Produced in cell cultures rather than eggs, making it suitable for those with severe egg allergies. Recombinant Influenza Vaccine (RIV): Contains recombinant hemagglutinin proteins and is completely egg-free. Live Attenuated Influenza Vaccine (LAIV): A nasal spray form containing a weakened virus, suitable for healthy individuals aged 2-49, not recommended for high-risk groups.

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- Vaccination Schedule: Annual vaccination is recommended due to frequent mutations (antigenic drift) in the virus. Immunity lasts 6-12 months, with optimal vaccination time before the flu season onset.
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- Annual vaccination is recommended due to frequent mutations (antigenic drift) in the virus.
- Immunity lasts 6-12 months, with optimal vaccination time before the flu season onset.

Prescribe supportive care only (Option B): While supportive care (hydration, antipyretics) is important, antiviral therapy is recommended in high-risk patients, even if initiated after 48 hours, due to potential benefits in reducing complications.

Administer high-dose azithromycin (Option C): Azithromycin targets bacterial infections, not viral. It is not indicated unless there is a secondary bacterial infection, which is not suggested in this scenario.

Recommend the influenza vaccine now (Option D): Vaccination is a preventive measure and would not be effective for an already symptomatic patient.

Influenza can present with fever, chills, myalgia, and respiratory symptoms, especially in high-risk groups (e.g., elderly, COPD).

- Treatment includes antivirals (Oseltamivir) for reducing severity and complications, particularly effective within 48 hours of symptom onset.
- Vaccination is recommended annually for prevention, especially in high-risk populations.

Solution for Question 13:

Correct Answer: B) Influenza B is classified into subtypes based on its HA and NA proteins.

Explanation:

Influenza B does not have subtypes but has two main lineages (B/Yamagata and B/Victoria).

Influenza

Antigenic Subtypes and Nomenclature

- Genera: Influenza viruses are divided into types A, B, C, and D based on RNP and M proteins.
- Subtypes: Influenza A: Has 18 HA (H1–H18) and 11 NA (N1–N11) subtypes. (Option A ruled out) Six HA (H1, H2, H3, H5, H7, H9) and two NA (N1, N2) subtypes have been found in humans. Most subtypes primarily infect animals and birds, but some infect humans, causing epidemics/pandemics. (Option C ruled out) Influenza B: No subtypes but has two main lineages—B/Yamagata and B/Victoria. (Option B) Influenza C: Rarely causes infection, is mild in nature, and has low public health impact. (Option D ruled out) Influenza D: Primarily infects cattle and does not infect humans.
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- Influenza C: Rarely causes infection, is mild in nature, and has low public health impact. (Option D ruled out)
- Influenza D: Primarily infects cattle and does not infect humans.
- Nomenclature: Standard naming format includes virus type/host (for non-human origin)/geographical origin/strain number/year/(HA NA subtype). Human strain example: Influenza A/Hong Kong/03/1968 (H3N2). Non-human strain example: Influenza

A/swine/Iowa/15/1930 (H1N1).

• Human strain example: Influenza A/Hong Kong/03/1968 (H3N2).

• Non-human strain example: Influenza A/swine/Iowa/15/1930 (H1N1).

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• Influenza B: No subtypes but has two main lineages—B/Yamagata and B/Victoria. (Option B)

• Influenza C: Rarely causes infection, is mild in nature, and has low public health impact. (Option D ruled out)

• Influenza D: Primarily infects cattle and does not infect humans.

• Six HA (H1, H2, H3, H5, H7, H9) and two NA (N1, N2) subtypes have been found in humans.

• Most subtypes primarily infect animals and birds, but some infect humans, causing epidemics/pandemics. (Option C ruled out)

• Human strain example: Influenza A/Hong Kong/03/1968 (H3N2).

• Non-human strain example: Influenza A/swine/Iowa/15/1930 (H1N1).

Epidemiology

• Seasonal Epidemics: Occur almost annually, with varying severity.

• Incidence: Approximately 3–5 million severe cases and 300,000–600,000 deaths globally each year.

• Pandemics: Occur every 10–40 years with novel influenza A subtypes, often causing higher mortality.

• Seasonality: Common in winter, with dominant strains differing by season and location.

• Annual Attack Rate: 5–10% in adults and 20–30% in children.

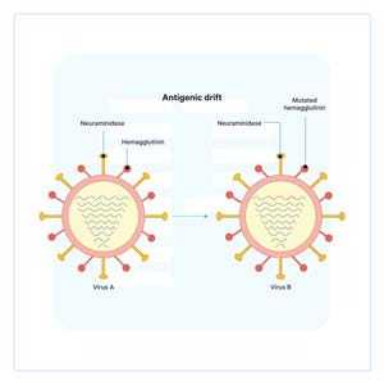
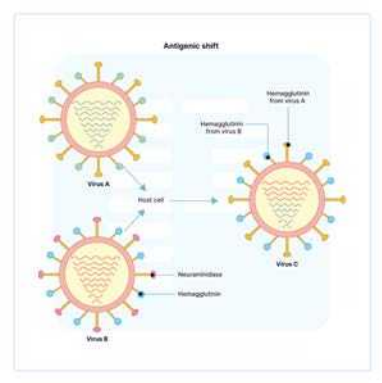
• Pattern: Influenced by the nature of antigenic variation in influenza types (antigenic drift and shift).

Solution for Question 14:

Correct Answer: B) Drift involves minor mutations; shift involves genetic re-assortment.

Explanation:

Antigenic drift involves minor changes due to point mutations, while antigenic shift is an abrupt change due to genetic re-assortment.

Antigenic Drift	Antigenic Shift
Minor change due to point mutations in HA/NA genes, altering amino acid sequences and helping the virus partially evade immune recognition. (Option D) Requires two or more mutations for significant impact. Occurs in both influenza A and B viruses. (Option C) Leads to outbreaks and minor epidemics, occurring every 2–3 years. (Option A)	An abrupt, major change in HA/NA genes due to genetic reassortment between influenza viruses infecting the same host. Produces a new virus strain, making prior antibodies ineffective. (Option D) Occurs only in influenza A virus. (Option C) Causes pandemics and major epidemics, with shifts happening every 10–20 years (e.g., the H1N1 pandemic of 2009). (Option A)
	

Solution for Question 15:

Correct Answer: C) The immune response mainly involves antibodies against the hemagglutinin protein.

Explanation:

- The patient's symptoms of high fever, severe headache, muscle aches, and dry cough, along with the rapid onset of these symptoms, strongly suggest influenza caused by an

orthomyxovirus. The presentation aligns with typical flu manifestations, particularly in thecontext of recent respiratory illness.

• The immune response to orthomyxoviruses primarily involves humoral immunity with antibodies targeting HA and neuraminidase (NA).

Orthomyxovirus

• Pathogenesis: Transmission: Through inhalation of respiratory droplets within 1 meter or contact with infected surfaces followed by touching the nose, eyes, or mouth. Target Cell Entry: Viral HA binds to sialic acid receptors on respiratory mucosa for viral entry. Replication and Spread: The virus replicates locally in infected cells, spreading to adjacent cells; rarely spreads to the lower respiratory tract or bloodstream. Local Damage: Causes cellular destruction and desquamation of respiratory mucosa, increasing susceptibility to secondary bacterial infections. Immune Response: Humoral immunity predominates with antibodies against HA and NA providing subtype-specific protection, but reinfection can still occur. (Option D ruled out)

• Transmission: Through inhalation of respiratory droplets within 1 meter or contact with infected surfaces followed by touching the nose, eyes, or mouth.

• Target Cell Entry: Viral HA binds to sialic acid receptors on respiratory mucosa for viral entry.

• Replication and Spread: The virus replicates locally in infected cells, spreading to adjacent cells; rarely spreads to the lower respiratory tract or bloodstream.

• Local Damage: Causes cellular destruction and desquamation of respiratory mucosa, increasing susceptibility to secondary bacterial infections.

• Immune Response: Humoral immunity predominates with antibodies against HA and NA providing subtype-specific protection, but reinfection can still occur. (Option D ruled out)

• Clinical Manifestations: Incubation Period: Typically 18–72 hours, depending on viral load and host immunity. (Option B ruled out) Uncomplicated Influenza (Flu Syndrome): Often presents as chills, headache, dry cough, high-grade fever, myalgia, and anorexia; usually self-limiting.

• Incubation Period: Typically 18–72 hours, depending on viral load and host immunity. (Option B ruled out)

• Uncomplicated Influenza (Flu Syndrome): Often presents as chills, headache, dry cough, high-grade fever, myalgia, and anorexia; usually self-limiting.

• Complications: Pneumonia: Secondary bacterial pneumonia (common agents include staphylococci, pneumococci, Haemophilus influenzae) or, rarely, primary influenza pneumonia. Other Respiratory Complications: Exacerbation of COPD, chronic bronchitis, bronchiolitis, otitis media, parotitis, asthma. Extrapulmonary Complications: Myositis, rhabdomyolysis, myocarditis, encephalitis, and Guillain-Barre syndrome. Reye’s Syndrome:

Liver degeneration and acute encephalopathy in children following aspirin use with influenzaB, varicella-zoster, or rarely influenza A.

• Pneumonia: Secondary bacterial pneumonia (common agents include staphylococci, pneumococci, Haemophilus influenzae) or, rarely, primary influenza pneumonia.

• Other Respiratory Complications: Exacerbation of COPD, chronic bronchitis, bronchiolitis, otitis media, parotitis, asthma.

• Extrapulmonary Complications: Myositis, rhabdomyolysis, myocarditis, encephalitis, and Guillain-Barre syndrome.

• Reye's Syndrome: Liver degeneration and acute encephalopathy in children following aspirin use with influenza B, varicella-zoster, or rarely influenza A.

• Treatment (Option A ruled out) Neuraminidase Inhibitors: Drugs like oseltamivir, zanamivir, and peramivir are effective for influenza A and B; start within 48 hours of symptom onset. Schedule: Twice daily for 5 days (treatment); once daily (chemoprophylaxis for 7 days). M2 Inhibitors: Amantadine and rimantadine were used for some influenza A strains; resistance was noted in certain strains like A/H1N1 and A/H5N1.

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• Transmission: Through inhalation of respiratory droplets within 1 meter or contact with infected surfaces followed by touching the nose, eyes, or mouth.

• Target Cell Entry: Viral HA binds to sialic acid receptors on respiratory mucosa for viral entry.

• Replication and Spread: The virus replicates locally in infected cells, spreading to adjacent cells; rarely spreads to the lower respiratory tract or bloodstream.

• Local Damage: Causes cellular destruction and desquamation of respiratory mucosa, increasing susceptibility to secondary bacterial infections.

• Immune Response: Humoral immunity predominates with antibodies against HA and NA providing subtype-specific protection, but reinfection can still occur. (Option D ruled out)

• Incubation Period: Typically 18–72 hours, depending on viral load and host immunity. (Option B ruled out)

• Uncomplicated Influenza (Flu Syndrome): Often presents as chills, headache, dry cough, high-grade fever, myalgia, and anorexia; usually self-limiting.

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• Other Respiratory Complications: Exacerbation of COPD, chronic bronchitis, bronchiolitis, otitis media, parotitis, asthma.

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• Schedule: Twice daily for 5 days (treatment); once daily (chemoprophylaxis for 7 days).

Solution for Question 16:

Correct Answer: C) 1-D, 2-C, 3-B, 4-F, 5-E, 6-A

Explanation:

Orthomyxovirus (Influenza Virus) Segments and Structure:

• Genera: Includes influenza A, B, C, and D viruses.

• Genome Segments: Influenza A and B: Eight segments of negative-sense, single-stranded RNA. Influenza C and D: Seven segments (lacking the segment coding for neuraminidase).

• Influenza A and B: Eight segments of negative-sense, single-stranded RNA.

• Influenza C and D: Seven segments (lacking the segment coding for neuraminidase).

• Shape and Size: Spherical, approximately 80–120 nm in diameter. Helical symmetry with a helical nucleocapsid enclosed in an envelope.

• Spherical, approximately 80–120 nm in diameter.

• Helical symmetry with a helical nucleocapsid enclosed in an envelope.

• Replication Site: Unique among RNA viruses, replication occurs in the nucleus.

• Unique among RNA viruses, replication occurs in the nucleus.

• Viral Proteins: Structural Proteins: Polymerase proteins: PB1, PB2, and PA are responsible for RNA transcription and replication. Nucleoprotein (NP): Forms a ribonucleoprotein complex with viral RNA. Matrix proteins: M1 forms a shell beneath the envelope; M2 forms ion channels in the envelope. Glycoproteins: Hemagglutinin (HA) binds sialic acid receptors for viral entry; Neuraminidase (NA) aids in viral release and prevents virion aggregation. Non-structural Proteins: NS1 acts as an interferon antagonist. NS2 aids in nuclear export.

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• Non-structural Proteins: NS1 acts as an interferon antagonist. NS2 aids in nuclear export.

• NS1 acts as an interferon antagonist.

• NS2 aids in nuclear export.

• Envelope: Lipid envelope containing HA and NA glycoproteins, which are essential for host cell entry and virus release. Hemagglutinin: Triangular-shaped peplomer. Binds to mucin or sialic acid receptors on respiratory epithelial cells. Facilitates viral entry into host cells. Neuraminidase: Mushroom-shaped and present in fewer numbers than HA. Sialidase enzyme that degrades sialic acid receptors on host cells. Facilitates release of virus particles from infected cell surfaces during budding. Prevents self-aggregation of virions to host cells. Helps the virus pass through the mucin layer in the respiratory tract to reach target epithelial cells.

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• Facilitates release of virus particles from infected cell surfaces during budding.

• Prevents self-aggregation of virions to host cells.

• Helps the virus pass through the mucin layer in the respiratory tract to reach target epithelial cells.

• Influenza A and B: Eight segments of negative-sense, single-stranded RNA.

• Influenza C and D: Seven segments (lacking the segment coding for neuraminidase).

• Spherical, approximately 80–120 nm in diameter.

• Helical symmetry with a helical nucleocapsid enclosed in an envelope.

• Unique among RNA viruses, replication occurs in the nucleus.

• Structural Proteins: Polymerase proteins: PB1, PB2, and PA are responsible for RNA transcription and replication. Nucleoprotein (NP): Forms a ribonucleoprotein complex with viral RNA. Matrix proteins: M1 forms a shell beneath the envelope; M2 forms ion channels in the envelope. Glycoproteins: Hemagglutinin (HA) binds sialic acid receptors for viral entry; Neuraminidase (NA) aids in viral release and prevents virion aggregation.

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• NS1 acts as an interferon antagonist.

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• Mushroom-shaped and present in fewer numbers than HA.

• Sialidase enzyme that degrades sialic acid receptors on host cells.

• Facilitates release of virus particles from infected cell surfaces during budding.

• Prevents self-aggregation of virions to host cells.

• Helps the virus pass through the mucin layer in the respiratory tract to reach target epithelial cells.

• Triangular-shaped peplomer.

• Binds to mucin or sialic acid receptors on respiratory epithelial cells.

• Facilitates viral entry into host cells.

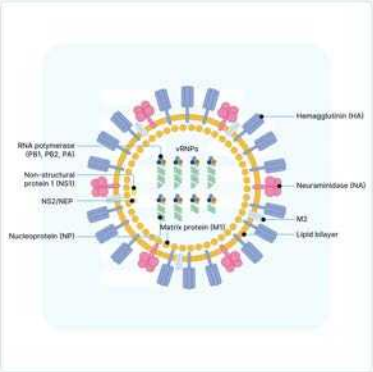
• Mushroom-shaped and present in fewer numbers than HA.

• Sialidase enzyme that degrades sialic acid receptors on host cells.

• Facilitates release of virus particles from infected cell surfaces during budding.

• Prevents self-aggregation of virions to host cells.

• Helps the virus pass through the mucin layer in the respiratory tract to reach target epithelial cells.



Solution for Question 17:

Correct Answer: B) Bunyavirus has segmented RNA with 7 segments.

Explanation:

Bunyavirus has 3 segments of RNA, not 7. All other statements about the respective virus families and their properties are true.

Some examples of:

Positive sense	Negative sense
Picornaviridae Flaviviridae Togaviridae Hepeviridae Coronaviridae (Option A ruled out)	M: Myxovirus R: Rhabdovirus (Option D ruled out) F: Filovirus A: Arenavirus B: Bunyavirus

Segmented RNA - BIRA

- Bunyavirus- 3 segments
- Orthomyxovirus (Influenza)- 7-8 segments (Option C ruled out)
- Reovirus- 10-12 segments
- Arenavirus- 2 segments

Picornaviridae and Rhabdoviridae

1. During an outbreak investigation in a labor colony, health officials are tracking a viral infection. The causative virus has these characteristics: exclusively human reservoir, fecal-oral transmission, and three distinct serotypes. The outbreak is showing these features: predominantly affects unvaccinated children, begins with fever and headache, and in some cases progresses to acute flaccid paralysis. The investigation team is discussing viral characteristics. Which of the following statements about this virus is INCORRECT?

A. Virus can be isolated from feces for up to 6 weeks after symptom onset

B. Type 2 commonly causes endemic infections

C. Type 1 is the least common cause of epidemics

D. Virus shedding begins 2 days before symptom onset

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2. A 4-year-old unvaccinated child presents with a biphasic illness. The initial phase included fever, headache, and sore throat lasting 5 days, followed by 2 days of improvement. Now the child has returned with high fever, neck stiffness, and weakness in the right leg. The CSF analysis shows aseptic meningitis. About the pathogenesis of this viral infection, which of the following sequences is correct?

A. Direct neural invasion → Viremia → CNS symptoms → Muscular weakness

B. Primary viremia → RES (reticuloendothelial system) multiplication → Secondary viremia → CNS invasion → Anterior horn cell destruction

C. Respiratory infection → Direct neural spread → CNS invasion → Secondary viremia

D. Primary viremia → Direct CNS invasion → Muscular weakness → Secondary viremia

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3. A 5-year-old child presents to the clinic with fever, headache, and neck stiffness. Two days later, he developed weakness in his left leg, with loss of reflexes but no sensation changes. A cerebrospinal fluid (CSF) analysis shows mild pleocytosis and a slight increase in protein levels. What is the most likely diagnosis, and which test should be done first?

A. Guillain-Barré Syndrome; perform nerve conduction studies.

B. Nonparalytic poliovirus infection; obtain a throat swab for viral PCR.

C. Viral meningitis; order a PCR for enterovirus on CSF.

D. Paralytic poliomyelitis; obtain stool and throat samples for poliovirus PCR.

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4. You're seeing a 6-year-old patient who's had a fever, sore throat, and painful sores in the mouth for the past few days. He now has small red spots and blisters on his palms and soles. During the exam, as shown, you notice vesicles on the soft palate and pinkish lesions on his hands and feet. Given these symptoms, what's the most likely diagnosis and causative agent?



A. Coxsackie B; Pleurodynia

B. Coxsackie A; Hand, Foot, and Mouth Disease

C. Coxsackie A; Herpangina

D. Coxsackie B; Hand, Foot, and Mouth Disease

---

5. A 2-year-old child presents with fever, headache, and neck stiffness. A physical exam shows a mild skin rash, but no other remarkable findings. Cerebrospinal fluid (CSF) analysis suggests aseptic meningitis. Which of the following is the most likely causative agent, and what is another common manifestation associated with this virus type?

A. Enterovirus; Acute Flaccid Myelitis (AFM)

B. Coxsackie B virus; Hand, Foot, and Mouth Disease (HFMD)

C. ECHO virus; Upper Respiratory Infections

D. Poliovirus; Myocarditis

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6. A 25-year-old patient with a history of asthma comes to your clinic with complaints of a runny nose, sore throat, mild cough, and a low-grade fever that started a few days ago. On examination, you note nasal congestion but no wheezing or serious respiratory distress. Which of the following is the most likely cause of his symptoms, and what's the underlying mechanism?

A. Rhinovirus infection, attaching to ICAM-1 receptors in the nasal lining and triggering inflammation.

B. Influenza virus infection causes a systemic cytokine storm by binding to respiratory epithelial cells.

C. Adenovirus infection binds to lower respiratory tract cells, leading to cell damage and high fever.

D. Streptococcus pneumoniae infection colonizes the throat and releases toxins that cause severe pain and fever.

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7. A 30-year-old wildlife researcher, who received a full pre-exposure rabies vaccination series two years ago, is bitten by a stray dog in a rural area with known rabies cases. Which of the following is the most appropriate management for this

patient?

A. Administer a full post-exposure prophylaxis (PEP) series with Rabies Immunoglobulin (RIG) and four doses of rabies vaccine on Days 0, 3, 7, and 14.

B. Give a booster dose of rabies vaccine on Days 0 and 3, without Rabies Immunoglobulin (RIG).

C. No additional vaccination or treatment is needed, as the patient completed the pre-exposure series.

D. Administer a single booster dose of rabies vaccine with Rabies Immunoglobulin (RIG) for rapid immunity.

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8. Carlos, a 40-year-old father and avid hiker, arrives at the emergency department after being bitten by a stray dog on his forearm while out on a rural trail. The bite is deep enough that it has broken the skin, causing some bleeding. Carlos mentions he's never received a rabies vaccination before and is anxious about the possibility of infection. What is the most appropriate next step in managing Carlos's injury?



A. Clean the wound and reassure Carlos that no further treatment is necessary.

B. Clean the wound and administer a series of rabies vaccines on Days 0, 3, 7, and 14, without giving Rabies Immunoglobulin (RIG).

C. Thoroughly wash the wound, administer rabies vaccines on Days 0, 3, 7, and 14, and provide Rabies Immunoglobulin (RIG) around the wound site.

D. Give Carlos a single dose of rabies vaccine and Rabies Immunoglobulin (RIG) for rapid immunity.

9. Which of the following statements is true regarding the structure and characteristics of the Rabies virus?

A. Spherical structure and is best visualized by light microscopy.

B. Single-stranded RNA genome and is best visualized using electron microscopy.

C. Has a hexagonal structure, observed under a compound microscope.

D. There are three types of antigens in Rabies

10. Consider the following statements regarding the Rabies virus and its clinical features. Mark the correct answer. After entry, the Rabies virus travels directly to the bloodstream, where it causes widespread viremia. Rabies virus initially replicates at the site of infection, then travels along peripheral nerves to the central nervous system Spreads centrifugally via nerves. The virus binds to the acetylcholine receptors.

A. Statement 1 is true.

B. Statement 2 is false.

C. Statement 3 is true.

D. Statement 4 is false.

11. Which of the following is the most sensitive and specific test for laboratory diagnosis of rabies?

A. Viral RNA PCR

B. Antigen detection

C. Antibody detection

D. Virus Isolation

12. In the case of re-exposure to rabies, what is the recommended re-exposure prophylaxis regimen if the initial post-exposure prophylaxis (PEP) was completed 2 months ago?

A. 4 doses on days 0, 3, 7, and 14

B. 3 doses on days 0, 7, and 21

C. 2 doses on days 0 and 3

D. Wound toilet only, without additional vaccine doses

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	4
Question 4	2
Question 5	3
Question 6	1
Question 7	2
Question 8	3
Question 9	2
Question 10	3
Question 11	1
Question 12	4

Solution for Question 1:

Correct Answer: C) Type 1 is the least common cause of epidemics

Explanation: Type 1 poliovirus is the most common and causes most epidemics.

Poliovirus	
Types and Strains	Type 1 (Brunhilde, Mahoney): Most common, causes epidemics Type 2 (Lansing, MEF1): Endemic infections (Option B) Type 3 (Leon, Saukett): Can cause epidemics

Viral Antigens	D antigen: Found in whole virion, type-specific, protective C antigen: Found in empty virus, not protective No common group antigen Heat converts D to C antigen (56°C)
Transmission Factors	Route: Primarily fecal-oral Source: Human feces, throat secretions Reservoir: Only humans Environmental survival: 6 months in sewage Resistant to: pH 3, ether, detergents
Time Course	Incubation: Range 4-28 days (average 10) Communicable period: -2 to +5 days from symptoms Fecal shedding: Up to 6 weeks (Option A) Initial symptoms: 1-5 days Asymptomatic period: 2-3 days Major illness onset: 3-4 days after minor illness
Clinical-Epidemiological Aspects	Subclinical:Clinical = 1000:1 or 100:1 Risk enhancers: Pregnancy, tonsillectomy, injections Cross-immunity: Between 1&2, 2&3; minimal 1&3 Peak season: Late winter and early spring No carrier state exists

Viral shedding in feces continues for several weeks: 80% in first week, 50% till third week, 25% till sixth week (Option A)

Type 2 typically causes endemic infections rather than epidemics (Option B)

Infectivity begins 2 days before symptom onset and continues until 5 days after (Option D)

Solution for Question 2:

Correct Answer: B) Primary viremia → RES multiplication → Secondary viremia → CNS invasion → Anterior horn cell destruction

Explanation: The clinical presentation is indicative of Poliomyelitis. The correct sequence of pathogenesis of Poliomyelitis is represented below, which follows the virus's natural progression:

Poliomyelitis- Pathogenesis and Clinical Features	

Pathogenesis Stages	1. Entry: Oral route 2. Initial replication: Nasopharynx and GI tract 3. Primary viremia: Virus enters bloodstream 4. RES multiplication: In lymphoid tissues 5. Secondary viremia: More extensive spread 6. CNS invasion: Via bloodstream or neural route 7. Anterior horn cell destruction: Causes paralysis
Clinical Presentations	Inapparent Infection (90-95%): - Asymptomatic - Only seroconversion Abortive Poliomyelitis (4-8%): - Fever - Headache - Sore throat - Malaise - Duration: 1-5 days Non-paralytic (1-2%): - Aseptic meningitis - Stiff neck - Spinal rigidity Paralytic (0.1-1%): - Asymmetric paralysis - Muscle weakness - Respiratory involvement possible
CNS Changes	Chromatolysis of Nissl bodies Nuclear degeneration Cell lysis Inflammatory response Anterior horn predominantly affected
Recovery Patterns	Recovery begins: 2-3 weeks after paralysis Maximum recovery: Within 6 months Residual paralysis: May persist Death rate: 5-10% (respiratory involvement) Muscle atrophy: Common in affected limbs

Solution for Question 3:

Correct Answer: D) Paralytic poliomyelitis; obtain stool and throat samples for poliovirus PCR.

Explanation:

- Paralytic polio may begin with flu-like symptoms and progresses to asymmetric limb weakness and loss of reflexes without sensory loss.
- Testing for poliovirus in stool and throat samples is preferred, as poliovirus can often be isolated early in the illness.
- In this child’s case, the asymmetric limb weakness and reflex loss, coupled with the mild CSF findings, align most closely with paralytic polio.
- Spinal: Limb paralysis.
- Bulbar: Difficulty breathing, and swallowing.

• Bulbospinal: Combination of both.

• Often asymmetrical muscle weakness and loss of reflexes, with respiratory failure in severe cases.

• Clinical Examination: Neurological assessment for muscle weakness and paralysis.

• Laboratory Testing: Stool/Throat Samples: Virus isolation. CSF Analysis: Detects inflammation. PCR: Detects poliovirus RNA. Serology: Identifies antibodies if virus isolation is unsuccessful.

• Stool/Throat Samples: Virus isolation.

• CSF Analysis: Detects inflammation.

• PCR: Detects poliovirus RNA.

• Serology: Identifies antibodies if virus isolation is unsuccessful.

• Imaging (MRI): For severe cases to assess nerve involvement.

• Stool/Throat Samples: Virus isolation.

• CSF Analysis: Detects inflammation.

• PCR: Detects poliovirus RNA.

• Serology: Identifies antibodies if virus isolation is unsuccessful.

Vaccination

• Two types of vaccines are available.

	OPV (Sabin)	IPV (Salk)
Type	Live attenuated virus	Killed formalised virus
Route	Oral	ID (preferred) or SC or IM
Dose	2 drops or as stated on the label	Intradermally, 0.1ml (fractional IPV or fIPV)
Schedule	1st dose: 6 weeks 2nd dose: 10 weeks 3rd dose: 14 weeks Booster dose: 16-24 months	1st dose: 6 weeks 2nd dose: 14 weeks 3rd dose: 9 completed months
Immunity provided	Humoral + Intestinal	Humoral
Prevents	Paralysis Intestinal re-infection	Paralysis
Storage & transport	Requires sub-zero temperatures	Less stringent conditions
Shelf life	Short	Longer

Guillain-Barré Syndrome; perform nerve conduction studies (Option A), is incorrect as while Guillain-Barré also causes weakness and loss of reflexes, it is typically symmetric (both sides of the body are affected equally) and often follows a respiratory or gastrointestinal illness.

Diagnosis involves nerve conduction studies and lumbar puncture for CSF analysis, but thischild's symptoms and asymmetry make polio more likely.

Nonparalytic poliovirus infection; obtaining a throat swab for viral PCR (Option B) is incorrect as paralytic polio causes fever, headache, and neck stiffness without progressing to muscle weakness. A PCR for polio in throat swabs is correct, but this doesn't fully explain the new weakness.

Viral meningitis; order a PCR for enterovirus on CSF (Option C), is incorrect as viral meningitis often presents with fever, headache, and neck stiffness, and enterovirus is a common cause. However, it does not typically cause focal (one-sided) muscle weakness or reflex loss.

Solution for Question 4:

Correct Answer: B) Coxsackie A; Hand, Foot, and Mouth Disease

Explanation: With this child's fever, mouth sores, and characteristic rash on the hands and feet, Hand, Foot, and Mouth Disease caused by Coxsackie A is the most fitting answer.





Coxsackie virus: Types and clinical manifestations

Coxsackie Virus Type	Associated Conditions	Clinical Manifestations
Coxsackie A	Primarily skin and mucosal infections	Hand, Foot, and Mouth Disease (HFMD): Fever, painful red spots/blisters on hands, feet, and inside the mouth; may spread to elbows, knees, or buttocks. (Option B) (Option D ruled out) Herpangina: Sudden high fever, painful ulcers/vesicles on soft palate, uvula, and tonsils; sore throat, headache, difficulty swallowing. (Option C ruled out)
Coxsackie B	Primarily systemic infections	Pleurodynia (Bornholm Disease): Sudden, severe, stabbing chest or abdominal pain; intermittent, lasting days, with fever and muscle tenderness. (Option A ruled out) Myocarditis and Pericarditis: Chest pain, fatigue, palpitations, irregular heartbeat; severe cases may lead to heart failure. Pancreatitis (rare): Inflammation of the pancreas. Neonatal Sepsis (rare): Severe infection in newborns, sepsis-like symptoms, potential multi-organ failure.

Solution for Question 5:

Correct Answer: C) ECHO virus; Upper Respiratory Infections

Explanation: ECHO virus is a frequent cause of aseptic meningitis, particularly in children. It can also cause mild skin rashes, respiratory infections, and other less severe symptoms, making this the best answer for the presentation described.

Enterovirus and ECHO virus:

Virus	Transmission	Common Infections	Severe Manifestations

Enterovirus	Fecal-oral, respiratory	Hand, Foot, and Mouth Disease (HFMD): Usually caused by Enterovirus A (e.g., EV-A71). Aseptic Meningitis: Fever, headache, neck stiffness, photophobia; multiple Enterovirus types. Herpangina: Sudden fever, painful throat ulcers; Enterovirus A. Acute Hemorrhagic Conjunctivitis: Red eyes with hemorrhage; Enterovirus D.	Myocarditis and Pericarditis: Heart inflammation, and chest pain, caused by Enterovirus B. Acute Flaccid Myelitis (AFM): Neurological condition causing limb weakness, often Enterovirus D68. Neonatal Sepsis-like Illness: Severe illness in newborns, fever, irritability, potential organ failure.
ECHO Virus	Fecal-oral	Aseptic Meningitis: Fever, headache, neck stiffness; most common ECHO virus manifestation. Exanthems: Non-specific rashes, often in children. Upper Respiratory Infections: Cough, sore throat, mild fever. Gastroenteritis: Vomiting, diarrhea, abdominal pain.	Myocarditis and Pericarditis: Can lead to chest pain, and palpitations. Neonatal Infections: Severe in newborns, presenting as sepsis-like illness or hepatitis.

Enterovirus; Acute Flaccid Myelitis (Option A) is incorrect. While Enterovirus (especially EV-D68) can cause AFM, this rare and severe presentation is not as common in the age group described and is usually associated with paralysis rather than a mild rash and meningitis.

Coxsackie B virus; HFMD (Option B), is incorrect as Coxsackie B is known for causing myocarditis and pericarditis, not HFMD, which is typically associated with Coxsackie A. HFMD also commonly presents with mouth sores, which are not seen here.

Poliovirus; Myocarditis (Option D) is incorrect as Poliovirus can cause flaccid paralysis but is not known to cause myocarditis. Additionally, polio is now rare due to vaccination and does not fit the presentation of fever with aseptic meningitis.

Solution for Question 6:

Correct Answer: A) Rhinovirus infection, attaching to ICAM-1 receptors in the nasal lining and triggering inflammation.

Explanation:

- Rhinovirus, a common cold culprit, binds to ICAM-1 receptors in the upper respiratory tract, particularly in the nose.
- This process triggers cytokines, which cause congestion, a sore throat, and mild fever.
- His asthma might make him a little more sensitive, but rhinovirus usually stays in the upper airways, rarely moving deeper.

Rhinovirus classification: Family Picornaviridae, over 100 serotypes in three species (A, B, and C), with type C often causing more severe symptoms.

- Transmission: Spread by respiratory droplets and contact with contaminated surfaces.
- Attachment: Binds to ICAM-1 receptors on upper respiratory epithelial cells, where it replicates, causing cell damage.
- Immune Response: Infected cells release cytokines, causing inflammation and typical cold symptoms.
- Temperature Sensitivity: Thrives at 33-35°C, usually limited to the upper respiratory tract.

Influenza virus infection causes a systemic cytokine storm by binding to respiratory epithelial cells (Option B), is incorrect as Influenza, although viral, causes more severe symptoms like high fever, body aches, and fatigue. The patient is not experiencing the intense symptoms flu often brings.

Adenovirus infection, binding to cells in the lower respiratory tract, leading to cell damage and high fever (Option C), is incorrect as Adenovirus can also cause respiratory symptoms but typically involves more intense fever and possibly conjunctivitis (eye irritation). This patient's symptoms do not fit that picture.

Streptococcus pneumoniae infection colonizes the throat and releases toxins that cause severe pain and fever (Option D): This is incorrect as this bacterium is behind conditions like pneumonia and sometimes throat infections, but it is much more aggressive. Fever tends to be high, and there would likely be sharp pain or significant chest symptoms, which does not align with this patient’s profile.

Solution for Question 7:

- Human Diploid Cell Vaccine (HDCV)
- Purified Chick Embryo Cell Vaccine (PCECV)

• Vero Cell Rabies Vaccine (VCRV) (These vaccines are interchangeable and have similar efficacy).

Protocol type	Recommended to	Schedule
Pre-exposure prophylaxis (PrEP)	Individuals at higher risk (e.g., veterinarians, animal handlers, travelers to rabies-endemic areas)	3 doses on Day 0, Day 7, and Day 21 or 28
Post-exposure prophylaxis (PEP)	Individuals exposed to potentially rabid animals (e.g., bites or scratches)	Unvaccinated Individuals: 4 doses (Day 0, 3, 7, and 14) with Rabies Immunoglobulin (RIG) administered on Day 0 for immediate passive immunity. Previously Vaccinated Individuals: 2 doses (Day 0 and Day 3), no RIG required

• Route: Intramuscular injection in the deltoid muscle; thigh for infants.

• Effectiveness: Near 100% if given promptly and correctly post-exposure.

• Common: Pain at the injection site, mild fever, headache, and muscle soreness.

• Rare: Severe allergic reactions.

• Pregnancy: Safe for use if there's exposure or high risk.

• Immunocompromised Individuals: May need additional monitoring or doses.

Administer a single booster dose of rabies vaccine with Rabies Immunoglobulin (RIG) for rapid immunity (Option D), is incorrect as a single dose is insufficient. Previously vaccinated individuals need two doses (Days 0 and 3) after exposure to ensure adequate immunity.

Solution for Question 8:

Correct Answer: C) Thoroughly wash the wound, administer rabies vaccines on Days 0, 3, 7, and 14, and provide Rabies Immunoglobulin (RIG) around the wound site.

Explanation:

• Carlos's injury is classified as a Category III exposure because the bite broke the skin and caused bleeding.

• For Category III exposures, particularly in unvaccinated individuals, treatment requires thorough wound cleaning, a full series of rabies vaccinations, and Rabies Immunoglobulin (RIG) around the wound to provide immediate passive immunity.

Categories of dog bite and management:

Category	Exposure Type	Management
Category I	Touching or feeding animals Licking of intact skin	Ensure proper hygiene No treatment required
Category II	Nibbling on uncovered skin Minor scratches or abrasions without bleeding	Immediate wound washing with soap and water Rabies vaccination: Unvaccinated: Days 0, 3, 7, and 14 Previously vaccinated: Booster doses on Days 0 and 3 No Rabies Immunoglobulin (RIG) required
Category III	Bites or scratches with bleeding Licks on broken skin Exposure to mucous membranes (mouth, eyes, nose)	Immediate wound washing with soap and water, flush for at least 15 minutes. Suture: Do not suture the wound immediately, as it may spread the virus, instead delay it for 24 to 48 hours. Rabies vaccination: Unvaccinated: Full PEP series on Days 0, 3, 7, and 14 Previously vaccinated: Booster doses on Days 0 and 3 Rabies Immunoglobulin (RIG): Equine immunoglobulin or ERIG Human rabies Immunoglobulin or HRIG Preferred (free from the danger of sensitisation, HIV and hepatitis viruses) Dose: – 40 IU / kg IM Started on Day 0 along with vaccination (can be given up to 7 days). 70% is given around the wound and, the rest is given distant from the wound. Required for unvaccinated individuals ... (content truncated)

Simply cleaning the wound is not enough for rabies prevention after a deep bite from a potentially rabid animal. (Option A)

RIG is required for unvaccinated individuals with significant exposure to provide immediate antibodies at the wound site (Option B)

A single dose of rabies vaccine is not sufficient to confer lasting immunity, and the full vaccination series is needed along with RIG for adequate post-exposure prophylaxis (Option D)

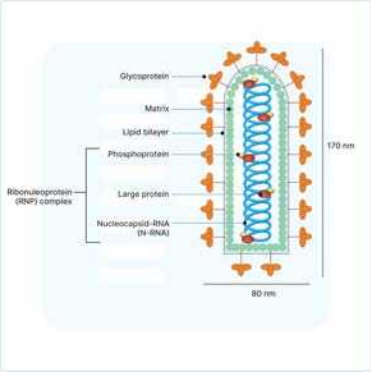
Solution for Question 9:

Correct Answer: B) Single-stranded RNA genome and is best visualized using electron microscopy.

Explanation: Rabies virus is a single-stranded, negative-sense RNA virus that belongs to the Rhabdoviridae family. Its structure is best visualized using electron microscopy due to its small size and unique bullet-like shape.

Rabies virus-general features:

- The Rabies virus belongs to the family Rhabdoviridae and has a characteristic bullet shape, making it unique among many other viruses.
- Its genome is single-stranded RNA with negative-sense polarity, meaning it cannot directly translate into proteins and requires RNA-dependent RNA polymerase for replication.
- The structure of the virus is a helical nucleocapsid (Option C) encased in a lipid envelope with embedded glycoprotein spikes, which play a critical role in binding to host cells and inducing an immune response.
- Two types of antigens- Glycoprotein G and nucleoprotein are present. (Option D)
- Due to the small size of viruses, visualization of their structure is best achieved using electron microscopy rather than light microscopy, (Option A) as electron microscopy provides the resolution required to observe such fine structural details.

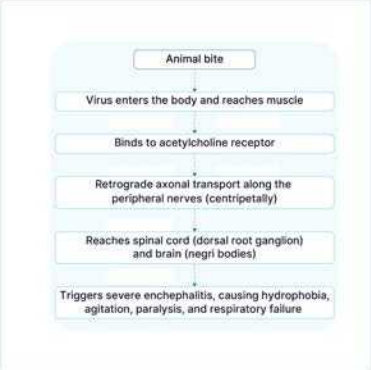


Solution for Question 10:

Correct Answer: C) Statement 3 is true.

Explanation:

• Rabies virus, a bullet-shaped, negative-sense RNA virus in the Rhabdoviridae family, initially infects muscle cells near the bite site. It binds enters peripheral nerves, traveling retrograde to the CNS. In the brain, it triggers severe encephalitis, causing hydrophobia, agitation, and paralysis, ultimately leading to respiratory failure.



• Statement 1 is incorrect because the rabies virus does not cause viremia (the presence of the virus in the blood). Instead, it travels via the nervous system to reach the CNS. Therefore ruling out option A.

• Statement 2 is true. (Option B) Rabies virus enters the body through a bite or scratch from an infected animal. After entry, it initially replicates in muscle cells at the site of infection. The virus then travels along peripheral nerves by retrograde axonal transport to the central nervous system (CNS), where it primarily affects neurons. This neuronal invasion leads to classic clinical manifestations, such as hydrophobia (fear of water), agitation, hallucinations, and, in later stages, paralysis and coma.

• Statement 3 is indeed true. (Option C) The virus spreads centrifugally, meaning it moves outward from the CNS to the peripheral nervous system and various organs through efferent nerves. This spread is responsible for the virus reaching organs such as the salivary glands, skin, and even the cornea, which is why Rabies can be transmitted through saliva.

• Statement 4 is true. (Option D)The affinity for acetylcholine receptors contributes to the virus's neurotropism, or preference for nervous tissue, making Rabies a neuroinvasive pathogen. The presence of these receptors in various neural tissues allows the virus to

spread efficiently within the CNS, leading to the extensive neuronal infection seen in Rabies.

Solution for Question 11:

Correct Answer: A) Viral RNA PCR

Explanation:

Viral RNA PCR is highly sensitive and specific for diagnosing Rabies in antemortem cases (diagnosing the disease before death), and where sample types like saliva, cerebrospinal fluid (CSF) or skin biopsies are used.

Antigen detection (Option B) requires brain tissue and, hence, is not a viable option for diagnosis in living patients (obtaining brain tissue is impractical while a patient is alive), so this limits antigen detection use mainly to post-mortem diagnostics or euthanized animals.

Antibody detection (Option C):

- Rabies antibodies are usually only detectable in the blood or CSF after the infection has progressed to a certain stage, which may be too late for timely intervention in active infections.
- Antibodies may only appear in the final stages of Rabies or after vaccination (in post-exposure prophylaxis cases). This delay in antibody production makes it a poor choice for early or definitive diagnosis, especially in acute or urgent settings where rapid detection is critical.

Virus Isolation (Option D) is very specific yet technically demanding, time-consuming, and less sensitive than PCR. Rabies virus does not grow readily in standard cell culture, so this method is often unsuccessful or requires prolonged incubation. It has a lower sensitivity (may fail to detect low levels of the virus), which PCR can often pick up more reliably and quickly.

Solution for Question 12:

Correct Answer: D) Wound toilet only, without additional vaccine doses

Explanation:

- For individuals re-exposed to rabies who previously completed a full PEP course, the recommended re-exposure prophylaxis consists of: Within 3 months of PEP: Wound toilet only done (Option D) After 3 months of PEP (Option C) 2 visits 2 doses Day 0, 3
- Within 3 months of PEP: Wound toilet only done (Option D)
- After 3 months of PEP (Option C) 2 visits 2 doses Day 0, 3
- 2 visits
- 2 doses
- Day 0, 3
- Within 3 months of PEP: Wound toilet only done (Option D)
- After 3 months of PEP (Option C) 2 visits 2 doses Day 0, 3
- 2 visits
- 2 doses
- Day 0, 3
- 2 visits
- 2 doses
- Day 0, 3

4 doses on days 0, 3, 7, and 14 (Option A): This regimen describes the standard post-exposure prophylaxis (PEP) for individuals who are receiving rabies vaccination for the first time (unvaccinated individuals). The PEP protocol for unvaccinated individuals typically includes 4 doses of the rabies vaccine administered on days 0, 3, 7, and 14, along with rabies immune globulin (RIG) if indicated.

3 doses on days 0, 7, and 21 (Option B): This regimen is typically used for a primary vaccination series in individuals who have not been vaccinated against rabies (e.g., pre-exposure prophylaxis). In the context of re-exposure prophylaxis after completing a previous PEP course, it is not the recommended approach.

RNA Virology- Arboviridae and Retroviridae

1. A 28 y/o woman presents to the clinic with a high fever, severe muscle aches, and a headache that has persisted for three days. She has recently travelled to a dengue-endemic region. Which of the following symptoms is NOT considered a warning sign in dengue for this patient?

A. Abdominal tenderness

B. Persistent vomiting

C. Tourniquet test positive

D. Liver enlargement >2 cm

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2. Which of the following findings would be most indicative of HTLV-1-associated adult T-cell leukaemia (ATL)?

A. High levels of interleukin-2 receptors in peripheral blood

B. Detection of p24 antigen in serum

C. Presence of CD4+ T cells in the peripheral blood

D. Detection of GLUT1 on the surface of leukemic cells

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3. A 30 y/o man presents to the clinic with flu-like symptoms 12 days after potential exposure to HIV. The clinician suspects acute HIV infection and orders a diagnostic test. Which test would be most appropriate to confirm the infection at this early stage and provide insights into the viral load?

A. p24 Antigen Detection

B. HIV Antibody Test

C. RT-PCR

D. HIV DNA PCR

4. A 32-year-old woman receives a positive HIV screening test result. To confirm the diagnosis, the healthcare provider decides to use a supplemental test for validation due to its high specificity in detecting individual antibodies against various HIV antigens. Which of the following tests would be most appropriate for confirming the diagnosis in this case?

A. p24 Antigen Detection

B. HIV RNA Detection (RT-PCR)

C. Western Blot

D. HIV DNA Detection

5. A 34 y/o patient with recent high-risk exposure tests positive for HIV-1 antibodies and negative for HIV-2 antibodies on an HIV-1/HIV-2 differentiation immunoassay. The initial fourth-generation antigen/antibody immunoassay had also shown positive results for HIV-1. What is the next step in confirming the diagnosis?

A. Order a Western Blot to confirm HIV-1 antibodies

B. Perform an HIV-1 NAT (Nucleic Acid Test)

C. Perform a repeat fourth-generation immunoassay

D. No further testing is needed; confirm HIV-1 infection

6. A 40 y/o HIV-positive patient who used to weigh 80kg, now weighs only 64kg and reports experiencing chronic diarrhoea for the last 5 weeks and a persistent fever that has lasted over a month. He has no other signs or symptoms. According to the WHO AIDS case definition, which of the following most accurately describes his condition?

A. The patient does not meet the criteria for AIDS diagnosis.

B. The patient meets the criteria for AIDS diagnosis based on major signs only.

C. The patient meets the criteria for AIDS diagnosis based on minor signs only.

D. The patient meets the criteria for AIDS diagnosis based on both major and minor signs.

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7. Match the following opportunistic infections with their respective CD4 counts Cerebral toxoplasmosis a) <50 Pneumocystis jiroveci pneumonia b) <200 Kaposi sarcoma c) <100 Mycobacterium avian complex (MAC) d) <500

Cerebral toxoplasmosis	a) <50
Pneumocystis jiroveci pneumonia	b) <200
Kaposi sarcoma	c) <100
Mycobacterium avian complex (MAC)	d) <500

A. 1-a,2-c,3-b,4-d

B. 1-c, 2-b, 3-d, 4-a

C. 1-d,2-c,3-b,4-a

D. 1-d, 2-b, 3-c, 4-a

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8. A 30 y/o woman visits the clinic with complaints of swollen lymph nodes in her neck and armpits. She reports that these have been present for the past four months, but she feels generally well otherwise. A laboratory evaluation shows no other significant findings. Based on this information, which of the following CDC Classifications is the most appropriate for her condition?

A. Group I

B. Group II

C. Group III

D. Group IV

9. A 40 y/o man presents to the clinic after having unprotected sex with multiple partners. He is concerned about his risk of contracting HIV and asks how the virus enters the body and its modes of transmission. Which of the following statements regarding HIV transmission and pathogenesis is CORRECT?

A. The least common mode of HIV transmission is through perinatal transmission.

B. HIV binds its gp41 to the CD4 receptor on host cells.

C. Viral load is the least in saliva and CSF.

D. The CCR5 Delta 32 mutation can confer resistance to HIV in homozygous individuals.

10. Identify and group the structural (a) and non-structural (b) genes of HIV from the given: Gag Vpr Tat Pol Env Nef Vif

A. a- 2,3,4,7 ; b- 1,5,6

B. a- 1,4,5 ; b- 2,3,6,7

C. a- 3,5,6 ; b- 1,2,4,7

D. a- 1,4,6,7 ; b- 2,3,5

11. A 35 y/o man visits the clinic after testing positive for HIV. He is concerned about the type of virus he may have and how it affects his treatment options. The clinician explains the structure of HIV and its classification into different serotypes. Which of the following statements regarding the structure and serotyping of HIV is NOT correct?

A. HIV has an envelope with glycoproteins gp120 and gp41.

B. HIV-1 is classified into groups M, N, O, and P, with group M being the most dominant.

C. The capsid has an complex symmetry and contains core proteins.

D. HIV-2 has eight subtypes, mainly found in Africa, with Group A being the most common.

12. A 30-year-old man presents to the clinic with a high fever, severe headache, and joint pain for the past 4 days. O/E, he has mild abdominal tenderness, and his laboratory results show leukopenia. Which of the following tests would be most appropriate to confirm his diagnosis at this stage of his illness?

A. NS1 Antigen Detection

B. IgG Antibody Detection

C. Virus Isolation

D. Rapid Diagnostic Test

13. A 30 y/o farmer from Central Asia presents with high fever, muscle pain, and significant bleeding from his gums and nose. He mentions being bitten by a tick while working on his farm. Laboratory tests reveal elevated liver enzymes. Which virus is most likely responsible?

A. Sin Nombre virus

B. Rift Valley fever virus

C. Nairovirus

D. Hantaan virus

14. A 26-y/o patient presents with fever, headache, muscle and joint pains, and maculopapular rashes over the chest and upper limbs for five days. The patient reports lymphadenopathy, retro-orbital pain, and loss of appetite. Lab results show haemoglobin at 10 g/dL, WBC count of 10,000 cells/mm^3, platelet count of 15,000 cells/mm^3, and prolonged bleeding time with normal clotting time. What is the most likely diagnosis?

A. Dengue

B. Malaria

C. Cholera

D. Typhoid

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15. A 26 y/o pregnant woman is concerned about Zika virus exposure after returning from Brazil, where she experienced mild fever and rash. What infant complication is associated with the Zika virus?

A. Cataract

B. Microcephaly

C. Heart defects

D. Deafness

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16. A 27-y/o employee from the forest department presents with high fever, malaise, and frontal headaches, followed by bleeding from the nasal cavity, throat, gums and hematemesis. Recently, he worked in a forest area in Karnataka that has a significant population of monkeys. What is the vector responsible for the disease?

A. Sandfly

B. Hard ticks

C. Flea

D. Louse

---

17. A 45-year-old male returns from South America and presents to the ED with severe abdominal pain, jaundice, and bleeding manifestations. O/E his liver is

enlarged, and lab tests show elevated liver enzymes. Despite receiving medical care, he did not survive. An autopsy shows the presence of intranuclear Torre bodies in the liver. What is the likely diagnosis, and which vaccine could have prevented it?

A. Yellow fever, 17D

B. Measles, Edmonston Zagreb

C. Mumps, Jeryll-Lynn

D. Rubella, RA 27/3

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18. A 35-y/o male presents with fever, chills, headache, dizziness, myalgia, and backache, followed by nausea and vomiting. O/E hemorrhagic manifestations, thrombocytopenia, jaundice, relative bradycardia and elevated creatinine levels are seen. Histopathology findings show mid-zonal necrosis with Councilman bodies in the liver. What is the vector responsible for transmitting the disease?

A. Aedes aegypti

B. Female anopheles

C. Culex

D. Mansonia

---

19. Which of the following vaccines is included in India's National Immunization Program to prevent Japanese encephalitis in children?

A. Inactivated Nakayama strain vaccine

B. Live attenuated SA 14-14-2 vaccine

C. Formalin-inactivated mouse brain vaccine

D. Live attenuated JE-CV vaccine

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20. A 12-year-old girl from Bihar presents with high fever, confusion, and seizures. Her family reports she frequently plays near rice fields. O/E there is nuchal rigidity. Which of the following is the primary amplifier host in the transmission cycle of the suspected infection?

A. Humans

B. Cattle

C. Pigs

D. Birds

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21. What laboratory test would be most helpful in confirming a diagnosis of an active rubella infection in a pregnant patient?

A. IgG Avidity Test

B. Virus Isolation

C. IgM Antibody Detection

D. TORCH Panel

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22. A 6 y/o boy presents to the clinic with a rash that started on his face and spread to his trunk and limbs. His mother reports that he has been experiencing mild fever and has developed swollen lymph nodes behind his ears. What additional finding would most likely support a diagnosis of rubella?

A. Blueberry muffin rash

B. Forchheimer spots

C. Ocular defects

D. Patent ductus arteriosus

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23. A 58-year-old woman presents with fever, severe joint pain in the wrist and ankle, and a faint rash. She reports mosquito bites over the past week while visiting a tropical region. Given these symptoms and recent exposure history, which test result would most likely confirm the suspected viral infection?

A. Detection of IgG antibodies in the first week

B. Presence of IgM antibodies in serum in the first two days

C. Detection of viral RNA by RT-PCR

D. Positive blood culture for chikungunya virus

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	3
Question 4	3
Question 5	4
Question 6	2
Question 7	2
Question 8	3
Question 9	4
Question 10	2
Question 11	3
Question 12	1
Question 13	3
Question 14	1
Question 15	2
Question 16	2
Question 17	1
Question 18	1
Question 19	2
Question 20	3
Question 21	3
Question 22	2

Question 23	3

Solution for Question 1:

Correct Answer: C) Tourniquet test positive

Explanation:

The tourniquet test positive indicates increased capillary permeability, which is not a warning sign. It is a feature of probable dengue fever.

Positive tourniquet test (>20 petechial spots per square inch area in cubital fossa and spontaneous bleeding from skin, nose, mouth and gums.

Positive tourniquet test

Dengue Case Classification based on the severity (WHO 2009)		
Criteria for Dengue ± Warning Signs	Criteria for Severe Dengue	
Probable dengue Live in/travel to dengue endemic area. Fever and two of the following criteria: Nausea, vomiting Rash Aches and pains Tourniquet test positive Leukopenia Any warning sign Laboratory-confirmed dengue (important when no sign of plasma leakage)	Warning signs* Abdominal pain or tenderness (Option A ruled out) Persistent vomiting (Option B ruled out) Clinical fluid accumulation Mucosal bleed Lethargy, restlessness Liver enlargement >2 cm (Option D ruled out) Laboratory: increase in HCT concurrent with rapid decrease in platelet count *Requiring strict observation and medical intervention	Severe plasma leakage leading to: Shock (Dengue shock syndrome) Fluid accumulation with respiratory distress Severe bleeding As evaluated by the clinician (causing hemodynamic instability and may require a blood transfusion) Severe organ involvement Liver: AST or ALT ≥1000 IU/mL CNS: Impaired consciousness Heart and other organs

Solution for Question 2:

Correct Answer: A) High levels of interleukin-2 receptors in peripheral blood

Explanation:

HTLV-1 infection leads to elevated interleukin-2 receptors on infected T cells, which is associated with ATL.

Human T-Cell Leukemia Virus Type 1 (HTLV-1)

- Retroviral Classification: Classified as a retrovirus with antigens on its envelope (type-specific glycoproteins) and core (group-specific nucleoproteins).
- Classified as a retrovirus with antigens on its envelope (type-specific glycoproteins) and core (group-specific nucleoproteins).
- Associated Conditions: Linked to adult T-cell leukaemia/lymphoma, cutaneous T-cell lymphoma, and tropical spastic paraparesis, a demyelinating condition. Contains an oncogenic Tax gene, promoting malignancy.
- Linked to adult T-cell leukaemia/lymphoma, cutaneous T-cell lymphoma, and tropical spastic paraparesis, a demyelinating condition.
- Contains an oncogenic Tax gene, promoting malignancy.
- Target Cells and Infection: Preferentially infects T4 (CD4) cells.uses GLUT1 (glucose transporter protein-1) on CD4 cells for attachment. Infected cells express high levels of interleukin-2 receptors. (Option A)
- Preferentially infects T4 (CD4) cells.uses GLUT1 (glucose transporter protein-1) on CD4 cells for attachment.
- Infected cells express high levels of interleukin-2 receptors. (Option A)
- Transmission: Primarily mother-to-child via breast milk, but also sexual contact (especially male-to-female) and infected blood.
- Primarily mother-to-child via breast milk, but also sexual contact (especially male-to-female) and infected blood.
- Genomic Structure and Antigens: Contains gag, pol, and env genes: Gag: Codes for nucleocapsid core proteins (group-specific antigens). Pol: Encodes RNA-dependent DNA polymerase. Env: Encodes envelope glycoproteins. Has a transactivating gene (tax/tat) that regulates viral gene expression.
- Contains gag, pol, and env genes: Gag: Codes for nucleocapsid core proteins (group-specific antigens). Pol: Encodes RNA-dependent DNA polymerase. Env: Encodes envelope glycoproteins.
- Gag: Codes for nucleocapsid core proteins (group-specific antigens).
- Pol: Encodes RNA-dependent DNA polymerase.

• Env: Encodes envelope glycoproteins.

• Has a transactivating gene (tax/tat) that regulates viral gene expression.

• Host Specificity: Infects mainly one host species, depending on specific receptors on host cells.

• Infects mainly one host species, depending on specific receptors on host cells.

• Classified as a retrovirus with antigens on its envelope (type-specific glycoproteins) and core (group-specific nucleoproteins).

• Linked to adult T-cell leukaemia/lymphoma, cutaneous T-cell lymphoma, and tropical spastic paraparesis, a demyelinating condition.

• Contains an oncogenic Tax gene, promoting malignancy.

• Preferentially infects T4 (CD4) cells.uses GLUT1 (glucose transporter protein-1) on CD4 cells for attachment.

• Infected cells express high levels of interleukin-2 receptors. (Option A)

• Primarily mother-to-child via breast milk, but also sexual contact (especially male-to-female) and infected blood.

• Contains gag, pol, and env genes: Gag: Codes for nucleocapsid core proteins (group-specific antigens). Pol: Encodes RNA-dependent DNA polymerase. Env: Encodes envelope glycoproteins.

• Gag: Codes for nucleocapsid core proteins (group-specific antigens).

• Pol: Encodes RNA-dependent DNA polymerase.

• Env: Encodes envelope glycoproteins.

• Has a transactivating gene (tax/tat) that regulates viral gene expression.

• Gag: Codes for nucleocapsid core proteins (group-specific antigens).

• Pol: Encodes RNA-dependent DNA polymerase.

• Env: Encodes envelope glycoproteins.

• Infects mainly one host species, depending on specific receptors on host cells.

Cloverleaf cells or Flower cells -pathognomonic of HTLV-induced adult T cell leukaemia (ATL)

Detection of p24 antigen in serum (Option B): Incorrect, as p24 antigen detection is specific for HIV, not HTLV-1.

Presence of CD4+ T cells in the peripheral blood (Option C): While HTLV-1 infects CD4+ T cells, their presence alone does not indicate ATL; rather, it's the transformation and malignancy that are of concern.

Detection of GLUT1 on the surface of leukemic cells (Option D): While GLUT1 is used by HTLV-1 for attachment to CD4 cells, its presence alone does not confirm ATL and is not a specific diagnostic marker.

Solution for Question 3:

Correct Answer: C) RT-PCR

Explanation:

RT-PCR is the most appropriate test as it detects HIV RNA as early as 10-14 days post-exposure and quantifies viral load, providing critical information for early diagnosis and monitoring.

RT-PCR for HIV Diagnosis and Monitoring

- Principle: Reverse transcription–polymerase chain reaction (RT-PCR) quantifies HIV RNA, measuring the viral load by directly estimating the number of HIV virions in a blood sample.
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- Applications: Confirmation During Window Period: Detects HIV earlier than other methods, as early as 10–14 days post-exposure. Viral Load Monitoring: Used after an initial diagnosis to monitor HIV progression and response to antiretroviral therapy (ART). Typing and Subtyping: Differentiates between HIV-1 and HIV-2 and can detect specific genotypes/subtypes. Detection of Drug Resistance Genes: Assesses viral mutations associated with resistance to ART.
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- Limitations: Costly and Technically Demanding: Not used routinely for screening due to high cost and complexity.
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p24 Antigen Detection (Option A): While this test can detect HIV earlier than antibody tests (12-26 days post-infection), it may not be as reliable in quantifying viral load as RT-PCR.

HIV Antibody Test (Option B): This test typically detects antibodies that develop later (usually 2-8 weeks post-infection), making it unsuitable for confirming early infection.

HIV DNA PCR (Option D): This test is primarily used for pediatric HIV diagnosis and not for early detection in adults, making it less relevant in this scenario.

Solution for Question 4:

Correct Answer: C) Western Blot

Explanation:

The Western Blot test is a supplemental HIV test recommended for confirming positive HIV screening results. It detects individual antibodies to HIV-specific proteins, offering a high level of specificity in confirming HIV infection, especially in cases where the initial screening test was positive.

Western Blot for HIV Diagnosis

- Principle: Uses an immunoblot technique to detect antibodies in serum against specific HIV antigenic fragments (e.g., gag, pol, env gene products). Antigen-antibody complexes appear as distinct bands on a nitrocellulose strip.

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• Antigen-antibody complexes appear as distinct bands on a nitrocellulose strip.

• Detection of HIV Antibodies: Gag gene products: p55, p40, p24, p18. Pol gene products: p65/66, p55/51, p31. Env gene products: gp120, gp160, gp41.

• Gag gene products: p55, p40, p24, p18.

• Pol gene products: p65/66, p55/51, p31.

• Env gene products: gp120, gp160, gp41.

• Criteria for Interpreting Results: WHO Criteria: Requires the presence of at least two envelope bands (gp120, gp160, or gp41) with or without gag or pol bands. CDC Criteria: Requires presence of any two of the following bands: p24, gp120, gp160, or gp41.

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• CDC Criteria: Requires presence of any two of the following bands: p24, gp120, gp160, or gp41.

• Applications: Highly specific antibody detection method. Used for validation after initial screening. Expensive, laborious and may give indeterminate results.

• Highly specific antibody detection method.

• Used for validation after initial screening.

• Expensive, laborious and may give indeterminate results.

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• Highly specific antibody detection method.

• Used for validation after initial screening.

• Expensive, laborious and may give indeterminate results.

Confirmatory Tests for HIV Diagnosis

• p24 Antigen Detection: Detectable 12–26 days post-infection, elevated in late-stage HIV/AIDS, used for diagnosis during the window period, late-stage HIV, infants, and monitoring progression.

• Viral RNA Detection (Gold Standard): Detects HIV RNA 10–14 days post-exposure, highly sensitive and specific, used for confirmation during the window period, viral load monitoring, differentiation of HIV-1 and HIV-2, and drug resistance detection.

• HIV DNA Detection: Identifies proviral DNA, which is especially useful for pediatric diagnosis and differentiating latent from active infections.

• Virus Isolation: This time-consuming method is used mainly for research, involving the co-cultivation of patient's cells with healthy donor cells to detect viral RNA or antigen.

• Non-specific Tests: CD4 T cell count assesses immune function and monitors response to antiretroviral therapy; abnormal protein markers indicate immune response to HIV.

p24 Antigen Detection (Option A): The p24 antigen test is generally used in early detection and may assist in diagnosis when antibodies are not yet detectable. However, it lacks the specificity needed for confirmatory testing, as p24 levels can fluctuate and are often undetectable once antibodies have formed.

HIV RNA Detection-RT-PCR (Option B): While extremely sensitive, it is typically not used as a confirmatory test for HIV antibodies because it focuses on viral RNA rather than antibody presence, which is what screening tests initially detect.

HIV DNA Detection (Option D) is particularly useful in pediatric cases or to identify latent infection in adults. However, like RNA detection, it is not routinely used for confirmatory antibody testing following a positive HIV screening result in adults.

Solution for Question 5:

Correct Answer: D) No further testing is needed; confirm HIV-1 infection.

Explanation:

The combination of a positive fourth-generation HIV-1/2 immunoassay and a positive HIV-1 result on the differentiation immunoassay confirms an established HIV-1 infection without the need for further testing.

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CDC Guidelines for 4th Generation ELISA

Screening Assays for HIV Detection

- General Detection Methods: Detect antibodies (via ELISA/EIA, CIA, particle agglutination). Detect p24 antigen or viral nucleic acid.
- Detect antibodies (via ELISA/EIA, CIA, particle agglutination).
- Detect p24 antigen or viral nucleic acid.
- Generations of EIA: First Generation: Detects antibodies against whole-cell lysate from infected cells. Second Generation: Uses recombinant-produced HIV antigens instead of whole-cell lysate. Third Generation: Utilizes the "antigen sandwich" method to detect both IgG and IgM, allowing for earlier antibody detection. Fourth Generation: Coats wells with p24 antibodies and HIV-1 antigens; detects p24 antigen or antibodies with enzyme-labeled antibodies. Fifth Generation: Quantifies HIV antibodies and p24 separately.
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- Fourth Generation: Coats wells with p24 antibodies and HIV-1 antigens; detects p24 antigen or antibodies with enzyme-labeled antibodies.
- Fifth Generation: Quantifies HIV antibodies and p24 separately.

Order a Western Blot for HIV-1 to confirm antibodies (Option A): Incorrect, as Western Blot is no longer recommended for confirmation; the HIV-1 differentiation immunoassay has replaced it in current protocols.

Perform an HIV-1 NAT (Nucleic Acid Test) (Option B): Incorrect, since NAT is generally used to confirm acute infection when initial results are inconclusive, which isn’t the case here.

Perform a repeat fourth-generation immunoassay (Option C): Incorrect, as repeating the initial screening test isn’t necessary when the antibody differentiation assay has already confirmed

HIV-1 antibodies.

Solution for Question 6:

Correct Answer: B) The patient meets the criteria for AIDS diagnosis based on major signs only.

Explanation:

According to the WHO AIDS case definition, a diagnosis can be made with at least two major signs and one or more minor signs. This patient meets the major sign criteria with weight loss >10%, chronic diarrhoea>1 month, and prolonged fever >1 month. However, he does not display any minor signs, so he meets the criteria based only on major signs.

CDC-AIDS defining conditions:

- Candidiasis of the bronchi, trachea, oesophagal or lungs.
- Cryptosporidiosis with diarrhoea (for longer than 1 month)
- Cytomegalovirus disease (other than liver, spleen, or nodes), beginning when older than one month
- Herpes simplex: chronic ulcers (lasting longer than 1 month); or bronchitis, pneumonitis, or esophagitis (beginning when older than one month)
- Kaposi sarcoma
- Lymphoma, primary, of brain
- Mycobacterium avium complex (type of bacterial infection) or M. kansasii, disseminated or extrapulmonary
- Pneumocystis jirovecii pneumonia (PCP)
- Progressive multifocal leukoencephalopathy
- Toxoplasmosis of the brain, beginning when older than one month

WHO AIDS definition

Major signs (2 signs or more)	Minor signs (1 or more)

Weight loss of at least 10% of body weight Chronic diarrhoea for >1 month Prolonged fever for >1 month (intermittent or constant)	Persistent cough for >1 month Generalized pruritic dermatitis History of herpes zoster Oropharyngeal candidiasis Chronic progressive or disseminated herpes virus infection Generalized lymphadenopathy.

The patient does not meet the criteria for AIDS diagnosis (Option A): Incorrect, as the patient does meet the criteria with three major signs.

The patient meets the criteria for AIDS diagnosis based on minor signs only (Option C): Incorrect, since the patient has no minor signs and meets only major criteria.

The patient meets the criteria for AIDS diagnosis based on both major and minor signs (Option D): Incorrect, as the patient has major signs but does not display any minor signs.

Solution for Question 7:

Correct Answer: B) 1-c, 2-b, 3-d, 4-a

Explanation:

CD4 count	Opportunistic Infection
<500	Tuberculosis, Bacterial pneumonia, Herpes zoster, oropharyngeal candidiasis, non-Typhoid Salmonellosis, Kaposi sarcoma, non-Hodgkin lymphoma.
<200	Pneumocystis Jiroveci Pneumonia, Herpes Simplex Ulcer, Oesophogeal Candidiasis, Isospora Bell diarrhoea, HIV wasting syndrome, HIV-associated dementia.
<100	Cerebral Toxoplasmosis, Cryptococcal meningitis, Cryptosporidiosis, Microsporidiosis, CMV, MAC, PMLE
<50	CMV, MAC, Toxoplasma gondi

Solution for Question 8:

Correct Answer: C) Group III

Explanation:

The patient has palpable lymphadenopathy that has persisted for over three months without significant symptoms, which fits the criteria for Group III- Persistent generalized lymphadenopathy.

Group	Description
Group I	Acute infection Defined by a mononucleosis-like syndrome with seroconversion for antibodies. (Option A ruled out) Antibody seroconversion is essential; viral isolation methods lack sensitivity.
Group II	Asymptomatic infection No signs or symptoms of infection; no prior classification in Groups III or IV. (Option B ruled out) Laboratory evaluations, including blood counts, are key for subclassification.
Group III	Persistent generalized lymphadenopathy Palpable lymphadenopathy (≥1 cm) at two or more extra-inguinal sites for over 3 months. (Option C) Laboratory evaluations can subclassify patients.
Group IV	Other disease Classified into subgroups (A-E) independent of lymphadenopathy presence. (Option D ruled out) Includes a range of symptoms, with potential for further specificity in subgroups.
Subgroup A	Constitutional disease
Subgroup B	Neurologic disease
Subgroup C	Secondary infectious diseases
Category C-1	Specified secondary infectious diseases listed in the CDC surveillance definition for AIDS**
Category C-2	Other specified secondary infectious diseases
Subgroup D	Secondary cancers**
Subgroup E	Other conditions

Solution for Question 9:

Correct Answer: D) The CCR5 Delta 32 mutation can confer resistance to HIV in homozygous individuals.

Explanation:

Individuals who are homozygous for the CCR5 Delta 32 mutation have a genetic resistance to HIV, making it difficult for the virus to enter their cells.

Modes of Transmission of HIV

- Sexual Transmission: Most common (75% of cases). Heterosexual Route: Male to female via vaginal intercourse is typical; risk per encounter is low (0.1–1%). Higher risk with anal intercourse.
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- Heterosexual Route: Male to female via vaginal intercourse is typical; risk per encounter is low (0.1–1%). Higher risk with anal intercourse.
- Blood Transfusion: Least common (5% of cases) but has a high transmission risk (90–95%). (Option A ruled out)
- Percutaneous/Mucosal Transmission: Includes needle stick injuries and sharing razors; less effective compared to sexual transmission.
- Perinatal Transmission: 20–40% risk of mother-to-fetus transmission without intervention; highest risk during delivery.
- Casual Contact: No evidence of transmission through casual contact, kissing, or insect bites.
- Viral Load: Highest in blood, genital secretions, and cerebrospinal fluid; minimal in saliva and other body fluids. (Option C ruled out)
- Saliva: Contains substances that may inhibit HIV transmission.
- Most common (75% of cases).
- Heterosexual Route: Male to female via vaginal intercourse is typical; risk per encounter is low (0.1–1%). Higher risk with anal intercourse.

Pathogenesis:

- Receptor Attachment Main Receptor: HIV binds its gp120 to the CD4 receptor on host cells, primarily expressed on helper T cells and other cells like monocytes and macrophages. (Option B ruled out) Co-Receptors: Fusion requires a second receptor: CXCR4: Found on T lymphocytes. CCR5: Present on macrophage lineage cells. DC-SIGN: A dendritic cell-specific receptor that binds HIV-1 but does not mediate entry; facilitates transport to

lymphoid organs. CCR5 Delta 32 Mutation: (Option D) Mutation can block HIV entry:Homozygous individuals are resistant to HIV (1% of Northern Europeans). Heterozygousindividuals may have delayed progression to AIDS (10–15% of Europeans).

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Solution for Question 10:

Correct Answer: B) a- 1,4,5 ; b- 2,3,6,7

Explanation:

HIV has three main structural genes and it also contains six non-structural or regulatory genes that play roles in viral replication and pathogenesis.

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Solution for Question 11:

Correct Answer: C) The capsid has an complex symmetry and contains core proteins.

Explanation:

HIV is a spherical virus and its capsid has an icosahedral symmetry also contains two copies of single-stranded positive-sense RNA in its core.

Structure of HIV

• HIV is a spherical virus measuring 80–110 nm in diameter and belongs to the lentivirus family. Its structure includes:

• Envelope: Composed of: (Option A ruled out) Lipid Component: Derived from the host cell membrane. Protein Component: gp120: Glycoprotein forming knob-like spikes on the surface. gp41: Glycoprotein that acts as anchoring transmembrane pedicles.

• Lipid Component: Derived from the host cell membrane.

• Protein Component: gp120: Glycoprotein forming knob-like spikes on the surface. gp41: Glycoprotein that acts as anchoring transmembrane pedicles.

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• gp41: Glycoprotein that acts as anchoring transmembrane pedicles.

• Nucleocapsid: The capsid has an icosahedral symmetry and contains core proteins. (Option C) Inside, there is a dense cylindrical inner core that houses: RNA: Two identical copies of single-stranded positive-sense linear RNA. Viral Enzymes: Including reverse transcriptase, integrase, and proteases, which are closely associated with the RNA.

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• RNA: Two identical copies of single-stranded positive-sense linear RNA.

• Viral Enzymes: Including reverse transcriptase, integrase, and proteases, which are closely associated with the RNA.

HIV Serotyping

• Serotypes: HIV is classified into two main serotypes: HIV-1 and HIV-2.

• Divided into groups M, N, O, and a proposed group P.

• Group M: Dominant globally with eleven subtypes (A-K) and circulating recombinant forms (CRFs). (Option B ruled out)

• Quasispecies: Hosts may have multiple closely related viral subtypes or CRFs.

• Geographical Distribution: Subtype A: Common in West Africa. Subtype B: Predominant in Europe, America, Japan, and Australia. Subtype C: Most common worldwide (47%); dominant in Southern/Eastern Africa, India, and China. Cameroon: High diversity and possible origin of the virus.

• Subtype A: Common in West Africa.

• Subtype B: Predominant in Europe, America, Japan, and Australia.

• Subtype C: Most common worldwide (47%); dominant in Southern/Eastern Africa, India, and China.

• Cameroon: High diversity and possible origin of the virus.

• Transmission: Subtypes C and E are more often transmitted heterosexually. Subtype B spreads primarily through blood and homosexual contact.

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• Cameroon: High diversity and possible origin of the virus.

• Subtypes C and E are more often transmitted heterosexually.

• Subtype B spreads primarily through blood and homosexual contact.

2.HIV-2: Comprises eight subtypes (A–H), mainly found in Africa and parts of India, with Group A being the most common. (Option D ruled out)

Solution for Question 12:

Correct Answer: A) NS1 Antigen Detection

Explanation:

The patient's recent travel to a dengue-endemic area, combined with high fever, severe headache, and joint pain, suggests dengue fever. The presence of mild abdominal tenderness and leukopenia further supports the diagnosis, indicating potential complications associated with the disease.

NS1 Antigen Detection is the most appropriate test at this stage, as it can be detected from day 1 of fever and is specific to dengue. It helps in early diagnosis before antibodies have developed.

Dengue Fever:

Investigations:

- IgM (Primary Infection): Appears after 5 days; MAC-ELISA is the recommended test.
- IgG (Secondary Infection): Rapid rise; may cross-react with other flaviviruses.

Treatment:

- There is no specific antiviral therapy; supportive care includes plasma replacement, correction of electrolyte imbalances, and platelet transfusions as needed.
- Outpatient Management (no warning signs): Administer acetaminophen. Encourage sufficient oral fluid intake.
- Administer acetaminophen.
- Encourage sufficient oral fluid intake.
- Inpatient Management (with warning signs or severe dengue): Initiate IV crystalloids; adjust the fluid rate based on patient response. For shock not responding to crystalloid boluses, consider colloids.
- Initiate IV crystalloids; adjust the fluid rate based on patient response.
- For shock not responding to crystalloid boluses, consider colloids.
- Blood Transfusion: Indicated in cases of severe or suspected bleeding if unstable despite fluid resuscitation and hematocrit falls. Consider platelet transfusion if the platelet count is below 20,000 cells/µL with high bleeding risk.
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- Encourage sufficient oral fluid intake.
- Initiate IV crystalloids; adjust the fluid rate based on patient response.
- For shock not responding to crystalloid boluses, consider colloids.
- Indicated in cases of severe or suspected bleeding if unstable despite fluid resuscitation and hematocrit falls.
- Consider platelet transfusion if the platelet count is below 20,000 cells/µL with high bleeding risk.

Vaccine (CYD-TDV - Dengvaxia):

- Live-attenuated, tetravalent vaccine; indicated for ages 9-45 with prior dengue infection.
- Administered in 3 doses over six months; effective (~80%) in reducing hospitalized cases in high-prevalence regions.
- WHO advises use only in previously infected individuals due to potential risks for severe dengue in dengue-naive recipients.

IgG Antibody Detection (Option B): IgG antibodies indicate a secondary infection and appear later than the NS1 antigen. This test would not be suitable for early diagnosis in this acute phase of illness.

Virus Isolation (Option C): While virus isolation can confirm dengue infection, it is generally performed in specialized research centres and is not practical for immediate diagnosis in clinical settings.

Rapid Diagnostic Test (Option D): Although RDTs can detect IgM or NS1 antigen, they have limited sensitivity and require confirmation with ELISA. Therefore, they are less reliable for initial diagnosis compared to NS1 antigen detection

Solution for Question 13:

Correct answer: C) Nairovirus

Explanation:

Nairovirus, which causes Crimean-Congo hemorrhagic fever, is common in Central Asia, transmitted by tick bites, and is known for causing severe hemorrhagic symptoms.

Bunyaviridae		
Bunyaviridae is a large family of over 300 arbovirus species, mostly spread by mosquitoes and ticks. Structure: Virions are spherical, enveloped, and are about 100 nm in size. They have a segmented, single-stranded RNA genome. Key Genera: Four genera are medically significant—Bunyavirus, Phlebovirus, Nairovirus, and Hantavirus. Hantaviruses are rodent-borne rather than transmitted by arthropods.		
Name	Features	Transmission
Genus Bunyavirus	Encephalitis, aseptic meningitis, and fever. California encephalitis viruses are endemic to the USA. Oropouche virus causes fever with aseptic meningitis in Brazil.	California encephalitis virus - Mosquito Oropouche virus - Midges
Genus Phlebovirus	Causes sandfly fever (Pappataci fever) in Central Asia and Rift Valley fever in Africa. Fever and myalgia	Sandfly fever - Phlebotomus papatasii sandflies Rift Valley fever - Mosquitoes (Option B ruled out)
Genus Nairovirus (Option C)	Nairobi sheep disease virus and Crimean-Congo hemorrhagic fever (CCHF).	CCHF - Hyalomma ticks
Genus Hantavirus	Causes hemorrhagic fever with renal syndrome (HFRS). Hantaan virus: Severe HFRS in Asia and Russia. (Option D ruled out) Seoul virus: Worldwide mild disease Puumala virus: Mild nephropathia in Europe. Sin Nombre virus: Hantavirus pulmonary syndrome. (Option A ruled out)	Rodents

Solution for Question 14:

Correct Answer: A) Dengue

Explanation:

The given clinical manifestations of fever, headache, muscle and joint pains, and maculopapular rashes and the corresponding lab results confirm the diagnosis of Dengue fever.

Dengue virus:

- Dengue virus (DENV) is the most common arbovirus in India, belonging to the Flaviviridae family.
- It is an enveloped virus with single-stranded RNA (ssRNA).
- DENV has four serotypes (DEN-1 to DEN-4), with a fifth serotype (DEN-5) discovered in 2013 in Bangkok.

Vector:

- Vectors: Aedes aegypti (main) and Aedes albopictus; both bite during the day.
- Feeding Behavior: A. aegypti is a nervous feeder, bites multiple hosts for blood meals, and resides in domestic areas, making it more efficient.
- Infectivity: Aedes becomes infective after feeding on viremic patients (1 day before to end of the febrile period, ~5 days).
- Extrinsic Incubation: Takes 8–10 days for Aedes to become infective; remains infective for life once infected.
- Transovarial Transmission: Aedes can pass dengue virus to offspring.
- Transmission Cycle: Humans and Aedes are the main reservoirs; the four serotypes do not involve other animals, while the fifth serotype follows a sylvatic cycle.
- The sylvatic cycle of dengue involves the transmission of the dengue virus between non-human primates (monkeys) and forest mosquitoes (e.g., Aedes aegypti, Aedes albopictus) in tropical jungles. It acts as a natural reservoir, ensuring the virus's survival even during low human transmission periods. Human encroachment into forested areas can cause spillover, initiating urban transmission.

Pathogenesis:

- Primary Dengue Infection: Occurs upon first infection with any one dengue serotype.
- Secondary Dengue Infection: Results from infection with a different serotype than the one involved in the primary infection.

Antibody Response Against Dengue Virus:

- Dengue virus infection induces both neutralizing and non-neutralizing antibodies.

- Neutralizing Antibodies: Protective and produced against the infective serotype (lasting for life) and other serotypes (temporary cross-protection).
- Protection Duration: Lifelong protection against the infective serotype, while cross-protection to other serotypes diminishes over a few months.
- Non-Neutralizing Antibodies: Last for life and are heterotypic, produced against other serotypes but not the infective ones. During a secondary dengue infection, antibodies produced from the first serotype can bind to the second serotype but fail to neutralize it. Instead, these antibodies enhance the infection by inhibiting immune responses, preventing B cell activation against the second serotype.
- Antibody-Dependent Enhancement (ADE): This phenomenon leads to the recruitment of mononuclear cells and cytokine release, contributing to increased severity of secondary dengue infections.

Dengue fever (DF): It is characterized by:

- Abrupt onset of high fever (also called biphasic fever, break bone fever or saddleback fever)
- Maculopapular rashes over the chest and upper limbs
- Severe frontal headache
- Muscle and joint pains
- Lymphadenopathy
- Retro-orbital pain
- Loss of appetite, nausea and vomiting

Malaria (Option B): The patient's symptoms, including a maculopapular rash, retro-orbital pain, and joint pain, indicate dengue fever, supported by low platelet count and prolonged BT. In contrast, malaria presents with cyclical fever and splenomegaly.

Cholera (Option C): Cholera presents with severe watery diarrhoea, vomiting, and dehydration, but does not cause low platelet count or prolonged BT, which indicates dengue fever. Additionally, cholera is transmitted via contaminated water and food, while dengue is spread by Aedes mosquitoes.

Typhoid (Option D): Typhoid presents with sustained high fever and abdominal pain, but usually has normal platelet counts and does not cause prolonged BT, which indicates dengue fever. Additionally, typhoid is caused by Salmonella typhi and transmitted through contaminated food and water, unlike dengue, which is spread by Aedes mosquitoes.

Solution for Question 15:

Correct Answer: B) Microcephaly

Explanation:

Among the given options the complication associated with Congenital Zika syndrome (CZS) is microcephaly.

Zika virus:

Zika virus (ZIKV) is an arbovirus from the Flaviviridae family, characterized by its single-stranded RNA.

Transmission:

- ZIKV is primarily transmitted by the Aedes aegypti mosquito.
- Other transmission modes include mother-to-child (especially in the first trimester), and rarely through sexual contact, blood transfusion, and organ transplantation.

The life cycle of Zika virus

Clinical Manifestations:

- Incubation: A few days to 1 week; >80% of infections are asymptomatic.
- Zika Fever: Symptoms include mild fever, rash, and conjunctivitis.
- Congenital Zika Syndrome (CZS): Characterized by microcephaly (Option B) and neurological issues.
- Neurological Complications: Rarely includes Guillain-Barré syndrome, neuropathy, and myelitis.

Laboratory Diagnosis:

- Nucleic Acid Testing (NAT): RT-PCR is the preferred method, detecting ZIKV RNA in blood and urine up to 7 days after symptom onset.
- IgM Antibody Detection: Appears after 1 week of symptoms and can remain positive for several months; may cross-react with dengue antibodies.
- Enzyme Immunoassays (EIAs) and Immunofluorescence Assays (IFA): Used if symptoms last ≥ 7 days, utilizing viral lysate or recombinant proteins.

Congenital Rubella Syndrome:

- The most serious consequence of rubella virus infection is congenital rubella syndrome.
- Rubella is highly teratogenic and affects:
- Ear (deafness) (Option D ruled out)

- Eyes (cataract) (Option A ruled out)
- Heart (patent ductus arteriosus) (Option C ruled out)
- The severity is maximum in the first trimester.

Solution for Question 16:

Correct Answer: B) Hard ticks

Explanation:

The given clinical scenario is Monkey fever/Kyasanur forest disease, which is caused by Hard ticks. Endemic in five districts of Karnataka: Shimoga, North Kannada, South Kannada, Chikkamagaluru, and Udupi.

Monkey fever/Kyasanur forest disease:

General features:

- Virus Characteristics: Enveloped virus belonging to the family Flaviviridae; contains single-stranded RNA (ssRNA).
- Vector: Hard ticks (Haemaphysalis spinigera)
- Hosts: Monkeys, rodents, and squirrels are common hosts; monkeys act as amplifier hosts where the virus multiplies exponentially.
- Seasonality: KFD cases peak during dry months (November to June), coinciding with increased human activity in forests.

The life cycle of Kyasanur forest disease

Clinical features:

- Incubation Period: 3-8 days.
- First Stage (Hemorrhagic Fever): Acute high fever with malaise and frontal headaches. Hemorrhagic symptoms: bleeding from the nasal cavity, throat, gums, and gastrointestinal tract.
- Acute high fever with malaise and frontal headaches.
- Hemorrhagic symptoms: bleeding from the nasal cavity, throat, gums, and gastrointestinal tract.
- Second Stage: Meningoencephalitis may occur 7–21 days after the first stage.

• Acute high fever with malaise and frontal headaches.

• Hemorrhagic symptoms: bleeding from the nasal cavity, throat, gums, and gastrointestinal tract.

Lab diagnosis:

• Diagnosis Methods: Virus isolation from blood. IgM antibody detection by ELISA. RT-PCR.

• Virus isolation from blood.

• IgM antibody detection by ELISA.

• RT-PCR.

• Virus isolation from blood.

• IgM antibody detection by ELISA.

• RT-PCR.

Vaccination:

• Vaccine Type: Killed KFD vaccine (formalin-inactivated chick embryo) developed at Haffkine Institute, Mumbai.

• Vaccination Schedule: Initial two doses at 2-month intervals. Booster doses at 6–9 months, then every 5 years.

• Initial two doses at 2-month intervals.

• Booster doses at 6–9 months, then every 5 years.

• Target Area: Recommended for endemic areas of Karnataka.

• Initial two doses at 2-month intervals.

• Booster doses at 6–9 months, then every 5 years.

Vector	Diseases transmitted
Sandfly (Option A ruled out)	Kala-azar Oriental sore Sandfly fever (Papatasi fever)
Flea (Option C ruled out)	Hymenolepis diminuta Hymenolepis nana Bubonic plague Endemic typhus
Louse (Option D ruled out)	Epidemic relapsing fever Epidemic typhus Trench fever

Hard tick	Babesiosis Kyasanur Forest disease virus (Option B) Colorado tick fever virus Russian spring-summer encephalitis virus Omsk hemorrhagic fever virus Crimean Congo hemorrhagic fever virus Tularemia Lyme disease Rocky Mountain spotted fever Indian tick typhus Ehrlichiosis
Soft tick	Endemic relapsing fever
Reduvid bug	Chagas' disease

Solution for Question 17:

Correct Answer: A) Yellow fever, 17D

Explanation:

The patient's clinical features and lab investigations suggest acute viral hepatitis with hemorrhagic features, which is characteristic of Yellow Fever. Vaccination with the 17D yellow fever vaccine can prevent the disease.

Yellow fever diagnosis:

- Serology Test: IgM ELISA can be performed 3 days after symptom onset. Limitations: May yield false positives in other flavivirus infections (e.g., dengue, West Nile, Zika). False positives can occur in individuals vaccinated within the last 30 days. Confirmation: Positive results should be confirmed with a specific test (e.g., plaque-reduction neutralization test).
- Limitations: May yield false positives in other flavivirus infections (e.g., dengue, West Nile, Zika). False positives can occur in individuals vaccinated within the last 30 days.
- Confirmation: Positive results should be confirmed with a specific test (e.g., plaque-reduction neutralization test).
- Molecular Method Test: RT-PCR detects specific viral RNA (NS5 region) in blood. It is more reliable than serology. Timing: This should be performed within 10 days of symptom onset; RNA becomes undetectable beyond this period.
- Timing: This should be performed within 10 days of symptom onset; RNA becomes undetectable beyond this period.

- Limitations: May yield false positives in other flavivirus infections (e.g., dengue, West Nile, Zika). False positives can occur in individuals vaccinated within the last 30 days.
- Confirmation: Positive results should be confirmed with a specific test (e.g., plaque-reduction neutralization test).
- Timing: This should be performed within 10 days of symptom onset; RNA becomes undetectable beyond this period.

Yellow fever 17D vaccine:

- Live attenuated vaccine prepared in the allantoic cavity of chick embryos.
- Safety: No risk of encephalitis
- Cold Chain Requirements: A strict cold chain must be maintained during transport at temperatures between -30°C and +5 °C. Available in lyophilized form, requiring reconstitution before use. Typically reconstituted with physiological saline. Should be used within 30 minutes of reconstitution.
- Dosage and Efficacy: Administered as a single subcutaneous dose. Effective within 7 days of administration, with immunity lasting up to 35 years.
- Vaccine Interaction: Cholera and yellow fever vaccines should not be administered together; maintain a 3-week interval.
- Contraindications: Children under 9 months (under 6 months during epidemics) Pregnancy (except during outbreaks) Individuals with HIV People with egg allergies
- Children under 9 months (under 6 months during epidemics)
- Pregnancy (except during outbreaks)
- Individuals with HIV
- People with egg allergies
- Children under 9 months (under 6 months during epidemics)
- Pregnancy (except during outbreaks)
- Individuals with HIV
- People with egg allergies

Torres bodies: Intracytoplasmic inclusion bodies seen inside hepatocytes in yellow fever on HPE.

Measles, Edmonston Zagreb (Option B): The diagnosis is yellow fever (YF), not measles, due to factors such as YF's endemicity in South America and the patient's severe abdominal pain, jaundice, and bleeding typical of YF, whereas measles causes respiratory symptoms and a rash. The measles vaccine uses the Edmonston strain, with variations like the Moraten and Schwartz strain, underscoring that the clinical presentation aligns with YF rather than measles.

Mumps, Jeryll-Lynn (Option C): YF is endemic to South America, and the patient's severe abdominal pain, jaundice, and bleeding indicate liver dysfunction typical of YF, while mumps

primarily cause parotitis and respiratory symptoms. The mumps vaccine uses the Jeryl Lynnstrain, and the YF vaccination is specifically recommended for travellers to endemic regions,making YF the appropriate diagnosis.

Rubella, RA 27/3 (Option D): YF is endemic to South America, while rubella is not. The patient's severe abdominal pain, jaundice, and bleeding indicate liver dysfunction typical of YF, whereas rubella primarily presents with a mild rash and fever. The rubella vaccine uses the RA 27/3 strain, and the YF vaccination is recommended for travellers to endemic areas, reinforcing YF as the correct diagnosis.

Solution for Question 18:

Correct Answer: A) Aedes aegypti

Explanation: The given clinical scenario is Yellow fever (YF) and the vector responsible for transmitting the disease is Aedes aegypti.

General points:

- YF is an acute febrile illness.
- Virus Taxonomy: The YF virus is an enveloped ssRNA arbovirus in the Flaviviridae family.
- Geography: Endemic in West Africa and Central South America; not found in India.
- Genotypes: At least seven genotypes were identified, but only one serotype exists antigenically.

Transmission of YF:

- Vector: Aedes aegypti (tiger mosquito).
- Transmission Cycles: Jungle Cycle: Involves transmission between monkeys and forest mosquitoes. Urban Cycle: Involves transmission between humans and urban mosquitoes (Aedes aegypti).
- Jungle Cycle: Involves transmission between monkeys and forest mosquitoes.
- Urban Cycle: Involves transmission between humans and urban mosquitoes (Aedes aegypti).
- Jungle Cycle: Involves transmission between monkeys and forest mosquitoes.
- Urban Cycle: Involves transmission between humans and urban mosquitoes (Aedes aegypti).

Clinical Manifestations of YF:

- Incubation Period: 3-6 days.
- Early Stage (Febrile Illness): Symptoms: Fever, chills, headache, dizziness, myalgia, backache, followed by nausea, vomiting, and relative bradycardia. Viremic stage: The patient can infect mosquitoes.
- Symptoms: Fever, chills, headache, dizziness, myalgia, backache, followed by nausea, vomiting, and relative bradycardia.
- Viremic stage: The patient can infect mosquitoes.
- Severe Cases: Hemorrhagic manifestations. Platelet dysfunction. Liver involvement: Mid-zonal necrosis, Presence of councilman bodies, Jaundice. Renal dysfunction. Rarely, encephalitis.
- Hemorrhagic manifestations.
- Platelet dysfunction.
- Liver involvement: Mid-zonal necrosis, Presence of councilman bodies, Jaundice.
- Renal dysfunction.
- Rarely, encephalitis.
- Mortality Rate: High (>20%), especially among children and the elderly.
- Symptoms: Fever, chills, headache, dizziness, myalgia, backache, followed by nausea, vomiting, and relative bradycardia.
- Viremic stage: The patient can infect mosquitoes.
- Hemorrhagic manifestations.
- Platelet dysfunction.
- Liver involvement: Mid-zonal necrosis, Presence of councilman bodies, Jaundice.
- Renal dysfunction.
- Rarely, encephalitis.

Female anopheles (Option B): The symptoms indicate yellow fever, transmitted by Aedes aegypti mosquitoes, which thrive in urban settings. In contrast, Anopheles mosquitoes primarily transmit Malaria and are not involved in yellow fever transmission. The specific clinical features, such as hemorrhagic manifestations and liver necrosis, confirm the role of Aedes aegypti as the vector for this disease.

Culex (Option C): The clinical features suggest yellow fever, which is transmitted by Aedes aegypti mosquitoes, not Culex. Culex primarily transmits diseases like Japanese encephalitis and West Nile virus. Additionally, Aedes aegypti feeds during the day and breeds in clean, standing water found in artificial containers, whereas Culex mosquitoes are more nocturnal and prefer polluted, stagnant water.

Mansonia (Option D): The clinical features indicate yellow fever, transmitted by Aedes aegypti, not Mansonia. Aedes aegypti is specifically known for yellow fever transmission, while Mansonia primarily transmits filarial infections. Thus, Aedes aegypti is the vector responsible for yellow fever.

Solution for Question 19:

Correct answer: B) Live attenuated SA 14-14-2 vaccine

Explanation:

Live attenuated SA 14-14-2 vaccine is produced from the SA 14-14-2 strain and is currently used in India's National Immunization Program. Given its effectiveness, it is administered to children in JE-endemic districts across several Indian states, reducing the disease burden significantly.

- It is given at 9–12 months, with a second dose at 16–24 months.
- Another vaccine used is the Inactivated Vero cell culture-derived vaccine.

Treatment of JE: No specific antiviral drug is available. It is primarily supportive and includes hospitalisation, close monitoring for fever, seizure control and nutritional and respiratory support.

Inactivated Nakayama strain vaccine (Option A): This inactivated vaccine, based on the Nakayama strain, was previously used but has been discontinued in India.

Formalin-inactivated mouse brain vaccine (Option C): This vaccine, derived from the mouse brain, was once utilised in Japan and India but has been phased out.

Live attenuated JE-CV vaccine (Option D): While live attenuated vaccines exist for JE, only the SA 14-14-2 strain is used in India's immunization efforts.

Solution for Question 20:

Correct answer: C) Pigs

Explanation:

The clinical features of fever, confusion, seizures, and geographical information of the rural area in India and rice fields, with an age below 15 years point to probable Japanese Encephalitis infection.

Pigs are considered the main amplifier hosts for the JE virus. The virus can replicate extensively in them, resulting in high levels of viraemia. This abundance of the virus in pigs makes them critical in transmitting it to mosquitoes, spreading it to other hosts.

Two transmission cycles are predominant in Japanese encephalitis:

- Pigs→Culex mosquito→Pigs
- Ardeid birds→Culex mosquito→Ardeid birds
- Humans, when afflicted are considered to be a dead end and there is no "Man→mosquito→man" cycle

Japanese Encephalitis Virus	
Vector	Clinical features
Culex mosquitoes, particularly Culex tritaeniorhynchus and Culex vishnui (common in India). Breeding Habitats: These mosquitoes thrive in rice fields and stagnant water, especially in rural regions.	Incubation Period: Typically 5-15 days. Many infections are asymptomatic or subclinical. Stages of Disease: Prodromal Stage: Initial fever and flu-like symptoms. Acute Encephalitis Stage: High fever, neurological symptoms (confusion, seizures, stiff neck). Late Stage and Sequelae: Potential for full recovery or long-term neurological complications. Age Distribution: Primarily affects children under 15, significantly impacting young children. Geographical Distribution: Endemic across South East Asia

JE Life cycle

Humans (Option A): Humans act as incidental and dead-end hosts for the JE virus. The virus does not replicate to levels sufficient for transmission through mosquitoes, meaning infected individuals do not contribute to the spread of the virus.

Cattle (Option B): Cattle can become infected with the JE virus, however, they do not play a role in the virus's natural transmission cycle and are not amplifier hosts.

Birds (Option D): Ardeid birds, such as herons and egrets, are part of the natural transmission cycle of the JE virus, as they can become infected and allow mosquitoes to acquire the virus. However, they are not the primary amplifier hosts.

Solution for Question 21:

Correct answer: C) IgM Antibody Detection

Explanation:

IgM Antibody Detection is most helpful for diagnosing active rubella infection, as the presence of IgM antibodies indicates a recent infection. In pregnant women, a positive IgM result raises concerns about potential congenital rubella syndrome in the foetus, making it a primary diagnostic tool.

RT PCR is also another method for detecting rubella-specific RNA.

Vaccine and Prophylaxis

- Vaccine Type: Live attenuated vaccines - RA 27/3 strain is standard for rubella immunization.
- Administration: Given subcutaneously, either alone or as part of the MMR (measles, mumps, rubella) vaccine.
- Target Populations: All children should be vaccinated to prevent rubella spread. Women of childbearing age should be prioritized for vaccination to avoid congenital rubella syndrome (CRS).
- Contraindications: Pregnant women should not be vaccinated due to risks to the fetus. Pregnancy should be avoided up to 3 months after vaccination. Immunocompromised individuals should avoid vaccination due to potential complications.
- Pregnant women should not be vaccinated due to risks to the fetus. Pregnancy should be avoided up to 3 months after vaccination.
- Immunocompromised individuals should avoid vaccination due to potential complications.
- Pregnant women should not be vaccinated due to risks to the fetus. Pregnancy should be avoided up to 3 months after vaccination.
- Immunocompromised individuals should avoid vaccination due to potential complications.

IgG Avidity Test (Option A): This test assesses the strength of binding between IgG antibodies and rubella antigens, helping to differentiate between recent and past infections, the older the Ab, higher the avidity. However, it is not the first-line test for diagnosing an acute rubella infection.

Virus Isolation (Option B): While this method can confirm rubella infection, it is not commonly employed due to the complexities and time delays involved in isolating the virus. It is more practical for specific cases, such as congenital rubella, rather than acute scenarios.

TORCH Panel (Option D): The TORCH panel screens for various congenital infections, including rubella. While it can help identify multiple infections, it is not the first choice for diagnosing current rubella infection.

Solution for Question 22:

Correct answer: B) Forchheimer spots

Explanation:

Forchheimer spots are small, pinhead-sized petechiae that may appear on the soft palate around the time the rubella rash develops. Their presence can serve as a clinical indicator to support the diagnosis of rubella in a child.

Rubella, or German measles, is a mild, self-limiting illness caused by the rubella virus from the Rubivirus genus in the Togaviridae family. It is a ss-RNA virus acquired through inhalation, the incubation period is 2-3 weeks. Symptoms generally include low-grade fever, rash, upper respiratory symptoms and lymphadenopathy.

Blueberry muffin rash (Option A): This rash is associated with congenital rubella syndrome, which presents in newborns as petechiae, particularly those that resemble blueberries, due to thrombocytopenia. It is not a characteristic finding in older children.

Image showing Blueberry muffin rash

Ocular defects (Option C): Ocular issues su6ts can be seen in congenital rubella syndrome but are not relevant to diagnosing rubella in postnatal cases.

Patent ductus arteriosus (Option D): This cardiac condition is a potential complication of congenital rubella syndrome, which arises due to maternal infection during pregnancy. It does not relate to the clinical features seen in postnatal rubella.

Solution for Question 23:

Correct answer: C) Detection of viral RNA by RT-PCR

Explanation:

The clinical features of fever, joint pain in small joints, and rash point to chikungunya and it is best confirmed by RT-PCR.

Chikungunya virus		
Vector	Clinical Manifestation	Diagnosis
Chikungunya virus belongs to A the Togaviridae family. Transmitted by Aedes aegypti, which are day-biting insects often found in human dwellings.	Incubation Period: Around 5 (3–7) days. High fever and severe joint pain (arthralgia) causing a stooped posture. Other symptoms: headache, muscle pain, skin rashes, and tenosynovitis. Joint pain is intense, affecting small joints like wrists and ankles and it can persist for months.	Viral Isolation: Viruses can be isolated from mosquito cell lines, but are only done in specialized labs. Serum Antibodies: IgM antibodies: Appear around 4 days after infection, lasting up to 3 months. (Option B ruled out) IgG antibodies: Detected after 2 weeks and persist for years, significant when there is a fourfold rise. (Option A ruled out) Molecular Diagnosis: RT-PCR is effective for detecting viral RNA. (Option C)

Togaviridae Family:

- Alphavirus: Includes arboviruses transmitted by mosquitoes, such as chikungunya, O'nyong-nyong, Mayaro, Sindbis, Ross River, and various equine encephalitis viruses.
- Rubivirus: Contains the rubella virus, known for causing congenital rubella syndrome.
- Pestivirus: Primarily infects animals (e.g., mucosal disease virus, hog cholera virus) and does not typically affect humans.

Positive blood culture for chikungunya virus (Option D): Chikungunya virus is not usually grown or detected in routine blood cultures, as this technique is designed to identify bacteria, not viruses.

Other RNA viruses and COVID-19

1. A 35-year-old man from a rural area presents with fever, malaise, and bleeding gums. He reports recent exposure to rodent droppings in his home. Which virus is most likely responsible for his symptoms?

A. Astrovirus

B. Rotavirus

C. Lassa fever virus

D. Coronavirus

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2. A 2 y/o child presents with acute onset of watery diarrhoea, vomiting, and fever. Given the symptoms and risk of dehydration, stool analysis is conducted to identify the causative virus. What shape would likely be observed in the virus detected from this child's stool sample?

A. Bullet-shaped

B. Brick-shaped

C. Wheel-shaped

D. Helical

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3. A healthcare worker who recently returned from the Democratic Republic of Congo develops fever, severe muscle pain, and sore throat. A few days later, symptoms progress to abdominal pain, vomiting, and a maculopapular rash. What is the likely morphology of the virus expected in this case?

A. Icosahedral and spherical

B. Filamentous and pleomorphic

C. Spherical with envelope

D. Brick-shaped

---

4. Match each virus to its corresponding category of disease. Choose the correct option from the choices below. Viruses Categories Coronavirus Crimean Congo haemorrhagic fever virus Rotavirus Arenavirus Hantavirus Calicivirus Ebola Norwalk virus A. Hemorrhagic Fever B. Gastroenteritis

Viruses	Categories
Coronavirus Crimean Congo haemorrhagic fever virus Rotavirus Arenavirus Hantavirus Calicivirus Ebola Norwalk virus	A. Hemorrhagic Fever B. Gastroenteritis

A. A: 1, 4, 5, 7 | B: 2, 3, 6, 8

B. A: 2, 4, 5, 7 | B: 1, 3, 6, 8

C. A: 1, 2, 5, 6 | B: 3, 4, 7, 8

D. A: 3, 5, 7, 8 | B: 1, 2, 4, 6

---

5. A 68 y/o man with a history of diabetes and hypertension presents with fever, shortness of breath, and ground-glass peripheral opacities on chest imaging as shown in image below. Which mechanism most likely explains the severe respiratory symptoms he is experiencing?

A. Viral attachment to ACE-2 receptors

B. Increased pulmonary surfactant production

C. Enhanced membrane glycoprotein activity

D. Cytokine storm causing alveolar and endothelial damage

6. A 70 y/o woman with a history of diabetes presents with fatigue, decreased alertness, diarrhoea, and loss of appetite but no fever or cough. Her oxygen saturation (SpO■) is 92% on room air. What is the most appropriate classification of her COVID-19 illness severity?

A. Mild

B. Moderate

C. Severe

D. Asymptomatic

7. A 55 y/o man presents to the ED with a high fever, cough, and difficulty breathing for the past week. He has a history of COPD. The physician orders a chest X-ray and CT scan to assess his lung condition. The chest X-ray of the patient is given below. What are the expected CT findings commonly associated with viral pneumonia in this patient?



A. Isolated pleural effusion

B. Ground glass opacities and consolidation in a patchy peripheral distribution

C. Nodular lesions with cavitation

D. Widespread bronchiectasis

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8. A 70 y/o patient presents to the ER with severe respiratory distress. Blood tests were obtained. Additionally, a viral transport medium is prepared for potential viral testing, containing HBSS with Ca and Mg ions, gentamycin, amphotericin B, and fetal bovine serum, stored at 4°C. Which of the following components does NOT correlate with the blood findings observed in this patient?

A. Decreased IL-6 Levels

B. Elevated D-dimer

C. Raised Serum Ferritin

D. Increased C-reactive Protein

---

9. Which of the following statements is FALSE regarding COVID-19 diagnostic methods?

A. Real-time RT-PCR can process up to 90 samples in a single run.

B. Automated systems like Truenat and CBNAAT provide results in 4-5 hours.

C. Antigen detection tests are highly specific but have moderate sensitivity.

D. Real-time RT-PCR detects specific gene targets for screening and confirmation.

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	2
Question 5	4
Question 6	2
Question 7	2
Question 8	1
Question 9	2

Solution for Question 1:

Correct Answer: C) Lassa fever virus

Explanation: Lassa fever virus, part of the Arenaviridae family, is transmitted through rodent exposure and is known to cause hemorrhagic fever, fitting the scenario.

- Organisms & Diseases: Includes Lymphocytic choriomeningitis virus, Lassa fever virus, and South American hemorrhagic fever viruses. It is known to cause severe hemorrhagic illness in humans.
- Includes Lymphocytic choriomeningitis virus, Lassa fever virus, and South American hemorrhagic fever viruses.
- It is known to cause severe hemorrhagic illness in humans.
- Structure: Spherical or pleomorphic, 50–300 nm in size, with a unique sandy appearance due to ribosome-like particles. Composed of single-stranded, negative-sense RNA in two circular segments; has a helical structure.

• Spherical or pleomorphic, 50–300 nm in size, with a unique sandy appearance due to ribosome-like particles.

• Composed of single-stranded, negative-sense RNA in two circular segments; has a helical structure.

• Vector: Primarily infects rodents as their natural host.

• Primarily infects rodents as their natural host.

• Reservoir: Rodents serve as the primary reservoir

• Rodents serve as the primary reservoir

• Transmission: Occurs rarely in humans, often through contact with rodent excretions or materials contaminated by infected rodents.

• Occurs rarely in humans, often through contact with rodent excretions or materials contaminated by infected rodents.

• Diagnosis: Diagnosed through clinical recognition of hemorrhagic illness symptoms and laboratory testing specific to the virus type like RT-PCR and ELISA

• Diagnosed through clinical recognition of hemorrhagic illness symptoms and laboratory testing specific to the virus type like RT-PCR and ELISA

• Includes Lymphocytic choriomeningitis virus, Lassa fever virus, and South American hemorrhagic fever viruses.

• It is known to cause severe hemorrhagic illness in humans.

• Spherical or pleomorphic, 50–300 nm in size, with a unique sandy appearance due to ribosome-like particles.

• Composed of single-stranded, negative-sense RNA in two circular segments; has a helical structure.

• Primarily infects rodents as their natural host.

• Rodents serve as the primary reservoir

• Occurs rarely in humans, often through contact with rodent excretions or materials contaminated by infected rodents.

• Diagnosed through clinical recognition of hemorrhagic illness symptoms and laboratory testing specific to the virus type like RT-PCR and ELISA

Astrovirus (Option A) generally causes mild gastroenteritis, primarily in children, and does not lead to hemorrhagic symptoms.

Rotavirus (Option B) is associated with gastroenteritis in children and does not present with hemorrhagic symptoms or rodent transmission.

Coronavirus (Option D) mainly causes respiratory symptoms and is not typically linked to hemorrhagic fever or rodent transmission.

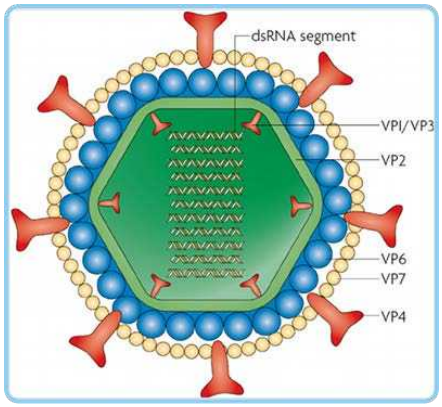
Solution for Question 2:

Correct Answer: C) Wheel-shaped

Explanation: The symptoms of watery diarrhoea, vomiting, and fever, especially in a young child, strongly suggest rotavirus infection, a leading cause of pediatric gastroenteritis. Rotavirus is distinctively wheel-shaped, making it the expected appearance in stool analysis for this clinical presentation.

- Leading cause of childhood diarrhoea; belongs to the Reoviridae family (the only family with double-stranded RNA).

Structure:



- Size: 60–80 nm, icosahedral symmetry with a triple-layered, wheel-shaped capsid.
- Genome: Segmented dsRNA (11 segments).
- Proteins: Structural (VP1–VP7, except VP5); VP6 is group-specific. VP7 (outer capsid) and VP4 (spikes) induce neutralizing antibodies.
- Structural (VP1–VP7, except VP5); VP6 is group-specific.
- VP7 (outer capsid) and VP4 (spikes) induce neutralizing antibodies.
- Structural (VP1–VP7, except VP5); VP6 is group-specific.
- VP7 (outer capsid) and VP4 (spikes) induce neutralizing antibodies.

Classification:

- Traditional: Seven groups (A–G), with Group A causing most human diarrhoea.

Epidemiology:

- The major cause of pediatric gastroenteritis worldwide is outbreaks, which are common in areas with poor sanitation and seasonally in temperate zones.

Pathogenesis:

- Transmission: Fecal-oral route.
- Action: Infects small intestine enterocytes, causing secretory diarrhoea.
- NSP4 protein acts as an enterotoxin, altering epithelial function.

Clinical Manifestations:

- Incubation: 1–3 days.
- Manifestations: Vomiting, watery diarrhoea, fever, and abdominal pain. Severe dehydration risk in children.

Diagnosis:

- Specimen: Stool collected early in the illness.
- Tests: Immunoelectron microscopy (IEM): Wheel-shaped appearance. Antigen detection: ELISA, latex agglutination. RT-PCR: Most sensitive method. G serotype and P genotype detection via RNA typing.
- Immunoelectron microscopy (IEM): Wheel-shaped appearance.
- Antigen detection: ELISA, latex agglutination.
- RT-PCR: Most sensitive method.
- G serotype and P genotype detection via RNA typing.
- Immunoelectron microscopy (IEM): Wheel-shaped appearance.
- Antigen detection: ELISA, latex agglutination.
- RT-PCR: Most sensitive method.
- G serotype and P genotype detection via RNA typing.

Vaccines:

- Rotavac: Live attenuated G9P[10] strain; three oral doses at 6, 10, and 14 weeks.
- Rotarix: Live attenuated G1P[8] strain; two doses at 6 and 10 weeks.

Bullet-shaped (Option A): The bullet shape is characteristic of the rabies virus, which does not cause gastroenteritis and is not detected in stool samples.

Brick-shaped (Option B): The brick shape is associated with poxviruses, such as the smallpox virus, which does not cause diarrhoea.

Helical (Option D): A helical structure is typical of viruses like the influenza virus, which does not cause gastrointestinal symptoms.

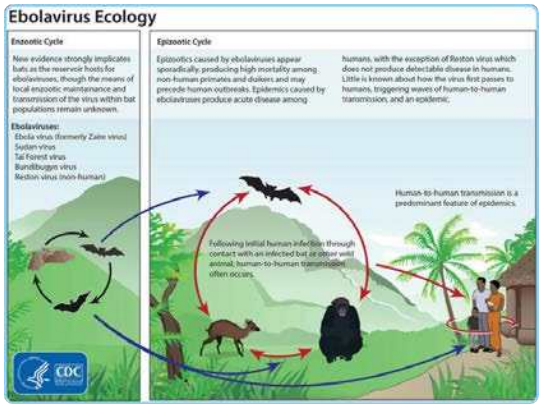
Solution for Question 3:

Correct Answer: B) Filamentous and pleomorphic

Explanation: The clinical presentation of progressive hemorrhagic symptoms, including rash and gastrointestinal signs and recent exposure in the DRC, aligns with the Ebola virus, known for its filamentous and pleomorphic morphology, especially in high-risk individuals like healthcare workers.

Ebola Virus

- Family: Belongs to the Filoviridae family. Causes African hemorrhagic fever with high mortality.
- Belongs to the Filoviridae family.
- Causes African hemorrhagic fever with high mortality.
- Morphology: Pleomorphic, filamentous threads, 80–1000 nm (average 805 nm)
- Pleomorphic, filamentous threads, 80–1000 nm (average 805 nm)
- Lifecycle & Transmission:
- Belongs to the Filoviridae family.
- Causes African hemorrhagic fever with high mortality.
- Pleomorphic, filamentous threads, 80–1000 nm (average 805 nm)



- Reservoir hosts: Suspected to be fruit bats or primates.
- Initial Transmission: Close contact with blood or body fluids of infected animals (e.g., bats, monkeys).

• Human-to-Human Spread: Direct contact with blood, bodily fluids, or contaminated surfaces/materials.

• Risk Groups: Health-care workers, close contacts, and family members of infected individuals.

• The Ebola virus can remain in semen for up to 3 months.

• Clinical Features: Incubation Period: 2–12 days. Early Symptoms: Fever, headache, muscle pain, and sore throat. Progressive Symptoms: Abdominal pain, vomiting, severe diarrhoea, and maculopapular rash. Severe Cases: Shock, petechiae, ecchymosis, and high fatality (~50%;).

• Incubation Period: 2–12 days.

• Early Symptoms: Fever, headache, muscle pain, and sore throat.

• Progressive Symptoms: Abdominal pain, vomiting, severe diarrhoea, and maculopapular rash.

• Severe Cases: Shock, petechiae, ecchymosis, and high fatality (~50%;).

• Diagnosis: Antibody Detection: ELISA for IgM and IgG; immunofluorescence. Antigen Detection: Capture ELISA for NP, VP40, and GP proteins; immunohistochemical staining. Molecular Testing: RT-PCR for viral RNA; detectable after 3 days of fever onset.

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• Molecular Testing: RT-PCR for viral RNA; detectable after 3 days of fever onset.

• Treatment & Prevention: Treatment: Supportive care (rehydration, symptomatic management). Inmazeb (atoltivimab, maftivimab, and odesivimab-ebgn) – FDA-approved as the first monoclonal antibody treatment for Zaire ebolavirus in adult and pediatric patients. Preventive Measures: Infection control, PPE (e.g., N95 masks, coveralls), patient isolation, avoiding contact with infected materials, and 21-day monitoring post-exposure.

• Treatment: Supportive care (rehydration, symptomatic management).

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Icosahedral and spherical (Option A): This morphology is typical of viruses like adenoviruses, which do not cause hemorrhagic fever.

Spherical with envelope (Option C): While common in respiratory viruses (e.g., influenza), this morphology is unrelated to hemorrhagic fevers.

Brick-shaped (Option D): Seen in poxviruses, which cause skin lesions but not hemorrhagic fever symptoms like those described.

Solution for Question 4:

Correct Answer: B) A: 2, 4, 5, 7 | B: 1, 3, 6, 8

Explanation: Crimean Congo haemorrhagic fever virus, Arenavirus, Hantavirus, and Ebola are associated with Hemorrhagic Fever; Coronavirus, Rotavirus, Calicivirus, and Norwalk virus are linked to Gastroenteritis.

Viruses Causing Hemorrhagic Fever

• Arenavirus

• Bunyavirus Crimean Congo haemorrhagic fever virus Rift valley fever Hantavirus

- Crimean Congo haemorrhagic fever virus
- Rift valley fever
- Hantavirus
- Flavivirus Dengue Yellow fever KFD
- Dengue
- Yellow fever
- KFD
- Filovirus Ebola Marburg virus
- Ebola
- Marburg virus
- Crimean Congo haemorrhagic fever virus
- Rift valley fever
- Hantavirus
- Dengue
- Yellow fever
- KFD
- Ebola
- Marburg virus

Viruses causing Gastroenteritis

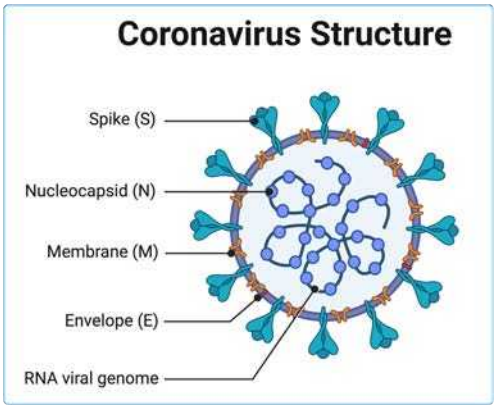
- Calicivirus
- Adenovirus
- Rotavirus
- Coronavirus
- Astrovirus
- Norwalk virus

Solution for Question 5:

Correct Answer: D) Cytokine storm causing alveolar and endothelial damage

Explanation: The diagnosis is likely COVID-19 due to the patient's symptoms and characteristic ground-glass opacities on chest imaging, which are commonly associated with COVID-19 pneumonia.

Severe COVID-19 can trigger a cytokine storm, which damages alveolar cells and endothelial tissues, leading to fluid retention, impaired gas exchange, and severe respiratory symptoms, especially in high-risk individuals.



- Type: Virus, classified as a coronavirus.
- Size: Approximately 120 nm with helical symmetry.
- Components: Nucleocapsid: Positive-sense single-stranded RNA (~30 kb) surrounded by nucleocapsid protein (N). Envelope: Lipoprotein structure with embedded proteins, host-derived.
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- Structural Proteins: Spike Protein (S): Facilitates attachment to host cells; contains two subunits: S1 Subunit: Contains receptor-binding domain (RBD) for host cell attachment. S2 Subunit: Mediates virus-cell membrane fusion. Membrane Glycoprotein (M): Most abundant, provides shape to the virus. Envelope Protein (E): Transmembrane protein with ion channel activity.
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- S2 Subunit: Mediates virus-cell membrane fusion.
- Membrane Glycoprotein (M): Most abundant, provides shape to the virus.
- Envelope Protein (E): Transmembrane protein with ion channel activity.

• Non-structural Proteins: Include enzymes like RNA-dependent RNA polymerase (RdRp) crucial for viral replication.

• Nucleocapsid: Positive-sense single-stranded RNA (~30 kb) surrounded by nucleocapsid protein (N).

• Envelope: Lipoprotein structure with embedded proteins, host-derived.

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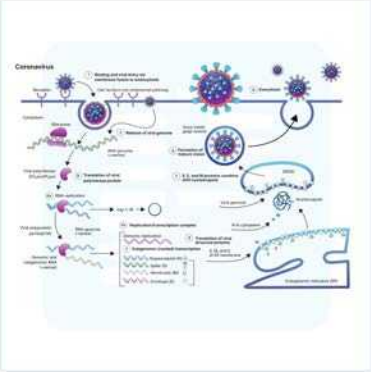
• Membrane Glycoprotein (M): Most abundant, provides shape to the virus.

• Envelope Protein (E): Transmembrane protein with ion channel activity.

• S1 Subunit: Contains receptor-binding domain (RBD) for host cell attachment.

• S2 Subunit: Mediates virus-cell membrane fusion.

Pathogenesis of COVID-19	
Host Cell Entry	Binding: Spike glycoprotein (S) attaches to angiotensin-converting enzyme-2 (ACE-2) receptors on lung type II alveolar cells. Cleavage: Host proteases (e.g., TMPRSS2) cleave the spike protein, facilitating viral entry via the endosomal pathway.
Initial Inflammatory Response	Influenza-like Illness (ILI): Infection of the pharyngeal epithelium initiates ILI.
Acute Respiratory Distress Syndrome (ARDS)	Damage to Alveolar Cells: Impaired type II alveolar cells decrease pulmonary surfactant, causing alveolar collapse and fluid retention. (Option B ruled out) Cytokine Storm: Increased immune cytokine release leads to: Tissue damage and necrosis. Impaired gas exchange and hypoxia. Endothelial damage and microvascular thrombosis. Pulmonary oedema and ground-glass opacities on imaging. Fibrosis and respiratory failure.
Multiorgan Failure	Elevated cytokines may harm other organs, resulting in sepsis, septic shock, and acute kidney/cardiac injury.
Risk Factors for Severe Disease	Demographics: Age > 60 years. Comorbidities: Diabetes, hypertension, COPD, CKD and cancer.



Viral attachment to ACE-2 receptors (Option A): While viral attachment is a crucial initial step for infection, it does not directly explain the severe respiratory symptoms observed in the patient.

Enhanced membrane glycoprotein activity (Option C): Though the membrane glycoprotein plays a role in viral structure, it is not responsible for the severe respiratory symptoms experienced by the patient.

Solution for Question 6:

Correct Answer: B) Moderate

Explanation: The patient's symptoms—fatigue, decreased alertness, and lack of typical COVID-19 fever or cough—align with atypical COVID-19 presentations in older adults. Her oxygen saturation of 92% and the presence of dyspnea fit the criteria for moderate disease (SpO■ < 94%).

- Incubation Period: Average of 5-6 days (up to 14 days).

Common Symptoms	Atypical Symptoms
Fever Cough with expectoration Fatigue Shortness of breath Myalgia Rhinorrhea (runny nose) Sore throat Diarrhea Loss of smell or taste (may precede respiratory symptoms)	Common in older adults and immunosuppressed patients: Fatigue Reduced alertness and mobility Diarrhea Loss of appetite Delirium Absence of fever Children may have less frequent fever or cough.

Mild Disease	Influenza-like illness (ILI) with mild respiratory symptoms (fever, cough, sore throat, malaise). No breathlessness or hypoxia. (Option A ruled out)
Moderate Disease	Pneumonia without severe signs: Dyspnea, fever, cough. Hypoxia (SpO2 < 94%); respiratory rate ≥ 24/min.
Severe Disease (SARI)	Severe Pneumonia: Respiratory distress (respiratory rate > 30/min or SpO2 < 90%). (Option C ruled out) ARDS: New/worsening respiratory symptoms; bilateral opacities on imaging; PaO2/FiO2 < 300. Sepsis: Acute multiorgan dysfunction (SOFA score). Septic Shock: Persistent hypotension despite resuscitation; requires vasopressors (MAP ≥ 65 mm Hg).

Asymptomatic (Option D): Asymptomatic COVID-19 patients lack observable symptoms, while this patient presents with multiple symptoms, including fatigue and reduced SpO■, indicating she is symptomatic.

Solution for Question 7:

Correct Answer: B) Ground glass opacities and consolidation in a patchy peripheral distribution

Explanation:

In this scenario, the diagnosis is likely viral pneumonia, potentially caused by COVID-19 given the symptoms of high fever, cough, and difficulty breathing, along with the chest X-ray findings of bilateral multifocal alveolar opacities.

The expected CT findings of ground glass opacities and patchy consolidation are characteristic of viral pneumonia, particularly associated with SARS-CoV or COVID-19.

Radiographic Imaging in COVID-19

- Chest X-Ray: Typically shows bilateral multifocal alveolar opacities. May demonstrate pleural effusions. Not sensitive; 20% of patients may have no abnormalities during the illness.
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- May demonstrate pleural effusions.

• Not sensitive; 20% of patients may have no abnormalities during the illness.

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• May demonstrate pleural effusions.

• Not sensitive; 20% of patients may have no abnormalities during the illness.



• Chest Computed Tomography (CT): More sensitive than chest X-rays but not specific for COVID-19. Common findings include: Multifocal bilateral ground glass opacities. Consolidation changes, usually in a patchy peripheral distribution. No imaging findings can definitively rule in or rule out COVID-19. ACR(American College of Radiology) advises against the routine use of chest CT for screening or diagnosis.

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• ACR(American College of Radiology) advises against the routine use of chest CT for screening or diagnosis.

• Multifocal bilateral ground glass opacities.

- Consolidation changes, usually in a patchy peripheral distribution.

Isolated pleural effusion (Option A): While pleural effusions can be present, they are not the primary finding associated with viral pneumonia.

Nodular lesions with cavitation (Option C): Nodular lesions with cavitation are more indicative of bacterial infections or certain malignancies, not typical of viral pneumonia.

Widespread bronchiectasis (Option D): Bronchiectasis is a chronic condition and not a typical finding in acute viral pneumonia.

Solution for Question 8:

Correct Answer: A) Decreased IL-6 Levels

Explanation:

In this case, the 70-year-old patient presents with severe respiratory distress. The viral transport medium, containing HBSS with Ca and Mg ions, gentamycin, amphotericin B, and fetal bovine serum, is prepared for potential viral testing, mostly COVID-19, which is appropriate for preserving viral samples.

Blood Findings:

- Increased IL-6 Levels: This suggests a cytokine storm.
- Elevated D-dimer: Indicates high levels of fibrin degradation products, suggesting an underlying coagulopathy. (Option B ruled out)
- Raised Serum Ferritin: This is a marker of inflammation. (Option C ruled out)
- Severe Lymphopenia: Indicates a significant reduction in lymphocyte count.
- Increased C-reactive Protein (CRP): This serves as a marker of acute inflammation. (Option D ruled out)

Viral Transport Medium:

- HBSS with Ca, Mg ions
- Gentamycin, Amphotericin B
- Fetal Bovine Serum
- Storage is at 4■

Solution for Question 9:

Correct Answer: B) Automated systems like Truenat and CBNAAT provide results in 4-5 hours.

Explanation:

Automated systems like Truenat and CBNAAT have a quick turnaround time of 30-60 minutes, not 4-5 hours, which is the turnaround time for standard real-time RT-PCR.

Real-Time RT-PCR for COVID-19 Diagnosis

- Real-time reverse transcriptase PCR (RT-PCR) is the gold standard for ko diagnosing COVID-19.
- Turnaround Time: The average time from sample receipt to result generation is approximately 4-5 hours.
- Capacity: Can process up to 90 samples in a single run, making it efficient for frontline testing. (Option A ruled out)
- Gene Targets: (Option D ruled out) Screening (Genus Specific): Spike protein (S) Envelope protein (E) Membrane protein (M) Nucleocapsid protein (N) Confirmation (Species Specific for SARS-CoV-2): RNA-dependent RNA polymerase (RdRp) Open reading frames (ORF1a/b) N2 nucleocapsid
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- RNA-dependent RNA polymerase (RdRp)
- Open reading frames (ORF1a/b)
- N2 nucleocapsid
- Principle of Testing: Qualitative real-time PCR amplifies target genes in the specimen using a thermocycler. Fluorescence is generated when the amplicon binds to the probe, with the cycle threshold (Ct) indicating positivity. A sample is considered positive if both screening

and confirmatory genes are detected with a Ct value ≤ 40 cycles.

• Qualitative real-time PCR amplifies target genes in the specimen using a thermocycler.

• Fluorescence is generated when the amplicon binds to the probe, with the cycle threshold (Ct) indicating positivity.

• A sample is considered positive if both screening and confirmatory genes are detected with a Ct value ≤ 40 cycles.

• Detectability: NAAT can be positive as early as day 1 of symptom onset (usually 5 days post-infection) and generally declines by the third week, often becoming undetectable thereafter.

• Screening (Genus Specific): Spike protein (S) Envelope protein (E) Membrane protein (M) Nucleocapsid protein (N)

• Spike protein (S)

• Envelope protein (E)

• Membrane protein (M)

• Nucleocapsid protein (N)

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• RNA-dependent RNA polymerase (RdRp)

• Open reading frames (ORF1a/b)

• N2 nucleocapsid

• Spike protein (S)

• Envelope protein (E)

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• Nucleocapsid protein (N)

• RNA-dependent RNA polymerase (RdRp)

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• Fluorescence is generated when the amplicon binds to the probe, with the cycle threshold (Ct) indicating positivity.

• A sample is considered positive if both screening and confirmatory genes are detected with a Ct value ≤ 40 cycles.

Automated Real-Time RT-PCR

• Systems include Truenat and CBNAAT (Cartridge-based nucleic acid amplification test, e.g., GeneXpert).

• Widely accessible

• Turnaround Time: Quick results in 30-60 minutes.

• Automation: Fully automated with minimal handling, reducing biosafety hazards.

• Gene Targets: CBNAAT: E gene for screening, N2 gene for confirmation. Truenat: E gene for screening, RdRp gene for confirmation.

• CBNAAT: E gene for screening, N2 gene for confirmation.

• Truenat: E gene for screening, RdRp gene for confirmation.

• Disadvantages: Limited to 1-4 samples per run, suitable for low sample loads (24-48 samples/day).

• CBNAAT: E gene for screening, N2 gene for confirmation.

• Truenat: E gene for screening, RdRp gene for confirmation.

Antigen Detection

• Rapid chromatographic immunoassay (SD Biosensor) for qualitative detection of SARS-CoV-2 nucleocapsid protein.

• Sample Collection: Nasopharyngeal swab immersed in viral extraction buffer to inactivate the virus and release the antigen.

• Point-of-Care Testing: Conducted at the bedside within one hour; antigen stability is limited to one hour.

• Performance: Highly specific (99–100%) but moderate sensitivity (50–84%); symptomatic but negative patients should be referred for RT-PCR testing. (OptionC ruled out)

Previous Year Questions

1. Patient came to the OPD with complaints of pain in the geniital area. On examination there was a painful vesicle associated with painful Lymphadenopathy . Diagnosis?

A. HSV 2

B. MOLLUSCUM

C. HPV

D. HZV

---

2. A 33-year-old male comes with complaints of dry cough, fever and shortness of breath for 2 days along with a loss of taste and smell. He is a social worker who has been traveling a lot. What would be the investigation of choice for this patient to confirm the diagnosis of COVID-19?

A. PCR

B. ELISA

C. Serology

D. Real-time RT-PCR

---

3. Which of the following virus does need Reverse transcription for PCR?

A. RSV

B. EBV

C. CMV

D. Adenovirus

4. A patient presented with a 3 day history of fever. CBC showed a platelet count of <90,000. for a suspicion of dengue which of the following cannot be done on day 3 of infection?

A. PCR

B. NS1

C. IgM ELISA

D. Xenodiagnosis using culture in aedes mosquito

5. HIV genome needs which of the following to enter the membrane?

A. GP120

B. CCR5

C. GP41

D. CXCR4

6. A patient presents with fever and retro-orbital pain after 5 days, suggestive of dengue hemorrhagic fever. Which investigation is appropriate for confirmation?

A. NS1 antigen test

B. Viral culture

C. Circulating IGM

D. IgM ELISA

7. A pregnant woman presents with a history of painful genital ulcers during pregnancy. Her newborn also has similar ulcers. Examination of a smear reveals the presence of Tzanck cells. What is the likely diagnosis?

A. HSV2

B. VZV

C. HPV

D. Klebsiella

---

8. Match the following viruses with their corresponding nucleic acid type: 1. Rotavirus A. ssRNA 2. SARS-CoV-2 (COVID-19 virus) B. dsRNA 3. Parvovirus C. dsDNA 4. Human papillomavirus (HPV) D. ssDNA

1. Rotavirus	A. ssRNA
2. SARS-CoV-2 (COVID-19 virus)	B. dsRNA
3. Parvovirus	C. dsDNA
4. Human papillomavirus (HPV)	D. ssDNA

A. 1- C, 2- A, 3- D, 4- B

B. 1- B, 2- A, 3- D, 4- C

C. 1- B, 2- C, 3- D, 4- A

D. 1- D, 2- A, 3- B, 4- C

---

9. A 25-year-old patient presents to the hospital with symptoms of hydrophobia, which is often associated with rabies infection. The clinician decides to perform further diagnostic tests on the patient's corneal scraping to confirm the diagnosis. Which of the following tests is the most appropriate and accurate for confirming the diagnosis in this patient?

A. RT- PCR

B. Viral culture

C. Antibody detection by ELISA

D. Direct fluorescent antibody test

---

10. A 13 year old girl manifested with fever and retro-orbital headache for 2 days. Tourniquet test and NS1 Ag were positive. Which of the following come under severe criteria?

A. Leucopenia plus thrombocytopenia

B. Persistent vomiting

C. Hepatomegaly >1 cm

D. AST and ALT >1000

---

11. 18 days old baby was brought with fever and cough and has vesicles over trunk and upper limb. Mother had genital ulcers 1 yr back but they have resolved now. The biopsy of vesicles shows multinucleate giant cells. The diagnosis is :

A. Herpes simplex

B. MCV

C. HPV

D. Varicella

---

12. Which of the following has the least teratogenicity?

A. CMV

B. Chickenpox

C. HSV - 2

D. Rubella

13. HIV patient with white patch on the tongue or buccal mucosa and it can not be removed on rubbing. What is the diagnosis?

A. EBV

B. CANDIDA

C. HSV

D. HHV8

14. A 45-year-old male who had a history of domestic animal bite has come for a follow-up post 90 days after receiving rabies post-exposure prophylaxis. Now what will you use for rapid diagnosis of rabies in this case?

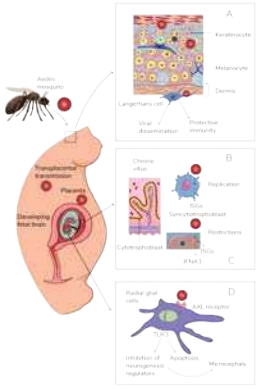
A. Skin biopsy with fluorescent antibody testing

B. Rabies virus specific antibodies

C. Inoculation in mice

D. Corneal impression smear

15. The transplacental transmission of the organism given in the image which causes microcephaly is ?



A. Nipah

B. Zika virus

C. CMV

D. Rubella

---

16. SSPE is a complication of?

A. Measles

B. Mumps

C. Rubella

D. Rabies

---

17. Which subtype of human papilloma virus has maximum chances of causing carcinoma cervix:

A. HPV 16 & 18

B. HPV 6 & 11

C. HPV 31 & 32

D. HPV 1 & 2

18. A patient is diagnosed with Hepatitis B infection. Which of the following markers will always be present in his serum, even during recurrences, or if he develops chronic hepatitis?

A. HBcAg

B. Anti-HBs Ab

C. Anti-HBc Ab

D. HBe Ag

19. What is the vector responsible for transmitting the Zika virus among the options listed?

A. Aedes australis

B. Aedes albopictus

C. Aedes mitchellae

D. Aedes polynesiensis

20. Receptor used for the entry of HIV virus into the host cell is?

A. CCR7

B. CCR5

C. CD6

D. CD8

21. Which of the following viruses are associated with the development of nasopharyngeal carcinoma?

A. Adenovirus

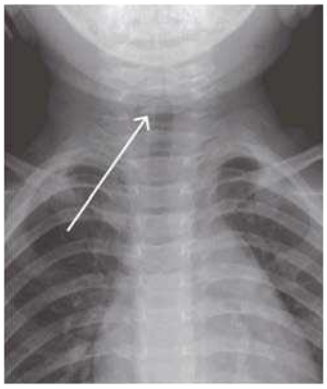
B. EBV

C. Herpes virus

D. Papillomavirus

---

22. A 5 year old boy was suffering from low grade fever, inspiratory stridor and barking cough for past 5 days. His chest X-ray is given below. What is the cause of this condition?



A. Moraxella

B. Parainfluenza

C. Ebola virus

D. Staphylococcus aureus

---

23. The following are true about chancroid : A. School of fish appearance B. Bleeds on touch C. Painless D. Cullen sign

A. C & D only

B. A, B, C & D

C. A, B & C

D. A & B only

---

24. Which of the following virus is from Herpes virus family?

A. Rubella

B. Measles

C. Rabies

D. EBV

---

25. Which of the following statements about Corona virus is incorrect?

A. ssRNA genome

B. Icosahedral capsid

C. Host cell entry occurs by attachment of viral spike

D. Human corona virus 229 E is alpha type

---

26. Molluscum contagiosum is caused by?

A. Adenovirus

B. Flavivirus

C. Rubivirus

D. Poxvirus

---

27. Which of the following virus is from Herpes virus family?

A. Rubella

B. Measles

C. Rabies

D. EBV

---

28. Antemortem diagnosis of rabies is made with?

A. Rabies virus specific antibodies

B. Inoculation in cell lines

C. Negri bodies in hippocampus

D. Corneal impression smear

---

29. Which protein is present in the HPV vaccine ?

A. E6 & E7

B. L1

C. E1& E2

D. L2

---

30. False about the Ebola virus is:

A. It is filovirus

B. Transmitted by mosquito

C. Bats acts as reservoir

D. Human-to-human transmission is seen

31. Which of the following method is employed to measure the viral load?

A. ELISA

B. Western Blot

C. Real time RT-PCR

D. Immunoelectrophoresis

32. A photographer who recently returned from Africa presents to the emergency department with abdominal pain, hepatomegaly, and hemorrhagic manifestations. He died despite treatment, and an autopsy revealed intranuclear Torres bodies in the liver. Which of the following vaccines with which strain could have prevented it?

A. 17D

B. Nakayama vaccine

C. Weigl’s vaccine

D. Jeryl Lynn strain

33. A PLHIV came with multiple dog bites with a punctured wound. Choose the correctmanagement:

A. Local wound cleaning

B. Vaccine

C. Local Treatment + RIG

D. Local Treatment + RIG + Vaccine

34. A patient sample positive for Infectious Hepatitis B. Choose the correct marker for thiscondition.

A. HBe Ag

B. HBsAg + HBe

C. HBcAg

D. HBs + Anti Hbe

---

35. A Tzanck smear prepared from a vesicle shows multinucleated giant cells. What is the diagnosis?

A. HIV

B. HSV

C. HPV

D. EBV

---

36. A patient is positive for HBsAg and anti-HBc IgM. What is the most likely diagnosis?

A. Acute hepatitis

B. Chronic hepatitis

C. Recovery phase

D. Vaccination

---

37. A patient presents with painful vesicles in the genital region. Identify the lesion shown in the image and choose the correct diagnosis:



A. Herpes

B. Chancroid

C. Syphilis

D. Candidiasis

---

38. Which of the following is the diagnostic test for monkeypox?

A. ELISA

B. Hemagglutination assay

C. PCR

D. Hybrid Capture assay

---

39. Parvo B19 associated with all except?

A. Fever with petechial rash

B. Transient aplastic crisis with hemolytic anemia

C. Aplastic anemia in immunocompromised

D. Hydrops fetalis

---

40. A 45-year-old HIV patient presents with persistent headaches, confusion, and a gradual onset of fever. A lumbar puncture is performed, revealing an elevated opening pressure and a cerebrospinal fluid analysis shows a positive india ink stain. What is the most likely diagnosis?

A. Bacterial meningitis

B. Viral meningitis

C. Cryptococcus neoformans

D. Tuberculous meningitis

---

41. Vaccine that require strain change due to evolving organism is?

A. Influenza

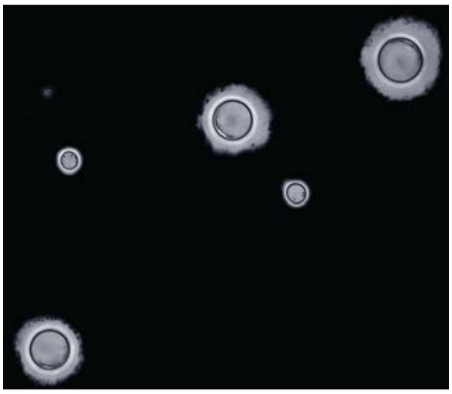
B. Diphtheria

C. Polio

D. Typhoid

---

42. Identify the fungus shown in the image given below which is stained by India ink dye.



A. Aspergillus

B. Penicillium

C. Cryptococcus

D. Candida

---

43. In the time frame of 1 to 4 months after renal transplantation, which pathogen is most commonly responsible for infecting recipients of the transplant?

A. Cytomegalovirus

B. JC virus

C. BK virus

D. Epstein-Barr virus

---

44. What is the most likely to be transmitted by a needle stick injury?

A. HIV

B. Dengue

C. Hepatitis C

D. Hepatitis B

45. A 6-months-old child presented with diarrhea and vomiting for three days. Which of the following enterotoxins is most likely responsible for the condition?

A. NSP4

B. NSP6

C. VP3

D. VP7

46. Which of the statements below is false regarding Japanese encephalitis?

A. Transmitted by Culex vishnui

B. Pig is the amplifier host

C. Man acts as the reservoir

D. Human is a dead end host

47. Which of the following statements about the influenza virus is false?

A. Secondary attack rate is 5 to 15%

B. Intranasal Live attenuated vaccine is used

C. 5-12 years is a High Risk Group

D. Aquatic birds act as reservoir

48. Which subtype of human papilloma virus has maximum chances of causing carcinoma cervix?

A. HPV 16 & 18

B. HPV 6 & 11

C. HPV 31 & 32

D. HPV 1 & 2

---

49. Which of the following is false about influenza ?

A. Intranasal live attenuated vaccine is used

B. 5-12 years is a high risk group

C. Aquatic birds act as reservoir

D. Secondary attack rate is 5-15%

---

50. Which of the following is amplified in an RT-PCR test for the COVID-19 virus?

A. RNA

B. DNA

C. Structural protein

D. Envelope

---

51. Which of the following viruses is responsible for this presentation?



A. Human herpesvirus 8

B. Human herpesvirus 4

C. Human herpesvirus 6

D. Human herpesvirus 7

---

52. A 13-year-old girl has had 4 episodes of meningitis over the last 4 years, with the disease lasting a few days at a time, slowly resolving neurological symptoms such as visual and memory loss. A CSF cytology shows lymphocytic pleocytosis, and a diagnosis of Mollaret's meningitis is made. Which organism is implicated here?

A. Measles virus

B. Mumps virus

C. Herpes simplex virus

D. Rubella virus

---

53. A 12-year-old boy is brought to the OPD with complaints of fever, ear pain, and difficulty in swallowing for 1 week. On examination, there is testicular swelling, and his jaw is shown below. What is the most likely diagnosis?



A. Croup

B. Mumps

C. Infectious mononucleosis

D. Herpangina

---

54. Condyloma Accuminata is most commonly caused by?

A. HPV 31, 33

B. HPV 6, 11

C. HPV 16, 18

D. HPV 16, 11

---

55. A 4-year-old male child was brought to your clinic with an erythematous rash on the cheek which later spread downwards to involve his trunk and limbs. He has also had a low-grade fever, headache, and mild upper respiratory tract infection for the past few days. The suspected organism contains single-stranded DNA. What is the most likely diagnosis?



A. Scarlet fever

B. Erythema infectiosum

C. Exanthema subitum

D. Rubeola

---

56. Zika virus is transmitted by:

A. Aedes aegypti

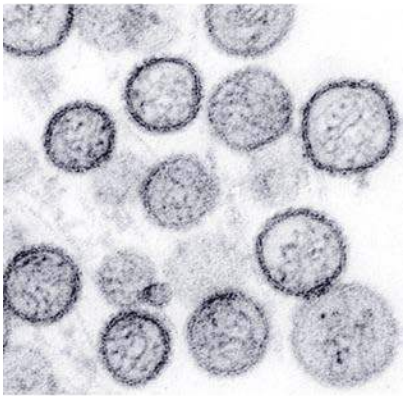
B. Culex

C. Anopheles

D. Phlebotomus

---

57. Identify the organism?



A. Influenza virus

B. Hantavirus

C. Adenovirus

D. None of above

---

58. Koplik spots are seen in?



A. Measles

B. Mumps

C. Diptheria

D. Rubella

---

59. An unimmunized 2-year-old child presented with cough, coryza, conjunctivitis, and bluish-white spots in his buccal mucosa near the lower molar teeth. A day later, he developed a maculopapular rash on the face and neck. What is the nature of the causative virus?

A. Enveloped ssRNA

B. Naked ssRNA

C. Naked dsRNA

D. Enveloped ds RNA

---

60. Match the Following: Organism Incubation Period a. Syphilis 1. 6 Days b. Chicken pox 2. 90 Days c. SARS COVID 3. 16 Days d. Hepatitis A 4. 28 Days

Organism	Incubation Period
a. Syphilis	1. 6 Days
b. Chicken pox	2. 90 Days
c. SARS COVID	3. 16 Days
d. Hepatitis A	4. 28 Days

A. a-2, b-3, c-1, d-4

B. a-3, b-4, c-2, d-1

C. a-3, b-4, c-1, d-2

D. a-1, b-4, c-2, d-3

---

61. A person is bitten by a dog. The dog and the person are fully immunized against rabies. There is a laceration on the site of the bite. What would you advise to the person?

A. Metronidazole

B. Ciprofloxacin

C. Amoxiclav

D. Observation

---

62. Which statement about rubella is incorrect?

A. It is one of the teratogenic TORCH infections

B. Causes maculopapular rash that starts on the face and spreads caudally to involve the trunk and extremities

C. It is a Flavivirus

D. Symptoms include “Blueberry muffin” appearance

---

63. A child appeared with the following appearance can present with which of the following options?



A. Aplastic crises in sickle cell disease

B. Immunodeficiency

C. Lymphoproliferative disorder

D. BK Nephropathy

64. What is the underlying factor responsible for inducing aplastic crisis in individuals with hereditary spherocytosis?

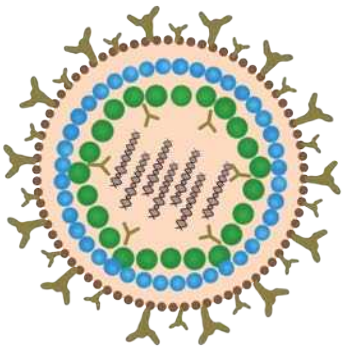
A. Poxvirus

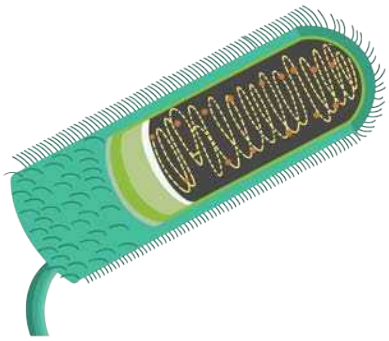
B. Ebstein barr virus

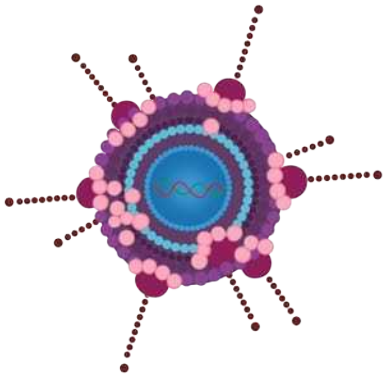
C. Adenovirus

D. Parvovirus

65. Which of the following viruses exhibits genetic re-assortment similar to that of the influenza virus?







A. 1

B. 2

C. 3

D. 4

---

66. The viruses of the filoviridae family like Ebola and Marburg resemble which of the following morphologies?

A.

B.

C.

D.

---

67. Which of the following pairs of diseases and their corresponding populations related to parvovirus is incorrect? Aplastic crises in sickle cell disease Viral envelop - Naked virus Erythema infectiosum – infants Non-erosive arthritis – Patients of SLE

A. 1

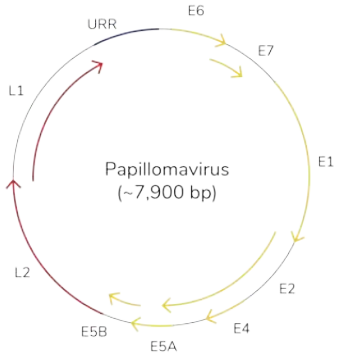
B. 2

C. 3

D. 4

---

68. The image below shows the structure of a virus. Which of the proteins of the virus act as oncogenes?



A. E1, E2, E5

B. E6, E7

C. L1, L2

D. E1, E2

---

69. Petechial rash is seen in all of the following, except?

A. Kyasanur forest disease

B. Dengue

C. Zika fever

D. Roseola infantum

---

70. Which of the following is true about anti-CMV IgG antibodies?

A. Denotes latent CMV infection

B. Denotes chronic CMV infection with immunity to other serotypes

C. Indicates acute CMV infection

D. IgG avidity assay helps in differentiating past and primary infection

---

71. Which of the following tests can be employed to validate the diagnosis of HSV encephalitis in a patient who exhibits symptoms such as high fever, altered sensorium, headache, neck rigidity, and seizures?

A. IgM in CSF

B. Tzanck smear of CSF

C. CSF PCR for viral DNA

D. CSF culture on chick embryo lines

---

72. What laboratory test is utilized to diagnose rotavirus-induced diarrhea?

A. Antigen detection in the stool sample

B. Antigen detection in blood

C. Antibody detection in serum

D. Light microscopy of stool specimen

---

73. A child presents with a fever and a rash. Urine examination showed cells with owl eye appearance. What is the most likely diagnosis?

A. Cytomegalovirus

B. Epstein-Barr virus

C. Herpes simplex virus

D. Toxoplasma gondii

---

74. A patient hailing from Delhi presents with high fever, arthralgia, retrobulbar pain and extensive petechial rash for 3 days. Lab investigations revealed a hemoglobin of 9 g/dL, a white blood cell count of 9000 cells/mm3, a platelet count of 20000 cells/mm3, and a positive tourniquet test. What is the most likely diagnosis?

A. Dengue fever

B. Dengue hemorrhagic fever

C. Dengue shock syndrome

D. Malaria

---

75. Which of the following hepatitis viruses is likely to get transmitted via feco-oral route?

A. Hepatitis A

B. Hepatitis B

C. Hepatitis C

D. Hepatitis D

---

76. A patient presented with meningitis with a history of fever for 4 days and erythematous rashes on the chest. There was no history of travel and tick bite. His CSF findings showed increased protein, normal glucose, mild lymphocytosis and few RBCs. Which of the following is the most likely organism to cause this clinical picture?

A. West nile virus

B. Neisseria meningitidis

C. Streptococcus pneumoniae

D. Enterovirus

---

77. Which of the following statements is accurate regarding antigenic drift?

A. There is a gradual sequential change in antigenic structure

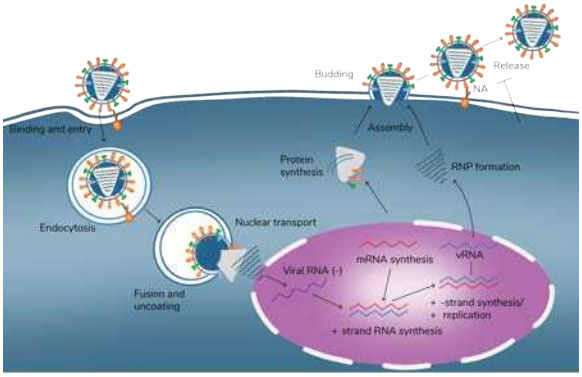
B. Causes a sudden increase in the number of cases

C. The new antigens formed are unrelated to the previous antigens

D. Antibodies to the predecessor viruses do not neutralize the new variants

---

78. Which of the following organism is represented by the given form of replication ?



A. Hepatitis virus

B. Influenza virus

C. HIV

D. Herpes virus

---

79. Choose the correct statement regarding CMV.

A. Ganciclovir resistance is due to mutation in UL97 phosphotransferase gene

B. CMV can be cultured in vitro in any mammalian cell

C. Commonest manifestation of CMV associated disease is watery diarrhea

D. UL54 mutated CMV strains do not show cross resistance to Foscarnet

---

80. Which of the following investigative procedures does not require the use of real-time PCR to determine viral load?

A. Person with hepatitis B on tenofovir therapy

B. Detection of HSV causing temporal encephalitis

C. BK virus in patient of allograft renal transplant

D. CMV in patient of liver transplant

---

81. Northern blotting is used for detection of which of the following ?

A. DNA

B. RNA

C. Proteins

D. None

---

82. Which type of infection is more likely to occur following the transfusion of Solvent/Detergent (SD) plasma, given its ability to destroy lipid-enveloped viruses?

A. Human immunodeficiency Virus (HIV)

B. Hepatitis A Virus (HAV)

C. Hepatitis B Virus (HBV)

D. Hepatitis C Virus (HCV)

---

83. The association of human herpesvirus 8 is observed in all of the following conditions except:

A. Kaposi Sarcoma

B. Castleman's Disease

C. Adult T-cell Leukemia

D. Primary Effusion Lymphoma

---

84. An elderly patient was rushed to the casualty following bouts of severe breathlessness. He reported a history of fever and cough for the past 3 days. Lab tests were done, and target sequences for the RdRp gene, E gene, N gene, and ORF1 were detected. What was the test done?

A. CLIA

B. ELISA

C. CBNAAT

D. Real-time RT-PCR

---

85. What is the term used to refer to the antigen known as Australia antigen among the options provided?

A. HBsAg

B. HBeAg

C. HBcAg

D. HBX

86. During the period between 1 to 4 months following renal transplantation, which pathogen is most commonly responsible for infecting renal transplant recipients?

A. Cytomegalovirus

B. JC virus

C. BK virus

D. Epstein-Barr virus

87. Among the given options, which disease has the highest mortality rate once it becomes clinically evident?

A. Nipah

B. Rabies

C. Ebola

D. Zika

88. Antibody testing for SARS-CoV-2 depends upon?

A. Time of incubation

B. Presence of acute viral load

C. A high risk of reinfection

D. That the patient is in window period

89. Specimen handling for molecular testing of SARS CoV 2 would require?

A. BSL 1

B. BSL 2

C. BSL 3

D. BSL4

---

90. Oral hairy leukoplakia is associated with which of the following infections?

A. Cytomegalovirus

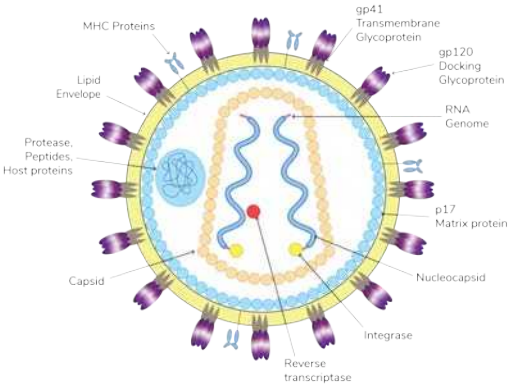
B. Human papilloma virus

C. Herpes simplex 1 virus

D. Epstein-Barr virus

---

91. Identify the virus given in the image below.



A. HIV

B. EBV

C. CMV

D. Coronavirus

---

92. Which type of virus is rubella?

A. Flavivirus

B. Togavirus

C. Paramyxovirus

D. Orthomyxovirus

---

93. Cowdry Type A inclusions are seen in:

A. Herpes simplex

B. Rabies

C. Molluscum contagiosum

D. Adenovirus

---

94. Viral infection mimicking erythroblastosis fetalis is:

A. HSV

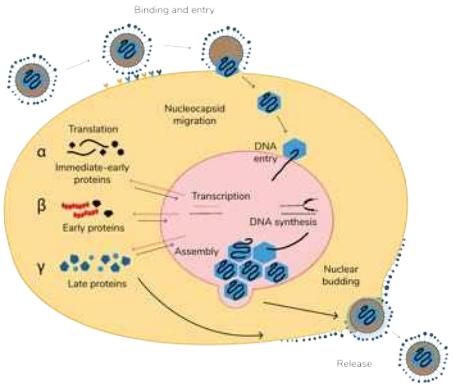
B. Epstein-Barr virus

C. Picornavirus

D. Parvovirus B19

---

95. Identify the virus given in the following replication cycle.



A. Human immunodeficiency virus

B. Herpes simplex virus

C. Hepatitis B virus

D. Coronavirus

---

96. A 24-year-old male patient presented to the outpatient department with cervical lymphadenopathy. As a part of the evaluation, the Paul-Bunnell test was done. What is the principle of this test?

A. Complement fixation

B. Superantigens

C. Heterophile antibodies

D. Haptens

---

97. Which of the following viruses cannot integrate with the human genome to initiate carcinogenesis?

A. HPV

B. SV 40

C. HAV

D. EBV

---

98. Reinfection with which of the following virus can lead to more severe disease with complications?

A. Dengue

B. Rabies

C. Measles

D. Mumps

---

99. Match the following in the Japanese encephalitis virus life cycle 1. Accidental host 2. Amplifier 3. Reservoir 4. Vector a. Ardeid birds b. Culex c. Human d. Pigs

1. Accidental host 2. Amplifier 3. Reservoir 4. Vector	a. Ardeid birds b. Culex c. Human d. Pigs

A. 1-d, 2-c, 3-a, 4-b

B. 1-c, 2-a, 3-d, 4-b

C. 1-c, 2-d, 3-a, 4-b

D. 1-d, 2-c, 3-b, 4-a

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	1
Question 4	3
Question 5	1
Question 6	4
Question 7	1
Question 8	2
Question 9	1
Question 10	4
Question 11	1
Question 12	3
Question 13	1
Question 14	1
Question 15	2
Question 16	1
Question 17	1
Question 18	3
Question 19	2
Question 20	2
Question 21	2
Question 22	2

Question 23	4
Question 24	4
Question 25	2
Question 26	4
Question 27	4
Question 28	4
Question 29	2
Question 30	2
Question 31	3
Question 32	1
Question 33	4
Question 34	2
Question 35	2
Question 36	1
Question 37	1
Question 38	3
Question 39	3
Question 40	3
Question 41	1
Question 42	3
Question 43	1
Question 44	4
Question 45	1
Question 46	3
Question 47	3
Question 48	1

Question 49	2
Question 50	2
Question 51	1
Question 52	3
Question 53	2
Question 54	2
Question 55	2
Question 56	1
Question 57	2
Question 58	1
Question 59	1
Question 60	1
Question 61	3
Question 62	3
Question 63	1
Question 64	4
Question 65	1
Question 66	1
Question 67	4
Question 68	2
Question 69	4
Question 70	4
Question 71	3
Question 72	1
Question 73	1
Question 74	2

Question 75	1
Question 76	4
Question 77	1
Question 78	2
Question 79	1
Question 80	2
Question 81	2
Question 82	2
Question 83	3
Question 84	4
Question 85	1
Question 86	1
Question 87	2
Question 88	1
Question 89	3
Question 90	4
Question 91	1
Question 92	2
Question 93	1
Question 94	4
Question 95	2
Question 96	3
Question 97	3
Question 98	1
Question 99	3

Solution for Question 1:

Correct Option A- HSV-2 (Herpes Simplex Virus Type 2):

- Presentation: Painful vesicles in the genital area are characteristic of genital herpes. The virus can cause painful lymphadenopathy in the affected region.
- Clinical Features: Recurrent episodes of genital lesions, and flu-like symptoms during initial infection.
- Diagnosis: Often diagnosed through clinical presentation and confirmed by laboratory tests, including PCR and serology.

Incorrect Options:

Option B- MOLLUSCUM:

- Presentation: Molluscum contagiosum typically presents as small, painless, raised lesions with a central dimple.
- Clinical Features: Flesh-colored or pearly papules.
- Diagnosis: Clinical examination; a biopsy may be performed in certain cases.

Option C- HPV (Human Papillomavirus):

- Presentation: Genital warts are associated with certain strains of HPV. They may be painless or cause minimal discomfort.
- Clinical Features: Warts in the genital or anal area.
- Diagnosis: Clinical examination; HPV testing in certain cases.

Option D- HZV (Herpes Zoster Virus):

- Presentation: Herpes zoster (shingles) typically presents as a painful vesicular rash along a dermatome, not limited to the genital area.
- Clinical Features: Unilateral rash with grouped vesicles.
- Diagnosis: Clinical examination; laboratory tests like PCR.

Solution for Question 2:

Correct Option D - Real-time RT-PCR:

• It is a gold standard test for the diagnosis of coronavirus.

• It can even detect 10 copies/ml and can run 90 samples in a single run.

• The average time taken is around 4-6 hrs.

• Gene targets for screening are genus-specific Envelope protein (E. gene) Nucleocapsid protein (N. gene) Spike protein (S. gene)

• Envelope protein (E. gene)

• Nucleocapsid protein (N. gene)

• Spike protein (S. gene)

• Confirmatory genes RNA-dependent RNA polymerase gene (RdRp) Open reading frames (ORF1a/b)

• RNA-dependent RNA polymerase gene (RdRp)

• Open reading frames (ORF1a/b)

• Principle: The target gene/s in the specimen is amplified in the thermocycler When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value (Cyclic Threshold) is inversely proportional to the load. A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.

• The target gene/s in the specimen is amplified in the thermocycler

• When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value (Cyclic Threshold) is inversely proportional to the load.

• A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.

• Envelope protein (E. gene)

• Nucleocapsid protein (N. gene)

• Spike protein (S. gene)

• RNA-dependent RNA polymerase gene (RdRp)

• Open reading frames (ORF1a/b)

• The target gene/s in the specimen is amplified in the thermocycler

• When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value (Cyclic Threshold) is inversely proportional to the load.

• A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 3:

Correct Option A- RSV (Respiratory Syncytial Virus):

- Genome Type: RSV has a negative-sense single-stranded RNA genome.
- Need for Reverse Transcription: Reverse transcription is required to convert the RNA genome into complementary DNA (cDNA) before PCR amplification.

Incorrect Options:

Option B- EBV (Epstein-Barr Virus):

- Genome Type: EBV has a double-stranded DNA genome.
- Need for Reverse Transcription: Reverse transcription is not required for DNA viruses like EBV. PCR can directly amplify DNA targets

Option C- CMV (Cytomegalovirus):

- Genome Type: CMV has a double-stranded DNA genome.
- Need for Reverse Transcription: Reverse transcription is not required for DNA viruses like CMV. PCR can directly amplify DNA targets.

Option D- Adenovirus:

- Genome Type: Adenovirus has a double-stranded DNA genome.
- Need for Reverse Transcription: Reverse transcription is not required for DNA viruses like Adenovirus. PCR can directly amplify DNA targets.

Solution for Question 4:

Correct Option C- IgM ELISA (Enzyme-Linked Immunosorbent Assay):

• IgM antibodies start to appear a few days after the onset of symptoms, making IgM ELISA less sensitive in the very early stages of infection. It might not be as reliable on day 3 but can become more useful after 6 days.

Incorrect Options:

Option A- PCR (Polymerase Chain Reaction):

• PCR is often most effective during the early viral phase of the infection when viral RNA is present in the bloodstream. Therefore, PCR is feasible and can be performed on day 3 to detect the viral genetic material.

Option B- NS1 Antigen Test:

• The NS1 antigen test detects a viral protein present in the serum of infected individuals. It is usually reliable from the early days of infection, making it a suitable option on day 3.

Option D- Xenodiagnosis using Culture in Aedes Mosquito:

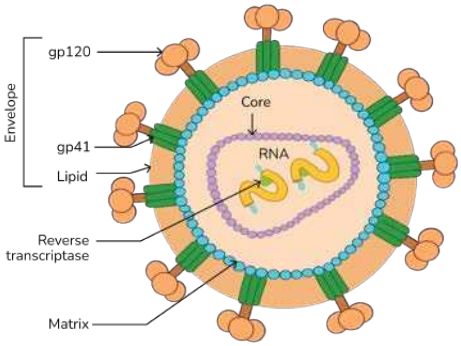
• Xenodiagnosis involves allowing Aedes mosquitoes to feed on the patient's blood and then testing the mosquitoes for the presence of the virus. This technique is generally not performed, but can be done in the early stage of infection.

Solution for Question 5:

Correct Option A- GP120: The HIV genome requires the collaborative action of various components, including GP120, CCR5 or CXCR4 co-receptors, and GP41, for entry into the host cell membrane. The initiation of this complex process is triggered by the binding of GP120 to the CD4 receptor on the host cell surface.

• Function: GP120 is a glycoprotein on the surface of the HIV envelope.

• Role in Entry: GP120 binds to CD4 receptors on the surface of host cells (usually CD4+ T cells), initiating the attachment of the virus to the cell membrane. This interaction exposes a binding site for a co-receptor.



Incorrect Options:

Option B- CCR5 (Chemokine Receptor 5):

- Function: CCR5 is a chemokine receptor found on the surface of certain immune cells, including CD4+ T cells.
- Role in Entry: After GP120 binds to the CD4 receptor, it undergoes a conformational change, allowing it to interact with a co-receptor. CCR5 is a primary co-receptor used by some strains of HIV for membrane fusion and entry into the host cell.

Option C- GP41 (Glycoprotein 41):

- Function: GP41 is another glycoprotein on the HIV envelope.
- Role in Entry: GP41 facilitates the fusion of the viral envelope with the host cell membrane. It forms a hairpin-like structure (six-helix bundle) during this process, bringing the viral and cellular membranes in close proximity.

Option D- CXCR4 (Chemokine Receptor 4):

- Function: CXCR4 is another chemokine receptor found on the surface of immune cells.
- Role in Entry: Some strains of HIV can use CXCR4 as an alternative co-receptor to CCR5 during the entry process.

Solution for Question 6:

Correct Option D - IgM ELISA:

- IgM ELISA test offers a reliable diagnosis, usually positive after the 5th day.

Incorrect Options:

Option A - NS-1 antigen test:

- The mainstay of diagnosis is the detection of a non-structural viral protein antigen (NS-1).
- This can be detected in the blood for up to 1-4 days.

Option B - Viral culture:

- Viral culture is rarely done in detection of the dengue virus.

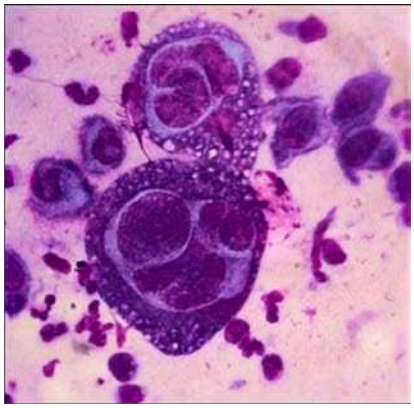
Option C - Circulating IGM:

- Circulating IGM is detected from day 7.

Solution for Question 7:

Correct Option A - HSV2:

- Genital ulcer with child affected with similar lesions points towards diagnosis of HSV2.
- HSV2 is transmitted vertically and sexually affecting below the waist.
- Clinical features include genital lesions below waist and Neonatal herpes (mother to child transmission).
- Other features include: Gingivostomatitis, Encephalitis, Meningitis, and Dendritic ulcers.
- Tzanck smear shows multinucleated giant cells with faceted nuclei and homogeneously stained ‘ground glass’ chromatin (Tzanck cells) constitute a positive smear.



Incorrect Options:

Option B - VZV:

• VZV is a virus similar in morphology to HSV but it causes chicken pox which presents with skin lesions not ulcers.

Option C - HPV:

• Human papilloma virus causes plantar wart, Epidermodysplasia, Laryngeal papillomatosis, Condyloma accuminata.

Option D - Klebsiella:

• Klebsiella is a gram negative bacteria which is a common cause of community acquired pneumonia.

Solution for Question 8:

Correct Option B - 1- B, 2- A, 3- D, 4- C:

1. Rotavirus	B. dsRNA
2. SARS-CoV-2 (COVID-19 virus)	A. ssRNA
3. Parvovirus	D. ssDNA
4. Human papillomavirus (HPV)	C. dsDNA

Incorrect Options:

• Option A, C & D: These options are incorrect options

Solution for Question 9:

Correct Option A - RT- PCR

• This is the correct answer. Reverse transcription polymerase chain reaction (RT-PCR) is a highly sensitive and specific method for detecting viral RNA, including the RNA of the rabies virus. It can be used in various types of samples, including saliva, cerebrospinal fluid, skin biopsies, and corneal scrapings. It is therefore the most appropriate test for confirming a

diagnosis of rabies.

Incorrect Options:

Option B - Viral culture: While viral culture can be used to diagnose many types of viral infections, it is not typically used for rabies due to the slow growth of the virus and the biohazard risk associated with culturing the rabies virus.

Option C - Antibody detection by ELISA: ELISA (enzyme-linked immunosorbent assay) can be used to detect antibodies against the rabies virus in the patient's blood. However, it can take 1-2 weeks after the onset of symptoms for these antibodies to develop, so a negative result does not rule out rabies in the early stages of the disease.

Option D - Direct fluorescent antibody test: This test is used to detect the presence of rabies virus in a sample (often a skin biopsy from the back of the neck) by staining it with fluorescent antibodies that bind to the virus. While it's a common method for diagnosing rabies, its sensitivity can be lower than RT-PCR, especially in corneal scrapings.

Solution for Question 10:

Correct Option D - AST and ALT >1000:

Retro-orbital headache and Tourniquet test is a typical point towards dengue.

- NS1 Ag is positive and confirms the diagnosis of dengue.
- AST and ALT >1000 come under severe criteria in dengue.
- Dengue virus can directly affect the liver, leading to hepatic injury. The virus may replicate in the liver cells, causing inflammation and damage. This liver involvement is known as dengue hepatitis.
- The liver may also be affected by the reduced blood supply, contributing to elevated liver enzymes.

Incorrect Options:

Option A - Leucopenia plus thrombocytopenia: It is incorrect as Leucopenia plus thrombocytopenia comes under probable dengue.

Option B - Persistent vomiting: This option is incorrect as Persistent vomiting is a warning sign of dengue.

Option C - Hepatomegaly >1 cm: This option is incorrect as Hepatomegaly >1 cm is a warning sign of dengue.

Solution for Question 11:

Option A - Herpes simplex virus (HSV):

- The history of the mother having had genital ulcers in the past is significant, as the transmission of HSV to the baby can occur during childbirth if the mother is shedding the virus. The presence of vesicles and multinucleated giant cells on biopsy further supports the diagnosis of HSV infection in the newborn.

Incorrect Options:

Option B - Molluscum Contagiosum (MCV):

- Molluscum contagiosum is a viral skin infection caused by the Molluscipoxvirus.
- It typically presents as small, flesh-colored, dome-shaped papules with a central dimple.
- It is not associated with fever, cough, or multinucleated giant cells on biopsy.

Option C - Human Papillomavirus (HPV):

- HPV is a DNA virus that causes warts. Genital warts are associated with certain strains of HPV.
- While HPV can cause skin lesions, it doesn't typically present with systemic symptoms like fever and cough.
- The biopsy findings of multinucleated giant cells are not characteristic of HPV.

Option D - Varicella (Chickenpox):

- Varicella-zoster virus (VZV) causes chickenpox, characterized by a generalized vesicular rash.
- Chickenpox can occur in infants, but at 18 days old, the timing seems less likely for typical chickenpox.
- The multinucleated giant cells on biopsy are more suggestive of herpes simplex virus.

Solution for Question 12:

Correct Option C - HSV-2:

- HSV-2 has a low risk of congenital infection because the transmission of herpes simplex virus to the newborn occurs most commonly (85%) during transvaginal birth in a woman with an active lesion of herpes in the genital tract of the mother. This is prevented by preferring a caesarian section for delivery of the baby and reducing the risk of transmission to the baby.
- The transplacental route of transmission is uncommon (5%); thus, it is least teratogenic for the baby.
- If a mother acquires primary infection in the last trimester, then the baby may present with IUGR.

Incorrect Options:

Option A – CMV:

- Cytomegalovirus - CMV has a high risk of congenital infections and is transmitted tranplacentally resulting in microcephaly along with periventricular calcification.

Option B – Chickenpox:

- Chickenpox is transmitted transplacentally and can result in congenital varicella syndrome characterized by hypoplasia of the limbs, IUGR, and microcephaly.

Option D - Rubella:

- The transplacental spread of the rubella virus results in congenital rubella syndrome characetrized by patent ductus arteriosus, cataract, and SNHL.

Solution for Question 13:

Correct Option A- EBV (Epstein-Barr Virus):

- Oral Hairy Leukoplakia (OHL): This condition is strongly associated with EBV, and it presents as white, corrugated, or hairy-appearing patches on the lateral borders of the tongue or other oral mucosal surfaces. The white patches in OHL cannot be scraped off easily and are often asymptomatic.

Incorrect Options:

Option B- Candida:

• Candidiasis: Candida infections in the oral cavity can cause white patches, commonly referred to as oral thrush. However, these patches can usually be rubbed off, revealing an erythematous (red) base. In the case described, where the white patch cannot be removed by scraping, candidiasis seems less likely.



Option C- HSV (Herpes Simplex Virus):

• Herpetic Ulcers: HSV infections can cause painful ulcers in the oral cavity, but they are not typically described as white patches that cannot be removed by scraping. HSV ulcers often have a yellowish base surrounded by an erythematous halo.

Option D- HHV8 (Human Herpesvirus 8):

• Kaposi's Sarcoma: HHV8 is associated with Kaposi's sarcoma, which can present with lesions in the oral cavity. However, these lesions are more commonly reddish or purple rather than white, and they may not exactly fit the description given.

Solution for Question 14:

Correct Option A - Skin Biopsy with Fluorescent Antibody Testing:

• Skin biopsy for rabies antigen detection via direct fluorescent antibody testing is a commonly used method for rapid diagnosis of rabies. A skin biopsy sample taken from the nape of the neck with at least 10 hair follicles is the best specimen, and it is stained with fluorescent antibodies specific to the rabies virus. If the virus is present, it will bind to the antibodies and produce a fluorescence, confirming the diagnosis of rabies.

Incorrect Options:

Option B - Rabies Virus Specific Antibodies: Testing for rabies virus-specific antibodies in the patient's blood is not a suitable method for diagnosis of rabies. The presence of antibodies in blood appears late and is present in both rabies-infected and immunized individuals. CSF antibodies appear early, and they are produced only in rabies-infected individuals but not in response to vaccination

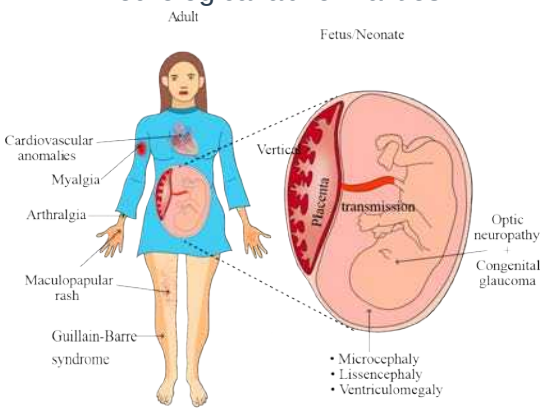
Option C - Inoculation in Mice: Inoculation in mice helps in the isolation of the virus. It involves injecting the patient's tissue sample into a live mouse and observing for signs of rabies infection in the mouse. This method is time-consuming and may not provide rapid results.

Option D - Corneal Impression Smear: Corneal impression smear is another method used for diagnosing rabies via direct fluorescent antibody testing, but with a low sensitivity of 30%.

Solution for Question 15:

Correct Option B - Zika virus:

- Zika virus is a zoonotic disease transmitted to humans via vector Aedes aegypti or albopticus.
- Zika virus is transmitted from human to human transmission through body fluids and transplacental transmission.
- In vertical transmission it leads to various adverse effects on the developing fetus, including microcephaly and ventriculomegaly. Microcephaly refers to an abnormal smallness of the head, which is often associated with incomplete brain development and can result in neurological abnormalities.



Incorrect Options:

Option A - Nipah: It is a zoonotic disease that is transmitted to humans via food and water contaminated by fruit bats' urine and can cause fever and encephalitis.

Option C - CMV: While CMV infection during pregnancy can lead to various congenital abnormalities, including microcephaly, it is not a vector-borne disease and can be transmitted through body secretions.

Option D - Rubella: Rubella virus infection during pregnancy can result in congenital rubella syndrome, which can cause a range of birth defects like patent ductus arteriosus, sensorineural hearing loss, cataracts, and blueberry muffin syndrome. It is transmitted by air droplets and transplacental route.

Solution for Question 16:

Correct Option A - Measles:

- SSPE (Subacute Sclerosing Panencephalitis) is a rare, slowly progressive neurological disorder that occurs as a late complication of measles virus infection.
- It occurs many years after measles, around 7- 13 years. But once it occurs, within 1-3 years, it becomes fatal.
- It manifests as progressive deterioration of motor and cognitive functions, focal/generalized seizures, myoclonus, ataxia, and visual disturbances.

Incorrect Options:

Options B, C, and D are incorrect. SSPE is not a complication of these viruses.

Solution for Question 17:

Correct Option A - HPV 16 & 18:

- High-risk subtypes of human papillomavirus (HPV), particularly HPV 16 and HPV 18, are known to have the highest association with the development of cervical carcinoma.

• Other high-risk human papillomaviruses associated with cervical carcinoma are HPV 31, 33, and 35.

Incorrect Options:

Option B - HPV 6 & 11: HPV 6 and HPV 11 are classified as low-risk subtypes and are commonly associated with condyloma acuminatum and laryngeal papillomatosis.

Option C - HPV 31 & 32: HPV 31 is a high-risk subtype associated with cervical cancer, but HPV 32 is not associated with cervical carcinoma.

Option D - HPV 1 & 2: HPV 1 and HPV 2 are low-risk subtypes associated with plantar warts/verruca and filiform warts.

Solution for Question 18:

Correct Option C - Anti HBc Ab:

• Anti-HBcAb (Antibodies against Hepatitis B core antigen) is present during and after the acute phase of Hepatitis B infection. It persists in the serum even during recurrences and in individuals with chronic hepatitis.

Incorrect Options:

Option A - HBc Ag: Hepatitis B core antigen is not detectable in the serum.

Option B - Anti HBs Ab: Antibodies against Hepatitis B surface antigen indicates immunity to Hepatitis B following vaccination.

Option D - HBe Ag: Hepatitis B pre-core antigen typically suggests the infectivity of the virus; presence of this antigen suggests high infectivity, and the absence suggests low infectivity.

Solution for Question 19:

Correct Option B - Aedes Albopictus:

• Zika virus is primarily transmitted to humans through the bites of infected mosquitoes, specifically Aedes aegypti and Aedes albopticus.

Incorrect Options:

Options A, C, and D are incorrect. These spp. of Aedes do not transmit the Zika virus.

Solution for Question 20:

Correct Option B - CCR5:

• Receptors essential for the entry of HIV into the host cell are: CD4: HIV enters into the target cells by binding its gp120 to the CD4 receptor on the host cell surface. CD4 molecules are mainly expressed on helper T cells and also on the surface of monocytes, macrophages, Langerhans cells, astrocytes, keratinocytes, and glial cells. A second co-receptor is necessary for fusion of HIV to gain entry into the host cell. CXCR4 molecules present on T lymphocytes. CCR5 molecules present on cells of macrophage.

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• CXCR4 molecules present on T lymphocytes.

• CCR5 molecules present on cells of macrophage.

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• CCR5 molecules present on cells of macrophage.

Incorrect Options:

Options A, C, and D are incorrect. These are not used as a receptor for the entry of HIV into the host cells.

Solution for Question 21:

Correct Option B - EBV:

- Epstein-Barr virus is associated with the following malignancies: Carcinoma The most common is Gastric Carcinoma. Nasopharyngeal Carcinoma Lymphoma Hodgkin (mixed cellularity): Classical Non-Hodgkin lymphoma Burkitt's DLBCL (Diffuse Large B Cell Lymphoma) Post Transplant Lymphoma T/Natural Killer cell Lymphoma Sarcoma: Leiomyosarcoma
- Carcinoma The most common is Gastric Carcinoma. Nasopharyngeal Carcinoma
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- Nasopharyngeal Carcinoma
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• Post Transplant Lymphoma
• Sarcoma: Leiomyosarcoma
• The most common is Gastric Carcinoma.
• Nasopharyngeal Carcinoma
• Hodgkin (mixed cellularity): Classical
• Non-Hodgkin lymphoma
• Burkitt's
• DLBCL (Diffuse Large B Cell Lymphoma)
• Post Transplant Lymphoma

Incorrect Options:

Options A, C, and D are incorrect. These viruses are not associated with the formation of the nasopharyngeal carcinoma.

Solution for Question 22:

Correct Option B - Parainfluenza:

• The given symptoms of low-grade fever, inspiratory stridor, and barking cough, along with the characteristic chest X-ray showing the steeple sign, are indicative of croup.
• Croup is an infection caused by parainfluenza virus that primarily affects the upper airway, causing inflammation and swelling of the larynx and trachea.
• The hallmark symptom of croup is the barking cough, which is often worse at night and may be accompanied by inspiratory stridor (a high-pitched, crowing sound during inhalation).
• The steeple sign on the chest X-ray is a characteristic finding in croup, representing the narrowed subglottic region.
• Other organisms causing croup: Influenza virus Respiratory syncytial virus Adenoviruses Rhinoviruses H. influenza
• Influenza virus
• Respiratory syncytial virus
• Adenoviruses
• Rhinoviruses

- H. influenza
- Influenza virus
- Respiratory syncytial virus
- Adenoviruses
- Rhinoviruses
- H. influenza

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 23:

Correct Option D - A & B only:

- "School of fish" appearance: This is a characteristic finding seen in chancroid caused by H. ducreyi, a pleomorphic gram-negative coccobacillus that occurs in groups or in parallel chains. They frequently take bipolar staining, and the arrangement has been described as a school of fish or railroad track appearance
- Bleeds on touch: Chancroid ulcers are often fragile and can easily bleed upon touch or manipulation.
- Painless: Chancroid is typically associated with painful genital ulcers.
- Cullen sign: It is not a feature of chancroid. It is an exam finding of ecchymoses or bruising around the umbilicus.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 24:

Correct Option D - EBV:

• EBV (Epstein-Barr virus) EBV, also known as Human herpesvirus 4 (HHV-4), is a member of the Herpesviridae family. It is associated with infectious mononucleosis (glandular fever) and is also linked to several other diseases, including certain cancers, such as Burkitt's lymphoma and nasopharyngeal carcinoma.

• Other members of Herpesviridae are

Type	Virus	Associated Diseases
HHV 1	Herpes Simplex Virus 1	Oral herpes (cold sores), genital herpes, and encephalitis.
HHV 2	Herpes Simplex Virus 2	Genital herpes.
HHV 3	Herpes Zoster Virus	Chickenpox and shingles.
HHV 4	Epstein–Barr virus	Infectious mononucleosis, Burkitt's lymphoma, nasopharyngeal carcinoma.
HHV 5	Cytomegalovirus (largest member)	Congenital infections, mononucleosis, organ transplant complications.
HHV 6	Human Herpesvirus 6	Roseola infantum, febrile illness
HHV 7	Human Herpesvirus 7	Febrile illness and a few cases of roseola infantum
HHV 8	Human Herpesvirus 8	Kaposi's sarcoma

Incorrect Options:

Option A - Rubella: Rubella is caused by the Rubella virus, which belongs to the Togaviridae family, not the Herpesviridae family.

Option B - Measles: Measles, also known as rubeola, is caused by the Measles virus, which belongs to the Paramyxoviridae family.

Option C - Rabies: Rabies is caused by the Rabies virus, which belongs to the Rhabdoviridae family.

Solution for Question 25:

Correct Option B - Icosahedral Capsid:

- Coronavirus has a Helical capsid.

Incorrect Options:

Option A - ssRNA genome: Coronavirus is a single-stranded RNA virus. All RNA viruses have ssRNA genomes except Reoviridae (dsRNA)

Option C - Host cell entry occurs by attachment of viral spike: Spreads primarily through respiratory droplets and aerosols. Host cell entry occurs by attachment of viral spike protein to angiotensin-converting enzyme 2 receptor on cell membranes.

Option D - Human coronavirus 229E is an alpha-type: Human coronavirus 229E is indeed classified as an alpha-type coronavirus.

Solution for Question 26:

Correct Option D - Poxvirus:

- Molluscum contagiosum is caused by the Molluscum contagiosum virus (MCV), which belongs to the Poxvirus family.
- MCV infects the skin and mucous membranes, resulting in the formation of small, flesh-colored, or pinkish bumps with a central indentation. These bumps are typically painless and may resolve on their own over time.
- Other members of poxvirus are the variola virus, causing smallpox, and the vaccinia virus, causing chicken pox.

Incorrect Options:

Option A - Adenovirus: Adenovirus is a group of human adenovirusesA to F that can cause respiratory infections, and conjunctivitis.

Option B - Flavivirus: Flaviviruses are a family of viruses including dengue virus, zika virus, and yellow fever virus.

Option C - Rubivirus: Rubivirus refers to the Rubella virus, which causes Rubella (German measles).

Solution for Question 27:

Correct Option D - EBV:

- EBV (Epstein-Barr virus) EBV, also known as Human herpesvirus 4 (HHV-4), is a member of the Herpesviridae family. It is associated with infectious mononucleosis (glandular fever) and is also linked to several other diseases, including certain cancers, such as Burkitt's lymphoma and nasopharyngeal carcinoma.
- Other members of Herpesviridae are

Type	Virus	Associated Diseases
HHV 1	Herpes Simplex Virus 1	Oral herpes (cold sores), genital herpes, and encephalitis.
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HHV 4	Epstein–Barr virus	Infectious mononucleosis, Burkitt's lymphoma, nasopharyngeal carcinoma.
HHV 5	Cytomegalovirus (largest member)	Congenital infections, mononucleosis, organ transplant complications.
HHV 6	Human Herpesvirus 6	Roseola infantum, febrile illness
HHV 7	Human Herpesvirus 7	Febrile illness and a few cases of roseola infantum
HHV 8	Human Herpesvirus 8	Kaposi's sarcoma

Incorrect Options:

Option A - Rubella: Rubella is caused by the Rubella virus, which belongs to the Togaviridae family, not the Herpesviridae family.

Option B - Measles: Measles, also known as rubeola, is caused by the Measles virus, which belongs to the Paramyxoviridae family.

Option C - Rabies: Rabies is caused by the Rabies virus, which belongs to the Rhabdoviridae family.

Solution for Question 28:

Correct Option D - Corneal Impression Smear:

- The antemortem diagnosis of rabies is challenging as it requires specialized laboratory techniques.
- Antemortem diagnosis of rabies can be made by rabies antigen detection by performing a corneal impression smear, skin biopsy from the nape of the neck containing at least 10 hair follicles, and saliva. The viral antigens are present in these preparations and are examined for their presence using techniques like direct fluorescent antibody (DFA) testing.

Incorrect Options:

Option A - Rabies Virus Specific Antibodies: While the presence of rabies virus-specific antibodies can indicate a past infection or immune response, it is not a reliable method for antemortem diagnosis. Antibodies may take time to develop and may not be detectable during the early stages of infection.

Option B - Inoculation in Cell Lines: Inoculation of samples in cell lines is a method used for viral isolation and growth but is not commonly used for antemortem diagnosis of rabies. It is a time-consuming process and requires specific laboratory facilities.

Option C - Negri Bodies in Hippocampus: Negri bodies are characteristic microscopic inclusion bodies found in certain neurons of the central nervous system affected by rabies virus infection. However, their detection usually requires brain tissue examination postmortem (after death) and is not feasible for antemortem diagnosis.

Solution for Question 29:

Correct Option B - L1:

• HPV (Human Papillomavirus) vaccines primarily target the major capsid protein called L1 (Late 1).

• L1 protein forms the viral capsid and is responsible for inducing protective immune responses against HPV infection.

Incorrect Options:

Option A - E6 & E7: E6 and E7 are viral oncoproteins produced by high-risk HPV types, which are associated with cervical, penile, anal, and oral cancer.

Option C - E1 & E2: E1 (controller) and E2 are viral proteins involved in viral replication and regulation

Option D - L2: L2 is another minor capsid protein of HPV, but it is not the primary antigen targeted by the HPV vaccine.

Solution for Question 30:

Correct Option B - Transmitted by Mosquito:

• Ebola virus is a Filovirus that spreads among people through direct contact with bodily fluids or contaminated surfaces/materials.

Incorrect Options:

Option A - It is filovirus: Ebola virus belongs to the Filoviridae family and is characterized by its filamentous or bowl of spaghetti shape under electron microscopy.

Option C - Bats act as reservoir: Bats are considered natural reservoir hosts for the Ebola virus.

Option D - Human-to-human transmission is seen: Ebola virus spreads among people through direct contact with bodily fluids or contaminated surfaces/materials.

Solution for Question 31:

Correct Option C - Real time RT-PCR:

- Real-time RT-PCR is the method employed to measure the viral load in a sample. It is a sensitive and specific molecular technique that allows for nucleic acid amplification with qualitative and quantification of viral genetic material (RNA) present in a sample. Real-time PCR can detect and quantify the amount of viral nucleic acid, providing information about the viral load.

Incorrect Options:

Option A - ELISA (Enzyme-Linked Immunosorbent Assay): ELISA is an antibody-based assay used to detect and measure the presence of specific antibodies or antigens in a sample.

Option B - Western blot: Western blot is a technique used to detect and identify specific proteins in a sample. It is commonly used to confirm the presence of specific viral proteins or antibodies in a sample.

Option D - Immunoelectrophoresis: Immunoelectrophoresis is a laboratory technique used to separate and identify proteins based on their antigen-antibody interactions.

Solution for Question 32:

Correct Option A - 17D:

- The presence of the Torres body (cowdry A intranuclear inclusion bodies) in the liver of the photographer on autopsy is suggestive of yellow fever, which is caused by the yellow fever virus. Yellow fever is a viral hemorrhagic fever that is transmitted by infected mosquitoes in tropical regions of Africa and South America.
- The 17D strain is a live attenuated vaccine that provides long-lasting immunity against yellow fever. It is recommended for travelers to areas where yellow fever is endemic.

Incorrect Options:

Option B - Nakayama Strain: The Nakayama strain is used for the preparation of the formalin-inactivated mouse brain vaccine against Japanese encephalitis.

Option C - Weigl’s Strain: Weigl's strain is a type of typhus vaccine against R. prowazekii.

Option D - Jeryl Lynn Strain: The Jeryl Lynn strain is used for the preparation of the live virus vaccine against mumps.

Solution for Question 33:

Correct Answer: D) Local Treatment + RIG + Vaccine

Explanation: For a PLHIV with category III dog bites (multiple deep wounds), the correct management is local treatment, Rabies Immune Globulin (RIG), and the rabies vaccine.

Management of Dog Bite:

Category	Exposure Type	Management
Category I	Touching or feeding animals Licking of intact skin	Ensure proper hygiene No treatment required
Category II	Nibbling on uncovered skin Minor scratches or abrasions without bleeding	Immediate wound washing with soap and water Rabies vaccination: Unvaccinated: Days 0, 3, 7, and 14 Previously vaccinated: Booster doses on Days 0 and 3 No Rabies Immunoglobulin (RIG) required

Category III	Bites or scratches with bleeding Licks on broken skin Exposure to mucous membranes (mouth, eyes, nose)	Immediate wound washing with soap and water, flush for at least 15 minutes. Suture: Do not suture the wound immediately, as it may spread the virus, instead delay it for 24 to 48 hours. Rabies vaccination: Unvaccinated: Full PEP series on Days 0, 3, 7, and 14 Previously vaccinated: Booster doses on Days 0 and 3 Rabies Immunoglobulin (RIG): Equine immunoglobulin or ERIG Human rabies Immunoglobulin or HRIG Preferred (free from the danger of sensitisation, HIV and hepatitis viruses) Dose: – 40 IU / kg IM Started on Day 0 along with vaccination (can be given up to 7 days). 70% is given around the wound and, the rest is given distant from the wound. Required for unvaccinated individuals ... (content truncated)

Solution for Question 34:

Correct Answer: B) HBsAg + HBe

Explanation: The correct marker for Infectious Hepatitis B is HBsAg + HBe, as HBsAg indicates active infection, and HBeAg signifies high infectivity.

Stages and Associated Markers of Hepatitis B Virus (HBV):

HBsAg	HBeAg	Anti-HBc	Anti-HBs	Anti-HBe	Interpretation
+	+	IgM	-	-	Acute HBV infection Highly infectious
+	+	IgG	-	-	Chronic HBV infection, highly

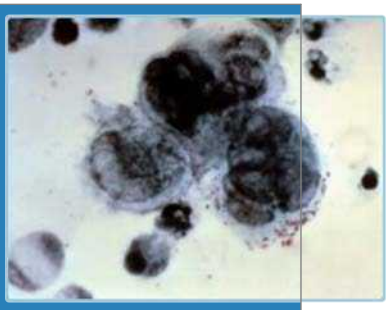
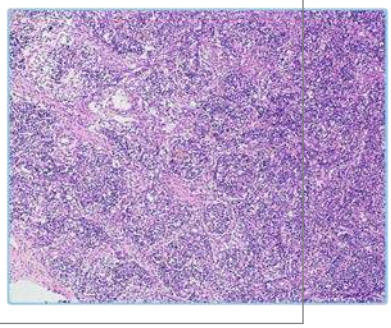
					infectious
+	-	IgG	-	-/+	Chronic HBV infection, low infectivity
-	-	IgG	-/+	-/+	Recovery
-	-	IgM	-	-	Window period
-	-	-	+	-	Post-vaccination

Solution for Question 35:

Correct Answer: B) HSV

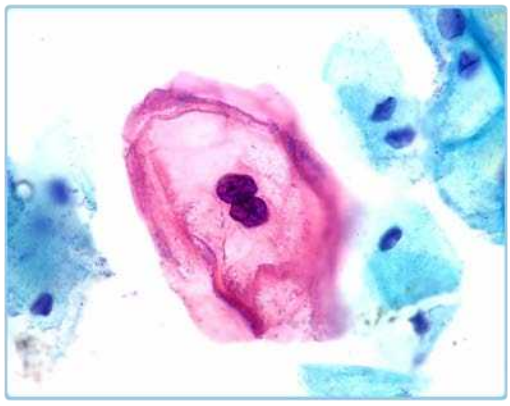
Explanation: A Tzanck smear showing multinucleated giant cells is characteristic of Herpes simplex virus (HSV) infections.

Microscopic Features of the Herpes Virus:

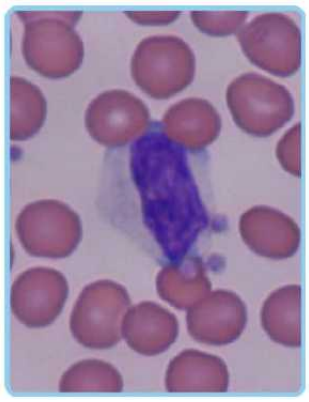
Herpes Simplex Virus: Tzanck smear: Multinucleated giant cells with faceted nuclei and homogeneously stained 'ground-glass' chromatin (Tzanck cells). Formation of Lipschutz bodies (Cowdry type A).	
HHV 8 (human herpes virus type 8): Castleman disease: Onion skinning of lymphocytes and blood vessels, giving the appearance of “Lollipop follicles”.	

HIV (Option A) does not cause multinucleated giant cells on a Tzanck smear; it is diagnosed with serological tests (e.g., ELISA, Western blot) or PCR.

HPV (Option C) is associated with koilocytes (perinuclear halos in epithelial cells), not multinucleated giant cells.



EBV (Option D) is associated with Downey cells, which are atypical lymphocytes with a ballerina skirt appearance, seen in peripheral blood smears. Diagnosed using the Monospot test or serology, not a Tzanck smear.



Solution for Question 36:

Correct Answer: A) Acute hepatitis

Explanation: A patient positive for HBsAg and anti-HBc IgM points toward Acute hepatitis B.

Stages and Associated Markers of Hepatitis B Virus (HBV):

HBsAg	HBeAg	Anti-HBc	Anti-HBs	Anti-HBe	Interpretation

+	+	IgM	-	-	Acute HBV infection Highly infectious
+	+	IgG	-	-	Chronic HBV infection, highly infectious (Option B)
+	-	IgG	-	-/+	Chronic HBV infection, low infectivity
-	-	IgG	-/+	-/+	Recovery (Option C)
-	-	IgM	-	-	Window period
-	-	-	+	-	Post-vaccination (Option D)

Solution for Question 37:

Correct Option A - Herpes:

- Bilateral painful vesicles, along with painful inguinal lymphadenopathy, are seen in HSV-2 infection.

Incorrect Options:

Option B - Chancroid: Chancroid causes painful ulcers but typically presents as a single, larger ulcer with ragged edges, not multiple vesicles.

Option C - Syphilis: Syphilis causes painless genital ulcers.

Option D - Candidiasis: Candidiasis does not cause painful vesicular lesions.

Solution for Question 38:

Correct Option C - PCR:

- PCR (Polymerase Chain Reaction) is the diagnostic test of choice for monkeypox.

Incorrect Options:

- Options A, B, and D are not the preferred tests for diagnosing monkeypox.

Solution for Question 39:

Correct Option C - Aplastic anemia in immunocompromised:

- Persistent pure red cell aplasia in immunocompromised is associated with Parvo B-19 virus not Aplastic anemia.

Incorrect Options:

Option A - Fever with petechial rash:

- Parvo B19 virus is associated with Erythema infectiosum which presents with macular erythematous rash present on the cheek

Option B - Transient aplastic crisis with hemolytic anemia:

- Transient aplastic crisis with hemolytic anemia is associated with Parvo B-19 virus.

Option D - Hydrops fetalis:

- Hydrops fetalis is associated with Parvo B-19 virus.

Solution for Question 40:

Correct Option C - Cryptococcus neoformans:

- Clinical presentation given in question is consistent with Cryptococcus neoformans.
- Infection occurs by inhalation of spores (from soil with pigeon droppings)
- It affects immunocompromised patients
- It affects usually whole body,but it affects brain more(Meningitis)

• Diagnosis: CSF Sample: positive India ink stain Biopsy: Mucicarmine SDA : shows mucoid colony Niger or bird or caffeic seed agar: shows brownish-yellow pigment. Latest agglutination test(confirmatory test)

• CSF Sample: positive India ink stain

• Biopsy: Mucicarmine

• SDA : shows mucoid colony

• Niger or bird or caffeic seed agar: shows brownish-yellow pigment.

• Latest agglutination test(confirmatory test)

• Treatment: Liposomal amphotericin B + Flucytosine.

• CSF Sample: positive India ink stain

• Biopsy: Mucicarmine

• SDA : shows mucoid colony

• Niger or bird or caffeic seed agar: shows brownish-yellow pigment.

• Latest agglutination test(confirmatory test)

Incorrect Options:

Option A - Bacterial meningitis:

• CSF analysis of patient with bacterial meningitis shows polymorphonuclear predominace, low glucose and high protein levels.

Option B - Viral meningitis:

• Viral meningitis characteristically has a CSF mononuclear pleocytosis, although there may be initially be a neutrophilic predominance.

Option D - Tuberculous meningitis:

• Characteristic CSF findings of Tuberculous meningitis shows lypmphocytic predominant pleiocytosis, elevated protein and normal glucose levels.

Solution for Question 41:

Correct Option A - Influenza:

• Influenza is caused by influenza viruses, which have the ability to undergo frequent genetic changes through antigenic drift (seen in influenza A & B, point mutation occurs) and

antigenic shift (seen in Influenza A, genetic reassotment occurs) .

• This means that the viral strains circulating in the population can change over time, making the previously developed vaccines less effective against the new strains. Therefore, annual updates and strain changes are required in the influenza vaccine to ensure its effectiveness against the evolving strains.

Incorrect Options:

Option B - Diphtheria: The diphtheria vaccine is based on the toxoid produced by Corynebacterium diphtheriae and does not require frequent strain changes.

Option C - Polio: The polio vaccine targets the poliovirus, and while there are different serotypes of the virus, the vaccine does not require frequent strain changes as the available vaccines offer protection against all known serotypes.

Option D - Typhoid: The typhoid vaccine is based on specific strains of Salmonella typhi bacteria and does not require frequent strain changes.

Solution for Question 42:

Correct Option C - Cryptococcus:

• Cryptococcus is a genus of fungi that includes pathogenic species, most notably Cryptococcus neoformans and Cryptococcus gattii. These fungi are encapsulated yeast-like organisms and can cause serious infections, particularly in individuals with weakened immune systems.

• India ink staining is a diagnostic technique used to visualize Cryptococcus species in clinical samples, such as cerebrospinal fluid (CSF). The ink particles do not penetrate the capsule surrounding the Cryptococcus cells, resulting in a clear halo around the unstained organisms against the dark background. This "halo effect" is a characteristic feature observed in India ink preparations for Cryptococcus.

Incorrect Options:

Option A - Aspergillus: These are filamentous fungi characterized by their acute dichotomous branching, septate hyphae, and conidial structures. Specific staining techniques such as lactophenol cotton blue or potassium hydroxide (KOH) mounts are commonly employed to examine their hyphal structures and conidia.

Option B - Penicillium: These are dimorphic fungi characterized by brush/broom conidial structures. Specific staining techniques such as lactophenol cotton blue or potassium hydroxide (KOH) mounts are commonly employed to examine their hyphal structures and conidia.

Option D - Candida: Candida is a yeast-like fungus that forms budding cells rather than hyphae. Instead, techniques such as Gram staining, wet mounts with KOH, or specialized fungal stains like methylene blue or calcofluor white are commonly used to visualize Candida cells.

Solution for Question 43:

Correct Option A - Cytomegalovirus:

- Cytomegalovirus (CMV), is the predominant pathogen to infect renal (solid organ) transplant recipients in the period around 1 to 4 months following transplantation because of reactivation of the virus in immunocompromised individuals.
- The immunosuppressive medications taken by transplant recipients increase their susceptibility to CMV infection, which can lead to significant morbidity and complications in the early post-transplant period.
- It presents in various forms, such as: Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients Febrile leukopenia is seen among solid organ transplant recipients. Obliterative bronchiolitis in lung transplants. Graft atherosclerosis in heart transplants. Rejection of renal allografts.
- Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients
- Febrile leukopenia is seen among solid organ transplant recipients.
- Obliterative bronchiolitis in lung transplants.
- Graft atherosclerosis in heart transplants.
- Rejection of renal allografts.
- Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients
- Febrile leukopenia is seen among solid organ transplant recipients.
- Obliterative bronchiolitis in lung transplants.
- Graft atherosclerosis in heart transplants.
- Rejection of renal allografts.

Incorrect Options:

Options B, C, and D are incorrect. They are not the most common organism to cause infection in an organ transplant recipient between 1 to 4 months.

Solution for Question 44:

Correct Option D - Hepatitis B:

- Blood-borne diseases that could be transmitted by a needlestick injury include Hepatitis B - 30% Hepatitis C - 3% HIV - 0.3%
- Hepatitis B - 30%
- Hepatitis C - 3%
- HIV - 0.3%
- The risk of transmission is highest for hepatitis B virus infection.
- Hepatitis B - 30%
- Hepatitis C - 3%
- HIV - 0.3%

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 45:

Correct Option A - NSP4:

- Diarrhea and vomiting in a 6-month-old child are commonly associated with viral gastroenteritis, particularly caused by rotavirus infection. Rotavirus is a leading cause of severe diarrhea in infants and young children worldwide.
- Rotavirus produces several viral proteins, including enterotoxins, that contribute to the pathogenesis of gastroenteritis. Among these, NSP4 (Nonstructural Protein 4) is an

enterotoxin primarily responsible for causing diarrhea by altering epithelial cell function andpermeability.

• Structural proteins of Rotavirus: Structural protein: VP1-4, VP6, VP7. Non-structural protein: NSP1-6.

• Structural protein: VP1-4, VP6, VP7.

• Non-structural protein: NSP1-6.

• Structural protein: VP1-4, VP6, VP7.

• Non-structural protein: NSP1-6.

Incorrect Options:

Option B - NSP6: NSP6 is another nonstructural protein produced by rotavirus. However, it is not primarily associated with causing diarrhea or vomiting in infected individuals.

Option C - VP3: VP3 is a structural protein of the rotavirus and is involved in the capping of the virus particle.

Option D - VP7: VP7 is another structural protein of the rotavirus, known as the glycoprotein, that forms the capsid layer of the virus.

Solution for Question 46:

Correct Option C - Man Acts as the Reservoir:

• Japanese encephalitis is a vector-borne disease transmitted by Culex tritaeniorhynchus (most common worldwide) and Culex vishnui (most common in India).

• Host Reservoir - Ardeid birds Amplifier - Pigs Humans - Dead-end host

• Reservoir - Ardeid birds

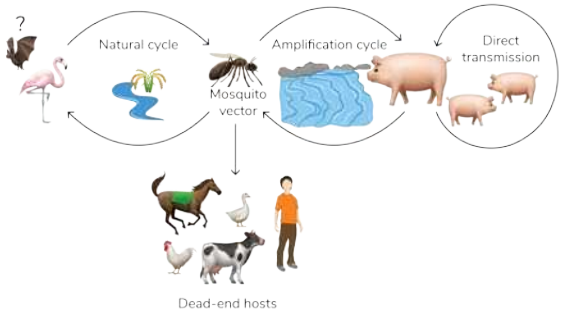
• Amplifier - Pigs

• Humans - Dead-end host

• Reservoir - Ardeid birds

• Amplifier - Pigs

• Humans - Dead-end host



Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 47:

Correct Option C - 5-12 years is a High Risk Group:

- The high-risk groups for severe complications of influenza typically include young children (especially those under 2 years), the elderly (65 years and older), pregnant women, individuals with underlying health conditions, and immunocompromised individuals. The age group of 5-12 years is generally considered to have a lower risk compared to these high-risk groups.

Incorrect Options:

Option A - Secondary Attack rate is 5-15%: The secondary attack rate refers to the proportion of susceptible individuals who develop the disease after being exposed to an infected individual. In the case of influenza, the secondary attack rate can vary depending on factors such as the specific strain, population density, and level of immunity. Generally, the secondary attack rate for influenza is estimated to be around 5 to 15%.

Option B - Intranasal Live Attenuated Vaccine is used: Intranasal live attenuated vaccine, a trivalent vaccine H1N1 (A), H3N2 (A), type B is indicated for individuals with influenza infection, contraindicated in low immunity.

Option D - Aquatic Birds act as Reservoir: Aquatic birds, particularly wild waterfowl, are considered the natural reservoir for influenza viruses. They can carry different strains of influenza viruses, including avian influenza viruses, without showing symptoms. These viruses

can occasionally cross over to infect other species, including humans.

Solution for Question 48:

Correct Option A - HPV 16 & 18:

- High-risk subtypes of human papillomavirus (HPV), particularly HPV 16 and HPV 18, are known to have the highest association with the development of cervical carcinoma.
- Other high-risk human papillomaviruses associated with cervical carcinoma are HPV 31, 33, and 35.

Incorrect Options:

Option B - HPV 6 & 11: HPV 6 and HPV 11 are classified as low-risk subtypes and are commonly associated with condyloma acuminatum and laryngeal papillomatosis.

Option C - HPV 31 & 32: HPV 31 is a high-risk subtype associated with cervical cancer, but HPV 32 is not associated with cervical carcinoma.

Option D - HPV 1 & 2: HPV 1 and HPV 2 are low-risk subtypes associated with plantar warts/verruca and filiform warts.

Solution for Question 49:

Correct Option B - 5-12 years is a High Risk Group:

- The high-risk groups for severe complications of influenza typically include young children (especially those under 2 years), the elderly (65 years and older), pregnant women, individuals with underlying health conditions, and immunocompromised individuals. The age group of 5-12 years is generally considered to have a lower risk compared to these high-risk groups.

Incorrect Options:

Option A - Intranasal Live attenuated vaccine is used: Intranasal live attenuated vaccine, a trivalent vaccine H1N1 (A), H3N2 (A), type B is indicated for individuals with influenza infection, contraindicated in low immunity.

Option C - Aquatic birds act as reservoir: Aquatic birds, particularly wild waterfowl, are considered the natural reservoir for influenza viruses. They can carry different strains of influenza viruses, including avian influenza viruses, without showing symptoms. These viruses can occasionally cross over to infect other species, including humans.

Option D - Secondary attack rate is 5-15%: The secondary attack rate refers to the proportion of susceptible individuals who develop the disease after being exposed to an infected individual. In the case of influenza, the secondary attack rate can vary depending on factors such as the specific strain, population density, and level of immunity. Generally, the secondary attack rate for influenza is estimated to be around 5 to 15%.

Solution for Question 50:

Correct Option B - DNA:

- In an RT-PCR test for the COVID-19 virus, the viral RNA is first converted into complementary DNA (cDNA) using an enzyme called reverse transcriptase. This cDNA is then amplified through the polymerase chain reaction (PCR) using DNA primers and DNA polymerase. The amplification and detection steps target the viral DNA to confirm the presence of the virus.

Incorrect Options:

Option A -RNA: In an RT-PCR test for COVID-19, the viral RNA is converted into complementary DNA (cDNA) using reverse transcriptase enzyme. This cDNA is then amplified through PCR.

Option C - Structural Protein: Structural proteins are components of the virus, but they are not the target of RT-PCR testing. RT-PCR is designed to detect and amplify the genetic material of the virus.

Option D - Envelope: The envelope is another component of the virus, but it is not specifically amplified in RT-PCR testing. The viral envelope is primarily composed of lipid and protein molecules.

Solution for Question 51:

Correct Option A - Human herpesvirus 8:

- Human herpesvirus 8 is also known as Kaposi sarcoma-associated herpesvirus. In immunocompromised individuals HHV 8 is associated with: Kaposi sarcoma, a soft tissue sarcoma of vascular origin, is characterized by dark/violaceous plaques or nodules, as seen in the image, represent vascular proliferations. Primary effusion lymphoma. Castleman disease.
- Kaposi sarcoma, a soft tissue sarcoma of vascular origin, is characterized by dark/violaceous plaques or nodules, as seen in the image, represent vascular proliferations.
- Primary effusion lymphoma.
- Castleman disease.

Human herpesvirus 8 is also known as Kaposi sarcoma-associated herpesvirus. In immunocompromised individuals HHV 8 is associated with:

- Kaposi sarcoma, a soft tissue sarcoma of vascular origin, is characterized by dark/violaceous plaques or nodules, as seen in the image, represent vascular proliferations.
- Primary effusion lymphoma.
- Castleman disease.

Kaposi sarcoma, a soft tissue sarcoma of vascular origin, is characterized by dark/violaceous plaques or nodules, as seen in the image, represent vascular proliferations.

Primary effusion lymphoma.

Castleman disease.

Incorrect Options:

Option B - Human herpesvirus 4: HHV 4, also known as Epstein-Barr virus, is associated with infectious mononucleosis (kissing disease) characterized by organomegaly, gastric carcinoma, Classical Hodgkin's lymphoma, and Burkitt lymphoma.

Option C - Human herpesvirus 6: HHV 6 causes the Sixth disease, aka roseola infantum, aka exanthem subitum characterized by fever and Nagayama spots (erythematous rose pink, non-pruritic papules).

Option D - Human herpesvirus 7: HHV 7 is also associated with few cases of Sixth disease, aka roseola infantum, aka exanthem subitum characterized by fever and Nagayama spots

(erythematous rose pink, non-pruritic papules).

Solution for Question 52:

Correct Option C - Herpes Simplex Virus:

- Mollaret's meningitis, aka recurrent benign lymphocytic meningitis, is a rare condition where there are recurrent episodes of meningitis lasting for 2 to 4 days and associated with spontaneous resolution and intermittent relapses occurring weeks to years apart. The most common pathophysiology is the reactivation of the Herpes simplex virus (HSV 2 >> HSV 1). It can also be caused by the Epstein-Barr virus.

Incorrect Options:

Options A, B, and D are incorrect. These organisms do not cause Mollarets meningitis.

Solution for Question 53:

Correct Option B - Mumps:

- Mumps is a paramyxovirus. It presents with non-specific symptoms like fever, myalgia, and anorexia, and specific symptoms: Bilateral parotitis, the most common presentation, is characterized by acute non-suppurative enlargement of the gland. Second most common manifestation - unilateral orchitis (male) and oophoritis (female), develops in 15–30% of cases in postpubertal males. Orchitis is unilateral in most of the cases.
- Bilateral parotitis, the most common presentation, is characterized by acute non-suppurative enlargement of the gland.
- Second most common manifestation - unilateral orchitis (male) and oophoritis (female), develops in 15–30% of cases in postpubertal males. Orchitis is unilateral in most of the cases.
- Bilateral parotitis, the most common presentation, is characterized by acute non-suppurative enlargement of the gland.

• Second most common manifestation - unilateral orchitis (male) and oophoritis (female), develops in 15–30% of cases in postpubertal males. Orchitis is unilateral in most of the cases.

Incorrect Options:

Option A - Croup: Croup is a common respiratory illness of the trachea, larynx, and bronchi that can lead to inspiratory stridor and barking cough. The parainfluenza virus and H. influenza.

Option C - Infectious mononucleosis: Infectious mononucleosis aka kissing disease is most frequently caused by the Epstein-Barr virus (EBV), characterized by fever, sore throat, arthralgia, lethargy, and organomegaly (splenomegaly).

Option D - Herpangina: Herpangina is an acute febrile fever accompanied by tiny ulcerative or vesicular lesions in the pharynx caused by Coxsackie A virus, and enterovirus 71.

Solution for Question 54:

Correct Option B - HPV 6, 11:

• Condyloma acuminata means anogenital warts characterized by small, light brown, gray, or skin-colored bumps caused by human papillomavirus (HPV). They are usually painless but can grow large and cause itching and bleeding.

• The most common strains of HPV that cause anogenital warts are 6 and 11. It is a double-stranded DNA virus that spreads through sexual contact.

• The virus can infect the skin of the genitals, anus, mouth, and throat.

Incorrect Options:

Option A - HPV 31, 33: Human papillomavirus (HPV) types 31 and 33 are high-risk HPV types and are associated with squamous cell carcinomas like cervical cancer, anal cancer, penile cancer, oral cancer, laryngeal cancer, and esophageal cancer.

Option C - HPV 16,18: HPV 16 and 18 are both high-risk types and are associated with squamous cell carcinomas like HPV 31 and 33.

Option D - HPV 16,11: HPV 16 is a high-risk type and is associated with squamous cell carcinomas. HPV 11 is a low risk and is associated with causing benign conditions, particularly genital warts or condyloma acuminata.

Solution for Question 55:

Correct Option B - Erythema Infectiosum:

- Erythema infectiosum, aka fifth disease, is caused by parvovirus B19, the only DNA virus to possess a single-stranded DNA.
- Symptoms include a mild fever, erythematous rash on cheek (slapped-cheek appearance), and a lacy rash on the arms, legs, and trunk.

Incorrect Options:

Option A - Scarlet fever: Group A Streptococcus is the bacterium that causes scarlet fever, aka 2nd disease.

Option C - Exanthema subitum: Roseola infantum, also known as exanthema subitum or sixth disease, is caused by human herpesvirus 6 (HHV-6) or, less frequently, by human herpesvirus 7 (HHV-7). HHV-6 is a double-stranded deoxyribonucleic acid (DNA) virus.

Option D - Rubeola: Rubeola, aka measles, belongs to paramyxovirus, a single-stranded RNA. Measles is characterized by fever, cough, coryza, conjunctivitis, Koplik spots and maculopapular rash.

Solution for Question 56:

Correct Option A - Aedes Aegypti:

- Aedes aegypti mosquitoes are known as the primary vector for Zika virus transmission.
- These mosquitoes are also responsible for transmitting other diseases such as Eastern equine encephalitis, California encephalitis, Chikungunya, Rift valley fever, Yellow fever, and dengue..

Incorrect Options:

Option B - Culex: Culex mosquitoes are known to transmit Japanese encephalitis, West Nile fever, Western equine encephalitis, and Bancroftian filariasis.

Option C - Anopheles: Anopheles mosquitoes are primarily known for transmitting malaria and Brugia timori.

Option D - Phlebotomus: Phlebotomus is a genus of sandflies which are primarily associated with transmitting diseases like leishmaniasis.

Solution for Question 57:

Correct Option B - Hantavirus:

- Hantavirus is a single-stranded RNA virus responsible for causing Hantavirus pulmonary syndrome. The main target site for Hantavirus is the lungs.

Incorrect Options:

Option A - Influenza Virus: Influenza is caused by Orthomyxovirus and is a single-stranded RNA virus. It has neuraminidase and hemagglutinin spike proteins.

Option C - Adenovirus: It is a double-stranded DNA virus that is responsible for causing respiratory illness.

Option D - None of the Above: This option cannot be the answer since the Hantavirus is an organism shown in the picture.

Solution for Question 58:

Correct Option A - Measles:

- Koplik spots are pathognomonic features characterized by small, white-colored bumps on the erythematous base that appear in the mouth in the prodromal stage of measles.
- Prodromal phase of measles Fever - 1st manifestation around 10th day. 3 C's - Cough, coryza, and conjunctivitis. Pathognomonic feature - Kopliks spot near molar appears after 2 days of fever.
- Fever - 1st manifestation around 10th day.

• 3 C's - Cough, coryza, and conjunctivitis.

• Pathognomonic feature - Kopliks spot near molar appears after 2 days of fever.

• Fever - 1st manifestation around 10th day.

• 3 C's - Cough, coryza, and conjunctivitis.

• Pathognomonic feature - Kopliks spot near molar appears after 2 days of fever.

Incorrect Options:

Options B, C, and D are incorrect. Koplik spots are not seen in these conditions.

Solution for Question 59:

Correct Option A - Enveloped ssRNA:

• The characters stated in question point towards measles infection.

• Measles is caused by a single-stranded, enveloped RNA virus with 1 serotype. It is classified as a member of the genus Morbillivirus in the Paramyxoviridae family. Humans are the only natural hosts of the measles virus.

• Measles presents with Fever - 1st manifestation around 10th day. 3 C's - Cough, coryza, and conjunctivitis. Pathognomonic feature - Kopliks spot near molar, white color dots on erythematous base. Rash - appears on the 14th day; the maculopapular rash starts from the ear to the face & follows the rest of the body.

• Fever - 1st manifestation around 10th day.

• 3 C's - Cough, coryza, and conjunctivitis.

• Pathognomonic feature - Kopliks spot near molar, white color dots on erythematous base.

• Rash - appears on the 14th day; the maculopapular rash starts from the ear to the face & follows the rest of the body.

• Fever - 1st manifestation around 10th day.

• 3 C's - Cough, coryza, and conjunctivitis.

• Pathognomonic feature - Kopliks spot near molar, white color dots on erythematous base.

• Rash - appears on the 14th day; the maculopapular rash starts from the ear to the face & follows the rest of the body.



Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 60:

Correct Option A - a-2, b-3, c-1, d-4:

The incubation period of

- Syphilis- 90 days (9 to 90 days).
- Chickenpox- 16 days (14 to 21 days).
- SARS-COVID -6 days (4 to 14days).
- Hepatitis A- 28 days (15 to 50 days).

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 61:

Correct Option C - Amoxiclav:

• Antimicrobial prophylaxis is recommended for various scenarios, including primary wound closure, moderate to severe bite wounds, puncture wounds, facial, hand or foot, and genital area bites, and in immunocompromised or asplenic individuals.

• For high-risk dog wounds in healthy patients, the first-line antimicrobial prophylaxis is typically amoxicillin-clavulanate for 3 to 5 days. Alternatively, ampicillin-sulbactam may be administered intravenously if oral antibiotics are not feasible. In cases of penicillin allergy, extended-spectrum cephalosporins or trimethoprim-sulfamethoxazole plus clindamycin are recommended.

• Amoxiclav is a broad-spectrum antibiotic used for a combination of Gram-positive, Gram-negative, and anaerobic bacteria.

Incorrect Options:

Option A - Metronidazole: Metronidazole is an antibiotic commonly used to treat infections caused by anaerobic bacteria and certain parasites. It needs to be added with others for aerobic bacteria. However, it is not the recommended treatment for a dog bite.

Option B - Ciprofloxacin: Ciprofloxacin is an antibiotic effective against many bacteria. While it may be used to treat certain bacterial infections, it is not appropriate for a dog bite in this case. If used, it needs to be given with metronidazole for anaerobic cover.

Option D - Observation: In this case, both the dog and the person are fully immunized, and there is a laceration on the bite site. The person should clean the wound thoroughly with soap and water, apply antiseptic, and use broad-spectrum antibiotics like amoxiclav.

Solution for Question 62:

Correct Option C - Flaviviruses:

• This statement is incorrect as rubella is a Togaviruses with a linear and single-stranded RNA.

Incorrect Options:

Option A - It is one of the teratogenic TORCH infections: Rubella is indeed one of the teratogenic TORCH infections. TORCH infections refer to a group of infections that can be transmitted from mother to fetus during pregnancy. The acronym TORCH stands for Toxoplasmosis, Other (syphilis, varicella-zoster, parvovirus B19), Rubella, Cytomegalovirus (CMV), and Herpes simplex virus (HSV).

Option B - Causes maculopapular rash that starts on the face and spreads caudally to involve the trunk and extremities: Rubella does cause a maculopapular rash that typically begins on the face and then spreads downward to involve the trunk and extremities. This characteristic rash is often one of the key clinical features of rubella infection.

Option D - Symptoms include “Blueberry muffin” appearance: Symptoms of rubella infection can indeed include a "Blueberry muffin" appearance in newborns who acquire the infection congenitally. This appearance refers to purpuric or petechial skin lesions resembling the appearance of blueberries, which can occur due to widespread hematopoiesis in various organs.

Solution for Question 63:

Correct Option A - Aplastic crises in sickle cell disease:

- Conditions associated with Parvovirus B19 Erythema Infectiosum (5th disease) Presents as respiratory infection with erythematous rash present on the cheek (slapped cheek appearance) and lacy rash on the rest of the body Followed by lymphadenopathy and arthralgia In Adults: papular Purpuric Gloves & socks syndrome (PPGSS) Aplastic Crisis in patients with underlying hemolytic anemia Pure red cell aplasia in patients with underlying immunosuppression. Hydrops Fetalis.
- Erythema Infectiosum (5th disease) Presents as respiratory infection with erythematous rash present on the cheek (slapped cheek appearance) and lacy rash on the rest of the body Followed by lymphadenopathy and arthralgia
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- Followed by lymphadenopathy and arthralgia
- In Adults: papular Purpuric Gloves & socks syndrome (PPGSS)
- Aplastic Crisis in patients with underlying hemolytic anemia
- Pure red cell aplasia in patients with underlying immunosuppression.
- Hydrops Fetalis.

Conditions associated with Parvovirus B19

- Erythema Infectiosum (5th disease) Presents as respiratory infection with erythematous rash present on the cheek (slapped cheek appearance) and lacy rash on the rest of the body Followed by lymphadenopathy and arthralgia

• Presents as respiratory infection with erythematous rash present on the cheek (slapped cheek appearance) and lacy rash on the rest of the body

• Followed by lymphadenopathy and arthralgia

• In Adults: papular Purpuric Gloves & socks syndrome (PPGSS)

• Aplastic Crisis in patients with underlying hemolytic anemia

• Pure red cell aplasia in patients with underlying immunosuppression.

• Hydrops Fetalis.

• Presents as respiratory infection with erythematous rash present on the cheek (slapped cheek appearance) and lacy rash on the rest of the body

• Followed by lymphadenopathy and arthralgia

Incorrect Options:

• Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 64:

Correct Option D - Parvovirus:

• Conditions associated with Parvovirus B19 Erythema Infectiosum (5th disease/slapped cheek appearance) In Adults: papular Purpuric Gloves & socks syndrome (PPGSS) Aplastic Crisis in patients with underlying hemolytic anemia. E.g., Hereditary Spherocytosis and Sickle cell anemia. Pure red cell aplasia in patients with underlying immunosuppression. Hydrops Fetalis.

• Erythema Infectiosum (5th disease/slapped cheek appearance)

• In Adults: papular Purpuric Gloves & socks syndrome (PPGSS)

• Aplastic Crisis in patients with underlying hemolytic anemia. E.g., Hereditary Spherocytosis and Sickle cell anemia.

• E.g., Hereditary Spherocytosis and Sickle cell anemia.

• Pure red cell aplasia in patients with underlying immunosuppression.

• Hydrops Fetalis.

• Erythema Infectiosum (5th disease/slapped cheek appearance)

• In Adults: papular Purpuric Gloves & socks syndrome (PPGSS)

• Aplastic Crisis in patients with underlying hemolytic anemia. E.g., Hereditary Spherocytosis and Sickle cell anemia.

• E.g., Hereditary Spherocytosis and Sickle cell anemia.

• Pure red cell aplasia in patients with underlying immunosuppression.

• Hydrops Fetalis.

• E.g., Hereditary Spherocytosis and Sickle cell anemia.

Incorrect Options:

Option A - Pox virus: Pox virus is a large and complex virus known to cause two major diseases smallpox and molluscum contagiosum.

Option B - Epstein-Barr virus: Epstein-Barr virus (EBV) is a member of the herpesvirus family. EBV can cause majorly infectious mononucleosis, gastric carcinoma, Burkitt's lymphoma, and nasopharyngeal carcinoma:

Option C - Adenovirus: Adenovirus is known to cause various benign diseases like upper and lower respiratory tract infections, gastroenteritis in children, urinary tract infections, and conjunctivitis.

Solution for Question 65:

Correct Option A - 1:

• Virus which have segmented genome will show genetic reassortment in which co-infection of a host cell with multiple viruses results in the shuffling of gene segments to generate progeny viruses with novel genome combinations. Hence Option A shows virus is having segmented genome.

• Reassortment is exclusively seen in viruses with a segmented genome. It is defined as the exchange of intact genes within the entire segment, which occurs during coinfection.

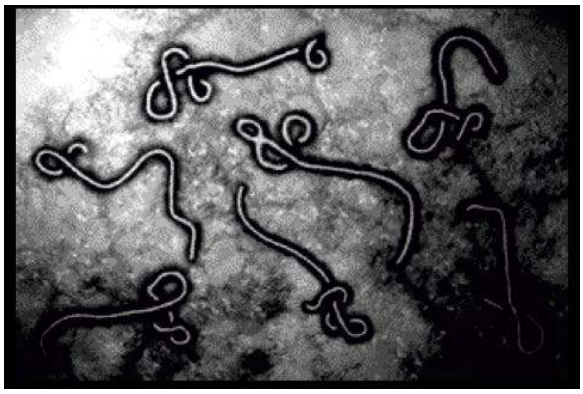
Virus	Segments
Bunyavirus	3 segments
Influenza	8 segments
Reovirus/rotavirus	11 segments
Arenavirus	2 segments

Incorrect Options:

- Options B, C, and D do not possess a segmented genome

Solution for Question 66:

Correct Option A:



- Filoviruses are filamentous, enveloped structures with a single-stranded RNA. A single glycoprotein forms the spikes on the virion surface. The nucleocapsid contains the RNA and four viral structural proteins, including the virus-encoded polymerase.

Incorrect Options:

Option B: It is incorrect as the given image is of Adenovirus, giving the appearance of a space vehicle.

Option C: It is incorrect as given image is of Influenza A virus which is an RNA virus showing 2 proteins on the surface of the viral envelope i.e hemagglutinin, neuraminidase.

Option D: It is incorrect as the given image is of Coronavirus showing club-shaped surface projections surrounding the periphery (crown with peplomers).

Solution for Question 67:

Correct Option D - 4:

Conditions associated with Parvovirus B19

- Erythema Infectiosum (5th disease; slapped cheek appearance) Macular erythematous rash present on cheek Lacy rash on the rest of the body Self-limiting
- Macular erythematous rash present on cheek
- Lacy rash on the rest of the body
- Self-limiting
- In Adults: papular Purpuric Gloves & socks syndrome (PPGSS)
- Arthritis, Arthralgia
- Aplastic Crisis in patients with underlying hemolytic anemia
- Pure red cell aplasia in patients with underlying immunosuppression.
- Hydrops Fetalis.
- Macular erythematous rash present on cheek
- Lacy rash on the rest of the body
- Self-limiting

Incorrect Options:

Option A - 1: Aplastic anemia is associated with parvovirus infection as parvovirus has an affinity towards erythroid precursor cells, leading to an aplastic crisis in sickle cell anemic patients.

Option B - 2: Parvovirus Viral envelop - Naked virus. This option is the correct pair.

Option C - 3: Parvovirus is associated with Erythema infectiosum – infants, also known as the 5th disease, present a Slapped cheek appearance.

Solution for Question 68:

Correct Option B - E6, E7:

- HPV has two oncoproteins, E6 and E7, which are responsible for the development of HPV-induced carcinogenesis. E6 will inactivate p53, and E7 will inactivate the RB gene, which leads to cancer. They act as oncogenes.

Incorrect Options:

Option A - E1, E2, E5: E1 is responsible for cell fusion, E2 is a controller, and E5 is responsible for koilocytosis.

Option C - L1, L2: The capsid protein L2 plays an important role in both papillomavirus assembly and the infectious process.L1 forms the majority of the capsid and can self-assemble into empty virus-like particles (VLPs)

Option D - E1, E2: E1 is responsible for cell fusion, and E2 is a controller.

Solution for Question 69:

Correct Option D - Roseola infantum:

- Roseola infantum/Sixth disease/exanthem subitum is caused by herpes virus 6. It is characterized by fever and characteristic nagayama spots.
- Nagayama spots Erythematous, rose pink, non-pruritic papules after fever.
- Erythematous, rose pink, non-pruritic papules after fever.
- Erythematous, rose pink, non-pruritic papules after fever.

Incorrect Options:

Options A, B, and C are incorrect. A petechial rash is seen in all these cases.

Solution for Question 70:

Correct Option D - IgG avidity assay helps in differentiating past and primary infection:

- Positive CMV IgG results indicate past or recent CMV infection. IgG avidity assays are the test that shows the binding strength between IgG antibodies and CMV. In primary CMV infection, IgG antibodies will show low binding strength (low avidity) initially (acute infection), then after 4 months, affinity increases suggesting past infection.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 71:

Correct Option C - CSF PCR for viral DNA:

- PCR (Polymerase Chain Reaction) is a highly sensitive diagnostic technique used to detect the presence of viral DNA or RNA.
- HSV type 1 is the most common cause of encephalitis, and PCR can also differentiate between the two species, so recommended for HSV encephalitis.

Incorrect Options:

Option A - IgM in CSF: IgM antibodies are produced during the acute phase of an infection and can be detected in the cerebrospinal fluid (CSF). A four fold increase in the antibodies is suggestive of herpes and cannot differentiate HSV 1 and HSV 2..

Option B - Tzanck smear of CSF: Tzanck smear is a microscopic examination of cells obtained from a skin lesion to detect the presence of viral infections. Tzanck smear is has low sensitivity and cannot differentiate between HSV 1 and HSV 2.

Option D - CSF culture on chick embryo lines: CSF culture involves the growth and isolation of microorganisms in a laboratory setting. HSV is a fastidious virus that requires specialized cell cultures, such as human embryonic fibroblasts, for optimal growth. Culturing HSV from CSF is technically challenging and time-consuming, making it less practical for rapid diagnosis.

Solution for Question 72:

Correct Option A - Antigen detection in the stool sample:

- Rotavirus is a common cause of viral gastroenteritis, particularly in infants and young children. The virus is primarily transmitted through the fecal-oral route, and the infection leads to symptoms such as watery diarrhea, vomiting, fever, and abdominal pain.

• Laboratory Diagnosis of rotavirus Direct detection of virus: Feces collected early in the illness are the most ideal specimen. Rotaviruses can be demonstrated in stool by Immunoelectron microscopy (IEM). They have a sharp-edged triple-shelled capsid that looks like the spokes grouped around the hub of a wheel. Detection of viral antigen in stool by ELISA and latex agglutination-based methods RT-PCR is the most sensitive detection method for the detection of rotavirus from the stool. Typing methods: G serotypes and P genotypes of rotaviruses can be detected by RNA sequence typing and neutralization test, respectively Serologic tests (ELISA) can be used to detect the rise of antibody titer.

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Laboratory Diagnosis of rotavirus

• Direct detection of virus: Feces collected early in the illness are the most ideal specimen. Rotaviruses can be demonstrated in stool by Immunoelectron microscopy (IEM). They have a sharp-edged triple-shelled capsid that looks like the spokes grouped around the hub of a wheel.

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• Serologic tests (ELISA) can be used to detect the rise of antibody titer.

• RT-PCR is the most sensitive detection method for the detection of rotavirus from the stool.

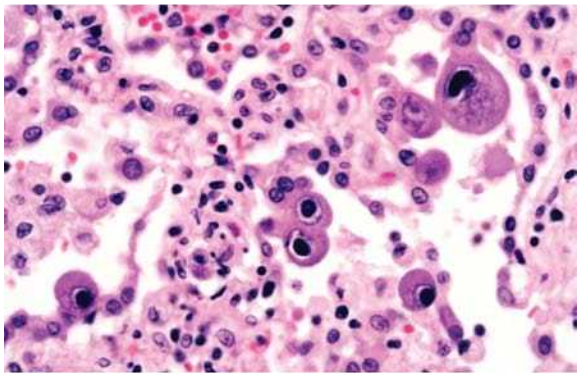
Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an example.

Solution for Question 73:

Correct Option A - Cytomegalovirus:

- Cytomegalovirus is a member of the Herpesviridae family and is a common viral infection.
- It can cause a wide range of symptoms, from mild or asymptomatic infections to severe manifestations, especially in immunocompromised individuals and newborns.
- The key feature mentioned in the case that points towards CMV infection is the presence of cells with an owl eye appearance in the urine. These cells are characteristic of CMV infection and represent enlarged cells with intranuclear inclusions surrounded by a clear halo, giving them an owl's eye-like appearance.
- In children, it can be associated with a mononucleosis-like syndrome, which includes symptoms such as fever, fatigue, sore throat, and lymphadenopathy. The presence of a rash further supports the possibility of a viral infection.



Incorrect Options:

Options B, C, and D are incorrect. These organisms do not show owl eye appearance as intranuclear inclusion bodies.

Solution for Question 74:

Correct Option B - Dengue Hemorrhagic Fever:

- Dengue fever is a viral infection transmitted by mosquitoes, particularly the Aedes aegypti mosquito. It is prevalent in tropical and subtropical regions, including Delhi.

• The typical clinical features of dengue fever include fever (aka break bone fever), arthralgia (joint pain), retrobulbar pain, and a characteristic petechial rash.

• The laboratory findings, in this case, are consistent with dengue fever. The hemoglobin level of 9 g/dL indicates mild anemia, which can be seen in dengue due to the destruction of red blood cells. The white blood cell count of 9000 cells/mm3 is within the normal range, which is also typical in dengue.

• However, the platelet count of 20000 cells/mm3 and a positive tourniquet test are common findings seen in dengue hemorrhagic fever.

WHO Grading Dengue Infections:

DF/DHF	Grade	Symptoms	Labs
DF	-	Fever with two or more the following features Headache Retro orbital pain Myalgia Arthralgia	Leukopenia Thrombocytopenia (≤100000) No evidence of plasma loss
DHF	I	Above signs plus Positve tourniquet test	Thrombocytopenia (<100000) HCT rises (>20%)
DHF	II	Above signs plus Spontaenous bleeding	Thrombocytopenia (<100000) HCT rises (>20%)
DHF	III	Above sign plus circulatory failure (weak pulse, hypotension, restlessness)	Thrombocytopenia (<100000) HCT rise >20%
DSS	IV	Profound shock with undetectable blood pressure and pulse	Thrombocytopenia (<100000) HCT rises (>20%)

Incorrect Options:

Options A and C are incorrect. Refer to Option B for an explanation.

Option D - Malaria: Malaria is also a mosquito-borne infectious disease prevalent in tropical and subtropical regions. It can cause fever, but the presentation usually includes periodic fevers with chills and rigors rather than a continuous fever. Additionally, the laboratory findings in malaria typically show a normal or elevated platelet count, whereas this patient has thrombocytopenia (low platelet count).

Solution for Question 75:

Correct Option A - Hepatitis A:

- Hepatitis A spreads through fecal-oral route, i.e., water and food that has been contaminated by stool containing the virus. This is more common in areas that have poor sanitation.

Incorrect Options:

Option B - Hepatitis B: It is transmitted through the parenteral route, like bodily fluids, such as saliva, blood, and semen, or through a contaminated object, such as a toothbrush or razor, where the virus can live for days.

Option C - Hepatitis C: It is transmitted from person to person, primarily through contact with contaminated blood, such as sharing needles with IV drug abusers and unprotected sex with multiple partners.

Option D - Hepatitis D: It is transmitted via bodily fluids like the hepatitis B virus.

Solution for Question 76:

Correct Option D - Enterovirus:

- CSF findings revealing normal glucose, lymphocytosis, and increased protein is highly suggestive of Viral etiology.

	White Cell Count	Biochemistry		
Neutrophils (x 106/L)	Lymphocytes (x 106/L)	Protein (g/L)	Glucose (CSF: blood ratio)	
Normal	0	≤ 5	< 0.4	≥ 0.6
Bacterial meningitis	100-10,000	Usually < 100	>1.0	<0.4
Viral meningitis	Usually <100	10-1000	0.4-1	Usually normal
TB meningitis	Usually <100	50-1000	1-5	< 0.3

• Virus causing meningitis are Enteroviruses (ECHO, Coxsackie, Polio, Paramyxoviruses (Mumps, measles), Herpesviruses (Herpes simplex, Varicella zoster), Adenovirus, and Arboviruses (Flavivirus, Buniyavirus).

• Enteroviruses, including coxsackieviruses, echoviruses, and poliovirus can cause viral meningitis. They are commonly associated with fever, headache, and meningitis symptoms.

Incorrect Options:

Option A - West Nile virus: West Nile virus is a flavivirus, primarily transmitted through mosquito bites and can cause a range of symptoms, including fever, headache, and meningitis.

Option B - Neisseria meningitidis: It is a common cause of bacterial meningitis.

Option C - Streptococcus pneumoniae: Streptococcus pneumoniae is a common bacterial cause of meningitis.

Solution for Question 77:

Correct Option A - There is a gradual sequential change in antigenic structure:

• Antigenic drift is a phenomenon that occurs in influenza viruses. It involves a gradual sequential change in the antigenic structure of the virus due to mutations in the genes that code for its surface proteins.

• These surface proteins, hemagglutinin (HA) and neuraminidase (NA) are responsible for the recognition and binding of the virus to host cells and the release of newly formed virus particles from infected cells, respectively.

• The accumulation of these mutations in the HA and NA genes over time can lead to the emergence of new strains.

Incorrect Options:

Option B - Causes a sudden increase in the number of cases: This is inaccurate because antigenic drift is less likely to cause major epidemics and pandemics.

Option C - The new antigens formed are unrelated to the previous antigens: The new antigens formed due to antigenic drift are related to the previous antigens, so they react with antisera to the predecessor virus strains to varying degrees.

Option D - Antibodies to the predecessor viruses do not neutralize the new variants: Antibodies to the predecessor viruses still provide some degree of cross-protection against the new

variants, but the degree of protection may vary depending on the degree of antigenic similaritybetween the old and new strains.

Solution for Question 78:

Correct Option B - Influenza Virus:

- The correct choice for the given form of replication is the Influenza virus.
- Influenza virus is an RNA virus that belongs to the Orthomyxoviridae family. Its replication process involves several distinct steps: Attachment: The spikes of hemagglutinin bind to sialic acid found on the surface of the host cell’s membrane. Entry: The virus enters the host cell by endocytosis, which involves the engulfment of the virus particle by the cell membrane and the formation of a vesicle called an endosome. Fusion: The viral envelope fuses with the endosomal membrane, and the M2 ion channel allows entry of H+ ions - H+ influx, resulting in uncoating and releasing the viral genetic material into the host cell's cytoplasm. Replication: The viral RNA serves as a template for the synthesis of new viral RNA segments (turning negative sense RNA into positive sense RNA), as well as viral proteins. The viral RNA polymerase, which is carried by the virus, is responsible for replicating and transcribing the viral RNA. Assembly: Newly synthesized viral RNA segments and proteins are assembled into new virus particles within the host cell. Budding: The assembled virus particles bud from the host cell's plasma membrane, acquiring an envelope derived from the host cell membrane. This process allows the virus to be released by neuraminidase and potentially infect other cells.
- Attachment: The spikes of hemagglutinin bind to sialic acid found on the surface of the host cell’s membrane.
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• Assembly: Newly synthesized viral RNA segments and proteins are assembled into new virus particles within the host cell.

• Budding: The assembled virus particles bud from the host cell's plasma membrane, acquiring an envelope derived from the host cell membrane. This process allows the virus to be released by neuraminidase and potentially infect other cells.

• Attachment: The spikes of hemagglutinin bind to sialic acid found on the surface of the host cell’s membrane.

• Entry: The virus enters the host cell by endocytosis, which involves the engulfment of the virus particle by the cell membrane and the formation of a vesicle called an endosome.

• Fusion: The viral envelope fuses with the endosomal membrane, and the M2 ion channel allows entry of H+ ions - H+ influx, resulting in uncoating and releasing the viral genetic material into the host cell's cytoplasm.

• Replication: The viral RNA serves as a template for the synthesis of new viral RNA segments (turning negative sense RNA into positive sense RNA), as well as viral proteins. The viral RNA polymerase, which is carried by the virus, is responsible for replicating and transcribing the viral RNA.

• Assembly: Newly synthesized viral RNA segments and proteins are assembled into new virus particles within the host cell.

• Budding: The assembled virus particles bud from the host cell's plasma membrane, acquiring an envelope derived from the host cell membrane. This process allows the virus to be released by neuraminidase and potentially infect other cells.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 79:

Correct Option A - Ganciclovir resistance is due to mutation in UL97 phosphotransferase gene:

• Ganciclovir is a nucleoside DNA polymerase inhibitor an antiviral drug commonly used for the treatment of CMV infections. Resistance to ganciclovir can occur due to mutations in the UL97 and UL54 gene of CMV.

• Ganciclovir is converted by viral phosphotransferase encoded by the CMV UL97 gene to form ganciclovir triphosphate that inhibits CMV DNA polymerase. Mutations in the UL97

gene can result in reduced phosphorylation of ganciclovir, leading to decreased efficacy ofthe drug and the development of resistance.

Incorrect Options:

Option B - CMV can be cultured in vitro in any mammalian cell: Human fibroblasts are the most specific and ideal cell lines for the isolation of the cytomegalovirus.

Option C - The commonest manifestation of CMV-associated disease is watery diarrhea: CMV is the most common infection associated with congenital defects. In Immunocompetent individuals, after transplantation, it causes mononucleosis-like disease, and in immunocompromised individuals (HIV/transplantation), it causes, most commonly, CMV retinitis.

Option D - UL54 mutated CMV strains do not show cross-resistance to Foscarnet: UL54 refers to the DNA polymerase gene of CMV, which is the target of antiviral drugs like ganciclovir and foscarnet, and mutation in this gene can lead to resistance against ganciclovir, foscarnet, and cidofovir.

Solution for Question 80:

Correct Option B - Detection of HSV Causing Temporal Encephalitis:

• Real-time PCR is used for the quantification, hence real-time PCR is not preferred for detection of HSV causing temporal encephalitis, instead conventional PCR is preferred for the detection of the HSV DNA in cerebrospinal fluid. It can also differentiate between HSV 1 and HSV 2. Monitoring viral load is not routinely performed.

Incorrect Options:

Option A - Person with Hepatitis B on Tenofovir Therapy: In individuals with hepatitis B receiving tenofovir therapy, monitoring the viral load using real-time PCR is essential. It helps determine the response to treatment, assess viral suppression, and evaluate the risk of drug resistance.

Option C - BK Virus in a Patient with Allograft Renal Transplant: BK virus infection can occur in patients who have undergone renal transplantation. Monitoring the viral load using real-time PCR in urine or blood samples is important to detect and manage BK virus-associated complications, such as nephropathy.

Option D - CMV in a Patient with Liver Transplant: Cytomegalovirus infection is a significant concern in patients undergone solid organ transplantation. Real-time PCR for CMV viral load in

blood is commonly performed in these patients to monitor for active infection, guide antiviraltherapy, and assess response to treatment.

Solution for Question 81:

Correct Option B - RNA:

- Northern blotting is a laboratory technique used to study gene expression by detecting and analyzing mRNA. It involves the separation of RNA molecules based on their size using gel electrophoresis, followed by their transfer onto a solid support membrane (usually nylon or nitrocellulose). The transferred RNA molecules are then immobilized on the membrane.
- To detect specific RNA molecules, the membrane is hybridized with a labeled nucleic acid probe that is complementary to the target RNA sequence of interest. The probe binds specifically to its complementary RNA sequence on the membrane, allowing the detection and visualization of the target RNA molecule.

Incorrect Options:

Option A - DNA: Southern blotting is used for the detection of DNA molecules.

Option C - Proteins: Western blotting, immunohistochemistry, or enzyme-linked immunosorbent assay (ELISA) are commonly used for protein detection and analysis.

Option D - None: Northern blotting is specifically used for the detection of RNA molecules.

Solution for Question 82:

Correct Option B - Hepatitis A Virus (HAV):

- Hepatitis A virus (HAV) infection is likely to occur with transfusion of solvent/detergent (SD) plasma. Solvent/detergent treatment is a method used to inactivate lipid-enveloped viruses in plasma products. It involves the use of organic solvents and detergents to disrupt the lipid envelope of viruses, rendering them non-infectious. This treatment method is effective against lipid-enveloped viruses.

• Non-enveloped DNA viruses - parvovirus, adenovirus and papovavirus

• Non-enveloped RNA viruses - picornavirus, reovirus, calicivirus, hepatitis A virus, and hepatitis E virus.

Incorrect Options:

Options A, C, and D are incorrect. These are enveloped viruses that can be destroyed by the solvent/detergent method.

Solution for Question 83:

Correct Option C - Adult T-cell Leukemia:

• Human herpesvirus 8 (HHV-8) is not associated with Adult T-cell leukemia. Adult T-cell leukemia is caused by human T-cell leukemia virus type 1 (HTLV-1), a different virus belonging to the Retroviridae family.

• Human herpesvirus 8, also known as Kaposi's sarcoma-associated herpesvirus (KSHV), is a member of the herpesvirus family. It is primarily associated with certain malignancies and diseases, including: Kaposi sarcoma: A soft tissue sarcoma of vascular origin characterized by dark/violaceous plaques or nodules in immunocompromised individuals. Primary effusion lymphoma: A rare type of non-Hodgkin lymphoma characterized by the accumulation of lymphoma cells in body cavities, such as the pleural or peritoneal cavity. Castleman disease: A rare lymphoproliferative disorder characterized by enlarged lymph nodes.

• Kaposi sarcoma: A soft tissue sarcoma of vascular origin characterized by dark/violaceous plaques or nodules in immunocompromised individuals.

• Primary effusion lymphoma: A rare type of non-Hodgkin lymphoma characterized by the accumulation of lymphoma cells in body cavities, such as the pleural or peritoneal cavity.

• Castleman disease: A rare lymphoproliferative disorder characterized by enlarged lymph nodes.

• Kaposi sarcoma: A soft tissue sarcoma of vascular origin characterized by dark/violaceous plaques or nodules in immunocompromised individuals.

• Primary effusion lymphoma: A rare type of non-Hodgkin lymphoma characterized by the accumulation of lymphoma cells in body cavities, such as the pleural or peritoneal cavity.

• Castleman disease: A rare lymphoproliferative disorder characterized by enlarged lymph nodes.

Incorrect Options: Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 84:

Correct Option D - Real-time RT-PCR:

- The detection of the target sequences against the RdRp gene, E gene, N gene, and ORF1 confirms the diagnosis of coronavirus.
- Real-time RT-PCR is a gold standard test for the diagnosis of coronavirus through the target sequence against these screening and confirmatory genes.
- It can even detect 10 copies/ml and can run 90 samples in a single run.
- The average time taken is around 4-6 hrs.
- Gene targets for screening are Envelope protein (E. gene) Nucleocapsid protein (N. gene) Spike protein (S. gene)
- Envelope protein (E. gene)
- Nucleocapsid protein (N. gene)
- Spike protein (S. gene)
- Confirmatory genes are RNA-dependent RNA polymerase gene (RdRp) Open reading frames (ORF1a/b)
- RNA-dependent RNA polymerase gene (RdRp)
- Open reading frames (ORF1a/b)
- Principle: The target gene/s in the specimen is amplified in the thermocycler When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value is inversely proportional to the load. A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.
- The target gene/s in the specimen is amplified in the thermocycler
- When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value is inversely proportional to the load.
- A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.
- Envelope protein (E. gene)

• Nucleocapsid protein (N. gene)

• Spike protein (S. gene)

• RNA-dependent RNA polymerase gene (RdRp)

• Open reading frames (ORF1a/b)

• The target gene/s in the specimen is amplified in the thermocycler

• When the amplicon binds with the probe, a fluorescence is generated. The point at which the fluorescence starts is the cycle threshold (Ct) of the run. CT value is inversely proportional to the load.

• A sample is considered positive when both screenings, as well as confirmatory genes, are detected with a Ct value ≤ 40 cycles.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

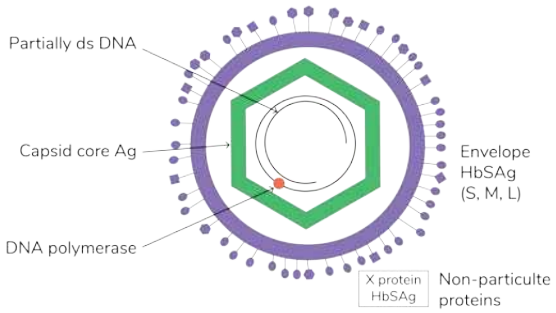
Solution for Question 85:

Correct Option A - HBsAg:

• The Australia antigen, also known as the Australia antigen marker, is represented by the HBsAg.

• The presence of HBsAg in the blood indicates an active Hepatitis B infection. It is an important marker used for screening and diagnosis of Hepatitis B.

Structural proteins of HBV	Description
P – DNA Polymerase	Largest Reverse transcriptase RNAse
C – Core gene	Has a core area and a pre-core area Core area coats for C antigen Pre-core area codes for E antigen
S – Surface Antigen (Australia Antigen)	These have different sizes (Small, medium and large)
X – HBX responsible for carcinogenesis	Liver (Hepatocellular carcinoma)



Incorrect Options:

Option B - HBeAg: HBeAg is another antigen produced by the Hepatitis B virus during active viral replication. Its presence indicates high viral replication and infectivity. HBeAg is used as a marker to assess the severity and prognosis of Hepatitis B infection.

Option C - HBcAg: Hepatitis B core antigen (HBcAg) is an antigen that is located in the inner core of the Hepatitis B virus.

Option D- HBX: HBX is an oncoprotein responsible for hepatocellular carcinoma.

Solution for Question 86:

Correct Option A - Cytomegalovirus:

- Cytomegalovirus (CMV), is the predominant pathogen to infect renal (solid organ) transplant recipients in the period around 1 to 4 months following transplantation because of reactivation of the virus in immunocompromised individuals.
- The immunosuppressive medications taken by transplant recipients increase their susceptibility to CMV infection, which can lead to significant morbidity and complications in the early post-transplant period.
- It presents in various forms, such as: Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients Febrile leukopenia is seen among solid organ transplant recipients. Obliterative bronchiolitis in lung transplants. Graft atherosclerosis in heart transplants. Rejection of renal allografts.
- Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients

• Febrile leukopenia is seen among solid organ transplant recipients.

• Obliterative bronchiolitis in lung transplants.

• Graft atherosclerosis in heart transplants.

• Rejection of renal allografts.

• Bilateral interstitial pneumonia, a life-threatening complication seen in 15–20% of bone marrow and solid organ transplant recipients

• Febrile leukopenia is seen among solid organ transplant recipients.

• Obliterative bronchiolitis in lung transplants.

• Graft atherosclerosis in heart transplants.

• Rejection of renal allografts.

Incorrect Options:

Options B, C, and D are incorrect. They are not the most common organism to cause infection in an organ transplant recipient between 1 to 4 months.

Solution for Question 87:

Correct Option B - Rabies:

• Rabies is a viral disease that affects the central nervous system characterized by hypersalivation, arrhythmia, hydrophobia, and hyperexcitability. It is usually transmitted through the bite of an infected animal, commonly dogs, bats, or other mammals. Once symptoms of Rabies appear, the disease is almost always fatal (100% mortality rate).

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 88:

Correct Option A - Time of Incubation:

• The presence of antibodies depends on the incubation period, presence of antibody against SARS-CoV-2 in the blood indicates that the individual has been exposed to the virus and their immune system has produced an immune response.

• IgM antibody can be detected after the incubation period of 7 days and disappears by 21 days.

• IgG antibody starts appearing in the blood after the incubation period of 14 days and remains in blood providing immunity.

Incorrect Options:

Option B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 89:

Correct Option C - BSL 3:

• SARS-CoV-2 causes serious or potentially lethal diseases via inhalation and comes under biosafety level 3.

Biosafety level	Characteristics	Pathogens/Disease
Biosafety level I	The level of precaution at this level is minimal and usually involves wearing a face mask and no close contact.	Escherichia coli
Biosafety level 2	Involves organisms responsible for mild infections in humans and are often difficult to contract via aerosolized particles	Hepatitis A, B, and C, Lyme disease, Salmonella, measles, mumps, HIV, and dengue.
Biosafety level 3	Includes organisms that can be fatal to humans but for which vaccines and other treatments are available.	Mycobacterium tuberculosis, Bacillus anthracis, yersinia, SARS-Cov-2, hantavirus, Rift Valley fever, Malaria, Rocky Mountain spotted fever, and Yellow fever.
Biosafety level 4	Includes dangerous viruses that may cause serious or lethal infections with no treatment.	Ebola virus, Marburg virus, Lassa fever, Bolivian hemorrhagic fever, Nipah virus, and Hendra virus.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 90:

Correct Option D - Epstein-Barr Virus:

- Oral hairy leukoplakia is a condition characterized by wart-like growth of epithelial cells of the tongue. It is typically seen in individuals with compromised immune systems, such as those with HIV/AIDS or transplant recipients. The underlying cause of oral hairy leukoplakia is Epstein-Barr virus.

Incorrect Options:

Options A, B, and C are incorrect. These organisms are not associated with oral hairy cell leukoplakia.

Solution for Question 91:

Correct Option A - HIV:

- In the image above, two copies of the ssRNA genome and reverse transcriptase point toward HIV.
- Generally, all RNA viruses have a single copy of RNA except the HIV and HTLV viruses, both of which have 2 copies of genomic RNA. Both of them also contain reverse transcriptase enzymes and are thus called retroviruses.
- Additionally, all RNA viruses replicate inside the cytoplasm, except Retroviruses (HIV and HTLV). Retroviruses with the help of reverse transcriptase convert their RNA to DNA and replicate in the nucleus.
- 3 structural genes of HIV:

Env (envelope) (gp 160)	Surface: gp120 Transmembrane: gp41

Gag	P6 – Budding protein (BP) P7 – Nucleocapsid (NC) P17 – Matrix (MA) P24 – Capsid (CA), most important [Needle stick injury]
Pol (Pnemonic – RIP)	Reverse transcriptase: p66, p51 IN (Integrase, p32) PR – Protease, p12

Incorrect Options:

Option B - EBV: Epstein-Barr virus also called the HHV-4 (human herpesvirus 4) belongs to the family of herpes viruses. It is an enveloped ds DNA virus and thus does not have a reverse transcriptase enzyme.

Option C - CMV: Cytomegalovirus is also called the HHV-5 (Human herpes virus 5) and is also an enveloped ds DNA virus and does not have a reverse transcriptase enzyme.

Option D - Coronavirus: Coronavirus is an RNA virus that consists of a single RNA copy and does not contain reverse transcriptase.

Solution for Question 92:

Correct Option B - Togavirus:

- Rubella virus is classified as a Togavirus. Togaviruses are a family of enveloped, positive-sense, single-stranded RNA viruses. Rubella virus, also known as the German measles virus, belongs to the genus Rubivirus within the Togaviridae family.
- Other viruses included in Togavirus are: Chikungunya virus. Western equine encephalitis virus. Eastern equine encephalitis virus.
- Chikungunya virus.
- Western equine encephalitis virus.
- Eastern equine encephalitis virus.
- Chikungunya virus.
- Western equine encephalitis virus.
- Eastern equine encephalitis virus.

Incorrect Options:

Option A - Flavivirus: Flaviviruses are another family of RNA viruses and include viruses such as Japanese encephalitis, Yellow fever, Zika, Dengue, West Nile fever, and Hepatitis C.

Option C - Paramyxovirus: Paramyxoviruses are yet another family of RNA viruses known for causing respiratory infections such as respiratory syncytial virus measles, mumps, parainfluenza, and Nipah.

Option D. Orthomyxovirus: Orthomyxoviruses are a family of RNA viruses that include influenza viruses.

Solution for Question 93:

Correct Option A - Herpes simplex:

Inclusion Bodies and Viruses Producing Them
Intracytoplasmic inclusion bodies Negri bodies – rabies virus Paschen body – variola virus Guamieri bodies – vaccinia virus Bollinger bodies – fowlpox virus Molluscum bodies – molluscum contagiosum virus Perinuclear cytoplasmic body – reovirus
Intranuclear Inclusion Bodies
Cowdry type A inclusions Torres body – yellow fever virus Lipschutz body – herpes simplex virus Cowdry type B inclusions Poliovirus Adenovirus
Intracytoplasmic and Intranuclear Inclusion Bodies Owl's eye appearance – cytomegalovirus Measles virus

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 94:

Correct Option D - Parvovirus B19:

Parvovirus B19 infection is known to cause a condition called “fifth disease” or erythema infectiosum. In some cases, Parvovirus B19 can lead to a transient aplastic crisis in a patient with an underlying hemolytic condition and hydrops fetalis, which mimics erythroblastosis fetalis. The virus infects erythroid progenitor cells, leading to temporary suppression of red blood cell production. Parvovirus B19 is, therefore, a viral infection that mimics erythroblastosis fetalis.

Incorrect Options:

Option A - HSV (Herpes Simplex Virus): HSV infection is primarily associated with oral or genital ulcers, and it does not typically mimic erythroblastosis fetalis.

Option B - Epstein-Barr Virus: Epstein-Barr virus (EBV) is the causative agent of infectious mononucleosis. Although EBV can cause abnormal liver function and anemia, it does not specifically mimic erythroblastosis fetalis.

Option C - Picornavirus: Picornaviruses are a large family of viruses that include common viruses such as rhinoviruses and enteroviruses. While certain picornaviruses can cause mild respiratory or gastrointestinal symptoms, they do not mimic erythroblastosis fetalis.

Solution for Question 95:

Correct Option B - Herpes Simplex Virus:

- The virus shown in the picture is a DNA virus. In the above image, the virus first binds to the cell and enters inside the cell by uncoating.
- After the entry of the virus inside the cell, three types of proteins, i.e., alpha, beta, and gamma, are formed via transcription and replication, which are pathognomonic of the herpes virus.
- It has two cycles: Lytic cycle: Immediate-early genes are expressed first, initiating the lytic life cycle of the virus, leading to the production of infectious virions and lysis of the host cell. Latent cycle: There is limited gene expression and a lack of infectious virus particles.
- Lytic cycle: Immediate-early genes are expressed first, initiating the lytic life cycle of the virus, leading to the production of infectious virions and lysis of the host cell.

- Latent cycle: There is limited gene expression and a lack of infectious virus particles.
- Lytic cycle: Immediate-early genes are expressed first, initiating the lytic life cycle of the virus, leading to the production of infectious virions and lysis of the host cell.
- Latent cycle: There is limited gene expression and a lack of infectious virus particles.

Incorrect Options:

Option A - Human Immunodeficiency Virus: It belongs to Retroviridae, which is a family of single-stranded RNA viruses.

Option C - Hepatitis B Virus: HBV replication cycle involves the following steps: attachment and entry into hepatocytes, the release of the viral genome into the nucleus, conversion of the viral DNA into a covalently closed circular DNA (cccDNA) form, transcription of viral genes, and replication of the viral genome, assembly of viral particles in the endoplasmic reticulum, and release through budding.

Option D - Coronavirus: It belongs to Coronaviridae, which is a family of single-stranded RNA viruses.

Solution for Question 96:

Correct Option C - Heterophile Antibodies:

- The principle of the Paul-Bunnell test, also known as the Monospot test is done for Infectious mononucleosis caused by Epstein-Barr virus, is based on the detection of heterophile antibodies.
- Heterophile antibodies are a type of antibody that can react with multiple antigens, including those that are not specific to the infecting pathogen.
- In the Paul-Bunnell test, sheep or horse red blood cells (RBCs) that have been coated with antigens obtained from the EBV-infected cells are combined with the patient's serum.
- The coated RBCs will agglutinate (clump together), suggesting a positive reaction if the patient has heterophile antibodies specific to the EBV.

Incorrect Options:

Option A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 97:

Correct Option C - HAV:

- Hepatitis A virus (HAV) is an RNA virus that causes acute liver infection but does not integrate into the host genome or cause cancer. Unlike other hepatitis viruses (such as HBV and HCV), HAV does not have a chronic phase and is not associated with hepatocellular carcinoma.

Incorrect Options:

Option A - HPV: Human Papillomavirus (HPV) is a DNA virus that primarily presents with cutaneous and mucosal lesions. High-risk HPV types, such as HPV-16 and HPV-18, are strongly associated with the development of cervical, anal, vulvar, vaginal, penile, and oropharyngeal cancers. E6 and E7 viral oncoproteins are expressed by high-risk HPV types and can inactivate the p53 and RB proteins, respectively, leading to uncontrolled cell growth and increased risk of malignancy.

Option B - SV 40: Simian vacuolating virus 40 is a Polyomavirus associated with mesothelioma. Large T antigen and small T antigen are the two viral oncoproteins that are expressed by SV 40 and can inactivate p53, RB, and PP2A, leading to malignancy.

Option D - EBV: Epstein-Barr Virus (EBV) is a herpesvirus that infects B lymphocytes and epithelial cells. EBV is associated with the development of several malignancies, including Burkitt's lymphoma, Hodgkin's lymphoma, nasopharyngeal carcinoma, and some gastric cancers. LMP 1 viral oncoproteins are expressed by EBV and can inactivate TRAFs (tumor necrosis factor receptor-associated factor), leading to malignancy.

Solution for Question 98:

Correct Option A - Dengue:

- Dengue is caused by four different serotypes of the dengue virus (DEN-1, DEN-2, DEN-3, and DEN-4). If a person is infected with one serotype and later gets infected with a different serotype, they are at a higher risk of developing severe dengue, also known as dengue

hemorrhagic fever (DHF) or dengue shock syndrome (DSS). This phenomenon is known asantibody-dependent enhancement (ADE).

• ADE occurs when the antibodies produced during the first infection bind to the virus particles of the second serotype but fail to neutralize them effectively. ADE was shown to enhance viral entry into immune cells via their Fcγ receptors, which promotes viral replication, leading to increased viremia and pro-inflammatory responses.

Incorrect Options:

Option B - Rabies: Rabies is a viral infection transmitted through the bite or scratch of an infected animal. Once the symptoms of rabies develop, it is almost always fatal.

Option C - Measles: Measles is a highly contagious viral infection. Although reinfection with measles is possible, it does not typically lead to more severe disease or complications. However, being unvaccinated or having a compromised immune system can increase the risk of severe measles and its complications.

Option D - Mumps: Mumps is a viral infection that primarily affects the parotid gland and salivary glands, leading to swelling and pain. Reinfection with mumps is rare, as most people develop long-lasting immunity after recovering from the initial infection.

Solution for Question 99:

Correct Option C - 1-c, 2-d, 3-a, 4-b:

• Accidental host: Human (c)

• Amplifier: Pigs (d)

• Reservoir: Ardeid birds (c)

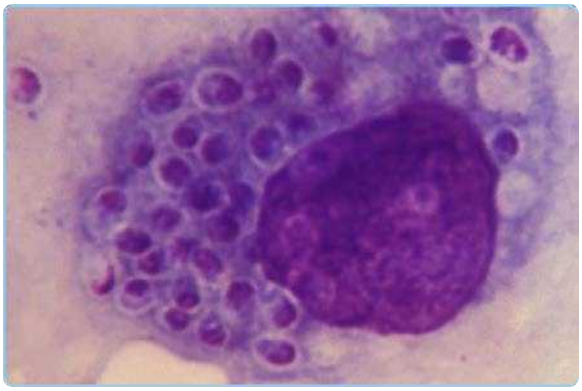
• Vector: Culex (Culex tritaeniorhynchus - worldwide and Culex vishnui in India).

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Introduction to Mycology and its Classification

1. A 45-year-old male presents with a persistent cough and fever for the past month. He has a history of working in construction and has been exposed to soil and bird droppings. A bronchoalveolar lavage is performed, and microbial elements are observed under the microscope. His urine antigen test for the causative agent is positive and a histopathological image is shown as below. Which of the following structures is most likely observed?



A. Peptidoglycan cell wall

B. Chitin and glucan in the cell wall

C. Lipopolysaccharide outer membrane

D. Teichoic acid

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2. A 34-year-old female presents with a persistent cough and fever after returning from a trip. A laboratory workup reveals a fungal infection. The fungus is dimorphic in nature. Which of the following cannot be the causative organism?

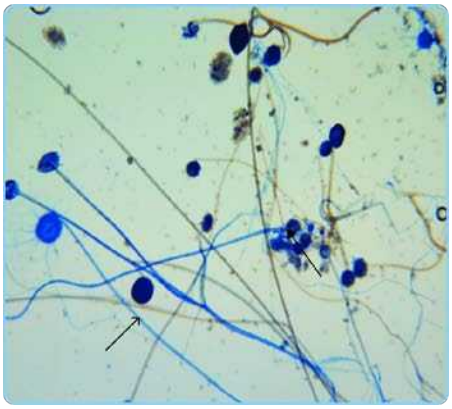
A. Histoplasma capsulatum

B. Blastomyces dermatitidis

C. Coccidioides immitis

D. Aspergillus fumigatus

3. You are presented with an image of fungal spores. Which of the following is the incorrect statement regarding the image given?



A. Formed within a sac

B. Seen in Rhizopus

C. Seen in Mucor

D. Formed by budding

4. A 35-year-old woman presents with a chronic ulcer on her leg that has been slowly expanding over several months. She has a history of gardening and often works with soil. The physician suspects a subcutaneous mycosis. Which of the following conditions is not classified in the same group as the suspected infection?

A. Mycetoma

B. Chromoblastomycosis

C. Tinea Versicolor

D. Spirotrichosis

5. Which of the following statements about KOH preparation for identifying fungal elements is incorrect?

A. Suitable for samples from skin, hair, and nails.

B. Percentage of KOH can be increased depending on the nature of the specimen.

C. Procedure involves placing the specimen in a drop of 10% KOH

D. KOH can be used for urine samples

---

6. A 30-year-old male patient with HIV presents with symptoms of fever, headache, and neck stiffness. A lumbar puncture is performed, and cerebrospinal fluid (CSF) is sent for analysis. Which stain is most appropriate for identifying specific organisms in this CSF sample?

A. Periodic Acid-Schiff

B. India Ink

C. Lactophenol cotton blue

D. Calcofluor white

---

7. Which of the following statements correctly describes the role of mucicarmine in diagnosing Cryptococcus neoformans?

A. Mucicarmine is the only stain that can identify Cryptococcus species in all clinical samples

B. Mucicarmine specifically stains the polysaccharide capsule, aiding in the identification of encapsulated Cryptococcus neoformans

C. Mucicarmine is equally effective for both encapsulated and capsule-deficient Cryptococcus strains

D. Mucicarmine is only useful when Cryptococcus neoformans cultures are positive

8. Which of the following features of Sabouraud Dextrose Agar (SDA) makes it particularly useful for culturing dermatophytes in clinical mycology?

A. Neutral pH and presence of antibacterial agents

B. Acidic pH and addition of cycloheximide

C. Presence of growth factors to support bacterial and fungal growth

D. Neutral pH and enriched with glucose for rapid bacterial growth

9. A sample from a patient with suspected cryptococcosis is cultured on birdseed agar. What is the main purpose of using birdseed agar in this case?

A. To inhibit bacterial growth and favour fungal growth

B. To selectively isolate Candida species from other fungi

C. To promote the identification of Cryptococcus neoformans through melanin production

D. To enhance the growth rate of all fungal species by providing additional nutrients

10. What is the primary purpose of using potassium hydroxide (KOH) preparation in clinical specimens?

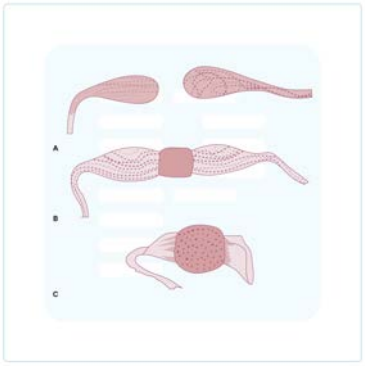
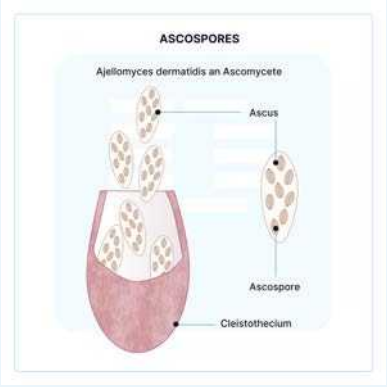
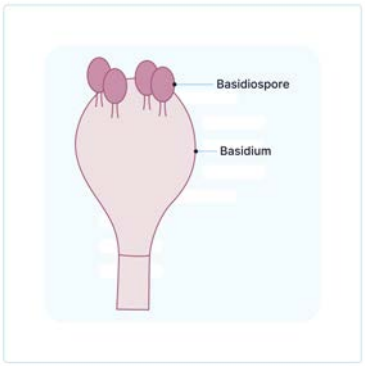
A. Enhance the visibility of fungi by staining

B. Digest keratin and reveal fungal structures

C. Detect antibodies against fungal infections

D. Amplify fungal DNA for identification

11. Match the following sexual spores with their organisms Sexual spores Organisms a. 1. Mucor b. 2. Penicillium c. 3. Filobasidiella d. NONE 4. Coccidioides

Sexual spores	Organisms
 ... (content truncated)	1. Mucor
 ... (content truncated)	2. Penicillium
 ... (content truncated)	3. Filobasidiella
d. NONE	4. Coccidioides

A. a-1; b-2; c-3; d-4

B. a-2; b-1; c-4; d-3

C. a-4; b-2; c-3; d-1

D. a-3; b-2; c-1; d-4

---

12. Which of the following statements are correct? 1. Sterols are present in the Cytoplasmic Membrane of bacteria except Mycoplasma. 2. Sterols are present in Cytoplasmic Membrane of fungi except in Pneumocystis carinii. 3. Bacteria use both asexual and sexual spores for reproduction. 4. Bacterial cell wall contains Peptidoglycan, LPS and teichoic acid. 5. Fungi have no nucleus or nuclear membrane.

A. 1, 2, 3 and 4

B. 2,3 and 4

C. 3 and 5 only

D. 2 and 4 only

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	4
Question 4	3
Question 5	4
Question 6	2
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	1
Question 12	4

Solution for Question 1:

Correct Answer: B) Chitin and glucan in the cell wall

Explanation:

• This is a case of Histoplasma capsulatum infection which is a fungus and fungi have cell walls composed of chitin and glucans. The patient has a history of working in construction and being exposed to soil and bird droppings. His symptoms, including a persistent cough and fever, align with pulmonary histoplasmosis. Histoplasma can be detected by urine antigen test. Histopathological examination shows a Giemsa-stained tissue smear with macrophages containing phagocytosed yeast forms of the fungus.

• The patient has a history of working in construction and being exposed to soil and bird droppings.

• His symptoms, including a persistent cough and fever, align with pulmonary histoplasmosis.

• Histoplasma can be detected by urine antigen test.

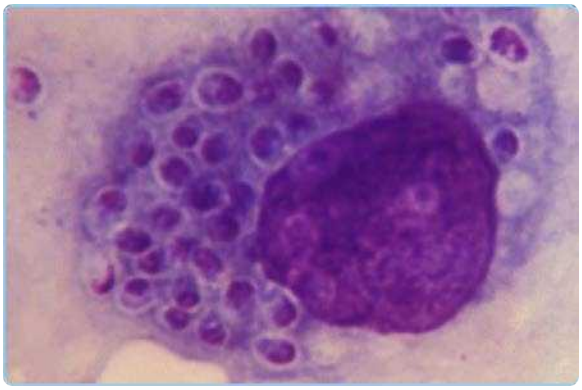
• Histopathological examination shows a Giemsa-stained tissue smear with macrophages containing phagocytosed yeast forms of the fungus.

• The patient has a history of working in construction and being exposed to soil and bird droppings.

• His symptoms, including a persistent cough and fever, align with pulmonary histoplasmosis.

• Histoplasma can be detected by urine antigen test.

• Histopathological examination shows a Giemsa-stained tissue smear with macrophages containing phagocytosed yeast forms of the fungus.



Giemsa-stained tissue smear shows a macrophage with phagocytosed H. capsulatum yeast forms.

Property	Fungi	Bacteria
Cell Structure	Eukaryote	Prokaryote
Cell Wall Composition	Multilayered; chitin, mannan, and glucans	Peptidoglycan, LPS, teichoic acid (Options A, C, and D)
Cytoplasmic Membrane	Sterols are present except in Pneumocystis carinii	Sterols are absent except in Mycoplasma
Cytoplasmic Contents	Membrane-bound organelles (e.g., mitochondria, Golgi apparatus, ER)	Membrane-bound organelles absent
Nucleus	True nucleus with nuclear membrane; paired chromosomes	No nucleus or nuclear membrane; single circular dsDNA chromosome
Diameter	5 – 10 µm; may be > 100 µm	< 2 µm
Morphology	Yeast: ovoid, round; Moulds: filamentous	Cocci, bacilli, some branching

Spores	Both asexual and sexual for reproduction	Asexual (endospores) primarily for survival

Clinical implications:

- Beta 1,3 glucan Antigenin serum for invasive fungal infections except Cryptococcus and Mucor/Rhizopus
- Galactomannan test for invasive aspergillosis.

Solution for Question 2:

Correct Answer: D) Aspergillus fumigatus

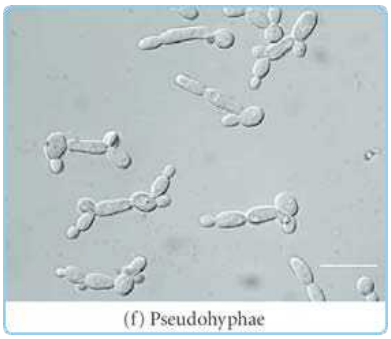
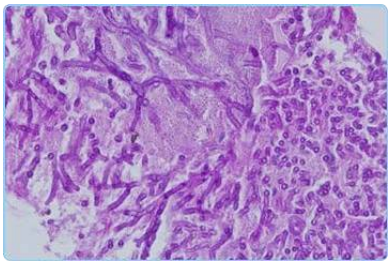
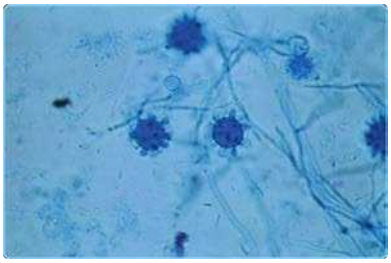
Explanation:

- Aspergillus species do not exhibit dimorphism.
- They exist only as moulds in their environmental form and do not switch to a yeast form at body temperature as is seen in dimorphic fungi.

(Mnemonic: bodY temperature -Yeast)

Fungal classification based on morphology:

Classification	Characteristics	Examples
Yeast	 ... (content truncated)	Cryptococcus Saccharomyces cerevisiae

Yeast-like (Pseudohyphae)	 Elongated buds, known as pseudohyphae {{caption_text}}	Candida Malassezia Trichosporon Hortae
Mold/Hyphae	 Aseptate (non-segmented): Hyphae is Broad ribbon like without segments {{caption_text}}	Zygomycetes (e.g., Mucor)
 Septate (segmented): Phaeoid - Masson Fontanna silver stain+ (melanin-pigmented) Hyaline hyphae {{caption_text}}	Phaeoid (Piedra, Alternaria, Madurella, Curvularia) Hyaline (Fusarium, Acremonium, Aspergillus, Penicillium, Dermatophytes)	
Dimorphic Fungi	 Two morphologies: Mold at 25°C	Blastomyces Histoplasma Penicillium marneffei (Talaromyces) Coccidioides Sporothrix (Option A, B, and C)

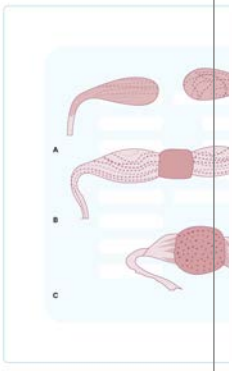
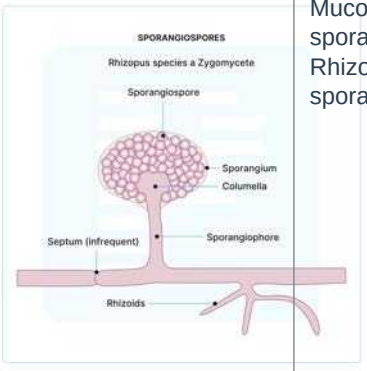
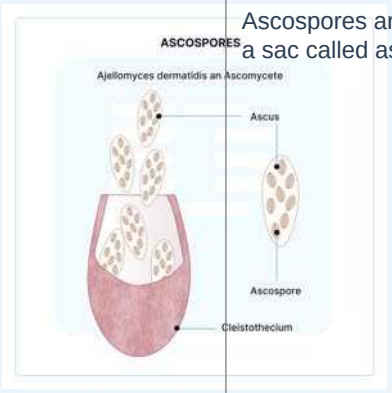
	Yeast at 37°C {{caption_text}}	

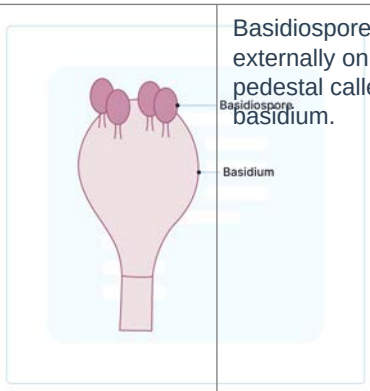
Solution for Question 3:

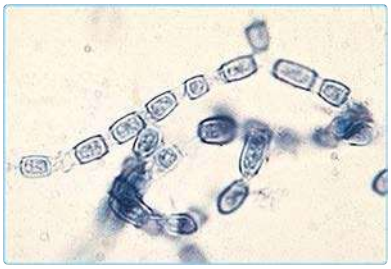
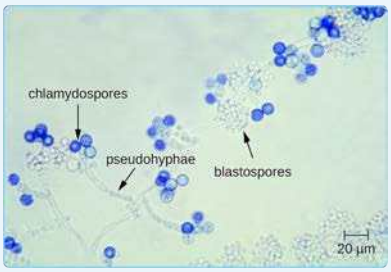
Correct Answer: D) Formed by budding

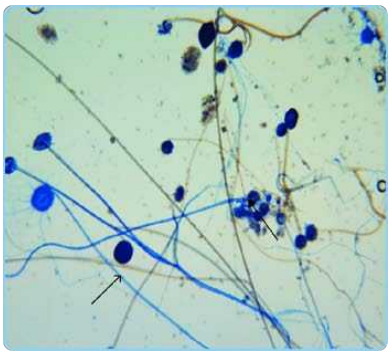
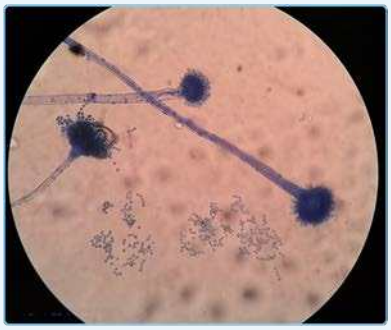
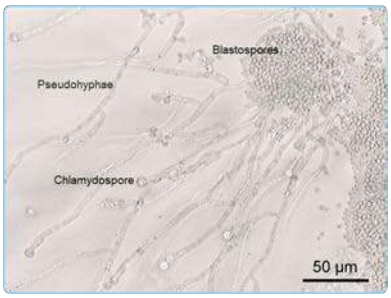
Explanation:

- The given image is of a sporangiospore.
- Sporangiospores are not formed by budding; they are produced within a sac (sporangium) by molds like Rhizopus and Mucor.

Classification	Sexual Spore	Description	Examples
Zygomycetes	 ... (content truncated)	 ... (content truncated)	Mucor (with aerial sporangiospores) Rhizopus (Rhizoid with sporangiospores above)
Ascomycetes	 ... (content truncated)	Ascospores are formed in a sac called ascus.	Penicillium Aspergillus Piedra hortae

Basidiomycetes	 ... (content truncated)	Basidiospores are formed externally on the tip of a pedestal called a basidium.	Filobasidiella Trichosporon species
Deuteromycetes (Fungi Imperfecti)	None	Fungi that lack a known sexual state.	Coccidioides

Asexual Spore	Description	Example
Arthrospores	 Arise by fragmentation of the ends of hyphae {{caption_text}}	Coccidioides immitis
Chlamydospores	 Rounded, thick-walled spores {{caption_text}}	Candida albicans

Sporangiospores	 Formed within a sac (sporangium) on a stalk by molds ... (content truncated)	Rhizopus, Mucor
Conidiospores	 Arise directly from a conidiophore {{caption_text}}	Aspergillus
Blastospores	 Formed by the budding process {{caption_text}}	Candida albicans

Solution for Question 4:

Correct Answer: C) Tinea Versicolor

Explanation:

Tinea versicolor is a superficial fungal infection caused by Malassezia and primarily affects the skin's outer layers, not classified as a subcutaneous mycosis.

Classification	Examples
Superficial Mycosis	Tinea Versicolor (Malassezia species) Dermatophytes (Trichophyton, Microsporum, Epidermophyton)
Subcutaneous Mycosis (Option A, B, and D)	Mycetoma (Madurella species) Rhinosporidiosis (Rhinosporidium seeberi) Chromoblastomycosis (Fonsecaea pedrosi, Phialophora, Cladophialophora) Sporotrichosis (Sporothrix schenckii)
Deep Mycosis/Systemic Mycosis	Dimorphic fungi Histoplasma Penicillium marneffei Coccidioides Sporothrix Blastomyces Opportunistic infections Candida Cryptococcus Aspergillus Mucormycosis)

Solution for Question 5:

Correct Answer: D) KOH can be used for urine samples.

Explanation:

KOH preparation is not suitable for urine samples; it is primarily used for specimens from skin, hair, and nails, which contain keratin.

Direct Examination of Fungal Specimens

1. KOH Preparation

- Samples: All samples except urine, CSF, blood, serum
- Procedure: Place the specimen in a drop of 10% potassium hydroxide (KOH) on a microscope slide (Option C) Cover with a coverslip and gently heat the slide. Examine under

a microscope.

• Place the specimen in a drop of 10% potassium hydroxide (KOH) on a microscope slide (Option C)

• Cover with a coverslip and gently heat the slide.

• Examine under a microscope.

• Purpose: KOH digests keratin Direct demonstration of fungal morphology Hyphae, pseudohyphae, yeast cells

• KOH digests keratin

• Direct demonstration of fungal morphology Hyphae, pseudohyphae, yeast cells

• Hyphae, pseudohyphae, yeast cells

• Key points: Ideal for samples from skin, hair, and nails (Option A) After digestion, hyphae become visible under the microscope 10% clears tissue materials and makes the background clear The percentage of KOH can be increased according to the nature of the specimen (20%,40% such as nails, takes longer time to dissolve) (Option B)

• Ideal for samples from skin, hair, and nails (Option A)

• After digestion, hyphae become visible under the microscope

• 10% clears tissue materials and makes the background clear

• The percentage of KOH can be increased according to the nature of the specimen (20%,40% such as nails, takes longer time to dissolve) (Option B)

• Place the specimen in a drop of 10% potassium hydroxide (KOH) on a microscope slide (Option C)

• Cover with a coverslip and gently heat the slide.

• Examine under a microscope.

• KOH digests keratin

• Direct demonstration of fungal morphology Hyphae, pseudohyphae, yeast cells

• Hyphae, pseudohyphae, yeast cells

• Hyphae, pseudohyphae, yeast cells

• Ideal for samples from skin, hair, and nails (Option A)

• After digestion, hyphae become visible under the microscope

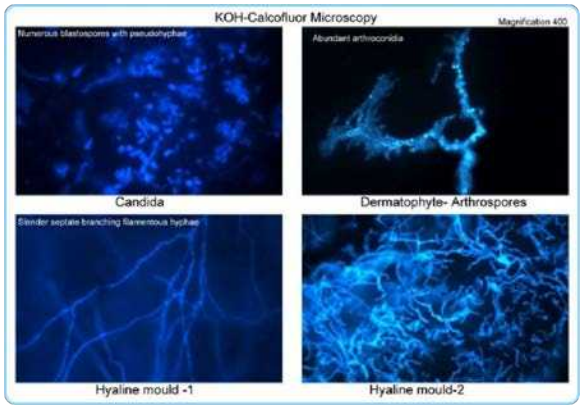
• 10% clears tissue materials and makes the background clear

• The percentage of KOH can be increased according to the nature of the specimen (20%,40% such as nails, takes longer time to dissolve) (Option B)



2. KOH with Calcofluor White

- Procedure: Add a drop of Calcofluor White solution to the KOH preparation before placing the coverslip.
- Add a drop of Calcofluor White solution to the KOH preparation before placing the coverslip.
- Purpose: Calcofluor White binds to fungal components, causing them to fluoresce bright green when exposed to UV light.
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- Benefits: Enhances visibility of fungal elements, making identification easier.
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- Calcofluor White binds to fungal components, causing them to fluoresce bright green when exposed to UV light.
- Enhances visibility of fungal elements, making identification easier.

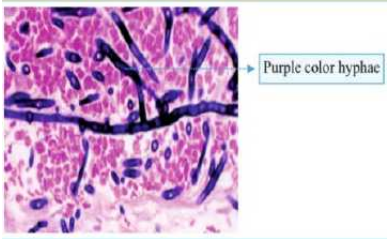


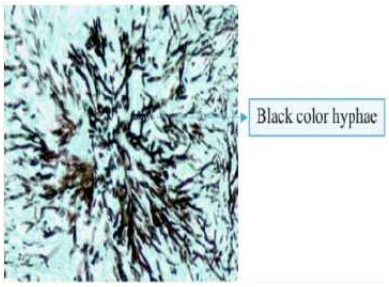
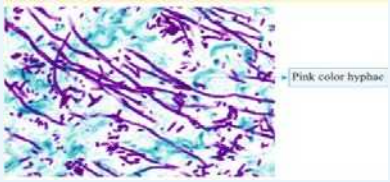
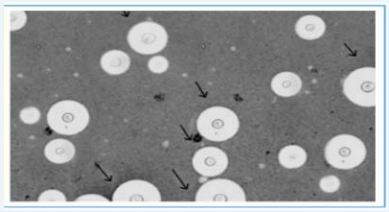
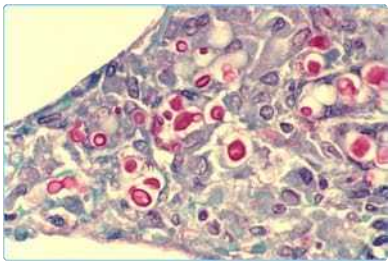
Solution for Question 6:

Answer: B) India Ink

Explanation:

- The likely diagnosis in this case is cryptococcal meningitis.
- In patients with HIV, cryptococcal meningitis is a common opportunistic infection, making India Ink particularly relevant for diagnosis.

Stain	Description
	Mycology lab
Calcofluor White	Fluorochrome binds to glucans and chitin in the fungal cell wall, enhancing visualization of fungal elements.
Lactophenol Cotton Blue (LCB)	 ... (content truncated)
	Pathology
Hematoxylin and Eosin (H&E)	 Used primarily in pathology to visualize host cell responses to fungi Does not stain most fungi, except some like Aspergillus spp. and Mucorales.

Grocott Methenamine Silver (GMS)	 Stains fungal cell walls black, providing clear contrast for identifying fungi in tissue samples.
Periodic Acid-Schiff (PAS)	 Stains the polysaccharide components of the fungal cell walls magenta Useful for detecting fungal elements in histological samples.
	Selected specific stains
India Ink	 A "negative" stain outlining the polysaccharide capsule of Cryptococcus spp Less sensitive than antigen detection for cryptococcal meningitis. {{caption_text}}
Mucicarmine	 Stains the mucopolysaccharide capsule of Cryptococcus spp.; can also stain Blastomyces and Rhinosporidium. {{caption_text}}
Fontana-Masson	Silver stain is used to detect melanin pigment in cell walls of Cryptococcus and pheoid moulds.

Solution for Question 7:

Correct Answer: B) Mucicarmine specifically stains the polysaccharide capsule, aiding in the identification of encapsulated Cryptococcus neoformans.

Explanation:

Mucicarmine is used to stain the polysaccharide capsule of C. neoformans, which is essential for identifying encapsulated forms in clinical samples.

Role of Mucicarmine in Diagnosing Cryptococcus neoformans:

- Primary Use: Stains the polysaccharide capsule of Cryptococcus neoformans, aiding diagnosis in immunocompromised patients.
- Importance: Essential for identifying encapsulated forms, especially when other diagnostic methods are insufficient.

Key Diagnostic Roles of Mucicarmine:

Role	Description
Capsule Visualization	Stains the Cryptococcus capsule, which is a major virulence factor and critical for organism identification.
Comparison with Other Stains	Less effective with capsule-deficient strains; alternative stains like Fontana-Masson silver may be needed.
Clinical Relevance	Valuable when fungal cultures are negative or unavailable, especially when combined with other fungal stains.
Limitations	Not reliable for all Cryptococcus strains; complementary tests are necessary for accurate diagnosis.

Summary:

- Diagnostic Aid: Useful in confirming cryptococcosis, particularly for encapsulated C. neoformans.
- Limitations: Capsule-deficient strains may yield negative results with mucicarmine, requiring other stains for verification.

Solution for Question 8:

Correct Answer: B) Acidic pH and addition of cycloheximide

Explanation:

SDA has an acidic pH, favouring fungal growth over bacterial growth. The addition of cycloheximide further increases specificity by inhibiting non-dermatophytic fungi, making it ideal for dermatophytes.

Sabouraud Dextrose Agar (SDA) in Mycology:

- Purpose: Used to culture fungi, especially dermatophytes (e.g., Trichophyton rubrum). (Option C ruled out)
- pH: Acidic (pH 5.4), favouring fungal growth. (Options A&D ruled out)
- Additives: Chloramphenicol: Prevents bacterial contamination. Cycloheximide: Inhibits growth of non-dermatophytic fungi, enhancing specificity.
- Chloramphenicol: Prevents bacterial contamination.
- Cycloheximide: Inhibits growth of non-dermatophytic fungi, enhancing specificity.
- Temperature for Culturing: Incubation at room temperature (22°C) and 37°C to observe growth characteristics.
- Incubation at room temperature (22°C) and 37°C to observe growth characteristics.
- Chloramphenicol: Prevents bacterial contamination.
- Cycloheximide: Inhibits growth of non-dermatophytic fungi, enhancing specificity.
- Incubation at room temperature (22°C) and 37°C to observe growth characteristics.

Fungal Identification on SDA:

Identification Method	Description
Macroscopic Features	Observed for colony colour, texture, growth rate, and appearance on both sides of the culture plate Cryptococcus: Off-white mucoid colony Candida: Creamy white pasty colony Histoplasma: White cotton growth
Microscopic Examination	Techniques include teased mounts, cellophane tape mounts, and slide cultures; lactophenol cotton blue stain is used.

Solution for Question 9:

Correct Answer: C) To promote the identification of Cryptococcus neoformans through melanin production

Explanation:

Bird seed agar is used to identify Cryptococcus neoformans, as this species produces melanin on the medium, resulting in a characteristic dark brown or black colour.

Other Cultures in Mycology:

Czapek-Dox Medium:

- Used for isolating and cultivating a wide range of fungi, particularly nutrient-demanding species.
- Chemically defined, allowing for precise study of fungal physiology and metabolism.

Bird Seed Agar:

- Specifically designed for isolating and identifying Cryptococcus neoformans & gatti. (Options A&B ruled out)
- Contains Niger seeds, which facilitate melanin production in Cryptococcus neoformans, leading to brown to black pigmented colonies. (Option D ruled out)

Corn Meal Agar:

- Stimulates sporulation and reveals morphological characteristics of fungi, especially yeasts like Candida albicans and Candida dubinensis.
- Simple medium made from corn meal extract; Tween 80 can be added to enhance chlamydospore production for identification.

Note: Non-specific media, like blood agar, can be adapted for fungal use with antibacterials.

Incubation Temperatures: Cultures are incubated at 22°C (room temperature) and 37°C to observe growth patterns and aid in species differentiation.

Fungal Culture Techniques:

Technique	Purpose
Macroscopic Clues	Growth rate, texture, and colour help in identifying fungal species.
Microscopic Analysis	Reveals fungal morphology through mounts and stains, detailing hyphae, conidia.

Importance of Fungal Cultures:

Aspect	Description
Definitive Diagnosis	Confirms fungal species responsible for infection, guiding specific treatments.
Species Differentiation	Distinguishes species by growth and microscopic traits, aiding in understanding fungi's role.
Antifungal Susceptibility Testing	Determines fungal sensitivity to antifungals, guiding effective therapy.

Solution for Question 10:

Answer: B) Digest keratin and reveal fungal structures

Explanation:

KOH preparation dissolves keratin, allowing fungal elements to be clearly visible under the microscope.

Detection Methods for Fungi Beyond Culture Media:

Method	Description	Use
Direct Microscopic Examination		
KOH Preparation	Dissolves keratin in clinical specimens (e.g., skin scrapings, nail clippings) to visualise fungal elements.	Diagnosing dermatophyte infections
Calcofluor White Staining	Enhances fungal visibility by binding to chitin in cell walls, causing them to fluoresce under UV light. (Option A ruled out)	Observing fungal morphology
Gram Stain	Traditionally for bacterial identification, also useful for yeasts, which typically stain Gram-positive (purple).	Detecting certain fungi
India Ink Preparation	Negative staining method for encapsulated yeasts (e.g., Cryptococcus neoformans), revealing a clear halo around cells.	Identifying encapsulated yeasts

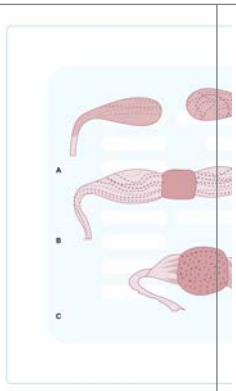
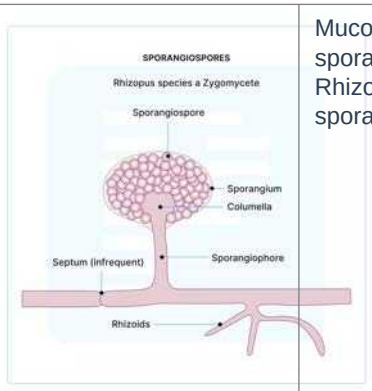
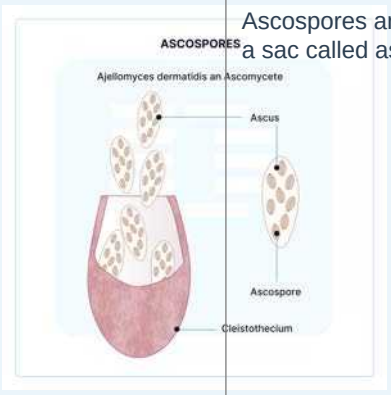
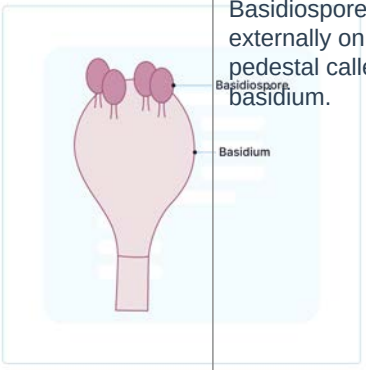
Methenamine Silver and PAS Staining	Histochemical techniques that stain fungi black (methenamine) or magenta (PAS) in tissue sections.	Highlighting fungi in tissues
Lactophenol Cotton Blue (LCB) Mount	Enhances contrast in teased fungal colonies for morphological observation.	Microscopic examination of fungi
Serology		
Antibody Detection	Identifies antibodies produced by the host in response to fungi using methods like ELISA. (Option C ruled out)	Tracking immune response to fungal infections
Antigen Detection	Detects fungal antigens in body fluids using techniques such as latex agglutination tests.	Identifying fungal presence in the body
Skin Tests		
Delayed-Type Hypersensitivity (DTH)	Assesses previous exposure to fungi through localised reactions to injected antigens.	Evaluating past fungal exposure
Newer Rapid Diagnostic Tests		
Nucleic Acid Hybridisation	Uses labelled DNA probes for rapid fungal DNA detection in clinical samples.	Quick identification of fungal pathogens
Polymerase Chain Reaction (PCR)	Amplifies specific fungal DNA sequences, enabling early diagnosis even with low fungal loads. (Option D ruled out)	Early detection of fungal infections

Solution for Question 11:

Correct Answer: A) a-1; b-2; c-3; d-4

Explanation:

Classification	Sexual Spore	Description	Examples

Zygomycetes	 ... (content truncated)	 ... (content truncated)	Mucor (with aerial sporangiospores) Rhizopus (Rhizoid with sporangiospores above)
Ascomycetes	 ... (content truncated)	Ascospores are formed in a sac called ascus.	Penicillium Aspergillus Piedra hortae
Basidiomycetes	 ... (content truncated)	Basidiospores are formed externally on the tip of a pedestal called a basidium.	Filobasidiella Trichosporon species
Deuteromycetes (Fungi Imperfecti)	None	Fungi that lack a known sexual state.	Coccidioides

Solution for Question 12:

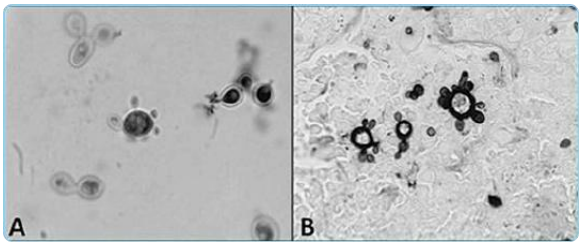
Correct answer: D) 2 and 4 only

Explanation:

Property	Fungi	Bacteria
Cell Structure	Eukaryote	Prokaryote
Cell Wall Composition	Multilayered; chitin, mannan, and glucans	Peptidoglycan, LPS, teichoic acid
Cytoplasmic Membrane	Sterols present except in Pneumocystis carinii	Sterols absent except in Mycoplasma
Cytoplasmic Contents	Membrane-bound organelles (e.g., mitochondria, Golgi apparatus, ER)	Membrane-bound organelles absent
Nucleus	True nucleus with nuclear membrane; paired chromosomes	No nucleus or nuclear membrane; single circular dsDNA chromosome
Diameter	5 – 10 μm; may be > 100 μm	< 2 μm
Morphology	Yeast: ovoid, round; Moulds: filamentous	Cocci, bacilli, some branching
Spores	Both asexual and sexual for reproduction	Asexual (endospores) primarily for survival

Superficial and subcutaneous mycosis, and dimorphic fungi

1. A 45-year-old male presents with cough, nasal discomfort and weight loss. He has a travel history to South America in the past. On examination, he is found to have ulcerative granulomas on his nasal and buccal mucosa. A microscopic examination of the lesion is shown below. Which is the most likely causative organism?



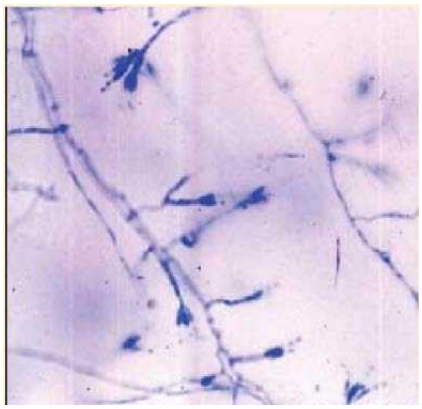
A. Histoplasma

B. Blastomyces

C. Paracoccidioides

D. Cryptococcus

2. A 45-year-old male with a history of AIDS presents to the clinic with persistent fever, significant weight loss over the past month, and cough that has worsened over the last two weeks. He resides in a rural area of Southeast Asia, where he frequently encounters bamboo rats. On physical examination, he has cervical lymphadenopathy and hepatomegaly. A sputum sample is collected for microbiological examination. An image is shown below. Which of the following is characteristic of the colony that is likely to grow in culture from this patient's sputum sample?



A. White, cottony, mycelial growth

B. Green, granular colonies with a distinctive red pigment

C. Smooth, mucoid, cream-coloured colonies

D. Creamy white, smooth colonies and with a yeasty odour

---

3. What microscopic features are not observed when examining cultures of Histoplasma capsulatum?

A. Tuberculate macroconidia

B. Thick-walled spherical spores

C. Smooth-walled oval microconidia

D. Intracellular yeast cells

---

4. A 35-year-old male from a tropical region presents with nasal obstruction, epistaxis, and a soft, fleshy nasal mass. Histopathological examination reveals thick-walled sporangia filled with endospores and hyperplastic tissue without eosinophilic infiltration. Which of the following statements regarding this condition is correct?

A. It affects the mucosal membrane of the nose only.

B. The recommended course of primary treatment is Dapsone

C. The causative agent cannot be cultured in vitro

D. It is not associated with contaminated water exposure

---

5. A biopsy from a cutaneous lesion in a patient with suspected sporotrichosis is examined under a microscope. Which of the following findings would strongly support the diagnosis?

A. Thick-walled spherules filled with endospores

B. Mariner’s wheel-shaped yeast cells

C. Cigar-shaped yeast cells

D. Broad-based budding yeast cells

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6. Which of the following statements best describes the typical transmission route of Coccidioides immitis?

A. Ingestion of contaminated water

B. Inhalation of dust containing arthrospores

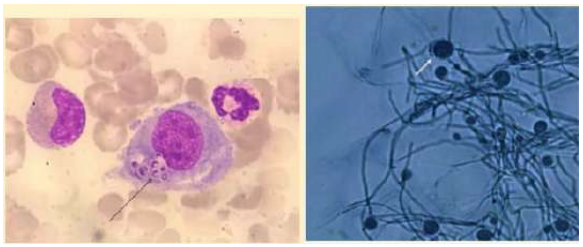
C. Minor trauma or thorn prick

D. Insect bite from an endemic vector

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7. A 50-year-old immunocompromised patient presents with cough, fever, weight loss, hepatosplenomegaly, and lymphadenopathy. She reports having recently cleaned out an old barn where she encountered large amounts of bird droppings. Upon examination, she has mild tachycardia and her lungs exhibit scattered crackles on auscultation. A chest X-ray shows patchy infiltrates and several calcified nodules. Histopathology and (LPCB) mount of colonies grown in culture

are shown below. Which of the following is the most appropriate treatment?



A. Itraconazole

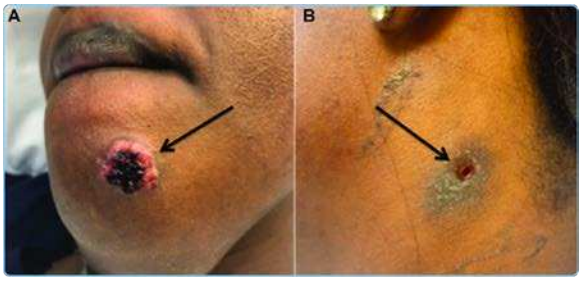
B. Amphotericin-B

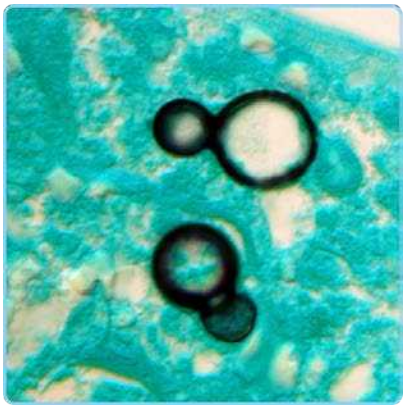
C. Fluconazole

D. Surgical resection

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8. A 40-year-old male from North America presents with cough, fever, and weight loss. Chest imaging shows focal consolidation in the lungs, and he has developed ulcerative skin lesions on his face as shown below. A skin biopsy is performed, and the sample is sent for GMS staining. An image of the GMS stain is shown below. Which of the following is the most likely diagnosis?





A. Blastomycosis

B. Chromoblastomycosis

C. Mycotic Mycetoma

D. Sporotrichosis

---

9. Dimorphic fungi can evade the host immune system primarily through which of the following mechanisms?

A. Formation of biofilms

B. Conversion to a yeast form in the host

C. Production of endotoxins

D. Direct cell-to-cell transmission

---

10. A 45-year-old male agricultural worker from a rural area presents with an extensive slowly growing, warty, nodular lesion on his lower leg as shown in the image below. The lesion is dark, and rough, and resembles the surface of cauliflower. He recalls injuring his leg on a thorn while working barefoot in the fields several months ago. A KOH mount of the skin specimen is shown below. What is the most likely diagnosis?





A. Rhinosporidiosis

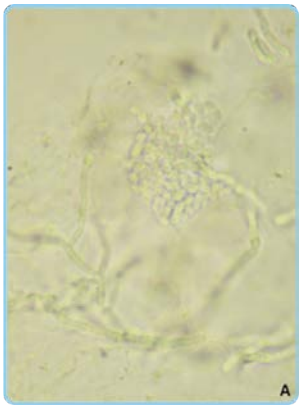
B. Chromoblastomycosis

C. Mycotic Mycetoma

D. Sporotrichosis

---

11. A 25-year-old male presents with multiple light-coloured patches on his chest and back, which are more noticeable after sun exposure. The lesions are asymptomatic but have not resolved with moisturizers. An image of the Microscopic examination of skin scrapings is shown below. What is the most likely diagnosis?



A. Tinea corporis

B. Tinea nigra

C. Tinea versicolor

D. Candidiasis

---

12. Which of the following statements is TRUE regarding the Splendore–Hoeppli phenomenon?

A. It is exclusively seen in functional infections

B. It is never associated with parasitic infestations

C. It can occur in both infectious and noninfectious conditions

D. It does not involve any immune cells

---

13. A 45-year-old man presents with a chronic, painless nodule on his foot, which has progressively enlarged over the past year. Upon examination, he has draining sinuses with black grains in the exudate. Microscopic examination of the tissue reveals thick-walled grains and hyphae. Which of the following is the most likely causative agent of this condition?

A. Actinomadura madurae

B. Nocardia spp.

C. Madurella mycetomatis

D. Curvularia lunata

---

14. Match the following: Column A (Infection category) Column B (Disease) Superficial (i) Dermatophytosis Subcutaneous (ii) White piedra Systemic (iii) Mycetoma Cutaneous (iv) Blastomycosis

Column A (Infection category)	Column B (Disease)
Superficial	(i) Dermatophytosis
Subcutaneous	(ii) White piedra
Systemic	(iii) Mycetoma
Cutaneous	(iv) Blastomycosis

A. a-(ii), b-(iii), c-(iv), d-(i)

B. a-(iii), b-(iv), c-(i), d-(ii)

C. a-(i), b-(ii), c-(iv), d-(iii)

D. a-(iv), b-(iii), c-(i), d-(ii)

---

15. Which statement is correct regarding the use of Wood’s lamp in dermatology?

A. It is not useful for diagnosing scalp infections as hair blocks fluorescence

B. It is done in a bright room using ultraviolet light

C. Normal skin will fluoresce bright pink under Wood’s lamp

D. Microsporum infections commonly fluoresce bright greenish-yellow under Wood's lamp

---

16. A 28-year-old patient presents with hair loss and scalp lesions. A hair perforation test is performed to differentiate between Trichophyton rubrum and Trichophyton mentagrophytes. Which of the following statements about the hair perforation test is true?

A. Incubation at 37°C for results

B. T. rubrum shows wedge-shaped hair

C. Microscopically hair is examined with Grocott's methenamine silver (GMS) stain

D. T. mentagrophytes exhibits surface erosion of the hair shaft.

---

17. A 28-year-old male presents with a pruritic rash in the groin area. Upon examination, the lesion is erythematous with raised borders and has a clear centre. Which of the following is the most likely diagnosis based on the anatomical classification of dermatophytosis?

A. Tinea barbae

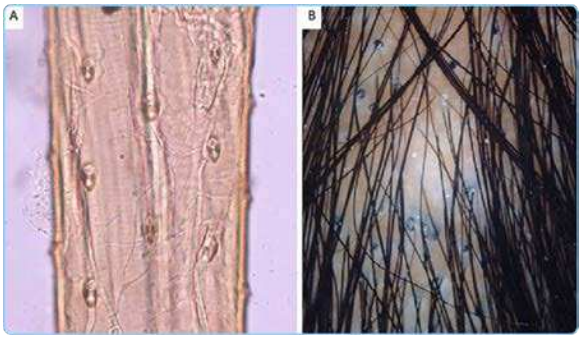
B. Tinea capitis

C. Tinea cruris

D. Tinea corporis

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18. A 10-year-old child presents with hair loss and black dot tinea capitis. Microscopic examination of the hair reveals elongated hyphae within the hair shaft. Which of the following statements is true regarding the likely condition the child has?



A. Trichoscopy shows perifollicular scaling

B. It is commonly associated with Microsporum canis

C. It typically involves Trichophyton tonsurans and Trichophyton violaceum

D. The treatment of choice is typically 2 weeks

19. A 25-year-old male presents with dark, hard nodules along his scalp hair, which he mistakenly believes to be lice nits. Upon microscopic examination, branched hyphae and arthrospores are observed. Which of the following is the most likely cause of his condition?

A. Trichosporon inkin

B. Piedraia hortae

C. Candida albicans

D. Malassezia furfur

20. A 25-year-old woman from a tropical region presents with a brown, macular lesion on the palm of her hand. There is no itching or scaling. Microscopic examination of skin scrapings reveals brownish, branched, septate hyphae along with budding cells. What is the most likely causative agent?

A. Malassezia furfur

B. Exophiala werneckii

C. Trichophyton rubrum

D. Candida albicans

Correct Answers

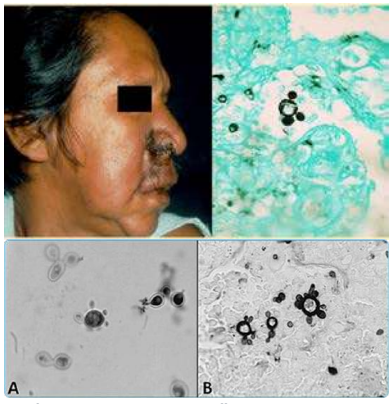
Question	Correct Answer
Question 1	3
Question 2	2
Question 3	4
Question 4	3
Question 5	3
Question 6	2
Question 7	2
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	3
Question 13	3
Question 14	1
Question 15	4
Question 16	4
Question 17	3
Question 18	3
Question 19	2
Question 20	2

Solution for Question 1:

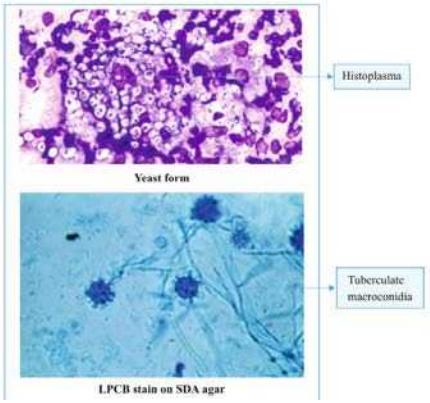
Answer: C) Paracoccidioides

Explanation:

Based on the travel history, clinical manifestation and characteristic mariner's or pilot's wheel appearance on microscopy, the most probable diagnosis is Paracoccidioidomycosis. It is caused by Paracoccidioides brasiliensis.

Paracoccidioidomycosis	
Causative agent	Thermally dimorphic fungus Paracoccidioides brasiliensis. (Option C)
Epidemiology	Confined to South America, it is called South American blastomycosis.
Route of transmission	Inhalation of spores
Clinical features	Chronic granulomatous disease of the skin, mucosa, lymph nodes and internal organs. Primary pulmonary infection spreads by the hematogenous route to the mucosa of the nose, mouth, gastrointestinal tract, skin and lymphatics. Ulcerative granulomas of the buccal, and nasal mucosa.
Diagnosis	 ... (content truncated)
Treatment	Itraconazole is the drug of choice

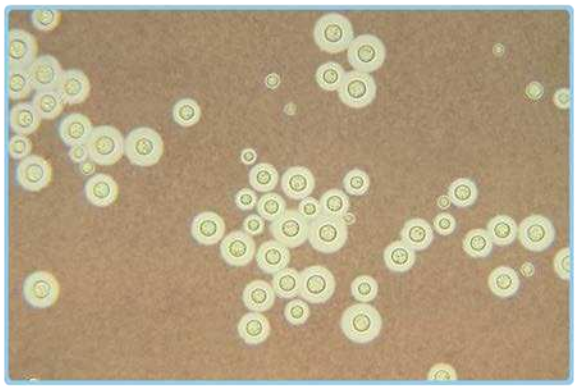
Histoplasma (Option A): It causes histoplasmosis, especially in immunocompromised patients. It can cause pulmonary and systemic infections. Microscopy shows small intra- and extracellular yeast cells. This option is incorrect.



Blastomyces (Option B): It causes blastomycosis. It primarily causes pulmonary symptoms and may disseminate to the skin, bones, and other organs. The yeast form of Blastomyces is typically large, thick-walled, and appears as single broad-based buds. This option is incorrect.



Cryptococcus (Option D): It causes cryptococcosis, primarily in immunocompromised individuals. It commonly presents as a pulmonary or central nervous system infection, but mucosal involvement is rare. The yeast cells of Cryptococcus are typically round or oval, single-celled, and surrounded by a thick capsule, with no multiple budding patterns. This option is incorrect.



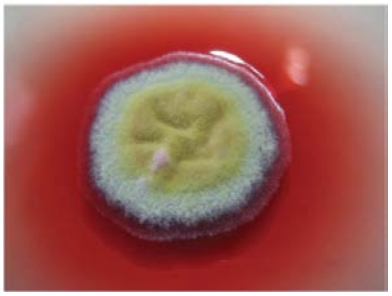
Solution for Question 2:

Answer: B) Green, granular colonies with a distinctive red pigment

Explanation:

Given the patient's symptoms, history of exposure to bamboo rats, and immunocompromised status due to AIDS, microscopy showing septate hyphae with branched conidiophores, the likely pathogen is Talaromyces marneffei. This fungus produces green, granular colonies with a distinctive red pigment that diffuses into the agar when cultured on Sabouraud agar, which is a characteristic feature of this organism.

Penicillium marneffei	
Characteristics	Currently, the name is changed to Talaromyces marneffei. Thermally dimorphic fungus
Causes	Talaromycosis (Penicilliosis)
Epidemiology	Common in Southeast Asia, southern China, and northeastern India. High incidence in AIDS and immunocompromised patients.
Reservoir	Bamboo rat
Route of transmission	Inhalation of fungal spores No person-to-person transmission
Clinical manifestations	It affects the reticuloendothelial system, causing lymphadenopathy, hepatomegaly, and splenomegaly. Lung involvement is common. Increased risk of dissemination in AIDS and immunocompromised individuals.
Specimens for Diagnosis	Bone marrow, blood, sputum, lymph nodes, skin lesions

Diagnosis	 ... (content truncated)
Treatment	Amphotericin B (IV for two weeks), followed by Itraconazole. Others: Voriconazole, Posaconazole.

White, cottony, mycelial growth (Option A): It is typically associated with Histoplasma capsulatum. This option is incorrect.

Smooth, mucoid, cream-coloured colonies (Option C): It is typically associated with Cryptococcus neoformans. This option is incorrect.

Creamy white, smooth colonies and with a yeasty odour (Option D): It is typically associated with Candida albicans. This option is incorrect.

Solution for Question 3:

Answer: D) Intracellular yeast cells

Explanation:

Intracellular yeast cells are seen in stained tissue samples or body fluids during microscopic examination and not in cultures.

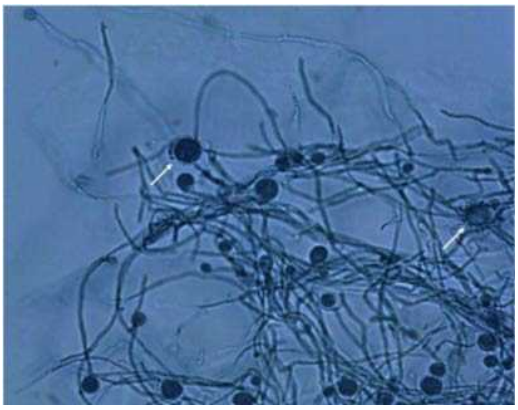
Microscopic examination of Yeast in histoplasmosis:

- Culture: Blood agar and Sabouraud agar show White, cottony growth
- Blood agar and Sabouraud agar show White, cottony growth
- Microscopy: From culture colonies: Mycelial growth with large thick-walled spherical spores with tuberculate (Tuberculate macroconidia) and smooth-walled oval microconidia (Options A, B & C ruled out)

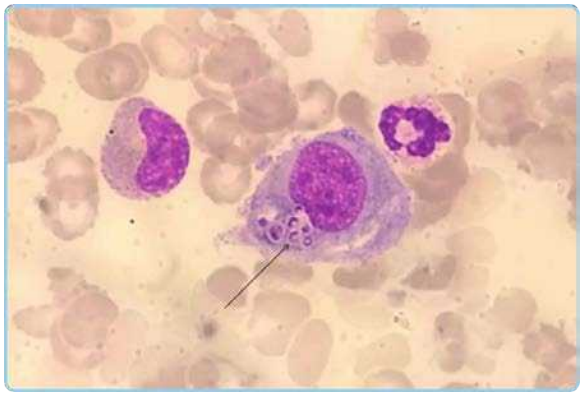
• From culture colonies: Mycelial growth with large thick-walled spherical spores with tuberculate (Tuberculate macroconidia) and smooth-walled oval microconidia (Options A, B & C ruled out)

• Blood agar and Sabouraud agar show White, cottony growth

• From culture colonies: Mycelial growth with large thick-walled spherical spores with tuberculate (Tuberculate macroconidia) and smooth-walled oval microconidia (Options A, B & C ruled out)



• Stained smears of the specimens: Show small intra- and extracellular yeast cells (Option D)



Solution for Question 4:

Answer: C) The causative agent cannot be cultured in vitro.

Explanation:

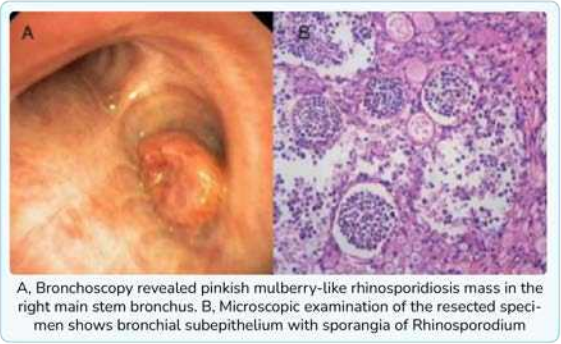
Rhinosporidium seeberi is a eukaryotic pathogen that exhibits characteristics of both fungi and protozoa, making it difficult to culture in laboratory settings. MO

Rhinosporidiosis

- Chronic granulomatous infection caused by Rhinosporidium seeberi
- Primarily affects mucosal membranes of: (Option A ruled out) Nose Nasopharynx Conjunctiva Urethra
- Nose
- Nasopharynx
- Conjunctiva
- Urethra
- Endemic in tropical and subtropical regions; associated with contaminated water exposure (Option D ruled out)
- Nose
- Nasopharynx
- Conjunctiva
- Urethra

Microbiology of Rhinosporidiosis

Aspect	Details
Causative Agent	Rhinosporidium seeberi - eukaryotic pathogen exhibiting features of fungi and protozoa; cannot be cultured in vitro (Option C)
Pathogenesis	Occurs via endospores infecting traumatized mucosal surfaces, leading to hyperplastic growths and rhinosporidial polyps
Histopathology	Presence of thick-walled sporangia (50-1000 µm) filled with endospores (5-10 µm) Surrounding tissue shows hyperplasia and infiltration by inflammatory cells (lymphocytes, plasma cells) The lack of eosinophilic infiltration distinguishes it from similar infections.
Staining Techniques	H&E: Identifies sporangia and inflammatory response PAS: Stains endospores positively GMS: Visualizes fungal elements



A, Bronchoscopy revealed pinkish mulberry-like rhinosporidiosis mass in the right main stem bronchus. B, Microscopic examination of the resected specimen shows bronchial subepithelium with sporangia of Rhinosporodium

Clinical Presentation

- Nasal Masses: Soft, fleshy lesions, pedunculated or flat Red or granular appearance with yellowish-white dots (mature sporangia)
- Soft, fleshy lesions, pedunculated or flat
- Red or granular appearance with yellowish-white dots (mature sporangia)
- Symptoms: Nasal obstruction Epistaxis (nosebleeds) Discharge
- Nasal obstruction
- Epistaxis (nosebleeds)
- Discharge
- Soft, fleshy lesions, pedunculated or flat
- Red or granular appearance with yellowish-white dots (mature sporangia)
- Nasal obstruction
- Epistaxis (nosebleeds)
- Discharge

Diagnosis

- Histopathology: Definitive through examination of excised tissue showing characteristic sporangia and inflammation.
- Definitive through examination of excised tissue showing characteristic sporangia and inflammation.
- Clinical History: Important to note exposure to stagnant water or endemic areas.
- Important to note exposure to stagnant water or endemic areas.
- Definitive through examination of excised tissue showing characteristic sporangia and inflammation.
- Important to note exposure to stagnant water or endemic areas.

Treatment

Treatment Option	Details
Surgical Intervention	Wide surgical excision of lesions. Cauterization of base is recommended to minimize recurrence.
Adjunctive Therapy	Postoperative Dapsone is used to prevent recurrence, though efficacy is controversial due to resistance. (Option B ruled out)

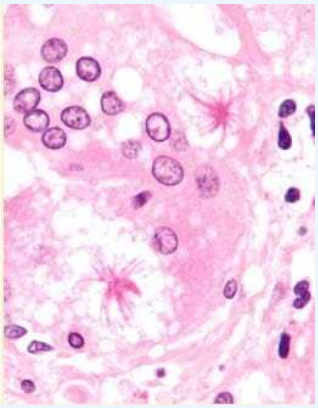
Solution for Question 5:

Answer: C) Cigar-shaped yeast cells

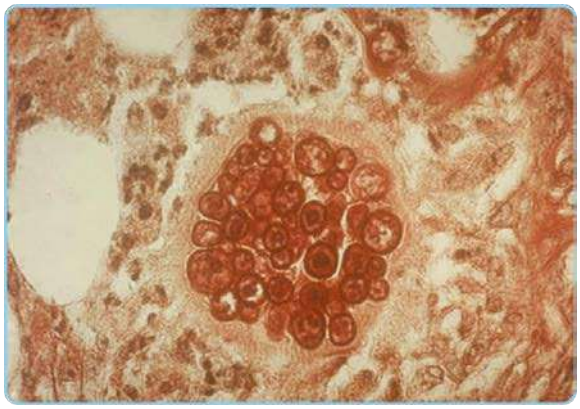
Explanation:

Sporothrix schenckii in the yeast phase is often described as cigar-shaped, which helps differentiate it from other fungi.

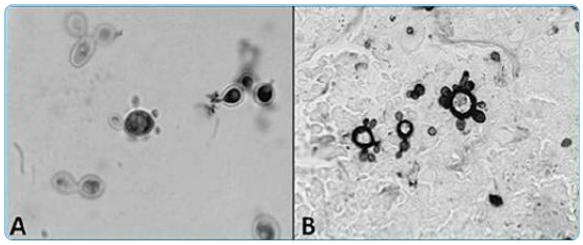
Sporotrichosis	
Causative agent	Dimorphic fungus Sporothrix schenckii.
Epidemiology	Common in gardeners, forest workers, and manual labourers.
Route of transmission	Minor trauma or thorn prick.
Clinical features	 Chronic infection affects cutaneous, subcutaneous, and lymphatic tissue. Lesions form at sites of minor trauma or thorn pricks, starting as nodules. ... (content truncated)
Specimens	Aspiration from nodules and biopsy material.

Diagnosis	 ... (content truncated)
Treatment	Medications: Itraconazole Terbinafine Amphotericin B: For disseminated infection Cryotherapy.

Thick-walled spherules filled with endospores (Option A): It is characteristic of Coccidioides immitis, the causative agent of coccidioidomycosis. This option is incorrect.



Mariner’s wheel-shaped yeast cells (Option B): It is characteristic of Paracoccidioides brasiliensis, which causes paracoccidioidomycosis. This option is incorrect.



Broad-based budding yeast cells (Option D): It is characteristic of Blastomyces dermatitidis, the causative organism of blastomycosis. This option is incorrect.



Skin biopsy demonstrated Blastomyces dermatitidis yeast forms, 8-15 mm in diameter with broad based budding pattern. Grocott's methanamine silver stain was used to visualize fungal elements that cannot be viewed as easily

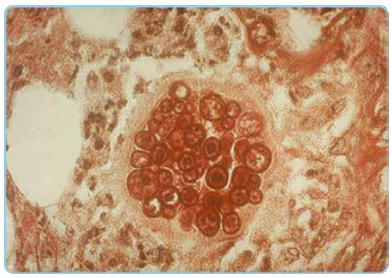
Solution for Question 6:

Answer: B) Inhalation of dust containing arthrospores

Explanation:

Coccidioides immitis is transmitted through the inhalation of dust containing fungal arthrospores, especially in arid, dusty environments.

Coccidioidomycosis	
Causative agent	Thermally dimorphic fungus Coccidioides immitis.
Route of transmission	Inhalation of dust containing fungal arthrospores. (Option B)
Clinical features	Self-limited, flu-like fever ("valley fever" or "desert rheumatism"). Primary pulmonary infection. Severe cases: Less than 1% develop chronic disseminated disease (coccidioidal granuloma), which can be fatal.

Diagnosis	 In tissue: Spherules with a thick, doubly refractile wall, filled with endospores. Culture: Mycelial phase with hyphae that fragment into arthrospores. Arthrospores are infective. Skin test: Positive delayed hypersensitivity with " coccidioidin."
Treatment	Mild cases: Fluconazole or itraconazole. Severe cases: Amphotericin B.

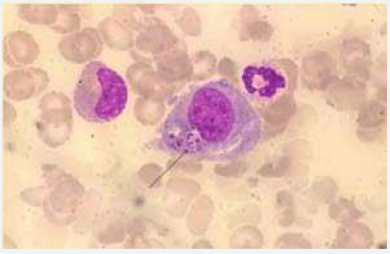
Solution for Question 7:

Answer: B) Amphotericin-B by itraconazole

Explanation:

- Based on the patient's history, histopathology showing intramacrophagic yeast form and (LPCB) mount showing numerous hyaline hyphae with few tuberculate macroconidia (arrows) and many smooth-walled oval microconidia are pointing towards a diagnosis of disseminated histoplasmosis.
- The recommended treatment for disseminated histoplasmosis, especially in immunocompromised patients, is Amphotericin-B. After stabilization, itraconazole is prescribed usually for a year to prevent recurrence and manage the infection.

Histoplasmosis	
Causative Agent	Histoplasma capsulatum; thermally dimorphic fungus. H. capsulatum var. capsulatum: Causes classical histoplasmosis. H. capsulatum var. duboisii: Causes African histoplasmosis.
Epidemiology	Endemic to central and eastern USA; also found in Africa.

Route of transmission	Thrives in soil, decaying trees, and bird droppings. Inhalation of fungal spores.
Clinical manifestations	Classical histoplasmosis is often asymptomatic. Calcified lesions. Pulmonary disease mimics tuberculosis. Disseminated histoplasmosis can occur. Infection of the reticuloendothelial system causes: Lymphadenopathy, hepatosplenomegaly, fever, and anaemia Skin and mucous membrane: Granulomatous and ulcerative lesions. African Histoplasmosis: It primarily affects the skin, subcutaneous tissues, and bones Lungs are rarely involved, and disseminated cases are infrequent
Specimens for Diagnosis	Blood, urine, bone marrow, lesions
Diagnosis	 ... (content truncated)
Treatment	Amphotericin-B for two to four weeks followed by a year of itraconazole (Option A ruled out)

Fluconazole (Option C): It is usually reserved for less severe infections or maintenance therapy after initial treatment with more potent antifungals like amphotericin-B. This option is incorrect.

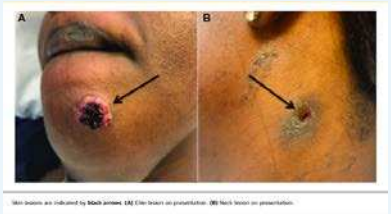
Surgical resection (Option D): It may be considered for localized infections or complications like abscesses, but it is not appropriate as a primary treatment strategy for disseminated histoplasmosis.

Solution for Question 8:

Answer: A) Blastomycosis

Explanation:

Blastomycosis is caused by Blastomyces dermatitidis, a dimorphic fungus commonly found in North America. It often presents with respiratory symptoms (cough, fever, weight loss) and may disseminate to the skin, forming ulcerative lesions. Microscopically, the spherical yeast cells with a thick double-contoured black-coloured wall and a broad-based bud, visible with GMS staining, are classic for blastomycosis, making this the most likely diagnosis.

Blastomycosis	
Causative Agent:	Blastomyces dermatitidis, a thermally dimorphic fungus.
Epidemiology	Primarily found in North America; cases are also reported in Africa and Delhi, India.
Route of transmission	Inhalation of fungal spores from contaminated soil.
Clinical manifestations	 Acute or chronic infection, characterised by the formation of suppurative and granulomatous lesions with a marked preference for lungs and skin. Primary Lung Infection: Mimics tuberculosis or histoplasmosis; symptoms vary from asymptomatic to severe with focal consolidation, miliary lesions, or abscesses. Disseminated Disease: Spreads through the bloodstream, forming multiple abscesses; high fatality rate. Cutaneous Disease: Begins as a papule, forming large, ulcerative lesions, often on the face or other exposed skin.
Diagnosis	Microscopy: Large, thick double-contoured wall, spherical yeast with a single broad-based bud. Grocott's methenamine silver (GMS) stain: Black staining of fungal cell walls. GMS answer" data-author="" data-hash="10191" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/VJTV83IbwnalK0pMJ51y1744477904.png" /> Culture: Filamentous growth with septate hyphae and round/oval conidia.
Treatment	Itraconazole: For mild to moderate cases. Amphotericin B: For severe and life-threatening cases.

Chromoblastomycosis (Option B): It is a chronic fungal infection, that typically presents as warty, nodular skin lesions, not as ulcerative lesions. It is commonly found in tropical regions and is not associated with lung involvement or systemic symptoms like weight loss. Microscopic examination shows sclerotic bodies. This option is incorrect.

Mycotic Mycetoma (Option C): It is caused by the fungi Madurella mycetomatis. It presents as a chronic infection of the skin, subcutaneous tissue, and sometimes bone, usually in the feet. It manifests as nodules that can develop into draining sinuses and does not typically form warty lesions or display sclerotic bodies in microscopy. This option is incorrect.

Sporotrichosis (Option D): It is caused by Sporothrix schenckii, often occurs after a traumatic implantation and typically presents with nodular lesions that can ulcerate, forming a lymphatic spread pattern. Microscopy of sporotrichosis shows cigar-shaped yeast cells instead. This option is incorrect.

Solution for Question 9:

Answer: B) Conversion to a yeast form in the host

Explanation:

In the environment, dimorphic fungi exist in mold form but convert to yeast form in the host (at 37°C). The yeast form is typically more resistant to phagocytosis and immune responses, allowing the fungi to survive and multiply in the host.

Dimorphic Fungus

Have the ability to convert between two specific morphologic forms, yeast and hyphae.

Types:

- Thermally dimorphic fungi: Converts between hyphae (22–25°C) and yeast (37°C), based on the temperature Thermally dimorphic fungi infect humans and other mammals Conversion to yeast within healthy and immunocompromised mammalian hosts is essential for virulence (Option B) Eg. Blastomyces dermatitidis, Histoplasma capsulatum, Coccidioides immitis and posadasii, Paracoccidioides brasiliensis and lutzii, Sporothrix schenckii, Talaromyces marneffei, Emmonsia spp., Lacazia loboi.
- Converts between hyphae (22–25°C) and yeast (37°C), based on the temperature

• Thermally dimorphic fungi infect humans and other mammals

• Conversion to yeast within healthy and immunocompromised mammalian hosts is essential for virulence (Option B)

• Eg. Blastomyces dermatitidis, Histoplasma capsulatum, Coccidioides immitis and posadasii, Paracoccidioides brasiliensis and lutzii, Sporothrix schenckii, Talaromyces marneffei, Emmonsia spp., Lacazia loboi.

• Nonthermal dimorphic fungi: Typically infects plants

• Typically infects plants

• Converts between hyphae (22–25°C) and yeast (37°C), based on the temperature

• Thermally dimorphic fungi infect humans and other mammals

• Conversion to yeast within healthy and immunocompromised mammalian hosts is essential for virulence (Option B)

• Eg. Blastomyces dermatitidis, Histoplasma capsulatum, Coccidioides immitis and posadasii, Paracoccidioides brasiliensis and lutzii, Sporothrix schenckii, Talaromyces marneffei, Emmonsia spp., Lacazia loboi.

• Typically infects plants

dimorphic fungus.png" data-author="" data-hash="10188" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/UvujaelLftKr86JvtFT61744477430.png" />

Formation of biofilms (Option A): It is a common immune evasion mechanism for Candida, but is not a primary characteristic of dimorphic fungi. This option is incorrect.

Production of endotoxins (Option C): Endotoxins are lipopolysaccharides found in the outer membrane of Gram-negative bacteria, not in fungi. Dimorphic fungi do not produce endotoxins. This option is incorrect.

Direct cell-to-cell transmission (Option D): Dimorphic fungi are primarily transmitted through inhalation of spores from the environment and then converted to the yeast form within the host, rather than spreading directly from cell to cell. This option is incorrect.

Solution for Question 10:

Answer: B) Chromoblastomycosis

Explanation:

Chromoblastomycosis is a fungal infection often seen in individuals from tropical regions who have sustained trauma, such as an injury from a thorn. It commonly presents as slowly growing, warty, nodular lesions resembling cauliflower. The presence of sclerotic bodies (round, dark brown, yeast-like bodies with septate) in KOH mounts is characteristic, making it the most likely diagnosis in this case.

Chromoblastomycosis (Also known as verrucous dermatitis)	
Epidemiology	It is the most common form of chromomycosis. Common among barefoot agricultural workers and woodcutters. Tropical region.
Causative agents	Fonsecaea pedrosoi, Fonsecaea compacta, Exophiala dermatitidis, Phialophora, Cladophialophora.
Route of transmission	Infecting fungi enter the skin by traumatic implantation.
Clinical manifestation	Warty, cutaneous nodular lesions (resemble the florets of cauliflower). Usually confined to subcutaneous tissue of the feet and lower legs Slow-developing lesions around the site of implantation. Chromoblastomycosis foot" data-author="" data-hash="10182" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/XDM2CdVMChdG0KscAB1w1744476678.png" />
Diagnosis	Microscopy: The lesions show the presence of the fungus as round or irregular, dark brown, yeast-like bodies with septate, called sclerotic bodies (other names: muriform bodies, copper pennies, medlar bodies). Sclerotic cells" data-author="" data-hash="10185" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/vw70kUv1HY15bZZnjgC31744476801.png" /> Demonstration of these sclerotic bodies in KOH mounts or tissue sections, and by culture on Sabouraud agar.
Treatment	Medical: Amphotericin B, thiabendazole, 5-fluorocytosine, itraconazole and (new drug) voriconazole. Surgical excision.

Rhinosporidiosis (Option A): It is a chronic infection caused by Rhinosporidium seeberi. It typically presents as polypoid masses in the nasal cavity, conjunctiva, or other mucosal surfaces, not as warty, nodular skin lesions. KOH mounts do not show sclerotic bodies. This option is incorrect.

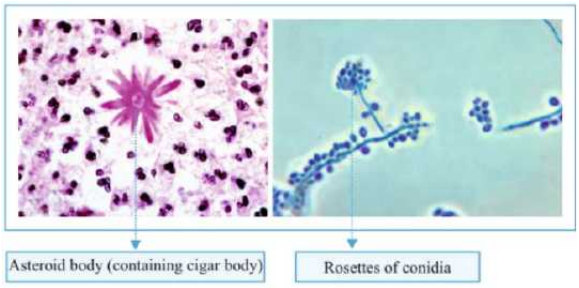
Rhinosporidiosis nose and histopath" data-author="" data-hash="10187" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/uqLTrwGoVmKDuHix8pzM1744476900.jpg" />

Mycotic Mycetoma (Option C): It is caused by the fungi Madurella mycetomatis. It presents as a chronic infection of the skin, subcutaneous tissue, and sometimes bone, usually in the feet. It manifests as nodules that can develop into draining sinuses and does not typically form warty lesions or display sclerotic bodies in microscopy. This option is incorrect.



KOH mount of M. mycetoma

Sporotrichosis (Option D): It is caused by Sporothrix schenckii, often occurs after a traumatic implantation and typically presents with nodular lesions that can ulcerate, forming a lymphatic spread pattern. Microscopy of sporotrichosis shows cigar-shaped yeast cells instead. This option is incorrect.



Solution for Question 11:

Answer: C) Tinea versicolor

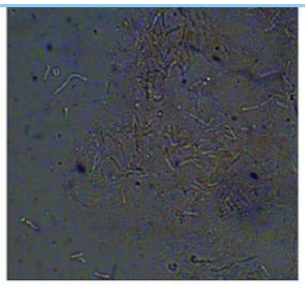
Explanation:

Tinea versicolor is caused by Malassezia furfur and presents as hypopigmented or hyperpigmented macular lesions, typically on the chest, back, or upper limbs. It is asymptomatic and diagnosed by finding yeast cells grouped in a “grape bunch” format with short, branched filaments on microscopy.

Tinea Versicolor (Pityriasis Versicolor)

- A superficial fungal infection of the stratum corneum (outer skin layer)
- Mostly asymptomatic but may lead to visible skin discolouration
- Prevalent in tropical climates, commonly affecting young adults
- Causative agent: Malassezia furfur (previously called Pityrosporum orbiculare)
- Appearance: Presents as discrete or confluent macular lesions Affected areas show pigmentation changes (hypopigmentation or hyperpigmentation)
- Presents as discrete or confluent macular lesions
- Affected areas show pigmentation changes (hypopigmentation or hyperpigmentation)
- Commonly affected sites: Chest, abdomen, upper limbs, and back
- Chest, abdomen, upper limbs, and back
- Presents as discrete or confluent macular lesions
- Affected areas show pigmentation changes (hypopigmentation or hyperpigmentation)
- Chest, abdomen, upper limbs, and back

Diagnostic Tools:

Test/Tool	Findings
Microscopy (Skin Scrapings)	 Image showing Spaghetti and meatball appearance in pityriasis versicolor under KOH microscopy Abundant yeast cells and short, branched filaments (yeast cells grouped in a “grape bunch” format with short, branched filaments): Tinea versicolor KOH." data-author="" data-hash="10095" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com /notes/RrVjSljkSpVrKoq2sYTO1744469150.png" /> ... (content truncated)

Culture	Grown on Sabouraud agar with an olive oil overlay. Versicolor SDA" data-author="" data-hash="10097" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/rHhlfhaSGBz1aVg5YagT1744469378.png" />
Wood’s Lamp Exam	Infected areas fluoresce golden-yellow under UV light. Versicolor Wood’s Lamp " data-author="" data-hash="10098" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/EVtP4p0OtNCrspsH4Gs61744469541.jpg" />

versicolor on face" data-author="" data-hash="10099" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/mHm11m1bRBeDqREyedFM1744469645.png" />

Tinea corporis (Option A): A dermatophyte infection characterized by annular, scaly lesions with raised borders. This option is incorrect.



Tinea nigra (Option B): Caused by Hortaea werneckii and characterized by black or brown macular lesions. This option is incorrect.



Candidiasis (Option D): Presents as erythematous plaques or white patches in moist areas (like skin folds) or mucosal surfaces (oral thrush, vaginal candidiasis). This option is incorrect.



Cutaneous Candidiasis



Oral thrush

Solution for Question 12:

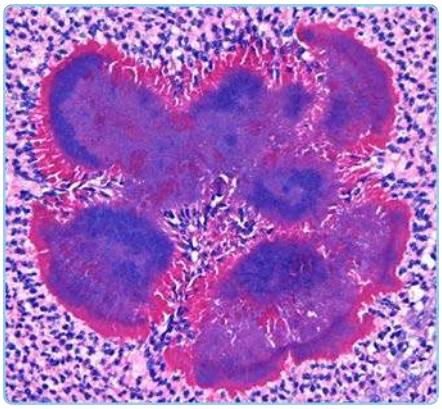
Answer: C) It can occur in both infectious and noninfectious conditions.

Explanation:

The Splendore–Hoeppli phenomenon is seen in a variety of conditions, both infectious (fungal, bacterial, or parasitic infections) and noninfectious (Hypereosinophilic syndrome and allergic conjunctival granulomas).

Splendore-Hoeppli phenomenon

- Splendore–Hoeppli phenomenon (asteroid bodies) involves microorganisms or inert substances surrounded by radiating eosinophilic material
- Microscopically, it appears as strongly eosinophilic, amorphous material with star-like or club-shaped formations.



- This phenomenon surrounds or is adjacent to the causative agent
- It is often accompanied by an inflammatory reaction rich in eosinophils, histiocytes, epithelioid cells, and giant cells (Option D ruled out)

Conditions where the phenomenon is seen	
Fungal infections	Sporotrichosis, pityrosporum folliculitis, Zygomycosis, candidiasis, aspergillosis blastomycosis.
Bacterial infections	Botryomycosis, nocardiosis actinomycosis.
Parasitic infestations	Orbital pythiosis, strongyloidiasis, schistosomiasis cutaneous larva migrans.
Noninfective pathologies	Hypereosinophilic syndrome allergic conjunctival granulomas.

Solution for Question 13:

Answer: C) Madurella mycetomatis

Explanation:

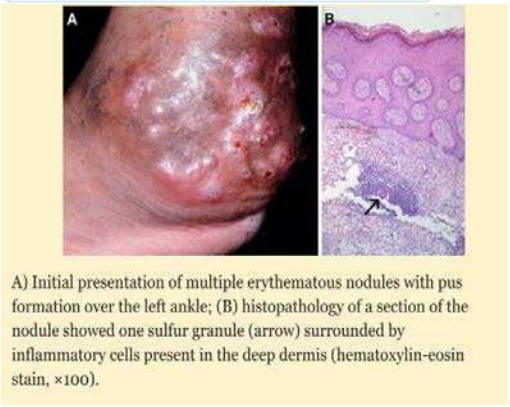
Madurella mycetomatis is a common fungal agent associated with eumycetoma and is characterized by the presence of thick-walled grains and hyphae in microscopic examination.

Mycetoma (Madura Foot)

- Definition: Chronic granulomatous infection with subcutaneous masses, draining sinuses, and grains in exudates
- Types: Eumycetoma (fungal) Actinomycetoma (bacterial)
- Eumycetoma (fungal)
- Actinomycetoma (bacterial)
- Eumycetoma (fungal)
- Actinomycetoma (bacterial)



A case of eumycetoma showing
A. A large plaque with atrophic surface studded with multiple sinuses on the lower leg.
B. A deep dermal granulomatous infiltrate with multiple colonies.
C. Gram-negative staining
D. Periodic acid-Schiff-positive staining



A) Initial presentation of multiple erythematous nodules with pus formation over the left ankle; (B) histopathology of a section of the nodule showed one sulfur granule (arrow) surrounded by inflammatory cells present in the deep dermis (hematoxylin-eosin stain, ×100).

Clinical Presentation

- Initial Lesion: Painless nodule; may be asymptomatic for years
- Painless nodule; may be asymptomatic for years
- Progression: Leads to swelling, ulceration, and pus discharge from sinuses containing grains
- Leads to swelling, ulceration, and pus discharge from sinuses containing grains

• Complications: Severe deformities, osteomyelitis, potential amputation due to extensive tissue damage

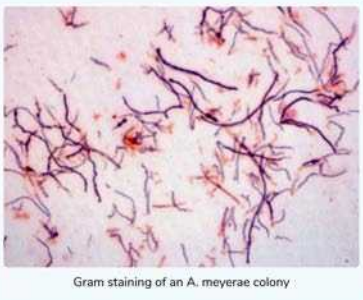
• Severe deformities, osteomyelitis, potential amputation due to extensive tissue damage

• Painless nodule; may be asymptomatic for years

• Leads to swelling, ulceration, and pus discharge from sinuses containing grains

• Severe deformities, osteomyelitis, potential amputation due to extensive tissue damage

Microscopic Examination

Type	Causative Agents	Microscopic Features
Eumycetoma	Madurella mycetomatis, (Option C) Curvularia lunata, Scedosporium spp., Acremonium spp.	 KOH wet mount direct microscopic examination of M. mycetomatis grains showing its hyphal structure ... (content truncated)
Actinomycetoma	Actinomadura madurae, Nocardia spp. (Options A & B ruled out)	 Gram staining of an A. meyerae colony Thin, branching filaments visible with Gram staining. Similar granulomas to eumycetoma but distinct morphology.

Diagnostic Techniques

• Biopsy and Culture: Tissue biopsy for visualization of pathogens. Culturing exudates for specific organism identification. Special stains to differentiate fungal and bacterial infections.

• Tissue biopsy for visualization of pathogens.

• Culturing exudates for specific organism identification.

• Special stains to differentiate fungal and bacterial infections.

- Imaging Studies: X-rays and MRI to assess bone involvement in advanced cases.
- X-rays and MRI to assess bone involvement in advanced cases.
- Tissue biopsy for visualization of pathogens.
- Culturing exudates for specific organism identification.
- Special stains to differentiate fungal and bacterial infections.
- X-rays and MRI to assess bone involvement in advanced cases.

Treatment

Type	Treatment Approach
Eumycetoma	Surgical excision with antifungals (itraconazole, ketoconazole). High recurrence rates.
Actinomycetoma	Prolonged antibiotic therapy (e.g., trimethoprim-sulfamethoxazole, dapsone) along with surgical intervention.

Trichophyton rubrum (Option D): It is a foot fungus responsible for tinea pedis (athlete's foot). It does not produce the thick-walled grains characteristic of eumycetoma. This option is incorrect.

Solution for Question 14:

Answer: A) a-(ii), b-(iii), c-(iv), d-(i)

Explanation:

Fungal infection classification depending on the tissue affected:

Category	Diseases
Superficial (localised to the stratum corneum)	Pytriasis versicolor, Tinea nigra, white piedra, black piedra.
Cutaneous	Dermatophytosis, Candidosis of skin, nail and mucosa.
Subcutaneous	Mycetoma, Chromomycosis Sporothicosis, Phaeohyphomycosis Entomophthoromycoses.

Deep or systemic (often endemic)	Histoplasmosis, Coccidiodomycosis Blastomycosis, Paracoccidiodomycosis.
Opportunistic	Candidosis, Cryptococcosis Aspergillosis, Penicilliosis, Zygomycosis, Pneumocystis pneumonia.

Solution for Question 15:

Answer: D) Microsporum infections commonly fluoresce bright greenish-yellow under Wood's lamp.

Explanation:

Microsporum infections are known for their bright greenish-yellow fluorescence, under Wood's lamp.

Wood's lamp examination for detecting superficial fungal infections

- Procedure: It is done in a dark room using ultraviolet light on the area of interest (Option B ruled out)
- Findings: Normal skin does not glow under UV light (Option C ruled out) Infected skin will fluoresce under light
- Normal skin does not glow under UV light (Option C ruled out)
- Infected skin will fluoresce under light
- Clinical significance: It helps detect fungal infections on the skin, or scalp (Option A ruled out) Microsporum infections glow bright greenish-yellow (Option D) Malassezia (Pityriasis versicolor) infections glow golden-yellow Tinea capitis will reveal areas of baldness.
- It helps detect fungal infections on the skin, or scalp (Option A ruled out)
- Microsporum infections glow bright greenish-yellow (Option D)
- Malassezia (Pityriasis versicolor) infections glow golden-yellow
- Tinea capitis will reveal areas of baldness.
- Normal skin does not glow under UV light (Option C ruled out)

- Infected skin will fluoresce under light
- It helps detect fungal infections on the skin, or scalp (Option A ruled out)
- Microsporum infections glow bright greenish-yellow (Option D)
- Malassezia (Pityriasis versicolor) infections glow golden-yellow
- Tinea capitis will reveal areas of baldness.

Infection	Cause	Color
Tinea capitis	Fungal	Blue-green
Tinea (Pityriasis) versicolor	Fungal	Yellowish green or orange
Erythrasma	Bacterial	Red
Vitiligo	Cell death	White
Porphyria cutanea tarda	Genetic or exposure	Red-pink
Pseudomonas infections	Bacterial	Green

Solution for Question 16:

Answer: D) T. mentagrophytes exhibits surface erosion of the hair shaft.

Explanation:

T. mentagrophytes causes a characteristic wedge-shaped appearance due to surface erosion of the hair shaft.

Hair Perforation Test

- Purpose: Differentiate between Trichophyton rubrum and Trichophyton mentagrophytes.

Procedure

Test Results

- If T. mentagrophytes is present: The hair shaft appears wedge-shaped due to surface erosion. (Option B ruled out) (Option D)
- The hair shaft appears wedge-shaped due to surface erosion. (Option B ruled out) (Option D)
- The hair shaft appears wedge-shaped due to surface erosion. (Option B ruled out) (Option D)

Characteristics of Fungal Infections:

- Both T. rubrum and T. mentagrophytes cause ectothrix hair infections, where arthrospores form a sheath around the hair.
- T. rubrum is the most common dermatophyte in humans, often leading to chronic, treatment-resistant lesions.

Identification Methods

Method	Description
Macroscopic examination	Assess the growth rate, texture, and colour on media
Urea hydrolysis test	Determines ability to hydrolyze urea
Growth on polished rice or special media	Tests the ability to grow in specific conditions
Temperature tolerance test	Assesses resistance to elevated temperatures
Nutritional requirements test	Checks requirements for specific amino acids and vitamins



The in vitro hair perforation test (lactophenol cotton blue preparation): a negative control without fungi; b positive control with Trichophyton mentagrophytes; c Penicillium glandicola was negative. Scale bars 100 µm

Solution for Question 17:

Answer: C) Tinea cruris

Explanation:

Tinea cruris is a condition, commonly known as jock itch, specifically affects the groin and perineum, characterized by erythema and raised borders.

Classification of Dermatophytosis

Anatomic Classification

Dermatophytosis can be classified based on the affected anatomical site:



Tinea pedis Image C" data-author="" data-hash="10136" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/1DRJ57u3V7TKsS5PkcNN1744473577.png" />







Clinical Type	Description
Tinea barbae	Infection of the bearded areas of the face and neck (barber's itch). (Option A ruled out)
Tinea capitis	Infection of the scalp; variants include favus and kerion. (Option B ruled out)
Tinea corporis	Infection of smooth, non-hairy skin (Tinea glabrosa). (Option D ruled out)
Tinea cruris	Infection of the groin and perineum (jock itch). (Option C)
Tinea imbricata	Tropical variant of Tinea corporis with concentric scaly patches.
Tinea manuum	Infection of the hand.
Tinea pedis	Infection of the foot (athlete's foot).
Tinea unguium	Infection of the nails.

Classification by Genus

Genus	Key Characteristics
Microsporum	Predominantly macroconidia; spindle-shaped, multicellular structures; infects hair and skin.
Trichophyton	More abundant microconidia in clusters; macroconidia are thin, and elongated; include species like T. rubrum which is common in humans. Infects hair, nails and skin.
Epidermophyton	Lacks microconidia; club-shaped, multicellular macroconidia; infects skin and nails but not hair.

Features	Trichophyton	Microsporum	Epidermophyton
1. Site of infection	Hair, nail, and skin	Hair and skin only	Skin and nail only
2. Colony	Powdery pigmented	Cotton-like pigmented	Powdery greenish yellow
3. Spores			

a. Microconidia	Abundant	Relatively scanty	Absent
b. Macroconidia	Pencil or cylindrical-shaped	Spindle-shaped	Club or pear-shaped



Types of Conidia

- Microconidia: Small, unicellular conidia.
- Macroconidia: Larger, multicellular, septate conidia.

Classification by Habitat

Dermatophytes are further classified by their natural habitats:

Habitat Type	Characteristics	Examples
Anthropophilic	Primarily found in humans; causes mild, chronic lesions.	T. rubrum, E. floccosum, M. audouinii
Zoophilic	Natural parasites of animals; cause severe inflammation but are treatable.	T. verrucosum, M. canis
Geophilic	Found in soil; less pathogenic to humans.	M. gypseum, T. ajelloi

Solution for Question 18:

Answer: C) It typically involves Trichophyton tonsurans and Trichophyton violaceum.

Explanation:

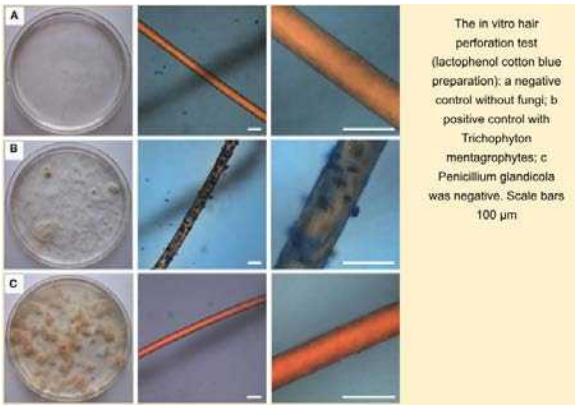
• Trichophyton tonsurans and Trichophyton violaceum are the primary fungi responsible for endothrix infections, where the fungus penetrates the hair shaft, leading to less contagious forms of tinea capitis, such as black dot tinea capitis.

• This distinguishes them from Microsporum canis, which causes ectothrix infections characterized by superficial hair involvement and is more contagious.

Hair perforation test:

• The hair perforation test is a diagnostic procedure used in dermatology to evaluate the presence of certain fungal infections, particularly Trichophyton species, which can cause scalp or skin conditions such as ringworm.

• This test assesses the ability of a hair to be penetrated or perforated by fungal elements, offering insight into whether a fungal infection may be affecting the hair



The in vitro hair perforation test (lactophenol cotton blue preparation): a negative control without fungi; b positive control with Trichophyton mentagrophytes; c Penicillium glandicola was negative. Scale bars 100 µm

Dermatophyte Infections: Endothrix vs. Ectothrix

Endothrix

• Characteristics Invasion: Fungi invade and grow inside the hair shaft, with arthroconidia present within the hair; the cuticle remains intact. Common Species: Trichophyton tonsurans, Trichophyton violaceum (non-inflammatory tinea capitis). (Option C) Clinical Presentation: Presents as black dot tinea capitis; less contagious than ectothrix.

• Invasion: Fungi invade and grow inside the hair shaft, with arthroconidia present within the hair; the cuticle remains intact.

• Common Species: Trichophyton tonsurans, Trichophyton violaceum (non-inflammatory tinea capitis). (Option C)

• Clinical Presentation: Presents as black dot tinea capitis; less contagious than ectothrix.

• Diagnosis Microscopy: Shows elongated hyphae within the hair. Trichoscopy: Features comma-shaped and corkscrew hairs.

• Microscopy: Shows elongated hyphae within the hair.

• Trichoscopy: Features comma-shaped and corkscrew hairs.

• Treatment

• Invasion: Fungi invade and grow inside the hair shaft, with arthroconidia present within the hair; the cuticle remains intact.

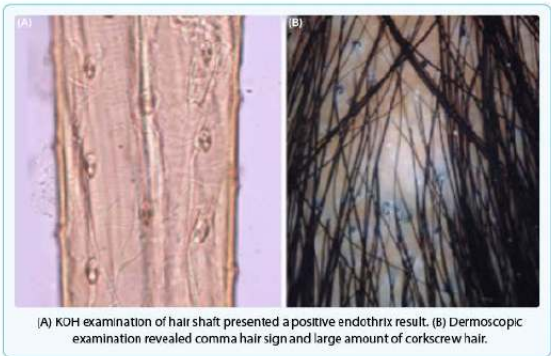
• Common Species: Trichophyton tonsurans, Trichophyton violaceum (non-inflammatory tinea capitis). (Option C)

• Clinical Presentation: Presents as black dot tinea capitis; less contagious than ectothrix.

• Microscopy: Shows elongated hyphae within the hair.

• Trichoscopy: Features comma-shaped and corkscrew hairs.

Antifungal Therapy: Griseofulvin; treatment duration: 6-12 weeks.(Option D ruled out)



(A) KOH examination of hair shaft presented a positive endothrix result. (B) Dermoscopic examination revealed comma hair sign and large amount of corkscrew hair.

Ectothrix

• Characteristics Invasion: Fungi grow on the hair shaft surface, degrading the cuticle; hyphae and arthroconidia are on both the surface and inside the hair. Common Species: Microsporum canis (inflammatory tinea capitis). (Option B ruled out) Clinical Presentation: Appears as grey patch tinea capitis; spreads rapidly.

• Invasion: Fungi grow on the hair shaft surface, degrading the cuticle; hyphae and arthroconidia are on both the surface and inside the hair.

• Common Species: Microsporum canis (inflammatory tinea capitis). (Option B ruled out)

• Clinical Presentation: Appears as grey patch tinea capitis; spreads rapidly.

• Diagnosis Microscopy: Fungal structures on the hair shaft's exterior. Trichoscopy: Features perifollicular scaling. (Option A ruled out)

• Microscopy: Fungal structures on the hair shaft's exterior.

• Trichoscopy: Features perifollicular scaling. (Option A ruled out)

• Treatment Antifungal Therapy: Griseofulvin; treatment duration may be longer.

• Antifungal Therapy: Griseofulvin; treatment duration may be longer.

• Invasion: Fungi grow on the hair shaft surface, degrading the cuticle; hyphae and arthroconidia are on both the surface and inside the hair.

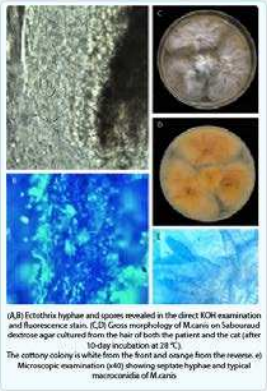
• Common Species: Microsporum canis (inflammatory tinea capitis). (Option B ruled out)

• Clinical Presentation: Appears as grey patch tinea capitis; spreads rapidly.

• Microscopy: Fungal structures on the hair shaft's exterior.

• Trichoscopy: Features perifollicular scaling. (Option A ruled out)

• Antifungal Therapy: Griseofulvin; treatment duration may be longer.



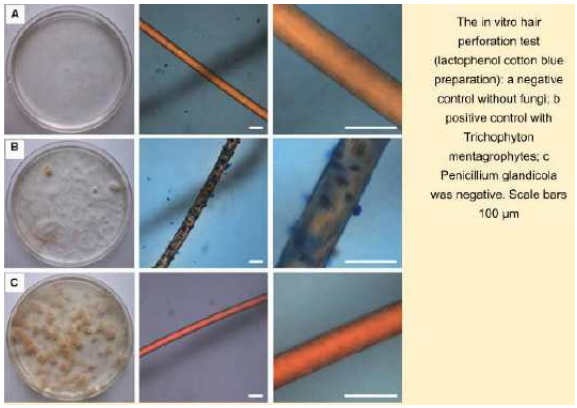
(A,B) Ectothrix hyphae and spores revealed in the direct KOH examination and fluorescence stain. (C,D) Gross morphology of M.canis on Sabouraud dextrose agar cultured from the hair of both the patient and the cat (after 10-day incubation at 28 °C). The cottony colony is white from the front and orange from the reverse. e) Microscopic examination (x40) showing septate hyphae and typical macroconidia of M.canis

The hair perforation test is a diagnostic tool used to differentiate dermatophyte species. It involves incubating hair with a fungal isolate in a sterile solution and examining for wedge-shaped perforations under a microscope.

Interpretation:

• Positive test: Presence of wedge-shaped perforations in the hair shaft (e.g., Trichophyton mentagrophytes).

• Negative test: Absence of perforations (e.g., Trichophyton rubrum).



The in vitro hair perforation test (lactophenol cotton blue preparation): a negative control without fungi; b positive control with Trichophyton mentagrophytes; c Penicillium glandicola was negative. Scale bars 100 µm

Solution for Question 19:

Answer: B) Piedraia hortae

Explanation:

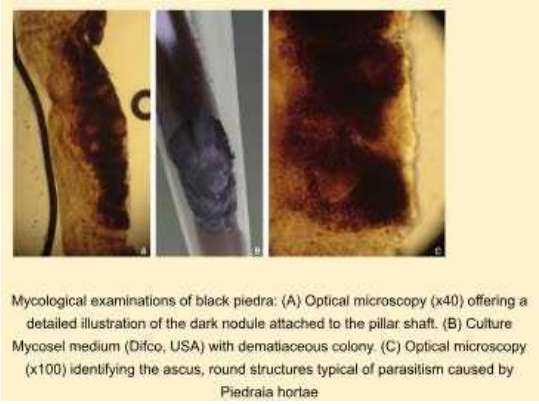
Piedraia hortae is the causative agent of black piedra, characterized by dark nodules on scalp hair and the presence of branched hyphae and arthrospores upon microscopic examination.

Piedra Infection

- Superficial fungal infection affecting hair shaft.
- Characterized by nodules.
- Two main types: Black Piedra and White Piedra.

Black Piedra

Aspect	Details
Etiology	Caused by Piedraia hortae (dematiaceous fungus) (Option B).
Characteristics	Affects scalp hair; it is common in individuals with long hair and poor hygiene.
Clinical Features	Hard, black nodules that may resemble lice nits; complex structure of filamentous hyphae and spores; examined using KOH shows branched hyphae and arthrospores.
Diagnosis	Direct microscopic examination: Branched hyphae and arthrospores Culture: Sabouraud dextrose agar shows dark colonies.
Treatment	Antifungal shampoos or systemic antifungals; treatment efficacy may be compromised by nodules' compact nature.



Mycological examinations of black piedra: (A) Optical microscopy (x40) offering a detailed illustration of the dark nodule attached to the pillar shaft. (B) Culture Mycosel medium (Difco, USA) with dematiaceous colony. (C) Optical microscopy (x100) identifying the ascus, round structures typical of parasitism caused by Piedraia hortae

White Piedra

Aspect	Details
Etiology	Caused by Trichosporon species, commonly Trichosporon beigelli, Trichosporon inkin (Option A ruled out).
Characteristics	Affects facial, axillary, and pubic hair; typically asymptomatic.
Clinical Features	Whitish, loosely adherent nodules consisting of arthroconidia and blastoconidia.
Diagnosis	KOH examination and culture reveal yeast-like colonies.
Treatment	Topical antifungals such as ketoconazole shampoo.



Macroscopic and microscopic aspect (400X) of White Piedra nodules. The white arrows indicate the nodules in the hair of each of the patients.

Mixed Piedra

- Rare condition involving both black and white piedra
- Presents with both dark and light nodules
- Requires careful mycological examination for accurate diagnosis

Candida albicans (Option C): Candida albicans do not cause superficial fungal infections of the hair shaft like piedra. Microscopy shows Candida yeast with pseudohyphae. This option is incorrect.

Candida albicans" data-author="" data-hash="10118" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/GEtQ8WqVhTuPIQMfSKII1744471180.png" />

Malassezia furfur (Option D): Malassezia furfur do not cause piedra. It primarily affects the skin and is involved in conditions related to skin oiliness rather than hard nodules along the hair shaft. Microscopy shows yeast cells grouped in a “grape bunch” format with short, branched filaments or spaghetti and meatball appearance. This option is incorrect.



Solution for Question 20:

Answer: B) Exophiala werneckii

Explanation:

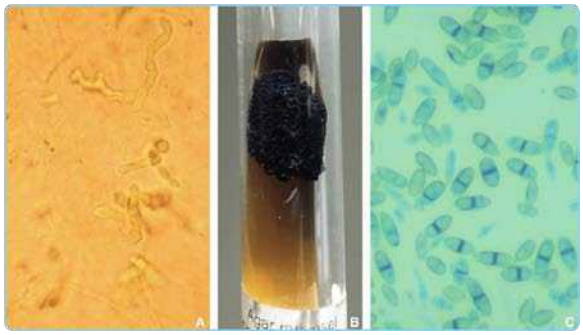
Exophiala werneckii causes Tinea nigra, characterized by black or brown macular lesions on the palms, with microscopy showing brown septate hyphae and budding cells.

Tinea Nigra

- Definition: A superficial fungal infection limited to the stratum corneum
- Geography: Common in tropical climates
- Clinical Presentation: Black or brown macular lesions on the skin Frequently affects the palms
- Black or brown macular lesions on the skin
- Frequently affects the palms
- Causative Agents: Exophiala werneckii (previously Cladosporium wernickii / Hortaea wernickii) Exophiala castellanii
- Exophiala werneckii (previously Cladosporium wernickii / Hortaea wernickii)
- Exophiala castellanii
- Black or brown macular lesions on the skin
- Frequently affects the palms
- Exophiala werneckii (previously Cladosporium wernickii / Hortaea wernickii)
- Exophiala castellanii

Diagnosis of Tinea Nigra:

- Microscopy (skin scrapings): Brownish, branched, septate hyphae Budding cells
- Brownish, branched, septate hyphae
- Budding cells
- Brownish, branched, septate hyphae
- Budding cells



- Culture: Sabouraud agar: Colonies appear grey or black
- Sabouraud agar: Colonies appear grey or black
- Sabouraud agar: Colonies appear grey or black

nigra SDA" data-author="" data-hash="10106" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/9ACDTn9nTdEX7AYlTtjX1744470147.png" />

Comparison: Tinea Nigra vs. Tinea Versicolor

Feature	Tinea Nigra	Tinea Versicolor
Affected Area	Palms	Chest, abdomen, upper limbs, back
Lesion Appearance	Black/brown macules	Hypopigmented or discolored macules
Causative Agent	Exophiala werneckii, Exophiala castellanii	Malassezia furfur (formerly Pityrosporum orbiculare) (Option A ruled out)
Microscopy	Brown hyphae and budding cells	Yeast-like cells and short, branched filaments
Culture	Grey/black colonies on Sabouraud agar	Requires Sabouraud agar with olive oil
Wood's Lamp Findings	None	Golden-yellow fluorescence

Tinea nigra" data-author="" data-hash="10109" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/DnUBv7geDnspXcfBdRD317444

70334.png" />

Trichophyton rubrum (Option C): Trichophyton rubrum causes dermatophytosis (e.g., tinea corporis), with scaly lesions, not brown macules. This option is incorrect.

Trichophyton rubrum" data-author="" data-hash="10111" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/slSLV6M0rdQZ8JKoEABu1744470434.png" />

Candida albicans (Option D): Candida albicans typically affects mucosal areas and causes candidiasis, not macular skin lesions. This option is incorrect.

albicans intertrigo" data-author="" data-hash="10112" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/2iSx43R2bkJiPoda9dVN1744470512.png" />

Opportunistic infections

1. A 55-year-old immunocompromised patient presents with fever with chills, diarrhea and new onset cardiac murmur. Candida species are isolated from her blood culture and IV fluconazole is considered for treatment. Which of the following Candida species is least likely to respond to azole therapy?

A. Candida albicans

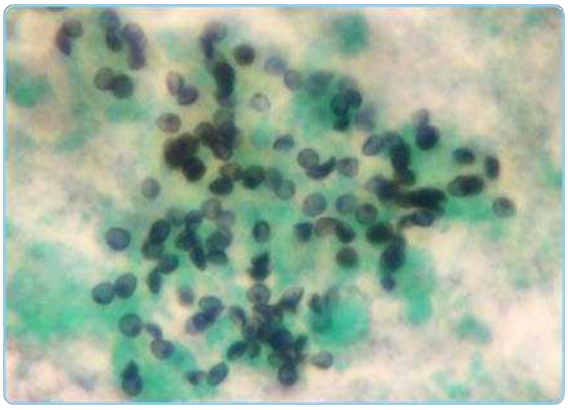
B. Candida glabrata

C. Candida dubliniensis

D. Candida krusei

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2. A 32-year-old HIV-positive male presents to the emergency department with a 2-week history of worsening shortness of breath, dry cough, and fever. He reports significant weight loss over the past month and has a CD4 count of 150 cells/mm³. HRCT Thorax shows ground-glass opacities. A bronchoalveolar lavage (BAL) is performed, and a histological examination shows organisms with a characteristic appearance, as shown in the image below. What is the most likely diagnosis?



A. Bacterial pneumonia

B. Tuberculosis

C. Pneumocystis Carinii pneumonia (PCP)

D. Fungal pneumonia

---

3. Which statements about the β-1,3-D-glucan test are incorrect?

A. β-1,3-D-glucan is a polysaccharide cell wall component.

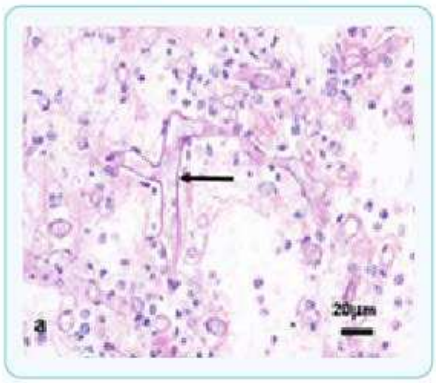
B. β-1,3-D-glucan is negative for Blastomyces, Mucor, Cryptococcus

C. β-1,3-D-glucan is only specific for Aspergillus.

D. None of the above

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4. A 55-year-old diabetic male presents to the emergency department with facial swelling, nasal discharge, and fever. He has a history of poorly controlled diabetes mellitus. On examination, there is necrosis of the nasal turbinates and blackish discharge. A CT scan of the sinuses reveals extensive necrosis. A tissue biopsy shows non-septate, wide-angle hyphae similar to the image given below. What is the most common causative organism of the above condition?



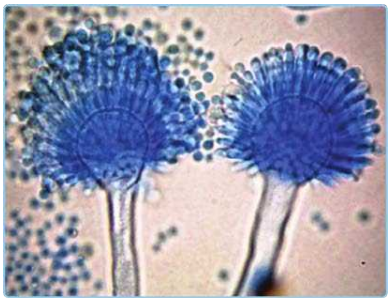
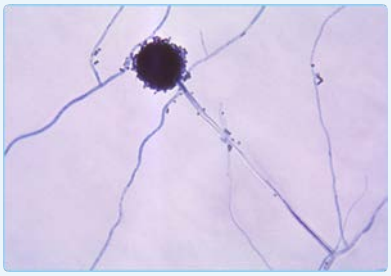
A. Aspergillus

B. Mucor

C. Absidia species

D. Rhizopus arrhizus

5. Match the following image with its corresponding Aspergillus species. a) Aspergillus flavus 1. b) Aspergillus niger 2. c) Aspergillus fumigatus 3.

a) Aspergillus flavus	 1.
b) Aspergillus niger	 2.
c) Aspergillus fumigatus	 3.

A. a-1, b-2, c-3

B. a-2, b-3, c-1

C. a-3, b-1, c-2

D. a-2, b-1, c-3

6. In which of the following infections can galactomannan in serum be seen?

A. Candidiasis

B. Aspergillus infection

C. Cryptococcosis

D. Mucormycosis

7. A 55-year-old male with a history of COPD presents to the emergency department with 1-week history of fever, cough, and worsening shortness of breath. Colonies exhibiting a yellow-green color and a velvety texture were isolated from the culture. He is immunocompromised due to long-term corticosteroid therapy. The attending physician wants to initiate antifungal therapy. Which of the following is the drug of choice for this patient?

A. Voriconazole

B. Amphotericin B

C. Itraconazole

D. Surgical resection

8. A 12-year-old female patient with a history of cystic fibrosis presents to the clinic with complaints of worsening facial pain, nasal congestion, and a brownish purulent nasal discharge for the past 2 weeks. A nasal endoscopy is performed, and a sample of the discharge is sent for microscopy and culture. If the patient is diagnosed with rhinosinusitis of fungal etiology, which of the following pathogens is most likely to be responsible?

A. Mucor

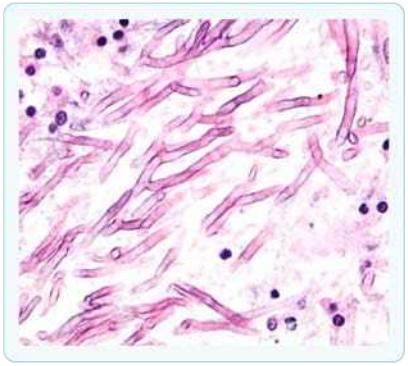
B. Rhizopus

C. Pneumocystis

D. Aspergillus

---

9. Based on the microscopic image shown below, identify the organism?



A. Mucor

B. Rhizopus

C. Candida

D. Aspergillus

---

10. Which of the following statements about Cryptococcus are correct? i. It is a mould type of fungi ii. Polypeptide capsule provides significant protection against phagocytosis. iii. Exposure to bird droppings can increase the likelihood of infection. iv. Cryptococcus neoformans is the principle pathogenic species v. Cryptococcus gattii is associated with B, C serotypes.

A. i, ii and iv

B. i, ii and v

C. iii, iv and v

D. ii, iii and iv

11. Which of the following is false regarding Candida dubliniensis?

A. Culture on cornmeal agar can produce chlamydospores

B. The germ tube test is positive

C. It can grow in hypertonic saline

D. It cannot grow at temperatures above 45 degrees

12. A 54-year-old woman with a history of type 2 diabetes mellitus presents to her primary care physician with complaints of intense vaginal itching, a thick, white, curd-like vaginal discharge, and irritation. Per vaginal examination reveals the following finding. The vaginal pH is found to be acidic (approximately 4.5). What is the most appropriate management for this patient's condition?



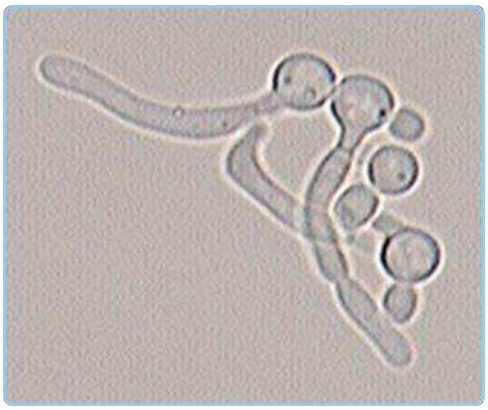
A. Topical nystatin

B. Oral fluconazole

C. Topical metronidazole

D. Topical hydrocortisone

13. Identify the test performed and the organism implicated from the image given below



A. Germ tube test - Candida albicans

B. Chrom agar test - Candida glabrata

C. SDA Culture media test - Candida tropicalis

D. Corn meal agar test - Candida krusei

---

14. A 50-year-old male with a history of poorly controlled diabetes presents with white patches on the tongue and inner cheeks. On examination, the lesions can be scraped off, revealing a red, inflamed base. Based on the clinical presentation, what is the most likely diagnosis?

A. Oral Hairy leukoplakia

B. Oral candidiasis

C. Lichen planus

D. Herpes simplex stomatitis

---

15. All of the following are true about Cryptococcus gattii except?

A. C. gattii is commonly found in Tropical and subtropical parts

B. It corresponds to the serotypes B and C

C. C. gattii inhabits around eucalyptus tree

D. It primarily affects immunocompromised individuals

---

16. A 40-year-old female presents to the clinic with a persistent cough, shortness of breath, and chest pain for the past month. She is currently on immunosuppressive therapy for an autoimmune condition. A chest CT reveals a solitary pulmonary nodule in the right upper lobe. What is the most likely diagnosis?

A. Bacterial pneumonia

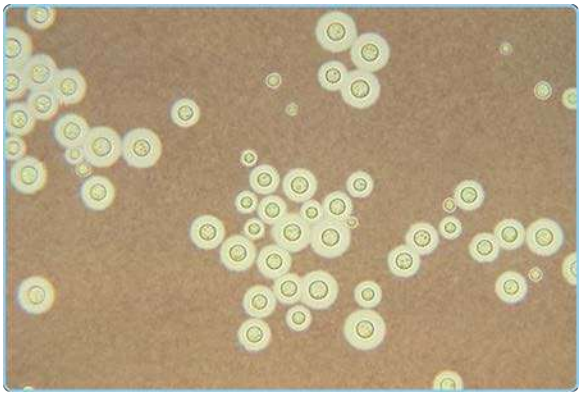
B. Pulmonary tuberculosis

C. Pulmonary cryptococcosis

D. Lung abscess

---

17. A 35-year-old male, known to be HIV positive, presents with fever, severe headache, neck stiffness, and altered mental status. A lumbar puncture is performed to analyze his cerebrospinal fluid (CSF), and following microscopic findings are observed. What stain has been used here to visualize the organism?



A. Gram stain

B. India ink stain

C. Acid-fast stain

D. Giemsa stain

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	1
Question 8	4
Question 9	4
Question 10	3
Question 11	3
Question 12	1
Question 13	1
Question 14	2
Question 15	4
Question 16	3
Question 17	2

Solution for Question 1:

Correct answer: D) Candida krusei

Explanation:

Candida krusei is resistant to azole antifungals, especially fluconazole.

This patient is suffering from systemic mycosis and is infected with Candida species.

The treatment of choice for invasive candidiasis is

- Echinocandins (caspofungin, micafungin, anidulafungin) Used as first line agents They can also be used if azole resistance is present or a contraindication to amphotericin B is present - renal disorders.
- Used as first line agents
- They can also be used if azole resistance is present or a contraindication to amphotericin B is present - renal disorders.
- Alternative therapy - Liposomal amphotericin B
- If the patient is stable and the Candida spp is identified as Azole sensitive - step down therapy with Fluconazole Candida krusei is resistant to azoles CNS or ophthalmic involvement - Voriconazole (Better tissue penetration)
- Candida krusei is resistant to azoles
- CNS or ophthalmic involvement - Voriconazole (Better tissue penetration)
- Used as first line agents
- They can also be used if azole resistance is present or a contraindication to amphotericin B is present - renal disorders.
- Candida krusei is resistant to azoles
- CNS or ophthalmic involvement - Voriconazole (Better tissue penetration)

Candida albicans, Candida glabrata, and Candida dubliniensis are sensitive and respond well to azoles (Options A,B, and C).

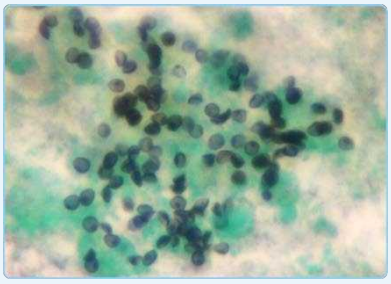
Solution for Question 2:

Correct Answer: C) Pneumocystis Carinii pneumonia (PCP)

Explanation:

Pneumocystis carinii pneumonia (PCP), now known as Pneumocystis jirovecii pneumonia, is a common opportunistic infection in immunocompromised individuals, particularly those with

HIV/AIDS.

Pneumocystis jirovecii	
Morphology	Trophozoite - thin-walled, irregularly shaped, 1-5 µm in size Precyst - an intermediate stage of the sexual phase, 5-8 µm in size Cyst - thick-walled, spherical, containing up to 8 intracystic bodies and up to 8 µm in size
Risk factors	HIV: CD4 < 200 Primary immunodeficiency states Transplant recipients Hematological malignancies
Clinical features	transmitted by respiratory droplets and is asymptomatic in immunocompetent individuals. In immunocompromised patients, life-threatening pneumonia develops. PCP - Pneumocystis Carinii pneumonia Dyspnoea Dry cough Fever Weight loss B/L interstitial pneumonia
Diagnosis	 ... (content truncated)
Treatment	The cornerstone for both prevention and treatment is trimethoprim-sulfamethoxazole. Alternative therapies have been used in AIDS patients; they include pentamidine, trimethoprim-dapsone, clindamycin-primaquine, atovaquone, and trimetrexate.

Bacterial pneumonia (Option A): Is caused by various bacteria (e.g., Streptococcus pneumoniae, Staphylococcus aureus) and typically presents with symptoms such as productive cough, high fever, and localized crackles on auscultation. Chest X-rays often show lobar consolidation rather than the diffuse ground-glass opacities characteristic of PCP.

Tuberculosis (TB) (Option B): Is a chronic infection caused by Mycobacterium tuberculosis. It usually presents with a persistent cough, night sweats, weight loss, and hemoptysis.

Fungal pneumonia (Option D): Can be caused by various fungi, such as Histoplasma, Coccidioides, or Cryptococcus. These infections typically present differently from PCP, often with more localized symptoms and findings on imaging

Solution for Question 3:

Correct Answer: C) β-1,3-D-glucan is only specific for Aspergillus.

Explanation:

The β-1,3-D-glucan test is not specific for Aspergillus; it can also be positive in infections caused by other fungi, such as Candida.

Clinical studies have demonstrated the utility of this assay for diagnosing invasive fungal infections.

Difference between β-1,3-D-glucan and Galactomannan test:

β-1,3-D-glucan	Galactomannan
Polysaccharide cell wall component (Option A)	Major polysaccharide component of aspergillus cell wall.
Routinely used for : Candida, Aspergillus	Routinely used for : Aspergillus
Positive in: PJP, Histoplasma, Fusarium, Scedosporium, dermatophytes, Cocci	Positive in: Fusarium, Penicillium, Histoplasma, Blastomyces, Paecilomyces, cryptococcus
Negative in: Blastomyces, Mucor, Cryptococcus (Option B)	Negative in: Mucorales

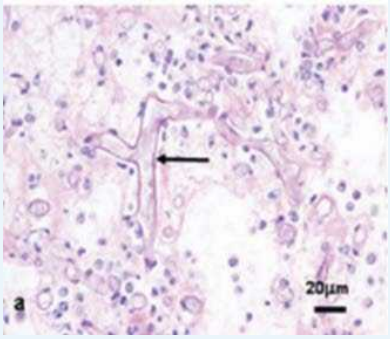
Solution for Question 4:

Correct Answer: D) Rhizopus Arrhizus

Explanation:

Rhizopus Arrhizus, also known as R. oryzae, is the most common causative organism of mucormycosis, especially in diabetic patients. It features non-septate hyphae that branch at right

angles.

Mucormycosis	
 ... (content truncated)	
Pathogenesis	Begins in the respiratory tract or nasal cavity, where spores germinate and form mycelia → invade adjacent tissues, leading to local destruction. Fungi invade blood vessels, causing thrombosis and infarction in affected areas, which contributes to its high mortality rate (Angio-invasiveness) Common Sites Rhinocerebral Form: Involves the nose, nasal sinuses, and orbit; frequently seen in diabetic patients. Pulmonary Infections: When the lungs are the primary site, the fungi can invade lung arteries, leading to serious complications. Others: Cutaneous mucormycosis, GI mucormycosis, Disseminated mucormycosis
Clinical Features	Rhinocerebral Zygomycosis: Facial swelling, nasal discharge (often black), fever, and mucosal necrosis. Rapid progression can lead to severe complications, including visual loss and neurological symptoms. Pulmonary Zygomycosis: Symptoms may include cough, fever, and hemoptysis
Diagnosis	 ... (content truncated)
Treatment:	Intravenous amphotericin B combined, where appropriate, with surgical drainage is the recommended treatment. Medical control of diabetes is required.

Aspergillus species (Option A), such as Aspergillus fumigatus, typically produce septate hyphae that branch at acute angles. The presence of non-septate, wide-angle hyphae in this case makes Aspergillus an unlikely causative organism.

Mucor species (Option B): can cause mucormycosis. It has similar characteristics to Rhizopus, including non-septate hyphae. However, it is less frequently implicated compared to Rhizopus arrhizus in cases of sinus infections. More in Covid-19 due to use of steroids.

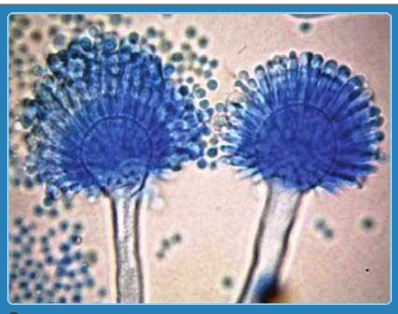
Features	Aspergillus	Mucor
Branching	<45°	>90° or 90°
Hyphae	Septate	Aseptate
Shape	Thin	Broad ribbon-like

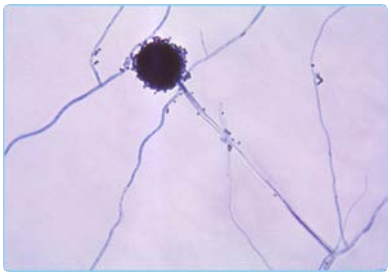
Absidia species (Option C) : can cause mucormycosis. It typically has a similar morphology, with non-septate hyphae, but is less common than Rhizopus.

Solution for Question 5:

Correct Answer: B) a-2, b-3, c-1

Explanation:

a) Aspergillus flavus Conidia: All around the vesicles (Flower has petals all around)	 2. {{caption_text}}

b) Aspergillus niger Blackish colour	 3. {{caption_text}}
c) Aspergillus fumigatus Conidia: All around the vesicles (Flower has petals all around)	 1. {{caption_text}}

Solution for Question 6:

Correct Answer: B) Aspergillus infection

Explanation:

Galactomannan is primarily associated with Aspergillus infections, particularly invasive aspergillosis.

- Galactomannan is a component of the Aspergillus cell wall
- Detection of Galactomannan in serum or bronchoalveolar lavage fluid is a marker for the diagnosis of invasive aspergillosis in adults and children, in hematopoietic stem cell transplant recipients or in patients with hematologic malignancies.

Galactomannan Test: Detects galactomannan, a carbohydrate antigen specific to Aspergillus species.

- Positive for: Aspergillus species
- False-negative for: Most fungi except Aspergillus; not reliable for Candida, Pneumocystis jirovecii, or Mucorales.

Candidiasis (Option A): Candidiasis is caused by Candida species, such as Candida albicans. While Candida infections can be serious, they do not typically produce galactomannan. Diagnosis is usually made through culture.

Cryptococcosis (Option C): Cryptococcosis is caused by Cryptococcus neoformans.The diagnosis usually involves detecting cryptococcal antigen (CrAg) in serum or cerebrospinal fluid or by Indian ink stain microscopy of CSF, not by detection of serum galactomannan.

Mucormycosis (Option D): It does not produce galactomannan. Diagnosis is typically through culture or histopathological examination, rather than serological tests for galactomannan.

Solution for Question 7:

Correct Answer: A) Voriconazole

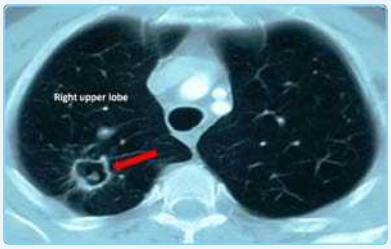
Explanation:

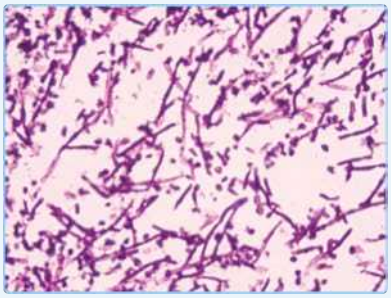
In this case, the patient is diagnosed with disseminated aspergillosis, presenting with pneumonia. Given his immunocompromised status from long-term corticosteroid treatment, the risk of disseminated fungal infection is high.

Voriconazole is the preferred first-line treatment for disseminated aspergillosis

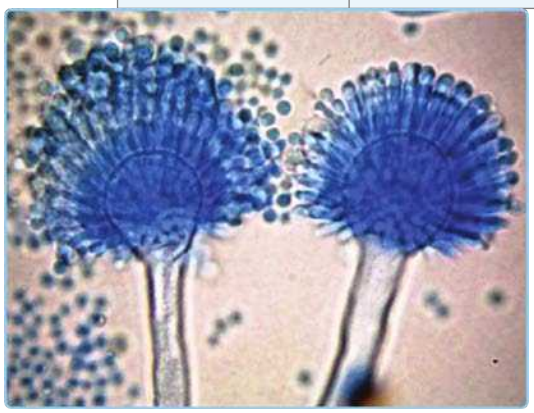
It is specifically effective in patients with compromised immune systems, making it suitable for this patient's condition, given his history of chronic obstructive pulmonary disease (COPD) and long-term corticosteroid therapy.

Aspergillosis		
Type	Pathology	Features

1. ABPA (allergic bronchopulmonary aspergillosis)	Inhaled spores provoke hypersensitivity reaction: Type 1: asthma Type 3: extrinsic alveolitis Combined type 1 and type 3	bronchopulmonary aspergillosis" data-author="Avani R. Patel, Amar R. Patel, Shivank Singh, Shantanu Singh, Imran Khawaja" data-hash="10294" data-license="CC BY 3.0" data-source="https://en.wikipedia.org/wiki/Allergic_bronchopulmonary_aspergillosis" data-tags="" src="https://image.prepladder.com/notes/4rRR0YCtfPGVmEIXHDRk1744601812.png" /> {{caption_text}} Chest X-ray showing: left-sided perihilar opacity (blue arrow) along with non-homogeneous infiltrates (transient pulmonary infiltrates indicated by red arrows) in all zones of both lung fields
2. Aspergilloma	Fungal ball grows within and is restricted to an existing lung cavity, for example, due to old tuberculosis infection or bronchiectasis.	 {{caption_text}} CT of a simple aspergilloma of the right lung
3. Invasive aspergillosis	Fungus first causes pneumonia and later disseminates to involve other organs, for example, the brain, kidney or heart. Patients who develop this disease are usually immunocompromised or debilitated due to prolonged treatment with antibiotics, steroids and cytotoxic drugs.	Invasive Fungal Diseases in Hematological Malignancies: A Critical Review of Evidence and Turkish Expert Opinion " data-hash="10296" data-license="CC BY 2.5" data-source="https://www.researchgate.net/figure/nvasive-Aspergillus-infection-showing-a-nodule-with-a-halo-sign-at-the-periphery-arrows_fig1_270001675" data-tags="" src="https://image.prepladder.com/notes/UxYH3uoM3RToNd0Ftz0a1744602021.png" /> {{caption_text}} Invasive Aspergillus infection showing a nodule with a halo sign at the periphery (arrows).

4. Superficial	Superficial infections of the external ear (otomycosis), the eye (mycotic keratitis) and the nasal sinuses.	 ... (content truncated)
Diagnosis		
Culture	 ... (content truncated)	
Histopathology	 Stain used - Lactophenol cotton blue It demonstrates septate hyphae and acute angled (<45°) / Dichotomous branching. {{caption_text}}	
Others	Detection of galactomannan Tissue sections: biopsy (stained with GMS/PAS)	
Treatment		

Medical	Drug of choice: Voriconazole (Option A) 2nd line drugs: Itraconazole (Option C), Posaconazole Other drugs: Amphotericin B (Option B) / Liposomal amphotericin B	
Surgical resection (Option D)	May be indicated for localized fungal infections, such as aspergillomas or when there are complications such as abscess formation.	



Solution for Question 8:

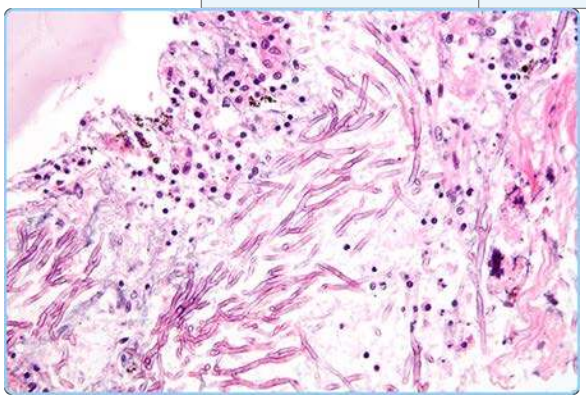
Correct answer: D) Aspergillus

Explanation:

Based on the patient's clinical presentation and her history of cystic fibrosis, she is most likely suffering from aspergillus rhinosinusitis.

Aspergillus rhinosinusitis	
Risk factor	Asthma No mi Cystic fibrosis Immunocompromised state

Clinical features	Facial pain or pressure, particularly in the sinus areas (maxillary or frontal sinuses), gradual onset Nasal congestion and obstruction. Purulent nasal discharge.
Microscopy	Acute angle dichotomous branching hyphae + septate hyphae

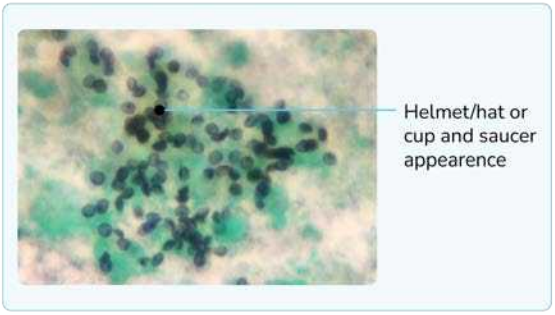


Aspergillus-Acute angled branched septate hyphae

Mucor and Rhizopus (Options A, B): Members of the Zygomycetes group typically group, typically feature broad, non-septate hyphae that branch at 90 degrees or more. They cause mucormycosis, which presents rapidly and aggressively, often with black necrotic tissue and more severe systemic symptoms. It is associated with diabetes mellitus and other immunocompromised states.

Mucor-Ribbon-like hyphae which branch at 90°

Pneumocystis (Option C) causes pneumonia with a dry cough. It is associated with HIV/AIDS (CD4+ <200 cells/μL). On microscopy, it appears like a crushed ping-pong ball (helmet/hat or cup and saucer appearance).



Solution for Question 9:

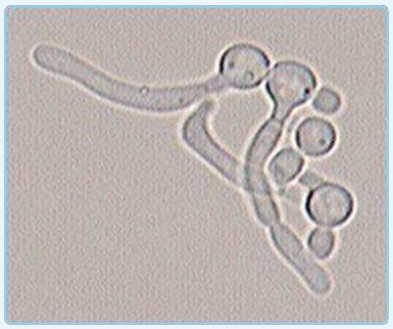
Correct answer: D) Aspergillus

Explanation:

The given image shows:

- Acute angle dichotomous branching hyphae
- Septate hyphae

These are the characteristic findings of Aspergillus.

Morphology of different fungi		
FUNGI	MORPHOLOGY	MICROSCOPY
Aspergillus	Acute angle dichotomous branching hyphae + septate hyphae	dichotomous branching" data-author="" data-hash="10287" data-license="" data-source="" data-tags="" height="317" src="https://image.prepladder.com/notes/6sFVv19Vcm4XzlzCQsPO1744600969.png" width="317" />
Mucor and Rhizopus species (Option A and B)	Broad, aseptate hyphae branching at 90 degrees or more	Rhizopus species" data-author="Yale Rosen" data-hash="10288" data-license="CC BY SA 2.0" data-source="https://commons.wikimedia.org/wiki/File:Zygomycosis,_mucormycosis_2.jpg" data-tags="" height="243" src="https://image.prepladder.com/notes/04nT6wems3zlRw3qFjQw1744601044.png" width="364" /> {{caption_text}}
Candida (Option C)	Single-cell with or without pseudohyphae	 {{caption_text}}

Solution for Question 10:

Correct answer: C) iii, iv and v only

Explanation:

Cryptococcus Spp.	
Cryptococcus is a true yeast. (Statement i)	
Capsule	Polysaccharide capsule - helps in serotyping and protection against phagocytosis . (Statement ii)
Serotypes	A, D, AD serotypes - Cryptococcus variety neoformans (Statement iv) B, C serotypes -Cryptococcus variety gattii (Statement v)
Virulence factors	Polysaccharide Capsule: The primary virulence factor. It helps the pathogen evade phagocytosis and suppresses the host's local immune responses, enabling the organism to survive and spread. Melanin Production: Cryptococcus produces melanin via the enzyme phenol oxidase, which contributes to its virulence by protecting the pathogen from oxidative stress and immune defenses. Enzyme Production: The fungus secretes several enzymes, including: Phospholipase: Aids in tissue invasion and damage. Urease: Helps the fungus survive in the host environment by neutralizing acidic conditions, enhancing its ability to colonize and spread.
Risk of infection	Immunocompromised States: HIV/AIDS, cancer, organ transplant recipients, those on immunosuppressive therapies, extremes of age are at higher risk. Geographic Location: Exposure to environments where Cryptococcus gattii is endemic, such as tropical and subtropical regions Environmental Exposure: Exposure to bird droppings or decaying wood can increase the likelihood of infection, especially with C. neoformans. (Statement iii)

Solution for Question 11:

Correct answer: C) It can grow in hypertonic saline

Explanation:

Feature	Candida albicans	Candida dubliniensis
Germ tube test	Positive	Positive (Option B)
Chlamydospores	Positive	Positive (Option A)
Growth above 45 degrees	Yes	No (Option D)
Growth in hypertonic saline	Yes	No

Solution for Question 12:

Correct answer: A) Topical nystatin

Explanation:

Given the patient's symptoms, risk factors, and the acidic vaginal pH, the presentation is consistent with vaginal candidiasis and the treatment of choice is Topical nystatin

Vaginal candidiasis	
Risk factors	Old age Diabetes mellitus Corticosteroid use Pregnancy Immunocompromised states - HIV Low neutrophil count
Presentation	Intense vaginal itching and irritation Splash dysuria seen Thick, white, curd-like vaginal discharge
Treatment	Topical nystatin - DOC for uncomplicated cases Oral fluconazole (Option B) used for more severe or recurrent infections that do not respond to standard

	treatment.

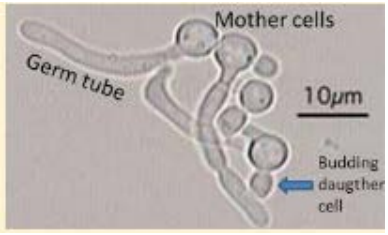
Topical metronidazole (Option C): Metronidazole is an antibiotic primarily used for bacterial infections or protozoal infections(Bacterial vaginosis, Trichomoniasis) and does not treat fungal infections like vaginal candidiasis.

Topical hydrocortisone (Option D): Although hydrocortisone can help reduce inflammation and itching, it does not treat the fungal infection itself.

Solution for Question 13:

Correct answer: A) Germ tube test - Candida albicans

Explanation:

Laboratory diagnosis of Candida		
Germ tube test	The germ tube test is a specific diagnostic test for Candida albicans. Germ tube is also seen in Candida dubliniensis Candida tropicalis- pseudo germ tube formation Also known as Reynolds Braude Phenomenon Candida + Human serum → Incubation → Big tube-like structure (Pseudohyphae).	
Gram stain	Yeast-like organism - Has both buds and pseudohyphae.	stain candida" data-author="Graham Beards" data-hash="10281" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Candida_Gram_stain.jpg" data-tags="" src="https://image.prepladder.com/notes/b5kBTbV6WBLglro1CrCl1744564319.png" /> {{caption_text}}

SDA Culture media test	Colonies are creamy white, smooth and with a yeasty odor	 ... (content truncated)
Corn meal agar test	Forms chlamydospores on corn meal agar cultures at 20°C -identified by double wall configuration.	agar test" data-author="Radoslaw Spiewak" data-hash="10283" data-license="NA" data-source="https://www.researchgate.net/figure/Microscopy-of-Candida-albicans-growing-on-corn-meal-agar-Courtesy-of-Dr-Michael-R_fig9_13426624" data-tags="" height="206" src="https://image.prepladder.com/notes/v9biNGpEOjV5tBktAySh1744564487.png" width="323" /> {{caption_text}}
Chrom agar test	Colour differentiation test Each type of organism is seen differently	agar test" data-author="Mikael Häggström" data-hash="10284" data-license="CC0" data-source="https://commons.wikimedia.org/wiki/File:CHROMAgar_with_N_glabratus,_P_kudriavzevii,_Candida_albicans_and_Candida_tropicalis,_annotated.jpg" data-tags="" height="227" src="https://image.prepladder.com/notes/yQtAIjSopHUVdFy6Zh8Z1744564562.png" width="321" /> {{caption_text}}

Solution for Question 14:

Correct answer: B) Oral candidiasis

Explanation:

The diagnosis is Oral candidiasis/ Moniliasis, indicated by the presence of white patches that can be scraped off and the patient's risk factors, including poorly controlled diabetes.

Oral candidiasis	
Oral candidiasis, commonly known as thrush, is an infection caused by an overgrowth of Candida albicans, a type of yeast.	
Presentation	 Presence of white patches that can be scraped off, leaving behind a red inflamed base {{caption_text}}
Risk factors	Diabetes: Poorly controlled diabetes creates a favorable environment for Candida growth due to elevated blood sugar levels. Neutropenia Immunosuppression: HIV/AIDS, cancer, or immunosuppressive medications increase susceptibility. Use of inhaled corticosteroids Antibiotic Use: Broad-spectrum antibiotics can disrupt normal oral flora, allowing Candida to overgrow. Poor Oral Hygiene: Leads to an increase in fungal colonization. Dry Mouth: Conditions that lead to reduced saliva production can promote candidiasis.
Treatment	Antifungal medications, such as nystatin or fluconazole, along with addressing any underlying risk factors. C. krusei- Resistant to azoles

Oral leukoplakia (Option A) presents as white patches that cannot be scraped off and may indicate precancerous changes.

Lichen planus (Option C) presents with white, lacy patches and ulcerations in the mouth. The lesions are typically painful and not easily scraped off.

Herpes simplex stomatitis (Option D) caused by HSV, presents as gingivostomatitis or pharyngitis, with painful vesicles on the buccal mucosa. Recurrent infections often lead to herpes labialis. Other lesions can include ulcerative stomatitis, tonsillitis, and eyelid vesicles. Many cases are asymptomatic but may predispose to secondary bacterial infections.

Solution for Question 15:

Correct answer: D) It primarily affects immunocompromised individuals

Explanation:

Cryptococcus gattii is known for causing infections primarily in immunocompetent individuals, which distinguishes it from Cryptococcus neoformans, which mainly affects immunocompromised patients.

- Commonly found in tropical and subtropical regions, particularly around eucalyptus trees (Options A and C)
- It corresponds to the serotypes B and C (Option B)
- C. gattii can cause severe respiratory illnesses and can disseminate to the CNS, leading to cryptococcal meningitis.
- Diagnosis is typically made through microscopy, culture, serological tests for cryptococcal antigen, and imaging studies for lung lesions.
- Treatment usually involves antifungal medications, such as amphotericin B and fluconazole, depending on the severity of the infection.

Solution for Question 16:

Correct answer: C) Pulmonary cryptococcosis

Explanation:

Pulmonary cryptococcosis is the most likely diagnosis due to the patient's immunocompromised state and the solitary pulmonary nodule on imaging.

Clinical features of cryptococcosis	

Respiratory system	Cryptococcus typically enters the body via inhalation and can cause pulmonary infection/pneumonitis - primary site of infection Patients may present with cough, pleuritic chest pain, and respiratory distress. Imaging shows a solitary pulmonary nodule Dissemination of pulmonary infection leads to involvement of other organs
Central Nervous System	Meningitis: Characterized by severe headaches, fever, altered mental status, sensory and memory loss, cranial nerve paresis and loss of vision It is the most serious type of cryptococcal infection
Skin	Cutaneous Cryptococcosis: Causes skin lesions, particularly in immunocompromised patients. Associated with C.neoformans var neoformans (serotype D)
Other Organs	Liver, spleen, and bones may be involved in disseminated infections

Bacterial pneumonia (Option A) typically presents with acute symptoms like high fever and productive cough, and Imaging usually shows diffuse infiltrates or lobar consolidation, not a solitary nodule

Pulmonary tuberculosis (Option B) often involves systemic symptoms (weight loss, night sweats) and typically presents as multiple nodules or cavitary lesions on imaging

Lung abscess (Option D) presents with localized symptoms and is associated with bacterial infections. Imaging typically reveals a cavity filled with pus, not a solitary nodule.

Solution for Question 17:

Correct answer: B) India ink stain

Explanation:

The given scenario describes meningitis and India ink stain is used to visualize the organism in this scenario.

- The India ink stain (A type of negative stain) is specifically used to visualize the capsule surrounding Cryptococcus neoformans in CSF samples.

• A clear halo around the yeast cells indicates the presence of the capsule, which is an important virulence factor.

• Urease test positive

• Maltese cross appearance under polarising microscope

Diagnosis of Cryptococcosis	
Microscopy	India ink- Demonstrates the capsule, which appears as clear space surrounding the round budding yeast cells Mucicarmine stain: It stains the carminophilic cell wall of C. neoformans
Culture	 ... (content truncated)
Antigen detection tests	Demonstration of the capsular antigen by Precipitation and Latex agglutination test
Treatment	Initial Treatment with liposomal Amphotericin B and Flucytosine Maintenance Therapy with Fluconazole for 6-12 months, depending on immune status. Monitoring: Regularly check CSF for fungal clearance, renal function, and electrolytes. Management of Underlying Conditions: Optimize ART to reduce recurrence risk.

Gram stain (Option A) is used to classify bacteria into Gram-positive or Gram-negative, and it does not visualize the capsule of Cryptococcus.

Acid-fast stain (Option C) is used primarily to identify Mycobacterium species, such as Mycobacterium tuberculosis, but it does not detect Cryptococcus. Acid-fast bacteria have mycolic acid in their cell wall, which can be stained with acid-fast / Ziehl Nielsen stain.

Giemsa stain (Option D) contains both acidic and basic dyes, which helps identify certain blood parasites and cellular components.

• Eosin is an acidic dye that stains the cytosol and cytosolic granules red

• Methylene blue stain that stains the nucleus, cytosolic RNA and basophilic granules blue

• Azure dye stains the acidic components of the cell,s such as phosphate groups of DNA

Previous Year Questions

1. A patient diagnosed with Cryptococcal meningitis. What is the treatment?

A. Flucytosine

B. Fluconazole

C. L.Amp

D. Flucytosine + L.Amp

---

2. An AIDS-positive patient came with a history of fever, vomiting, and meningismus. Which of the following tests help in the rapid diagnosis of cryptococcal meningitis?

A. India ink preparation of CSF

B. Blood culture

C. CSF culture

D. Urine lateral flow assay for cryptococcal antigen

---

3. In patients with diabetic ketoacidosis, which type of fungus is frequently linked to orbital cellulitis?

A. Candida

B. Rhizopus

C. Aspergillus

D. Trichophyton

---

4. Patient recovered from COVID-19, given steroids, 3 weeks complaints of nasal discharge, nasal image of fungus?



A. Mucor

B. Aspergillus

C. Candida

D. Trichosporon

---

5. A 30-year-old patient presents with a skin infection characterized by itching, redness, and scaling. Microscopic examination of a skin sample reveals a pencil-shaped appearance. Which of the following is the most likely causative organism responsible for this skin infection?

A. Trichophyton

B. Microsporum

C. Epidermophyton

D. None of the above

---

6. A 46-year-old farmer present with a traumatic ulcer in the eye. Microscopic examination reveals septate hyphae at an acute angle. Which of the following will be the causative organism?

A. Mucor

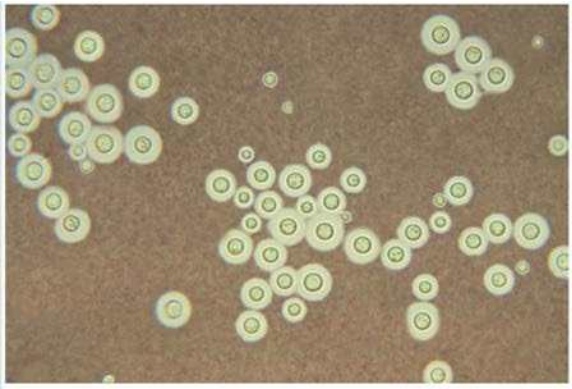
B. Aspergillosis

C. Candida

D. Rhizopus

---

7. Select the correct statement regarding the organism shown in the image:



A. Humans are the only reservoir

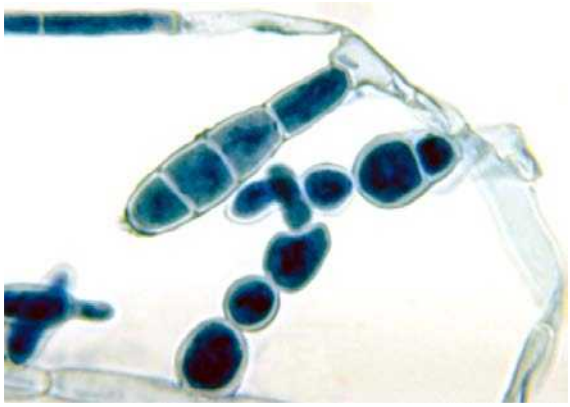
B. Patient is at high risk with CD4<900

C. Lung is the primary site of infection

D. Pseudohyphae and budding is seen

---

8. Identify the given organism?



A. Microsporum Gypseum

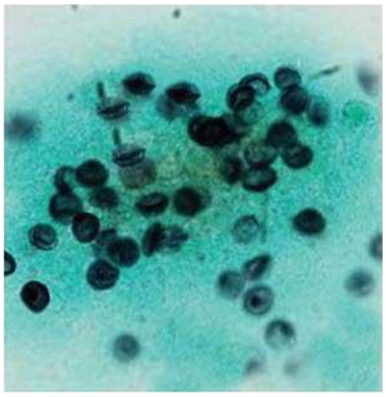
B. Microsporum audouinii

C. Epidermophyton Floccosum

D. Trichophyton

---

9. A patient with pneumonia provides a BAL sample, as shown below. What is the likely pathogen responsible, and what is the relative management?



A. Mucormycosis, Terbinafin

B. Aspergillus, Amphotericin B

C. Pneumocystis Carinii, Cotrimoxazole

D. Candida,Itraconazole

10. Which disease is commonly referred to as "Desert Rheumatism"?

A. Coccidomycosis

B. Histoplasmosis

C. Paracoccidioidomycosis

D. Mucormycosis

11. Which organism is associated with broad-based budding?

A. Sporothrix schenckii

B. Blastomyces

C. Talaromyces marneffei

D. Paracoccidioidomycosis

12. A central line infection is noted in an ICU patient, and staining shows a gram-positive oval-shaped organism that reproduces by budding. Identify the organism:

A. Staphylococcus aureus

B. Candida

C. Staphylococcus epidermidis

D. Escherichia coli

13. A 39-year-old patient with HIV and a CD4 count of 139 cells/μL presents with altered sensorium and impaired consciousness. CSF examination using an India ink preparation reveals a positive result. Which organism is most likely responsible?

A. Cryptococcus neoformans

B. Aspergillus fumigatus

C. Histoplasma capsulatum

D. Candida albicans

---

14. Which of the following microorganisms is likely responsible for causing bilateral infiltrates in an HIV-positive patient?

A. Cryptococcus

B. Histoplasmosis

C. Pneumocystis Jirovecii

D. Aspergillus

---

15. A person came with the H/o thorn prick a week ago. A few days later, he developed ulcers along lymphatic drainage. Choose the correct organism.

A. Coccidioides immitis

B. Aspergillus flavus

C. Sporothrix schenckii

D. Trichophyton rubrum

---

16. Which among the following is a subcutaneous fungal agent ?

A. Cryptococcus neoformans

B. Histoplasma capsulatum

C. Sporothrix schenckii

D. Talaromyces marneffei

---

17. Which of the following is a subcutaneous fungal agent?

A. Cryptococcus neoformans

B. Histoplasma capsulatum

C. Sporothrix schenckii

D. Talaromyces marneffei

---

18. Which of the following exists in yeast form at 37ºC and hyphal form at 25ºC?

A. Malassezia furfur

B. Cryptococcus neoformans

C. Aspergillus

D. Histoplasma capsulatum

---

19. Which of the following conditions cannot be diagnosed using the β 1-3 glucan assay?

A. Candidiasis

B. Aspergillosis

C. Pneumocystis pneumonia

D. Mucormycosis

20. A truck driver with diabetes visits you complaining of cough and difficulty breathing. Pneumonia has been diagnosed, and histopathological examination indicates the presence of nonseptate hyphae branching at wide angles, as demonstrated below. What is the probable cause of this condition?



A. Cryptococcus neoformans

B. Mucor

C. Candida

D. Aspergillus

21. A COVID patient presented with fever, nasal congestion, and unilateral headache. Later, he developed facial pain and numbness, along with blackish lesion. On SDA, cotton wooly growth is seen. H&E staining shows non-septate hyphae at the right angle/obtuse angle. What is the most probable diagnosis?

A. Aspergillosis

B. Bacterial orbital cellulitis

C. Mucormycosis

D. Cavernous sinus thrombosis

22. Which of the following is the probable cause in a case of tinea capitis where white powdery colonies were obtained from the hair specimen and the hair perforation test was positive on the isolate?

A. Trichophyton rubrum

B. Trichophyton schoenleinii

C. Trichophyton mentagrophytes

D. Trichophyton violaceum

23. Which of the subsequent microorganisms leads to illness after being introduced into the skin through trauma?

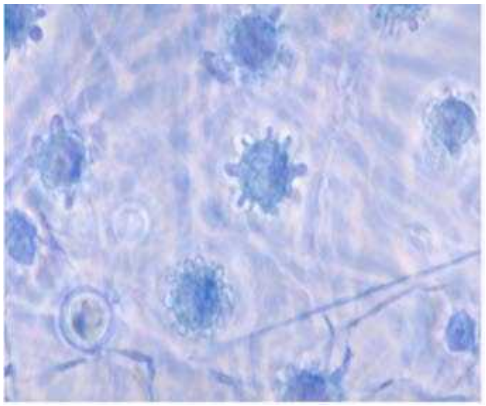
A. Sporothrix schenckii

B. Blastomyces dermatitidis

C. Coccidioides immitis

D. Paracoccidioides brasiliensis

24. A 54-year-old farmer from Ohio presents to a physician with a chronic cough. A chest x-ray demonstrates a cavitary lesion with hilar lymphadenopathy. Biopsy of the lymph node and staining with lactophenol cotton blue demonstrates the following characteristics as shown in the image. Which of the following is the most likely diagnosis?



A. Histoplasmosis

B. Coccidioidomycosis

C. Cryptococcus

D. Blastomycosis

---

25. A chronic asthmatic patient who is on inhaled steroids presents with a history of burning sensation on eating hot or spicy food, and the following findings are seen. What is the most possible diagnosis?



A. Candidiasis

B. Burning mouth syndrome

C. Geographical tongue

D. Oral lichen planus

26. A patient presents with a mobile, warm-to-touch nodule in the neck. The anaerobic culture of the biopsy material from this swelling showed colonies, as shown in the image, which had gram-positive branching rods on staining. Which of the following is the drug of choice in this condition?



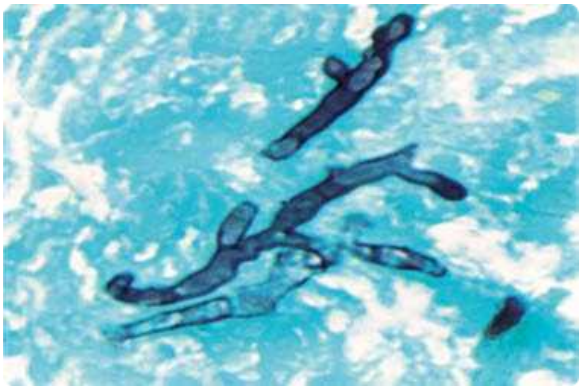
A. Doxycycline

B. Cotrimoxazole

C. Penicillin

D. Surgical drainage

27. The below slide represents a gomori methenamine silver stained slide with acute angle branching and septate hyphae. What is the organism?



A. Blastomycosis

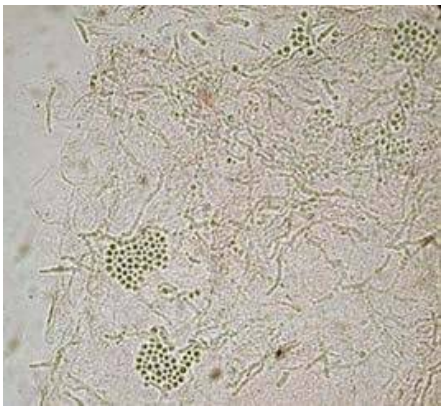
B. Candida

C. Aspergillus

D. Mucor

---

28. A 24-year-old woman presents with hypopigmented macules and confluent patches over the chest. Skin scraping and KOH preparation show an abundance of yeast-like cells and short branch filaments like a 'Spaghetti and meatball appearance' as shown in the image below. What is the most likely organism?



A. Malassezia

B. Trichophyton

C. Chlamydospores

D. Candida

29. Which of the following medium can be utilized to determine the probable species of the causative agent in a female patient presenting with symptoms of thick curdy white vaginal secretions and microscopic examination of vaginal scraping shows budding yeast and pseudohyphae?

A. Birds seed agar

B. Brain heart infusion agar

C. CHROM agar

D. SDA

30. A patient with cervical lymphadenopathy and is found to be retropositive. A fungal culture depicts a velvety growth with red diffusible pigment on the underside, as shown below. Which is the most likely causative organism?



A. Talaromyces marneffei

B. Blastomyces

C. Aspergillus

D. Pneumocystis jirovecii

31. A farmer presents with a swollen foot and multiple sinuses as shown in the below image. What is the most likely diagnosis?



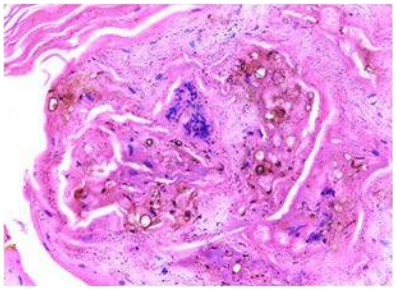
A. Mycetoma

B. Squamous cell carcinoma

C. Nocardiosis

D. Sporotrichosis

32. What is the most probable diagnosis for a farmer who has a wart-like skin growth on his foot and histopathology of the growth is shown in the below image?



A. Sporotrichosis

B. Mycetoma

C. Chromoblastomycosis

D. Phaeohyphomycosis

---

33. Which of the following dimorphic fungi causes subcutaneous mycosis?

A. Sporothrix schenckii

B. Histoplasma capsulatum

C. Blastomyces dermatitidis

D. Penicllium marneffei

---

34. A patient presented with complaints of persistent dry cough, fever, and weight loss. He was diagnosed as HIV positive 6 months back and is poorly adherent to his medications. Gomori methamine silver staining of BAL reveals a crushed ping pong ball appearance. What is the most likely causative organism?

A. Paracoccidiodes brasiliensis

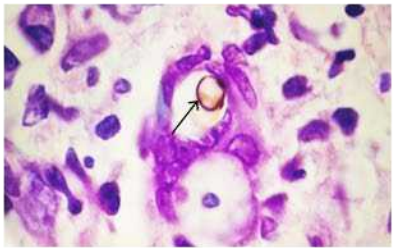
B. Histoplasma capsulatum

C. Coccidiodes immitis

D. Pneumocystis jirovecii

---

35. Identify the pathognomonic structure in the slide given below.



A. Sclerotic body

B. Budding yeast

C. Negri body

D. Asteroid body

36. A forest worker developed skin lesions over the forearm, which initially started as macules but then became nodules. Histology of the nodule shows the following findings. Which of the following is true regarding this condition?



A. Infection does not involve underlying fascia, tendons and muscle

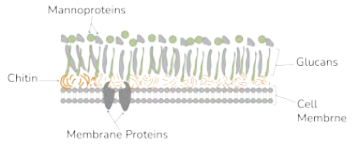
B. Infection can spread and involve underlying subcutaneous tissue, tendon, muscle or joints

C. Angioinvasion can occur in patients of hematological malignancies

D. These bodies are fungi engulfed by macrophages

---

37. The cell wall depicted below is most likely to belong to which of the following organisms?



A. Aspergillus fumigatus

B. Mycobacterium tuberculosis

C. Mycoplasma pneumoniae

D. Rhinosporidum seeberi

---

38. A man has undergone renal transplant and is taking immunosuppressant drug. On biopsy there was presence of budding cells with pseudohyphae. Identify the organism?

A. Candida

B. Aspergillus

C. Pneumocystis jirovecii

D. Histoplasma capsulatam

---

39. A 35-year-old HIV-positive patient presents with mucosal lesions in the mouth as shown in the image below. On microscopy, budding yeasts and pseudohyphae are seen. What is the likely diagnosis?



A. Oral candidiasis

B. Hairy leukoplakia

C. Lichen planus

D. Diphtheria

---

40. Which medication is preferred for treating invasive pulmonary aspergillosis?

A. Itraconazole

B. Voriconazole

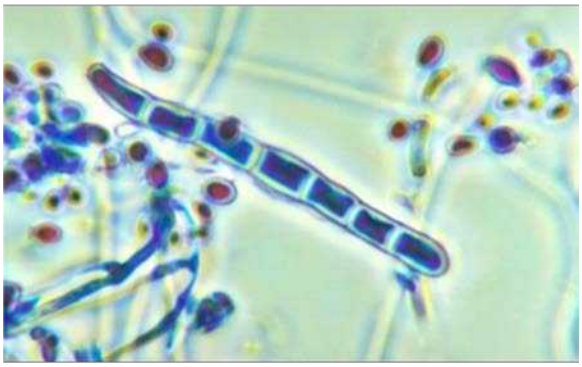
C. Lyophylised Amphotericin B

D. Posaconazole

---

41. A patient presents with itching in the axilla. On microscopic examination of the skin scrapings, a red pigment-producing fungus with pencil–shaped macroconidia

is seen. Which of the following is the most likely organism?



A. Trichophyton violaceum

B. Trichophyton rubrum

C. Trichophyton mentagrophytes

D. Trichophyton schoenleinii

---

42. What is the primary risk factor for the disease in a patient who exhibits symptoms of a black, malodorous nasal discharge, along with the presence of a black, necrotic mass that fills the nasal cavity and causes erosion of the septum and hard palate during examination?

A. Hypothyroidism

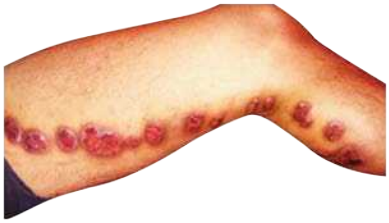
B. Diabetes

C. Hypertension

D. Foreign body

---

43. A man presents with the following lesion after traumatic inoculation. Identify the causative organism.



A. Mycobacterium scrofulaceum

B. Sporothrix schenckii

C. Madurella mycetomatis

D. Phialophora verrucosa

---

44. Which one of these options is classified as a dematiaceous fungus?

A. Mucor species

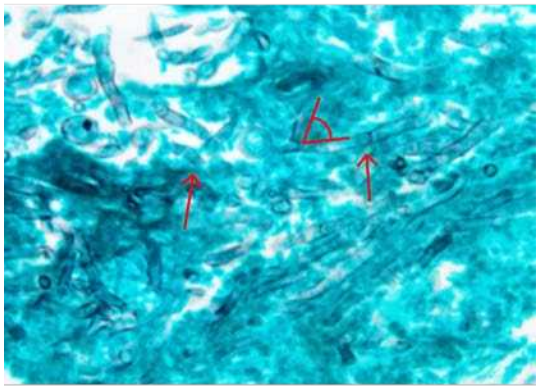
B. Aspergillus niger

C. Cladosporium

D. Blastomyces dermatitidis

---

45. A patient presented with fever, dyspnea, palpitations, and tachycardia. The echocardiography revealed vegetation in the septal leaflets. Tissue sections from them reveal the following findings as shown in the image. What is the most likely implicated agent?



A. Pigmented septate hyphae at 90 degrees, Mucormycosis

B. Right angle branching septate hyphae, Fusarium

C. Acute angle branching hyphae, Rhizopus

D. Acute angle branching septate hyphae, Aspergillus

---

46. What is the stain used for identifying Cryptococcus?

A. Giemsa stain

B. Ziehl-Nielsen stain

C. Gram staining

D. Mucicarmine

---

47. Identify the dermatophyte based on the macroconidia shown in the image below



A. Trichophyton

B. Microsporum

C. Epidermophyton

D. None of the above

---

48. Galactomannan in serum is seen in : Invasive aspergillosis Systemic candidiasis Mucormycosis Histoplasma capsulatm

A. 1 & 2 only

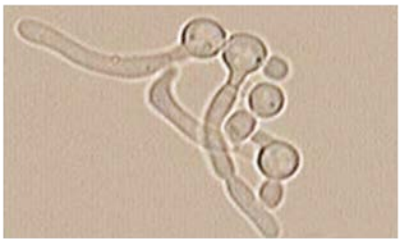
B. 1 & 3 only

C. 1 only

D. 1, 2 & 3 only

---

49. The following test is positive for identification of ……..



A. Candida albicans

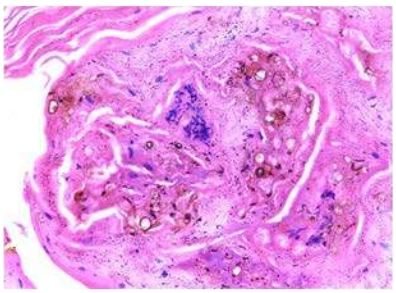
B. Candida glabrata

C. Candida parapsilosis

D. Candida tropicalis

---

50. A patient presented with wart-like lesions on exposed areas of the skin, such as the legs or forearms. Histopathology of the wart revealed findings as shown in the image. What is the possible diagnosis?



A. Histoplasmosis

B. Chromoblastomycosis

C. Coccidioidomycosis

D. Candida albicans

51. Stain used to demonstrate fungal elements are?

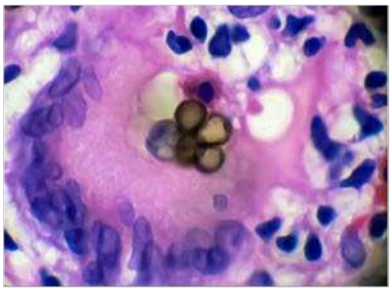
A. Acid fast stain

B. Gabbet's stain

C. Methenamine silver

D. Leifson & Ryu's stain

52. A skin biopsy sample was taken and sent for histopathological examination. The characteristic microscopic body, as shown in the image, was seen. Which of the following could be the most likely diagnosis?



A. Blastomycosis

B. Pheohyphomycosis

C. Chromoblastomycosis

D. Sporotrichosis

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	2
Question 4	1
Question 5	1
Question 6	2
Question 7	3
Question 8	3
Question 9	3
Question 10	1
Question 11	2
Question 12	2
Question 13	1
Question 14	3
Question 15	3
Question 16	3
Question 17	3
Question 18	4
Question 19	4
Question 20	2
Question 21	3
Question 22	3

Question 23	1
Question 24	1
Question 25	1
Question 26	3
Question 27	3
Question 28	1
Question 29	3
Question 30	1
Question 31	1
Question 32	3
Question 33	1
Question 34	4
Question 35	1
Question 36	1
Question 37	1
Question 38	1
Question 39	1
Question 40	2
Question 41	2
Question 42	2
Question 43	2
Question 44	3
Question 45	4
Question 46	4
Question 47	3
Question 48	3

Question 49	1
Question 50	2
Question 51	3
Question 52	3

Solution for Question 1:

Correct Answer: D) Flucytosine + L.Amp

Explanation: The treatment for Cryptococcal meningitis is Flucytosine + Liposomal Amphotericin B (L.Amp).

Fungal Meningitis:

Causes	Cryptococcus neoformans (most common) Candida Histoplasma Blastomyces Coccidioides Sporothrix
Risk factors	Immunocompromised patients
Clinical features	Headache Altered mental status, lethargy Fever Neck stiffness Nausea and vomiting.
Investigations	Lumbar Puncture: Low glucose and high protein levels. White cell count can be normal or higher than 20 microL and have a lymphocyte predominance. India ink stain shows encapsulated yeast cells. CSF ELISA for CRAG (Cryptococcal Antigen).
Treatment	LAMB (Liposomal Amphotericin-B) and 5-Fluorocytosine (Flucytosine) for 2 weeks Fluconazole is given subsequently for 8-10 weeks.

Solution for Question 2:

Correct Option D - Urine lateral flow assay for cryptococcal antigen:

- Urine lateral flow assay for cryptococcal antigen is a rapid diagnostic test for cryptococcal meningitis. It detects the presence of cryptococcal antigen in the blood, providing results within 20 minutes. The test is based on an immunochromatographic assay, where a positive result indicates the presence of cryptococcal infection. This test is highly recommended for rapid diagnosis, especially in resource-limited settings.
- Other rapid diagnostic tests are:

Sample	Antigen	Test	Sensitivity, %	Specificity, %
Serum	GXM	LA, EIA	90	95-100
		LFA	87-91	87-100
CSF		LA, EIA	97	86-100
		LFA	99	99
Urine (Rapid, 20-minute test)		LFA	70-80	92
GXM, Capsular glucuronoxylomannan; EIA, enzyme immunoassay; LA, latex agglutination; LFA, Lateral flow assay.				

Incorrect Options:

Option A, B, and C are incorrect. These are not the rapid diagnostic tests for demonstrating Cryptococcus.

Solution for Question 3:

Correct Option B - Rhizopus:

• Rhizopus is a genus of fungi, and the species most commonly associated with orbital cellulitis in patients with diabetic ketoacidosis is Rhizopus.

• This fungal infection is known as rhinocerebral mucormycosis.

• Rhizopus can invade the sinuses and orbit, leading to orbital cellulitis. This infection can spread rapidly and is considered a medical emergency.

• Predisposing factors for mucormycosis: Iron is essential for mucormycosis growth. High iron levels increase the risk of invasive mucormycosis. Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum. End-stage renal disease. Patients taking iron therapy or deferoxamine (iron chelator) Defects in phagocytic functions (e.g., neutropenia or steroid therapy).

• Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum.

• End-stage renal disease.

• Patients taking iron therapy or deferoxamine (iron chelator)

• Defects in phagocytic functions (e.g., neutropenia or steroid therapy).

• Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum.

• End-stage renal disease.

• Patients taking iron therapy or deferoxamine (iron chelator)

• Defects in phagocytic functions (e.g., neutropenia or steroid therapy).

Incorrect Options:

Option A - Candida: Candida is a genus of yeast that can cause oral thrush, vulvovaginitis, esophagitis, endocarditis, and UTI in individuals with weakened immune systems (HIV)

Option C - Aspergillus: Aspergillus is a genus of filamentous fungi commonly found in the environment. It can cause invasive fungal infections in immunocompromised individuals, but it is more commonly associated with lung infections such as invasive pulmonary aspergillosis.

Option D - Trichophyton: Trichophyton is a genus of fungi that primarily causes infections of the skin, hair, and nails. It is not commonly associated with orbital cellulitis or invasive fungal infections in the orbit in patients with diabetic ketoacidosis.

Solution for Question 4:

Correct Option A- Mucor:

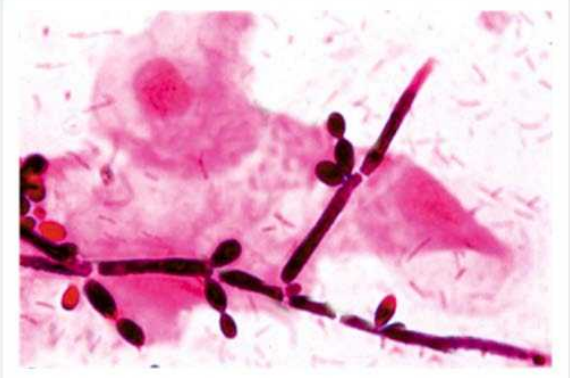
- In this clinical context, both COVID-19 and the administration of immunosuppressive agents, particularly steroids, can weaken the immune system.
- The accompanying image depicts fungal characteristics, including hyphae with aseptate morphology and branching angles exceeding or equal to 90 degrees. Additionally, the hyphae exhibit a broad ribbon-like shape. This is suggestive of the presence of mucor.

Incorrect Options:

Option B- Aspergillus: Aspergillus typically has septate hyphae with acute-angle branching. The branching angles are not broad, and the hyphae are divided by septa.



Option C- Candida: Candida is a yeast that may form pseudohyphae and true hyphae, but it does not typically present with aseptate hyphae. Candida hyphae usually have septa.



Option D- Trichosporon: Trichosporon species may show septate hyphae. However, the characteristic branching angles of Trichosporon hyphae are not typically broad, as seen in Mucor.

Solution for Question 5:

Correct Option A - Trichophyton

- Trichophyton species are dermatophytic fungi that commonly cause fungal skin infections known as dermatophytoses or ringworm. These infections are characterized by itching, redness, and scaling of the skin. The term "pencil-shaped appearance" refers to the appearance of the fungal hyphae under the microscope, where the hyphae have a tapered, narrow appearance resembling a pencil. Trichophyton species can cause various types of dermatophytoses, including tinea corporis (ringworm of the body), tinea cruris (jock itch), tinea pedis (athlete's foot), and others.

Incorrect Options:

Option B - Microsporum: Microsporum species are another group of dermatophytic fungi that can cause fungal skin infections. They also lead to symptoms like itching, redness, and scaling. However, unlike the "pencil-shaped appearance" mentioned in the question, Microsporum species are known for their characteristic appearance under the microscope, where they produce large, multicellular, club-shaped structures called macroconidia. Therefore, this is not the correct answer in this case.

Option C - Epidermophyton: Epidermophyton species are also dermatophytic fungi responsible for causing fungal skin infections. They can cause similar symptoms as described in the question, but the "pencil-shaped appearance" is not typically associated with Epidermophyton species. Instead, they produce smooth, thin-walled, and irregularly branched hyphae. Therefore, this is not the correct answer in this case.

Option D - None of the above: This option is not the correct answer because the presentation of itching, redness, scaling, and the specific "pencil-shaped appearance" of hyphae in the skin sample is indicative of a dermatophytic fungal infection, and the most likely causative organism among the given options is Trichophyton.

Solution for Question 6:

Correct Option B - Aspergillosis

• Aspergillus is a fungus that can cause various infections collectively referred to as aspergillosis. The hyphae of Aspergillus are septate, meaning they have cross-walls that divide the hyphae into distinct segments. These septate hyphae often branch at acute angles, forming a characteristic "V" or "Y" shape. This description matches the findings in the question.

Incorrect Options:

Option A - Mucor: Mucor is a type of fungus that can cause mucormycosis, also known as zygomycosis. The hyphae of Mucor are wide, ribbon-like, and non-septate (aseptate). In other words, they lack cross-walls or septa.

Option C -Candida: Candida is a yeast that can cause candidiasis. Yeasts typically do not form septate hyphae. Instead, they form pseudohyphae, which are chains of elongated yeast cells. These pseudohyphae do not show the acute-angle branching seen in the question's description.

Option D - Rhizopus: Rhizopus is another type of fungus that can cause mucormycosis. Similar to Mucor, the hyphae of Rhizopus are wide, ribbon-like, and aseptate. They lack the cross-walls seen in septate hyphae.

Solution for Question 7:

Correct Option C - Lung is the primary site of infection:

• The above image shows capsulated, budding yeast cells in the India-ink stain which is characteristic of Cryptococcus neoformans.

• It is a fungus abundant particularly in the feces of pigeons and other bird droppings.

• The infection is usually acquired by inhalation leading to mild pneumonitis, but it affects the brain more causing meningitis.

Incorrect Options:

Option A - Humans are the only reservoir:

• Cryptococcus is found in the feces of pigeons and bird droppings. Hence this statement is false.

Option B - Patient is at high risk with CD4<900:

• Patients with a CD4 <200 are at the risk of infection.

Option D - Pseudohyphae and budding is seen:

- Pseudohyphae and budding is seen in candida species.

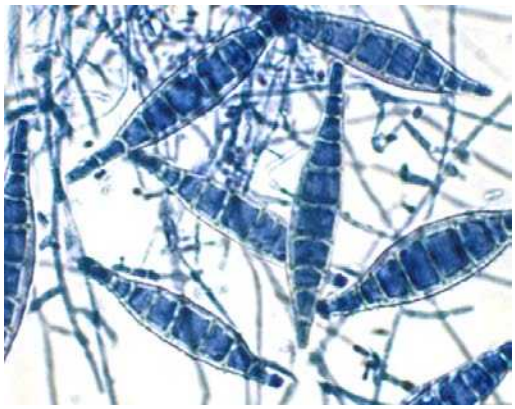
Solution for Question 8:

Correct Option C - Epidermophyton Floccosum:

- These are club shaped and represent the Macroconidia of Epidermophyton

Incorrect Options:

Option A, B:



- The above image represents the Macroconidia representing Microsporum (Lesser Microconidia)
- These are spindle shaped

Option D - Trichophyton:



• These are pencil shaped

Solution for Question 9:

Correct Option C - Pneumocystis Carinii, Cotrimoxazole:

• The above image shows Helmet/Hat or Cup and Saucer appearance of Pneumocystis Carinii

• It causes Pneumocystis Carinii Pneumonia (PCP).

• Appearance -Crushed Ping Pong ball (PCP). (Also called Helmet/Hat or cup and saucer appearance)

• Clinical features - Pneumonia features, Dry cough.

• It is treated with Cotrimoxazole

Incorrect Options:

Options A, B, D:

• The above image does not indicate these organisms

Solution for Question 10:

Correct Option A - Coccidomycosis:

- Coccidioidomycosis, also known as "Valley fever" or "Desert rheumatism," is a fungal infection caused by Coccidioides immitis. It is commonly found in desert regions and can cause joint pain, hence the name "Desert Rheumatism."

Incorrect Options:

- Options B, C, and D are not referred to as Desert Rheumatism.

Solution for Question 11:

Correct Option B - Blastomyces:

- Blastomyces dermatitidis is a dimorphic fungus that shows broad-based budding.

Incorrect Options:

Option A - Sporothrix schenckii: Sporothrix schenckii appears as rosettes of conidia on microscopy.

Option C - Talaromyces marneffei: Talaromyces marneffei appear as brush/broom conidia on microscopy.

Option D - Paracoccidioidomycosis: Paracoccidioidomycosis is characterized by a "pilot's wheel" or "Mickey Mouse" appearance of the yeast form.

Solution for Question 12:

Correct Option B - Candida:

- Candida species are yeast-like fungi that reproduce by budding and can cause central line infections, especially in immunocompromised patients.

Incorrect Options:

Option A - Staphylococcus aureus: Staphylococcus aureus are gram-positive cocci in clusters, not oval-shaped.

Option C - Staphylococcus epidermidis: Staphylococcus epidermidis are gram-positive cocci in clusters, not oval-shaped.

Option D - Escherichia coli: Escherichia coli are gram-negative rods.

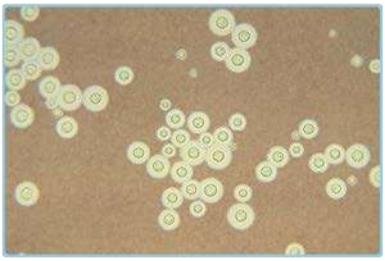
Solution for Question 13:

Correct Answer: A) Cryptococcus neoformans

Explanation: A patient with HIV and a CD4 count of 139 cells/µL, altered sensorium, and a positive India ink preparation of CSF points toward Cryptococcus neoformans, a common cause of opportunistic meningitis in immunocompromised individuals.

Clinical Features of Cryptococcosis	
Respiratory System	Cryptococcus typically enters the body via inhalation and can cause pulmonary infection / pneumonitis - primary site of infection Patients may present with cough, pleuritic chest pain, and respiratory distress. Imaging shows a solitary pulmonary nodule Dissemination of pulmonary infection leads to the involvement of other organs
Central Nervous System	Meningitis: Characterized by severe headaches, fever, altered mental status, sensory and memory loss, cranial nerve paresis and loss of vision It is the most serious type of cryptococcal infection
Skin	Cutaneous Cryptococcosis: Causes skin lesions, particularly in immunocompromised patients. Associated with C.neoformans var neoformans (serotype D)
Other Organs	Liver, spleen, and bones maybe involved in disseminated infections

Diagnosis of Cryptococcosis	

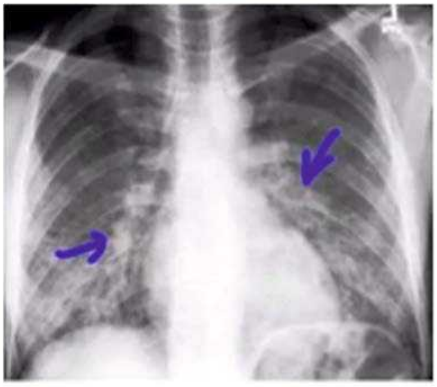
Microscopy	 India ink: Demonstrates the capsule, which appears as clear space surrounding the round budding yeast cells Mucicarmine stain: It stains the carminophilic cell wall of C. neoformans
Culture	 Sabouraud Dextrose Agar, forming smooth, mucoid, cream-coloured colonies. ... (content truncated)
Antigen detection tests	Demonstration of the capsular antigen by Precipitation and Latex agglutination test
Treatment	Initial Treatment with liposomal Amphotericin B and Flucytosine Maintenance Therapy with Fluconazole for 6-12 months, depending on immune status. Monitoring: Regularly check CSF for fungal clearance, renal function, and electrolytes. Management of Underlying Conditions: Optimize ART to reduce recurrence risk.

Solution for Question 14:

Correct Option C - Pneumocystis Jiroveci infection:

• In AIDS-positive patients with CD count less than 200 with perihilar infiltrates, nonproductive 4 cough (7-10 days):Pneumocystis Jiroveci

• In the later part of the illness,Chest X ray shows bilateral perihilar diffuse infiltrates



Incorrect Options:

Option A, B, D:

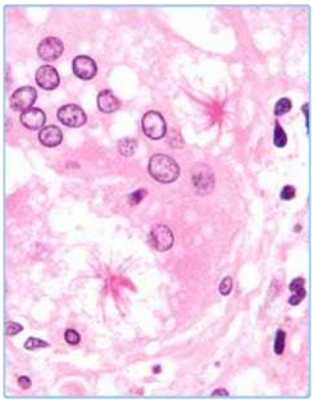
• Cryptococcus, Histoplasmosis and Aspergillus infection are not associated with bilateral peri hilar infiltrates

Solution for Question 15:

Correct Answer: C) Sporothrix schenckii

Explanation: The patient's history of a thorn prick followed by ulcerative lesions along lymphatic drainage suggests sporotrichosis, a subcutaneous fungal infection caused by Sporothrix schenckii.

Sporotrichosis	
Causative agent	Dimorphic fungus Sporothrix schenckii.
Epidemiology	Common in gardeners, forest workers, and manual labourers.
Route of Transmission	Minor trauma or thorn prick.

Clinical features	 Chronic infection affects cutaneous, subcutaneous, and lymphatic tissue. Lesions form at sites of minor trauma or thorn pricks, starting as nodules. ... (content truncated)
Specimens	Aspiration from nodules and biopsy material.
Diagnosis	 ... (content truncated)
Treatment	Medications: Itraconazole Terbinafine Amphotericin B: For disseminated infection Cryotherapy.

Coccidioides immitis (Option A) causes valley fever (coccidioidomycosis), primarily a respiratory illness, not associated with lymphatic ulcer spread.

Aspergillus flavus (Option B) causes invasive aspergillosis, primarily affecting immunocompromised individuals. It does not cause lymphatic ulcers following trauma. Instead, it is associated with lung infections and aflatoxin production (hepatotoxicity).

Trichophyton rubrum (Option D) is a dermatophyte that causes superficial fungal infections such as tinea, onychomycosis, and athlete's foot, not deep tissue involvement with lymphatic spread.

Solution for Question 16:

Correct Option C - Sporothrix Schenckii:

- Sporothrix schenckii is a dimorphic fungus that causes sporotrichosis, a subcutaneous fungal infection. Sporotrichosis typically presents as nodules or ulcers along the lymphatic channels and is commonly acquired through traumatic inoculation of the fungus, such as by thorns or splinters.

Incorrect Options:

Option A - Cryptococcus Neoformans: Cryptococcus neoformans is a fungal pathogen primarily associated with respiratory and central nervous system infections in immunocompromised individuals. It is classified as an opportunistic infection.

Option B - Histoplasma Capsulatum: It is a dimorphic fungus that causes histoplasmosis. Histoplasmosis is an intracellular infection of the reticuloendothelial system. It is classified under systemic mycoses.

Option D - Talaromyces Marneffei: It is a dimorphic fungus that causes opportunistic infections in an individual infected with AIDS. It causes a disseminated infection with multiple organ involvement. It is classified as an opportunistic infection.

Solution for Question 17:

Correct Option C - Sporothrix Schenckii:

- Sporothrix schenckii is a dimorphic fungus that causes sporotrichosis, a subcutaneous fungal infection. Sporotrichosis typically presents as nodules or ulcers along the lymphatic channels and is commonly acquired through traumatic inoculation of the fungus, such as by thorns or splinters.

Incorrect Options:

Option A - Cryptococcus Neoformans: Cryptococcus neoformans is a fungal pathogen primarily associated with respiratory and central nervous system infections in immunocompromised

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Option D - Talaromyces Marneffei: It is a dimorphic fungus that causes opportunistic infections in an individual infected with AIDS. It causes a disseminated infection with multiple organ involvement. It is classified as an opportunistic infection.

Solution for Question 18:

Correct Option D - Histoplasma Capsulatum:

- Dimorphic fungi exist as molds in the environment at 25°C and as yeasts in human tissues at 37°C.
- Several medically important fungi are thermally dimorphic, such as: Histoplasma capsulatum Blastomyces dermatitidis Coccidioides immitis Paracoccidioides brasiliensis Penicillium marneffei Sporothrix schenckii.
- Histoplasma capsulatum
- Blastomyces dermatitidis
- Coccidioides immitis
- Paracoccidioides brasiliensis
- Penicillium marneffei
- Sporothrix schenckii.
- Histoplasma capsulatum
- Blastomyces dermatitidis
- Coccidioides immitis
- Paracoccidioides brasiliensis
- Penicillium marneffei
- Sporothrix schenckii.

Incorrect Options:

Options A, B, and C are incorrect. They are not dimorphic fungus.

Solution for Question 19:

Correct Option D - Mucormycosis:

- β-1,3-D Glucan is a polysaccharide cell wall component of fungi. Beta 1,3 D glucan assay is commonly used for diagnosis of Candida and Aspergillus. Other organisms showing positive β-1,3-D Glucan assay are Pneumocystis jirrovecii pneumonia, Histoplasma, Fusarium, Scedosparium, and dermatophytes. Organisms with negative β-1,3-D Glucan assay are Cryptococcus, mucorales (rhizopus and mucor), and Blastomyces dermatitidis.
- Beta 1,3 D glucan assay is commonly used for diagnosis of Candida and Aspergillus.
- Other organisms showing positive β-1,3-D Glucan assay are Pneumocystis jirrovecii pneumonia, Histoplasma, Fusarium, Scedosparium, and dermatophytes.
- Organisms with negative β-1,3-D Glucan assay are Cryptococcus, mucorales (rhizopus and mucor), and Blastomyces dermatitidis.
- Mucorales are responsible for causing mucormycosis and cannot be diagnosed by using β-1,3-D Glucan assay.
- Beta 1,3 D glucan assay is commonly used for diagnosis of Candida and Aspergillus.
- Other organisms showing positive β-1,3-D Glucan assay are Pneumocystis jirrovecii pneumonia, Histoplasma, Fusarium, Scedosparium, and dermatophytes.
- Organisms with negative β-1,3-D Glucan assay are Cryptococcus, mucorales (rhizopus and mucor), and Blastomyces dermatitidis.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 20:

Correct Option B - Mucor:

• The histopathological finding showing aseptate hyphae branching at wide angles and suggests a fungal infection with the order Mucorales.

• The most common genus of Mucorales causing infection is Rhizopus, but the presence of diabetes and the clinical presentation of pneumonia suggests that the causative agent is more likely to be Mucor.

• The aseptate hyphae of mucor branches at an angle of 90° or more and have a broad ribbon shape.

• Mucor infections are known as Zygomycosis, and they can cause a range of infections, including pneumonia, sinusitis, and cutaneous infections.

• Mucorales are opportunistic fungi that cause disease in immunocompromised individuals, such as those with poorly controlled diabetes.

Incorrect Options:

Option A - Cryptococcus neoformans: Cryptococcus neoformans is a capsulated yeast that causes cryptococcal meningitis in immunocompromised individuals.

Option C - Candida: Candida is a genus of yeasts that can cause a range of infections involving skin, mucosa, and organs.

Option D - Aspergillus: Aspergillosis is caused by aspergillus, a septate fungi branching at less than 45°, commonly seen in patients on prolonged steroid use.

Solution for Question 21:

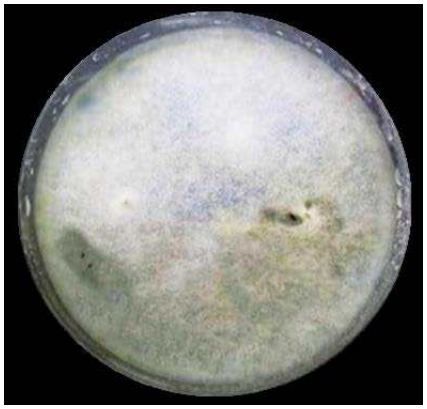
Correct Option C - Mucormycosis:

• The most probable diagnosis based on the given symptoms, findings, and H&E staining is Mucormycosis caused by mucor.

• Mucormycosis is predominant in COVID-19 patients because of the usage of steroids presenting with a black lesion.

• Mucor on SDA agar shows cotton wooly growth, giving an appearance of salt &pepper.

• Mucor shows aseptate hyphae branching at 90° or an obtuse angle (>90°) on H&E staining.



Incorrect Options: Option A - Aspergillosis: Aspergillosis is a fungal infection caused by Aspergillus. Aspergillus is a septate fungus showing acute angle (< 45°.) dichotomous branching. It is commonly seen in immunocompromised individuals presenting with aspergilloma in pre-existing cavitary lesions (tuberculosis) in the lung, allergic bronchopulmonary aspergillosis (ABPA). The growth on the SDA agar is species-dependent, A. fumigatus (smoky-green colony), A. niger (Black colonies), A. flavus (yellow-green colonies).

Option B - Bacterial orbital cellulitis: Bacterial orbital cellulitis is a bacterial infection that affects the soft tissues surrounding the eye. It can cause symptoms such as fever, nasal congestion, and facial pain. It will not show hyphae and SDA agar is not used for the culture.

Option D - Cavernous sinus thrombosis: Cavernous sinus thrombosis is a rare condition characterized by the formation of a blood clot within the cavernous sinus, a large vein at the base of the skull. It can cause symptoms such as headache with or without vomiting, papilledema, and visual problems, focal deficits, seizures, stupor, or coma.

Solution for Question 22:

Correct Option C - Trichophyton Mentagrophytes:

- Tinea capitis is a scalp infection characterized by scaly patches produced on the scalp, in which hair shafts are broken off right above the skin caused by Trichophyton spp, and some Microsporum spp.. The positive hair perforation test and white powder colonies obtained in culture from the specimen of hair is suggestive of Trichophyton mentagrophytes as a causative organism.
- Hair perforation test is positive only for T. mentagrophytes and M. canis.

• Macroscopic appearance of other trichophyton species are:

Dermatophytes	Macroscopic appearance on culture
T. rubrum	Velvety, red pigment on reverse
T. mentagrophytes	White to tan powdery and waxy
T. schoenleinii	Smooth, waxy
T. violaceum	Slow-growing, waxy Violet pigment on reverse

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 23:

Correct Option A - Sporothrix Schenckii:

- The organism that commonly causes disease following traumatic inoculation into the skin is Sporothrix schenckii.
- Sporothrix schenckii is a dimorphic fungus that causes a condition known as sporotrichosis or rose gardener's disease.
- It is found in soil and vegetation, particularly in environments associated with decaying plant matter.
- Traumatic inoculation occurs when the fungus enters the skin through thorns, splinters, or other puncture wounds, leading to localized infections.

Incorrect Options:

Option B - Blastomyces Dermatitidis: Blastomyces dermatitidis is the causative agent of blastomycosis. While it primarily affects the lungs through inhalation of spores, it is not commonly associated with traumatic inoculation.

Option C - Coccidioides Immitis: Coccidioides immitis is responsible for causing coccidioidomycosis or Valley fever. It is primarily acquired through inhalation of fungal spores from contaminated soil, rather than traumatic inoculation.

Option D - Paracoccidioides Brasiliensis: Paracoccidioides brasiliensis is the causative agent of paracoccidioidomycosis, a systemic fungal infection. It is typically acquired through inhalation of

fungal spores and is not commonly associated with traumatic inoculation.

Solution for Question 24:

Correct Option A - Histoplasmosis:

- The clinical presentation of cough, lymphadenopathy, cavitary lesion, and biopsy of a lymph node shows tuberculate macroconidia, which is a mold form of the fungus, as shown in the image, is suggestive of pulmonary histoplasmosis, also known as Cave's disease/Ohio disease, Mississippi disease & darling disease.
- Histoplasmosis is an intracellular infection of the reticuloendothelial system caused by the thermally dimorphic fungus Histoplasma capsulatum associated with farmers and individuals who are exposed to bird or bat droppings
- It presents with a mild flu-like illness and pulmonary infiltrates in the chest X-ray with hilar or mediastinal lymphadenopathy.

Incorrect Options:

Option B - Coccidioidomycosis: It is also known as Valley fever and is caused by the fungus Coccidioides immitis or Coccidioides posadasii. Pulmonary coccidioidomycosis presents with pneumonia, cavities, pleural effusion, or nodule formation. On microscopic examination, barrel-shaped arthrospores are characteristic of the infection.

Option C - Cryptococcus: It is a yeast-like fungus that can cause pulmonary infections, particularly in individuals with compromised immune systems. Halo around the organism on negative staining is characteristic of this fungus.

Option D - Blastomycosis: It is caused by the fungus Blastomyces dermatitidis and is commonly found in regions with moist soil, such as the Mississippi and Ohio River valleys. While it can cause lung involvement, mycelial form containing hyphae with small pear-shaped conidia are characteristic of this fungi.

Solution for Question 25:

Correct Option A - Candidiasis:

- The most possible diagnosis in this case is oral thrush, aka oropharyngeal candidiasis.
- Immunocompromised states (on steroid or immunosuppressive drugs, post-transplantation, malignancy, HIV-infected people) are predisposed to candida infection.
- Oral thrush is a common finding seen in patients suffering from Candida albicans. It is characterized by the presence of a white, adherent, painless patch in the oral cavity, as shown in the image.

Incorrect Options:

Option B - Burning mouth syndrome: Burning mouth syndrome is a condition characterized by a burning or scalding sensation in the mouth, often without any visible abnormalities. It is not typically associated with white patches or lesions.

Option C - Geographical tongue: Geographical tongue, also known as benign migratory glossitis, is a benign chronic relapsing recurring inflammatory condition of the oral cavity of unknown etiology.

Option D - Oral lichen planus: Oral lichen planus is a chronic inflammatory condition that can affect the oral mucosa and skin. It is seen clinically as reticular, papular, plaque-like, erosive, atrophic, or bullous types.

Solution for Question 26:

Correct Option C - Penicillin:

- Actinomyces israelii is an anaerobic bacteria that are normally present in the mouth, intestine, and vagina and is responsible for causing actinomycosis.
- A. israelii, when culture on solid media (brain-heart infusion agar), produces small 'spidery colonies' in 48-72 hours that become heaped up, white, and irregular or smooth, large colonies in 10 days, as shown in the image. Penicillin is the drug of choice to treat the infection.

Incorrect Options:

Options A, B, and D are not the drug of choice for treating actinomyces infection.

Solution for Question 27:

Correct Option C - Aspergillus:

- Aspergillus when stained with GMS stain shows acute angle dichotomous branching (<45°, two branches) with septate hyphae.

Incorrect Options:

Option A- Blastomycosis: On GMS stain shows a black outline with a classic figure of 8 appearance.

Option B- Candida: Under microscope shows both buds and pseudohyphae

Option D- Mucor: Is the opposite of Aspergillus, shows obtuse angle branching and aseptate hyphae.

Solution for Question 28:

Correct Option A - Malassezia:

- The clinical presentation and KOH mount is sugggestive of Pytriasis versicolor caused by malassezia furufur.
- It is a chronic recurrent condition involving the superficial layer (stratum corneum) of the skin characterized by flat-round scaly patches of hypo- to hyperpigmentation of the skin in areas rich in sebaceous glands commonly involved, such as the neck, chest, or upper arms.
- Malassezia is recognized by the detection of characteristic unipolar budding yeasts and accompanied by short hyphae, thus the so-called spaghetti and meatballs appearance.
- Other manifestation of malassezia are dermatitis, seborrheic dermatitis, and Malassezia folliculitis.

Incorrect Options:

Option B - Trichophyton: Trichophyton is a genus of fungi, which includes the parasitic varieties that cause tinea, including athlete's foot, ringworm, jock itch, and similar infections of the nail, beard, skin, and scalp. Trichophyton is white and cottony on the surface, and the underside is usually red. Microscopically pencil-shaped or cigar-shaped macronidia are sen.

Option C - Chlamydospores: Chlamydospores are produced by candida and represent enlarged, thick-walled vegetative cells with varied forms and condensed cytoplasm that forms within hyphae or at hyphal tips. Chlamydospores cause diseases like Aspergillosis, Blastomycosis, Coccidioidomycosis, and Histoplasmosis.

Option D - Candida: Candida is normally found in mouth, skin, and intestines. In immunocompromised individuals they are repsonsible for causing candidiasis including vaginal yeast infection, diaper rash, and oral thrush. On miscroscpic examination it reveals pseudihyphae.

Solution for Question 29:

Correct Option C - CHROM Agar:

• Thick white vaginal secretions and microscopic examination showing pseudohyphae and yeast-like organisms are suggestive candidiasis.

• CHROM agar is a differential medium that can differentiate between different species of Candida based on their ability to metabolize specific sugars.

• Candida albicans, the most common cause of candidiasis, produces colonies that are typically pink or purple on CHROM agar, while other species of Candida produce different-colored colonies.

• After incubation for 1–4 days on CHROMagar, colonies of C. albicans are green C. tropicalis is blue C. glabrata is dark purple C. parapsilosis, C. lusitaniae, C. guilliermondii, and C. krusei acquire a pinkish hue.

• C. albicans are green

• C. tropicalis is blue

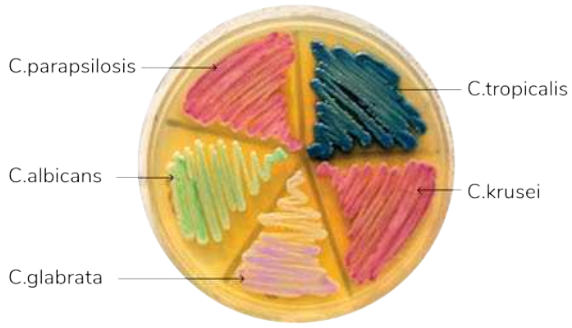
• C. glabrata is dark purple

• C. parapsilosis, C. lusitaniae, C. guilliermondii, and C. krusei acquire a pinkish hue.

• C. albicans are green

• C. tropicalis is blue

- C. glabrata is dark purple
- C. parapsilosis, C. lusitaniae, C. guilliermondii, and C. krusei acquire a pinkish hue.



Incorrect Options:

Option A - Bird seed Agar: It is a selective medium used to grow Cryptococcus neoformans, a fungus that causes cryptococcosis, a potentially fatal infection of the central nervous system.

Option B - Brain Heart Infusion Agar: Brain heart infusion agar is a nutrient-rich medium used to grow fastidious microorganisms like Bacteroides and actinomyces israelii.

Option D - Sabouraud Dextrose Agar: Sabouraud dextrose agar is a general-purpose medium used to grow a variety of fungi. While it is commonly used to isolate Candida species, it is not differential and cannot differentiate between different species of Candida.

Solution for Question 30:

Correct Option A - Talaromyces Marneffei:

- Talaromyces marneffei is a dimorphic fungus causing an opportunistic infection characterized by fever, weight loss, dyspnea, lymphadenopathy, hepatosplenomegaly, and warty lesions mimicking that of molluscum contagiosum.
- When grown on culture media, the mold forms a red diffusible pigment, as shown in the image.

Incorrect Options:

Option B - Blastomyces: Blastomyces is a dimorphic fungus that typically causes pulmonary most commonly and verrucous or ulcerative type of skin lesions, and osteomyelitis. It is characterized by the figure of 8 appearance on staining.

Option C - Aspergillus: Aspergillus is a common fungal pathogen that typically causes invasive pulmonary disease in immunocompromised patients. When grown on SDA, agar colonies have a velvety to powdery surface and are characteristically colored: A.fumigatus - dark green, A.niger - black, and A.flavus-yellow-green.

Option D - Pneumocystis Jirovecii: Pneumocystis jirovecii is a fungal organism that commonly causes pneumonia in immunocompromised patients, but it does not grow in culture and is not associated with cervical lymphadenopathy or red diffusible pigment.

Solution for Question 31:

Correct Option A - Mycetoma:

- The swollen foot and multiple discharging sinuses are highly suggestive of Mycetoma aka madura foot.
- Mycetoma is a chronic granulomatous infection of the subcutaneous tissues and bones that can cause a swollen foot and multiple sinuses.
- It is of two types Eumycetoma is caused by fungi such as Madurella mycetomatis. Actinomycetoma caused by Actinomadura, Nocardia, and streptomyces.
- Eumycetoma is caused by fungi such as Madurella mycetomatis.
- Actinomycetoma caused by Actinomadura, Nocardia, and streptomyces.
- Eumycetoma is caused by fungi such as Madurella mycetomatis.
- Actinomycetoma caused by Actinomadura, Nocardia, and streptomyces.

Incorrect Options:

Option B - Squamous cell carcinoma: Squamous cell carcinoma is a type of malignant skin cancer that usually appears as a firm, red nodule or a flat lesion with a scaly, crusty surface.

Option C - Nocardia: Nocardia is a type of bacteria that can cause cutaneous infections like actinomycetoma immunocompetent and involves the lungs and brain in immunocompromised individuals.

Option D - Sporotrichosis: Sporotrichosis is a fungal infection that usually affects the skin and subcutaneous tissue. It can cause nodules, ulcers, and lymphangitis.

Solution for Question 32:

Correct Option C - Chromoblastomycosis:

- The histopathological image shows a characteristic "copper penny" appearance of sclerotic bodies, and warty growth on the foot of a farmer is consistent with the diagnosis of chromoblastomycosis, aka verrucous dermatitis (cauliflower-like wart).
- Chromoblastomycosis is a chronic fungal infection caused by dematiaceous pigmented fungi Phialophora verrucosa, Fonsecaea pedrosoi, commonly affecting the skin of the feet and legs of farmers and agricultural workers.

Incorrect Options:

Option A - Sporotrichosis: Sporotrichosis is a subcutaneous fungal infection caused by the dimorphic fungus Sporothrix schenckii. It typically presents as a painless nodule or ulcer at the site of inoculation, which can progress to form a chain of subcutaneous nodules or abscesses along lymphatic channels. Histopathological examination shows suppurative granulomas with asteroid bodies.

Option B - Mycetoma: Mycetoma is a chronic bacterial or fungal infection that typically affects the subcutaneous tissue of the feet or hands. It presents as a painless swelling of the sinus tract with multiple discharging sinuses and granules.

Option D - Phaeohyphomycosis: It is a rare fungal infection caused by dematiaceous fungi such as Exophiala and Cladophialophora, which can affect various tissues, including the skin and subcutaneous tissue but do not form sclerotic bodies.

Solution for Question 33:

Correct option A - Sporothrix Schenckii:

- Sporothrix schenckii is a dimorphic fungus that causes chronic infection of the cutaneous, subcutaneous, and lymphatic tissue. It is frequently encountered in gardeners after a prick of a thorn of a plant which becomes the port of entry for the fungus.

Incorrect Options:

Option B - Histoplasma Capsulatum: It is a dimorphic fungus that causes histoplasmosis. Histoplasmosis is an intracellular infection of the reticuloendothelial system. It is classified under systemic mycoses.

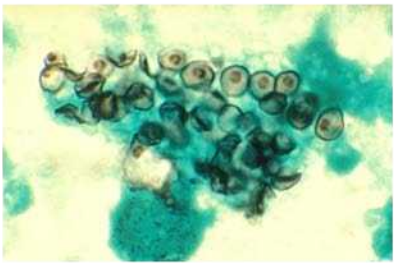
Option C - Blastomyces Dermatitidis: It is a dimorphic fungus that causes blastomycosis. Blastomycosis is a suppurative and granulomatous condition that is predominantly present in the lungs. It is classified under systemic mycoses.

Option D - Penicillium Marneffei: It is a dimorphic fungus that causes opportunistic infections in an individual infected with AIDS. It causes a disseminated infection with multiple organ involvement. It is classified as an opportunistic infection.

Solution for Question 34:

Correct Option D - Pneumocystis Jirovecii:

- Pneumocystis jirovecii is a fungal organism that commonly causes Pneumocystis carinii pneumonia (PCP) in immunocompromised individuals, including those with HIV infection.
- It presents with persistent dry cough, fever, and weight loss.
- Gomori methenamine silver staining of the bronchoalveolar lavage shows a black crushed ping-pong appearance against the green background.



Incorrect Options:

Option A - Paracoccidioides Brasiliensis: Paracoccidioides brasiliensis is a dimorphic fungus that causes paracoccidioidomycosis, a systemic fungal infection. It typically presents with respiratory

symptoms such as cough, fever, skin and mucosal lesions and cervical lymphadenoapthy.Histopathology staining reveals Mickey mouse or pilot wheel appearance.

Option B - Histoplasma Capsulatum: Histoplasma capsulatum is a dimorphic fungus that causes histoplasmosis, an infection primarily affecting the lungs. On gomori methenamine silver staining of the specimens (sputum) reveals tiny oval yeast cells with narrow-based budding within the macrophages with an underlying granulomatous response.

Option C - Coccidioides Immitis: Coccidioides immitis is a dimorphic fungus that causes coccidioidomycosis, commonly known as Valley fever. While respiratory symptoms and weight loss can occur in coccidioidomycosis, the Gomori methamine silver staining specimens demonstrates spherules which are large sac like structures, have thick, double refractile wall, and are filled with endospores.

Solution for Question 35:

Correct Option A - Sclerotic Body:

- This is a micrograph of chromoblastomycosis caused by Phialophora verrucosa and Fonsecaea pedrosoi, showing the sclerotic bodies aka copper penny bodies, or muriform bodies.
- The fungi produce thick-walled, single, or multicellular clusters called sclerotic or muriform bodies. There is remarkable pseudoepitheliomatous hyperplasia of the epidermis.

Incorrect Options:

Option B - Budding Yeast: Budding yeast is Saccharomyces cerevisiae, which forms a bud from the mother cell during reproduction.

Option C - Negri Body: Negri bodies are pathognomonic inclusion bodies, eosinophilic, found in the cytoplasm of nerve cells in patients affected with the rabies virus.

Option D - Asteroid Body: Asteroid bodies are found in sporotrichosis and sarcoidosis; these are extracellular eosinophilic bodies and are located within abscesses.

Solution for Question 36:

Correct Option A - Infection does not involve underlying fascia, tendons, and muscle:

- A forest worker has a history of trauma while working in the forest, which then appears as macules and later on becomes nodules, causing a Verrucous lesion (a warty lesion on the skin) and the image revealing pathognomonic finding of Medlar bodies/sclerotic bodies is suggestive of chromoblastomycosis caused by dematiaceous fungi like Fonsecaea pedrosi, F.compacta, and exophiala.
- The infection manifests only as a cutaneous or subcutaneous infection and does not involve deeper structures, such as tendons, muscles, fascia, and osteoarticular sites.

Incorrect Options:

Options B, C, and D are incorrect. These statements are incorrect about chromoblastomycosis.

Solution for Question 37:

Correct Option A - Aspergillus Fumigatus:

- Aspergillus fumigatus is a fungus which differ from bacteria by having rigid cell walls containing chitin, mannan and polysaccharides. Cytoplasmic membrane contains sterols.

Incorrect Options:

Option B - Mycobacterium Tuberculosis: M. tuberculosis is an acid fast bacilli, cell wall is made up of peptidoglycan (murein layer), arabinogalactan layer, mycolic acid layer and outer peptidoglycolipids layer.

Option C - Mycoplasma Pneumoniae: Mycoplasmas have no cell walls, and they are placed in a separate class known as Mollicutes.

Option D - Rhinosporidum Seeberi: Rhinosporidium seeberi is an aquatic protist, which do not have rigid cell wall, on biopsy and shows endospores typical of fungus, but it is not a fungus as it is not cultured on SDA.

Solution for Question 38:

Correct Option A - Candida:

- Immunosuppressant drugs are commonly prescribed to prevent organ rejection after transplantation and treat autoimmune diseases like rheumatoid arthritis or lupus. These drugs can weaken the immune response, reducing the body's ability to fight off infections.
- Candidiasis can manifest as oral thrush (white patches in the mouth and throat), vaginal yeast infections, or invasive candidiasis affecting the bloodstream, organs, or other body sites.
- In Candida, the bud remains attached to the mother cell, elongates, and undergoes repeated budding to form chains of elongated cells known as pseudohyphae.

Incorrect Options:

Option B - Aspergillus: Aspergillus has a filamentous growth form consisting of acute branching dichotomous hyphae. Hyphae are thin, thread-like structures that make up the fungal body.

Option C - Pneumocystis Jirovecii: Pneumocystis primarily affects individuals with weakened immune systems and causes pneumonia-like symptoms. Microscopy shows crushed ping pong ball appearance.

Option D - Histoplasma Capsulatum: Histoplasma capsulatum is a dimorphic fungus showing both hyphal and yeast forms, causing Histoplasmosis, which is an intracellular infection of the reticuloendothelial system.

Solution for Question 39:

Correct Option A - Oral Candidiasis:

- The likely diagnosis for a 35-year-old HIV-positive patient with mucosal lesions in the mouth and the presence of budding yeasts and pseudohyphae on microscopy is oral candidiasis.
- Oral candidiasis, also known as oral thrush, is a fungal infection caused by the overgrowth of Candida species, particularly Candida albicans, in the oral cavity.
- It is common in individuals with compromised immune systems, such as those with HIV/AIDS.

• Microscopic examination of a sample from the mucosal lesions would reveal the presence of budding yeasts (small circular structures) and pseudohyphae (elongated structures with constrictions or constrictions between the budding cells). This characteristic appearance confirms the diagnosis of oral candidiasis.

Incorrect Options:

Option B - Hairy Leukoplakia: Hairy leukoplakia is a condition caused by the Epstein-Barr virus (EBV) and is commonly seen in individuals with HIV infection. It presents as white, corrugated lesions on the lateral borders of the tongue.

Option C - Lichen Planus: Lichen planus is a chronic inflammatory condition that can affect the skin and mucous membranes, including the mouth. It typically presents as white, lacy patches or plaques.

Option D - Diphtheria: Diphtheria is a bacterial infection caused by Corynebacterium diphtheriae. It primarily affects the throat and can lead to the formation of grayish-white pseudomembranes.

Solution for Question 40:

Correct Option B - Voriconazole:

• The drug of choice for invasive pulmonary aspergillosis is Voriconazole.

• Voriconazole exhibits potent antifungal activity, good tissue penetration, and a broad spectrum of activity against various Aspergillus species.

• Voriconazole, a second-generation triazole antifungal agent, inhibits ergosterol synthesis by inhibiting cytochrome P 450, which is essential for fungal cell membrane integrity.

Incorrect Options:

Options A, C, and D are incorrect. These are not the drugs of choice to treat invasive aspergillosis.

Solution for Question 41:

Correct Option B - Trichophyton Rubrum:

Dermatophytes	Macroscopic appearance	Microscopic appearance
T. rubrum	Velvety, red pigment on reverse	Microconidia - teardrop shaped, plenty Macroconidia – few, long pencil-shaped
T. mentagrophytes	White to tan Powdery Pigment variable	Microconidia-numerous, round to pyriform Macroconidia - cigar-shaped Spiral hyphae seen
T. Schoenleinii	Smooth, waxy	Microconidia and macroconidia-rare or absent Chlamydospores seen Hyphae- hyphal swelling and favic chandelier seen
T. violaceum	Slow-growing, waxy Violet pigment on reverse	Microconidia and macroconidia – rare or absent Chlamydospores seen

Incorrect Options: Options A, C, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 42:

Correct Option B - Diabetes:

• The clinical presentation of foul smell and black necrotic mass in the nose invading the septum is suggestive of mucormycosis caused by Mucor.

• Predisposing factors for mucormycosis: Iron is essential for mucormycosis growth. High iron levels increase the risk of invasive mucormycosis. Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum. End-stage renal disease. Patients taking iron therapy or deferoxamine (iron chelator) Defects in phagocytic functions (e.g., neutropenia or steroid therapy).

• Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum.

• End-stage renal disease.

- Patients taking iron therapy or deferoxamine (iron chelator)
- Defects in phagocytic functions (e.g., neutropenia or steroid therapy).
- Diabetic ketoacidosis is the most important risk factor. Acidosis causes the release of iron from the sequestered proteins in the serum.
- End-stage renal disease.
- Patients taking iron therapy or deferoxamine (iron chelator)
- Defects in phagocytic functions (e.g., neutropenia or steroid therapy).

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 43:

Correct Option B - Sporothrix Schenckii:

- Sporothrix schenckii is a dimorphic fungus that causes a condition known as sporotrichosis or rose gardener's disease. It is commonly found in soil and vegetation and can infect humans and animals through traumatic introduction into the skin, such as via thorns or splinters.
- Sporotrichosis typically presents as a localized skin infection characterized by a painless, nodular lesion that can ulcerate and progress along lymphatic channels as shown in the image.
- These lesions or bumps are pink or purple and often appear on the finger or hand. The infection can spread to other parts of the body through the lymphatic system.

Incorrect Options:

Option A - Mycobacterium Scrofulaceum: Mycobacterium scrofulaceum is a nontuberculous mycobacterium and is not commonly associated with traumatic skin lesions.

Option C - Madurella Mycetomatis: Madurella mycetomatis is a fungus that causes a chronic granulomatous infection known as mycetoma, characterized by swelling and draining sinuses, typically affecting the extremities.

Option D - Phialophora Verrucosa: Phialophora verrucosa is a dematiaceous fungus associated with chromoblastomycosis, a chronic subcutaneous fungal infection characterized by slowly developing granulomatous lesions.

Solution for Question 44:

Correct Option C - Cladosporium:

- The term dematiaceous refers to the presence of melanin, which gives these fungi characteristic dark coloration.
- Cladosporium is melanin-producing fungi. Hence, it is a dematiaceous fungus.
- Cladosporium is a common environmental mold found in soil, plants, and decaying organic matter. It is one of the recognized fungal agents that can cause chromoblastomycosis.
- Other dematiaceous fungi are P. verrucosa, F. pedrosoi, Fonsecaea compacta, and Rhinocladiella aquaspersa.

Incorrect Options:

Option A - Mucor Species: Mucor species belong to the group of zygomycetes; it is not dematiaceous fungi.

Option B - Aspergillus Niger: Aspergillus niger is an ascomycete fungus known to cause aspergillosis.

Option C - Blastomyces Dermatitidis: Blastomyces dermatitidis is a dimorphic fungus known to cause the disease blastomycosis.

Solution for Question 45:

Correct Option D - Acute angle branching septate hyphae, Aspergillus:

- The clinical presentation of fever, dyspnea, palpitations, and tachycardia, along with echocardiographic findings of vegetation in the septal leaflets, suggests infective endocarditis.
- The histopathological findings show uniform, dichotomous branching at acute angles and septate hyphae of Aspergillus in tissue stained with Gomori methenamine silver is consistent

with Aspergillus.

• Aspergillus can manifest as cardiac aspergillosis manifesting as endocarditis and percarditis.

Incorrect Options:

Option A - Pigmented septate hyphae at 90 degrees, Mucormycosis: Mucormycosis is caused by mucor, characterized by aseptate hyphae branching at 90° or more than 90° and is not associated with endocarditis.

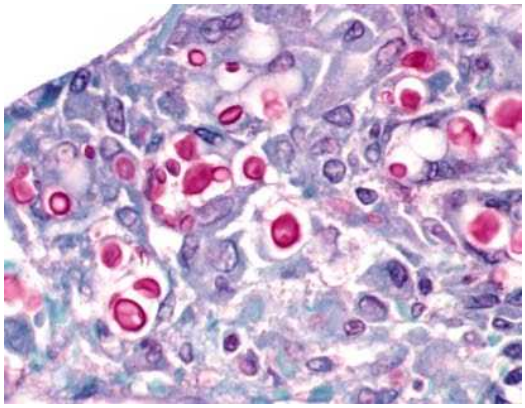
Option B - Right angle branching septate hyphae, Fusarium: Fusarium is a filamentous characterized by hyaline septate hyphae bearing round microconidia, sickle-shaped large macroconidia, and chlamydospores fungi. It is more commonly associated with endophthalmitis and keratitis.

Option C - Acute angle branching hyphae, Rhizopus: Rhizopus is characterized by broad aseptate hyphae, from which sporangiophore arises and then ends at sporangium, and root-like growth from hyphae called rhizoid. It commonly manifests as Rhino-oculo-cerebral disease or may involve the lungs.

Solution for Question 46:

Correct Option D - Mucicarmine Stain:

• Cryptococcus can be identified using a special stain called mucicarmine stain. This stain specifically targets the polysaccharide capsule of Cryptococcus and stains it pink.



• Other stains used to demonstrate cryptococcus neoformans are: Masson-Fontana stain Alcian blue stain

- Masson-Fontana stain
- Alcian blue stain
- Masson-Fontana stain
- Alcian blue stain

Incorrect Options:

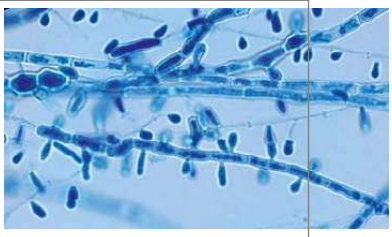
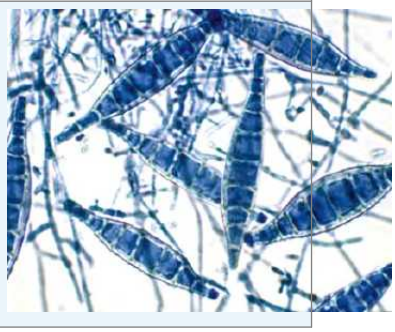
Option A - Giemsa Stain: Giemsa stain is commonly used for demonstrating Yersinia pestis, Borrelia duttoni, and rickettsia infections.

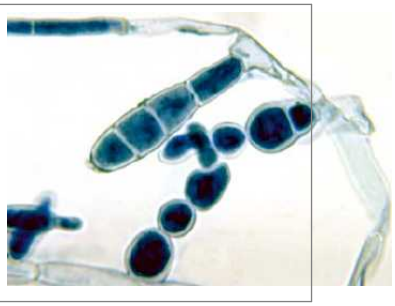
Option B - Ziehl-Nielsen Stain: Ziehl-Nielsen stain is used for acid-fast staining to detect acid-fast bacteria like Mycobacterium tuberculosis, coccidian parasites, and M. leprae.

Option C - Gram Staining: Gram staining is a commonly used staining method to differentiate bacteria into Gram-positive and Gram-negative based on their cell wall properties

Solution for Question 47:

Correct Option C - Epidermophyton:

Organisms	Macroconidia	Microconidia	LPCB
Trichophyton	Pencil-shaped	+++ (maximum)	
Microsporum	Spindle-shaped	++	

Epidermophyton	Club-shaped	+/- (least)	

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 48:

Correct Option C - 1 only:

- Galactomannan is a major polysaccharide constituent of Aspergillus cell walls. It is a useful diagnostic marker for detecting this fungal infection.
- It is commonly used for the diagnosis of Aspergillus.
- Other pathogens positive for galactomannan are: Fusarium, Penicillium, Histoplasma, Blastomyces, and Paecilomyces.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

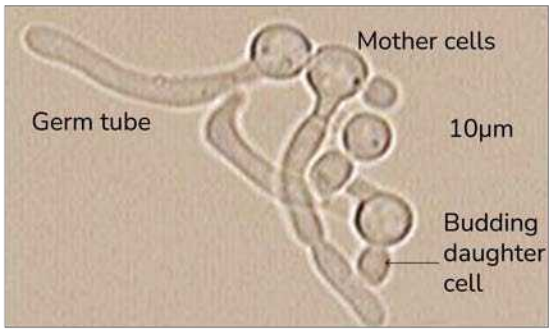
Solution for Question 49:

Correct Option A - Candida Albicans:

- The given image shows the positive germ tube test, aka Reynolds Braude phenomenon.
- The germ tube test is a laboratory diagnostic test used to identify Candida albicans. This test is based on the ability of Candida albicans to produce germ tubes, which are elongated

outgrowths from yeast cells when mixed with human serum and incubated for 2 hours.

• Another species of candida which shows a positive germ tube test is C. dubliniensis.



Incorrect Options:

Options B, C, and D are incorrect. These species of candida do not show positive germ tube tests.

Solution for Question 50:

Correct Option B - Chromoblastomycosis:

• The histopathological image shows a characteristic "copper penny" appearance of sclerotic bodies, and warty growth on the foot is consistent with the diagnosis of chromoblastomycosis, aka verrucous dermatitis (cauliflower-like wart).

• Chromoblastomycosis is a chronic fungal infection caused by dematiaceous pigmented fungi Phialophora verrucosa, Fonsecaea pedrosoi, commonly affecting the skin of the feet and legs of farmers and agricultural workers.

Incorrect Options:

Option A - Histoplasmosis: Histoplasmosis is caused by dimorphic fungus Histoplasma capsulatum. It is characterized by tuberculate macronidia on LPCB (Lactophenol cotton blue) wet mount and commonly affects respiratory symptoms (Tuberculosis like clinical features are seen).

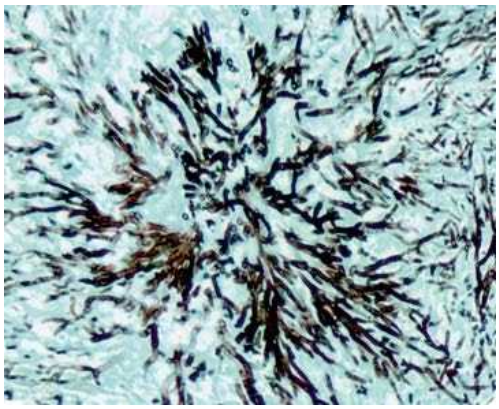
Option C - Coccidioidomycosis: Coccidioidomycosis is caused by the fungus Coccidioides immitis or Coccidioides posadasii. The microscopic examination typically shows barrel-shaped arthrospores on LPCB.

Option D - Candida Albicans: Candida albicans is a yeast-like fungus that can cause various types of infections, including oral thrush and vaginal yeast infections. On microscopic examination, it shows pseudohyphae.

Solution for Question 51:

Correct Option C - Methenamine Silver:

- Gomori methenamine silver stain is commonly used to visualize fungal elements in histopathological specimens. It stains the fungal hyphae in black color, allowing for their identification under a microscope.



Incorrect Options:

Option A - Acid-Fast Stain: It is used to detect acid-fast bacteria such as Mycobacterium species and coccidian parasites, not fungal elements.

Option B - Gabbet's Stain: It is also known as kinyoun stain, used for demonstrating the coccidian parasites (isosopora, cyclospora, and cryptosporidium).

Option D - Leifson & Ryu's Stain: This stain is used to demonstrate the flagella.

Solution for Question 52:

Correct Option C - Chromoblastomycosis:

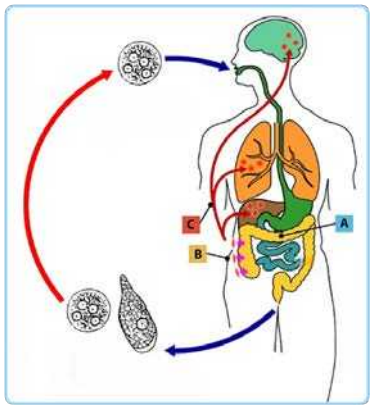
- This is a micrograph of chromoblastomycosis caused by Phialophora verrucosa and Fonsecaea pedrosoi, showing the sclerotic bodies, aka copper penny bodies, or muriform bodies.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Parasitology - Classification & Amoeba

1. Identify which organism this life cycle belongs to.



A. Naegleria fowleri

B. Entamoeba histolytica

C. Acanthamoeba

D. Giardia lamblia

---

2. A 40-year-old woman presents with severe abdominal pain and diarrhea after travelling to an area with poor sanitation. A histology of the intestinal mucosa obtained via colonoscopy revealed the following. What is the infective form of the organism that causes this condition?



A. Trophozoite

B. Tetrad

C. Cyst

D. Oocyst

---

3. All of the following have a low infective dose except:

A. Giardia lamblia

B. Entamoeba histolytica

C. Shigella

D. Vibrio cholerae

---

4. A 35-year-old man presents with severe abdominal pain, dysentery, and intestinal ulcers. He reports recent travel to an area with poor sanitation. Stool examination reveals a protozoan parasite. All of the following are the features of this condition except:

A. Trophozoites have a cartwheel nucleus

B. Chromatid bar of the cyst is composed of ribonucleoprotein

C. Trophozoites contain engulfed WBCs

D. Trophozoites produce characteristic flask-shaped ulcerative lesions

5. Which of the following is the most common extraintestinal complication of amoebiasis?

A. Hepatic abscess

B. Cutaneous amoebiasis

C. Splenic abscess

D. Renal failure

6. A 45-year-old male presents with a month-long history of abdominal discomfort, intermittent bloody diarrhea, fatigue, weight loss (8 kg), and low-grade fevers. He recently traveled to a rural area with limited access to clean drinking water. Physical examination reveals right upper quadrant tenderness. Laboratory findings show mild anemia and elevated white blood cell count. What is the gold standard test to diagnose active infection in this patient?

A. Stool culture

B. Serologic antibody testing

C. Stool microscopy for trophozoites

D. PCR

7. A 35-year-old male presents to the emergency department with a history of cramping abdominal pain, dysentery, and intermittent fever over the past week. Stool analysis confirms the presence of Entamoeba histolytica trophozoites. Which of the following is the primary virulence factor responsible for E. histolytica's ability to adhere to and invade the colonic mucosa?

A. Gal/GalNAc lectin antigen

B. CP47

C. M protein

D. Hyaluronidase

8. A 42-year-old male presents to the OPD with complaints of fever, abdominal pain, bloody diarrhea and weight loss after returning from a trip to a region with poor sanitation. Stool microscopy reveals the presence of Charcot Leyden crystals. Which of the following is the most appropriate first-line treatment for this condition?

A. Tinidazole

B. Nitazoxanide

C. Metronidazole

D. Praziquantel

9. A previously healthy 15-year-old boy presents to the emergency department with fever, severe headache, vomiting, and rapidly progressing confusion. He was diagnosed with meningitis, and his parents report he recently went swimming during a camping trip. Despite treatment, his condition worsens rapidly. Which of the following organisms is most likely causing this condition?

A. Acanthamoeba species

B. Naegleria fowleri

C. Cryptococcus neoformans

D. Listeria monocytogenes

10. A 45-year-old man presents with 6 weeks of headaches, confusion, and cognitive decline after swimming in freshwater lakes during travel to an area with poor sanitation. Lumbar puncture shows elevated WBCs (predominantly lymphocytes) and trophozoites with distinctive spikes on their surface. Which of the following statements about the causative organism is not true?

A. It is associated with both central nervous system infections and ocular infections.

B. Both cysts and trophozoites are infective forms.

C. CT scan shows space-occupying lesions in the brain.

D. It is seen in immunocompetent individuals.

---

11. What is the suggested treatment for the infection caused by this organism?



A. Tetracycline

B. Albendazole

C. Ivermectin A

D. Amphotericin B

---

12. Match the following organisms with their infective forms. a. Entamoeba histolytica 1. Trophozoite b. Naegleria fowleri 2. Cyst c. Acanthamoeba 3. Both

cysts and trophozoites d. Balantidium coli

a. Entamoeba histolytica	1. Trophozoite
b. Naegleria fowleri	2. Cyst
c. Acanthamoeba	3. Both cysts and trophozoites
d. Balantidium coli	

A. a-1, b-2, c-3, d-1

B. a-2, b-1, c-3, d-2

C. a-2, b-1, c-3, d-1

D. a-1, b-2, c-3, d-2

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	3
Question 5	1
Question 6	3
Question 7	1
Question 8	3
Question 9	2
Question 10	4
Question 11	1
Question 12	2

Solution for Question 1:

Correct Answer: B) Entamoeba histolytica

Explanation: The given diagram depicts the life cycle of Entamoeba histolytica.

E.histolytica causes amoebic dysentery and a wide range of other invasive diseases, including amoebic liver abscesses.

Morphology: Has three stages:

Life Cycle:

- Host: Single host, i.e. man.

• Infective form: Mature quadrinucleate cyst

• Mode of transmission: E. histolytica is transmitted by Feco-oral route (most common) Sexual (anogenital or orogenital) contact Very rarely by vectors (mechanical transmission from faeces).

• Feco-oral route (most common)

• Sexual (anogenital or orogenital) contact

• Very rarely by vectors (mechanical transmission from faeces).

• Development in Human Intestine: Small intestine: Undergoes excystation Large intestine: Multiplication and colonization.

• Small intestine: Undergoes excystation

• Large intestine: Multiplication and colonization.

• Manifestations: Asymptomatic cyst passers Amoebic dysentery Invasive amoebiasis

• Asymptomatic cyst passers

• Amoebic dysentery

• Invasive amoebiasis

• Encystation: When the lesions start healing, and the patient improves, trophozoites transform into cysts and then into cysts in the large intestine, which are liberated in faeces. These cysts are infective.

• Feco-oral route (most common)

• Sexual (anogenital or orogenital) contact

• Very rarely by vectors (mechanical transmission from faeces).

• Small intestine: Undergoes excystation

• Large intestine: Multiplication and colonization.

• Asymptomatic cyst passers

• Amoebic dysentery

• Invasive amoebiasis

Naegleria fowleri (Option A): The life cycle of Naegleria fowleri includes an infective trophozoite stage that enters through the nasal mucosa and a cyst stage that helps it survive in harsh conditions outside the host. Causes primary amoebic meningoencephalitis (PAM).

Acanthamoeba (Option C): Man acquires infection by inhalation of aerosols contaminated with cyst or trophozoite and causes granulomatous amoebic encephalitis in immunocompromised and keratitis in contact lens users.

Giardia lamblia (Option D): The life cycle of Giardia lamblia involves ingestion of cysts from contaminated food or water. These cysts transform into trophozoites in the small intestine, multiply, and eventually form new cysts that are excreted in the feces to infect new hosts. This

causes giardiasis.

Solution for Question 2:

Correct Answer: C) Cyst

Explanation: The given clinical history and the histology specimen showing a flask-shaped ulcer suggest amoebiasis caused by Entamoeba histolytica. This organism's infective form is the cyst.

Entamoeba histolytica:

Life Cycle:

- Host: Single host, i.e. man.
- Infective form: Mature cyst containing 1 to 4 nucleus – Quadrinucleate Thick Chromatoid body/ bar
- 1 to 4 nucleus – Quadrinucleate
- Thick Chromatoid body/ bar
- 1 to 4 nucleus – Quadrinucleate
- Thick Chromatoid body/ bar
- Mode of transmission: Feco-oral route (most common)
- Development in Human Intestine: Small intestine: Undergoes excystation. Large intestine: Multiplication and colonization.
- Small intestine: Undergoes excystation.
- Large intestine: Multiplication and colonization.
- Manifestations: Asymptomatic cyst passers Amoebic dysentery Invasive amoebiasis
- Asymptomatic cyst passers
- Amoebic dysentery
- Invasive amoebiasis
- Small intestine: Undergoes excystation.
- Large intestine: Multiplication and colonization.
- Asymptomatic cyst passers
- Amoebic dysentery
- Invasive amoebiasis

Solution for Question 3:

Correct Answer: D) Vibrio cholerae

Explanation

Vibrio cholerae is highly acid-labile and hence requires a large infective dose of approximately 108 bacteria to overcome the gastric barrier.

Infective dose refers to the minimum dose required to initiate the infection. For example,

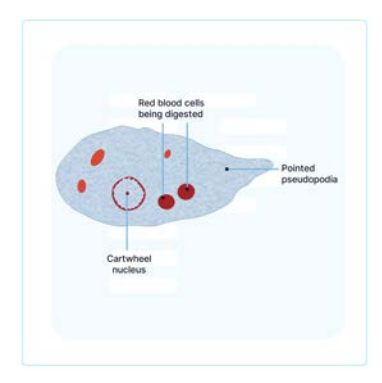
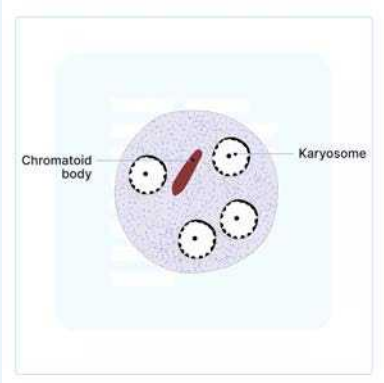
- Infective dose of 10-100 bacteria or cysts (Mnemonic - CHHota SmaLL): Cryptosporidium Entamoeba Histolytica EnteroHemorrhagic E. coli Shigella Giardia Lamblia
- Cryptosporidium
- Entamoeba Histolytica
- EnteroHemorrhagic E. coli
- Shigella
- Giardia Lamblia
- Cryptosporidium
- Entamoeba Histolytica
- EnteroHemorrhagic E. coli
- Shigella
- Giardia Lamblia
- Infective dose of 103 – 105 bacilli: Salmonella
- Infective dose of 105 – 108 bacilli: Vibrio cholerae

Solution for Question 4:

Correct Answer: C) Trophozoites contain engulfed WBCs

Explanation:

The given clinical history suggests amoebiasis caused by Entamoeba histolytica. These organisms' trophozoites contain engulfed RBCs, not WBCs.

Entamoeba histolytica	
Trophozoite	Cyst
 ... (content truncated)	 ... (content truncated)

Entamoeba histolytica	Entamoeba coli
Trophozoite Pointed pseudopodia 1 nucleus Central karyosome Cartwheel nucleus Erythrophagocytosis	Trophozoite Blunt pseudopodia 1 nucleus Eccentric karyosome No cartwheel nucleus No erythrophagocytosis
Cyst 1 to 4 nuclei- Quadrinucleate Thick chromatoid body/bar	Cyst 1 to 8 nuclei- Octanucleate Thin filamentous chromatoid body/bar

Trophozoites produce characteristic flask-shaped ulcerative lesions (Option D): These ulcers are most commonly found in the cecum. Ulcers are usually small, round to oval, scattered with intervening normal mucosa.



Solution for Question 5:

Correct Answer: A) Hepatic abscess

Explanation:

The most common extraintestinal complication of amoebiasis is a hepatic abscess, often presenting with a distinctive "anchovy sauce" appearance due to the reddish-brown, pus-like material in the abscess cavity.

Complications of intestinal amoebiasis:

- Fulminant amoebic colitis
- Amoebic appendicitis
- Intestinal perforation and amoebic peritonitis
- Toxic megacolon and intussusception
- Amoebiasis cutis or cutaneous amoebiasis (Option B)
- Ameboma (amoebic granuloma)
- Chronic amoebiasis

Extraintestinal complications of amoebiasis:

- Amoebic liver abscess - most common The posterior superior surface of the right lobe is affected. Gross: Anchovy sauce appearance (brown colour fluid) Microscopy: Trophozoites in Pus
- The posterior superior surface of the right lobe is affected.
- Gross: Anchovy sauce appearance (brown colour fluid)
- Microscopy: Trophozoites in Pus
- Rarely, it can also infect the heart, brain, kidneys, spleen, and skin
- The posterior superior surface of the right lobe is affected.
- Gross: Anchovy sauce appearance (brown colour fluid)
- Microscopy: Trophozoites in Pus

Solution for Question 6:

Correct Answer: C) Stool microscopy for trophozoites

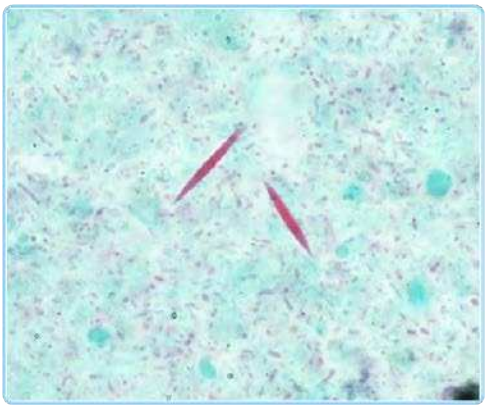
Explanation

The patient's signs and symptoms, recent travel history, and exposure to potentially contaminated water raise suspicion of an intestinal infection, particularly of Entamoeba histolytica. Demonstration of trophozoite in freshly passed stool is considered the gold standard microscopic test to diagnose active infection. Trophozoites are absent in asymptomatic carriers.

Diagnosis of E. histolytica:

- Sample collection: Stool is the specimen of choice. Minimum of 3 stool samples should be collected on alternate days (within 10 days) as amoebae are shed intermittently.
- Stool macroscopy: Foul smelling, copious in amount, dark red in colour mixed with blood and mucus and not adherent to the container
- Stool microscopy: Direct examination of stool by saline and iodine mount to demonstrate trophozoites and cysts. Demonstration of trophozoites is considered the gold standard microscopic test for active infection. If Trophozoites are seen - Active disease. If only Cyst is seen - Carrier stage.
- If Trophozoites are seen - Active disease.
- If only Cyst is seen - Carrier stage.
- Histology: Intestinal biopsies stained with PAS or H&E stains reveal trophozoites
- Stool culture
- Stool Antigen Detection: Fresh/frozen samples are used. ELISA Immunochromatographic test (ICT)
- ELISA
- Immunochromatographic test (ICT)
- Molecular Diagnosis: Nested multiplex PCR Real-time PCR BioFire FilmArray
- Nested multiplex PCR
- Real-time PCR
- BioFire FilmArray
- Colonoscopy: May show collar button or flask-shaped amoebic ulcers
- Miscellaneous finding on stool microscopy: Charcot Leyden crystal
- If Trophozoites are seen - Active disease.
- If only Cyst is seen - Carrier stage.
- ELISA
- Immunochromatographic test (ICT)

- Nested multiplex PCR
- Real-time PCR
- BioFire FilmArray



Solution for Question 7:

Correct Answer: A) Gal/GalNAc lectin antigen

Explanation:

The Gal/GalNAc lectin antigen is a key virulence factor that facilitates adhesion by binding to glycoprotein receptors on the epithelial cells of the large intestine and the vascular endothelium.

Trophozoite of E. histolytica is the invasive form and possesses many virulence factors such as:

- Enzymes Metallo-collagenase Neuraminidase
- Metallo-collagenase
- Neuraminidase
- Metallo-collagenase
- Neuraminidase

CP47 (Option B), or the Cryptosporidium protein, is the virulence factor of the intestinal coccidian parasites (Cryptosporidium, Cyclospora, and Cystoisospora).

M protein (Option C) is the principal virulence factor of group A Streptococcus. It inhibits complement-mediated opsonisation by phagocytes.

Hyaluronidase (Option D) is an enzyme produced by some bacteria such as Staphylococcus aureus, Streptococcus and Clostridia species, which breaks down hyaluronic acid in connective tissues. It aids in the spread of infections, but is not a virulence factor of E. histolytica.

Solution for Question 8:

Correct Answer: C) Metronidazole

Explanation:

The given clinical history along with the findings of Charcot Leyden crystals on stool microscopy is suggestive that the patient has intestinal amoebiasis. Metronidazole is the first-line treatment for intestinal amoebiasis followed by a luminal agent.

Treatment:

- Active Metronidazole x 10 days Followed by luminal amoebicide - Paromomycin
- Metronidazole x 10 days
- Followed by luminal amoebicide - Paromomycin
- Asymptomatic Paromomycin - For prevention if the patient is a carrier.
- Paromomycin - For prevention if the patient is a carrier.
- Metronidazole x 10 days
- Followed by luminal amoebicide - Paromomycin
- Paromomycin - For prevention if the patient is a carrier.

Tinidazole is used in the treatment of Giardiasis and Trichomoniasis. (Option A)

Nitazoxanide is used in the treatment of Cryptosporidiosis. (Option B)

Praziquantel is used to treat cysticercosis and diphyllobothrium infections. (Option D)

Solution for Question 9:

Correct Answer: C) Naegleria fowleri

Explanation:

The clinical history described is most likely suggestive of primary amoebic meningoencephalitis (PAM), which is caused by Naegleria fowleri.

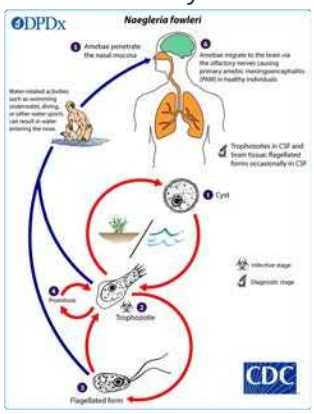
Naegleria fowleri:

Naegleria is a free-living amoeba commonly found in warm freshwater environments. In nature, it exists in two forms: cysts and trophozoites.

- Cyst: Round
- Trophozoite: Can occur in two forms – amoeboid and flagellated forms.

Life cycle:

- Infective form: Trophozoite.
- Mode of transmission: Nasal contamination occurs during swimming in fresh, hot water bodies like ponds, rivers, swimming pools, or lakes.
- CNS invasion: Invades nasal mucosa → Cribriform plate → Olfactory nerve → Brain → Primary Amoebic Meningoencephalitis.



Primary Amoebic Meningoencephalitis

- Causative organism: Naegleria fowleri
- Occurs in healthy children or young adults with a recent history of swimming in fresh, hot water.
- Initial symptoms include altered taste and smell (due to involvement of the olfactory nerve), followed by nausea, vomiting, high fever, and meningeal signs such as a stiff neck and positive Kernig's sign.
- Secondary symptoms include confusion, hallucinations, lack of attention, ataxia, and seizures.
- Death occurs within 7–14 days after exposure.

• Diagnosis: CSF analysis: Thick, purulent CSF with pus cells, elevated protein and reduced sugar level (mimic bacterial meningitis). CSF microscopy: Detection of characteristic trophozoites in CSF (confirmatory). Culture: NNA (Non-Nutrient Agar) - lawn cultured with Escherichia coli.

• CSF analysis: Thick, purulent CSF with pus cells, elevated protein and reduced sugar level (mimic bacterial meningitis).

• CSF microscopy: Detection of characteristic trophozoites in CSF (confirmatory).

• Culture: NNA (Non-Nutrient Agar) - lawn cultured with Escherichia coli.

• Treatment: Amphotericin B.

• CSF analysis: Thick, purulent CSF with pus cells, elevated protein and reduced sugar level (mimic bacterial meningitis).

• CSF microscopy: Detection of characteristic trophozoites in CSF (confirmatory).

• Culture: NNA (Non-Nutrient Agar) - lawn cultured with Escherichia coli.

Acanthamoeba species (Option A): Acanthamoeba is a free-living amoeba that can cause Granulomatous Amoebic Encephalitis (GAE), a rare and often chronic infection of the brain. GAE usually occurs in immunocompromised individuals. It also causes keratitis in contact lens users.

Cryptococcus neoformans (Option C): It is a fungal pathogen that primarily affects immunocompromised individuals, such as those with HIV/AIDS. It causes chronic meningitis, typically presenting with headache, fever, and neurological changes. Infection is acquired by inhalation of aerosolized forms of Cryptococcus.

Listeria monocytogenes (Option D): It is a bacterial cause of meningitis, often linked to foodborne transmission. It generally affects immunocompromised individuals, the elderly, neonates, or pregnant women (extremes of ages).

Solution for Question 10:

Correct Answer: D) It is seen in immunocompetent individuals.

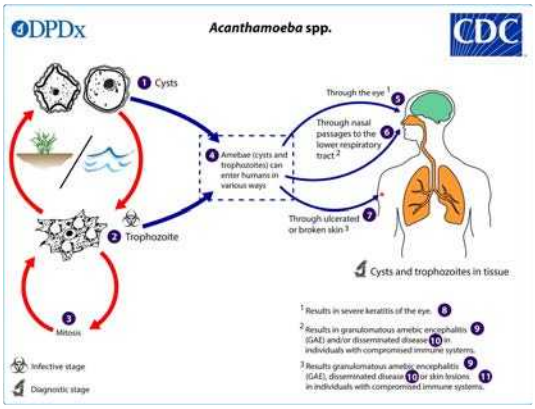
Explanation:

The clinical presentation is indicative of granulomatous amoebic encephalitis (GAE). The microscopic examination of cerebrospinal fluid revealing trophozoites with spikes points to Acanthamoeba species as the causative organism. Infections caused by Acanthamoeba are typically observed in immunocompromised rather than immunocompetent individuals.

Acanthamoeba:

These species are free-living amoebae that infect CNS, skin and eye.

- Morphology: Cyst and trophozoite forms. There is no flagellated form.
- Infective form: Both cysts and trophozoites (Option B).
- Mode of transmission: Entry through inhalation, ingestion, and cutaneous/ ocular contaminated lens solution.
- It causes (Option A): GAE (granulomatous amoebic encephalitis) in immunocompromised patients. Keratitis among contact lens users.
- GAE (granulomatous amoebic encephalitis) in immunocompromised patients.
- Keratitis among contact lens users.
- GAE (granulomatous amoebic encephalitis) in immunocompromised patients.
- Keratitis among contact lens users.



Granulomatous Amoebic Encephalitis (GAE)

GAE has an insidious onset, and the incubation period varies from several weeks to months.

- Symptoms: Confusion, dizziness, nausea, headache, stiff neck and rarely seizure and hemiplegia.
- In HIV/immunocompromised patients: In addition to GAE, Acanthamoeba produces nasal ulcers, cutaneous ulcers and musculoskeletal abscesses.
- Pathology: Focal granulomatous lesions in the brain.
- CSF analysis: Lymphocytosis is seen. However, in patients with AIDS, no cells are seen in CSF.
- Diagnosis: Brain biopsy: shows trophozoites and cysts.
- Brain biopsy: shows trophozoites and cysts.
- Brain biopsy: shows trophozoites and cysts.

Trophozoite	Cyst

Show spikes (Acantho - Spikes)	Cyst " data-author="Julia Walochnik, Ute Scheikl, Eva■Maria Haller■Schober" data-hash="9555" data-license="CC BY 3.0" data-source="https://www.researchgate.net/figure/Acanthamoeba-cysts-A-morphological-group-II-polygonal-with-wrinkled-outer-cyst-wall_fig4_264124933" data-tags="" src="https://image.prepladder.com/notes/TckGxw2AhSdyjwRW10GS1744390291.png" /> {{caption_text}} Double-walled, with an outer wrinkled cyst wall.

• Culture: NNA (Non-Nutrient Agar) – lawn culture of Escherichia coli.

• Molecular methods: Multiplex real-time PCR.

• Imaging method: CT scan or MRI reveals space-occupying or ring-enhancing lesions in the brain. (Option C)

• Treatment: Pentamidine + Azole + Sulfonamide + Flucytosine.

Vs Balamuthia mandrillaris Infection:

• Also causes Granulomatous Amoebic Encephalitis (GAE).

• Life cycle is similar to Acanthamoeba, and has trophozoites and cysts.

• Unlike Acanthamoeba, Balamuthia encephalitis has been found in immunocompetent individuals (Option D).

• Differences from Acanthamoeba Trophozoite is fish-shaped. The cyst is smoother. It can occur in both immunocompetent and immunocompromised.

• Trophozoite is fish-shaped.

• The cyst is smoother.

• It can occur in both immunocompetent and immunocompromised.

• Trophozoite is fish-shaped.

• The cyst is smoother.

• It can occur in both immunocompetent and immunocompromised.

Solution for Question 11:

Correct Answer: A) Tetracycline

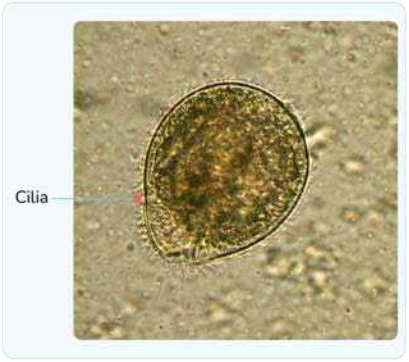
Explanation:

The image shows a ciliated parasite, which suggests a trophozoite of Balantidium coli.

Balantidiasis:

Intestinal infection caused by a protozoan parasite, Balantidium coli. It is the only ciliated parasite of humans.

- Host: Pig (main reservoir). Man is an accidental host.
- Morphology: It exists in two forms – Trophozoites and cysts.
- Infective form: Cyst
- Life cycle:
- Clinical features: Profuse diarrhea and dysentery with mucus and blood, fever, nausea and vomiting.
- Diagnosis (Stool microscopy): Trophozoites: Detected in acute disease. Oval-shaped, exhibits rotatory motility with the help of cilia.
- Trophozoites: Detected in acute disease. Oval-shaped, exhibits rotatory motility with the help of cilia.
- Trophozoites: Detected in acute disease. Oval-shaped, exhibits rotatory motility with the help of cilia.

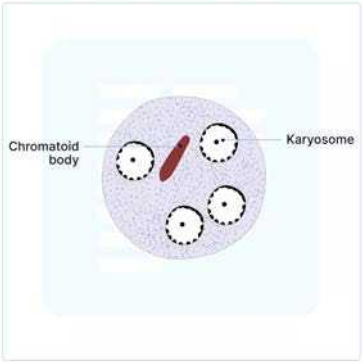
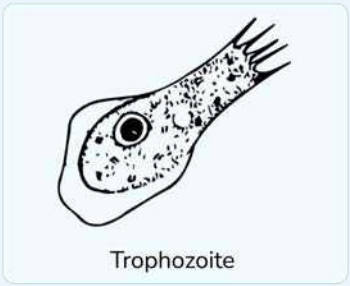


- Cysts: Found in carriers and chronic cases. Round-shaped, infective form.
- Both forms are binucleated, and have a large macronucleus and a small micronucleus.
- Treatment: Tetracycline.

Solution for Question 12:

Correct Answer: B) a-2, b-1, c-3, d-2

Explanation:

Organism	Infective form
Entamoeba histolytica	 ... (content truncated)
Naegleria fowleri	 Trophozoite {{caption_text}}
Acanthamoeba	Both cysts and trophozoites Cyst " data-author="Julia Walochnik, Ute Scheikl, Eva■Maria Haller■Schober" data-hash="9555" data-license="CC BY 3.0" data-source="https://www.researchgate.net/figure/Acanthamoeba-cysts-A-morphological-group-II-polygonal-with-wrinkled-outer-cyst-wall_fig4_264124933" data-tags="" height="366" src="https://image.prepladder.com/notes/TckGxw2AhSdyjwRW10GS1744390291.png" width="339" /> {{caption_text}}
Balantidium coli	Cyst coli Cyst (Q11 and Q12)" data-author="B. Roy, M. Mondal, M. Talukder, S. Majumder" data-hash="9560" data-license="NA" data-source="https://www.semanticscholar.org/paper/Prevalence-of-Balantidium-coli-in-Buffaloes-at-of-Roy-Mondal/3ed3ea49661685f4a3b7bbb88a5db5fffd4ee006" data-tags="" height="273" src="https://image.prepladder.com/notes/ytuE7sYCSlsI1w1wkYbB1744390628.png" width="330" /> {{caption_text}}

Flagellates & Hemoflagellates

1. A 28-year-old woman presents to her primary care physician with complaints of vaginal discharge and itching for 1 week. Upon examination, a profuse, thin, frothy yellow-green discharge and a strawberry cervix are noted. Which of the following statements regarding the causative organism is NOT true?

A. Flagellated trophozoite is the infective form.

B. It has a characteristic twitching motility.

C. It has one anterior and 4 posterior flagella.

D. It has a single nucleus.

---

2. A 42-year-old man from a rural area in Africa presents to the clinic with a two-week history of fever, headaches, and general malaise. He reports experiencing night sweats and often feels excessively tired during the day. On examination, there is notable posterior cervical lymphadenopathy, as shown below. What is the most likely diagnosis based on the patient's symptoms and examination findings?



A. Sleeping sickness

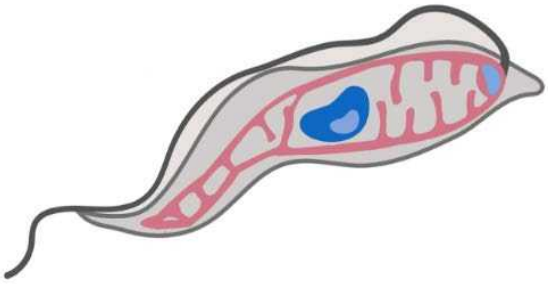
B. Leptospirosis

C. Malaria

D. Dengue fever

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3. Which of the following statements is TRUE regarding the marked structure in the image below?



A. Point of origin of cilia

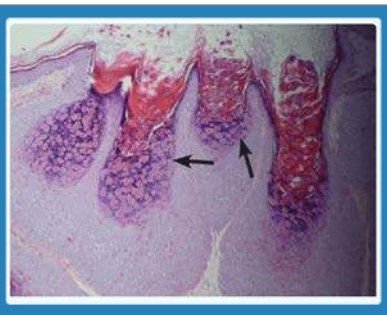
B. Present in organisms that live in the intestines

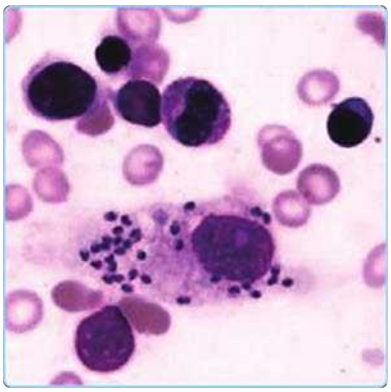
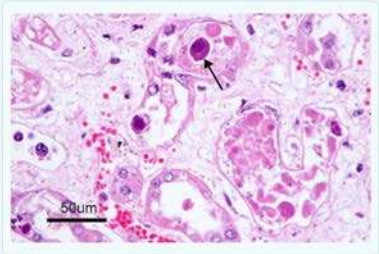
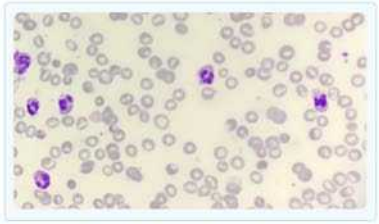
C. Present in Giardia and Trichomonas

D. Contains nonnuclear DNA found in addition to the nucleus

---

4. Match the following 1. Leishman Donovan bodies a) 2. Dohle bodies b) 3. Henderson Petersen bodies c) 4. Cowdry Type A bodies d)

1. Leishman Donovan bodies	 a)

2. Dohle bodies	 ... (content truncated)
3. Henderson Petersen bodies	 c)
4. Cowdry Type A bodies	 d)

A. 1-c, 2-b, 3-d, 4-a

B. 1-d, 2-c, 3-b, 4-a

C. 1-b, 2-d, 3-a, 4-c

D. 1-a, 2-c, 3-b, 4-d

5. Match the following Darting motility a) Listeria monocytogenes Swarming motility b) Trichomonas vaginalis Tumbling motility c) Proteus Jerky motility d) Vibrio

Darting motility	a) Listeria monocytogenes
Swarming motility	b) Trichomonas vaginalis
Tumbling motility	c) Proteus
Jerky motility	d) Vibrio

A. 1-c, 2-b, 3-d, 4-a

B. 1-d, 2-c, 3-a, 4-b

C. 1-b, 2-d, 3-c, 4-a

D. 1-a, 2-c, 3-b, 4-d

---

6. Match the following: Naegleria fowleri a) Flagella Trichomonas cruzi b) Cilia Plasmodium c) Pseudopodia Balantidium coli d) Non-motile

Naegleria fowleri	a) Flagella
Trichomonas cruzi	b) Cilia
Plasmodium	c) Pseudopodia
Balantidium coli	d) Non-motile

A. 1-d, 2-b, 3-c, 4-a

B. 1-a, 2-c, 3-b, 4-d

C. 1-b, 2-c, 3-d, 4-a

D. 1-c, 2-a, 3-d, 4-b

---

7. A 28-year-old male returns to his rural village after spending time in a forested area known for sandfly infestations. He presents to the clinic with a painless ulcer on his forearm that has gradually increased in size over the past month, as shown in the image below. The ulcer has raised, indurated borders and a necrotic base. The patient has no fever or systemic symptoms. What is the most likely diagnosis

for this patient's condition?



A. Mucocutaneous Leishmaniasis

B. Cutaneous Leishmaniasis

C. Leishmaniasis Recidivans

D. Tuberculous Ulcer

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8. A 30-year-old female, previously treated for visceral leishmaniasis six months ago, now presents with multiple erythematous, hypopigmented, slightly scaly papules and plaques on her face, neck, and arms. The lesions are painless, gradually increasing in number, and there are no associated systemic symptoms. What is the most likely diagnosis?



A. Post-Kala-Azar Dermal Leishmaniasis

B. Leprosy

C. Cutaneous Leishmaniasis

D. Psoriasis

9. A 25-year-old male presents to the outpatient clinic with a 2-week history of fever, abdominal distension, fatigue, and significant weight loss. A rapid diagnostic test using the rK39 dipstick comes back positive. What is the recommended first-line drug for treating this patient's condition?

A. Liposomal Amphotericin B

B. 10% Charcoal in Sulfuric Acid

C. Glucocorticoids

D. Pentamidine

10. A 20-year-old female from Africa presents with hepatosplenomegaly, pallor, and generalised lymphadenopathy. Given her clinical features, there is a high suspicion of kala-azar. Which laboratory tests can be used to confirm the diagnosis? rk39 serological test Bone marrow aspirate Splenic aspirate Liver biopsy

A. 1 and 4

B. 1,2,3

C. 2,3,4

D. Only 1

11. A 10-year-old male with a three-month history of fever, weight loss, fatigue, night sweats, and abdominal distention, along with splenomegaly, anemia, leukopenia, and thrombocytopenia, recently returned from rural India, where he

was exposed to the vector given below. What is the most likely diagnosis?



A. Malaria

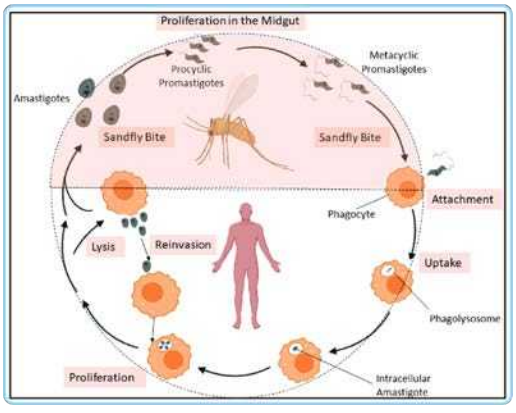
B. Dengue Fever

C. Kala Azar

D. Tuberculosis

---

12. What is the infective form of Leishmania donovani?



A. Amastigote

B. Promastigote

C. Epimastigote

D. Trypomastigote

13. Which of the following statements accurately describes the forms of Trypanosoma brucei that can be found in the blood of an infected individual?

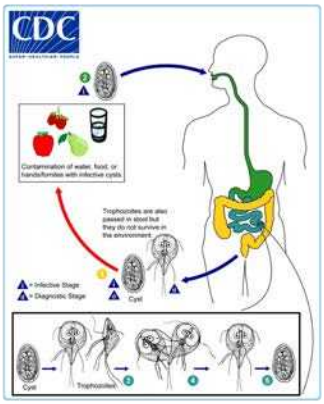
A. The trypomastigote form is the only stage present in the blood.

B. The epimastigote form is commonly found in human blood during acute infection.

C. The procyclic form is typically present in the bloodstream of infected humans.

D. Both the trypomastigote and epimastigote forms can be found in human blood.

14. Which organism does the following life cycle correspond to?



A. Giardia lamblia

B. Entamoeba histolytica

C. Trichomonas vaginalis

D. Balantidium coli

15. A 35-year-old woman with a travel history to South America presents to the clinic with complaints of fatigue, mild chest pain, and palpitations. She reports a history of fever, malaise, and swelling around her right eye 11 months ago, which resolved spontaneously. On examination, her heart rate is irregular, and she has mild cardiomegaly on chest X-ray. Which treatment options are most appropriate for her at this stage?

A. Benznidazole

B. Nifurtimox

C. Supportive care and monitoring

D. Antiarrhythmic medications only

---

16. Which of the following culture media is used to diagnose T. cruzi?

A. Sabouraud Dextrose Agar

B. Blood Agar

C. Yager's LIT Medium (Liver Infusion Tryptose)

D. MacConkey Agar

---

17. A 50-year-old man presents with persistent fatigue, chest pain, and difficulty breathing. He grew up in rural northern Argentina, where a childhood bug bite caused left eye swelling. Examination reveals leg swelling, an S3 gallop, and fatigue. IgG antibodies to Trypanosoma cruzi are detected by ELISA. Which of the following statements are true regarding this condition? Autoimmune hypothesis is responsible for tissue injury seen in the chronic form of disease. Benznidazole is considered a drug of choice for the chronic form of disease. Chronic form of disease has cardiac and gastrointestinal involvement. Peripheral blood microscopy can be used to diagnose chronic disease. The acute form of disease presents with a generalised rash.

A. 1, 3

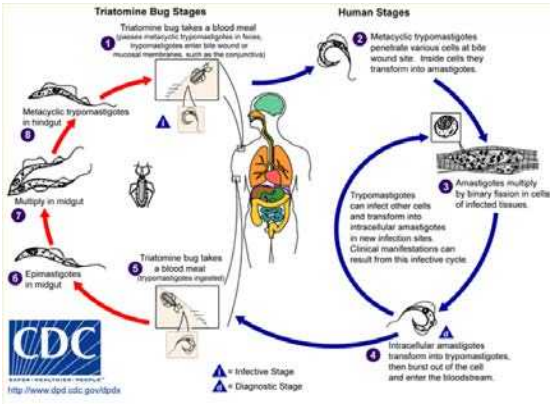
B. 2, 4

C. 1, 5

D. 3, 5

---

18. Which organism does this life cycle belong to?



A. Trypanosoma cruzi

B. Leishmania donovani

C. Plasmodium falciparum

D. Entamoeba histolytica

---

19. What is the drug of choice to treat Colpitis macularis caused by Trichomonas vaginalis?

A. Metronidazole 2g single dose

B. Tinidazole 5g single dose

C. Metronidazole 200 mg for 7 days

D. Topical Nystatin

---

20. A 32-year-old woman presents to her primary care physician with vaginal discharge and itching for 1 week. Examination findings are provided below. Which of the following culture media cannot be used to identify the causative organism in this case?



A. Diamond medium

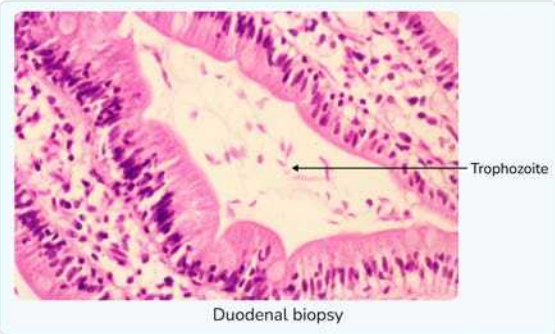
B. Trussell and Johnson medium

C. Novy Nicolle McNeal medium

D. Feinberg Whittington medium

---

21. A 30-year-old HIV-positive male patient presented with a chronic history of watery diarrhoea. The histopathology of the duodenum shows trophozoites in the duodenal wall, as shown in the picture below. Which of the following features is not seen in this condition?





A. Blood in stool

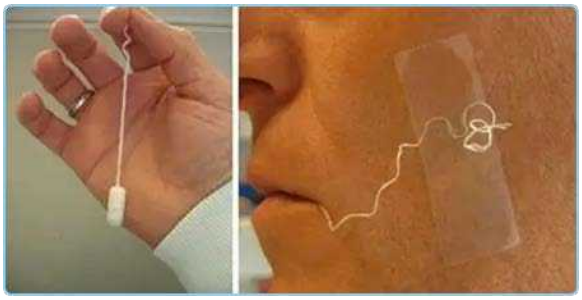
B. Iron deficiency

C. Steatorrhea

D. Flatulence and belching

---

22. A 28-year-old male patient is undergoing the test in the image below. Which of the following statements is true regarding the given image?



A. The Entero Test is used to diagnose giardiasis by obtaining duodenal specimens.

B. The String Test is used to diagnose amoebiasis by detecting trophozoites in the stool.

C. The Entero Test involves swallowing a coiled thread to diagnose echinococcosis by examining the liver.

D. The String Test is still widely used today due to its low cost and high diagnostic accuracy.

---

23. A 28-year-old patient presents with watery diarrhea, abdominal cramps, and dehydration. A string test reveals a Giardia lamblia infection. After being treated with metronidazole, the patient does not fully respond, raising concerns about resistance. What would be the most appropriate next-line treatment?

A. Tinidazole

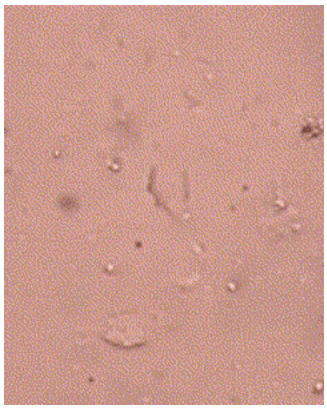
B. Paromomycin

C. Doxycycline

D. Trimethoprim-sulfamethoxazole

---

24. The microscopic examination of a wet mount preparation of stool from a patient with diarrhoea is shown below. Which organism is likely to exhibit this type of motility?



A. Vibrio cholera

B. Giardia lamblia

C. Campylobacter jejune

D. Listeria monocytogenes

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	3
Question 5	2
Question 6	4
Question 7	2
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	2
Question 13	1
Question 14	1
Question 15	1
Question 16	3
Question 17	1
Question 18	1
Question 19	1
Question 20	3
Question 21	1
Question 22	1

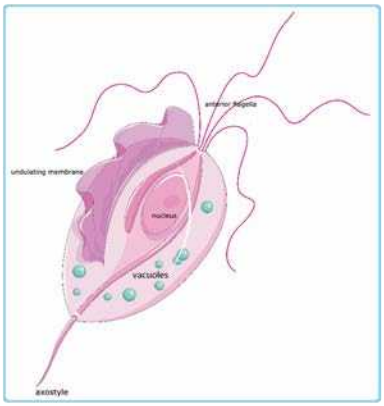
Question 23	1
Question 24	2

Solution for Question 1:

The given clinical scenario, with a profuse, thin, frothy yellow-green discharge and a strawberry cervix, describes a case of Trichomoniasis, with 4 anterior flagella and 1 lateral flagellum (5 flagella).

Trichomoniasis:

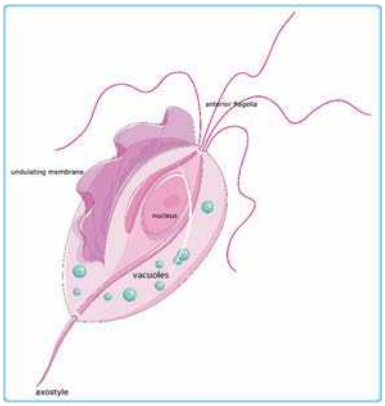
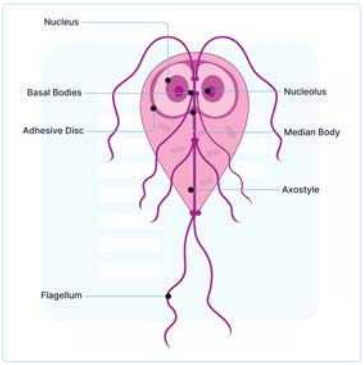
- It is the most common parasitic sexually transmitted infection (STI) caused by a flagellated parasite, Trichomonas vaginalis.
- It has only the trophozoite stage; there is no cyst stage.
- Trophozoite has two forms: Flagellated trophozoite: Infective and diagnostic form (Option A). Amoeboid trophozoite: Actively replicating form (found in the tissue feeding stage).
- Flagellated trophozoite: Infective and diagnostic form (Option A).
- Amoeboid trophozoite: Actively replicating form (found in the tissue feeding stage).
- Flagellated trophozoite: Infective and diagnostic form (Option A).
- Amoeboid trophozoite: Actively replicating form (found in the tissue feeding stage).



Trophozoite of Trichomonas vaginalis

- It is a pear-shaped organism.
- It shows characteristic jerky or twitching motility in saline mount preparation (Option B).
- It bears five flagella—four anterior flagella and one lateral flagellum called as recurrent flagellum.

• It has a single nucleus containing a central karyosome (Option D).

Trophozoite		
Feature	Trichomonas vaginalis	Giardia lamblia
Shape	Pear-shaped or ovoid	Heart-shaped or pyriform (tennis racket-shaped)
Size	10-30 µm in length, 5-10 µm in breadth	15 µm long, 9 µm wide, 4 µm thick
Flagella	Five flagella- Four anterior flagella and one along the undulating membrane	Four pairs of flagella arising from the blepharoplast
Axostyle	Prominent, runs the full length and projects posteriorly like a tail	One axostyle running along the midline
Cytoplasm	Contains prominent siderophilic granules along the axostyle and costa	It has two parabasal bodies
Movement	Rapid, jerky, or twitching motion	Slow oscillation resembling a "falling leaf" movement
Other Structures	Undulating membrane which reaches up to the middle of the body	Concave sucking disk on the ventral side for attachment to mucosa
Images	 ... (content truncated)	 ... (content truncated)

Solution for Question 2:

Sleeping sickness is a parasitic disease caused by protozoan parasites of the genus Trypanosoma, transmitted to humans through the bite of infected tsetse flies. It primarily affects rural populations in sub-Saharan Africa.

• Vector: Both forms are transmitted by tsetse flies (Glossina species), which bite humans to feed on their blood. The flies are typically found in rural areas near rivers and lakes.

• Reservoir Hosts: T. brucei rhodesiense is often associated with wild and domestic animals serving as reservoir hosts, while T. brucei gambiense is primarily human-centric.

• Clinical features: Symptoms may appear weeks to months after infection and include: Early stage: Fever, Headaches, fatigue and Joint pain Winterbottom sign- Swelling of lymph nodes (particularly posterior cervical lymph nodes in T. brucei gambiense). Late stage: As the disease progresses, especially in T. brucei gambiense, symptoms may include - Neurological involvement leads to confusion, personality changes, and sleep disturbances (insomnia or excessive daytime sleepiness) Ataxia (loss of coordination) Coma

• Early stage: Fever, Headaches, fatigue and Joint pain

• Winterbottom sign- Swelling of lymph nodes (particularly posterior cervical lymph nodes in T. brucei gambiense).

• Late stage: As the disease progresses, especially in T. brucei gambiense, symptoms may include -

• Neurological involvement leads to confusion, personality changes, and sleep disturbances (insomnia or excessive daytime sleepiness)

• Ataxia (loss of coordination)

• Coma

• Early stage: Fever, Headaches, fatigue and Joint pain

• Winterbottom sign- Swelling of lymph nodes (particularly posterior cervical lymph nodes in T. brucei gambiense).

• Late stage: As the disease progresses, especially in T. brucei gambiense, symptoms may include -

• Neurological involvement leads to confusion, personality changes, and sleep disturbances (insomnia or excessive daytime sleepiness)

• Ataxia (loss of coordination)

• Coma

In T. brucei rhodesiense, the disease can progress more rapidly, with severe symptoms developing in weeks.

• Serological Tests: Detection of antibodies against the parasites (e.g., ELISA).

• Microscopy: Examination of blood, lymphatic fluid, or cerebrospinal fluid (CSF) for the presence of the parasites.

- PCR

Treatment

- Early Stage (no CNS involvement): Pentamidine: Used for T. brucei gambiense. Suramin: Used for T. brucei rhodesiense.
- Pentamidine: Used for T. brucei gambiense.
- Suramin: Used for T. brucei rhodesiense.
- Late Stage (CNS involvement): Eflornithine: Preferred for T. brucei gambiense, particularly if the patient cannot tolerate melarsoprol. Melarsoprol: Used for late-stage T. brucei rhodesiense and can be toxic, leading to side effects like encephalopathy.
- Eflornithine: Preferred for T. brucei gambiense, particularly if the patient cannot tolerate melarsoprol.
- Melarsoprol: Used for late-stage T. brucei rhodesiense and can be toxic, leading to side effects like encephalopathy.
- Pentamidine: Used for T. brucei gambiense.
- Suramin: Used for T. brucei rhodesiense.
- Eflornithine: Preferred for T. brucei gambiense, particularly if the patient cannot tolerate melarsoprol.
- Melarsoprol: Used for late-stage T. brucei rhodesiense and can be toxic, leading to side effects like encephalopathy.

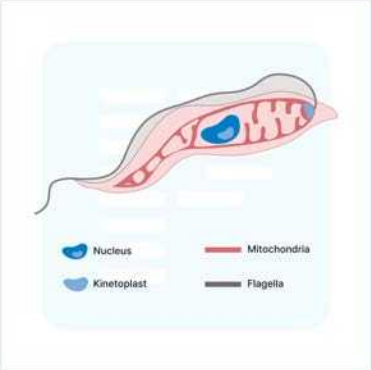
Leptospirosis (Option C): This can present with fever and systemic symptoms, but it usually does not cause significant cervical lymphadenopathy or distinctive sleep pattern changes.

Malaria (Option B): While malaria presents with fever and malaise, it typically involves symptoms such as chills and rigours, and does not usually lead to the sleep disturbances seen in sleeping sickness.

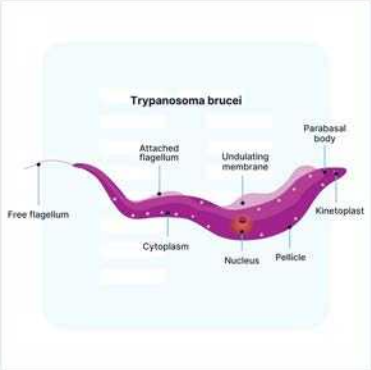
Dengue fever(Option D): This can cause fever and malaise but is less likely to produce the specific symptoms of sleep disturbances and lymphadenopathy described.

Solution for Question 3:

The given picture depicts a Kinetoplast.

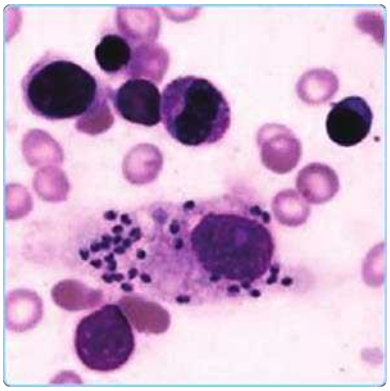
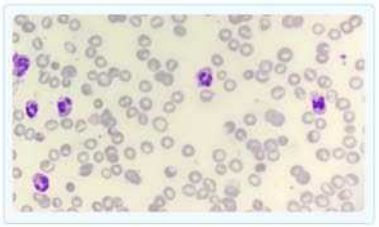
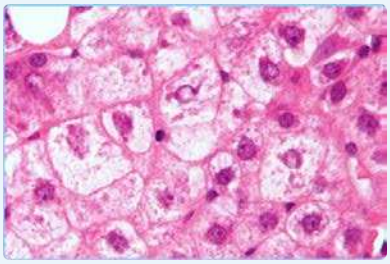


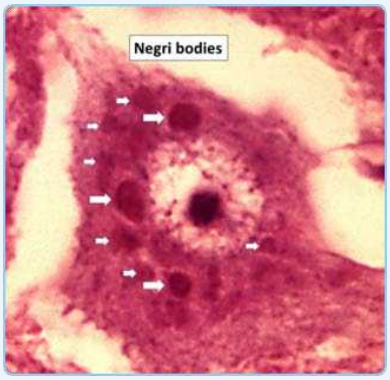
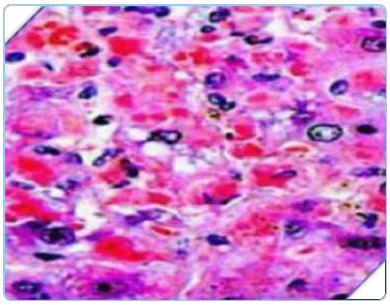
The kinetoplast is nonnuclear DNA found in addition to the nucleus, seen in trypanosomes. The flagellum originates near the kinetoplast, and its point of origin is called the basal body. (Option A ruled out)



Characteristics	Kinetoplastida	Flagellates without Kinetoplast
Possess a kinetoplast	Yes	No
Flagella	Single flagellum	Multiple flagella
Examples	Trypanosomes Leishmania	Giardia (Option C ruled out) Trichomonas
Transmission	Transmitted by blood-sucking insects. Causes systemic or local infections.	Live in the intestine. (Option B ruled out) Called intestinal flagellates.

Solution for Question 4:

Histopathological Bodies in Microbiology	
Leishman Donovan (LD) bodies Found in untreated leishmaniasis. They are light blue, ellipsoid structures measuring 2-4 µm in length. LD bodies have an eccentric nucleus and a smaller kinetosome at the opposite pole. These bodies are located within large macrophages (also known as Wright's cells).	 ... (content truncated)
Dohle bodies Light blue-grey, oval, basophilic staining inclusions were found in the cytoplasm of neutrophils in the peripheral smear. Seen in bacterial infections like tuberculosis, diphtheria, and typhoid, Indicate defective neutrophil production and maturation.	 {{caption_text}}
Hendersen Petersen bodies Molluscum contagiosum, caused by a poxvirus, is characterised by the presence of large, ellipsoid intracytoplasmic inclusions in the stratum spinosum and corneum of infected skin known as Henderson-Peterson bodies. They are made up of nuclear and cytoplasmic protein aggregates and mark the site of viral replication.	 {{caption_text}}
Councilman Bodies Acidophilic inclusion bodies are found in the cytoplasm of hepatocytes in viral infections like hepatitis and yellow fever. These bodies result from hepatocyte apoptosis and are associated with ballooning degeneration of liver cells.	 {{caption_text}}
Cowdry Type A bodies Cowdry bodies are eosinophilic intranuclear inclusions made of nucleic acid and protein, seen in infections like herpes, yellow fever, polio, and adenovirus. Type A (found in herpes and yellow fever) appears as droplet-like masses with clear halos and chromatin	Type A bodies" data-author="Abe, Takashi & Nishi, Shinichi & Furukawa, Tatsuo & Ajioka, Yoichi & Masuko, Masayoshi & Fuse, Ichiro." data-hash="10038" data-license="NA" data-source="https://www.researchgate.net/figure/The-three-types-of-inclusion-bodies-Cowdry-Along-arrow-full-type-intermediatelength_fig3_22191982

margination around the nucleus. Type B (associated with polio and adenovirus) lacks nuclear changes.	6" data-tags="April2025" height="206" src="https://image.prepladder.com/notes/fTePgILjjSVvNo vlp8iL1744458752.png" width="300" /> {{caption_text}}
Negri bodies Negri bodies are eosinophilic, cytoplasmic inclusion bodies found in neurons, primarily in rabies infection. They represent sites of viral replication and are considered a hallmark of rabies in histopathological examination.	 ... (content truncated)
Torres bodies The intranuclear inclusion body seen in liver cells infected with the yellow fever virus is composed of host proteins.	 {{caption_text}}

Solution for Question 5:

MOTILITY	MICROORGANISM
Darting motility	Vibrio, Campylobacter
Swarming motility	Proteus
Tumbling motility	Listeria monocytogenes
Stately motility	Clostridia
Corkscrew motility	Treponema pallidum
Falling leaf motility	Giardia lamblia
Twitching motility	Eiknella corrodens

Jerky motility	Trichomonas vaginalis
Gliding motility	Mycoplasma
Lashing motility	Saprophytic spirochetes

Solution for Question 6:

Organs of Locomotion in Protozoans	
Amoebae Entamoeba histolytica Naegleria fowleri Acanthamoeba	Pseudopodia
Flagellates Trypanosoma Leishmania Giardia Trichomonas	Flagella
Apicomplexa (Sporozoans) Cryptosporidium Isospora Sarcocystis Toxoplasma Plasmodium Babesia	Non-motile
Ciliophora Balantidium coli	Cilia

Solution for Question 7:



Cutaneous leishmaniasis:

Most commonly caused by L.Tropica.

Mode of transmission:

- The most common mode of infection is through the bite of sandflies.
- Infection may be transmitted from man to man or animal to man by direct inoculation or amastigotes.

Incubation period: The incubation period varies from 2 to 8 months.

Pathology:

- Amastigote forms are found in histiocytes and endothelial cells.
- There is an inflammatory granulomatous reaction with infiltration of lymphocyte and plasma cells.
- Early lesions are papular, followed by ulceration necrosis. Papules and ulcers are the main pathological lesions. They heal over months to years, leaving scars.

Clinical features:

L. tropica causes old-world cutaneous leishmaniasis.

Three distinct patterns of old-world CL have been recognised.

- Causing painless dry ulcerating lesions, leading to disfiguring scars. The most important vector is P. sergenti. - It is seen mainly in children in endemic areas and is called as oriental sore or Delhi boil.
- Causes moist ulcers which are inflamed, often multiple, caused by L. major.
- Lesions due to L. major heal more rapidly than L. tropica. P. papatasi is the most important vector.
- The nonulcerative and often diffuse lesions caused by L. aethiopica

Laboratory diagnosis:

• Microscopy: Smear is made from the material obtained from the indurated edge of nodule or sore and stained by Giemsa or Leishman stain.

• Amastigotes are found in large numbers inside the macrophages.

• A definitive diagnosis is made by demonstrating amastigote in the smear collected from the lesion.

• Culture: Promastigote forms can be isolated by culture of the aspirate material in NNN medium.

• Montenegro skin test (MST)

Treatment:

• The specific treatment of CL is the same as VL.

• Antimony-resistant diffuse CL can be treated with pentamidine.

• Topical treatment consists of a paste of 10% charcoal in sulfuric acid or liquid nitrogen.

Mucocutaneous Leishmaniasis (Option B) :

• Mucocutaneous leishmaniasis is a more severe form of leishmaniasis that usually follows cutaneous leishmaniasis.

• L. braziliensis complex causes both mucocutaneous leishmaniasis (espundia) and "CL".

• It involves the nose, mouth and larynx.

• The patient experiences a nodule at the site of the sand fly bite with symptoms consistent with oriental sore.

• Subsequent mucocutaneous involvement leads to nodules inside the nose, perforation of the nasal septum, and enlargement of the nose and lips (espundia)

• The absence of any mucosal involvement in this patient, along with the focus on skin ulcers, suggests that this is not the correct diagnosis.

Leishmaniasis Recidivans (Option C) :

• Leishmaniasis recidivans is a type of lesion seen in persons with a high degree of cell-mediated immunity to the parasite.

• The lesions are chronic with alternating periods of activity and healing, characterised by a central scar with peripheral activity.

• The lesions resemble those of lupus or tuberculoid leprosy.

• Parasites are very scanty in the lesions.

• The Leishmanin test is strongly positive.

• Chemotherapy is not very useful.

• Better results follow the local application of heat.

• However, in this case, the patient presents with an acute, new ulcer rather than a recurrence of a previous lesion, making this option less likely.

Tuberculous Ulcer (Option D): A tuberculous ulcer is caused by Mycobacterium tuberculosis and usually presents as a chronic, painless ulcer, which may occur in patients with systemic tuberculosis. While it can present similarly to cutaneous leishmaniasis, the clinical history (recent travel to an endemic area and the appearance of the ulcer) strongly supports a diagnosis of cutaneous leishmaniasis rather than a tuberculous ulcer.

Solution for Question 8:

Post-kala-azar dermal leishmaniasis is a non ulcerative lesion of the skin seen in India, Nepal, Bangladesh and Sudan.

The lesions are of three types:

Diagnosis of post-kala-azar dermal leishmaniasis:

- The nodular lesions are biopsied, and amastigote forms are demonstrated in stained sections.
- The biopsy material can be cultured, or animal inoculation can be done.
- Immunodiagnosis has no role in the diagnosis of PKDL.

Treatment of post-kala-azar dermal leishmaniasis:

- Liposomal amphotericin-8 (AmBisome) 2.5 mg/ kg/ day for 20 days

or

- sodium stibogluconate (SSG) 20 mg/ kg/day for 40- 60 days is given.

Leprosy (Option B) : Leprosy, caused by Mycobacterium leprae, presents with skin lesions, peripheral nerve damage, and is characterized by numbness and weakness. Although some skin manifestations can resemble those of PKDL, the systemic symptoms and characteristic nerve involvement help differentiate it.

Cutaneous Leishmaniasis (Option C): Cutaneous leishmaniasis presents with ulcerative lesions or nodules, typically following a sandfly bite. Unlike PKDL, these lesions do not usually appear months to years after visceral leishmaniasis treatment and are associated with the presence of Leishmania parasites in the skin.

Psoriasis (Option D): Psoriasis is a chronic autoimmune skin condition characterised by red, scaly patches and is often itchy. While it can present with similar erythematous plaques, the patient's history of recent visceral leishmaniasis and the specific characteristics of her lesions (hypopigmented and non-itchy) make psoriasis an unlikely diagnosis in this case.

Solution for Question 9:

Liposomal Amphotericin B, is the recommended first-line treatment for visceral leishmaniasis.

Treatment for Kala-azar:

- Sodium stibogluconate (100 mg of Sbv/mL) (SSG)
- Meglumine antimoniate (85 mg of Sbv/mL).
- Dosage: The daily dose is 20 mg/ kg by rapid intravenous (TV) infusion or intramuscular (IM) injection for 20- 30 days.
- Dosage: 0.75-1.0 mg/ kg on alternate days for a total of 15 infusions.
- Note: Fever with chills is almost seen in all patients, using amphotericin-8 infusions.
- Dose being 3 mg/ kg daily.
- By using liposomal amphotericin-B, higher doses can be given, improving the cure, without toxicity.
- Current recommendation in India is 10 mg/ kg single dose.
- Dosage: It is given in a dose of 11 mg/ kg daily for 21 days.
- Dosage: 50 mg daily for 28 days for patients weighing less than 25 kg, and twice daily for patients weighing more than 25 kg.

Prophylaxis:

- Early detection and treatment of all cases.
- Integrated insecticidal spraying to reduce sandfly population.
- Destruction of animal reservoir host in cases or zoonotic kala-azar.

Regime of coadministered drug include:

- AmBisome + Paromomycin
- AmBisome + Miltefosine
- Paromomycin + Miltefosine

10% Charcoal in Sulfuric Acid (Option B) is administered topically for Cutaneous leishmaniasis not for visceral leishmaniasis.

Glucocorticoids (Option C) : Glucocorticoids are not used to treat visceral leishmaniasis. They may be used in respiratory complications associated with cutaneous leishmaniasis.

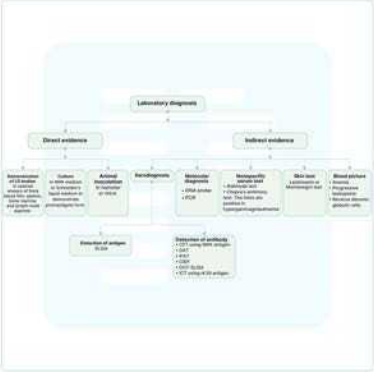
Pentamidine (Option D) : Pentamidine is used for certain infections but is not a first-line treatment for visceral leishmaniasis. It can be used for antimony resistant diffuse cutaneous leishmaniasis.

Solution for Question 10:

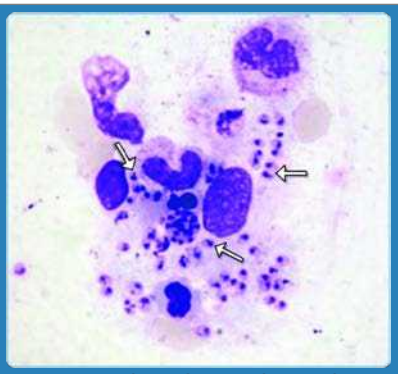
rk39 serological test, bone marrow aspirate and splenic aspirate are 3 important investigations done for the diagnosis of kala-azar.

Diagnosis of Kala-azar :

Depends upon direct and indirect evidences:



CFT:complement fixation test; CIEP: counter immunoelectrophoresis; DAT: direct agglutination test; DNA: deoxyribonucleic acid; ELISA: enzyme linked immunosorbent assay; ICT: immunochromatographic test; IFAT: indirect immunofluorescent antibody test; LD: Leishman-Donovan; NNN: Novy, MacNeal and Nicolle; PCR: polymerase chain reaction; rK39: recombinant kinesin 39

Microscopy	 Demonstration of amastigotes in smears of tissue aspirates is the gold standard for diagnosis of kala-azar / VL ... (content truncated)
Culture	Media: NNN (Novy, MacNeal and Nicolle).
Serodiagnosis	Detection of antigen: ELISA and PCR Detection of antibody: rk39 test: A specific rapid immunochromatographic test (ICT) method for antibodies using a recombinant leishmanial antigen rk 39. The sensitivity of the test is 98% and specificity is 90%. Complement fixation test Indirect immunofluorescent antibody test (IFAT) Counter immunoelectrophoresis (CIEP) ELISA and DOT-ELISA Direct agglutination test (DAT)
Non-specific serum tests	Napier aldehyde test Chopra's antimony test
Skin test	 Leishmanin skin test- Montenegro test: Delayed hypersensitivity test. Positive result is indicated by induration and erythema of 5 mm or more after 48-72 hours. Positive result indicates prior exposure to the leishmanial parasites. In active kala-azar, this test is negative and becomes positive usually 6-8 weeks after cure from the disease. ... (content truncated)

Complete blood count	It shows normocytic normochromic anemia and thrombocytopenia. Leukocyte count reveals leukopenia. The ratio of leukocyte to erythrocyte is greatly altered and may be about 1:200 to 1:100 (normal 1:750). The serum shows hypergammaglobulinemia and a reversal of the albumin: globulin (A: G) ratio. Liver function tests show mild elevations of liver enzymes. Erythrocyte sedimentation rate is elevated.

Liver biopsy is not a safe procedure and carries a risk of haemorrhage.

Solution for Question 11:

The given image of the vector is a sandfly. The patient has a persistent fever, significant weight loss, splenomegaly, anaemia, and leukopenia, and has a history of exposure in an endemic area with exposure to sandflies, making Kala Azar the most likely diagnosis, also known as Visceral leishmaniasis.

Clinical features of kala-azar:

- Onset: Insidious.
- The clinical illness begins with high-grade fever which may be remittent with twice daily spikes or intermittent or less commonly continuous.
- Night sweats
- Splenomegaly starts early and is progressive and massive. It is usually soft and nontender.
- Hepatomegaly is moderate.
- Lymphadenopathy is common in most endemic areas except the Indian subcontinent.
- Skin becomes dry, rough and darkly pigmented (hence, the name kala-azar).
- The hair becomes thin and brittle.
- Cachexia with marked anaemia, emaciation and weight loss is seen.
- Hematological abnormalities: Anemia is most always present and is usually severe leukopenia , thrombocytopenia is associated with epistaxis, gum bleeding, gastrointestinal (GI) bleeding.
- Ascites and edema may occur due to hypoalbuminemia.
- Renal involvement is also common.

- In the late stage of human immunodeficiency virus (HIV) infection, VL can present as an opportunistic infection. HIV coinfection rate is 5% in India and 20% in African countries.
- Secondary infections such as herpes, measles, pneumonia, tuberculosis, and bacillary dysentery may occur.
- Most untreated patients die in about two years due to some intercurrent disease such as dysentery, diarrhoea and tuberculosis.

Child with visceral leishmaniasis with hepatosplenomegaly

Diseases caused by Sandflies:

- Leishmania
- Sandfly fever caused by the Toscano virus
- Bunya virus causing ‘Three-day fever’

Malaria (Option A): While malaria can present with fever and splenomegaly, it typically has more intermittent fever patterns and does not usually cause significant weight loss or leukopenia to the extent described in this case.

Dengue Fever (Option B): Dengue often causes fever and can lead to thrombocytopenia; however, it typically presents with severe joint and muscle pain ("breakbone fever"), and significant splenomegaly is not common.

Tuberculosis (Option D) : While tuberculosis can cause fever, weight loss, and systemic symptoms, it usually presents with respiratory symptoms (cough, hemoptysis) and does not typically lead to splenomegaly or the blood findings seen in this case.

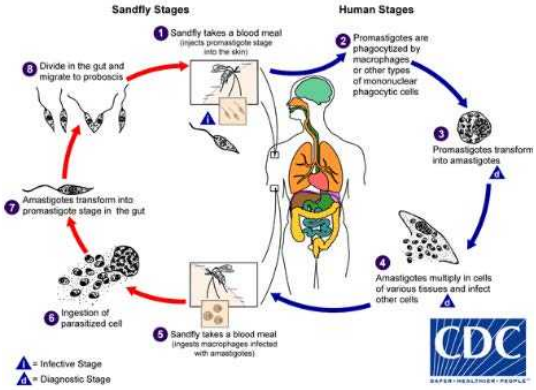
Solution for Question 12:

Promastigote is the infective form of Leishmania donovani.

- A female sandfly (e.g., Phlebotomus, Lutzomyia) bites an infected mammal and ingests blood containing amastigotes.
- Inside the sandfly’s midgut, amastigotes transform into promastigotes, which multiply and migrate to the proboscis.
- When the sandfly bites a new host, it injects the infective promastigotes.
- Promastigotes are engulfed by macrophages and transform into amastigotes, which are non-flagellated and replicate intracellularly.

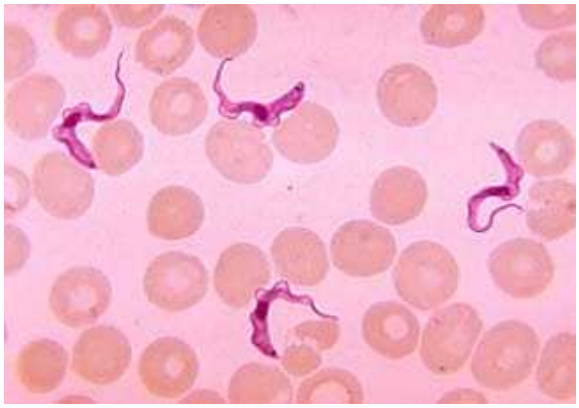
• Amastigotes multiply within macrophages, causing cell lysis and releasing new amastigotes to infect other macrophages.

• If an uninfected sandfly bites the infected host, it ingests amastigotes, and the cycle continues.



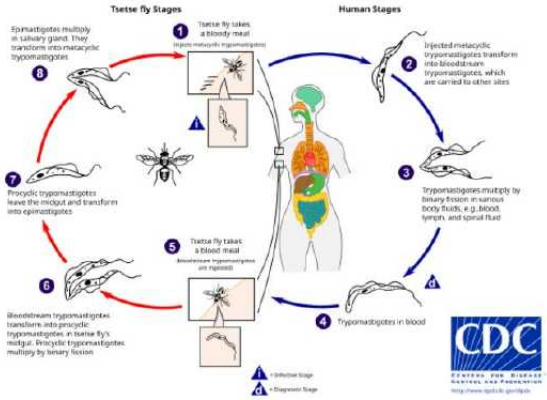
Solution for Question 13:

• Trypomastigote Form: This is the infective form of Trypanosoma brucei found in the bloodstream of infected humans. It is elongated and flagellated.



• Epimastigote form: This form is primarily found in the tsetse fly (not present in human blood).

• Procyclic form: This form occurs in the tsetse fly's midgut (not found in the human bloodstream).



• Serological Tests: Detection of antibodies against the parasites (e.g., ELISA).

• Microscopy: Examination of blood, lymphatic fluid, or cerebrospinal fluid (CSF) for the presence of the parasites.

• PCR

Treatment

• Early Stage (no CNS involvement): Pentamidine: Used for T. brucei gambiense. Suramin: Used for T. brucei rhodesiense.

• Pentamidine: Used for T. brucei gambiense.

• Suramin: Used for T. brucei rhodesiense.

• Late Stage (CNS involvement): Eflornithine: Preferred for T. brucei gambiense (if the patient cannot tolerate melarsoprol). Melarsoprol: Used for late-stage T. brucei rhodesiense and can be toxic, leading to side effects like encephalopathy.

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• Melarsoprol: Used for late-stage T. brucei rhodesiense and can be toxic, leading to side effects like encephalopathy.

	West African Trypanosomiasis	East African Trypanosomiasis
Parasite	Trypanosoma gambiense	Trypanosoma rhodesiense
Vectors	Glossina palpalis	Glossina morsitans
Primary reservoirs	Man-to-man transmissio n	Wild game animals

Illness	Chronic	Acute & rapidly progressive Febrile paroxysms are more frequent Myocarditis is more common & death may occur due to congestive heart failure
Duration of illness	Months to years	Less than 9 months
Lymphadenopathy (Winterbottom's sign)	Prominent	Minimal
CNS involvement	Late	Early
Parasitemia	Low	High
Investigations	Microscopy- Trypomastigote in blood Rodent inoculation- Not useful	Microscopy- Trypomastigote in blood Rodent inoculation-Diagnostic
Treatment	Pentamidine Eflornithine Melarsoprol	Suramin Melarsoprol

Solution for Question 14:

• The life cycle provided belongs to Giardia lamblia.

• It includes 2 stages: Trophozoite (or vegetative form) Cysts (or cystic form).

• Trophozoite (or vegetative form)

• Cysts (or cystic form).

• Trophozoite (or vegetative form)

• Cysts (or cystic form).

• Infective form: Mature cyst

• Transmission: Feco oral route.

• Habitat: Mucosa of duodenum and upper ileum.

• Ingesting a minimum of 10 cysts can cause an infection in a person.

• Within half an hour of ingestion, the cyst hatches out into two trophozoites, which multiply successively by binary fission and colonise the duodenum.

• During unfavorable condition , encystment occurs and cyst are passed in stool and remain viable in soil and water for several weeks.

• Life cycle: Host (only human) → Contaminated food and water (faeco-oral route) → Cyst (ingested form) → excystation occurs → Trophozoite.

• Diagnosis: Sample: Stool Microscopy: Cyst and trophozoite are seen. Trophozoite: 2 nuclei and four pairs of flagella with pear-shaped or tennis racket-shaped. Cyst: 1-4 nuclei and no flagella. String test: A capsule will have a string, and the capsule is swallowed, then the capsule goes to duodenum and dissolves, then the string is pulled back and, a microscopic examination is done. Duodenal biopsy: Giardia lamblia is a luminal organism Organisms hanging in the lumen Sickle shaped organism is seen. Falling leaf motility is shown.

• Sample: Stool

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• Duodenal biopsy: Giardia lamblia is a luminal organism Organisms hanging in the lumen Sickle shaped organism is seen. Falling leaf motility is shown.

• Giardia lamblia is a luminal organism

• Organisms hanging in the lumen

• Sickle shaped organism is seen.

• Falling leaf motility is shown.

• Sample: Stool

• Microscopy: Cyst and trophozoite are seen. Trophozoite: 2 nuclei and four pairs of flagella with pear-shaped or tennis racket-shaped. Cyst: 1-4 nuclei and no flagella.

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• Giardia lamblia is a luminal organism

• Organisms hanging in the lumen

- Sickle shaped organism is seen.
- Falling leaf motility is shown.
- Trophozoite: 2 nuclei and four pairs of flagella with pear-shaped or tennis racket-shaped.
- Cyst: 1-4 nuclei and no flagella.
- Giardia lamblia is a luminal organism
- Organisms hanging in the lumen
- Sickle shaped organism is seen.
- Falling leaf motility is shown.
- Treatment: Metronidazole Tinidazole (single dose of 2g)
- Metronidazole
- Tinidazole (single dose of 2g)
- Metronidazole
- Tinidazole (single dose of 2g)

Solution for Question 15:

This patient's presentation is consistent with chronic Chagas disease, caused by the protozoan parasite Trypanosoma cruzi. The history of fever and eyelid swelling points to a possible acute phase of the infection, while her current symptoms of fatigue, chest pain, palpitations, and cardiomegaly are typical complications of chronic Chagas disease.

Benznidazole is one of the two mainstays of treatment for Chagas disease. It is effective in both the acute and chronic phases, particularly when cardiac symptoms are present. Early treatment can help reduce the risk of progression to severe cardiac disease and complications. The recommended treatment duration is typically 60 days.

Nifurtimox (Option B) is another option for treating Chagas disease, but it is often associated with more significant side effects, including gastrointestinal issues and neurotoxicity. It is reserved for cases where benznidazole is contraindicated or not tolerated.

Supportive care and monitoring (Option C): While essential for managing symptoms and potential complications of Chagas disease (such as arrhythmias), this approach does not address the underlying infection.

Antiarrhythmic medications only (Option D) can help manage the arrhythmias associated with Chagas-related cardiomyopathy, E.g., beta-blockers and digoxin.

Supportive treatment such as pacemakers to manage arrhythmias and surgery for correction of megaesophagus and megacolon may be useful.

Solution for Question 16:

Yager's LIT Medium (Liver Infusion Tryptose)- LIT medium is specifically designed to support the growth of Trypanosoma cruzi in laboratory settings, making it crucial for its diagnosis.

The diagnosis of Trypanosoma cruzi, the causative agent of Chagas disease, involves several methods.

- Clinical Evaluation Symptoms: Fever, fatigue, body aches, and chagoma (swelling at the infection site). Chronic symptoms affect the heart and digestive system. History of travel or residence in endemic areas.
- Symptoms: Fever, fatigue, body aches, and chagoma (swelling at the infection site). Chronic symptoms affect the heart and digestive system.
- History of travel or residence in endemic areas.
- Serological Tests ELISA: Detects antibodies against T. cruzi. IHA: Measures antibodies in the blood. Western Blot: Confirms positive serological results.
- ELISA: Detects antibodies against T. cruzi.
- IHA: Measures antibodies in the blood.
- Western Blot: Confirms positive serological results.
- Microscopic Examination Blood smears can detect the trypomastigote form, especially during the acute phase. Giemsa or Wright's stain enhances visibility.
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- Giemsa or Wright's stain enhances visibility.
- Culture Methods Yager's LIT Medium: Cultures T. cruzi from blood or tissue samples, helpful in chronic cases.
- Yager's LIT Medium: Cultures T. cruzi from blood or tissue samples, helpful in chronic cases.
- Molecular Methods PCR: Used for DNA detection.

• PCR: Used for DNA detection.

• Symptoms: Fever, fatigue, body aches, and chagoma (swelling at the infection site). Chronic symptoms affect the heart and digestive system.

• History of travel or residence in endemic areas.

• ELISA: Detects antibodies against T. cruzi.

• IHA: Measures antibodies in the blood.

• Western Blot: Confirms positive serological results.

• Blood smears can detect the trypomastigote form, especially during the acute phase.

• Giemsa or Wright's stain enhances visibility.

• Yager's LIT Medium: Cultures T. cruzi from blood or tissue samples, helpful in chronic cases.

• PCR: Used for DNA detection.

Sabouraud Dextrose Agar (Option A): This medium is primarily used for cultivating fungi and some bacteria. It is not suitable for growing Trypanosoma cruzi.

Blood Agar (Option B): Blood agar is an enriched medium used for the cultivation of a wide variety of organisms, particularly fastidious bacteria. While it supports some protozoan growth, it is not specifically used for Trypanosoma cruzi diagnosis.

MacConkey Agar (Option D): This medium is selective for Gram-negative bacteria and differentiates lactose fermenters from non-fermenters. It is not suitable for protozoan cultures like Trypanosoma cruzi.

Solution for Question 17:

The clinical presentation with cardiac involvement leading to fatigue, difficulty breathing, and Trypanosoma cruzi antibody detection in the peripheral blood is consistent with a chronic form of Chagas disease.

Chagas disease is caused by the protozoan parasite Trypanosoma cruzi, transmitted by triatomine bugs.

Chagas' disease occurs in four stages.

1. Early Stage Disease: It is characterised by cutaneous and ocular manifestations

• Chagoma: a painful subcutaneous nodule at the site of deposition of bug's faeces, commonly seen on the face.

• Romana's sign: It causes unilateral painless oedema of the eyelid and conjunctivitis.



2. Acute Chagas' disease:

• About 1% of patients, progress to acute-stage disease, characterised by fever, hepatosplenomegaly, and lymphadenopathy (Statement 5)

3. Indeterminate stage:

• It is the asymptomatic phase; lasting for years to decades, before progressing into the symptomatic chronic stage

4. Chronic Chagas' disease:

• The parasite multiples in the muscles (cardiac and GIT) and nervous tissue, producing various types of manifestations.

• Cardiac form: Occurs in 30% of the patients. The patient develops dilated cardiomyopathy, rhythm disturbances and thromboembolism.

• Gastrointestinal form: Megaesophagus and megacolon.

• Autoimmune hypothesis: T. cruzi antigens cross-react with mammalian antigens (molecular mimicry), producing autoantibodies that may be responsible for tissue injury seen in chronic Chagas' disease

Treatment of Chagas disease

• Therapy for Chagas' disease is still unsatisfactory.

• In acute disease, Benznidazole is the drug of choice in Latin America. Nifurtimox or allopurinol is given alternatively.

• In chronic disease, these drugs lack efficacy and may cause many side effects. Supportive treatment, such as pacemakers to manage arrhythmias and surgery to correct megaesophagus and megacolon, may be useful (Statement 2).

In acute Chagas disease, the trypomastigotes are frequently found in peripheral blood, which can be detected by:

- Wet mount preparation
- Thick and thin smear examination.

Antibody detection: Chronic Chagas disease is diagnosed by detecting specific IgG antibodies using ELISA, Western blot and Chagas RIPA (radioimmunoprecipitation assay) (Statement 4).

Solution for Question 18:

This is the life cycle of Trypanosoma cruzi, the causative agent of Chagas disease.

The life cycle involves several stages and hosts:

In the Reduvid bug: Epimastigote and trypomastigote are seen.
In adults: Amastigote and trypomastigote are seen.

T. cruzi trypomastigotes in peripheral blood smear

Solution for Question 19:

Strawberry appearance of vaginal mucosa or cervix (Colpitis macularis) is observed in 2% of patients with Trichomoniasis. It is characterised by small punctate hemorrhagic spots on vaginal or cervical mucosa.

Metronidazole or tinidazole is the drug of choice for Trichomonas vaginalis.

- Standard therapy: 2 g, single dose is usually effective.
- Both sexual partners must be treated simultaneously to prevent reinfection, especially asymptomatic males.
- Resistance to metronidazole is rare but reported (2.5-10%). If the standard therapy fails (with 5-day therapy), consider repeating the treatment course.

Trichomoniasis can be prevented by:

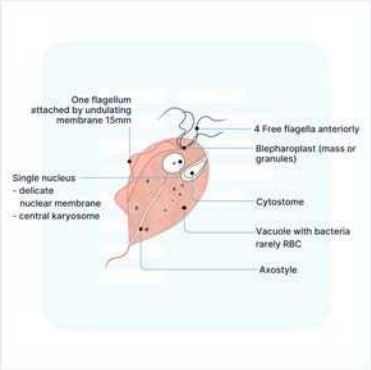
- Treatment of both sexual partners.
- Safe sex practices like the use of condoms.
- Avoidance of sex with an infected person.

Solution for Question 20:

The given scenario, featuring vaginal discharge, itching, and a strawberry cervix, is consistent with a diagnosis of Trichomoniasis caused by Trichomonas vaginalis. This organism cannot be detected using Novy-MacNeal-Nicolle (NNN) medium, which is typically used for detecting Trypanosoma cruzi and Leishmania.

Diagnosis

- Microscopy Wet mount reveals, The twitching (jerky or wobbling or rotatory) motility. Pear shape with a single nucleus 4 anterior flagella, 1 posterior flagella with an undulating membrane.
- Wet mount reveals, The twitching (jerky or wobbling or rotatory) motility. Pear shape with a single nucleus 4 anterior flagella, 1 posterior flagella with an undulating membrane.
- The twitching (jerky or wobbling or rotatory) motility.
- Pear shape with a single nucleus
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- Wet mount reveals, The twitching (jerky or wobbling or rotatory) motility. Pear shape with a single nucleus 4 anterior flagella, 1 posterior flagella with an undulating membrane.
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• Permanent stain Acridine orange, Papanicolaou and Giemsa stains can be used.

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• Acridine orange, Papanicolaou and Giemsa stains can be used.

• DFA (Direct fluorescent antibody) More sensitive than wet mount

• More sensitive than wet mount

• Culture Most sensitive, gold standard investigation. Following culture media can be used: Lash cysteine hydrolysate serum Diamond medium (Option A) Trussell and Johnson medium (Option B) Feinberg Whittington medium (Option D)

• Most sensitive, gold standard investigation.

• Following culture media can be used: Lash cysteine hydrolysate serum Diamond medium (Option A) Trussell and Johnson medium (Option B) Feinberg Whittington medium (Option D)

• Lash cysteine hydrolysate serum

• Diamond medium (Option A)

• Trussell and Johnson medium (Option B)

• Feinberg Whittington medium (Option D)

• Serology Tests like ELISA and Rapid immunochromatographic tests (ICTs) are also introduced.

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• Molecular method DNA hybridisation, PCT, and nucleic acid amplification tests (NAAT) are also available now.

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• More sensitive than wet mount

• Most sensitive, gold standard investigation.

• Following culture media can be used: Lash cysteine hydrolysate serum Diamond medium (Option A) Trussell and Johnson medium (Option B) Feinberg Whittington medium (Option D)

• Lash cysteine hydrolysate serum

• Diamond medium (Option A)
• Trussell and Johnson medium (Option B)
• Feinberg Whittington medium (Option D)
• Lash cysteine hydrolysate serum
• Diamond medium (Option A)
• Trussell and Johnson medium (Option B)
• Feinberg Whittington medium (Option D)
• Tests like ELISA and Rapid immunochromatographic tests (lCTs) are also introduced.
• DNA hybridisation, PCT, and nucleic acid amplification tests (NAAT) are also available now.

Solution for Question 21:

• The clinical scenario and histopathological findings showing pear-shaped trophozoites with two equally sized nuclei and four pairs of flagella strongly suggest Giardia lamblia infection.
• The small intestine morphology is usually normal, however, in heavy infections, there may be villous blunting and intraepithelial infiltrates.
• Clinical features of giardiasis include:
• Diarrhea
• Steatorrhea
• Dull epigastric pain.
• Belching and flatulence.
• Iron deficiency
• Stool will have excess mucus and fat but no blood

Solution for Question 22:

• The given image depicts a String Test or Entero Test, which is used to diagnose giardiasis.

• It is a useful method for obtaining duodenal specimens in giardiasis. (Option B & C ruled out)

• In this test, the patient swallows a coiled thread inside a small, weighted gelatin capsule after attaching the free end of a thread to the cheek.

• The capsule passes through the stomach to the duodenum. After 2 hours, the thread is withdrawn, placed in saline, and shaken.

• The centrifuged deposit of saline is then examined for giardia.

• Nowadays, the Entero test is not performed due to high cost. (Option D ruled out)

Solution for Question 23:

• For Giardia lamblia infection, Metronidazole is a drug of choice. However, some individuals may show resistance to metronidazole, leading to suboptimal treatment outcomes.

• In such cases, an alternative treatment option is tinidazole, which is effective against giardia and offers the advantage of being administered as a single dose.

• Treatment (DOC): Metronidazole Tinidazole (Single dose of 2g)

• Metronidazole

• Tinidazole (Single dose of 2g)

• Metronidazole

• Tinidazole (Single dose of 2g)

Paromomycin (option B) is used to treat intestinal protozoan infection but is not the next-line treatment after metronidazole resistance.

Doxycycline (option C) is an antibiotic commonly used for bacterial infections, but it is not a recommended treatment for giardia.

Trimethoprim-sulfamethoxazole (TMP-SMX)(Option D): Antibiotic combination used for bacterial infection, not giardia.

Solution for Question 24:

The wet mount stool preparation displayed shows an organism exhibiting falling leaf-like motility, characteristic of Giardia lamblia (trophozoite is motile with a slow oscillation along its axis, resembling a falling leaf).

Types of motilities of microorganism:

Motility	Microorganism
Darting motility	Vibrio, Campylobacter
Swarming motility	Proteus
Tumbling motility	Listeria monocytogenes
Stately motility	Clostridia
Corkscrew motility	Treponema pallidum
Falling leaf motility	Giardia
Twitching motility	Eiknella corrodens
Jerky motility	Trichomonas vaginalis
Gliding motility	Mycoplasma
Lashing motility	Saprophytic spirochetes

Coccidian parasites, Plasmodium & Babesia

1. A 30-year-old man diagnosed with Plasmodium falciparum malaria presents with fever and chills after travelling to an endemic area. Which phase of the Plasmodium life cycle occurs in the human host?

A. Formation of sporozoites in the liver

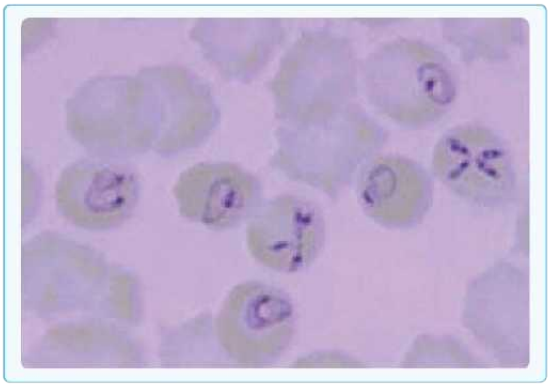
B. Asexual reproduction of merozoites in red blood cells

C. Fertilization of gametocytes in the bloodstream

D. Development of oocysts in RBC

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2. A 38-year-old man presents with a 7-day history of fever, chills, fatigue, and dark urine after a weekend hiking trip in rural New England. He has a temperature of 102°F (38.9°C), mild splenomegaly, and jaundice. Microscopic findings are noted below. Which of the following statements about this condition is true?



A. Rodents are definitive hosts

B. Presence of schizonts in blood smear

C. Sporozoites invade RBCs

D. Sporzoites are the infective form

3. A 28-year-old woman presents to the emergency department drowsy and confused, with a 104°F (40°C) fever, severe headache, and neck stiffness. Recently returned from Southeast Asia, she has tachycardia, hypotension, thrombocytopenia, and elevated liver enzymes. A rapid diagnostic test is positive. What is the drug of choice?

A. Chloroquine

B. Artesunate

C. Primaquine

D. Sulfadoxine

4. A 32-year-old man comes to the emergency department with fever, chills, headache, and muscle pain. His examination shows a temperature of 102°F (39°C), fatigue, splenomegaly, and mild jaundice. A rapid diagnostic test detects the presence of HRP2. Which species is responsible for this finding?

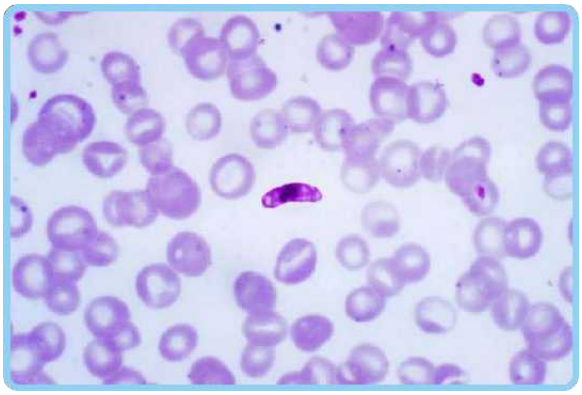
A. Plasmodium vivax

B. Plasmodium falciparum

C. Plasmodium malariae

D. Plasmodium ovale

5. A 28-year-old woman was recently diagnosed with malaria. A microscopic examination of her blood smear showed infected red blood cells is given below. Based on these findings, which of the following plasmodium species has most likely caused malaria in this patient?



A. Plasmodium vivax

B. Plasmodium ovale

C. Plasmodium malariae

D. Plasmodium falciparum

---

6. An 18-year-old man presents to the hospital with a three-day history of high fever along with severe headaches, muscle aches, and chills every 48 hours. After lab testing, he was started on Artesunate-based combination therapy for malaria. Which of the following tests was most likely used as a gold standard for diagnosing malaria?

A. Quantitative buffy coat test

B. Microscopy with Romanowsky stains

C. Dual antigen test

D. Parasite-F test

---

7. A 40-year-old male presents to the emergency department with a high fever, chills, and dark-coloured urine. He recently returned from a trip to a malaria-endemic region in West Africa and self-treated with antimalarial medication. On examination, he is febrile (39.5°C), tachycardic, and has jaundice.

Laboratory investigations show: Hemoglobin: 8 g/dL LDH: Elevated Urinalysis: Dark brown urine, positive for haemoglobin. Total Bilirubin: 4 mg/dL (direct: 1 mg/dL, indirect: 3 mg/dL) Which of the following best explains the diagnosis?

A. Benign malaria

B. Septicemic malaria

C. Algid malaria

D. Blackwater fever

---

8. A 30-year-old male presents to the outpatient clinic with a history of malaria treated six months prior. He reports a sudden onset of fever, chills, and fatigue over the past three days, with symptoms similar to his previous malaria episode. He states that he has been symptom-free since the initial treatment. Which of the following species is most likely responsible for the relapse of Malaria?

A. Plasmodium Knowlesi

B. Plasmodium falciparum

C. Plasmodium malariae

D. Plasmodium vivax

---

9. A 35-year-old man returns from a trip to a malaria-endemic region with a high fever, chills, and headache. Laboratory tests confirm he is infected with Plasmodium falciparum. What is the incubation period for Plasmodium falciparum, and which receptor does it use to bind to red blood cells?

A. 12-17 days; Duffy antigen

B. 12 days; Basigin

C. 28 days; CD147

D. 14-17 days; Glycophorin

---

10. A 28-year-old woman presents with a very high fever, chills, and shivering after returning from a tropical region. Laboratory tests confirm infection with a Plasmodium species. It is noted that infected red blood cells adhere to the endothelium of blood vessels. In which species of Plasmodium is this phenomenon primarily observed?

A. Plasmodium vivax

B. Plasmodium ovale

C. Plasmodium malariae

D. Plasmodium falciparum

---

11. A 28-year-old pregnant woman presents for a routine check-up with mild flu-like symptoms, fatigue, and swollen lymph nodes. She has a cat at home and frequently handles its litter. Serology shows positive IgM and IgG antibodies for a coccidian parasite. Which of the following statements about this condition is correct?

A. It cannot be transmitted vertically.

B. Sporulated oocysts are a non-infective stage.

C. Cats are the intermediate host.

D. Bradyzoites spread via undercooked meat.

---

12. A 28-year-old man returns from a malaria-endemic region and presents with fever, chills, and fatigue. In the context of malaria, which of the following statements correctly identifies the host and the infective form?

A. Female anopheles mosquito; Trophozoites

B. Human; Sporozoites

C. Host: Human; Gametocytes in case of transfusion

D. Host: Human; Merozoites

---

13. A 35-year-old man presents with two weeks of diarrhoea and abdominal pain after working on a pig farm. His stool sample reveals a ciliate protozoan. Which of the following statements is true about the protozoan?

A. It is the smallest protozoan parasite of humans

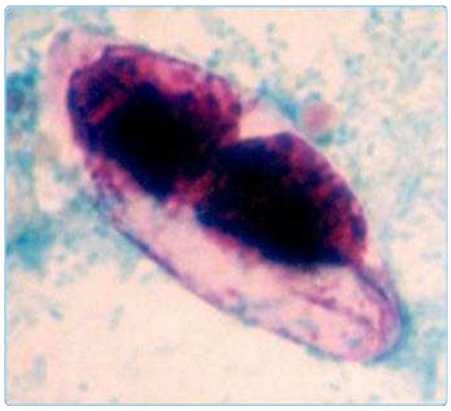
B. The infective form is the trophozoite

C. Tetracycline is the drug of choice

D. Transmission is primarily through direct contact

---

14. A 42-year-old man presents with severe diarrhoea, abdominal cramps, and weight loss. He recently returned from a tropical region and has a history of HIV with a low CD4 count. The stool analysis finding is given in the image. Which of the following statements about the identified organism is not true?



A. Cotrimoxazole is used for the treatment

B. Oocysts are usually between 20-33 µm

C. The organism shows variable autofluorescence

D. The incubation period is 2-14 days

15. Which of the following statements is not true about the life cycle of Cyclospora?

A. Oocyst shed in the faeces sporulate outside the host

B. During excystation, 8 sporozoites are released from 2 sporocyst

C. Sporozoites reproduce asexually in the enterocytes

D. Autoinfection does not occur with Cyclospora

16. Which of the following statements about the life cycle of Cryptosporidium is not true?

A. It completes sexual and asexual cycles in one host

B. Oocysts released in faeces are fully mature and infective

C. Excystation of ingested cyst occurs in the small intestine

D. Thick-walled oocysts can lead to autoinfection within the host

17. Which of the following coccidia parasites is the most common cause of diarrhoea in HIV patients?

A.

B.

C.

D.

18. A 26-year-old pregnant woman in her second trimester presents with eye pain, photophobia, and headache. Fundoscopy reveals a fluffy white lesion and optic disc swelling. To rule out toxoplasmosis, her physician orders a serological test. Which of the following tests is the gold standard for diagnosis of toxoplasmosis?

A. Animal inoculation of amniotic fluid

B. USG of Fetus

C. Sabin - Feldman dye test

D. PCR of amniotic fluid for toxoplasma DNA

---

19. An asymptomatic newborn is evaluated for congenital infections soon after birth as his mother was diagnosed with toxoplasmosis during her third trimester. Serologic tests of the infant also confirm congenital toxoplasmosis. Which of the following findings is least likely to develop in this patient?

A. Chorioretinitis

B. Hydrocephalus

C. Convulsions

D. Periventricular calcifications

---

20. A 28-year-old woman who handles her pet cat and is now pregnant is concerned about serious infections that could affect her fetus, including toxoplasmosis. During which stage of pregnancy is the risk of transplacental infection with toxoplasmosis highest?

A. First trimester

B. Infection before conception

C. Third trimester

D. Throughout the gestation

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	2
Question 4	2
Question 5	4
Question 6	2
Question 7	4
Question 8	4
Question 9	2
Question 10	4
Question 11	4
Question 12	2
Question 13	3
Question 14	4
Question 15	2
Question 16	4
Question 17	2
Question 18	3
Question 19	4
Question 20	3

Solution for Question 1:

In the human host, after sporozoites enter the bloodstream and reach the liver, they develop into merozoites. These merozoites invade red blood cells, where they reproduce asexually, leading to the symptoms of malaria.

Life Cycle of the Malaria Parasite:

- The malaria parasite completes its life cycle in two hosts: Definitive Host: Female Anopheles mosquito Intermediate Host: Humans
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- Intermediate Host: Humans
- The life cycle consists of two main phases: Asexual Phase (in Humans): Aka schizogony, this occurs in two locations: Erythrocytic Schizogony: Inside red blood cells (RBCs) Exoerythrocytic Schizogony: In liver cells (also called the tissue phase) Sexual Phase (in Mosquito): Sexual forms (gametocytes) develop in human RBCs, followed by maturation and fertilisation in the mosquito, leading to the formation of sporozoites through sporogony.
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- Asexual Phase (Schizogony): Human Infection: Infection occurs via the bite of an infective female Anopheles mosquito. Sporozoites: These infective forms are present in the mosquito's salivary glands and are injected into the human bloodstream during feeding. Pre-erythrocytic Stage (Liver Phase): Sporozoites reach the liver, invade hepatocytes, and enlarge. They undergo multiple nuclear divisions, forming multinucleated schizonts (meronts) that contain 2,000 to 50,000 merozoites. Release of Merozoites: Mature liver schizonts rupture after 6-15 days, releasing merozoites into the bloodstream. Erythrocytic Stage: Merozoites invade RBCs via a process of invagination Development in RBCs: Inside the RBC, merozoites lose internal organelles and transform into ring forms (young trophozoites), which feed on haemoglobin and produce malaria pigment (hemozoin). Maturation: Ring forms develop into late trophozoites and then into mature schizonts containing 8-32 merozoites, which are released upon rupture of the RBC

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• Maturation: Ring forms develop into late trophozoites and then into mature schizonts containing 8-32 merozoites, which are released upon rupture of the RBC

• Definitive Host: Female Anopheles mosquito

• Intermediate Host: Humans

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Malaria Parasite" data-author="" data-hash="9625" data-license="" data-source="" data-tags="" height="640" src="https://image.prepladder.com/notes/91TwpN2CgHbfM9zseW4R1744394484.png" width="936" />

• Gametogony: After several erythrocytic cycles, some merozoites develop into sexually differentiated forms, the gametocytes.

• Sexual Phase (Sporogony): When a female Anopheles mosquito consumes parasitised blood, the gametocytes are released in the midgut: Development of gametes: Male gametocytes undergo division to produce microgametes, while female gametocytes mature into macrogametes. Fertilization: A microgamete fertilizes a macrogamete, forming a zygote. Ookinete formation: The zygote transforms into a motile ookinete penetrating the mosquito's stomach wall. Oocyst development: The ookinete develops into an oocyst, where sporozoites are produced. Mature oocyst: Once mature, the oocyst ruptures, releasing sporozoites into the mosquito’s hemocoel and eventually into the salivary glands.

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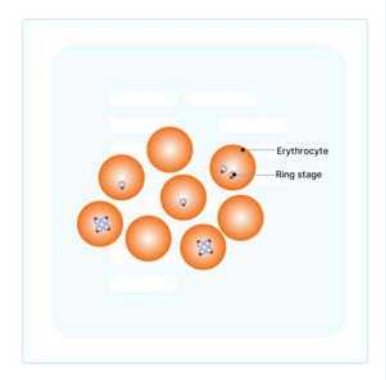
Formation of sporozoites in the liver(Option A): Formation of sporozoites occurs in the mosquito, not in the human host.

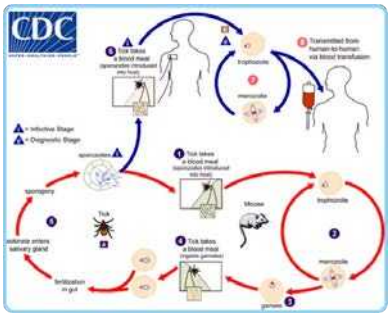
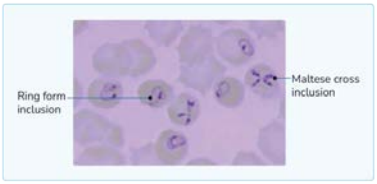
Fertilization of gametocytes in the bloodstream(Option C): Fertilization happens in the mosquito's midgut after a blood meal, not in the human bloodstream.

Development of oocysts in RBCs(Option D): Oocysts develop in the mosquito, not red blood cells; this option is incorrect as oocysts do not form in humans.

Solution for Question 2:

The clinical features of fever, chills, dark urine after hiking (with tick exposure), splenomegaly, and jaundice, along with the Maltese cross appearance in the image, indicate a diagnosis of babesiosis caused by Babesia, where sporozoites are the infective form.

Babesia		
Description	Intraerythrocytic sporozoan parasite	
Morphology	 ... (content truncated)	

Life cycle	 Definitive hosts: Hard ixodid tick (Option A) Intermediate hosts: Rodents or mammal Accidental, dead-end hosts: Human ... (content truncated)	
Clinical features	Malaise, fatigue, fever, chills, myalgia, arthralgia, dry cough and anorexia Neck stiffness, sore throat, abdominal pain, jaundice, and anaemia are less common. Severe babesiosis requires hospitalisation.	
Types	Babesia microti	Babesia divergens
Host: Rodent Mild disease with normal spleen	Host: Cattle Severe disease, splenomectised	
Microscopy	 Tetrad of rings (known as Maltese cross appearance) are observed. {{caption_text}}	
Management	Azithromycin + Atovaquone Clindamycin + Quinine (severe cases)	
Complications	Renal failure Disseminated intravascular coagulation (DIC) Acute respiratory distress syndrome (ARDS) Congestive cardiac failure (CCF)	

Presence of schizonts in blood smear (Option B) indicates a malarial parasite (Plasmodium), whereas they are absent in babesiosis.

Solution for Question 3:

The scenario presenting with fever, headache, neck stiffness, disorientation, hypotension, tachycardia, and a positive rapid test suggests a diagnosis of complicated malaria, for which the drug of choice is intravenous artesunate.

Management of Uncomplicated Malaria

As per NVBDCP guidelines (National Vector Borne Disease Control Programme), the following regimen is followed,

Plasmodium vivax or ovale	Plasmodium falciparum	Mixed infection
Chloroquine + Primaquine Chloroquine 25 mg for 3 days as 10 mg/kg on day 1, 10 mg/kg on day 2 and 5 mg/kg on day 3. Primaquine 0.25 mg/kg body wt for 14 days	Primaquine + ACT ACT (Artemisinin-based combination therapy): All Indian states (except northeastern): Artesunate (4 mg/kg body weight) + sulfadoxine (25 mg/kg body weight) or pyrimethamine (1.25 mg/kg body weight) Northeastern states: Artemether (20 mg) + Lumefantrine (120 mg)	ACT+ Primaquine 0.25 mg per kg body weight daily for 14 days

• ACT for chloroquine-resistant malaria: Artesunate + Mefloquine Artesunate + Amodiaquine Artesunate + Pyronaridine (used in China) Arterolane + Piperaquine

• Artesunate + Mefloquine

• Artesunate + Amodiaquine

• Artesunate + Pyronaridine (used in China)

• Arterolane + Piperaquine

• Artesunate + Mefloquine

• Artesunate + Amodiaquine

• Artesunate + Pyronaridine (used in China)

• Arterolane + Piperaquine

• In Artemisinin allergic individuals: QDC regimen: Quinidine + Doxycycline + Clindamycin

• QDC regimen: Quinidine + Doxycycline + Clindamycin

• QDC regimen: Quinidine + Doxycycline + Clindamycin

Management of Complicated malaria (acute severe malaria)

• Supportive and emergency treatment as needed

• IV Artesunate (drug of choice) (Option B) (Only Artemisinin to be given intravenously; rest are given intramuscularly)

• IM Artemether

• IM Arteether

• IV Quinine in 5% dextrose solution (quinine cause hypoglycemia)

Solution for Question 4:

• The scenario includes fever, chills, splenomegaly, and jaundice, with the detection of HRP-2 (histidine-rich protein-2) in the Rapid Diagnostic Test (RDT), which suggests a diagnosis of malaria caused by Plasmodium falciparum, as HRP-2 is specific to this species.

• It is an immuno-chromatographic antigen detection test.

• Sensitivity: 50-100 parasites per microlitre

• It detects the presence of: Aldolase: All plasmodium spp. (Option A, C, D) pLDH (parasite lactate dehydrogenase): All plasmodium spp. (Option A, C, D) HRP-2 (histidine-rich protein-2): Present only in Plasmodium falciparum (Option B)

• Aldolase: All plasmodium spp. (Option A, C, D)

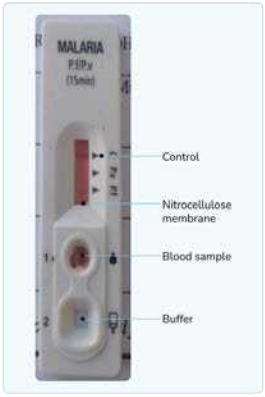
• pLDH (parasite lactate dehydrogenase): All plasmodium spp. (Option A, C, D)

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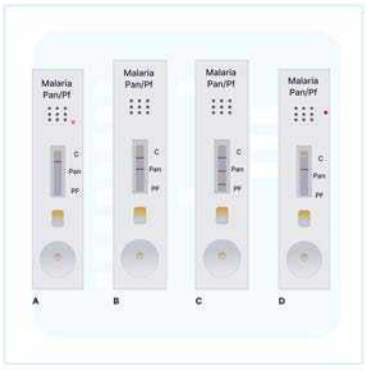
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• Interpretation:

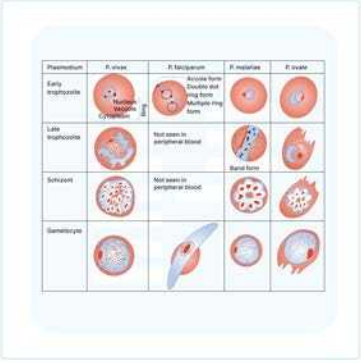


Test	Test “A”	Test “B”	Test “C”	Test “D”
Control line	Present	Present	Present	INVALID (no control line)
Pan malaria	Absent	Present	Present	
Plasmodium falciparum	Absent	Absent	Present	
Interpretation	Negative test	Positive test	Positive test (Mixed infection or Plasmodium falciparum only)	

Solution for Question 5:

The microscopic findings described—infected red blood cells of various sizes, multiple ring forms within single red blood cells, and the presence of crescentic (banana-shaped) gametocytes—are characteristic of Plasmodium falciparum.

	Vivax	Falciparum	Malariae	Ovale
RBC size	Enlarged	Normal	Normal	Enlarged with frayed margins
Pigments	Yellowish brown	Dark brown		
Schizont	Large (9-10 µm) Fills the RBC Contains 12-24 mero zoite	Not seen in the periphery	Small (6 µm), almost fills the RBC and contains 8 merozoites	
Gametocytes	Spherical	Crescentic Larger than RBC	Spherical	
Ring form	2.5 µm Occupies ■ rd of RBC size Cytosol opposite to the nucleus is thick	1.5 µm Occupies ■ of RBC size It may be Multiple forms Accole (attached to RBC membran e) Double dot/headp hone shaped with fragment nucleus	Band form	2.5 µm Occupies ■ rd of RBC size Cytosol opposite to the nucleus is thick
In Plasmodium falciparum, only gametocyte and ring forms are demonstrated in the PBS, as later stages of the erythrocytic cycle occur in deep vessels and not in peripheral blood.				



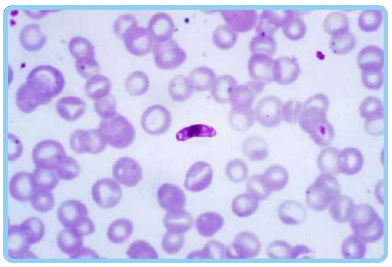
Solution for Question 6:

• The gold standard test used for the diagnosis of malaria is microscopy using the Romanowsky stain.

• This method allows direct visualisation of malaria parasites within red blood cells, providing information about the species, parasite density, and stage of development.

• It is highly specific and widely used, especially in resource-limited settings.

LAB DIAGNOSIS OF MALARIA	
Microscopy (Gold standard)	Thick smear - for rapid detection of malarial parasite Quantification is done under an oil immersion lens + - 1 to 10 parasites per 100-thick film field ++ - 11 to 100 parasites per 100 thick film field +++ - 10 to 10 per thick film field ++++ - more than 10 per thick film field At least 200 oil immersion fields should be examined before a film is declared negative. It can be used for the detection of malarial parasites as low as 20 parasites / μL Thin smear - for determination of plasmodium species Stains used - Romanowsky stain (Leishman, Giemsa or JSB stain)

Quantitative buffy coat test (Option A)	 RBCs with parasites are less dense; thus, concentrate just below the buffy coat of WBC on top of the RBC column Acridine orange is used to induce fluorescence on the parasite, giving a fluorescent greenish-yellow appearance on a red background {{caption_text}} Microscopic examination of a smear made using the buffy coat is done.
Kawamoto fluorescence microscopy	 ... (content truncated)
Rapid antigen detection test	Immunochromatography is done using monoclonal antibody directed against the parasite antigen
Parasite F test (Option D)	Monoclonal antibody is used to detect histidine-rich protein 2 (HRP-2), which is produced by asexual stages of plasmodium falciparum High sensitivity and specificity Can detect low parasitemia (40 parasite / μL of blood) Can detect only the asexual stage of plasmodium falciparum
Dual antigen test (Option C)	Detects Parasite lactate dehydrogenase (pLDH) and Plasmodium falciparum histidine-rich protein (PfHRP-2) pLDH is produced by trophozoites and gametocytes of all plasmodium spp. PfHRP-2 is produced only by the asexual stage of Plasmodium falciparum.

Molecular method	PCR: Used for species determination and detection of drug resistance. DNA probe: Highly sensitive and can detect less than 10 parasites / μL of blood.

Solution for Question 7:

• Blackwater fever is a severe form of malaria primarily associated with Plasmodium falciparum infection; it commonly occurs in patients with repeated malaria infections or inadequate treatment with quinine.

• It is characterised by hemoglobinuria (dark-coloured urine), severe hemolytic anaemia, and jaundice.

• It results from the severe hemolysis of red blood cells, which releases free haemoglobin into the bloodstream and its subsequent excretion in the urine.

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species	P.vivax	P.falciparum	P.malariae	P.ovale	P. Knowlesi
Type of fever	Benign tertian fever (fever occurs every 3rd day)	Malignant tertian fever (severe form of malaria- cerebral malaria)	benign quartan fever (fever occurs every 4th day)	Benign tertian fever	Benign quotidian or simian fever (fever occurs every 24 hrs & not seen in India)

Solution for Question 8:

Plasmodium vivax and Plasmodium ovale are the two species capable of causing malaria relapses. They can form hypnozoites, which are dormant forms in the liver that can reactivate months or even years after the initial infection, leading to recurrent symptoms.

Relapse in malaria	Recrudescence in malaria

Occurs due to the reactivation of dormant forms of underdeveloped sporozoites, known as hypnozoites, in the liver.	Occurs due to inadequate treatment or resistance to it.
This is seen in (Option A) P. vivax P. ovale	This is seen in (Options B and C) P. falciparum P. malariae
This can happen after 24 weeks to 5 Years of initial attack.	This can happen in 1 to 2 years in P. falciparum and around 50 to 60 years in P. malariae.

Solution for Question 9:

• For Plasmodium falciparum, the average incubation period is typically around 12 days from the time of infection until the onset of symptoms.

• The parasite binds to RBCs using the Basigin receptor (CD147), facilitating its entry into the cells.

Properties of different species of Plasmodium:

Properties	P.vivax	P.ovale	P.falciparum	P.malariae
Relapse (Hypnozoites)	Seen	Seen	Not seen	Not seen
Recrudescence	Not seen	Not seen	Seen	Seen
Preference	Reticulocytes	Reticulocytes	Young erythrocytes	Old erythrocytes
Erythrocytic Cycle	48 hours	48 hours	36–48 hours	72 hours
Prepatent Period	8 days	9 days	5 days	13 days
Incubation Period	12- 17 days	14-17 days	12 days	28-30 days
Stages found in peripheral blood	Trophozoite, schizonts, gametocytes	As in vivax	Ring form, accole form, gametocytes	Band forms

Infected erythrocyte	Enlarged pale with Schuffner's dots	Pear-shaped, fimbriated, Schuffner's dots (Jame's dots)	Normal size, Maurer's cleft	Normal size, Ziemann's stippling
Gametocyte	Spherical, fills the cell	Similar to but smaller than vivax	Crescent, Banana shaped	Similar to but smaller than vivax
Colour of pigment	Yellowish-brown	Yellowish-brown	Black	Dark brown

Prepatent period: The time from the entry of the parasite into the host until it is detectable in peripheral blood.

Incubation period: Time interval from the entry of the parasite until the appearance of the first clinical symptoms.

Receptors on RBC:

- Plasmodium vivax: Binds to Duffy antigen (also known as Duffy blood group antigen) on the surface of red blood cells (RBCs)

Plasmodium falciparum: Binds to Basigin (also known as CD147), a protein expressed on the surface of RBCs

Solution for Question 10:

Plasmodium falciparum is the primary species associated with cytoadherence, where infected red blood cells adhere to endothelial cells in blood vessels. This leads to complications like cerebral malaria due to vessel blockage and reduced blood flow.

Pathogenesis of Malignant Malaria:

Cytoadherence:

- In Plasmodium falciparum malaria, late-stage schizonts secrete proteins that create knob-like protrusions on the surfaces of infected RBCs. This results in the expression of the adhesive protein PfEMP-1 (P. falciparum Erythrocyte Membrane Protein), which makes the infected RBCs sticky.

• Inflammatory cytokines, particularly IFN-γ, enhance the expression of endothelial receptors such as thrombospondin, E-selectin, VCAM-1, and ICAM-1 in the brain, chondroitin sulfate B in the placenta, and CD36 in most other organs. These receptors promote the cytoadherence of infected RBCs to the endothelium, leading to blockage of capillaries and venules.

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Rosette formation: Infected RBCs can adhere to uninfected RBCs, forming rosettes.

Auto-Agglutination: The phenomenon where parasitised red blood cells (RBCs) clump together

These mechanisms cause capillary plugging and reduced microcirculation in vital organs, including the brain, kidneys, lungs, spleen, intestines, bone marrow, and placenta, resulting in severe complications like cerebral malaria.

Other notable virulence factors:

• Histidine-rich protein II (HRP II)

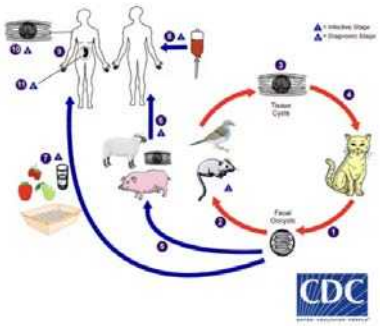
• Glycosylphosphatidylinositol (GPI)

Solution for Question 11:

• The patient's symptoms (flu-like illness, fatigue, and lymphadenopathy), with a history of handling cat litter and positive IgG and IgM antibodies, suggest toxoplasmosis caused by Toxoplasma gondii.

• Bradyzoites (tissue cyst form) are a transmissible stage of Toxoplasma and can infect humans through the consumption of undercooked or raw meat from infected animals.

Toxoplasma gondii	
Host	Definitive host: Cat and other cat species (family Felidae) (Option C) Intermediate host: Humans and animals like sheep, rodents, pigs etc
Infective stages	Sporulated oocyst (Option B) Tachyzoites Bradyzoites (tissue cyst)
Modes of transmission	Infective form

Blood Transfusion	Tachyzoites
Mother to child (Option A)	
Undercooked or raw meat	Sporulated Oocyst
Ingestion of contaminated products	Bradyzoites
	
	

Solution for Question 12:

• Humans are the definitive host for malaria parasites.

• The infective form introduced into the bloodstream by the female Anopheles mosquito is the sporozoite.

Malaria:

• Human malaria is caused by five main Plasmodium species: P.vivax: Benign tertian malaria; fever every 48 hours. P. falciparum: Malignant tertian malaria; severe; fever every 48 hours. P.malariae: Benign quartan malaria; fever every 72 hours. P.ovale: Ovale tertian

malaria; fever every 48 hours. P.knowlesi: Quotidian or simian malaria; fever daily; primarilyinfects monkeys but can affect humans.

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• P. falciparum: Malignant tertian malaria; severe; fever every 48 hours.

• P.malariae: Benign quartan malaria; fever every 72 hours.

• P.ovale: Ovale tertian malaria; fever every 48 hours.

• P.knowlesi: Quotidian or simian malaria; fever daily; primarily infects monkeys but can affect humans.

• Plasmodium completes its life cycle in two hosts: Definitive Host: Female Anopheles mosquito (sexual cycle). Intermediate Host: Humans (asexual cycle).

• Definitive Host: Female Anopheles mosquito (sexual cycle).

• Intermediate Host: Humans (asexual cycle).

• Transmission and Infective form: Intermediate Host: Humans Transmission through the bite of an infected female Anopheles mosquito. Infective form: Sporozoites are introduced from the mosquito's salivary glands into the bloodstream during the bite. Blood transfusion and Transplacental transmission: Infective form: Trophozoites / Merozoites Definitive Host: Mosquitos Infective form: Gametocytes

• Intermediate Host: Humans Transmission through the bite of an infected female Anopheles mosquito. Infective form: Sporozoites are introduced from the mosquito's salivary glands into the bloodstream during the bite.

• Transmission through the bite of an infected female Anopheles mosquito.

• Infective form: Sporozoites are introduced from the mosquito's salivary glands into the bloodstream during the bite.

• Blood transfusion and Transplacental transmission: Infective form: Trophozoites / Merozoites

• Infective form: Trophozoites / Merozoites

• Definitive Host: Mosquitos Infective form: Gametocytes

• Infective form: Gametocytes

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• Intermediate Host: Humans (asexual cycle).

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• Infective form: Trophozoites / Merozoites

• Definitive Host: Mosquitos Infective form: Gametocytes

• Infective form: Gametocytes

• Transmission through the bite of an infected female Anopheles mosquito.

• Infective form: Sporozoites are introduced from the mosquito's salivary glands into the bloodstream during the bite.

• Infective form: Trophozoites / Merozoites

• Infective form: Gametocytes

Mosquito-Borne Malaria and Blood Transfusion Malaria:

Feature	Mosquito-Borne Malaria	Blood Transfusion Malaria
Mode of Transmission	Mosquito bite	Blood or blood product transfusion
Infective Stage	Sporozoite	Trophozoite
Incubation Period	Long	Short
Severity	Comparatively less	More complications seen
Relapse	May occur	Does not occur
Radical Treatment	Required	Not required

Solution for Question 13:

The given scenario, with symptoms of diarrhoea, a history of working in a pig farm, and ciliate protozoan in the stool sample, points towards Balantidium coli (the only ciliate protozoan

parasite of humans), and tetracycline is the drug of choice used in the management.

Balantidium coli: (Phylum: Ciliophora)

- Largest protozoan parasite (Option A) residing in the human large intestine
- Resides in the large intestine of humans, pigs, and monkeys

Morphology:

- Trophozoite: Ovoid cell, 60-70 µm in length Two nuclei: a large kidney-shaped macronucleus and a small micronucleus. Motile due to cilia covering the surface. Features include a groove (peristome) leading to the mouth (cytostome) and a short funnel-shaped cytopharynx and cytopyge (analpore) posteriorly.
- Two nuclei: a large kidney-shaped macronucleus and a small micronucleus.
- Motile due to cilia covering the surface.
- Features include a groove (peristome) leading to the mouth (cytostome) and a short funnel-shaped cytopharynx and cytopyge (analpore) posteriorly.
- Cyst: Description: Spherical, 40-60 µm in diameter, with a thick double-layered wall. Content: Granular cytoplasm with macronucleus, micronucleus, and vacuoles. Role: Infective stage found in chronic cases and carriers.
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- Description: Spherical, 40-60 µm in diameter, with a thick double-layered wall.
- Content: Granular cytoplasm with macronucleus, micronucleus, and vacuoles.
- Role: Infective stage found in chronic cases and carriers.
- Life Cycle: Type: Monoxenous (single-host). Habitat: Large intestine Natural Host: Pigs. Accidental Host: Humans. Reservoirs: Pigs, monkeys, rats. Infective Form: Cyst (Option B) Transmission: Infection occurs through ingestion of contaminated food/water containing cysts (Option D)
- Type: Monoxenous (single-host).
- Habitat: Large intestine
- Natural Host: Pigs.
- Accidental Host: Humans.

• Reservoirs: Pigs, monkeys, rats.

• Infective Form: Cyst (Option B)

• Transmission: Infection occurs through ingestion of contaminated food/water containing cysts (Option D)

• Type: Monoxenous (single-host).

• Habitat: Large intestine

• Natural Host: Pigs.

• Accidental Host: Humans.

• Reservoirs: Pigs, monkeys, rats.

• Infective Form: Cyst (Option B)

• Transmission: Infection occurs through ingestion of contaminated food/water containing cysts (Option D)

• Clinical Features: Most infections are asymptomatic Trophozoites can invade the intestinal mucosa, leading to inflammation, mucosal ulcers, and submucosal abscesses, resembling amebic lesions & presets with: Diarrhea or dysentery Abdominal colic Tenesmus Nausea and vomiting Complications can include secondary bacterial infections, intestinal perforation, and rare involvement of the genital or urinary tracts.

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• Diarrhea or dysentery

• Abdominal colic

• Tenesmus

• Nausea and vomiting

• Complications can include secondary bacterial infections, intestinal perforation, and rare involvement of the genital or urinary tracts.

• Laboratory Diagnosis: The diagnosis was confirmed by identifying trophozoites in diarrheal stools and cysts in formed stools. The trophozoites are large, with distinct nuclei and rapid motility. If stool examination is negative, biopsies from intestinal ulcers may reveal trophozoites and cysts.

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• Treatment: Drug of Choice: Tetracycline (500 mg, four times daily for 10 days). Alternatives include doxycycline, metronidazole, and nitroimidazole.

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• Drug of Choice: Tetracycline (500 mg, four times daily for 10 days).

• Alternatives include doxycycline, metronidazole, and nitroimidazole.

Solution for Question 14:

• The clinical scenario of diarrhoea in an immunocompromised host and a stool sample showing oval oocysts suggests cystoisospora belli.

• The incubation period of cystoisospora is 1-4 days, not 2-14 days.

Intestinal Coccidian Parasites:

• Cryptosporidium species: C.parvum (mammals) & C.hominis (primarily humans)

• Cyclospora cayetanensis

• Cystoisospora belli

Clinical Features:

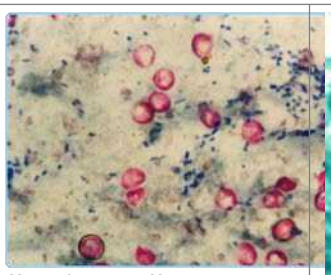
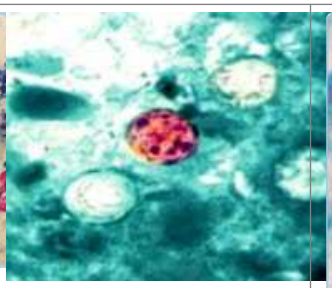
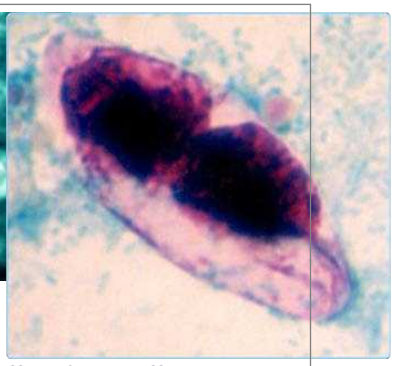
• Immunocompetent Hosts: Symptoms are often mild and may include watery, non-bloody diarrhoea, abdominal pain, nausea, and fever. They are typically self-limiting.

• Immunocompromised Hosts (e.g., AIDS): Severe, persistent diarrhoea (up to 25 litres/day), significant weight loss & extraintestinal manifestations (Biliary infections) Severity increases with lower CD4+ T cell counts

• Severity increases with lower CD4+ T cell counts

• Severity increases with lower CD4+ T cell counts

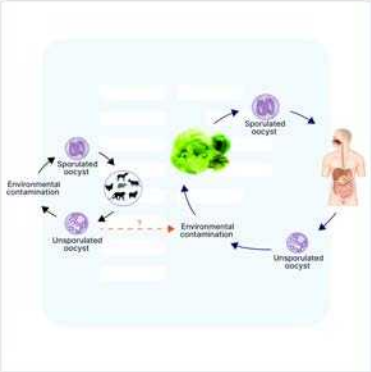
Features	Cryptosporidium	Cyclospora	Cystoisospora
Transmission	Ingestion of contaminated food/water, autoinfection	Ingestion of contaminated food/water	Ingestion of sporulated oocysts
Infective form	Sporulated oocyst	Sporulated oocyst	Sporulated oocyst
Diagnostic form	Sporulated oocyst	Unsporulated oocyst	Unsporulated oocyst
Incubation Period	2-14 days	1-7 days	1-4 days
Oocyst shape	Round	Round	Oval
Oocyst size	4-6 µm	8-10 µm	20–33 µm
Sporulated oocyst	Four sporozoites	Two sporocysts, each having two sporozoites	Two sporocysts, each having four sporozoites
Acid-fastness	Uniformly acid-fast	Variable acid-fast	Uniformly acid-fast
Autofluorescence	–	++	+/-
Resistance to chlorination	+	–	–
Sporulation of the oocyst	Inside the host cells (enterocytes)	soil	soil
Zoonotic potential	Yes	Uncertain	No
Treatment	Nitazoxanide	Cotrimoxazole	Cotrimoxazole

	 {{caption_text}}		 {{caption_text}}

Solution for Question 15:

In Cyclospora, each oocyst contains 2 sporocysts, but each sporocyst contains 2 sporozoites . Therefore, a total of 4 sporozoites are released upon excystation (not 8 sporozoites).

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Solution for Question 16:

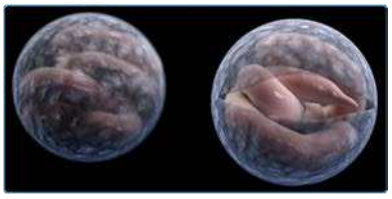
• While Cryptosporidium produces both thick-walled and thin-walled oocysts, the thin-walled oocysts contribute to autoinfection within the host.

• Thick-walled oocysts are typically excreted in the faeces and are primarily responsible for transmission to other hosts.

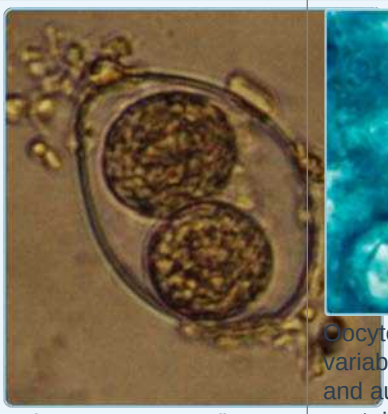
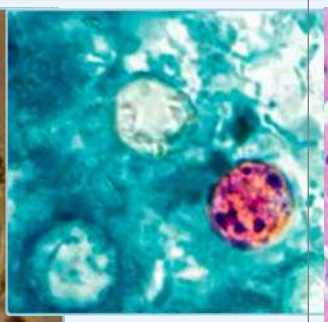
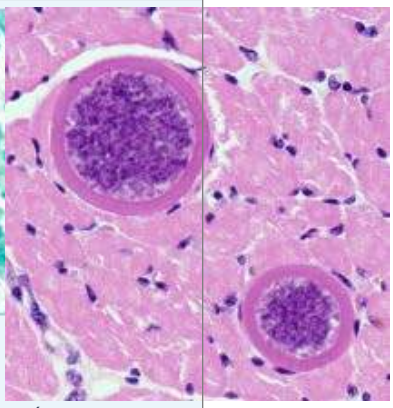
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Solution for Question 17:

Image B shows an oocyst of cryptosporidium, which is the most common opportunistic parasitic infection in HIV-positive / immunocompromised patients.

Cryptosporidium	
Infective form	Sporulated oocyst in contaminated water
Clinical features	In Immunocompetent patients: Self-limited watery diarrhoea. In Immunocompromised patients: Chronic persistent watery diarrhoea.
Microscopy	 Appearance: Acid-fast, round oocyst Size: 4–6 microns Contents: Each oocyst contains four sporozoites Types of oocyst: Thick-walled - excreted out from host in faeces. Thin-walled - involved in autoinfection. {{caption_text}}

	Isospora (Option A)	Cyclospora (Option D)	Sarcocystis (Option C)
Infective form	Sporulated oocyst	Sporulated oocyst	Sporulated oocyst

Clinical features	Mild fever with diarrhoea Can cause chronic diarrhoea in the immunocompromised host	Formation of sarcocysts within the muscle fibres (Muscular sarcocystosis) Nausea, abdominal pain and diarrhoea (Intestinal sarcocystosis)	
Microscopy	 ... (content truncated)	 Oocyte is round, acid variable, 8-10 microns and autofluorescent, containing 2 - 2 sporozoites in each wheel. {{caption_text}}	 ... (content truncated)

Pathogens causing diarrhoea in HIV-positive patients	
Parasites	Cryptosporidium parvum (Most common)
Isospora belli	
Cyclospora	
Giardia lamblia	
Entamoeba histolytica	
Microsporidia spp	
Bacteria	Shigella species (Most common)
Salmonella species	
Campylobacter jejuni	
Viruses	Cytomegalovirus
Adenovirus	
Fungi	Histoplasma
Candida	

Solution for Question 18:

• The Sabin-Feldman dye test is the gold standard for diagnosing Toxoplasma gondii infection due to its high specificity.

• It is a complement fixation test.

• Based on the principle that antibodies specific to Toxoplasma gondii prevent the parasite from absorbing alkaline methylene blue. Antibody present: Colour will be absent. Antibody absent: Colour will be present.

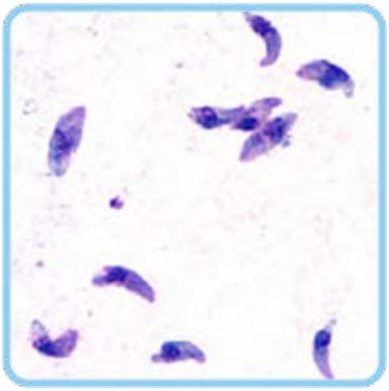
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Diagnosis of Toxoplasmosis	
Antenatal	
USG	Done at 20-24 weeks of gestation and repeated every 2-4 weeks It helps detect lesions of congenital infection, such as Hydrocephalus Cerebral calcifications Ascites
Serology	ELISA (Enzyme-Linked Immunosorbent Assay) or IFA (Immunofluorescence Assay) is done using maternal serum to detect IgM and IgG
PCR	Amniotic fluid can be used to detect Toxoplasma DNA, which can confirm the diagnosis. It is not done routinely as it is invasive and expensive, and other safer options are available.
Postnatal diagnosis	
Serology	Newborn Sera and maternal sera are both used to detect IgG >= 1000: Possible infection IgM >= 1:4 after 2 weeks of age: Indicates probable diagnosis IgA: Disappears within 10 days of birth; thus, persistence indicates postnatal infection Sabin-Feldman test - Complement fixation test

Other tests	 ... (content truncated)

Solution for Question 19:

In congenital toxoplasmosis, common findings include chorioretinitis, hydrocephalus, and intracranial calcifications (typically diffuse or scattered). However, periventricular calcifications are more characteristic of congenital cytomegalovirus (CMV) infection rather than toxoplasmosis.

	Congenital Toxoplasmosis	Congenital CMV
Clinical features	Classical triad: Chorioretinitis + Hydrocephalus/Microcephaly + Intracerebral calcification Hepatosplenomegaly Intrauterine growth restriction Ocular manifestations: Scotoma Glaucoma Strabismus It usually causes B/L chorioretinitis. May lead to blindness. Toxoplasmosis Clinical" data-author="Kamal EL Bissati, Pauline Levigne, Joseph Lykins, El Bachir Adlaoui, Amina Barkat, Amina Berraho, Majda Laboudi, Bouchra El mansouri Mansouri, Azeddine Ibrahimi, Mohamed Rhajaoui, Fred Quinn, Manoradhan Murugesan, Fouad Seghrouchni, Jorge Gomez-Marin, François Peyron, Rima Mcleod" data-hash="9576" data-license="CC	Jaundice Petechial rash Hepatosplenomegaly Chorioretinitis Cataracts Microcephaly Mental retardation Sensorineural hearing loss (SNHL) CMV Clinical" data-author="Sarra Benmiloud, Ghizlane Elhaddou, Zoubida Alaoui Belghiti, Moustapha Hida, Abdelhak Bouharrou" data-hash="9579" data-license="CC BY 2.0" data-source="https://commons.wikimedia.org/wiki/File:Blueberry_muffin_baby.jpg" data-tags="" height="151" src="https://image.prepladder.com/notes/duFZUGAWnQEhWAGZTzrS1744391769.png" width="170" /> {{caption_text}}

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Imaging	Intracerebral calcifications Toxoplasmosis Radiology" data-author="Jmarchn" data-hash="9577" data-license="CC BY SA 3.0" data-source="https://commons.wikimedia.org/wiki/File:BrainToxoplasmosis_MRI_4_10.png" data-tags="" src="https://image.prepladder.com/notes/3dP6qM6T6wGkbn073Rj01744391705.png" /> {{caption_text}}	Periventricular calcifications CMV Radiology" data-author="Ilana Neuberger, Jacquelyn Garcia, Mariana L Meyers, Tamara Feygin, Dorothy I. Bulas, David Michael Mirsky" data-hash="9580" data-license="NA" data-source="https://www.researchgate.net/figure/Congenital-cytomegalovirus-infection-in-a-14-year-old-girl-Axial-non-contrast-head-CT_fig1_323831656" data-tags="" height="183" src="https://image.prepladder.com/notes/RXi2OPDEMDi7dOXghGxg1744391849.png" width="153" /> {{caption_text}}
Treatment	Oral pyrimethamine + sulfadiazine	Ganciclovir Valganciclovir Foscarnet
Clinical features common in both		
Chorioretinitis Microcephaly Mental retardation Hepatosplenomegaly		

The Gregg triad is a classic set of three symptoms associated with rubella, seen in infants whose mothers contracted the rubella virus during pregnancy. The Gregg triad consists of:

1. Deafness
2. Cataract
3. Cardiac anomalies

Solution for Question 20:

In Toxoplasma gondii infections during pregnancy, the risk of transplacental transmission to the fetus is highest in the third trimester.

Toxoplasmosis in pregnancy	
Mother	Usually asymptomatic but can transmit the disease to the infant.
Infective form	Trophozoite crosses the placenta to infect the neonate.
Risk of transplacental infection based on time of infection of the mother	Transplacental infection can occur at any time during the pregnancy, but the gestational age influences the risk of transmission (Option D). As the gestational age increases, the risk of transmission increases, but the severity of neonatal disease decreases. Pre-pregnancy (Option B) Very low chance of transmission (immunocompromised mother) First trimester Minimal risk of transmission (Option A). If transmitted, it results in more severe fetal complications. Third trimester Maximum risk of transmission (Option C). If transmitted, it is usually asymptomatic in newborns.

• First Trimester: Toxoplasmosis: High risk of severe fetal consequences if infection occurs early. Rubella: Greatest risk of congenital defects if contracted during the first trimester. CMV: Risk of transmission increases with gestational age but can lead to severe outcomes if contracted early.

• Toxoplasmosis: High risk of severe fetal consequences if infection occurs early.

• Rubella: Greatest risk of congenital defects if contracted during the first trimester.

• CMV: Risk of transmission increases with gestational age but can lead to severe outcomes if contracted early.

• Second Trimester: Syphilis: Can be transmitted at any stage, but early detection and treatment are crucial to prevent fetal complications. Varicella Zoster Virus (VZV): Risk of congenital varicella syndrome is present if the mother contracts the virus during this period.

• Syphilis: Can be transmitted at any stage, but early detection and treatment are crucial to prevent fetal complications.

• Varicella Zoster Virus (VZV): Risk of congenital varicella syndrome is present if the mother contracts the virus during this period.

• Third Trimester: Herpes Simplex Virus (HSV): Primarily transmitted during delivery; however, maternal infection late in pregnancy poses a significant risk for neonatal herpes. Group B Streptococcus (GBS): Commonly associated with infections during labor and delivery rather than in utero transmission.

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• Rubella: Greatest risk of congenital defects if contracted during the first trimester.

• CMV: Risk of transmission increases with gestational age but can lead to severe outcomes if contracted early.

• Syphilis: Can be transmitted at any stage, but early detection and treatment are crucial to prevent fetal complications.

• Varicella Zoster Virus (VZV): Risk of congenital varicella syndrome is present if the mother contracts the virus during this period.

• Herpes Simplex Virus (HSV): Primarily transmitted during delivery; however, maternal infection late in pregnancy poses a significant risk for neonatal herpes.

• Group B Streptococcus (GBS): Commonly associated with infections during labor and delivery rather than in utero transmission.

Cestodes

1. A 30-year-old patient presents with gastrointestinal symptoms after consuming poorly cooked meat. A stool examination reveals the presence of eggs or gravid proglottids, suggesting a diagnosis of taeniasis. Which of the following statements regarding the causative agents of this infection is correct?

A. The eggs of Taenia saginata are infective to humans

B. The infective stage of Taenia saginata is Cysticercus bovis

C. The eggs of Taenia solium are not infective to humans.

D. Taenia solium is acquired through poorly cooked beef

---

2. A 30-year-old man presents with abdominal pain and diarrhea. A stool sample is analyzed, revealing the presence of eggs that are oval in shape, have a smooth outer surface, and lack an operculum similar to the image given below. Given these findings, which of the following treatments is the most appropriate for this condition?



A. Albendazole

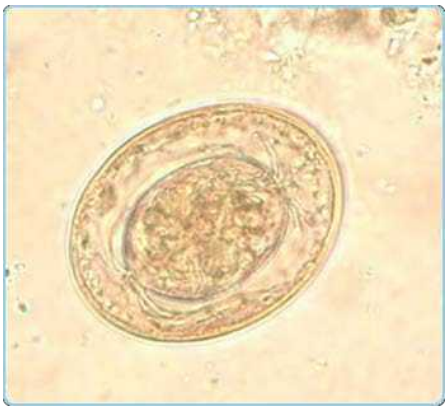
B. Niclosamide

C. Ivermectin

D. Praziquantel

---

3. An egg was extracted and observed under the microscope. Which of the following species does it belong to?



A. Taenia spp.

B. Diphyllobothrium latum

C. Hymenolepis nana

D. Hymenolepis diminuta

---

4. Which of the following statements correctly differentiates between Hymenolepis nana and Hymenolepis diminuta?

A. H. nana has hooks on its rostellum; H. diminuta does not

B. H. nana requires two intermediate hosts; H. diminuta requires one

C. H. diminuta exhibits autoinfection; H. nana does not

D. H. nana is the rat tapeworm; H. diminuta is the dwarf tapeworm

---

5. A 45-year-old patient presents with liver imaging that reveals a cystic mass with an invasive, infiltrative growth pattern spreading into surrounding tissues. Serology confirms the diagnosis. Which of the following species is most likely responsible for this presentation?

A. E. granulosus

B. E. multilocularis

C. E. vogeli

D. E. oligarthrus

---

6. A 39-year-old male presents with abdominal pain and a palpable abdominal mass. Ultrasound imaging shows findings depicted in the accompanying image. Based on the WHO classification of hydatid cysts, to which stage does this cyst correspond?



A. CE 1

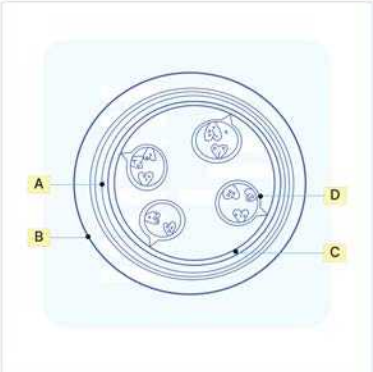
B. CE 2

C. CE 3

D. CE 4

---

7. A 35-year-old patient presents with abdominal pain, nausea, and occasional fever. On examination, a palpable mass is detected in the right upper quadrant. The patient mentions a recent trip to a region known for livestock farming. An ultrasound reveals a cystic structure with characteristic internal septations. Based on these findings, which layer of the cyst is primarily involved in protecting the parasite from the host's immune system?



A. A

B. B

C. C

D. D

---

8. A 30-year-old male is diagnosed with hydatid cysts. Which of the following serves as the accidental intermediate host in this infection?

A. Dog

B. Fox

C. Sheep

D. Human

---

9. A 34-year-old patient presents with gastrointestinal discomfort and reports passing segments in stool. Stool microscopy shows Taenia eggs, but further tests are needed for species identification. Which test best differentiates Taenia saginata from Taenia solium, and what is the drug of choice?

A. MRI to detect non-calcified cysts; Praziquantel

B. ELISA for anti-cysticercus antibodies; Albendazole

C. Microscopy of gravid proglottids to count lateral uterine branches; Praziquantel

D. Stool concentration using the formol-ether sedimentation method; Albendazole

---

10. A 35-year-old patient presents with a history of multiple painless subcutaneous nodules (shown below), seizures, and recent neurological symptoms. The patient has a history of consuming undercooked pork. Microscopy of a biopsy from one of the subcutaneous nodules reveals larval forms with a scolex. Which of the following is the most likely diagnosis?



A. Neurocysticercosis

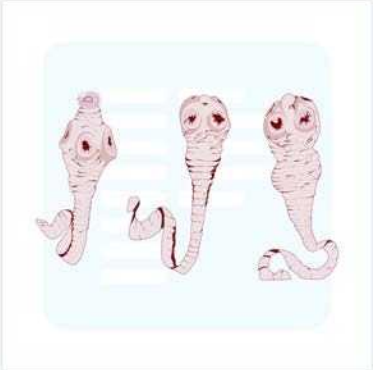
B. Cutaneous amoebiasis

C. Cutaneous leishmaniasis

D. Cutaneous larva migrans

---

11. Identify the correct feature of the given group of organisms.



A. They have a complete alimentary canal

B. They lack hooks and suckers

C. They are hermaphrodites

D. They possess a body cavity

---

12. A 32-year-old patient presents with gastrointestinal symptoms, including abdominal pain and diarrhea. A stool examination reveals the presence of tapeworm segments. Which of the following statements correctly differentiates between Taenia saginata and Taenia solium?

A. T. saginata has a small globular scolex, while T. solium has a large quadrate scolex.

B. T. saginata has fewer than 1000 proglottids, whereas T. solium has more than 1000

C. T. saginata lacks a rostellum and hooks, while T. solium possesses a rostellum with hooks

D. T. saginata has thin, dichotomous uterine branches, while T. solium has thick, dendritic uterine branches

---

13. A 65-year-old woman presents with complaints of tingling in her hands and feet, along with weakness and fatigue. She has a history of consuming

undercooked fish. Laboratory tests reveal macrocytic anemia and elevated mean corpuscular volume (MCV). A stool examination shows operculated eggs. What is the most likely diagnosis?

A. Schistosoma haematobium infection

B. Diphyllobothrium latum infection

C. Ancylostoma duodenale infection

D. Fasciola hepatica infection

---

14. A medical student is studying various parasitic infections and learns that certain parasites are characterized by operculate eggs. Which of the following parasites is not associated with operculate eggs?

A. Fasciola hepatica

B. Diphyllobothrium latum

C. Clonorchis sinensis

D. Schistosoma mansoni

---

15. A 28-year-old female with a history of consuming locally caught fish presents with complaints of weakness and mild abdominal pain. Her stool sample reveals eggs, as shown in the figure. Which form of the parasite would have been present in the fish she consumed?



A. Coracidium

B. Procercoid

C. Plerocercoid

D. Adult worm

---

16. A 38-year-old female from a rural farming community presents with upper abdominal pain, weight loss, and fatigue. Physical examination reveals hepatomegaly, and ultrasound reveals a well-defined cystic lesion in the liver with internal septations. What is the most likely pathogen causing this condition?

A. Taenia solium

B. Hymenolopis nana

C. Echinococcus

D. Spirometra

---

17. Match the following cestodes with the right features. Organism Features i) Taenia solium 4 suckers, no hooks ii) Taenia saginata 4 suckers, single row of 20-30 hooks iii) Hymenolopis nana 2 suctorial grooves or bothria, no suckers, no hooks iv) Diphyllobothrium latum 4 suckers, 2 rows of hooks

Organism	Features
i) Taenia solium	4 suckers, no hooks
ii) Taenia saginata	4 suckers, single row of 20-30 hooks
iii) Hymenolopis nana	2 suctorial grooves or bothria, no suckers, no hooks
iv) Diphyllobothrium latum	4 suckers, 2 rows of hooks

A. i-a, ii-b, iii-c, iv-d

B. i-a, ii-d, iii-c, iv-b

C. i-d, ii-b, iii-a, iv-c

D. i-d, ii-a, iii-b, iv-c

---

18. A microbiologist is examining the stool sample of a patient suspected to have a helminth infestation. Which of the following methods would be the best to assess bile staining property of helminth eggs?

A. Iodine mount

B. Sedimentation technique

C. Saline mount

D. Floatation technique

---

19. Which of the following parasites does not have humans as its definitive host?

A. Enterobius vermicularis

B. Ancylostoma duodenale

C. Echinococcus granulosus

D. Taenia solium

Correct Answers

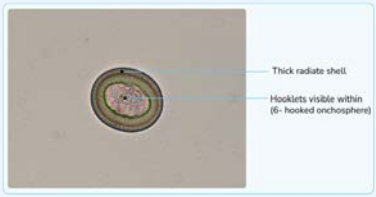
Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	1
Question 5	2
Question 6	3
Question 7	1
Question 8	4
Question 9	3
Question 10	1
Question 11	3
Question 12	4
Question 13	2
Question 14	4
Question 15	3
Question 16	3
Question 17	4
Question 18	3
Question 19	3

Solution for Question 1:

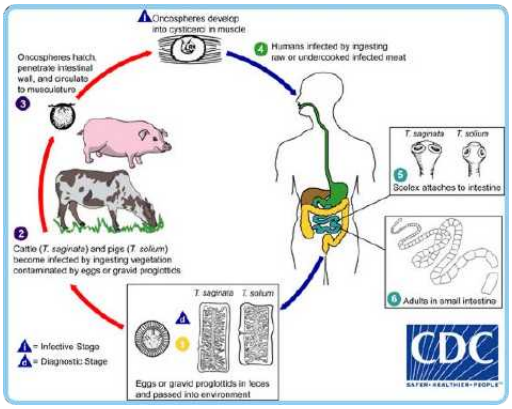
Correct Answer: B) The infective stage of Taenia saginata is Cysticercus bovis.

Explanation:

Cysticercus bovis is the infective larval stage found in cattle, which can be transmitted to humans through the consumption of poorly cooked beef.

Taenia Species	T. saginata	T.solium
Definitive Host	Man	Man
Intermediate Host	Cattle (cow or buffalo)	Pig
Larva	Cysticercus bovis; present in cow not in man	Cysticercus cellulosae; present in pig and also in man
Egg	 Not infective to man (Option A) {{caption_text}}	 ... (content truncated)
Mode of Infection	Ingestion of the larval stage in poorly cooked cattle (cow or buffalo) meat.	Ingestion of the larval stage in poorly cooked pig meat. (Option D)
Diagnostic Form	Eggs or gravid proglottids in feces	
Infective Form	Cysticercus bovis (Option B)	Cysticercus cellulosae

Life Cycle:



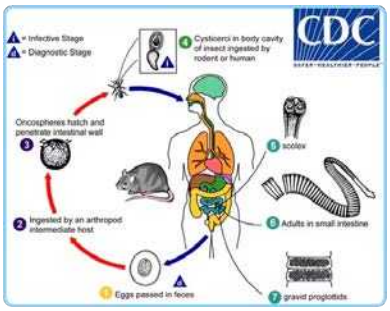
Solution for Question 2:

Correct Answer: D) Praziquantel

Explanation:

Praziquantel is the drug of choice for treating infections caused by Hymenelopis dimunata. It works by increasing the permeability of the parasite's cell membrane to calcium ions, leading to paralysis and death of the parasite.

Hymenolepis diminuta	
Hymenolepis diminuta, or rat tapeworm, is a parasitic tapeworm that primarily infects rodents but can occasionally infect humans through ingestion of infected insects.	
Morphology	 ... (content truncated)

Life Cycle	 Intermediate Host (Insects): Eggs ingested by insects (e.g., beetles, fleas) develop into cysticercoid larvae. Definitive Host (Rodent/Human): Infection occurs when an insect with cysticercoid larvae is ingested. The larvae develop into adults in the host's small intestine. Human Accidental Host: Humans are infected through accidental ingestion of infected insects; direct human-to-human transmission is rare. ... (content truncated)
Clinical Features	Asymptomatic: Many human infections are asymptomatic, especially when there is a low parasite load. Symptoms (in heavy infections): Abdominal pain, nausea, vomiting, diarrhea, and occasional weight loss.
Investigations	Stool Microscopy: Key diagnostic tool that reveals characteristic eggs that lack polar filaments. Serology/PCR: Molecular confirmation in research or doubtful cases.
Management	Anthelmintics: Praziquantel (preferred) - single dose.

Solution for Question 3:

Correct Answer: C) Hymenolepis nana

Explanation:

This is an egg of Hyenopelsis nana.

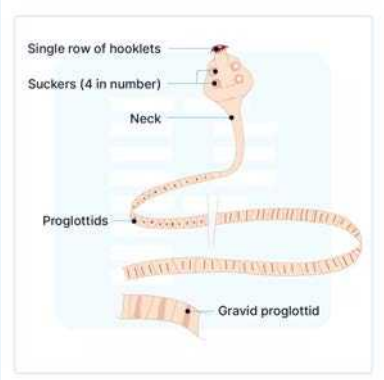


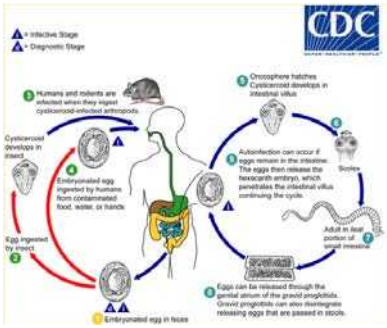
• Size: 30–50 µm in diameter.

• Shape: Oval or spherical with a thin shell.

• Inner membrane/embryophore: Characterized by polar filaments extending into the space between the inner and outer shell.

• Embryo: Contains an oncosphere with three pairs of hooklets.

Hymenolepis nana (Dwarf Tapeworm)	
Hymenolepis nana, or dwarf tapeworm, is a parasitic cestode primarily infecting humans and rodents, commonly causing infection in children. It can complete its life cycle in a single host without requiring an intermediate host.	
Morphology	 ... (content truncated)

Life Cycle	 Infective Stage: Eggs are excreted in feces and can survive in the environment. Autoinfection: Ingested eggs hatch in the intestine, allowing direct development into adults, leading to reinfection. ... (content truncated)
Clinical features	Asymptomatic in many cases. Symptomatic individuals may experience: Abdominal pain and cramps Diarrhea or loose stools Loss of appetite and weight loss Irritability (especially in children) Nausea and vomiting (less common)
Investigations	Stool Examination: Microscopic detection of eggs using methods like Kato-Katz or concentration techniques. PCR: Used for specific identification of H. nana DNA in stool samples.
Management	Antiparasitic Medication: Praziquantel: First-line treatment, typically a single oral dose of 25 mg/kg. Alternative: Nitazoxanide.

Solution for Question 4:

Correct Answer: A) H. nana has hooks on its rostellum; H. diminuta does not. Explanation:

Hymenolepis nana and Hymenolepis diminuta		
Feature	Hymenolepis nana (H. nana)	Hymenolepis diminuta (H. diminuta)

Common Name (Option D)	Dwarf tapeworm	Rat tapeworm
Definitive Hosts	Humans, rodents	Primarily rodents, occasionally humans
Intermediate Host (Option B)	Optional, can complete life cycle in a single host	Required, typically insects (e.g., beetles, fleas)
Egg	Oval and 37 to 47 µm in diameter, and has a double-layered shell that encloses filaments arising from bipolar thickenings	Rounder and 60 to 85 µm in diameter and have no polar filaments.
Adult Size	15-40 mm long	20-60 cm long
Scolex (Option A)	Four suckers, retractable rostellum with hooks	Four suckers, no hooks
Infective Stage	Embryonated eggs (directly infective)	Cysticercoid larva in the intermediate host
Mode of Transmission (Option C)	Fecal-oral, autoinfection possible	Ingestion of infected intermediate host
Life Cycle	Simple, can complete cycle within one host	Complex, requires two hosts
Diagnostic Method	Identification of eggs in stool samples	Identification of eggs in stool samples
Treatment	Praziquantel or niclosamide	Praziquantel

Solution for Question 5:

Correct Answer: B) E. multilocularis

Explanation:

E. multilocularis is responsible for alveolar echinococcosis, known for its invasive, infiltrative growth pattern that lacks a well-defined capsule, allowing the infection to spread similarly to malignant tumors, especially within the liver.

Echinococcus multilocularis

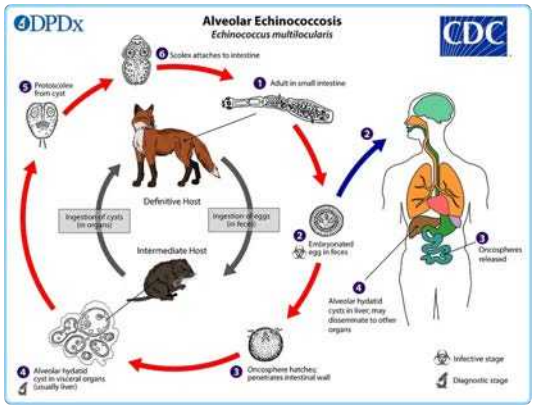
Echinococcus multilocularis is a parasitic tapeworm causing alveolar echinococcosis (AE) in humans. It primarily infects the liver, forming tumour-like lesions that can infiltrate surrounding tissues and spread to other organs.

Epidemiology: Found in Europe, Asia, and North America, with rural residents in contact with foxes or dogs at higher risk.

Pathology:

- Multiloculated, invasive liver cysts lacking a true capsule, resembling malignant tumors.
- It also affects the lungs, spleen, and brain by metastasis.

Life Cycle:



- Definitive hosts: Foxes, dogs, and other canids host adult worms in their intestines, shedding eggs in feces.
- Intermediate hosts: Rodents ingest eggs, which hatch, migrate to the liver, and form multilocular cysts.
- Accidental hosts (humans): Humans ingest eggs from contaminated sources, with larvae forming invasive cysts primarily in the liver.

Clinical presentation:

- It is often asymptomatic initially but can progress to symptoms like abdominal pain, jaundice, and hepatomegaly.
- If untreated, the disease may spread to other organs, such as the lungs, brain, and bones.

Investigations:

Imaging:

- Ultrasound: Detects irregular liver cysts.
- CT/MRI: Defines extent and spread of cysts.

Serology:

ELISA/Immunoblot: Detect antibodies against E. multilocularis.

Histopathology:

Confirms the presence of larval elements when a biopsy is performed.

Management:

Surgery:

- Radical Resection: Preferred for curative treatment.
- Liver Transplantation: Considered for extensive, unresectable disease.

Pharmacotherapy:

Albendazole: The main drug used, often requiring prolonged or lifelong treatment.

Echinococcus multilocularis (Alveolar Echinococcosis) and Echinococcus granulosus (Cystic Echinococcosis):

Feature	Echinococcus multilocularis (Alveolar Echinococcosis)	Echinococcus granulosus (Cystic Echinococcosis) (Option A)
Cyst Structure	Multi-chambered, invasive, tumor-like growth	Single, large fluid-filled cyst with thick laminated wall
Growth Pattern	Infiltrative, resembles malignancy, spreads to nearby tissues	Expansive, non-invasive, contained within a single cyst
Common Sites	Primarily liver Can metastasise (e.g., lungs, brain)	Mostly liver, sometimes lungs
Calcification	Common, granular appearance on imaging	Less common, cyst typically has a " water lily" sign
Treatment	Complex; surgery and long-term albendazole, poor prognosis	Surgery is often effective, sometimes with albendazole

E. vogeli (Option C): E. vogeli causes polycystic formations, usually forming multiple cysts within a compact mass that mainly affects the liver but without the aggressive invasive behaviour of E. multilocularis.

E. oligarthrus (Option D): E. oligarthrus is a rare cause of polycystic formations in humans, primarily affecting animals, and does not display the invasive growth pattern seen with E. multilocularis.

Solution for Question 6:

Correct Answer: C) CE 3

Explanation:

CE 3 is characterized by the presence of the "water lily sign," indicating a cyst with a detached membrane floating within the cyst cavity, which is a hallmark of this stage.

Investigations for Echinococcus granulosus

Imaging Studies

- Ultrasound (USG): USG is the first-line imaging modality, particularly for liver involvement. Features: Echinococcal cyst wall shows double echogenic lines separated by a hypoechoic layer (double contour).
- USG is the first-line imaging modality, particularly for liver involvement.
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- USG is the first-line imaging modality, particularly for liver involvement.
- Features: Echinococcal cyst wall shows double echogenic lines separated by a hypoechoic layer (double contour).

WHO Classification	Cyst Type	Ultrasound Findings
CL	Unilocular Cyst	Cystic lesion and no visible cyst wall
CE1 (Option A)	Simple Active Cyst	Double-line sign; well-defined anechoic cyst
CE2 (Option B)	Multivesicular Cyst	Honeycomb appearance: Visible cyst wall and internal septations
CE3	Transitional Cyst	Have detached laminar membranes or may be partially collapsed, and floating within the cyst cavity (Water-lily sign)
CE4 (Option D)	Inactive Cyst	Non-homogenous mass
CE5	Inactive Cyst	Cyst with a thick calcified wall

- CT Scan: For detection of extrahepatic disease.
- MRI: For detection of cyst in difficult sites such as spinal vertebrae and cardiac cysts.

Serological Tests:

• ELISA: Primary serological test sensitive for antibodies against E. granulosus.

• Immunoblotting: Highly specific for confirming equivocal cases and differentiating Echinococcus from other parasitic infections.

Casoni Intradermal Test:

• Obsolete due to anaphylaxis risk from cyst fluid.

• Represents a Type 1 hypersensitivity reaction.

ARC5 Immunoelectrophoresis

• This method involves separating proteins from patient serum or cyst fluid via electrophoresis, followed by immunodiffusion with antibodies targeting ARC5, a distinct antigen from hydatid cyst fluid.

• The formation of precipitin lines confirms the presence of hydatid disease, although more technically demanding, ARC5 IEP offers greater specificity and sensitivity than traditional tests like the Casoni test./

• Management of Echinococcus granulosus Management is guided by the WHO classification (CE1-CE5) and depends on cyst stage, location, symptoms, and complication risk. Pharmacotherapy Albendazole and Praziquantel:

• Albendazole is the drug of choice for uncomplicated cysts, used to prevent recurrence and to reduce the size of the cyst before surgery or PAIR.

• Praziquantel serves as an adjunct therapy to enhance treatment efficacy.

• PAIR Technique (Puncture, Aspiration, Injection, Re-aspiration): Best mode of treatment of cysts Steps:

• Puncture: Insert needle into cyst under ultrasound or CT guidance.

• Aspiration: Partially aspirate cyst fluid to reduce pressure.

• Injection: Inject scolicidal agent (e.g., hypertonic saline, ethanol, cetrimide) into cyst.

• Reaspiration: Re-aspirate injected fluid to remove cyst contents and scolicidal agent.

Management of Echinococcus granulosus

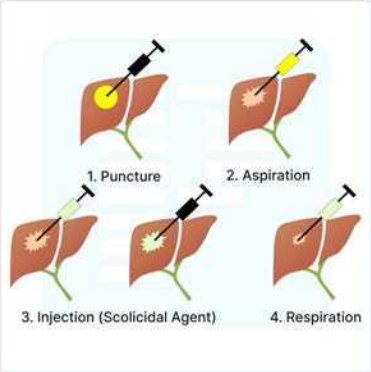
Management is guided by the WHO classification (CE1-CE5) and depends on cyst stage, location, symptoms, and complication risk.

Pharmacotherapy

Albendazole and Praziquantel:

PAIR Technique (Puncture, Aspiration, Injection, Re-aspiration): Best mode of treatment of cysts

Steps:



- Minimally Invasive: Effective for unilocular cysts (CE1 and CE3a); involves puncturing and injecting a scolicidal agent.
- Advantages: Lower recurrence risk and shorter hospital stays.
- Contraindications: Not suitable for Cysts with extensive daughter cysts or at risk of rupture Calcified cyst
- Cysts with extensive daughter cysts or at risk of rupture
- Calcified cyst
- Cysts with extensive daughter cysts or at risk of rupture
- Calcified cyst

Percutaneous Drainage (PERC): An alternative to PAIR for draining cysts without scolicidal agents.

Surgical Management

Cystectomy or Pericystectomy:

For large, complicated, or ruptured cysts; it is preferred for CE2 and CE3b cysts or challenging anatomical locations.

Watch and Wait:

Inactive or Calcified Cysts (CE4 and CE5) can often be observed safely with regular imaging follow-up.

Solution for Question 7:

Correct Answer: A) A

Explanation:

Ectocyst (A), or laminated membrane, serves as a barrier that prevents the host’s immune system from recognizing and attacking the cyst. Its unique structure minimizes immune detection and enhances parasite survival.

hydatid cyst Labelled" data-author="" data-hash="10181" data-license="" data-source="" data-tags="April2025" height="400" src="https://image.prepladder.com/notes/FdOozmbtODTEKYe6di6O1744476623.png" width="400" />

Hydatid Cyst

Hydatid cysts are parasitic cysts caused by Echinococcus granulosus.

Commonly Affected Organs

- Liver (most common site, about 60-70% of cases)
- Lungs (second most common site)
- Other organs (rare but possible): Brain, spleen, kidneys, bones, and peritoneum

Clinical Features

Asymptomatic Phase: Common due to slow cyst growth.

Symptomatic Phase:

- Liver Involvement: Mostly right lobe; Abdominal pain, hepatomegaly, obstructive jaundice, portal hypertension.
- Lung Involvement: Cough, chest pain, dyspnea, hemoptysis, pneumothorax.

Layers of a hydatid cyst

Layer	Name	Origin	Description	Main Function
Outer Layer	Pericyst	Host-derived	Thick fibrous layer created by the host's inflammatory response around the cyst. Eventually it develops into a dense fibrous capsule which may even calcify.	Provides structural support and isolates the cyst from surrounding tissues.

Middle Layer	Ectocyst (Laminated Membrane)	Parasite-derived	Acellular, chitinous, laminated hyaline layer secreted by the parasite. Has the appearance of the white of a hard-boiled egg.	Acts as a barrier against host immune cells and protects the inner germinal layer.
Inner Layer	Endocyst (Germinal Layer)	Parasite-derived	Thin cellular layer lining the cyst interior. Consists of a number of nuclei embedded in a protoplasmic mass. Vital layer of the cyst.	Site of asexual reproduction giving rise to brood capsules with scolices. Also secrete hydatid fluid, which fills the cyst.

Solution for Question 8:

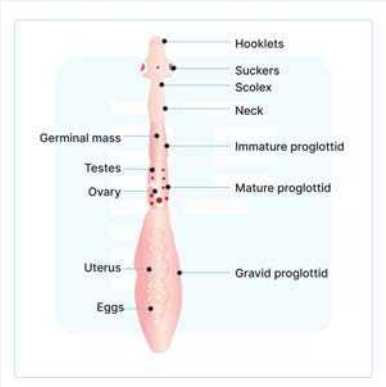
Correct Answer: D) Human

Explanation:

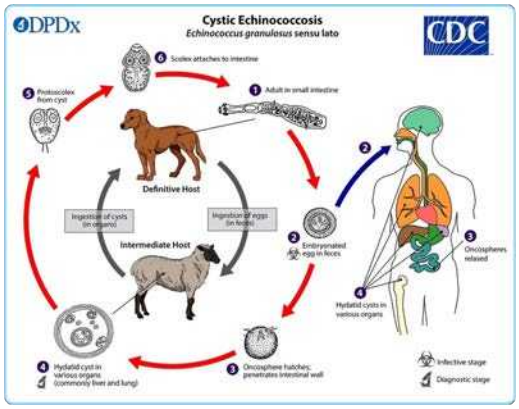
Humans act as accidental intermediate hosts for hydatid cysts by ingesting eggs from contaminated sources, often due to close contact with infected dogs.

Echinococcus granulosus:

Other name	Dog tapeworm
Definitive Host	Dog (optimal host), wolf, jackal and fox (Option A & B)
Intermediate Host	Sheep and cattle (Option C)
Accidental Intermediate Host	Man (Harbors larval stage, not adult form) (Option D)
Infective Form	Embryonated egg
Mode of Infection	Ingestion of the eggs due to: Intimate handling of infected dogs Eating raw vegetables or other food items contaminated with dog feces
Eggs	Ovoid in shape and brown in color Contains an embryo with three pairs of hooklets

Larva	Represent the structure of the scolex of the adult worm and remains invaginated within a vesicular body.
Adult Worm	 ... (content truncated)

Life Cycle:



Solution for Question 9:

Correct Answer: C) Microscopy of gravid proglottids to count lateral uterine branches; Praziquantel

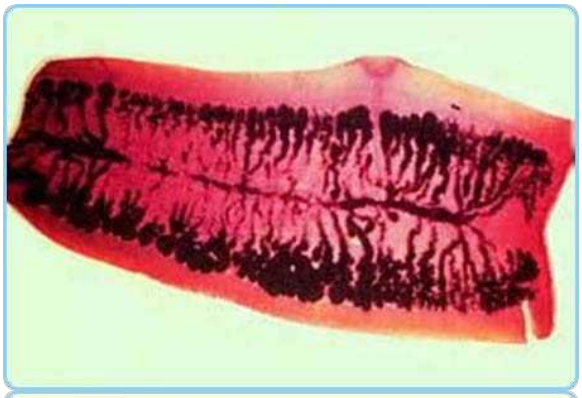
Explanation:

To differentiate T. saginata from T. solium, microscopy of the gravid proglottids is essential, with T. saginata having 15-20 lateral uterine branches and T. solium having fewer than 13. Praziquantel is the drug of choice for treating intestinal taeniasis.

Laboratory Diagnosis:

Intestinal Taeniasis:

- Stool Microscopy: Shows characteristic eggs of Taenia but can't differentiate T. solium and T.saginata Concentration method: Formol ether sedimentation method (Option D)
- Shows characteristic eggs of Taenia but can't differentiate T. solium and T.saginata
- Concentration method: Formol ether sedimentation method (Option D)
- Microscopy of Gravid Proglottids: (Option C) Can be done for species identification. 15-20 lateral uterine branches: T. saginata <13 lateral uterine branches: T. solium
- Can be done for species identification.
- 15-20 lateral uterine branches: T. saginata
- <13 lateral uterine branches: T. solium
- Shows characteristic eggs of Taenia but can't differentiate T. solium and T.saginata
- Concentration method: Formol ether sedimentation method (Option D)
- Can be done for species identification.
- 15-20 lateral uterine branches: T. saginata
- <13 lateral uterine branches: T. solium





Cysticercosis:

• Biopsy of Lesion: Show invaginated scolex with suckers and hooks

• X-ray: For detecting calcified cysticerci in subcutaneous tissue and muscles

• CT scan of the Brain: For detecting dead calcified cysts

• MRI scan: Detection of noncalcified cysts, spinal cysticerci and ventricular cysts. (Option A)



MRI scan: Multiple small cystic lesions of neurocysticercosis.

• ELISA: For detection of anti-cysticercus antibodies in serum and CSF. (Option B)

Treatment:

• Intestinal Taeniasis: Single dose of Praziquantel (drug of choice)

• Cysticercosis: Excision is the best treatment, wherever possible. Neurocysticercosis: Praziquantel + Albendazole Glucocorticoids must be started before starting albendazole, as the killed larval forms released from the cyst can cause inflammation.

• Excision is the best treatment, wherever possible.

• Neurocysticercosis: Praziquantel + Albendazole

• Glucocorticoids must be started before starting albendazole, as the killed larval forms released from the cyst can cause inflammation.

• Excision is the best treatment, wherever possible.

• Neurocysticercosis: Praziquantel + Albendazole

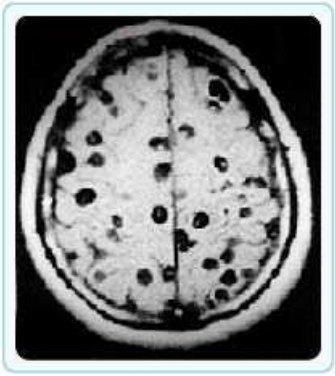
• Glucocorticoids must be started before starting albendazole, as the killed larval forms released from the cyst can cause inflammation.

Solution for Question 10:

Correct Answer: A) Neurocysticercosis

Explanation:

The combination of subcutaneous nodules, seizures, and neurological symptoms in a patient with a history of consuming undercooked pork suggests neurocysticercosis.

Cysticercosis	
Infection	direct consumption of T. solium eggs via contaminated food Caused by the larval stage (cysticercus cellulosae) of Taenia solium.
Pathology	Can affect any organ or tissue, with the most common sites being subcutaneous tissues and muscles. Other affected areas include Eyes and the brain Less often, heart, liver, lungs, abdominal cavity and spinal cord. Cysticercus is surrounded by a fibrous capsule except in the eye and brain ventricles. The larvae trigger a cellular reaction, initially involving neutrophils, eosinophils, lymphocytes, plasma cells, and occasionally giant cells. This reaction leads to fibrosis, death of the larva, and eventual calcification.
Clinical Features	 ... (content truncated)
Intestinal taeniasis	
Infection	Via consumption of undercooked infected meat of beef (T. saginata) or pork(T. solium)
Clinical features	Adult worms, despite their large size, cause little discomfort to the patient. When symptoms do occur, they may include: Vague abdominal discomfort Indigestion Nausea Diarrhea Weight loss

	Rarely, complications such as acute intestinal obstruction, acute appendicitis, and pancreatitis.

Cutaneous amoebiasis (Option B) typically presents with painful, foul-smelling ulcers and is associated with dysenteric diarrhea and liver abscesses. The absence of these symptoms and the specific microscopy findings in this case make cutaneous amoebiasis unlikely.

Cutaneous leishmaniasis (Option C) is caused by protozoan parasites of the Leishmania genus, transmitted by sandflies, and typically presents as a painless lesion that evolves into a chronic ulcer. The absence of such lesions, along with the patient's history of consuming undercooked pork and the specific microscopy findings, make cutaneous leishmaniasis unlikely.

Cutaneous larva migrans (Option D) is caused by hookworms such as Ancylostoma braziliense or Ancylostoma caninum, entering the skin, presenting as a small reddish papule that progresses to a serpiginous, pruritic rash. The absence of this characteristic rash and the patient's history of consuming undercooked pork make cutaneous larva migrans unlikely.

Solution for Question 11:

Correct Answer: C) They are hermaphrodites

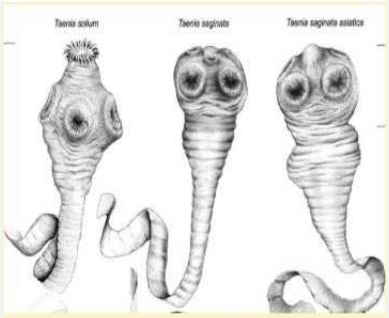
Explanation:

The image given is that of cestodes, which include Taenia solium, T. saginata, and other long flat worms. Cestodes are hermaphrodites (having both male and female sex organs).

Intestinal helminths:

- Can be classified as cestodes, trematodes, and nematodes.
- These cause many gastrointestinal infections.

Cestodes	Trematodes	Nematodes
These are long and segmented worms.	These are leaf-like and unsegmented.	These are elongated unsegmented worms.
They are dorsoventrally flattened and have no body cavity. (Option D)	These are also flatworms without a body cavity.	These are cylindrical worms and have a body cavity with well-differentiated organs.
They have no alimentary canal. (Option A)	They have an incomplete alimentary canal without an anus.	They have a complete alimentary canal.

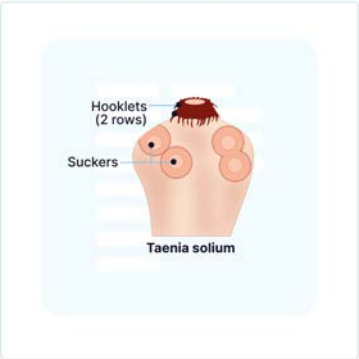
They have suckers for attaching to the host. Some have hooks as well. (Option B)	Suckers are present. Hooks are absent.	Hooks and suckers are absent.
These are hermaphrodites (monoecious) (Option C)	Hermaphrodites except Schistosoma.	The sexes are separate (diecious)
Examples: Diphyllobothrium Taenia saginata Taenia solium Echinococcus granulosus Hymenolepis nana Dipylidium caninum	Examples: Intestinal flukes: Fasciolopsis buski Blood flukes: Schistosoma mansoni	Examples: Small intestinal nematodes: Ascaris Hookworm Strongyloides Large intestinal: Trichuris Enterobius
 Cestodes ... (content truncated)		

Solution for Question 12:

Correct Answer: D) T. saginata has thin, dichotomous uterine branches, while T. solium has thick, dendritic branches.

Explanation:

Feature	Taenia saginata No	Taenia solium

Image		
Length	5-10 meters	2-3 meters
Scolex (Option A & C)	Large quadrate Rostellum and hooks are absent Suckers may be pigmented	Small and globular Rostellum and hooks are present Suckers not pigmented
Neck	Long	Short
Proglottids (Option B)	1000-2000	Below 1000
Measurement (gravid segment)	20 mm X 5 mm	12 mm X 6 mm
Expulsion	Expelled singly	Expelled passively in chains of 5 or 6
Uterus (Option D)	Lateral branches 15-30 on each side; thin and dichotomous	Lateral branches 5-10 on each side; thick and dendritic
Vagina	Present	Absent
Accessory lobe of ovary	Absent	Present
Testes	300-400 follicles	15-200 follicles

Solution for Question 13:

Correct Answer: B) Diphyllobothrium latum infection

Explanation:

The patient’s history of consuming undercooked fish along with Macrocytic anemia and elevated MCV indicate a vitamin B12 deficiency, strongly suggesting a Diphyllobothrium latum infection,

as this parasite is associated with infected freshwater fish. Additionally, the stool examinationreveals operculated eggs, which are characteristic of Diphyllobothrium latum.

Diphyllobothrium latum		
Clinical Features	Most infections are asymptomatic; only a few individuals develop symptoms. Minor Symptoms: These may include abdominal discomfort, diarrhoea, nausea, vomiting, weakness, and weight loss. Rarely, patients may experience acute abdominal pain due to intestinal obstruction, cholangitis, or cholecystitis.	
Vitamin B12 Deficiency: The adult worm interferes with the vitamin B12-intrinsic factor complex, leading to decreased absorption in the ileum, resulting in megaloblastic anemia (bothriocephalus anemia) and neurological sequelae (e.g., paresthesia). (Option B)		
Laboratory Diagnosis	Stool Examination: Identifies characteristic operculated eggs; they are: Bile-stained, Operculated with a knob. Do not float in the saturated salt solution.	Diphyllobothrium latum" data-author="Stefan Walkowski " data-hash="10154" data-license="CC BY-SA 4.0" data-source="https://en.wikipedia.org/wiki/Diphyllobothrium#/media/File:Diphyllobothrium_latum_egg.jpg" data-tags="April2025" height="274" src="https://image.prepladder.com/notes/4BLEhQxCHDe2wAJGjkU81744474648.png" width="400" /> {{caption_text}}
Molecular Testing: PCR and sequencing allow for precise species identification. Blood Tests: Eosinophilia and megaloblastic anaemia indicators include: Increased MCV & MCH Normal MCHC Macrocytes		

Treatment	Drug of Choice: Praziquantel (5–10 mg/kg once). Alternative Option: Niclosamide. Parenteral vitamin B12 should be administered if B12 deficiency is evident.	

Schistosoma haematobium infection (Option A) is linked to freshwater exposure and typically presents with urinary symptoms. Eggs are found in urine, not stool.

Hookworm (Option C) causes iron deficiency anemia rather than vitamin B12 deficiency. The stool would not reveal operculated eggs, as hookworm eggs are not operculated.

Fasciola / Liver fluke (Option D) has a round and operculated egg similar to the one in this scenario. However, it does not lead to vitamin B12 deficiency but rather causes fever, diarrhea, jaundice, right upper quadrant pain, hepatomegaly, myalgias, and urticarial rash.

Solution for Question 14:

Correct Answer: D) Schistosoma mansoni

Explanation:

The eggs of Schistosoma mansoni are oval and possess a distinctive spine, but they lack an operculum. In contrast, other trematodes have operculate eggs, which are characterised by a lid-like structure at one end.

Egg of Diphyllobothrium latum	
Diphyllobothrium latum" data-author="NA" data-hash="10156" data-license="NA" data-source="NA" data-tags="April2025" height="300" src="https://image.prepladder.com/notes/qzKU3MBch8O4ONGO3vi51744474826.png" width="300" /> {{caption_text}}	A single D. latum worm can lay approximately one million eggs daily. Egg is broadly ovoid, about 65 µm by 45 µm, with a thick, light brown shell. It has an operculum at one end and often a small knob at the other. Freshly passed eggs contain an immature embryo surrounded by yolk granules, with the embryo referred to as an oncosphere, which has six hooklets. Eggs are resistant to chemicals but are vulnerable to drying. Eggs do not float in saturated salt solution and exhibit bile staining. These eggs are not infectious to humans.

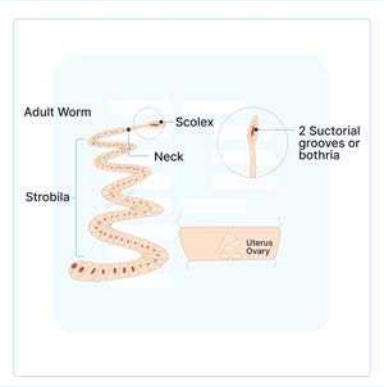
Parasites with operculate eggs	
Fasciola hepatica (Option A) Fasciola gigantica Fasciolopsis buski Clonorchis sinensis (Option C) Paragonimus westermani Gastrodiscoides hominis	Opisthorchis felineus Opisthorchis viverrini Heterophyes heterophyes (All Trematodes except schistosoma) Diphyllobothrium latum (Option B) Spirometra

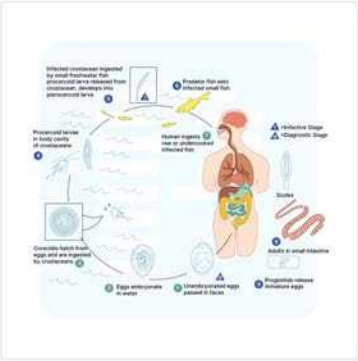
Solution for Question 15:

Correct Answer: C) Plerocercoid

Explanation:

This patient likely has a fish tapeworm infection, supported by her history of consuming locally caught fish and finding operculated eggs in her stool. Diphyllobothrium latum infects humans via the consumption of fish containing the infective L3 larva (plerocercoid).

Diphyllobothrium latum Also known as Fish tapeworm/Broad tapeworm, it is the largest and longest cestode found infecting the human intestine.	
Morphology	
	Adult Worm: Can grow >15 meters in length with more than 3,000 segments (proglottids); has a scolex (head) with two grooves (bothria). No suckers and hooklets. Egg: Operculated structure. Larvae: Develops in three stages: Coracidium (L1) Procercoid (L2) Plerocercoid (L3)
Life Cycle	

 ... (content truncated)	

Solution for Question 16:

Correct Answer: C) Echinococcus

Explanation:

The cestode Echinococcus is known to cause hydatid cysts in its intermediate host (humans) and hence, it can be called a somatic cestode in man.

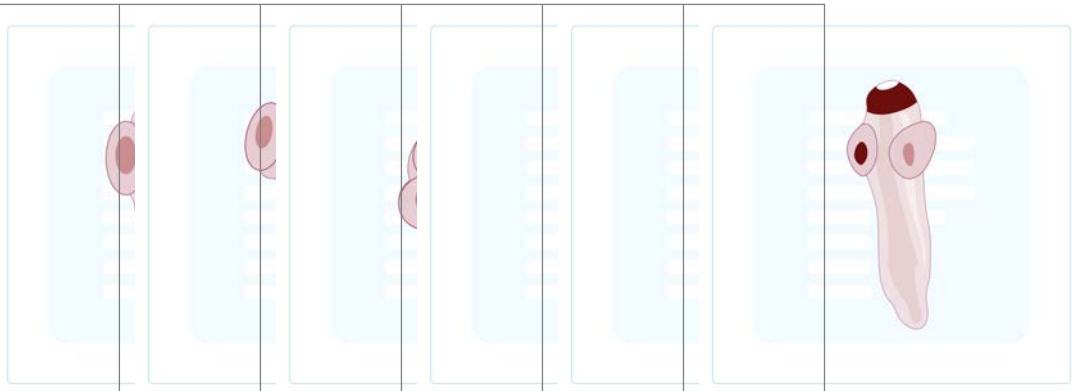
Intestinal cestodes	Somatic cestodes
These are commonly found infesting the intestines.	These are found in other parts of the body, like muscles, the brain, etc.
Taenia saginata: The beef tapeworm Man is the definitive host.	Echinococcus: The dog tapeworm Larva causes hydatid disease. Man is an accidental intermediate host.
Taenia solium: The pork tapeworm Man is the definitive host.	Taenia solium (Option A) Larva causes cysticercus cellulosae. Man is the definitive host.
Hymenolopis nana: The dwarf tapeworm (Option B) It is the smallest intestinal cestode. Man is the definitive host. No intermediate host.	Taenia multiceps Humans are accidental intermediate hosts.
Diphyllobothrium latum: The fish tapeworm Man is a definitive host.	Spirometra (Option D) Larva causes sparganosis. Man is an accidental intermediate host.

Solution for Question 17:

Correct Answer: D) i-d, ii-a, iii-b, iv-c

Explanation:

Morphology of cestodes						
Cestodes are dorsoventrally flat and segmented intestinal worms that may vary in size from a few millimetres to several meters. They are made of three parts: Head or scolex Neck Trunk						
Distinguishing features of the different cestodes						
Cestode	Taenia solium	Taenia saginata	Hymenolepis nana	Hymenolepis diminuta	Diphyllobothrium latum	Echinococcus granulosis
Head features	4 suckers 2 rows of hooks	4 suckers No hooks	4 suckers Single row of 20-30 hooks	4 suckers No hooks	2 suctorial grooves or bothria No suckers No hooks	4 suckers 2 rows of hooks

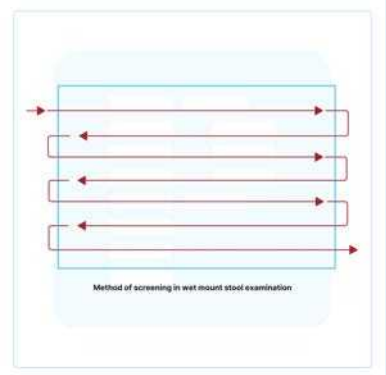
Images						
						
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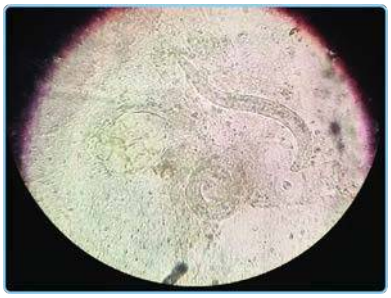
Solution for Question 18:

Correct Answer: C) Saline mount

Explanation:

The best method for assessing bile staining property (bile-stained eggs appear golden brown) and observing the motility of the trophozoites and larvae is the saline mount method.

Stool sample processing	
Microscopic examination of stool samples	This is done by two methods: Direct wet mount examination Permanent staining methods
 ... (content truncated)	Preparation: Drops of saline and Lugol's iodine are used on the left and right sides of the slide. Mix ~2 mg of feces to form a smooth suspension. Place a coverslip on the mount for examination under a microscope. Observation: Trophozoites Normal constituents of stool like plant fibers and starch cells can be seen Cellular Elements: Pus cells (in inflammatory diarrhea) Red blood cells (in dysentery) Charcot-Leyden Crystals

 ... (content truncated)	Types of wet mounts: Saline mount: Uses: Detects trophozoites, cysts, helminth eggs, and larvae. Advantages: Demonstrates motility of trophozoites and larvae. Bile staining property can be assessed (bile-stained eggs appear golden brown). Iodine mount: Advantages: Better visualization of nuclear details in cysts, larvae, and eggs. Disadvantages: Iodine immobilizes and kills parasites, hindering motility observation. (Option A) Bile staining property cannot be assessed.
 ... (content truncated)	This method stains the internal structures. Purpose: For accurate detection of protozoan cysts and trophozoites. Common Stains: Iron-hematoxylin stain. Trichrome stain. Modified acid-fast stain for coccidian parasites (e.g., Cryptosporidium).

Sedimentation techniques and Flotation techniques (Options B & D) are concentration techniques used to recover low numbers of parasites in stool samples.

- Useful for epidemiological studies and assessing treatment response.
- Detects eggs, cysts, and larvae but does not detect trophozoites, as these get destroyed in the process.
- Sedimentation techniques: Eggs and cysts settle at the bottom due to higher density. Example: Formalin-ether concentration increases detection sensitivity by 8–10 times. Flotation techniques: Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float. Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).
- Eggs and cysts settle at the bottom due to higher density.
- Example: Formalin-ether concentration increases detection sensitivity by 8–10 times.
- Flotation techniques: Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float. Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).

• Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float.

• Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).

• Eggs and cysts settle at the bottom due to higher density.

• Example: Formalin-ether concentration increases detection sensitivity by 8–10 times.

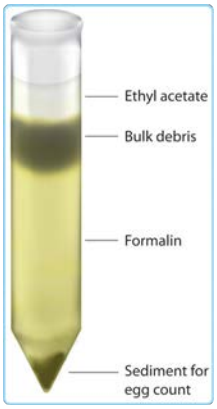
• Flotation techniques: Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float. Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).

• Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float.

• Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).

• Specimens are suspended in a denser medium (e.g., saturated salt solution) in which the eggs and cysts float.

• Limitations: Cannot concentrate non-floating parasites (e.g., unfertilized Ascaris eggs).



Solution for Question 19:

Correct Answer: C) Echinococcus granulosus

Explanation:

Humans are the definitive host of all of the above parasites except for Echinococcus granulosus

Hosts	

A host is an organism that supports a parasite, providing it with nourishment and shelter, and is typically larger than the parasite.	
Definitive host	This is the host where the adult stage of the parasite resides and undergoes sexual reproduction. Examples: Mosquito: Acts as the definitive host for the malaria parasite. Humans: Often serve as definitive hosts for various parasites like Taenia solium (pork tapeworm)(Option D) Taenia saginata (beef tapeworm) Enterobius vermicularis (pinworm)(Option A) Ascaris lumbricoides (roundworm) Ancylostoma duodenale (hookworm)(Option B) Necator americanus (hookworm) Trichuris trichiura (whipworm) Schistosoma species (blood flukes) Strongyloides stercoralis (threadworm)
Intermediate host	This host is where the larval stage of the parasite lives or where asexual reproduction occurs. Some parasites may require two different intermediate hosts to complete their life cycle, referred to as the first and second intermediate hosts. Example: Humans are intermediate hosts in: Babesia spp. Echinococcus granulosus (Option C) Echinococcus multiocularis Toxoplasma gondii Plasmodium spp.
Paratenic host	This type of host allows the larval stage of the parasite to remain viable without further development. It acts as a carrier that can transmit the infection to another host. Example: Fish can serve as a paratenic host for the plerocercoid larva of Diphyllobothrium.
Reservoir host	In endemic areas, a reservoir host continually maintains the presence of a parasite, serving as a significant source of infection for other susceptible hosts. Example: Dogs are reservoir hosts for hydatid disease.
Accidental host	This host is not typically part of the parasite's life cycle. Example: Humans can be accidental hosts for cystic echinococcosis.

Trematodes

1. A microbiologist was studying worms infecting the human body and discovered a flat, leaf-like species with an incomplete digestive system. To which of the following classes does this organism most likely belong?

A. Cestode

B. Nematode

C. Trematode

D. Annelid

---

2. Match the following trematode species with their primary habitat in the primary host. Trematode Species Primary Habitat 1. Fasciolopsis buski a. Bile ducts 2. Schistosoma mansoni b. Small intestine 3. Clonorchis sinensis c. Mesenteric veins 4. Paragonimus westermani d. Lungs

Trematode Species	Primary Habitat
1. Fasciolopsis buski	a. Bile ducts
2. Schistosoma mansoni	b. Small intestine
3. Clonorchis sinensis	c. Mesenteric veins
4. Paragonimus westermani	d. Lungs

A. 1-b, 2-c, 3-a, 4-d

B. 1-c, 2-d, 3-a, 4-b

C. 1-d, 2-a, 3-b, 4-c

D. 1-a, 2-b, 3-c, 4-d

---

3. Which of the following trematode species does not produce operculated eggs?

A. Fasciola hepatica

B. Clonorchis sinensis

C. Schistosoma haematobium

D. Paragonimus westermani

---

4. A 28-year-old man presents with complaints of abdominal pain, bloody diarrhea, and a recent onset of fatigue. He mentions that he has been working near freshwater lakes in an endemic region. A stool sample reveals Schistosoma eggs. Which of the following statements about Schistosoma species is incorrect?

A. The intermediate host for all Schistosoma species is a snail.

B. The life cycle of Schistosoma includes redia and metacercaria larvae.

C. Schistosoma japonicum primarily inhabits the superior mesenteric vein.

D. Schistosoma infections occur through skin penetration by cercaria larvae

---

5. A 32-year-old male presents with painless hematuria and dysuria after returning from a trip to the Middle East. He reports swimming in freshwater lakes during his stay. Laboratory tests reveal eggs with a characteristic terminal spine in the urine. Which of the following is a potential complication of this condition?

A. Severe epileptic seizures

B. Adenocarcinoma of bladder

C. Squamous cell carcinoma of the bladder

D. Intestinal bilharziasis with bloody diarrhea

---

6. A 35-year-old male presents with abdominal pain and bloody diarrhea after frequent exposure to freshwater while traveling in Africa. Stool examination reveals the presence of eggs with a characteristic lateral spine. Which of the following is inconsistent with this condition?

A. Painless hematuria and dysuria

B. Portal hypertension and periportal fibrosis

C. Swimmer’s itch due to cercarial penetration

D. Neurological deficit due to egg embolization

---

7. A patient is suspected to have schistosomiasis. Microscopic examination of urine and stool is conducted, including acid-fast staining. Which of the following Schistosoma species is not acid-fast positive?

A. Schistosoma haematobium

B. Schistosoma intercalatum

C. Schistosoma japonicum

D. Schistosoma mansoni

---

8. The given image was seen during a microscopic examination of a sputum sample from a patient with a cough with associated hemoptysis. Identify the false statement.



A. The eggs are operculated

B. It is the infective form of the causative agent

C. The mode of transmission is via ingestion

D. Crabs are second intermediate hosts, and humans are the definitive hosts

---

9. Match the following parasites with their correct common names and source of infection: Parasite Common Name and Source of Infection Fasciolopsis buski a) Small intestinal fluke; Freshwater fish Metagonimus yokogawai b) Chinese liver fluke; Freshwater fish Fasciola hepatica c) Giant intestinal fluke; Freshwater plants Clonorchis sinensis d) Colonic fluke; Freshwater plants Gastrodiscoides hominis e) Sheep liver fluke; Freshwater plants

Parasite	Common Name and Source of Infection
Fasciolopsis buski	a) Small intestinal fluke; Freshwater fish
Metagonimus yokogawai	b) Chinese liver fluke; Freshwater fish
Fasciola hepatica	c) Giant intestinal fluke; Freshwater plants
Clonorchis sinensis	d) Colonic fluke; Freshwater plants
Gastrodiscoides hominis	e) Sheep liver fluke; Freshwater plants

A. 1-a, 2-b, 3-c, 4-d, 5-e

B. 1-b, 2-c, 3-a, 4-e, 5-d

C. 1-c, 2-a, 3-e, 4-b, 5-d

D. 1-a, 2-d, 3-b, 4-c, 5-e

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10. A 56-year-old male patient presents with jaundice, weight loss, and pruritus. Imaging studies reveal a mass in the bile duct, and ERCP demonstrates strictures and dilated bile ducts. Further investigations suggest an infectious etiology. Which of the following evolutionary stages of an organism and secondary intermediate host is responsible for this presentation?

A. Cercaria, freshwater fish

B. Metacercaria, freshwater fish

C. Cercaria, crabs

D. Metacercaria, crabs

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11. A 35-year-old male presents with right upper quadrant abdominal pain, jaundice, nausea, and a history of consuming contaminated water and aquatic vegetation while camping near a river. Physical examination reveals tenderness in the right upper quadrant and mild hepatomegaly. Laboratory tests show elevated liver enzymes and bilirubin, and stool examination reveals operculated eggs (as shown in the image). What is the best management strategy for this patient?



A. Praziquantel

B. Mebendazole

C. Triclabendazole

D. Albendazole

---

12. A 35-year-old man presents with complaints of abdominal pain, diarrhea, and significant weight loss over the past month. He recently traveled to a rural area where he ingested aquatic plants while swimming in a freshwater lake. On examination, he exhibits pallor and mild abdominal tenderness. A stool examination reveals the presence of eggs. Considering his symptoms and recent travel history, what is the most likely diagnosis?



A. Opisthorchis viverrini

B. Schistosoma japonicum

C. Fasciolopsis buski

D. Entamoeba histolytica

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	3
Question 4	2
Question 5	3
Question 6	1
Question 7	1
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	3

Solution for Question 1:

Trematodes, or flukes, are flat, leaf-shaped worms with an incomplete digestive system, meaning they have a mouth and a simple gut but no anus.

Mode of Infection	Examples	Infective Form	Hosts
Ingestion of contaminated water/vegetation	Liver and intestinal flukes (e.g., Fasciola hepatica, Fasciolopsis buski)	Metacercaria	Definitive: Humans Intermediate: Snails (primary), vegetation (secondary)

Consumption of raw/undercooked fish/crustaceans	Lung and liver flukes (e.g., Clonorchis sinensis, Paragonimus westermani)	Metacercaria	Definitive: Humans Intermediate: Snails (primary), fish/crustaceans (secondary)
Skin penetration	Blood flukes (e.g., Schistosoma species)	Cercaria	Definitive: Humans Intermediate: Snails

Cestode (Option A): Cestodes, or tapeworms, are flatworms as well, but they are typically segmented and lack a digestive system entirely. Instead, they absorb nutrients directly through their body surface.

Nematode (Option B): Nematodes, also known as roundworms, have a cylindrical, unsegmented body and a complete digestive system.

Annelid (Option D): Annelids, or segmented worms, include earthworms and leeches. They have a segmented body and a complete digestive system.

Solution for Question 2:

Trematode Species	Primary Habitat
1. Fasciolopsis buski	b. Small intestine
2. Schistosoma mansoni	c. Mesenteric veins
3. Clonorchis sinensis	a. Bile ducts
4. Paragonimus westermani	d. Lungs

Habitats of common trematode species

Trematode Group	Species	Primary Habitat
Intestinal Trematodes	Fasciolopsis buski	Small intestine
Liver Trematodes	Fasciola hepatica	Liver and bile ducts
Clonorchis sinensis	Bile ducts	
Opisthorchis viverrini	Bile ducts	
Lung Trematodes	Paragonimus westermani	Lungs
Blood Trematodes	Schistosoma mansoni	Mesenteric veins

Schistosoma haematobium	Veins around the bladder	
Schistosoma japonicum	Mesenteric veins	
Pancreatic Trematodes	Eurytrema pancreaticum (rare)	Pancreatic ducts

Solution for Question 3:

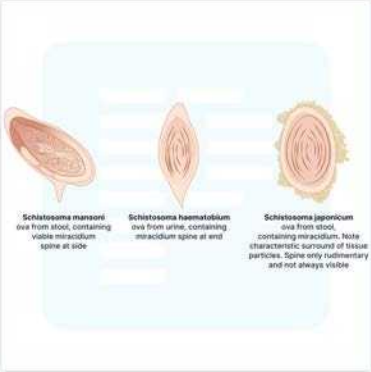
Schistosoma is unique among trematodes as it does not produce operculated eggs. Instead, its eggs are oval and have a characteristic spine, making them easily distinguishable from those of other trematodes.

General Characteristics of Trematode Eggs

- Size and Shape: Trematode eggs vary in size, typically ranging from small (20-30 µm) to large (up to 150 µm). The shape can be oval, round, or slightly asymmetrical.
- Operculum: Many trematode eggs have an operculum, a lid-like structure, that facilitates the release of larvae (miracidia) upon contact with water.
- Spines or Knobs: Certain species, particularly Schistosoma, have distinctive spines (terminal or lateral) used for species identification.
- Embryonation: Some eggs, such as those of Clonorchis and Opisthorchis, are embryonated upon excretion, meaning they contain a developed miracidium ready to hatch.
- Shell Color: Eggs often have a golden-brown appearance due to bile staining as they pass through the liver and intestines in many host species.

Trematode	Key Features	Sample Type
Schistosoma haematobium	Terminal spine at one end; found in urine	Urine
Schistosoma mansoni	Lateral spine on one side	Stool
Schistosoma japonicum	Rudimentary spine, often difficult to see	Stool
Fasciola hepatica	Operculum at one end; golden-brown	Stool
Paragonimus westermani	Thick shell, operculum present	Sputum/Stool

Clonorchis sinensis	Thickened rim around operculum; knob opposite operculum; embryonated	Stool
Opisthorchis spp.	Prominent operculum, embryonated	Stool

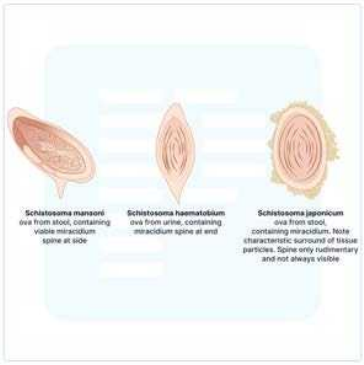
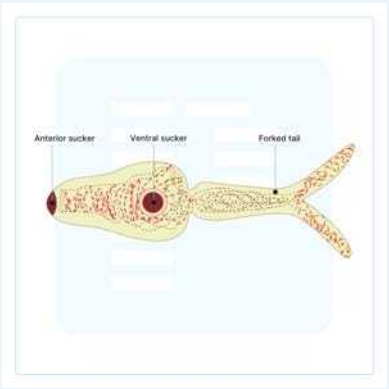


Solution for Question 4:

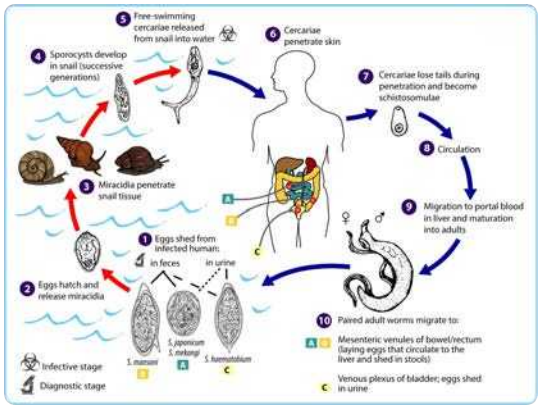
Life cycle of Schistosoma species lacks redia and metacercaria larvae. Instead, it consists of stages such as miracidia and sporocysts followed by cercariae.

Schistosoma (Blood Fluke):

Species	Schistosoma haematobium (Africa, Near East, Middle East, India) Schistosoma mansoni (Africa, South America) Schistosoma japonicum (China, Japan, Far East)
Habitat	S. haematobium: Veins of vesical and pelvic plexuses S. mansoni: Inferior mesenteric vein S. japonicum: Superior mesenteric vein (Option C)
Definitive Host	Man
Intermediate Host	Snail (Option A)
Mode of Infection	Skin penetration by cercaria (Option D)
Infective form	Cercaria larva

Eggs	 ... (content truncated)
Larva	 ... (content truncated)
Adult worm	Sexes are separate. Has 2 muscular suckers: Oral and ventral sucker Gynecophoric canal begins behind the ventral sucker, where the female worm is held.

Life Cycle:



Solution for Question 5:

Clinical features of painless hematuria and dysuria, combined with laboratory findings of eggs with a terminal spine, point toward a diagnosis of schistosomiasis caused by Schistosoma haematobium, which is associated with an increased risk of squamous cell carcinoma of the bladder.

Clinical Features of Varoius Schistosoma Species:

S. Haematobium	S. Mansoni	S. Japonicum
Swimmer's itch Painless hematuria, dysuria, proteinuria Urinary bladder carcinoma: Squamous cell carcinoma > Transitional cell carcinoma (Option C) Calculus formation Chronic cystitis	Swimmer's itch Katayama fever: High fever, rash, arthralgia, hepatosplenomegaly, lymphadenopathy Intestinal bilharziasis: Abdominal pain, bloody diarrhea (Option D) Portal hypertension and periportal fibrosis.	Severe swimmer's itch Katayama fever in acute phase Colicky abdominal pain, bloody diarrhea, anemia CNS involvement: Severe epileptic seizures (Option A) Lung involvement: Cor pulmonale,pulmonary hypertension.

Parasites associated with Malignancy:

Table rendering failed

Solution for Question 6:

The diagnosis of intestinal schistosomiasis is supported by abdominal pain and bloody diarrhea, with stool examination revealing eggs with a lateral spine which is characteristic of schistosoma mansoni . Painless hematuria and dysuria are more associated with urogenital schistosomiasis (Schistosoma haematobium), not intestinal schistosomiasis.

Schistosomiasis/ Bilharziasis:

Intestinal Schistosomiasis	Urogenital Schistosomiasis

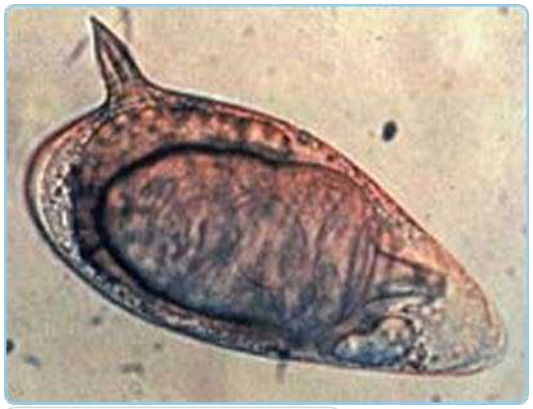
Caused by S. intercalatum, S. japonicum, S. mansoni and S. mekongi Swimmer's itch/Cercarial dermatitis (Option C) Katayama fever (S. Japonicum > S. Mansoni): Serum sickness like illness with high fever, rash, arthralgia, hepatosplenomegaly, lymphadenopathy Abdominal pain, bloody dysentery Portal hypertension Microabscess, granuloma, hyperplasia and eventually fibrosis of intestinal wall Periportal fibrosis (clay pipestem fibrosis) (Option B) Schistosomiasis - Periportal fibrosis (clay pipestem fibrosis)" data-author="" data-hash="9483" data-license="" data-source="" data-tags="" src="https://image.prepladder.com/notes/Dbk5bdKRHSfgJIqIBP9d1744385948.png" /> Lung involvement: Cor pulmonale CNS involvement: Severe epileptic seizures Neurological deficit: Egg embolization (Option D)	Caused by S. haematobium Swimmer's itch/Cercarial dermatitis Pulmonary symptoms: Cough, mild fever Painless hematuria Dysuria (Option A) Proteinuria Pseudoabscess in bladder Fibrosis of vesical mucosa and formation of egg granuloma (sandy patches) Recurrent UTI Hydronephrosis Hydroureter Urinary bladder carcinoma: Squamous cell carcinoma > Transitional cell carcinoma Chronic cystitis ... (content truncated)

Solution for Question 7:

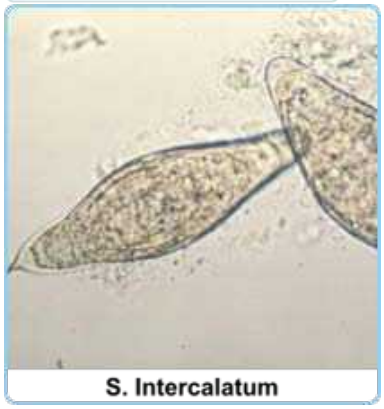
Schistosoma haematobium is acid-fast negative.

Microscopic Examination:

- Diagnostic form of Schistosoma: Non-operculated egg
- Stool Microscopy: S. mansoni (114 to 180 µm long): Ova with a lateral spine, acid-fast stain positive (Option D) S. japonicum (70-100 µm long): Ova with a lateral knob, acid-fast stain positive (Option C) S. intercalatum (140-240 µm): Ova with a terminal spine, acid-fast stain positive (Option B)
- S. mansoni (114 to 180 µm long): Ova with a lateral spine, acid-fast stain positive (Option D)
- S. japonicum (70-100 µm long): Ova with a lateral knob, acid-fast stain positive (Option C)
- S. intercalatum (140-240 µm): Ova with a terminal spine, acid-fast stain positive (Option B)
- S. mansoni (114 to 180 µm long): Ova with a lateral spine, acid-fast stain positive (Option D)
- S. japonicum (70-100 µm long): Ova with a lateral knob, acid-fast stain positive (Option C)
- S. intercalatum (140-240 µm): Ova with a terminal spine, acid-fast stain positive (Option B)







• Urine Microscopy of S. Haematobium (110-170 µm long): Ova with a terminal spine Acid fast stain negative (Option A)

• Ova with a terminal spine

• Acid fast stain negative (Option A)

• Ova with a terminal spine

• Acid fast stain negative (Option A)



S. haematobium

Treatment:

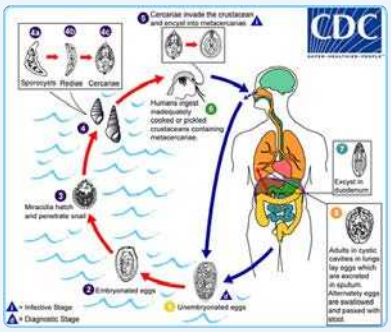
- Praziquantel (single oral dose 40 mg/kg): Drug of choice

Solution for Question 8:

The diagnosis based on the clinical scenario and the image is paragonimiasis caused by the parasite Paragonimus westermani also known as the lung fluke in which the infective form is the metacercarea (larva) and not the operculated eggs.

Paragonimus westermani / Lung fluke		
This is a trematode which is also known as the oriental lung fluke. It causes endemic hemoptysis in humans. Epidemiology: Global Prevalence: Approximately 22 million cases and 10 species infect humans. Key Regions: World: Predominantly found in the Far East (Korea, Japan, Taiwan, China, Philippines). India: Endemic in Northeast states, particularly Manipur (prevalence: 6.7%).		

Morphology	Adult worms: Leaf-like shape. Eggs: Operculated eggs. (Option A) They are oval and golden brown. Size: 80–120 µm × 45–65 µm. Eggs are unembryonated when laid. Larvae: Five developmental stages.	 An adult specimen stained with carmine ... (content truncated)
Hosts (Option D)	Definitive host: Humans. Intermediate hosts: First: Snails. Second: Crabs or crayfish.	
Transmission	Infection occurs through ingestion of infected crabs/crayfishes carrying metacercariae (infective larvae). Larvae penetrate the intestinal wall, enter the peritoneal cavity, and migrate to the lungs. Adult worms produce eggs, which are expelled through sputum (commonly) or feces.	

Pathogenicity and clinical features	Pulmonary paragonimiasis: It causes eosinophilic granulomatous inflammation and cyst formation around worms, often in the right lung. Symptoms: Productive cough with brownish, blood-tinged sputum (offensive, fishy odor). Possible frank hemoptysis and peripheral blood eosinophilia. Chronic infections may lead to bronchitis, bronchiectasis, pneumonia, or lung abscesses. Extrapulmonary paragonimiasis: Worms may migrate to sites like the liver, spleen, abdominal wall, and occasionally the brain. Forms: Cerebral Paragonimiasis: this is a severe form. Cutaneous Paragonimiasis: Migratory subcutaneous nodules and larva migrans.	
 Life cycle: Definite host: Humans. ... (content truncated)		
Diagnosis	Sputum microscopy: Collect early morning sputum samples. Examine saline mount for operculated eggs. Formalin-ether sedimentation technique is used for concentration.	

Egg characteristics: Shape: Oval, operculated, golden-brown. Size: 80–120 µm × 45–65 µm. Eggs are unembryonated when laid.	 {{caption_text}}	
Serological tests: Useful in early disease stages when microscopy is negative. Antibody detection: ELISA Western Blot Antigen Detection: Dot ELISA		
Other Diagnostic Tests: Peripheral blood eosinophilia. Radiological tests: MRI and CT scans for cyst localization; chest X-ray for pulmonary lesions.		
Management	Treatment	
Drug of Choice: Praziquantel (25 mg/kg/dose, three doses per day for 2 days). Surgical Management: This may be necessary for severe pulmonary or cerebral lesions.		
Prevention strategies		
Sanitary disposal of sputum. Control of snail populations. Treatment of infected individuals. Health education to raise awareness.		

Solution for Question 9:

Liver Flukes	Common Name	Habitat	Definitive Host	First Intermediate Host	Second Intermediate Host

Fasciola hepatica	Sheep liver fluke	Liver and biliary passages of definitive host	Sheep, goat, cattle and man	Snail	Freshwater plants
Fasciola gigantica	Giant liver fluke or tropical fluke				
Clonorchis sinensis	Chinese liver fluke	Biliary tract, sometimes pancreatic duct of humans	Human	Snail	Freshwater fish

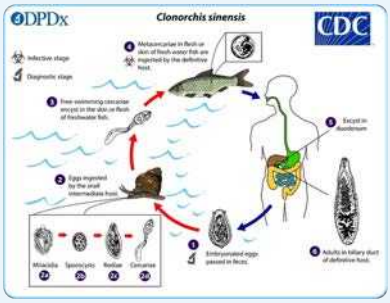
Intestinal Flukes	Common Name	Habitat	Definitive Host	First Intermediate Host	Second Intermediate Host
Fasciolopsis buski	Giant intestinal fluke	Duodenum or jejunum of pigs and man	Man and pigs	Snails	Freshwater plants
Metagonimus yokogawai	Small intestinal fluke	Small intestine of definitive host	Humans, pigs, dogs, cats and pelicans.	Snail	Freshwater fish
Heterophyes heterophyes	Small intestinal fluke	Small intestine of definitive host	Humans, cats, dogs, foxes and other fish-eating mammals	Snail	Freshwater fish
Gastrodiscoides hominis	Colonic fluke	Human large intestine	Man, pigs and monkey	Snail	Freshwater plants

Solution for Question 10:

Metacercariae are the larval stage of trematodes like Clonorchis sinensis, which infects the bile ducts. These larvae encyst in fish and are transmitted to humans through the consumption of undercooked fish, leading to inflammation and strictures in the biliary system.

Clonorchis sinensis " Chinese liver fluke ":

Parasite	Clonorchis sinensis (Chinese liver fluke) is a parasitic fluke that causes clonorchiasis in humans.
Transmission	Acquired by consuming undercooked or raw freshwater fish contaminated with the parasite.

Morphology	 ... (content truncated)
Life Cycle	 {{caption_text}} 1. Egg Release: Infected individuals excrete eggs in feces, contaminating freshwater sources. 2. Snail Host: Eggs ingested by freshwater snails (first intermediate host) hatch and develop through stages: miracidia, sporocysts, rediae, and finally cercariae. ... (content truncated)
Presentation Features	Acute Infection: Symptoms are often mild or absent. When present, it may include right upper quadrant abdominal pain, fever, fatigue, nausea, diarrhea, and eosinophilia. Chronic Infection: Long-term infections may lead to biliary obstruction, cholangitis, cholecystitis, or gallstones. Severe cases can cause liver cirrhosis, bile duct cancer (cholangiocarcinoma), and pancreatitis.
Investigations	1. Stool Examination: Microscopy for characteristic eggs in stool samples (primary diagnostic method; eggs may be scarce in light infections). 2. Duodenal Aspirate/Bile Examination: Detects eggs in bile samples or duodenal aspirates, helpful when stool examination is inconclusive. 3. Serology: Blood tests to detect antibodies or antigens specific to Clonorchis sinensis. 4. Imaging: Ultrasound may show biliary dilatation or thickening, suggestive of chronic infection. CT/MRI useful for identifying biliary obstruction, liver abnormalities, or cholangiocarcinoma in severe cases.
Management	Praziquantel (Drug of Choice): Dosage 25 mg/kg orally, three times daily for 2 days. Alternative: Albendazole can be used if Praziquantel is contraindicated.

Solution for Question 11:

This patient likely has a liver fluke infection caused by Fasciola hepatica, which is commonly acquired through the consumption of contaminated water or aquatic vegetation. The operculated eggs found in the stool confirm this diagnosis. Fasciola hepatica infection is typically treated with Triclabendazole, a broad-spectrum anthelmintic that is effective against various liver flukes.

Fasciola hepatica

Fasciola hepatica, commonly known as the liver fluke, is a parasitic trematode (flatworm) that primarily infects the livers of various mammals, including humans. It causes the disease fascioliasis, leading to liver damage, biliary obstruction, and inflammatory reactions.

Morphology

Adult Fluke:



- Shape: Leaf-shaped, flat, and broad.
- Size: Approximately 2-3 cm in length and 1 cm in width.
- Color: Typically brown or grey.
- Distinctive Features: Cone-shaped anterior end. Two suckers, an oral sucker at the front and a ventral sucker close to it, used for attachment to host tissues. The fluke's body is covered by a tegument (a protective outer layer) with spines aiding in attachment.
- Cone-shaped anterior end.
- Two suckers, an oral sucker at the front and a ventral sucker close to it, used for attachment to host tissues.
- The fluke's body is covered by a tegument (a protective outer layer) with spines aiding in attachment.

• Cone-shaped anterior end.

• Two suckers, an oral sucker at the front and a ventral sucker close to it, used for attachment to host tissues.

• The fluke’s body is covered by a tegument (a protective outer layer) with spines aiding in attachment.

Eggs:



• Size: Large, oval, and approximately 130-150 µm in length.

• Color: Yellowish-brown.

• Characteristics: Each egg has an operculum (a lid-like structure) at one end, which opens to release the miracidium (the larval stage). Eggs are unembryonated when passed out in feces.

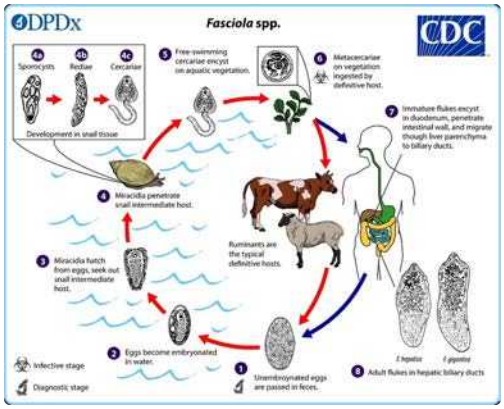
• Each egg has an operculum (a lid-like structure) at one end, which opens to release the miracidium (the larval stage).

• Eggs are unembryonated when passed out in feces.

• Each egg has an operculum (a lid-like structure) at one end, which opens to release the miracidium (the larval stage).

• Eggs are unembryonated when passed out in feces.

Life Cycle



- Eggs are expelled in the feces of infected mammals (definitive hosts).
- Under suitable environmental conditions, miracidia are released from eggs after 9–15 days.
- Miracidia penetrate specific freshwater snails, such as Lymnaea spp.
- Inside the snail, they progress through stages: sporocyst, redia, and eventually cercaria.
- Cercariae exit the snail and encyst on aquatic vegetation, transforming into metacercariae, the infective stage.
- Mammals, including humans, become infected by consuming contaminated water or vegetation with metacercariae.
- In the duodenum, metacercariae excyst, penetrate the intestinal wall, migrate to the liver, and lodge in the bile ducts, maturing into adult flukes.

Clinical Presentation

Acute Phase (Invasive or Hepatic Stage):

- Duration: Begins about 2-3 weeks post-infection and may last for several months.
- Symptoms: Fever, right upper quadrant abdominal pain. Hepatomegaly (enlarged liver), and occasionally jaundice. Nausea, vomiting, and weight loss. Eosinophilia (elevated eosinophil count in blood).
- Fever, right upper quadrant abdominal pain.
- Hepatomegaly (enlarged liver), and occasionally jaundice.
- Nausea, vomiting, and weight loss.
- Eosinophilia (elevated eosinophil count in blood).
- Fever, right upper quadrant abdominal pain.
- Hepatomegaly (enlarged liver), and occasionally jaundice.
- Nausea, vomiting, and weight loss.
- Eosinophilia (elevated eosinophil count in blood).

Chronic Phase (Biliary Stage):

• Duration: Occurs months to years post-infection.

• Symptoms: Biliary colic, obstructive jaundice, and cholangitis (inflammation of the bile ducts). Gallbladder symptoms, liver abscesses, and secondary bacterial infections in severe cases. Possible fibrosis of the bile ducts with prolonged infection.

• Biliary colic, obstructive jaundice, and cholangitis (inflammation of the bile ducts).

• Gallbladder symptoms, liver abscesses, and secondary bacterial infections in severe cases.

• Possible fibrosis of the bile ducts with prolonged infection.

• Biliary colic, obstructive jaundice, and cholangitis (inflammation of the bile ducts).

• Gallbladder symptoms, liver abscesses, and secondary bacterial infections in severe cases.

• Possible fibrosis of the bile ducts with prolonged infection.

Investigations

• ELISA for detection of anti-Fasciola antibodies (sensitive in both acute and chronic infections).

• Immunoblotting can also confirm the diagnosis.

• Microscopic detection of eggs in stool samples; however, eggs are typically not present in the acute phase and may be difficult to detect in chronic cases.

• Ultrasound: Useful in detecting hepatobiliary abnormalities such as bile duct dilation or liver abscesses.

• Eosinophilia is common in the acute phase.

• Liver function tests: Elevated liver enzymes, especially in chronic disease.

Management

Antiparasitic Treatment:

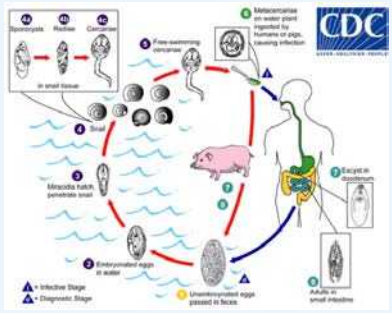
• Triclabendazole: The drug of choice, effective against both adult and larval stages. Dosage: Oral dose of 10-12 mg/kg, with a repeat dose after 12 hours if needed.

Alternative Medications:

Bithionol: Sometimes used if triclabendazole is unavailable.

Solution for Question 12:

The clinical presentation and history of the patient are suggestive of an infection caused by Fasciolopsis buski, a intestinal fluke commonly associated with aquatic plants ingestion.

Parasite	Fasciolopsis buski (Giant Intestinal Fluke) is a parasitic trematode infecting the small intestine of humans and pigs, causing fasciolopsiasis, leading to gastrointestinal symptoms and malabsorption.
Morphology	 ... (content truncated)
Life Cycle	 Egg Stage: Eggs released in feces of infected humans or pigs, reach freshwater sources. Miracidium Stage: Eggs hatch into miracidia in water, which seek freshwater snails. ... (content truncated)
Presentation Features	Acute Phase: Symptoms appear 2-3 months after ingestion. Abdominal Pain: Common, usually around the upper abdomen. Diarrhea: Intermittent or prolonged. Nausea and Vomiting: Frequent in heavy infections. Malabsorption and Nutritional Deficiencies: Severe cases lead to growth retardation and edema, especially in children. Chronic Phase: Prolonged infection causes chronic diarrhea, weakness, and malabsorption due to the large size of flukes. Eosinophilia: Elevated eosinophil count common in parasitic infections.

Investigations	Stool Examination: Primary diagnostic tool for detecting large, operculated eggs in fecal samples. Serologic Tests: ELISA and immunodiagnostic tests, though less commonly used in endemic areas. Imaging: Ultrasound or CT scans to detect adult flukes in cases of heavy infection or suspected obstruction.
Management	Pharmacological Treatment: Praziquantel: Drug of choice for treating fasciolopsiasis, with a dosage of 25 mg/kg in a single dose. Alternative Treatment: Triclabendazole, also effective but not universally available.

Nematodes

1. A 32-year-old male presents with intense itching and a rash on his lower legs and arms after camping near a river. Examination reveals multiple papules and thickened, scaly skin. He reports blurred vision and seeing "floaters," with an ophthalmological exam showing keratitis and punctate keratopathy. Which vector should you be particularly vigilant for to prevent further transmission of this infection?

A. Mansonia

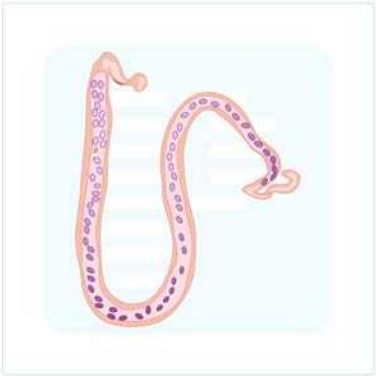
B. Chrysops

C. Simulium

D. Culex

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2. A microbiologist is studying various tissue nematodes and comes across a specimen characterized by a single nucleus located at the terminal position & shape as shown in the image. Which of the following nematodes is most likely to be identified from this specimen?



A. Wuchereria bancrofti

B. Brugia malayi

C. Onchocerca volvulus

D. Loa loa

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3. A 35-year-old patient presents with intermittent fever, lymphadenopathy, and swelling in the right leg. Blood tests show eosinophilia, and the patient has a history of travel to endemic regions. Which of the following statements about the laboratory tests for diagnosing filariasis is incorrect?

A. Nucleopore filtration is used to concentrate microfilaria in blood samples.

B. USG scrotum can identify motile adult worms within dilated lymphatics

C. Immunochromatographic test (ICT) detects W. bancrofti antigen in serum

D. Antibody demonstration distinguishes between current and past infections

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4. Which of the following statements regarding classical and occult filariasis is incorrect?

A. Microfilaria is present in the blood in both classical and occult filariasis

B. Classical filariasis is due to adult worms; occult filariasis is from hypersensitivity

C. Classical filariasis doesn't respond to treatment; occult filariasis responds to DEC

D. Classical filariasis has lymphangitis; occult filariasis shows eosinophilic granulomas

---

5. A 38-year-old patient presents with dry nocturnal cough, dyspnea, and wheezing. Blood tests show marked eosinophilia and elevated IgE, with no microfilaria in the blood. A chest X-ray reveals mottled shadows. The patient has a history of travel to endemic areas. What is the most likely diagnosis?

A. Loeffler’s syndrome

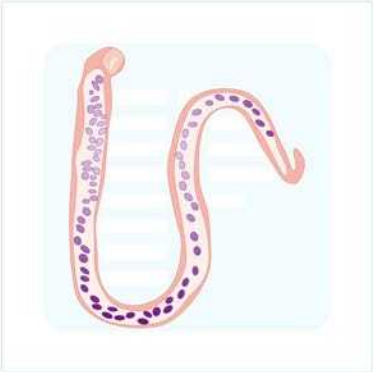
B. Miliary tuberculosis

C. Weingarten's Syndrome

D. Chronic eosinophilic pneumonia

---

6. Based on the morphological pattern shown in the image, which parasite is most likely being represented?



A. Loa loa

B. Mansonia perstans

C. Onchocerca volvulus

D. Wuchereria bancrofti

---

7. A healthcare team in an endemic region is educating the community on ways to prevent Dracunculus medinensis infection. Which of the following would be the best advice to reduce the risk of infection?

A. Use bed nets to avoid mosquito bites

B. Avoid consuming raw vegetables and fruits

C. Boil or filter drinking water to eliminate Cyclops

D. Take prophylactic anti-helminthic medication regularly

---

8. A 29-year-old female presents with symptoms including fever, myalgia, and periorbital swelling. She mentions consuming undercooked pork during a recent camping trip. On physical examination, she exhibits noticeable edema around the eyes, tenderness in the muscles, and a palpable rash. Which diagnostic test is the most definitive for confirming the suspected condition?

A. Serologic tests for antibodies

B. Stool examination for ova and parasites

C. Muscle biopsy

D. Complete blood count with differential

---

9. A 30-year-old man presents with complaints of muscle pain, fever, and swelling. He reports recently consuming pork meat during a camping trip. On examination, he has a finding as shown in the image & muscle tenderness. Laboratory investigations reveal elevated eosinophil counts and creatine kinase (CK) levels. What is the most likely diagnosis?



A. Leptospirosis

B. Toxoplasmosis

C. Taeniasis

D. Trichinellosis

10. A 30-year-old man who recently traveled to a tropical region presents with nausea, intermittent epigastric pain, occasional loose stools, and a maculopapular rash on his perineum. Stool culture identifies Strongyloides stercoralis rhabditiform larvae. Which treatment should be administered?

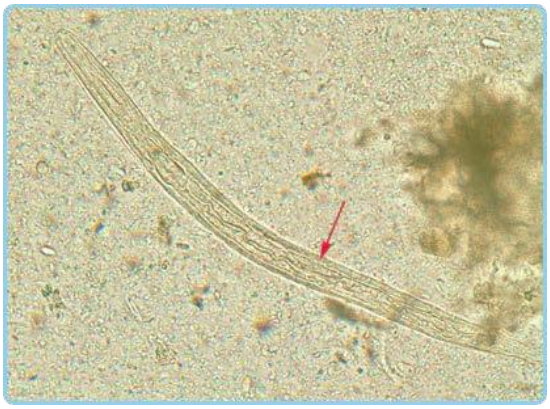
A. Albendazole

B. Mebendazole

C. Ivermectin

D. Praziquantel

11. A 30-year-old woman presents with intermittent abdominal pain, diarrhea, and weight loss over the past month. She has a history of walking barefoot in a rural tropical region. A stool sample reveals the following findings. Which larval stage of the parasite is most likely identified in the stool sample?



A. Rhabditiform larva

B. Filariform larva

C. Procercoid larva

D. Plerocercoid larva

12. A 40-year-old man from a rural village in South India presents with a chronic cough, intermittent wheezing, and vague abdominal pain. His symptoms have been present for several months, and a recent stool examination reveals rhabditiform larvae. Which of the following is the most likely route of infection in this patient?

A. Ingestion of food contaminated with an embryonated egg

B. Skin penetration by filariform larvae

C. Ingestion of food contaminated with rhabditiform larvae

D. Skin penetration by rhabditiform larvae from contaminated soil

13. A male patient comes with complaints of rashes on the skin that appear as follows. He gives a history of walking barefoot in soil in the fields near his house. Which organism could be the most probable cause of this condition?



A. Gnathostoma spinigerum

B. Ascaris suum

C. Ancylostoma braziliense

D. Toxocara canis

14. Match the following nematodes with their respective characteristics. Nematode Characteristics a. Trichinella spiralis 1. Ovoviviparous b. Onchocerca volvulus 2. Oviparous c. Enterobius vermicularis 3. Viviparous d. Strongyloides stercoralis 4. Causes cutaneous filariasis e. Trichuris trichiura 5. Found in large intestine

Nematode	Characteristics
a. Trichinella spiralis	1. Ovoviviparous
b. Onchocerca volvulus	2. Oviparous
c. Enterobius vermicularis	3. Viviparous
d. Strongyloides stercoralis	4. Causes cutaneous filariasis
e. Trichuris trichiura	5. Found in large intestine

A. a - 3, b - 1, c - 4, d - 5, e - 2

B. a - 1, b - 3, c - 4, d - 2, e - 5

C. a - 2, b - 4, c - 3, d - 1, e - 5

D. a - 5, b - 2, c - 6, d - 3, e - 4

---

15. Identify the distinguishing features of Old World hookworms from New World hookworms.

A. The presence of four ventral teeth and two dorsal knob-like teeth in the adult worm

B. A lesser incidence of pulmonary reactions associated with Old World hookworms

C. The presence of two ventral and two dorsal chitinous cutting plates in the adult worm

D. The head of the larva bends in the opposite direction to the curvature of the body

---

16. A group of doctors conduct a health checkup camp in a village. A screening for hookworm infection in the community is done. The Chandler’s index of the

community is 400 eggs/gram. What does this observation signify?

A. Indicates potential damage

B. Indicates minor public health problem

C. Not of much importance

D. Indicates major public health problem

---

17. A patient comes with mild epigastric pain, weakness, and diarrhea. On further evaluation, he is found to have anemia and eosinophilia. Stool examination reveals eggs of Ancylostoma. Which of the following differentiates a male hookworm from a female?

A. Presence of copulatory bursa

B. Tapering posterior end

C. Larger size

D. Genital opening at the junction of the middle and posterior 3rd of the body

---

18. An 8-year-old child is brought by her parents with complaints of passing urine in bed at night for a month. She also complains of itching around her perianal region. On examination, she has perianal pruritis. The NIH swab method is done. Which of the following is false about this method and its findings?

A. The sample is collected from the perianal region in this method for microscopy

B. The eggs are observed to have a planoconvex shape

C. Eggs carry tadpole-shaped larvae

D. A piece of cellophane tape is applied to the perianal region to collect the sample

---

19. A 5-year-old child is brought to the hospital by her parents complaining of her waking up during the night with intense itching in the perianal region. The parents also noticed worms in the child's perianal region, especially during the night. Which of the following stages of this organism is infectious to humans?

A. Larvae

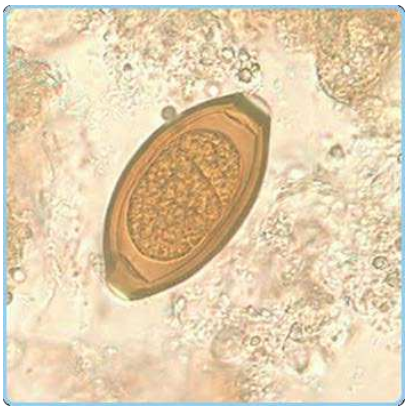
B. Embryonated eggs

C. Adult female worms

D. Adult male worms

---

20. A patient presents with abdominal pain and anorexia. She also complains of bloody diarrhea and generalized weakness. During stool examination, the following was observed: Which of the following drugs is the best choice for treatment?



A. Mebendazole

B. Albendazole

C. Ivermectin

D. Metronidazole

---

21. A 6-year-old child is brought with abdominal pain. On further evaluation, he is anemic and has growth retardation, and the physician suspects trichuriasis. Which of the following is true regarding the lifecycle of the organism? The eggs hatch into the larvae in the small intestine Man is the intermediate host in its lifecycle The larva is the infective form The development of the egg takes place in the soil

A. 1 and 4

B. 1,2, and 3

C. 3 and 4

D. 2 and 3

---

22. A 5-year-old boy is brought to the clinic with a 2-week history of fever, cough, and abdominal pain. His mother mentions that he often plays in a garden where their dog defecates. A blood test reveals eosinophilia and elevated liver enzymes. A chest X-ray shows bilateral pulmonary infiltrates. Which of the following is the most likely cause of his symptoms?

A. Toxocara cati

B. Anisakis

C. Toxocara canis

D. Ascaris lumbricoides

---

23. A 35-year-old woman presents with abdominal pain, diarrhea, and recent weight loss after consuming unwashed vegetables in a rural area. Upon examination, a diagnostic test reveals the presence of eggs in her stool. Which of the following best describes the egg shown below?



A. It floats in saturated salt solution

B. It does not float in saturated salt solution

C. It has translucent shell with three layers

D. It has a large unsegmented ovum with coarse lecithin granules

24. A 7-year-old boy presents with a 2-week history of cough, chest discomfort, and fever. His parents report that he has been playing in a community garden and often eats unwashed vegetables. Laboratory tests reveal eosinophilia and pulmonary infiltrates. What is the most likely diagnosis for this child?

A. Strongyloides stercoralis

B. Toxocara canis

C. Ascaris lumbricoides

D. Schistosoma mansoni

25. A 7-year-old child presents with abdominal discomfort, growth retardation, and a cough. The clinician suspects a parasitic infection. Upon examination, stool analysis shows cylindrical worms as given below. Which of the following features is most characteristic of the female worm?



A. Ventral curvature of the posterior end

B. Presence of two copulatory spicules

C. A straight conical posterior end

D. Smaller size compared to the male

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	4
Question 4	1
Question 5	3
Question 6	4
Question 7	3
Question 8	3
Question 9	4
Question 10	3
Question 11	1
Question 12	2
Question 13	3
Question 14	3
Question 15	1
Question 16	4
Question 17	1
Question 18	4
Question 19	2
Question 20	2
Question 21	1
Question 22	3

Question 23	2
Question 24	3
Question 25	3

Solution for Question 1:

Correct Answer: C) Simulium

Explanation:

The clinical presentation suggests onchocerciasis, also known as "river blindness," which is caused by the parasite Onchocerca volvulus. This disease is transmitted by the Simulium fly, commonly known as the blackfly, which breeds in fast-flowing rivers and streams, consistent with the patient's recent camping near a river.

Onchocerca volvulus

Onchocerca volvulus is a parasitic nematode responsible for onchocerciasis, also known as " river blindness." It primarily affects the skin and eyes, leading to severe itching, disfigurement, and potential blindness.

Transmission: Simulium blackflies, commonly found near fast-flowing rivers and streams.

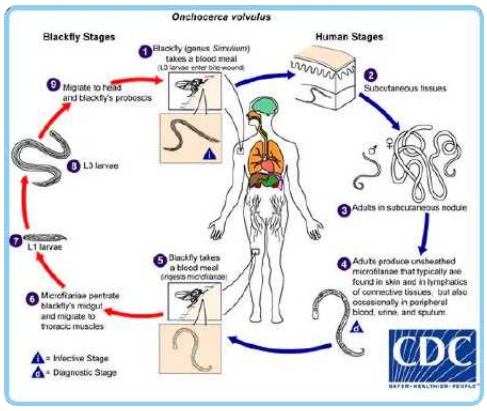
Morphology

Adult Worms: Long and thin; males are around 3–5 cm, while females are significantly larger, reaching up to 50 cm. Both sexes have a cuticle with annular grooves.



Microfilariae: Small, thread-like, measuring around 220–250 μm in length. They lack a sheath and migrate mainly to the skin and eyes.

Life Cycle



Presentation

Skin: Severe itching, dermatitis, and skin depigmentation ("leopard skin"). Chronic cases may show skin thickening and loss of elasticity.

Eyes: Microfilariae in the eye leads to punctate keratitis, sclerosing keratitis, chorioretinitis, and eventual blindness ("river blindness").

Nodules: Subcutaneous nodules (onchocercomas) often form over bony prominences.

Investigations

- Skin snip: A small sample of skin is examined microscopically to detect microfilariae.
- Mazzotti test: An oral dose of diethylcarbamazine (DEC) causes a pruritic reaction in the affected skin, but due to severe side effects, it’s rarely used now.
- Serology: ELISA and rapid antibody tests can aid in diagnosis, especially in non-endemic areas.
- Slit-Lamp Examination: For detecting microfilariae in the cornea and anterior chamber of the eye.

Management

Ivermectin:

- Primary treatment.
- Kills microfilariae and reduces symptoms, though it does not kill adult worms.
- Administered every 6-12 months to control infection spread and symptoms.

Doxycycline:

- Targets Wolbachia, an endosymbiotic bacteria essential for the survival of adult worms, which helps reduce microfilariae production.

Solution for Question 2:

Correct Answer: D) Loa loa

Explanation:

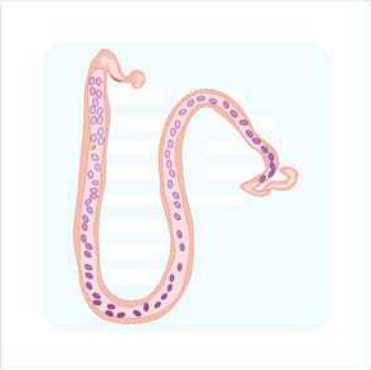
Loa loa is characterized by a long, slender shape, a curved tail with a prominent terminal bulb, and a single nucleus at the tail's terminal position.

Loa loa

Loa loa, commonly known as the African eye worm, is a parasitic nematode that causes loiasis, a disease primarily characterized by the migration of adult worms through subcutaneous tissues and occasionally into the conjunctiva of the eye.

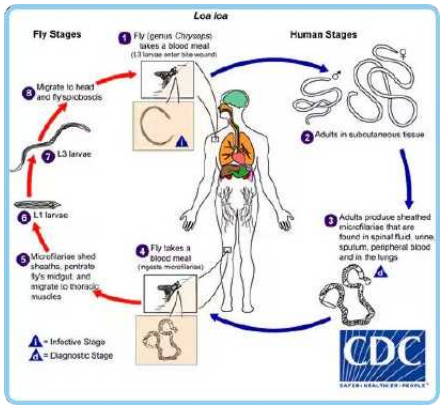
Vector: Deer flies from the genus Chrysops.

Morphology



- Shape: Long and slender, typically measuring about 50-70 mm in length.
- Tail Features: The posterior end is characterized by a curved tail with a prominent terminal bulb.
- Nucleus in Tail: Contains one nucleus located at the terminal position of the tail.
- Sheathing: Loa loa does not have a sheath, which distinguishes it from some other filarial species.

Life Cycle



Investigations and Diagnosis

Blood Smear:

- Identifies microfilariae in the blood.
- Best detected during the day when they are more prevalent.

Serological Tests:

- Utilizes serological assays and PCR for confirmation.
- Particularly useful when microfilariae are hard to detect.

Clinical Examination: Ocular symptoms, such as the presence of the worm in the conjunctiva, are key indicators of infection.

Management

Primary Treatment: Diethylcarbamazine (DEC) to eliminate microfilariae and adult worms.

Alternatives: Ivermectin may be used in cases of severe allergic reactions or significant eye involvement.

Solution for Question 3:

Correct Answer: D) Antibody demonstration distinguishes between current and past infections.

Explanation:

Antibody demonstration cannot effectively distinguish between current and past infections of filariasis.

Lab Diagnosis:

1. Demonstration of Microfilaria:

- In blood, chylous urine, exudate of lymph varix or hydrocele fluid.
- Best collect night blood samples between 10 pm and 4 am.
- Microfilaria can be seen in unstained as well as stained specimens.
- Staining is performed using Giemsa, Leishman, or polychrome methylene blue.



- nuclei not going till tip
- sheathed

2. Concentration Techniques: Done if microfilaria density is low

- Knott's concentration technique
- Nucleopore filtration (Option A)
- Diethylcarbamazine provocation test (DCT): Small dose of DEC given (2 mg/kg body weight) Forces microfilaria to appear in peripheral blood during daytime.
- Small dose of DEC given (2 mg/kg body weight)
- Forces microfilaria to appear in peripheral blood during daytime.
- Quantitative buffy coat
- Small dose of DEC given (2 mg/kg body weight)
- Forces microfilaria to appear in peripheral blood during daytime.

3. Ultrasonography:

- Identify motile adult worms in the scrotum and female breast within dilated lymphatics, known as the "filaria dance sign."(Option B)

4. Hemogram:

- Eosinophilia
- Elevated serum IgE levels

5. Demonstration of Antibody: Cannot distinguish between current and past infections. (Option D)

6. Demonstration of Circulating Antigen:

- Highly sensitive and specific
- Trop-bio test: Semiquantitative sandwich ELISA
- ELISA: To detect specific IgG4 antibody against W.bancrofti antigen (WbSXP-1)
- Immunochromatographic test (ICT): Detects soluble W.bancrofti antigen using monoclonal antibody (AD12) in serum. (Option C)

Treatment:

- Drug of Choice: Diethylcarbamazine (DEC)
- Mass drug therapy is given as either: DEC alone DEC + Albendazole DEC + Ivermectin
- DEC alone
- DEC + Albendazole
- DEC + Ivermectin
- DEC alone
- DEC + Albendazole
- DEC + Ivermectin

Solution for Question 4:

Correct Answer: A) Microfilaria is present in the blood in both classical and occult filariasis.

Explanation:

Microfilaria is present in the blood in classical filariasis but is found in tissues, not blood, in occult filariasis.

Aspect	Classical filariasis	Occult filariasis
Cause (Option B)	Due to adult and developing worms	Hypersensitivity to microfilarial antigen
Basic Lesion (Option D)	Lymphangitis Lymphadenitis	Eosinophilic granuloma formation
Organs Involved	Lymphatic vessels and lymph nodes	Lymphatic system, lung, liver, spleen, joints

Microfilaria (Option A)	Present in blood	Present in tissues but not in blood
Serological Tests	Complement fixation test not so sensitive	Complement fixation test highly sensitive
Therapeutic Response (Option C)	No response	Prompt response to diethylcarbamazine (DEC)

Solution for Question 5:

Correct Answer: C) Weingarten's Syndrome

Explanation: The clinical presentation with marked eosinophilia, elevated IgE, and a history of travel to endemic areas points toward a diagnosis of Weingarten's Syndrome (Tropical Pulmonary Eosinophilia).

Clinical Features:

- Acute Filariasis: Fever Lymphangitis Lymphadenitis Lymphedema Lymphangiovarix
- Fever
- Lymphangitis
- Lymphadenitis
- Lymphedema
- Lymphangiovarix
- Chronic Filariasis: Hydrocele: Occurs due to obstruction of lymph vessels of the spermatic cord Lymphorrhagia: Rupture of lymph varices resulting in: Chyluria: Milkish white urine Chylous diarrhea Chylous ascites Chylothorax
- Hydrocele: Occurs due to obstruction of lymph vessels of the spermatic cord
- Lymphorrhagia: Rupture of lymph varices resulting in: Chyluria: Milkish white urine Chylous diarrhea Chylous ascites Chylothorax
- Chyluria: Milkish white urine
- Chylous diarrhea
- Chylous ascites
- Chylothorax
- Fever
- Lymphangitis

• Lymphadenitis

• Lymphedema

• Lymphangiovarix

• Hydrocele: Occurs due to obstruction of lymph vessels of the spermatic cord

• Lymphorrhagia: Rupture of lymph varices resulting in: Chyluria: Milkish white urine Chylous diarrhea Chylous ascites Chylothorax

• Chyluria: Milkish white urine

• Chylous diarrhea

• Chylous ascites

• Chylothorax

• Chyluria: Milkish white urine

• Chylous diarrhea

• Chylous ascites

• Chylothorax



• Elephantiasis: Delayed sequel due to repeated lymphangitis, obstruction, and lymphedema.



Tropical Pulmonary Eosinophilia/ Weingarten's Syndrome: (Option C)

• Manifestation of occult filariasis.

• Cause: Hypersensitivity reaction to microfilaria antigens → Massive destruction of microfilaria

• Clinical Manifestation: Low-grade fever Loss of weight Dry nocturnal cough Dyspnea Asthmatic wheezing

• Low-grade fever

• Loss of weight

• Dry nocturnal cough

• Dyspnea

• Asthmatic wheezing

• Investigations: Chest X-ray: Mottled shadows Blood: Increased eosinophil count (>3000 μm) High level of serum IgE and filarial antibodies No microfilaria in peripheral blood. Serology: Positive filarial antigens

• Chest X-ray: Mottled shadows

• Blood: Increased eosinophil count (>3000 μm) High level of serum IgE and filarial antibodies No microfilaria in peripheral blood.

• Increased eosinophil count (>3000 μm)

• High level of serum IgE and filarial antibodies

• No microfilaria in peripheral blood.

• Serology: Positive filarial antigens

• Low-grade fever

• Loss of weight

• Dry nocturnal cough

• Dyspnea

• Asthmatic wheezing

• Chest X-ray: Mottled shadows

• Blood: Increased eosinophil count (>3000 μm) High level of serum IgE and filarial antibodies No microfilaria in peripheral blood.

• Increased eosinophil count (>3000 μm)

• High level of serum IgE and filarial antibodies

• No microfilaria in peripheral blood.

• Serology: Positive filarial antigens

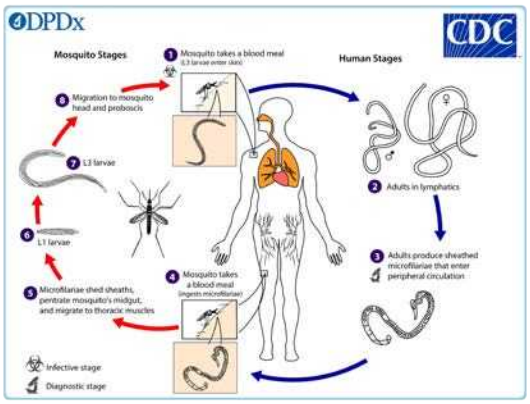
• Increased eosinophil count (>3000 μm)

• High level of serum IgE and filarial antibodies

• No microfilaria in peripheral blood.

• Treatment: Diethylcarbamazine (DEC)

Life Cycle of Filaria (Wuchereria bancrofti, Brugia malayi):



Loeffler’s syndrome (Option A) is less likely because it typically presents with systemic symptoms such as fever, urticaria, and wheezing, which are not mentioned in the case. Additionally, Loeffler's syndrome is characterized by migratory pulmonary infiltrates on imaging, which does not align with the described findings of mottled shadows.

Miliary tuberculosis (Option B) is less likely because it usually presents with systemic symptoms such as fever, night sweats, and weight loss, which are not indicated in this case. The presence of marked eosinophilia and elevated IgE levels also points away from tuberculosis, as this condition is generally associated with lymphocytic responses rather than eosinophilia.

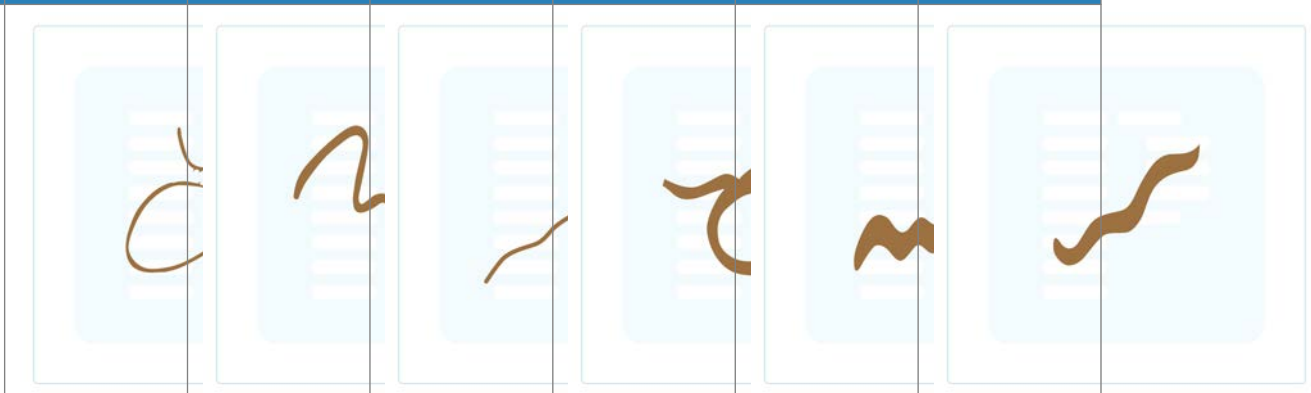
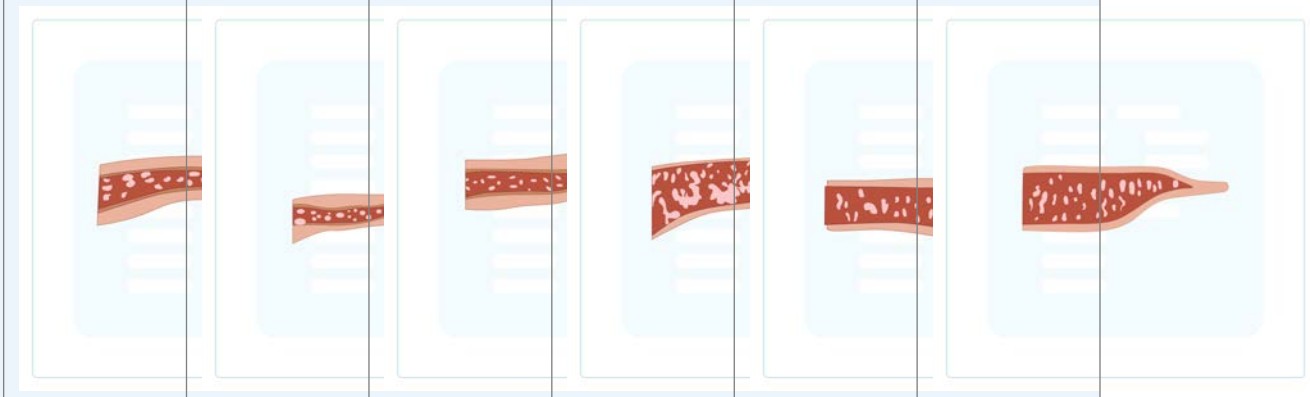
Chronic eosinophilic pneumonia (Option D) is characterized by a triad of pulmonary symptoms, eosinophilia, and specific radiographic findings, including bilateral peripheral opacities; however, the presence of mottled shadows on the chest X-ray and a history of travel to endemic areas suggest a diagnosis more consistent with Weingarten's syndrome.

Solution for Question 6:

Correct Answer: D) Wuchereria bancrofti

Explanation: The parasite shown is Wuchereria bancrofti, which is sheathed and has nuclei that do not extend to the tip of the tail.

Species	Wuchereria bancrofti (Option D)	Brugia malayi	Loa loa (Option A)	Mansonella perstans (Option B)	Mansonella ozzardi	Onchocerca volvulus (Option C)

Shape						
Posterior end						
Tail nuclei	Nuclei do not extend to the tip of the tail	2 nuclei at the tip of the tail	Nuclei form continuous rows in the tip of the tail	Nuclei extend to the tip of the tail	Nuclei do not extend to the tip of the tail	Nuclei do not extend to the tip of the tail
Sheath	Sheathed	Unsheathed				
Periodicity	Nocturnal	Nocturnal	Diurnal	Nonperiodic		
Primary Vector	Culex, Anopheles, Aedes	Mansonia spp., Anopheles	Chrysops spp.	Simulium spp.	Culicoides	Culicoides

Brugia timori:

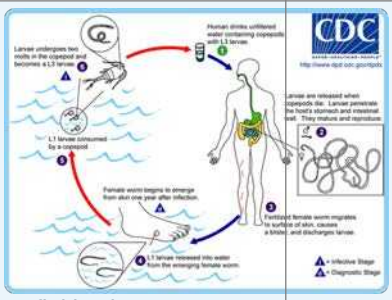
- Periodicity: Nocturnal
- Vector: Anopheles barbirostris

Solution for Question 7:

Correct Answer: C) Boil or filter drinking water to eliminate Cyclops

Explanation:

Boiling or filtering water is the most effective prevention strategy against Dracunculus medinensis infection, as it removes Cyclops, the water flea that serves as the intermediate host and carries the infective L3 larvae. By eliminating Cyclops, individuals prevent the ingestion of larvae, which breaks the transmission cycle. This approach is essential in endemic areas with limited access to safe water.

Dracunculus Medinensis		
Life Cycle		
 Definitive host: Human Intermediate host: Cyclops Transmission: Ingestion of water-containing Cyclops with L3 larvae. ... (content truncated)		
Pathogenesis & Symptoms	Symptoms appear ~1 year after infection when the female worm emerges. Symptoms: painful blisters with erythema, fever, nausea, pruritus; common sites include lower leg, ankle, foot. Secondary bacterial infections possible at blister sites.	
Laboratory Diagnosis	Detection of Adult Worm: Identify female worms in blisters; X-ray for calcified worms. Detection of L1 Larvae: Microscopic examination after placing the leg in cold water. Antibody Detection: ELISA for antibodies against D. medinensis.	

	Eosinophilia: Peripheral blood eosinophilia may be present.	
Treatment		Worm Removal: Gentle extraction over 15-20 days with a stick; avoid heavy pressure. Symptomatic Treatment: Wet compresses, analgesics, topical antibiotics; no known effective anti-helminthic drugs. (Option D)
Eradication Efforts in India	Initiatives: Launched in 1984 with WHO assistance, focused on: Safe drinking water provision (filtration, hand pumps) Controlling Cyclops populations by using larvicides (e.g., abate). Educating the public on boiling or filtering drinking water. (Option C) Treating identified cases to prevent further transmission.	

Bed nets (Option A) prevent mosquito-borne diseases but are ineffective against Dracunculus medinensis, which is transmitted through contaminated water, not by mosquitoes.

Avoid consuming raw vegetables and fruits (Option B): While helpful for foodborne parasites, avoiding raw produce does not prevent Dracunculus infection, as transmission occurs via water contaminated with Cyclops.

Solution for Question 8:

Correct Answer: C) Muscle biopsy

Explanation:

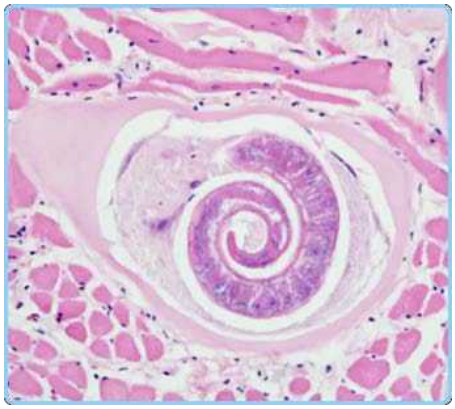
Presentation of fever, myalgia, periorbital swelling, and a history of consuming undercooked pork, which raises suspicion for trichinellosis (caused by the parasite Trichinella spiralis).

Muscle biopsy is the most definitive test for confirming trichinellosis. A muscle biopsy can reveal the presence of encysted Trichinella larvae within muscle tissues. This method provides direct evidence of the infection and is particularly helpful in cases where serological tests are inconclusive.

Investigations and Management of Trichinella spiralis

Investigations and Tests:

- Serology (Option A): ELISA and Western blot tests, which detect specific antibodies against the parasite's antigens. These tests typically yield positive results by the second week of infection. However, they may not be definitive, especially in the early stages, as antibodies can take time to develop, leading to false negatives.
- Muscle Biopsy:



- If serology is inconclusive, a biopsy can reveal encysted larvae in muscle tissue. This method provides direct evidence of the infection and is particularly helpful in cases where serological tests are inconclusive.
- Sites of biopsy: deltoid, gastrocnemius, and biceps.
- Bachman Intradermal Test: This historical diagnostic skin test was used to identify trichinellosis by detecting a delayed hypersensitivity reaction (Type IV hypersensitivity).

Management

Antiparasitic therapy: Albendazole or Mebendazole.

Stool examination for ova and parasites (Option B): Trichinella spiralis larvae do not mature in the intestines to produce eggs that can be found in stool samples.

Complete blood count with differential (Option D): Although a complete blood count may show eosinophilia, which can occur in parasitic infections, it is not a definitive test for diagnosing trichinellosis.

Solution for Question 9:

Correct Answer: D) Trichinellosis

Explanation:

Trichinellosis, or trichinosis, is an infection caused by the Trichinella spiralis, typically contracted from eating undercooked pork containing the larvae. Symptoms include muscle pain, fever, and periorbital edema, resulting from the larvae migrating into muscle tissues.

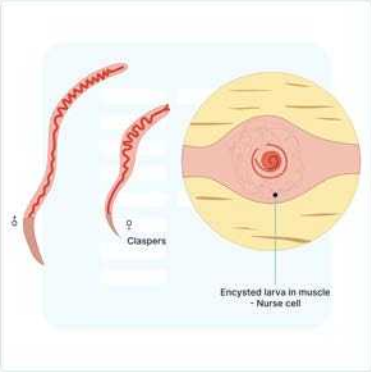
Trichinella spiralis

Trichinella spiralis is a parasitic roundworm responsible for trichinellosis (or trichinosis), an infection contracted through eating undercooked or raw meat containing infective larvae, primarily pork.

Transmission: Acquired by eating contaminated meat with Trichinella cysts.

Morphology

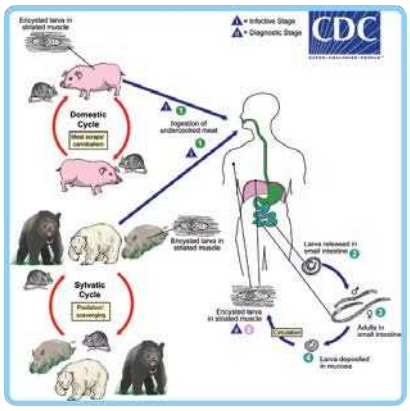
- Adult Worms: Small, thread-like worms in the intestines of the host.



- Larvae and Nurse Cells: Larvae encapsulate in muscle fibers, forming specialized nurse cells that support larval survival within muscle tissue.



Life Cycle



- Ingestion: Infection begins when undercooked meat with cysts is eaten.
- Larval Release and Intestinal Phase: Stomach acids release larvae from cysts. These larvae then invade the intestinal lining, where they mature and reproduce.
- Migration Phase: Female worms release new larvae that enter the bloodstream, spreading to striated muscles, where they encyst.
- Encystment and Muscle Phase: Larvae mature within muscle tissue, becoming infective in 2-3 weeks, and can persist for months or years in muscle fibers.

Clinical Presentation

- Intestinal Phase (Initial Symptoms): 1-2 days post-ingestion, gastrointestinal symptoms such as nausea, vomiting, diarrhea, and abdominal pain may occur.
- Muscle Phase: Symptoms include fever, muscle pain (especially in the neck and calves), periorbital edema (swelling around the eyes), and generalized weakness. Severe cases may result in myocarditis, encephalitis, or respiratory distress as larvae migrate to the heart, brain, or diaphragm.
- Late Phase: Chronic inflammation in muscles may lead to long-lasting pain, weakness, and muscle dysfunction as larvae encapsulate and eventually die.

Leptospirosis (Option A): Usually presents with muscle pain and fever but is often associated with exposure to contaminated water, not undercooked meat.

Toxoplasmosis (Option B): This infection typically presents with mild flu-like symptoms or lymphadenopathy and lacks the periorbital swelling and muscle involvement seen here.

Taeniasis (Option C): Caused by Taenia tapeworms, this infection is more likely to present with gastrointestinal symptoms rather than muscle pain and periorbital edema.

Solution for Question 10:

Correct Answer: C) Ivermectin

Explanation:

- Ivermectin is the treatment of choice for Strongyloides stercoralis infection, even in mild or asymptomatic cases, to prevent potential complications.
- Mechanism of action: It causes hyperpolarisation by enhancing chloride influx by targeting GABA receptors resulting in paralysis of the worm.

Pathogenesis and Clinical Features of Strongyloides Stercoralis Infection	
Effect of Migrating Larvae	Asymptomatic Infection: Over 50% may be asymptomatic. Rashes: Recurrent maculopapular or urticarial rashes on buttocks, perineum, and thighs. Cutaneous Larva Migrans: Serpiginous rash (larva currens) advancing up to 10 cm/hour. Pulmonary Symptoms: Rare, mainly in chronic obstructive lung disease.
Effects of Adult Worms & Filariform Larvae	Mild to Moderate Load: Epigastric pain, nausea, diarrhea, constipation, and blood loss. Heavy Larval Load: Risk of hyperinfection syndrome and disseminated strongyloidiasis.
Hyperinfection Syndrome	
Cause: Repeated autoinfection leads to increased filariform larvae. Risk Factors: Glucocorticoid therapy, immunosuppressive conditions, HTLV-1/HIV coinfection. Features: Colitis, enteritis, malabsorption, possible progression to disseminated strongyloidiasis.	

Disseminated Strongyloidiasis	
Pathophysiology: Larvae invade GIT and other organs. Disrupted mucosa may allow enteric flora to cause bacterial sepsis, pneumonia, or meningitis. Clinical Manifestations: CNS invasion, brain abscesses, meningitis; larvae in CSF. Eosinophilia, often absent in severe cases. Mortality Rate: Nearing 100% untreated; over 25% even with treatment.	
Laboratory Diagnosis	Stool Microscopy: Rhabditiform larvae; requires multiple samples (4+) for accurate diagnosis.
Microscopy of Other Specimens: Entero-test for duodenal aspirates; high filariform larvae are found in stool, sputum, and tissue biopsies for disseminated cases.	
Stool Culture: Differentiates rhabditiform from hookworm larvae.	
Serology: ELISA for antibodies; may show cross-reactivity, less effective for recent infections.	
Coproantigen Detection: Antigen capture ELISA specific to Strongyloides.	
Molecular Diagnosis: Real-time PCR targeting specific genes.	
Treatment	Asymptomatic Treatment: Strongyloidiasis should be treated regardless of symptoms due to the risk of hyperinfection. Ivermectin (200 mg/kg daily for 2 days) (Option C)

Albendazole and mebendazole (Option A & B) are not effective against Strongyloides stercoralis, a tissue nematode, as they primarily target intestinal nematodes.

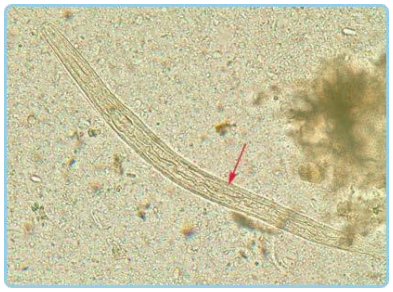
Praziquantel (Option D) Is indicated for trematode infections (flukes) and schistosomiasis but ineffective for Strongyloides stercoralis.

Solution for Question 11:

Correct Answer: A) Rhabditiform larva

Explanation:

The stool sample reveals slender larvae with a short, muscular double-bulb esophagus, which are indicative of Rhabditiform larvae. This larval stage is typically present in feces and serves as the diagnostic form for infections like Strongyloides stercoralis, particularly in patients with gastrointestinal symptoms following exposure from walking barefoot in contaminated soil.

Rhabditiform larva (Option A)	Filariform larva (Option B)
	
1st stage of the larva (L1).	3d stage of the larva (L3).
Formed when eggs hatch in the small intestine.	Developed from two molts of the L1 larva.
Measures approximately 0.25 mm in length with a short muscular double-bulb esophagus.	Measures about 0.55 mm in length, characterized by a long, slender shape, uniform esophagus & notched tail.
Found in feces.	Seen in hyperinfection & disseminated infection, may show in sputum, stool and body fluids.
Diagnostic form	Infective form

Procercoid larvae (Option C) are larval stages of certain cestodes (tapeworms) and are found in first intermediate hosts, such as aquatic insects.

Plerocercoid larvae (Option D) are associated with cestodes like Diphyllobothrium latum and are typically found in second intermediate hosts (e.g., fish).

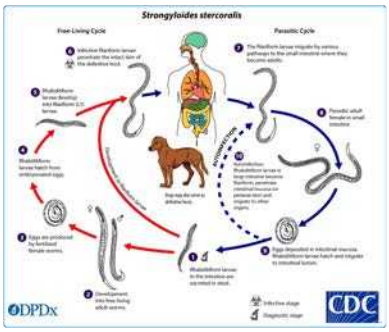
Solution for Question 12:

Correct Answer: B) Skin penetration by filariform larvae.

Explanation:

The scenario indicates an infection with Strongyloides stercoralis, characterized by the patient's chronic cough, wheezing, abdominal pain, and the presence of rhabditiform larvae in the stool.

Humans are infected when filariform (L3) larvae penetrate the skin, often through barefootcontact with contaminated soil. After entering the bloodstream, these larvae migrate to the lungs,are swallowed, and eventually reach the gastrointestinal tract, where they mature.

Strongyloides Stercoralis (Threadworm)	
Morphology	Smallest nematode known to cause human infection. Forms: Exists in three forms: adult worms, larvae (4 stages), & eggs. Adult Worm: Only females inhabit the human small intestine (2–3 mm long). Males are free-living in the environment. Eggs: Ovoviviparous, hatch into larvae immediately upon laying. (Option A)
Life Cycle	
 Hosts: Primarily humans; occasionally domestic pets act as reservoirs. Infective Form: L3 (filariform) larva. ... (content truncated)	
Phases of Infection	Migratory Phase: L3 larvae enter subcutaneous venules, reach lungs via venous circulation, migrate to alveolar spaces, ascend to the pharynx, and are swallowed to enter the GIT.
Intestinal Phase: L3 larvae mature into adult females in the small intestine mucosa (males are absent in humans). Female worms lay eggs through parthenogenesis, & rhabditiform (L1) (Option C & D) larvae hatch & are either excreted in feces or mature to L3 for autoinfection.	
Autoinfection & Chronicity	L1 larvae in the intestine can transform into L3 larvae within the host, penetrate the intestinal wall or perianal skin, enter circulation, and migrate to the lungs. This can sustain infection for up to 30–40 years and may lead to disseminated disease.
Environmental Development	Direct Development: L1 larvae molt to L3 in moist, warm soil, becoming directly infective to humans (common in temperate climates).

Indirect Development: L1 larvae molt into adults (both sexes) in tropical climates. These adults reproduce, producing eggs that hatch into L1 larvae, which then develop into infective L3 larvae.	

Solution for Question 13:

Correct Answer: C) Ancylostoma braziliense

Explanation:

The diagnosis in the given clinical scenario is cutaneous larva migrans and it is usually caused by non-human nematodes (hookworms) like Ancylostoma braziliense that migrate through the skin in human hosts whereas the other organisms cause visceral larva migrans.

 Cutaneous larva migrans	
It is also known as the creeping eruption (also known as ground itch). This condition is caused by an infection by a nematode larvae through skin penetration.	
Etiology	Common causes: Nonhuman hookworm species: Ancylostoma braziliense A. caninum

Pathogenesis	Lifecycle: Parasite eggs are excreted in the feces of infected animals and hatch in soil. Infection occurs when contaminated soil comes into contact with human skin. Larvae penetrate the skin, causing infection, and migrate beneath the epidermis. In human hosts: Humans are accidental hosts; hence, the larvae cannot penetrate the deeper layers of the skin (basement membrane). So, the disease remains confined to the outer skin layers.
Clinical features	Creeping eruptions: Itchy papules develop into serpiginous tunnels in the epidermis. The lesions shift with larval movements, leading to the name "creeping eruption." Secondary bacterial infections may arise from scratching. Transient creeping eruptions: Occasionally caused by the human hookworm, Necator americanus. Other conditions like gnathostomiasis and sparganosis may produce similar symptoms. The lesions are deeper and subcutaneous or in the muscles. Loeffler's syndrome may develop in many cases.
Clinical diagnosis	Eosinophilia is rare and occurs primarily with Loeffler's syndrome. Serological tests are not developed. Larvae are rarely found on biopsy; hence, diagnosis relies on clinical features.
Treatment	Medications: Thiabendazole is effective for treatment. Lesion management: Freezing the advancing part of the eruption with ethyl chloride is effective for localized lesions.

Difference between cutaneous and visceral larva migrans:

	Cutaneous larva migrans	Visceral larva migrans
Definition	This is a serpiginous skin eruption due to burrowing larvae of animal hookworms.	It is caused by non-human nematodes that are unable to develop in human hosts and instead migrate through tissues as larvae. The migration of the larvae through the skin causes an eosinophilic inflammation.

Tissue involved	It involves the skin	Involves various organs of the body like the liver, lungs, and eyes.
Infecting organisms	Mostly by non-human hookworms like A.caninum A.braziliense Uncinaria stenocephala Bunostomum phlebotomum	Mainly by dog and cat roundworms like Toxocara canis (Option D) Toxocara cati Ascaris suum (Option B) Angiostrongylus cantonensis Gnathostoma spinigerum (Option A) Anisakis simplex Baylisascaris procyonis
Portal of entry	Penetration of the skin by larva	Ingestion of infected eggs
Eosinophilia	Mild	Persistent high
Serodiagnosis	Not developed	Well developed
Treatment	Thiabendazole	Diethylcarbamazine and prednisolone

Solution for Question 14:

Correct answer: C) a - 2, b - 4, c - 3, d - 1, e - 5

Explanation:

Nematode	Characteristics
a. Trichinella spiralis	2. Oviparous
b. Onchocerca volvulus	4. Causes cutaneous filariasis
c. Enterobius vermicularis	3. Viviparous
d. Strongyloides stercoralis	1. Ovoviviparous
e. Trichuris trichiura	5. Found in large intestine

General properties of Nematodes

- Developmental Stages: Nematodes progress through six stages—adult, egg, and four larval stages (L1–L4)

• Body is bilaterally symmetrical and encased in a body wall.

• Contains several organ systems: alimentary canal, body cavity, nervous system, excretory system, and reproductive organs.

• Sex: Dioecious (separate male and female worms).

Classification of nematodes based on habitat		
Intestinal	Somatic	
Small intestine	Large intestine	Trichinella spiralis Dracunculus medinensis Filarial Lymphatic filariasis: Wuchereria bancrofti & Brugia malayi Cutaneous filariasis: Loa loa, Onchocerca volvulus, & Mansonella
Roundworm Hookworm Strongyloides	Enterobius Trichuris	

Classification of nematodes based on whether they lay eggs or larvae		
Oviparous	Viviparous	Ovoviviparous
Female lays eggs that develop into larvae outside the body.	Female directly gives birth to larvae.	Female lays eggs with larvae that hatch immediately.
Filaria Trichinella Dracunculus	Roundworm Hookworm Enterobius Trichuris	Strongyloides

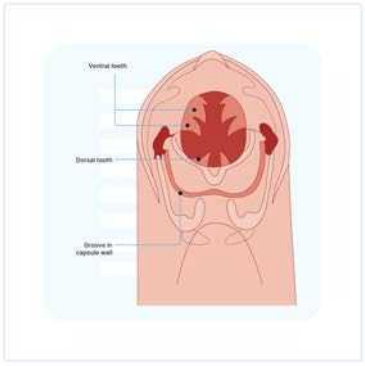
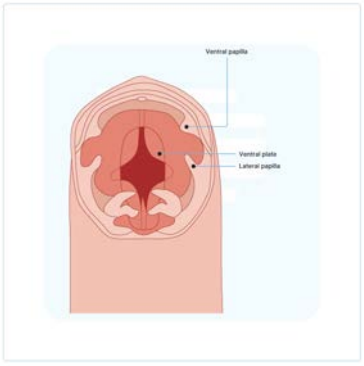
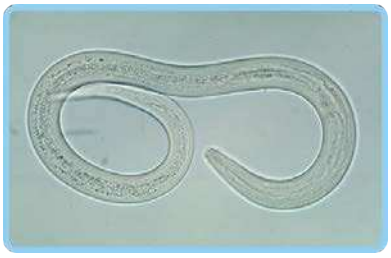
Solution for Question 15:

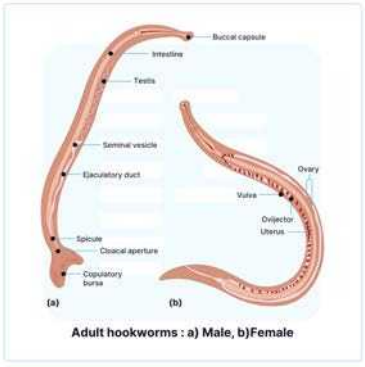
Correct Answer: A) The presence of four ventral teeth and two dorsal knob-like teeth in the adult worm

Explanation:

The Old World hookworm is Ancylostoma duodenale, which has four ventral teeth and 2 dorsal knob-like teeth, whereas the New World hookworm (Necator americanus) has 2 ventral and 2 dorsal chitinous cutting plates.

The two species of Ancylostoma are known as Old World and New World hookworms.

Features	Ancylostoma duodenale (Old World)	Necator americanus (New World)
Eggs	Similar	Similar
Larva (1st & 2nd stage)	Similar	Similar
Adult worms		
Size	Larger and thicker	Small and slender
Buccal capsule	 ... (content truncated)	 ... (content truncated)
Pathogenicity	More	Comparatively less
Caudal spine in female	Present	Absent
Filariform worms	The anterior end bends in the same direction as the curvature of the body (vs Option D)	The anterior end is bent in the opposite direction to the curvature of the body
Pulmonary reaction	More common (vs Option B)	Less common
Life span	2-7 years	4-20 years
Filariform larvae images		

Adult worms	 Adult hookworms : a) Male, b)Female	

Solution for Question 16:

Correct Answer: D) Indicates major public health problem.

Explanation:

The Chandler's index is used for measuring the severity of the public health problem of ancylostomiasis and a value of more than 300 eggs/gram of stool indicates a major public health problem.

Chandler's index:

- This is an index used to measure the severity of the public health problem of ancylostomiasis.
- It is the average number of hookworm eggs per gram of stool for the entire community.
- It also helps evaluate the control measures after mass treatments.

Average no. of eggs per gm of stool	Public health importance of hookworm infection
<200	Not of much importance (Option C)
200-250	Indicates potential damage (Option A)
250-300	Indicates minor public health problem (Option B)
>300	Major public health issue

Pathogenicity

- The hookworm attaches to the intestinal wall using its buccal capsule and teeth, cutting the tissue to suck blood through muscular contractions of the esophagus.

• Hydrolytic enzymes are secreted to aid in tissue degradation.

• They release anticoagulants to ensure continuous blood flow from the attachment site.

Clinical features	
Affect due to migrating larvae	Affect due to adult worms in the intestine
 Cutaneous lesions: Pruritic maculopapular dermatitis and rashes (referred to as "ground itch") at the site of skin penetration. ... (content truncated)	Asymptomatic infections: Most hookworm infections show no symptoms. Early intestinal phase: Infections with a lower worm load can lead to epigastric pain, inflammatory diarrhea, and eosinophilia. Late intestinal phase: Heavy worm loads result in iron deficiency anemia and protein-energy malnutrition, with symptoms like weakness, shortness of breath, and, rarely, cognitive impairments. Wakana disease: Caused by ingestion of L3 larvae of A.duodenale, leading to both gastrointestinal and pulmonary symptoms, such as nausea, vomiting, pharyngeal irritation, cough, dyspnea, and hoarseness.

Laboratory diagnosis:

Stool microscopy:

• The diagnosis is confirmed by identifying characteristic eggs in feces.

• Stool concentration may be necessary for detecting lighter infections.

• In non-fresh stool samples, eggs may hatch to release rhabditiform larvae.

• The number of eggs per gram of stool helps estimate individual and community disease burden and can be measured by the following methods: Kato-Katz technique Direct smear method (Beaver method) Modified Stoll’s dilution egg count method

• Kato-Katz technique

• Direct smear method (Beaver method)

• Modified Stoll’s dilution egg count method

• Stool examination may also show occult blood and Charcot Leyden crystals.

• Kato-Katz technique

• Direct smear method (Beaver method)

• Modified Stoll’s dilution egg count method

Stool culture:

• Fresh stool samples can be cultured to hatch eggs into L3 stage filariform larvae within 5–7 days.

• Culture techniques: Harada-Mori filter paper tube method Baermann funnel technique Charcoal culture method Agar plate technique (more sensitive)

• Harada-Mori filter paper tube method

• Baermann funnel technique

• Charcoal culture method

• Agar plate technique (more sensitive)

• Harada-Mori filter paper tube method

• Baermann funnel technique

• Charcoal culture method

• Agar plate technique (more sensitive)

Molecular Diagnosis:

• PCR and real-time PCR assays target mitochondrial cytochrome oxidase genes.

• They are species-specific and highly sensitive.

Other investigations:

• Blood analysis: Hypochromic microcytic anemia Eosinophilia Hypoalbuminemia

• Hypochromic microcytic anemia

• Eosinophilia

• Hypoalbuminemia

• Chest X-ray may show pulmonary infiltrates during the migratory phase

• Hypochromic microcytic anemia

• Eosinophilia

• Hypoalbuminemia

Treatment:

Antiparasitic Medications:

• Albendazole (400 mg once)

• Mebendazole (500 mg once)

• Pyrantel pamoate (11 mg/kg for 3 days)

• Resistance: Decreased efficacy reported due to widespread use.

• Symptomatic treatment: Mild iron-deficiency anemia: Treat with oral iron and folic acid. Severe hookworm disease: Requires nutritional support and oral or parenteral iron replacement.

• Mild iron-deficiency anemia: Treat with oral iron and folic acid.

• Severe hookworm disease: Requires nutritional support and oral or parenteral iron replacement.

• Mild iron-deficiency anemia: Treat with oral iron and folic acid.

• Severe hookworm disease: Requires nutritional support and oral or parenteral iron replacement.

Prevention:

• Improved personal hygiene

• Proper disposal of feces

• Enhanced nutrition with dietary iron

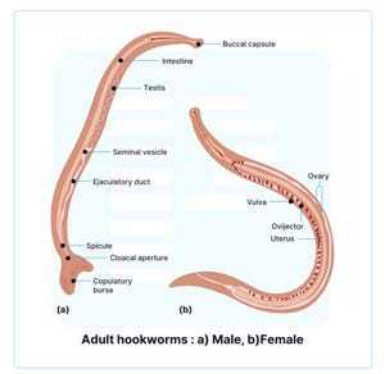
• Treatment of infected individuals

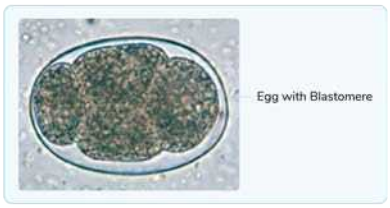
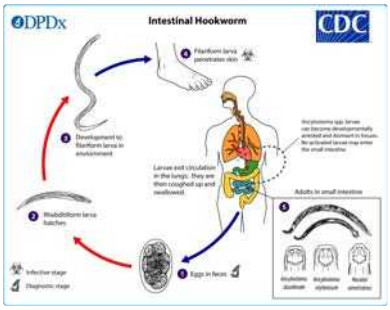
Solution for Question 17:

Correct Answer: A) Presence of copulatory bursa

Explanation:

The diagnosis is of hookworm (Ancylostoma duodenale) infection and in hookworms, the male hookworm has a copulatory bursa at the posterior end which the female does not possess.

Ancylostoma duodenale / Hookworm	
These are mostly found in the jejunum of infected humans > duodenum > ileum.	
 Adult hookworms : a) Male, b)Female ... (content truncated)	The adult worms are stout and cylindrical. The body is curved (concave dorsally and convex ventrally). The anterior end is constricted and bent dorsally, following the general curvature of the body, which is why they are called hookworms. A prominent buccal capsule, reinforced with a hard chitin-like substance, features six teeth: Four hook-like teeth ventrally Two knob-like teeth with a median cleft dorsally
Female adult worm	Male adult worm

Larger in size	Smaller in size (vs Option C)
Copulatory bursa is absent	The copulatory bursa is present
Genital opening is at the junction of the middle and posterior 3rd of the body	The genital opening is at the cloaca along with the anus (vs Option D)
The posterior end is tapering	The posterior end is expanded like an umbrella (vs Option B)
 Eggs	Shape: The egg is oval or elliptical. Color: Colorless and not bile-stained. Shell: Surrounded by a thin, transparent hyaline shell membrane. Floats in saturated salt solution. Development: Initially, the egg contains an unsegmented ovum when released into the intestine. As it travels down the intestine, the ovum develops. When passed in feces, it contains a segmented ovum, typically with four or eight blastomeres. There is a clear space between the segmented ovum and the eggshell.
Epidemiology	Prevalence in India: N. americanus is predominant in South India, while A. duodenale is more common in North India. Age and sex distribution: Males and young adults are more commonly affected. Anemia due to iron loss is particularly severe in children and pregnant women. Endemic index: Chandler's index is utilized in epidemiological studies to estimate the morbidity and mortality associated with hookworm infection, which is influenced by the worm load in the community.
 Lifecycle: Ancylostoma has only one host, which is man. The infective form is the third-stage (L3) filariform larva. ... (content truncated)	

Solution for Question 18:

Correct Answer: D) A piece of cellophane tape is applied to the perianal region to collect the sample

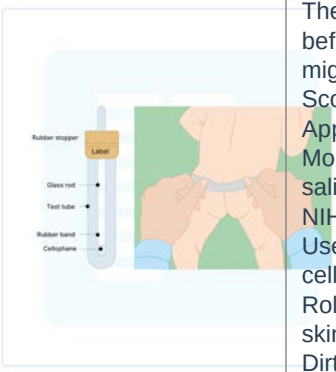
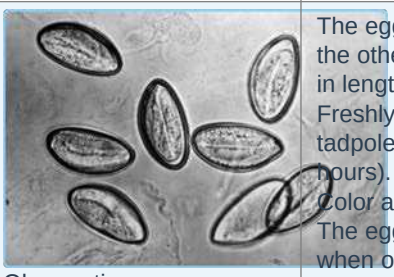
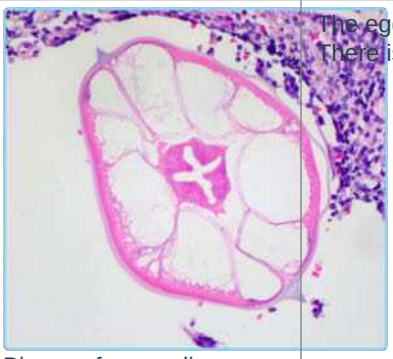
Explanation:

The diagnosis here is enterobiasis, where perianal pruritis and nocturnal enuresis are commonly seen in children, and the sample collection can be done using a cellophane tape applied to the anal region and later observed under a microscope. But, in this scenario, the NIH swab method, which consists of a glass rod with cellophane tape attached by a rubber band that is rolled over the perianal region, is used to collect the samples.

Clinical features:

- About 1/3rd of cases are asymptomatic.
- Intense itching in the perianal and perineal region due to the female worm laying eggs is known as pruritis ani.
- Nocturnal enuresis is seen as the worm moves outside to lay eggs mostly at night.
- The worm may crawl into the vulva and vagina causing irritation and mucoid discharge.
- The worm may migrate into the uterus, fallopian tubes, and the peritoneum, leading to symptoms of: Salpingitis Cervicitis Peritonitis Recurrent urinary tract infections
- Salpingitis
- Cervicitis
- Peritonitis
- Recurrent urinary tract infections
- It may cause appendicitis.
- Abdominal pain and weight loss are seen in heavy infections.
- No eosinophilia or elevated IgE.
- Salpingitis
- Cervicitis
- Peritonitis
- Recurrent urinary tract infections

Diagnosis:

Microscopy	Microscopic examination of perianal skin samples is the preferred method for detecting characteristic pinworm eggs.
 ... (content truncated)	The perianal samples are collected early in the morning before going to the toilet or bathing, as the female worm migrates to lay eggs in the perianal region at night. Scotch tape method: Apply a piece of cellophane tape to the perianal region. Mount the tape on a clear glass slide with a drop of saline for microscopic examination. NIH swab method: Use the NIH swab, which consists of a glass rod with cellophane tape attached by a rubber band. Roll the cellophane part over the perineal and perianal skin to collect the sample. (Option A) Dirt from under fingernails can sometimes show eggs in infected children.
 Observation	The eggs are planoconvex, with one side convex and the other side flat, measuring approximately 50–60 µm in length. (Option B) Freshly passed eggs are embryonated and contain a tadpole-shaped larva inside (become infectious after 6 hours). (Option C) Color and Staining: The eggs are colorless and do not exhibit bile staining when observed in a saline mount. Floatation: These eggs are capable of floating in a saturated salt solution.
 Biopsy of appendix ... (content truncated)	The eggs of E.vermiocularis are seen. There is extensive lymphoid hyperplasia of the mucosa.

Management:

- Medications can be any of the following: Mebendazole: 100 mg once. Albendazole: 400 mg once. Pyrantel pamoate: 11 mg/kg once (maximum dose of 1 g).
- Mebendazole: 100 mg once.
- Albendazole: 400 mg once.

- Pyrantel pamoate: 11 mg/kg once (maximum dose of 1 g).
- Follow-Up Treatment: Repeat the same treatment after 2 weeks.
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- Household Treatment: Treating all household members is recommended to eliminate asymptomatic carriers and reduce the risk of reinfection.
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- Repeat the same treatment after 2 weeks.
- Treating all household members is recommended to eliminate asymptomatic carriers and reduce the risk of reinfection.

Prevention strategies:

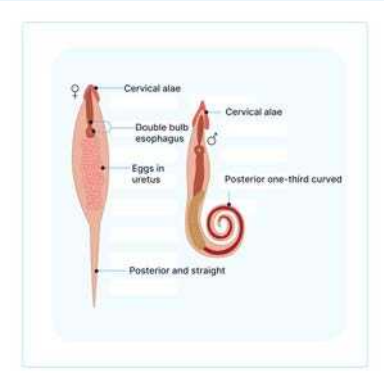
- Realistic Expectations: Complete prevention is difficult due to the common transmission and potential for autoinfection.
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- Key Preventive Measures: Improve personal hygiene by: Washing bed linens regularly. Keeping nails short and clean. Practicing frequent hand washing.
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- Practicing frequent hand washing.
- Washing bed linens regularly.
- Keeping nails short and clean.
- Practicing frequent hand washing.

Solution for Question 19:

Correct Answer: B) Embryonated eggs

Explanation: The parasite infestation in the given scenario is of Enterobius vermicularis or the pinworm, which causes intense itching in the perianal region due to the female worm laying eggs in this region at night. The infective stage of this is the embryonated egg.

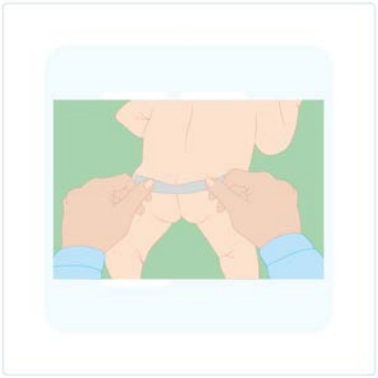
Enterobius vermicularis / Pinworm / Threadworm

Epidemiology: The most common age group are school children. Infections can persist for years due to auto-infective cycles. Factors contributing to infection include: Overcrowding Impaired hygiene Poor personal care practices (such as nail-biting and inadequate handwashing).	
Morphology	
Adult worm	 ... (content truncated)
Female worm (Option C)	Females are long. The posterior third tapers into a thin, pointed, pin-like tail. Females are oviparous.
Male worm (Option D)	The posterior end is tightly curved ventrally, sharply truncated, and has a prominent copulatory spicule.

Eggs	 The eggs are colorless and not bile-stained They float in saturated salt solution and have an elongated ovoid shape. Each egg contains a tadpole-shaped, coiled embryo that becomes infectious only 6 hours after being deposited.

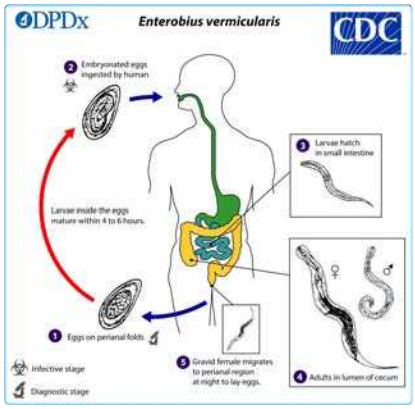
Scotch-tape test:

A piece of clear tape is pressed against the skin around the anus to collect eggs, typically overnight. The tape is then examined under a microscope for pinworm eggs.



Life cycle:

- Enterobius vermicularis spends the entire lifecycle in humans.
- It has no intermediate hosts and does not undergo systemic migration.
- The eggs hatch in the intestine, and the larvae enter the caecum and mature into adults.
- The male worm does not migrate and most often dies after mating and is passed in the feces.
- The female worm migrates to the rectum from the colon and travels to the perianal skin at night to lay its sticky eggs and dies after laying all eggs or gets crushed due to scratching.



Pathogenicity:

- The embryonated egg is the infective form.
- The disease thus caused is called enterobiasis, which is more common in children than in adults.
- It affects females > males.
- The infection occurs in two ways: Exogenous autoinfection is where the eggs are transferred from the perianal region to the mouth by nails after scratching the region. Endogenous autoinfection/retro-infection is where the larva migrates retrograde and travels back into the colon after hatching in the perianal skin.
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- Endogenous autoinfection/retro-infection is where the larva migrates retrograde and travels back into the colon after hatching in the perianal skin.
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- Endogenous autoinfection/retro-infection is where the larva migrates retrograde and travels back into the colon after hatching in the perianal skin.

Larvae (Option A): The larvae mature in the cecum into adult worms and are not infectious forms of Enterobius.

Solution for Question 20:

Correct Answer: B) Albendazole

Explanation:

The image given here is of the bile-stained eggs of Trichuris trichiura in a wet mount and the best treatment choice is administering albendazole as it is both safe and effective compared to the other choices of drugs.

Diagnosis of trichuriasis

Clinical features:

- Generally asymptomatic unless the worm load is heavy.
- Potential issues arise from mechanical effects or allergic reactions.
- Abdominal pain and anorexia.
- Trichuris dysentery syndrome: bloody or mucoid.
- Iron deficiency anemia: due to blood loss.
- Recurrent rectal prolapse: due to heavy worm load in the rectum.
- Malnutrition causes growth retardation.
- Cognitive impairment: due to inflammatory cytokines induced by Trichuris species.

Laboratory diagnosis:

- Stool Examination: Conducted through direct wet mount or stool concentration methods. A single stool specimen is typically sufficient for diagnosing symptomatic cases due to high egg output. Egg characteristics: Measure approximately 50 × 22 µm. Barrel-shaped with mucus plugs at both ends. Bile-stained and capable of floating in a saturated salt solution.
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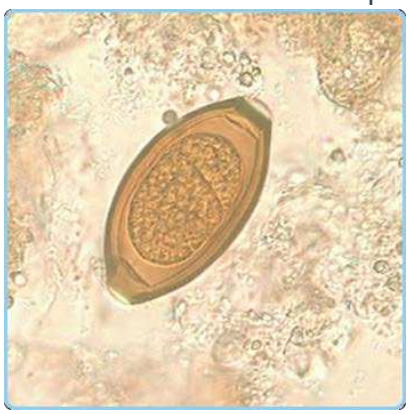
• Barrel-shaped with mucus plugs at both ends.

• Bile-stained and capable of floating in a saturated salt solution.

• Measure approximately 50 × 22 µm.

• Barrel-shaped with mucus plugs at both ends.

• Bile-stained and capable of floating in a saturated salt solution.



• Adult worm characteristics: Whip-shaped, measuring 3–5 cm long. The anterior three-fifths is thin and coiled, resembling a rope, while the posterior two-fifths is short and thick.

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• Other findings: Peripheral blood eosinophilia (typically less than 15%). Increased serum IgE levels.

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• Increased serum IgE levels.

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• Peripheral blood eosinophilia (typically less than 15%).

• Increased serum IgE levels.

• Colonoscopy: The adult worms are seen attached to the mucosa, giving a characteristic “coconut cake rectum” appearance.

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• The adult worms are seen attached to the mucosa, giving a characteristic “coconut cake rectum” appearance.



Management of trichuriasis:

- Albendazole: 400 mg daily for three doses. It is safe and effective.
- 400 mg daily for three doses.
- It is safe and effective.
- Mebendazole: (Option A) 500 mg as a single dose It is only moderately effective, with cure rates of 70-90%.
- 500 mg as a single dose
- It is only moderately effective, with cure rates of 70-90%.
- Ivermectin: (Option C) 200 mg/kg daily for three doses. It is safe but less effective.
- 200 mg/kg daily for three doses.
- It is safe but less effective.
- 400 mg daily for three doses.
- It is safe and effective.
- 500 mg as a single dose
- It is only moderately effective, with cure rates of 70-90%.
- 200 mg/kg daily for three doses.
- It is safe but less effective.

Metronidazole (Option D): It is a 5-nitroimidazole derivative that has both antibacterial and antiprotozoal properties. It is used for amebiasis and other protozoal diseases such as giardiasis and trichomoniasis, but not used for the treatment of trichuriasis.

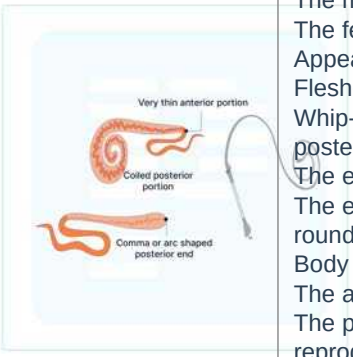
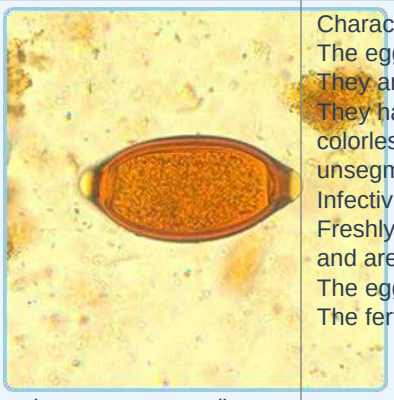
Solution for Question 21:

Correct Answer: A) 1 and 4

Explanation:

Trichuriasis is caused by the organism Trichuris trichiura, in which the development of the eggs takes place in the soil, and when swallowed by man, hatches into larvae in the small intestine.

Trichuris trichiura / Whip worm

Morphology	
 ... (content truncated)	The male worm is 30-45 mm long. The female worm is 40-50 mm long (slightly larger). Appearance: Flesh-colored. Whip-shaped: anterior (3/5) thin and thread-like; posterior (2/5) thick and fleshy (like a handle). The end of the male worms is coiled ventrally. The end of the female worm is straight, blunt, and rounded. Body structure: The anterior portion contains a capillary esophagus. The posterior part contains the intestines and reproductive organs. Lifespan: 5-10 years.
 ... (content truncated)	Characteristics: The eggs appear brown as they are bile-stained. They are barrel-shaped. They have a triple shell with a brown outer layer and a colorless mucus plug at each pole containing an unsegmented ovum. Infectivity: Freshly passed eggs contain an unsegmented ovum and are not infective for humans. The egg floats in a saturated salt solution. The fertilized female lays about 5,000 eggs per day.
Epidemiology	Distribution: Worldwide, it has a higher prevalence in tropical regions. Common in tropical Africa, South America, and Southeast Asia. Widespread in India. Prevalence: An estimated 800 million people are infected globally. While whipworm infections are frequent, whipworm disease is relatively rare.

Pathogenicity:

• The incubation period is 70 to 90 days.

• Transmission: The infective stage is the embryonated eggs, which carry the rhabditiform larvae. (vs statement 3) The mode of transmission is by contaminated food or water. The swallowed eggs hatch into larvae in the small intestine. Adult female worms live in the cecum and lay eggs that are excreted in the feces.

• The infective stage is the embryonated eggs, which carry the rhabditiform larvae. (vs statement 3)

• The mode of transmission is by contaminated food or water.

• The swallowed eggs hatch into larvae in the small intestine.

• Adult female worms live in the cecum and lay eggs that are excreted in the feces.

• The infective stage is the embryonated eggs, which carry the rhabditiform larvae. (vs statement 3)

• The mode of transmission is by contaminated food or water.

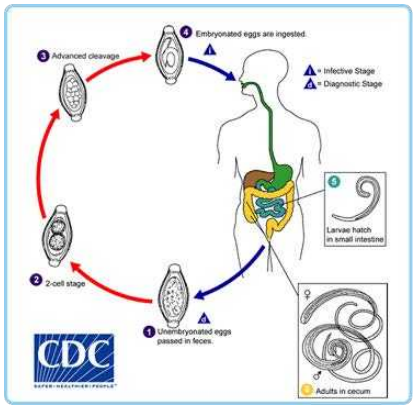
• The swallowed eggs hatch into larvae in the small intestine.

• Adult female worms live in the cecum and lay eggs that are excreted in the feces.

The life cycle of Trichuris trichiura:

• Man is the only natural host for T.trichiura and the egg develops in the soil.

• There is no intermediate host. (vs statement 2)

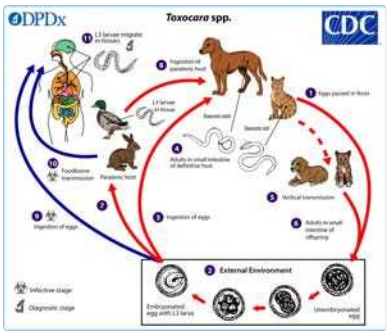


Solution for Question 22:

Correct Answer: C) Toxocara canis infection

Explanation:

The child's symptoms of fever, cough, and abdominal pain, combined with eosinophilia and elevated liver enzymes, strongly suggest Visceral Larva Migrans. The exposure to a garden where a dog defecates indicates a likely infection from Toxocara canis, as this is the most common cause of Visceral Larva Migrans in children.

Visceral larva migrans	
Visceral Larva Migrans is caused by the migration of larvae from nonhuman nematodes that infect humans via the oral route.	
Etiology	Most commonly caused by Toxocara canis (Option C) (dog ascarid) and less frequently by Toxocara cati (Option A) (cat ascarid). Other potential causes include: Anisakis (marine animal ascarids) (Option B) Gnathostoma spinigerum Angiostrongylus cantonensis
Pathogenesis	 Infective eggs in the soil contaminated by dog & cat feces are ingested. Larvae hatch in the small intestine, penetrate the gut wall, and migrate to the liver. ... (content truncated)
Clinical Features	More common in children due to soil ingestion.

Common findings include: Fever Hepatomegaly Pneumonitis Hyperglobulinemia Pica	 Possible neurological disturbances (neural larva migrans) Endophthalmitis (ophthalmic larva migrans) Marked leukocytosis ... (content truncated)
Diagnosis	Diagnosed through serological tests such as: Passive hemagglutination Bentonites flocculation Microprecipitation Enzyme-linked immunosorbent assay (ELISA) is more specific for diagnosing toxocariasis.
Treatment	Diethylcarbamazine (DEC): 100 mg TDS for 3 weeks in adults to kill larvae and halt the disease. Prednisolone: Should be administered concurrently.
Prophylaxis	Deworming of household pets is essential for preventing soil contamination and reducing transmission risk.

Ascaris (Option D) primarily causes intestinal symptoms and is linked to fecal-oral transmission through contaminated food, not specifically associated with dog feces.

Solution for Question 23:

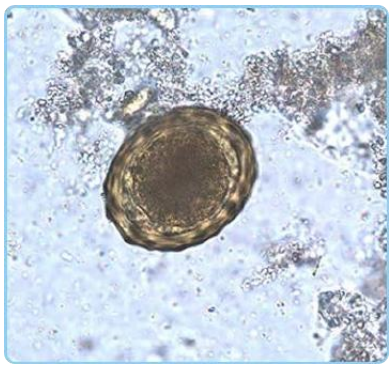
Correct Answer: B) It does not float in saturated salt solution

Explanation:

The patient's symptoms of abdominal pain, diarrhea, and weight loss align with gastrointestinal Ascaris infection. Her consumption of unwashed vegetables in a rural area suggests exposure to contaminated soil. And the image shows an unfertilized egg of Ascaris lumbricoides. It is elliptical, narrow, and long, measuring approximately 80 µm x 55 µm, and it does not float in saturated salt solution.

Laboratory diagnosis & Treatment of Ascaris lumbricoides	
Stool Examination	Detects both fertilized & unfertilized eggs using saline and iodine wet mount. If negative, concentration by sedimentation is recommended (floatation is ineffective as unfertilized eggs do not float).
Adult Worm Detection	Occasionally visible in stool or sputum. Barium X-ray may show worms in the intestine as "trolley car lines" if aligned in parallel. Ultrasound or cholangiopancreatography helps detect worms in extraintestinal sites.
Larva Detection	During early pulmonary migration, larvae may appear in sputum or gastric aspirates before stool detection.
Serology	Antibody testing (e.g., ELISA) is sensitive but cannot distinguish recent from past infections and may cross-react with other helminths.
Molecular Methods	PCR can identify Ascaris in stool, and differentiate between Ascaris, Trichuris, and hookworm. Real-time PCR quantifies parasite load.
Other Indicators	Eosinophilia is common in early lung stages. Charcot-Leyden crystals may appear in sputum and stool.
Treatment	Albendazole: 400 mg as a single dose. Mebendazole: 100 mg twice daily for 3 days or 500 mg as a single dose.

Unfertilized Egg	Fertilized Egg
It is Elliptical, narrow, & long Size: 80 µm x 55 µm Shell: Thin with an irregular albumin coating Contents: Small, atrophied ovum with disorganized refractile granules Does not Float in saturated salt solution. (Option B)	Round or oval in shape. Size: 60-75 µm x 40-45 µm Color: Golden brown, bile-stained Shell: Thick, smooth, translucent with three layers Outer: Coarsely mammillated albuminous coat Middle: Thick transparent layer Inner: Lipoidal vitelline membrane (Option C) Decorticated Variant: Some eggs lack the outer mammillated coat. Contents: Large unsegmented ovum with coarse lecithin granules, leaving clear areas at poles (Option D) Floats in saturated salt solution. (Option A)

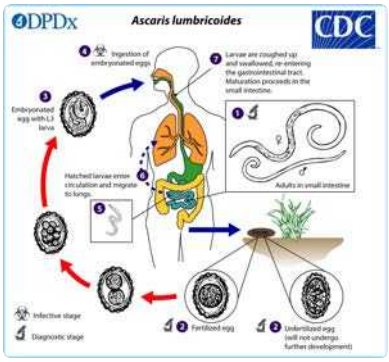
 ... (content truncated)	

Solution for Question 24:

Correct Answer: C) Ascaris lumbricoides

Explanation:

The symptoms of cough, chest discomfort, and fever, along with laboratory findings of eosinophilia and pulmonary infiltrates, indicate Loeffler’s syndrome, which is associated with the larval migration of Ascaris lumbricoides. The boy's exposure to contaminated soil and unwashed vegetables supports this diagnosis, as it is the typical transmission route for this parasite.

Life Cycle of Ascaris	
 Host: Humans only. ... (content truncated)	
Life Cycle Phases	
Migratory Phase	L2 larvae hatch, molt to L3, penetrate the intestine and travel to the lungs via the heart.

	In the lungs, they molt to L4, migrate to the pharynx, and are swallowed back into the intestine.
Intestinal Phase	L4 larvae become adult worms in the small intestine. Females lay fertilized eggs in feces; unfertilized eggs may also be laid.
Development in Soil	Fertilized eggs become infective embryonated (L2 larvae) in 2 weeks under suitable conditions. Unfertilized eggs disintegrate and are non-infective. Embryonated eggs can survive for up to 15 years due to a thick eggshell and ascaroside, a resistant lipoprotein.
Pathogenesis and Clinical Features	
Caused by the host immune response, larval migration, adult worm obstruction, and nutritional deficiencies. Incubation: 60–70 days.	
Pulmonary Phase	Symptoms appear within two weeks post-ingestion: cough, chest discomfort, and fever. Severe cases may lead to Loeffler's syndrome, characterized by eosinophilic pneumonia and transient lung infiltrates.
Intestinal Phase	Most infections are asymptomatic, but malnutrition and growth retardation can occur, particularly in children <5 years. Complications include intestinal obstruction, perforation, intussusception, or volvulus in older children.
Extraintestinal Complications	Larger worms may obstruct the biliary tree, causing biliary colic or pancreatitis.
Allergic Reactions	Reactions such as fever, urticaria, angioedema, and conjunctivitis may arise due to toxins released by adult worms.

Strongyloides (Option A) is more often associated with chronic gastrointestinal issues and hyperinfection in immunocompromised patients.

Toxocara canis (Option B) causes visceral larva migrans and the specific exposure history suggest Ascaris lumbricoides is more likely.

Schistosoma mansoni (Option D) generally leads to intestinal and hepatic symptoms.

Solution for Question 25:

Correct Answer: C) A straight conical posterior end

Explanation:

The given image shows Ascaris lumbricoides. The female Ascaris lumbricoides is identified by a straight conical posterior end & a vulvar groove, distinguishing it from the male, which has a ventrally curved posterior end. Females are also larger, measuring 20-40 cm, compared to males at 15-30 cm.

Roundworm (Ascaris lumbricoides)	
Habitat	Adult worms reside in the small intestine (85% in the jejunum and 15% in the ileum) and are the largest nematode parasites in humans.
Morphology	Adult Worms: Large, cylindrical, with tapering ends and pale pink color that turns white outside the body. Male Worms: Smaller than females (15-30 cm long, 2-4 mm thick) A ventrally curved posterior end and two copulatory spicules. (Options A & B) Female Worms: Larger than males (Option D) Straight, conical posterior end (Option C) & a vulvar groove aiding in mating.
Reproduction	A single female can lay up to 200,000 eggs daily, which are expelled in the feces.
Host	Humans only
Infective Form	Embryonated eggs with L2 larvae.
Transmission	Ingestion of eggs from contaminated soil, food, or water.

Previous Year Questions

1. A 35-year-old female with a long-standing history of fever presents with discoloration of the skin over the lower limb and edema. Microscopy reveals a sheathed organism with no nuclei at the tip of the tail. What is the most likely causative organism?

A. Wuchereria bancrofti

B. Brugia malayi

C. Loa Loa

D. Mansonelia

---

2. Which of the following is a xenodiagnostic method ?

A. Rabbit ileal loop assay for ETEC

B. Injecting Aedes thorax with blood of a patient with suspected dengue

C. Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis

D. Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria

---

3. The egg shown below belongs to?



A. Hookworm

B. Whipworm

C. Pinworm

D. Round worm

---

4. Which of the following diseases is not transmitted by the vector shown below?



A. Orientia tsutsugamushi infection

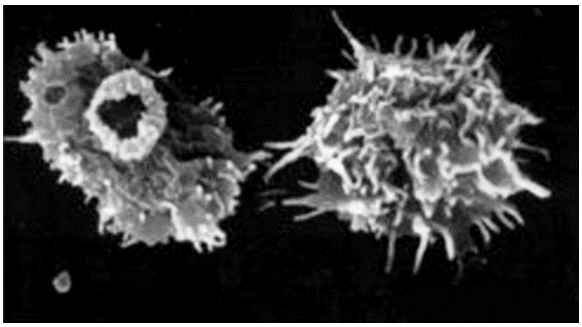
B. Crimean congo fever

C. Babesia

D. Ehrlichia

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5. A person from a rural area experiencing CNS symptoms and vision loss for the past 12 days, the electron microscopic examination reveals the presence of which organism?



A. Naegleria Floweri

B. Acanthamoeba

C. Balmuthia

D. None of the above

---

6. Which of the following is/are an advantage(s) of a saline mount compared to an iodine mount? Motility of trophozoites Internal structure of cysts Differentiate bile stained vs non bile stained

A. 1

B. 1 and 3

C. 2 and 3

D. 1,2 and 3

---

7. A 28-year-old female presents to the clinic with complaints of frequent diarrhea, abdominal cramps, bloating, and weight loss over the past two weeks. She

mentions that she recently traveled to a rural area for a camping trip and drank untreated stream water during her stay. On physical examination, no fever is noted, but mild tenderness is present in the right upper quadrant of the abdomen. Stool examination reveals the presence of trophozoites and cysts with characteristic " falling leaf" motility under microscopy. Which organism is most likely responsible for this patient's symptoms?

A. Salmonella typhi

B. Entamoeba histolytica

C. Giardia lamblia

D. Cryptosporidium parvum

---

8. Which of the following parasites demonstrates parthenogenesis?

A. Trichuris

B. Enterobius

C. Strongyloides

D. Ancylostoma

---

9. What is the best investigation for identifying malaria species?

A. Quantitative buffy coat

B. Giemsa stain with thin smear

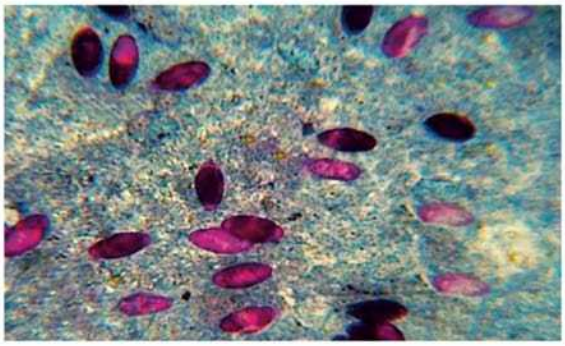
C. Giemsa stain with thick smear

D. Acridine orange stain with thin smear

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10. An HIV patient presented with watery diarrhea. Stool examination revealed oocysts of size 20-30 microns and oval shaped organisms, as shown in the image

below. Which among the following is the correct answer?



A. Cryptosporidium

B. Cystoisospora

C. Cyclospora

D. Microsporidia

---

11. Consider the following assertion and reason regarding Plasmodium falciparum infection: Assertion: P. falciparum doesn't show schizont stage in peripheral blood. Reason: They get sequestered in small capillaries due to attachment to membranes. Which of the following statements is correct?

A. Both the assertion and reason are correct, and the reason explains the assertion.

B. Both the assertion and reason are correct, but the reason does not explain the assertion.

C. The assertion is correct, but the reason is incorrect.

D. The assertion is incorrect, but the reason is correct.

---

12. Infection in humans occurs by ingestion of quadrinucleate cysts through contaminated food or water. Which of the following is likely the cause of this infection?

A. Giardia lamblia

B. Entamoeba Histolytica

C. Cryptosporidium parvum

D. Cyclospora Cayetanensis

---

13. A patient presents with perianal itching. A perianal swab was taken, and the image below shows the findings. What is the most likely diagnosis?



A. H Nana

B. Ancylostoma

C. Enterobius vermicularis

D. Necator Americanus

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14. An HIV-positive patient with a history of gastrointestinal helminth infection has a stool sample showing motile 200-micron larvae but no eggs. Identify the parasite:

A. Ascaris

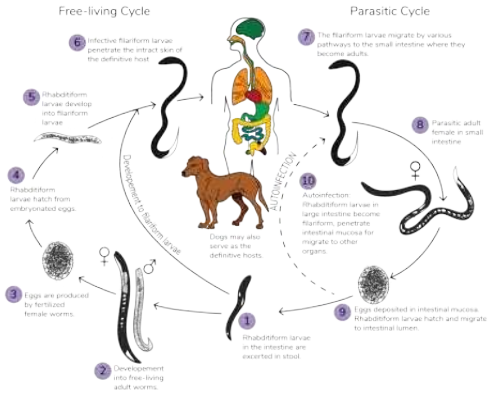
B. Strongyloides

C. Ancylostoma

D. Schistostoma

---

15. Identify the helminth from the life cycle given below.



A. Ancylostoma duodenate

B. Strongyloides stercoralis

C. Ascaris lumbricoides

D. Enterobius vermicularis

---

16. Which among the following occupations is a risk factor for this presenting illness?



A. Kennel Worker

B. Lifeguard in a swimming pool

C. Veterenarian

D. Poultry worker

17. Choose the wrong statement about Plasmodium species:

A. Drug resistant falciparum is seen in India

B. Presence of duffy blood group antigen is protective against vivax

C. Severe malaria with shock is called algid malaria

D. Relapse is seen in P. ovale

18. Identify the organism shown in this image.



A. Strongyloides sterocoralis

B. Enterobius vermicularis

C. Ancylostoma duodenale

D. Ascaris lumbricoides

19. All of the following can be seen intracellularly in hepatocytes except.

A. Toxoplasma

B. Plasmodium

C. Babesia

D. Leishmania

---

20. Which organism is the cause of perianal itching in a child who experiences symptoms primarily at night, leading to sleep disturbances? The eggs of the responsible organism can be observed in the accompanying image.



A. Enterobius vermicularis

B. Trichuris trichiura

C. Trichinella spiralis

D. Strongyloides stercoralis

---

21. What is the causative organism for the condition shown in the image below ?



A. Echinococcus granulosus

B. Schistsoma japonicum

C. Echinococcus multilocularis

D. Schistosoma mansoni

---

22. Which of the following organisms causes multiple alveolar-like lesions in the liver?

A. Echinococcus multilocularis

B. Echinococcus granulosus

C. Entamoeba histolytica

D. Cysticercus cellulosae

---

23. Transmission assessment survey is carried out for?

A. P. vivax

B. Syphilis

C. Filariasis

D. Leishmania

---

24. A 29-year-old middle school teacher presented to the OPD with lower abdominal pain, laterally radiating, and dysuria. In urine examination, eggs with terminal spines were seen. Computerized tomography (CT) scans revealed extensive calcifications of the urinary tract. A probable diagnosis of squamous cell carcinoma of the urinary bladder was made on histopathological examination. Which of the following parasites lives in the bladder plexus?

A. Schistosoma mansoni

B. Schistosoma japonicum

C. Schistosoma hematobium

D. Fasciola hepatica

---

25. What is the most common mode of transmission for babesiosis?

A. Reduviid bugs

B. Tsetse fly

C. Sand fly

D. Ticks

---

26. Match the following parasites with their respective hosts: 1. Diphylobothrium 2. Paragonimus 3. Brugia 4. Fasciola a. Sheep b. Crab c. Cyclops d. Mosquitoes

1. Diphylobothrium 2. Paragonimus 3. Brugia 4. Fasciola	a. Sheep b. Crab c. Cyclops d. Mosquitoes

A. 1-c, 2-b,3-d, 4-a

B. 1-b, 2-d, 3-a, 4-c

C. 1-a, 2-c, 3-b, 4-d

D. 1-d, 2-a, 3-c, 4-b

---

27. Neurocysticercosis is caused by?

A. Cysticercus bovis

B. Taenia saginata asiatica

C. Cysticercus cellulose

D. Taenia saginata

---

28. Which is correct about the larval stage of Taenia solium?

A. Cysticercus bovis

B. Cysticercus cellulosae

C. Chiggers

D. None of the above

---

29. A patient presents with seizures. His CT head reveals multiple calcified parenchymal cysts. Which of the following is correct regarding management in this case?

A. Steroids followed by antiparasitic drugs

B. Antiparasitic drugs followed by steroids

C. Steroids before antiepileptics

D. Continue antiepileptics

---

30. Which is the agent used to treat the infection caused by the cestode egg in the image?



A. Praziquantel

B. Ivermectin

C. Albendazole

D. Diethylcarbamazine

---

31. A 6-year-old boy presents with lower abdominal pain and diarrhea. The stool examination finding is shown below. Which of the following is most likely causative organism?



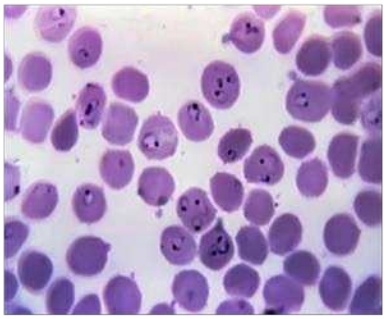
A. Giardia lamblia

B. Shigella dysenteriae

C. Entamoeba histolytica

D. Campylobacter jejuni

32. What is the underlying cause of the symptoms exhibited by a 55-year-old male, including fever, fatigue, and abnormal vital signs, along with the identification of the causative agent based on the microscopic examination of his peripheral blood smear?



A. Plasmodium vivax

B. Plasmodium falciparum

C. Plasmodium ovale

D. Plasmodium malariae

33. Following vector is not involved in transmission of which of the following disease ?



A. Kyasanur forest disease

B. Visceral leishmaniasis

C. Oriental sore

D. Oroya fever

---

34. Assertion: Malathion is used as ultra – low volume (ULV) during epidemics
Reason: Ultra low volume fogger has residual effects

A. Both assertion and reason are true and the reason is the correct explanation for the assertion

B. Both assertion and reason are true, but the reason is not the correct explanation for the assertion

C. Assertion is true, but the reason is false

D. Both assertion and reason are false

E. The assertion is false but reason is true

---

35. Flask shaped ulcers in the colon are caused by:

A. Entamoeba histolytica

B. Giardia lamblia

C. Helicobacter pylori

D. Entamoeba vermicularis

36. Match the following eggs to the organism. 1. Hymenolepis nana a. 2. Ascaris lumbriocoides b. 3. Ancylostoma duodenale c. 4. Taneia solium d.

1. Hymenolepis nana	 ... (content truncated)
2. Ascaris lumbriocoides	 ... (content truncated)
3. Ancylostoma duodenale	 c.

4. Taneia solium	 ... (content truncated)

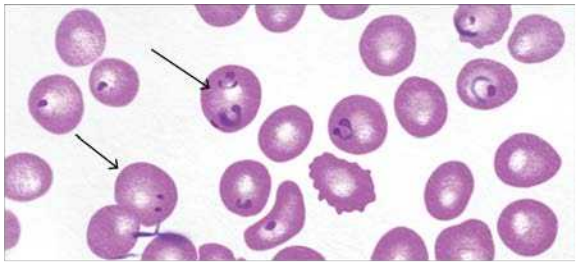
A. 1-d, 2-c, 3-a, 4-b

B. 1-c, 2-d, 3-a, 4-b

C. 1-d, 2-c, 3-b, 4-a

D. 1-c, 2-d, 3-b, 4-a

37. A 30-year-old female from tribal area presented with fever for the last 3 days. On examination of the peripheral smear, a diagnosis of malaria was made. Which of the following is most likely to be the causative agent?



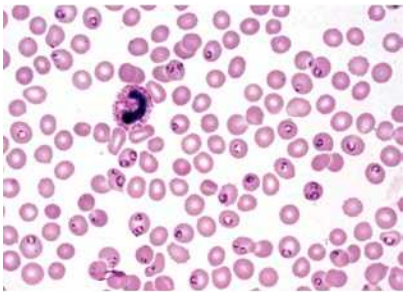
A. P. falciparum

B. P. malariae

C. P. ovale

D. P. vivax

38. A 55 year old man comes with complaints of fever and increasing fatigue for the past week. On examination, he is febrile at 40 degree C, hypotensive, and has tachycardia. The blood smear of this patient is given below. What is the causative agent?



A. Plasmodium vivax

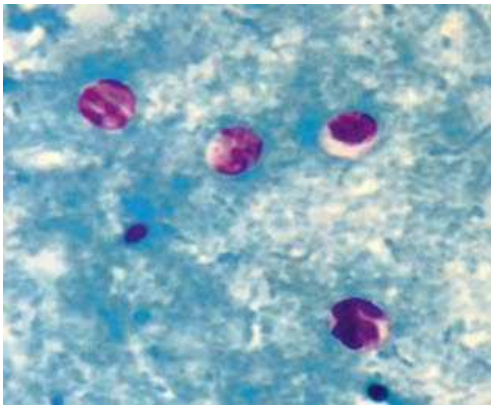
B. Plasmodium falciparum

C. Plasmodium ovale

D. Plasmodium malariae

---

39. A patient who underwent a renal transplant 3 months ago comes with complaints of diarrhea for the last 9 days. The stool sample was subject to Kinyoun stain revealing 4-6 μm structures shown below. Which of the following is true about the organism ?



A. Pathogen does not shows autoinfection

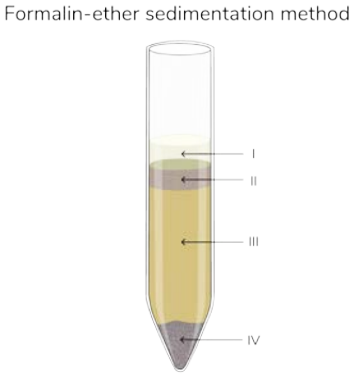
B. Sporozoites are infective form

C. Sporozoites adhere to intestinal mucosa

D. High infectivity dose is required

---

40. A stool sample is collected. The specimen is processed for examination and is labeled at different levels as A,B,C, and D. Identify the level at which parasitic ova will be seen when viewed under a microscope.



A. A

B. B

C. C

D. D

---

41. Match the following: Parasite Organ of locomotion 1. Balamuthia mandirillaris 2. Balantidium coli 3. Trichomonas vaginalis 4. Toxoplasma gondi a. Flagella b. Cilia c. Pseudopod d. No particular organ

Parasite	Organ of locomotion
1. Balamuthia mandirillaris 2. Balantidium coli 3. Trichomonas vaginalis 4. Toxoplasma gondi	a. Flagella b. Cilia c. Pseudopod d. No particular organ

A. 1-b, 2-a, 3-c, 4-d

B. 1-c, 2-b, 3-a, 4-d

C. 1-d, 2-a, 3-b, 4-c

D. 1-a, 2-b, 3-d, 3-c

---

42. Which of the following is a xenodiagnostic method?

A. Rabbit ileal loop for enterotoxigenic Escherichia coli

B. Injecting Aedes thorax with blood of a suspected dengue patient

C. Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis

D. Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria

---

43. What is the average incubation period of plasmodium falciparum?

A. 12

B. 16

C. 30

D. 28

---

44. A Patient who has a pet dog at home complains of hypochondrial pain. Radiological imaging shows a multilocular cyst within the liver. The organism involved is?

A. Echinococcus granulosus

B. Entamoeba histolytica

C. Strongyloides stercoralis

D. Ancylostoma duodenale

---

45. Which of the following statements accurately describes Taenia solium?

A. Eggs are not infective to man

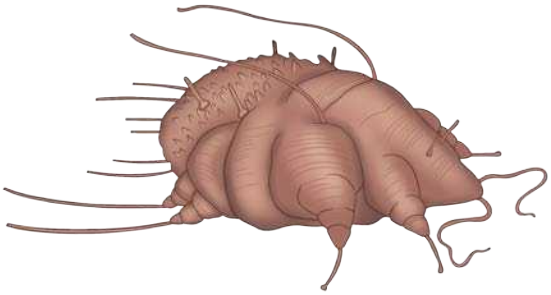
B. Hooks are absent

C. The eggs do not float in saturated salt solution

D. Intermediate host is cow

---

46. A 30-year-old female developed pruritic papules with excoriated plaques in the interdigital web spaces. Later the lesion spread to the groin and under the surface of the breasts. The lady gives a history of intense pruritus, which is more severe at night. Given below is the organism responsible for this condition. Identify the wrong statement about the same.



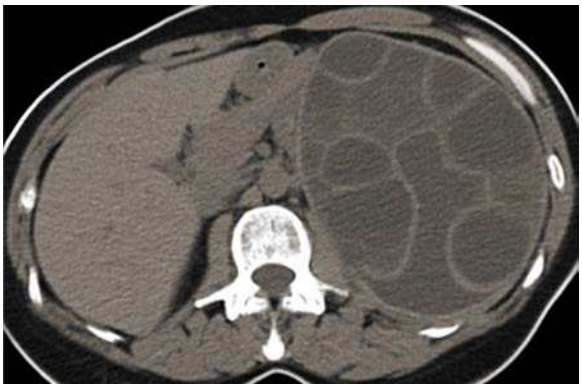
A. Life history has 4 stages

B. Larvae have 4 pairs of legs

C. Classic scabies have a lower mite burden than crusted scabies

D. Male parasite burrows into the epidermis

47. A patient presents to you with left-sided flank pain. A CT scan was done which is shown below. Which of the following statements regarding the condition is incorrect?



A. Albendazole can be started preoperatively

B. Cyst contents can produce anaphylactic reaction

C. A segmented dog tapeworm is the causative agent

D. FNAC or biopsy to be done before nephrectomy

48. Which of the following causative agents is commonly transmitted by soil and is characterized by the egg in stool, as shown in the image below?



A. Trichuris trichiura

B. Ancyclostoma duodenale

C. Necator americanus

D. Ascaris lumbricoides

49. Identify parasite?



A. E. histolytica

B. E. coli

C. E. gingivalis

D. E. polecki

---

50. Smallest cestode is?

A. Echinococcus

B. T. solium

C. H. nana

D. D. latum

---

51. Two nuclei present at the tail tip are seen in?

A. Wuchereria bancrofti

B. Brugia malayi

C. Loa loa

D. Onchocerca volvulus

---

52. Which of the following parasite is transmitted by reduviid bug?

A. T. brucei

B. T. cruzi

C. Leishmania donovani

D. Both A & B

---

53. A patient who was on steroids for the treatment of chronic urticaria presents to the emergency department with acute worsening of respiratory symptoms. He says that he has been having a nocturnal cough for the last several weeks. A hemogram revealed elevated eosinophils. Bronchoalveolar lavage (BAL) specimen following bronchoscopy showed the presence of certain larvae as shown in the image below. Identify the causative organism.



A. Ancylostoma caninum

B. Capillaria philippinensis

C. Strongylodes stercoralis

D. Enterobius vermicularis

---

54. Name the parasite whose microfilaria have a sheath with no nuclei at the tail end?

A. Wuchereria bancrofti

B. Brugia malayi

C. Brugia timori

D. Mansonella prestans

---

55. What is the likely vector for the disease in a patient from Uttar Pradesh who presented with symptoms of fever, pallor, hepatosplenomegaly, pancytopenia, peripheral smear examination showing macrophages laden with organisms containing a kinetoplast?

A. Sandfly

B. Tse-Tse fly

C. Triatomine bug

D. Female anopheles mosquito

---

56. Which of the following is the infective stage of the organism being transmitted through cat feces?

A. Tachyzoite

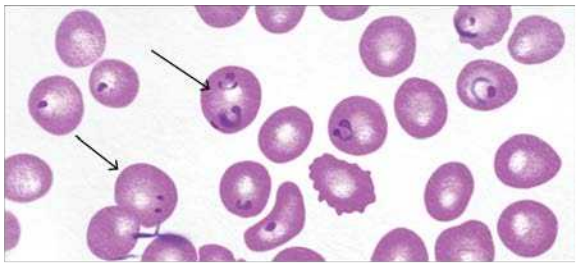
B. Bradyzoite

C. Oocyst

D. Gametocyte

---

57. A patient comes with a history of high fever for 5 days and altered sensorium. On further probing, it was found that he had two seizures one day ago. Peripheral smear shows the following findings. How will you manage this patient?



A. Doxycycline 100 mg intravenously for 7 days

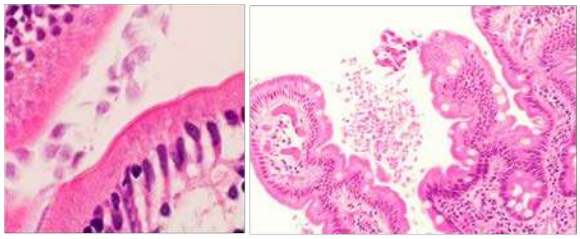
B. Clindamycin 600 mg intravenously 8 hourly for 7 days

C. Artesunate 2.4 mg/kg intravenously at 0, 12 and 24 hours, followed by daily once

D. Chloroquine intramuscularly 8 hourly for 3 days

---

58. A 30-year-old patient presented with a chronic history of watery diarrhea. The histopathology of the duodenum is shown in the image below. Which of the following is the causative organism?



A. Giardia lamblia

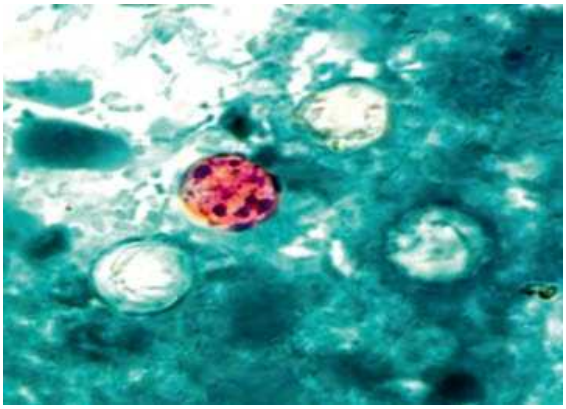
B. Entamoeba histolytica

C. Microsporidia

D. Cryptosopridia

---

59. On Cold Ziehl–Neelsen (ZN) staining, oocysts of size 8-10 μm cysts are visible. Identify the organism.



A. Isospora

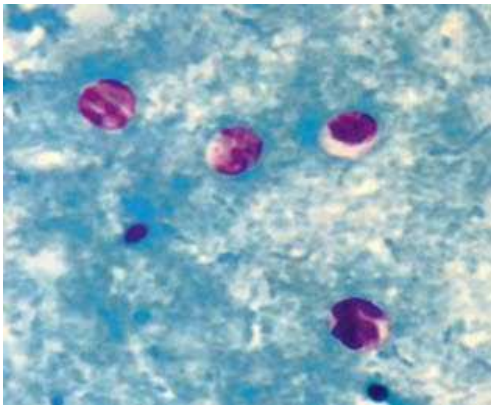
B. Cryptosporidium

C. Cyclospora

D. Microsporidia

---

60. An HIV patient came to the clinic with a history of diarrhea. Stool microscopy showed the oocysts that were 4-8 μm in size. The image is shown below. Identify the organism.



A. Isospora

B. Cyclospora

C. Cryptosporidium

D. Microsporidia

---

61. Which of the following statements regarding the given image is correct?



A. Albendazole is the drug of choice

B. Majority of infections are asymptomatic in humans

C. Infection is acquired by ingestion of aquatic plants

D. No scolex and suckers

---

62. What should be the subsequent course of action in managing a patient who presents with fever and a positive result on the filarial antigen test?

A. Bone marrow biopsy

B. DEC provocation test

C. Detection of microfilariae in the blood smear

D. Ultrasound of the scrotum

---

63. A male patient exhibited symptoms of fever and chills. The rapid test indicated a positive result for a specific antigen, HRP-2. Based on this information, which species of Plasmodium is most likely responsible for causing the infection?

A. Plasmodium falciparum

B. Plasmodium malariae

C. Plasmodium vivax

D. Plasmodium ovale

---

64. Infection with Clonorchis sinensis is associated with an increased risk of?

A. Cervical cancer

B. Cholangiocarcinoma

C. Gastric carcinoma

D. Bladder carcinoma

---

65. What is the probable cause of an ulcer with erythema on the right leg, observed in a woman from West Rajasthan, based on the findings of dark staining nuclei and kinetoplast in the biopsy's microscopic examination?

A. Leishmania tropica

B. Babesia

C. Trypanosoma cruzi

D. Histoplasma

---

66. Which of the following statements accurately describes Trichomonas vaginalis?

A. Trophozoite has 2 nucleus

B. Twitching motility is seen on wet saline mount

C. Cysts are seen on wet saline mount of vaginal secretions

D. Infective form is cyst

---

67. Which of the following parasites invade the skin directly?

A. Schistosoma mansoni

B. Clonorchis sinensis

C. Wuchereria bancrofti

D. Diphyllobothrium latum

---

68. A patient came to the gynecology OPD with complaints of foul-smelling frothy vaginal discharge, and intense itching. On examination, the cervix appears strawberry-like. Which of the following statements is false regarding the causative organism?

A. It exists in cyst and trophozoite forms

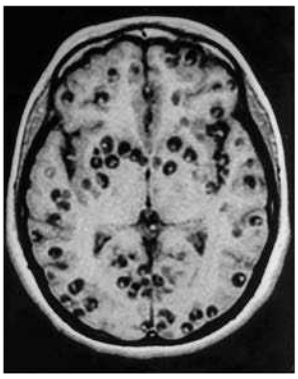
B. It is an obligate parasite

C. The gold standard for diagnosis is culture on CPLM

D. The treatment of choice is metronidazole

---

69. A patient is brought to the hospital, with complaints of seizures and the MRI image of the brain is given below. The disease is most probably caused by?



A. Cysticercus cellulosae

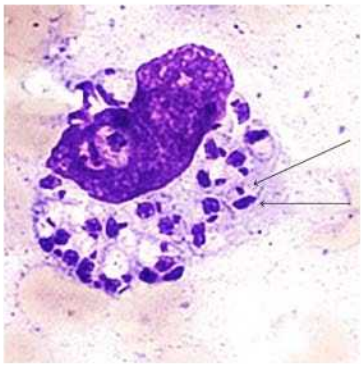
B. T. saginata

C. Cysticercus bovis

D. T. asiatica

---

70. The bone marrow aspirate from a patient revealed the following findings as seen in the image. Which of the following statements is correct about the disease?



A. Buffy coat is used to diagnose the parasite

B. It is an obligate extracellular parasite

C. It is transmitted by Culex

D. Stibogluconate is the first line of treatment in India.

71. Which infection is characterized by the presence of bloody diarrhea, liver abscess, and right upper quadrant pain, and histology of colon biopsy shows flask-shaped ulcers?

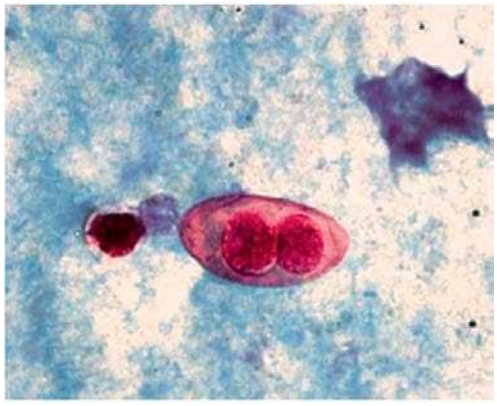
A. Giardia lamblia

B. Entamoeba histolytica

C. Campylobacter jejuni

D. Staphylococcus aureus

72. A kinyoun positive oocyst (25 μm) was isolated from an HIV patient with diarrhea. Identify the organism.



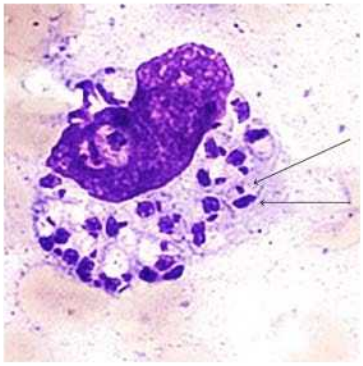
A. Cystoisospora

B. Cyclospora

C. Cryptosporidium

D. Microsporidia

73. What is the vector for the condition diagnosed by the image shown below?



A. Anopheles mosquito

B. Ixodes Tick

C. Sandfly

D. Tsetse fly

---

74. In which of the following cases is the adult form of echinococcus observed?

A. Dog

B. Cat

C. Swine

D. Sheep

---

75. For which condition is the Sabin-Feldman dye test utilized in diagnosing?

A. Paragonimiasis

B. Hydatid cyst

C. Toxoplasmosis

D. Cryptococcus

76. In the given image, which of the following specimens would not contain filarial worms, considering the presented disease?



A. Blood

B. Saliva

C. Urine

D. Hydrocele fluid

77. Identify the parasite.



A. Giardia lamblia

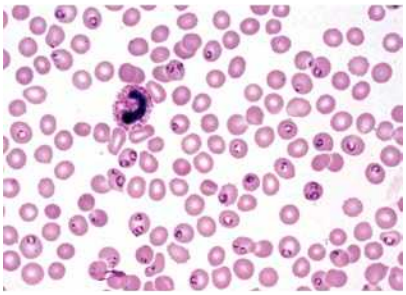
B. Trichomonas vaginalis

C. Trypanosoma

D. Cyclospora

---

78. A 55-year-old man comes with complaints of fever and increasing fatigue for the past week. On examination, he is febrile, hypotensive and tachycardic. The findings of microscopic examination of his peripheral blood smear is given below. What is the causative agent?



A. Plasmodium vivax

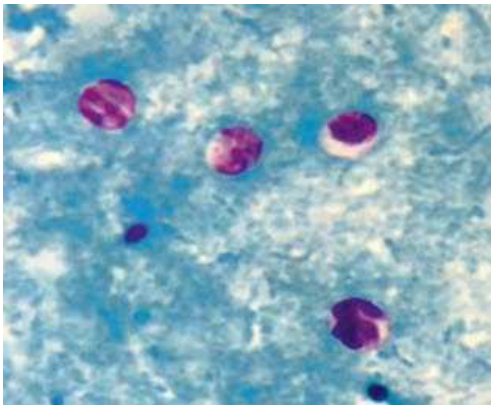
B. Plasmodium falciparum

C. Plasmodium ovale

D. Plasmodium malariae

---

79. An HIV patient presents with persistent diarrhea. Microscopic examination of the stool reveals oocyst of size 4 to 8 µm as shown below. Identify the causative organism:



A. Cryptosporidium

B. Cyclospora

C. Microspora

D. Cystoisospora

---

80. Microfilariae with sheathed tail and two nuclei at the tail is suggestive of?

A. Wuchereria bancrofti

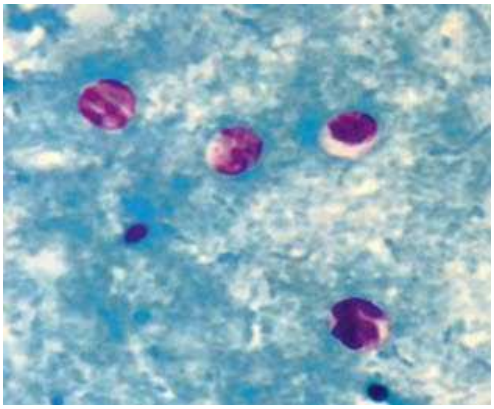
B. Brugia malayi

C. Loa loa

D. Onchocerca volvulus

---

81. An HIV patient presents with persistent diarrhea. Microscopic examination of the stool reveals oocyst of size 4 to 8 µm as shown below. Identify the causative organism:



A. Cryptosporidium

B. Cyclospora

C. Microspora

D. Cystoisospora

82. A 23-year-old patient with a history of travel to China and the consumption of pickled crab now presents with complains of rusty sputum and cough. What is the most likely causative organisms?

A. Fasciola buski

B. Fasciola hepatica

C. Paragonimus westermani

D. Schistosoma japonicum

83. Which of the following organisms follow a bird-arthropod–human transmission route?

A. Malaria

B. Japanese encephalitis

C. Plague

D. Paragonimiasis

84. A patient presents with gastrointestinal problems including abdominal pain and distension, bloody and mucus-filled diarrhea, and tenesmus, with rectal prolapse. A stool examination reveals the presence of typical barrel-shaped eggs. What is the possible causative agent?

A. Giardia lamblia

B. Trichuris trichiura

C. Ascaris lumbricoides

D. Ancylostoma duodenale

85. Identify the organism shown in this image.



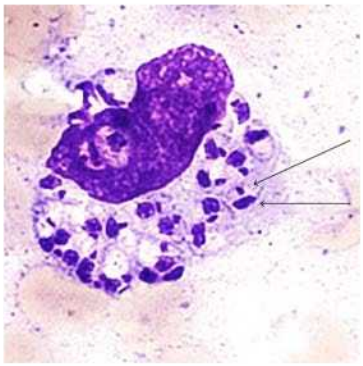
A. Strongyloides sterocoralis

B. Enterobius vermicularis

C. Ancylostoma duodenale

D. Ascaris lumbricoides

86. A man with a fever of unknown origin has a job that requires traveling around the world. The histology of the bone marrow is shown below. Which of the following investigations would be the most appropriate in this case?



A. rk 39 antigen test

B. pLDH

C. Galactose lectin antigen

D. HRP-2 antigen

---

87. Which of the given statements accurately reflects the characteristics of trematodes?

A. Fasciola hepatica are seen in the intrahepatic biliary radicles

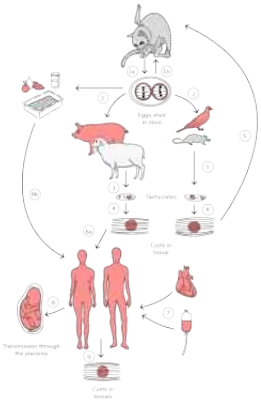
B. Schistosoma are monoecious

C. Fasciola hepatica is not acquired through contaminated water

D. Triclabendazole is the treatment of choice for all trematode infections

---

88. Identify the organism from the life cycle shown in the image given below



A. Cystolsospora

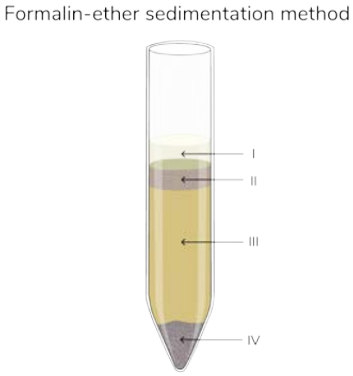
B. Cryptosporidium

C. Plasmodium knowlesi

D. Toxoplasma gondii

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89. A stool sample is collected. The specimen is processed for examination and is labeled at different levels as A, B, C, and D. Identify the level at which parasitic ova will be seen when viewed under a microscope.



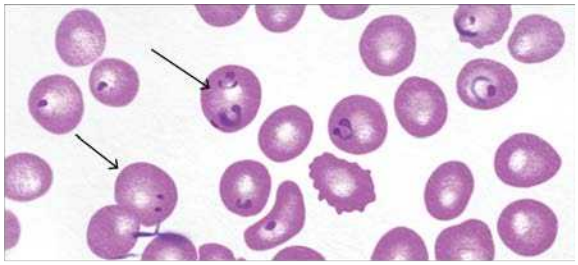
A. I

B. II

C. III

D. IV

90. A 55-year-old man comes with complaints of fever and increasing fatigue for the past week. On examination, he is febrile, hypotensive, and tachycardic. The findings of the microscopic examination of his peripheral blood smear are given below. What is the causative agent?



A. Plasmodium vivax

B. Plasmodium falciparum

C. Plasmodium ovale

D. Plasmodium malariae

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	1
Question 5	2
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	2
Question 11	1
Question 12	2
Question 13	3
Question 14	2
Question 15	2
Question 16	1
Question 17	2
Question 18	4
Question 19	3
Question 20	1
Question 21	1
Question 22	1

Question 23	3
Question 24	3
Question 25	4
Question 26	1
Question 27	3
Question 28	2
Question 29	1
Question 30	1
Question 31	3
Question 32	2
Question 33	1
Question 34	3
Question 35	1
Question 36	4
Question 37	1
Question 38	2
Question 39	3
Question 40	4
Question 41	2
Question 42	2
Question 43	1
Question 44	1
Question 45	3
Question 46	4
Question 47	4
Question 48	1

Question 49	1
Question 50	3
Question 51	2
Question 52	2
Question 53	3
Question 54	1
Question 55	1
Question 56	3
Question 57	3
Question 58	1
Question 59	3
Question 60	3
Question 61	2
Question 62	3
Question 63	1
Question 64	2
Question 65	1
Question 66	2
Question 67	1
Question 68	1
Question 69	1
Question 70	1
Question 71	2
Question 72	1
Question 73	3
Question 74	1

Question 75	3
Question 76	2
Question 77	1
Question 78	2
Question 79	1
Question 80	2
Question 81	1
Question 82	3
Question 83	2
Question 84	2
Question 85	4
Question 86	1
Question 87	1
Question 88	4
Question 89	4
Question 90	2

Solution for Question 1:

Correct Option A - Wuchereria bancrofti:

- Wuchereria bancrofti is the organism responsible for causing lymphatic filariasis. The characteristic finding on microscopy for Wuchereria bancrofti is a sheathed microfilaria with no nuclei at the tip of the tail.

Incorrect Options:

Option B - Brugia malayi: Brugia malayi's microfilariae have two nuclei that extend to the tip of the tail.

Option C - Loa Loa: The microfilariae of Loa Loa are sheathed, but the nuclei form a continuous row at the tip of the tail.

Option D - Mansonelia: The microfilariae of Mansonelia are unsheathed and extend to the tip of the tail.

Solution for Question 2:

Correct Option B - Injecting Aedes thorax with blood of a patient with suspected dengue:

- In the Xenodiagnostic method, arthropod vectors are fed on infected patients for 2-4 weeks, after which the vectors are examined for parasite presence.
- It involves injecting the thorax of Aedes mosquitoes (the vector for dengue virus) with blood from a patient suspected to have dengue. The mosquitoes are then observed for viral replication and transmission, which can help confirm the diagnosis.

Incorrect Options:

Option A - Rabbit ileal loop assay for ETEC: It is not a xenodiagnostic method as it does not involve the use of a vector and the organism is being detected for the disease is bacteria.

Option C - Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis: This is not a xenodiagnostic method as the vector for leishmaniasis is sandfly.

Option D - Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria: This is not a xenodiagnostic method as it does not involve the use of a vector, and the organism is being detected for the disease is bacteria.

Solution for Question 3:

Correct Option B - Whipworm:

- The image shows the egg of Whipworm aka Trichuris trichiura.

• Microscopic examination of the stool sample is used to identify the presence of Trichuris trichiura eggs. The eggs are characterized by Bomb barrel-shaped. Brown colored spots: Bile stained. Mucous plugs at both ends are non-bile stained.

• Bomb barrel-shaped.

• Brown colored spots: Bile stained.

• Mucous plugs at both ends are non-bile stained.

• Bomb barrel-shaped.

• Brown colored spots: Bile stained.

• Mucous plugs at both ends are non-bile stained.

Incorrect Options:

Option A - Hookworm: The eggs of hookworm/Ancylostoma duodenale are oval or elliptical, colorless, not bile stained, and surrounded by a thin transparent hyaline shell membrane. When passed in feces, the egg contains a segmented ovum, usually with four or eight blastomeres.

Option C - Pinworm: The egg of pinworm/Enterobius vermicularis is colorless, i.e., not bile stained, and is ovoid, flattened on one side, and convex on the other.

Option D - Roundworm: Fertilized eggs of roundworm/Ascaris lumbricoides are round or oval in shape, and the middle of the egg is a large unsegmented ovum containing a mass of coarse lecithin granules. It nearly fills the egg, except for a clear crescentic area at either pole.

Solution for Question 4:

Correct Option A- Orientia tsutsugamushi:

• Transmission: Scrub typhus, caused by Orientia tsutsugamushi infection, is conveyed through the bite of chigger larvae belonging to the Trombiculidae mite family, rather than hard tick.

Incorrect Options:

Option B- Crimean-Congo Fever:

• Transmission: Crimean-Congo Hemorrhagic Fever is primarily transmitted by ticks, including hard ticks.

Option C- Babesia:

• Transmission: Babesiosis is mainly transmitted by Ixodes ticks, which are hard ticks

Option D- Ehrlichia:

• Transmission: Ehrlichiosis can be transmitted by different types of ticks, including hard ticks such as Amblyomma and Dermacentor.

Solution for Question 5:

Correct Option B- Acanthamoeba:

• Given the symptoms of CNS (Central Nervous System) involvement, loss of vision, and the identification of Acanthamoeba under electron microscopy, the likely causative agent is Acanthamoeba. The patient is experiencing a disseminated Acanthamoeba infection - Granulomatous Amebic Encephalitis.

Incorrect Options:

Option A- Naegleria fowleri:

• Infection: Known for causing primary amoebic meningoencephalitis (PAM).

• Transmission: Usually occurs when contaminated water enters the nasal passages, often during activities like swimming in warm freshwater.

• Symptoms: Rapidly progressive CNS symptoms, within 5 days

Option C- Balamuthia:

• Infection: Causes granulomatous amoebic encephalitis (GAE).

• Transmission: occurs through the skin wounds or cuts when individuals come into contact with soil containing Balamuthia or when dust containing Balamuthia is breathed into the lung

• Symptoms: CNS symptoms, but the disease progresses more slowly than Naegleria fowleri

Solution for Question 6:

Correct Option B- 1 and 3:

Motility of Trophozoites:

- Advantage of Saline Mount: Saline mounts are suitable for observing the motility of trophozoites (the active, motile form of some protozoa). In Iodine mount - trophozoites are immobilized.

Internal Structure of Cysts:

- Advantage of Iodine Mount: Iodine mounts are often preferred for the examination of internal structures, including the nuclear and cytoplasmic details of cysts. Iodine enhances the contrast and visibility of internal features.

Differentiation Between Bile-Stained and Non-Bile-Stained:

- Advantage of Saline Mount: In iodine mounts, both bile- stained and non-bile-stained helminthic ova are stained brown.

Solution for Question 7:

Correct Option C - Giardia lamblia

- Giardia lamblia is a protozoan parasite that commonly causes gastrointestinal symptoms such as diarrhea, abdominal cramps, bloating, and weight loss. The "falling leaf" motility pattern is a classic finding seen in Giardia infections. This motility pattern is due to the unique flagella arrangement of Giardia trophozoites. The fact that the patient drank untreated stream water during her camping trip makes Giardia infection a likely culprit, as it is transmitted via contaminated water sources.

Incorrect Options:

Option A - Salmonella typhi: Salmonella typhi is a bacterium that causes typhoid fever. It typically presents with fever, abdominal pain, headache, and constipation rather than diarrhea. The symptoms described in the scenario, including frequent diarrhea, abdominal cramps, and " falling leaf" motility in the stool, are not characteristic of Salmonella typhi infection.

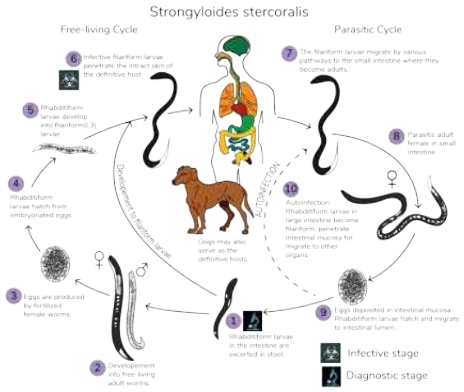
Option B - Entamoeba histolytica: Entamoeba histolytica is a protozoan parasite that causes amoebic dysentery. While it can lead to diarrhea, abdominal pain, and weight loss, the "falling leaf" motility is not a characteristic finding for Entamoeba histolytica. Additionally, there is no mention of bloody diarrhea or the presence of colonic ulcers, which are often associated with this infection.

Option D - Cryptosporidium parvum: Cryptosporidium parvum is another protozoan parasite that causes diarrhea. However, the symptoms of cryptosporidiosis are generally more severe in immunocompromised individuals. The stool examination findings in cryptosporidiosis typically reveal oocysts, which are not described as "falling leaf" motility.

Solution for Question 8:

Correct Option C - Strongyloides:

- Parthenogenesis is the ability of the females to reproduce asexually.
- Strongyloides stercoralis is the smallest nematode that is capable of Parthenogenesis.
- Egg is not seen in Strongyloides stercoralis- Ovoviviparous



Incorrect Options:

- Option A, B and D: These options are incorrect as these parasites are not capable of Parthenogenesis.

Solution for Question 9:

Correct Option B - Giemsa stain with thin smear:

• Thin smear with Giemsa stain under light microscopy is gold standard in the identification of malarial species.

• Sensitivity: 200 parasites/microlitre.

• Jaswant singh and Bhattacharya staining is preferred as per National malaria program.

Incorrect Options:

Option A - Quantitative buffy coat:

• Quantitative buffy coat is the microhematocrit method of diagnosing malaria. But it is not the best test for malaria speciation identification.

Option C - Thick smear with Giemsa stain:

• Thick smear with Giemsa stain is the gold standard for quantification. Sensitivity is 5 parasites/microlitre

Option D - Acridine orange stain with thin smear:

• Acridine orange stain with thin smear is expensive and not advised as per National malaria program.

Solution for Question 10:

Correct Option B - Cystoisospora:

• Cystoisospora belli are long and oval shaped organisms, measuring 20-33 micron in size.

• These organisms affect immunosuppressed patients, particularly those with AIDS, often presenting with profuse diarrhea associated with weakness, anorexia, and weight loss.

Incorrect Options:

Option A - Cryptosporidium:

• Cryptosporidium is 4-6 micron in size and round in shape.

Option C - Cyclospora:

• Cyclospora is 8-12 micron in size and round in shape.

Option D - Microsporidia:

• Microsporidia is 1-4 micron in size and possesses a polar tubule.

Solution for Question 11:

Correct Option A - Both the assertion and reason are correct, and the reason explains the assertion.:

- Schizonts of P. falciparum are not seen in the peripheral blood, they are seen only in the brain vasculature.
- This is due to the cytoadherence property of P. falciparum.

Schizonts of P. falciparum are not seen in the peripheral blood, they are seen only in the brain vasculature.

This is due to the cytoadherence property of P. falciparum.

Red blood cell
↓
Infected by P.falciparum (infects brain)
↓
P.falciparum starts making knobs - PF-Erythrocyte Membrane Protein
↓
These knobs will bind to ICAM present in endothelial cells
↓
The cell will get adhered - Cytoadherence

Incorrect Options:

- Option B, C and D: These options are incorrect

Option B, C and D: These options are incorrect

Solution for Question 12:

Correct Option B - Entamoeba Histolytica:

- Quadrinucleate cysts are associated with Entamoeba Histolytica

Incorrect Options:

Options A, C, D:

- Giardia lamblia,Cryptosporidium Parvum and Cyclospora Cayetanensis are not associated with Quadrinucleate cysts

Solution for Question 13:

Correct Option C - Enterobius vermicularis:

- Enterobius vermicularis (pinworm) causes perianal itching, particularly at night, due to the female worm laying eggs around the anus. Diagnosis is typically made by identifying the characteristic D-shaped eggs using the scotch tape test or NIH swab.

Incorrect Options:

Option A - Hymenolepis nana: The dwarf tapeworm typically causes abdominal pain, nausea, vomiting, and diarrhea but not perianal itching.

Options B and D - Ancylostoma and Necator americanus: Ancylostoma, now known as Necator americanus, is a hookworm that primarily causes iron deficiency anemia, ground itch, and creeping eruption.

Solution for Question 14:

Correct Option B - Strongyloides:

- Rhabditiform larvae of Strongyloides stercoralis hatch and migrate to the intestinal lumen and are excreted in stool. No eggs are seen.

Incorrect Options:

- Options A, C, and D: Eggs are present in the stool sample.

Solution for Question 15:

Correct Option B - Strongyloides Stercoralis:

- The image represents the life cycle of Strongyloides stercoralis. Points for identification.
- Females are capable of asexual reproduction (Parthenogenesis)
- Ovoviviparous larva comes out immediately from the egg
- Infective form: Filariform larva penetrates the intact skin of the definitive host.
- Route: By penetration of the Skin/autoinfection
- Rhabditiform larva comes first and transforms into filariform larva in 3 ways: Autoinfection: Penetrate intestine, produce Filariform larva Direct cycle: Rhabditiform larvae exit the body and transform into Filariform larvae, and re-enters the body. Indirect cycle Rhabditiform larva forms adult(male and female), lays eggs Eggs develop into Rhabditiform larvae Rhabditiform larva transforms into Filariform larva
- Autoinfection: Penetrate intestine, produce Filariform larva
- Direct cycle: Rhabditiform larvae exit the body and transform into Filariform larvae, and re-enters the body.
- Indirect cycle Rhabditiform larva forms adult(male and female), lays eggs Eggs develop into Rhabditiform larvae Rhabditiform larva transforms into Filariform larva
- Rhabditiform larva forms adult(male and female), lays eggs
- Eggs develop into Rhabditiform larvae
- Rhabditiform larva transforms into Filariform larva
- Autoinfection: Penetrate intestine, produce Filariform larva
- Direct cycle: Rhabditiform larvae exit the body and transform into Filariform larvae, and re-enters the body.
- Indirect cycle Rhabditiform larva forms adult(male and female), lays eggs Eggs develop into Rhabditiform larvae Rhabditiform larva transforms into Filariform larva
- Rhabditiform larva forms adult(male and female), lays eggs
- Eggs develop into Rhabditiform larvae
- Rhabditiform larva transforms into Filariform larva
- Rhabditiform larva forms adult(male and female), lays eggs
- Eggs develop into Rhabditiform larvae
- Rhabditiform larva transforms into Filariform larva

Incorrect Options:

Option A - Ancylostoma Duodenale: Ancylostoma duodenale is a hookworm that infects humans by penetrating the skin and migrating to the intestines. Its life cycle does not involve autoinfection.

Option C - Ascaris Lumbricoides: Ascaris lumbricoides is a large roundworm that infects humans. Its life cycle does not involve autoinfection.

Option D - Enterobius Vermicularis: Enterobius vermicularis, also known as pinworm, is a small nematode that commonly infects humans. Its life cycle involves the ingestion of eggs, hatching of larvae in the intestines, and migration to the colon and rectum for maturation and reproduction.

Solution for Question 16:

Correct Option A - Kennel Worker:

- The reddish serpiginous tunnels in the epidermis, as shown in the image, are suggestive of Cutaneous larva migrans caused by Ancylostoma braziliense.
- Kennel workers who handle and care for animals, including dogs and cats, are at an increased risk of exposure to the larvae of hookworms (Ancylostoma braziliense), which can cause cutaneous larva migrans (CLM). Ancylostoma braziliensis is transmitted by the larva which penetrates the skin.

Incorrect Options:

Options B, C, and D are incorrect. Kennel workers are at maximum risk of developing cutaneous larva migrans.

Solution for Question 17:

Correct Option B - Presence of duffy blood group antigen is protective against vivax:

- The invasion of red cells by merozoites requires specific glycoprotein receptors, i.e., Duffy blood group antigen, to invade red blood cells. Individuals who lack this antigen are generally resistant to P. vivax infection, while those who have it are susceptible to the parasite.

Incorrect Options:

Option A - Drug-resistant falciparum is seen in India: Drug-resistant Plasmodium falciparum, the most virulent species of malaria, is indeed seen in India.

Option C - Severe malaria with shock is called algid malaria: Algid malaria is characterized by peripheral circulatory failure, rapid thready pulse with low blood pressure, and cold, clammy skin. There may be severe abdominal pain, vomiting, diarrhea, and profound shock.

Option D - Relapse is seen in P. Ovale: Relapse is a unique characteristic of Plasmodium ovale and vivax infection. After the initial infection is treated, the parasite can remain dormant in the liver and cause relapses of malaria symptoms months or even years later.

Solution for Question 18:

Correct Option C - Ascaris Lumbricoides:

- The shown worms have characteristics of Ascaris lumbricoides.
- They are large cylindrical worms with tapering ends, the anterior end being more pointed than the posterior.
- They are pale pink or flesh-colored when freshly passed in stools but become white outside the body.
- The adult male worm is a little smaller than the female. It measures 15-30 cm in length and 2-4 mm in thickness. Its posterior end is curved ventrally to form a hook and carries two copulatory spicules.
- The female is larger than the male, measuring 20-40 cm in length and 3-6 mm in thickness. Its posterior extremity is straight and conical.

Incorrect Options:

Option A - Stongyloides Stercoralis: The female worm is thin, transparent, about 2.5 mm long and 0.05 m wide, and the male worm is shorter and broader than the female measuring 0.6- 1 mm in length and 40-50 mm in width.

Option B - Enterobius Vermicularis: Enterobius vermicularis is known as Pinworm. The adults are short, white, fusiform worms with pointed ends, looking like bits of white thread and the female worm is 8-13 mm long and 0.3-0.5 mm thick, its posterior third is drawn into a thin pointed pin-like tail.

Option D - Ancylostoma Duodenale: They are known as hookworms. The male worm is smaller than the female worm, 8-11 mm in length and 0.4 mm thick. The posterior end of the male is expanded into a copulatory bursa, which consists of three lobes. The female worm is larger, 10-13 mm long and 0.6 mm thick. Its hind end is conoid, with a subterminal anus situated ventrally.

Solution for Question 19:

Correct Option C - Babesia:

- Babesia species are tick-borne parasites that primarily infect red blood cells. Sporozoites in the salivary gland of the ticks are transmitted to humans after bite and enters bloodstream. They change to trophozoites, invade red blood cells, and multiply asexually. Newly formed trophozoites are released by rupturing erythrocytes and invade new erythrocytes.

Incorrect Options:

Option A - Toxoplasma: The tachyzoites of T. gondii formed in the intestine can infect the brain, eye, liver, spleen, lung, and skeletal muscles and form tissue cysts.

Option B - Plasmodium: Plasmodium species are the causative agents of malaria. The pre-erythrocytic schizogony takes place in the liver.

Option D - Leishmania: The amastigote of the Leishmania donovani multiply enormously in the macrophages of the liver, bone marrow, and spleen.

Solution for Question 20:

Correct Option A - Enterobius Vermicularis:

- Enterobius vermicularis, also known as the pinworm, is the causative organism for the perianal itching, with eggs shown in the provided image.
- Eggs are colorless D-shaped Plano-convex and have a tadpole larva inside them.

• It is transmitted by the ingestion of the embryonated egg-containing larva by a contaminated finger or autoinfection.

Incorrect Options:

Option B - Trichuris Trichiura: This parasite, also known as the whipworm, can cause dysentery and iron deficiency anemia. The eggs of this parasite are bile-stained, barrel-shaped with non-bile stained mucus plugs at the end that are characteristic.

Option C - Trichinella Spiralis: It is a parasitic worm that causes trichinellosis, a disease acquired by consuming undercooked or raw meat infected with the parasite. It is viviparous do not show eggs in the life cycle.

Option D - Strongyloides Stercoralis: It is a parasitic worm that causes pulmonary and small intestine infection. It is ovoviviparous i.e., larva comes out immediately out of the egg.

Solution for Question 21:

Correct Option A - Echinococcus granulosus:

• The causative organism for the condition shown in the image of a hydatid cyst in the liver is Echinococcus granulosus.

• Hydatid cyst is characterized by multiple pearly white cysts filled with clear or pale yellow fluid, the pH of the fluid is acidic (6.5), and contains salts and proteins. It is antigenic in nature and can cause anaphylaxis.

• The cysts formed by Echinococcus granulosus can grow quite large and can lead to symptoms such as abdominal pain, hepatomegaly (enlarged liver), and cyst rupture.

Incorrect Options:

Option B - Schistosoma japonicum: Schistosoma japonicum is a parasitic trematode or fluke that causes oriental schistosomiasis/Katayama disease characterized by intestinal mucosal hyperplasia, hepatosplenomegaly, and portal hypertension. The liver is hard and shows periportal fibrosis.

Option C - Echinococcus multilocularis: Echinococcus multilocularis causes alveolar echinococcosis characterized by multiple cysts and a single cyst containing multiple locules leading to multiple alveolar-like lesions in the liver

Option D - Schistosoma mansoni: Schistosoma mansoni is another parasitic trematode or fluke that causes schistosomiasis characterized by swimmer's itch, Katayama fever (high fever, rash, arthralgia), and affects the intestine (dysentery, abscess formation on the wall, hyperplasia and fibrosis).

Solution for Question 22:

Correct Option A - Echinococcus Multilocularis:

- Echinococcus multilocularis is a parasitic tapeworm that causes alveolar echinococcosis. It can form multiple cyst and inside single cyst multiple locules are present leading to multiple alveolar-like lesions in the liver, hence called alveolar hydatid cyst/multilocularis. The lesions are invasive and can infiltrate surrounding tissues.

Incorrect Options:

Option B - Echinococcus Granulosus: Echinococcus granulosus is another parasitic tapeworm that causes hydatid cysts. Multiple cysts are present, filled with fluid, which may cause anaphylaxis if drained in the body.

Option C - Entamoeba Histolytica: Entamoeba histolytica is a protozoan parasite that causes amoebic dysentery and liver abscesses. It affects the right lobe of the liver and presents with an abscess described as anchovy sauce.

Option D - Cysticercus Cellulosae: Cysticercus cellulosae is the larval stage of the pork tapeworm Taenia solium responsible for causing cysticercosis, characterized by multiple calcified cysts an multiple organs.

Solution for Question 23:

Correct Option C - Filariasis:

- The Transmission Assessment Survey (TAS) is carried out for filariasis, which is caused by parasitic worms transmitted through mosquito bites. The TAS helps assess the transmission

of filariasis, monitor the progress of elimination programs, and determine if interventionshave successfully reduced the prevalence of the disease in endemic areas or if mass drugadministration is required or not.

Incorrect Options:

Options A, B, and D are incorrect. Transmission assessment survey is one for Filariasis.

Solution for Question 24:

Correct Option C - Schistosoma Hematobium:

Schistosoma Haematobium	Schistosoma Mansoni	Schistosoma Japonicum
Resides in pelvic vesical venous plexus.	Resides in Inferior mesenteric venous plexus draining sigmoid rectal region	Resides in Superior mesenteric venous plexus draining ilecocecal region
Hematuria Squamous Cell Carcinoma>> Transitional Cell Carcinoma of urinary bladder.	Swimmers Itch	Severe swimmer's itch Genital schistosomiasis CNS schistosomiasis
 ... (content truncated)	 ... (content truncated)	 ... (content truncated)
Seen in urine sample	Seen in stool sample	Seen in stool sample

Incorrect Options:

Options A and B are incorrect. Refer to Option C for an explanation.

Option D - Fasciola Hepatica: It is a parasitic liver fluke that infects the liver and bile ducts, leading to bile duct obstruction, jaundice, and pain right upper quadrant pain. It has operculated eggs.

Solution for Question 25:

Correct Option D - Ticks:

- Babesiosis is a tick-borne disease. Ticks (Ixodid) act as vectors and can transmit the Babesia parasites to humans during a blood meal.

Incorrect Options:

Option A - Reduviid Bugs: Chagas disease caused by Trypanosoma cruzi transmitted by reduviid bugs (triatomine).

Option B - Tsetse Fly: Sleeping sickness caused by Trypanosoma brucei transmitted by tsetse fly.

Option C - Sand Fly: Leishmaniasisis caused by leishmania spp. is transmitted by sand fly.

Solution for Question 26:

Correct Option A - 1-c, 2-b,3-d, 4-a:

Parasites with their respective hosts:

- Diphyllobothrium spp. Definitive host – Man, 1st Intermediate host – cyclops, 2nd Intermediate host – Fish.
- Definitive host – Man, 1st Intermediate host – cyclops, 2nd Intermediate host – Fish.
- Paragonimus spp. Definitive host – Man, 1st Intermediate host – Snail, 2nd Intermediate host – Crab and Fish.
- Definitive host – Man, 1st Intermediate host – Snail, 2nd Intermediate host – Crab and Fish.

• Brugia spp. Definitive host – Sheep and Man, Intermediate host – Vector mosquito (Mansonia).

• Definitive host – Sheep and Man, Intermediate host – Vector mosquito (Mansonia).

• Fasciola spp. Definitive host – Man, 1st intermediate host – Snail, 2nd intermediate host – Aquatic plant

• Definitive host – Man, 1st intermediate host – Snail, 2nd intermediate host – Aquatic plant

• Definitive host – Man, 1st Intermediate host – cyclops, 2nd Intermediate host – Fish.

• Definitive host – Man, 1st Intermediate host – Snail, 2nd Intermediate host – Crab and Fish.

• Definitive host – Sheep and Man, Intermediate host – Vector mosquito (Mansonia).

• Definitive host – Man, 1st intermediate host – Snail, 2nd intermediate host – Aquatic plant

Solution for Question 27:

Correct Option C - Cysticercus Cellulose:

• Neurocysticercosis is caused by the larva of Taenia solium called cysticercus cellulosae.

Incorrect Options:

Option A - Cysticercus Bovis: It is the larval stage of Taenia saginata responsible for causing intestinal taeniasis.

Option B - Taenia Saginata Asiatica: Taenia saginata asiatica is another species of tapeworm that resembles Taenia saginata, and cysticerci are located in liver.

Option D - Taenia Saginata: Taenia saginata is responsible for causing intestinal taeniasis.

Solution for Question 28:

Correct Option B - Cysticercus Cellulosae:

• The larval stage of Taenia solium, a parasitic tapeworm, is known as cysticercus cellulose, aka bladder worm, is responsible for causing cysticercosis.

• In this condition, the larval cysts of the tapeworm can be found in various tissues of the body, including the muscles, brain, and subcutaneous tissues.

Incorrect Options:

Option A - Cysticercus Bovis: Cysticercus bovis is the larval stage of Taenia saginata responsible for causing intestinal taeniasis.

Option C - Chigger: Chigger is a 6-legged larval stage of trombiculid mite responsible for causing Scrub typhus/Chigger-borne disease.

Option D - None of the Above: Cysticercus cellulosae is the larval stage of Taenia solium.

Solution for Question 29:

Correct option A - Steroids followed by antiparasitic drugs:

• The management of a patient with seizures and multiple parenchymal cysts seen on the CT head depends on the underlying cause of the cysts, which is suggestive of neurocysticercosis, which is caused by larval cysts (Cysticercus cellulosae) of the pork tapeworm (Taenia solium).

• The appropriate management for neurocysticercosis is steroids followed by antiparasitic drugs. Steroids: In cases of neurocysticercosis, the use of corticosteroids is recommended as an adjunctive therapy to reduce inflammation and edema around the cysts, which can help alleviate symptoms and prevent complications. Antiparasitic drugs: Antiparasitic treatment with the combination of albendazole (400 mg twice daily for 30 days) and praziquantel (50 mg/kg in three divided doses for 20-30 days) is administered after starting steroids to minimize the risk of the inflammatory response caused by the death of parasites.

• Steroids: In cases of neurocysticercosis, the use of corticosteroids is recommended as an adjunctive therapy to reduce inflammation and edema around the cysts, which can help alleviate symptoms and prevent complications.

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Incorrect Options:

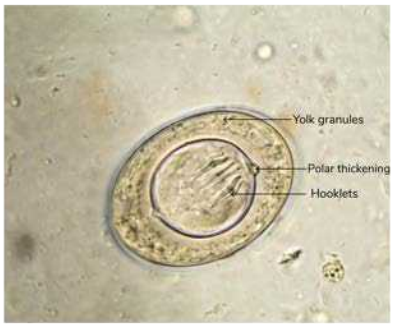
• Option B, C and D are incorrect. Refer to the explanation of correct answer.

Solution for Question 30:

Correct Option A - Praziquantel:

• The given image is the Egg of Hymenolepis nana. The egg is round to slightly oval in shape. It has two membranes that surround an embryo with six hooklets. The space between the two membranes is filled by yolk granules. Both the poles of the embryophore are thickened.

• The drug of choice to treat the infection is praziquantel works by causing severe spasms and paralysis of the tapeworm's muscles, leading to the detachment of the worm from the intestinal wall. It also increases the permeability of the tapeworm's cells to calcium ions, which ultimately results in the worm's death.



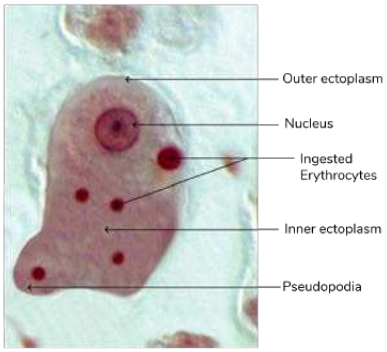
Incorrect Options:

Options B, C, and D are incorrect. These drugs are not the drug of choice to treat infection caused by H. nana.

Solution for Question 31:

Correct Option C - Entamoeba Histolytica:

- The given image is of E. histolytica trophozoites. Entamoeba histolytica is a parasitic amoeba that can cause amoebic dysentery. Points for identification of trophozoite are:
- It is 12 to 60 μm and actively motile in freshly passed dysenteric stool.
- Cytoplasm – Outer ectoplasm is clear, transparent, and refractile. Inner endoplasm is finely granular, having a ground glass appearance.
- Pseudopodia are finger-like projections formed by sudden jerky movements of ectoplasm in one direction.
- Nucleus is spherical 4-6 μm in size and contains central karyosome, surrounded by clear halo and anchored to the nuclear membrane by fine radiating fibrils, giving a cartwheel appearance.
- Ingested erythrocytes may also be present within the cytoplasm of E. histolytica trophozoites.



Incorrect Options:

Option A - Giardia Lamblia: Infection with Giardia lamblia is charactereized by steatorrhea, due to malabsorption. The characteristic finding in stool examination for Giardia infection is the presence of cysts.

Option B - Shigella Dysenteriae: It is a bacterial pathogen that causes bacillary dysentery. Stool examination findings in Shigella infection typically show leukocytes (pus cells) and red blood cells.

Option D - Campylobacter Jejuni: It is a bacterial pathogen commonly associated with gastroenteritis. Stool examination findings may show an increase in leukocytes and occasionally blood.

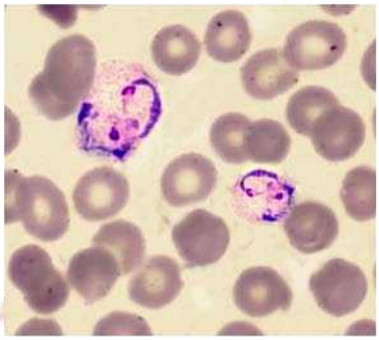
Solution for Question 32:

Correct Option B - Plasmodium Falciparum:

- The image showing multiple ring forms/headphone-shaped late trophozoites in erythrocytes, the most likely causative agent of malaria in this case is Plasmodium falciparum.
- Plasmodium falciparum is one of the species of Plasmodium that causes cerebral malaria in humans characterized by high fever, headache, anorexia, nausea, anemia, hypotension, tachycardia, hepatosplenomegaly, thrombocytopenia, confusion, increased muscle tone, seizures, paralysis, and coma.

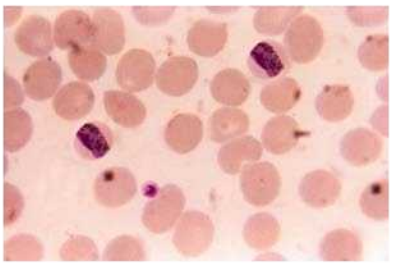
Incorrect Options:

Option A - Plasmodium Vivax: P. vivax infection usually shows single ring forms or single dots or amoeboid-shaped late trophozoite. This parasite is associated with the relapse and usually caused mild disease than P. falciparum.



Option C - Plasmodium Ovale: P. Ovale infection shows similar characteristics to P. vivax, including ring forms. However, P. ovale is less common and has a more restricted geographic distribution, primarily in Africa.

Option D - Plasmodium Malariae: P. Malariae infection typically exhibits band form late trophozoites and usually caused mild disease than P. falciparum.



Solution for Question 33:

Correct Option A - Kyasanur Forest Disease:

- The vector shown in the image is a sandfly. Points for identification. Light or dark brown flies, smaller than mosquitoes. Body and wings covered by dense hair. Head contains a pair of long, slender, and hairy antennae, palpi, and a proboscis. Thorax contains a pair of wings and three pairs of legs.
- Light or dark brown flies, smaller than mosquitoes.
- Body and wings covered by dense hair.
- Head contains a pair of long, slender, and hairy antennae, palpi, and a proboscis.
- Thorax contains a pair of wings and three pairs of legs.
- Diseases caused by sandflies are Kala-azar (visceral leishmaniasis), Sandfly fever, Oriental sore, and Oroya fever.
- Kyasanur Forest Disease (KFD), also known as monkey fever, is caused by the Kyasanur Forest Disease virus (KFDV). Hard ticks are the main vectors involved in the transmission of Kyasanur Forest Disease.
- Light or dark brown flies, smaller than mosquitoes.
- Body and wings covered by dense hair.
- Head contains a pair of long, slender, and hairy antennae, palpi, and a proboscis.
- Thorax contains a pair of wings and three pairs of legs.

Incorrect Options:

Options B, C, and D are incorrect. These diseases are transmitted by sandfly.

Solution for Question 34:

Correct option C - Assertion is true, but the reason is false:

- The assertion states that Malathion is used as an ultra-low volume (ULV) during epidemics, which is true, and the reason that ultra-low volume defoggers have residual effects is wrong. Malathion is the least toxic organophosphorus compound commonly used as an insecticide during disease outbreaks, particularly for dengue-hemorrhagic fever and mosquito-borne encephalitis epidemics. ULV refers to the method of applying a small volume of insecticide as fine droplets, which allows for effective coverage and control of mosquitoes or other vector populations.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 35:

Correct Option A - Entamoeba Histolytica:

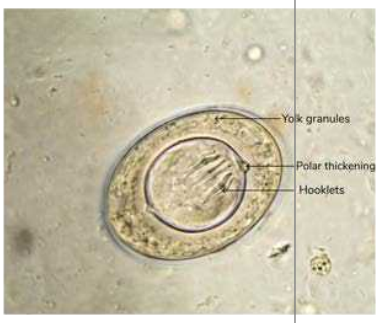
- Entamoeba histolytica is a protozoan parasite that can cause intestinal infections, specifically amoebic dysentery.
- One of the characteristic features of amoebic dysentery is the presence of flask-shaped ulcers in the colon.
- The trophozoites of E. histolytica damage the mucosa and submucosa but cannot damage the muscular as it is difficult to break. Thus, it starts breaking the submucosa on the sides, resulting in a flask-shaped ulcer.

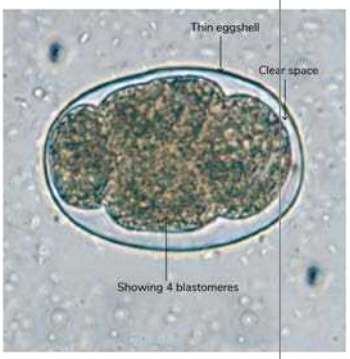
Incorrect Options:

Options B, C, and D do not cause flask-shaped ulcers in the colon.

Solution for Question 36:

Correct Option D - 1-c, 2-d, 3-b, 4-a:

Organism	Egg
Hymenolepis nana: Egg is round to slightly oval in shape. It has two membranes that surround an embryo with six hooklets. The space between the two membranes is filled by yolk granules. Both the poles of the embryophore are thickened.	
Ascaris lumbriocoides: Fertilized eggs are round to oval in shape, surrounded by a thick albuminous coat. Bile stained, appears golden brown. Unfertilized egg are elongated, their albuminous coat is thin, contains nsegmented, small atrophied ovum.	

Ancylostoma duodenale: Eggs are oval in shape and surrounded by a thin eggshell; the embryo is segmented comprising 4 – 32 blastomeres. There is a clear space between the eggshell and the embryo.	
Taneia solium: Egg contains an embryo or oncosphere, which contains three pairs of hooklets surrounded by an embryophore.	

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

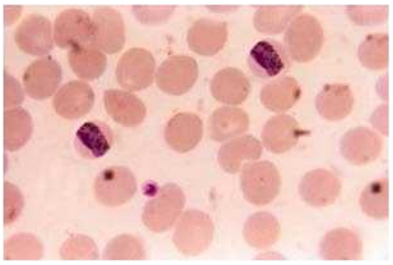
Solution for Question 37:

Correct Option A- P. Falciparum:

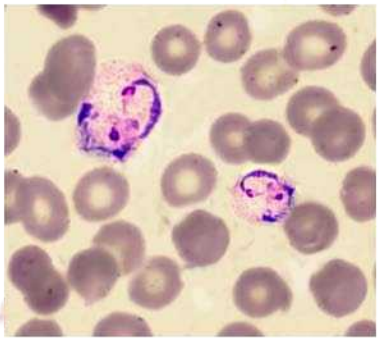
- The image showing multiple ring forms/headphone-shaped late trophozoites in erythrocytes, the most likely causative agent of malaria in this case is Plasmodium falciparum.
- Plasmodium falciparum is one of the species of Plasmodium that causes cerebral malaria in humans characterized by high fever, headache, anorexia, nausea, anemia, hypotension, tachycardia, hepatosplenomegaly, thrombocytopenia, confusion, increased muscle tone, seizures, paralysis, and coma.

Incorrect Options:

Option B - P. Malariae: P. malariae infection typically exhibits band form late trophozoites and usually caused mild disease than P. falciparum.



Option C - P. Vivax: P. vivax infection usually shows single ring forms or single dots or amoeboid-shaped late trophozoite and usually caused mild disease than P. falciparum.



Option D - P. Ovale: P. ovale infection shows similar characteristics to P. vivax, including ring forms. However, P. ovale is less common and has a more restricted geographic distribution, primarily in Africa.

Solution for Question 38:

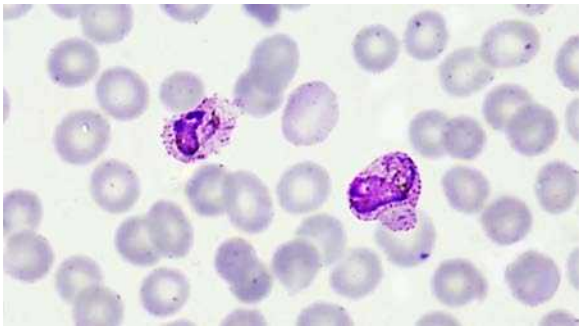
Correct Option B - Plasmodium Falciparum:

• The presence of "headphone-shaped" trophozoites in the blood smear is characteristic of Plasmodium falciparum, which is the causative agent of malaria.

• Plasmodium falciparum is one of the species of Plasmodium that causes cerebral malaria in humans characterized by high fever, headache, anorexia, nausea, anemia, hypotension, tachycardia, hepatosplenomegaly, thrombocytopenia, confusion, increased muscle tone, seizures, paralysis, and coma.

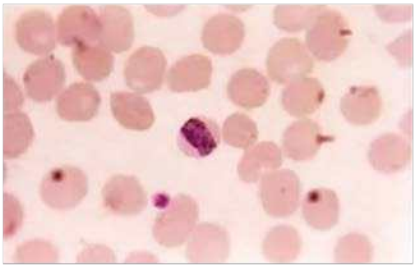
Incorrect Options:

Option A - Plasmodium Vivax: P. vivax infection usually shows single ring forms or single dots or amoeboid-shaped trophozoite.



Option C - Plasmodium Ovale: P. ovale infection shows similar characteristics to P. vivax, including ring forms. However, P. ovale is less common and has a more restricted geographic distribution, primarily in Africa.

Option D - P. Malariae: P. malariae infection typically exhibits band forms trophozoite.



Solution for Question 39:

Correct Option C - Sporozoites adhere to intestinal mucosa:

• Cryptosporidium parvum is a protozoan parasite that causes gastrointestinal infections, particularly in immunocompromised individuals and are acid fast as shown in the image.

• Oocysts are the infective form, which is ingested; following ingestion, excystation occurs. The sporozoites are released and hide in the tunnel, which is known as Parasitophorous vacuoles which hide along the brush border cells of the intestine. In these cells, the parasites develop into sporulated oocysts in the infected host.

• Two different types of oocysts are produced, the thick-walled, which is commonly excreted from the host, and the thin-walled oocyst, which is primarily involved in autoinfection.

• Low infective dose of Cryptosporidium (10–100 oocysts can initiate the infection)

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 40:

Correct Option D - IV:

• In the given information where the formol-ether sedimentation technique is mentioned and level D is labeled at the base of the tube, it indicates that parasitic ova will be seen at this level when viewed under a microscope.

• Formol-ether sedimentation is a commonly used technique for concentrating parasites and ova from stool samples when the egg burden is low. In this technique, the stool sample is mixed with a formalin-ether solution. It takes advantage of the high specific gravity of protozoan cysts and helminth eggs compared to water. Formalin fixes the eggs, larvae, oocysts, and spores so that they are no longer infectious, as well as preserves their morphology. Fecal debris is extracted into the ethyl acetate phase of the solution.

• The different levels represented by markings are: Level 1: represents the topmost layer of the sample and typically contains ether. Level II: contain some debris. Level III: contains formalin. Level IV: protozoan cyst and eggs.

• Level 1: represents the topmost layer of the sample and typically contains ether.

• Level II: contain some debris.

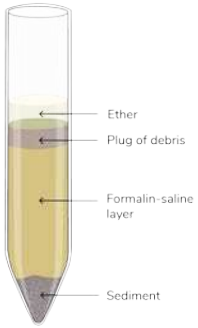
• Level III: contains formalin.

• Level IV: protozoan cyst and eggs.

• Level 1: represents the topmost layer of the sample and typically contains ether.

- Level II: contain some debris.
- Level III: contains formalin.
- Level IV: protozoan cyst and eggs.

Formalin-ether sedimentation method



Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 41:

Correct Option B - 1-c, 2-b, 3-a, 4-d:

1. Balamuthia mandirillaris - Pseudopod
2. Balantidium coli - Cilia
3. Trichomonas vaginalis - Flagella
4. Toxoplasma gondii - No particular organ

Solution for Question 42:

Correct Option B - Injecting Aedes thorax with blood of a patient with suspected dengue:

• In the Xenodiagnostic method, arthropod vectors are fed on infected patients for 2-4 weeks, after which the vectors are examined for parasite presence.

• It involves injecting the thorax of Aedes mosquitoes (the vector for dengue virus) with blood from a patient suspected to have dengue. The mosquitoes are then observed for viral replication and transmission, which can help confirm the diagnosis.

Incorrect Options:

Option A - Rabbit ileal loop assay for ETEC: It is not a xenodiagnostic method as it does not involve the use of a vector and the organism is being detected for the disease is bacteria.

Option C - Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis: This is not a xenodiagnostic method as the vector for leishmaniasis is sandfly.

Option D - Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria: This is not a xenodiagnostic method as it does not involve the use of a vector, and the organism is being detected for the disease is bacteria.

Solution for Question 43:

Correct Option A - 12 days:

• The average incubation period of Plasmodium falciparum, the most severe and potentially fatal malaria parasite, is approximately 12 days.

• The average incubation period of other plasmodium species are: P. vivax - 14 days. P. malariae - 28 days. P. ovale - 17 days.

• P. vivax - 14 days.

• P. malariae - 28 days.

• P. ovale - 17 days.

• P. vivax - 14 days.

• P. malariae - 28 days.

• P. ovale - 17 days.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 44:

Correct Option A - Echinococcus granulosus:

- History and symptoms in given question is consistent with Echinococcus granulosus.
- Hydatid disease is caused by Echinococcus granulosus/Dog tapeworm..
- Definitive host: Dog
- Intermediate host: sheep and other animals
- Accidental host: Man
- Hydatid cyst Common site - right side of liver(presents with hypochondrial pain,hepatomegaly,may complicate as portal hypertension) 2nd most common site- lungs(presents with cough with sputum/chest pian)
- Common site - right side of liver(presents with hypochondrial pain,hepatomegaly,may complicate as portal hypertension)
- 2nd most common site- lungs(presents with cough with sputum/chest pian)
- Diagnosis Imaging: water lily sign FNAC is contraindicated Cyst fluid: hooklets present which are ZN stain positive.
- Imaging: water lily sign
- FNAC is contraindicated
- Cyst fluid: hooklets present which are ZN stain positive.
- Common site - right side of liver(presents with hypochondrial pain,hepatomegaly,may complicate as portal hypertension)
- 2nd most common site- lungs(presents with cough with sputum/chest pian)
- Imaging: water lily sign
- FNAC is contraindicated
- Cyst fluid: hooklets present which are ZN stain positive.
- Treatment:PAIR technique(Puncture the cyst, Aspirate, Injection with 95% ethanol,Reaspirate once the scolices are killed)

Incorrect Options:

Option B - Entamoeba histolytica:

• Entamoeba histolytica affects intestine(causes asymptomatic intraluminal amoebiasis(50-80%),amoebic colitis,appendicitis,bloody diarrhoea),liver (causes amoebic liver abscess).

Option C- Strongyloides stercoralis:

• Strongyloides stercoralis is smallest nematode known to cause human infection which presents with clinical features of Cutaneous larva currens Pulmonary lung infection Small intestinal infection Hyperinfection syndrome

• Cutaneous larva currens

• Pulmonary lung infection

• Small intestinal infection

• Hyperinfection syndrome

• Cutaneous larva currens

• Pulmonary lung infection

• Small intestinal infection

• Hyperinfection syndrome

Option D - Ancylostoma duodenale:

• Ancylostoma duodenale/hookworm infestation presents with clinical feature of skin(Ground itch,Creeping eruption) Iron deficiency anemia Pulmonary symptoms.

• skin(Ground itch,Creeping eruption)

• Iron deficiency anemia

• Pulmonary symptoms.

• skin(Ground itch,Creeping eruption)

• Iron deficiency anemia

• Pulmonary symptoms.

Solution for Question 45:

Correct Option C - The eggs do not float in saturated salt solution:

• Among the statements you provided, the true statement regarding Taenia solium is: The eggs do not float in the saturated salt solution.

Incorrect Options:

Option A - Eggs of Taenia solium are non-infective to humans: Taenia solium eggs can lead to human infection if ingested, usually through undercooked pork containing tapeworm larvae (cysticerci). Once ingested, the eggs hatch in the intestine and migrate, causing cysticercosis.

Option B - Lack of hooks on Taenia solium: Taenia solium does possess hooks on its scolex, aiding its attachment to the intestinal lining of its host.

Option D - Intermediate host is a cow: While Taenia solium does have an intermediate host, it's not a cow. Pigs serve as the intermediate host, where tapeworm larvae develop in their muscles after ingesting eggs. Humans can become definitive hosts by consuming undercooked pork containing these larvae, leading to adult tapeworm development in the intestines.

Solution for Question 46:

Correct Option D - Male Parasite Burrows into the Epidermis:

- The clinical presentation and image showing the causative organism Sarcoptes scabiei is characterized by a round body dorsally and a flattened body ventrally, covered with short bristles, and has two pairs of legs in front and two pairs behind are suggestive of scabies. Classic scabies typically manifest as an intensely pruritic eruption with a characteristic distribution. The sides and webs of the fingers, wrists, axillae, areolae, and genitalia are among the common sites of involvement.
- Scabies is a parasitic infestation of the skin caused by the burrowing of female mites into the epidermis facilitated by the secretion of proteolytic enzymes that cause keratinocyte damage.

Incorrect Options:

Option A - Life History has 4 Stages: The life cycle of Sarcoptes scabiei involves four stages: egg, larva, nymph, and adult.

Option B - Larvae have 4 Pairs of Legs: The larvae of Sarcoptes scabiei have four pairs of legs.

Option C - Classic Scabies have a Lower Mite Burden than Crusted Scabies: The mite burden in a patient with classic scabies is 10 to 15 mites, and for crusted scabies, the mite burden goes up to millions.

Solution for Question 47:

Correct Option D - FNAC or biopsy to be done before nephrectomy:

- Given CT scan is that of Renal Hydatid Cyst showing cystic structures surrounded by a rim of calcifications. FNAC or biopsy should never be done before nephrectomy in case of a hydatid cyst as it contains lots of hydatid fluid, which can produce an anaphylactic reaction in the patient.

Incorrect Options:

Option A - Albendazole can be started preoperatively: Albendazole can be started preoperatively as it will reduce viability of cyst. Started 1-2 month before surgery and continued post operatively.

Option B - Cyst contents can produce anaphylactic reaction: Cyst contents can produce anaphylactic reaction in patient as cyst contain hydatid fluid which can be released into the vasculature following trauma and can lead to complication from mild urticaria to life-threatening circulatory shock.

Option C - A segmented dog tapeworm is the causative agent: A segmented dog tapeworm (Echinococcus granulosus) is the causative agent.

Solution for Question 48:

Correct Option A - Trichuris Trichiura:

- The helminthic infection commonly transmitted by soil and characterized by the egg in stool, as shown in the image, is Trichuris trichiura (whipworm) infection.
- Transmission: Trichuris trichiura is primarily transmitted through ingestion of food or water contaminated with soil containing infective eggs.
- Microscopic examination of the stool sample is used to identify the presence of Trichuris trichiura eggs. The eggs are characterized by Bomb barrel-shaped. Brown colored spots: Bile stained. Mucous plugs at both ends are non-bile stained.

- Bomb barrel-shaped.
- Brown colored spots: Bile stained.
- Mucous plugs at both ends are non-bile stained.
- Bomb barrel-shaped.
- Brown colored spots: Bile stained.
- Mucous plugs at both ends are non-bile stained.

Incorrect Options:

Option B - Ancyclostoma Duodenale: Ancyclostoma duodenale is a hookworm species that primarily infects humans through skin penetration by larvae present in contaminated soil. Eggs are oval or elliptical, colorless, not bile stained, and surrounded by a thin transparent hyaline shell membrane. When passed in feces, the egg contains a segmented ovum, usually with four or eight blastomeres.

Option C - Necator Americanus: Necator americanus is another hookworm species that infects humans through skin penetration by larvae in contaminated soil. Similar to Ancyclostoma duodenale, the eggs of Necator americanus are oval or elliptical, colorless, not bile stained, and surrounded by a thin transparent hyaline shell membrane. When passed in feces, the egg contains a segmented ovum, usually with four or eight blastomeres.

Option D - Ascaris Lumbricoides: Ascaris lumbricoides is a roundworm that is commonly transmitted through ingestion of contaminated food or water with eggs. An unfertilized egg is elliptical in shape and contains a small atrophied ovum with a mass of disorganized, highly refractile granules of various sizes. Fertilized eggs are round or oval in shape, and the middle of the egg is a large unsegmented ovum containing a mass of coarse lecithin granules. It nearly fills the egg, except for a clear crescentic area at either poles.

Solution for Question 49:

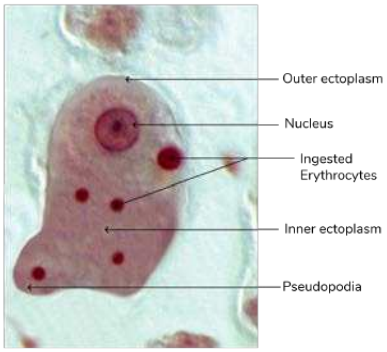
Correct Option A - E. Histolytica:

- The given image is of E. histolytica trophozoites. Points for identification are:
- It is 12 to 60 μm and actively motile in freshly passed dysenteric stool. Typical ameboid motility is a crawling or gliding movement.
- Cytoplasm – Outer ectoplasm is clear, transparent, and refractile. Inner endoplasm is finely granular, having a ground glass appearance.

• Pseudopodia are finger-like projections formed by sudden jerky movements of ectoplasm in one direction.

• Nucleus is spherical 4-6 µm in size and contains central karyosome, surrounded by clear halo and anchored to the nuclear membrane by fine radiating fibrils, giving a cartwheel appearance.

• Ingested erythrocytes may also be present within the cytoplasm of E. histolytica trophozoites.



Incorrect Options:

Option B - Entamoeba Coli: E. coli trophozoites are larger than E. histolytica, ranging from 20-50 micrometers in size. They have multiple nuclei (up to 8) with a small central karyosome and a coarse, irregularly granular cytoplasm. E. coli trophozoites often contain ingested bacteria, cell debris, or leukocytes.

Option C - Entamoeba Gingivalis: Trophozoites are 10 to 20 µm actively motile with multiple pseudopodia and have a single nucleus with a small eccentric karyosome, and contain engulfed epithelial cells or leukocytes.

Option D - Entamoeba Polecki: E. polecki trophozoites are similar in size to E. histolytica but have multiple nuclei (up to 16) with a small central karyosome and a coarsely granular cytoplasm. They may also contain ingested bacteria, yeast cells, or debris.

Solution for Question 50:

Correct Option C - H. Nana:

• Hymenolepsis nana is the smallest cestode, hence known as a dwarf cestode.

Incorrect Options:

Options A, B, and D are incorrect. H. nana is the smallest cestode.

Solution for Question 51:

Correct Option B - Brugia Malayi:

- Sheathed microfilariae with 2 nuclei at the tip of the tail is suggestive of Brugia malayi.

Incorrect Options:

Option A - Wuchereria Bancrofti: Microfilariae of W. bancrofti are sheathed, but nuclei do not extend to the tail.

Option C - Loa Loa: Microfilariae of Loa loa are sheathed and bear a column of nuclei extending till the tail tip

Option D - Onchocerca Volvulus: Microfilariae of Onchocerca volvulus are unsheathed, and nuclei do not extend to the tail tip.

Solution for Question 52:

Correct Option B - T. Cruzi:

- This parasite causes Chagas disease, also known as American trypanosomiasis. It is transmitted by the triatomine (reduviid) bug, commonly known as the "kissing bug." When the bug bites a person, it can defecate near the bite wound, and if the person scratches the area, the parasite can enter the bloodstream through the broken skin.

Incorrect Options

Option A - T. Brucei: It is known for causing African Trypanosomiasis, commonly known as Sleeping Sickness transmitted by tsetse fly.

Option C - Leishmania Donovani: It is known for causing visceral leishmaniasis, transmitted by sandflies.

Option D - Both A & B: Although both parasites belong to the genus Trypanosoma, T. Cruzi is transmitted by reduviid bug, and T. brucei is transmitted by the tsetse fly.

Solution for Question 53:

Correct Option C - Strongylodes Stercoralis:

- Strongyloides stercoralis is a parasitic nematode that can cause strongyloidiasis in humans. The larvae of Strongyloides stercoralis can migrate through the lungs and may enter the alveoli, leading to respiratory symptoms like asthma.
- The chronic use of steroids for the treatment of urticaria can result in an impaired immune system, leading to hyperinfection syndrome. In this the filariform larvae may enter in to arterial circulation and lodge in various organs.
- The presence of rhabditiform larvae in the bronchoalveolar lavage (BAL) specimen characterized by tapering end is diagnostic of Strongyloides stercoralis.

Incorrect Options:

Option A - Ancylostoma Caninum: Ancylostoma caninum is a hookworm species that primarily affects dogs. While it can cause skin manifestations in humans, it does not typically involve the respiratory system.

Option B - Capillaria Philippinensis: Capillaria philippinensis is a parasitic nematode that causes intestinal infection characterized by non-specific abdominal pain, watery diarrhea in humans.

Option D - Enterobius Vermicularis: Enterobius vermicularis, also known as the pinworm, is a common intestinal parasite in humans. It is mostly asymptomatic and some cases may complaints of perianal pruritis worse in night and excoriation of perianal skin.

Solution for Question 54:

Correct Option A - Wuchereria Bancrofti:

- Wuchereria bancrofti is a filarial worm that causes lymphatic filariasis in humans. Its microfilariae have a sheath, and nuclei do not extend to the tip of the tail.

Incorrect Options:

Option B - Brugia Malayi: It has sheathed microfilariae with 2 nuclei at the tip of the tail.

Option C - Brugia Timori: It has sheathed microfilariae with 2 nuclei at the tip of the tail

Option D - Mansonella Prestans: It has unsheathed microfilariae, and nuclei do not extend to the tip of the tail.

Solution for Question 55:

Correct Option A - Sandfly:

- The clinical presentation of fever and hepatosplenomegaly with a peripheral smear showing pancytopenia and organism-laden macrophages with kinetoplast is highly suggestive of visceral leishmaniasis, aka kala-azar caused by Leishmania donovani.
- It is transmitted by a sandfly that releases promastigote in the blood, leading to leishmaniasis.

Incorrect Options:

Option B - Tse-Tse fly: Tsetse fly is a vector of trypanosoma it causes African sleeping sickness. The clinical features are characterized by lymphadenopathy, fever, and encephalitis.

Option C - Triatomine bug: Triatomine bug is a vector of Trypanosoma cruzi, which causes Chagas disease. Infection is most commonly acquired through contact with the feces of an infected triatomine bug (or "kissing bug"), a blood-sucking insect that feeds on humans and animals. The disease mainly affects the nervous system, digestive system, and heart.

Option D - Female anopheles: Female anopheles is a vector of plasmodium species. Infections caused by P. falciparum are the most likely to progress to severe, potentially fatal forms with central nervous system involvement (cerebral malaria), acute renal failure, severe anemia, or acute respiratory distress syndrome.

Solution for Question 56:

Correct Option C - Oocyst Stage:

- The cat, specifically the definitive host, is responsible for the sexual reproduction of Toxoplasma gondii. Inside the cat's intestines, the parasite produces oocysts. These oocysts are then shed in the cat's feces. After being shed in the environment, the oocysts require a period of time, usually 1 to 5 days, to undergo sporulation. Sporulation makes the oocysts infective and is transmitted by ingestion of the contaminated food and water.

Incorrect Options:

Option A - Tachyzoite: Once inside the intermediate host, the oocysts release sporozoites that invade the host's cells. The sporozoites rapidly multiply and transform into tachyzoites, which disseminate throughout the host's body. Tachyzoites are responsible for transmitting the disease through blood transfusion or vertical transmission.

Option B - Bradyzoite: The tachyzoites can invade various tissues and organs, where they transform into a dormant, slow-growing form called bradyzoites. These bradyzoites form tissue cysts primarily in the brain, muscle tissue, and other organs of the intermediate host. Bradyzoite can transmit infection by eating undercooked meat and organ transplantation.

Option D - Gametocyte: Inside the cat's intestines, the bradyzoites develop into male and female gametes, which fuse to form zygotes. The zygotes develop into oocysts, which are then shed in the cat's feces, completing the life cycle.

Solution for Question 57:

Correct Option C - Artesunate 2.4 mg/kg intravenously at 0, 12 and 24 hours, followed by daily once:

- The clinical presentation of seizures, fever, and peripheral smear reveals a headphone-shaped or multiple-ring form that is suggestive of cerebral malaria caused by Plasmodium falciparum.

• Artemisinin derivatives are the first-line agent in case of cerebral malaria. Artesunate 2.4 mg/kg is given intravenously at 0, 12, and 24 hours, followed by daily once given in case of cerebral malaria.

Incorrect Options:

Options A, B, and D are incorrect. These are not the treatment of choice in patients with complicated falciparum malaria.

Solution for Question 58:

Correct Option A - Giardia Lamblia:

• The clinical presentation of diarrhea and histopathology of the duodenal biopsy reveals trophozoites in the duodenal lumen showing falling leaf motility as shown in the image, the most likely causative organism in this case is Giardia lamblia.

• The characteristic appearance of trophozoites showing falling leaf motility in the duodenal lumen is consistent with Giardia lamblia infection.

• In the duodenum, where the histopathology findings were observed, Giardia lamblia trophozoites typically attach to the intestinal epithelium using a sucking disk mechanism This attachment disrupts the normal microvillus architecture and impairs the absorption of nutrients, leading to malabsorption and watery diarrhea.

• This attachment disrupts the normal microvillus architecture and impairs the absorption of nutrients, leading to malabsorption and watery diarrhea.

• This attachment disrupts the normal microvillus architecture and impairs the absorption of nutrients, leading to malabsorption and watery diarrhea.

Incorrect Options:

Option B - Entamoeba Histolytica: This protozoan parasite is known to cause amoebic dysentery, which is characterized by bloody diarrhea. The histopathology findings in amoebic dysentery typically show flask-shaped ulcers in the intestinal mucosa with trophozoites and ingested red blood cells.

Option C - Microsporidia: Microsporidia are unicellular parasites that can infect various tissues, including the intestine. Microsporidia infection is often observed in HIV-positive individuals with severe immunosuppression and can present with chronic diarrhea, but the histopathological appearance would typically show intracellular spores rather than trophozoites.

Option D - Cryptosporidia: Cryptosporidium is another protozoan parasite that causes diarrhea, particularly in immunocompromised individuals. However, the histopathology findings associated with Cryptosporidium infection include intracellular acid-fast parasites within epithelial cells, which appear as round or oval structures.

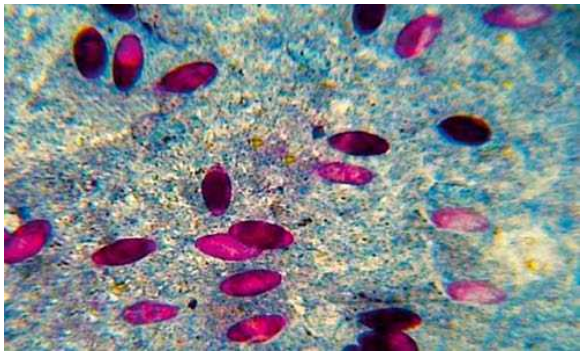
Solution for Question 59:

Correct Option C - Cyclospora:

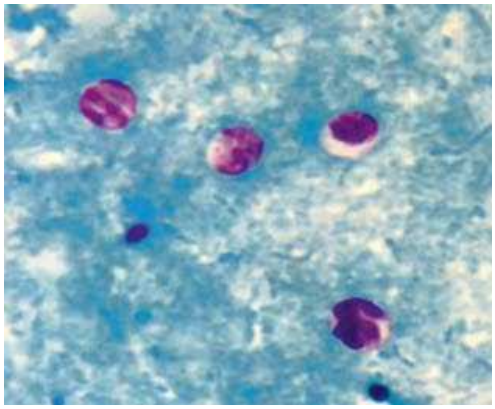
- This Cold ZN staining reveals the presence of 4 cyclospora round oocyst-bearing sporozoites of size 8-10 µm, showing variability in staining in the field. Cyclosporiasis is a disease caused by Cyclospora cayetanensis, transmitted by feces or feces-contaminated food and water

Incorrect Options:

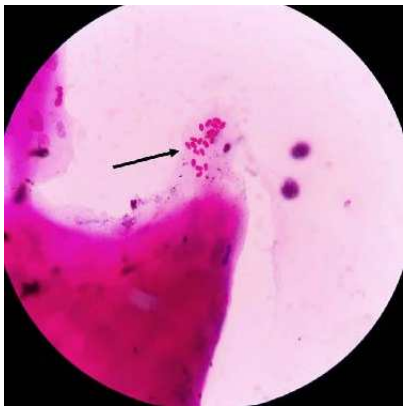
Option A - Isospora: Isospora belli oocyst stained with the modified acid-fast technique reveals oval oocysts of size 23-36 µm that stain red to reddish-orange.



Option B - Cryptosporidium: Cold ZN staining will show acid-fast round oocysts of size 4-6 µm.



Option D - Microsporidia: Modified Zeihl Neelsen (ZN) stain shows bright red in color.



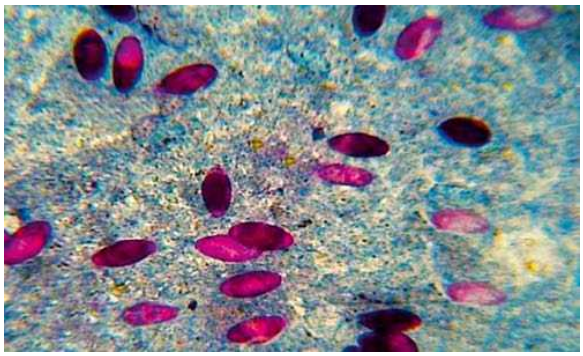
Solution for Question 60:

Correct Option C - Cryptosporidium:

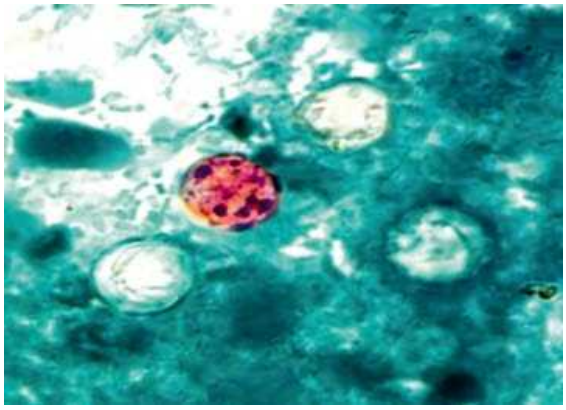
- Coccidian parasites are responsible for causing diarrhea in an immunocompromised individual (HIV). Cryptosporidium is the most common among coccidian parasites to cause persistent diarrhea. The oocyst with the size of 4-6 μm and an image showing acid fastness confirms the causative parasite to be Cryptosporidium parvum. It is stained with modified acid-fast staining using 0.2%-1% of acid alcohol, showing a pinkish-red color on a uniformly green background.

Incorrect Options:

Option A - Isospora: Oocyst of isospora are also acid-fast, but the size of the oocyst is large, i.e., 23-36 μm in size, and is oval in shape.



Option B - Cyclospora: Cyclospora oocysts on modified acid-fast technique, stain red to reddish-orange, 8-12 μm in size, round, variably acid-fast, contains two sporocysts each bearing sporozoites.



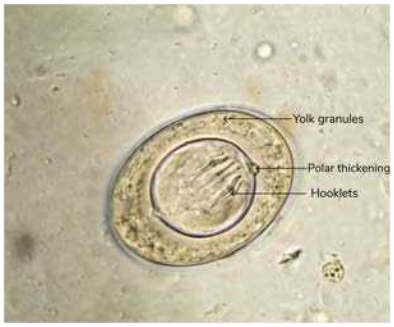
Option D - Microsporidia: Microsporidia are intracellular parasites characterized by a unicellular spore containing a coiled spring-like tubular polar filament.

Solution for Question 61:

Correct Option B - Majority of infections are asymptomatic in humans:

• The given image is the Egg of Hymenolepis nana. The egg is round to slightly oval in shape. It has two membranes that surround an embryo with six hooklets. The space between the two membranes is filled by yolk granules. Both the poles of the embryophore are

thickened.



• Hymenolepis nana infections are mostly asymptomatic. If there is a heavy burden of worms (1000-2000 worms), it manifests as anorexia, abdominal pain, headache, dizziness, and diarrhea.

Incorrect Options:

Option A - Albendazole is the drug of choice: Albendazole is not a drug of choice for H. nana treatment. Praziquantel (25mg/kg once) is the drug of choice for infection with H. nana.

Option C - Infection is acquired by ingestion of aquatic plants: Infection is acquired most commonly from eggs in the feces of another infected individual, which are transferred in food, by contamination, or by autoinfection.

Option D - No scolex and suckers: The H. nana scolex has four suckers and up to 30 hooks.

Solution for Question 62:

Correct Option C - Detection of microfilariae in the blood smear:

• The detection of microfilariae in the blood smear is an essential diagnostic step in the management of filarial infections.

• The standard method for diagnosing active infection is the identification of microfilariae in a blood smear by microscopic examination. Microfilariae can be detected microscopically on blood smears obtained at night (10 PM–2 AM), and a thick smear should be made and stained with Giemsa or hematoxylin and eosin.

• Other methods of detecting microfilariae microscopically are: By examination of an unstained mount of blood under a microscope. By acridine orange - microhematocrit tube technique.

• By examination of an unstained mount of blood under a microscope.

• By acridine orange - microhematocrit tube technique.

• By examination of an unstained mount of blood under a microscope.

• By acridine orange - microhematocrit tube technique.

Incorrect Options:

Option A - Bone Marrow Biopsy: Bone marrow biopsy is not a routine procedure for the diagnosis or management of filarial infections. It is typically performed to evaluate disorders of the bone marrow, such as hematological malignancies.

Option B - DEC Provocation Test: DEC (diethylcarbamazine) is an antiparasitic medication used for the treatment of filarial infections. The DEC provocation test involves giving a small dose of DEC and observing for the presence of adverse reactions, such as fever, rash, or swelling, which can occur in patients with active filarial infections.

Option D - Ultrasound of the Scrotum: Ultrasound of the scrotum in individuals with filariasis reveals a dance sign (motile adult worm in the scrotal lymphatics) in active infection, not confirmatory.

Solution for Question 63:

Correct Option A - Plasmodium Falciparum:

• Plasmodium falciparum is the most common and deadly species responsible for malaria infections worldwide, particularly in tropical and subtropical regions.

• The HRP-2 (Histidine-rich protein 2) antigen is specific to Plasmodium falciparum and can be detected in the blood using rapid diagnostic tests (RDTs).

Incorrect Options:

Options B, C, and D are incorrect. Parasite lactate dehydrogenase (pLDH) and aldolase are common to all other species.

Solution for Question 64:

Correct Option B - Cholangiocarcinoma:

- Cholangiocarcinoma refers to cancer that arises from the bile ducts.
- Clonorchis sinensis is one of the major causes of cholangiocarcinoma in areas where the parasite is endemic, such as parts of East Asia, including China, Korea, and Vietnam.
- When a person ingests raw or undercooked freshwater fish infected with the larvae of Clonorchis sinensis, the parasite can establish itself in the bile ducts of the liver.
- Over time, chronic infection and inflammation caused by the presence of the parasite can lead to bile duct damage, scarring, and the development of cholangiocarcinoma.

Incorrect Options:

Option A - Cervical cancer: Cervical cancer is primarily caused by persistent infection with high-risk types of human papillomavirus (HPV).

Option C - Gastric carcinoma: Gastric carcinoma, or stomach cancer, is primarily linked to factors such as Helicobacter pylori, dietary factors, and genetic predisposition.

Option D - Bladder carcinoma: Bladder carcinoma is primarily linked to factors such as Schistosoma haematobium and smoking.

Solution for Question 65:

Correct Option A - Leishmania Tropica:

- Class Kinetoplastida have a special type of extrachromosomal DNA called kinetoplast, which is only present in hemoflagellates like Leishmania and Trypanosoma.
- This is a case of Cutaneous leishmaniasis or oriental sore or delhi boil characterized by a localized ulcer surrounded by erythema. The parasite invades the skin, causing tissue destruction and the formation of an ulcer.
- Leishmania tropica is a protozoan parasite that causes cutaneous leishmaniasis, a vector-borne disease transmitted by the sandfly. It is prevalent in various regions of the

world, including parts of Rajasthan and India.

• Microscopic examination of a biopsy from the edge of the ulcer plays a crucial role in identifying the causative agent. The presence of organisms with dark staining nuclei and kinetoplast is consistent with Leishmania spp.

Incorrect Options:

Option B - Babesia: Babesia is a protozoan parasite that causes babesiosis, a tick-borne disease. It typically presents with symptoms such as fever, fatigue, hemolytic anemia and does not show kinetoplast.

Option C - Trypanosoma Cruzi: Trypanosoma cruzi causes Chagas disease characterized by chagoma at the site of entry, periorbital edema (Romana sign), hepatosplenomegaly and lymphadenopathy; chronic cases may show shoe myocarditis, megacolon, and meningoencephalitis.

Option D - Histoplasma: Histoplasma is a fungus that causes histoplasmosis, a respiratory infection. It does not typically present with skin ulcers and does not show kinetoplast.

Solution for Question 66:

Correct Option B - Twitching motility is seen on wet saline mount:

• Trichomonas vaginalis is the most common cause of sexually transmitted disease and non-gonococcal urethritis.

• The infective and diagnostic stage is trophozoite; the cyst form does not exist.

• It is diagnosed by a pap smear of the cervix showing a pear-shaped trophozoite with one nucleus, 4 anterior and 1 posterior flagella with undulating membrane.

• The parasite exhibits rapid, jerky movements, which are described as "twitching." This motility is due to the presence of flagella on the surface of the protozoan, allowing it to move in a unique and distinctive manner.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 67:

Correct Option A - Schistosoma Mansoni:

- Infection from Schistosoma mansoni occurs when the skin comes in contact with contaminated freshwater where cercariae released from the snails are present. These cercariae penetrate the skin and lead to infection in humans

Incorrect Options:

Option B - Clonorchis Sinensis - Infection from Clonorchis sinensis is due to ingestion of the metacercariae present in the raw freshwater fish.

Option C - Wuchereria Bancrofti: It is transmitted to man by many different mosquitoes like Aedes spp., Anopheles spp., and Culex spp., depending on geographical distribution. The mosquito injects the larvae into the man.

Option D - Diphyllobothrium Latum - Infection from Diphyllobothrium latum occurs by ingestion of larvae in raw freshwater fish.

Solution for Question 68:

Correct Option A - It exists in cyst and trophozoite forms:

- The clinical presentation of foul-smelling frothy-vaginal discharge with strawberry cervix is suggestive of trichomoniasis caused by Trichomonas vaginalis.
- Trichomonas vaginalis is an obligate parasite that exists only in trophozoite form, which is both diagnostic and infective stage.
- Diagnosis by culture is the gold standard test. It is cultured in Cysteine peptone liver maltose media (CPLM).
- The drug of choice for trichomoniasis is a single dose of 2-gram metronidazole for both partners..

Incorrect Options:

- Options B, C, and D: These statements are accurate about Trichomonas vaginalis.

Solution for Question 69:

Correct Option A - Cysticercus cellulosae:

- The given image of the CT scan of the brain shows multiple calcified cysts of cysticercus cellulosae in the brain parenchyma and a history of seizures is suggestive of neurocysticercosis caused by the larval stage (cysticercus cellulosae) of the pork tapeworm, Taenia solium.

Incorrect Options:

Option B - T. saginata: Taenia saginata is another species of tapeworm that infects humans, but it causes a different condition known as intestinal taeniasis. Taeniasis occurs when a person ingests the larval stage (cysticercus bovis) of Taenia saginata found in undercooked beef.

Option C - Cysticercus bovis: Cysticercus bovis is the larval stage of Taenia saginata found in undercooked beef and responsible for causing intestinal taeniasis.

Option D - T. asiatica: Taenia saginata asiatica is another species of tapeworm that resembles Taenia saginata, and cysticerci are located in the liver.

Solution for Question 70:

Correct Option A - Buffy coat is used to diagnose the parasite:

- The given image is Leishman Donovan bodies, which are macrophages-laden amastigotes, which are ovoid or rounded cells, about 2-4 μm in size, suggesting Leishmaniasis. Leishmaniasis is a parasitic disease caused by the protozoan parasites of the Leishmania genus. These parasites infect various tissues of the body, including the bone marrow.
- The gold standard test to diagnose leishmaniasis is by demonstration of the amastigote in peripheral blood, bone marrow aspirate, spleen aspirate, and large lymph nodes. Examination of buffy coat is best to examine for amastigote in peripheral blood smear. Buffy coat smears show a diurnal periodicity, with more smears being positive when collected during the day than at night.

Incorrect Options:

Option B - It is an obligate extracellular parasite: Leishmania parasites are obligate intracellular parasites, specifically residing inside macrophages and other immune cells. They are obligate intracellular parasites and require host cells for their survival and replication.

Option C - It is transmitted by culex: Leishmaniasis is primarily transmitted to humans through the bites of infected female sandflies.

Option D - Stibogluconate is the first line of treatment in India: The choice of treatment for leishmaniasis depends on various factors, including the species of Leishmania causing the infection and the geographical location. The drug of choice to treat leishmaniasis is liposomal amphotericin B, which is also used as a first-line treatment in different regions.

Solution for Question 71:

Correct Option B - Entamoeba Histolytica:

- The clinical presentation of bloody diarrhea, liver abscess, and flask-shaped ulcer is suggestive of Amebiasis caused by Entamoeba histolytica.
- The most commonly affected organ is the intestine leading to bloody diarrhea (traveler's diarrhea) and appendicitis. The most common extraintestinal infection occurs in liver which may lead to amebic liver abscess (anchovy sauce).
- A colon biopsy shows a flask-shaped ulcer is characteristic of Entamoeba histolytica. The trophozoites of E. histolytica damage the mucosa and submucosa but cannot damage the muscular as it is difficult to break. Thus, it starts breaking the submucosa on the sides, resulting in a flask-shaped ulcer.

Incorrect Options:

Option A - Giardia Lamblia: It is responsible for causing giardiasis characterized by steatorrhea due to malabsorption, not dysentery.

Option C - Campylobacter Jejuni: This is a gram-negative bacteria that causes fever, bloody diarrhea, and abdominal pain. It is diagnosed by gram staining showing darting motility or by growing in culture.

Option D - Staphylococcus Aureus: Staphylococcus aureus is a gram-positive bacteria that can cause various infections, including food poisoning characterized by nausea, vomiting, and

abdominal cramps. It can be diagnosed by gram staining and culture medium.

Solution for Question 72:

Correct Option A - Cystoisospora:

- Cystoisospora (formerly known as Isospora) is a genus of intracellular parasites that infect the gastrointestinal tract, leading to diarrhea in an immunocompromised individual. The oocysts of Cystoisospora are acid fast, 23-36 μm in size and are oval in shape.

Incorrect Options:

Option B - Cyclospora: The oocysts of Cyclospora are variably acid-fast, 8-12 μm in size, and round in shape.

Option C - Cryptosporidium: The oocyst of Cryptosporidium is acid-fast, 4-6 μm in size, and round in shape.

Option D - Microsporidia: Microsporidia are intracellular parasites characterized by a unicellular spore containing a coiled spring-like tubular polar filament.

Solution for Question 73:

Correct Option C - Sandfly:

- The given image is Leishman Donovan bodies, which are macrophages-laden amastigotes, which are ovoid or rounded cells, about 2-4 μm in size, suggesting typical Leishmaniasis.
- The vector for the transmission of leishmaniasis is sandfly, particularly Phlebotomous argentipes and orientalis.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 74:

Correct Option A - Dog:

- The adult worm of Echinococcus lives in the jejunum and duodenum of dogs and other canine carnivora (wolf and fox).
- The larval stage (hydatid cyst) is found in humans and herbivorous animals like sheep, goats, cattle, and horses.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 75:

Correct Option C - Toxoplasmosis:

- The Sabin-Feldman dye test is a gold standard test for the diagnosis of toxoplasmosis. It is a type of complement fixation test that detects the presence of specific antibodies, known as Toxoplasma-specific antibodies, in a person's blood.
- It involves mixing the patient's serum with live tachyzoites from Toxoplasma gondii and then adding a dye (methylene blue). If antibodies are present, the tachyzoites will be colorless, and if antibodies are absent, the tachyzoites will show color. The presence of antibodies will activate the complement fixation and destroy the tachyzoites, and lead to distorted and colorless tachyzoites.

Incorrect Options:

Options A, B, and D are incorrect. The Sabin-Feldman dye test is specific and sensitive to toxoplasmosis.

Solution for Question 76:

Correct Option B - Saliva:

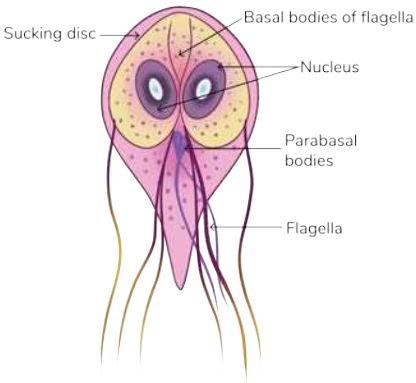
- The given image is of lymphedema of left leg seen in lymphatic filariasis aka elephantiasis occurs as a result of chronic infection of filariasis, due to lymphatic obstruction ultimately leading to swelling of the limbs. The causative parasite belongs to the class of lymphatic filarial nematodes such as Wucheria Bancrofti.
- Filarial worms can be found in blood, urine, exudate of lymph varix, and hydrocele fluid, but not in saliva.

Incorrect Options: Options A, C, and D are incorrect. They can be used for demonstration of the microfilaria.

Solution for Question 77:

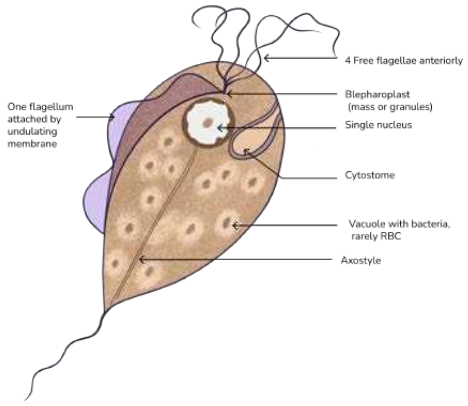
Correct Option A - Giardia lamblia:

- The above image shows the trophozoite form of Giardia Lamblia. It is a pyriform/tennis racket-shaped organism, and a sucking disc is present n the ventral side. It has 2 nuclei, 4 pairs of flagella responsible for falling leaf motility, a blepharoplast, from which the flagella arise, one pair of axosystole running along the midline, and 2 sausage-shaped parabasal or median bodies.

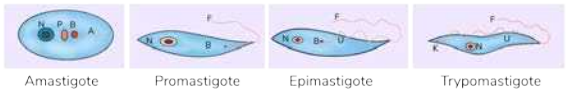


Incorrect Options:

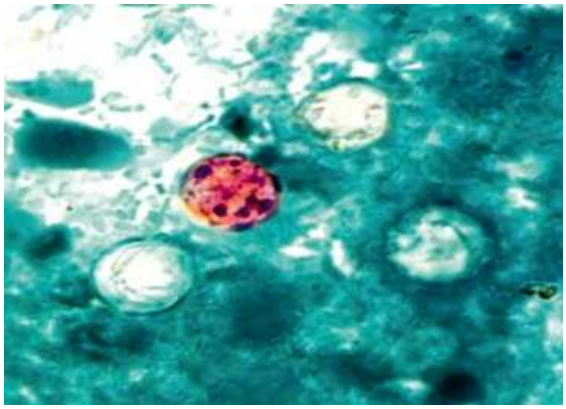
Option B - Trichomonas vaginalis: The trophozoite of Trichomonas is a pyriform or oval-shaped organism with 4 flagella on the anterior end, and 5th flagella remain folded on the outer surface of the undulating membrane. An axosystole runs throughout the length of the body.



Option C - Trypanosoma: Trypanosoma is a hemoflagellate with 4 different morphological forms all of them containing 1 nucleus, and kinetoplast. Different forms are Amastigote (round or ovoid in shape without an external flagellum), Promastigote (lanceolate shaped, with emerging flagellum without undulating membrane), Epimastigote (short undulating membrane is present), and Trypomastigote (spindle shaped with nucleus at one end, kinetoplast is anterior to nucleus, and long undulating membrane is present).



Option D - Cyclospora: It is a coccidian Protozoa, which is usually identified by its oocysts. Unsporulated oocyst is excreted in the stools that convert to sporulated oocysts. Sporulated oocysts of cyclospora are round in shape as well as acid-fast, but contain 2 sporocysts, with each sporocyst containing two sporozoites.



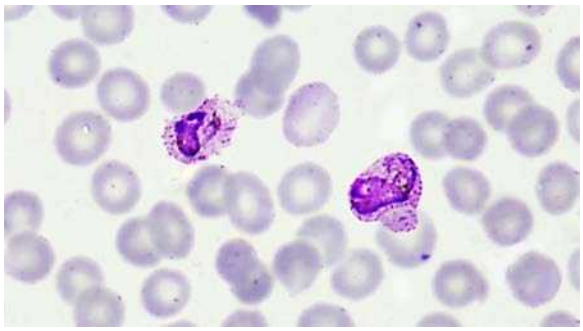
Solution for Question 78:

Correct Option B - Plasmodium Falciparum:

- The presence of "headphone-shaped" trophozoites in the blood smear is characteristic of Plasmodium falciparum, which is the causative agent of malaria.
- Plasmodium falciparum is one of the species of Plasmodium that causes cerebral malaria in humans characterized by high fever, headache, anorexia, nausea, anemia, hypotension, tachycardia, hepatosplenomegaly, thrombocytopenia, confusion, increased muscle tone, seizures, paralysis, and coma.

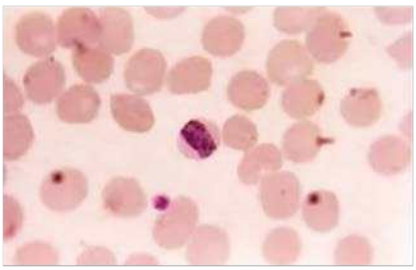
Incorrect Options:

Option A - Plasmodium Vivax: P. vivax infection usually shows single ring forms or single dots or amoeboid-shaped trophozoite.



Option C - Plasmodium Ovale: P. ovale infection shows similar characteristics to P. vivax, including ring forms. However, P. ovale is less common and has a more restricted geographic distribution, primarily in Africa.

Option D - P. Malariae: P. malariae infection typically exhibits band forms trophozoite.



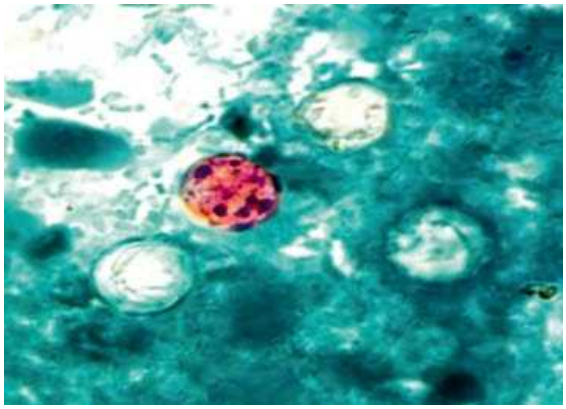
Solution for Question 79:

Correct Option A - Cryptosporidium:

• Coccidian parasites are responsible for causing diarrhea in an immunocompromised individual (HIV). Cryptosporidium is the most common among coccidian parasites to cause persistent diarrhea. The image showing an acid-fast oocyst round in shape confirms the causative parasite to be Cryptosporidium parvum. It is stained with modified acid-fast staining using 0.2%-1% of acid alcohol, showing a pinkish-red color.

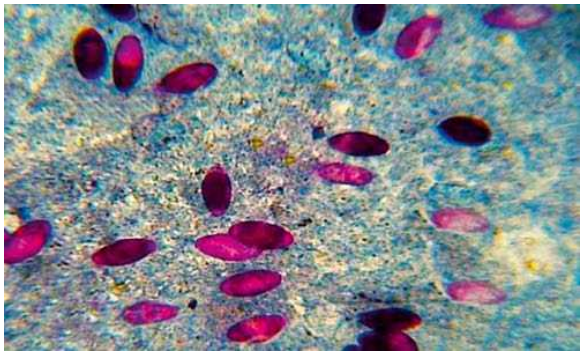
Incorrect Options:

Option B - Cyclospora: Cyclospora oocysts on modified acid-fast technique, stain red to reddish-orange, 8-12 μm in size, round, variably acid-fast, contains two sporocysts each bearing sporozoites.



Option C- Microspora: Microspora is a group of spore-forming parasites that appear as ovoid spores with bright red walls on the trichrome stain, they are associated with skin and hair dermatophytoses.

Option D - Cystoisospora: Cystoisospora, formerly known as Isospora, is another protozoan parasite that can cause diarrhea in immunocompromised individuals. Oocyst of isospora are also acid-fast, but the size of the oocyst is large, i.e., 23-36 μm in size, and is oval in shape.



Solution for Question 80:

Correct Option B - Brugia malayi:

- Sheathed microfilariae with 2 nuclei at the tip of the tail is suggestive of Brugia malayi.

Incorrect Options:

Option A - Wuchereria Bancrofti: Microfilariae of W. bancrofti are sheathed, but nuclei do not extend to the tail.

Option C - Loa Loa: Microfilariae of Loa loa are sheathed and bear a column of nuclei extending till the tail tip

Option D - Onchocerca Volvulus: Microfilariae of Onchocerca volvulus are unsheathed, and nuclei do not extend to the tail tip.

Solution for Question 81:

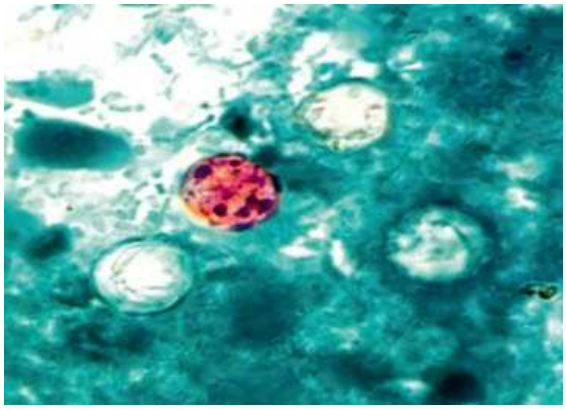
Correct Option A - Cryptosporidium:

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causative parasite to be Cryptosporidium parvum. It is stained with modified acid-faststaining using 0.2%-1% of acid alcohol, showing a pinkish-red color.

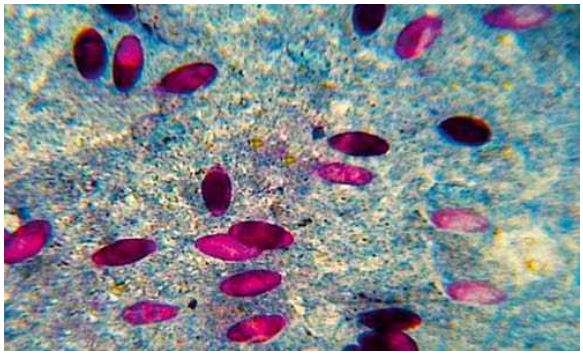
Incorrect Options:

Option B - Cyclospora: Cyclospora oocysts on modified acid-fast technique, stain red to reddish-orange, 8-12 μm in size, round, variably acid-fast, contains two sporocysts each bearing sporozoites.



Option C- Microsporum: Microsporum are associated with skin and hair dermatophytoses.

Option D - Cystoisospora: Cystoisospora, formerly known as Isospora, is another protozoan parasite that can cause diarrhea in immunocompromised individuals. Oocyst of isospora are also acid-fast, but the size of the oocyst is large, i.e., 23-36 μm in size, and is oval in shape.



Solution for Question 82:

Correct Option C - Paragonimus Westermani:

• Paragonimus westermani is a lung fluke that causes paragonimiasis, a parasitic infection characterized by chronic cough and rusty or bloody sputum containing eggs. The infection is commonly acquired by consuming raw or undercooked freshwater crustaceans, such as crabs and crayfish, which serve as second intermediate hosts for the parasite. The patient's history of travel to China and consumption of pickled crab aligns with the potential exposure to Paragonimus westermani.

• Humans are the definitive hosts, and snails act as the first intermediate hosts.

Incorrect Options:

Option A - Fasciola Buski: Fasciola buski is a trematode that causes a parasitic infection of the intestine characterized by diarrhea, abdominal pain, and malabsorption. It is acquired by consuming contaminated water plants, such as watercress, which are the second intermediate hosts. Humans and sheep are the definitive hosts, and snails act as the first intermediate hosts.

Option B - Fasciola Hepatica: Fasciola hepatica is another trematode that causes a parasitic infection of the liver characterized by right upper quadrant pain, jaundice, pain, and bile duct obstruction. It is also acquired by consuming contaminated water plants, such as watercress, which are the second intermediate hosts.

Option D - Schistosoma Japonicum: Schistosoma japonicum is a blood fluke that causes schistosomiasis characterized by intestinal mucosa hyperplasia, hepatosplenomegaly, and portal hypertension. It is acquired by penetration of the skin by cercariae released by snails, the intermediate host contaminating the freshwater sources.

Solution for Question 83:

Correct Option B - Japanese Encephalitis:

• Japanese encephalitis is a viral infection caused by the Japanese encephalitis virus (JEV), which is transmitted through a bird-arthropod-human transmission route. The primary transmission cycle involves mosquito Culex as the arthropod vector and ardeid birds as the reservoir hosts. Mosquitoes acquire the virus by feeding on infected birds and then transmit it to humans through their bites. Humans are accidental hosts, and pigs are amplifiers.

Incorrect Options:

Option A - Malaria: Malaria is transmitted through the bite of infected female Anopheles mosquitoes that acquire the causative organism plasmodium by feeding on infected individual.

Option C - Plague: Plague is primarily transmitted through the bites of infected fleas that acquire the causative organism Yersinia by feeding on wild rodents.

Option D - Paragonimiasis: Paragonimiasis is a parasitic infection caused by lung flukes of the genus Paragonimus. It is acquired by consuming raw or undercooked freshwater crustaceans, such as crabs and crayfish.

Solution for Question 84:

Correct Option B - Trichuris Trichiura:

- Trichuris trichiura also known as whipworm, is a parasitic roundworm that infects the large intestine of humans. The characteristic symptoms of a Trichuris trichiura infection include abdominal pain, distension, bloody and mucus-filled diarrhea (dysentery), tenesmus (the feeling of incomplete evacuation), and rectal prolapse (in severe cases).
- The presence of typical barrel-shaped eggs in the stool examination is a key diagnostic feature of Trichuris trichiura infection and can be identified microscopically in the feces of infected individuals. These eggs are bile stained, and mucous plugs at both ends that are non-bile stained.



Incorrect Options:

Option A - Giardia Lamblia: Giardia lamblia is a protozoan parasite that causes a diarrheal illness known as giardiasis. However, the characteristic symptom of giardiasis is watery diarrhea, not bloody and mucus-filled diarrhea.

Option C - Ascaris Lumbricoides: Ascaris lumbricoides is a large parasitic roundworm that primarily infects the small intestine and manifests as protein-energy malnutrition, vitamin A

deficiency, urticaria, angioneurotic edema and may cause mechanical obstruction.

Option D - Ancylostoma Duodenale: Ancylostoma duodenale is a hookworm that infects humans by penetrating the skin, usually through the feet. Hookworm infections can cause ground itch, creeping eruptions, respiratory symptoms like pneumonitis, and bronchitis, vomiting, and diarrhea. Eggs are oval or elliptical, colorless, not bile stained, and surrounded by a thin transparent hyaline shell membrane. When passed in feces, the egg contains a segmented ovum, usually with four or eight blastomeres.

Solution for Question 85:

Correct Option D - Ascaris Lumbricoides:

- The shown worms have characteristics of Ascaris lumbricoides.
- They are large cylindrical worms with tapering ends, the anterior end being more pointed than the posterior.
- They are pale pink or flesh-colored when freshly passed in stools but become white outside the body.
- The adult male worm is a little smaller than the female. It measures 15-30 cm in length and 2-4 mm in thickness. Its posterior end is curved ventrally to form a hook and carries two copulatory spicules.
- The female is larger than the male, measuring 20-40 cm in length and 3-6 mm in thickness. Its posterior extremity is straight and conical.

Incorrect Options:

Option A - Stongyloides Stercoralis: The female worm is thin, transparent, about 2.5 mm long and 0.05 m wide, and the male worm is shorter and broader than the female measuring 0.6- 1 mm in length and 40-50 mm in width.

Option B - Enterobius Vermicularis: Enterobius vermicularis is known as Pinworm. The adults are short, white, fusiform worms with pointed ends, looking like bits of white thread and the female worm is 8-13 mm long and 0.3-0.5 mm thick, its posterior third is drawn into a thin pointed pin-like tail.

Option C - Ancylostoma Duodenale: They are known as hookworms. The male worm is smaller than the female worm, 8-11 mm in length and 0.4 mm thick. The posterior end of the male is expanded into a copulatory bursa, which consists of three lobes. The female worm is larger,

10-13 mm long and 0.6 mm thick. Its hind end is conoid, with a subterminal anus situatedventrally.

Solution for Question 86:

Correct Option A - rk 39 antigen test:

- The given histology image of bone marrow picture histology reveals Leishman Donovan bodies, suggesting typical Leishmaniasis.
- Leishman Donovan bodies are macrophages-laden amastigotes, which are ovoid or rounded cells, about 2-4 μm in size.
- So, the most appropriate investigation, in this case, would be the rk 39 micro ELISA antigen test for leishmaniasis.
- The rk39 antigen is a qualitative immunochromatographic assay for the detection of antibodies.
- Antibodies against leishmania donovani can be detected in the serum of infected individuals using serological tests, such as the rk39 rapid diagnostic test (RDT) or WKK (Witebsky, Klingenstein, Kuhn) antigen.

Incorrect Options:

Option B - pLDH: Plasmodium lactate dehydrogenase (pLDH) is a diagnostic test for malaria caused by Plasmodium species.

Option C - Galactose lectin antigen: Galactose lectin antigen is a diagnostic test for Entamoeba histolytica.

Option D - HRP-2 antigen: HRP-2 antigen is a diagnostic criteria for malaria caused by Plasmodium falciparum.

Solution for Question 87:

Correct Option A - Fasciola Hepatica are seen in the Intrahepatic Biliary Radicles:

- Fasciola hepatica is a species of trematode commonly known as the liver fluke.
- It is primarily found in the liver and bile ducts of infected animals and humans.
- The adult flukes reside in the intrahepatic biliary radicles, which are the smallest branches of the bile ducts within the liver. They cause inflammation and damage to the liver tissue.

Incorrect Options:

Option B - Schistosoma are Monoecious: Schistosoma is dioecious, i.e., males and females are separate.

Option C - Fasciola Hepatica is not acquired through Contaminated Water: Fasciola hepatica is primarily acquired through ingestion of contaminated water or food that contains metacercariae, the infective stage of the fluke.

Option D - Triclabendazole is the Treatment of Choice for all Trematode Infections: While triclabendazole is an effective treatment for Fasciola hepatica infections, it is not the treatment of choice for all trematode infections. Other medications, such as praziquantel, are commonly used for the treatment of various trematode infections, including schistosomiasis.

Solution for Question 88:

Correct Option D - Toxoplasma gondii:

- Toxoplasma gondii has a complex life cycle involving definitive hosts (felids) and intermediate hosts, including humans.
- In the definitive host, such as domestic cats, the cycle begins when they ingest tissue cysts from intermediate hosts or oocysts shed in the feces of infected cats.
- Once inside the cat's small intestine, the parasite undergoes asexual replication, forming rapidly dividing tachyzoites.
- Sexual replication then occurs within the intestinal cells, and fertilization results in the production of zygotes, which develop into oocysts. These oocysts are shed in the cat's feces, contaminating the environment.
- Mode of Transmission Blood Transfusion, Mother to child: Tachyzoites Ingestion: Sporulated oocyst via ingestion of contaminated soil, food, or water. Bradyzoites occur via undercooked meat.

• Blood Transfusion, Mother to child: Tachyzoites

• Ingestion: Sporulated oocyst via ingestion of contaminated soil, food, or water. Bradyzoites occur via undercooked meat.

• Sporulated oocyst via ingestion of contaminated soil, food, or water.

• Bradyzoites occur via undercooked meat.

• Blood Transfusion, Mother to child: Tachyzoites

• Ingestion: Sporulated oocyst via ingestion of contaminated soil, food, or water. Bradyzoites occur via undercooked meat.

• Sporulated oocyst via ingestion of contaminated soil, food, or water.

• Bradyzoites occur via undercooked meat.

• Sporulated oocyst via ingestion of contaminated soil, food, or water.

• Bradyzoites occur via undercooked meat.

Incorrect Options:

Option A - Cystoisospora: It has only one host, that is, humans. Here, both the sexual and asexual cycle takes place inside the lumen of the human.

Option B - Cryptosporidium: It has only one host, which is humans. Here both the sexual and asexual cycle takes place inside the lumen of the human.

Option C - Plasmodium knowlesi: The asexual cycle takes place in humans. The sexual cycle takes place inside the mosquito.

Solution for Question 89:

Correct Option D - IV:

• In the given information where the formol-ether sedimentation technique is mentioned and level D is labeled at the base of the tube, it indicates that parasitic ova will be seen at this level when viewed under a microscope.

• Formol-ether sedimentation is a commonly used technique for concentrating parasites and ova from stool samples when the egg burden is low. In this technique, the stool sample is mixed with a formalin-ether solution. It takes advantage of the high specific gravity of protozoan cysts and helminth eggs compared to water. Formalin fixes the eggs, larvae, oocysts, and spores so that they are no longer infectious, as well as preserves their

morphology. Fecal debris is extracted into the ethyl acetate phase of the solution.

• The different levels represented by markings are: Level 1: represents the topmost layer of the sample and typically contains ether. Level II: contain some debris. Level III: contains formalin. Level IV: protozoan cyst and eggs.

• Level 1: represents the topmost layer of the sample and typically contains ether.

• Level II: contain some debris.

• Level III: contains formalin.

• Level IV: protozoan cyst and eggs.

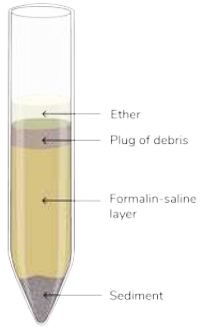
• Level 1: represents the topmost layer of the sample and typically contains ether.

• Level II: contain some debris.

• Level III: contains formalin.

• Level IV: protozoan cyst and eggs.

Formalin-ether sedimentation method



Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 90:

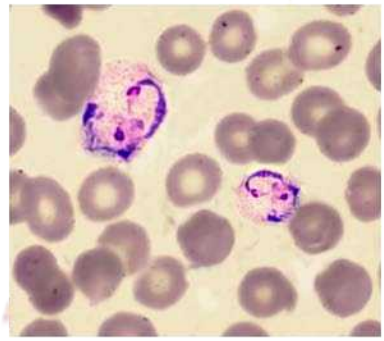
Correct Option B - Plasmodium Falciparum:

• The image showing multiple ring forms/headphone-shaped late trophozoites in erythrocytes, the most likely causative agent of malaria in this case is Plasmodium falciparum.

• Plasmodium falciparum is one of the species of Plasmodium that causes cerebral malaria in humans characterized by high fever, headache, anorexia, nausea, anemia, hypotension, tachycardia, hepatosplenomegaly, thrombocytopenia, confusion, increased muscle tone, seizures, paralysis, and coma.

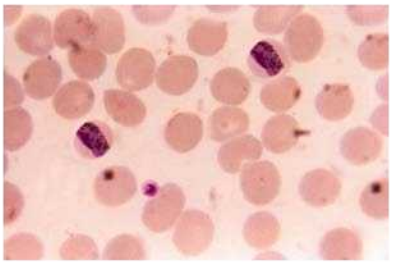
Incorrect Options:

Option A - Plasmodium Vivax: P. vivax infection usually shows single ring forms or single dots or amoeboid-shaped late trophozoite. This parasite is associated with the relapse and usually caused mild disease than P. falciparum.



Option C - Plasmodium Ovale: P. ovale infection shows similar characteristics to P. vivax, including ring forms. However, P. ovale is less common and has a more restricted geographic distribution, primarily in Africa.

Option D - Plasmodium Malariae: P. malariae infection typically exhibits band form late trophozoites and usually caused mild disease than P. falciparum.



Immunity

1. A 30-year-old woman comes to the clinic with a severe respiratory infection. The physician explains the body's immune response, detailing the roles of different immune systems in fighting infections. Which of the following statements about innate and acquired immunity, as explained by the physician, is false?

A. Innate immunity provides a rapid response to infection without prior exposure to the antigen.

B. Acquired immunity develops over time and requires prior exposure to the antigen.

C. Innate immunity has immunological memory and can recognize specific antigens.

D. Acquired immunity involves T cells and B cells, which are specific to particular pathogens.

---

2. A 45-year-old man presents to the emergency department with signs of infection. The physician discusses the components of his immune system that are involved in the initial defence against pathogens. Which of the following is NOT a component of innate immunity, as explained by the physician?

A. A) Alternative and mannose-binding pathway

B. B) Neutrophils and macrophages

C. C) Natural Killer (NK) cells

D. D) Classical complement pathway

---

3. A 60-year-old woman with a history of cancer presents to the clinic for follow-up. She has been receiving treatment that involves the use of antibodies to target her tumor cells. The physician explains how her immune system can enhance the destruction of these cells. Which of the following cells are not involved in the

mechanism described by the physician?

A. A) Eosinophils

B. B) Macrophages

C. C) NK cells

D. D) Basophils

---

4. A 35-year-old man presents to the clinic with symptoms of a severe bacterial infection. The physician discusses how the body's immune system recognizes and responds to pathogens. Which of the following statements about Pattern Recognition Receptors (PRRs) is TRUE?

A. A) PRRs are primarily found in the bloodstream and not on cell surfaces.

B. B) Toll-like receptors (TLRs) are specific for identifying viral organisms.

C. PRRs recognize microbial-associated molecular patterns (MAMPs) on pathogens.

D. RIG-I-like receptors (RLRs) are the main receptors for fungal recognition.

---

5. A 9-year-old girl presents with recurrent fevers, skin rashes, and neurological symptoms. The physician is considering various autoinflammatory disorders associated with dysregulated type I interferon signalling. Which of the following is NOT a type of interferonopathy?

A. CANDLE syndrome

B. SAVI

C. Aicardi-Goutières Syndrome

D. Familial Mediterranean Fever

---

6. A 7-year-old boy is brought to the clinic after experiencing frequent infections. The physician explains the roles of various lymphoid organs in his immune system. Which of the following statements about primary and secondary lymphoid organs is FALSE?

A. The thymus is responsible for the maturation of T cells.

B. The spleen filters blood-borne antigens and activates T and B cells.

C. Bone marrow is the primary site for B cell maturation.

D. Lymph nodes are primarily involved in hematopoiesis.

---

7. During T cell development in the thymus, immature T cells undergo a selection process. Which of the following statements correctly describes this selection process?

A. Positive selection allows survival of T cells that cannot recognize MHC molecules

B. Negative selection eliminates T cells that react with self-antigens

C. Both CD4 and CD8 markers are retained on mature T cells

D. Double-negative T cells directly transform into single-positive T cells

---

8. A research student is studying cytokine profiles of different T helper cell subsets. In the experiment, naive CD4+ T cells were activated and differentiated into different subsets. Which of the following correctly matches the T helper cell subset with its primary cytokine secretion pattern and major function? a. TH-1 1. IL-17, IL-22 x. Recruit neutrophils b. TH-2 2. IFN-γ, TNF-β y. Promote antibody production c. TH-17 3. IL-4, IL-5, IL-10 z. Enhance macrophage activation

a. TH-1	1. IL-17, IL-22	x. Recruit neutrophils
b. TH-2	2. IFN-γ, TNF-β	y. Promote antibody production
c. TH-17	3. IL-4, IL-5, IL-10	z. Enhance macrophage activation

A. a-1-x, b-3-y, c-2-z

B. a-1-z, b-2,x, c-3-y

C. a-2-x, b-3-y, c-1-z

D. a-2-z, b-3-y, c-1-x

---

9. Which of the following correctly describes the mechanism by which CD8+ T cells eliminate virus-infected target cells?

A. Secretion of antibodies that neutralize viral particles

B. Release of perforins and granzymes leading to target cell apoptosis

C. Production of IL-4 and IL-5 to enhance eosinophil recruitment

D. Activation of complement cascade to lyse infected cells

---

10. Which of the following represents the correct sequence of events in B cell-T cell interaction leading to antibody production?

A. B cells undergo class switching after CD40-CD40L interaction without requiring cytokine signals

B. T cells present antigens to B cells via MHC-II molecules

C. B cells process antigen, present via MHC-II to T cells, receive CD40L and cytokine signals

D. CD28-B7 interaction between B and T cells is sufficient for plasma cell differentiation

---

11. Which of the following cytokines and mechanisms are correctly associated with this isotype switch?

A. IFN-γ stimulation leading to direct IgM to IgE switch

B. IL-4 and CD40-CD40L interaction leading to IgE production

C. IL-2 stimulation alone causes isotype switching

D. TGF-β inducing direct IgE production without prior IgM synthesis

---

12. A laboratory analysis reveals an increase in B cells with the expression of CD19 and CD20. Which of the following best describes the role of these B cells?

A. They produce antibodies in response to T-cell activation.

B. They directly kill infected cells.

C. They are involved in cell-mediated immunity.

D. They secrete cytokines to modulate the immune response.

---

13. A patient's immune system encounters a new antigen, leading to the initial production of IgM antibodies by B cells. Over time, these antibodies become more effective at targeting the antigen. Which process in the germinal centres of lymph nodes primarily accounts for this improved antibody effectiveness?

A. Class switching

B. Affinity maturation

C. Clonal selection

D. Somatic recombination

---

14. A 5-year-old child presents with a sore throat, fever, and swollen lymph nodes. A throat swab confirms a bacterial infection. As part of the immune response, cells in the lymph nodes begin processing and presenting bacterial antigens to activate CD4+ T helper cells. Which of the following cells is primarily responsible for presenting antigens to CD4+ T helper cells?

A. Erythrocytes

B. Macrophages

C. Dendritic cells

D. Neutrophils

---

15. A 24-year-old male presents with muscle weakness, difficulty swallowing, blurred vision, and drooping eyelids. Blood tests reveal antibodies against acetylcholine receptors, suggesting myasthenia gravis. The physician explains that the patient's immune system may be reacting to "self" proteins. Which of the following statements best explains the immunological basis for this autoimmune reaction?

A. The antigen is presented by MHC Class I molecules to CD8+ T-cells, leading to cytotoxic T-cell-mediated destruction.

B. The antigen is presented by MHC Class II molecules to CD4+ T-cells, stimulating an immune response involving helper T-cells and autoantibody production.

C. MHC Class I molecules are found only on professional antigen-presenting cells, which present the antigen to helper T-cells.

D. MHC Class II molecules are present on all nucleated cells and help in presenting intracellular antigens.

---

16. Which of the following statements about Gamma Delta ($\gamma\delta$) T cells is not correct?

A. They are present in the mucus membranes of the GIT and genitourinary tract.

B. They represent 5-10% of all peripheral blood T cells.

C. They recognize antigens without the need for MHC presentation.

D. They process lipids and peptides only through MHC-dependent pathways.

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	4
Question 4	3
Question 5	4
Question 6	4
Question 7	2
Question 8	4
Question 9	2
Question 10	3
Question 11	2
Question 12	1
Question 13	2
Question 14	3
Question 15	2
Question 16	4

Solution for Question 1:

Innate immunity lacks immunological memory and does not recognize specific antigens.

Table rendering failed

Solution for Question 2:

Classical complement pathways are part of acquired immunity, while the alternative and mannose-binding pathways are part of innate immunity.

Components of Innate Immunity

- Anatomical and Physiological Barriers: Skin, mucosal surfaces, body temperature, pH, and secretions act as first-line defenses against microbes.
- Phagocytes: Neutrophils, macrophages, and monocytes engulf and destroy microbes in three steps—engulfment, lysosome fusion, and microbial killing. (Option B ruled out)
- Natural Killer (NK) Cells: These lymphocytes target and kill virus-infected and tumor cells. (Option C ruled out)
- Other Lymphocyte Classes: Includes $\gamma\delta$ T cells, NK-T cells, B-1 cells, and marginal-zone B cells, which bridge innate and adaptive immunity.
- Mast Cells: Found in mucosal linings; release histamine and cytokines upon activation to trigger inflammation.
- Dendritic Cells: Initiate inflammation and transport antigens to lymph nodes to activate T cells, linking innate and adaptive immunity.
- Complement Pathways: Alternative and mannose-binding pathways mediate microbial lysis, inflammation, and stimulate acquired immunity. (Option A ruled out)
- Inflammatory Response: Involves vasodilation, plasma leakage, phagocyte recruitment, and destruction of microbes at infection sites.
- Normal Resident Flora: Competes with pathogens for resources and produces antibacterial substances.
- Cytokines: Includes TNF-α, interleukins, interferons, and TGF-β, which mediate immune responses to infections.
- Acute Phase Reactant Proteins (APRs): Proteins like C-reactive protein, complement proteins, and fibrinogen increase during inflammation to support immune defense.

Solution for Question 3:

• The physician describes a process in which antibodies bind to tumor cells, facilitating their destruction by immune cells. This is called Antibody-dependent Cellular Cytotoxicity (ADCC).

• Neutrophils and macrophages can engage in this response, while eosinophils are particularly effective against larger targets like parasites.

• Additionally, Natural Killer (NK) cells are critical players, binding to the Fc region of antibodies attached to the target cells, leading to their destruction.

Antibody-dependent Cellular Cytotoxicity (ADCC)

• Neutrophils

• Macrophages

• Eosinophils

• NK cells

• Mechanism: IgG antibodies bind to target cells via the Fab region, while the Fc region binds to Fc receptors on effector cells (e.g., NK cells), leading to target cell destruction.

• Target: ADCC targets cells that are too large for phagocytosis, such as parasites, tumor cells, or transplanted tissue.

• Antibody Types: Primarily mediated by IgG antibodies; in some cases, such as eosinophil-mediated killing of parasites, IgE antibodies are involved.

Basophils are primarily involved in allergic responses and do not play a role in ADCC.

Solution for Question 4:

PRRs recognize MAMPs, which are crucial for the immune response.

Pattern Recognition Receptors (PRRs)

• Present on the surface of host cells (e.g., phagocytes). (Option A ruled out)

• Recognize MAMPs (microbial-associated molecular patterns).

• Generally consists of conserved regions encoded by germline genes.

• Five types based on protein domain homology: Toll-like receptors (TLRs) identify bacterial organisms. (Option B ruled out) Nucleotide oligomerization domain (NOD)-like receptors (NLRs) It is also associated with pyroptosis Retinoic acid-inducible gene-I (RIG-I) -like receptors (RLRs) identify viruses. (Option D ruled out) C-type lectin receptors (CLRs) identify

fungal organisms. Absent in melanoma-2 (AIM2)-like receptors (ALRs) recognize intracellularDNA.

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• Toll-like Receptors (TLRs) Classical examples of PRRs, named after the fruit fly (Drosophila). Main receptors for inducing innate immunity. Binding of TLRs to MAMPs generates signals that: Activate transcription factors. Stimulate expression of genes encoding cytokines and enzymes. Involved in antimicrobial activities of innate immune cells.

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Location of pattern recognition receptors:

- Toll-like and C-type lectin receptors: Present on the cell membrane and membranes of organelles like endosomes.
- RIG and NOD receptor: They are cytoplasmic Floating around
- They are cytoplasmic
- Floating around
- They are cytoplasmic
- Floating around



Solution for Question 5:

Familial Mediterranean Fever is an autoinflammatory disorder (AA type of amyloid deposition associated with IL-6) but not classified as interferonopathy.

Interferonopathies

• Interferonopathies are a group of rare, genetically defined autoinflammatory disorders characterized by dysregulation of the type I interferon (IFN) signalling pathway.

• These conditions often manifest with systemic inflammation and a distinct IFN signature detectable in peripheral blood. The most notable types include: CANDLE/PRAAS (Chronic Atypical Neutrophilic Dermatosis with Lipodystrophy and Elevated Temperature / Proteasome-Associated Autoinflammatory Syndrome) (Option A ruled out) SAVI (Stimulator of Interferon Genes-Associated Vasculopathy with Onset in Infancy) (Option B ruled out) AGS (Aicardi-Goutières Syndrome) (Option C ruled out)

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• SAVI (Stimulator of Interferon Genes-Associated Vasculopathy with Onset in Infancy) (Option B ruled out)

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• These disorders typically arise from mutations in specific genes that affect the immune response, leading to an overactive or improperly regulated interferon response.

• Etiopathogenesis The pathogenesis of interferonopathies involves genetic mutations that disrupt normal interferon signalling pathways. These mutations can lead to: Constitutive activation of the interferon pathway due to defective regulation. Altered responses to nucleic acid stimuli, which may involve intracellular receptors like cGAS and STING. Phenotypic heterogeneity, where different mutations can result in varying clinical presentations and disease severity.

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• Genetic Mutations PSMB8, PSMB9, and PSMA3 are associated with CANDLE syndrome. IFIH1 and DDX58 are linked to other interferonopathies. These mutations often lead to clinical features such as skin rashes, neurological symptoms, and systemic inflammation.

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• Clinical Features Dermatological manifestations: Rashes, lipodystrophy, chilblains, and livedo reticularis. Neurological involvement: Symptoms can range from developmental delays to severe neurological deficits. Systemic symptoms: Fever, myalgia, hepatomegaly, splenomegaly, and signs of chronic inflammation. The clinical presentation is often diverse and can overlap with other autoimmune or inflammatory diseases, complicating diagnosis.

• Dermatological manifestations: Rashes, lipodystrophy, chilblains, and livedo reticularis.

• Neurological involvement: Symptoms can range from developmental delays to severe neurological deficits.

• Systemic symptoms: Fever, myalgia, hepatomegaly, splenomegaly, and signs of chronic inflammation.

• The clinical presentation is often diverse and can overlap with other autoimmune or inflammatory diseases, complicating diagnosis.

• Diagnosis: Confirmed through clinical evaluation, genetic testing for specific mutations, inflammatory marker levels, and neuroimaging if neurological symptoms are present.

• Management: Focuses on symptom alleviation, complication prevention, and targeted therapies to inhibit the overactive interferon response.

• CANDLE/PRAAS (Chronic Atypical Neutrophilic Dermatosis with Lipodystrophy and Elevated Temperature / Proteasome-Associated Autoinflammatory Syndrome) (Option A ruled out)

• SAVI (Stimulator of Interferon Genes-Associated Vasculopathy with Onset in Infancy) (Option B ruled out)

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• The clinical presentation is often diverse and can overlap with other autoimmune or inflammatory diseases, complicating diagnosis.

Solution for Question 6:

Lymph nodes are not involved in hematopoiesis; they filter antigens and activate lymphocytes.

Table rendering failed

Solution for Question 7:

During T cell development in the thymus:

• Initially, double-negative T cells (CD4- CD8-) express CD3 molecule

• These cells then express both CD4 and CD8 to become double-positive T cells

• During positive selection, only 5% of cells whose αβ receptors recognize MHC molecules survive

• During negative selection, self-reacting T cells (2-5%) are eliminated by apoptosis

• Finally, surviving cells become single positive (either CD4+ or CD8+) mature T cells

• Defects in the AIRE gene impair the negative selection of T cells, leading to Autoimmune Polyendocrinopathy due to the survival of self-reactive T cells.

T-Lymphocytes	

T Cell Development Stages	1. Double Negative Stage (CD4- CD8-) • Express CD3 molecule • 95% express TCR αβ • 5% express TCR γδ 2. Double Positive Stage (CD4+ CD8+) • Undergo positive selection (5% survive) • Undergo negative selection (2-5% eliminated) 3. Single Positive Stage • Either CD4+ or CD8+ T cells • Released into circulation
Major T Cell Types	1. Helper T Cells (CD4+) • TH1: Secrete IL-2, IFN-γ, TNF-β • TH2: Secrete IL-4, IL-5, IL-6, IL-10, IL-13 • TH17: Produces IL-17, IL-22 2. Cytotoxic T Cells (CD8+) • Kill virus-infected cells and tumor cells 3. Regulatory T Cells • Express CD4, CD25, Foxp3 - Causes IPEX syndrome(Immune dysregulation , Polyendocrinopathy , Enteropathy, X linked disorder. • Suppress immune response
Key Surface Markers	• CD3: Universal T cell marker • TCR: Antigen recognition • CD4: Helper T cells, MHC-II restricted • CD8: Cytotoxic T cells, MHC-I restricted • Memory markers: CD45RO (memory), CD45RA (naive)
Functions	• Helper T cells: Coordinate immune response • Cytotoxic T cells: Direct cell killing • Regulatory T cells: Maintain self-tolerance • Memory T cells: Long-term immunity

Option A: Positive selection retains T cells that can recognize MHC molecules

Option C: Mature T cells express either CD4 or CD8, not both

Option D: Double-negative cells first become double-positive before becoming single-positive

Solution for Question 8:

Different T helper cell subsets have distinct cytokine profiles and functions:

T Helper Subset	Key Cytokines	Major Functions	Role in Immunity

TH1 Cells	• IL-2 • IFN-γ • TNF-β	• Activate macrophages • Enhance cell-mediated immunity • Stimulate cytotoxic T cells	• Protection against intracellular pathogens • Mediate delayed-type hypersensitivity
TH2 Cells	• IL-4 • IL-5 • IL-6 • IL-10 • IL-13	• Promote B cell proliferation • Enhance antibody production • Stimulate IgE class switching • Promote eosinophil growth	• Protection against extracellular pathogens • Mediate allergic responses • Involved in helminth immunity
TH17 Cells	• IL-17 • IL-22	• Recruit neutrophils • Enhance inflammation • Promote tissue inflammation	• Protection at mucosal surfaces • Involved in autoimmune diseases • Contribute to inflammatory conditions

Solution for Question 9:

CD8+ T cells (cytotoxic T cells) eliminate virus-infected cells through direct cell-mediated cytotoxicity by release of perforins and granzymes.

CD8+ T cells (cytotoxic T cells)	
CD8+ T-Cell Activation	1. Antigen-specific signal: • TCR binds to peptide-MHC I complex • CD8 interacts with α3 domain of MHC-I 2. Costimulatory signal: • CD28 binds to B7 on target cells 3. Cytokine signal: • IL-2 (from TH1 cells)
Mechanism of Action	1. Recognition Phase: • Binds to virus-infected or tumor cells • Recognizes antigenic peptides with MHC-I 2. Killing Phase: • Releases perforins (form pores) • Releases granzymes (enter through pores) • Induces apoptosis through caspase pathway 3. Resolution Phase: • Detachment from target cell • Can kill multiple target cells sequentially

Types	1. Naive CD8+ T cells: • Circulating, unexposed to antigen 2. Effector CD8+ T cells: • Activated, cytotoxic function 3. Memory CD8+ T cells: • Long-lived • Rapid response to re-exposure
Primary Functions	• Elimination of virus-infected cells • Destruction of tumor cells • Defense against intracellular pathogens • Generation of immunological memory

Option A: Antibody production is a function of B cells, not CD8+ T cells

Option C: IL-4 and IL-5 production is associated with TH2 cells

Option D: Complement activation is part of humoral immunity, not CD8+ T cell function

Solution for Question 10:

B cell-T cell interaction requires multiple signals for effective antibody production:

Stage	Process	Key Components
Initial B Cell Activation	1. Antigen Recognition: • B cell captures antigen via surface IgM/IgD • Antigen internalization 2. Antigen Processing: • Processed into peptides • Loaded onto MHC-II molecules	• B cell receptor (BCR) (IgM and IgD) • Surface immunoglobulins • MHC-II molecules
B Cell-T Cell Interaction	1. Signal Generation: • Signal 1: Antigen presentation to TH cells • Signal 2: CD40-CD40L interaction • Signal 3: Cytokine stimulation 2. T Cell Help: • TH1: IFN-γ (IgG production) • TH2: IL-4 (IgE production)	• TCR-MHC-II complex • CD40-CD40L • Cytokine receptors • Co-stimulatory molecules

B Cell Response	1. Germinal Center Formation: • B cell proliferation • Somatic hypermutation • Class switching 2. Differentiation: • Into plasma cells • Into memory B cells	• Antibody production • Affinity maturation • Immunological memory
Outcomes	1. Immediate Response: • Antibody secretion • Different Ig classes based on cytokines 2. Long-term Response: • Memory formation • Enhanced secondary response	• IgG, IgA, IgE production • High-affinity antibodies • Long-lived plasma cells • Memory B cells



Solution for Question 11:

IL-4, produced by TH2 cells, specifically directs class switching to IgE. This switch requires both IL-4 and CD40-CD40L interaction between T and B cells. This dual signal requirement ensures proper regulation of IgE production. Immunoglobulin synthesis and isotype switching is a multi-step process:

Immunoglobulin Synthesis and Class Switching

Process	Key Features	Requirements
Initial Antibody Synthesis	• First antibody produced is IgM • Later co-expression of IgD • Occurs in bone marrow • No T cell help needed	• Gene rearrangement • VDJ recombination • Expression of RAG genes

Class Switch Requirements	• Occurs in germinal centers • Requires T-cell help • Involves DNA recombination • Irreversible process	• CD40-CD40L interaction • Specific cytokines • AID enzyme activity • Switch recombination
Cytokine Control of Class Switching	• IFN-γ → IgG2a, IgG3 • IL-4 → IgE, IgG4 • TGF-β + IL-5 → IgA • IL-2,4,5 → IgM • IL-4,5,6 + IFN-γ → IgG	• TH1 cells (IFN-γ) • TH2 cells (IL-4) • Specific transcription factors • CD40-CD40L required for all switches
Outcomes	• Changed effector functions • Same antigen specificity • Different tissue distribution • Different half-lives	• New constant region • Same variable region • Different Fc receptors • Different complement activation

Option A: IFN-γ promotes switching to IgG2a and IgG3, not IgE

Option C: IL-2 alone cannot cause isotype switching; CD40-CD40L interaction is also req No uired

Option D: TGF-β promotes IgA production, not IgE; all B cells first produce IgM

Solution for Question 12:

• B cells differentiate into plasma cells upon activation, producing specific antibodies. B cells are a type of lymphocyte involved in the humoral immune response.

• They originate in the bone marrow and are characterized by surface markers such as CD19 and CD20.

• Upon encountering an antigen, B cells can undergo clonal expansion and differentiation into plasma cells, which produce antibodies. They can also present antigens to helper T cells (CD4+) via MHC class II molecules, enhancing the immune response.

Formation of B Cells:

• B cells originate from hematopoietic stem cells in the bone marrow.

• Differentiation Process: Hematopoietic stem cells differentiate into lymphoid progenitor cells. These progenitors develop into B cell precursors, initiating B cell lineage.

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• Stages in Bone Marrow: Pro-B Cell Stage: Initial recombination of the heavy chain gene. Pre-B Cell Stage: Expression of μ heavy chain, starting light chain recombination. Immature B Cell Stage: Successful expression of IgM on the cell surface; undergoes selection for self-tolerance.

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• Mature B Cell Development: Immature B cells that pass selection express both IgM and IgD, becoming mature naive B cells.

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• Peripheral Circulation: Mature B cells leave the bone marrow and circulate in peripheral blood, ready for antigen encounter.

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Features of B Cells:

• Antigen-Specific Receptors: B cells express B cell receptors (BCRs) on their surface, primarily IgM and IgD, which bind specific antigens.

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• Differentiation into Effector Cells: Plasma Cells: Specialized B cells that secrete antibodies targeting the specific antigen. Memory B Cells: Long-lived cells that quickly respond to future exposures to the same antigen.

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B Cell Markers and Their Functions:

CD19	Present throughout B cell development and maturity; used as a universal B cell marker.
CD20	Expressed from the late pro-B cell stage onward; target for certain therapeutic antibodies.
CD21	Complement receptor involved in B cell activation by enhancing BCR signaling.
CD40	It is critical for interactions with T cells, enables antibody class switching for adaptive immune responses, and is also responsible for Anergy and self-tolerance.
Immunoglobulin Markers: IgM and IgD	Primary B cell receptors on mature, naive B cells.

They directly kill infected cells (Option B): This function is primarily performed by cytotoxic T cells (CD8+ T cells).

They are involved in cell-mediated immunity (Option C): B cells are primarily responsible for humoral immunity, while T cells mediate cellular immunity.

They secrete cytokines to modulate the immune response (Option D): Although B cells do release cytokines, this is a secondary function. Cytokine production is more characteristic of T helper cells and certain innate immune cells, which coordinate and modulate the immune response extensively.

Solution for Question 13:

Affinity maturation is a process where B cells undergo somatic hypermutation in germinal centers, leading to the production of antibodies with higher affinity for the antigen.

This occurs after the initial IgM response, resulting in the production of IgG (or other classes) with enhanced specificity and effectiveness.

Antibody formation involves B cell activation upon encountering an antigen.

- The initial response produces IgM, which can then undergo class switching to produce other antibody types like IgG, IgA, and IgE.
- This process is crucial for mounting an effective immune response against pathogens, with IgG being the most abundant antibody in serum and providing long-term immunity.

The formation of antibodies, or immunoglobulins involves multiple cellular components and steps:

- The process begins when B cells encounter a specific antigen. This antigen must be recognized as foreign to activate the adaptive immune response.
- B cell receptors (BCRs) on the surface of B cells bind to the antigen. This recognition is highly specific and depends on the unique variable regions of the immunoglobulin on BCRs.
- Upon binding with the antigen, B cells become activated, which requires help from T helper (Th) cells.
- T cells are required to release cytokines and co-stimulatory signals, which aid in the full activation of B cells, especially in the case of protein antigens.
- Activated B cells undergo clonal expansion, where they proliferate to create identical cells that can produce the same specific antibody.
- This clonal expansion amplifies the immune response, ensuring that a sufficient number of cells are available to target the pathogen.
- Following expansion, B cells differentiate into two main cell types: Plasma Cells: These are the effector cells that actively secrete antibodies. Plasma cells are responsible for producing large quantities of antibodies, which are then released into the bloodstream. Memory B Cells:

These cells are long-lived and retain the memory of the specific antigen. Upon subsequentexposure to the same antigen, memory B cells can quickly initiate a secondary immuneresponse, leading to a faster and more robust production of antibodies.

• Plasma Cells: These are the effector cells that actively secrete antibodies. Plasma cells are responsible for producing large quantities of antibodies, which are then released into the bloodstream.

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• Memory B Cells: These cells are long-lived and retain the memory of the specific antigen. Upon subsequent exposure to the same antigen, memory B cells can quickly initiate a secondary immune response, leading to a faster and more robust production of antibodies.

• Initially, B cells produce IgM antibodies. However, with the help of Th cells, B cells can undergo a process called "class switching" to produce other types of antibodies like IgG, IgA, or IgE.

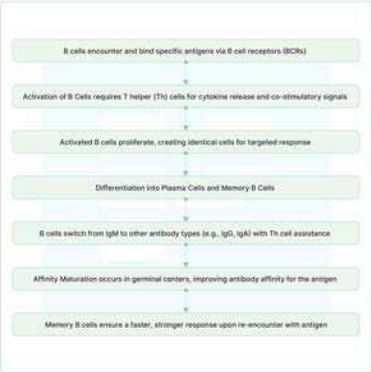
• Class switching allows the immune system to adapt the type of antibody produced to the specific requirements of the immune response, depending on the nature of the pathogen and the location of the infection.

• Affinity maturation occurs in the germinal centers of lymphoid tissues and involves the somatic hypermutation of antibody genes.

• This process leads to the production of antibodies with increased affinity for the antigen, allowing for a more effective immune response.

• Memory B cells generated during the primary response ensure long-term immunity. If the same antigen is encountered again, these cells quickly respond by differentiating into plasma cells and producing antibodies.

• This secondary immune response is usually faster and more intense due to the Spresence of these memory cells.



Class switching (Option A) and Somatic recombination (Option D) are important processes in the immune response but don't directly improve antibody affinity over time.

Clonal selection (Option C) ensures that the immune system selects the right B cell, but affinity maturation is the process that improves antibody affinity during the immune response.

Solution for Question 14:

- Antigen-presenting cells (APCs) are important for initiating T-cell responses.
- Dendritic cells capture and process antigens and present them on MHC class II molecules to CD4+ T helper cells. This interaction is essential for T cell activation and subsequent adaptive immune responses.
- Other APCs include macrophages and B cells, but dendritic cells are the most potent in activating naïve T cells.

Features of Antigen-Presenting Cells:

- APCs are specialized immune cells that capture, process, and present antigens on their surface to T cells, facilitating the initiation of an immune response.
- Types of APCs: The primary types include dendritic cells (most potent), macrophages, and B cells, each with unique roles in antigen processing.
- Location: APCs are located in peripheral tissues, lymph nodes, and the spleen, where they encounter pathogens.

Mechanism of Antigen Processing and Presentation:

• APCs ingest pathogens or pathogen-derived proteins via phagocytosis, endocytosis, or pinocytosis.

• Dendritic cells are particularly efficient in capturing antigens in peripheral tissues and migrating to lymph nodes.

• The internalized antigen undergoes enzymatic processing within the APC, where it is fragmented into peptide segments.

• These peptide fragments are then loaded onto major histocompatibility complex (MHC) molecules within the APC.

• The processed antigen-MHC complex is transported to the cell surface.

• MHC Class I molecules present antigens to CD8+ T cells (cytotoxic T cells), while MHC Class II molecules present to CD4+ T cells (helper T cells).

Functions:

• Activation of T Cells: By presenting antigens, APCs activate T cells, a crucial step in adaptive immunity. CD4+ T cells assist other immune cells, while CD8+ T cells directly destroy infected cells.

• Immune Memory Formation: APCs are integral in the formation of immunological memory, allowing for a faster response upon re-exposure to the same pathogen.

• Cytokine Production: APCs release cytokines that further modulate immune responses, enhancing the activation and proliferation of T cells.

APC Types and Their Unique Roles:

Professional APC's:

• Dendritic Cells: Most potent APCs. They are essential for initiating T-cell responses and are found in nearly all tissues.

• Macrophages: These cells primarily act as phagocytes, engulfing pathogens and presenting antigens, especially in inflammatory sites.

• B Cells: Unique among APCs as they can bind specific antigens through their surface immunoglobulin receptors and are more specialized in interacting with helper T cells.

Non-Professional APC's:

• Thymic epithelial cells

• Endothelial cells

• Fibroblasts

Erythrocytes (Option A) lack a nucleus and MHC molecules, making them incapable of antigen presentation.

Macrophages (Option B) can present antigens via MHC class II but are less efficient than dendritic cells at activating naive CD4+ T cells.

Neutrophils (Option D) are involved in innate immunity and lack MHC class II molecules, so they cannot present antigens to T cells.

Solution for Question 15:

• In autoimmune diseases like myasthenia gravis, an immune response is mistakenly directed against the body's own tissues.

• This reaction often involves MHC Class II molecules on antigen-presenting cells (APCs) that present "self" antigens (such as acetylcholine receptors in this case) to CD4+ T-cells.

• The activation of CD4+ T-helper cells can then stimulate B-cells to produce autoantibodies against these "self" proteins, leading to the symptoms observed.

Major Histocompatibility Complex (MHC) and Human Leukocyte Antigen (HLA)

• MHC Class I: Found on the surface of almost all nucleated cells, MHC I molecules present endogenous (intracellular) antigens, such as viral or tumor proteins, to CD8+ T-cells (cytotoxic T-cells). The three primary HLA genes for MHC I are HLA-A, HLA-B, and HLA-C.

• MHC Class II: Primarily expressed on antigen-presenting cells (APCs) like macrophages, dendritic cells, and B-cells, MHC II molecules present exogenous (extracellular) antigens to CD4+ T-cells (helper T-cells). HLA genes for MHC II include HLA-DP, HLA-DQ, and HLA-DR.

Feature	MHC Class I	MHC Class II
Expression	Found on almost all nucleated cells	Found mainly on antigen-presenting cells (APCs) such as macrophages, dendritic cells, and B-cells
Antigen Source	Presents intracellular antigens (e.g., viral or tumor proteins)	Presents extracellular antigens (e.g., bacterial peptides)
Interacts With	CD8+ T-cells (cytotoxic T-cells)	CD4+ T-cells (helper T-cells)
Main HLA Genes	HLA-A, HLA-B, HLA-C	HLA-DP, HLA-DQ, HLA-DR

Structure	Consists of one α (heavy) chain and β2-microglobulin; peptide-binding groove formed by α1 and α2	Consists of α and β chains; peptide-binding groove formed by α1 and β1 domains
Function	Identifies and helps eliminate infected or malignant cells by presenting endogenous antigens to CD8+ T-cells	Activates immune response by presenting exogenous antigens to CD4+ T-cells
Peptide Length	Presents shorter peptides (8-10 amino acids)	Presents longer peptides (10-30 amino acids)

The antigen is presented by MHC Class I molecules to CD8+ T-cells, leading to cytotoxic T-cell-mediated destruction (Option A): MHC Class I molecules present intracellular antigens to CD8+ T-cells, not CD4+ T-cells. MHC I is not primarily involved in antibody-mediated autoimmune diseases like myasthenia gravis, which involve MHC Class II.

MHC Class I molecules are found only on professional antigen-presenting cells, which present the antigen to helper T-cells (Option C): MHC Class I molecules are found on all nucleated cells, not just APCs. Additionally, MHC I interacts with CD8+ T-cells rather than helper T-cells (CD4+). The statement here mischaracterizes both the distribution and the function of MHC Class I molecules.

MHC Class II molecules are present on all nucleated cells and help in presenting intracellular antigens (Option D): MHC Class II is restricted to antigen-presenting cells, not all nucleated cells, and presents extracellular antigens, not intracellular ones. In autoimmune reactions, extracellular antigens are generally presented by MHC II to CD4+ T-cells, which is key to the autoantibody production pathway in myasthenia gravis.

Solution for Question 16:

Gamma delta T cells:

- γδ T cells represent a small subset of T cells (5-10 %) that possess a distinct TCR composed of γ and δ chains; instead of α/β chains.
- They lack both CD4 and CD8 molecules.
- They differ from the conventional α/β T cells by the fact that they do not require antigen processing and MHC presentation of peptides.

• Instead, they recognize non-peptide antigens like lipids, phosphoantigens, and stress-induced molecules directly through their γδ T-cell receptors.

• γδ T cells are abundant in mucosal tissues, including the gastrointestinal (GIT) and genitourinary tracts, where they play an important role in tissue surveillance, immune defense, and maintaining homeostasis.

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Hypersensitivity reactions

1. A 30-year-old woman presents with a rash and difficulty breathing after being administered penicillin. Which of the following types of hypersensitivity reactions is most likely responsible for her symptoms according to the Gell and Coombs classification?

A. Type I

B. Type II

C. Type III

D. Type IV

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2. A 30-year-old healthcare worker develops a red, itchy rash on her hands two days after using a new brand of latex gloves. She reports no immediate reaction at the time of exposure, but 48 hours later, she noticed redness, swelling, and itching at the site of contact. She has no known history of allergies or similar reactions in the past. Which type of hypersensitivity reaction is most likely responsible for her symptoms?

A. Type I hypersensitivity

B. Type II hypersensitivity

C. Type III hypersensitivity

D. Type IV hypersensitivity

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3. A 28-year-old woman with a history of asthma experiences severe wheezing, shortness of breath, and facial swelling minutes after accidentally ingesting peanuts. She is treated promptly in the emergency department with epinephrine, antihistamines, and corticosteroids, and her symptoms improve. However, 8 hours

later, she began to develop increased chest tightness and wheezing, requiring additional medical intervention. Which of the following best explains her delayed respiratory symptoms?

A. Incomplete clearance of histamine from the bloodstream

B. Continuous allergen exposure leading to sustained mast cell degranulation

C. Recruitment of eosinophils and other inflammatory cells, causing a delayed inflammatory response

D. Development of IgG antibodies against the allergen, causing a delayed hypersensitivity reaction

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4. A 28-year-old woman develops sudden hives, facial swelling, and shortness of breath after taking an antibiotic. She is rushed to the emergency room, where she receives epinephrine and improves quickly. Which of the following hypersensitivity reactions best explains her symptoms?

A. Type I Hypersensitivity (Early)

B. Type I Hypersensitivity (Delayed)

C. Type III Hypersensitivity

D. Type IV Hypersensitivity

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5. A 5-year-old boy experiences facial swelling, hives, and difficulty breathing after accidentally eating a peanut cookie. He is treated with epinephrine and antihistamines and stabilizes, but later develops wheezing and difficulty breathing. Which pair of Type I hypersensitivity reactions best explains his experience?

A. Anaphylaxis followed by allergic asthma

B. Urticaria followed by allergic rhinitis

C. Atopic dermatitis followed by urticaria

D. Allergic rhinitis followed by food allergy

6. A 45-year-old patient receives an iodine-based contrast agent for a CT scan. Within minutes, she develops hives, facial swelling, and difficulty breathing. She reports no prior allergies to iodine or contrast agents. Which of the following best describes her reaction and its mechanism?

A. Anaphylaxis, caused by IgE-mediated mast cell activation

B. Anaphylaxis, caused by T-cell-mediated hypersensitivity

C. Anaphylactoid reaction, caused by direct mast cell activation without IgE

D. Anaphylactoid reaction, requiring prior exposure for mast cell sensitization

7. A 30-year-old man comes to the ER with coughing up blood and difficulty breathing. He's scared and tells you he's never had symptoms like this before. Lab results reveal he has antibodies against the basement membrane in his lungs and kidneys. His immune system is attacking these tissues, and cell destruction is being triggered through a specific mechanism. Which of the following best explains what's happening inside his body?

A. Antigen-specific IgE binding to mast cells

B. Antibody binding to cell surface antigens with complement activation

C. Immune complexes depositing in blood vessels

D. Direct T-cell attack on self-antigens

8. A 45-year-old woman presents with fatigue, jaundice, and shortness of breath. Laboratory tests reveal a hemoglobin level of 7.2 g/dL, elevated bilirubin, and a positive direct Coombs test. Peripheral blood smear shows spherocytes. Which of the following is the most likely diagnosis and its corresponding type of hypersensitivity?

A. Autoimmune Hemolytic Anemia (AIHA); Type 1

B. Rheumatoid Arthritis; Type 1

C. Autoimmune Hemolytic Anemia (AIHA); Type 2

D. Myasthenia Gravis; Type 4

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9. A 30-year-old woman presents with joint pain, fever, and a malar rash. Laboratory tests show proteinuria, low complement levels, and the presence of anti-nuclear antibodies (ANA) and anti-dsDNA antibodies. Which of the following mechanisms best explains the pathology of her disease?

A. Direct binding of IgE to mast cells, leading to histamine release

B. Antibody-mediated cytotoxicity targeting red blood cell antigens

C. Deposition of immune complexes in tissues

D. T-cell-mediated cytotoxicity targeting self-antigens on tissue cells

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10. A 25-year-old male presents with a recent history of sore throat, followed by dark-colored urine, facial swelling, and high blood pressure. Laboratory findings show proteinuria, hematuria, and elevated anti-streptolysin O (ASO) titers. Which of the following best describes the patient's condition?

A. Goodpasture syndrome

B. Post-streptococcal glomerulonephritis

C. Myasthenia gravis

D. Contact dermatitis

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11. A 35-year-old man presents with a red, itchy rash on his hands that developed two days after using a new pair of latex gloves. Physical examination reveals

erythematous, scaly patches on both hands. Which of the following best describes the mechanism of this patient's condition, a Type IV hypersensitivity reaction?

A. Activation of IgE antibodies bound to mast cells

B. Formation of antigen-antibody complexes

C. Direct binding of antibodies to cell surface antigens

D. Activation of sensitized T cells

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12. A 40-year-old male presents with a rash on his arms and neck that developed several days after he worked in his garden, where he handled several plants. He mentions he has never had a rash before but recently experienced severe itching and redness in the affected areas. A patch test reveals an allergic reaction to a specific plant resin. What type of hypersensitivity does it represent?

A. Type II hypersensitivity

B. Type IV hypersensitivity

C. Type III hypersensitivity

D. Type I hypersensitivity

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	3
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	3
Question 9	3
Question 10	2
Question 11	4
Question 12	2

Solution for Question 1:

Correct Answer: A) Type I

Explanation: The Gell and Coombs classification categorizes hypersensitivity reactions into four types:

Type	Description	Examples
Type I (Immediate) (Option A)	IgE-mediated reactions, leading to mast cell degranulation and histamine release.	Examples include allergic rhinitis and anaphylaxis.

Type II (Cytotoxic) (Option B)	Mediated by IgG or IgM antibodies directed against antigens on cell surfaces or matrix, causing cell lysis.	Conditions include autoimmune hemolytic anemia and rheumatic fever.
Type III (Immune Complex) (Option C)	Caused by the formation of antigen-antibody complexes that deposit in tissues, leading to inflammation.	Examples include systemic lupus erythematosus and post-streptococcal glomerulonephritis.
Type IV (Delayed) (Option D)	Mediated by T cells, occurring hours to days after exposure to an antigen.	Seen in contact dermatitis and tuberculin reactions.

Solution for Question 2:

Correct Answer: D) Type IV

Explanation:

Duration of the types of hypersensitivity reactions:

Type I (Immediate): Occurs within 2–30 minutes due to IgE-mediated release of histamine.

Type II (Cytotoxic): Can occur within 5–8 hours; mediated by antibody binding to cell antigens, but does not fit the delayed category.

Type III (Immune Complex): Symptoms can develop 2–8 hours later but are generally not as delayed as Type IV.

Type IV (Delayed): Characterized by delayed hypersensitivity, mediated by T lymphocytes, with symptoms appearing 24 to 72 hours after antigen exposure, as seen in tuberculosis skin tests.

Solution for Question 3:

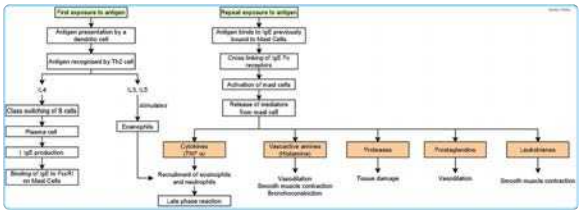
Correct Answer: C) Recruitment of eosinophils and other inflammatory cells, causing a delayed inflammatory response

Explanation: In the late phase of a Type I hypersensitivity reaction, inflammatory cells like eosinophils are recruited to the site, releasing additional mediators that lead to a delayed inflammatory response, which explains the patient's recurrent symptoms hours after the initial reaction.

Phases of type I hypersensitivity reaction:

1. Early Phase:

- Occurs within 2 hours.
- Mediators: Performed: Histamine, Tryptase Newly formed: Prostaglandins, Leukotrienes, Platelet Activating Factor (PAF).
- Performed: Histamine, Tryptase
- Newly formed: Prostaglandins, Leukotrienes, Platelet Activating Factor (PAF).
- Effects: Histamine causes vasodilation (blood vessels) and bronchoconstriction (lungs). Other mediators lead to bronchoconstriction.
- Histamine causes vasodilation (blood vessels) and bronchoconstriction (lungs).
- Other mediators lead to bronchoconstriction.
- Performed: Histamine, Tryptase
- Newly formed: Prostaglandins, Leukotrienes, Platelet Activating Factor (PAF).
- Histamine causes vasodilation (blood vessels) and bronchoconstriction (lungs).
- Other mediators lead to bronchoconstriction.



2. Late Phase (Delayed Reaction)

- Onset: Begins 4–12 hours after the initial reaction, peaks at 24–48 hours, and can last for several days.
- Mechanism: Cytokine and Chemokine Release: Mast cells and basophils release cytokines (e.g., IL-4, IL-5, IL-13) and chemokines that attract and activate other inflammatory cells, particularly eosinophils, neutrophils, and T-helper cells (Th2). Infiltration of Inflammatory Cells: These cells migrate to the site of allergen exposure and release additional mediators like leukotrienes, prostaglandins, and proteases. Sustained Inflammation: The continued release of these mediators maintains the inflammatory response, causing tissue damage and prolonged symptoms.

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• Cytokine and Chemokine Release: Mast cells and basophils release cytokines (e.g., IL-4, IL-5, IL-13) and chemokines that attract and activate other inflammatory cells, particularly eosinophils, neutrophils, and T-helper cells (Th2).

• Infiltration of Inflammatory Cells: These cells migrate to the site of allergen exposure and release additional mediators like leukotrienes, prostaglandins, and proteases.

• Sustained Inflammation: The continued release of these mediators maintains the inflammatory response, causing tissue damage and prolonged symptoms.

Incomplete clearance of histamine from the bloodstream (Option A), is incorrect because histamine, released in the early phase, is typically cleared from the system within minutes to hours, and delayed symptoms are not due to prolonged histamine activity alone.

Continuous allergen exposure leading to sustained mast cell degranulation (Option B), is incorrect, as the patient's symptoms are not due to continuous allergen exposure but rather to the late-phase inflammatory response triggered by cytokines and chemokines.

Developing IgG antibodies against the allergen, causing a delayed hypersensitivity reaction (Option D), is incorrect because IgG antibodies are not primarily involved in Type I hypersensitivity. The patient's reaction is mediated by IgE and mast cells in the early phase, with eosinophil involvement in the late phase.

Solution for Question 4:

Correct Answer: A) Type I Hypersensitivity (Early)

Explanation:

• Type I Hypersensitivity (Early), is an immediate reaction triggered by IgE-mediated mast cell degranulation, causing symptoms like hives, swelling, and respiratory distress within minutes of allergen exposure.

• Other options describe features inconsistent with her presentation and typical of delayed hypersensitivity types (I (Delayed), III, and IV), which have different mechanisms and symptom timelines.

Type I hypersensitivity reaction phases and their features:

Phase	Timing	Mechanism	Clinical Features
Early Phase	Seconds to minutes	Mast cell/basophil degranulation: IgE cross-linking leads to the release of histamine and other mediators.	Skin: Urticaria (hives), itching (pruritus), erythema (redness), localized angioedema (swelling around lips, eyes) Respiratory: Bronchoconstriction leading to wheezing, shortness of breath, chest tightness, sneezing, nasal congestion (allergic rhinitis) Cardiovascular: Vasodilation and hypotension (drop in blood pressure), dizziness, syncope (fainting) Gastrointestinal: Nausea, vomiting, abdominal pain, diarrhea due to smooth muscle contraction Systemic Reaction: Anaphylaxis in severe cases, with generalized urticaria, angioedema, severe hypotension, and respiratory distress (life-threatening)

Late Phase	4–12 hours, peaks at 24–48 hours, lasting up to days	Inflammatory cell recruitment: Cytokines/chemokines attract eosinophils, neutrophils, and macrophages, causing sustained inflammation.	Skin: Persistent redness, swelling, intense itching, and development of an eczema-like rash (seen in atopic dermatitis) Respiratory: Recurrent bronchoconstriction, prolonged wheezing, coughing, chest tightness, mucus production, and airway hyperresponsiveness (chronic asthma exacerbation) Eyes: Persistent redness and irritation in conjunctiva (allergic conjunctivitis) Chronic Changes: In prolonged or recurrent reactions, there may be tissue remodeling, fibrosis, and thickening of affected areas (e.g., airways in asthma, skin in chronic dermatitis)

Types of hypersensitivity reactions and their features:

Type	Mechanism	Clinical Features	Onset
Type I Early	IgE-mediated mast cell degranulation	Hives, swelling, respiratory distress	Immediate (minutes)
Type I Delayed	IgE-mediated mast cell degranulation	Wheezing, chest tightness, localized itching (Option B ruled out)	Hours to days
Type II	Antibody-mediated cell destruction	Hemolytic anemia, thrombocytopenia	Delayed (hours to days)
Type III	Immune complex deposition	Joint pain, fever, vasculitis (Option C ruled out)	Delayed (hours to days)
Type IV	T-cell mediated response	Skin rash, contact dermatitis (Option D ruled out)	Delayed (48-72 hours)

Solution for Question 5:

Correct Answer: A) Anaphylaxis followed by allergic asthma

Explanation:

- The child's immediate symptoms—facial swelling, hives, and difficulty breathing—are classic signs of anaphylaxis, a severe and potentially life-threatening Type I hypersensitivity reaction triggered by the peanuts.
- His subsequent wheezing and chest tightness later that night indicate the development of allergic asthma, a common late-phase reaction that can occur after an initial allergic response.

Condition	Trigger	Early Phase Symptoms	Late Phase Symptoms
Anaphylaxis	Foods (e.g., peanuts, shellfish), insect stings, medications (e.g., penicillin)	Sudden onset of hives, facial swelling, difficulty breathing, low blood pressure, rapid progression to shock	Persistent wheezing, chest tightness, difficulty breathing, sustained low blood pressure; can last hours after initial exposure
Allergic Rhinitis (Hay Fever)	Pollen, dust mites, pet dander	Sneezing, itchy and watery eyes, runny nose, nasal congestion	Prolonged nasal congestion, sinus pressure, and coughing due to postnasal drip
Allergic Asthma	Pollen, mold, animal dander, air pollutants	Coughing, wheezing, chest tightness, shortness of breath	Persistent wheezing, chest tightness, and airway hyperresponsiveness lead to chronic symptoms if untreated
Food Allergy	Common allergens include peanuts, tree nuts, eggs, milk, seafood	Hives, itching, swelling of lips/face, nausea, abdominal pain, vomiting, and, in severe cases, anaphylaxis	Persistent abdominal pain, diarrhea, skin rash (eczema) in some chronic cases, worsening of asthma or other respiratory symptoms
Atopic Dermatitis (Eczema)	Environmental allergens, certain foods, skin irritants	Red, itchy rash, often in flexural areas (e.g., inside elbows, behind knees)	Chronic dry, thickened skin due to scratching, possible development of secondary infections due to skin barrier breakdown
Urticaria (Hives)	Foods, medications, insect stings, temperature changes	Raised, red, itchy welts (hives) on the skin, can appear within minutes of exposure	If chronic, repeated episodes of itching and welts lasting hours to days after initial exposure

Urticaria followed by allergic rhinitis (Option B), is incorrect because while urticaria (hives) can appear in allergic reactions, it does not typically cause significant breathing issues, and allergic rhinitis primarily involves nasal symptoms like sneezing and congestion.

Atopic dermatitis followed by urticaria (Option C), is incorrect because atopic dermatitis (eczema) is a chronic skin condition and does not directly lead to the acute symptoms seen in this case.

Allergic rhinitis followed by food allergy (Option D), is incorrect because allergic rhinitis is not a match for the child's initial acute symptoms, and food allergy alone wouldn't explain the subsequent wheezing.

Solution for Question 6:

Correct Answer: C) Anaphylactoid reaction, caused by direct mast cell activation without IgE

Explanation: The patient's reaction is anaphylactoid, directly activating mast cells without needing IgE antibodies or prior sensitization.

Features of Anaphylaxis and Anaphylactoid Reactions

Feature	Anaphylaxis	Anaphylactoid Reaction
Mechanism	IgE-mediated (Type I hypersensitivity)	Non-IgE mediated (direct mast cell activation)
Prior Sensitization Needed	Yes	No (Option D ruled out)
Trigger	Specific allergens (e.g., foods, insect stings)	Direct stimuli (e.g., radiocontrast, opioids)
First-Line Treatment	Epinephrine	Epinephrine
Additional Symptoms	Hives, swelling, respiratory distress, hypotension	Same as anaphylaxis
Examples	Peanuts, shellfish, penicillin	Radiocontrast dyes, NSAIDs, opioids

Anaphylaxis, caused by IgE-mediated mast cell activation (Option A), is incorrect as anaphylaxis would require IgE-mediated activation after prior exposure.

Anaphylaxis, caused by T-cell-mediated hypersensitivity (Option B), is incorrect, as Type I hypersensitivity is IgE-mediated, not T-cell-mediated.

Solution for Question 7:

Correct Answer: B) Antibody binding to cell surface antigens with complement activation
Explanation: The patient's symptoms, involving basement membrane antibodies, indicate a Type II hypersensitivity reaction. Here, IgG or IgM antibodies target cell surfaces, activating the complement system and leading to inflammation and cell destruction.

Mechanism of Type-2 Hypersensitivity reaction:

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1. Antibody-Antigen Interaction

- In Type II HS reactions, IgG or IgM antibodies are produced in response to specific antigens on the surfaces of target cells (such as red blood cells or tissues). These antigens can be: Intrinsic: Naturally present on the cell, like blood group antigens in hemolytic reactions. Extrinsic: Foreign antigens that have attached to cell surfaces, often from medications or infections.
- Intrinsic: Naturally present on the cell, like blood group antigens in hemolytic reactions.
- Extrinsic: Foreign antigens that have attached to cell surfaces, often from medications or infections.
- Intrinsic: Naturally present on the cell, like blood group antigens in hemolytic reactions.
- Extrinsic: Foreign antigens that have attached to cell surfaces, often from medications or infections.

2. Activation of Complement Cascade

- When antibodies bind to these cell surface antigens, they form antibody-antigen complexes. This binding activates the complement system (a group of proteins in the blood that assist immune responses).
- The complement pathway progresses through a series of steps, ultimately forming the Membrane Attack Complex (MAC), which punctures the cell membrane, leading to direct cell lysis (rupture).

3. Opsonization and Phagocytosis

- Complement activation also enhances opsonization—the process of marking cells with " tags" (complement components and antibodies) to attract phagocytes.
- Phagocytes (like macrophages and neutrophils) recognize these "tags," engulf them and digest the targeted cells.

4. Antibody-Dependent Cellular Cytotoxicity (ADCC)

- Natural Killer (NK) cells bind to the Fc region of IgG antibodies attached to the target cell, releasing cytotoxic substances that further contribute to cell death
- Type 5 Hypersensitivity (ADCC) involves antibodies binding to cell surface receptors, leading to altered cell function rather than direct cell destruction. Graves' Disease: Autoantibodies stimulate the TSH receptor on thyroid cells, causing excess thyroid hormone production (hyperthyroidism). Myasthenia Gravis: Autoantibodies block acetylcholine receptors at the neuromuscular junction, leading to muscle weakness.
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- Myasthenia Gravis: Autoantibodies block acetylcholine receptors at the neuromuscular junction, leading to muscle weakness.
- Graves' Disease: Autoantibodies stimulate the TSH receptor on thyroid cells, causing excess thyroid hormone production (hyperthyroidism).
- Myasthenia Gravis: Autoantibodies block acetylcholine receptors at the neuromuscular junction, leading to muscle weakness.

5. Tissue Damage and Inflammation: The localized immune response and subsequent cell destruction lead to tissue damage, inflammation, and the clinical symptoms associated with Type II HS reactions.

Antigen-specific IgE binding to mast cells (Option A), is incorrect as this describes a Type I hypersensitivity reaction (like allergies), where IgE antibodies bind to allergens and cause mast cell degranulation, releasing histamine and leading to symptoms like hives and bronchoconstriction.

Immune complexes depositing in blood vessels (Option C), is incorrect as this describes a Type III hypersensitivity reaction, where immune complexes (antigen-antibody clusters) circulate and lodge in tissues, especially blood vessels.

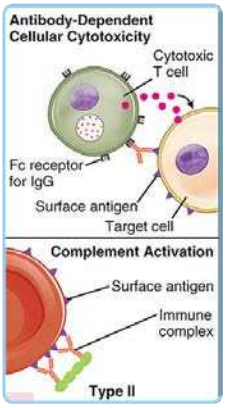
Direct T-cell attack on self-antigens (Option D), is incorrect as this describes a Type IV hypersensitivity reaction, which is T-cell mediated and does not involve antibodies. It causes delayed immune responses, as seen in conditions like contact dermatitis and tuberculosis.

Solution for Question 8:

Correct Answer: C) Autoimmune Hemolytic Anemia (AIHA): Type 2 Hypersensitivity

Explanation:

- In AIHA, antibodies (IgG or IgM) target red blood cell antigens, leading to RBC destruction via complement activation or phagocytosis, typical of Type II hypersensitivity reactions. The direct Coombs test is positive in AIHA, indicating antibodies attached to RBCs, consistent with Type II hypersensitivity, which is an antibody-mediated reaction against cell surface antigens.
- Type II hypersensitivity reactions, also known as cytotoxic hypersensitivity, occur when antibodies (primarily IgG or IgM) bind to antigens on the surface of cells or tissues. This binding activates the complement system or mediates cellular destruction via phagocytosis and cellular cytotoxicity, leading to tissue damage and inflammation. The damage is specific to the target cell or tissue expressing the antigen.
- Examples of Type 2 Hypersensitivity reactions: Myasthenia Gravis Graves Disease Good pastures syndrome Autoimmune hemolytic anemia
- Myasthenia Gravis
- Graves Disease
- Good pastures syndrome
- Autoimmune hemolytic anemia
- Myasthenia Gravis
- Graves Disease
- Good pastures syndrome
- Autoimmune hemolytic anemia



Autoimmune Hemolytic Anemia (AIHA): Type 1 Hypersensitivity (Option A) AIHA does not involve Type I hypersensitivity, which is IgE-mediated and usually linked to allergies.

Rheumatoid Arthritis: Type 1 Hypersensitivity (Option B) Rheumatoid arthritis is primarily associated with Type III hypersensitivity (immune complex-mediated).

Myasthenia Gravis: Type 4 Hypersensitivity (Option D) Myasthenia gravis is a Type II hypersensitivity reaction.

Solution for Question 9:

Correct Answer: Option C) Deposition of immune complexes in tissues

Explanation:

- The patient's symptoms and lab findings suggest Systemic Lupus Erythematosus (SLE), a classic Type III hypersensitivity reaction.
- In Type III hypersensitivity, immune complexes (e.g., anti-dsDNA antibodies bound to DNA) are formed in circulation and deposit in tissues such as the kidneys, skin, and joints.
- These deposits activate the complement system, leading to inflammation and tissue damage. This explains the joint pain, proteinuria, and low complement levels commonly seen in SLE.

Direct binding of IgE to mast cells, leading to histamine release (Option A) describes the mechanism of Type I hypersensitivity (e.g., allergies).

Antibody-mediated cytotoxicity targeting red blood cell antigens(Option B) relates to Type II hypersensitivity, where antibodies bind directly to cell surface antigens.

T-cell-mediated cytotoxicity targeting self-antigens on tissue cells (Option D) is characteristic of Type IV hypersensitivity, which involves T-cell-mediated cellular destruction.

Solution for Question 10:

Correct Answer: B) Post-streptococcal glomerulonephritis

Explanation:

- The patient’s symptoms and laboratory findings suggest post-streptococcal glomerulonephritis (PSGN), a Type III hypersensitivity reaction. In PSGN, immune

complexes formed in response to streptococcal antigens deposit in the glomeruli of thekidneys, leading to complement activation, inflammation, and tissue damage.

• Type III hypersensitivity reactions, also known as immune complex-mediated hypersensitivity, involve the formation of immune complexes (antigen-antibody complexes) that circulate in the bloodstream.

• Examples of Type 3 HS reactions include: Serum Sickness: occurs after exposure to foreign serum proteins, with immune complexes causing fever, rash, and joint pain. SLE: immune complexes deposit in various organs, causing multi-organ inflammation. Henoch Schonelien Purpura: deposition of IgA immune complexes in blood vessels Arthus reaction: localized reaction from immune complex deposition at the injection site, causing swelling and redness. PSGN: immune complexes from streptococcal infection deposit in the kidneys.

• Serum Sickness: occurs after exposure to foreign serum proteins, with immune complexes causing fever, rash, and joint pain.

• SLE: immune complexes deposit in various organs, causing multi-organ inflammation.

• Henoch Schonelien Purpura: deposition of IgA immune complexes in blood vessels

• Arthus reaction: localized reaction from immune complex deposition at the injection site, causing swelling and redness.

• PSGN: immune complexes from streptococcal infection deposit in the kidneys.

• Serum Sickness: occurs after exposure to foreign serum proteins, with immune complexes causing fever, rash, and joint pain.

• SLE: immune complexes deposit in various organs, causing multi-organ inflammation.

• Henoch Schonelien Purpura: deposition of IgA immune complexes in blood vessels

• Arthus reaction: localized reaction from immune complex deposition at the injection site, causing swelling and redness.

• PSGN: immune complexes from streptococcal infection deposit in the kidneys.

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Type III (Immune Complex)	Caused by the formation of antigen-antibody complexes that deposit in tissues, leading to inflammation.	Examples include systemic lupus erythematosus and post-streptococcal glomerulonephritis.
Type IV (Delayed)	Mediated by T cells, occurring hours to days after exposure to an antigen.	Seen in contact dermatitis and tuberculin reactions.

Goodpasture syndrome (Option A) involves Type II hypersensitivity, where antibodies target basement membrane antigens in the kidneys and lungs.

Myasthenia gravis (Option C) is a Type II hypersensitivity reaction targeting acetylcholine receptors at the neuromuscular junction.

Contact dermatitis (Option D) is a Type IV hypersensitivity reaction mediated by T cells in response to allergens on the skin.

Solution for Question 11:

Correct Answer: D) Activation of sensitized T cells

Explanation:

- The patient's presentation is consistent with allergic contact dermatitis due to latex exposure, a classic Type IV hypersensitivity reaction.
- In this delayed-type hypersensitivity, T cells are sensitized to an antigen (latex in this case). Upon re-exposure, these T cells release cytokines, which attract macrophages and other immune cells, leading to inflammation and skin changes.
- Type IV hypersensitivity is characterized by a T cell-mediated immune response that results in delayed inflammation and tissue damage upon re-exposure to an antigen.
- Unlike other types of hypersensitivity that involve antibodies, Type IV is purely cell-mediated and plays a crucial role in both protective immunity and autoimmune diseases.
- The key players in this response are T cells, macrophages, and cytokines, which work together to produce inflammation and tissue injury.
- Seen in contact dermatitis and tuberculin reactions.

Activation of IgE antibodies bound to mast cells (Option A) describes the mechanism of Type I hypersensitivity, involving IgE and mast cells.

Formation of antigen-antibody complexes (Option B) relates to Type III hypersensitivity, where immune complexes deposit in tissues.

Direct binding of antibodies to cell surface antigens (Option C) is characteristic of Type II hypersensitivity, involving antibodies targeting cell surface antigens.

Solution for Question 12:

Correct Answer: B) Allergic contact dermatitis: Type IV hypersensitivity

Explanation:

- The patient's presentation is indicative of allergic contact dermatitis, a classic example of Type IV hypersensitivity. This occurs when sensitized T cells respond to contact with specific allergens, such as plant resins, leading to delayed local inflammation.
- Type IV hypersensitivity is characterized by a T cell-mediated immune response that results in delayed inflammation and tissue damage upon re-exposure to an antigen.
- Some examples include: Granuloma formation - chronic immune response seen in tuberculosis, sarcoidosis, and leprosy. Contact dermatitis - skin reaction to allergens like poison ivy, nickel, or latex. Multiple Sclerosis - T-cell attack on myelin in the central nervous system. Rheumatoid arthritis - T-cell-mediated attack on joint tissues. Tuberculin skin test (Mantoux test) – reaction to injected tuberculin antigen in previously sensitized individuals. Hashimoto’s thyroiditis – T-cell-mediated destruction of thyroid cells.
- Granuloma formation - chronic immune response seen in tuberculosis, sarcoidosis, and leprosy.
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- Hashimoto’s thyroiditis – T-cell-mediated destruction of thyroid cells.

Systemic lupus erythematosus (Option A) is an autoimmune condition primarily associated with Type III hypersensitivity.

Serum sickness (Option C) is a Type III hypersensitivity reaction involving immune complexes.

Anaphylaxis (Option D) is a Type I hypersensitivity reaction mediated by IgE antibodies and mast cells.

Transplant immunology

1. A 45-year-old man with end-stage renal disease is scheduled for a kidney transplant. The best match for a donor kidney is from his identical twin brother. Which type of graft will this procedure involve?

A. Autograft

B. Isograft

C. Allograft

D. Xenograft

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2. In pre-transplant testing, which of the following methods is most commonly used to assess HLA compatibility between a kidney transplant donor and recipient?

A. A) Mixed Lymphocyte Reaction (MLR)

B. B) Microcytotoxicity Test

C. C) Histocompatibility Testing

D. D) ABO Blood Group Compatibility

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3. A 42-year-old woman is being considered for a liver transplant. Her doctors have found a donor with an identical ABO blood group but only partial HLA matching. Which additional test should be performed to assess compatibility further and reduce graft rejection risk?

A. Coombs cross-match

B. Microcytotoxicity testty

C. Restriction Fragment Length Polymorphism (RFLP)

D. ABO blood group compatibility

4. A 55-year-old woman is undergoing a kidney transplant. During the pre-transplant evaluation, it is discovered that she has pre-existing antibodies against the donor's HLA antigens due to a previous organ transplant. Which of the following is most likely to occur immediately after transplantation?

A. Acute cellular rejection

B. Hyperacute rejection

C. Chronic rejection

D. Graft-versus-host disease

5. A 45-year-old male recipient of a kidney transplant presents with a sudden increase in serum creatinine levels three days post-transplant. Laboratory tests reveal the presence of donor-specific antibodies (DSAs) and linear C4d staining in peritubular capillaries on biopsy. Which of the following statements best describes the condition?

A. It is primarily mediated by T-cell activation.

B. It is characterized by the presence of pre-existing antibodies against donor antigens.

C. It is commonly associated with chronic rejection mechanisms.

D. It usually resolves spontaneously without intervention.

6. A patient presents with increased serum creatinine levels and graft tenderness shortly after a kidney transplant. A biopsy reveals lymphocytic infiltration in the graft tissue. Which of the following best describes the underlying mechanism of acute cellular rejection (ACR)?

A. Antibody-mediated destruction of the graft's vascular endothelium

B. Activation of recipient T-lymphocytes recognizing donor antigens

C. Activation of the complement system leading to tissue necrosis

D. Direct damage from pre-existing antibodies against donor cells

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7. Which factor is NOT typically associated with chronic transplant rejection?

A. A) Persistent low-grade immune response despite immunosuppressive therapy.

B. Inflammation during the first year post-transplant.

C. Antibody-mediated damage leading to tissue injury.

D. Non-immune factors like infections and ischemia.

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8. Which of the following statements about acute graft-versus-host disease (aGVHD) is true?

A. aGVHD typically manifests within the first 30 days post-transplant.

B. The primary target organs affected are the skin, liver, and lungs.

C. Corticosteroids are the first-line treatment for managing aGVHD.

D. aGVHD is solely caused by mismatches in blood type between donor and recipient.

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9. Which of the following statements about chronic graft-versus-host disease (cGVHD) is true?

A. cGVHD can present with symptoms resembling those of systemic lupus erythematosus.

B. Skin manifestations may include erythema, hyperkeratosis, and desquamation.

C. The condition is primarily associated with infections rather than immune dysregulation.

D. Second-line treatments for cGVHD typically include high-dose corticosteroids.

10. Which of the following statements about post-transplant complications is true?

A. Allograft rejection primarily involves the activation of B lymphocytes leading to antibody production.

B. Hyperacute rejection occurs immediately due to pre-existing antibodies in the recipient.

C. Graft-versus-host reaction is less likely to occur in immunocompromised recipients.

D. Hemolytic disease of the newborn is caused by maternal incompatibility with paternal blood group antigens.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	2
Question 5	2
Question 6	2
Question 7	2
Question 8	3
Question 9	2
Question 10	2

Solution for Question 1:

An isograft is a type of transplant where the donor and recipient are genetically identical, such as in the case of identical twins. Since there is no genetic difference between them, the transplant is not rejected by the recipient's immune system.

Types of Grafts:

- Autograft (Option A ruled out) Graft taken from and used on the same individual.
- Graft taken from and used on the same individual.
- Isograft Graft between genetically identical individuals (e.g., identical twins). No rejection due to identical genetic makeup.
- Graft between genetically identical individuals (e.g., identical twins).
- No rejection due to identical genetic makeup.

• Allograft (formerly Homograft) (Option C ruled out) Graft between two genetically different individuals of the same species. Commonly used in clinical transplantation. Requires immunosuppressive therapy to prevent rejection.

• Graft between two genetically different individuals of the same species.

• Commonly used in clinical transplantation.

• Requires immunosuppressive therapy to prevent rejection.

• Xenograft (formerly Heterograft) (Option D ruled out) Graft between individuals of different species. Higher rejection rate due to significant genetic differences.

• Graft between individuals of different species.

• Higher rejection rate due to significant genetic differences.

• Graft taken from and used on the same individual.

• Graft between genetically identical individuals (e.g., identical twins).

• No rejection due to identical genetic makeup.

• Graft between two genetically different individuals of the same species.

• Commonly used in clinical transplantation.

• Requires immunosuppressive therapy to prevent rejection.

• Graft between individuals of different species.

• Higher rejection rate due to significant genetic differences.

Solution for Question 2:

The Microcytotoxicity Test is commonly used for HLA typing. It identifies specific HLA antigens on an individual's lymphocytes and uses antibodies against HLA antigens to determine compatibility.

Pre-Transplant Testing and Matching:

• Histocompatibility Testing (Option C ruled out) Assesses compatibility between donor and recipient to reduce graft rejection risk.

• Assesses compatibility between donor and recipient to reduce graft rejection risk.

• ABO Blood Group Compatibility (Option D ruled out) Essential for transplantation. Recipient’s plasma must not contain antibodies against the donor's RBC antigens.

• Essential for transplantation.

• Recipient’s plasma must not contain antibodies against the donor's RBC antigens.

• HLA Compatibility Key to allograft success. Human Leukocyte Antigen (HLA) System: Major histocompatibility complex in humans. Matching HLA antigens between donor and recipient improves graft acceptance.

• Key to allograft success.

• Human Leukocyte Antigen (HLA) System: Major histocompatibility complex in humans.

• Matching HLA antigens between donor and recipient improves graft acceptance.

• HLA Typing Methods Microcytotoxicity Test: Uses antibodies to target specific HLA antigens. Destruction of targeted cells indicates HLA type. Molecular Methods: Techniques like RFLP (Restriction Fragment Length Polymorphism) and PCR for precise HLA gene identification.

• Microcytotoxicity Test: Uses antibodies to target specific HLA antigens. Destruction of targeted cells indicates HLA type.

• Uses antibodies to target specific HLA antigens.

• Destruction of targeted cells indicates HLA type.

• Molecular Methods: Techniques like RFLP (Restriction Fragment Length Polymorphism) and PCR for precise HLA gene identification.

• Techniques like RFLP (Restriction Fragment Length Polymorphism) and PCR for precise HLA gene identification.

• Tissue Matching After identifying HLA-compatible donors, tissue matching selects the most suitable donor. Mixed Lymphocyte Reaction (MLR): (Option A ruled out) Donor and recipient lymphocytes are cultured together. Incompatibility triggers the recipient’s T cells to undergo blast transformation, indicating rejection risk.

• After identifying HLA-compatible donors, tissue matching selects the most suitable donor.

• Mixed Lymphocyte Reaction (MLR): (Option A ruled out) Donor and recipient lymphocytes are cultured together. Incompatibility triggers the recipient’s T cells to undergo blast transformation, indicating rejection risk.

• Donor and recipient lymphocytes are cultured together.

• Incompatibility triggers the recipient’s T cells to undergo blast transformation, indicating rejection risk.

• Assesses compatibility between donor and recipient to reduce graft rejection risk.

• Essential for transplantation.

• Recipient’s plasma must not contain antibodies against the donor's RBC antigens.

• Key to allograft success.

• Human Leukocyte Antigen (HLA) System: Major histocompatibility complex in humans.

• Matching HLA antigens between donor and recipient improves graft acceptance.

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• Donor and recipient lymphocytes are cultured together.

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Solution for Question 3:

Coombs cross-match is correct because it detects any pre-formed antibodies in the recipient's serum against the donor's cells, ensuring better compatibility and reducing the risk of rejection.

Ideal Versus Practical Matching in Transplantation:

• Histocompatibility Importance Minimising antigenic differences reduces graft rejection. Perfect antigen matching is rarely achievable in practice.

• Minimising antigenic differences reduces graft rejection.

• Perfect antigen matching is rarely achievable in practice.

• Ideal Matching Ideally, the donor and recipient are genetically identical (isograft), eliminating rejection risk. Rarely achievable except in identical twins.

• Ideally, the donor and recipient are genetically identical (isograft), eliminating rejection risk.

• Rarely achievable except in identical twins.

• Practical Matching ABO Blood Group: Essential to match to avoid severe haemolytic reactions. HLA System: Priority is given to matching HLA-A, -B, and -DR antigens, crucial for graft survival. Sibling Donors: Best potential donors due to shared HLA haplotypes. Unrelated Donors: Large registries help find matches, though perfect matches are less likely than with siblings.

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• Sibling Donors: Best potential donors due to shared HLA haplotypes.

• Unrelated Donors: Large registries help find matches, though perfect matches are less likely than with siblings.

• Cross-Matching Final cross-match detects recipient antibodies against donor cells. Techniques include saline cross-match and Coombs cross-match.

• Final cross-match detects recipient antibodies against donor cells.

• Techniques include saline cross-match and Coombs cross-match.

• Immunosuppressive Therapy Lifelong therapy is essential to prevent graft rejection. Uses a combination of drugs targeting the immune system.

• Lifelong therapy is essential to prevent graft rejection.

• Uses a combination of drugs targeting the immune system.

• Minimising antigenic differences reduces graft rejection.

• Perfect antigen matching is rarely achievable in practice.

• Ideally, the donor and recipient are genetically identical (isograft), eliminating rejection risk.

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• ABO Blood Group: Essential to match to avoid severe haemolytic reactions.

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• Final cross-match detects recipient antibodies against donor cells.

• Techniques include saline cross-match and Coombs cross-match.

• Lifelong therapy is essential to prevent graft rejection.

• Uses a combination of drugs targeting the immune system.

Microcytotoxicity test (Option B) is incorrect as it is primarily used for determining HLA typing rather than assessing pre-formed antibodies.

Restriction Fragment Length Polymorphism (RFLP) (Option C) is incorrect because it is a molecular method used for detailed genetic analysis, not for immediate compatibility assessment.

ABO blood group compatibility (Option D) is incorrect; while it is essential for safety, it has already been established as identical, and the focus should be on assessing antibodies to prevent rejection.

Solution for Question 4:

Hyperacute rejection occurs rapidly within hours of transplantation when the recipient has pre-existing antibodies against the donor's antigens. This leads to immediate graft failure.

Hyperacute Transplant Rejection:

• Definition Rapid and severe rejection occurring within hours of transplantation. Mediated by pre-existing antibodies in the recipient against donor antigens.

• Rapid and severe rejection occurring within hours of transplantation.

• Mediated by pre-existing antibodies in the recipient against donor antigens.

• Mechanism Antibodies bind to antigens on the graft's endothelial cells. Activation of the complement cascade leads to: Thrombosis: Blood clots form in graft vessels, blocking blood flow and causing ischemia. Inflammation: Neutrophil influx causes further graft damage. Necrosis: Tissue death due to ischemia and inflammation.

• Antibodies bind to antigens on the graft's endothelial cells.

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• Thrombosis: Blood clots form in graft vessels, blocking blood flow and causing ischemia.

• Inflammation: Neutrophil influx causes further graft damage.

• Necrosis: Tissue death due to ischemia and inflammation.

• Causes of Pre-existing Antibodies Prior Transplantation: Previous rejection can lead to antibody formation. Blood Transfusion: Exposure to foreign antigens can sensitize recipients. Pregnancy: Exposure to fetal antigens can trigger antibody production.

• Prior Transplantation: Previous rejection can lead to antibody formation.

• Blood Transfusion: Exposure to foreign antigens can sensitize recipients.

• Pregnancy: Exposure to fetal antigens can trigger antibody production.

• Characteristics White graft response: Graft remains pale and is rapidly rejected without vascularisation.

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• Prior Transplantation: Previous rejection can lead to antibody formation.

• Blood Transfusion: Exposure to foreign antigens can sensitize recipients.

• Pregnancy: Exposure to fetal antigens can trigger antibody production.

• White graft response: Graft remains pale and is rapidly rejected without vascularisation.

Prevention of Hyperacute Rejection:

• Pre-transplant Screening Essential to test for pre-existing antibodies against donor antigens (HLA and blood group).

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• Cross-Matching Final cross-match using recipient serum and donor cells to detect last-minute incompatibilities.

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• Essential to test for pre-existing antibodies against donor antigens (HLA and blood group).

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Treatment of Hyperacute Rejection:

• Graft Removal Difficult to treat; usually results in graft loss. Prompt removal of the graft is necessary to prevent complications.

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Type of rejection	Mechanism	Time after transplant
Hyperacute rejection	Mediated by preformed antibodies specific for antigens on graft endothelial cells, these may be natural IgM antibodies specific for blood group antigens, or may be antibodies specific for allogeneic MHC molecules that were induced by prior exposure of the organ recipient to allogeneic cells through blood transfusions, pregnancy, or transplantation of another organ	Minutes to hours
Acute cellular rejection	CD8+ cells may directly destroy graft cells, or CD4+ cells secrete cytokines and induce inflammation, which damages the graft	Anytime (most commonly first 6 months)
Acute antibody-mediated (vascular, or humoral) rejection	Antibodies bind to vascular endothelium and activate complement via the classical pathway. The resultant inflammation and endothelial damage cause graft failure.	
Chronic rejection	Chronic rejection is an indolent form of graft damage that occurs over months or years, leading to	Usually after 1st year

	progressive loss of graft function. Chronic rejection manifests as interstitial fibrosis and gradual narrowing of graft blood vessels (graft arteriosclerosis).	

Acute cellular rejection (Option A) is incorrect as it typically occurs days to weeks after transplantation, not immediately.

Chronic rejection (Option C) is incorrect because it develops over months to years and is a slow process.

Graft-versus-host disease (Option D) is incorrect, as it occurs when donor immune cells attack the recipient's tissues, usually associated with bone marrow transplants, not kidney transplants.

Solution for Question 5:

Acute Humoral Rejection (AHR) is diagnosed through the presence of donor-specific antibodies (DSAs) in the serum, along with histological evidence of glomerulitis or peritubular capillaritis observed in biopsy samples. Additionally, linear C4d staining in peritubular capillaries serves as a critical marker for confirming the diagnosis.

Acute Humoral Rejection (AHR):

- Definition Significant complication in organ transplantation, particularly in kidney transplants. Characterized by donor-specific antibodies (DSAs) targeting the vascular endothelium, leading to tissue injury and graft dysfunction.
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Mechanisms of AHR:

- Pathophysiology Binding of DSAs to donor HLA molecules activates the complement system, resulting in: Capillary Injury: Damage to peritubular capillaries is a hallmark of AHR. C4d Deposition: A key diagnostic marker; approximately 30% of acute rejection biopsies

show widespread C4d staining. Histological Features: May include neutrophilic infiltration,glomerulitis, and fibrinoid necrosis in severe cases.

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• Histological Features: May include neutrophilic infiltration, glomerulitis, and fibrinoid necrosis in severe cases.

Diagnosis:

• Diagnostic Criteria Combination of clinical presentation, serological testing, and histopathological findings: Histologic Evidence: Glomerulitis or peritubular capillaritis. Antibody Interaction: Linear C4d staining indicates recent antibody interaction with vascular endothelium. Detection of DSAs: Presence of DSAs in serum confirms diagnosis.

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Clinical Presentation:

• Symptoms Patients may experience: Acute Elevation in Creatinine: Noted within days to weeks post-transplant. Fever and Malaise: General systemic inflammation signs. Graft Tenderness: Possible tenderness over the transplant site.

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• Fever and Malaise: General systemic inflammation signs.

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Treatment:

• Management Approaches Aim to reduce antibody-mediated injury: Plasmapheresis: Removes circulating antibodies. Intravenous Immunoglobulin (IVIG): Modulates immune responses. Rituximab: Targets CD20 on B cells to reduce antibody production. Eculizumab: A complement inhibitor effective in severe AHR cases.

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• Plasmapheresis: Removes circulating antibodies.

• Intravenous Immunoglobulin (IVIG): Modulates immune responses.

• Rituximab: Targets CD20 on B cells to reduce antibody production.

• Eculizumab: A complement inhibitor effective in severe AHR cases.

It is primarily mediated by T-cell activation (Option A) is incorrect because AHR is primarily mediated by antibody interactions, not T-cell responses.

It is commonly associated with chronic rejection mechanisms (Option C) is incorrect since AHR is an acute process, distinct from chronic rejection.

It usually resolves spontaneously without intervention (Option D) is incorrect; AHR typically requires immunosuppressive treatment for resolution.

Solution for Question 6:

Acute cellular rejection (ACR) occurs when the recipient's T-lymphocytes recognize foreign antigens on the donor organ through direct and indirect pathways, leading to T-cell activation and infiltration into the graft tissue, causing damage.

Acute Cellular Rejection (ACR):

- ACR is a significant concern in organ transplantation, characterized by T-lymphocyte infiltration into transplanted tissue.
- Occurs within days to weeks post-transplant, mediated by the recipient’s T-lymphocytes recognizing foreign antigens on the donor organ.

Mechanisms of Acute Cellular Rejection:

- T-cell Activation: Direct Pathway: Donor dendritic cells present antigens directly to recipient T-cells. Indirect Pathway: Recipient antigen-presenting cells process and present donor antigens after donor cell death.
- Direct Pathway: Donor dendritic cells present antigens directly to recipient T-cells.
- Indirect Pathway: Recipient antigen-presenting cells process and present donor antigens after donor cell death.
- Cytokine Release: Activated T-cells release cytokines (e.g., IL-2, TNF-alpha, IFN-gamma), promoting inflammation and immune responses.
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- Direct Pathway: Donor dendritic cells present antigens directly to recipient T-cells.
- Indirect Pathway: Recipient antigen-presenting cells process and present donor antigens after donor cell death.
- Activated T-cells release cytokines (e.g., IL-2, TNF-alpha, IFN-gamma), promoting inflammation and immune responses.

Histopathological Features:

Feature	Description
Lymphocytic Infiltration	Predominantly CD4+ and CD8+ T-lymphocytes damage graft tissue.
Grading System	Severity graded based on histology:
Grade 0	No rejection
Grade 1R	Mild infiltration with minimal damage
Grade 2R	Moderate infiltration with significant damage
Grade 3R	Severe diffuse infiltration with extensive damage

Clinical Presentation:

- Increased Serum Creatinine Levels: Indicative of kidney dysfunction.
- Graft Tenderness or Pain: Common in kidney and heart transplants.
- Fever and Malaise: Signs of systemic inflammation.

Diagnosis:

- Serum Creatinine Measurement: To assess renal function.
- Biopsy of the Transplanted Organ: Evaluates for lymphocytic infiltration and histopathological changes.

Treatment:

Treatment	Description
Corticosteroids	Administered intravenously to reduce inflammation.
Anti-T-cell Antibodies	E.g., antithymocyte globulin (ATG), and monoclonal antibodies (OKT3) targeting T-cells.
Supportive Therapies	Prophylactic antibiotics to prevent infections during immunosuppression.

Antibody-mediated destruction of the graft's vascular endothelium (Option A) is characteristic of acute humoral rejection, not ACR.

Activation of the complement system leading to tissue necrosis (Option C): While the complement system is involved in some rejection mechanisms, it is primarily associated with humoral rejection rather than the T-cell-mediated process of ACR.

Direct damage from pre-existing antibodies against donor cells (Option C): Pre-existing antibodies play a role in hyperacute rejection, not in the T-cell mediated process of ACR.

Solution for Question 7:

Inflammation during the first year post-transplant is a common occurrence and is usually associated with acute rejection. Chronic rejection is a slower, more insidious process that develops over time and is characterized by persistent immune response, antibody-mediated damage, and non-immune factors.

Chronic Transplant Rejection:

• Definition: Chronic transplant rejection is a slow, progressive form of organ rejection occurring years post-transplantation.

Characteristics of Chronic Transplant Rejection:

Characteristic	Description
Gradual Decline in Graft Function	Slow deterioration of transplanted organ function over time.
Fibrosis and Scarring	Chronic inflammation leads to fibrous tissue deposition, causing scarring and loss of normal organ structure.
Vascular Changes	Thickening and narrowing of blood vessels in the graft, reducing blood flow and causing tissue damage.

Potential Contributing Factors:

• Chronic, Low-Grade Immune Response: Persistent immune response despite immunosuppressive therapy, causing ongoing inflammation and tissue damage. (Option A ruled out)

• Antibody-Mediated Damage: Antibodies bind to graft cells, activating complement, leading to cell lysis and inflammation. (Option C ruled out)

• Non-Immune Factors: Infections, ischemia (lack of blood flow), and medication toxicity contribute to chronic graft damage. (Option D ruled out)

Challenges in Understanding and Managing Chronic Rejection:

Challenge	Description
Complex and Multifactorial	Involves multiple factors, complicating cause identification and therapy development.
Limited Treatment Options	Existing therapies focus on acute rejection prevention; less effective for chronic rejection.
Impact on Long-Term Graft Survival	Major cause of long-term graft failure.

Solution for Question 8:

Corticosteroids, such as methylprednisolone, are the standard first-line treatment for managing acute graft-versus-host disease, with about 50% of patients responding adequately.

Acute Graft-Versus-Host Disease (aGVHD):

• Definition: aGVHD is a significant complication following allogeneic hematopoietic stem cell transplantation (HSCT) where donor immune cells attack the recipient's tissues.

• Onset: Typically occurs within the first 100 days post-transplant, but late-onset cases may arise. (Option A ruled out)

Pathophysiology:

• Mechanism:

Primary Target Organs: (Option B ruled out)

Organ	Symptoms
Skin	Maculopapular rash
Liver	Elevated bilirubin levels and liver enzymes
Gastrointestinal Tract	Abdominal pain, diarrhea, nausea, vomiting

Phases of Pathophysiology:

Diagnosis:

• Key Diagnostic Criteria: Clinical Symptoms: Development of skin rash, gastrointestinal distress, or liver dysfunction within 100 days post-transplant. Biopsy: Histological examination helps differentiate aGVHD from other conditions (e.g., viral infections, drug reactions).

• Clinical Symptoms: Development of skin rash, gastrointestinal distress, or liver dysfunction within 100 days post-transplant.

• Biopsy: Histological examination helps differentiate aGVHD from other conditions (e.g., viral infections, drug reactions).

• Clinical Symptoms: Development of skin rash, gastrointestinal distress, or liver dysfunction within 100 days post-transplant.

• Biopsy: Histological examination helps differentiate aGVHD from other conditions (e.g., viral infections, drug reactions).

Staging of Acute GVHD:

Grade	Description
I	Mild involvement
II	Moderate involvement
III	Severe involvement, poor prognosis
IV	Very severe involvement, high mortality risk

Risk Factors:

- Donor-Recipient Compatibility: Mismatches in HLA and minor histocompatibility antigens increase risk.
- Sex Mismatch: Higher risk when female donors are paired with male recipients, especially if previously pregnant or transfused.
- Age of Recipient: Increased risk in older patients.

Treatment:

- Management Focus: Immunosuppression to mitigate the immune response.
- First-Line Treatment: Corticosteroids (e.g., methylprednisolone). Approximately 50% of patients respond adequately.
- Corticosteroids (e.g., methylprednisolone).
- Approximately 50% of patients respond adequately.
- Second-Line Treatments: Janus kinase inhibitors (e.g., ruxolitinib) for steroid-resistant cases. Lack of universally accepted standards for refractory cases.
- Janus kinase inhibitors (e.g., ruxolitinib) for steroid-resistant cases.
- Lack of universally accepted standards for refractory cases.
- Supportive Care: Essential for managing infections and nutritional needs related to gastrointestinal involvement.
- Corticosteroids (e.g., methylprednisolone).
- Approximately 50% of patients respond adequately.
- Janus kinase inhibitors (e.g., ruxolitinib) for steroid-resistant cases.
- Lack of universally accepted standards for refractory cases.

Solution for Question 9:

Skin manifestations in chronic graft-versus-host disease (cGVHD) often include erythema, hyperkeratosis, and desquamation. These symptoms arise due to the immune-mediated attack on the skin by donor T cells, leading to chronic inflammation and tissue damage.

Chronic Graft-Versus-Host Disease (cGVHD):

Definition: cGVHD is a significant complication following allogeneic hematopoietic stem cell transplantation (HSCT) and is a leading cause of morbidity and mortality in transplant recipients.

It occurs when donor immune cells recognize recipient tissues as foreign.

Pathophysiology:

- Mechanism: Immune Dysregulation: Donor T cells attack host tissues, leading to chronic inflammation and fibrosis. (Option C ruled out) Tissue Damage: Results in damage to various organs, resembling autoimmune diseases.
- Immune Dysregulation: Donor T cells attack host tissues, leading to chronic inflammation and fibrosis. (Option C ruled out)
- Tissue Damage: Results in damage to various organs, resembling autoimmune diseases.
- Onset: Typically occurs between 80 days and 2 years post-transplant; approximately 30% are classified as de novo cases.
- Immune Dysregulation: Donor T cells attack host tissues, leading to chronic inflammation and fibrosis. (Option C ruled out)
- Tissue Damage: Results in damage to various organs, resembling autoimmune diseases.

Clinical Manifestations:

Organ System	Symptoms
Skin	Erythema, hyperkeratosis, sclerodermatous changes, desquamation
Oral Cavity	Xerostomia, oral ulcers, lichenoid changes
Eyes	Sicca syndrome (dry eyes) and discomfort
Liver	Cholestatic changes (elevated liver enzymes and bilirubin)
Lungs	Bronchiolitis obliterans syndrome (respiratory complications)
Gastrointestinal Tract	Anorexia, dysphagia, malabsorption, weight loss

Other Manifestations: Polymyositis, serositis, and autoimmune features (less common).

Diagnosis:

- Key Diagnostic Aspects: Clinical Examination: Assess symptoms across multiple organ systems. Biopsy: Skin or other tissue biopsies may confirm the diagnosis and exclude other conditions.
- Clinical Examination: Assess symptoms across multiple organ systems.
- Biopsy: Skin or other tissue biopsies may confirm the diagnosis and exclude other conditions.
- NIH Criteria: Standardized criteria established for diagnosing and classifying cGVHD based on clinical manifestations.

• Clinical Examination: Assess symptoms across multiple organ systems.

• Biopsy: Skin or other tissue biopsies may confirm the diagnosis and exclude other conditions.

Treatment:

• Management Focus: Immunosuppressive therapies to control the immune response.

• First-Line Therapy: Corticosteroids are typically the first line of treatment; initial responses are common, but recurrences may occur upon tapering. (Option D ruled out)

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Second-Line Therapies: (Option D ruled out)

Options	Description
Calcineurin Inhibitors	e.g., tacrolimus
Mycophenolate Mofetil	Commonly used immunosuppressant
Newer Agents	ibrutinib or ruxolitinib for steroid-refractory cases

Supportive Care: Essential for managing symptoms and complications, including infection treatment due to immunosuppression.

cGVHD can present with symptoms resembling those of systemic lupus erythematosus (Option A): While cGVHD can present with a variety of symptoms, it does not primarily resemble systemic lupus erythematosus; it is characterized by immune dysregulation affecting multiple organs.

Solution for Question 10:

Hyperacute rejection occurs immediately after transplantation due to pre-existing antibodies in the recipient. These antibodies recognize and bind to the antigens on the donor graft, leading to an acute immune response that results in rapid graft failure.

Post-Transplant Complications:

Immune-Mediated Complications:

• Allograft Rejection: Occurs when the recipient's immune system attacks the transplanted organ, recognizing it as foreign. Mediated primarily by T lymphocytes, leading to tissue damage and graft failure. (Option A ruled out)

• Occurs when the recipient's immune system attacks the transplanted organ, recognizing it as foreign.

• Mediated primarily by T lymphocytes, leading to tissue damage and graft failure. (Option A ruled out)

• Occurs when the recipient's immune system attacks the transplanted organ, recognizing it as foreign.

• Mediated primarily by T lymphocytes, leading to tissue damage and graft failure. (Option A ruled out)

Types of Allograft Rejection:

Type	Description
First-Set Rejection	Initial rejection episode after transplantation.
Second-Set Rejection	More rapid and aggressive response upon re-transplantation from the same donor due to immunological memory.
Hyperacute Rejection	Immediate graft failure due to pre-existing antibodies in the recipient targeting the graft.

• Graft-versus-Host (GVH) Reaction: Transplanted immune cells (T cells) attack the recipient's tissues, the reverse of typical transplant rejection. More likely in immunocompromised recipients (e.g., newborns, those on immunosuppressive therapy). (Option C ruled out) Can lead to severe conditions, such as runt disease in animals.

• Transplanted immune cells (T cells) attack the recipient's tissues, the reverse of typical transplant rejection.

• More likely in immunocompromised recipients (e.g., newborns, those on immunosuppressive therapy). (Option C ruled out)

• Can lead to severe conditions, such as runt disease in animals.

• Factors Influencing Rejection: Histocompatibility antigens (particularly the HLA system) play a crucial role. Matching donor and recipient HLA types minimizes rejection risk.

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• Matching donor and recipient HLA types minimizes rejection risk.

• Transplanted immune cells (T cells) attack the recipient's tissues, the reverse of typical transplant rejection.

• More likely in immunocompromised recipients (e.g., newborns, those on immunosuppressive therapy). (Option C ruled out)

• Can lead to severe conditions, such as runt disease in animals.

• Histocompatibility antigens (particularly the HLA system) play a crucial role.

• Matching donor and recipient HLA types minimizes rejection risk.

• Hemolytic Transfusion Reactions: Occur when incompatible ABO and Rh blood groups are transfused. Can trigger an immune response, leading to red blood cell destruction, kidney failure, or death.

• Occur when incompatible ABO and Rh blood groups are transfused.

• Can trigger an immune response, leading to red blood cell destruction, kidney failure, or death.

• Hemolytic Disease of the Newborn: (Option D ruled out) Rh incompatibility between mother and fetus can lead to fetal red blood cell destruction. Maternal sensitization to fetal Rh antigens during pregnancy leads to antibody production that crosses the placenta.

• Rh incompatibility between mother and fetus can lead to fetal red blood cell destruction.

• Maternal sensitization to fetal Rh antigens during pregnancy leads to antibody production that crosses the placenta.

• Infectious Complications: Risks include transmission of infectious agents (e.g., HIV, hepatitis B and C, syphilis, malaria) through transfusions, despite screening procedures.

• Risks include transmission of infectious agents (e.g., HIV, hepatitis B and C, syphilis, malaria) through transfusions, despite screening procedures.

• Occur when incompatible ABO and Rh blood groups are transfused.

• Can trigger an immune response, leading to red blood cell destruction, kidney failure, or death.

• Rh incompatibility between mother and fetus can lead to fetal red blood cell destruction.

• Maternal sensitization to fetal Rh antigens during pregnancy leads to antibody production that crosses the placenta.

• Risks include transmission of infectious agents (e.g., HIV, hepatitis B and C, syphilis, malaria) through transfusions, despite screening procedures.

Antigen, antibody and their reactions

1. In which clinical test is the "Ring test" precipitation reaction applied, and what is its primary use?

A. Ascoli’s thermo precipitin test for detecting anthrax antigens

B. VDRL test for detecting syphilis

C. Lancefield technique for grouping mycobacteria

D. Kahn test for detecting lupus antibodies

---

2. A 30-year-old patient presents with recurrent infections and a history of autoimmune disease. Laboratory tests reveal a deficiency in complement component C3. Which of the following is the most likely consequence of this deficiency?

A. A) Increased susceptibility to infections by Neisseria meningitidis

B. B) Impaired opsonization and phagocytosis, leading to severe bacterial infections

C. C) Development of hereditary angioneurotic oedema

D. D) Enhanced immune complex clearance and reduced inflammation

---

3. Which of the following statements correctly describes the pathways of complement activation?

A. A) The classical pathway is triggered by the spontaneous hydrolysis of C3 and does not require antibodies.

B. B) The alternative pathway is initiated by the binding of mannose-binding lectin (MBL) to mannose residues on pathogens.

C. C) The lectin pathway involves the binding of C1q to antigen-antibody complexes, leading to the formation of the membrane attack complex (MAC).

D. D) All three pathways converge at the formation of C3 convertase, which cleaves C3 into C3a and C3b.

---

4. Which of the following characteristics of Immunochromatographic Card Tests is NOT accurate?

A. A) They provide rapid results, often within 15-30 minutes.

B. B) The tests for Hepatitis B can detect both antibodies and antigens.

C. C) They are primarily used in well-equipped laboratory settings only.

D. D) The sensitivity and specificity of these tests can be greater than 90%.

---

5. Which of the following statements best describes the indirect immunofluorescence test?

A. A) It requires specific fluorescent conjugates for each antigen being tested.

B. It uses an antiglobulin fluorescent conjugate to detect any human antibody, allowing for broader application.

C. It directly labels antigens with a fluorescent dye for visualisation.

D. It is primarily used to detect antigens in brain smears, such as in rabies diagnosis.

---

6. Which of the following statements about Chemiluminescent Immunoassays (CLIA) is TRUE?

A. A) CLIA exclusively uses fluorescent dyes to detect antibodies in clinical samples.

B. The sensitivity of the VITROS® SARS-CoV-2 Ag Test is reported to be 94.3%.

C. CLIA methods have lower diagnostic accuracy compared to traditional serological tests.

D. Neutralising antibody levels against SARS-CoV-2 can only be detected using RT-PCR

---

7. Which type of ELISA involves an antigen in the sample competing with a known antigen for a limited amount of antibody, resulting in an inverse colour change curve?

A. A) Sandwich ELISA

B. B) Indirect ELISA

C. C) Competitive ELISA

D. D) Capture ELISA

---

8. In the Complement Fixation Test (CFT), what indicates a positive result?

A. Lysis of sheep red blood cells (SRBCs)

B. Formation of an immune complex between antigen and antibody

C. No lysis of sheep red blood cells (SRBCs)

D. Presence of active complement in the serum

---

9. A 30-year-old Rh-negative G2P1L1 woman is evaluated during her routine prenatal visit. What would be the significance of performing an indirect Coombs test at this stage?

A. To determine the mother’s ABO blood type

B. To check for antibodies against Rh-positive blood

C. To confirm the sensitization of red cells with antibodies in the fetus

D. All of the above

---

10. Which type of agglutination test is typically used for blood typing?

A. Heterophile Agglutination test

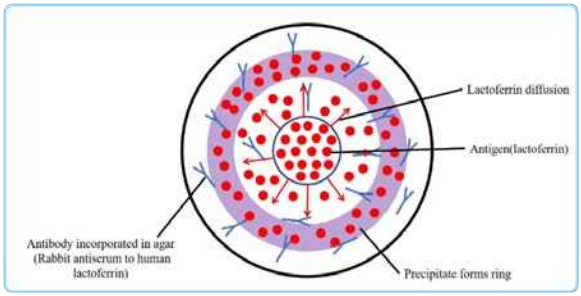
B. Co-agglutination test

C. Direct Agglutination test

D. Passive Agglutination test

---

11. The image below shows a precipitation test in a gel medium in which a circular precipitation pattern is formed due to an antigen-antibody reaction. What type of precipitation test in the gel does this image depict?



A. Ouchterlony Procedure

B. Radial immunodiffusion

C. Oakley-Fulthorpe Procedure

D. Counterimmunoelectrophoresis

---

12. Which of the following statements best describes a pro-antigen?

A. A molecule that binds to carrier proteins to become immunogenic and trigger an immune response.

B. A substance that directly triggers an immune response without any modification.

C. A molecule that requires metabolic conversion into reactive intermediates to become immunogenic.

D. A compound that can bind to antibodies but does not provoke a full immune response without a carrier protein.

---

13. Which immunoglobulin is most closely associated with the early stages of B cell response and is found alongside IgM on unstimulated B lymphocytes?

A. IgA

B. IgD

C. IgE

D. All of the above

---

14. Which of the following is true about IgG, but not IgM?

A. It requires a J chain for polymerisation.

B. It has a valency of 10.

C. It is the only antibody that can cross the placenta.

D. It is the first antibody produced by the fetus.

---

15. Which of the following statements regarding the enzymatic digestion of immunoglobulins is incorrect?

A. Papain digestion results in the formation of two Fab fragments and one Fc fragment.

B. Mercaptoethanol irreversibly cleaves disulfide bonds, separating heavy and light chains.

C. Pepsin digestion produces one F(ab')2 fragment and several smaller peptide fragments.

D. Pepsin digestion cleaves the Ig molecule above the disulfide bridge in the hinge region.

16. A 25-year-old man is preparing for a trip to Africa and South America. The healthcare provider informs him that he needs a yellow fever vaccine, which is important for preventing the disease. Which of the following statements about the yellow fever vaccine is incorrect?

A. The vaccine is effective in providing long-lasting immunity

B. Immunity from the vaccine develops immediately after vaccination.

C. The vaccine contains a live attenuated virus

D. It is a form of artificially acquired active immunity.

17. A 68-year-old man presents with complaints of persistent fatigue, bone pain, and frequent infections. Laboratory tests reveal anemia, hypercalcemia, and elevated serum protein levels. A peripheral blood smear shows rouleaux formation. Which of the following statements regarding the types of antibodies produced is correct?

A. The antibodies recognize different epitopes and show variability.

B. The antibody population includes both kappa and lambda light chains.

C. The antibodies are identical across batches and highly cross-reactive.

D. The antibodies are homogeneous from a single clone of B cells.

18. A 19-year-old college student presents to the clinic with a 2-week history of fatigue, sore throat, fever, and swollen lymph nodes. On examination, he has significant posterior cervical lymphadenopathy and hepatosplenomegaly. The physician suspects a viral infection and decides to perform a specific agglutination test. Which of the following statements regarding this test is correct?

A. The test detects heterophile antibodies that agglutinate sheep red blood cells.

B. An agglutination titer of 50 or below is indicative of a positive result.

C. The test requires fresh serum samples without any prior heating.

D. Heterophile antibodies are produced only in response to one type of virus

---

19. A 35-year-old man presents with high fever, hypotension, and a diffuse rash. He reports having eaten at a barbecue the previous day, where he consumed undercooked food. He is also experiencing confusion and severe muscle pain. Laboratory tests reveal elevated inflammatory markers. Which of the following statements regarding the mechanism of his condition is correct?

A. The active proteins bind directly to the lateral aspect of the T cell receptor β chain.

B. The immune response involves specific antigen recognition by TCR α chains and MHC molecules.

C. The proteins involved are typically small peptides with a molecular weight under 10 kDa.

D. They induce massive release of cytokines, such as IL-2, leading to lymphocyte apoptosis.

---

20. A 40-year-old woman visits the clinic for evaluation of her immune response following a recent vaccination. The physician discusses the differences between T cell-dependent and T cell-independent antigens. Which of the following statements is incorrect?

A. T cell-independent antigens can activate complement

B. T cell-dependent antigens result in a limited range of antibody isotypes.

C. T cell-independent antigens primarily lead to the production of IgM and IgG3.

D. T cell-dependent antigens generate immunologic memory.

---

21. Which of the following factors is most likely to decrease the immunogenicity of a substance?

A. Large molecular size of the antigen

B. Metabolic resistance to degradation by enzymes

C. Parenteral route of administration

D. Presence of adjuvants

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	3
Question 5	2
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	3
Question 11	2
Question 12	3
Question 13	2
Question 14	3
Question 15	4
Question 16	2
Question 17	4
Question 18	1
Question 19	1
Question 20	2
Question 21	2

Solution for Question 1:

The Ring test is utilized in Ascoli's thermo precipitin test to detect the presence of anthrax antigens. This reaction is based on a precipitation reaction in a liquid medium. In this test, the suspected antigen is layered over an antiserum in a narrow tube, and a precipitate forms at the interface if anthrax antigens are present.

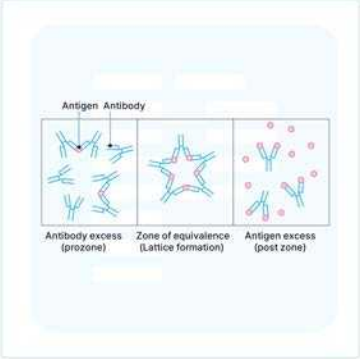
Precipitation Reactions

Soluble antigen and antibody combine to form antigen-antibody complex in the presence of electrolytes (e.g., NaCl) at the right temperature and pH to form an insoluble precipitate.

Mechanism of precipitation

Lattice hypothesis by Marrack:

- Precipitation results when a large lattice forms from alternating antigen and antibody molecules in the zone of equivalence
- In the Zones of antigen or antibody excess, the lattice does not enlarge, as the valencies of the antibody and the antigen, respectively, are fully satisfied. Prozone phenomenon occurs when there is too much antigen in relation to the amount of antibody in the test. Postzone phenomenon occurs when there is too much antigen in relation to the amount of antibody in the test.
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Types:

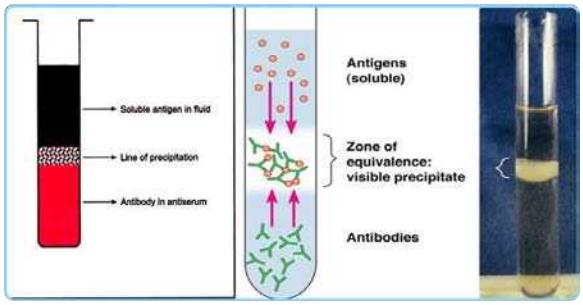
- Precipitation reaction in liquid
- Precipitation reaction in the gel

Precipitation reaction in liquid

- Ring test
- Flocculation tests

Ring test:

- Antigen layered over antiserum in a narrow tube; precipitate forms at the interface
- Application: Ascoli's thermo precipitin test - diagnosis of anthrax (Option A) Grouping of streptococci by the Lancefield technique (Option C ruled out)
- Ascoli's thermo precipitin test - diagnosis of anthrax (Option A)
- Grouping of streptococci by the Lancefield technique (Option C ruled out)
- Ascoli's thermo precipitin test - diagnosis of anthrax (Option A)
- Grouping of streptococci by the Lancefield technique (Option C ruled out)



Flocculation tests

- Slide test: Antigen and the antiserum (antibody) are placed on a slide and mixed by shaking; floccules appear Application: VDRL test for syphilis (Option B ruled out)
- Antigen and the antiserum (antibody) are placed on a slide and mixed by shaking; floccules appear
- Application: VDRL test for syphilis (Option B ruled out)
- Tube test: Tube is used instead of a slide Application: Kahn test for syphilis (Option D ruled out)
- Tube is used instead of a slide
- Application: Kahn test for syphilis (Option D ruled out)
- Antigen and the antiserum (antibody) are placed on a slide and mixed by shaking; floccules appear
- Application: VDRL test for syphilis (Option B ruled out)

• Tube is used instead of a slide

• Application: Kahn test for syphilis (Option D ruled out)

Solution for Question 2:

Impaired opsonization and phagocytosis due to C3 deficiency increases susceptibility to severe bacterial infections, particularly from pyogenic bacteria.

Complement Deficiency:

• Definition: Complement deficiencies arise from genetic defects in complement components or regulatory proteins, compromising both innate and acquired immunity.

• Clinical Manifestations: Vary from mild recurrent infections to severe life-threatening conditions.

Consequences of Complement Deficiency:

• Impaired Immune Clearance: Reduced ability to clear microbial antigens and immune complexes leads to: Recurrent Infections: Increased susceptibility to bacterial and fungal infections, especially with C3 deficiencies. Common infections include pneumonia, skin infections, and sepsis due to pyogenic bacteria (e.g., Staphylococcus aureus, Streptococcus pneumoniae). Neisserial Infections: Terminal complement deficiencies (C5-C9) increase susceptibility to Neisseria meningitidis and Neisseria gonorrhoeae, leading to meningitis, septicaemia, and gonorrhoea. (Option A ruled out) Autoimmune Diseases: It is associated with conditions like systemic lupus erythematosus (SLE) due to impaired clearance of immune complexes, resulting in tissue deposition, inflammation, and damage.

• Recurrent Infections: Increased susceptibility to bacterial and fungal infections, especially with C3 deficiencies. Common infections include pneumonia, skin infections, and sepsis due to pyogenic bacteria (e.g., Staphylococcus aureus, Streptococcus pneumoniae).

• Increased susceptibility to bacterial and fungal infections, especially with C3 deficiencies.

• Common infections include pneumonia, skin infections, and sepsis due to pyogenic bacteria (e.g., Staphylococcus aureus, Streptococcus pneumoniae).

• Neisserial Infections: Terminal complement deficiencies (C5-C9) increase susceptibility to Neisseria meningitidis and Neisseria gonorrhoeae, leading to meningitis, septicaemia, and gonorrhoea. (Option A ruled out)

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• Neisserial Infections: Terminal complement deficiencies (C5-C9) increase susceptibility to Neisseria meningitidis and Neisseria gonorrhoeae, leading to meningitis, septicaemia, and gonorrhoea. (Option A ruled out)

• Terminal complement deficiencies (C5-C9) increase susceptibility to Neisseria meningitidis and Neisseria gonorrhoeae, leading to meningitis, septicaemia, and gonorrhoea. (Option A ruled out)

• Autoimmune Diseases: It is associated with conditions like systemic lupus erythematosus (SLE) due to impaired clearance of immune complexes, resulting in tissue deposition, inflammation, and damage.

• It is associated with conditions like systemic lupus erythematosus (SLE) due to impaired clearance of immune complexes, resulting in tissue deposition, inflammation, and damage.

• Increased susceptibility to bacterial and fungal infections, especially with C3 deficiencies.

• Common infections include pneumonia, skin infections, and sepsis due to pyogenic bacteria (e.g., Staphylococcus aureus, Streptococcus pneumoniae).

• Terminal complement deficiencies (C5-C9) increase susceptibility to Neisseria meningitidis and Neisseria gonorrhoeae, leading to meningitis, septicaemia, and gonorrhoea. (Option A ruled out)

• It is associated with conditions like systemic lupus erythematosus (SLE) due to impaired clearance of immune complexes, resulting in tissue deposition, inflammation, and damage.

Specific Deficiencies and Associated Syndromes:

Deficiency	Condition/Clinical Manifestation
C1 Inhibitor Deficiency (Option C ruled out)	Hereditary angioneurotic oedema: Uncontrolled activation of classical pathway causes episodic swelling

	(can be life-threatening).
C3 Deficiency	Increased susceptibility to severe recurrent pyogenic infections; impaired opsonization, phagocytosis, and immune complex clearance.
C5-C9 Deficiencies	Increased susceptibility to bacteraemia, especially with Gram-negative diplococci; impaired formation of the membrane attack complex (MAC).

Enhanced immune complex clearance and reduced inflammation (option D) are incorrect, as C3 deficiency leads to impaired clearance and increased inflammation due to the accumulation of immune complexes.

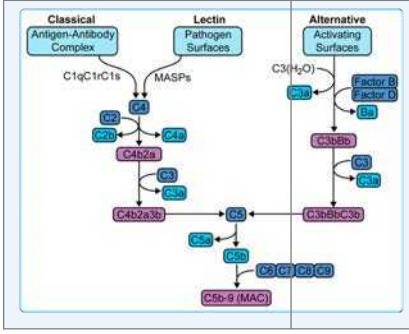
Solution for Question 3:

The convergence of the pathways at C3 convertase is crucial for activating the complement cascade, regardless of the pathway triggered. This is a key feature of the complement system, allowing for a unified response to pathogen invasion.

Complement System:

- Definition: A complex network of at least 30 proteins in the blood, crucial for innate and acquired immunity.
- Functions: Lysis: Directly kills pathogens by forming membrane pores. Opsonization: Coats pathogens to enhance recognition and uptake by phagocytes. Immune Clearance: Removes immune complexes from circulation. Inflammation: Triggers release of mediators, recruiting immune cells.
- Lysis: Directly kills pathogens by forming membrane pores.
- Opsonization: Coats pathogens to enhance recognition and uptake by phagocytes.
- Immune Clearance: Removes immune complexes from circulation.
- Inflammation: Triggers release of mediators, recruiting immune cells.
- Lysis: Directly kills pathogens by forming membrane pores.
- Opsonization: Coats pathogens to enhance recognition and uptake by phagocytes.
- Immune Clearance: Removes immune complexes from circulation.
- Inflammation: Triggers release of mediators, recruiting immune cells.

Pathways of Complement Activation:

Pathway	Trigger Mechanism	Key Components Involved
Classical Pathway	Binding of C1q to antigen-antibody complexes (Option A ruled out)	C1q, C3 convertase, Membrane Attack Complex (MAC)
Alternative Pathway	Direct binding of complement proteins to pathogens (Option B ruled out)	C3, factors B and D, C3 convertase
Lectin Pathway	Binding of mannose-binding lectin (MBL) to mannose (Option C ruled out)	MBL, MASP1, MASP2, C4, C2, C3 convertase
		

Key Points:

- Convergence: All pathways converge at the formation of the C3 convertase.
- C3 Convertase: Cleaves C3 into C3a and C3b, leading to the formation of C5 convertase and, ultimately, MAC.
- Regulation: The complement system is tightly regulated to prevent excessive activation and damage to host tissues.
- Regulatory Proteins: Include inhibitors and inactivators that control the cascade.

Solution for Question 4:

These tests are designed for use in resource-limited settings and can be used in various environments.

Immunochromatographic Card Tests:

- Purpose: Rapid detection of infectious diseases including Hepatitis B, Hepatitis C, COVID-19, and malaria.

• Advantages: Quick results (15-30 minutes) (Option A ruled out) High sensitivity and specificity Useful in resource-limited settings

• Quick results (15-30 minutes) (Option A ruled out)

• High sensitivity and specificity

• Useful in resource-limited settings

• Quick results (15-30 minutes) (Option A ruled out)

• High sensitivity and specificity

• Useful in resource-limited settings

Disease-Specific Findings:

Disease	Type of Detection	Sensitivity (%) (Option D ruled out)	Specificity (%) (Option D ruled out)	Note
Hepatitis B Virus (HBV)	Antibodies (anti-HBs) or antigens (HBsAg) (Option B ruled out)	> 95	> 99	Reliable for initial screening
Hepatitis C Virus (HCV)	Antibodies against HCV	90 - 98	95 - 100	Critical for quick identification
COVID-19	IgM and IgG antibodies	98.68 (IgM), 98.72 (IgG)	93.10 (IgM), 100 (IgG)	Effective for serological diagnosis
Rapid antig en tests	94.3	100	Effective for early diagnosis	
Malaria	Specific antigens from malaria parasites	90 - 95	~95	Timely treatment decisions are critical

Solution for Question 5:

The indirect immunofluorescence test uses a single anti-human globulin fluorescent conjugate, making it versatile and reducing the need for multiple conjugates.

Immunofluorescence:

• Technique to visualise antigen-antibody interactions using fluorescent dyes.

• Developed by Coons et al. in 1942.

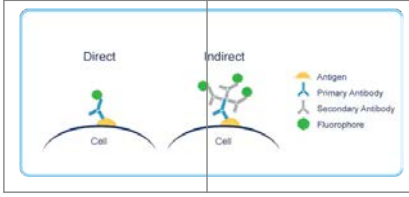
• Diagnostic and research applications for identifying antigens in tissues.

Principle of Immunofluorescence:

• Fluorescence: A substance absorbs specific light wavelengths and emits light at a longer wavelength.

• Fluorescent Dyes: Appear bright when exposed to UV light, allowing visualisation of antibody-bound antigens under a microscope.

Types of Immunofluorescence Tests:

Type	Description	Example	Limitation
Direct Immunofluorescence	Detects antigen directly using a fluorescent-labelled antibody. (Option C ruled out)	Rabies diagnosis via brain smears (Option D ruled out)	Requires separate fluorescent conjugates for each antigen. (Option A ruled out)
Indirect Immunofluorescence	It uses antiglobulin fluorescent conjugate to detect human antibodies, enhancing versatility.	Fluorescent treponemal antibody test for syphilis	Allows the use of a single anti-human conjugate.
			

Procedure for Indirect Immunofluorescence (e.g., Treponemal Antibody Test):

Applications of Fluorescent Dyes in Immunology:

• Conjugation with Complement Proteins: Enables detection of antigens/antibodies.

• Sandwich Technique: Used to detect antibodies involving a labelled secondary antibody.

• Immunohistochemistry: Combines serology specificity with histology for tissue visualisation.

• Flow Cytometry: Fluorescence-based cell identification and counting with specific markers.

Advantages and Disadvantages of Immunofluorescence:

Advantages	Disadvantages
High sensitivity for detecting small amounts of antigen/antibody.	Non-specific fluorescence due to autofluorescence or non-specific binding.
High specificity with target-specific antibodies.	Photobleaching (loss of fluorescence over time with light exposure).
Versatile: applicable to various samples (tissues, cells, microorganisms).	High cost of equipment and reagents.

Allows direct visualisation, aiding interpretation.	

Solution for Question 6:

The sensitivity of the VITROS® SARS-CoV-2 Ag Test is reported to be 94.3% is true, as it accurately reflects the performance metrics of this CLIA-based antigen detection kit.

Chemiluminescent Immunoassays (CLIA):

- Definition: Advanced diagnostic tools using chemiluminescence to detect specific antibodies or antigens in clinical samples.
- Mechanism: Utilises a chemiluminescent reaction that emits light due to a chemical reaction. (Option A ruled out) Highly sensitive; capable of detecting low analyte concentrations in serum or plasma. (Option C ruled out)
- Utilises a chemiluminescent reaction that emits light due to a chemical reaction. (Option A ruled out)
- Highly sensitive; capable of detecting low analyte concentrations in serum or plasma. (Option C ruled out)
- Utilises a chemiluminescent reaction that emits light due to a chemical reaction. (Option A ruled out)
- Highly sensitive; capable of detecting low analyte concentrations in serum or plasma. (Option C ruled out)

Applications in SARS-CoV-2 Detection:

Application	Details	Performance Metrics
Neutralising Antibodies	Automated CLIA was developed for detecting neutralising antibody (NAb) levels against SARS-CoV-2.	Sensitivity: 87.4% Specificity: 78.1%
Antigen Detection	VITROS® SARS-CoV-2 Ag Test evaluated against RT-PCR and rapid antigen tests (RATs).	Sensitivity: 94.3% Specificity: 100%

Comparative Performance:

• Serological Assays: CLIA methods effectively detected SARS-CoV-2 antibodies over time. Some assays maintained high sensitivity, while others showed reduced performance after several months. Indicates the necessity for periodic reassessment of assay parameters.

• Some assays maintained high sensitivity, while others showed reduced performance after several months.

• Indicates the necessity for periodic reassessment of assay parameters.

• Some assays maintained high sensitivity, while others showed reduced performance after several months.

• Indicates the necessity for periodic reassessment of assay parameters.

Diagnostic Accuracy:

• Clinical Settings: Different CLIA methodologies showed diagnostic accuracies. Sensitivities exceeding 85% for previous COVID-19 infections. High specificity across various tests, confirming CLIA's reliability in serological testing.

• Sensitivities exceeding 85% for previous COVID-19 infections.

• High specificity across various tests, confirming CLIA's reliability in serological testing.

• Sensitivities exceeding 85% for previous COVID-19 infections.

• High specificity across various tests, confirming CLIA's reliability in serological testing.

Neutralising antibody levels against SARS-CoV-2 can only be detected using RT-PCR (Option D), which is incorrect because specific automated CLIA assays are developed to detect neutralising antibodies, not just RT-PCR.

Solution for Question 7:

Competitive ELISA involves competition between the unknown antigen (sample) and a known antigen for binding to a limited amount of antibody. Higher antigen levels result in less colour change, creating an inverse curve.

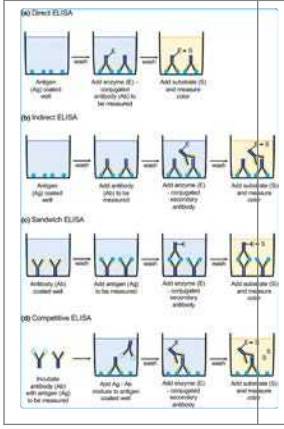
ELISA Types:

ELISA (Enzyme-linked immunosorbent assay) is a popular serological method known for its versatility, sensitivity, simplicity, and cost-effectiveness. It is commonly used in diagnostic labs due to commercial test kits and automation options.

Types of ELISA:

Type	Description	Application Example
Homogeneous ELISA	One-step procedure; no separation of bound and free fractions needed. Suitable for analysing haptens like drugs.	Enzyme-multiplied immunoassay technique (EMIT) for drug detection
Heterogeneous ELISA	Multi-step process requiring separation of bound and free fractions (e.g., by centrifugation). Commonly used for microbial antigen and antibody detection.	Standard ELISA

ELISA Methods:

Method	Description	Example
Sandwich ELISA (Option A ruled out)	Used to detect antigens by capturing them between two antibodies (capture and detection antibodies).	Detection of rotavirus antigen in faeces
Indirect ELISA (Option B ruled out)	Used to detect antibodies in the sample.	Anti-HIV antibody test
Competitive ELISA	Antigen in the sample competes with known antigen for a limited amount of antibody; produces an inverse curve (less colour at higher antigen levels).	Hapten detection
Capture ELISA (Option D ruled out)	Highly specific for detecting specific antigens.	Various immunometric tests
		

Sandwich ELISA Procedure:

ELISA Modifications:

Modification	Description and Application	Example

Cylinder/Cassette ELISA	For rapid, small-scale testing using disposable cassettes. Results are read visually without special equipment.	Dot Blot Assay for HIV detection
Sandwich ELISA	Used specifically for antigen detection.	Detection of food toxins and adulterants

Key Applications of ELISA:

- Disease Diagnosis: HIV, hepatitis, and other infectious diseases.

Food Safety: Detection of toxins and adulterants.

Solution for Question 8:

No lysis of sheep red blood cells (SRBCs) indicates a positive CFT. In a positive test, the complement is fixed by the immune complex formed between antigen and antibody, leaving no free complement to lyse the SRBCs.

Complement Fixation Test (CFT):

- Purpose: Detects specific antigens or antibodies in serum by measuring complement fixation.
- Principle: Immune complex (antigen-antibody binding) fixes complement, which is then detected using sensitised sheep red blood cells (SRBCs).

Procedure of CFT:

Step	Description
Step 1: Complement Fixation	1. Heat patient serum to inactivate native complement. 2. Add known antigen or antibody and complement. 3. Immune complex fixes the complement if a specific antigen or antibody is present.
Step 2: Indicator Stage	1. Add sensitised SRBCs (coated with amboceptor) to the mixture. 2. Positive CFT: Complement fixed, no SRBC lysis. 3. Negative CFT: Free complement lyses SRBCs.

Applications:

- Sensitivity: Detects 0.04 mg of antibody nitrogen and 0.1 mg of antigen.

• Example: Wassermann reaction for syphilis diagnosis.

Limitations:

• Requires precise standardisation of reagents.

• Sera may contain non-specific complement inhibitors, leading to false positives.

Indirect Complement Fixation Test:

• Modification: Used for sera that cannot fix guinea pig complement (e.g., avian/mammalian sera).

• Method: Add standard antiserum that can fix complement after the first CFT stage. Positive Result: Hemolysis occurs if antigen-antibody binding has consumed the complement.

• Positive Result: Hemolysis occurs if antigen-antibody binding has consumed the complement.

• Positive Result: Hemolysis occurs if antigen-antibody binding has consumed the complement.

Other Complement-Dependent Serological Tests:

Test Type	Description
Immune Adherence	Bacteria aggregate and adhere to cells (e.g., erythrocytes) in the presence of complement and specific antibodies.
Immobilisation Test	Antibodies inhibit bacterial motility with complement (e.g., Treponema pallidum immobilisation test).
Cytolytic/Cytocidal Tests	Bacteria are lysed or killed when mixed with specific antibodies and complement.

Lysis of sheep red blood cells (SRBCs) (Option A) indicates a negative result, as it shows that the complement was not fixed, and the SRBCs are free to be lysed.

Formation of an immune complex (Option B) occurs in both positive and negative results but does not directly indicate a positive outcome.

Presence of active complement in the serum (Option C) before heating could interfere with the test, but it is not an indicator of a positive result.

Solution for Question 9:

The indirect Coombs test checks for antibodies in the mother's serum that may attack Rh-positive red blood cells, which is important at this stage in assessing the risk of Rh incompatibility in the fetus.

Coombs (Antiglobulin) Test

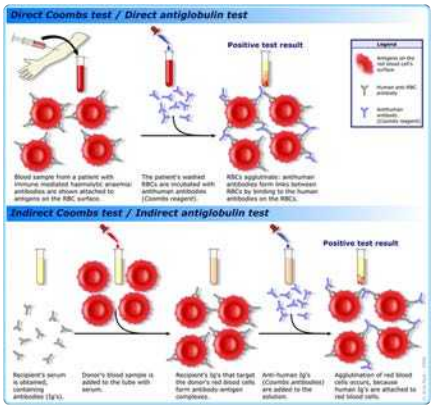
Originally developed for the detection of anti-Rh antibodies that do not agglutinate Rh-positive erythrocytes in saline.

Principle:

- Incomplete anti-Rh antibodies coat Rh-positive red cells without causing agglutination.
- When these coated cells are washed and treated with Coombs serum (anti-human globulin), agglutination occurs

Types:

- Direct Coombs Test: Detects in vivo sensitization of red cells with antibodies, used in cases like hemolytic disease of the newborn due to Rh incompatibility (sample for test is blood from the baby) (Option C ruled out)
- Indirect Coombs Test: Detects in vitro sensitization by adding patient serum to test red cells, useful for blood transfusion compatibility and antibody screening (sample for test is blood from the mother)



Applications:

- Initially, for anti-Rh antibody detection.
- It also detects other incomplete or non-agglutinating antibodies, such as in brucellosis.

To determine the mother's ABO blood type (Option A): It is important to know the mother's ABO blood type for managing potential blood group incompatibilities. However, it is not the primary purpose of the indirect Coombs test at this stage, especially considering she is Rh-negative and having a second pregnancy. This option is incorrect.

Solution for Question 10:

Direct agglutination refers to the clumping of particles (such as cells) that have specific antigens on their surface in the presence of corresponding antibodies. When blood is mixed with anti-A or anti-B antibodies, direct agglutination occurs if the specific antigens are present in the red blood cells.

Agglutination Reactions

- Clumping of particulate antigens with specific antibodies in the presence of electrolytes at the right temperature and pH
- More sensitive than precipitation for antibody detection

Principle:

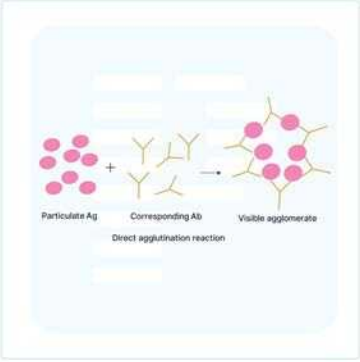
- Agglutination occurs optimally when antigens and antibodies react in equivalent proportions
- The zone phenomenon may be seen when either an antibody or an antigen is in excess
- Incomplete or monovalent antibodies do not cause agglutination, though they combine with the antigen

Types

Direct (Active) Agglutination: Antibody reacts directly with particulate antigen

- Slide Agglutination: Used for bacterial identification, blood grouping, and cross-matching (Option C)
- Tube Agglutination: Quantitative; used in diagnosing brucellosis, typhus, and for the Widal test
- Heterophile Agglutination Tests: (Option A ruled out) Weil-Felix Reaction: Diagnoses typhus fevers Streptococcus MG Test: Diagnoses primary atypical pneumonia Paul-Bunnell Test: For infectious mononucleosis
- Weil-Felix Reaction: Diagnoses typhus fevers
- Streptococcus MG Test: Diagnoses primary atypical pneumonia
- Paul-Bunnell Test: For infectious mononucleosis
- Coombs (Antiglobulin) Test: Detects anti-Rh antibodies and antibodies in brucellosis
- Weil-Felix Reaction: Diagnoses typhus fevers
- Streptococcus MG Test: Diagnoses primary atypical pneumonia

• Paul-Bunnell Test: For infectious mononucleosis



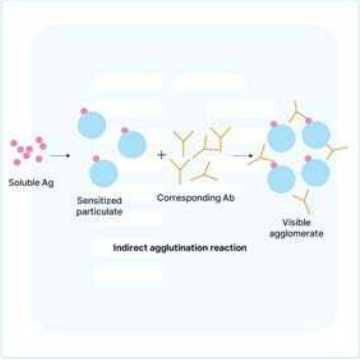
Indirect (passive) Agglutination: Soluble antigens attached to carrier particles make agglutination tests more sensitive and convenient (Option D ruled out)

• Carrier Particles: Red blood cells, latex particles, or bentonite

• Hemagglutination: Detects rheumatoid arthritis (RA) factor

• Latex Agglutination: Detects anti-streptolysin O (ASO), C-reactive protein (CRP), RA factor, and human chorionic gonadotropin (HCG)

• Co-agglutination: Uses Staphylococcus aureus with Protein A to bind antibodies, causing visible clumping when reacting with specific antigens (Option B ruled out)



Solution for Question 11:

In radial immunodiffusion (single diffusion two-dimension), the antigen is placed in a central well and diffuses outward through a gel containing evenly distributed antibodies. As the antigen

diffuses radially, it forms a circular precipitation ring. The size of this ring is proportional to theantigen concentration.

Precipitation Reactions in Gel

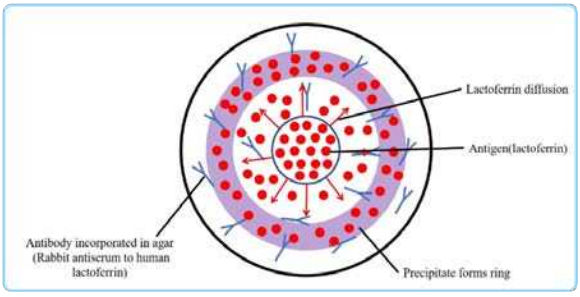
- Immunodiffusion in a soft agar or agarose gel forms visible, stable bands of precipitation that can be stained
- Advantages: Distinct, preservable precipitation bands allow easy observation of different antigens

Types and Modifications:

- One-Dimension: Agar gel in a test tube
- Two-Dimension: Agar gel on a flat surface (slide or petri dish)

Single Diffusion: Antigen diffuses towards antibody

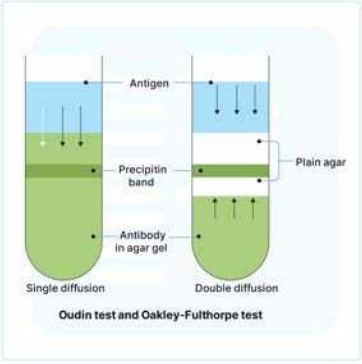
- One-Dimension (Oudin Procedure)
- Two-Dimension (Radial Immunodiffusion): Antigen diffuses radially, forming ring-shaped bands (Option B) Application: Screening sera for antibodies (e.g., influenza viruses)
- Application: Screening sera for antibodies (e.g., influenza viruses)
- Application: Screening sera for antibodies (e.g., influenza viruses)



Double Diffusion: Antigen and antibody move toward each other.

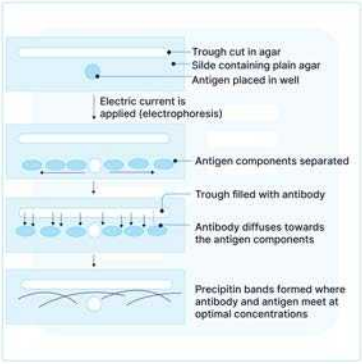
- One-Dimension (Oakley-Fulthorpe Procedure) (Option C ruled out)
- Two-Dimension (Ouchterlony Procedure): (Option A ruled out) Compares different antigens and antisera directly. Application: Elek test for diphtheria toxigenicity.
- Compares different antigens and antisera directly.
- Application: Elek test for diphtheria toxigenicity.
- Compares different antigens and antisera directly.
- Application: Elek test for diphtheria toxigenicity.





Immunoelectrophoresis:

• Involves the electrophoretic separation of a composite antigen (such as serum) into its constituent proteins, followed by immunodiffusion against its antiserum, resulting in separate precipitin lines, indicating a reaction between each individual protein with its antibody.



Electroimmunodiffusion:

Combines electrophoresis and immunodiffusion by electrically driving antigen and antibody diffusion.

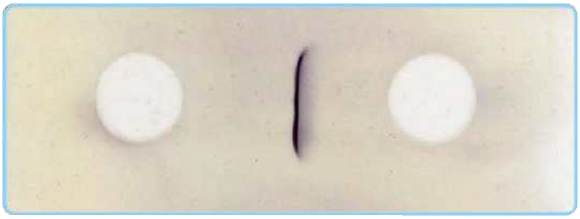
• Counterimmunoelectrophoresis (CIE): (Option D ruled out) Electrophoresis of antigen and antibody in opposite directions; precipitation forms between them. Application: Detects antigens like alpha-fetoprotein and specific antigens of Cryptococcus and Meningococcus in cerebrospinal fluid.

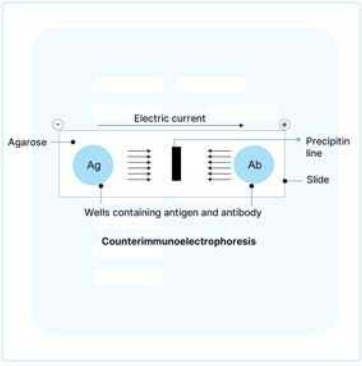
• Electrophoresis of antigen and antibody in opposite directions; precipitation forms between them.

• Application: Detects antigens like alpha-fetoprotein and specific antigens of Cryptococcus and Meningococcus in cerebrospinal fluid.

• Electrophoresis of antigen and antibody in opposite directions; precipitation forms between them.

• Application: Detects antigens like alpha-fetoprotein and specific antigens of Cryptococcus and Meningococcus in cerebrospinal fluid.





• Rocket Electrophoresis (One-Dimension Single Diffusion): Antigen, in increasing concentrations, is placed in wells punched in the set gel. The antigen is then electrophoresed into the antibody-containing agarose Application: Quantitative estimation of antigens

• Antigen, in increasing concentrations, is placed in wells punched in the set gel.

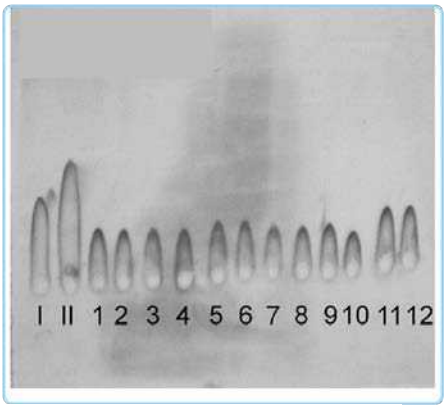
• The antigen is then electrophoresed into the antibody-containing agarose

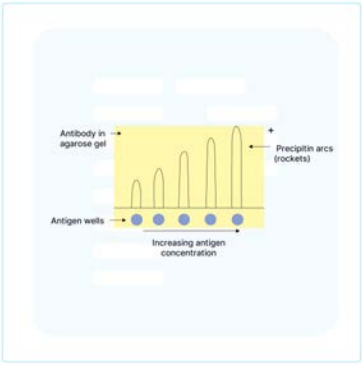
• Application: Quantitative estimation of antigens

• Antigen, in increasing concentrations, is placed in wells punched in the set gel.

• The antigen is then electrophoresed into the antibody-containing agarose

• Application: Quantitative estimation of antigens





Solution for Question 12:

Pro-antigens need metabolic conversion to form reactive intermediates, which then bind to proteins to trigger immune responses.

Haptens, Pro-Antigens, and Incomplete Antigens:

Term	Definition	Mechanism	Clinical Relevance
Haptens	LMW compounds (<1000 Da) that require binding to carrier proteins to trigger immune response	Bind to proteins to form conjugates, recognized by T cells, stimulating humoral and cellular responses (Option A ruled out)	Associated with allergic contact dermatitis (e.g., nickel, penicillin)
Pro-Antigens	Substances needing metabolic conversion to become immunogenic	Metabolized into reactive intermediates, then bind proteins, forming	Example: Urushiol in poison ivy, requiring activation to cause

	(Option B ruled out)	immune-triggeringcomplexes	allergy
Incomplete Antigens	Molecules that bind to antibodies or T-cell receptors but don't provoke full immune response alone (Option D ruled out)	Require attachment to larger carrier molecules to be immunogenic	Involved in immune tolerance and sensitization mechanisms

Solution for Question 13:

IgD and IgM occur on the surface of unstimulated B lymphocytes. It acts as a recognition receptor for antigen and causes specific stimulation of the B cell - either activation or cloning to produce antibodies or suppression.

IgA: (Option A ruled out)

- Two types – IgA1 and IgA2
- Major immunoglobulin in colostrum, saliva, and tears
- Two Forms: Monomeric serum IgA and dimeric secretory IgA
- Secretory IgA: Two monomer units joined together at their carboxy terminals by a glycopeptide termed the J chain Produced by plasma cells near mucosal or glandular epithelium Secretory piece or secretory component in the IgA protects IgA from bacterial proteases, especially in areas with abundant bacterial flora (e.g., intestines) Role in Local Immunity: Concentrated in secretions, it forms an "antibody paste" on mucosal surfaces to protect against respiratory and intestinal pathogens. Resistance: Relatively resistant to digestive enzymes and reducing agents Activates the alternative complement pathway
- Two monomer units joined together at their carboxy terminals by a glycopeptide termed the J chain
- Produced by plasma cells near mucosal or glandular epithelium
- Secretory piece or secretory component in the IgA protects IgA from bacterial proteases, especially in areas with abundant bacterial flora (e.g., intestines)
- Role in Local Immunity: Concentrated in secretions, it forms an "antibody paste" on mucosal surfaces to protect against respiratory and intestinal pathogens.
- Resistance: Relatively resistant to digestive enzymes and reducing agents
- Activates the alternative complement pathway

• Two monomer units joined together at their carboxy terminals by a glycopeptide termed the J chain

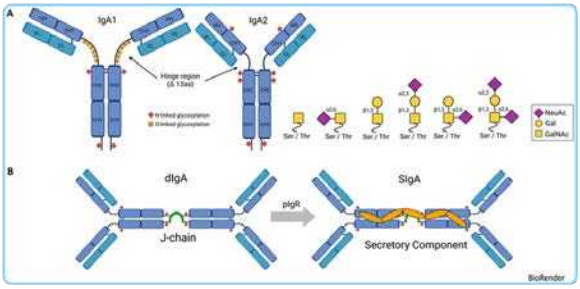
• Produced by plasma cells near mucosal or glandular epithelium

• Secretory piece or secretory component in the IgA protects IgA from bacterial proteases, especially in areas with abundant bacterial flora (e.g., intestines)

• Role in Local Immunity: Concentrated in secretions, it forms an "antibody paste" on mucosal surfaces to protect against respiratory and intestinal pathogens.

• Resistance: Relatively resistant to digestive enzymes and reducing agents

• Activates the alternative complement pathway



IgD: (Option B)

• IgD and IgM occur on the surface of unstimulated B lymphocytes

• Function: Acts as recognition receptors for antigens It causes specific stimulation of the B cell - either activation or cloning, to produce antibodies or suppression.

• Acts as recognition receptors for antigens

• It causes specific stimulation of the B cell - either activation or cloning, to produce antibodies or suppression.

• Acts as recognition receptors for antigens

• It causes specific stimulation of the B cell - either activation or cloning, to produce antibodies or suppression.

IgE: (Option C ruled out)

• Heat-labile (sensitive to heat)

• It has an affinity for the surface of tissue cells (particularly mast cells) of the same species (homocytotropism)

• Mediates Prausnitz-Küstner Reaction

• Distribution: Primarily found in extravascular spaces

• Elevated Levels: In atopic conditions (type I allergies) like asthma, hay fever, eczema, and parasitic infections

• Function: Protects against pathogens by triggering mast cell degranulation and releasing inflammatory mediators

Immunoglobulin Functions:

• IgG: Protects body fluids

• IgA: Protects body surfaces

• IgM: Guards the bloodstream

• IgE: Mediates allergic hypersensitivity

• IgD: Present on the surface of B lymphocytes

Solution for Question 14:

IgG is the only maternal antibody transported across the placenta, providing passive immunity to newborns.

IgG

• Four types – IgG1, IgG2, IgG3, and IgG4

• It is the main serum immunoglobulin, constituting 80% of total serum antibodies

• Distribution: Both intravascular and extravascular

• Only maternal antibody transported across the placenta, providing passive immunity to newborns (Option C) mop

• Function: Binds to microorganisms, enhancing their phagocytosis Participates in complement fixation, precipitation, and neutralisation of toxins/viruses Part of chronic inflammation

• Binds to microorganisms, enhancing their phagocytosis

• Participates in complement fixation, precipitation, and neutralisation of toxins/viruses

• Part of chronic inflammation

• Clinical Application: Used in isoimmunisation for Rh-negative mothers with anti-Rh(D) IgG during delivery.

• Binds to microorganisms, enhancing their phagocytosis

• Participates in complement fixation, precipitation, and neutralisation of toxins/viruses

• Part of chronic inflammation

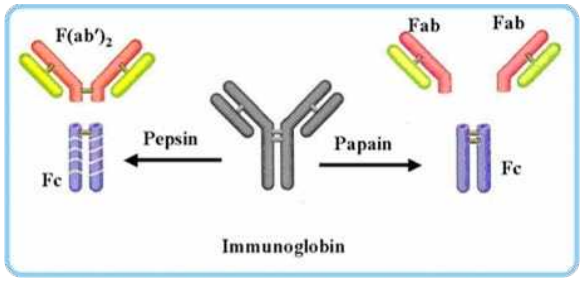
IgM

- It is a heavy molecule, hence called 'the millionaire molecule.'
- Structure: Made of five four-peptide subunits; requires a J chain for polymerisation (Option A ruled out)
- Valency: 10 (Option B ruled out)
- Distribution: Primarily intravascular (80% of IgM)
- Early Production: First antibody synthesized by the fetus, starting around 20 weeks of gestation (Option D ruled out)
- Short-Lived: IgM antibodies degrade sooner than IgG
- Immune Hemolysis: One IgM molecule can induce hemolysis, unlike IgG, which requires 1,000 molecules for similar effect
- Opsonization Efficiency: 500-1,000 times more effective in opsonization than IgG

Solution for Question 15:

Pepsin digestion actually cleaves below the disulfide bridge in the hinge region, leading to the formation of an F(ab')2 fragment and additional smaller fragments.

Enzymatic Digestion of Immunoglobulins



Papain Digestion

- Mechanism: Cleaves Ig above the disulfide bridge in the hinge region.
- Cleaves Ig above the disulfide bridge in the hinge region.
- Results: Generates three fragments: (Option A) Two Fab Fragments: Soluble fragments that bind to antigens (Ag). One Fc Fragment: Insoluble, crystallizes when cold.
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• Two Fab Fragments: Soluble fragments that bind to antigens (Ag).

• Soluble fragments that bind to antigens (Ag).

• One Fc Fragment: Insoluble, crystallizes when cold.

• Insoluble, crystallizes when cold.

• Cleaves Ig above the disulfide bridge in the hinge region.

• Generates three fragments: (Option A) Two Fab Fragments: Soluble fragments that bind to antigens (Ag). One Fc Fragment: Insoluble, crystallizes when cold.

• Two Fab Fragments: Soluble fragments that bind to antigens (Ag).

• Soluble fragments that bind to antigens (Ag).

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• Two Fab Fragments: Soluble fragments that bind to antigens (Ag).

• Soluble fragments that bind to antigens (Ag).

• One Fc Fragment: Insoluble, crystallizes when cold.

• Insoluble, crystallizes when cold.

• Soluble fragments that bind to antigens (Ag).

• Insoluble, crystallizes when cold.

Pepsin Digestion

• Mechanism: Cleaves the Ig molecule below the disulfide bridge in the hinge region.

• Cleaves the Ig molecule below the disulfide bridge in the hinge region.

• Results: Produces: (Option C) One F(ab')2 Fragment: Two Fab subunits are still bound together. Many Smaller Fragments: Additional peptide fragments result from further cleavage.

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• One F(ab')2 Fragment: Two Fab subunits are still bound together.

• Two Fab subunits are still bound together.

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• Cleaves the Ig molecule below the disulfide bridge in the hinge region.

• Produces: (Option C) One F(ab')2 Fragment: Two Fab subunits are still bound together. Many Smaller Fragments: Additional peptide fragments result from further cleavage.

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- Additional peptide fragments result from further cleavage.
- One F(ab')2 Fragment: Two Fab subunits are still bound together.
- Two Fab subunits are still bound together.
- Many Smaller Fragments: Additional peptide fragments result from further cleavage.
- Additional peptide fragments result from further cleavage.
- Two Fab subunits are still bound together.
- Additional peptide fragments result from further cleavage.

Mercaptoethanol (Option B)

- Function: Irreversibly cleaves disulfide bonds.
- Irreversibly cleaves disulfide bonds.
- Outcome: Separates all heavy chains and light chains of the immunoglobulin.
- Separates all heavy chains and light chains of the immunoglobulin.
- Irreversibly cleaves disulfide bonds.
- Separates all heavy chains and light chains of the immunoglobulin.

Solution for Question 16:

This statement is incorrect because immunity typically takes a few weeks to develop after vaccination.

Active immunity

- The most common type of immunity developed in response to infection or vaccination.
- Characteristics: Resistance developed as a result of antigenic stimulation. Used for prophylaxis to enhance body resistance.
- Resistance developed as a result of antigenic stimulation.
- Used for prophylaxis to enhance body resistance.
- Types: Natural Active Immunity Artificial Active Immunity
- Natural Active Immunity
- Artificial Active Immunity

Naturally acquired active Immunity

- Development: Forms after exposure to antigens via natural processes (e.g., infection). The body's immune system produces antibodies in response.
- Forms after exposure to antigens via natural processes (e.g., infection).
- The body's immune system produces antibodies in response.
- Duration: It can be long-lasting (e.g., smallpox, measles, chickenpox, yellow fever). May diminish over time (e.g., diphtheria, tetanus, influenza).
- It can be long-lasting (e.g., smallpox, measles, chickenpox, yellow fever).
- May diminish over time (e.g., diphtheria, tetanus, influenza).
- Premunition: A special type of immunity observed in syphilis, is also called infection immunity. Rapidly acquired, short-lived, and partially effective. Immunity lasts only as long as the original infection is active.
- A special type of immunity observed in syphilis, is also called infection immunity.
- Rapidly acquired, short-lived, and partially effective.
- Immunity lasts only as long as the original infection is active.
- Forms after exposure to antigens via natural processes (e.g., infection).
- The body's immune system produces antibodies in response.
- It can be long-lasting (e.g., smallpox, measles, chickenpox, yellow fever).
- May diminish over time (e.g., diphtheria, tetanus, influenza).
- A special type of immunity observed in syphilis, is also called infection immunity.
- Rapidly acquired, short-lived, and partially effective.
- Immunity lasts only as long as the original infection is active.

Artificially acquired active immunity

- Definition: Induced by vaccination.
- Induced by vaccination.
- Process: Carefully chosen antigens (e.g., vaccines, toxoids) are intentionally introduced. The immune system synthesizes antibodies in response.
- Carefully chosen antigens (e.g., vaccines, toxoids) are intentionally introduced.
- The immune system synthesizes antibodies in response.
- Immunity duration: Typically provides long-term immunity.
- Typically provides long-term immunity.
- Vaccines available: Against diseases such as cholera, tuberculosis, plague, pneumonia, Rocky Mountain spotted fever, smallpox, polio, tetanus, influenza, measles, rabies, and yellow fever.

• Against diseases such as cholera, tuberculosis, plague, pneumonia, Rocky Mountain spotted fever, smallpox, polio, tetanus, influenza, measles, rabies, and yellow fever.

• Toxoids: Available for protection against diphtheria and tetanus, where major effects are due to toxins.

• Available for protection against diphtheria and tetanus, where major effects are due to toxins.

• Induced by vaccination.

• Carefully chosen antigens (e.g., vaccines, toxoids) are intentionally introduced.

• The immune system synthesizes antibodies in response.

• Typically provides long-term immunity.

• Against diseases such as cholera, tuberculosis, plague, pneumonia, Rocky Mountain spotted fever, smallpox, polio, tetanus, influenza, measles, rabies, and yellow fever.

• Available for protection against diphtheria and tetanus, where major effects are due to toxins.

Options(A, C, and D) are accurate. The vaccine provides long-lasting immunity, contains a live attenuated virus, and represents a form of artificially acquired active immunity.

Learning Objective:

Category	Active Immunity	Passive immunity
Source	Produced by the individual's own immune system.	Acquired from an external source
Mechanism	The immune system responds to the antigen by producing its own antibodies	Pre-formed antibodies are transferred into the individual.
Onset of Immunity	It takes time to develop	Immediate protection after administration.
Duration of Immunity	Long-lasting	Short term
Quality	Has high quality and is more effective	This is transient and less effective
Memory Formation	The immune system retains the memory of the pathogen.	No memory is formed; immunity is not long-term.
On subsequent dose	Shows booster effect	Is less effective
In Immunodeficient	It may be ineffective or insufficient in patients with severe immunodeficiencies	Critical for immunodeficient individuals

Solution for Question 17:

• The clinical features of this patient suggest multiple myeloma, which is characterized by the production of monoclonal antibodies from a single clone of B cells.

• This results in a homogeneous population of antibodies targeting a specific epitope.

Feature	Monoclonal Antibodies	Polyclonal Antibodies
Definition	Homogeneous population produced by a single clone of plasma B cells	The mixture of immunoglobulin molecules from multiple clones of plasma B cells
Antibody Population	Homogeneous	Heterogeneous
Epitope Interaction	Interact with a particular epitope on antigen	Recognize different epitopes on the same antigen (Option A)
Cross Reactivity	Low cross-reactivity	Increased likelihood for cross-reactivity with similar antigens (Option C)
Lot Variability	Identical lots	Lot-to-lot variability
Light Chains	Same type of light chain is produced (e.g., in multiple myeloma)	May have both Kappa (K) and Lambda (λ) light chains (e.g., in infections) (Option B)

Solution for Question 18:

• This scenario is indicative of infectious mononucleosis (IM).

• This statement accurately describes the principle behind the Paul-Bunnell test, which is used to diagnose infectious mononucleosis caused by the Epstein-Barr virus (EBV).

• The test identifies heterophile antibodies present in the patient's serum that can agglutinate sheep red blood cells, which is a characteristic immune response associated with this viral infection.

Heterophile Antigens

- Definition: A type of hetero-antigen present in different species that share epitopes.
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- Cross-Reactivity: Antibodies produced against an antigen from one species can react with the heterophile antigen from another species and vice versa (Option D)
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- Diagnostic Applications: Heterophile antigens are used in various serological tests to detect antibodies in patient serum. Examples of Tests: Weil-Felix Reaction: Used to diagnose certain infections, such as rickettsial infections. Paul-Bunnell Test: Primarily used for diagnosing infectious mononucleosis. Cold Agglutination Test: This test is used to detect specific antibodies that cause agglutination at low temperatures. This test is primarily used to identify mycoplasma pneumonia. Streptococcus MG Test: Used to detect antibodies associated with streptococcal infections.
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- A type of hetero-antigen present in different species that share epitopes.
- Antibodies produced against an antigen from one species can react with the heterophile antigen from another species and vice versa (Option D)
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• Weil-Felix Reaction: Used to diagnose certain infections, such as rickettsial infections.

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• Weil-Felix Reaction: Used to diagnose certain infections, such as rickettsial infections.

• Paul-Bunnell Test: Primarily used for diagnosing infectious mononucleosis.

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• This test is primarily used to identify mycoplasma pneumonia.

• Streptococcus MG Test: Used to detect antibodies associated with streptococcal infections.

• This test is primarily used to identify mycoplasma pneumonia.

Paul-Bunnell Test

• Purpose: Diagnosis of infectious mononucleosis caused by the Epstein-Barr virus.

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• Method: Utilizes sheep red blood cells (RBCs) as antigens to detect heterophile antibodies in the patient’s serum.

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• Principle: In infectious mononucleosis, heterophile antibodies agglutinate sheep erythrocytes. These antibodies can also occur after serum injections and in some normal individuals.

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• These antibodies can also occur after serum injections and in some normal individuals.

• Procedure: The inactivated serum is prepared by heating at 56°C for 30 minutes (Option C) The serum is mixed in doubling dilutions with an equal volume of a 1% sheep erythrocyte suspension. Incubate at 37°C for 4 hours.

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• Incubate at 37°C for 4 hours.

• Interpretation: Examine for agglutination. An agglutination titer of 100 or above suggests infectious mononucleosis (Option B)

• Examine for agglutination.

• An agglutination titer of 100 or above suggests infectious mononucleosis (Option B)

• Diagnosis of infectious mononucleosis caused by the Epstein-Barr virus.

• Utilizes sheep red blood cells (RBCs) as antigens to detect heterophile antibodies in the patient's serum.

• In infectious mononucleosis, heterophile antibodies agglutinate sheep erythrocytes.

• These antibodies can also occur after serum injections and in some normal individuals.

• The inactivated serum is prepared by heating at 56°C for 30 minutes (Option C)

• The serum is mixed in doubling dilutions with an equal volume of a 1% sheep erythrocyte suspension.

• Incubate at 37°C for 4 hours.

• Examine for agglutination.

• An agglutination titer of 100 or above suggests infectious mononucleosis (Option B)

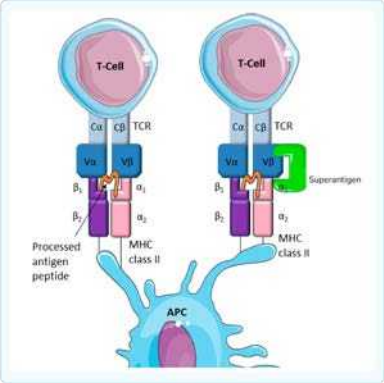
Note: The Paul-Bunnell test uses sheep erythrocytes; the Monospot test, horse red cells.

Solution for Question 19:

• This is a case of toxic shock syndrome (TSS), likely due to a superantigen produced by Staphylococcus aureus.

• These active proteins bind directly to the lateral aspect of the T cell receptor (TCR) β chain, bypassing the need for specific binding to major histocompatibility complex (MHC) molecules.

• This mechanism allows superantigens to engage a large number of T cells regardless of their antigen specificity, leading to extensive T cell activation. As a result, there is a massive release of cytokines.



Superantigens:

- Definition: Superantigens are protein molecules that activate large numbers of T cells regardless of their antigenic specificity.
- Binding Mechanism: Bind directly to the lateral aspect of the T cell receptor (TCR) β chain. Unlike conventional antigens, which bind to the TCR α/β heterodimer via the MHC molecules (Option B)
- Bind directly to the lateral aspect of the T cell receptor (TCR) β chain.
- Unlike conventional antigens, which bind to the TCR α/β heterodimer via the MHC molecules (Option B)
- Molecular Weight: Typically medium-sized proteins (22-29 kDa) (Option C)
- Stability: Highly resistant to proteases and denaturation.
- Target Cells: Primarily activate CD4+ T cells.
- Cytokine Release: Induce massive release of cytokines, such as IL-2.
- Effects on Lymphocytes: Lead to significant proliferation of lymphocytes (Option D)
- Cytokine Cascade: Result in further release of various cytokines, profoundly affecting the immune system.
- Bind directly to the lateral aspect of the T cell receptor (TCR) β chain.
- Unlike conventional antigens, which bind to the TCR α/β heterodimer via the MHC molecules (Option B)

Examples:

Source	Superantigens	Associated Diseases
Staphylococcus aureus	Enterotoxins	Food poisoning, Toxic shock syndrome, Multiple sclerosis
Group A streptococci	Pyrogenic exotoxins	Shock, Rheumatic heart disease, Psoriasis

Mycobacterium tuberculosis	Not identified	Tuberculosis
HIV	Nef (negative regulatory factor)	AIDS
Rabies virus	Nucleocapsid protein	Rabies
Epstein-Barr virus	Not identified	B cell lymphoma

Solution for Question 20:

T cell-dependent antigens typically result in a full range of antibody isotypes, including IgM, IgG, IgA, and IgE, rather than a limited range.

Criteria	T cell-dependent antigens	T cell-independent antigens
T cell involvement	Yes	No
Antigen interaction	Involves tertiary complex (TCR, MHC, peptide)	Involves binary complex (membrane Ig and receptor)
Antigen processing by macrophages	Yes	No
Chemical nature of the antigen	Mostly soluble proteins	Type 1: LPS; Type 2: Polysaccharides & polymers
Degradability	Easily degradable	Type 2: Poorly degradable
Complement activation	No	Type 2: can activate complement (Option A)
Isotype switching	Yes	Limited (Type 1: no; Type 2: limited)
Immunologic memory	Yes (Option D)	No
Polyclonal activation	No	Yes
Effect on antibody production	Full range (IgM, IgG, IgA, IgE)	Limited (mainly IgM and IgG3) (Option C)

Solution for Question 21:

Immunogenicity requires antigens to be broken down by phagocytic cells. Substances resistant to enzymatic degradation, like polystyrene latex, cannot be processed effectively, reducing their immunogenicity.

Factors Affecting Immunogenicity:

Factor	Description	Details
Molecular Size	Larger molecules are more immunogenic. (Option A ruled out)	High MW molecules (e.g., hemocyanins) are strongly antigenic. Small molecules (<5000 MW) are weak or non-antigenic.
Chemical Nature	Composition affects immunogenicity.	Proteins and polysaccharides are highly antigenic. Lipids and nucleic acids are weak antigens, but binding to proteins enhances their antigenicity.
Foreignness	Non-self substances are immunogenic.	"Self" substances are non-immunogenic. The degree of foreignness matters; antigens from the same species are less immunogenic than those from other species.
Susceptibility to Enzymes	Immunogenic substances must be metabolically degradable.	Antigens require a breakdown by phagocytes for immune cell recognition. Substances resistant to degradation (e.g., polystyrene latex) are non-antigenic.
Route of Administration	Route influences the immune response.	Parenteral administration (e.g., injection) elicits a stronger response than oral/nasal. (Option C ruled out) Injection site matters (e.g., hepatitis B vaccine is more effective in the deltoid than the gluteal muscle).
Antigen Dose	Dose impacts immune response strength.	An optimal dose is required for a strong response. High doses can cause immunological paralysis. Repeated exposure enhances response.

Adjuvants	Enhance immunogenicity through various mechanisms. (Option D ruled out)	Depot adjuvants (e.g., aluminium hydroxide) increase antigen persistence. Immunostimulatory adjuvants (e.g., Freund's complete) activate immune cells.
Genetic Factors	Genes influence immune response to specific antigens.	Immune response (Ir) genes control individual immune responses to particular antigens.
Age and Nutritional Status	Age and nutrition affect immune system maturity.	Immune development is gradual; infants and young children are less responsive. Malnutrition suppresses immune responses.

Previous Year Questions

1. Which blood is transfused in a patient with a Bombay blood group?

A. A

B. B

C. AB

D. O negative

2. Which test is used in the detection of long term glucose tolerance?

A. Fasting blood glucose

B. Random blood glucose

C. HbA1C

D. CBC

3. Penicillin was given to the patient and immune hemolytic anemia develops. Which type of hypersensitivity reaction is this?

A. Type 1

B. Type 2

C. Type 3

D. Type 4

4. If a patient's plasma contains elevated levels of interferon beta, it is indicative of a probable infection caused by...

A. Virus

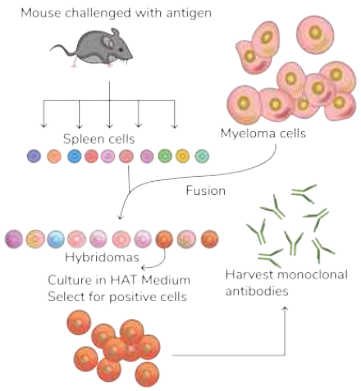
B. Bacteria

C. Fungi

D. Mycoplasma

---

5. The following image depicts :



A. Production of monoclonal antibodies

B. Cell lines for viral culture

C. Vaccine synthesis

D. Process of genetic engineering

---

6. Which one of the options below represents a site that has immunological privilege?

A. Seminiferous tubule

B. Area postrema

C. Optic nerve

D. Loop of Henle

---

7. Deficiency of which complement system increases the risk of N. meningitidis infection?

A. C1-C4 deficiency

B. C5-C9 deficiency

C. C3 deficiency

D. C2 deficiency

---

8. Which of the subsequent options triggers the classical complement cascade?

A. C1

B. C3 Convertase

C. IgA

D. Ag - Ab complex

---

9. Affinity maturation is due to:

A. Isotype switching

B. Somatic hypermutation

C. mRNA splicing

D. VDJ rearrangement

---

10. Which of the following is a crucial element involved in activating the complement through the alternative pathway?

A. C1

B. C2

C. C3

D. C4

---

11. In a case of chronic allergy, which immunoglobulin demonstrates increased levels?

A. IgA

B. IgM

C. IgE

D. IgG

---

12. The classical complement pathway is activated by the binding of ..... ?

A. C1

B. C3 convertase

C. Antibody

D. Ag-Ab complex

---

13. An unknown disease of suspected infectious origin has come up in your town. As a microbiologist, you collect the relevant samples and detect the etiological cause. Which of the following types of PCR would you do?

A. Real time PCR

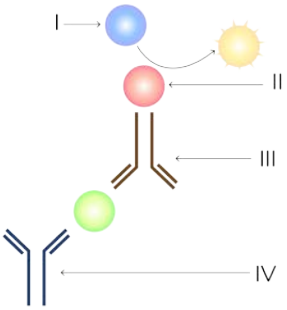
B. Nested PCR

C. Reverse transcriptase PCR

D. Multiplex

---

14. Which of the following is the enzyme conjugate in the given ELISA technique?



A. I

B. II

C. III

D. IV

---

15. A patient was suspected of having acute brucellosis. A serum sample was sent for a standard agglutination test. It came out to be negative initially, but after serial dilutions of the serum sample, the test turned out to be positive. Which of the following is responsible for the initial negative test?

A. Incomplete antibodies

B. Complement inactivation

C. Prozone phenomenon

D. Post zone phenomenon

16. Which of the subsequent organisms is responsible for the production of superantigens?

A. Staphylococcus epidermidis

B. Streptococcus pyogenes

C. Clostridium perfringens

D. Vibrio cholera

17. N. meningitidis can be caused by deficiency of ..... ?

A. C1, C2, and C4 deficiency

B. C5-C9 deficiency

C. C3 deficiency

D. C1 esterase deficiency

18. Affinity maturation is due to:

A. Isotype switching

B. Somatic hypermutation

C. mRNA splicing

D. VDJ rearrangement

19. A 10-year-old boy is brought to the ED after being stung by a bee while playing outside. Within minutes of the sting, he developed shock, respiratory failure, and

vascular collapse. What type of hypersensitivity reaction is most likely responsible?

A. IgG-mediated reaction

B. IgA-mediated hypersensitivity

C. IgE-mediated reaction

D. T cell-mediated response

---

20. A 6-month-old boy presents with recurrent bacterial and fungal infections, chronic diarrhea, and failure to thrive. He is diagnosed with severe combined immunodeficiency due to an autosomal recessive inheritance pattern. Which enzyme deficiency is responsible?

A. Phosphomannose isomerase

B. Ornithine transcarbamylase

C. Hypoxanthine-guanine phosphoribosyltransferase

D. Adenosine deaminase

---

21. The test kit shown in the image below works on the principle of:



A. ELISA

B. Immunochromatography

C. Immunofluroscence

D. Chemiluminiscence

---

22. Which of the following immunoglobulins shows the highest concentration in serum?

A. IgG

B. IgM

C. IgA

D. IgE

---

23. Antigen binding to MHC class I activates which of the following cells?

A. Cytotoxic T cells

B. B cells

C. NK cells

D. Helper cells

---

24. In a healthy adult, what is the proportion between T cells and B cells?

A. 1:1

B. 3:1

C. 1:2

D. 1:3

---

25. Vaccination against which of the following is recommended in the elderly?
Varicella zoster virus Rota virus Influenza vaccine Pneumococcal Meningitis

A. 1,2,3,4

B. 1,3

C. 1,3 and 4

D. 2,3

---

26. Which of the following is not a part of innate immunity?

A. Monocytes

B. B lymphocytes

C. Gamma delta T lymphocytes

D. Lectin pathway

---

27. In which of the following disorders is vaccination not contraindicated?

A. Digeorge syndrome

B. Wiskott Aldrich syndrome

C. Complement Deficiency

D. Ataxia telangiectasia

---

28. CD 40 marker is absent in a person. Which of the following would be seen?

A. Absent T cells

B. Absent B cells

C. Absent NK cells

D. Absent Macrophages

29. Which of the following methods can detect deoxynucleotide incorporation in a DNA?

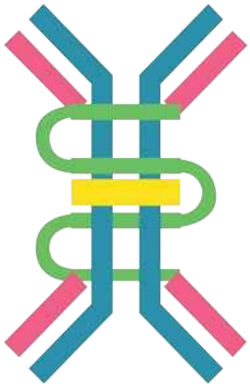
A. Flow cytometry

B. Pyrosequencing

C. FISH

D. RFLP analysis

30. Identify the class of immunoglobulin shown in the image below.



A. Ig A

B. Ig G

C. Ig M

D. Ig E

31. Anaphylaxis occurred in a boy who was eating peanut butter. This scenario exemplifies:

A. T cell-mediated cytotoxicity

B. IgE-mediated reaction

C. IgG-mediated reaction

D. Antigen-antibody complex mediated

---

32. Which specific cell plays a role in type 4/delayed type hypersensitivity?

A. Cytotoxic T Cell

B. Macrophage

C. Dendritic cell

D. Neutrophil

---

33. What is the likely diagnosis for an 11-year-old boy who presented at the pediatric outpatient department with multiple abscesses on his face, chest, and back, along with a history of frequent respiratory infections. Uponexamination, the child showed signs of atopic excoriating skin and multiple cold abscesses on his back.

A. Hyper IgM syndrome

B. Wiskott-Aldrich syndrome

C. Hyper IgE syndrome

D. Carcinoid syndrome

---

34. Which of the following statements is incorrect regarding MHC molecules?

A. MHC class II presents peptide antigen to CD4 T cells

B. The action of MHC II occurs via endocytic pathway

C. MHC Class II presents endogenous antigens

D. MHC II is present on antigen presenting cells

---

35. Which of the following investigations does not involve the use of antibody labeling?

A. ELISA

B. Radioimmunoassay

C. Immunofluorescence

D. Hemagglutination Test

---

36. In the case of a 3-year-old child with a history of recurrent sinopulmonary infections caused by encapsulated organisms, which of the following is the most probable deficiency?

A. IgG1

B. IgG2

C. IgG3

D. IgG4

---

37. Vaccine that require strain change due to evolving organism is?

A. Influenza

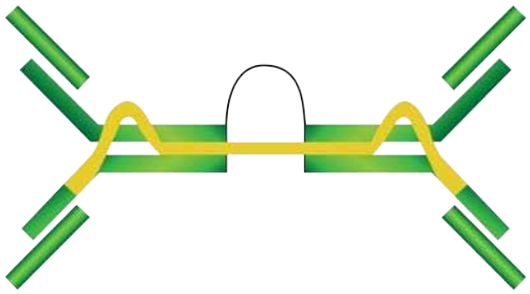
B. Diphtheria

C. Polio

D. Typhoid

---

38. Identify the class of immunoglobulin shown in the image below.



A. Ig A

B. Ig G

C. Ig M

D. Ig E

---

39. Type IV hypersensitivity is due to ?

A. Cell mediated response

B. Cell mediated and antibody mediated response

C. Antibody mediated response

D. Innate response

---

40. Due to the lack of _____, nude mice do not exhibit resistance to xenografts.

A. B cell

B. T cell

C. Both B and T cell

D. NK cell

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	2
Question 4	1
Question 5	1
Question 6	1
Question 7	2
Question 8	4
Question 9	2
Question 10	3
Question 11	3
Question 12	4
Question 13	1
Question 14	2
Question 15	3
Question 16	2
Question 17	2
Question 18	2
Question 19	3
Question 20	4
Question 21	2
Question 22	1

Question 23	1
Question 24	2
Question 25	3
Question 26	2
Question 27	3
Question 28	2
Question 29	2
Question 30	1
Question 31	2
Question 32	1
Question 33	3
Question 34	3
Question 35	4
Question 36	2
Question 37	1
Question 38	1
Question 39	1
Question 40	2

Solution for Question 1:

Correct Option D - O negative:

- The blood group that is transfused in a patient with a Bombay blood group is O negative
- Bombay blood group: No A antigen, B antigen or AB Antigen and even H antigen is not present but anti H-Ab is present
- Safest blood group for transfusion in emergency = O

Incorrect Options:

Option A, B, C:

- These are not the blood groups transfused in a patient with a Bombay blood group

Solution for Question 2:

Correct Option C - HbA1C:

- Detection of long term glucose intolerance is done by HbA1C

Incorrect Options:

Option A, B, D:

- These are not used for the detection of long term glucose intolerance

Solution for Question 3:

Correct Option B - Type 2

- Type 2 hypersensitivity reaction, also known as cytotoxic hypersensitivity, involves the activation of the immune system against specific cells or tissues. In this case, penicillin can induce the production of antibodies that bind to the patient's own red blood cells, causing them to be targeted and destroyed by the immune system. This is what happens in immune hemolytic anemia.

Incorrect Options:

Option A - Type 1: Type 1 hypersensitivity reaction involves the release of histamine and other inflammatory mediators from mast cells and basophils. This leads to immediate allergic reactions, such as hay fever, asthma, and anaphylaxis. It is not associated with immune hemolytic anemia.

Option C - Type 3: Type 3 hypersensitivity reaction occurs when immune complexes (antigen-antibody complexes) deposit in various tissues, leading to inflammation and tissue

damage. This is typically seen in diseases like systemic lupus erythematosus (SLE) or certaintypes of vasculitis. It is not directly associated with immune hemolytic anemia.

Option D - Type 4: Type 4 hypersensitivity reaction is a delayed-type hypersensitivity involving T cells and macrophages. It's associated with cell-mediated immune responses, such as contact dermatitis or granulomatous inflammation. This type of reaction is not directly related to immune hemolytic anemia caused by penicillin.

Solution for Question 4:

Correct Option A - Virus:

- Interferon beta is a type of interferon that is produced by cells in response to viral infections. Interferons are proteins that play a crucial role in the body's immune response against viral infections. They help to inhibit viral replication, activate immune cells, and enhance the overall antiviral defense mechanisms.
- The elevated levels of interferon beta in the patient's plasma suggest that the patient is experiencing a viral infection.

Incorrect Options:

Options B, C, and D are incorrect. Interferon beta shows antiviral activity.

Solution for Question 5:

Correct Option A - Production of monoclonal antibodies:

- Image depicting the process shown, which is the production of monoclonal antibodies
- Monoclonal antibodies are antibodies that are produced against an antigen with single epitope that stimulates only single clone of B cells, known as hybridoma cells.

The image likely represents the steps involved in the hybridoma technology, which is a widely used method for generating monoclonal antibodies. This process typically includes the following

steps:

Incorrect Option:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 6:

Correct Option A - Seminiferous tubule:

- Immunological privilege refers to the ability of certain anatomical sites or tissues to exhibit reduced immune surveillance and regulation. These privileged sites are characterized by the presence of specialized mechanisms that protect them from immune-mediated damage or inflammation.
- The seminiferous tubules, which are structures found in the testes, are considered immunologically privileged sites. They play a crucial role in the production of sperm cells. The seminiferous tubules are protected from immune responses because they are enclosed within the blood-testis barrier, a specialized structure formed by tight junctions between the cells lining the tubules.

Examples of immunologically privileged sites include:

Incorrect Options:

Options B, C, and D are incorrect. They are not immunologically privileged sites

Solution for Question 7:

Correct Option B - C5-C9 deficiency:

- Neisseria meningitidis is a bacterium that can cause meningococcal meningitis. The complement system plays an important role in the immune response against N. meningitidis.
- Deficiency of the complement components C5-C9, also known as the membrane attack complex (MAC), impairs the ability of the immune system to form the MAC and lyse bacterial

cells and increases the susceptibility to infections caused by this bacterium.

Incorrect Options:

Option A - C1-C4 deficiency: Deficiencies in C1-C4 components of the complement system are associated with an increased susceptibility to immune complex-mediated diseases like systemic lupus eryhtematosus,and glomerulonephritis.

Option C - C3 deficiency: C3 deficiency is associated with an increased susceptibility to recurrent pyogenic infections.

Option D - C2 deficiency: C2 deficiency is associated with an increased risk of systemic lupus erythematosus and immune complex diseases.

Solution for Question 8:

Correct Option D - Ag - Ab complex:

- The first step is the binding of C 1 to the antigen-antibody complex.
- The recognition unit of C1 is C1q, which reacts with the Fc piece of bound IgM or IgG.
- C1q has six combining sites. Effective activation occurs only when C 1 q is attached to immunoglobulins by at least two of its binding sites.
- One molecule of IgM or two molecules of IgG can, therefore, initiate the process.

Incorrect Options:

Option A - C1: C1 is the first component of the classical complement pathway and is activated by binding to antigen-antibody complexes.

Option B - C3 convertase: C3 convertase is an enzyme complex formed during the activation of the complement system but does not specifically activate the classical pathway.

Option C - IgA: Immunoglobulin A (IgA) is an antibody isotype involved in mucosal immunity and is not directly involved in activating the classical complement pathway.

Solution for Question 9:

Correct Option B - Somatic hypermutation:

- Affinity maturation is a process that occurs during the immune response, specifically in the development of B cells. It is responsible for improving the binding affinity of antibodies to their target antigens. During affinity maturation, the B cells undergo somatic hypermutation.
- Somatic hypermutation is a genetic process in which the variable region genes of the antibody undergo random mutations. These mutations introduce changes in the amino acid sequence of the antibody's antigen-binding site. The mutated B cells then produce a variety of antibody variants with slightly different binding affinities to the antigen.

Incorrect Options:

Option A - Isotype Switching: It is a process in which B cells change the class of antibody they produce (e.g., switching from IgM to IgG).

Option C - mRNA Splicing: It is a process involved in the removal of introns and joining of exons during the production of mature mRNA, but it is not directly related to affinity maturation.

Option D - VDJ Rearrangement: It is a process in which gene segments for the variable region of antibodies are rearranged to generate antibody diversity.

Solution for Question 10:

Correct Option C - C3:

- C3 is a major component in the activation of the complement system via the alternative pathway. The alternative pathway is an innate immune response that can be activated independently of antibodies. C3 undergoes spontaneous hydrolysis, resulting in the production of C3b, which initiates the alternative pathway.

Incorrect Options:

Options A, B, and D are incorrect. These are not involved in the activation of the alternate pathway.

Solution for Question 11:

Correct Option C - IgE:

- IgE is the immunoglobulin that plays a central role in allergic reactions. It is responsible for triggering the release of histamine and other inflammatory substances in response to an allergen. Elevated levels of IgE are commonly seen in individuals with chronic allergies and are associated with conditions like allergic rhinitis, asthma, and atopic dermatitis.

Incorrect Option:

Option A - IgA: IgA is primarily involved in local or mucosal immunityand provides protection against pathogens by cross-linking bigger antigens with multiple epitopes and preventing entry through mucosal surface.

Option B - IgM: IgM is the first antibody produced in response to an infection and is the most potent activator of complement pathway.

Option D - IgG: IgG provides immunity to the fetus and newborn, mediates precipitation and neutralization reactions, and activates complement pathway.

Solution for Question 12:

Correct Option D - Ag-Ab complex:

- The classical pathway is one of the three pathways that activate the complement system, an important part of the immune response. The classical pathway is primarily activated by the formation of antigen-antibody (Ag-Ab) complexes.
- The first step is the binding of C1 to the antigen-antibody complex. The recognition unit of C1 is C1 q, which reacts with the Fc piece of bound lgM or IgG, leading to activation of the classical complement pathway.

Incorrect Options:

Option A - C1: C1 is a component of the complement system and is involved in the activation of the classical pathway. However, C1 is not the specific activator of the classical pathway; it requires binding to Ag-Ab complexes to initiate the pathway.

Option B - C3 convertase: C3 convertase is an enzyme involved in the cleavage of complement component C3 into its active forms. While C3 convertase is an important component in the complement system, it is not the activator of the classical pathway.

Option C - IgA: Classical complement pathway is activated by the IgG and IgM class of antibody, not IgA.

Solution for Question 13:

Correct Option A - Real-time PCR:

- The most appropriate type of PCR in this scenario is Real-time PCR.
- As Real-time PCR is used for - Quantitative: Real time -PCR can quantify the DNA or RNA present in the specimen. Takes less time: In real time PCR, the amplification can be visualized simultaneously during the process of amplification, unlike conventional PCR, where there is an extra step of gel electrophoresis to detect the amplicons. The contamination rate is extremely low. The sensitivity and specificity of real timePCR assays are much more than those of conventional PCR.
- Quantitative: Real time -PCR can quantify the DNA or RNA present in the specimen.
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- Quantitative: Real time -PCR can quantify the DNA or RNA present in the specimen.
- Takes less time: In real time PCR, the amplification can be visualized simultaneously during the process of amplification, unlike conventional PCR, where there is an extra step of gel electrophoresis to detect the amplicons.
- The contamination rate is extremely low.
- The sensitivity and specificity of real timePCR assays are much more than those of conventional PCR.

Incorrect Options:

Option B - Nested PCR: Nested PCR is a modification of conventional PCR where two rounds of PCR amplification are carried out by using two primers that are targeted against two different DNA sequences of same organism. It is more specific and sensitive than conventional PCR but, risk of contamination of the PCR tubes is high and it is not a quantitative test.

Option C - Reverse Transcriptase PCR (RT-PCR): RT-PCR is a variation of PCR that is used to amplify RNA instead of DNA. It involves the conversion of RNA to complementary DNA (cDNA) using the enzyme reverse transcriptase. It is also not a quantitative test, and like conventional PCR false positive and false negative are high.

Option D - Multiplex PCR: Multiplex PCR is a technique that allows the amplification of multiple target DNA sequences simultaneously in a single reaction. It involves using multiple sets of primers, each specific to a different target sequence. It is also not a quantitative test.

Solution for Question 14:

Correct Option B - II:

- The given image is of Sandwich ELISA, which detects antigens in a patient's sample by sandwiching the antigen between two antibodies
- Steps of sandwich ELISA: Step 1: The well is pre-coated with the capture antibody targeted against the test antigen. Step 2: The test serum (antigen) is added to the wells. Ag gets attached to the captured antibody and coated on the well. Step 3: After washing, an enzyme-labeled primary detector antibody specific to the antigen is added. The detector antibody can be the same as the capture antibody. Step 4: After washing, a substrate–chromogen is added, and color is developed.
- Step 1: The well is pre-coated with the capture antibody targeted against the test antigen.
- Step 2: The test serum (antigen) is added to the wells. Ag gets attached to the captured antibody and coated on the well.
- Step 3: After washing, an enzyme-labeled primary detector antibody specific to the antigen is added. The detector antibody can be the same as the capture antibody.
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- Step 2: The test serum (antigen) is added to the wells. Ag gets attached to the captured antibody and coated on the well.

• Step 3: After washing, an enzyme-labeled primary detector antibody specific to the antigen is added. The detector antibody can be the same as the capture antibody.

• Step 4: After washing, a substrate–chromogen is added, and color is developed.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 15:

Correct Option C - Prozone Phenomenon:

• According to the Marrack's Lattice hypothesis explains the concept agglutination test.

• According to this hypothesis there are zones: Zone of Equivalence: Antigen is equal to the antibody, so Ag-Ab reaction is maximum giving positive agglutination test. Prozone phenomenon: Antibodies are very high compared to antigen, inhibits Ag-Ab reaction giving false negative test. Postzone phenomenon: Antigen are very high compared to antibodies, Ag-Ab reaction fail to occur giving false negative test.

• Zone of Equivalence: Antigen is equal to the antibody, so Ag-Ab reaction is maximum giving positive agglutination test.

• Prozone phenomenon: Antibodies are very high compared to antigen, inhibits Ag-Ab reaction giving false negative test.

• Postzone phenomenon: Antigen are very high compared to antibodies, Ag-Ab reaction fail to occur giving false negative test.

• Diluting the serum sample reduces the antibody concentration, allowing for proper antigen-antibody interactions and the resulting positive test.

• Zone of Equivalence: Antigen is equal to the antibody, so Ag-Ab reaction is maximum giving positive agglutination test.

• Prozone phenomenon: Antibodies are very high compared to antigen, inhibits Ag-Ab reaction giving false negative test.

• Postzone phenomenon: Antigen are very high compared to antibodies, Ag-Ab reaction fail to occur giving false negative test.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 16:

Correct Option B - Streptococcus Pyogenes:

- Streptococcus pyogenes, it produces a group of superantigens known as streptococcal pyrogenic exotoxins (SPEs).
- Superantigens are a class of antigens that binds to the lateral Vβ part of toll-like receptors inducing activation of a large number of T-cells.
- Other examples of superantigens are: Staphylococcal antigen/ Toxic Shock Syndrome Toxin (TSST)/ Enterotoxin F. Streptococcal toxin (SPE- A&C). Yersinia enterocolitica. Yersinia pseudotuberculosis. M.tuberculosis EBV CMV Rabies Malassezia furfur
- Staphylococcal antigen/ Toxic Shock Syndrome Toxin (TSST)/ Enterotoxin F.
- Streptococcal toxin (SPE- A&C).
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- Streptococcal toxin (SPE- A&C).
- Yersinia enterocolitica.
- Yersinia pseudotuberculosis.
- EBV
- CMV
- Rabies
- Malassezia furfur

Incorrect Options:

Option A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 17:

Correct Option B - C5-C9 Deficiency:

- Neisseria meningitidis is a bacterium that can cause meningococcal meningitis. The complement system plays an important role in the immune response against N. meningitidis. Specifically, the deficiency of the terminal complement components C5-C9 increases the susceptibility to infections caused by this bacterium.
- Deficiency of the complement components C5-C9, also known as the membrane attack complex (MAC), impairs the ability of the immune system to form the MAC and lyse bacterial cells.

Incorrect Options:

Option A - C1, C2, and C4 Deficiency: Deficiency of the complement components C1, C2, and C4 is associated with an increased susceptibility to systemic lupus erythematosus.

Option C - C3 Deficiency: C3 deficiency is a common complement deficiency associated with glomerulonephritis and pyogenic infections.

Option D - C1 Esterase Deficiency: C1 esterase deficiency leads to overactivity of the classical pathway leading to Hereditary angioneurotic edema.

Solution for Question 18:

Correct Options B - Somatic Hypermutation:

- Affinity maturation is a process that occurs during the immune response, specifically in the development of B cells. It is responsible for improving the binding affinity of antibodies to their target antigens. During affinity maturation, the B cells undergo somatic hypermutation.
- Somatic hypermutation is a genetic process in which the variable region genes of the antibody undergo random mutations. These mutations introduce changes in the amino acid sequence of the antibody's antigen-binding site. The mutated B cells then produce a variety

of antibody variants with slightly different binding affinities to the antigen.

• Through subsequent rounds of selection, B cells with higher affinity antibodies are preferentially chosen for survival and proliferation. This process leads to the production of antibodies with improved binding specificity and higher affinity for the target antigen.

Incorrect Options:

Option A - Isotype Switching: It is a process in which B cells change the class of antibody they produce (e.g., switching from IgM to IgG).

Option C - mRNA Splicing: It is a process involved in the removal of introns and joining of exons during the production of mature mRNA, but it is not directly related to affinity maturation.

Option D - VDJ Rearrangement: It is a process in which gene segments for the variable region of antibodies are rearranged to generate antibody diversity.

Solution for Question 19:

Correct Option C- IgE-mediated reaction:

• IgE-mediated hypersensitivity reactions, also known as Type I hypersensitivity reactions, occur rapidly (within minutes to hours) after exposure to an allergen.

• In this scenario, the bee venom acts as the allergen, triggering the release of histamine and other mediators from mast cells and basophils.

• This results in vasodilation, increased vascular permeability, smooth muscle contraction (leading to bronchoconstriction), and ultimately shock and respiratory failure.

• The clinical features described, including shock, respiratory failure, and vascular collapse, are classic manifestations of an IgE-mediated anaphylactic reaction.

Incorrect Options:

Option A- IgG-mediated reaction: IgG-mediated reactions are also delayed and involve the formation of immune complexes that can lead to complement activation and tissue damage. They are associated with conditions like serum sickness or certain autoimmune diseases.

Option B- IgA-mediated hypersensitivity: IgA anaphylactic reactions are rare. These reactions occur in response to blood transfusions in individuals who are IgA-deficient.

Option D- T cell-mediated response: This type of reaction is typically delayed (occurring over 24-48 hours) and is mediated by cytotoxic T cells. It is associated with conditions like contact

dermatitis and drug-induced hypersensitivity reactions.

Solution for Question 20:

Correct Option D- Adenosine deaminase:

- Adenosine deaminase (ADA) deficiency is the underlying cause of adenosine deaminase-deficient severe combined immunodeficiency.
- This condition results in the accumulation of toxic metabolites, particularly deoxyadenosine, which leads to lymphocyte depletion and impairs both T-cell and B-cell function.
- Both cell-mediated and humoral immunity are impaired.
- The clinical presentation of SCID typically includes severe infections, chronic diarrhea, and failure to thrive.
- ADA-SCID is inherited in an autosomal recessive manner, often presenting in infancy with life-threatening infections due to severe immunodeficiency.

Incorrect Options:

Option A- Phosphomannose isomerase: Phosphomannose isomerase deficiency is a rare disorder known as CDG1b (congenital disorder of glycosylation type 1b). It affects glycoprotein synthesis and can result in various symptoms, including developmental delays, liver dysfunction, and coagulation abnormalities.

Option B- Ornithine transcarbamylase: Ornithine transcarbamylase deficiency is associated with a urea cycle disorder. It leads to hyperammonemia, metabolic acidosis, and neurologic symptoms.

Option C- Hypoxanthine-guanine phosphoribosyltransferase: HGPRT deficiency is responsible for Lesch-Nyhan syndrome, a disorder characterized by neurologic abnormalities, self-injurious behavior, and uric acid overproduction.

Solution for Question 21:

Correct Option B - Immunochromatography:

The image showing HbsAg rapid card test, the test kit works on the principle of immunochromatography.

- Principle: Immunochromatography uses a liquid sample to carry the target substance through the components of a test strip. The test strip contains a capture antibody that is immobilized on the nitrocellulose membrane, as well as a labeled antibody that typically contains colored particles. This labeled antibody specifically binds to the target substance, and when it interacts with it, produces a visible line or color change that indicates a positive result.

Incorrect Options:

Option A - ELISA (enzyme-linked immunosorbent assay): ELISA is a different immunoassay technique that relies on the interaction between enzymes and their substrates to produce a detectable signal.

Option C - Immunofluorescence: Immunofluorescence is a technique that uses fluorescently labeled antibodies to detect the presence of antigens or antibodies.

Option D - Chemiluminescence: Chemiluminescence is an assay method that utilizes a chemical reaction to generate light, which is then detected to determine the presence of the target substance. It typically involves the use of enzymes and substrates that produce a chemiluminescent signal upon interaction.

Solution for Question 22:

Correct Option A - IgG:

- The decreasing order of antibody concentration in the body is IgG > IgA > IgM > IgD > IgE.
- The serum concentration of different antibodies are: IgG – 12 mg/ml IgA – 2 mg/ml IgM – 1.2 mg/ml IgD – 0.03 mg/dl IgE – 0.00004
- IgG – 12 mg/ml
- IgA – 2 mg/ml
- IgM – 1.2 mg/ml
- IgD – 0.03 mg/dl
- IgE – 0.00004

- IgG – 12 mg/ml
- IgA – 2 mg/ml
- IgM – 1.2 mg/ml
- IgD – 0.03 mg/dl
- IgE – 0.00004

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 23:

Correct Option A - Cytotoxic T Cells:

- HLA genes 1 produce MHC class I molecules, which are present on the surface of all nucleated cells and platelets. The cytotoxic T cells (CD8) recognizes MHC class I antigens.

Incorrect Options:

Option B - B Cells: B cells are a type of lymphocyte that is responsible for humoral immunity. These white blood cells produce antibodies, which play an important part in immunity.

Option C - NK Cells: Natural killer cells are cytolytic cells involved for virally transformed target cells, certain tumor lines and are involved in allograft rejection. NK cells are present in the spleen and peripheral blood. Action of NK cells is not dependent on antibody.

Option D - Helper T-Cells: Helper T-cells/CD 4 cells recognize MHC II molecules formed by HLA gene II. They help in activation of cytotoxic T-cells, or they may activate B-cells, which produce antibodies to fight for the infection.

Solution for Question 24:

Correct Option B - 3 : 1:

• The percentage of T cells in a normal individual is about 60 to 70% as compared to B cells, with a percentage of 20 to 30%. T cells outnumber the B cells at least 3 to 5 times. Hence, only option B shows more T cells than B cells, so the best option is 3:1.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 25:

Correct Option C - 1, 3, and 4:

• Vaccination against Varicella zoster virus is recommended as latent VZV infection can reactivate as herpes zoster or shingles. As age advances, cell-mediated immune responses (CMI) to VZV decline, with resultant increased incidence of reactivation as herpes zoster among older people.

• The Influenza vaccine is highly recommended in the elderly to prevent severe disease.

• Pneumococcal polysaccharide vaccine (PPV) is given to people aged 65 and over. It helps prevent meningitis and other illnesses caused by 23 strains of pneumococcal bacteria

Incorrect Options:

Options A, B and D are incorrect.

Solution for Question 26:

Correct Option B - B Lymphocytes:

Components of innate immunity	Components of Acquired Immunity

Anatomical barriers such as skin and mucosa Physiological barriers (e.g., body temperature). Phagocytes (neutrophils, macrophages, and monocytes). Other classes of lymphocytes: Gamma delta T cells, NK-T cells β-1 cells and marginal-zone B cells Mast cells, macrophages, and dendritic cells Complement pathways – alternative and mannose-binding pathways Fever and inflammatory responses Normal resident flora Cytokines: TNF-α, interleukins (IL-1, IL-6, IL-8, IL-12, IL-16, IL-18), IFN-α, β and TGF-β Acute phase reactant proteins (APRs). Lectin pathway	T cell B cell Classical complement pathway Cytokines (IL-2, IL-4, IL-5, IFN-gamma)
Abbreviations: TNF, tumor necrosis factor; TGF-b, transforming growth factor-b; IFN, interferon; IL, interleukin	

Incorrect Options: Options A, C, and D are incorrect. These are components of innate immunity.

Solution for Question 27:

Correct Option C - Complement Deficiency:

Immune deficiency	Diseases	Vaccine contraindication	
Primary immune deficiencies	B lymphocyte (humoral)	Severe antibody deficiencies (X-linked agammaglobulinemia, variable immune deficiency)	OPV, live influenza (LAIV), live bacteria (BCG, typhoid fever), MMR, varicella and rotavirus
Mild antibody deficiency (selective IgA deficiency, Ig subgroup deficiency)	OPV, BCG. All other live vaccines can be administered.		
T lymphocyte (cellular)	Complete deficiency (severe combined immune deficiency, complete DiGeorge syndrome)	All live vaccines	
Partial deficiencies (partial Di George syndrome,	All live vaccines		

ataxia-telangiectasia,Wiskott-Aldrich syndrome)			
Complement deficiency	Persistent complement deficiency, C5–C9, C3, properdin, factor B deficiency	None	
Phagocyte dysfunction	Chronic granulomatous disease, leukocyte adhesion defect, myeloperoxidase deficiency	Live bacterial vaccines (BCG, typhoid fever)	
Secondary immune deficiencies	Asplenia/functional asplenia	None	
Chronic renal diseases	LAIV		
Cancer, organ transplantation, autoimmune disease, immunosuppressive treatment, radiotherapy	Depending on the individual's immune status, all live viral/bacterial vaccines.		
HIV/AIDS	OPV, BCG, MMRV, LAIV, if severe immunosuppression is present, MMR and varicella vaccines are not administered		
OPV: oral polio vaccine; LAIV: live attenuated influenza vaccine; HIV: human immunodeficiency virus; AIDS: acquired immunodeficiency syndrome; BCG: live bacterial vaccines; Hib: hemophilus influenza type B; MMR: measles-mumps-rubella;			

Incorrect Options:

Option A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 28:

Correct Option B - Absent B cells:

- B cells cannot respond to T-dependent antigens without a co-stimulatory signal from TH cells. These antigens bind to B cells and are presented to T cells.
- Once TH cells recognize these complexes, they secrete cytokines and start expressing the CD40 ligand, which interacts with the CD40 receptor on B cells, providing the stimulus for B cell activation.
- CD 40 signaling is essential for B cell activation, maturation, and production of antibodies. In the absence of CD40, B cells would not receive the necessary signals for proper activation and differentiation. This would result in impaired B cell function and reduced production of antibodies.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 29:

Correct Option B - Pyrosequencing:

- Pyrosequencing is a DNA sequencing technique that uses light generation to detect nucleotide incorporation. The reaction mixture includes a single-stranded DNA template, sequencing primer, and several enzymes like DNA polymerase, ATP sulfurylase, luciferase, and apyrase. Two substrates, APS (adenosine 5‘ phosphosulfate) and luciferin, are also added.
- DNA polymerase adds deoxynucleotides to the DNA strand, forming a phosphodiester bond and releasing pyrophosphate. The enzyme ATP sulfurylase turns PPi into ATP, which is then used by luciferase to convert luciferin into oxyluciferin, emitting light proportional to the amount of ATP used/ number of nucleotides incorporated.

Incorrect Options

Option A - Flow cytometry: It is commonly used for cell counting, cell sorting, and assessing cell characteristics such as size, granularity, and surface markers.

Option C - FISH (Fluorescence in situ hybridization): FISH is a molecular cytogenetic technique used to detect and visualize specific DNA sequences on chromosomes or within cells for detecting chromosomal abnormalities .

Option D - RFLP analysis (Restriction Fragment Length Polymorphism): RFLP analysis is a technique used to detect variations in DNA sequences by digesting DNA samples with specific restriction enzymes. RFLP analysis can provide information about genetic variations.

Solution for Question 30:

Correct Options A - IgA:

- The image given is of dimeric IgA aka secretory IgA.
- IgA (Immunoglobulin A) is a class of antibody that plays a critical role in mucosal immunity, protecting the mucous membranes in various parts of the body. It exists in two main forms: Monomeric IgAr/Serum IgA Dimeric IgA/Secretory IgA.
- Monomeric IgAr/Serum IgA
- Dimeric IgA/Secretory IgA.
- Monomeric IgA: Each subunit of IgA monomer consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.
- Each subunit of IgA monomer consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.
- Dimeric IgA: IgA monomers can further polymerize to form IgA dimers. The joining of two IgA monomers occurs through the J chain, a polypeptide that links the monomers together. Additionally, a secretory component (also known as the polymeric immunoglobulin receptor or pIgR) is associated with the IgA dimer.
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- Monomeric IgAr/Serum IgA
- Dimeric IgA/Secretory IgA.
- Each subunit of IgA monomer consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.
- IgA monomers can further polymerize to form IgA dimers.

• The joining of two IgA monomers occurs through the J chain, a polypeptide that links the monomers together.

• Additionally, a secretory component (also known as the polymeric immunoglobulin receptor or pIgR) is associated with the IgA dimer.

Incorrect Options:

Option B - IgG: Each subunit of IgG consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.

Option C - IgM: IgM can be monomeric (when present as a membrane-bound antibody on B cells) or pentameric (secreted) in nature, i.e., five IgM monomeric units are joined with each other by a J chain to form a complete IgM pentamer having 10 Fab regions and 10 valencies.

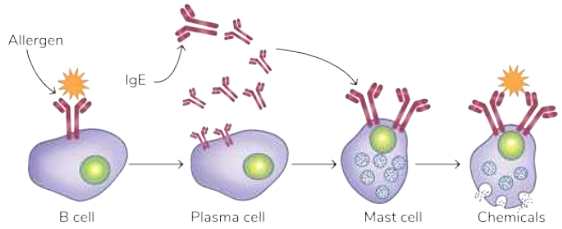
Option D - IgE: Each subunit of IgE consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.

Solution for Question 31:

Correct Option B - IgE-Mediated Reaction:

• Anaphylaxis is a Type 1 hypersensitivity reaction/IgE-mediated hypersensitivity reaction.

• In response to the allergic food, the immune system produces large amounts of IgE antibodies that bind to mast cells and basophils. This triggers the release of histamine and other inflammatory mediators that cause the symptoms of anaphylaxis.



• This reaction occurs in 2 phases: Sensitization Phase Effector Phase

• Sensitization Phase

• Effector Phase

• Sensitization Phase: Allergen is processed by APC & peptides of it are presented to CD4 T-cells → activates TH cells → IL4 is released → activates B cells → IgE production → homocytotropism (movement towards the mast cells) and sensitize the mast cells.

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• Effector Phase: Same allergen presented second time → Mast cell degranulation → Primary and secondary mediators.

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• Sensitization Phase

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• Same allergen presented second time → Mast cell degranulation → Primary and secondary mediators.

Incorrect Options:

Option A - T Cell-Mediated Cytotoxicity: It is a type 4 hypersensitivity reaction, for example, Mantoux test and multiple sclerosis.

Option C - IgG-Mediated Reaction: IgG-mediated reactions are seen in type 2 hypersensitivity reactions, for example, Rh incompatibility and blood transfusion reactions.

Option D - Antigen-Antibody Complex Mediated: It is a type 3 hypersensitivity reaction, for example, Systemic lupus erythematosus and Henoch schonlein purpura.

Solution for Question 32:

Correct Option A - Cytotoxic T cell:

• Type 4 hypersensitivity reaction aka delayed type of hypersensitivity reaction is cell-mediated, and the characteristic cells called TDTH cells (delayed type of hypersensitivity T cells) are the principal mediators of type IV reactions.

• The antigen-presenting cells process and present the antigenic peptides along with MHC-II to the helper T cells. TH cells are differentiated to form TDTH cells.

• Most TDTH cells are derived from TH1 cells, but occasionally, other T cells, such as CD8+ T cells and CD4+ TH17, can also act as TDTH cells.

• The TDTH cells, on subsequent contact with the antigen, secrete a variety of cytokines that attract and recruit various inflammatory cells.

Incorrect Options:

Option B - Macrophage: In type 4 hypersensitivity reaction, macrophages are activated as an endpoint of the reaction but are mediated through T cell activation.

Option C - Dendritic cells: They are the macrophages found in the epidermis and act as APCs. They have large dendritic projections used for capturing various pathogens.

Option D - Neutrophils: They are primary immune cells and act as first-response cells during acute inflammation. They are not involved in hypersensitivity reactions.

Solution for Question 33:

Correct Option C - Hyper IgE syndrome:

• Hyper-IgE syndrome aka Job's syndrome is a rare primary immunodeficiency disease characterized by eczema, recurrent staphylococcal skin abscesses, recurrent lung infections (pneumatocele), eosinophilia, and high serum levels of IgE.

• The underlying mechanism is a defect in neutrophil chemotaxis due to mutations in either STAT3 or DOCK8 genes.

Incorrect Options:

Option A - Hyper IgM syndrome: Hyper-IgM syndrome is an X-linked disorder characterized by increased recurrent sinopulmonary infections and diarrhea (reduced IgG), autoimmune hemolytic anemia, thrombocytopenia, or neutropenia due to Increased IgM levels.

Option B - Wiskott-Aldrich syndrome: Wiskott-Aldrich syndrome is a rare X-linked immunodeficiency disorder characterized by eczema, recurrent infections, and thrombocytopenia (low platelet count).

Option D - Carcinoid syndrome: Carcinoid syndrome is a set of symptoms caused by carcinoid tumors, which are neuroendocrine tumors that produce excessive serotonin. The symptoms include flushing, diarrhea, bronchoconstriction, and cardiac abnormalities.

Solution for Question 34:

Correct option C - MHC Class II presents endogenous antigens:

Differences between MHC Class I and MHC Class II Molecules		
	MHC Class I	MHC Class II
Present on	All nucleated cells (except sperms) and platelets	Antigen-presenting cells (APCs)
Peptide antigen is	Presented to CD8 T cells	Presented to CD4 T cells
Nature of peptide antigen	Endogenous or intracellular (viral/tumor antigen)	Exogenous
Peptide antigen (size)	8-10 amino acid long	13-18 amino acid long
Antigen presentation pathways	Cytosolic pathway	Endocytic pathway
Peptide-binding site	alpha1/alpha2 groove	alpha1/beta1 groove
CD4 or CD 8 binding site	alpha 3 binds to CD8 molecules on Tc cells	beta 2 binds to CD4 on TH cells

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 35:

Correct Option D - Hemagglutination Test:

- The hemagglutination test is a technique used to detect certain viral or bacterial infections by observing the agglutination (clumping) of red blood cells (RBCs). In this test, the antigen of interest (usually a virus or bacteria) is mixed with RBCs (containing antibodies). If the antigen interacts with the antibodies in RBCs, it can cause them to clump together or

agglutinate.

• The hemagglutination test does not involve the use of labeled antibodies. Instead, it relies on the interaction between the antigen and the RBCs.

• Examples of hemagglutination test are: Paul bunnel test (for EBV), Coomb's test, and cold agglutination test for mycoplasma.

Incorrect Options:

Options A, B, and D are incorrect. These tests require labeled antibody to detect the presence of antigens.

Solution for Question 36:

Correct Option B - IgG2:

• IgG2 deficiency is associated with an increased susceptibility to infections caused by encapsulated bacteria. IgG2 plays a crucial role in opsonization, which is the process of tagging pathogens for destruction by phagocytes. It facilitates the phagocytosis of encapsulated bacteria by immune cells, leading to their elimination.

Incorrect Options:

Option A - IgG1: IgG1 is important for opsonization and immune response against protein antigens. While a deficiency in IgG1 can increase susceptibility to certain infections, it is not specifically associated with an increased susceptibility to encapsulated bacterial infections.

Option C - IgG3: IgG3 is an important antibody subclass involved in complement activation and immune response against certain bacterial pathogens.

Option D - IgG4: IgG4 is involved in immune tolerance and is associated with allergic responses and autoimmune conditions.

Solution for Question 37:

Correct Option A - Influenza:

- Influenza is caused by influenza viruses, which have the ability to undergo frequent genetic changes through antigenic drift (seen in influenza A & B, point mutation occurs) and antigenic shift (seen in Influenza A, genetic reassotment occurs) .
- This means that the viral strains circulating in the population can change over time, making the previously developed vaccines less effective against the new strains. Therefore, annual updates and strain changes are required in the influenza vaccine to ensure its effectiveness against the evolving strains.

Incorrect Options:

Option B - Diphtheria: The diphtheria vaccine is based on the toxoid produced by Corynebacterium diphtheriae and does not require frequent strain changes.

Option C - Polio: The polio vaccine targets the poliovirus, and while there are different serotypes of the virus, the vaccine does not require frequent strain changes as the available vaccines offer protection against all known serotypes.

Option D - Typhoid: The typhoid vaccine is based on specific strains of Salmonella typhi bacteria and does not require frequent strain changes.

Solution for Question 38:

Correct Options A - IgA:

- The image given is of dimeric IgA, aka secretory IgA.
- IgA (Immunoglobulin A) is a class of antibody that plays a critical role in mucosal immunity, protecting the mucous membranes in various parts of the body. It exists in two main forms: Monomeric IgA /Serum IgA Dimeric IgA/Secretory IgA
- Monomeric IgA /Serum IgA
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• Monomeric IgA /Serum IgA

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• Additionally, a secretory component (also known as the polymeric immunoglobulin receptor or pIgR) is associated with the IgA dimer.

Incorrect Options:

Option B - IgG: Each subunit of IgG consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.

Option C - IgM: IgM can be monomeric (when present as a membrane-bound antibody on B cells) or pentameric (secreted) in nature, i.e., five IgM monomeric units are joined with each other by a J chain to form a complete IgM pentamer having 10 Fab regions and 10 valencies.

Option D - IgE: Each subunit of IgE consists of two heavy chains (alpha chains) and two light chains (kappa or lambda chains) connected by disulfide bonds.

Solution for Question 39:

Correct Option A - Cell mediated response:

• Type IV hypersensitivity, also known as delayed-type hypersensitivity or cell-mediated hypersensitivity, is primarily mediated by TDTH cells (delayed type of hypersensitivity T cells) formed from the differentiation of TH cells and CD8+ cells.

• It is a form of immune response that occurs when sensitized T cells encounter an antigen and release cytokines, leading to an inflammatory reaction. This type of hypersensitivity is characterized by a delayed onset, typically taking 24 to 72 hours to develop.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 40:

Correct Option B - T Cell:

• T cells are another type of lymphocyte that are essential for cell-mediated immunity.

• They play a central role in recognizing and eliminating foreign antigens.

• Nude mice(lack body hair), which lack functional T cells due to absence of thymus, are unable to mount an effective cell-mediated immune response.

• Therefore, the absence of T cells is the primary reason for the lack of resistance to xenografts in nude mice.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.